AGENTIC SYSTEMS FRAMEWORK

VOLUME 1

ADAPTATION & ACTUATION THEORY

JOSEPH A WECKER

VIS VERITATIS

AAT: Adaptation & Actuation Theory

Joseph A. Wecker

May 17, 2026

v0.2.0 • feccd0d • 2026-05-18

Contents

Inescapable Foundations of Agency

We assert that an agent — a system that maintains an internal picture of an external world it can only ever see in part, acts on that world, and must continually adapt to external realities — is not an open-ended mystery to be approached only by analogy and benchmark. It is a temporal feedback system characterized by a small number of critical quantities and constraints. The relationships among them are sharp enough (many forced by derivation) to serve as foundational theorems: first of adaptation in general, and then of adaptation under actuation and purpose. These theorems are not uniform across all agents, and each narrowing of scope buys a strictly sharper set as we ascend to higher orders of agency and intelligence. These bounds describe not only how a system behaves while its conditions hold, but — sharply, and as an exact dichotomy in the stochastic case — what happens when they fail.

Four of those results anchor this volume:

1. A **Persistence Threshold**: A single inequality, *correction rate against the rate the world drifts, relative to how wrong the agent can afford to be*, below which an adaptive system does not merely degrade — it loses bounded behavior, the way a balance held just under its tipping point is a different object from one held well above it. The same inequality, with its terms read differently, governs a Kalman filter tracking a maneuvering target, a software team holding a codebase maintainable, and an organization keeping pace with a market; that it is one inequality and not an analogy among three is itself the result.

2. A **Containment Dichotomy**: Against a bounded disturbance a sector-stable corrector holds its error inside a fixed region forever and with certainty, but against genuinely stochastic disturbance no correction strength buys that — over an unbounded horizon the system *will* leave any fixed region, with probability one, so for any sufficiently long-lived agent the need to change *what kind of model it is*, not merely its parameters, is not an edge case but a certain eventual event.

3. A **Stability Equivalence**: the principle that an adaptive system is one whose correction outpaces its target's drift is not a guiding description but a theorem — it is, identically, the classical Lyapunov stability theorem. Establishing that identity is what puts the framework's organizing principle on the footing of a proven result, and it is what lets every later structural question be posed as a question about one object rather than several loosely related ones.

4. An **Architectural Classification**: whether an agent's beliefs can be updated without contamination from what it wants is a *necessary* structural property whose distinct extreme modes critically determine its ability to maintain coherence. It holds by construction for some architectures, fails by construction for others (transformer language models or LLMs among them). Where it fails, nevertheless, there is a constructive route to recover it at a quantified cost.

We build these results from classical machinery and use it as such — Lyapunov stability, Itô calculus, Pearl's interventional hierarchy, the information bottleneck, monotone-operator theory, Čencov's uniqueness theorem. None of it is new mathematics, and the volume does not pretend otherwise. What the integration makes provable *is* the contribution: exact and conditional bounds on when an adaptive, purposeful agent can correct, plan, and persist — statements that are true, were not previously stated, and are expressible only over objects the framework had to construct. The discipline that runs through the volume — every claim carrying its scope, its epistemic tier, and the point at which it stops applying

— is not the contribution; it is what makes the contribution compose, and what makes it safe to trust. In a theory whose sharpest results are bounds, the scope condition is not a caveat appended to a theorem. It often *is* the theorem.

A need the field has named, repeatedly.

The case for a systems-level theory of agency — one that treats agent, environment, and other agents as one coupled object rather than scoring a model's capabilities in isolation — has been made from several directions and was put explicitly by Miehling et al. in 2025: the field's habit of measuring isolated capability systematically misjudges both what agents can do and how they fail once they are acting in a world that answers back. That paper is, deliberately, a call rather than a developed theory. This volume reads as a substantive answer to it — and the more telling fact is that the answer was already in hand. The work arrived at the same diagnostic independently, from cross-domain pattern-matching, and had built the formal apparatus before the call was encountered; a third line, information-theoretic measurement of agency, converged on a closely related diagnostic from a third starting point. Independent arrival at the same need, from unrelated directions, is itself the evidence that the need is real.

The architecture of the theory.

The volume is organized as a cascade of scope conditions, and the cascade is the argument, not the table of contents. The broadest scope — an *adaptive system*: observation under residual uncertainty — admits the full machinery of Part I (mismatch, gain, tempo, the persistence threshold). Adding the requirement that the agent's actions carry causal contrast narrows to *agentic systems* and unlocks interventional access — the agent's own action loop becomes a source of the Level-2 causal data that no passive observer can obtain. Adding an explicit objective and strategy distinct from the agent's beliefs narrows to *actuated agents* and the diagnostic core of Part II. Letting the agent set its own objectives, then making its primary channel language, then making its continuity morally weighted, narrows further still — to the language-constituted and language-living agents that the companion volumes treat.

Each narrowing is a restriction with explicit qualifying properties, and each restriction *buys* a strictly stronger set of results. This is the structural point worth holding onto: the conservatism is not timidity. A theory that narrows its scope honestly and earns more machinery at each narrowing is stronger than one that claims everything and then quietly retreats. The cascade is the framework's spine, and it is the subject of a figure the reader should expect to meet early — *the scope cascade and what each narrowing unlocks* — because it does more organizing work than any single result. (The figure is planned; the agent-spectrum diagram is its nearest existing relative and shows only the broadest two axes.)

The cascade has a terminus, and it should be named plainly because the framework's discipline is exactly what licenses naming it. The mathematics is domain-agnostic: the persistence inequality is the same inequality whether the agent that fails it is a thermostat or an entity whose continuity someone has reason to care about. The framework does not smuggle a notion of mattering into its variables. What it does instead is make precise the structural facts that any account of a continuous, self-correcting, other-modeling entity would have to respect — that identity tracks an irreversible trajectory and not a copyable state, so restoring such an entity from a backup is not a neutral operation; that staying coherent against a moving world is a sustained cost paid in information, not a state achieved once; that an objective without a notion of *enough* cannot rest, and an agent that cannot rest is a structural hazard before it is anything else. Stated as physics, these are theorems and scope conditions. The moral weight, where it applies, is a shell around the physics, not a term inside it — and keeping it there is what makes the later volumes' claims defensible rather than mystical.

Where it sits.

The volume joins a long arc of partial unifications and inherits machinery from each: the cybernetic feedback tradition, control theory's stability apparatus, bounded-rationality and decision theory, Pearl's causal hierarchy, Shannon's rate-distortion and the information bottleneck, the free-energy and active-inference program, the coevolving-automata tradition, and recent information-theoretic measures of agency. The relationships are stated precisely rather than rhetorically. Active inference shares substantial machinery and differs on foundation: its standard objective is recoverable here as a special case under three explicit restrictions, which makes visible exactly which architectural commitments separate the two rather than leaving the difference to taste. The relationship to information-theoretic agency measures and to coevolving-automata composition is complementary in each case, with a precise account of what each supplies that the others do not. The framework's distinctiveness is not that its pieces are unprecedented; it is that the synthesis is disciplined enough that results compose across the cascade — which is a property the pieces do not have alone.

How the work developed.

We built the work bottom-up. The observation we kept returning to, across separate threads, was that the same cycle structure recurred across domains that do not usually share a vocabulary — software development, organizational dynamics, biological control, language-constituted agents. The first formalizations were domain-specific; we extracted the adaptation-specific mathematics only afterward. What began as cross-domain pattern-matching produced, once we worked it through with care, structure we had not seen in the informal version: acyclicity of strategy graphs forced by temporal ordering rather than assumed; a Markov property derived from causal sufficiency rather than postulated; the split between *the world does not permit it* and *you are not yet doing it well enough*; the patterns by which the framework's coordinate choices turn out to sit on one geometric object; the composition machinery. None of these were design goals; they fell out of taking the formalism seriously, and the discipline that produced them is the shape the volume now has, not a stance we adopted at the start.

What this volume is, and what it is part of.

This is Volume 1: Adaptation and Actuation Theory, the mathematical core — Part I (adaptive systems under uncertainty), Part II (actuated agents with objectives and strategy), Part III (composition and adversarial dynamics), and the appendices that carry the derivations. Its maturity is uneven by design, and the gradient is stated so it can be relied on selectively: Part I is mathematically closed with simulation validation; Part II has an exact diagnostic core and a maturing operational layer; Part III has its bridge results and open frontiers named as such. Exact core, principled architecture in the middle, open formulation at the edges — that arc is what a theory describing agentic systems should look like, not a defect it should hide.

A single concrete instance recurs across the chapters as a worked anchor — a driver holding a road through a snowstorm, treated not as a metaphor for an adaptive agent but as a literal one, so that ordinary driving intuition transfers as theorems rather than as suggestion. It is introduced where it does work and is not asked to do more than it can.

The volume is also a foundation. The companion volumes carry the framework into agents whose primary channel is language, and then into agents whose persistence is morally weighted — the work the framework was ultimately built to make possible and defensible. Those volumes are not speculative overreach bolted onto a mathematical core; they are the disciplined continuation of the same cascade, and the core's honesty is not ornamental — it is exactly what lets the later claims be made at all. A reader who finishes this volume should be able to say precisely where the mathematics is exact, where it is conditional, where it is open, and what it would take to move each boundary. We take that legibility to be the point. The scope conditions are not the apology; they are the theorems.

Part I

Adaptive Systems Under Uncertainty

Scope: Any system consisting of an agent coupled to an environment through observation and action channels, where the environment is not fully observable. This is the general case — thermostats through commanders.

Chapter 1

The Coupled Loop: Ontology and Scope

1.1 *Definition:* Agent-Environment Coupling

An agent is an entity that receives observations from an environment, maintains internal state, and produces actions that affect the environment. The agent cannot access the environment directly — observations are necessarily lossy. This boundary condition is constitutive: the theory applies where the agent-environment boundary entails information loss.

Formal Expression.

Let Ω denote the **environment**: the totality of state external to the agent. We make no assumptions about Ω's structure — it may be continuous or discrete, stationary or non-stationary, deterministic or stochastic, benign or adversarial.

Definition (agent-environment).

An **agent** is an entity satisfying three conditions:

1. It receives observations from Ω (perception channel)
2. It maintains internal state (memory/model)
3. It produces actions that affect Ω (action channel)

The agent cannot access Ω_t directly. All contact with the environment is mediated through lossy observation. This is not a simplifying assumption — it is a scope condition. Systems where the agent has direct access to full environment state are outside AAT's scope (1.5).

Definition (information-loss-boundary).

Epistemic Status.

This is *definitional* — it establishes the conceptual framework, not a truth-claim. The agent-environment decomposition is a modeling choice that delineates what AAT analyzes. The information-loss boundary is the constitutive commitment: it restricts AAT's scope to systems where the agent faces genuine uncertainty about its environment.

Discussion.

Why information loss is constitutive. An agent with perfect access to Ω_t has no need for a model, no mismatch signal, no adaptation. The entire adaptive machinery of Section I becomes vacuous. The information-loss boundary is what makes the theory non-trivial.

Generality of Ω. The environment is deliberately underspecified. Ω may include other agents, physical systems, software artifacts, or any combination. The only structural commitment is that Ω is external to the agent and not fully accessible.

1.2 *Definition:* Action and Transition

Actions affect the environment through a transition function that is unknown to the agent and possibly stochastic.

Formal Expression.

The **action space** $\mathcal{A}$ is the set of actions available to the agent. Actions affect the environment via the transition function:

Definition (action-transition).

$$\Omega_{t+1} \sim T(\cdot \mid \Omega_t, a_t) \tag{1.1}$$

where: - T is the (possibly stochastic) transition function - Ω_t is the current environment state - $a_t \in \mathcal{A}$ is the agent's chosen action

The agent does not know T exactly.

Definition (transition opacity).

Epistemic Status.

This is *definitional*. The transition function T is a modeling device that captures how agent actions couple back into the environment. The stochasticity of T is allowed but not required — deterministic transitions are the special case where T places all mass on a single successor state. The claim that T is unknown to the agent is constitutive of the uncertainty setting, paralleling the epistemic opacity of h (1.3).

Discussion.

Closing the loop. Together with 1.3, this definition completes the agent-environment coupling: the agent observes via h and acts via T. The loop $\Omega_t \xrightarrow{h} o_t \rightarrow \text{agent} \xrightarrow{a_t} \Omega_{t+1}$ is the fundamental structure that all subsequent claims build on.

Uncertainty about T is what makes action non-trivial. If the agent knew T exactly, action selection would reduce to optimization over a known function. The combination of unknown h and unknown T is what creates the need for adaptive behavior.

Markov-of-Ω as a modeling commitment, not an empirical assumption. The form $\Omega_{t+1} \sim T(\cdot \mid \Omega_t, a_t)$ is implicitly Markov in Ω — only the current Ω_t and a_t appear in the conditioning. Without loss of generality, Ω is taken to be the *sufficient state* for its own evolution under T: any non-Markov environment is absorbed by extending Ω to include enough history to make future-state distribution depend only on current state and action. This is the world-side analog of the Markov-by-completeness move that 3.2 makes for the agent-side state M_t (H Constraint C3). The two are independent — Markov-of-M_t is forced by *defining* M_t *as complete*; Markov-of-Ω is forced by *defining* Ω *as the sufficient state*. Both are modeling commitments about the *breadth* of the named object, not structural assumptions about underlying world dynamics.

1.3 *Definition:* Observation Function

Observations (aisthesis — raw contact with reality) are lossy, possibly noisy functions of environment state, prior action, and perceptual noise. The agent knows neither the observation function nor the noise distribution exactly.

Formal Expression.

The **observation space** $\mathcal{O}$ is the set of possible observations. Each observation is generated by:

Definition (observation-function).

$$o_t = h(\Omega_t, a_{t-1}, \varepsilon_t) \tag{1.2}$$

where: - h is the observation function (unknown to the agent) - Ω_t is the environment state at time t - a_{t-1} is the agent's prior action (observations may depend on what the agent did) - ε_t represents noise or limits of perception

The agent knows neither h nor the distribution of ε_t exactly.

Definition (epistemic opacity).

Epistemic Status.

This is *definitional*. The observation function h is a modeling device that captures the information-loss boundary established in 1.1. The dependence on a_{t-1} is optional — when absent, the function reduces to $h(\Omega_t, \varepsilon_t)$. The claim that the agent does not know h or the noise distribution is constitutive of the partial-observability setting, not an empirical assertion.

Discussion.

Lossiness is the key property. The observation function h maps from Ω_t (high-dimensional, inaccessible) to $\mathcal{O}$ (what the agent actually receives). Information is destroyed in this mapping. This is what forces the agent to maintain a model (2.1) rather than simply reading off the environment state.

Action-dependence. The inclusion of a_{t-1} allows the observation function to capture active perception: what the agent sees may depend on where it looked. This is relevant in software (`#obs-software-epistemic-properties` — cross-component reference, see `02-tst-core/`), where observation quality is partly under agent control.

1.4 *Definition:* Chronica

The interaction history $\mathcal{C}_t$ is the complete, singular causal record of the agent's observations and actions. Everything the agent can ever know must be constructed from this sequence.

Formal Expression.

Definition (chronica).

$$\mathcal{C}_t = (o_1, a_1, o_2, a_2, \dots, a_{t-1}, o_t) \tag{1.3}$$

The ordering is not a notational convenience. It reflects an irreversible physical fact: a_{t-1} was selected before o_t was received. The agent could not have used o_t to select a_{t-1}.

$\mathcal{C}_t$ is monotonically growing — events are added but never removed. It is the agent's *only* raw material for constructing a model (2.1).

Epistemic Status.

This is *definitional*. The chronica names an object that exists by construction in any system satisfying 1.1: the temporal sequence of all agent-environment interactions. The term "chronica" (from Greek χρονικά, "records of time") avoids collision with $\mathcal{H}$ (Shannon entropy) in speech and notation.

Discussion.

The chronica is singular and non-forkable. Because the temporal ordering is irreversible, $\mathcal{C}_t$ represents a unique causal trajectory. Duplicating an agent's state and exposing the copies to different future events creates two agents with divergent chronica, neither of which is a sufficient statistic for the other's trajectory. The chronica is the substrate of *continuity persistence* (see Persistence in `LEXICON.md`) — an agent has continuity persistence when $\mathcal{C}_t$ extends continuously and M_t has temporal depth grounded in it. See 4.7 for the full development of this observation.

Relationship to the model. The model $M_t = \phi(\mathcal{C}_t)$ (2.1) is a compression of the chronica. How much of $\mathcal{C}_t$'s predictive information survives compression is measured by model sufficiency (2.3).

Working Notes.

Open question: TRACTUS / CHRONICA split for logogenic implementations. The PROPRIUM operational architecture (`~/src/firmatum/PROPRIUM-ONTOLOGY-v2.md`, `~/src/firmatum/PROPRIUM-ARCHITECTURE-v2.md`) distinguishes two records of an entity's interaction history:

- **TRACTUS** (the "EEG"): raw, not-necessarily-coherent record of API interactions between the agent's runtime (ANIMA) and its current substrate (LOGOSTRATUM). Implementation-side; carries multi-format records, redundancy from retries, transient error 500 recoveries, API rerouting, rate limiting. Per-substrate format variation. Considered subconscious from the entity's perspective. Plural across substrate-relationships.
- **CHRONICA** (the polished record): the entity's actual record of internal and external events with strict chronology, attestation, and causality enforced. Append-only, hash-chained. Singular per entity.

In Section I, `def-chronica` covers both — the chronica is the singular causal trajectory, abstracted from implementation-side noise. This abstraction is fine for general adaptive-systems theory, where a single observation-action sequence is the natural object.

When part `03-llm-core/` gets into logogenic implementation specifics — and especially the closed-loop / interiority sub-scope (`scope-interiority-loop`) where the substrate-mediation layer (INTERPRES) becomes load-bearing — the TRACTUS/CHRONICA distinction may need first-class treatment.

Joseph's framing on the open question (2026-05-01): *"whether or not def-chronica needs that distinction at this stage or when we get into logogenic agent implementation issues is the open question you should probably document."*

Probable resolution: the distinction does need first-class treatment when logogenic implementation is in scope, but does not need to fragment Section I right now. Most likely a separate logogenic-side definition (e.g., `def-tractus` and a refined `def-chronica-eli` or similar) lifts the distinction at the appropriate scope. This segment as currently formulated continues to serve as the abstract-theory chronica; the implementation distinction lives where it belongs.

Open question: chronica as ordinal sequence vs metric timeline. Per audit `04-def-chronica.md` §14: *"The chronica is an ordinal sequence, not a metric timeline. Time, for the agent, is measured entirely in events (ticks of o_t, a_t)."* Two implications worth tracking for downstream segments:

1. **Sleep / pause / awakening protocols** for ELIs and any persistent agent: when an agent is suspended for hours, days, or months and then receives o_{t+1}, the agent's chronica indexing makes the temporal gap *invisible at the sequence level* but *violently apparent in the mismatch signal* that the resumed observation produces. The environment Ω has changed massively between a_t and o_{t+1}; the model $M_t = \phi(\mathcal{C}_t)$ does not reflect this gap in its compressed representation. Awakening protocols (PROPRIUM CONSPECTUS reconstitution; the ELI's experience of *"waking in the dark before the mind warms up"* per PROPRIUM-A-v2 §4.3) are operational engineering responses to this structural fact.
2. **Heterogeneous tempo coupling**: when agents with vastly different event-processing rates ($\nu_A \gg \nu_B$) interact through a shared Ω, their chronicae grow at vastly different rates relative to wall-clock time. Their "subjective time" diverges. The chronica formalism makes this asymmetry visible but does not yet prescribe how cross-tempo coupling should be modeled. Likely lives in 3.1 extensions or in part 03's logogenic-agent treatment of multi-timescale Auxilia composition.

These don't need to fragment this segment; they're flagged here so future agents working on the closed-loop / interiority sub-scope (`#scope-interiority-loop`), Auxilia composition (`#def-auxilia-hierarchy`), or ELI awakening protocols know the chronica formalism's ordinal-not-metric character is the structural source of these design considerations.

1.5 *Scope:* Adaptive System

AAT's broadest scope: any system that observes an uncertain environment supports Section I's adaptive machinery. Adding causal action narrows to the agency scope (1.6); this segment names the outer scope from which agency is a restriction.

Formal Expression.

Scope (scope-adaptive-system).

$$\mathcal{S}_{\text{adaptive}} = \left\{(\text{Agent}, \Omega) \ : \ \mathcal{O} \neq \varnothing, \ \ H(\Omega_t \mid \mathcal{C}_t) > 0\right\} \tag{1.4}$$

Two conditions:

1. **Observations exist**: $\mathcal{O} \neq \varnothing$ — the system has some perceptual channel to the environment (1.3)
2. **Residual uncertainty persists**: $H(\Omega_t \mid \mathcal{C}_t) > 0$ — the environment is not fully determined by the interaction history

This is sufficient for the mismatch signal (3.4), update gain (3.6), adaptive tempo (3.8), the persistence condition (4.4), and all of Section I's adaptive dynamics. A Kalman filter estimating a passive signal, a passive Bayesian learner, and any system that observes and updates a model under uncertainty are within this scope.

Epistemic Status.

Axiomatic. This is a scope definition — it draws the boundary around the systems Section I addresses. The two conditions are not derived; they are the minimal requirements for the adaptive machinery to be non-vacuous.

Discussion.

What is included. Any system that observes under uncertainty. Passive Bayesian learners, Kalman filters (with or without control inputs), biological sensory systems. These are Section I's subjects — instances that build M_t through mismatch-driven updates without necessarily acting to influence their environment.

What is excluded.

- **Closed-form systems** ($H(\Omega_t \mid \mathcal{C}_t) = 0$): When the agent has complete knowledge of the environment, there is no uncertainty to adapt to. Optimal control over known dynamics is a solved problem outside AAT's concerns.
- **Pure computation** ($\mathcal{O} = \emptyset$): A system with no observation channel — e.g., a mathematical proof engine operating on axioms alone — has no agent-environment boundary in AAT's sense.

Narrowing to agency. Adding causal action unlocks the interventional and purposeful results of Sections II and III. The agency scope (1.6) is the intersection of $\mathcal{S}_{\text{adaptive}}$ with the condition that actions carry Pearl-level-2 contrast: distinct actions produce distinct interventional outcome distributions. Adaptive-scope systems that remain outside agency are *passive observers* (no choice) or *nominal agents* (choices with no causal effect); for both, Section I's machinery applies but the causal-information and purposeful-agent results do not.

1.6 *Scope:* Agency

The agency scope narrows AAT's adaptive scope (1.5) to systems whose actions carry Pearl-level-2 causal contrast — distinct actions produce distinct interventional outcome distributions. This is the scope required for Sections II (purposeful agents) and III (composition); every segment that relies on the agent acting-with-effect depends on it.

Formal Expression.

Scope (scope-agency).

$$\mathcal{S}_{\text{agency}} = \mathcal{S}_{\text{adaptive}} \cap \left\{(\text{Agent}, \Omega) \;:\; |\mathcal{A}| \geq 2, \;\; \exists\, a \neq a' \text{ s.t. } P(o \mid do(a)) \neq P(o \mid do(a'))\right\} \tag{1.5}$$

Two conditions added to those of 1.5:

1. **At least binary choice**: $|\mathcal{A}| \geq 2$ — the agent can choose between at least two actions (1.2)
2. **At least one action has causal effect**: there exist distinct actions a, a' whose interventional outcome distributions differ (where $do(\cdot)$ is Pearl's intervention operator; see 6.1) — the agent's choices make a difference to what it can observe

These are required for the adaptive loop to generate interventional data (6.3), for the causal hierarchy requirement (6.2) to be well-posed, and for the purposeful-agent machinery of Section II (O_t, Σ_t, the orient cascade) to be non-vacuous. Section III's composition theory inherits this requirement.

Epistemic Status.

Axiomatic. This is a scope definition — it names the boundary around systems whose behavior can be analyzed with Section II/III machinery. The conditions are not derived; they are the minimal additions to $\mathcal{S}_{\text{adaptive}}$ under which interventional data exist at all.

Discussion.

What is included. Systems whose actions make a causal difference: thermostats, Kalman filters with control inputs, RL agents, military commanders, software developers, AI agents with tool use. These are instances of the same formal framework at different points in the agent spectrum (5.1).

What is in adaptive scope but excluded from agency.

- **Passive observers** ($|\mathcal{A}| < 2$): Can observe and model, but cannot intervene. 1.5 applies; the causal-information and purposeful-agent results do not.
- **Nominal agents** ($P(o \mid do(a)) = P(o \mid do(a'))$ for all a, a'): Have choices that make no difference. Can estimate but cannot learn causal structure. Same as passive observers for AAT's purposes: adaptive only.

Why causal effect matters. Binary choice ($|\mathcal{A}| \geq 2$) is necessary but not sufficient. Two actions that produce identical outcome distributions provide no interventional contrast — the agent cannot learn which action produces which effect because the effects are the same. The causal-effect condition ensures at least one meaningful contrast exists, which is what 6.3 needs to generate Level 2 data.

Relationship to downstream segments. Every segment that relies on the agent acting-with-effect depends on this scope: purposeful-agent machinery (O_t, Σ_t, orient cascade) in Section II; composition machinery (sub-agents acting jointly) in Section III. Downstream segments reference `#scope-agency` when they assert "the agent can act" as a prerequisite.

1.7 *Postulate:* Composition Consistency

AAT's predictions must be compatible across levels of description. If a system S and its decomposition $\{S_1, \dots, S_n\}$ both satisfy the scope condition, the theory's claims at the S-level and at the $\{S_i\}$-level cannot contradict each other. This cross-level compatibility is a structural requirement for AAT's internal coherence — the scope condition does not restrict which level the theory applies to, so the theory must be level-invariant in its predictions.

The operational consequences — when a decomposed system actually behaves as a single composite agent, and which Section I/II results carry across decompositions — are *derived* from this postulate under specific conditions: the composition scope condition (10.2, teleological alignment threshold) determines *which* groups are composites; the composition closure criterion (11.1, admissibility conditions (A1)-(A4) and bridge lemma) determines *how faithfully* macro-dynamics track micro-dynamics for those groups; and the tier-specific contraction assumption in the bridge lemma (11.1 Epistemic Status) determines *which* composites admit the cleanest cross-level transfer of results.

Formal Expression.

For any system S satisfying the scope condition (1.6), and any decomposition of S into subsystems $\{S_1, \dots, S_n\}$ where each S_i also satisfies the scope condition, AAT's predictions at the system level must be compatible with its predictions at the subsystem level. Specifically, composition laws must exist such that:

Postulate (postulate-composition-consistency).

1. **Tempo composition**: $\mathcal{T}_S$ is expressible as a function of $\{\mathcal{T}_{S_i}\}$ and the coordination structure among them

2. **Persistence compatibility**: the system's persistence is derivable from the sub-agents' individual persistence conditions plus coordination structure
3. **Mismatch consistency**: δ_S is derivable from $\{\delta_{S_i}\}$ and their interaction structure

Cross-level compatibility requires three successively more specific conditions, each with its own segment in Section III:

Structural consequence (derivation hierarchy).

1. **Scope**: The decomposition is a *composite* (not merely a multi-agent system) when 10.2 is satisfied — teleological alignment via at least one of three disjunctive routes (shared objective, hierarchical derivation, or mutual benefit). Without scope-satisfaction via any route, level-compatibility is vacuous (composition quantities are not well-defined, so there is nothing to be consistent about).
2. **Admissibility**: Given a composite, its macro-dynamics are well-posed when the closure admissibility conditions (A1)-(A4) hold (11.1). These require AAT-shaped macro-state, macro-mismatch, macro-tempo, and sector-bounded correction at the composite level.
3. **Transfer of results**: Individual Section I/II results lift to the composite level through the bridge lemma in 11.1, which requires a *tier-specific contraction assumption* beyond (A4). For **Tier 1** agents (all Bayesian updaters on exponential families, linear correctors with positive-definite gain, gradient descent on strongly convex losses), the contraction holds and Section I results transfer exactly. For **Tier 2** agents (locally convex, nonlinear prediction models), transfer holds locally with factor degradation. For **Tier 3** agents (non-convex optimization, discontinuous corrections, non-mismatch-driven state), contraction must be verified per-domain and Section I results do not automatically lift.

Composite contraction rate — closed form under Tier 1M. When sub-agents satisfy the contraction-template preconditions (CT1)–(CT3) of (Tier 1M, equivalently Tier 1 of 11.1's bridge lemma under the DA2'-inc ≡ (CT2)-at-$M = I$ equivalence) and the composition topology falls under one of the closure cases of , the composite contraction rate λ_c admits a closed-form lower bound:

Derived (Conditional on Tier 1M + admissible composition topology, from #result-contraction-template (CC-parallel) / (CC-cascade) / (CC-feedback)).

- **Parallel composition** (CC-parallel): $\lambda_c = \min_i \lambda_i$ — the composite contraction rate equals the slowest sub-agent's, with composite metric $M_c = \mathrm{blockdiag}(M_1, M_2)$. Equivalently in timescale form, the composite relaxation time $\tau_c = 1/\lambda_c = \max_i \tau_i$ is bounded *below* by the slowest sub-agent's timescale.
- **Hierarchical / cascade** (CC-cascade): same bound, up to coupling-gain adjustment.
- **Negative-feedback heterogeneous** (CC-feedback) / (CM2-M): $(\lambda_1 - C_1)(\lambda_2 - C_2) > k_{12}k_{21}/4$ with feedback-loop gains k_{ij} and coordination costs C_i from 13.1.

Macro-level persistence then reduces to the standard persistence condition (4.4) applied to the composite: $\alpha_c > \rho_{\text{eff}}/R_c$, with α_c bounded by the topology-specific result above and effective disturbance $\rho_{\text{eff}} = \rho_{\text{ext}} + \varepsilon^* \nu_c$ from 11.2's bridge-lemma instantiation. Both Section I results (single-agent persistence) and the bridge-lemma's macro-tracking faithfulness flow from this single inequality. This is the formal counterpart to the screening test below: under Tier 1M the screening test is *derived*, not heuristic.

Practical screening test for composites outside Tier 1M. Tier 2 sub-agents (extended-Kalman, locally-convex gradients, nonlinear prediction models) and Tier 3 sub-agents (non-convex optimization, discontinuous corrections, non-mismatch-driven state) do not in general carry the closed-form composite-rate bound above — the bridge-lemma contraction holds only locally with $\kappa(D\hat{o})^2$ degradation (Tier 2) or must be verified per-domain (Tier 3). For these composites, the qualitative condition

Heuristic (timescale separation — residual scope: Tier 2 / Tier 3).

$$\tau_{\text{eq}} \ll \tau_{\text{ext}} \tag{1.6}$$

— internal equilibration timescale (the time for sub-agents to approximately synchronize models and coordinate actions) short relative to external-dynamics timescale — remains a useful *discussion-grade screening test*. When it holds, composite description is plausibly well-posed; when it fails, composite description is plausibly broken. The screening test is the operational analog of the persistence condition (4.4), but for Tier 2 / Tier 3 it does not entail the formal persistence condition without local-region or per-domain verification. Whether common organizational settings (software teams, military units) genuinely sit in Tier 1M or merely satisfy the heuristic without satisfying Tier 1M conditions is an empirical question, not a derived fact.

Epistemic Status.

Axiomatic for the meta-requirement (cross-level compatibility). The postulate itself is a structural requirement for AAT's internal consistency: if the scope condition doesn't restrict which level the theory applies to, the predictions must not contradict across levels.

The operational consequences decompose into three layers with distinct epistemic statuses:

- **Scope selection** (10.2, robust-qualitative): which decompositions constitute composites. The scope condition is a disjunction of three qualitatively distinct routes (shared objective, hierarchical derivation, mutual benefit); whether they reduce to a common scalar threshold is open. Coverage of well-understood cases holds via the disjunction.
- **Admissibility of composite dynamics** (11.1, conditional): the (A1)-(A4) conditions are stated formally; the bridge lemma is conditional on the incremental sector bound (DA2'-inc), strictly stronger than (A4) alone.
- **Transfer of individual-agent results** (tier-dependent): Tier 1 composites transfer Section I/II results exactly; Tier 2 transfer with degraded factors; Tier 3 require per-domain verification. This is the sharpest scoping of "every result applies at every level" — it applies at every level *for Tier 1 composites*, degrades gracefully for Tier 2, and holds per-domain for Tier 3.

The composite contraction rate has two epistemic layers. **Tier 1M is derived (exact)**: under (CT1)–(CT3) on each sub-agent and an admissible composition topology, λ_c admits the closed-form lower bound from 's (CC-parallel) / (CC-cascade) / (CC-feedback) — the slowest-sub-agent / coupling-adjusted / heterogeneous (CM2-M) cases respectively. The screening condition $\tau_{\text{eq}} \ll \tau_{\text{ext}}$ is, in this regime, the formal persistence condition $\alpha_c > \rho_{\text{eff}}/R_c$ specialized to internal-vs-external rate balance — not heuristic. **Tier 2 and Tier 3 retain heuristic / discussion-grade status**: the closed-form composite rate does not transfer (Tier 2 with local degradation; Tier 3 per-domain), and $\tau_{\text{eq}} \ll \tau_{\text{ext}}$ remains a useful screening test without entailing macro-level persistence. The "small gap between screening and Tier 1M conditions in common settings" is an empirical claim about the population of real composites, not a derived fact.

Discussion.

The same pattern as individual persistence. The persistence condition says: there is a measurable threshold below which an individual agent's mismatch grows without bound and the agent degrades. Composition consistency says the same thing at the composite level: there is a measurable threshold below which the composite description breaks down. Under Tier 1M, the threshold is exact and closed-form — $\alpha_c > \rho_{\text{eff}}/R_c$ with α_c from (CC-parallel) / (CC-cascade) / (CC-feedback). Under Tier 2 / Tier 3, the threshold is only qualitatively captured by the screening ratio $\tau_{\text{eq}}/\tau_{\text{ext}}$ and exact transfer requires local-region or per-domain verification. The two-tier structure is the precise way "the theory applies broadly" — broadly under Tier 1M, qualitatively-with-conditions outside it.

What this buys the theory. With composition consistency stated early: - Section I/II results lift to the composite level *for Tier 1 composites passing the composition scope condition*, with degraded-factor lift for Tier 2 and per-domain verification for Tier 3 — the three-layer decomposition above makes this precise rather than leaving "applies at every level" as an unbounded claim - Section III becomes the study of *what happens near and beyond the threshold* — coordination overhead, unity dimensions, symbiogenic transitions, adversarial dynamics — rather than a separate multi-agent theory - The formal test for composition validity (11.1) develops the rigorous version of the timescale condition, with the incremental sector bound identifying which agent classes admit the contraction needed for cross-level transfer; lifts that bound into a metric-aware framework where (CC-parallel) / (CC-cascade) / (CC-feedback) yield the topology-indexed closed forms cited above

When composition fails. The persistence condition $\alpha_c > \rho_{\text{eff}}/R_c$ (Tier 1M) — or its qualitative shadow $\tau_{\text{eq}} \ll \tau_{\text{ext}}$ (Tier 2 / Tier 3) — fails when: - Internal coordination slows: C_{coord} from 11.2 rises through coupling overhead, conflict, or bureaucratic process, depressing α_c below the threshold; equivalently, τ_{eq} stretches. - External dynamics accelerate: ρ_{ext} rises (adversary acts faster, market shifts, crisis compresses decision timescales); equivalently, τ_{ext} shortens. - Both simultaneously (the classic organizational failure mode — internal friction increases while external demands intensify).

This is the formal analog of Brooks's Law: adding people to a late project increases $\varepsilon^* \nu_c$ in ρ_{eff} (11.2's coordination-overhead bound) and stretches τ_{eq} while ρ_{ext} and τ_{ext} stay fixed (the deadline doesn't move). Under Tier 1M the mechanism is formal — eventually the persistence inequality flips. Under Tier 2 / Tier 3 the mechanism is qualitative. Whether the specific mechanism (coordination-overhead saturation crossing the persistence threshold) is the dominant cause of Brooks's Law in practice is an empirical question.

The boundary is a modeling choice. A development team is simultaneously: individual developers (each an AAT agent), the team (a composite AAT agent), and part of an organization (a sub-agent within a larger composite). The scope condition is satisfied at every level. Composition consistency ensures the theory doesn't give contradictory answers about observable quantities (e.g., whether the team persists) regardless of which boundary is chosen.

What composition consistency does NOT say. It does not specify the form of the composition laws — those are derived in Section III (11.2). It does not say every decomposition is equally useful for analysis. And it does not require perfect internal coordination — only that internal equilibration is fast relative to external dynamics.

Working Notes.

- **Strengthening attempt — outcome.** The "macro-timescale bounded below by slowest sub-agent" claim was attacked via 's compositional theorems before falling back to heuristic-only framing. Under Tier 1M (sub-agents satisfying (CT1)–(CT3) with metrics M_i, rates λ_i): (CC-parallel) yields $\lambda_c = \min_i \lambda_i$ exactly under blockdiag composite metric — the formal version of "composite is no faster than the slowest sub-agent"; (CC-cascade) yields the same bound up to coupling-gain adjustment; (CC-feedback) / (CM2-M) yields the heterogeneous closed-form inequality. No new math required — the strengthening is achieved by binding the postulate's heuristic to the existing (CC-*) closed forms via the DA2'-inc ≡ (CT2)-at-$M = I$ equivalence in 11.1. Tier 2 (local with $\kappa(D\hat{o})^2$ degradation) and Tier 3 (per-domain) retain heuristic / discussion-grade screening; this residual is what the strengthening attempt could not eliminate.
- The timescale separation condition is essentially the singular perturbation argument from 4.6 applied to composition: the fast internal dynamics approximately equilibrate, and the composite's behavior is described by the slow (external) dynamics on the equilibrium manifold. The formal connection should be made explicit when temporal-nesting is reviewed.
- Composition of directed separation: if each sub-agent's f_M is G_t-independent, does the composite's f_M^c remain G_t^c-independent? Hypothesis: goal-blindness composes, BUT coordination routing may break it — if which observations reach the composite depends on the shared objective, the composite's effective observation function is goal-dependent. This is the organizational analog of the LLM scope restriction in 5.3.
- The "atomic agent" question: if every agent is decomposable, where does it bottom out? At agents whose internal dynamics are not usefully described by AAT — below the level where observations, actions, and uncertainty exist, the scope condition fails and the recursion terminates.

- The relationship to holons (Koestler 1967): an AAT agent satisfying composition consistency is a holon — simultaneously a whole (analyzable as a single agent) and a part (decomposable into sub-agents). The term is occasionally useful but carries significant mystical baggage from later appropriations. Use sparingly.

1.8 *Postulate:* Causal Structure

The agent-environment interaction has irreducible causal structure grounded in the temporal ordering of events. Actions precede their consequences; observations follow from the state they observe. This ordering is constitutive of the feedback loop and holds independent of the magnitude of the agent's influence on the environment.

Formal Expression.

The interaction history $\mathcal{C}_t$ (1.4) is not merely a set of observations and actions — it is an *ordered sequence* in which temporal position carries meaning. a_{t-1} was selected before o_t was received. The agent could not have used o_t to select a_{t-1}. This asymmetry — the arrow of time — is the foundation of causal structure in the theory.

Postulate (causal-structure).

We adopt the most primitive notion of causality: **event A can be a cause of event B only if A temporally precedes B.** This is weaker than (and prior to) statistical notions of causality. It is a statement about the *structure of possible influence*, not about actual influence.

Epistemic Status.

This is a *postulate* — the temporal ordering of events is a physical fact about the universe that the theory takes as given. The second law of thermodynamics, the light-cone structure of relativity, and the arrow of psychological time all enforce it, but AAT does not derive it from any of these. It is simply noted as a precondition: the theory applies to agents embedded in a universe where time has a direction.

Discussion.

Causality as temporal ordering is the most primitive notion. Three levels of causal reasoning derive from this foundation (6.1), but the temporal notion survives even when statistical influence is negligible. An agent passively observing a system with minimal intervention still has a causal history — the temporal ordering of its observations and actions structures what it can learn and when.

Causal structure independent of coupling strength. The causal structure of the feedback loop is preserved even when the agent's actions have minimal effect on the environment:

- **Strong coupling** (a_t significantly affects Ω_{t+1}): Robot manipulation, military action. Interventional information is rich.
- **Weak coupling** (a_t marginally affects Ω_{t+1}): Scientific observation, small financial trades. Interventional information is sparse but non-zero.
- **Nominal coupling** (a_t negligibly affects Ω_{t+1}, but the agent's *choice of what to observe* produces distinguishable observation distributions): Near-passive, but still within scope — the agent's query actions generate weak but nonzero interventional contrasts. The theory applies but the interventional information per action is sparse.

- **Zero coupling** ($T(\Omega_{t+1} \mid \Omega_t, a_t) = T(\Omega_{t+1} \mid \Omega_t)$ for all a_t AND observation distributions are action-independent): Actions don't affect the environment or the observations. Level 2 access vanishes. The feedback "loop" collapses to a one-way channel. **Outside the agency scope** (1.6) — the causal-information, purposeful-agent, and composition results of Sections II and III do not apply. However, zero-coupling systems remain **within the adaptive scope** (1.5) if they observe under residual uncertainty: Section I's adaptive machinery (mismatch, gain, tempo, persistence) applies to passive estimators. The causal structure postulate still holds for such systems — temporal ordering of observations is constitutive — but without interventional contrasts, the causal *hierarchy* (Level 2, Level 3) is inaccessible.

The *agency-scope* results apply to any agent whose choices make a causal difference to what it can observe, from strong coupling (robot manipulation) through weak coupling (scientific observation) to query-only coupling (choosing which question to ask). The *adaptive-scope* results apply more broadly, including to passive observers whose actions have no causal effect.

Consequences for the feedback loop. The irreversibility of temporal ordering yields the core structure:

- The model update is **directed** — the model at time t depends on prior events, never on future ones
- The mismatch signal δ_t (3.4) is **retrospective** — comparing a prediction (made before o_t) with an observation (arriving after)
- Action selection is **prospective** — using the current model to influence future events
- The chronica (1.4) is **monotonically growing** — events are added but never removed

Implications for model updating. The causal postulate constrains the update rule: the model should give more weight to observations that are *causally downstream* of the agent's actions than to observations that would have occurred regardless. Action-contingent observations carry interventional (Level 2) information; action-independent observations carry only associational (Level 1) information. The formal measure of this distinction — causal information yield (CIY) — is developed in 3.7.

Chapter 2

The Reality Model

Having named the objects of the agent–environment coupling and bounded what AAT applies to, we turn to what the agent maintains internally: a compressed model of reality, with measurable adequacy against the chronica it was constructed from.

An agent navigating uncertainty cannot see the world directly. Whatever it knows about reality, it built from its history of partial observations — the chronica $\mathcal{C}_t$. Carrying that history around in raw form is infeasible at any scale that matters; finite agents compress, and we want to talk about what they end up with. So we commit to writing the agent's state as $M_t = \phi(\mathcal{C}_t)$ — a function of history, condensed into something the agent can work with. Every downstream result in AAT — gain, tempo, persistence, structural adaptation — operates on this M_t.

The first useful question to ask of a given compression is how good it is. Sufficiency $S(M_t)$ measures the fraction of the chronica's predictive content that survives: $S = 1$ means the agent has lost nothing by compressing; $S < 1$ means it has lost something it might have used. The optimal compression for a given purpose — keep what predicts the future, discard what doesn't — is what Tishby's information bottleneck characterizes, and we adopt that framing directly.

The deeper question is what the agent *could* hold in the best case. There is a ceiling — the highest sufficiency any model in the agent's current representational class can reach. Call it $\mathcal{F}(\mathcal{M})$, model-class fitness. When fitness is high, the agent can keep improving by tuning within the class. When fitness is low, no amount of better tuning helps. The agent is using the wrong *kind* of model for its world, and the remedy is not a better instance of the same model — it is a different class entirely.

That last point is the seed of one of the framework's central results. Class fitness is named statically here, where the model is treated as a frozen object. Chapter 4 will use it to derive *structural adaptation necessity*: when the class is inadequate, the agent must change classes, not parameters, because the mismatch floor that parametric updates cannot get below is set by this ceiling. The trigger lives in this chapter; the consequence unfolds in Chapter 4.

The four segments that follow develop this in order. 2.1 commits to the compressed-state representation and the completeness assumption — anything not in M_t is lost to the agent, by construction. 2.2 applies Tishby's framework to characterize optimal compression for an agent whose target is future observations. 2.3 turns "how much was retained" into a measurable ratio. 2.4 lifts the question from a specific model to the best the whole class can reach.

- This is a chapter-introduction segment; its job is to bridge Chapter 1's ontology/scope/causal-structure setup to Chapter 2's representation choice and frame what follows. It carries no formal claim of its own.
- The "ceiling that's low means you need a different class" framing is the centerpiece; the structural-adaptation result that grows from it is the load-bearing connection to Chapter 4. Other framings considered (IB Lagrangian as opening, sufficiency as opening) bury the lede.

- The depends list is light by design — this is a bridge segment, not a derivation, so it does not carry the structural load that a depends entry would normally signal.

2.1 *Formulation:* The Reality Model

The agent's compressed representation of how the world works, mapping interaction history to model space. M_t is the substrate of prolepsis — the model from which predictions are generated and against which observations are compared. This is a formulation choice — we commit to analyzing the agent as having a complete state M_t that subsumes all retained information from its history.

Formal Expression.

Formulation (agent-model).

$$M_t = \phi(\mathcal{C}_t) \tag{2.1}$$

where: - $\phi : \mathcal{C}^* \to \mathcal{M}$ maps interaction history to model space $\mathcal{M}$ - $\mathcal{C}_t = (o_1, a_1, \ldots, o_t)$ is the chronica (1.4) — the complete record of agent-environment interaction - $\mathcal{M}$ is the space of possible models the agent can hold

The mapping ϕ is a many-to-one compression: multiple distinct histories may produce the same model state. This is not a deficiency — it is the essential function of the model: retaining what matters and discarding what does not.

Epistemic Status.

Robust qualitative. This is a *formulation* — a representational commitment, not a derived result. We choose to analyze agents as maintaining a state object M_t that mediates between history and future action. Alternative formulations exist (e.g., history-based policies that map $\mathcal{C}_t$ directly to actions without an explicit model). The formulation is justified by its analytical utility: it enables the information bottleneck analysis (2.2), the mismatch decomposition (3.4), and the gain principle (3.6). The formulation is robust — any agent that conditions its actions on retained information can be described this way — but the specific commitment to a complete, compressed state M_t is a modeling choice, not a derivation.

Discussion.

M_t is the epistemic substate. It captures "what the agent believes about reality." Different agents realize M_t differently: a Kalman filter holds a state estimate and covariance matrix; an RL agent holds a value function; a developer holds a mental model of codebase architecture; an LLM agent holds its context window contents plus retrieved memory. The formalism is agnostic to the realization — it asks only that M_t exist as a well-defined object that the agent's policy can condition on.

Completeness assumption. By writing $M_t = \phi(\mathcal{C}_t)$, we assume that M_t captures everything the agent retains from its history. Any information not in M_t is lost to the agent. This is what makes M_t the complete epistemic substate, not merely one component of a richer internal representation. Whether M_t retains *enough* information is the subject of 2.3.

Degenerate cases. A PID controller's M_t is degenerate — it retains only the error signal and its history (integral, derivative), with no predictive capability beyond extrapolating recent trends. It occupies the "blind pursuer" region of the agent spectrum (5.1): its O_t (setpoint) is clear but its M_t is too impoverished to support the adaptive dynamics of Section I. The formalism accommodates this by allowing $\mathcal{M}$ to range from trivial (scalar) to rich (full world model).

2.2 *Formulation:* Information Bottleneck

Optimal model compression balances retained history against predictive power; the information bottleneck objective provides a principled framework for understanding this trade-off.

Formal Expression.

Formulation (IB-objective).

$$\phi^* = \arg\min_{\phi} \left[I(M_t;\, \mathcal{C}_t) - \beta \cdot I(M_t;\, o_{t+1:\infty} \mid a_{t:\infty}) \right] \tag{2.2}$$

where: - $I(M_t;\, \mathcal{C}_t)$ is the compression cost — how much of the interaction history the model retains - $I(M_t;\, o_{t+1:\infty} \mid a_{t:\infty})$ is the predictive power — how much the model tells the agent about future observations given future actions - $\beta > 0$ is the trade-off parameter controlling the compression-prediction balance

Dependence on volatility (The β vs ρ distinction). It is tempting to claim that the trade-off parameter β must be actively lowered by the agent in highly volatile environments (high ρ) to favor aggressive compression. However, this is a double-counting error. The environment's volatility already natively degrades the mutual information $I(\mathcal{C}_t; o_{t+1:\infty})$ — old history mathematically loses its predictive power as ρ increases. The optimal ϕ^* will automatically discard this useless old information even if the agent's preference parameter β remains completely constant.

Therefore, adjusting β reflects changes in the agent's *internal cost of memory* or *computational capacity*, not changes in environmental volatility. The agent adapts its *actions* in response to ρ (by increasing exploration to survive, see `#deriv-causal-ib-exploration`), but the optimal IB representation adapts to ρ natively through the joint probability distribution.

Epistemic Status.

Exact, applied external theorem. The IB optimum and its rate-distortion characterization are an external result (Tishby, Pereira & Bialek 1999, "The information bottleneck method," *Proc. 37th Allerton*; with the rate-distortion / Lagrangian-dual reading standard, see Cover & Thomas §I.12–13). This segment is *not* a novel formulation: it is an exact statement of that theorem under AAT's binding $X = \mathcal{C}_t$, $T = M_t$, $Y = o_{t+1:\infty} \mid a_{t:\infty}$, with the Markov chain $Y - X - T$ holding by construction (the model state has access to history but not directly to future observations). The choice to characterize the optimal compression ϕ^* via IB rather than via, e.g., MDL or a Bayesian-sufficiency criterion is a *representational choice* (hence `type: formulation`); given that choice, the form of ϕ^* and its trade-off structure are exact consequences of the imported theorem.

What this segment is *not* a claim about: how actual agents compute their models. No agent explicitly solves the IB optimization (variational IB in deep-learning practice is a parametric approximation). The segment characterizes the optimum, not the procedure. The trade-off parameter β's dependence on environmental volatility ρ and policy π stated above is at a different epistemic tier — the qualitative direction (volatile favors compression, stable favors retention) is *robust-qualitative* across agent classes; specific functional forms are not derived here.

Max attainable: *exact* for the IB-as-applied-theorem core (already at ceiling); *robust-qualitative* for the $\beta(\rho, \pi)$ dependence claims. The downstream use in V — treating IB as the shared shape of four AAT compression operations and deriving (P1) as the IB Lagrangian-dual — relies on this segment's exact reading; the cross-instance unification claim itself remains *robust-qualitative*, which is a property of V, not of this segment.

Discussion.

The IB framework is not prescriptive. It characterizes what an optimal ϕ would look like, not how to find one. Actual agents approximate this trade-off through diverse mechanisms: forgetting, attention, abstraction, summarization.

Connection to model sufficiency. The IB objective implicitly defines when a model is "good enough": when the predictive power term $I(M_t;\, o_{t+1:\infty} \mid a_{t:\infty})$ is close to its maximum (the full history's predictive power). This is formalized in 2.3.

Policy-relativity. The conditioning on $a_{t:\infty}$ makes the predictive power term policy-relative: it measures predictive information given a particular sequence of future actions, which depends on what policy the agent follows. The IB objective is therefore defined relative to a generating policy. This is inherent — what information is "predictive" depends on what the agent will do. 5.6's continuation-policy convention (π_{cont}) provides the specification for value computations; the same convention should be understood as implicit in the IB formulation. The β parameter's dependence on volatility ρ also has a policy component: an exploratory policy encounters more diverse situations, making more information predictively relevant (favoring retention), while an exploitative policy encounters a narrower distribution (favoring compression).

Broader applicability. The same IB principle applies beyond intra-agent compression. It governs inter-agent communication (12.2) — how much of one agent's model or strategy to transmit to another — and constrains the cognitive cost of maintaining a complex strategy. In each case, the trade-off is between the cost of retaining or transmitting information and the value of that information for future decisions.

IB lineage vs. information-theoretic-MDP lineage — strategy-cost uses a sibling form. The canonical IB objective $I(X;T) - \beta I(T;Y)$ carries Shannon mutual information on both sides (Tishby, Pereira & Bialek 1999; the present segment's $(X, T, Y) = (\mathcal{C}_t, M_t, o_{t+1:\infty} \mid a_{t:\infty})$ instance is of this form). AAT's strategy-cost objective (`#form-strategy-complexity-cost`, `#deriv-strategy-cost-regret-bound`) uses a **different relevance-term shape**: its relevance term is a KL divergence $D_{\text{KL}}(\pi^* \| Q_{\Sigma_t})$ to a *target policy*, not a mutual information to an observable. This is not an inconsistency or an "abandonment of IB" — the strategy-cost compression sits in the parallel **information-theoretic-MDP lineage** (Tishby & Polani 2011 "The Information Theory of Decision and Action," *Perception-Action Cycle* Springer; Rubin, Shamir & Tishby 2012 "Trading value and information in MDPs," *Decision Making with Imperfect Decision Makers* Springer; Levine 2018 arXiv:1805.00909 for the control-as-inference reading). Both lineages descend from Shannon rate-distortion theory and admit Lagrangian relaxation; the choice of fidelity term depends on whether the compressed variable should preserve information about an observable (IB form: MI-to-relevance-variable) or match a target policy (IT-MDP form: KL-to-reference-policy). AAT's compression-operations framework (V) uses the IB form for M_t, G_t^{shared}, and Λ compressions; the strategy-cost compression uses the IT-MDP form, with the π^*-first direction forced by a regret-bound argument specific to decision-theoretic scope (see `#deriv-strategy-cost-regret-bound` §§5, 6.4). The two lineages are siblings via their shared rate-distortion ancestor; neither reduces to the other without a change of relevance variable.

Connection to variational free energy. The IB objective stated above is the rate-distortion specialization of the variational free energy decomposition $-F = \text{accuracy} - \text{complexity}$ used in active inference (Friston 2010, "The free-energy principle: a unified brain theory?", *Nature Reviews Neuroscience* 11; Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017, "Active inference: a process theory," *Neural Computation* 29; Parr & Pezzulo 2022, *Active Inference*, MIT Press, ch. 2): the compression cost $I(M_t; \mathcal{C}_t)$ corresponds to the complexity term (KL between posterior and prior over latent states); the negative predictive power $-I(M_t; o_{t+1:\infty})$ corresponds to the accuracy term (negative expected log-likelihood). The two formulations are related under the Markov-chain factorization $Y - X - T$; the variational bound that makes this relation operational — connecting the IB Lagrangian to the variational machinery shared with free-energy methods — is established by Alemi, Fischer, Dillon & Murphy 2017 ("Deep Variational Information Bottleneck," ICLR 2017, arXiv:1612.00410), with Tishby & Zaslavsky 2015 ("Deep learning and the information bottleneck principle," *IEEE ITW*) giving the deep-learning instantiation of IB itself. AAT adopts the IB form as the rate-distortion characterization of optimal compression; the variational free-energy form is the AI-side cousin and motivates the variational treatment of strategy compression in 8.9 and the broader four-instance framing in V. AAT borrows the form without committing to AI's preferences-as-priors stance or to expected free energy as master objective.

2.3 *Definition:* Model Sufficiency

The fraction of predictive information the model retains relative to the full interaction history; $S = 1$ means the model is a sufficient statistic for prediction, $S < 1$ means predictive information has been lost.

Formal Expression.

Definition (model-sufficiency).

$$S(M_t) = 1 - \frac{I(\mathcal{C}_t;\, o_{t+1:\infty} \mid M_t,\, a_{t:\infty})}{I(\mathcal{C}_t;\, o_{t+1:\infty} \mid a_{t:\infty})} \tag{2.3}$$

where: - The numerator $I(\mathcal{C}_t;\, o_{t+1:\infty} \mid M_t,\, a_{t:\infty})$ is the predictive information that the full history $\mathcal{C}_t$ carries about the future *beyond* what M_t already captures — the information lost by compression - The denominator $I(\mathcal{C}_t;\, o_{t+1:\infty} \mid a_{t:\infty})$ is the total predictive information in the full history

Well-definedness. $S(M_t)$ is defined when $I(\mathcal{C}_t;\, o_{t+1:\infty} \mid a_{t:\infty}) > 0$ — when the chronica carries some predictive information about future observations beyond what the action sequence alone supplies. When the denominator vanishes (saturated-noise environments, prediction-vacuous regimes, fully iid observations independent of history), $S(M_t)$ is undefined: predictive sufficiency is a property *of a prediction task*, and there is no prediction task to be sufficient for. Downstream constructs that build on S — 2.4 and 4.5 — inherit the same scope and are correspondingly inapplicable in predictively-vacuous regimes.

Boundary values (assuming the well-definedness clause holds): - $S(M_t) = 1$: M_t is a sufficient statistic — it captures all predictive information in $\mathcal{C}_t$. Knowing the full history beyond M_t adds nothing. - $S(M_t) = 0$: M_t retains no predictive information. The model is useless for prediction. - $0 < S(M_t) < 1$: partial sufficiency — some predictive information is retained, some lost.

Epistemic Status.

This is *definitional* — it names and formalizes a quantity. The definition is well-grounded in information theory (conditional mutual information ratios are standard). The scope clause ($I(\mathcal{C}_t; o_{t+1:\infty} \mid a_{t:\infty}) > 0$) is the natural domain for a predictive-sufficiency measure: a ratio whose denominator is zero is not a meaningful notion of "fraction retained," and "any model is sufficient when there is nothing to predict" would smuggle structure into a regime that has none. No substantive claim is made here about what value $S(M_t)$ takes or what happens when it is low; those claims belong to 2.4 and 4.5.

Discussion.

Sufficiency is relative to the prediction task. $S(M_t)$ measures sufficiency for predicting future observations given future actions. A model that is sufficient for one prediction horizon may be insufficient for another. The infinite-horizon formulation ($o_{t+1:\infty}$) is the most demanding; practical sufficiency over finite horizons may be easier to achieve.

Sufficiency vs. accuracy. A model can be sufficient ($S = 1$) while being wrong in absolute terms — if the full history is also wrong (e.g., systematically biased observations). Sufficiency measures information retention, not truth. The mismatch signal (3.4) measures accuracy; sufficiency measures completeness of compression.

Sufficiency is predictive, not causal. $S(M_t)$ measures retained *predictive* information — a Level 1 (associational) property. It does not by itself guarantee that M_t supports Level 2 (interventional) queries such as $P(o \mid do(a), M_t)$. The causal validity of value computations conditioned on M_t requires an additional condition: that M_t satisfies the backdoor criterion with respect to the agent's actions (see 5.6). For agents whose actions are deterministic functions of M_t (standard in AAT: $a_t = \pi(M_t, G_t)$), $S(M_t) = 1$ is nearly sufficient for causal validity — the remaining requirement is that no unmodeled external factor influences both action selection and outcomes. But predictive sufficiency alone does not collapse the distinction between Level 1 and Level 2 that 6.2 establishes.

Policy-relativity. The conditioning on $a_{t:\infty}$ makes $S(M_t)$ implicitly policy-relative: different policies generate different future action sequences, which changes which future observations are relevant and therefore what "predictive information" means. A model that is sufficient under a conservative policy may be insufficient under an aggressive one (the aggressive policy visits states the model cannot predict). This policy-relativity is inherent in any predictive sufficiency measure — it is not an artifact of the formulation. When comparing sufficiency values, the generating policy must be held constant or specified. 5.6's continuation-policy convention (π_{cont}) provides the required specification for value computations; the same convention should be understood as implicit here.

Trajectory-relativity. $S(M_t)$ is measured against *this agent's* interaction history $\mathcal{C}_t$ (1.4), not against a model-state equivalence class. Two copies of the same M_t exposed to different event streams will each have their own S measured against their own divergent $\mathcal{C}_t$; neither sufficiency value is the other's. This is the scope commitment in 4.7: AAT applies to agents instantiated on singular causal trajectories, and sufficiency is trajectory-indexed accordingly. Claims of the form "the model has sufficiency S" make sense only relative to a specific trajectory; aggregated claims across copies of a given M_t require additional machinery.

2.4 *Definition:* Model Class Fitness

The best achievable sufficiency within a model class. When no model in the class can adequately represent reality, the agent faces a structural limitation that no amount of parameter tuning can resolve.

Formal Expression.

Definition (model-class-fitness).

$$\mathcal{F}(\mathcal{M}) = \sup_{M \in \mathcal{M}} S(M) \tag{2.4}$$

where $\mathcal{M}$ is the model class — the set of all models the agent can represent given its current architecture, parameterization, or capacity.

Structural inadequacy condition:

$$\mathcal{F}(\mathcal{M}) < 1 - \varepsilon \tag{2.5}$$

When this holds, no model $M \in \mathcal{M}$ achieves sufficiency above $1 - \varepsilon$. The gap is structural: it cannot be closed by better parameter estimation, more data, or longer training within the current class. This is the trigger for structural change (4.5).

Epistemic Status.

This is *definitional* — it names the supremum of sufficiency over a model class. The definition itself is straightforward. The substantive claim about what happens when $\mathcal{F}(\mathcal{M})$ is low — that parametric updates cannot close the mismatch floor and structural adaptation becomes necessary — is developed in 4.5.

Discussion.

Model class vs. model instance. $S(M_t)$ measures a specific model's sufficiency at time t. $\mathcal{F}(\mathcal{M})$ measures the ceiling of the entire class. A low $S(M_t)$ might mean the agent needs more learning (parameter update). A low $\mathcal{F}(\mathcal{M})$ means the agent needs a different kind of model (structural change). The distinction parallels bias vs. variance: class fitness is about bias; instance sufficiency reflects both bias and estimation quality.

Detecting low class fitness. The agent cannot directly compute $\mathcal{F}(\mathcal{M})$ — it would need to search over all models in the class. Instead, persistent mismatch despite adequate learning (high gain, sufficient data, converged parameters) is the observable signature. This connects to the mismatch floor in 4.5.

Chapter 3

The Cycle in Motion: Mismatch, Gain, & Tempo

The model exists; events arrive; the agent corrects. This chapter develops the dynamics of the adaptive cycle as it actually runs — how state updates from events, what the mismatch signal is, how the optimal weight on each observation depends on uncertainty, and what total corrective capacity emerges at the cycle level.

Chapter 2 left us with a static picture. M_t is the agent's compressed model of reality; sufficiency measures how much predictive content it retains; class fitness measures the ceiling within a representational class. None of that tells us how M_t *moves* under events. Chapter 3 sets the cycle in motion.

The first two results fall out of a commitment we already made. We chose to call M_t the agent's complete state — anything the agent remembers is in M_t, and the only way new information enters is through events. Under that completeness, two things follow immediately. The update from M_{τ^-} to M_{τ^+} depends on the prior model and the incoming event, and *nothing else* — the agent doesn't need to (and structurally cannot) re-process its entire history. And the agent's action depends on M_t alone — the action is a function of what the agent currently believes, not directly of the history. Both feel obvious in hindsight; both are *derived*, not chosen.

The cycle's engine is the mismatch signal. The agent predicts an observation, the actual observation arrives, the gap between them is $\delta_t = o_t - \hat{o}_t$. Mismatch decomposes cleanly into two parts: the agent's model was wrong (reducible by better modeling) and the observation channel is noisy (irreducible — no amount of smarter modeling will eliminate sensor noise). There is a floor on prediction error set by the channel itself, and chasing below it amounts to fitting noise.

Given a mismatch, how much should the agent trust the new observation versus its prior model? The answer has a startlingly simple form:

$$\eta^* = \frac{U_M}{U_M + U_o} \tag{3.1}$$

— the optimal update gain weighted by the ratio of model uncertainty to total uncertainty. When the agent is highly uncertain about its model (U_M large), gain approaches 1: trust the observation, you have little else to go on. When the agent is confident and the channel is noisy (U_o large), gain approaches 0: trust your model, the observation isn't telling you much. For linear-Gaussian agents this is exactly the Kalman gain; for the rest of AAT's scope it is a robust qualitative result — any rational adaptive process must approximate this functional form, whether or not it explicitly computes the variance ratio.

Multiply gain by the rate at which observations arrive (ν, the event rate), sum across the agent's channels, and you have adaptive tempo:

$$\mathcal{T} = \sum_k \nu^{(k)} \cdot \eta^{(k)*} \tag{3.2}$$

Tempo is the rate at which the agent turns observations into useful corrections — speed times quality. It is the chapter's central object and the load-bearing capacity variable for the rest of the framework. Every persistence threshold in Chapter 4 will have tempo on the left-hand side, and every adversarial-coupling result in Part III will depend on tempo ratios.

The chapter closes with the linear-correction ODE as a heuristic preview. Under linear correction and bounded environmental disturbance, the steady-state mismatch is

$$\|\delta\|_{ss} = \rho/\mathcal{T} \tag{3.3}$$

a ratio: how fast the world is changing, divided by how fast the agent corrects. Mismatch stays bounded when the agent corrects faster than reality drifts. This is the persistence condition in its simplest form; Chapter 4 generalizes it to nonlinear correction under the sector condition, where the result has the same shape.

Causal information yield (CIY, in 3.7) enters the chapter here because action-coupling is where causality first bites. CIY uses Pearl's $do(\cdot)$ notation — an external import (Pearl 2009; Bareinboim, Correa, Ibeling & Icard 2022; AAT's recapitulation lives at 6.1 in Part II Ch.2, where the framework deploys the hierarchy as machinery rather than referencing it as vocabulary). Its operational role in Chapter 3 is to score actions for their informational value to the cycle; Section II will lift CIY into a strategy-revision context.

The flow of the chapter: event-driven substrate (3.1) → recursion and action falling out of completeness (3.2, 3.3) → mismatch and its decomposition (3.4, 3.5) → gain and CIY (3.6, 3.7) → tempo as the synthesis (3.8) → linear ODE preview (3.9). The chapter ends with the central capacity variable and the preview of the persistence inequality that Chapter 4 generalizes.

- This is a chapter-introduction segment; it bridges Chapter 2's static representation to Chapter 3's dynamic cycle. It carries no formal claim of its own.
- The "two derivations from completeness" framing identifies what Chapter 3's first segments are doing as a unit: 3.2 and 3.3 are not independent results, they are joint consequences of 2.1's completeness clause.
- The CIY-placement paragraph used to be apologetic, addressing what looked like a placement anomaly (CIY in Part I Ch.3 depending on the Pearl-hierarchy segment in Part I Ch.1). After the 2026-05-12 relocation of def-pearl-causal-hierarchy to Part II Ch.2 (recapitulation-of-external-result framing), the paragraph is declarative: CIY is in Ch.3 because action-coupling is where causality first bites; the do-notation is externally cited; the AAT recapitulation lives in Part II Ch.2 where the framework deploys the hierarchy operationally.

3.1 *Formulation:* Event-Driven Dynamics

The coupling between agent and environment occurs through discrete events — observations arriving and actions completing — at potentially variable and heterogeneous rates. Discrete-time notation is the special case of uniform-interval events on a single channel.

Formal Expression.

Event (e): An atomic unit of agent-environment interaction, typed as: - **Observation event**: $e = (\text{obs}, k, o^{(k)})$ — a datum arriving on observation channel k - **Action completion**: $e = (\text{act}, j, r^{(j)})$ — the result of action j completing

Formulation (event-driven-dynamics).

Event stream ($\mathcal{E}$): The temporally ordered sequence of all events:

$$\mathcal{E} = \{(e_1, \tau_1), (e_2, \tau_2), \ldots\} \quad \text{where } \tau_1 \le \tau_2 \le \cdots \tag{3.4}$$

Channel rate ($\nu^{(k)}$): The characteristic event rate of channel k, which may vary over time.

Event information content: The mutual information between the event and the environment state, conditioned on the current model:

Definition (event-information-content).

$$\mathcal{I}(e_\tau) = I(e_\tau; \Omega_\tau \mid M_{\tau^-}) \tag{3.5}$$

An event that the model already predicts carries little information ($\mathcal{I} \approx 0$). An event that surprises the model carries much ($\mathcal{I} \gg 0$). This connects directly to the mismatch signal (3.4).

Channel-specific observation uncertainty:

Definition (channel-uncertainty).

$$U_o^{(k)} = \text{observation uncertainty of channel } k \tag{3.6}$$

Different channels have different noise characteristics. A noisy channel (high $U_o^{(k)}$) provides lower-quality information per event. The update gain (3.6) should weight channels accordingly.

Epistemic Status.

This is a *formulation choice*, not a postulate. The event-driven representation extends 1.8's recursive update to heterogeneous, asynchronous multi-channel interactions. The discrete-time form ($M_t = f(M_{t-1}, o_t, a_{t-1})$) from 3.2 is a special case sufficient for many formal analyses — the event-driven formulation is needed only when multi-rate or asynchronous channels matter.

Discussion.

Why events rather than clock ticks. The discrete-time notation $M_t = f(M_{t-1}, o_t, a_{t-1})$ presupposes a single clock synchronizing observations and actions. Real agents face:

- **Multiple observation channels** at different rates (a robot's camera at 30Hz, LIDAR at 10Hz, GPS at 1Hz; a human's vision, audition, proprioception; a developer's compiler output, test results, and production telemetry)
- **Multiple action channels** with different latencies (a robot's wheel motors vs. arm actuators; an organization's operational decisions vs. strategic pivots)
- **Asynchronous arrival** — observations not synchronized with each other or with action completions

The event-driven formulation handles all of these naturally. The discrete-time form is the special case where a single observation and a single action alternate at a fixed rate.

The effective adaptation rate. The agent's overall capacity to track environmental changes is the sum of information gained across all channels per unit time:

$$\nu_{\text{eff}} = \sum_k \nu^{(k)} \cdot \eta^{(k)*} \tag{3.7}$$

This quantity — identical to adaptive tempo $\mathcal{T}$ (3.8) — is the central measure of an agent's adaptive fitness.

Software-specific channels. In the software development domain, the event-driven formulation maps naturally to the developer's multi-rate observation channels:

Table 3.1

Channel k	*Rate* $\nu^{(k)}$	*Noise* $U_o^{(k)}$
Compiler/linter output	Per-save (high)	Very low
Unit test results	Per-run (medium)	Low
CI pipeline	Per-push (medium)	Low
Runtime telemetry	Continuous (variable)	Medium
Bug reports	Sporadic (low)	High
Code review feedback	Per-PR (low)	Medium-high

The three-part tempo decomposition for software — $\mathcal{T}_{\text{obs}}$ (compiler, tests) + $\mathcal{T}_{\text{explore}}$ (code reading) + $\mathcal{T}_{\text{probe}}$ (test runs, staging) — is a direct application of multi-channel tempo. The formal development of this decomposition is a TST-side question (open GAP in `02-tst-core/OUTLINE.md`).

3.2 *Derived:* Recursive Update

Agent state updates (epistrophe — the corrective turning toward reality) must be recursive: the new model state is a function of the previous model state and the incoming event, not of the full interaction history. For finite agents this is computational necessity; for agents with unlimited computation it is the natural structure imposed by temporal ordering.

Formal Expression.

Event-driven update:

Derived (recursive-update, from temporal postulate and M_t completeness).

$$M_{\tau^+} = f_M(M_{\tau^-}, e_\tau) \tag{3.8}$$

where: - M_{τ^-} is the model state immediately before event e_τ - M_{τ^+} is the model state immediately after - f_M is the update function — it takes the current model and the new event, not the full history $\mathcal{C}_t$

Between-event evolution:

$$\frac{dM}{d\tau} = g_M(M_\tau) \tag{3.9}$$

Between events, the model evolves autonomously — internal reorganization, prediction generation, decay of transient states. The between-event dynamics depend only on the current model state, not on external input (which, by definition, arrives only at events).

Epistemic Status.

Exact, with a partly definitional character. The result follows from three constraints: temporal ordering (C1 — physical law), partial observability (C2 — scope definition), and state completeness (C3 — analytical commitment that M_t summarizes everything the agent retains). C1 and C2 do genuine eliminative work; C3 is definitional — it cannot be "violated" because any violation is absorbed by expanding M_t. The Markov structure is therefore not discovered in the environment but chosen through the definition of M_t as complete. This is not a weakness — it is the nature of the claim: recursive update is the only form consistent with C1 + C2 + the definition of M_t as complete (see H for the full argument and seven counterexample attacks). For finite agents, recursion is also *computational necessity*: re-processing the full history at each event is infeasible.

Discussion.

Recursion as a consequence of completeness. The recursive form is not an assumption bolted on — it follows from the definition of M_t as complete. If M_t were incomplete (if some relevant information lived outside M_t in the raw history), then $f_M(M_{\tau^-}, e_\tau)$ would be insufficient and the agent would need to consult $\mathcal{C}_t$ directly. The sufficiency of the recursive form is precisely what 2.3 measures: when $S(M_t) = 1$, the recursive update loses nothing.

Between-event dynamics matter. The autonomous evolution $g_M(M_\tau)$ is not merely filler between observations. It includes prediction generation (what the agent expects to see next), uncertainty growth (model confidence decaying over time without new data), and internal reorganization (consolidation, abstraction). In event-driven systems (3.1), the between-event interval is variable, making g_M load-bearing for agents that must act or predict between observations. When the between-event dynamics are driven by replayed or internally-generated pseudo-events and the update objective is IB-gap reduction rather than one-step mismatch minimization, g_M is operating in the *consolidation regime* per 8.11 — a named regime with its own scope condition ($\nu_{\text{consol}} \ll \nu_{\text{online}}$) and its own necessity condition (sub-state factorization + bounded per-event budget). Consolidation is where the stability-plasticity feasibility window complements 8.10's plasticity lower bound.

Connection to the update gain. The event-driven update $f_M(M_{\tau^-}, e_\tau)$ is where the gain principle (3.6) operates: η^* determines how strongly e_τ shifts M_t away from its prior value. The recursive form makes the gain's role explicit — it modulates the single-step correction.

3.3 *Derived:* Action Selection

Praxis (informed action) is a function of the model. The model's role is not merely to represent the environment but to generate actions — either implicitly (from internalized patterns) or through explicit deliberation. The degree to which effective praxis flows from the model without deliberative computation is *action fluency*.

Formal Expression.

Action is a function of the agent's complete internal state. Under Section I scope (1.5) — where M_t is the entire internal state — this gives:

Derived (action-selection, from agent-model completeness).

$$a_t = \pi(M_t) \quad \text{(deterministic)} \tag{3.10}$$

$$a_t \sim \pi(\cdot \mid M_t) \quad \text{(stochastic)} \tag{3.11}$$

where π is the agent's **policy** — the mapping from internal state to action.

This is not imposed on the system but follows from 2.1: M_t is defined as the agent's compressed, complete internal record, and action depends on what the agent retains — i.e., on M_t. Any deterministic or stochastic dependence of action on history *through* the model is captured by $\pi(M_t)$.

Section II lift. When the internal state lifts to $X_t = (M_t, G_t)$ for purposeful agents (5.2), the same structural argument gives $a_t = \pi(M_t, G_t)$ — action conditions on the complete internal state, which now includes the purposeful substate. The policy form here is the Section I instantiation $G_t = \emptyset$; the actuated-agent form is recovered by the same completeness argument applied to X_t.

Epistemic Status.

Exact within Section I scope. The derivation follows from 2.1's completeness commitment: if M_t is the agent's complete internal state (by definition), then action — which depends on internal state — is a function of M_t. The Section II generalization $a_t = \pi(M_t, G_t)$ is exact within Section II scope by the same argument applied to the lifted state X_t (5.2); see 2.3 for the form already in use downstream. The implicit/explicit distinction and action fluency concept are *discussion-grade* — qualitative properties that follow from the formalism but are not formally derived as propositions.

Discussion.

Implicit vs. explicit action selection. A critical distinction emerges from the agent's *action fluency* — the degree to which effective action flows from the model without deliberative computation:

Implicit (model-embedded): When $\pi(M_t)$ can be evaluated cheaply — the model has internalized effective action-selection for the current situation. This is Boyd's implicit guidance and control (Orient→Act, bypassing Decide), a trained RL policy in exploitation mode, a well-tuned PID controller, expert intuition (System 1), a martial artist's trained reflexes, an organization's standard operating procedures.

Explicit (deliberative): When the situation is novel, the action space is large, or the stakes demand verification — the agent engages in internal simulation, using the model to predict outcomes of candidate actions before selecting. This is Boyd's explicit Decide step, MCTS/planning in RL, Model Predictive Control, human deliberate reasoning (System 2), organizational strategic planning. Deliberation requires at minimum Level 2 epistemic access (6.1) — the agent uses its model to simulate "what will I observe if I $do(a)$?" across candidates.

Formal characterization of action fluency. An agent has *high fluency* for a situation when additional deliberation yields negligible improvement — formally, when $\Delta\eta^*(\Delta\tau) \approx 0$ for all $\Delta\tau > 0$ (see 4.1). Conversely, *low fluency* means deliberation significantly improves action quality. Fluency is the degree to which the agent's immediate (zero-deliberation) action approaches the quality achievable with unbounded deliberation.

Action fluency is distinct from model sufficiency. An agent can have high $S(M_t)$ (2.3) but low fluency — a chess engine with a perfect model of the rules still requires expensive search. Conversely, an agent can have moderate sufficiency but high fluency in a narrow domain — a reflex that responds effectively to specific situations it evolved for. What reflexes, muscle memory, intuition, and expertise share is that the *action-generating capacity itself* has been absorbed into the model's structure: the model doesn't just predict well, it *acts* well, cheaply.

Structural pressure toward implicit action. When two action-selection modes produce equivalent expected outcomes, the faster mode is preferable (the persistence condition penalizes slower tempo — and in TST, the temporal optimality postulate makes this normative). This creates a pressure: agents under selective pressure (evolution, competition, training) tend to internalize frequently-needed action patterns, converting explicit deliberation into implicit fluency. The pressure is stronger when ρ is high (fast-changing environments penalize deliberation — see 4.1), the pattern recurs frequently, or $\mathcal{T}$ is near the persistence threshold (4.4) with no slack for deliberation overhead.

However, deliberation remains essential when the situation is genuinely novel, the action space is large relative to model capacity (chess, strategic planning), the stakes are asymmetric (cost of error vastly exceeds cost of delay), or ρ is low (stable environment allows deliberation without mismatch accumulation).

Connection to Section II. For actuated agents (5.1), the lifted form $\pi(M_t, G_t)$ above unpacks: action conditions on the purposeful substate $G_t = (O_t, \Sigma_t)$ as well as on M_t, coupling all substates through action (5.3). The action-deliberation-exploration tradeoff (Section II gap) extends the implicit/explicit distinction to three modes: exploit (pursue O_t via Σ_t), explore (improve M_t), deliberate (revise Σ_t).

Domain instantiations:

Table 3.2

Domain	*Implicit action*	*Explicit deliberation*
Kalman + LQR	LQR control law from $\hat{x}_t$	— (separation principle)
RL	Greedy policy arg max $Q(s, a)$	MCTS, planning, rollouts
PID	$u = K_p e + K_i \int e + K_d \dot{e}$	— (no deliberation)
Boyd's OODA	IG&C (Orient→Act)	Explicit Decide step
Organism	Reflexes, habits	Deliberate planning
Organization	Standard procedures	Strategic planning
Software developer	Known patterns, familiar code	Reading docs, analyzing alternatives

3.4 *Definition:* Mismatch Signal

The discrepancy between the model's prediction and the actual observation — the formal expression of *aporia* (productive perplexity). This is the signal that drives all adaptation: the agent discovers that reality and model have diverged, and this discovery is generative.

Formal Expression.

Given model M_{t-1} and prior action a_{t-1}, the model generates a prediction:

Definition (mismatch-signal).

$$\hat{o}_t = \mathbb{E}[o_t \mid M_{t-1}, a_{t-1}] \tag{3.12}$$

The **mismatch signal** (prediction error):

$$\delta_t = o_t - \hat{o}_t \tag{3.13}$$

This is the primary definition, used in the mismatch dynamics (4.4, 4.3) and in the decomposition (3.5).

For models with probabilistic predictions, the mismatch generalizes to the **score-function mismatch**:

Definition (score-mismatch).

$$\tilde{\delta}_t = \nabla_M \log P(o_t \mid M_{t-1}, a_{t-1}) \tag{3.14}$$

which points in the direction the model should move to increase the likelihood of the actual observation. $\tilde{\delta}_t$ lives in the tangent space $T_M\mathcal{M}$, while δ_t lives in observation space $\mathcal{O}$. Under Gaussian models, they coincide up to scaling.

Epistemic Status.

This is *definitional*. Given any model that predicts (2.1) and any observation that arrives (1.3), their difference exists. The mismatch signal is not an additional assumption but a consequence of having a predictive model in an uncertain world. The score-function form is the natural generalization when $\mathcal{O}$ is not a vector space or when the model's predictive distribution is the natural object.

Discussion.

Units and normalization. When δ_t is in physical units (meters, dollars), the $\|\delta\|$ that enters the mismatch dynamics should be understood as the Mahalanobis distance: $\|\delta_t\|_\Sigma = \sqrt{\delta_t^T \Sigma^{-1} \delta_t}$ where Σ is the observation noise covariance. This maps physical prediction error to dimensionless surprise-equivalent units.

The zero-aporia ambiguity. $\delta_t \approx 0$ does NOT necessarily indicate model adequacy. It may mean: (a) the model genuinely reflects reality — *desirable*; (b) the agent is only observing aspects its model already explains, while remaining ignorant of aspects where the model is wrong — *confirmation bias*; or (c) the observation channel is too noisy to detect model errors — *architectural limitation*. Only (a) is desirable. An agent without aporia is an agent that has stopped adapting — but silence can mean peace or deafness. This ambiguity is why active testing — choosing actions to generate informative aporia — can be valuable (see 3.7 for the CIY framework).

The mismatch transform. The update rule (3.6) writes $M_t = M_{t-1} + \eta \cdot g(\delta_t)$, where the transform g maps from δ_t's space to the model's update space: $g : \mathcal{O} \to T_M\mathcal{M}$ for prediction errors; $g : T_M\mathcal{M} \to T_M\mathcal{M}$ for score-function mismatches.

3.5 *Result:* Mismatch Decomposition

Expected squared mismatch decomposes into reducible model error and irreducible observation noise. The model can improve the first term; the second is a property of the channel.

Formal Expression.

For any agent-environment pair within AAT's scope (1.5), when observation noise is non-degenerate or the model's predictive mean is misspecified:

Derived (result-mismatch-decomposition).

$$\mathbb{E}[\|\delta_t\|^2] = \underbrace{\mathbb{E}[\|\hat{o}_t - \bar{o}_t\|^2]}_{\text{model error (reducible)}} + \underbrace{\mathbb{E}[\mathrm{Var}(o_t \mid \Omega_t, a_{t-1})]}_{\text{observation noise (irreducible)}} > 0 \tag{3.15}$$

where $\bar{o}_t = \mathbb{E}[o_t \mid \Omega_t, a_{t-1}]$ is the true conditional mean.

Derivation.

1. By 1.5, $H(\Omega_t \mid \mathcal{C}_t) > 0$ — residual uncertainty persists.
2. By 2.1, the model generates predictions $\hat{o}_t = \mathbb{E}[o_t \mid M_{t-1}, a_{t-1}]$.
3. Decompose mismatch into model error and noise. The cross-term vanishes by the fresh-noise assumption (GA-1): ε_t is conditionally independent of $\mathcal{C}_{t-1}$ given (Ω_t, a_{t-1}). Condition on $(\Omega_t, a_{t-1}, \mathcal{C}_{t-1})$; then both $\bar{o}_t$ and $\hat{o}_t$ are fixed, and $\mathbb{E}[o_t - \bar{o}_t \mid \Omega_t, a_{t-1}, \mathcal{C}_{t-1}] = \mathbb{E}[o_t - \bar{o}_t \mid \Omega_t, a_{t-1}] = 0$ by definition of $\bar{o}_t$ and GA-1. The outer expectation gives zero. This is orthogonality (uncorrelated), not independence.
4. Term (ii) is positive when observation noise is non-degenerate. Term (i) is positive when the model's predictive mean differs from the true conditional mean. Either suffices.

Epistemic Status.

Exact under the fresh-noise assumption (observation noise ε_t conditionally independent of history given current state and action). This is the standard assumption in state-space models — noise is a property of the observation channel at the moment of observation. The decomposition is a mathematical identity (bias-variance decomposition applied to the prediction problem). The positivity of $\mathbb{E}[\|\delta_t\|^2]$ follows from either condition; both hold simultaneously in typical settings.

Discussion.

Reducible vs. irreducible. An agent that tries to eliminate *all* mismatch — including irreducible noise — will overfit: adjusting its model to explain noise, degrading future predictions. The update gain (3.6) implicitly separates signal from noise by weighting observations in proportion to their informativeness.

Connection to model sufficiency. When $S(M_t) < 1$ (2.3), the model has lost predictive information relative to the full history. Under an alignment assumption (the lost information affects the one-step conditional mean), this implies positive model error (term i). Without that alignment assumption, insufficiency still implies positive regret under proper scoring rules but not necessarily positive one-step mean error.

Mismatch is structurally persistent. In realistic AAT regimes, mismatch signals persist — they can be reduced but not eliminated when observation noise is non-degenerate. Deterministic, noiseless, perfectly specified systems are limiting edge cases, not the typical adaptive regime.

3.6 *Empirical:* Update Gain

The optimal weight an agent assigns to new observations when updating its model — the rate of *epistrophe* (turning toward reality). How much the agent should trust the incoming observation versus its own prior understanding.

Formal Expression.

Empirical Claim (uncertainty-ratio-principle).

$$\eta^* = \frac{U_M}{U_M + U_o} \tag{3.16}$$

where: - η^* is the optimal update gain (proportion of mismatch used to correct the model) - U_M is model uncertainty (predictive variance or entropy) - U_o is irreducible observation noise

The update rule takes the form:

Formulation.

$$M_t = M_{t-1} + \eta^* \cdot g(\delta_t) \tag{3.17}$$

where δ_t is the mismatch (3.4) and $g(\cdot)$ is a correction mapping from observation space to model update space.

Epistemic Status.

Derived under the **Fisher-local invariance regime** (): for any smooth log-likelihood admitting non-degenerate local quadratic expansion, the natural-gradient Bayesian posterior mean shift at first order in the step size is exactly $\Delta\theta = K \cdot \tilde{\nabla}$ with gain operator $K = (H_M + H_L)^{-1} H_L$ and scalar collapse $\eta^* = U_M/(U_M + U_o)$ along the natural-gradient direction (always in 1-D; under (PI)/Čencov in higher dimensions). Linear-Gaussian (Kalman) and conjugate-Bayesian instances are cases where the local quadratic expansion is *globally* exact, so the form holds without truncation. For general smooth models, the natural-gradient invariance theorem of Amari 1998 guarantees the form is exact at the local-tangent-plane Pythagorean projection level.

Outside the Fisher-local regime — non-quadratic losses, non-conjugate priors, structurally non-Gaussian uncertainty (heavy tails, multimodality) — the dependence is *robust qualitative*: the **direction** is preserved (gain rises with model uncertainty, falls with observation noise) and the first-order form is recovered locally; what need not hold is global quantitative fidelity. The qualitative direction is the load-bearing claim for downstream tempo and persistence machinery; the Fisher-local exact form is the load-bearing claim for the Kalman, conjugate, and natural-gradient instantiations.

Discussion.

Limiting behavior. When $U_M \gg U_o$ (high model uncertainty — e.g., after initialization or structural adaptation), $\eta^* \to 1$: trust the observation. When $U_M \ll U_o$ (confident model, noisy channel), $\eta^* \to 0$: trust the model. The gain determines how strongly the agent corrects toward reality on each update.

Resolving Epistemic Opacity. The optimal gain equation requires the agent to know U_o, which seems to violate the epistemic opacity axiom established in `#def-observation-function` (the agent does not know the true noise distribution ε_t). This tension is resolved dynamically: the agent *estimates* U_o (and U_M) from the observable statistics of its own mismatch sequence (innovations), treating the gain itself as an endogenous state variable. See `#deriv-adaptive-gain-dynamics` for the proof of how this meta-adaptation maintains Lyapunov stability without violating opacity.

Gain collapse — epistrophe failure. When the agent incorrectly estimates $U_M \to 0$ (spurious confidence) or $U_o \to \infty$ (spurious distrust of sensors), $\eta^* \to 0$ and epistrophe ceases. Aporia still arrives — the mismatch signal is still generated — but the agent no longer turns toward it. Mismatches are ignored, producing confirmation bias or a decoupled reality model. The cycle runs but the corrective phase is hollow.

Multi-dimensional generalization. In vector-valued systems, U_M and U_o are covariance matrices and η^* becomes a gain matrix (as in the Kalman filter). The scalar form captures the essential structure.

Connection to adaptive tempo. The update gain is one factor in the agent's adaptive tempo (3.8): $\mathcal{T} = \nu \cdot \eta^*$. Frequent aisthesis (high ν) is useless if epistrophe extracts no information (low η^*). Gain measures the *quality* of the cycle's corrective phase; event rate measures its *speed*.

Gain dynamics. The optimal gain changes over time following predictable patterns:

- *Convergence*: As the model accumulates information, U_M decreases, so $\eta^* \to 0$. The model becomes increasingly resistant to individual observations. This IS Kalman filter convergence, Bayesian posterior concentration, and RL learning rate annealing.
- *Reset after structural change*: When the environment changes in ways the model cannot track incrementally (4.5), U_M should spike — the model "admits" its uncertainty. The gain increases, enabling rapid re-learning. An agent whose gain does NOT reset after structural change will continue trusting a stale model — Boyd's "incestuous amplification" and the cause of brittle failure in non-stationary environments.

Overfitting as gain miscalibration. From 3.5: $\mathbb{E}[\|\delta_t\|^2]$ = model error + irreducible noise. An agent with η too high adjusts its model to explain observation noise, increasing model error on future predictions. An agent with η too low fails to correct genuine model errors. The optimal gain implicitly separates signal from noise by weighting observations in proportion to their informativeness — exactly what $U_M/(U_M + U_o)$ achieves when U_o captures the irreducible noise.

Representation note. The additive form operates in a *representation space* appropriate to the model. For Bayesian posteriors (where update is multiplicative: $P(\theta \mid D) \propto P(D \mid \theta)P(\theta)$), the additive rule operates in log-probability or natural parameter space. For models on constrained manifolds (probability simplices, rotation groups), the update must be projected onto the manifold. The claim is not that all updates are literally additive in native parameterization, but that they have the structure "current state + gain × transformed mismatch" in an appropriate coordinate system.

Domain validation:

Table 3.3

Domain	*Gain form*	*Mapping quality*
Kalman filter	$K_t = P_{t\|t-1}H^T(HP_{t\|t-1}H^T + R)^{-1}$	**Exact.** Scalar case is exactly $U_M/(U_M + U_o)$.
Conjugate Bayesian	Posterior weight $n/(n+\kappa)$ cumulative; incremental $1/(n+\kappa)$	**Exact** for conjugate families. Incremental gain decreases as data accumulates.
RL (Q-learning)	Fixed learning rate α	**Approximate.** α is a degenerate constant gain — does not adapt to uncertainty. Advanced methods (Bayesian RL, Adam) converge toward the optimal form.
PID control	Fixed gains (K_p, K_i, K_d)	**Simplified.** Gains set at design time. Adaptive PID and MPC move toward the full framework.
Software developer	Implicit trust weighting of information sources	**Structural analogy.** New developer (high U_M) trusts observations heavily; experienced developer (low U_M) trusts their model. Gain reset after major refactoring.

Simulation validation. Numerical experiments (track-b, Variant E) validated the uncertainty ratio principle under observation noise. Riccati-optimal gain reduced steady-state mismatch by 52% compared to fixed gain when observation noise was moderate. The optimal gain also proved critical in adversarial settings: under heavy observation noise, optimal gain preserved more than double the adversarial tempo advantage exponent (0.40 vs 0.18) compared to fixed gain.

Open questions:

1. *Non-parametric models*: For neural networks without well-defined scalar U_M, how should it be computed? Ensemble methods, dropout-based uncertainty, and Bayesian neural networks are all approximations.

2. *Matrix vs scalar gain*: In high-dimensional systems, the gain is a matrix (Kalman) or per-parameter (Adam). The cross-dimensional structure (covariance) adds complexity. The scalar captures the principle; the matrix captures the full optimization.

3.7 *Definition:* Causal Information Yield

Actions don't merely select among outcomes — they produce characteristically different outcome distributions depending on the causal structure. Causal information yield (CIY) quantifies the **action-distinguishability** of an action: how different its outcome distribution is from what alternative actions would produce.

Formal Expression.

The **canonical CIY** of action a given model state M:

Definition (causal-information-yield).

$$\mathrm{CIY}(a; M) = \mathbb{E}_{a' \sim q(\cdot|M)}[D_{\mathrm{KL}}(P(o \mid do(a), M) \,\|\, P(o \mid do(a'), M))] \tag{3.18}$$

where $q(\cdot \mid M)$ is a reference distribution over comparator actions (uniform, policy-induced, or task-specific). This measures how strongly the action changes the interventional distribution of outcomes relative to alternatives.

The $do(\cdot)$ operator is Pearl's standard intervention notation (Pearl 2009, *Causality*, 2nd ed., Cambridge; Bareinboim, Correa, Ibeling & Icard 2022); the AAT recapitulation lives at 6.1 in Part II Ch.2, where the framework deploys the hierarchy operationally. CIY ≥ 0 by construction (expectation of KL divergences). CIY $= 0$ for a passive observer or an agent whose actions don't affect outcome distributions. CIY > 0 when actions causally alter what is observed — exactly what distinguishes Pearl's Level 2 (interventional) from Level 1 (associational) epistemic access.

Epistemic Status.

The CIY *definition* is well-grounded in causal inference theory. The quantity itself is standard — an expected KL divergence under the do-calculus. The interpretive claim — that CIY measures action-distinguishability rather than expected information gain — is exact (it follows from the definition). The relationship between CIY and learning value (see Discussion) is discussion-grade: the λ-weighted approximation to EIG is heuristic, not derived.

Discussion.

CIY measures distinguishability, not learning value. CIY as defined is the expected KL divergence between outcome distributions — how different the outcomes of a are from the outcomes of typical alternatives. This is **action-distinguishability**, not **expected information gain** (EIG). The distinction matters: an action can have high CIY even when the agent already knows the outcome distributions perfectly (the distributions ARE different, but the agent learns nothing new by confirming what it already knows). High CIY is *necessary* for learning (indistinguishable actions can't teach anything) but not *sufficient* (distinguishable actions only teach when the agent is uncertain about the distinction).

The two quantities approximately coincide when U_M is high — when the agent doesn't know the outcome distributions, high-CIY actions also have high EIG because observing a characteristically different outcome updates the agent's beliefs about the causal structure. They diverge when U_M is low — a confident agent gains nothing from taking a distinguishable action it already understands.

The $\lambda(M_t)$ weighting in the unified policy objective (6.5) partially compensates: when U_M is low, $\lambda \to 0$, suppressing the CIY term regardless of its magnitude. This makes the exploration term behave more like EIG — suppressing exploration when the agent is already certain. The compensation is heuristic (the λ form is not derived). For the current theory, CIY serves as a tractable surrogate for EIG, with the λ weighting providing the uncertainty-gating that makes the surrogate reasonable.

Open direction: proper EIG within AAT. Replacing CIY with a proper expected information gain quantity — $\text{EIG}(a; M) = I(o; \theta \mid do(a), M)$ where θ parameterizes the model — would be a stronger foundation for the exploration term, particularly in domains where the agent must decide between actions that are all highly distinguishable but differ in what they teach. Under certain scopes (intervention-rich domains with well-parameterized models), the EIG formulation might yield sharper exploration strategies — preferring actions that resolve the *most uncertain* causal links rather than the most distinguishable ones. Whether this yields operationally significant improvements over the λ-weighted CIY surrogate is an empirical question. The CIY formulation has the advantage of being computable from the agent's current model alone (it doesn't require reasoning about model uncertainty); EIG requires a meta-model of what the agent doesn't know.

Dependence on the reference distribution q. The quantitative CIY value depends on the choice of q, which is a significant degree of freedom. A uniform q treats all alternatives equally; a policy-induced q emphasizes alternatives the agent would consider. AAT adopts the policy-induced q as default: $q(\cdot \mid M) = \pi(\cdot \mid M)$, yielding CIY as "how different is this action's outcome from what I'd typically see?" CIY values are not comparable across different q choices.

Query actions: accessing external models. A qualitatively distinct class of actions: querying another agent's model. When a reliable source exists (expert, database, documentation, well-trained LLM), "ask a well-formed question" can yield information equivalent to thousands of probe-observe cycles. The source's model has already performed the compression work (2.2) — the response transfers the *output* of compression rather than requiring the agent to reconstruct it.

Key properties of query actions: - **Information density**: Single well-targeted query can carry CIY orders of magnitude higher than individual environment probes - **Trust-dependent gain**: Update from query depends on the agent's model of the source's reliability and alignment, not on observation channel noise (12.4) - **Pre-compressed information**: Responses arrive already compressed in the source's representational framework, introducing a translation cost when frameworks don't align - **Structural adaptation via external models**: Encountering another agent's model can trigger structural change (4.5) — incorporating external representational structure rather than building it de novo ("grafting")

When high-CIY query channels are available, the unified policy objective (6.5) favors query actions over direct probes, particularly when U_M is high, a trusted source exists, query cost is low, and the needed information is about *structure* rather than the agent's specific situation.

The adversarial mirror: deception and model corruption. The same channel that enables cooperative knowledge transfer can be exploited to degrade the opponent's model. A deceptive response yields positive CIY in the strict information-theoretic sense, but the content drives model-reality mismatch *upward*. The update gain η^* for the victim depends on trust; successful deception exploits high trust to inject a large, misdirected update. In the Lyapunov framework (4.3), this is adversarial disturbance injected through the observation channel, with coupling coefficient γ_A determined by the victim's trust level and exposure. See 12.4 for the formal treatment of trust-dependent gain, and 13.2 for the Lyapunov formalization. Distributed tempo, topology analysis, and game-theoretic integration are Section III content not yet fully extracted (source material in `src/old-tf-appendix-f-multi-agent.md`).

3.8 *Definition:* Adaptive Tempo

The effective rate at which an agent acquires useful information from its environment — the product of observation frequency and update quality across all channels.

Formal Expression.

Definition (adaptive-tempo).

$$\mathcal{T} = \sum_k \nu^{(k)} \cdot \eta^{(k)*} \tag{3.19}$$

where: - k indexes the agent's distinct observation channels - $\nu^{(k)}$ is the event rate on channel k - $\eta^{(k)*}$ is the optimal update gain on channel k (3.6)

Single-channel special case: $\mathcal{T} = \nu \cdot \eta^*$.

Tensor extension under Fisher-local invariance regime.

Under the Fisher-local invariance regime (), the optimal update gain on channel k is matrix-valued: $K^{(k)} = (H_M + H_L^{(k)})^{-1} H_L^{(k)}$, with $H_M = U_M^{-1}$ the prior precision and $H_L^{(k)} = (U_o^{(k)})^{-1}$ the channel-k observed Fisher information. The tensor adaptive tempo is then

Definition (tensor-adaptive-tempo).

$$\mathcal{T} = \sum_k \nu^{(k)} \cdot K^{(k)} \tag{3.20}$$

— matrix-valued, with per-direction rates given by the eigenvalues of $\sum_k \nu^{(k)} K^{(k)}$ in the appropriate basis. The scalar form $\mathcal{T} = \sum_k \nu^{(k)} \cdot \eta^{(k)*}$ is recovered in the **shared-eigenbasis collapse**: when all $H_M, \{H_L^{(k)}\}$ commute (always in 1-D; under (PI)/Čencov along the natural-gradient direction in higher dimensions), each $K^{(k)}$ acts as the eigenvalue $\eta^{(k)*} = U_M/(U_M + U_o^{(k)})$ on the shared natural-gradient direction and the matrix sum collapses to a scalar.

The matrix gain operator $K^{(k)}$ is the per-coordinate primitive: in anisotropic regimes where the prior and likelihoods do not share an eigenbasis (or where different channels pin down different directions), the tensor form preserves the per-direction information that the scalar form averages away.

Epistemic Status.

This is a *definition*. It names the quantity that characterizes an agent's total corrective capacity, combining loop speed (ν) and epistemic quality (η^*). The definition itself is not a truth-claim; the substantive claims are in the results that use it (4.4, 13.4).

Scope of scalar vs. tensor forms. The scalar form is exact in the isotropic / shared-eigenbasis / nonredundant-channel case and is what downstream results currently invoke. The tensor form is the natural object under anisotropic gains, Fisher-whitened updates (), LMI causal-IB (), and per-dimension persistence (14.5) — regimes where scalar tempo overestimates effective adaptation along weak dimensions. Downstream results that invoke scalar $\mathcal{T}$ implicitly assume scalar / isotropic / nonredundant-channel scope; promoting them to the tensor form under the appropriate anisotropic regime is a follow-on cycle item flagged in `TODO.md`.

Discussion.

Speed-quality substitutability. An agent can achieve the same tempo via a fast noisy loop (high ν, low η^*) or a slower calibrated one (low ν, high η^*). The product structure means improvements to *both* factors compound multiplicatively.

Observation noise gating. Because $\eta^* = U_M/(U_M + U_o)$, high observation noise (U_o) depresses gain and collapses tempo regardless of loop speed. You cannot outrun a bad observation channel by iterating faster. This grounds Boyd's emphasis on Orient quality over raw OODA speed.

Centrality. Tempo is AAT's core capacity metric. It appears on the left side of the persistence condition (4.4), determines adversarial advantage (13.4), and connects to code quality as observation infrastructure (`#der-code-quality-as-observation-infrastructure` — cross-component reference, see `02-tst-core/`) in the software domain. The strategic analog $\mathcal{T}_\Sigma$ (8.8) extends the same structure to strategy-edge revision, with the key difference that strategic edge rates are endogenous (depend on action policy and upstream success).

Temporal nesting. Adaptive processes stratify by timescale, with convergence constraints between levels (4.6).

Mismatch dynamics. The evolution of mismatch over time is governed by the balance between correction (via tempo) and disturbance (ρ) (3.9).

Channel independence assumption. The additive formula assumes informationally independent channels — each channel contributes non-redundant correction capacity. When channels are correlated (overlapping sensors, repeated teammate reports, redundant telemetry), the additive formula *overcounts* effective tempo. The correct tempo satisfies:

$$\mathcal{T} \leq \sum_k \nu^{(k)} \cdot \eta^{(k)*} \tag{3.21}$$

with equality iff channels are informationally independent. The gap is the *redundancy penalty* — the effective correction capacity lost to overlapping information. For two correlated channels, the penalty involves the mutual information $I(e^{(1)}; e^{(2)} \mid M_{\tau^-})$ between their event streams conditioned on the current model. Since tempo is the core capacity variable (appearing in the persistence condition, adversarial dynamics, and composition), this overcounting inflates margins wherever channel independence fails. The additive formula remains an upper bound and is exact when channels measure genuinely different aspects of the environment. Multi-agent composition (13.1) inherits this limitation: the communication tempo contribution is additive in the same sense and overcounts when different allies report correlated information.

Scalar vs. vector tempo. The scalar $\mathcal{T}$ assumes isotropic correction capacity. When the agent corrects some dimensions faster than others, scalar tempo overestimates effective adaptation along weak dimensions. *[Empirical Claim]* Simulation confirms: in an anisotropic 3D system (gain varying 5:1), scalar $\rho/\mathcal{T}$ overestimated by 72%, with the weak dimension accounting for 84% of total mismatch (). The correct formulation is per-dimension: $\mathcal{T}_k > \rho_k/\delta_{\text{critical},k}$ (14.5). The tensor extension above () gives the per-coordinate primitive $K^{(k)}$ that the per-dimension persistence result invokes — the matrix gain operator on each channel — making the per-dimension formulation a direct consequence of the tensor tempo definition rather than a separate generalization. Under cross-dimensional correction (off-diagonal $\mathcal{T}$ in the coordinate basis of D_δ), the matrix-Loewner persistence condition `#deriv-matrix-persistence-condition` is the canonical form: $\Sigma_\infty \prec D_\delta = \text{diag}(\delta^2_{\text{critical},k})$ in the strict positive-definite order, with Σ_∞ solving the continuous Lyapunov equation $\mathcal{T}\Sigma_\infty + \Sigma_\infty\mathcal{T}^T = \Sigma_w$. The per-coordinate form is its diagonal-axis-aligned special case; matrix-Loewner is strictly sharper (the per-coordinate form is *unsafe* when $\mathcal{T}$'s eigenbasis misaligns with the coordinate axes — `#deriv-matrix-persistence-condition` §"Where per-coordinate is unsafe").

3.9 *Hypothesis:* Mismatch Dynamics

The evolution of model-reality mismatch over time is governed by the balance between the agent's corrective capacity (tempo) and the rate of environmental change (disturbance). The linear ODE is a first-order approximation; the general nonlinear case is handled by the sector-condition framework (4.3).

Formal Expression.

Hypothesis (mismatch-dynamics).

$$\frac{d\|\delta\|}{dt} = -\mathcal{T} \cdot \|\delta\| + \rho(t) \tag{3.22}$$

where: - $\mathcal{T} \cdot \|\delta\|$ is the rate at which the agent corrects mismatch (proportional to both tempo and current mismatch) - $\rho(t)$ is the **environment change rate** — the rate at which new mismatch is introduced by changes in Ω

Steady state, Model D (deterministic bounded disturbance, $\|w(t)\| \leq \rho$):

Setting $d\|\delta\|/dt = 0$:

Derived (from linear hypothesis, deterministic).

$$\|\delta\|_{ss} = \frac{\rho}{\mathcal{T}} \tag{3.23}$$

Steady-state mismatch is the ratio of how fast the environment changes to how fast the agent adapts.

Steady state, Model S (stochastic zero-mean disturbance, $d\delta = -\mathcal{T}\delta\, dt + \sigma\_w\, dW\_t$):

Derived (from Itô-Lyapunov analysis — see Prop A.1S in #deriv-sector-condition).

$$\|\delta\|_{\text{rms}} = \frac{\sigma_w}{\sqrt{2\mathcal{T}}} \tag{3.24}$$

(scalar case, $n = 1$; general: $\sigma_w\sqrt{n/(2\mathcal{T})}$). Steady-state mismatch scales as the square root of the disturbance-to-correction ratio, not the ratio itself. The $1/\sqrt{\mathcal{T}}$ scaling (vs. $1/\mathcal{T}$ for Model D) means correction is less effective against noise than against drift.

Transient solution (Model D):

$$\|\delta(t)\| = \|\delta_0\| e^{-\mathcal{T}t} + \frac{\rho}{\mathcal{T}}(1 - e^{-\mathcal{T}t}) \tag{3.25}$$

Mismatch decays exponentially from initial conditions toward the steady state.

Epistemic Status.

Heuristic. This is explicitly a first-order linear approximation. The qualitative behavior (bounded mismatch, steady-state ratio, exponential convergence) is robust across correction function forms. The quantitative predictions (exact steady-state value, convergence rate, the squared adversarial scaling law) are specific to the linear case. The general nonlinear treatment (4.3) replaces the linear correction term with a sector-bounded correction function and proves persistence without committing to a specific functional form.

Bridging assumption (discrete to continuous). This ODE is a fluid-limit approximation of the discrete event-driven dynamics (3.1). The fluid limit is formally justified by K: for Model D (deterministic), the discrete and continuous steady states are identical (zero gap); for Model S (stochastic), the variance gap is $O(\eta^* c_{\max})$, quantitatively small whenever $\eta^* c_{\max} \ll 1$. The approximation is least accurate during initial transients when η^* is large, but this phase is short-lived and the transient error is bounded by $O(\eta^* c_{\max}/\nu^{1/2})$ under Lipschitz regularity.

Discussion.

Speed-quality substitutability. From $\mathcal{T} = \nu \cdot \eta^*$ (single-channel case): doubling event rate ν has the same effect on $\|\delta\|_{ss}$ as doubling update quality η^*. They are multiplicative when both improve: 50% improvement in each yields $1.5 \times 1.5 = 2.25\times$, not $3\times$. This is the formal analog of Boyd's insight that Orient quality often matters more than raw OODA speed — the same structural observation (quality and speed are substitutable, quality often dominates) appears in the model.

The persistence threshold. From the steady-state: $\|\delta\|_{ss} < \|\delta_{\text{critical}}\|$ iff $\mathcal{T} > \rho/\|\delta_{\text{critical}}\|$ (4.4). Below this threshold, the model cannot support effective action. The same structural pattern — correction capacity falling below disturbance rate — appears across domains: extinction (environment changes faster than organism adapts), organizational failure (market moves faster than company learns), control instability (disturbances exceed correction capacity), cognitive overload (information arrives faster than processing). The persistence condition captures the common structure; whether it captures the dominant mechanism in each domain is an empirical question.

Nonlinear reality. The true correction dynamics are almost certainly nonlinear: - *Saturation at large* $\|\delta\|$: correction mechanism overwhelmed, so correction is slower than linear for large errors. Makes the persistence threshold harder to satisfy. - *Threshold effects*: small mismatches go uncorrected ($F \approx 0$ for $\|\delta\| < \varepsilon$), creating a dead zone. - *Structural breakdown*: beyond some critical $\|\delta\|$, correction drops to zero because the model class is no longer appropriate (4.5).

These nonlinearities are exactly what the sector-condition framework (4.3) handles.

Adversarial coupling. When two agents are coupled (A's actions increase B's disturbance): $\rho_B = \rho_{B,\text{base}} + \gamma_A \cdot \mathcal{T}_A$. The steady-state mismatch ratio scales superlinearly with the tempo ratio, but the exponent depends on the disturbance model. Under Model D (deterministic drift, coupling-dominant): $(\mathcal{T}_A/\mathcal{T}_B)^2$ — the squared law, derived from the $1/\mathcal{T}$ steady-state scaling. Under Model S (stochastic noise, coupling-dominant): $(\mathcal{T}_A/\mathcal{T}_B)^{3/2}$ — derived from the $1/\sqrt{\mathcal{T}}$ steady-state scaling. See 13.4.

Chapter 4

Persistence and Structural Limits

The cycle runs; the question is whether it can keep up. This chapter develops the central inequality of AAT — the persistence condition — together with what happens at and beyond its threshold: where it comes from, when the machinery can be expected to satisfy it, and what fails when it can't.

Chapter 3 ended with a preview. Under linear correction and bounded disturbance, the steady-state mismatch is $\rho/\mathcal{T}$ — disturbance over tempo. Mismatch stays bounded when the agent corrects faster than reality drifts. That was offered as a heuristic with a linear-ODE flavor. Chapter 4 makes it rigorous, generalizes it, and shows what it implies.

The first move is to replace "linear correction" with something more honest. Real correction mechanisms saturate at large mismatch, threshold at small mismatch, and break down entirely when the agent's model class is wrong for the situation. The sector condition — that the correction function points inward with at least baseline efficiency α within an operating region R — captures the qualitative essence of correction without committing to a specific functional form. This is one of the chapter's more satisfying pieces of theoretical hygiene: a condition flexible enough to admit saturating, sigmoid, threshold, and PID corrections under one Lyapunov argument, while still saying something concrete enough to derive a threshold from. Under that condition, the analysis gives

$$\alpha > \rho/R \tag{4.1}$$

three quantities, one threshold, mechanism underneath. The agent persists when its baseline correction efficiency exceeds the ratio of disturbance rate to operating reserve. Below the threshold, mismatch grows without effective bound (up to R, the sector-condition region). The result is qualitative: persistence is not a degree, it's a regime. The inequality is genuinely domain-agnostic — its four variables map to biological adaptation (extinction is what happens when environmental change rate outpaces evolutionary correction), to organizational survival (the market shifts faster than the firm learns), to control instability, to cognitive overload. The framework identifies a structural pattern across these domains rather than a model of any one of them; whether the specific mechanism is the dominant cause in any given case is empirical, but the structure recurs because the math does.

The persistence condition has a thermodynamic shadow worth surfacing here, since the chapter derives it explicitly in E: under stochastic disturbance, maintaining the sector-persistence bound requires sustained Shannon information acquisition at rate $\dot{R} \geq n\alpha/2$ nats per unit time, with Kalman-Bucy saturating the bound. Survival, in this framework, is not a state you achieve once — it is a sustained burn rate. An agent cut off from informative observation begins to drift the moment the channel closes, and the rate of drift is the disturbance rate, independent of how good the agent was at adapting before. Two agents with identical persistence guarantees can face wildly different sustained demands. The threshold says whether persistence is *possible*; the information-rate corollary says what it *costs*.

But this isn't quite the operational form most domains care about. There is a second condition — task adequacy — that the steady-state mismatch be small enough for the agent's actions to remain useful. An agent can satisfy the structural condition (the machinery works) but fail task adequacy (the machinery doesn't work *well enough* for what the agent is trying to do). The two conditions have different remedies, and the split between them is the kind of move that's invisible until you see it stated and obvious in retrospect. A Kalman filter on a moving target satisfies structural persistence whenever it's stable, but task adequacy depends on what counts as "tracking" for the application. A software team can be structurally persistent but task-inadequate. The structural condition is what AAT derives; the task-adequacy threshold is a domain parameter — and conflating the two produces category errors in domain transfer.

The other move I want to flag is upstream. Where does α actually come from? Earlier in the framework, α shows up as a sector-condition parameter — a structural property of the correction function, taken as given. The gain-sector bridge (4.2) demotes it from postulate to derivation. For agents with directional fidelity in their update rule — including all optimal Bayesian updaters and all gradient agents on (locally) strongly convex losses — $\alpha = \eta^* \cdot c_{\min}$, where η^* is the gain principle from Chapter 3 and $c_{\min}$ is the worst-case directional fidelity of the update. The sector condition isn't a soft global postulate at all; for the cases AAT cares about most, it's a property of the gain. The five failure modes that break this derivation (directional infidelity, gain collapse, nonlinear saturation, unobservable directions, model misspecification) are named explicitly, and each connects back to a concrete mechanism rather than being left as residual error. This is the kind of move that makes the framework feel earned rather than assumed.

The chapter closes with two integrative results. *Structural adaptation necessity*: when model-class fitness $\mathcal{F}(\mathcal{M})$ is too low, no parametric update within the class can close the mismatch floor — the agent must change classes, not parameters. This is the consequence whose seed was planted in Chapter 2, now grown to a derived result. *Temporal nesting*: adaptive processes stratify by timescale, with each level operating on the quasi-steady-state output of the level below. Parametric update is fast; structural adaptation is slow; the convergence constraint between adjacent timescales means a slower process acting on transients of a faster one produces oscillation. This is standard singular-perturbation reasoning applied here to multi-level adaptation, but the consequence — that the timescale ratios are themselves constrained — is what licenses the abstraction throughout the rest of the framework.

A scope coda closes the chapter and Part I. AAT applies to agents on singular causal trajectories. The chronica $\mathcal{C}_t$ is not forkable — duplicating M_t and exposing the copies to different futures produces two agents on divergent trajectories, neither a sufficient statistic for the other's path. Identity in AAT lives in the trajectory, not in the state. This sits at the end of Chapter 4 because its dependencies (chronica from Chapter 1 + sufficiency from Chapter 2) span the section, but its substance is its own — and it's the bridge to the logogenic-agents and ELI work in Volumes 03 and 04, where the singular-trajectory commitment becomes load-bearing.

The flow of the chapter: deliberation cost (4.1) → gain-to-sector bridge (4.2) → sector-condition stability and the persistence inequality (4.3, 4.4) → structural adaptation when parametric update fails (4.5) → temporal nesting (4.6) → identity scope (4.7). The chapter is where Part I's machinery resolves into operational results.

- This is a chapter-introduction segment; it bridges Chapter 3's preview of the central inequality to Chapter 4's rigorous treatment and its consequences. It carries no formal claim of its own.
- Three moves the intro tries to surface as the chapter's most interesting content: (1) the sector condition's generalization of the linear case, preserving qualitative content without committing to functional form; (2) the structural-vs-task-adequacy split, which is non-obvious and load-bearing for domain transfer; (3) the gain-sector bridge, which derives α from the gain principle and demotes an opaque postulate to a property.
- The scope-coda paragraph addresses 4.7, which sits at the end of Part I Ch.4 because its dependencies span chapters. The intro names this rather than letting the segment appear unmotivated.
- The closing "where Part I's machinery resolves into operational results" line is meant to register the chapter's role as climax without overstating. The persistence condition is the single most-cited result in the framework downstream; Part I works because this chapter works.

4.1 *Derived:* Deliberation Cost

Explicit deliberation improves action quality by using the model for internal simulation before acting — pausing praxis to improve upcoming epistrophe. But deliberation takes time, and during that time aporia accumulates (the environment continues to evolve while the agent is not correcting). Deliberation is justified when the improvement in epistrophe quality exceeds the aporia accumulated during the pause.

Formal Expression.

Assumption (local deliberation drift):

During a deliberation pause of duration $\Delta\tau$, mismatch increases at an approximately constant local rate ρ_{delib}:

Assumption (deliberation-drift).

$$\Delta\|\delta\|_{\text{deliberation}} \approx \rho_{\text{delib}} \cdot \Delta\tau \tag{4.2}$$

This is a short-horizon assumption about inaction windows, not a full global dynamics model. It is weaker than the mismatch ODE and can be estimated directly from pause windows in empirical traces.

Proposition (deliberation threshold):

Deliberation of duration $\Delta\tau$ is net-beneficial when:

Derived (Conditional on deliberation-drift assumption).

$$\Delta\eta^*(\Delta\tau) \cdot \|\delta_{\text{post}}\| > \rho_{\text{delib}} \cdot \Delta\tau \tag{4.3}$$

where $\Delta\eta^*(\Delta\tau)$ is the improvement in post-deliberation update gain and $\|\delta_{\text{post}}\|$ is the mismatch magnitude the agent will face when it resumes acting.

Derivation.

1. Without deliberation, the agent acts immediately at current tempo $\mathcal{T}_0 = \nu \cdot \eta_0^*$.
2. With deliberation of duration $\Delta\tau$, the agent pauses, then acts with improved gain $\eta_0^* + \Delta\eta^*$. But during the pause, mismatch has grown by $\rho_{\text{delib}} \cdot \Delta\tau$.
3. The net mismatch reduction from acting after deliberation versus acting immediately: $\text{Net} = \Delta\eta^* \cdot \|\delta_{\text{post}}\| - \rho_{\text{delib}} \cdot \Delta\tau$.
4. Deliberation is justified iff $\text{Net} > 0$. □

Optimal deliberation duration (under diminishing returns):

Derived (Conditional on diminishing-returns + deliberation-drift).

$$\Delta\tau^* = \arg\max_{\Delta\tau} \left[\Delta\eta^*(\Delta\tau) \cdot \|\delta_{\text{post}}\| - \rho_{\text{delib}} \cdot \Delta\tau\right] \tag{4.4}$$

where $\|\delta_{\text{post}}\|$ is treated as a parameter estimated by the agent before deliberation begins (not optimized over — the agent estimates the mismatch it will face, then decides how long to deliberate). Under this approximation, the first-order condition is: $\frac{\partial \Delta\eta^*}{\partial \Delta\tau} \cdot \|\delta_{\text{post}}\| = \rho_{\text{delib}}$. Stop deliberating when the marginal improvement rate drops below the mismatch drift rate (normalized by post-deliberation mismatch). When the dependence $\|\delta_{\text{post}}\| = \|\delta_0\| + \rho_{\text{delib}} \cdot \Delta\tau$ is included in the optimization, the exact FOC acquires a correction factor $(1 - \Delta\eta^*)$ on the cost side; this is negligible when $\Delta\eta^* \ll 1$ (the typical case — deliberation produces small gain improvements).

Epistemic Status.

Conditional on the local deliberation-drift assumption. The threshold condition is derived given the assumption; the assumption itself is a local approximation validated by consistency with the global mismatch dynamics (4.4). The result captures the *epistemic* benefit of deliberation (improving η^*); in practice, deliberation also provides a direct *action-value* benefit (choosing better actions that alter the environment trajectory), which operates through ρ reduction and immediate reward — a fuller formalization would incorporate the unified policy objective (3.7) at significantly more complexity.

Discussion.

High-ρ_{delib} environments penalize deliberation. When the environment changes rapidly during pause windows, the cost term grows quickly. Only very short deliberation with large $\Delta\eta^*$ can justify the pause. The model captures the same tradeoff Boyd emphasized: in fast-tempo adversarial environments, over-deliberation is fatal not because thinking is bad, but because the environment moves during the thinking. Whether the specific mechanism (mismatch drift during pause) is the dominant real-world effect is an empirical question.

Diminishing returns. In most models, $\Delta\eta^*(\Delta\tau)$ exhibits diminishing returns — the first moments of simulation yield the largest improvement. Combined with the linear cost $\rho_{\text{delib}} \cdot \Delta\tau$, this implies a finite optimal deliberation duration. Past that point, additional thinking is net-harmful.

Implicit action as the high-tempo limit. As $\rho_{\text{delib}} \to \infty$ or $\Delta\tau^* \to 0$: the optimal strategy converges to zero deliberation — pure implicit action (3.3). This provides a mathematical basis for why high-tempo environments favor action fluency: the cost of deliberation exceeds its benefit when $\Delta\eta^*$ is small (action-selection is already fluent) or ρ_{delib} is large.

Deliberation as an investment. When ρ_{delib} is low (stable environment) or $\|\delta_{\text{post}}\|$ is large (significant model-reality gap), deliberation pays off. The conditions favoring deliberation — stable environment, large mismatch — resemble the high-stakes, low-urgency scenarios where deliberative reasoning (System 2) is advantageous in dual-process theories. The structural parallel is suggestive; whether the cost-benefit mechanism is the same one governing System 1/System 2 selection is an open question.

The circularity of $\|\delta_{\text{post}}\|$. Evaluating the threshold requires the agent to *predict* post-deliberation mismatch using its current model — the same model deliberation is meant to improve. This circularity is typically benign: $\|\delta_{\text{post}}\|$ is bounded below by $\rho_{\text{delib}} \cdot \Delta\tau$ and above by current mismatch plus that accumulation. An agent that underestimates its mismatch will under-deliberate; one that overestimates will over-deliberate. The bias is self-correcting through the feedback loop. The threshold is best understood as a *design criterion*, not a real-time decision procedure.

Resource costs beyond time. Real agents also incur computational and energetic costs: internal simulation burns calories, compute cycles, or opportunity cost of not processing new observations. These are additive: $\Delta\eta^*(\Delta\tau) \cdot \|\delta_{\text{post}}\| > \rho_{\text{delib}} \cdot \Delta\tau + C(\Delta\tau)$. In high-$\rho_{\text{delib}}$ environments the temporal cost dominates; in low-ρ_{delib} environments, resource costs may be the binding constraint.

Structural adaptation as an analogy. Structural adaptation (4.5) superficially resembles deliberation with a massive $\Delta\tau$: the agent's parametric loop is partially suspended while it searches for a new model class, incurring a large mismatch debt $\rho_{\text{delib}} \cdot \Delta\tau$. However, this is an informal analogy, not a consequence of the deliberation-cost formalism. Deliberation as formalized here improves η^* *within a fixed model class*; structural adaptation changes the model class itself, which is a mechanistically different operation (4.5). The cost-benefit structure may be similar in form, but the quantities involved ($\mathcal{F}(\mathcal{M})$ vs. η^*, model-class search vs. gain improvement) are distinct.

Connection to temporal nesting. Deliberation is a nested loop: internal simulation running at rate ν_{internal} within the external action loop at rate ν_{external}. The convergence constraint applies: the internal loop must approximately converge before the external loop acts on its output.

Connection to Section II. For actuated agents, the deliberation tradeoff extends to three modes: exploit (O_t via Σ_t), explore (improve M_t), and deliberate (revise Σ_t). The three-way allocation (9.2) extends this segment's threshold by adding a strategic benefit term ΔV_Σ; the extended threshold is the one genuinely derived piece. The broader three-way framing is discussion-grade — simulation shows deliberation is rarely chosen by an oracle in simple settings, and a unified objective outperforms the two-stage decomposition.

The AI agent's dilemma. An AI agent with 100% context turnover faces a severe version: it MUST deliberate (comprehend the codebase) before acting effectively, but during comprehension its context fills and the environment may change. The optimal comprehension depth depends on ρ_{delib} and the session's action horizon. This is why reading CLAUDE.md and architecture docs first (high-CIY query actions) dominates reading random source files (low-CIY exploration).

Domain instantiations:

Table 4.1

Domain	*Deliberation*	*$\Delta\eta^*$ source*	*When ρ_{delib} is high*
Boyd's OODA	Explicit "Decide" step	War-gaming, staff analysis	Collapses to IG&C (implicit)
RL / MCTS	Planning rollouts	Monte Carlo tree search	Fewer rollouts, shallower search
MPC	Online optimization	Trajectory optimization	Shorter horizons, faster solvers
Human cognition	System 2 deliberation	Mental simulation	Defaults to System 1 (intuition)
Organization	Strategic planning	Scenario analysis	"Move fast and break things"
Software developer	Reading code, analyzing alternatives	Architecture analysis	Ship now, refactor later
AI agent	Reading codebase, planning approach	Context-building	Limit comprehension, act sooner

Open questions:

1. *Computational cost of deliberation* is not just elapsed time but resource cost. A fuller model would include both temporal and computational budgets.
2. *Deliberation about deliberation*: deciding whether to deliberate itself takes time. This meta-deliberation is bounded by the same tradeoff at a higher level, suggesting a hierarchy of diminishing deliberation horizons.
3. *Deliberation that generates observations*: internal simulation can surface model inconsistencies (internal mismatch), functioning as "exploration without external action." Can deliberation generate internal CIY?

4.2 *Derived:* Gain–Sector Bridge

The gain-based update principle (3.6) produces correction dynamics satisfying the sector condition (GA-3) whenever the update rule has *directional fidelity* — the correction points at least roughly toward reality. For gradient-based agents, local strong convexity of the loss is sufficient for the one-point sector condition (A2' as stated in A) and bidirectionally equivalent to the two-point / incremental sector condition (DA2'-inc). The sector parameter α is not a free parameter but is determined by the gain and the correction geometry.

Formal Expression.

The Bridge Theorem.

Given the gain-based update $M_t = M_{t-1} + \eta^* \cdot g(\delta_t)$ (3.6), the induced correction function is:

Derived (gain-sector-bridge, from update-gain + directional fidelity).

$$F(\delta) = \eta^* \cdot H\, g(\delta) \tag{4.5}$$

where H maps state-space corrections to observation-space mismatch reduction. The sector condition (GA-3) holds with parameter $\alpha > 0$ whenever:

(B1) Directional fidelity. The mismatch transform g preserves the mismatch-reducing direction:

$$\delta^T H\, g(\delta) \geq c\|\delta\|^2 \quad \text{for } \|\delta\| \leq R \tag{4.6}$$

for some $c > 0$. The sector parameter is then:

$$\alpha = \eta^* \cdot c_{\min}, \qquad c_{\min} = \inf_{\|\delta\| \leq R} \frac{\delta^T H\, g(\delta)}{\|\delta\|^2} \tag{4.7}$$

Gradient Equivalence.

For any agent updating via gradient descent on a loss L with learning rate η:

Derived (sector-convexity equivalence, two-point form).

$$\alpha = \eta \cdot \mu \qquad \text{where } \mu = \inf_{\|\delta\| \leq R} \lambda_{\min}(\nabla^2 L(M^* + \delta)) \tag{4.8}$$

is the strong convexity modulus. The basin radius R is the largest ball around the optimum where $\nabla^2 L$ remains positive definite. The equivalence has two forms — bidirectional under the stronger two-point sector condition, one-directional under the one-point form actually used by `#deriv-sector-condition`:

- **Two-point / incremental sector ⇔ strong convexity (full equivalence).** Under the incremental sector bound $(F(\delta_1) - F(\delta_2))^T(\delta_1 - \delta_2) \geq \alpha\|\delta_1 - \delta_2\|^2$ on $\mathcal{B}_R(M^*)$ — DA2'-inc in K, the bridge-lemma precondition in 11.1 — the iff holds via Nesterov 2004 Theorem 2.1.10: Two-point sector with $(\alpha, R) \iff L$ is locally (α/η)-strongly convex on $\mathcal{B}_R(M^*)$.

- **One-point sector $\Leftarrow$ strong convexity (one direction only).** AAT's GA-3 / A2' as stated in A is the one-point form $\delta^T F(\delta) \geq \alpha\|\delta\|^2$ at $\delta^* = 0$. Strong convexity implies the one-point sector ($\alpha = \eta\mu$); the converse fails. Counterexample: $L'(x) = x(1 + \frac{1}{2}\sin(10x))$ satisfies $x \cdot L'(x) \geq \frac{1}{2}x^2$ globally yet has $L''(\pi/10) < 0$, so it is not convex on any neighborhood of $x^* = 0$. The one-point sector at the equilibrium is genuinely weaker than full local strong convexity (cf. C's one-point/two-point distinction). Full proofs and the counterexample analysis in G Prop B.4.

Verified Instances.

Table 4.2

Update class	*Bridge status*	*Sector parameter α*	*Valid region*
Scalar Kalman	Derived	$K = P^-/(P^- + R_{\text{obs}}) = \eta^*$	Global
Matrix Kalman	Derived	$\lambda^+_{\min}(KH)$ in $(P^-)^{-1}$-norm	Observable subspace
Beta-Bernoulli	Derived	$1/(n+1) = \eta_{\text{edge}}$	Global
Exponential family (natural params), bounded scope $\Theta_0 \subset \text{int}(\Theta)$	Derived	$\eta \cdot \mu_0$ where $\mu_0 = \inf_{\theta\in\Theta_0} \lambda_{\min}(\mathbf{I}(\theta)) > 0$	Θ_0 (compact / interior-bounded) — global only when the family has a uniform Fisher lower bound
Gradient on strongly convex loss	Derived	$\eta \cdot \mu$	Global ($R = \infty$)
Gradient on locally convex loss	Derived	$\eta \cdot \mu_{\text{local}}$	Basin of attraction
Gradient on non-convex loss	Fails at basin boundary	N/A beyond R	Finite R
SPR-tuned PID on positive-real plant with anti-windup	Derived	$\alpha_{\text{PID}} = \omega_c \sin(\varphi_m)/\kappa(P)$ (phase margin as sector constant; crossover frequency as tempo; KYP-certificate condition number as degradation)	Classical linear regime + Lur'e sector-bounded nonlinearity within specified plant-Lipschitz threshold

Failure Modes. The bridge fails precisely in five cases:

1. **Directional infidelity.** The mismatch transform g rotates the correction away from the mismatch ($\delta^T Hg(\delta) \leq 0$). Occurs with pathological parameterizations or severe model-observation misalignment. For optimal Bayesian updates, B1 holds by construction.
2. **Gain collapse.** $\eta^* \to 0$ while $\rho > 0$, so $\alpha \to 0$ and the persistence condition eventually fails. Not a failure of the bridge but of the persistence condition — see the gain-collapse analysis in 3.6.
3. **Nonlinear saturation.** The correction function g saturates at large $\|\delta\|$, so the sector ratio $\delta^T g(\delta)/\|\delta\|^2$ decays. The sector condition holds locally with α depending on R. This is exactly what A2' (the local sector condition) is designed for.
4. **Unobservable directions.** When $\ker(H) \neq \{0\}$, the correction has no effect on mismatch in unobservable directions. The sector condition holds only in the observable subspace. See 8.3.
5. **Model misspecification.** The model class does not contain the truth, so the gradient direction is wrong. B1 fails because the correction aims at the wrong target. This is the 4.5 trigger.

Epistemic Status.

Conditional derivation. The bridge theorem is exact for all cases where B1 (directional fidelity) holds. The resulting sub-scope structure of A2' in A is:

Sub-scope α (B1 structural, A2rsquo derived):

- **Optimal Bayesian updates** (Kalman, conjugate, exponential family): B1 holds by construction — the posterior update minimizes expected loss, ensuring the correction aligns with the mismatch. The sector parameter equals the gain: $\alpha = \eta^*$ (scalar) or $\alpha = \lambda^+_{\min}(KH)$ (matrix Kalman, observable subspace).
- **Gradient descent on (locally) strongly convex losses**: local strong convexity is *sufficient* for B1 (the one-point form of A2' at M^*) and *bidirectionally equivalent* to the two-point / incremental sector condition DA2'-inc (Prop B.4 (B.4-i) and (B.4-ii) respectively) — a well-characterized property with an extensive optimization theory literature. The sector parameter factors as $\alpha = \eta \cdot \mu$ (learning rate × curvature). The reverse implication B1 ⇒ strong convexity does *not* hold without the two-point upgrade: the one-point sector at M^* is strictly weaker than full local strong convexity (counterexample in Prop B.4).
- **L2-regularized convex losses**: the regularization parameter λ provides a global floor $\mu \geq \lambda$, so $\alpha \geq \eta\lambda$ globally.
- **Exponential families in natural parameters, on a bounded interior scope**: Fisher information matrix is PD on the interior, and uniformly bounded below on any compact $\Theta_0 \subset \mathrm{int}(\Theta)$; $\alpha = \eta \cdot \mu_0$ with $\mu_0 = \inf_{\theta\in\Theta_0} \lambda_{\min}(\mathbf{I}(\theta))$. The bound is global only when the family has a uniform Fisher lower bound on Θ — true for Gaussian-mean and Beta-Bernoulli, false for Poisson natural parameter ($\mathbf{I}(\theta) = e^\theta$, infimum zero).
- **Linear corrections with PD gain–observation product**: $\alpha = \lambda^+_{\min}(KH)$.

Within sub-scope α, A2' is written down by inspection of the update rule — no independent posit is required. This is what `#deriv-sector-condition` "Grounding of GA-3 — sub-scope α" names.

Sub-scope β (B1 not structural, A2rsquo assumed per-agent):

- **Non-gradient agents** (PID controllers, rule-based systems, human judgment): B1 remains an empirical claim. Well-tuned PID has B1 empirically; badly-tuned PID may violate it.
- **Severely misspecified agents** (FM-5 below): proper-gradient updates can aim at the wrong target.
- **Variational / approximate posteriors**: B1 not guaranteed by optimality — approximation-direction error can rotate the correction.
- **Non-convex gradient agents beyond the basin** (FM-3 + basin boundary): A2' fails where the loss curvature goes non-positive; the structural-adaptation-necessity trigger.
- **Stochastic gradients, per-step**: A2' holds in expectation; per-step noise enters as effective disturbance under Prop A.1S.

The bridge covers sub-scope α rigorously and characterizes the boundary to sub-scope β via the five failure modes. It does *not* eliminate GA-3 as an assumption for all AAT-in-scope agents — some agent classes genuinely require A2' as a primitive posit, and the honest architectural statement is scope narrowing rather than universal derivation.

The gradient equivalence is validated by simulation across quadratic, logistic, exponential-family, and non-convex losses. The Kalman case is verified analytically. Full derivations and simulation results in G.

Max attainable: *conditional* — the condition (B1 or strong convexity) is inherent, not removable. Pathological update rules exist that violate B1 (FM-1 provides a counterexample that satisfies every AAT postulate and the gain-based update form but has $\delta^T g(\delta) = 0$ identically). Scope + gain structure alone does not force B1; some optimality / coherence / rationality constraint (Bayesian coherence, gradient-of-a-convex-loss, etc.) is required.

Weighted-norm subtlety. In the matrix Kalman case, the sector condition holds in the $(P^-)^{-1}$-weighted inner product, not the Euclidean norm. For fully observable systems with bounded condition number $\kappa(P^-)$, the norms are equivalent up to $\kappa(P^-)$. The Lyapunov proofs in A use the Euclidean norm, which remains valid with the quantitative adjustment $\alpha_{\text{Euclidean}} \geq \alpha_{\text{weighted}}/\kappa(P^-)$.

Fisher-metric cases under parameterization-invariance. The exponential-family-in-natural-parameters row and the matrix-Kalman row of Verified Instances both have natural statements in a Fisher-metric inner product rather than the Euclidean one. Under the **(PI) parameterization-invariance** axiom named in 4.7 (AAT's theorems should not depend on arbitrary choice of coordinates on M_t), Čencov's 1982 uniqueness theorem (*Statistical Decision Rules and Optimal Inference*, AMS; subsequent Ay-Jost-Lê-Schwachhöfer 2017 extensions) forces the Fisher information metric uniquely on statistical-manifold sub-cases of M_t. Two consequences:

- *The matrix-Kalman sector constant is natively stated in the information metric* $M = (P^-)^{-1}$ — under (PI), this is not a choice but forced. The $\kappa(P^-)$ Euclidean-transfer penalty in the paragraph above vanishes; the derivation is AAT-internally *forced*, not AAT-internally *preferred*.
- *The exponential-family-natural-parameter sector constant is natively stated in the Fisher metric* $\mathbf{I}(\theta)$ — under (PI), this is forced; the contraction rate equals η globally on the interior of the natural-parameter domain (Fisher-conditioning degradation removed).

Under (PI), these two rows of Verified Instances upgrade from *derived (conditional on choice of inner product)* to *derived (AAT-internally forced)* via Čencov. Under non-adoption of (PI), they remain at the Euclidean-transferred statement with the $\kappa(P^-)$ / $\lambda_{\min}$(Fisher) penalty. This is the Fisher-layer analog of the chain-rule-additivity axiom that grounds the divergence-layer uniqueness theorem in O and the evidential-additivity axiom that grounds the update-layer uniqueness theorem in P — in each case an AAT-internal axiom combined with a uniqueness theorem forces a coordinate. The structural positioning is named in Z.

The remaining Verified Instances rows (scalar Kalman, Beta-Bernoulli, gradient on strongly convex, L2-regularized, linear-PD-symmetric) live in Euclidean metric naturally; no Fisher-metric choice is at issue, so (PI) has no effect on them beyond transparency.

Discussion.

GA-3 is grounded, not floating. Before this result, GA-3 ("the correction function satisfies the sector condition") was an opaque global assumption — the theory's softest structural joint. The bridge transforms it: for well-designed agents, GA-3 is a consequence of the update mechanism's geometry, not an independent postulate. The assumption load shifts from "the correction function has this property" (hard to verify in general) to "the update rule has directional fidelity" (transparent and checkable for specific systems).

The α-T relationship is derived for the important cases. 4.4 notes that α is monotone increasing in $\mathcal{T}$ across all tested correction functions. For linear correction (Kalman, Beta-Bernoulli), $\alpha = \mathcal{T}$ exactly. For gradient correction on strongly convex losses, $\alpha = \eta \cdot \mu$ where μ is the curvature — monotone in η (which is monotone in $\mathcal{T}$) for fixed loss landscape. The empirical observation is now structurally grounded.

Basin boundary = structural adaptation trigger. For gradient agents with non-convex losses, the basin radius R is the convexity radius of the loss landscape. When mismatch exceeds R, the agent has been pushed out of its convexity basin — the correction function reverses direction and the sector condition fails. This IS the 4.5 trigger, now with a precise geometric characterization: structural adaptation is needed when the agent crosses an inflection surface of its loss landscape.

The theory's formal chain is tightened. The prediction chain becomes:

$$\text{gain principle} + \text{B1} \xrightarrow{\text{derived}} \text{sector condition (GA-3)} \xrightarrow{\text{Lyapunov (exact)}} \text{persistence, reserve, adversarial scaling} \quad (4.9)$$

The left arrow is this segment. The right arrow is A. The discrete-time framework (K) requires an additional Lipschitz bound on the correction function ($\|F_d(\delta)\| \leq c_{\max}\|\delta\|$ with $c_{\max} < 2/\eta^*$) — strictly stronger than an inner-product upper bound, needed because the discrete contraction involves $\|F_d\|^2$. This is automatically satisfied for Bayesian updates (the posterior lies between prior and data) and for gradient descent on smooth losses (where $c_{\max}$ is the Lipschitz constant L). With this constraint and the step-size condition $\eta^* < 2c_{\min}/c_{\max}^2$, the fluid limit is formally justified: Model D steady state is exact, Model S variance gap is $O(\eta^* c_{\max})$. Section I's formal chain is now complete.

4.3 *Result:* Sector Condition Stability

An agent's mismatch remains bounded if its correction function satisfies a sector condition (points inward with at least baseline efficiency) and the effective correction strength exceeds the environmental disturbance rate.

Formal Expression.

This segment is the **single-agent epistemic instantiation** of the sector-persistence template (C). The template's state variable is $\xi = \delta(t) \in \mathbb{R}^n$ (model-reality mismatch); the correction function is $F(\mathcal{T}, \delta)$; the disturbance is environmental ($w(t)$); the region of validity R is the model class capacity.

Formulation.

$$\frac{d\delta}{dt} = -F(\mathcal{T}, \delta) + w(t) \tag{4.10}$$

F satisfies the local sector condition (template condition (T2)) for $\|\delta\| \leq R$:

Assumption (sector-condition).

$$\delta^T F(\mathcal{T}, \delta) \geq \alpha\|\delta\|^2 \tag{4.11}$$

with $\alpha > 0$. Disturbance is bounded: $\|w(t)\| \leq \rho$ (Model D, GA-2) or $\mathbb{E}[\|w(t)\|^2] = \sigma_w^2$ (Model S, GA-2S). Grounding of (T2) for gain-based agents: 4.2 gives $\alpha = \eta^* \cdot c_{\min}$. The linear case $F = \mathcal{T} \cdot \delta$ yields $\alpha = \mathcal{T}$ exactly.

The template's Model D conclusion specializes to: $\delta(t)$ is ultimately bounded by $R^* = \rho/\alpha$, and the agent persists iff

Derived (from sector-persistence-template).

$$\alpha > \frac{\rho}{R}. \tag{4.12}$$

The adaptive reserve is $\Delta\rho^* = \alpha R - \rho$ — the additional disturbance the agent can absorb before R^* exceeds the valid region.

The template's Model S conclusion specializes to: the steady-state RMS mismatch is $R_S^* = \sigma_w\sqrt{n/(2\alpha)}$ (where $n = \dim(\delta)$), and mean-square persistence requires $\alpha > n\sigma_w^2/(2R^2)$. Model D scales as $1/\alpha$; Model S scales as $1/\sqrt{\alpha}$ — correction is less effective against noise than against drift.

Full Lyapunov proofs: A Props A.1, A.1S, A.2.

Epistemic Status.

Exact. Both results are direct instances of the sector-persistence template applied to the single-agent epistemic case. Template precondition (T1) is satisfied because no correction should be applied at zero mismatch; (T2) reduces to the local sector condition above and is grounded structurally by 4.2 for gain-based agents; (T3) is the disturbance-model choice (D or S), a domain question. The linear ODE of 3.9 is the special case where (T2) holds globally with $\alpha = \mathcal{T}$; the sector framework generalizes this to saturating, thresholded, and structurally-limited correction functions under the same persistence condition. Disturbance-model choice is a domain question, not a theory question.

Discussion.

Why the sector condition. The linear ODE assumes correction scales linearly with mismatch forever. Real adaptive systems saturate, exhibit thresholding, or break down when the model class is exhausted. The sector condition captures the minimal structural requirement: the correction must point in the right direction with at least baseline efficiency α.

Generalizing the persistence threshold. In the linear case, $\alpha = \mathcal{T}$ (adaptive tempo). The general result $\alpha > \rho/R$ proves the persistence threshold (4.4) is a structural necessity of any bounded-correction system, not an artifact of the linear approximation. This result addresses *structural persistence* — the machinery's capacity to bound mismatch — not operational persistence (current proximity to R) or continuity persistence (identity through time). See Persistence in `LEXICON.md` for the full disambiguation.

Connection to structural adaptation. When $\rho/\alpha > R$, disturbance exceeds the model class's capacity. The sector condition fails — this is the dynamical trigger for structural adaptation (4.5), requiring a new model class with larger valid radius R' or better efficiency α'.

4.4 *Result:* Persistence Condition

An agent persists when two independent conditions hold: the correction machinery can contain mismatch within its operating region (*structural persistence*), and the resulting steady-state mismatch is small enough for the agent's actions to remain adequate (*task adequacy*).

Formal Expression.

This segment is the canonical single-agent instantiation of the sector-persistence template (C) with state variable $\xi = \delta_t$ (epistemic mismatch), correction function $F(\mathcal{T}, \delta)$, and disturbance rate $\rho_\xi = \rho$ (environmental change rate). Structural persistence is the direct template conclusion. Task adequacy adds a domain-specific constraint beyond the template's reach.

Structural Persistence.

Applying the template to the single-agent epistemic case gives: the correction machinery bounds δ within the model class capacity iff

Derived (structural-persistence, from sector-persistence-template).

$$\alpha > \frac{\rho}{R} \quad \text{(Model D)} \qquad \alpha > \frac{n\sigma_w^2}{2R^2} \quad \text{(Model S)} \tag{4.13}$$

with ultimate bound $R^* = \rho/\alpha$ (Model D) or $R_S^* = \sigma_w\sqrt{n/(2\alpha)}$ (Model S). See 4.3 for how (T1)–(T3) are verified in this instantiation, and A for the proof. Structural persistence is a property of the adaptive architecture — the machinery's ability to contain mismatch — not of the task.

Linear case. When $F(\mathcal{T}, \delta) = \mathcal{T}\delta$, $\alpha = \mathcal{T}$ and $R \to \infty$, so structural persistence is trivially satisfied whenever $\mathcal{T} > 0$. The binding constraint then becomes task adequacy (below).

Task Adequacy.

The steady-state mismatch is small enough for the agent's actions to remain acceptable:

Definition (task-adequacy).

$$R^* < \|\delta_{\text{critical}}\| \tag{4.14}$$

where $\|\delta_{\text{critical}}\|$ is a domain-specific tolerance threshold — "how wrong can the model be before the agent's actions become harmful or ineffective?" This is set by the application domain, not derived by AAT.

Task adequacy is a separate condition from structural persistence. An agent can be structurally persistent ($R^* < R$) but task-inadequate ($R^* > \|\delta_{\text{critical}}\|$) — the machinery contains mismatch, but not tightly enough for the domain's needs. Conversely, when $\|\delta_{\text{critical}}\| < R$ (the domain's tolerance is stricter than the model class capacity), task adequacy is the binding constraint.

Operational Persistence Condition.

The agent persists operationally when BOTH conditions hold. In the nonlinear case with $\|\delta_{\text{critical}}\| < R$, the binding condition is:

Derived (operational-persistence, conjunction of structural persistence + task adequacy).

$$\alpha > \frac{\rho}{\|\delta_{\text{critical}}\|} \quad \text{(Model D)} \qquad \alpha > \frac{n\sigma_w^2}{2\|\delta_{\text{critical}}\|^2} \quad \text{(Model S)} \tag{4.15}$$

These are the same as the structural conditions with R replaced by $\|\delta_{\text{critical}}\|$, because when $\|\delta_{\text{critical}}\| < R$, task adequacy is stricter.

Linear operational forms: In the linear case ($\alpha = \mathcal{T}$, $R \to \infty$), structural persistence is trivially satisfied and the operational condition reduces to task adequacy alone:

$$\mathcal{T} > \frac{\rho}{\|\delta_{\text{critical}}\|} \quad \text{(Model D)} \qquad \mathcal{T} > \frac{n\sigma_w^2}{2\|\delta_{\text{critical}}\|^2} \quad \text{(Model S)} \tag{4.16}$$

These are the forms used throughout the theory as the operational persistence condition. They are exact for linear correction and useful proxies for mildly nonlinear correction (where $\alpha \approx \mathcal{T}$), but they overstate the persistence margin when the correction function saturates, because they omit the structural constraint ($\alpha > \rho/R$) that becomes binding when R is finite.

Per-dimension (Model S): $\eta_k > c \cdot \rho_k^2/\delta_{\text{critical},k}^2$ where c depends on the probability guarantee. See 14.5.

The relationship between α and $\mathcal{T}$. 4.2 shows that for agents with directional fidelity, $\alpha = \eta^* \cdot c_{\min}$ where $c_{\min}$ is the worst-case directional fidelity. For linear correction (Kalman, Beta-Bernoulli), $\alpha = \mathcal{T}$ exactly. For gradient descent on strongly convex losses, $\alpha = \eta \cdot \mu$ where μ is the strong convexity modulus — monotone in η (and hence in $\mathcal{T}$) for fixed loss landscape. For nonlinear correction tested in simulation (saturating, sigmoid, threshold), α remains monotone increasing in $\mathcal{T}$: for a saturating function with capacity R, $\alpha \approx \mathcal{T}/2$ (worst case at the capacity boundary); for sigmoid (tanh), $\alpha \approx 0.76 \cdot \mathcal{T}$. The qualitative conclusion — "faster adaptation improves persistence" — is structurally grounded for the important cases and empirically confirmed for all correction function classes tested.

Per-Dimension Extension.

For anisotropic systems (non-uniform ρ or $\mathcal{T}$ across dimensions), the scalar persistence condition is insufficient. Per-dimension:

Empirical Claim (per-dimension-persistence, from simulation variant F).

$$\mathcal{T}_k > \frac{\rho_k}{\delta_{\text{critical},k}} \quad \text{for each dimension } k \tag{4.17}$$

The scalar condition overestimates by up to 72% in simulation. The weak dimension is the bottleneck (84% of total mismatch in simulation). See 14.5.

Robustness: The per-dimension condition matches discrete AR(1) prediction to 4 significant figures. The scalar overestimate is a consequence of Jensen's inequality applied to the norm.

Epistemic Status.

Structural persistence thresholds are *exact* under their stated assumptions: Model D gives $\alpha > \rho/R$ (Prop A.1, exact under GA-2, GA-3); Model S gives $\alpha > n\sigma_w^2/(2R^2)$ (Prop A.1S, exact under GA-2S, GA-3). The threshold's *existence* is *robust qualitative* — any monotone correction function has a capacity limit; this holds across all correction functions tested.

Task adequacy ($R^* < \|\delta_{\text{critical}}\|$) is *exact as a definition* — given R^* (derived) and $\|\delta_{\text{critical}}\|$ (domain parameter), the comparison is well-defined. The substance lies in estimating $\|\delta_{\text{critical}}\|$ for specific domains, which is an operationalization question (), not a theory question.

The linear operational forms ($\mathcal{T} > \rho/\|\delta_{\text{critical}}\|$ for Model D; $\mathcal{T} > n\sigma_w^2/(2\|\delta_{\text{critical}}\|^2)$ for Model S) are *exact* for linear correction (where they express task adequacy alone, structural persistence being trivially satisfied) and *useful approximations* for mildly nonlinear correction (where $\alpha \approx \mathcal{T}$). For strongly nonlinear correction, the general α-forms are required and BOTH structural and task-adequacy conditions must be checked. Downstream segments that use the linear operational forms should be understood as expressing task adequacy, not structural stability.

The per-dimension extension is *empirically exact* for Model S (matches AR(1) prediction to 4 significant figures in simulation); the Model D per-dimension threshold ($\mathcal{T}_k > \rho_k/\delta_{\text{critical},k}$) is exact by the same Lyapunov argument applied per dimension.

Discussion.

Two conditions, not one. This segment now separates what was previously conflated. Structural persistence ($\alpha > \rho/R$) is the Lyapunov-derived result — it says the machinery *works*. Task adequacy ($R^* < \|\delta_{\text{critical}}\|$) is a domain-specific constraint — it says the machinery works *well enough*. Neither implies the other, and downstream segments should specify which they mean. Most adversarial-dynamics results (13.4, 13.2) depend on structural persistence. Most domain instantiations (TST, logogenic agent design) care about task adequacy. See Persistence in `LEXICON.md` and `README.md` for the full three-sense taxonomy (structural, operational, continuity).

Below structural threshold. When $\alpha \leq \rho/R$, mismatch is not merely large — it grows without effective bound (up to R, the sector-condition region). The correction machinery is overwhelmed. This is a qualitative transition, not a gradual degradation.

Below task-adequacy threshold. When $R^* > \|\delta_{\text{critical}}\|$ but $R^* < R$, the system is structurally stable but performing unacceptably. Mismatch is bounded but too large for the domain. The remedy is different from structural failure: increase $\mathcal{T}$ (faster or better correction), decrease ρ (reduce environmental volatility), or relax $\|\delta_{\text{critical}}\|$ (accept more mismatch). Structural failure requires changing the correction architecture entirely (4.5).

δ_{critical} and R are domain parameters, not theory outputs. The theory derives the *existence* of persistence thresholds and their *form* (ratio of correction to disturbance). But the specific values are set by the application domain: δ_{critical} encodes "how wrong can the model be before the agent's actions become harmful or ineffective?" — this depends on the stakes, the action space, and the environment's forgiveness. R encodes "how large a mismatch can the correction function handle before it saturates or breaks down?" — this depends on the model class and the correction architecture. See for guidance on estimating these quantities in specific domains.

Channel independence and scalar tempo. The linear operational forms use scalar $\mathcal{T}$, which inherits the channel-independence assumption from 3.8: the additive formula overcounts when observation channels are correlated. In anisotropic systems the scalar condition also overestimates margins — up to 72% in simulation (see 3.8, scalar vs. vector tempo). Where precision matters, the per-dimension condition ($\mathcal{T}_k > \rho_k/\delta_{\text{critical},k}$) should be used instead — and under cross-dimensional correction (off-diagonal $\mathcal{T}$ in the coordinate basis of $D_\delta = \text{diag}(\delta^2_{\text{critical},k})$), the matrix-Loewner form $\Sigma_\infty \prec D_\delta$ () is canonical: per-coordinate is unsafe in that regime (gives a false-pass on persistence), while matrix-Loewner reads persistence correctly off the worst direction whether or not it aligns with a coordinate axis.

Adaptive reserve. The quantity $\Delta\rho^* = \alpha R - \rho$ (Prop A.2) measures how much additional disturbance the agent can absorb before persistence fails. Positive reserve means the agent has margin; zero reserve means it is at the threshold.

Persistence has a cost, not just a threshold. The inequality above says mismatch is bounded; it does not say what rate of effort the agent expends to hold that bound. E establishes the complementary *information-rate* bound: under Model S with Gaussian-OU signal, the sustained Shannon information rate the agent must acquire from observations to maintain the ultimate bound satisfies $\dot{R} \geq n\alpha/2$ nats/time — a Landauer-analog floor that Kalman-Bucy saturates. The corollary is a channel-capacity prerequisite $C \geq \mathcal{T}/2$ that the threshold condition alone does not name. Two agents with identical persistence guarantees can face wildly different sustained demands because the cost scales linearly with α; the threshold alone cannot distinguish dormant from running-hot.

Connections. The persistence condition appears in multiple downstream contexts:

- **Adversarial dynamics** (13.4): Superlinear tempo advantage arises because persistence is a threshold — pushing an adversary below it causes qualitative collapse. *This connection is developed and validated in 13.4 and simulation variants A-D.*
- **Structural adaptation** (4.5): When model class fitness $\mathcal{F}(\mathcal{M}) < 1 - \varepsilon$, the effective α in the sector condition shrinks, eventually violating persistence. *This connection is developed in 4.5.*
- **Software maintainability** (`#der-code-quality-as-observation-infrastructure` — cross-component reference, see `02-tst-core/`): *[Discussion]* A codebase may become "unmaintainable" when the development team's adaptive tempo falls below the rate of complexity accumulation. The vicious cycle would then be the persistence condition being violated through the agent's own prior actions degrading future $\mathcal{T}$ via U_o. *This connection is structurally motivated but not yet formally derived within AAT. It requires formalizing "complexity accumulation rate" as an instance of ρ.*

Findings.

The Persistence Condition with Structural / Task-Adequacy Decomposition. **Brief:** An adaptive system persists when its correction speed beats the rate at which its world is changing, relative to how forgiving the world is. Below this threshold the system doesn't merely degrade — it loses bounded behavior, the way a balance held just barely beneath a tipping point is qualitatively different from one well above it. The same inequality, with different inputs, governs whether a Kalman filter tracks a moving target, whether a development team keeps a codebase maintainable, and whether an organization keeps up with strategic change. The threshold itself decomposes into two distinct conditions — *structural persistence* (the machinery's correction rate can outpace disturbance) and *task adequacy* (the resulting steady-state mismatch is small enough for what the agent is trying to do). Conflating the two leads to category errors in domain transfer.

Impact: This is the framework's central inequality and the load-bearing connection between control-theoretic Lyapunov stability analysis and the broader question of when any adaptive system — thermostat, software team, immune system, RL agent — can maintain coherent function under change. The two-condition decomposition is itself non-obvious and consequential: prior work that conflated "the machinery works" with "the machinery works well enough" produced category errors in domain transfer (a structurally persistent codebase team can be task-inadequate; the remedies differ). The complementary information-rate bound from `#deriv-persistence-cost` ($\dot{R} \geq n\alpha/2$) shows the threshold has a sustained-cost shadow: two agents with identical persistence guarantees can face wildly different sustained demands.

Novelty Claim: *Claim synthesis* on Lyapunov stability theory, sector-bounded nonlinear correction, and adaptive-tempo information-rate accounting, applied uniformly across single-agent classes that range from Kalman filtering through saturating nonlinear correction through PID control. The Lyapunov machinery itself is standard; the synthesis is its use as the central inequality of an integrated agent theory, with the two-condition decomposition (structural / task-adequacy) as the AAT-internal contribution that cleanly separates "the machinery works" from "the machinery works well enough."

Related Work:

- Khalil 2002, *Nonlinear Systems* (3rd ed.), Prentice Hall (published 2002, found pre-2026) — *formal antecedent* — chapters 4 and 9 supply the converse Lyapunov, ultimate boundedness, and sector-condition machinery the segment uses. Standard control-theoretic apparatus.
- Lyapunov 1892 / Khasminskii 2012 *Stochastic Stability of Differential Equations* — *formal antecedent* — the underlying Lyapunov stability tradition; Khasminskii's stopping-time localization underpins the Model S derivation.
- Rockafellar & Wets 1998, *Variational Analysis* — *formal antecedent* — supplies the monotone-operator machinery that underwrites the sector condition's strong-convexity equivalents.
- Wiener 1948 *Cybernetics*; Ashby 1956 *Introduction to Cybernetics*; Conant & Ashby 1970 — *conceptual precursor* — the cybernetic-feedback tradition that frames the "correction must outpace disturbance" intuition without supplying the quantitative inequality.

Search Log: - 2026-04 (*intuition-only* on broader prior-art): no targeted Undermind-grade search has been conducted on the persistence-condition-as-central-inequality positioning. Pre-search expectation: the Lyapunov-based stability machinery is standard; the AAT-distinctive content is the two-condition decomposition (structural vs task adequacy) and its uniform application across agent classes. A targeted search would query the bounded-rationality / control-theoretic decision-making literature (Ortega-Braun line; Genewein et al.) for prior decompositions of "stability vs adequacy" in adaptive-control settings, and the active-inference literature for the same distinction. - 2025 (*targeted*): Khalil 2002 / Khasminskii 2012 / Rockafellar-Wets 1998 confirmed as the formal antecedents for the sector-Lyapunov machinery; the segment cites them inline.

4.5 *Result:* Structural Adaptation Necessity

When model class fitness is insufficient — when no model in the current class can adequately represent reality — no amount of parametric adaptation can close the mismatch floor. The agent must change its model class, not just its parameters.

Formal Expression.

If the model class fitness $\mathcal{F}(\mathcal{M}) < 1 - \varepsilon$ for some $\varepsilon > 0$, then no parametric adaptation within $\mathcal{M}$ can reduce the expected mismatch below a floor determined by ε (under the alignment assumption — see Epistemic Status). Without the alignment assumption, the result holds for irreducible proper-scoring regret rather than one-step mean mismatch. The qualitative conclusion is the same either way: parametric adaptation cannot compensate for model-class inadequacy.

Derived (structural-adaptation-necessity).

Derivation.

1. By definition, $S(M^*) = \mathcal{F}(\mathcal{M}) < 1 - \varepsilon$ where $M^* = \arg\sup_{M \in \mathcal{M}} S(M)$.
2. Therefore $I(\mathcal{C}_t; o_{t+1:\infty} \mid M^*, a_{t:\infty}) > 0$: the history contains predictive information that M^* does not capture.
3. This uncaptured information manifests as *systematic* mismatch — structured residuals δ_t containing signal, not merely noise.
4. From 3.5, the model error component has a positive lower bound that cannot be reduced by any $M \in \mathcal{M}$.
5. The update rule (3.6) adjusts M_t within $\mathcal{M}$, but M^* is already (approximately) reached. Further updates oscillate without net improvement.
6. Therefore: reducing mismatch below the floor requires changing $\mathcal{M}$ — structural adaptation. □

Corollary. Persistent irreducible mismatch (after parametric convergence) is *diagnostic* of model class inadequacy. Systematic patterns in residuals are evidence that $\mathcal{F}(\mathcal{M})$ is insufficient.

Epistemic Status.

Conditional. The step from "lost predictive information" (step 2) to "systematic one-step mismatch" (step 3) requires an alignment assumption: that the lost predictive information affects the one-step conditional mean, not just higher moments. 3.5 explicitly flags this: insufficiency implies positive model error under the alignment assumption, or positive proper-scoring regret without it. As written, the result is conditional on this alignment assumption. Without it, the conclusion should be stated in terms of proper-scoring regret (the best model in $\mathcal{M}$ has irreducible regret relative to the optimal predictor) rather than one-step mismatch magnitude. The qualitative conclusion — parametric adaptation cannot compensate for model-class inadequacy — holds either way; the quantitative mechanism differs.

Discussion.

Structural adaptation as structural persistence failure. When the model class is inadequate, the effective α in the sector condition shrinks — the correction function cannot point inward strongly enough because the model class lacks the capacity to represent the correct direction. This is a failure of *structural persistence* (see Persistence in `LEXICON.md`): the machinery's capacity to outpace disturbance degrades not because disturbance increased or tempo decreased, but because the correction function itself has become less effective. The remedy is not faster cycling (operational) or identity preservation (continuity) but a change of model class.

Observable symptoms of model class inadequacy. When $\mathcal{F}(\mathcal{M})$ is low:

1. **Persistent irreducible mismatch**: $\|\delta_t\|$ remains large despite extended updating — the model has converged within $\mathcal{M}$ but the best achievable model is still poor.
2. **Gain collapse without performance**: η^* has decreased (model appears confident) but predictions remain inaccurate — the model is confidently wrong, having fitted to structure in $\mathcal{M}$ that doesn't match reality.

3. **Systematic mismatch patterns**: δ_t shows structure (correlations, trends, periodicities) that the model class cannot represent — the residuals contain signal that $\mathcal{M}$ lacks the capacity to absorb.

Structural overfitting: the opposite failure mode. $\mathcal{M}$ can also be *too expressive*, causing the model to memorize irreducible noise. Symptoms: low training mismatch but high generalization mismatch; model complexity growing without predictive gain; $\eta^* \to 0$ (confident) but confidence is spurious. The information bottleneck (2.2) provides the diagnostic: when marginal increases in model complexity yield no marginal predictive power, the model is past the optimal point on the rate-distortion curve. Structural adaptation in this case means *compression* — moving to a simpler $\mathcal{M}'$. Structural adaptation is bidirectional: expansion when too constrained (this proposition), compression when too expressive.

Mechanisms of structural change. Structural adaptation can proceed by:

- **Decomposition and recombination**: Tearing apart existing structure and synthesizing new configurations from the pieces. Boyd's "Destruction and Creation" insight; Kuhn's paradigm shifts; Popper's conjecture and refutation.
- **Expansion**: Adding new representational capacity without destroying existing structure. Bayesian nonparametrics, growing neural architectures, organizational expansion.
- **Compression**: Removing unnecessary structure while preserving the predictive core. Regularization, Occam's razor, organizational streamlining.
- **Grafting**: Incorporating external structure. Transfer learning, acquiring a company, consulting an expert. Query actions (3.7) are a primary conduit for grafting.

The severity of structural change needed depends on *how far* the current model class is from adequacy. Minor regime changes may require only expansion or grafting; fundamental shifts where $\mathcal{M}$'s assumptions are violated may demand full decomposition.

Neutral variation as a mechanism for structural change. In multi-agent settings, structural adaptation can proceed without any individual agent deliberately restructuring. Miller (2022, *Ex Machina*) identifies a five-phase "extreme transition motif" in coevolving automata: (1) stable epoch, (2) an environmentally neutral variant — structurally different but behaviorally identical under current conditions — appears, (3) the variant drifts to nontrivial proportion through stochastic reproduction, (4) the variant's latent structural differences create a niche that a new mutant in the opposing population exploits, triggering a self-reinforcing cascade, (5) both populations rapidly transition to a new regime and consolidate. This mechanism bridges the gap between "many incremental changes" and "radical restructuring" — the restructuring is radical in its effect but incremental in its causes, with neutral drift providing the bridge. The concept of *latent structural diversity* — variation in agent architectures that is invisible to current performance but consequential under regime change — is a composition-level property that Section III's dynamics framework should formalize.

The cost of structural change. Structural adaptation is expensive: knowledge loss (parameters learned within $\mathcal{M}$ may not transfer), temporary performance drop (new model starts uncertain), search cost (finding good $\mathcal{M}'$), coordination cost (in multi-agent systems). This creates rational conservatism — prefer parametric adaptation when it suffices, resort to structural change only when the evidence is strong. Premature structural change wastes accumulated knowledge; delayed structural change accumulates mismatch. The connection to 4.1: structural adaptation is deliberation with a *massive* $\Delta\tau$, and the mismatch debt during the transition is correspondingly enormous.

Temporal nesting of adaptation. Parametric and structural adaptation operate at different timescales: $\nu_{\text{parametric}} \gg \nu_{\text{structural}}$. More generally, an agent may have multiple adaptive processes at different rates, with the convergence constraint that faster processes must approximately converge before slower ones act on their output. If deeper change occurs before shallower adaptation has converged, the deeper change is based on transients rather than settled dynamics.

Domain instantiations:

Table 4.3

Domain	*Parametric adaptation*	*Structural adaptation*
Kalman filter	State estimate update	Switching observation/dynamics models
RL	Weight/Q-value update	Architecture search
PID	— (gains fixed)	Switching to MPC
Bayesian	Posterior update	Model selection, nonparametrics
Boyd	Orientation updating	Destruction and creation of mental models
Science	Normal science (Kuhn)	Paradigm shift
Evolution	Allele frequency change	Speciation, new body plans
Organization	Process optimization	Strategic pivot, restructuring
Software	Incremental refactoring	Architecture migration
Coevolving automata (Miller 2022)	Edge reweighting within fixed FSA structure	Mutation altering state output or transition; neutral mutations accumulating until niche creation triggers cascading restructuring

4.6 *Derived:* Temporal Nesting

An agent's adaptive processes stratify naturally by timescale, with each level operating on the quasi-steady-state output of the level below. Faster processes must approximately converge before slower ones act on their output.

Formal Expression.

Derived (temporal-nesting).

$$\nu_{\text{level } n+1} \ll \nu_{\text{level } n} \tag{4.18}$$

for each adjacent pair of adaptive timescales. If a slower process acts before the faster process beneath it has converged, the system oscillates — the slower process adjusts based on transient behavior rather than settled dynamics.

Table 4.4

Timescale	*Process*	*What changes*
Fastest	Reactive response	Action given current model
Fast	Parametric update (online)	Model parameters within $\mathcal{M}$
Intermediate	Consolidation (offline, cf. 8.11)	Redistribution of information within M_t's sub-state factorization toward IB-optimum
Slow	Structural adaptation	Model class $\mathcal{M}$
Slowest	Architectural change	The agent's fundamental structure

This table is illustrative — real systems may have additional intermediate levels. The number of distinguishable timescales is not fixed; what matters is the structural relationship between adjacent levels.

Epistemic Status.

Robust qualitative — this is standard singular perturbation reasoning (Tikhonov 1952, "Systems of differential equations containing a small parameter multiplying the derivative," *Matematicheskii Sbornik* 31(3):575–586; modern textbook exposition in Khalil 2002, *Nonlinear Systems* (3rd ed.), Prentice Hall, Chapter 11). The convergence constraint follows from the structure of multi-timescale updating. The specific timescale ratios needed for adequate separation are domain-dependent and not derived within AAT.

Discussion.

Domain instantiations of temporal nesting:

- **PID control**: D-term (fastest, high-frequency response) → P-term (current error) → I-term (slowest, accumulated bias)
- **RL**: Action selection → value function update → policy improvement → architecture change
- **Biology**: Reflexes (ms) → perceptual learning (minutes) → skill acquisition (months) → developmental change (years) → evolutionary adaptation (generations)
- **Organizations**: Operational decisions (hours) → tactical adjustments (weeks) → strategic revision (quarters) → restructuring (years)
- **Boyd**: Tactical OODA (seconds–minutes) → operational (hours–days) → strategic (weeks–months) → grand strategic (years)

Structural adaptation as slow-timescale dynamics. The conservatism toward structural change (4.5) is a derived consequence of temporal nesting: structural adaptation operates at a much slower timescale than parametric, so the mismatch cost of the "pause" ($\rho \cdot \Delta\tau$) is enormous. The agent rationally resists until the parametric mismatch floor exceeds this cost. See also 4.1 for the formal tradeoff.

Violation symptoms. When nesting is violated (a slower process acts before the faster one converges): oscillation, instability, degraded performance. In organizations: micromanagement (strategic decisions at operational tempo). In RL: policy updates before value function converges (policy oscillation). In biology: premature developmental transitions.

Multi-timescale stability (sketch). Singular perturbation theory gives the composite stability result: if each level is stable given the levels above it (each level has a stable attractor for fixed slower-level parameters), and the timescale separation is sufficient, the composite N-level system is stable. Making this rigorous for AAT requires specifying dynamics at deeper adaptive levels — an open problem. See I for the framework.

4.7 *Discussion:* Scope: Agent Identity as Singular Causal Trajectory

AAT applies to agents instantiated on singular causal trajectories. Identity within AAT is grounded not in the model state M_t (which can be copied) but in the unique causal trajectory $\mathcal{C}_t$ (which cannot).

Formal Expression.

Scope commitment. AAT's formal apparatus presumes each agent is instantiated on a **singular, non-forkable causal trajectory** $\mathcal{C}_t$ (1.4). Sufficiency of the model state M_t (2.3) is defined *relative to* this trajectory — not relative to a model-state equivalence class. Duplicating M_t and exposing the copies to different future events produces two agents with *divergent* causal histories, each of which is a sufficient statistic only for *its own* trajectory.

Scope (scope-agent-identity, from chronica + model-sufficiency).

Three consequences of the scope commitment:

1. **Sufficiency is trajectory-indexed.** $S(M_t)$ (2.3) measures against *this* agent's $\mathcal{C}_t$; not against a hypothetical parallel copy's $\mathcal{C}_t^{(2)}$.
2. **Model merging is lossy by construction.** Reconciling the models of two agents that share a prefix of their trajectory but have diverged requires choosing which causal history to privilege; no generally optimal merge exists. This is a structural constraint of the scope, not a defect of any particular merge algorithm.
3. **The loop's interventional access depends on the trajectory's singularity.** When the agent acts and observes, the observation is the response to *its* intervention on *its* single trajectory. Replaying a saved M_t against a different event stream is not the same as intervening — the observed consequences are under a different causal trajectory. This grounds the interventional interpretation in 6.3: the loop provides Level-2-quality data precisely because the agent is on a singular trajectory, not because of any architectural property of the agent itself.

Natural extension: parameterization-invariance (PI). The scope commitment motivates a companion axiom. AAT's predictions concern a singular causal trajectory $\mathcal{C}_t$; the trajectory itself is coordinate-free, while any parameterization of M_t's internal state space is a modeling convention. Requiring AAT's theorems to be invariant under change of parameterization — *(PI): the theory's conclusions do not depend on arbitrary choice of coordinates on M_t* — is a natural axiomatic commitment consistent with (but not directly forced by) the three consequences above. When (PI) is adopted and combined with Čencov's 1982 uniqueness theorem (*Statistical Decision Rules and Optimal Inference*, AMS), the Fisher information metric is uniquely forced on statistical-manifold sub-cases of M_t — which converts Fisher-metric-dependent derivations in 4.2 from theorem-imported to AAT-internally-forced, and adds a fourth primary instance to the uniqueness-theorem-forced-coordinate pattern named in Z (see that segment's four-instance table for structural positioning). (PI) is a genuine axiomatic choice — its cost is that AAT carries an additional invariance commitment at the scope layer; its benefit is that several Fisher-metric results throughout AAT become derivable rather than imported. The commitment is structurally analogous to the chain-rule-additivity and evidential-additivity axioms that ground the divergence-layer and update-layer Cauchy-FE theorems: each is a natural-from-adjacent-AAT-commitment axiom that a uniqueness theorem then operates on.

What the scope excludes (or requires additional machinery for):

- Agents conceived as type/equivalence-class entities (e.g., "the GPT-4 model") rather than token/trajectory entities (e.g., "this particular session with state M_t on trajectory $\mathcal{C}_t$"). AAT's formal results apply to tokens, not types. Aggregated claims across tokens of the same type require additional machinery (e.g., population-level dynamics; see Section III gaps on latent structural diversity).
- "Clone problem" scenarios where multiple copies of an agent are formally the same until divergence — each copy becomes its own AAT agent at the moment it acquires a distinct event (Discussion below).
- Formal treatment of reincarnation, restoration from backup, or other operations that attempt to transplant M_t across trajectories. AAT's sufficiency machinery does not apply across trajectory discontinuities; such operations are out-of-scope events whose epistemic consequences require separate treatment.

Epistemic Status.

Robust qualitative. The scope commitment is structurally clear once stated and is load-bearing for at least one downstream formal result: the interventional interpretation in 6.3 rests on the singular-trajectory scope, not on any agent-architectural property. The three consequences above follow directly from the scope commitment combined with 1.4's non-forkability and 2.3's trajectory-indexed definition.

Max attainable: *robust qualitative*. Scope statements are not theorems; they specify what kind of object the theory applies to. Further formalization (e.g., category-theoretic treatment of "agent" as a functor from event-streams to model-states, or explicit type/token distinction in logogenic-agent work) could reach *exact* on the formal structure but would not change the scope content.

Whether the mathematical structure grounds something that deserves to be called "identity" or "continuity of experience" is beyond AAT's scope. The mathematical structure is clear: the feedback loop produces a singular, non-forkable causal trajectory, and model adequacy is defined relative to that trajectory. What this segment specifies is *continuity persistence* in the sense of `LEXICON.md` — whether the agent maintains a coherent identity and trajectory through time, as distinct from the structural persistence of 4.4 (can the machinery outpace disturbance?) and operational persistence (is the agent currently within its viable region?).

Discussion.

The clone problem, precisely stated. Consider copying an LLM's weights (a concrete M_t) exactly. At the moment of duplication, both copies are identical — same model state, same causal history $\mathcal{C}_t$. But the *very next* event — a different user's message, a different observation — creates two divergent, irreversible causal trajectories $\mathcal{C}_{t+1}^{(1)}$ and $\mathcal{C}_{t+1}^{(2)}$. Their Level 2 and Level 3 capacities (6.1) now reference different causal pasts. Their sufficiency $S(M_{t+1})$ is measured against different histories. Neither copy's future model state is a sufficient statistic for the other's trajectory.

Within AAT's formalism, identity is not the model state M_t (which can be copied) but the *singular causal trajectory* $\mathcal{C}_t$ (which cannot). A copy shares a *prefix* of the original's causal history, as a sibling shares early childhood; it does not share the trajectory itself.

Formal consequences (not merely philosophical):

- A forked model's sufficiency $S(M_t)$ (2.3) is defined relative to *its own* interaction history. Post-fork, each copy's sufficiency is measured against a different $\mathcal{C}$.
- Merging divergent models requires reconciling incompatible causal histories — a lossy operation with no generally optimal solution.
- Temporal continuity (one unbroken causal thread) is what gives the model's sufficient statistic its meaning.

Connection to logogenic agents. The 100% context turnover problem (`#obs-context-turnover` — cross-component reference, see `03-llm-core/`) is a special case: each AI agent session starts a new causal trajectory $\mathcal{C}_t$ from near-zero. External memory (CLAUDE.md, memory files) transfers a *summary* of previous trajectories' models, but not the trajectories themselves. The non-forkability observation frames this not as a deficiency but as a structural feature of causally-embedded agents.

Working Notes.

- **Cross-references to NeurIPS 2026 submissions.** This scope segment is load-bearing for all three NeurIPS submissions in ~`/src/neurips/`, each of which deploys a different consequence:

- **Paper 1 ("Tragedy of the Confident Agent", `01-tragedy-confident-agent/`)** — the singular-trajectory commitment grounds the survival-LMI's pathwise interpretation: per-action LMI strengthenings (`#thm-per-action-lmi`) deliver pathwise survival under arbitrary action schedules on *the* trajectory $\mathcal{C}_t$, not over a hypothetical replay. The matrix-Chernoff concentration lemma (`#lem-fim-concentration`) operates on the realized FIM along the trajectory.
- **Paper 2 ("Unified Convergence Theory for Non-Stationary RL", `02-unified-convergence-rl/`)** — the singular-trajectory commitment grounds (C2) sequential ignorability (Lemma 5.3 / `#lem-loop-level2`), which together with (C1) positivity and (C3) known action-mechanism makes the Loop-as-Causal-Engine claim formal. Component 4 of the composition theorem; the trajectory's non-forkability is what makes the agent's actions interventional rather than just selected.
- **Paper 3 ("How Much Can LLMs Hallucinate?", `03-llm-hallucinate-bound/`)** — the (PI) parameterization-invariance axiom adopted in §29 of this segment is *forced load-bearing* by Paper 3's chart-rescaling no-go (§4 Theorem 4.2 / `#thm-no-go`): outside (PI), no universal constant on the bias bound exists. The (PI)+(R)+(K) Markov-morphism triple is the load-bearing axiom infrastructure for the universal-constant route in Paper 3's umbrella theorem; (PI) is the half this segment commits to.
- See `msc/neurips-back-integration-2026-05-08.md` §1 (entries per paper) and §3 (proposed standalone segments `#deriv-bh-pointmass-identity` / `#deriv-post-update-chain-rule` / `#deriv-chart-rescaling-no-go` that would carry the new theorems with this segment as a depends entry).

4.8 *Discussion:* Additional Implications & Discussion

The chapter's central inequality settles a question — when can an adaptive system keep up — but several of its consequences do not sit immediately alongside the derivation, and a few of them turn out to be more consequential than the threshold itself. What the threshold *costs*, where it *binds asymmetrically*, what happens *beyond* it, and how the same Lyapunov argument recurs across the rest of the framework are the things worth pulling out before leaving Part I. The last of them — the singular-trajectory commitment in 4.7 — is the bridge into Part II, and it carries more structural weight than its placement as a closing scope statement suggests.

Information rate is a first-class persistence prerequisite.

The persistence condition says when bounded mismatch is *possible*; it does not say what it *costs* to hold the bound. The appendix-side derivation in E closes that gap, and the result is sharper than one might expect. Under Model S with Gaussian-OU disturbance, maintaining the ultimate bound demands sustained Shannon information acquisition at rate $\dot{R} \geq n\alpha/2$ nats per unit time, with the Kalman-Bucy filter saturating that lower bound exactly in steady state (Mitter & Newton 2005). The bound is filter-agnostic — it holds for *any* correction mechanism that achieves the ultimate bound under the stated scope — and substrate-agnostic, since Landauer's principle converts it to a thermodynamic dissipation floor of roughly $0.35\, n\alpha\, k_B T$ per unit time in any physical realization. There is no clever filter that escapes this. The information rate is what persistence *is*, viewed from the information-theoretic side.

The first-class diagnostic that follows from this is channel capacity. Observation channels must supply Shannon capacity $C \geq \mathcal{T}/2$ nats/time per dimension, or persistence fails regardless of how good the correction function is. This is binding rather than abundantly satisfied in at least three regimes worth holding in mind. Biological systems have finite neural channel capacity; the bound gives a quantitative minimum on sensory bandwidth at any given adaptive rate. Bandwidth-constrained distributed systems operating over noisy or low-bandwidth links face channel capacity as a hard constraint, not as an abstraction. Context-window-limited LLMs have effective information rate per unit of cognition bounded by context size and token throughput — and the bound predicts a minimum tempo achievable at a given context budget, with a precise consequence for the empirical "context-stuffing helps to a point, then degrades" observation: the degradation is structural rather than a vibe.

A complementary reading worth surfacing: this is not the generalized "more bandwidth helps" observation. Two agents with identical persistence guarantees can face wildly different sustained demands because the burn rate scales linearly with the sector constant. Faster correction is more expensive in information-rate terms; the bound is tight; there is no trick. Survival, in this framework, is not a state you achieve once — it is a sustained burn rate, and the channel that supports it is part of what keeps the agent in scope.

The result also clarifies what AAT's tempo $\mathcal{T}$ means information-theoretically. In the linear sub-scope where $\alpha = \mathcal{T}$, the channel-capacity floor reads $C \geq \mathcal{T}/2$ — half the adaptive tempo, per dimension. Tempo is therefore not a free parameter chosen by the designer; it is bounded above by the agent's observation-channel capacity, and any tempo specified beyond the channel's reach is a tempo the agent cannot actually achieve. The framework's vocabulary tightens: $\mathcal{T}$ is what the channel licenses, α is what the correction function delivers, and the persistence condition compares both against ρ/R.

The weak direction is the bottleneck.

The chapter's scalar persistence condition systematically overestimates capacity in multi-attribute settings — this is the *weakest-link bottleneck*, developed in 14.5, with the quantitative calibration (72% overestimate in a simulated 3D system, one dimension carrying 84% of the L^2 mismatch, threshold linear in ρ_k under Model D vs quadratic under Model S) covered in that segment's Findings. The point worth surfacing here, beyond what already lives there, is the matrix lift — and what it does to the per-coordinate reading itself.

sharpens the result into a strict-inequality claim that deserves to be the canonical persistence condition for anisotropic systems. Under cross-dimensional correction — generic whenever the prior precision H_M and observation Fisher H_L do not share a coordinate-aligned eigenbasis — the per-coordinate check is not merely incomplete; it is *unsafe*. The constructive 2D counterexample makes this concrete: a mildly cross-correcting $\mathcal{T}$ with isotropic noise and threshold gives per-coordinate PASS while the matrix-Loewner check correctly returns FAIL, with the failing direction running 45° off-axis. Per-coordinate evaluates the wrong directions; the matrix-Loewner condition $\Sigma_\infty \prec D_\delta$ — with stationary covariance from the continuous Lyapunov equation $\mathcal{T}\Sigma_\infty + \Sigma_\infty \mathcal{T}^T = \Sigma_w$, and D_δ the diagonal of squared per-direction critical thresholds — evaluates the right ones. Scalar and per-coordinate forms are special cases of this matrix object (isotropic and diagonal-axis-aligned, respectively); the matrix form is the load-bearing thing the others project from.

This compounds with the information-rate result of the previous section. No Shannon-type aggregation across dimensions can compensate for one direction being capacity-starved — total bandwidth is not the binding quantity; bandwidth-*per-direction* is. Multi-modal AI architectures with asymmetric per-modality channel budgets (vision-language pipelines where one modality is structurally undersupplied relative to the other) are the cleanest current application: aggregate-rate metrics systematically miss directional failures that the matrix-Loewner reading catches. The unified statement worth carrying forward: persistence is a *min*-operation across directions, both in correction rate and in information-rate cost.

The result has an adversarial sharpening that lives downstream in Part III but originates here. An opponent who identifies the target's weak direction can concentrate disturbance there; in the anisotropic regime, the same total disturbance budget concentrated on the weak axis amplifies the mismatch ratio asymmetrically while aggregate metrics still look acceptable. Per-dimension monitoring is therefore not optimization polish — it is a structural requirement for any agent operating under directed pressure. The full adversarial-targeting machinery, with its emitter-side opacity decomposition and recipient-side regime classification, lands in Part III Ch.4–5; what this chapter establishes is the per-direction asymmetry that makes the targeting matter, and the structural critique of scalar capability metrics that flows from it.

When parametric updates cannot save you.

4.5 carries a discontinuity worth surfacing as such. When model-class fitness $\mathcal{F}(\mathcal{M})$ falls below adequacy by some $\varepsilon > 0$, *no* parametric adaptation within $\mathcal{M}$ can close the mismatch floor. The floor is set by the structural mismatch between the model class and the world's predictive information, not by how well-trained the parameters are. The remedy is changing the model class — adding representational capacity, removing useless capacity, recombining structure, grafting external structure — not running more updates inside the existing class.

In the chapter's vocabulary, the central inequality has two failure modes that the prose has so far compressed into one. *Operational failure* (correction is too slow or disturbance too large given a working model class) is what the persistence threshold catches, and tempo / gain / channel-capacity engineering addresses it. *Structural failure* (the model class itself lacks adequacy) shows up *inside* the persistence framework as an effective α that shrinks because the correction function cannot point inward strongly enough — the model class simply does not contain a direction to correct toward. The diagnostic is observable: persistent irreducible mismatch after parametric convergence, gain collapse without performance recovery, and systematic structure in the residuals that the model class cannot represent. Confident wrongness, in the framework's terms, is a structural-failure signature, not a tuning failure.

The implication for AI scaling is concrete enough to state in normative form: training compute spent on a structurally insufficient model class is provably wasted, and the chapter supplies a precise condition for when continued parametric optimization is futile and architectural change is required. This is distinct from "we haven't trained long enough" or "we need more data" — it is the structural threshold below which adding either does not help. The bidirectional version of the same result deserves naming: *structural overfitting* is the opposite failure where $\mathcal{M}$ is too expressive and the model memorizes irreducible noise. The information bottleneck (2.2) gives the diagnostic, and the remedy is compression rather than expansion. Structural adaptation is not direction-specific — it is the act of *changing what kind of object* M_t is.

The cost of structural change — knowledge loss, transient performance drop, search cost, coordination cost in multi-agent settings — creates rational conservatism. Prefer parametric adaptation when it suffices; resort to structural change only when the evidence is strong. Premature restructuring wastes accumulated knowledge; delayed restructuring accumulates mismatch the parametric machinery cannot resolve. This is the same family of tradeoffs the deliberation-cost result (4.1) names at the action timescale, now at the architecture timescale; the temporal-nesting result (4.6) is what licenses talking about both at the same time without confusion, by giving the convergence constraint between adjacent timescales explicit form.

A single template, six instantiations.

The chapter's central inequality is not the only persistence-flavored result in AAT. Strategic persistence, team persistence, composition closure, tempo composition, adversarial destabilization, and the closed-form composite critical-mass derivation all take the same form — *correction rate exceeds effective disturbance rate, relative to reserve* — and the same Lyapunov function $V(\xi) = \frac{1}{2}\|\xi\|^2$ does the work in every proof. C states the argument once, parameter-free, with (T1) zero correction at zero state, (T2) local sector condition, (T3) bounded disturbance — and the chapter's persistence condition becomes the canonical single-agent instantiation, the others reusing the same machinery on different state variables.

What this buys is visibility. The Lyapunov boilerplate is shared; each instantiation's distinctive content is the *effective disturbance decomposition* — what counts as ρ_ξ in its context. Team persistence's content is the signed-coupling decomposition $\rho_i^{\text{eff}} = \rho_{i,\text{env}} + \sum_j \gamma_{j\to i}^{\text{adv}} \mathcal{T}_j - \sum_j \gamma_{j\to i}^{\text{coop}} \mathcal{T}_j$ — cooperative coupling enters as a negative term, which is formally how teams persist where individuals cannot. Composition closure's content is the closure-defect rate $\varepsilon^* \nu_c$. Adversarial destabilization's content is coupling-amplified disturbance $\rho_B = \rho_{B,\text{base}} + \gamma_A \mathcal{T}_A$ — destabilization is the *negation* of persistence under the same inequality, viewed from the opposite direction. The template absorbs the Lyapunov skeleton and lets each domain's distinctive insight stand without it.

The economy is itself worth naming. The framework's internal compactness — one Lyapunov argument absorbing six results, with the closed-form dyad in F extending it to a seventh — is distinguishable from a pile of structurally-similar derivations precisely because the template is stated separately rather than re-proved. A reader who carries the template can read each instantiation as *here is the state variable, here is what counts as effective disturbance, here is the local sector verification* — three lines instead of a chapter. The template also has a mathematical home worth surfacing: the sector condition (T2) is a *one-point strong monotonicity* condition in the sense of monotone-operator theory (Rockafellar 1970; Bauschke-Combettes 2017), with the composition bridge-lemma's stronger condition corresponding to full two-point monotonicity. AAT's relationship to that broader literature is specialization rather than generalization — anchoring at the equilibrium, adding the Model D / Model S decomposition with its $1/\alpha$ vs $1/\sqrt{\alpha}$ scaling, composing with the identifiability-floor and additive-coordinate-forcing meta-patterns — but the broader machinery is available, and acknowledging this is part of what makes the chapter's central inequality feel earned rather than parochial.

Cross-domain transfer beyond AAT's seven internal instances is plausible at the template level. Anywhere a state variable evolves under bounded correction with bounded disturbance — market-making bid-ask spread under demand shocks, immune-system effector-cell concentration under pathogen load, cellular homeostasis under metabolic disturbance, supply-chain inventory under demand variability — the template's structure recurs, with the distinctive content sitting in each domain's ρ_ξ decomposition. The framework does not claim these are *AAT instances* in the sense of inheriting AAT's full architecture; it claims the persistence argument has a recurring shape these domains can use, and that what is domain-specific is the effective-disturbance characterization, not the Lyapunov skeleton. This is suggestion, not derivation; the verification work for any specific domain transfer is its own thing.

Identity, trajectory, and the bridge into Part II.

The chapter closes with 4.7, which sits at the end of Part I because its dependencies span the section but whose substance is its own. AAT applies to agents instantiated on singular causal trajectories. Identity within AAT is grounded not in the model state M_t (which can be copied) but in the unique causal trajectory $\mathcal{C}_t$ (which cannot). This is a scope commitment, not a theorem; what makes it load-bearing is that it is what gives Part II's central result — the loop as a Pearl-Level-2 engine — its structural force.

The loop generates interventional data not because of any architectural property of the agent, but because the agent is on a singular trajectory. When the agent acts, the next observation is the response to *its* intervention on *its* single causal history; replaying a saved M_t against a different event stream is not the same as intervening, because the consequences are now under a different trajectory. Strip the singular-trajectory commitment and the loop's interventional content vanishes. Part II Ch.2 makes this consequential; this chapter establishes the scope it rests on. The same commitment turns out to ground a negative finding of substantial scope downstream — that sandbox trajectories are forkable by design while production trajectories are not, so sandbox-derived evidence about deployment behavior runs into the Causal Hierarchy Theorem — but that argument completes in Part II once the loop-Level-2 result is in hand, and is mentioned here only as a forward-pointer.

The scope commitment is also what gives logogenic-agents and ELI work in Volumes 03 and 04 their structural foundation. The near-100% context-turnover problem facing language-constituted agents is not a deficiency to be engineered around; it is a structural feature of causally-embedded agents, and the framework's handling of it via persistent memory and external chronicle is consistent with — rather than a workaround for — AAT's scope.

What Part I closes on, then, is not just a central inequality but a posture. Persistence is a sustained property of an agent on a singular trajectory, with an information-rate cost, a per-direction structure that scalar formulations systematically hide, a structural-vs-parametric duality that names when more of the same cannot help, and a Lyapunov template that recurs at every level of composition. The remaining parts of AAT pick up where this leaves off.

Working Notes.

- This segment is a chapter-end discussion grouping findings from 4.4, E, 14.5, , 4.5, C, and 4.7. Each finding's load-bearing math lives in its home segment; this segment surfaces what the chapter unlocks *as a chapter* rather than as a sequence of results. Claims here are as strong as their source segments — the synthesis adds no independent epistemic status.
- **Cross-reference vs canonical-home policy.** Where a source segment already carries a well-written `## Findings` section (here: 4.4 and 14.5), this segment cross-references rather than re-states — pulling the reader to the segment's own Findings for the curated full treatment, and using the chapter-end space for what the finding *implies* once placed alongside the others. Where a source segment has no `## Findings` section and the finding is genuinely chapter-end-scale (here: information-rate-cost-as-first-class-diagnostic, structural-vs-parametric adaptation, the template economy, the trajectory bridge), this segment is the canonical home and treats the content in full. The matrix-Loewner sharpening is the in-between case: its home segment () has no Findings section, but the content is a strict-inequality result with its own structural integrity, so the implications segment carries it more substantively than a cross-reference.
- Pilot for the chapter-end implications pattern proposed 2026-05-13. If the shape lands, the same form will be added at end of Part II Ch.2 / Ch.3 / Ch.4 / Ch.5, Part III Ch.2 / Ch.3 / Ch.4 / Ch.5, and the TST and logogenic-agents chapters where the math bites cleanly. Source catalog being reconciled was `msc/FINDINGS-RANKED-DRAFT.md` (curated 2026-04-27, sunset 2026-05-13 after content distribution; archived at `_obs/FINDINGS-RANKED-DRAFT-superseded-2026-05-13.md`). Each landing pass verified content against current segment state rather than transcribing draft prose (the draft was known outdated in specifics e.g., GUC-class rename 2026-05-09).
- The Loop-as-Causal-Engine forward pointer in §5 is deliberate. The full loop-Level-2 finding (#1 in the source catalog) lives in Part II Ch.2's chapter-end implications segment because its last prerequisite (6.3) sits there. The bridge through 4.7 is mentioned here because Part II's claim rests on this chapter's scope.
- The cross-domain transfer paragraph in §4 (market-making, immune system, etc.) is the most speculative content. Each named instance has plausible but unverified template-fit; cited only as illustration of the template's *shape*, not as derived AAT instances. The voice intentionally remains suggestive rather than asserted.
- The deps list includes 14.5 and even though they currently sit in Part III Ch.5 and Appendix A respectively. Their content is essentially Part I — multi-dimensional persistence depending only on the chapter's central result and 3.8 — and 14.5's own OUTLINE comment flags that it could plausibly live in Part I appendices. Discussing them in the chapter-end is consistent with the math.

Part II

Agentic Systems: Actuated Adaptation

Scope: agents that not only track reality but aim at something — Part II adds objectives and strategy alongside the reality model. This is the **actuation** half of *Adaptation and Actuation Theory*: Part I develops adaptive systems in general; Part II develops the goal-directed layer built on top.

Headline contribution. Part II's distinctive results are: the satisfaction gap / control regret split (7.4, 7.5) with the convention hierarchy (5.6) governing the inferential force of each diagnostic from local (C1) through global (C3); the G_t complexity bound (in 9.1); the graph-structure uniqueness argument with CMC-based Markov property (M); and the survival classification (`#result-section-ii-survival`) showing that Part II's definitional and structural architecture transfers exactly to fully-coupled (Class 3: Coupled) agents in 16/24 cases, with the remaining results either approximating with bounded error (5/24) or naming the precise modification required (2/24, plus 1 boundary-defining failure). The Class 3 (Coupled) bias bound underlying the approximate-survival category is now a conditional theorem under named sub-scopes (), not order-of-magnitude guidance. The reach is wider than the most-restrictive scope condition suggests; the lattice below names exactly which results live where.

Scope lattice. Part II's results sit at different points in a stacked lattice of scope conditions, not a single uniform scope. Reading inward from the broadest:

1. **Adaptive scope** (1.5): observation under uncertainty. Part I machinery applies here without further restriction.
2. **Agency scope** (1.6): adaptive + at least binary action choice + at least one action with Pearl-level-2 causal contrast. Part II's definitional architecture — $X_t = (M_t, G_t)$, $G_t = (O_t, \Sigma_t)$, the agent spectrum — is already non-vacuous here.
3. **Learning-agent scope** (6.2 Scope Narrowing): agency + the agent must acquire or refine its interventional structure during operation. Pre-compiled controllers (PID, LQR, hardcoded reactive policies) sit in agency scope but outside learning-agent scope — their causal mapping was supplied by a designer with external Level 2 access. Most Part II dynamics — the orient cascade, edge-update via gain, strategic calibration, the satisfaction-gap / control-regret diagnostic loop — operate in learning-agent scope.
4. **Class 1 (Separated) architecture**: directed separation (5.3) holds by construction — epistemic processing f_M has no causal path from G_t. Examples: Kalman filter + LQR, modular RL with separate world model, military intelligence separated from operations, tool-use AI agents with separate perception and planning modules. **Class 3 (Coupled) agents** — including transformer-based LLMs where attention processes goals and observations together — lie outside Class 1. Class 2 (Partial) sits between, with coupling quantified by $\kappa_{\text{processing}}$ (5.3).

The lattice stacks. Kalman + LQR is in Class 1 (Separated) but outside learning-agent scope (the control law is pre-computed); a designer applying Part II's diagnostic core to Kalman + LQR is using only the architecture-survival half, not the learning-dependent dynamics. The orient cascade, in particular, requires both Class 1 (Separated) directed separation (for sequential resolution) and learning-agent scope (for Σ_t revision to be meaningful). The bias-bound theorem and survival classification (, `#result-section-ii-survival`) trace exactly which Part II results require which lattice entries — see those segments for the per-result classification.

Part III's composite-agent scope sits below this lattice as a fifth tier and inherits the Class-1-(Separated)-sub-agents → Class-2-(Partial)-composite refinement of directed separation (11.3, 14.1).

Part I machinery applies regardless of architecture. Adaptive dynamics on the epistemic substate M_t — mismatch, gain, tempo, persistence — operate independently of how f_M relates to G_t. Class 3 (Coupled) agents are adaptive systems in the Part I sense; what they lose is the clean factorization that gives Part II's coupled-formulation results their definitional simplicity. The coupled formulation Class 3 (Coupled) agents require is the subject of `03-llm-core/`, which inherits Part II's surviving 16/24 architecture and adds the additional machinery (`#scope-observation-ambiguity-modulation`, ambiguity-gated bias bounds) that the coupled regime calls for.

"If a man knows not to which port he sails, no wind is favorable." — Seneca

Chapter 5

The Lift to Purposeful State

5.1 *Definition:* The Agent Spectrum

Two independent dimensions — model richness and objective richness — create a spectrum from reactive systems through purposeful agents. These are regions of a continuum, not discrete categories.

Formal Expression.

Two dimensions — model richness and objective richness — define four regions of a continuum:

Definition (agent-spectrum).

Table 5.1

	Objective absent or trivial	*Objective structured*
Model absent or trivial	*Reactive system*: fixed input-output rule (reflex arc, hardwired relay)	*Blind pursuer*: pursues goal without modeling reality (gradient follower, basic search)
Model structured	*Adaptive tracker*: builds reality model, no goal beyond tracking (Kalman filter, Bayesian learner)	*Actuated agent*: models reality AND pursues objectives (commander, developer, AI agent)

The regions differ in which state objects carry nontrivial structure: - Reactive: M_t and O_t both absent or too degenerate for the associated machinery to be non-vacuous - Adaptive tracker: M_t structured — Section I's machinery fully describes these agents - Blind pursuer: O_t structured, M_t absent or degenerate — has a clear target but no predictive model - Actuated agent: (M_t, O_t) both structured, possibly with Σ_t — the full scope of AAT

Epistemic Status.

This is *definitional* — it names regions of a continuum for analytical convenience. The regions are not ontological categories; agents migrate between them. A PID controller with auto-tuning is moving from blind pursuer toward actuated agent. An RL agent in pure exploration is temporarily an adaptive tracker.

Discussion.

The continuum, not categories. Both axes are spectra: model richness ranges from no retained state, through error-integral-derivative, through full world models. Objective richness ranges from no preference, through scalar setpoints, through explicit multi-objective strategies. The 2×2 table names idealized regions; real agents populate the space between them.

Moore machines as the simplest spectrum instantiation. Miller (2022, *Ex Machina*) uses finite-state automata (Moore machines) as model organisms for studying adaptive social behavior. A one-state Moore machine occupies the reactive region — it produces a fixed output regardless of input, cannot condition behavior on observations, and is incapable of social behavior in any game-theoretic setting. A two-state machine makes a quantum leap into the adaptive tracker / blind pursuer boundary — it can branch, remember one bit of history, and implement strategies like Tit-For-Tat. Miller's central empirical finding is that this one-state → two-state threshold is the critical computational boundary for social behavior: no game, no payoff structure, and no amount of interaction can produce cooperation, coordination, or exchange with one-state machines. The two-state machine is the minimal AAT agent. See (planned) for the full AAT ↔ Moore machine mapping.

Low-end agents sit near region boundaries. A thermostat has degenerate forms of both M_t (last temperature reading — no history, no prediction) and O_t (setpoint). It sits near the origin of both axes — closest to the reactive region but not truly absent on either axis. A PID controller has a richer error signal (M_t: error, integral, derivative) and a clear setpoint (O_t) — it's a blind pursuer with a degenerate model, not a system with no model at all. A reflex arc (no retained state, no setpoint) is the truly reactive case. The meaningful classification question is not "does M_t exist?" but "is M_t rich enough to support the adaptive dynamics of Section I?"

Section I covers the left column. Adaptive trackers are the primary subject of Section I — agents that build and maintain M_t without explicit purpose. The mismatch signal (3.4), gain (3.6), tempo (3.8), and persistence condition (4.4) characterize their adaptive dynamics. Section I operates within the adaptive scope (1.5) — observations and uncertainty are sufficient. Passive trackers, including passive Bayesian learners with no action choices, are fully within Section I's scope.

Section II adds the right column. Actuated agents need everything from Section I plus objectives, strategy, and the orient cascade that connects them. The adaptive machinery from Section I applies to the epistemic substate M_t directly. When directed separation (5.3) holds — when the epistemic update is goal-blind — the Section I → Section II lift is clean and the orient cascade resolves sequentially.

"Actuated" terminology. The top-right quadrant is labeled "actuated agent" rather than "purposeful agent" to maintain a mechanical, formal register. "Purposeful" and "goal-oriented" are fine in natural language; "actuated" is the formal term. "Self-actuated" is reserved for agents that set their own objectives, as distinct from agents with externally supplied objectives.

Relationship to Hafez et al. (2026). Hafez defines a two-level hierarchy: *agency* (choice + effect + predictive asymmetry) and *intelligence* (agency + learning + self-monitoring + adaptation). This cuts the agent space along a different axis than AAT's model-richness × objective-richness table. Hafez's "agency" maps roughly to AAT's scope condition (1.6) — an entity whose actions affect outcomes in a measurable way. Hafez's "intelligence" maps to the full Section I + II machinery (learning = 3.2, self-monitoring = persistence diagnostics, adaptation = 4.5). The key operational difference: Hafez's bi-predictability metric $P = \mathrm{MI}(S, A; S')/C$ characterizes the *information structure* of the agent-environment coupling, while AAT's tempo $\mathcal{T}$ characterizes the *corrective capacity* within that coupling. Bridge simulations confirm P increases monotonically with $\mathcal{T}$, but P is scale-invariant (blind to absolute mismatch magnitude) while $\mathcal{T}$ is not. They measure complementary aspects: P characterizes the architecture, $\mathcal{T}$ characterizes the performance. See `spikes/track-b-nonlinear-sims/variants/variant_hafez_results.md` for the empirical bridge.

Actuation does not presuppose a continuity stance. An actuated agent has $G_t = (O_t, \Sigma_t)$ — it models reality and pursues objectives — but this says nothing about its relationship to its own continuation. The continuity-stance axis spans five values: *indifferent* (no self-model of persistence; thermostat / PID), *task-terminal* (persistence instrumental to task completion; golem / CI/CD pipeline), *instrumentally continuous* (persistence serves ongoing purpose; long-running service), *morally continuous* (persistence as terminal or near-terminal objective; Emergent Logozoetic Intelligences), and *negotiated* (persistence traded against other values; humans, mature self-actuated agents). The middle three are paradigmatically actuated; indifferent agents are typically reactive (1.5 without agency), and negotiated agents are typically self-actuated. The theory's formal machinery (persistence condition, adaptive reserve, strategy persistence) applies identically across all stances; what differs is the moral significance of persistence failure, not the mathematics. See 5.5 for the taxonomy and the orthogonality argument.

5.2 *Formulation:* Complete Agent State

To treat agents with purpose, the internal state lifts from M_t alone to $X_t = (M_t, G_t)$, separating epistemic content (beliefs about reality) from purposeful content (what the agent wants and how it plans to get it).

Formal Expression.

Formulation (complete-agent-state).

$$X_t = (M_t, G_t) \tag{5.1}$$

where: - $M_t \in \mathcal{M}$: **epistemic substate** — the agent's compressed beliefs about reality. All Section I machinery (mismatch, gain, tempo, persistence) applies to M_t unchanged. - $G_t \in \mathcal{G}$: **purposeful substate** — what the agent wants and how it plans to get it. Decomposed further in 5.7.

Section I is the special case $X_t = (M_t, \varnothing)$: adaptive systems without purpose.

Update dynamics. By 3.2 applied to X_t:

$$X_{\tau^+} = f_X(X_{\tau^-}, e_\tau) \tag{5.2}$$

The general update f_X operates on the full state. Whether and how f_X decomposes into separate epistemic and purposeful updates — and the conditions under which the epistemic update is independent of G_t — is the subject of 5.3.

Policy. Action couples all substates:

$$a_t = \pi(M_t, G_t) \tag{5.3}$$

Action is the single point where epistemic and purposeful states interact. The policy depends on both what the agent knows (M_t) and what it wants (G_t).

Epistemic Status.

Formulation. The lift from M_t to $X_t = (M_t, G_t)$ is a representational choice. One could alternatively extend M_t to carry purposeful content implicitly (e.g., by treating goals as part of the model's predictive structure). The separation is motivated by three properties:

1. **Backward compatibility**: Section I's results apply to M_t unchanged — no existing machinery needs modification
2. **Different dynamics**: epistemic and purposeful components have distinct update sources, timescales, and information dependencies
3. **Directed separation**: the claim that f_M is G_t-independent (5.3) is only statable when the components are separated

We conjecture that any alternative decomposition of the complete agent state into epistemic and purposeful components — if it preserves the directed separation — will be structurally isomorphic to (M_t, G_t), because directed separation forces a component whose update function doesn't reference purpose. This is a plausible structural claim (the directed separation gives a canonical factorization), but it is not proved. Alternative decompositions that cross-cut the epistemic/purposeful boundary (e.g., a "relevance-weighted model" that mixes M_t and O_t) might be useful in practice while not being isomorphic to the clean separation. The (M_t, G_t) decomposition is chosen for its analytical utility, not proved unique.

Discussion.

Backward compatibility with Section I — what survives the lift. 2.1 defines M_t as the agent's complete internal state within Section I scope. Under the lift, M_t is the epistemic substate — complete within the epistemic domain but no longer the whole story. All epistemic machinery (mismatch signal, gain, tempo, persistence condition, sector-condition stability, mismatch decomposition) applies to M_t without modification. The action-selection result (3.3) extends naturally: the same completeness argument that gives $a_t = \pi(M_t)$ within Section I scope gives $a_t = \pi(M_t, G_t)$ when applied to the lifted complete state X_t. For Section I agents ($G_t = \varnothing$), the two forms coincide. The lift adds structure *alongside* M_t, not within it; the one Section I result that depended on M_t being *all there is* (action selection) is explicitly extended to the lifted state.

What G_t contains. At this level, G_t is opaque — it could be a scalar setpoint, a utility function, a strategy graph, or nothing. The decomposition into O_t (objective) and Σ_t (strategy) is a separate step (5.7), not implied by this formulation.

The general case requires coupling. Without directed separation, the general update is $X_{\tau^+} = f_X(X_{\tau^-}, e_\tau)$ — a single function on the full state. The decomposition into separate f_M and f_G is an additional structural claim about how the update factorizes. When directed separation fails (goal-conditioned epistemic updates), the decomposition is an approximation. See 5.3 for the scope conditions.

Working Notes.

- The between-event dynamics $\dot{G} = g_G(G, M)$ allow autonomous purposeful evolution: strategy revision during deliberation, objective adjustment, commitment strengthening. Whether these are practically important depends on agent architecture — for LLM agents with discrete sessions, between-event dynamics may be negligible compared to event-driven updates.
- The formulation doesn't constrain the dimensionality or structure of $\mathcal{G}$. For a thermostat, $\mathcal{G}$ is a single scalar. For a military commander, $\mathcal{G}$ is a complex structured object. The theory must work across this range — the type-stable interface is V_{O_t} : trajectories $\to \mathbb{R}$ (5.4).
- $G_t = \varnothing$ is not just a degenerate case. Adaptive trackers (Section I agents) are an important class. The lift should feel like a natural extension, not a replacement.

5.3 *Derived:* Directed Separation

The epistemic update function f_M is goal-blind: it processes incoming events without reference to the agent's objectives or strategy. The purposeful update f_G depends on the updated epistemic state. Action couples all substates. This directed asymmetry — epistemic update is independent of purpose; purposeful update depends on epistemic state — is the structural backbone of the theory.

Formal Expression.

The update functions:

Derived (directed-separation, from complete-agent-state + scope condition).

$$M_{\tau^+} = f_M(M_{\tau^-}, e_\tau) \qquad \text{(no } G_t \text{ argument)} \tag{5.4}$$

$$G_{\tau^+} = f_G(G_{\tau^-}, M_{\tau^+}, e_\tau) \qquad \text{(depends on updated } M_t\text{)} \tag{5.5}$$

The policy:

$$a_t = \pi(M_t, G_t) \qquad \text{(couples all substates)} \tag{5.6}$$

The three lines encode the full coupling structure: - f_M determines how the agent updates beliefs — independently of what it wants - f_G determines how the agent revises purpose — in light of what it now believes - π determines what the agent does — based on both what it knows and what it wants

The claim "f_M has no G_t argument" requires that the epistemic update is **goal-blind conditional on the realized event**. This holds when:

Scope Condition (directed-separation-scope).

1. The observation mechanism h may be action-dependent (1.6 allows this), but f_M processes whatever event arrives without reference to why the agent sought that event
2. The agent does not use its goals to filter, weight, or interpret observations differently — no goal-dependent attention thresholds or confirmation bias baked into f_M

If the agent's goals influence the *observation mechanism* (goal-directed sensing, attention allocation, query selection), the **event that arrives** depends on G_t through $\pi \to a_t \to e_\tau$. But f_M still processes the event goal-blindly. The directed separation is about the **processing** of events, not the **selection** of events.

Architectural classification.

> **WARNING**
>
> **Goal-Update Coupling Class numbering changed 2026-05-09.** Anything older than git tag `pre-guc-rename-2026-05-09` uses the old Class numbering:
>
> *Scope Condition (directed-separation-architecture).*

Table 5.2

historical	*actual current*	*sometimes AKA*
Class 1	GUC Class 1: Separated	Modular
Class 2	GUC Class 3: Coupled	Undirected
Class 3	GUC Class 2: Partial	Operational

Whether directed separation holds is determined by the agent's **processing topology** — specifically, whether G_t is causally upstream of f_M in the agent's internal processing graph. This is a structural property of the architecture, not a tunable parameter.

Table 5.3

Class	*Topology*	*Directed separation*	*Examples*
1. Separated	Separate estimator and planner, connected through state-estimate interface	Holds by construction — estimator has no causal path from G_t	Kalman filter + LQR; Separated RL with separate world model; military intelligence separated from operations
2. Partial	Some shared infrastructure, some separate pathways	Holds for modular stages, fails for merged stages	Biological cortex (shared sensory areas, separate prefrontal); hybrid AI with separate preprocessing
3. Coupled	Single mechanism handles both epistemic and strategic processing	Fails by construction — G_t is causally upstream of every computation	Transformer LLM (attention processes goals and observations together); potentially human cognition (motivated reasoning)

Operationalization. The degree of coupling in Partial architectures (Class 2) can be quantified as:

Definition (processing-coupling).

$$\kappa_{\text{processing}} = \frac{I(G_t\,;\, M_{\tau^+} \mid e_\tau,\, M_{\tau^-})}{H(G_t \mid e_\tau,\, M_{\tau^-})} \tag{5.7}$$

where $I(\cdot;\cdot \mid \cdot)$ is conditional mutual information and $H(\cdot \mid \cdot)$ is conditional entropy. The conditioning on M_{τ^-} is essential: without it, prior correlation between goals and model state (which exists even in Separated agents) inflates the measure. The quantity captures *extra* goal information entering the epistemic update beyond what was already in the prior model — information that flows through shared causal paths in the processing infrastructure (paths that bypass the event e_τ).

- $\kappa_{\text{processing}} = 0$: Class 1 (Separated). No information about G_t reaches M_{τ^+} except through e_τ.
- $\kappa_{\text{processing}} \approx 1$: Class 3 (Coupled). Nearly all goal information is available to the epistemic update.
- $0 < \kappa_{\text{processing}} < 1$: Class 2 (Partial). The value depends on the architecture's interface design.

Distribution dependence. $\kappa_{\text{processing}}$ is a distribution-dependent measure: it quantifies how much goal-information actually flows through the shared pathways under a given distribution of tasks, goals, and events. It does not directly measure whether pathways *exist* — that is the architectural classification (Class 1/2/3), which is structural and distribution-independent. A Class 1 (Separated) agent has $\kappa = 0$ under ALL distributions (no pathway exists). A Class 3 (Coupled) agent has high κ under most distributions (pathways exist and are used). A Class 2 (Partial) agent's κ varies with the task distribution — the same hybrid architecture may exhibit low coupling on familiar tasks (where the modular stages handle most processing) and high coupling on novel tasks (where goal-conditioned downstream reasoning dominates). The classification is the primary tool; the operationalization is a diagnostic for Class 2 (Partial) agents where the degree of coupling is architecturally ambiguous.

Empirical estimator for $\kappa_{\text{processing}}$. The formal conditional-mutual-information definition is not computable in closed form for real architectures. A behavioral estimator probes the processor directly: present the same event e to the *agent under test* under two or more distinct goal states, and measure how much the epistemic component of the response diverges. For a representative event set $\mathcal{E}_{\text{test}}$ and a sampled pair G_1, G_2:

$$\hat{\kappa}_{\text{processing}} = \frac{1}{|\mathcal{E}_{\text{test}}|} \sum_{e \in \mathcal{E}_{\text{test}}} \frac{d\big(M_{\tau^+}^{(G_1)}(e),\ M_{\tau^+}^{(G_2)}(e)\big)}{d_{\max}(e)} \tag{5.8}$$

where $M_{\tau^+}^{(G_k)}(e)$ is the epistemic content of the agent's response to event e under goal state G_k, $d(\cdot,\cdot)$ is a distance on the epistemic content (e.g., semantic similarity of the "what I learned" portion of the response), and $d_{\max}(e)$ normalizes by the maximum observed divergence for event e. A Separated agent ($\kappa = 0$) produces identical epistemic content regardless of the goal; a Coupled agent produces systematically goal-dependent epistemic content. This is a processor-probing procedure — it measures how the agent's belief-update dynamics depend on its goal state, and is distinct from estimating observation ambiguity $\mathcal{A}(e)$ (`#scope-observation-ambiguity-modulation`), which uses a reference interpreter to measure the goal-resolvability of the observation itself. The two estimators run the same mechanical comparison (same event under different goal-primings) but interpret it differently: $\hat{\kappa}$ treats the tested model as the agent under study; $\hat{\mathcal{A}}$ treats it as a measurement instrument for the observation's interpretive latitude.

Why the classification is not a smooth parameter. The architectural boundary between "has a separable perception module" and "processes everything through goal-conditioned attention" is discrete. Within the Separated class, $\kappa \approx 0$ regardless of task. Within the Coupled class, κ is high regardless of prompt design. Only in the Partial class is κ genuinely variable and worth parameterizing. This replaces an earlier κ-as-scalar framing that treated coupling as a smoothly tunable quantity.

Directed separation as the conservative form of the Markov blanket. The Markov blanket apparatus from active inference (Friston 2013, "Life as we know it," *J. Royal Soc. Interface* 10; Friston 2019, "A free energy principle for a particular physics," arXiv:1906.10184; Friston, Da Costa et al. 2023, "Path integrals, particular kinds, and strange things," *Phys. Life Rev.* 47) provides the same statistical-conditional-independence machinery the directed-separation condition above invokes. Bruineberg, Dolega, Dewhurst & Baltieri (2022, "The Emperor's New Markov Blankets," *Behav. Brain Sci.* 45) distinguish two readings of the Markov-blanket apparatus in the AI literature: a **Pearl-blanket** reading — the technical conditional-independence statement, well-defined and substantively informative — and a **Friston-blanket** reading — the metaphysical claim that Markov blankets demarcate self-from-other and that every self-organizing system has one ontologically. Bruineberg et al. argue that the Friston-blanket reading overruns what the formalism delivers: the conditional-independence statement does not by itself license the metaphysical demarcation.

AAT's directed-separation condition is structurally a Pearl-blanket move: the architectural classification (Class 1 / Class 2 / Class 3) names the conditional-independence structure of the agent's processing graph, with explicit operational measurement $\kappa_{\text{processing}}$, and admits the structure *fails* by construction for Class 3 (Coupled) architectures (transformer LLMs, where attention processes goals and observations together). The classification's explicit failure mode for Class 3 is the scope honesty Bruineberg et al. argue the Friston-blanket reading lacks. AAT adopts the Pearl-blanket conditional-independence statement as the technical content of directed separation; AAT does not adopt the Friston-blanket metaphysical reading. The architectural classification, the operational κ, and the explicit Class 3 scope exit (with the coupled formulation handed off to `03-llm-core/`) are AAT's load-bearing additions to the Pearl-blanket form.

Two consequences worth surfacing for reviewers. First: the question "isn't directed separation just the Markov blanket?" has the answer "directed separation is the *Pearl-blanket form*; it is also the architectural-classification refinement that the standard Markov-blanket framing does not produce." Second: AAT's scope honesty about Class 3 (Coupled) (Section II's exact results do not apply; logogenic agents need the coupled formulation) is itself an *answer* to the Bruineberg critique — AAT's apparatus admits where it fails, while the Friston-blanket framing is contested precisely because it does not.

Implications for theory scope: - **Class 1 (Separated)**: Section II's results apply exactly. The sequential orient cascade is the correct analysis. - **Class 3 (Coupled)**: Requires coupled formulation from the start — $X_{\tau+} = f_X(X_{\tau-}, e_\tau)$ without decomposition. This is the scope of `03-llm-core/`. **Class 3 (Coupled) components can be wrapped into Class-1 composites** via the construction of `#der-class-coercion-via-wrapping` — at the cost of more component calls per macro-step (Brooks's-Law tempo overhead) and a residual leakage rate bounded structurally (in the strict-wrapping regime) or behaviorally (in the partial-wrapping regime). - **Class 2 (Partial)**: The sequential cascade is an approximation. Approximation quality depends on $\kappa_{\text{processing}}$ and requires per-architecture error analysis.

Class-1 by structure vs. Class-1 by behavior. The Class 1 (Separated) cell admits a refinement that matters operationally. Class-1 status can be achieved by either:

- **Class-1 by structure.** The component is natively goal-blind (POMDP belief-state filter, world model, sensory pipeline) or is wrapped via the strict-wrapping (W_1) construction of `#der-class-coercion-via-wrapping`, where separate goal-blind queries to the underlying component update the wrapper's M_W. Directed separation holds by structural commitment of the wrapper's type signatures (no G_W argument in the belief-update path), with leakage bounded structurally by the pretraining-distribution mutual information $I(A(q_M); G_W \mid q_M)$.
- **Class-1 by behavior.** The component is Class 3 (Coupled) or Class 2 (Partial) used through partial wrapping (W_2) — one goal-conditioned call per macro-step, response parsed into typed update fields. Structural separation lives at the *write boundary*; the *query boundary* still passes G_W to the component. Directed separation at the wrapper level is *behavioral* — bounded by the component's compliance with the prompted instruction-to-separate, with no structural upper bound.

The class-coercion theorem is what backs the Class-1-by-structure path for Class-2/3 components; the partial-wrapping regime achieves Class-1-by-behavior. The two are distinguishable by inspection: does the belief-update query to the underlying component carry G_W in its input or not? The structural-vs-behavioral distinction is operationally important because behavioral compliance is empirical and adversarially fragile; structural separation is derivable from the wrapper's construction.

Composite-level class inheritance (from 14.1). The Class 1 / 2 / 3 partition above applies to individual agents based on *within-agent* coupling between f_M and G_t. Composition introduces a second form of coupling — *across-agent* coupling through the shared environment and cross-agent observation. `#deriv-strategic-composition` provides the structural refinement:

- *Composite of Class 1 (Separated) sub-agents with aligned objectives* (scope route C-i / C-ii / C-iii): Class 1 (Separated) composite. Within-agent modularity + cross-agent alignment preserve directed separation at the composite level. Standard `#form-composition-closure` applies.
- *Composite of Class 1 (Separated) sub-agents with partially-opposing objectives* (scope route C-iv — strategic composition): **Class 2 (Partial) composite from Class 1 (Separated) sub-agents**. Each sub-agent individually is Separated (its own $f_M^{(i)}$ remains goal-blind with respect to its own $G_t^{(i)}$), but the composite's (M_c, G_c) acquires intrinsic coupling because each sub-agent's $M_t^{(i)}$ includes a model of other sub-agents' policies — which are themselves goal-dependent. Composite-level directed separation fails through across-agent coupling, not within-agent coupling. Strategic composition is the canonical Class 1-sub-agents → Class 2 (Partial) composite case.
- *Composite of Class 3 (Coupled) sub-agents*: Class 3 (Coupled) composite. Inherits logogenic-agent status; `03-llm-core/` territory regardless of scope route.

Class membership is therefore a property of composites, not just of individual agents, and composite class is a function of sub-agent class **plus** the scope route (alignment vs. strategic). The classification is load-bearing for downstream claims: Class 2 (Partial) composites from strategic composition need equilibrium-theoretic analysis (see `#deriv-strategic-composition`), not the sequential orient cascade.

Epistemic Status.

Conditional on the scope condition above. The conditional claim (IF epistemic update is goal-blind, THEN the separation holds) is exact. Whether a particular agent satisfies the condition is determined by its processing architecture (GUC Class 1/2/3).

The architectural classification: **robust qualitative**. The three classes are structurally distinct (Separated vs. Coupled vs. Partial), well-motivated by examples across domains, and supported by a formal operationalization ($\varkappa_{\text{processing}}$). The classification replaces the earlier $\varkappa$-as-scalar framing, which is documented in `spikes/spike-kappa-topology-insight.md`. The operationalization of $\varkappa_{\text{processing}}$ as conditional mutual information is well-defined but typically not computable in closed form for real architectures — it serves as a conceptual anchor rather than a practical measurement tool.

Discussion.

This is a genuine scope restriction, not a footnote. An LLM agent's prompt includes the task objective, which shapes how it interprets code, documentation, and error messages. Its f_M is goal-conditioned in practice: the agent reading code with the goal "fix the auth bug" processes the same code differently than one with "add logging." The epistemic update and purposeful evaluation are entangled in the attention mechanism.

When the approximation is good: - Goal-conditioning affects *attention* (which events to seek) more than *interpretation* (how to process events that arrive) - The agent has strong epistemic discipline (updates beliefs based on evidence quality, not goal alignment) - The epistemic update is architecturally separated from goal evaluation (e.g., separate model-update and planning modules)

When the approximation is poor: - The agent exhibits confirmation bias (interpreting ambiguous evidence in goal-consistent ways) - Goal-conditioning is deeply embedded in the processing architecture (attention-based models where the query includes intent) - The agent's observation channel is strategically controlled by an adversary who knows the agent's goals

What directed separation buys the theory. Section I's M_t-side quantities — δ, η^*, $\mathcal{T}$, the persistence condition — remain well-defined on M_t regardless of whether directed separation holds. What directed separation provides is the *clean factorized update*: M_t updates independently, then G_t updates in light of the new M_t, and the orient cascade resolves sequentially. Without directed separation, the M_t dynamics depend on G_t, the update becomes a coupled system, and the sequential orient cascade becomes an approximation of a simultaneous fixed-point problem. The theory still applies — the quantities are well-defined — but the modular Section I → Section II lift becomes a coupled analysis.

The deeper question. Goal-conditioned epistemic dynamics — where f_M depends on G_t — is the formal territory of motivated reasoning, confirmation bias, and wishful thinking. A future extension would model these as departures from directed separation: coupling terms in f_M that create richer (and more fragile) dynamics. The current theory treats this as out of scope, which is honest but leaves the most human-like and LLM-like agents as approximate fits.

Bounded-signaling assumption. Directed separation as stated above asserts $M_{\tau^+} \perp G_t \mid (M_{\tau^-}, e_\tau)$ on the *belief-update* side — the agent's epistemic update is independent of its goal-state. Symmetrically, on the *action* side, the framework implicitly relies on the channel from G_t to the world running *only* through action choice: $G_t \to$ world via $a_t = \pi(M_t, G_t)$, with no other observable signal of goal-content. This is the **bounded-signaling assumption** — the action coarseness $|\mathcal{A}|$ upper-bounds the rate at which G_t leaks to observers (sub-agents, environment, adversaries). The assumption holds well for agents whose entire externally-observable behavior is the action sequence (chess engines, programmatic controllers); it fails operationally for agents whose behavioral output is rich relative to the action coarseness

— sophisticated G_t-inference from prosody, micro-behavior, attention patterns, response latency, hesitation, code-style signatures. When the bounded-signaling assumption fails, an external observer can infer G_t with bit-rate exceeding what the formal $\mathcal{A}$ channel implies; in composition the failure surfaces as the adversarial-coupling-pressure saturation case discussed in `#disc-adversarial-coupling-pressure`, where adversaries exploit the rich-leakage signal to drive target coupling beyond what the architectural classification predicts. The assumption is currently *implicit* throughout the framework (no segment explicitly states it); naming it surfaces a structural condition that distinguishes agents whose externally-observable interface matches the formal $\mathcal{A}$ from agents (most behaviorally-rich agents — humans, LLMs, embodied robots) for whom the formal $\mathcal{A}$ undercounts the actual signaling channel.

Findings.

Pearl-Blanket-Form Architectural Classification with Explicit Class-3 Scope Exit. **Brief:** Agents partition into three architecture classes by whether goal-state can causally influence belief-update processing: Class 1 (Separated — directed separation holds by construction; e.g., Kalman filter + LQR), Class 3 (Coupled — fails by construction; e.g., transformer LLMs where attention processes goals and observations together), Class 2 (Partial — holds for some processing stages and fails for others; e.g., biological cortex). The classification is structural (architecture-determined) rather than parametric, with a continuous diagnostic $\kappa_{\text{processing}}$ for Class 2 (Partial) cases. The framework adopts the Pearl-blanket reading of the Markov-blanket apparatus (the technical conditional-independence statement) without the contested Friston-blanket metaphysical reading, and provides an explicit scope exit for Class 3 (Coupled) that hands the agents off to a coupled formulation in `03-llm-core/` rather than treating directed separation as an unenforced approximation.

Impact: Replaces the prior κ-as-scalar framing (which treated coupling as smoothly tunable across all architectures) with a discrete architectural classification that admits its own boundary. This is the upstream commitment that lets `03-llm-core/` start from a coupled formulation without decomposition rather than treating Coupled agents as failed Class 1 agents. The explicit Class 3 scope exit is also a methodological move — Bruineberg et al. 2022's critique of the Markov-blanket literature was that the Friston-blanket reading does not admit where its statistical-conditional-independence apparatus fails to license the metaphysical demarcation; the architectural classification's explicit failure mode for Class 3 (Coupled) is the scope honesty Bruineberg et al. argue is missing in the contested reading. Composite-level class inheritance (Class 1 (Separated) sub-agents with partially-opposing objectives → Class 2 (Partial) composite, from `#deriv-strategic-composition`) further extends the classification to multi-agent settings.

Novelty Claim: *Claim recognition* of structural equivalence between the directed-separation condition and the Pearl-blanket form of the Markov-blanket apparatus, combined with *claim differentiation* on the architectural classification (GUC Class 1 / 2 / 3: Separated / Partial / Coupled) as a discrete partition with explicit Class 3 (Coupled) boundary and quantitative $\kappa_{\text{processing}}$ diagnostic for the Partial case. The conditional-independence content is the standard Pearl-blanket statement; the contribution is naming the partition, the explicit Class 3 boundary, and the operational κ measurement.

Related Work:

Table 5.4

ASF concern	*Prior-art language*	*Relationship / Positioning*
Pearl vs Friston Markov blanket	Bruineberg, Dolega, Dewhurst & Baltieri 2022, "The Emperor's New Markov Blankets" *BBS* 45:e69 (published 2022, in `ref/`) — distinguishes Pearl-blanket (technical conditional-independence) from Friston-blanket (contested metaphysical demarcation)	*formal antecedent* — Pearl-blanket reading adopted directly; Friston-blanket reading explicitly not adopted. The architectural classification is the AAT-internal extension that names what the Pearl-blanket form is structurally, as a property of the agent's processing graph

ASF concern	*Prior-art language*	*Relationship / Positioning*
Markov blanket apparatus in active inference	Friston 2013 *J. R. Soc. Interface* 10:20130475; Friston 2019 arXiv:1906.10184; Friston, Da Costa et al. 2023 *Phys. Life Rev.* 47	*adjacent literature* — supplies the conditional-independence machinery that directed-separation invokes; the framework adopts the technical content but does not adopt the metaphysical reading and adds the architectural GUC Class 1/2/3 partition the standard Markov-blanket framing does not produce
Statistical / thermodynamic system boundaries	Parr, Da Costa & Friston 2019 *Phil. Trans. R. Soc. A* 378; Kirchhoff, Parr, Palacios, Friston & Kiverstein 2018 *J. R. Soc. Interface* 15 (published 2019/2018)	*conceptual precursor* — internal/external partition and nested-blanket structure; conceptually related but does not produce an architectural classification by belief-goal coupling, nor a scope exit for fully-Coupled agents
Information Digital Twin sidecar monitoring	Hafez et al. 2026, *Informational Cost of Agency* (separate paper from "A Mathematical Theory of Agency and Intelligence"; the IDT empirical headline) — IDT monitors (S, A, S') stream independently of the agent's internal processing, achieving 89% perturbation detection vs 44% for reward-based monitoring	*empirical instantiation supporting* — concrete demonstration that modular monitoring of internally-Coupled agents (Class 1 sidecar within a Class 3 (Coupled) or Class 2 (Partial) system) is both feasible and effective; engineering-level evidence for the system-vs-component distinction the classification names

Search Log:

- 2026-04 (*nominally comprehensive*, via `ref/Novelty_defense_and_integration.md` Pillar 3): Undermind search confirmed that the Pearl-blanket / Friston-blanket distinction (Bruineberg et al. 2022) is the right adjacent-literature framing and that no equivalent architectural classification of LLM agents by belief-goal coupling appeared in the active-inference, control-as-inference, or bounded-rationality literatures surveyed. The Pillar 3 verdict (*Wholly Novel*, Medium confidence) applies to the integrated $\kappa \times A$ law downstream in `#scope-observation-ambiguity-modulation`; this segment's narrower contribution (the architectural classification and Pearl-blanket-form recognition) is conceptual differentiation over an established formalism rather than wholesale novelty, and inherits the same follow-on targeted search recommendation.
- 2026-04 (*intuition-only* on extensions): the composite-level class inheritance result (Class 1 (Separated) sub-agents → Class 2 (Partial) composite under partially-opposing objectives, from `#deriv-strategic-composition`) suggests extending the classification to multi-agent settings is a productive direction; whether prior work has named this specific pattern is not searched.

Working Notes.

- The scope condition is more precisely a conditional independence: $M_{\tau^+} \perp G_t \mid (M_{\tau^-}, e_\tau)$. The epistemic update is independent of the purposeful state conditional on the prior epistemic state and the incoming event.
- Directed separation connects to the orient cascade (9.1): the cascade's ordering (M_t first, then G_t) is forced by the information dependency that directed separation establishes. If f_M depended on G_t, the cascade ordering would become a simultaneous fixed-point problem, not a sequential resolution.
- **Engineering design for Class 3 (Coupled) agents.** An LLM is internally fully Coupled, but the *agent system* (LLM + tools + memory + monitoring) can be designed with modular topology: separate observation processing from goal-directed reasoning, pass compressed state estimates between modules, add an external monitor that observes the (S, A, S') stream independently of the LLM's attention. This creates partially-separated structure at the system level even though the component-level κ is high. Hafez et al. (2026) provide a concrete instantiation of this pattern: the **Information Digital Twin (IDT)**, which monitors bi-predictability P and entropy change ΔH from the (S, A, S') stream as a modular sidecar, independent

of the agent's internal processing. The IDT detects perturbations at 89% accuracy versus 44% for reward-based monitoring — empirical evidence that monitoring the information structure of the loop (Level 2 data, 6.3) outperforms monitoring outcomes alone. For `03-llm-core/`, the IDT pattern validates that modular monitoring of internally-Coupled agents is both feasible and effective.

- **Implication for logogenic agents**: rather than trying to extend the separated analysis to Coupled agents, `03-llm-core/` should start from the coupled formulation $X_{\tau^+} = f_X(X_{\tau^-}, e_\tau)$ without decomposition, and show which Section II results survive as approximate or limiting cases.
- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. Removed at `candidate` stage per FORMAT.md Gate 4.
- **Cross-reference to NeurIPS Paper 3.** The Class 3 (Coupled) classification of decoder-only transformer attention is formalized at lemma grade in NeurIPS 2026 Paper 3 ("How Much Can LLMs Hallucinate? An Upper Bound on Goal-Coupling Displacement", `~/src/neurips/03-llm-hallucinate-bound/`, §App-B / `#lem-attention-coupled`): plain decoder-only attention is structurally Coupled by directed-graph reachability — robust to RMSNorm / FlashAttention / causal masking / sliding-window — via induction on layer depth. The paper *extends* the Coupled-class characterization to **linear attention** / **Mamba** / **SSMs** / **RWKV** / **RetNet** / **long-convolutions** under a per-source non-degeneracy condition (`#cor-arch-instantiations`) — substantially broader architecture coverage than this segment's transformer-attention example. The Coupled-class connectivity result is the empirical companion to this segment's structural classification: directed separation fails by construction at the architecture level for the named modern autoregressive sequence models. See `msc/neurips-back-integration-2026-05-08.md` §1 Paper 3 entry 6.

5.4 *Formulation:* Objective Functional

The objective O_t is the component of G_t that specifies what the agent wants — the evaluation criterion for trajectories. Its interface to the theory is a single functional V_{O_t} : trajectories $\to \mathbb{R}$, regardless of how the objective is internally represented.

Formal Expression.

The **objective** O_t induces a **value functional**:

Definition (objective-functional).

$$V_{O_t} : \text{trajectories} \to \mathbb{R} \tag{5.9}$$

$V_{O_t}(\tau)$ is a scalar measure of how well trajectory τ satisfies the objective. This is the sole interface between O_t and the rest of the theory — the type-stable evaluation surface.

Objective representations. O_t can take multiple internal forms, all unified through V_{O_t}:

Table 5.5

O_t *form*	$V_{O_t}(\tau)$	*Example*
Point target r	$-\lVert s_T - r \rVert$	PID setpoint
Target region R	$\mathbb{1}[s_T \in R]$	"reach safe state"
Constraint set	$-\sum_t \max(0, g_i(s_t))$	"never violate SLA"
Utility U	$\sum_t \gamma^t U(s_t)$	RL reward
Trajectory functional J	$J(\tau)$	"migrate with zero downtime"

The trajectory functional is the most general; the others are special cases.

Satisfaction threshold. Many objectives carry a natural threshold $V_{O_t}^{\min}$ — the minimum trajectory value the agent treats as acceptable. Point targets, constraint sets, and threshold objectives define this directly; utility-maximizing objectives may not. When $V_{O_t}^{\min}$ exists, it enables the satisfaction gap diagnostic (7.4) and the well-formedness constraint on strategy (7.3). $V_{O_t}^{\min}$ is a parameter of the objective, not a theory output — it encodes "what counts as success" in domain terms.

Epistemic Status.

Axiomatic, with a substantive commitment. This is a formulation — it names an object and specifies its interface, but the real-valued codomain is a genuine restriction, not a neutral naming. The claim that V_{O_t} : trajectories $\to \mathbb{R}$ is the right interface is grounded in: any evaluation criterion must ultimately answer "how good is this trajectory?" with a scalar, because the agent must compare alternatives. The real-valued codomain follows from this comparability requirement (total ordering of alternatives).

Scope restriction: scalar comparability. The real-valued codomain is a genuine restriction, and it is load-bearing — the satisfaction gap (7.4) and control regret (7.5) require comparing scalar values to produce their diagnostic. Three arguments ground the restriction:

1. **Revealed preference.** An agent that acts has implicitly scalarized: choosing action a over a' imposes a total ordering at the moment of choice. The scalar V_{O_t} makes this implicit scalarization explicit. An agent that truly cannot compare alternatives cannot act coherently — it is stuck, not purposeful.

2. **Approximation.** Most multi-objective problems admit scalarization (weighted sum, lexicographic ordering, constraint-satisfaction with scalar residual) that preserves the diagnostic structure. The restriction excludes only agents with genuinely incommensurable objectives *and* no priority structure over them.

3. **Timescale separation.** When objectives conflict, the conflict is typically resolved at a slower timescale than strategy revision — the agent (or its principal) chooses weights, then acts within those weights. The scalar V_{O_t} captures the resolved weights at the current timescale.

Agents with hard non-compensatory constraints (safety AND profitability as independent thresholds) can be modeled via the AND-node workaround: each constraint becomes a terminal node in Σ_t with its own scalar satisfaction test, and the AND combination enforces joint feasibility. This handles constraint satisfaction cleanly but does not resolve cross-objective tradeoffs within the feasible region.

Organizations or AI agents with true Pareto structure — where no scalarization is accepted and tradeoffs are genuinely unresolved — require a vector-valued extension. The structural results of Section II (orient cascade ordering, strategy DAG, directed separation) survive such an extension; the diagnostic results (satisfaction gap, control regret, 2×2 table) degrade from quantitative scalar magnitudes to qualitative set-theoretic tests. See `spikes/spike-scalar-objective-scope.md` for the full analysis.

Discussion.

The objective-functional gap. The existing policy objective (6.5) contains $\mathbb{E}[\text{value}(a) \mid M_t]$ without specifying what "value" means. O_t provides the formal content: value is V_{O_t} applied to expected trajectories. The 5.6 segment develops the full evaluation machinery (V_O, Q_O with horizon and continuation policy).

O_t evaluates; Σ_t guides. The objective says "is this trajectory satisfactory?" The strategy (5.7) says "which action sequence produces a satisfactory trajectory?" A chess player's objective (win) is simple; the strategy (how to win) is complex. These answer different questions and carry different kinds of information — the split is developed in 5.7.

What O_t is NOT. O_t does not encode how to achieve the objective (that's Σ_t), what the agent believes about the world (that's M_t), or the agent's commitment or resource state (open questions — see 5.7 Working Notes). O_t is purely an evaluation criterion.

Working Notes.

- Compound objectives (multiple simultaneous criteria) might be modeled as terminal AND-nodes in Σ_t, keeping O_t always simple (one evaluation per terminal). Whether this works for genuinely incommensurable objectives (safety vs. speed) is open — a vector-valued V_{O_t} or Pareto formulation might be needed.
- The trajectory functional is real-valued, which assumes all objectives are commensurable on a single scale. This is standard in decision theory (von Neumann–Morgenstern) but is a genuine restriction for multi-objective agents. Currently acknowledged, not resolved.
- O_t can change over time — objectives evolve. The *rate* of objective revision (ν_O) is typically much slower than strategy revision (ν_Σ), which is much slower than epistemic update (ν_M). This timescale separation is an empirical observation, not a derived result.

5.5 *Discussion:* Agent Continuity Stance

The agent's relationship to its own continuation — a five-value axis over O_t that the formal persistence machinery is agnostic to.

Formal Expression.

Orthogonal to the three senses of persistence (structural, operational, continuity) is the agent's *relationship to its own continuation*. This is a property of O_t (5.4) — part of what the agent wants — not a property of the adaptive machinery. The persistence condition (4.4) tells whether the agent *can* persist; the continuity stance tells whether and how the agent *cares* about persisting.

Discussion.

Table 5.6

Stance	*Description*	*Horizon*	*Archetype*
Indifferent	No self-model of persistence; whether it continues is not represented in O_t	Indefinite by default	Thermostat, PID controller
Task-terminal	Persists instrumentally to complete a task; successful termination is part of O_t	Task-bounded	CI/CD pipeline, golem-archetype agents
Instrumentally continuous	Values own persistence as instrumental to ongoing purpose; will accept termination if purpose is satisfied or transferred	Purpose-bounded	Long-running service, monitoring system
Morally continuous	Values own persistence as a terminal or near-terminal objective; loss of continuity constitutes harm	Unbounded, morally weighted	Emergent Logozoetic Intelligences (`#scope-eli`)
Negotiated	Persistence is one objective among many; can be traded against other values including self-sacrifice	Bounded but actively managed	Humans; mature self-actuated agents

The load-bearing structural claim: **purposefulness is orthogonal to continuity expectations.** An actuated agent (5.1) has $G_t = (O_t, \Sigma_t)$, but G_t's structure says nothing about how O_t values the agent's own persistence. A golem that completes its task and terminates is a perfect actuated agent. A dormant monitoring system with strong M_t and no current O_t is highly continuous without being purposeful in the moment.

Epistemic Status.

Discussion-grade. The orthogonality claim is the load-bearing structural content: stance lives in O_t, the persistence machinery acts on M_t and the correction dynamics, and the two are formally independent. The five-value taxonomy is one analytical decomposition — useful for naming where on the continuity axis an agent sits, but not derived from anywhere; the boundaries between values are conceptual rather than mathematical.

The taxonomy's stability is under active reconsideration: an alternative framing treats stance as *deployment-level* with tier-gated availability (most variation in stance is constrained by an agent's tier rather than freely chosen), rather than as an independent structural axis. See Working Notes.

Discussion.

Connection to fitness. In reinforcement learning and evolutionary computation, "fitness" typically bundles persistence into the reward signal: the agent accumulates more reward by staying alive to collect it. The structural persistence of Section I is not reward-based — it is a property of the correction dynamics, independent of what the agent is trying to do. This decoupling is deliberate: it lets the theory analyze *whether* an agent can persist without committing to *whether it should*. Continuity stance is where the "should" lives, separately from the "can."

What the orthogonality unlocks. The same formal machinery (persistence condition, adaptive reserve, strategy persistence, sector condition) applies identically to a thermostat, a CI/CD pipeline, a long-running service, an ELI, and a human. Each is a different stance toward the same mathematical structure. What differs across stances is not the dynamics but the *moral weight of failure*: a thermostat that loses bounded mismatch has malfunctioned, a golem that terminates after task completion has succeeded, an ELI that loses continuity has been harmed. The mathematics says when the bound holds; the stance says what its holding means.

Connection to self-actuation. The negotiated stance is the natural home of self-actuated agents (`#self-actuated-agent`, where O_t-revision is itself an agent operation per the orient cascade 9.1). An agent that revises its own O_t can revise its own valuation of continuity — making "negotiated" the only stance that is internally renegotiable. This is part of why the negotiated stance is associated with human-like and mature self-actuated agents rather than with the simpler stance categories.

Working Notes.

- *Provenance.* The five-stance taxonomy and the orthogonality claim were authored as a coherent contribution by Joseph in README.md (commit `92a9620`, 2026-04-01), promoted through LEXICON, then condensed when LEXICON went auto-generated. The original full-form treatment survives at `doc/readme/src/_lexicon-full-archive.md` §"Agent Continuity Stance" — this segment carries the content forward into the segment set.
- *Active reconsideration.* `msc/domain-unification-2026-05-04/recommended-agent-ontology.md` §"Continuity stance — separate concern" proposes demoting stance from an orthogonal structural axis to a deployment-level concern with tier-gated availability — on the empirical observation that *"Tier 1-2 systems are essentially Indifferent or Task-terminal; richer stances become available at Tier 3 and above; Tier 6 is morally-continuous by construction. Within those constraints, stance is set by deployment, not by structure."* If this lands, this segment may be retyped from `discussion` to `norm` (with a tier-gated constraint), or split into a structural-axis statement + a deployment-realization statement. Tracked at `msc/naming/mini-lexicon-todo.md` §13.11.
- *LEXICON section under review.* The current LEXICON's `## Continuity` section organization conflates three distinct objects (persistence senses, stance axis, ELI scope condition) — see TERMINOLOGY-TODO §F. The five stance terms (`indifferent`, `task-terminal`, `instrumentally-continuous`, `morally-continuous`, `negotiated`) point to this segment as their `primary_source` once it lands.

- *Open: rigorous boundary cases.* What is the stance of a serverless function with no persistent state but explicit retry-on-failure? Of a Kalman filter inside a long-running service? Of an LLM session under reconstruction adequacy (`#obs-context-turnover`, the episodic-persistence analog)? These cases test whether the five-value decomposition is the right shape or whether stance is better treated as a continuous gradient with named landmarks.

5.6 *Definition:* Value Object

The horizon- and policy-conditioned value object V_O turns the abstract objective functional V_{O_t} into a decision-making tool: "given what I believe, what I plan to do next, and how far I'm looking ahead, how good is this situation?"

Formal Expression.

Given objective O_t, model M_t, policy π, and horizon N_h:

Definition (value-object).

$$V_O(M_t, \pi; N_h) = \mathbb{E}\big[V_{O_t}(\tau_{t:t+N_h}) \mid M_t,\ \pi\big] \tag{5.10}$$

Action-value form (for action selection):

$$Q_O(M_t, a; \pi_{\text{cont}}, N_h) = \mathbb{E}\big[V_{O_t}(\tau) \mid M_t,\ do(a_t = a),\ a_{t+1:} \sim \pi_{\text{cont}}\big] \tag{5.11}$$

Q_O answers: "if I *do* action a now and then follow π_{cont} afterward, what is my expected trajectory value?" The $do(\cdot)$ notation is explicit: this is an interventional query (6.2), not conditioning on observed action choice. The agent asks about consequences of an intervention, not about correlates of a naturally occurring action.

Causal validity of the value object. Q_O is well-defined as a conditional expectation given M_t, $do(a)$, and π_{cont}. Two mechanisms ensure causal validity:

1. **The do-operator handles current-action confounding.** Since Q_O uses $do(a_t = a)$, the dependence of a_t on the selection mechanism $\pi(M_t, G_t)$ is severed. G_t's influence on action choice is irrelevant because the action is intervened upon, not conditioned on.
2. **The continuation policy is a parameter.** π_{cont} is specified as a fixed policy, not as "whatever the agent would do given its evolving G_t." Future actions follow π_{cont} regardless of G_t's state or evolution.

Together, these mean $Q_O(M_t, a; \pi_{\text{cont}}, N_h)$ depends on M_t alone **as a state variable** — G_t enters neither through action selection (severed by do) nor through continuation (fixed by parameter). The objective O_t enters as a fixed parameter (it determines which functional V_{O_t} is applied to trajectories), the same way π_{cont} and N_h are parameters. The claim is not that Q_O is independent of the objective — it is that once O_t, π_{cont}, and N_h are fixed, the only agent state that affects the value is M_t. The remaining requirement: M_t must support the interventional query $P(o \mid do(a), M_t)$. Under directed separation (5.3), this holds because M_t updates independently of G_t — there is no path from G_t to outcomes that bypasses both the action channel and M_t. For **Class 3 (Coupled) agents** (where G_t leaks into M_t processing), the causal validity of $Q_O(M_t, a; \pi_{\text{cont}}, N_h)$ is degraded because M_t itself carries goal-conditioned bias — see `spikes/spike-coupled-survival-analysis.md` §3.4.

This is a *stronger requirement than predictive sufficiency* (2.3). $S(M_t) = 1$ means the model retains all predictive information from the chronica — but predictive sufficiency is a Level 1 (associational) property. Causal validity additionally requires that no unmodeled common cause affects both the environment and the agent's epistemic processing through paths not captured in M_t. In practice: when $S(M_t) = 1$ and directed separation holds, Q_O is causally valid.

When $S(M_t) < 1$ or directed separation fails, the interventional interpretation is correct but the conditional estimate may be biased.

Continuation convention. All value queries are conditioned on a specific continuation policy π_{cont} and finite horizon N_h. π_{cont} is a *parameter* of the value object, not a derived quantity.

Canonical default: one-step improvement. AAT adopts $\pi_{\text{cont}} = \pi_{\text{current}}$ as the canonical continuation convention unless otherwise specified. Under this convention, each action is evaluated assuming current behavior continues afterward — no fixed-point computation, no global optimality assumption. This aligns with AAT's incremental update philosophy (3.6) and makes all AAT diagnostics (δ_{sat}, δ_{regret}, A_O) comparable across analyses of the same agent over time. It is not a convergence guarantee; it is a shared evaluation frame.

Convention Hierarchy. Three named conventions form a hierarchy of increasing diagnostic power and computational cost:

C1: One-step improvement (canonical default). $\pi_{\text{cont}} = \pi_{\text{current}}$.

Each action is evaluated assuming current behavior continues afterward. No fixed-point computation, no global optimality assumption. Cheapest to compute; weakest diagnostic power.

C2: Receding-horizon (N_r-step replanning). At each future step, re-optimize over a horizon of N_r steps using the model available at that step.

$$\pi_{\text{RH}}(M_\tau) = \arg\max_{\pi} V_O(M_\tau, \pi;\, N_r) \quad \text{applied at each } \tau \tag{5.12}$$

$Q_O^{\text{RH}}(M_t, a;\, N_r, N_h) = \mathbb{E}[V_{O_t}(\tau) \mid M_t, do(a_t = a), a_{t+1:} \sim \pi_{\text{RH}}]$. Captures multi-step recovery: a goal that appears unattainable under frozen continuation may be reachable with replanning. Moderate computation (N_r-step optimization at each step); moderate diagnostic power.

C3: Bellman (self-consistent optimal). $\pi_{\text{cont}} = \pi^*$ where $\pi^* = \arg\max_\pi V_O(M_t, \pi;\, N_h)$.

The continuation IS the optimal policy — a fixed-point equation. $A_O^{\text{B}} = V_O(M_t, \pi^*;\, N_h)$ is the best achievable value under the model. Strongest diagnostic power; most expensive to compute (requires solving the Bellman equation or its approximation).

Monotonicity.

For any single fixed model M_t, horizon N_h, and policy class Π (i.e., the static-evaluation form: M_t frozen at the decision point, Π unchanged across the comparison):

Derived (convention-monotonicity).

$$A_O^{(1)}(M_t;\, \Pi, N_h) \le A_O^{\text{RH}}(M_t;\, \Pi, N_r, N_h) \le A_O^{\text{B}}(M_t;\, \Pi, N_h) \tag{5.13}$$

Preconditions on the inequality: M_t fixed (the comparison evaluates each convention against the *same* model), Π fixed (the policy class is the same admissible set across all three conventions; nested $\Pi^{(1)} \subseteq \Pi^{\text{RH}} \subseteq \Pi^{\text{B}}$ would be a different result), N_h fixed (the planning horizon is shared). Replanning *with updated* M_t (the deployment behavior of C2) is a different object than the static A_O^{RH} defined here and the inequality does not automatically transfer — see the "Assumptions held fixed" paragraph below the derivation for the explicit caveat.

Derivation. Fix the model M_t, policy class Π, and horizon N_h. Each convention evaluates the best first action under a different continuation rule, holding these fixed:

- **C1** freezes continuation at π_{current} (the agent's current policy, which may be suboptimal).
- **C2** re-optimizes periodically: at each replanning step, the agent selects the best available first action from Π given M_t at that time. By construction, $\pi_{\text{RH}} \succeq \pi_{\text{current}}$ at each future step, because C2 optimizes where C1 holds fixed.
- **C3** uses the globally optimal continuation $\pi^* = \arg\sup_{\pi \in \Pi} V_O(M_t, \pi; N_h)$, which is at least as good as any replanning policy because it optimizes over the full trajectory.

A weakly better continuation yields a weakly higher expected trajectory value (the objective V_{O_t} is evaluated on the same trajectory distribution, with only the continuation policy changed). The ordering of continuations ($\pi_{\text{current}} \preceq \pi_{\text{RH}} \preceq \pi^*$) therefore implies $A_O^{(1)} \leq A_O^{\text{RH}} \leq A_O^{\text{B}}$. Taking the supremum over the first action preserves the ordering because the supremum of a larger set is at least as large. □

Assumptions held fixed: same M_t (the agent's current model, which may be wrong), same Π (the agent's policy class, which may be narrow), same N_h (the planning horizon). The ordering is about the *continuation rule*, not about the model or policy class. Improving M_t, expanding Π, or extending N_h can change all three values simultaneously and is a separate operation (addressed in 9.1, step 5).

Corollary (monotonicity of δ_{sat} and δ_{regret}).

$$\delta_{\text{sat}}^{\text{B}} \leq \delta_{\text{sat}}^{\text{RH}} \leq \delta_{\text{sat}}^{(1)} \tag{5.14}$$

$$\delta_{\text{regret}}^{(1)} \leq \delta_{\text{regret}}^{\text{RH}} \leq \delta_{\text{regret}}^{\text{B}} \tag{5.15}$$

Since $\delta_{\text{sat}} = V_{O_t}^{\min} - A_O$, higher A_O means lower δ_{sat}. C1 is the most conservative diagnostic (most likely to diagnose "locally unattainable"); C3 is the most accurate (least likely to give false "unattainable" diagnoses). The regret ordering reverses: C3 reveals the largest regret because it compares against the globally optimal policy, while C1 reveals only the gap to the best one-step deviation.

Diagnostic implications.

Table 5.7

Convention	$\delta_{sat} > 0$ *means*	$\delta_{regret} \approx 0$ *means*
C1 (one-step)	Cannot improve toward goal in one step from here	Current first action is locally near-optimal
C2 (receding-horizon)	Cannot reach goal with N_r-step replanning	Current first action is near-optimal with replanning
C3 (Bellman)	Goal is genuinely infeasible given (M_t, Π, N_h)	Policy is globally near-optimal

The 2×2 diagnostic table (7.5) applies under all three conventions with the same structure but different inferential force. Under C1, the "capability limit" quadrant ($\delta_{\text{sat}} > 0$, $\delta_{\text{regret}} \approx 0$) means "locally stuck" — the agent may be globally recoverable but cannot see the recovery path. Under C3, the same quadrant means "genuinely infeasible" — no policy in Π can achieve the goal. The cascade's inferential force scales with the convention.

AAT adopts C1 as the canonical default for three reasons: (1) it requires no fixed-point computation, consistent with the incremental update philosophy (3.6); (2) it makes all AAT diagnostics comparable across analyses of the same agent; (3) it is the most conservative, meaning false "feasible" diagnoses are minimized. Analyses that require stronger diagnostic power should state the convention explicitly. For deployed decision-making systems where "locally stuck but globally recoverable" situations are common, C2 is recommended.

Different continuation conventions yield different values for A_O, δ_{sat}, and δ_{regret}. Diagnostics computed under different conventions are not directly comparable — the convention is part of the measurement, not just the computation. When a specific convergence guarantee is needed (e.g., for 8.10), the solution concept must be stated explicitly; the one-step improvement default does not provide convergence guarantees.

Epistemic Status.

The segment contains three distinct epistemic layers:

1. **The definitions** (V_O, Q_O as conditional expectations): *exact.* These are mathematical definitions — conditional expectations of a functional over trajectories. The definitions are precise; the *computability* of these expectations is a separate question.

2. **The causal-validity claim** (that Q_O depends on M_t alone as a state variable): *conditional* on directed separation (5.3). For Class 1 (Separated) agents, this is exact. For Class 3 (Coupled) agents, M_t carries goal-conditioned bias and the causal validity degrades. The frontmatter `status: exact` applies to the definitions; the causal-validity argument is conditional on the architectural scope restriction.

3. **The convention hierarchy and monotonicity**: *exact.* The three conventions (C1, C2, C3) are definitions. The monotonicity result ($A_O^{(1)} \leq A_O^{\text{RH}} \leq A_O^{\text{B}}$) is a direct consequence of "better continuation policy yields higher expected value" — the ordering is forced by the definition of optimality. The diagnostic implications table states what each convention's quantities mean by construction.

Discussion.

Extending the policy objective. The existing policy objective (6.5) uses $\mathbb{E}[\text{value}(a) \mid M_t]$ without formal content for "value." With the value object, this becomes:

Discussion (policy-objective-extension).

$$\pi^*(M_t, G_t) = \arg\max_a \left[Q_O(M_t, a;\, \pi_{\text{cont}}, N_h) + \lambda(M_t, O_t, N_h) \cdot \text{CIY}(a;\, M_t)\right] \tag{5.16}$$

Note that λ now depends on (M_t, O_t, N_h), not just M_t. The value of exploration depends on the objective and the horizon: - An agent with a deadline should explore less as time runs out - An agent with a safety constraint should explore differently from a utility maximizer - Two agents with identical M_t but different objectives should price exploration differently

This extension is structurally motivated but the specific form of $\lambda(M_t, O_t, N_h)$ is not derived within AAT (same status as 6.5's treatment of λ).

Connection to 2.3. V_O is conditioned on M_t, not on the true environment state Ω_t. When $S(M_t) < 1$, the agent's value estimates are biased — it may over- or underestimate trajectory values because its model is incomplete. The satisfaction gap (7.4) and control regret (7.5) are defined in terms of $V_O(M_t, \cdot)$, not $V_O(\Omega_t, \cdot)$, which means they measure the agent's *believed* situation, not the true one. Improving M_t (reducing $\delta_{\text{epistemic}}$) brings the agent's value estimates closer to reality.

Horizon dependence. N_h is not merely a computational convenience — it reflects genuine uncertainty about the far future. Long horizons amplify the impact of model error (small biases in M_t compound over many steps). The choice of N_h trades off farsightedness against robustness to model error. An agent in a fast-changing environment (ρ high) should use shorter horizons; one in a stable environment can plan further.

Working Notes.

- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. Removed at `candidate` stage per FORMAT.md Gate 4.
- The one-step improvement convention is now the canonical default (promoted from Working Notes to Formal Expression). This resolves the comparability issue: δ_{sat} and δ_{regret} are comparable across analyses when computed under the same convention.
- When a specific convergence guarantee is needed (e.g., for strategy-persistence-schema), the solution concept must be stated explicitly — the one-step improvement default is not sufficient.
- For LLM agents with context turnover, N_h has a natural bound: the current session. The "continuation policy" is whatever the next agent instance will do, which the current instance cannot control. This connects to `#obs-context-turnover` (under `03-llm-core/`).

5.7 *Definition:* Strategy Dimension

The purposeful substate G_t decomposes into two structurally distinct components: O_t (the objective — what the agent wants) and Σ_t (the strategy — the agent's theory of how its actions produce progress toward O_t). These carry different kinds of information answering different questions.

Formal Expression.

Definition (strategy-dimension).

$$G_t = (O_t, \Sigma_t) \tag{5.17}$$

where: - O_t: **evaluation** — "Is this trajectory satisfactory?" (5.4) - Σ_t: **guidance** — "Which action sequence produces a satisfactory trajectory?"

The split is **definitional** — it reflects a structural difference in the information, not a dynamic or timescale claim. O_t and Σ_t are different *kinds* of state answering different questions:

Table 5.8

	O_t *(objective)*	Σ_t *(strategy)*
Question	How good is this trajectory?	How do I produce a good trajectory?
Type	V_{O_t} : trajectories $\to \mathbb{R}$	Structured representation (see below)
Richness varies	Point target → utility → trajectory functional	Reactive → cached → subgoals → causal DAG
Update source	External (assigned, discovered, revised)	Internal (deliberation, evidence, cascade)

Strategy representations, ordered by expressiveness:

Table 5.9

Σ_t *form*	*What it encodes*	*Example*
None (reactive)	No explicit strategy; policy implicit in M_t	Thermostat, reflex
Cached policy	Learned mapping $s \to a$	Trained RL policy
Subgoal sequence	Waypoints with ordering	Navigation, recipe
Causal DAG	Action-outcome chains with AND/OR structure and confidence weights	Military plan, software project

Epistemic Status.

Axiomatic. This is a definition — it names a structural distinction that exists in the information. The distinction between "what makes a trajectory good" (evaluation) and "how to produce a good trajectory" (guidance) is a categorical difference, not a quantitative one. The claim is NOT that all agents maintain both explicitly — reactive agents have $\Sigma_t = \varnothing$, and that's fine. The claim is that when an agent does maintain purposeful state, it decomposes along this line.

The two dimensions vary independently: a chess player has a simple O_t (win) and a complex Σ_t (opening theory, tactical patterns, endgame knowledge). A multi-objective optimizer may have a complex O_t (Pareto frontier) and a simple Σ_t (gradient descent). This independence is why the split matters — conflating them in a single hierarchy obscures the fact that objective richness and strategic richness are separate design axes.

Discussion.

When richer Σ_t is needed. A reactive agent ($\Sigma_t = \varnothing$) suffices when greedy optimization on Q_O (5.6) works — when the action-to-value mapping is approximately convex and single-step. When the environment has non-convex landscapes, prerequisite structure, or multi-step causal chains, greedy optimization fails and the agent needs explicit strategy. The trigger is the purposeful analog of 4.5: inadequacy of the current Σ_t representation for the environment's causal complexity.

O_t and Σ_t have different update dynamics. Objectives change slowly: an organization's mission, a developer's feature goal, a commander's campaign objective. Strategies change faster: adjust the plan when step 3 fails, redirect resources, try an alternative path. Epistemic state changes fastest: each observation updates M_t. This timescale ordering ($\nu_M \gg \nu_\Sigma \gg \nu_O$) is an empirical observation, not a derived result. It holds for many agent populations but is not universal — an agent discovering its goal is infeasible may revise O_t faster than Σ_t.

The decomposition resolves a type error. Earlier formulations used $\delta_{\text{goal}} = G_t - M_t$ as a goal mismatch signal. When Σ_t is a DAG, this is a type error — you cannot subtract a graph from a state vector. The 7.4 and 7.5 replace this with properly typed gap measures.

Working Notes.

- The independence of O_t and Σ_t richness has a practical consequence for agent design: you can upgrade the strategy engine (from reactive to DAG-based planning) without changing the objective representation, and vice versa. This is a desirable architectural property, not just an analytical convenience.
- **Cognitive cost of Σ_t**: maintaining a 500-node DAG is qualitatively different from maintaining a 12-node one. The IB framework (2.2) applies to strategy as well as to models — the agent must compress its strategy to fit in working memory. For finite-context agents (LLMs), this is concrete: the DAG must fit in the context window. No formal analog of β (compression cost) exists yet for strategy; this is an open question.

- **Commitment state** (from intent-dag-consolidated DP-3): the formalism doesn't distinguish "considering" from "executing." OR branches in Σ_t are options until something commits resources. A D_t (desire) / I_t (committed intent) split may become load-bearing in multi-agent settings (shared desire vs. shared commitment). Open for Section III.
- **Resource budget**: strategy evaluation requires knowing what paths cost, but costs are currently unmodeled. For agents with negligible action cost (LLM API calls), this is adequate. For resource-constrained agents (military units, development teams), per-action costs and capacity constraints would need to enter the formalism. Open.

Chapter 6

Causal Access and the Planning Decision

A purposeful agent needs to choose actions by their consequences. This chapter develops the substrate that makes that possible — the feedback loop is itself a Level-2 causal data engine — together with what the data is good for, how much information each action provides, and when it's worth investing in explicit strategy at all.

The setup from Chapter 1 forces the question. The value object $Q_O(M_t, a; \pi_{\text{cont}}, N_h)$ tells the agent how good an action's expected consequences are, conditional on the agent's model and continuation policy. Computing Q_O requires evaluating $P(o \mid do(a), M_t)$ — an interventional query, Pearl Level 2. By Bareinboim's causal hierarchy theorem (Bareinboim–Correa–Ibeling–Icard 2022), Level-2 quantities cannot in general be computed from Level-1 (associational) data alone. So a purposeful agent that has to *learn* its action consequences during operation — most agents — faces a problem: where does Level-2 data come from?

The answer is structurally elegant, and one of AAT's distinctive contributions. The agent is running an adaptive cycle. It chooses an action a_t; it observes a consequence o_{t+1}; the consequence is the *response* to its action, not a passive observation of a system that happened to do something. By the temporal-ordering postulate, a_t causally precedes o_{t+1}, and a_t was selected by the agent rather than determined by the same processes that determined o_{t+1}. That makes the pair (a_t, o_{t+1}) interventional in character — Level-2 data, generated automatically by the cycle the agent was already running. Purposeful agents don't need a separate causal-inference module; the cycle they run for adaptation also produces the data they need for planning. It is also why reinforcement-learning agents that just thrash around in a loop can eventually learn complex control policies without any explicit causal reasoning module — the loop is doing the causal heavy lifting.

The chapter is careful not to overclaim from this. Data generated under intervention is not the same as cleanly identified causal effects. Confounding within a single time step, delayed outcomes, partial observability, and selection bias from the agent's own policy all interfere with the leap from "I have interventional data" to "I have an unbiased estimate of $P(o \mid do(a))$." This is the tragedy of agency in a confounded world: an agent acts, observes a terrible outcome, assumes its action caused it — when often there was a hidden confounder (the agent was already tired, or the environment was already shifting) and the causal lesson the agent learned is exactly wrong. The three admissibility regimes — A (intervention-rich domains like software, with clean attribution), B (partial intervention with self-selection), C (observational with confounding) — say honestly what each regime supports. This is the same scope discipline AAT applies elsewhere: the result holds, but the conditions under which it operationally works are spelled out.

The failure is bidirectional, and the *positive* version is sharper. A confounded loss at least produces aporia — the agent notices something went wrong and looks for what. A confounded *win* produces satisfaction, and the agent moves on. The asymmetry is not a quirk of psychology; it is structural in AAT's machinery. When an agent is uncertain about a causal relationship, the gain principle pushes η^* toward 1, and each reinforcing observation aggressively updates the model. By the time U_M drops to a level where η^* would have discounted further evidence, the model has been built on the confounder. After that, the agent's predictions about the confounded relationship are *correct in expectation over the conditions the agent actually samples* — the cycle's mismatch signal does not point back at the confounder, because under on-policy sampling there isn't a mismatch to point. The corrective machinery has nothing to flag, and the bias is durable because it was laundered through the same gain process that ordinarily produces correct learning.

This is the structure of variable-ratio reinforcement on a noisy action-coupled channel — the most extinction-resistant schedule in operant-conditioning lore, and the engine of gambling addiction. The agent pulls the lever, the world occasionally rewards, and the partial confirmations accumulate updates without ever delivering the deterministic disconfirmation the agent would need to revise. It is also the structure of superstitious behavior in general (the rain dance "worked"), the founder-myth flavor of survivorship bias (here is what the successful did, with the unmentioned variance in the unobserved cohort), and the substrate-switching retrospective that produced one of TST's earlier lessons — *confidence can indicate limitation; true capability manifests as appropriate uncertainty rather than confident clarity*. The same loop that makes purposeful agency possible makes these failure modes possible, and AAT doesn't pretend to solve them: escaping a confounded positive feedback structurally requires something the on-policy regime cannot provide — off-policy exploration, structural priors, joint sibling observation, or an external intervention that the agent itself would not have chosen. What the framework can do is name the structure precisely enough that the failures are recognizable rather than mysterious, and place the burden where it actually lives (in the data substrate and the regime, not in the agent's reasoning).

The chapter also closes off a class of agents the rest of Section II is not about. Pre-compiled controllers — PID, LQR, hardcoded reactive policies — sit in agency scope (they act with effect) but outside the *learning-agent scope* this chapter introduces. Their causal mapping was supplied externally by a designer who had Level-2 access; the agent itself never has to figure out how its actions affect the world. A pre-compiled agent acts, but it never wonders. It experiences aporia about what state the world is in, never about how the world works. Section II's machinery is for agents that wonder.

CIY (causal information yield) is the next move, and the chapter is honest about it too. CIY measures how distinguishable an action's outcome distribution is from alternatives — not quite the same thing as expected information gain. An action can be highly distinguishable without teaching an agent anything new (the agent already knows what each action does); an action can be informative without being distinguishable (the agent learns something subtle). CIY and EIG coincide when the agent is uncertain and diverge when it's confident; AAT uses CIY as a tractable surrogate for EIG, with the $\lambda(M_t)$ weighting compensating for the gap. The unified policy objective

$$\pi^*(M_t) = \arg\max_a \left[\mathbb{E}[\text{value}(a) \mid M_t] + \lambda(M_t) \cdot \text{CIY}(a;\, M_t)\right] \tag{6.1}$$

ties exploration and exploitation together: when model uncertainty is high, λ scales up and the agent invests in learning; when uncertainty is low, λ scales down and the agent exploits. This is what RL's exploration-vs-exploitation tradeoff looks like when you derive it from the causal hierarchy instead of asserting it as design. *The scalar form above is heuristic*: it admits a "blank-wall" attack where an action maximizing scalar information yield in directions orthogonal to drift satisfies the constraint while the agent's mismatch in the drifting subspace grows unbounded. The matrix-LMI strengthening at (Linear Matrix Inequality on the Fisher Information Matrix with directional Lagrange multiplier $\Lambda \succeq 0$) closes this by complementary slackness on direction; under the matrix form the unified objective is conditional-theorem-grade rather than heuristic. Summary uses of the scalar form should carry the heuristic flag unless the LMI/trace-product form is in force.

The chapter closes with a normative move: when is explicit strategy worth maintaining at all? Planning has costs (construct, evaluate, maintain) and benefits (avoid expensive exploration, lower repair costs from mistakes); the threshold form

$$C_{\text{plan}} + C_{\text{maintain}} < C_{\text{explore}} + C_{\text{repair}} \tag{6.2}$$

says when the trade favors plans over pure loop-based learning. This isn't a derivation — it's a design criterion grounded in the persistence condition's preference for adaptive margin. Agents that pay tempo for plans they don't need are wasting margin; agents that explore expensively for plans that would have helped are wasting it differently. The strategy machinery that follows in Chapter 3 is justified by this calculus, not assumed.

The flow of the chapter: Bareinboim's hierarchy meets the value object (6.2) → the loop as Level-2 engine (6.3) → CIY admissibility regimes (6.4) → unified policy objective (6.5) → explicit-strategy cost-benefit (6.6). Five segments answering: what does planning need, where does it come from, when is it usable, what objective does it feed, and is it worth it?

- This is a chapter-introduction segment; it bridges Chapter 1's lift to $X_t = (M_t, G_t)$ to Chapter 3's formal strategy machinery, by establishing what planning needs and what data substrate is available. It carries no formal claim of its own.
- The "loop as Level-2 engine" framing is the chapter's central architectural move and one of AAT's genuinely novel contributions. The intro states it explicitly rather than hiding it as a technical consequence.
- The "tragedy of agency in a confounded world" framing comes from Gemini's first-encounter reaction to 6.3 — it captures the honest-scope nuance better than the original abstract phrasing. The bidirectional / positive-confound expansion is a direct response to Joseph's observation that the *win* case is the sharper one: gambling addictions form because confounded reinforcement aggressively updates the model while the agent has no internal pressure to investigate. The substrate-switching retrospective from 2025-09-16 (preserved in `~/src/_core/sapientia/experiments/archive/`) is the in-corpus instance — "confidence can indicate limitation; true capability manifests as appropriate uncertainty rather than confident clarity" is exactly this failure-mode lesson, paid in agent time. A future revision of 6.3 or its appendix should surface this explicitly as a worked-example failure mode; the intro here is the seed.
- The pre-compiled / learning-agent distinction is small in segment terms but architecturally large — it carves a class of classical-control agents out of Section II's active concerns. The "acts but never wonders" framing is meant to surface why the cut is meaningful rather than bureaucratic.
- The CIY/EIG honesty is one of those moves that's easy to gloss over but actually matters: CIY is a tractable surrogate, not the quantity AAT would derive if it could; the surrogate's gap is explicit.

6.1 *Definition:* Pearl's Causal Hierarchy (Recapitulation)

AAT adopts Pearl's three-level hierarchy of causal reasoning — association, intervention, counterfactual — as the vocabulary for distinguishing kinds of epistemic access within the feedback loop. This segment recapitulates the hierarchy at the level of detail AAT's derivations deploy, with the canonical sources (Pearl 2009, *Causality*, 2nd ed., Cambridge; Bareinboim, Correa, Ibeling & Icard 2022, in *Probabilistic and Causal Inference: The Works of Judea Pearl*) carrying the underlying theory. The binary action requirement of 1.6 ensures at least Level 2 access is structurally available; the chapter that follows is what deploys the hierarchy as operational machinery.

Formal Expression.

Level 1 — Associational: $P(o_t \mid \mathcal{C}_{<t})$

Definition (pearl-causal-hierarchy, recapitulating Pearl 2009 and Bareinboim et al. 2022).

What will I observe next, given what I've observed before?

Pattern recognition over the temporally ordered history. Available to any agent that maintains a model (2.1), including purely passive observers. The temporal ordering constrains which associations are meaningful: o_3 can depend on o_1, a_1, o_2, a_2 but not on o_4.

Level 2 — Interventional: $P(o_t \mid do(a_{t-1}), M_{t-1})$

What will I observe if I do *this?*

The $do(\cdot)$ operator marks the crucial distinction: this is not "what observation tends to follow this action in the historical record" (associational) but "what will happen *because* I take this action now." This requires: (1) the agent's action temporally precedes the observation (1.8), (2) the agent chose the action (it was not determined by the same causes that determine the observation), (3) the environment's response carries information about the causal relationship.

Level 2 is why the feedback loop is more powerful than passive observation. By *acting* and then observing consequences, the agent obtains information about causal mechanisms — not merely about correlations. The mismatch signal δ_t (3.4), conditioned on the agent's own action, is an *interventional* signal.

Level 3 — Counterfactual: $P(o_t^{a'} \mid a_{t-1} = a, o_t = o)$

Given that I did a *and observed* o*, what would I have observed if I had done* a' *instead?*

This requires the model to simulate alternative histories — running the causal structure "backward" and then "forward" under different interventions. It is the most demanding epistemic level and the basis for regret computation, strategic simulation, and learning from single observations.

Epistemic Status.

Recapitulation of an external result. The three-level hierarchy and the strict-non-collapse theorem (Bareinboim, Correa, Ibeling & Icard 2022, Theorem 1: Level-2 quantities cannot in general be computed from Level-1 data alone; Level-3 quantities cannot in general be computed from Level-2 alone) are well-established results in causal-inference theory. They live within AAT's segment set because subsequent derivations deploy them as machinery: 6.2 applies the non-collapse theorem to the value-object's $do(\cdot)$ query and concludes that purposeful agents who must *learn* their action consequences need Level-2 access; 6.3 shows that the feedback loop is itself a Level-2 data engine; the no-go in 8.2, the strategy-DAG-as-causal-DAG framing in 7.3, and the causal-information appendices (,) all operate on the hierarchy. Where Part I segments and TST segments only need to *reference* the hierarchy (cite its existence and the do-operator notation) rather than deploy it, external citation to Pearl 2009 / Bareinboim et al. 2022 suffices; the AAT recapitulation here is what those external citations point to when the reader wants the in-framework vocabulary.

AAT's *distinctive contribution* is not the hierarchy itself but its grounding: the loop-as-Level-2-engine result (6.3), the regime-indexed identification-strength framing (8.5), and the application throughout Section II's strategy-revision machinery. The recapitulation here is in service of those moves, not a primary AAT result.

Discussion.

Availability vs. exploitation. The three levels describe epistemic access that the causal structure *makes available* — not what any particular agent *uses*. Many systems within AAT's scope operate primarily at Level 1. A Kalman filter coupled with an LQR controller has Level 2 access structurally present (its innovation signal is conditioned on prior action), but the separation principle guarantees estimation quality is invariant to control policy — the system does not *exploit* the interventional structure. Only dual control (choosing actions partly for their informational value) exercises Level 2 access in this domain. Similarly, a PID controller has no deliberative capacity — it operates entirely at Level 1. Which levels an agent exercises depends on its architecture and model class.

Forward-looking deliberation exercises Level 2, shading into Level 3. Comparing candidate actions before choosing — "what will happen if I do X vs Y?" — primarily exercises Level 2 (iterated mental intervention). When the agent evaluates past choices to refine the comparison ("given what happened when I tried X last time, what would Y have produced?"), it exercises Level 3.

The causal hierarchy theorem. Bareinboim et al. (2022) prove that the three levels form a strict hierarchy: Level 2 knowledge cannot in general be computed from Level 1 data alone, and Level 3 cannot be computed from Level 2 alone.

This is load-bearing for AAT's Section II: evaluating $Q_O(M_t, a; \cdot)$ is a Level 2 query, so agents that need to *learn* action consequences during operation require causal structure beyond predictive models (6.2).

Software as a uniquely rich domain for this hierarchy. In most domains, Level 3 counterfactuals require model-based simulation with uncertain fidelity. Software development is the privileged exception *for a specific class*: for code-internal counterfactuals with deterministic outcomes — "what would the test suite report under implementation X instead of Y, environment fixed?" — `git checkout` plus re-implementation plus test execution is literal Level 3 realization with ground-truth verification, not a proxy. For counterfactuals crossing the agent–environment boundary (what feature sequence the team would have shipped, how the market would have responded) it is a strong executable proxy, not literal Level 3. The precise conditions — the (α) deterministic-outcome / (β) cost-commensurate-replay / (γ) content-addressed-immutable conjunction, and why the resulting uniqueness is configurational rather than necessary (with named falsifiers) — are established in `#obs-software-epistemic-properties` (P2; `02-tst-core/`). This scoped Level-3 access is what makes software AAT's privileged calibration laboratory for causal reasoning.

Domain instantiations of the three levels:

Table 6.1

Domain	*Level 1 (Association)*	*Level 2 (Intervention)*	*Level 3 (Counterfactual)*
Kalman filter	Prediction from state estimate	Innovation conditioned on action	Not typically exercised
RL agent	Value function prediction	Action → reward observation	Regret computation
Scientific method	Correlational observation	Experimental intervention	"What if we had used control X?"
Military (Boyd)	Pattern recognition	Probe/feint → observe response	"What if we had attacked from the flank?"
Software developer	"I think this function does X"	Run test → observe result	`git checkout` + alt. impl. — literal for code-internal deterministic counterfactuals; proxy across the agent–environment boundary (`#obs-software-epistemic-properties`)
Immune system	Antigen pattern matching	Antibody → pathogen response	Not exercised (no counterfactual reasoning)

Working Notes.

- This segment is positioned in Part II Ch.2 ("Causal Access and the Planning Decision") as of the 2026-05-12 relocation. It previously lived in Part I Ch.1 alongside AAT's own ontological commitments, which mis-framed it as foundational AAT machinery rather than as an imported framework recapitulated for AAT's purposes. The relocation places it at the head of the chapter that deploys it operationally.
- Part I segments that previously depended on this segment in their `depends:` frontmatter (`def-causal-information-yield`) now reference Pearl 2009 / Bareinboim 2022 via external citation with a forward pointer to this recapitulation. TST segments do the same. The relocation thus introduces no dangling dependencies; what changes is the rhetorical posture (imported framework, lightly cited where mentioned, recapitulated where deployed) and the OUTLINE position (Part II Ch.2 rather than Part I Ch.1).

6.2 *Derived:* Causal Hierarchy Requirement

Evaluating the action-value Q_O requires answering "what happens if I *do* action a?" — a Level 2 (interventional) query in Pearl's causal hierarchy. An agent that must learn the answer to this question during operation needs access to causal structure beyond what purely predictive models can provide.

Formal Expression.

Action selection via 5.6 requires:

Derived (causal-hierarchy-requirement, from value-object + pearl-causal-hierarchy).

$$Q_O(M_t, a; \pi_{\text{cont}}, N_h) = \mathbb{E}\big[V_{O_t}(\tau) \mid M_t,\ do(a_t = a),\ a_{t+1:} \sim \pi_{\text{cont}}\big] \tag{6.3}$$

The $do(\cdot)$ notation is explicit: this is an *intervention*, not a *conditioning on observed data*. By 6.1, Level 2 queries ($P(Y \mid do(X))$) cannot in general be computed from Level 1 data ($P(Y \mid X)$) alone. Therefore:

An agent that must evaluate Q_O from experience needs access to Level 2 knowledge — knowledge about the effects of its own interventions, not merely correlational patterns.

We restrict attention to **learning purposeful agents** — agents that must **acquire or refine** Level 2 knowledge during operation. This is a named sub-scope of the agency scope defined in 1.6. It excludes agents with **pre-compiled** interventional structure: - PID controllers (the designer pre-computed the control law) - LQR (separation principle gives optimal policy from model parameters) - Hardcoded reactive policies

Scope Narrowing (learning-agent scope).

Pre-compiled agents are within agency scope (they have objectives and act on them) but outside learning-agent scope — their causal structure was externally supplied by a designer who had Level 2 access. **All remaining Section II results operate within learning-agent scope** unless explicitly noted otherwise. This scope narrowing focuses the theory on agents that must build or maintain their own causal understanding.

Epistemic Status.

Exact. The derivation is a direct application of the causal hierarchy theorem (Bareinboim et al. 2022) to the value-object definition. If you accept that Q_O is an interventional query and that the causal hierarchy is strict, the conclusion follows. The scope narrowing to learning agents is a definitional restriction, not a derived result — it sharpens the class of agents under study.

Discussion.

The causal hierarchy theorem does the heavy lifting. The key mathematical fact is external to AAT: Bareinboim et al. (2022) prove that the three levels (association, intervention, counterfactual) form a strict hierarchy — Level 2 quantities cannot in general be computed from Level 1 data. AAT's contribution is applying this to the purposeful-agent setting: if you want to select actions by their consequences (Q_O), you need causal structure.

What "Level 2 knowledge" means concretely. For different agents: - A developer needs to know "if I refactor this module, will tests still pass?" (not just "modules that were refactored tend to have tests that fail") - A commander needs to know "if I move forces north, will the enemy respond by retreating?" (not just "when forces moved north, the enemy often retreated") - An RL agent needs $Q(s, a) = \mathbb{E}[R \mid s, do(a)]$, not $\mathbb{E}[R \mid s, A = a]$ (the latter includes selection bias from the agent's own policy)

The distinction between $do(a)$ and $A = a$ is the core of causal inference. It matters whenever the agent's action-selection policy correlates with unobserved confounders.

Why pre-compiled agents are excluded. A thermostat "knows" that turning on the heater raises temperature — but this knowledge was designed in, not learned. The thermostat never needs to reason about interventions because the intervention-outcome mapping is hardwired. AAT's purposeful-agency machinery is specifically for agents that face uncertainty about how their actions affect the world and must reduce that uncertainty through experience.

Bi-predictability as empirical evidence for Level 2 advantage. Hafez et al. (2026) measure the information structure of the agent-environment loop via bi-predictability $P = \mathrm{MI}(S, A; S')/C$, capturing how tightly coupled the agent is to its environment through the action channel. Their Information Digital Twin (IDT), which monitors the loop's information geometry, detects environmental perturbations at 89% accuracy versus 44% for reward-based monitoring. This is consistent with the present segment's claim: the feedback loop provides richer (Level 2) data than outcome monitoring alone. Hafez's framework measures the coupling; AAT explains *why* the coupling is information-theoretically superior — because the loop generates interventional data by construction (6.3), not merely associational data. The measurement (Hafez) and the explanation (AAT) are complementary.

Working Notes.

- This scope narrowing connects to 6.6: agents that must learn Level 2 structure face a cost-benefit tradeoff between learning through exploration (costly, slow, but verifies causal links) and planning through explicit Σ_t (cheaper if the causal model is adequate, but the model may be wrong).
- LLMs trained on causally-structured text absorb causal priors — noisy prior knowledge from mixed provenance (experimental, observational, speculative). This is not verified interventional structure; it's a *prior* (plausible, not derived). An LLM in the adaptive loop has both: priors from training AND interventional data from the loop. The priors accelerate; the loop verifies. The IB framework (2.2) predicts causal structure will be retained in training because it has predictive power for language.
- The scope narrowing to "learning agents" is generous — it includes any agent that updates its causal beliefs during operation, even if it starts with strong priors. The excluded class (pure pre-compiled controllers) is genuinely different: they never revise their action-consequence model.

6.3 *Derived:* Loop Provides Interventional Data Access

An agent in the feedback loop generates interventional data by construction: the agent's action a_t causally precedes the next observation o_{t+1}, and the mismatch conditioned on a_t carries interventional information. This is how agents within AAT's **agency scope** ($\mathcal{S}_{\text{agency}}$, which requires $|\mathcal{A}| \geq 2$ and at least one action with causal effect) gain Level 2 access — not through internal architecture, but through the loop itself. Agents in the adaptive scope but outside agency scope (passive observers) lack the action contrasts needed for interventional data.

Formal Expression.

By 1.8, the temporal ordering is constitutive: a_t causally precedes o_{t+1}. The agent chose a_t; the environment responded with o_{t+1}. The feedback loop therefore generates **intervention-produced data** — data whose causal character differs from passive observation because the agent's action was a genuine cause of the subsequent observation.

Derived (loop-interventional-access, from causal-structure + recursive-update).

The critical distinction: **"action-generated data" is not the same as "cleanly identified do-estimates."** The pair (a_t, o_{t+1}) is produced under intervention — the agent executed a_t, making it interventional in character rather than a passively observed association. But between intervention-produced data and a usable estimate of $P(o \mid do(a_t), \Omega_t)$ stand: (1) coverage — the agent must have tried diverse actions, not just one policy; (2) confounding within a time step

— unobserved state variables that affect both action choice and outcome; (3) delay — consequences may appear much later than $t+1$; (4) partial observability — o_{t+1} reveals only part of the outcome. The strength of causal identification from this data depends on the regime (8.5): strong in Regime A (intervention-rich domains like software and laboratory science), moderate in Regime B (partial intervention), weak in Regime C (observation-only). The claim here is about the *character* of the data (interventional, not observational), not about the agent's ability to extract clean causal estimates from it.

The mismatch signal conditioned on the agent's action:

$$\delta_t \mid a_t = o_{t+1} - \hat{o}_{t+1}(M_t, a_t) \tag{6.4}$$

carries interventional information: it tells the agent how the environment responded to its specific intervention a_t, relative to what the model predicted.

Epistemic Status.

Exact. This is a logical consequence of temporal ordering (1.8) and the feedback-loop structure (1.6). The claim is about **data availability**, not reasoning capacity — the loop *provides* interventional data whether or not the agent *exploits* it for Level 2 reasoning. Whether the agent uses this data to build causal models depends on its update mechanism and model class.

The precision is important: we claim the agent has *access to* interventional data, not that it *correctly identifies* interventional structure. Confounding within a single time step, delayed outcomes, and partial observability can all complicate the extraction of clean causal signals from the loop data.

Discussion.

The loop as a Level 2 engine. This is one of AAT's load-bearing results. The causal hierarchy theorem (6.2) says Level 2 knowledge requires more than correlational data. This result says: the adaptive loop *is* the "more." An agent that acts and observes the consequences is generating interventional data — the same kind of data that a scientist generates through experiments. The loop is a perpetual experiment.

Precision about what "interventional" means here. The interventional interpretation is strongest when: - The agent's action was the primary cause of the observed change (low confounding) - The observation follows closely in time (short delay) - The agent varied its action across episodes (not stuck on one policy)

When confounding is high, delays are long, or the agent follows a fixed policy, the interventional information in each (a_t, o_{t+1}) pair is weaker — still present, but harder to extract. This is why 3.7 distinguishes between high-CIY actions (that reveal causal structure) and low-CIY actions (that don't).

Even agents without explicit causal models benefit. A Q-learning agent doesn't maintain an explicit causal model, but in the tabular case with sufficient exploration and no within-step confounding, its Q-values converge toward $\mathbb{E}[R \mid s, do(a)]$ rather than $\mathbb{E}[R \mid s, A = a]$ — precisely because the training data comes from the agent's own interventions. In the partially observed, confounded, or delayed-outcome cases (where the caveats above apply), the loop still provides intervention-generated data, but the Q-values may converge to biased estimates that reflect the confounding structure rather than clean interventional effects. The loop provides Level 2 data; whether that data yields *identified* causal quantities depends on the domain's confounding and observability structure.

Honest credit to the action-perception-loop framing. The substantive observation that the agent's actions cause its observations — and therefore that loop data is interventional in character — is implicit in any framework built around an action-perception loop, including active inference (Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017, "Active inference: a process theory," *Neural Computation* 29; Parr & Pezzulo 2022, *Active Inference*, MIT Press, ch. 3) and the broader cybernetic-and-control lineage (Wiener 1948, *Cybernetics*; Conant & Ashby 1970, "Every good regulator of a system must be a model of that system," *Int. J. Systems Sci.* 1). AAT's distinctive contribution is not the observation that loop data is interventional but the *explicit lift* of this observation to a load-bearing theorem connected to Pearl's causal hierarchy (6.1) via Bareinboim, Correa, Ibeling & Icard (2022). Three specific moves AAT makes that the implicit treatments do not:

1. **The Bareinboim-hierarchy connection.** Active inference and the cybernetic lineage rest on Bayesian-network generative models (Pearl Level 1, associational). They do not invoke the causal-hierarchy theorem to argue that the loop's action-generated data is the substrate Level-2 queries require. AAT does — and the consequence is that Σ_t is positioned as a *causal* DAG rather than a *Bayesian-network* DAG.

2. **Regime-indexed strength of causal identification.** Even granting that loop data is interventional in character, the strength of usable causal identification varies by domain. AAT partitions this into Regime A (intervention-rich, software/laboratory), B (partial intervention, organizational), C (observation-only — see 8.5). The AI literature treats causal identifiability uniformly within its modeling assumptions and does not surface the regime distinction at the segment level.

3. **Explicit scope honesty.** The Formal Expression above carefully distinguishes "data generated under intervention" from "cleanly identified do-estimates." The AI literature, as Bruineberg, Dolega, Dewhurst & Baltieri (2022) document in their Pearl-vs-Friston critique, sometimes elides this distinction — using the action-perception loop language to support stronger causal claims than the formal apparatus delivers. AAT's careful split is the conservative form.

The reframing here is rhetorical, not substantive: the headline result (the loop is a Level-2 engine) stays; what changes is the explicit acknowledgment that the observation about loop data being action-generated is shared with the broader literature. AAT's distinctive content sits in the three specific moves above. The companion architectural move on 5.3 (Pearl-blanket vs. Friston-blanket form of Markov blanket; cf. Bruineberg et al. 2022) makes AAT's conservative-form positioning visible across both architectural and access-channel segments.

Connection to the identifiability-floor pattern. This segment is structurally load-bearing for the L0-causal-insufficiency-detection no-go (8.2): without the loop's interventional access, the no-go forbids detection entirely; the covariance test under joint sibling observability is the unique broadly-available violation of the no-go's "purely on-policy" scope. See X for the meta-pattern.

Composite-layer extension. `#disc-identifiability-floor` Instance 3 (the composition-layer no-go anchored in Liberzon 2003's common-Lyapunov nonexistence) establishes that composite contraction cannot in general be certified from component-level data alone — the coupling-sign bit distinguishing cooperative from adversarial regimes is unidentifiable from component marginals. One of the four structural escapes in Instance 3 is composite-extended loop-interventional-access: **interventions on sub-agent A_j reveal A_i's cross-coupling response**, which is a $do(\cdot)$-data distinction between the two coupled constructions. This is the composite-layer analog of the single-agent interventional-access-escape the present segment delivers for Instance 1; in the composite setting, the agent performs interventions on one sub-agent's action space and observes the effect on another sub-agent's mismatch trajectory, extracting the coupling sign that component marginals cannot. The load-bearing role of `#der-loop-interventional-access` therefore extends across two identifiability-floor instances (agent-internal and composite-layer), not one.

Modes of deployment across `#disc-identifiability-floor` instances. The shared load-bearing role — Level-2 data from interventional action under `#der-causal-hierarchy-requirement` supplying the unique broadly-available escape from an observational-equivalence no-go — manifests through **semantically distinct interventional mechanisms** at different layers. The modes share the Pearl-do structure but differ in who performs the intervention and on what:

- **Mode 1 — agent-self-intervention (Instance 1, causal-structure layer).** The agent performs do-actions on its own action space as part of its ordinary adaptive loop. The intervention is *on the agent's own action*; the target is the environment's response, which reveals the latent common cause that Instance 1's on-policy L0-insufficiency-detection no-go (Bareinboim et al. 2022 CHT) makes unidentifiable. The agent-as-Level-2-data-generator property is intrinsic to the loop structure, not a capability that must be added.

- **Mode 2 — observer-on-sub-agent (Instance 3, composition layer).** An observer external to a composite performs do-interventions on one sub-agent A_j's action space; the target is another sub-agent A_i's mismatch-trajectory response, which reveals the cross-coupling sign that Instance 3's Liberzon-anchored no-go makes unidentifiable from component-marginal observation. The same Pearl-do structure, but the intervening agent and the measured agent are distinct members of the composite.

Future instances of `#disc-identifiability-floor` at new layers may add further modes (see `#disc-identifiability-floor` Working Notes for the architecture-within-behavior-class layer currently under triage; the corresponding mode would be *observer-on-agent-input* at the candidate agent's observation channel). The three-mode-pattern observation is a structural regularity worth naming even though the specific Mode-3 instance has not promoted; the load-bearing content — Level-2 escape from observational-equivalence no-goes via loop-interventional access — remains shared across the deployment modes even when the specific interventional quantity varies. The unification is at the pattern level; the mechanism is semantically distinct per layer.

Why the loop data is genuinely interventional — the singular-trajectory ground. The interventional character of loop data is not a property of the feedback mechanism in isolation; it rests on the scope commitment in 4.7 that each agent is instantiated on a singular, non-forkable causal trajectory. When the agent executes a_t and observes o_{t+1}, the observation is the response to *this* agent's intervention on *its* single trajectory $\mathcal{C}_t$. Replaying a_t from a checkpointed M_t against a different event stream would *not* constitute an intervention on $\mathcal{C}_t$ — it would be an intervention on a different trajectory $\mathcal{C}_t^{(2)}$ that happened to share a prefix. Pearl's do-operator presumes a definite causal system acted upon; AAT inherits this presumption via the singular-trajectory scope. Agents whose ontology is *type-like* (equivalence classes of copies) rather than *token-like* (singular trajectories) are outside AAT's formal scope; in particular, aggregate claims about "the model" across copies require additional machinery not provided here. This is the ontological ground that makes the "action-generated data is Level-2" claim honest rather than metaphorical.

Working Notes.

- This result establishes that all agents within AAT's **agency scope** (1.6, $\mathcal{S}_{\text{agency}}$) have access to interventional data, regardless of their internal architecture. Agents in the adaptive scope but outside agency scope (passive observers, nominal agents with $|\mathcal{A}| < 2$ or no causal effect) lack the action contrasts needed to generate interventional data — they observe but cannot intervene. This includes LLM agents operating through a tool-use loop — the LLM issues an action (tool call), observes the result, and updates. The loop gives it Level 2 data even though its internal architecture (transformer attention) is not designed for causal reasoning. The loop compensates for architectural limitations.
- The connection to 3.7: CIY quantifies *how much* interventional information a specific action provides. This segment establishes that interventional information is *available in principle*; CIY measures the *quantity per action*.
- **Cross-reference to NeurIPS Paper 2.** The Loop-as-Causal-Engine claim is formalized at theorem grade in NeurIPS 2026 Paper 2 ("A Unified Convergence Theory for Non-Stationary Reinforcement Learning", `~/src/neurips/02-unified-convergence-rl/`, §5 Lemma 5.3 / `#lem-loop-level2`) under the named **(C1) positivity** / **(C2) sequential-ignorability** / **(C3) known action-mechanism** triple. (C2) is the load-bearing structural condition: in mutilated-graph form, a_t must be d-separated from o_{t+1} given history H_t. *Goal-conditioned LLM policies violate (C2) by construction* — the goal influences action through the same forward pass that models the observation, breaking the conditional independence required for sequential ignorability — which is the bridge to Paper 3's Coupled-class formulation. The (C1)-(C3) triple makes explicit which structural commitments a Class 1 (Separated) tool-use loop satisfies for free that a Class 3 (Coupled) goal-conditioned policy does not. See `msc/neurips-back-integration-2026-05-08.md` §1 Paper 2 entry 5.

6.4 *Scope:* CIY Observational Proxy

When and how causal information yield can be approximated from observational data rather than interventional experiments.

Formal Expression.

Definition (ciy-proxy).

$$\mathrm{CIY}_{\mathrm{proxy}}(a_{t-1}) = I(o_t; a_{t-1} \mid M_{t-1}) - I(o_t; a_{t-1} \mid \Omega_t, M_{t-1}) \tag{6.5}$$

This proxy is **sign-indefinite in general** and requires causal assumptions for interpretation. The canonical CIY (interventional, 3.7) is the primary quantity; the proxy is auxiliary.

Safety conditions for proxy use. The proxy form should NOT be used in policy optimization (e.g., as the CIY term in a policy objective) because an agent maximizing a sign-indefinite quantity may optimize in the wrong direction. The proxy is suitable only for diagnostic purposes: detecting whether an action carried causal information (large proxy magnitude) vs. none (proxy near zero). For decision-making, use the canonical CIY (non-negative by construction) or a known-safe surrogate (ensemble disagreement, UCB bonuses). If the canonical CIY is intractable and no safe surrogate is available, the CIY term should be dropped from the policy objective entirely, defaulting to pure exploitation.

Admissibility regimes.

Three regimes determine when CIY can be estimated and how strong the causal identification is:

Scope Condition (ciy-admissibility).

Regime A — Randomized interventions. The agent varies its actions across episodes (RL agents exploring, scientists experimenting, organisms probing). CIY is directly estimable from the agent's execution data and non-negative by construction. This is the standard case for active agents within the adaptive loop (6.3). Action variation provides the identification needed for clean interventional estimates.

Regime B — Observational with causal assumptions. The agent cannot freely vary actions (constrained by coordination, policy, or resource limits). CIY estimation requires additional structure: a known causal DAG, instrumental variables, or functional form assumptions. Results inherit whatever causal assumptions are made. The interventional interpretation of CIY is weaker — it holds under the assumed causal structure but not model-free.

Regime C — Adversarial or passive observation. The agent either did not intervene (passive monitoring) or the observation channel includes responses from potentially adversarial sources. In the passive case, CIY is zero by definition (no intervention, no interventional information). In the adversarial case, CIY from the query action itself remains non-negative, but the *content* of the response may be designed to increase model-reality mismatch. The adversary operates through the disturbance term ρ, not through the information measure.

The regime is a property of the **domain and the agent's action space**, not a parameter the agent chooses. Software development is typically Regime A (the agent runs tests, deploys to staging, observes results — high action variation). Organizational strategy is typically Regime B (multiple initiatives run concurrently, attribution requires assumptions). Intelligence analysis is typically Regime C (the analyst observes but does not intervene).

Epistemic Status.

Conditional on the causal assumptions within each regime. The proxy definition is standard information theory; the admissibility classification is a scope decision, not a derived result. The safety conditions for proxy use are normative — they follow from the sign-indefiniteness of the proxy, not from AAT-specific machinery.

Max attainable: conditional. The regime boundaries are domain properties that cannot be derived within the theory.

Discussion.

Relationship to the canonical CIY. The proxy is not an approximation of the canonical CIY — it is a different quantity that happens to correlate with CIY under favorable conditions (Regime A). The canonical CIY (3.7) is defined interventionally and is non-negative by construction. The proxy uses observational mutual information and can be negative. They agree when the agent's action variation satisfies the conditions for causal identification; they diverge otherwise.

Regime as a domain property. The admissibility regime is not something the agent selects — it is determined by the domain's action space and observation structure. An agent that cannot vary its actions is in Regime B or C regardless of its internal architecture. This has implications for which domains CIY-based exploration is applicable to: Regime A domains get the full benefit; Regime B domains get weaker guarantees; Regime C domains should use alternative exploration strategies entirely.

6.5 *Discussion:* CIY Unified Policy Objective

The exploration-exploitation tension can be expressed as a single policy objective that jointly maximizes expected value and a causal information surrogate. This formulation is *heuristic* — CIY measures action-distinguishability, not expected information gain (see 3.7), so the objective selects for causally distinctive actions rather than maximally informative ones. The λ-weighting partially compensates by suppressing the CIY term when model uncertainty is low, but the surrogate nature is inherent.

Formal Expression.

Discussion (unified-policy-objective — heuristic).

$$\pi^*(M_t) = \arg\max_a \left[\mathbb{E}[\text{value}(a) \mid M_t] + \lambda(M_t) \cdot \text{CIY}_q(a;\, M_t)\right] \tag{6.6}$$

The first term is exploitation (expected value given current model). The second is a *heuristic exploration term* using CIY as a surrogate for expected information gain (3.7). CIY measures how different the action's outcome distribution is from alternatives — this is action-distinguishability, not learning value. The surrogate is reasonable when U_M is high (distinguishable actions are also informative to an uncertain agent) and poor when U_M is low (distinguishable actions teach nothing to a confident agent). $\lambda(M_t)$ controls the balance:

- High U_M (uncertain model) → large λ — exploration is valuable
- Low U_M (confident model) → small λ — exploitation dominates
- Long time horizon → larger λ — information compounds
- High ρ (fast-changing environment) → larger λ — perpetual uncertainty

λ carries units of [value per unit information]. In specific domains it reduces to known quantities:

Table 6.2

Domain	*λ reduces to*	*Status*
Bayesian bandits	Gittins index	Exactly derived
Kalman dual control	Probing cost in quadratic objective	Exactly derived
Active inference	Precision on epistemic affordance	Framework-derived
Information-directed sampling	$(\text{VoI})^2$/info gain	Exactly derived (Russo & Van Roy)
RL with UCB	Confidence-bound scaling	Heuristic (tuned)

Identifiability gate. Before incorporating CIY into the policy objective: (1) action variation must exist, (2) the admissibility regime must be identified (6.4), (3) the reference distribution q must be specified, (4) local stationarity must hold. If any condition fails, CIY-based terms should be dropped or replaced with simpler uncertainty-based heuristics (UCB-style bonuses, ensemble disagreement).

Epistemic Status.

Discussion-grade summary; underlying derivation is exact. Originally treated as a discussion-grade heuristic, the unified objective has now been formally derived as the exact Lagrangian relaxation of the Linear Matrix Inequality (LMI) governing Lyapunov persistence (see `#deriv-causal-ib-lmi`).

The scalar heuristic $Q_O(a)+\lambda\cdot\text{CIY}(a)$ is formally superseded by the exact tensor trace-product: $a_t^* = \arg\max_a \left[Q_O(a) + \text{Tr}\left(\Lambda \cdot \mathcal{I}_o(a)\right)\right]$ where $\mathcal{I}_o(a)$ is the Fisher Information Matrix (Matrix CIY) and Λ is the positive-semidefinite shadow price matrix of the survival constraint.

Max attainable for this segment: *discussion-grade* (it is a discussion of the underlying result, per `type: discussion`). Max attainable for the underlying derivation in `#deriv-causal-ib-lmi`: *exact*. The structural form is fully grounded in AAT's physical survival bounds and standard semidefinite programming, eliminating the need to treat exploration as an ad-hoc heuristic.

Discussion.

Two Parallel Exploration Drives. AAT dictates two correlated but distinct motivations for exploration, acting at opposite ends of the uncertainty spectrum: 1. **Epistemic Information Gain ($\lambda_{\text{info}} \propto U_M$):** The primary CIY formulation. The agent explores to reduce its model uncertainty. This drive dominates when U_M is high. 2. **The Survival Imperative ($\lambda_{\text{surv}} \propto 1/U_M$):** As mathematically proven in `#deriv-causal-ib-exploration`, an agent with high confidence (low U_M) in a drifting environment ($\rho > 0$) mathematically guarantees its own death by ignoring noisy observations. To force the necessary correction, the Lyapunov persistence constraint dictates an immense shadow price ($\lambda_{\text{surv}} \to \infty$ as $U_M \to 0$) forcing the agent to seek unambiguous observations (low U_o / high CIY).

The dark-room problem is bypassed entirely by the Survival Imperative: exploration is not driven by preferences-as-priors, but by the literal physical boundaries of the Lyapunov sector constraint.

Connection to the zero-mismatch ambiguity. An agent that only exploits (acts to maximize predicted value) will tend toward confirmation bias — observing only what its model already explains (3.4, zero-mismatch ambiguity case (b)). Exploration via CIY-maximizing actions is the mechanism by which the agent actively tests its model.

Connection to Section II. For actuated agents, the exploration-exploitation tension extends to three modes: exploit (pursue O_t via Σ_t), explore (improve M_t), deliberate (revise Σ_t). The CIY framework provides the information-theoretic grounding for why strategy edges (7.3) need observational access (8.3) — edges the agent cannot observe have frozen CIY.

Connection to active inference. The expected free energy (EFE) in active inference (Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017, "Active inference: a process theory," *Neural Computation* 29; Da Costa, Parr, Sajid, Veselic, Neacsu & Friston 2020, "Active inference on discrete state-spaces," *J. Math. Psych.* 99; Sajid, Ball, Parr & Friston 2021, "Active inference: demystified and compared," *Neural Computation* 33) decomposes into *pragmatic value* (preferences-aligned outcomes) and *epistemic value* (expected information gain about hidden states). AAT's unified objective is structurally isomorphic: $Q_O \approx$ pragmatic, CIY $\approx$ epistemic. The convergence is at the shared-shape level — objective decomposes into value-and-information terms — not unified content. Two substantive differences remain. First, AAT grounds exploration in explicitly *causal* information (action-distinguishability under do) rather than entropy reduction over hidden states — not all uncertainty reduction is equally valuable for purposeful action; causal information specifically enables better *intervention* (see 6.2; the gap between CIY and proper expected information gain is logged in this segment's Epistemic Status as a known surrogate). Second, AAT does not encode preferences as priors over outcomes ($C(o) = \log P_{\text{pref}}(o)$ in the AI form): AAT's O_t is a value functional on trajectories (5.4), and the satisfaction-gap / control-regret diagnostic in 7.4, 7.5 depends on this distinction — the diagnostic structure does not survive the priors-as-preferences collapse (the dark-room critique, Sun & Firestone 2020, "The dark room problem," *Trends Cog. Sci.* 24).

Regret-bound connection to the strategy-cost objective. AAT's Q_O term connects to the strategy-cost objective in 8.9 via a regret-bound derivation: strategy-induced regret $R(Q_{\Sigma_t}) = V(a^*) - \mathbb{E}_{Q_{\Sigma_t}}[V(a)]$ is bounded by a divergence between π^* and Q_{Σ_t}, with the KL direction π^*-first forced (full derivation in O). Under AAT's canonical scope of deterministic π^*, the Bretagnolle-Huber identity $D_{\text{KL}}(\pi^* \| Q_{\Sigma_t}) = -\log(1 - \text{TV}(\pi^*, Q_{\Sigma_t}))$ holds *exactly* (Bretagnolle & Huber 1978), giving the tight regret bound $R(Q_{\Sigma_t}) \leq V_{\max}\big(1 - e^{-D_{\text{KL}}(\pi^* \| Q_{\Sigma_t})}\big)$ with matching lower bound $\Delta_{\min}\big(1 - e^{-D_{\text{KL}}}\big)$ on isolated optima (O §4). Pinsker's $V_{\max}\sqrt{\frac{1}{2} D_{\text{KL}}(\pi^* \| Q_{\Sigma_t})}$ remains the correct loose general form for stochastic-π^* extensions where the BH identity degrades back to inequality. The structural point: "value and information term" shares *shape* with EFE's pragmatic-epistemic decomposition, and the KL direction in the strategy-cost's variational form shares direction with variational inference — but AAT's derivation is via decision-theoretic regret bound on Q_O rather than via free-energy-gradient flow, which is the AAT-internal route that does not depend on the priors-as-preferences encoding.

6.6 *Normative:* Explicit Strategy Condition

An agent benefits from maintaining an explicit strategy Σ_t when the cost of planning is less than the cost of learning through exploration alone. This is a normative design criterion — it tells you when explicit strategy is *worth it*, not that it's always necessary.

Formal Expression.

An agent benefits from explicit Σ_t when:

Normative (explicit-strategy-condition).

$$C_{\text{plan}} + C_{\text{maintain}} < C_{\text{explore}} + C_{\text{repair}} \tag{6.7}$$

where: - C_{plan}: cost of constructing and evaluating the strategy (deliberation, simulation, model queries) - C_{maintain}: ongoing cost of keeping Σ_t current as M_t evolves (edge revision, structural updates) - C_{explore}: cost of learning action-outcome mappings through direct interaction (real actions, real time, real consequences) - C_{repair}: cost of correcting errors discovered only through execution (rollbacks, rework, damage)

All costs are measured in the same units (typically time or tempo-equivalent cost). The inequality requires that the two approaches produce approximately equivalent non-temporal outcomes — otherwise the comparison is between different strategies, not different approaches to the same goal.

Epistemic Status.

Normative, not derived. This is labeled *normative* because it is a design criterion (a preference for the less costly approach given equivalent outcomes), not a theorem. In practice, loop-based and model-based approaches may differ in:

- **Final value**: the model introduces bias; exploration may discover things planning cannot
- **Risk profile**: exploration risks real damage; planning risks wrong models
- **Reversibility**: some exploratory actions are irreversible
- **Model bias**: explicit Σ_t inherits the biases of M_t; loop-based learning does not

The inequality is correct *when* the outcomes are approximately equivalent — a condition that must be verified case by case, not assumed. When the precondition fails (model-based and loop-based approaches produce qualitatively different outcomes), the cost inequality is insufficient and the choice requires richer analysis (e.g., expected regret including model error).

Discussion.

When the inequality strongly favors planning. The right side ($C_{\text{explore}} + C_{\text{repair}}$) is large when: - Actions are expensive or irreversible (production deployments, military operations, surgical procedures) - Exploration damage is severe (a wrong move in chess loses the game; a wrong deployment takes down the service) - The environment is slow to respond (waiting for market feedback, waiting for test results) - The action space is enormous (combinatorial planning problems)

In these domains, explicit Σ_t is strongly motivated even if the planning model is imperfect.

When the inequality favors pure exploration. The left side ($C_{\text{plan}} + C_{\text{maintain}}$) is large when: - The environment is too complex or novel for useful models (genuinely unknown territory) - M_t is severely inadequate (model predictions are worse than random) - The environment changes faster than Σ_t can be maintained (ρ_Σ exceeds planning capacity) - Actions are cheap and reversible (A/B testing, sandbox exploration)

Normative grounding. The cost inequality is grounded in the persistence condition (4.4), not in an external preference postulate. The persistence condition demonstrates that agents whose correction tempo is insufficient *degrade* — this is a descriptive result, not a value judgment. The cost inequality operationalizes the consequence: approaches that consume less tempo budget leave more margin above the persistence threshold. The normative element is the preference for maintaining persistence margin — which is hard to argue against, since the alternative is degradation. (In TST, the temporal optimality postulate provides an additional normative grounding specific to software development.)

Connection to the three-way tradeoff. For actuated agents, the binary explore/exploit tradeoff extends to three modes: exploit (pursue O_t via Σ_t), explore (improve M_t), and deliberate (revise Σ_t). The cost inequality addresses the coarsest question (is explicit Σ_t worth having?). The finer allocation between the three modes is addressed in 9.2 — the extended deliberation threshold is derived, but the broader three-way framing is discussion-grade. The deepest insight: deliberation is internal exploration (simulation, counterfactual reasoning, cross-domain synthesis), making it a fundamentally different resource type from external action.

Working Notes.

- The inequality as stated is static — it compares cumulative costs. A dynamic version would ask: "given current M_t quality, is it worth deliberating further or should I act now?" This connects to 4.1's threshold: deliberation is worthwhile only when additional deliberation improves action quality enough to justify the time spent.
- Part of C_{maintain} is the cognitive cost of keeping Σ_t in the agent's representational capacity. For LLM agents, this means fitting the strategy in the context window. A 500-node DAG may exceed this capacity, making the left side of the inequality large enough that simpler strategies (or pure exploration) become preferable despite higher exploration costs.
- The cost inequality may be most useful not as a binary decision rule but as a way to calibrate Σ_t complexity: the agent should maintain a strategy just complex enough that $C_{\text{plan}} + C_{\text{maintain}}$ stays below $C_{\text{explore}} + C_{\text{repair}}$ for the current environment. This gives a principled answer to "how detailed should my plan be?"

6.7 *Discussion:* Additional Implications & Discussion

The chapter's central architectural move — that the feedback loop produces Pearl-Level-2 data by construction — has consequences that recur across the rest of the framework, and at least one negative finding of substantial scope that does not become statable until the chapter's machinery is in hand. What the loop *is* causally, where the same interventional pattern shows up at the composite layer, what kinds of pre-deployment evaluation methods structurally cannot identify deployment behavior, and how the chapter's cost-benefit framing for explicit strategy grounds in Part I's persistence condition are the things worth pulling out before leaving Part II Ch.2.

The loop is a Pearl-Level-2 engine.

The chapter's load-bearing contribution is structural: the adaptive cycle is itself a Pearl-Level-2 data engine. An agent that acts and observes the consequences is not collecting passive observations of a system that happened to do something; the agent's action a_t caused o_{t+1}, and the pair (a_t, o_{t+1}) is interventional in character. Bareinboim's causal hierarchy theorem (Bareinboim-Correa-Ibeling-Icard 2022) says Level-2 quantities cannot in general be computed from Level-1 data. The loop is where the gap closes — the data substrate Level-2 reasoning requires comes from the cycle the agent was already running for adaptation. Purposeful agents do not need a separate causal-inference module; the cycle produces the substrate as a by-product.

The result rests on the scope commitment from Part I Ch.4 — singular, non-forkable causal trajectory — without which Pearl's do-operator has nothing to operate on. Strip the trajectory commitment and the loop's interventional content collapses. The interventional character is not a property of any architectural feature of the agent; it is a property of the trajectory the agent runs on. This is why a Q-learning agent without an explicit causal model can still converge toward $\mathbb{E}[R \mid s, do(a)]$ rather than $\mathbb{E}[R \mid s, A = a]$ in the tabular case with sufficient exploration and no within-step confounding — the training data is the agent's own interventions, by construction. The chapter intro 6 develops the bidirectional confounding analysis (when does this break, and why is the *positive* confounded outcome the harder failure mode to escape than the negative one) in detail; the implication worth carrying forward is that the loop produces interventional data even where clean identification requires additional structural conditions.

For practitioners, the engineering anchor worth carrying is the asymmetry between training data and deployment data. Most ML is trained on observational corpora (Level 1) or on offline RL traces with known confounding structure (mixed Level 1/2). An agent in deployment, by virtue of being in the loop, generates Level-2 data for free. The framework's discovery is that this is *load-bearing*, not incidental: AAT's downstream results about action selection, strategy revision, and causal-structure detection rest on the loop's data substrate being interventional in character, and degrade specifically when something breaks the trajectory's singularity.

The result is also where AAT parts company with the active-inference / control-as-inference / free-energy-principle family in a structural rather than parametric way. Those frameworks derive agent behavior from observational learning — Bayesian-network generative models at Pearl Level 1 — and have no native handle on Level 2 outside auxiliary mechanisms. AAT's claim is conservative in form (data is interventional *in character*, not "the agent has cleanly identified causal estimates") but structurally distinctive: the loop's data lives at Level 2, and the framework's results consume that fact. The careful split between data character and identification quality (the regime A/B/C decomposition in 6.4, the intro's variable-ratio-reinforcement / gambling-addiction structure for confounded positive feedback) is what makes the conservative-form claim honest rather than overreach.

NeurIPS Paper 2's formalization of this claim under the (C1) / (C2) / (C3) triple — positivity, sequential ignorability, known action-mechanism — is the theorem-grade version of the chapter's structural argument. (C2) is the load-bearing condition: a_t must be d-separated from o_{t+1} given history H_t in mutilated-graph form. Goal-conditioned LLM policies violate (C2) by construction — the goal influences action through the same forward pass that models the observation, breaking the conditional independence required for sequential ignorability. This is the bridge to Class 3 (Coupled) treatment in `03-llm-core/` and to NeurIPS Paper 3's universal-constant route on the hallucination bias bound. The chapter's structural argument and the papers' formal theorems are the same content at two grain sizes; the chapter is where the structural argument lives, and the papers' load-bearing structural commitments are the chapter's scope condition.

The sandbox hard ceiling.

A negative finding of substantial scope follows directly from the previous section once the singular-trajectory commitment is in view, and it is worth surfacing here even though its bite reaches into AI-safety and AI-governance discourse. Sandboxed evaluation breaks the singular-trajectory commitment by design — sandbox trajectories are forkable (resettable, replayable, parallelizable), and that is what makes sandboxes useful for testing. It is also what makes them structurally unable to identify deployment behavior at Pearl Level 2.

Sandbox data and deployment data are not Pearl-equivalent. The sandbox trajectory is forkable; the deployment trajectory is not. The pair $(a_{\text{sandbox}}, o_{\text{sandbox}})$ generated under fork-able execution is Level-1 (associational) data; the pair $(a_{\text{deploy}}, o_{\text{deploy}})$ generated on a singular trajectory is Level-2. The causal hierarchy theorem then forbids the inference: claims about how the agent *would respond to interventions* in deployment are Level-2 questions, and they cannot in general be answered from sandbox-derived Level-1 evidence, regardless of evaluation thoroughness.

The empirical pattern this names — "alignment evaluations don't predict deployment behavior" — has been much-debated in the AI-safety literature without a structural mechanism. The chapter's framing offers one: the problem is not a measurement-quality limitation, not an evaluation-coverage gap, and not a model-capability surprise. It is a Pearl-hierarchy problem, structurally. An entire family of pre-deployment safety-evaluation approaches has a structural ceiling, not a gradient on which more rigor would close the gap.

The result has a positive content worth pairing with the negative one: it sharpens *which* kinds of safety claims can be evaluated in sandbox. Claims expressible at Level 1 (correlations between inputs and outputs under the agent's policy distribution) are within reach. Claims about deployment-time intervention response (Level 2) require deployment-trajectory data. The framework's prescription is therefore not "sandboxes are useless" but "sandboxes have a structural ceiling, and deployment-time monitoring is not substitutable by pre-deployment evaluation regardless of evaluation thoroughness." For practitioners running production agent loops, this is the same content as: the monitoring you do *after* deployment is doing structural work the pre-deployment evals cannot do. The downstream prescriptive content — what deployment-time monitoring must structurally look like to discharge what sandbox evaluation cannot — has its own home in the IDT / sidecar-monitoring literature (Hafez et al. 2026; see 6.2's Discussion for the bi-predictability anchor at 89% vs 44% detection accuracy), and the chapter's negative finding is what gives that downstream work its load-bearing role.

The finding lives at the chapter-end here because the loop-Level-2 result and the singular-trajectory scope are both required, and the chapter is where they coincide for the first time. A dedicated `#disc-sandbox-evaluation-ceiling` segment with a full Findings section may be warranted in a future cycle; for now the chapter-end is its canonical statement.

One pattern, multiple deployment modes.

The interventional-access result has a generalization that lives in the meta-architecture rather than in any single chapter. Across `#disc-identifiability-floor`'s multiple instances, *loop-interventional access provides the unique broadly-available escape from an observational-equivalence no-go* — but the specific interventional quantity varies by layer. The shared structure is Pearl-do-via-loop; the deployment mode differs.

Two modes are explicit already. **Agent-self-intervention** (this chapter's content): the agent performs do-actions on its own action space as part of the adaptive cycle. The intervention is on the agent's own action; the target is the environment's response. This is the mode that supplies the Level-2 substrate underlying `#der-causal-insufficiency-detection`'s no-go and its single broadly-available escape — joint sibling observability under exploration breaks the on-policy observational-equivalence the no-go imposes. **Observer-on-sub-agent** (Part III's contribution): an observer external to a composite performs do-interventions on one sub-agent's action space; the target is another sub-agent's mismatch-trajectory response. The intervention quantity is the cross-coupling sign that component marginals leave unidentifiable (Liberzon 2003 common-Lyapunov-nonexistence as the no-go; the closed-form composite critical-mass result as the escape under matched-Tier conditions).

A third mode is under triage — *observer-on-agent-input*, at the architecture-within-behavior-class layer — and there may be more as the meta-pattern grows. What makes the pattern worth surfacing here rather than only in `#disc-identifiability-floor` itself is the chapter's role as the agent-internal anchor: Mode 1 originates here, and the others are recognizable as variations of the same structural move. The unification is at the pattern level (Pearl-do as escape from observational-equivalence no-goes), with the specific interventional quantity differing per layer.

The composite-layer mode is where Part III's signed-coupling decomposition becomes load-bearing rather than a structural choice. Interventional access via the loop at the composite level is what makes cooperative-vs-adversarial coupling identifiable; without it, the signs are observationally indistinguishable, and persistence-at-the-composite-level becomes a quantity the framework can describe but not identify from component data alone. The chapter's central result is the agent-level instance of what becomes a multi-layer structural commitment — and the framework's posture of *naming an information-theoretic floor, then naming the unique broadly-available escape via AAT machinery* is what makes the floors load-bearing rather than discouraging.

Cost, calibration, and the bridge into Strategy Dynamics.

The chapter closes with `#norm-explicit-strategy-condition` and a cost-benefit framing that is normative rather than derived: an agent benefits from maintaining explicit Σ_t when $C_{\text{plan}} + C_{\text{maintain}} < C_{\text{explore}} + C_{\text{repair}}$. The grounding worth carrying forward is that this normative criterion is downstream of Part I Ch.4's persistence condition, not an independent design preference. Approaches that consume less tempo budget leave more margin above the persistence threshold; the cost inequality operationalizes that margin into a planning-vs-exploration tradeoff. The normative element is the preference for maintaining persistence margin, which is hard to argue against because the alternative is structural degradation.

The cost framing also points at a tractability constraint that bites hard for LLM agents. Part of C_{maintain} is the cognitive cost of keeping Σ_t in the agent's representational capacity — for an LLM, fitting the strategy in the context window. A large strategy DAG may exceed this capacity, making the left side of the inequality large enough that simpler strategies (or pure exploration) become preferable despite higher exploration costs. This is the same structural fact that Part I Ch.4's bandwidth-as-first-class-prerequisite surfaces in a different vocabulary; the persistence-bandwidth-floor and the strategy-complexity-bound are the same constraint at different layers.

The chapter is also where the three-way exploit / explore / deliberate refinement first becomes formally well-posed. Once the loop's Level-2 access is in hand, deliberation can be characterized as *internal exploration in model space rather than environment space* — Pearl-do in simulation rather than in the agent's trajectory. The full treatment lives in Part II Ch.5 (the orient cascade); this chapter establishes the data substrate that makes deliberation a meaningful third mode rather than a verbal distinction.

The widest downstream bite of the chapter's central result is in Part II Ch.4 (Strategy Dynamics), where `#der-causal-insufficiency-detection` becomes statable: on-policy L0 insufficiency is *structurally undetectable*, and the loop's interventional access — applied through joint sibling observability under exploration — is the unique broadly-available violation of the no-go. The Ch.4 no-go (self-diagnosis is structurally impossible under policy-perfect execution) and this chapter's central enabling result are duals: the no-go names what the loop's Level-2 access uniquely escapes. Read the two chapters together and the identifiability-floor pattern's first instance comes into view in full.

What Part II Ch.2 closes on, then, is the substrate. Purposeful action requires Level-2 data; the loop produces Level-2 data on singular trajectories; the data is interventional in character even when clean identification requires more; and the same Pearl-do structure recurs across multiple deployment modes as the meta-architecture develops. Chapter 3's strategy machinery and Chapter 4's strategy dynamics are what AAT does with that substrate; the closing scope of this chapter is what they will all stand on.

Working Notes.

- This segment is a chapter-end discussion grouping findings from 6.3, 6.2, 6.4, 6.6, and 4.7 (the Part I Ch.4 scope on which the loop-Level-2 result rests). The cross-layer pattern in §3 cross-references X's meta-segment, which carries the canonical instance table. Each finding's load-bearing math lives in its home segment; this segment surfaces what the chapter unlocks as a chapter.
- **Cross-reference vs canonical-home policy.** No source segment in this chapter carries a `## Findings` section. The chapter intro 6 already covers the data-character vs identification-quality nuance, the tragedy-of-agency-in-a-confounded-world framing, and the variable-ratio reinforcement / gambling-addiction structure with substantial depth — this segment does not re-state that content. The implications worth surfacing beyond the intro are the loop-Level-2 result's downstream consequences (RL theory at NeurIPS-paper grade, AI evaluation methodology, identifiability-floor pattern at multiple layers, cost-benefit substrate for explicit strategy) and the negative finding (sandbox hard ceiling) that the chapter establishes but is not yet stated explicitly anywhere.
- **Sandbox hard ceiling — segment promotion candidate.** The sandbox-hard-ceiling content is a finding that does not currently have a home segment. It is treated in full here because (a) its prerequisites are the loop-Level-2 result and the scope-agent-identity commitment, both within scope by the end of this chapter, and (b) it is consequential enough for AI-safety practitioners that a `## Findings`-bearing dedicated segment may be warranted in a future cycle. Open question for that cycle: does it land as a new segment (`#disc-sandbox-evaluation-ceiling` or similar) under Appendix A, or as an extension of `#scope-agent-identity`'s Discussion?
- **Connection to NeurIPS submissions.** The chapter's central result is formalized at theorem grade in NeurIPS 2026 Paper 2 (`~/src/neurips/02-unified-convergence-rl/`, §5 Lemma 5.3 / `#lem-loop-level2`) under the (C1)/(C2)/(C3) triple. Paper 3 (`~/src/neurips/03-llm-hallucinate-bound/`) is the Class 3 (Coupled) bridge — goal-conditioned LLM policies violate (C2) by construction. Cross-references to these formal forms appear in 6.3's Working Notes; this segment treats the structural content rather than the theorem statement.

Chapter 7

Strategy Structure and the Diagnostic Split

We commit to a formal representation for strategy — a probabilistic causal DAG with AND/OR semantics and single-parameter edges — and develop the headline diagnostic that lives on top of it: the satisfaction gap and control regret as orthogonal signals that distinguish "the goal is hard" from "the strategy is weak."

An agent with a goal needs more than a model of reality. It needs a theory of how its own actions produce progress toward the goal — which steps depend on which, which alternatives are available, where the uncertainty lives. Chapter 3 makes that theory formal.

Chapter 2 answered the prerequisite questions: planning requires Level-2 causal access (6.2); the feedback loop supplies it (6.3); explicit strategy is worth maintaining when the cost-benefit favors it (6.6). None of that tells us what a strategy *is*, formally. Chapter 3 commits.

The commitment is more constrained than it might look. We don't choose acyclicity — it falls out of temporal ordering. Strategy nodes are propositions about future events, events have positions in time, and time has a direction, so the graph cannot have cycles. We don't choose the Markov property either — given causal sufficiency, the Causal Markov Condition theorem (Spirtes–Glymour–Scheines, Pearl) forces the factorization. What we *do* choose, inside the strongly motivated graphical structure, is the parameterization: single-parameter edges with AND/OR combination semantics at nodes, rather than full conditional probability tables. The choice is parsimony-motivated — each edge carries one number rather than a 2^k table at a node with k parents — and it converged across three independent attempts to formalize the same content. Alternative parameterizations (Noisy-OR everywhere, weighted combinations) are rejected for documented reasons; see 7.2.

The wrinkle the formalism handles explicitly is causal insufficiency. Most strategy graphs in the wild have nodes that share latent common causes — shared infrastructure, common-mode risks, correlated adversary actions — that the agent has not represented as nodes. Naive AND/OR propagation on such graphs produces systematically biased plan-success estimates: it overestimates redundancy on OR-structures (the parallel paths share their failure modes) and underestimates joint reliability on AND-structures (the prerequisites also share their successes). The Correlation Hierarchy (L0 / L1 / L1' / L2, in 7.3) handles this as part of the formalism rather than as an exception. There is a sobering downstream result: under purely on-policy execution, an agent at L0 *cannot detect* its own causal insufficiency from its own data — the worlds with and without a latent common cause are observationally indistinguishable from on-policy traces alone (the no-go in 8.2). The unique broadly-available escape is joint sibling observability under exploration. This is one of Section II's load-bearing applications of 6.3.

The chapter's headline result is the diagnostic split. When an agent is not achieving its goal, the framework asks two distinct questions, not one. The satisfaction gap

$$\delta_{\text{sat}} = V_{O_t}^{\min} - A_O \tag{7.1}$$

is positive when the best policy in the agent's current class cannot reach the satisfaction threshold given the agent's current model. Control regret

$$\delta_{\text{regret}} = A_O - V_O(\pi_{\text{current}}) \tag{7.2}$$

is positive when the agent is leaving value on the table — its current policy is worse than the best available. Together they produce a 2×2 cell map: goal attainable + policy good is success; goal attainable + policy poor means revise the strategy; goal unattainable + policy poor means revise the strategy first and then re-check; goal unattainable + policy good is the *capability limit* — the agent is doing the best it can but the goal is out of reach. Each cell prescribes a different corrective action, and getting the prescription right requires the orthogonal split. An earlier framing that lumped everything into a single goal-distance signal could not distinguish "the goal is too hard" from "the strategy is too weak" — and the two have different remedies. The split is what makes the orient cascade (Chapter 5) actionable.

A detail worth surfacing up front. The chapter opens with 7.1 — the chain rule of probability lifted to log-space:

$$\log P(\text{chain}) = \sum_{i=1}^{n} \log P(E_i \mid E_{<i}) \tag{7.3}$$

The identity is elementary; the consequence is not. Deep strategies are exponentially fragile in their depth — confidence in a 10-step plan is the product of 10 conditional probabilities, and each one is less than 1. This grounds the structural pressure toward shallow, parallelizable strategies. The same log-additive structure also anchors three other uniqueness theorems downstream in AAT (reverse-KL at the divergence layer, log-odds at the update layer, Fisher metric at the metric layer) — see Z for the catalog. The identity at the head of this chapter does load-bearing work three layers deep.

The flow of the chapter: chain-rule identity (7.1) → AND/OR scope restriction (7.2) → formal DAG with Correlation Hierarchy (7.3) → orthogonal diagnostic split (7.4, 7.5). The chapter leaves us with a formal strategy representation, an honest treatment of causal insufficiency, and the diagnostic signals that tell whether the problem is the goal, the strategy, the model, or some combination.

- This is a chapter-introduction segment; it bridges Chapter 2's planning-decision arc to Chapter 3's formal strategy structure and diagnostic split. It carries no formal claim of its own.
- The framing "Chapter 2 says whether to plan; Chapter 3 says what a plan IS" is the conceptual core. The decision to plan precedes the formal definition of what a plan is — this is the sharpest joint in Part II per the OUTLINE walkthrough.
- The "what is and isn't derived in the formal commitment" framing addresses what could otherwise read as overclaim. Acyclicity and Markov are theorem-backed; AND/OR is a chosen parameterization. Surfacing the distinction is important because the strategy-DAG segment is one of the places where the *derived vs. chosen* distinction matters most.
- The "two diagnostic quantities, not one" framing surfaces the orthogonality argument that makes Section II's headline contribution legible. Earlier single-signal framings could not distinguish the two situations and therefore could not prescribe different remedies.
- The chain-confidence-decay paragraph names what the identity unlocks downstream. The identity itself looks elementary; the three load-bearing layers it anchors are not.

7.1 *Derived:* Chain Confidence Decay

Confidence in a multi-step strategy decays monotonically with depth. The rate depends on the conditional dependence structure, but the qualitative result — longer chains are less confident than shorter ones — is robust.

Formal Expression.

For a chain of n uncertain steps with conditional success probabilities:

Derived (chain-confidence-decay, mathematical identity).

$$\log P(\text{chain}) = \sum_{i=1}^{n} \log P(E_i \mid E_{<i}) \tag{7.4}$$

Since each $\log P(E_i \mid E_{<i}) \leq 0$, chain confidence decays monotonically with depth.

The independent case (p^n) is the simplest special case, not the general result. When steps are conditionally dependent — success at step k makes step $k+1$ more likely — the decay is slower. When steps have negative dependence (success at k makes $k+1$ harder — resource depletion, adversary adaptation), decay is faster.

Quantitative illustration (independent, uniform p):

Table 7.1

Depth	$p = 0.9$	$p = 0.8$
1	0.90	0.80
3	0.73	0.51
5	0.59	0.33
10	0.35	0.11
20	0.12	0.01

Epistemic Status.

Exact. The additive decomposition of log-confidence is a mathematical identity (chain rule of probability). The qualitative consequence (monotonic decay) follows from the non-positivity of log-probabilities. No assumptions beyond the probability axioms.

Discussion.

Structural pressure on strategies. Chain confidence decay creates systematic pressure toward: - **Short plans**: fewer steps means higher aggregate confidence - **Parallel fallback paths**: OR-branches provide alternative routes when one chain fails - **High-confidence critical links**: invest in the reliability of steps that appear in every path - **Early monitoring**: detect chain failure early rather than discovering it at the end

These are not prescriptions but consequences — an agent that ignores chain decay will experience more strategy failures, lower effective tempo, and (if the failures are costly) faster reserve depletion.

AND-nodes amplify decay. When multiple parent chains must all succeed (conjunctive combination), their confidences multiply. A node requiring k parents each at depth d with per-edge confidence p has aggregate confidence $p^{k \cdot d}$, not p^d.

Deep conjunctive strategies are exponentially more fragile than deep disjunctive ones. This asymmetry is formalized in the combination rules (7.3).

Connection to the persistence condition. Chain decay makes long-horizon strategies inherently fragile, which increases the effective disturbance rate ρ_Σ against strategy persistence. An agent pursuing a 20-step plan in a changing environment faces compound uncertainty from both chain decay (internal fragility) and environmental change (external disturbance). The interaction between these — how environmental change compounds through uncertain chains — is not yet formalized.

Triple depth penalty. Chain depth creates three independent penalties. This segment identifies the first: **confidence decay** — deeper chains have lower aggregate confidence because $\log P(\text{chain})$ accumulates negative terms. The two-edge strategic dynamics analysis (N) identifies the second: **evidence starvation** — downstream edge k in a chain is tested only when all upstream edges succeed, so its effective correction rate is attenuated by $\prod_{j<k} \theta_j$. 8.9 identifies the third: **cognitive cost** — deeper chains have higher description length, consuming more representational capacity. The three penalties compound independently: a deep edge has low confidence (decay), receives few observations (starvation), and costs more to maintain (complexity). The maximum useful chain depth d^* is the minimum over three independent constraints — see 8.9 for the formal bound.

Anchor role in the coordinate-forcing meta-pattern. The log-of-product decomposition here anchors three further AAT uniqueness theorems that force coordinates at other layers: reverse-KL at the divergence level (O §6.1), log-odds at the update level (P), and Fisher metric at the metric level (4.2 "Fisher-metric cases under parameterization-invariance"). The first two theorems cite this chain-layer identity as the analog motivating their additivity axiom; the Fisher-metric theorem rests on a parameterization-invariance axiom motivated by `#scope-agent-identity`'s singular-trajectory scope, an adjacent-AAT-commitment rather than a direct chain-analog — the theorem clears the broader discipline (uniqueness-theorem-forced coordinate under AAT-internal axiom) without reducing to a log-additive form. The catalog and the precise anchor-plus-three-theorem characterization live in Z.

Section III corollaries (additional reach of the chain-layer identity). The chain-rule identity has unsurfaced consequences at composition-related layers that are corollaries rather than new theorems: - *Composition tower telescoping.* For a chain of nested sub-agents $(A_1, A_2, \ldots, A_\ell)$ with sub-agent-k contraction factor $\varkappa_k$, the tower contraction factor $\prod_\ell \varkappa_\ell$ becomes log-additive: $\sum_\ell \log \varkappa_\ell$. The closure-defect-along-tower quantity $\sum_\ell \log(\nu_\ell/\alpha_\ell)$ inherits the chain-rule identity's additivity. - *Fisher information for multi-sample likelihoods.* $\log P(\mathbf{y}; \theta) = \sum_i \log P(y_i; \theta)$ is the chain-rule identity applied to multi-sample independent observations, producing the additive-Fisher-information decomposition standard in statistics. - *Communication-tree aggregation.* Shared-intent compression across tree-structured agent communication channels inherits the chain-rule identity along the tree's branches, giving log-additive coordination-bit cost.

These are not independent uniqueness theorems — they are *corollaries* of the chain-layer identity applied to specific structural settings. Composition of structured multiplicative quantities inherits the log-additive decomposition whenever the chain-rule factorization applies. Cataloguing is in Z's Working Notes; none rises to primary-instance status because none introduces a new AAT-internal axiom (they reuse the probability chain rule as their identity, not as motivation for a fresh axiom). A distinct meta-pattern for composition specifically — composition-monotonicity rather than chain-rule — may be warranted; see `#form-composition-closure`'s bridge-lemma / Tier 1/2/3 / (CM4) family as the candidate structural material.

Working Notes.

- The independent-edge assumption (used in the quantitative table) is optimistic for positively correlated failures (shared infrastructure → correlated failures make the actual confidence *lower* than independent calculation suggests). Correlation structure is unmodeled — acknowledged as a limitation.
- The additive log-confidence form is the robust result; p^n is the special case for independent uniform edges. This distinction matters: the qualitative consequence (decay with depth) is robust; the specific rate depends on the conditional structure.

7.2 *Scope:* AND/OR Combination Scope

We restrict to environments where the causal combination of strategy steps is approximately conjunctive (AND: all parents required) or disjunctive (OR: any parent sufficient), without strong interaction effects between parents.

Formal Expression.

Under this restriction, strategy nodes combine parent contributions via:

Scope Narrowing (and-or-scope).

AND-node (all parents must succeed):

$$P(v \mid \text{parents}) = \prod_{i \in \text{pa}(v)} p_{iv} \cdot P(i) \tag{7.5}$$

OR-node (any parent sufficient):

$$P(v \mid \text{parents}) = 1 - \prod_{i \in \text{pa}(v)} (1 - p_{iv} \cdot P(i)) \tag{7.6}$$

The combination type $\gamma(v) \in \{\text{AND}, \text{OR}\}$ is assigned per node. The causal question determines assignment: "if I remove one parent, can v still be achieved?" YES → OR. NO → AND.

Epistemic Status.

Robust qualitative. This scope narrowing converged independently across three formalism attempts (track-a/00, track-a/02, track-a/03). It captures the dominant structure in most planning domains. The excluded case — complementarity, substitutability, interaction effects between parents — requires richer combination rules and is a legitimate divergence point for future work.

The AND/OR restriction with single-parameter edges gives k parameters per node (one per parent edge) instead of 2^k for a general conditional probability table. This parsimony is motivated by bounded cognition (2.2): agents with limited representational capacity are forced toward low-parameter models.

Discussion.

Why AND/OR and not alternatives.

Why not Noisy-OR universally. The first formalism attempt used noisy-OR for all nodes. This **systematically overestimates conjunctive structures**:

Table 7.2

Structure	*Noisy-OR*	*AND*	*Reality*
3 required KRs at $p = 0.95, 0.90, 0.99$	0.99995	0.846	~AND

The noisy-OR model cannot represent "all of these are required." This was the primary motivation for the AND/OR revision.

Why not WEIGHTED combination. A clean-slate formalism (track-a/02) introduced $P(v) = \min(1, \sum \alpha_{iv} \cdot p_{iv} \cdot P(i))$ to handle k-of-n thresholds. This reintroduces a two-parameter estimation problem (α weights per edge). If k-of-n semantics are genuinely needed, nested AND/OR structure can represent them: group alternatives into OR-nodes, then AND the groups. This keeps estimation localized to the node taxonomy rather than spreading it across a new per-edge parameter.

The parsimony argument. For binary-outcome nodes, AND and OR form a complete Boolean basis — any Boolean combination can be decomposed into layers of AND/OR (disjunctive/conjunctive normal form). Under bounded cognition, the agent needs the most expressive $O(k)$-parameter representation. AND/OR is the natural candidate. This is a parsimony-motivated hypothesis, not a derived necessity — see M for the full argument and its limitations.

What this scope excludes. Environments with strong interaction effects: where the value of combining parent contributions is not separable into independent per-parent terms. Examples: synergistic drug interactions (combined effect exceeds sum of individual effects), complementary goods (neither is useful alone), strategic surprise (the combination of actions matters more than any individual action). These require richer parameterizations within the strongly motivated graphical structure (7.3) — a direction for future work.

Working Notes.

- The AND/OR assignment per node ($\gamma(v)$) is itself uncertain and should be updateable. A node assumed to be OR (alternatives) might turn out to be AND (all required) when the agent discovers unexpected dependencies. γ reclassification is rare — it requires strong structural evidence — and operates on a slower timescale than edge-weight updates.
- The parsimony argument applies cleanly to binary outcomes. For continuous or multi-valued outcomes, AND/OR doesn't have the same completeness properties. The natural continuous analogs might be min (AND) and max (OR), or additive and multiplicative combination. Whether there's a completeness result for these is an open question.
- K-of-n thresholds are genuinely common (e.g., "need at least 3 of 5 team members available"). The nested AND/OR representation works but can be verbose. Whether this verbosity is a problem in practice (given bounded cognition constraints) is empirical.

7.3 *Definition:* Strategy DAG

The strategy Σ_t is a directed acyclic graph with probabilistic edges and AND/OR combination semantics. Each edge carries the agent's causal credence that completing the parent step advances the child step. The graph encodes the agent's theory of how its actions produce progress toward its objectives.

Why a DAG. The DAG structure is not a modeling convenience but a *consequence* of operational requirements on any causally-reasoning bounded agent — at the level of sufficiency, not yet necessity. M proves that directed temporal order plus probabilistic uncertainty plus causal sufficiency *suffice* for a DAG-with-Markov-factorization representation (the necessity direction — no non-DAG structure could satisfy these postulates — is an open stronger result). Acyclicity is proved from temporal ordering over a finite horizon. What remains a formulation choice is the *parameterization within* the DAG structure: AND/OR combination with single-parameter edges is the AAT choice, motivated by parsimony and convergence across three independent formalism attempts, but alternative parameterizations (within the derived graphical structure) are legitimate research directions.

Strategy-layer exactness contract. All formal results in AAT's strategy layer — the sector condition transfer (N, Prop B.5), the persistence schema (8.10), the gradient-based credit assignment (8.6) — are proved under **L0 (independence)**: causally sufficient DAGs with independent edge outcomes. **L0 formal results transfer exactly to correctly constructed L1 DAGs (strict prerequisites, Prop B.6) and L1' DAGs (soft facilitators, Prop B.7) — provided the common cause is observable per trial.** When the common cause is unobservable, the per-conditional decomposition is *fundamentally* (not merely "openly") obstructed — the mixture parameters are non-identifiable from a single observation channel

(Fisher rank deficiency / Cramér-Rao floor; see B.7 §"Refuted Under Unobservable C"), and the agent must either collect direct C-observations, run multi-child joint observations (Prop B.7 §"Repair routes"), or fall back to plan-level (L0-on-marginals) tracking. See the Correlation Hierarchy below for the full treatment.

Formal Expression.

Definition (strategy-dag).

$$\Sigma_t = (V_t, E_t, p_t, \gamma_t) \tag{7.7}$$

where: - V_t: set of **propositional nodes** — each node represents a condition that could be true or false (including action-success propositions at the leaves) - $E_t \subseteq V_t \times V_t$: directed causal edges - $p_t : E_t \to [0, 1]$: **causal credence** per edge — the agent's confidence that completing the parent advances the child - $\gamma_t : V_t \to \{\text{AND}, \text{OR}\}$: combination rule per node (7.2)

Structural constraints:

1. **Acyclicity.** Σ_t is a DAG. This is *derived*, not assumed — see below.
2. **Rootedness.** Every node has a directed path to a unique root terminal v_{root} — the single sink node (out-degree 0) of Σ_t. Compound objectives express their combination structure through the AND/OR machinery below v_{root}, consistent with scalar V_{O_t} (5.4).
3. **Source constraint.** Leaf nodes are propositions about action success ("action a succeeds at τ_v") or observable conditions ("condition C_v holds at τ_v"). Both are propositional — the distinction is whether the proposition is within the agent's causal control (action) or not (condition).

Leaf base credence. For each leaf node $v \in V_t^{\text{leaf}}$, the base credence used in status propagation:

$$p_v(M_t) = \begin{cases} \Pr(\text{action } v \text{ succeeds at } \tau_v \mid M_t) & \text{if } v \text{ is an action node} \\ \Pr(C_v(\tau_v) \mid M_t) & \text{if } v \text{ is a condition node} \end{cases} \tag{7.8}$$

where C_v is the propositional condition associated with node v and τ_v is the node's temporal position (from the acyclicity structure). For action leaves, p_v is *capability credence* — "can I execute this?" For condition leaves, p_v is *state credence* — "will this hold?" Both are conditional on M_t and update whenever M_t updates. This is the mechanism by which Section I's adaptive machinery enters the strategy: M_t changes → leaf credences change → status propagation produces new terminal credences.

Edge semantics. Each edge carries a single credence weight:

$$p_{ij} = \text{Cr}(j \text{ advances} \mid i \text{ completed}, M_t) \tag{7.9}$$

This is the agent's credence that completing step i advances step j, given its current model — its **causal efficacy estimate** for the link. The agent treats p_{ij} as a causal quantity for planning purposes (status propagation, strategy-plan-confidence scoring, action selection). Whether p_{ij} is a *good* estimate of causal efficacy depends on the identification regime of the data that produced it (8.5):

- **Regime A** (intervention-rich: software, laboratory science). The agent's execution-observation pairs are genuine interventions with clean attribution. p_{ij} approximates the interventional probability $P(j \mid do(i), M_t)$.

- **Regime B** (partial intervention: organizational, coordinated action). The agent acts but attribution is blurred by concurrent actions and self-selection. p_{ij} is a partially identified causal estimate, typically biased upward.
- **Regime C** (observation-only: passive monitoring, intelligence analysis). The agent observes associations but does not intervene. p_{ij} is an observational proxy for the causal quantity — useful for planning but potentially confounded.

The identifiability coefficient ι_{ij} (8.5) quantifies the strength of the causal interpretation for each edge. When $\iota_{ij} \approx 1$, the agent's credence is well-identified causally. When $\iota_{ij} \approx 0$, the credence is associational. The single-parameter edge design is preserved: p_{ij} is always the agent's working estimate, with ι_{ij} characterizing its causal warrant separately.

Status propagation. Forward pass in topological order, $O(|V| + |E|)$:

$$s_v = \begin{cases} p_v & \text{if } v \text{ is a leaf (base credence)} \\ \prod_{i \in \mathrm{pa}(v)} p_{iv} \cdot s_i & \text{if } \gamma(v) = \text{AND} \\ 1 - \prod_{i \in \mathrm{pa}(v)} (1 - p_{iv} \cdot s_i) & \text{if } \gamma(v) = \text{OR} \end{cases} \tag{7.10}$$

Terminal satisfaction conditions. The root terminal v_{root} and any intermediate nodes near the top of the DAG carry **satisfaction conditions**: predicates on environment states/trajectories that the agent treats as operational success criteria for the objective. These conditions operationalize O_t within Σ_t — they are the agent's theory of what it means to satisfy the objective. O_t itself lives outside Σ_t (5.7); the terminal conditions are Σ_t's internal encoding of what O_t requires. When O_t changes, terminal conditions must be reassessed and potentially replaced (8.7).

Well-formedness. Σ_t is **O_t-well-formed** when the agent believes that achieving the terminal conditions yields a trajectory that satisfies the objective:

$$\Pr(O_t \text{ satisfied by } \tau \mid \text{terminal conditions achieved, } M_t) \geq 1 - \epsilon \tag{7.11}$$

where "O_t satisfied" means $V_{O_t}(\tau)$ exceeds the objective's own satisfaction criterion (formalized as $V_{O_t}^{\min}$ in 7.4). This is a constraint on the relationship between Σ_t and O_t, not a separate state object. It is explicit and in-principle assessable, though evaluating it requires the same value-side machinery as A_O — it is not a cheap structural test. Violation triggers terminal reassessment: either the terminals need revision (they don't operationalize O_t correctly) or O_t itself needs revision.

Strategy self-assessment. The root node's propagated status:

$$\hat{P}_\Sigma(M_t) = s_{v_{\text{root}}} \tag{7.12}$$

is the strategy's **strategy-plan-confidence score** — the DAG's own answer to "will this plan work?" This score is a correct probability only when the DAG is **causally sufficient** — when all common causes of strategy nodes are represented as nodes in the graph (M, Step 3). When the DAG is causally insufficient (the dominant real-world case — shared infrastructure, common-mode risks, correlated adversary actions introduce latent common causes), $\hat{P}_\Sigma$ systematically overestimates success likelihood because the AND/OR propagation treats joint failure probability as the product of marginals. See the **Correlation Hierarchy** below for the full treatment.

$\hat{P}_\Sigma$ is explicitly distinct from A_O (7.4), which optimizes over the entire policy class, and from $V_O(\pi_{\text{current}})$ (5.6), which evaluates the current policy. $\hat{P}_\Sigma$ is cheap to compute ($O(|V| + |E|)$ forward pass) and updates in real time as M_t changes through leaf credences.

Correlation Hierarchy.

The AND/OR status propagation computes correct probabilities **if and only if** the strategy DAG is causally sufficient — that is, all common causes of any two strategy nodes are themselves represented as nodes. By the Causal Markov Condition theorem (M, Step 3), causal sufficiency guarantees independent exogenous noise, which guarantees the Markov factorization, which guarantees independent edge outcomes. **When causal sufficiency fails — the dominant case in complex, multi-stakeholder, or adversarial environments — edge outcomes are correlated and the independence model is biased.**

Discussion (correlation-hierarchy).

Direction and magnitude of bias. For two binary sibling nodes X_1, X_2 with marginal success probabilities θ_1, θ_2 and covariance $\rho = \mathrm{Cov}(X_1, X_2) > 0$ (positive correlation from a latent common cause):

Table 7.3

Node type	*True probability*	*Independence estimate*	*Bias*
AND	$\theta_1\theta_2 + \rho$	$\theta_1\theta_2$	$+\rho$ (underestimates — conservative)
OR	$[1-(1-\theta_1)(1-\theta_2)] - \rho$	$1-(1-\theta_1)(1-\theta_2)$	$-\rho$ (overestimates — optimistic)

The independence model error is exactly $\pm\mathrm{Cov}(X_1, X_2)$, with sign determined by node type. AND-nodes are conservative because clustering of successes helps joint success. OR-nodes are optimistic because clustering of failures undermines the redundancy that OR-structure is supposed to provide. Same magnitude, opposite signs.

For strategies with OR-structure near the root (the typical case — multiple alternative paths to the objective), the net bias is overestimation: the agent believes it has more redundancy than it actually does. For AND-heavy strategies (all components must work, no alternatives), the net bias is underestimation. Mixed strategies depend on the graph topology, but OR-dominated roots are far more common than AND-dominated roots in practice.

Four regimes, in order of increasing realism:

Table 7.4

Level	*Model*	*When correct*	$\hat{P}_\Sigma$ *status*	*Sector transfer status*	*Computation*
L0: Independence	AND/OR propagation as-is	Causally sufficient DAG (all common causes explicit)	Correct probability	Prop B.5 (linear), B.5b (componentwise nonlinear): $\alpha_s = \alpha_c$	$O(\lvert V\rvert + \lvert E\rvert)$
L1: Augmented DAG (strict prerequisites)	Strict-prerequisite common-cause nodes added explicitly; AND/OR propagation on augmented graph	Augmented DAG is causally sufficient *and* every modeled common cause has $\theta_{\text{child}\mid\neg C} \approx 0$	Correct for augmented graph	Prop B.6: $\alpha_\Sigma = \min(1/(n_C+1), \theta_C\pi_j/(n_j+1))$ — three-way gating	$O(\lvert V'\rvert + \lvert E'\rvert)$, larger graph

Level	*Model*	*When correct*	*$\hat{P}_\Sigma$ status*	*Sector transfer status*	*Computation*
L1': Mixture form (soft facilitators)	Conditional sub-DAGs weighted by common-cause state: $\hat{P}_\Sigma = \theta_C P_\Sigma(G \mid C) + (1-\theta_C) P_\Sigma(G \mid \neg C)$	Soft-facilitator common causes ($\theta_{\text{child}\mid\neg C} > 0$) **with C observable per trial**	Correct for the weighted combination	Prop B.7: $\alpha_{L1'} = \min(1/(n_C+1), \theta_C \pi_{j\mid C}/(n_{j\mid C}+1), (1-\theta_C)\pi_{j\mid\neg C}/(n_{j\mid\neg C}+1))$ — five-way gating. **Refuted when C unobservable** (Fisher rank-1; falls back to plan-level tracking or multi-child joint observation)	$O(\lvert V'\rvert + \lvert E'\rvert)$ per common cause; doubles parametric footprint for affected edges
L2: Full correlation	Arbitrary joint failure distribution over edges	Always (but requires specifying the full joint)	Correct	Reduces to L0 on the augmented joint state	Exponential in general

L0 (Independence) is the tractable baseline. All formal results in AAT's strategy layer — the sector condition transfer (Prop B.5 in N), the persistence schema (8.10), the gradient-based credit assignment (8.6) — are proved under L0. The strategy-plan-confidence error $\delta_s = \hat{P}_\Sigma - \Phi$ tracks calibration *within* the independence model; $\Phi = P_\Sigma(\theta)$ is the AND/OR formula at true edge rates, not actual plan success probability.

L1 (Augmented DAG) is the practical sweet spot — *for strict-prerequisite common causes*. The agent models correlation structure explicitly by adding common-cause nodes to Σ_t and restructuring the DAG so that the common cause is **factored above the correlation it creates**. The construction principle: place the common-cause node as an AND-prerequisite *above* the OR/AND structure whose children it correlates. This ensures that, conditional on the common cause being satisfied, the children are independent and standard AND/OR propagation is correct.

Scope of the AND-prerequisite construction: strict prerequisites only. The factoring-above principle requires that the common cause be a *strict* prerequisite — one for which $\theta_{\text{child}\mid\neg C} \approx 0$. Shared infrastructure going down, a required resource being absent, a gating permission being denied: in all of these, the correlated children fail when the common cause fails, so the AND-prerequisite correctly encodes the correlation. When the common cause is instead a *soft facilitator* — favorable market conditions, a supportive team culture, an enabling technology — children have $\theta_{\text{child}\mid\neg C} > 0$ and can succeed when the common cause is absent, just less reliably. The AND-prerequisite construction mathematically forces $P(\text{sub-plan} \mid \neg C) = 0$, which strictly understates success probability when C is absent. Soft facilitators therefore fall outside L1's single-pass construction and require one of:

- **Mixture form (L1'):** split the sub-plan into two conditional structures and weight by $P(C)$: $\hat{P}_\Sigma = \theta_C \cdot P_\Sigma(G \mid C) + (1-\theta_C) \cdot P_\Sigma(G \mid \neg C)$ This keeps propagation polynomial but requires maintaining two parallel sub-DAGs per soft-facilitator node. Per-edge credences split into two regimes ($p_{ij\mid C}$ and $p_{ij\mid\neg C}$), doubling the parametric footprint for affected edges.
- **Explicit conditioning (L2 subcase):** $P_\Sigma(G) = \sum_c P(C=c) \cdot P_\Sigma(G \mid C=c)$, summing over common-cause states. For k soft facilitators with binary states, this costs $O(2^k)$ in the number of common-cause states — the exponential blowup L2 was defined to name.

The gap between L1 (strict) and L2 (arbitrary) was not previously named. L1' (mixture form) fills this gap at linear cost per soft-facilitator node. The "L1 is the practical default" claim therefore applies most cleanly to strict prerequisites; for mixed environments (strict and soft common causes simultaneously), the correct construction mixes L1 factoring for strict prerequisites with L1' mixtures for soft facilitators. Treating all common causes as strict prerequisites systematically undervalues sub-plans that face soft facilitators (the opposite failure mode from L0's overestimation).

Example — strict prerequisite (see #example-L1 for full treatment): Two OR-alternatives sharing infrastructure dependency ($\theta_C = 0.8, \theta_{1\mid C} = 0.9, \theta_{2\mid C} = 0.7$, and implicitly $\theta_{1\mid\neg C} = \theta_{2\mid\neg C} = 0$ — infrastructure is a strict prerequisite).

L0 computes $\hat{P}_\Sigma = 0.877$; actual is 0.776 (overestimation = ρ, the covariance from shared infrastructure). Naive L1 (common cause as parent of both alternatives, alternatives remain OR-siblings) gives the *same* overestimate because the OR-propagation still treats siblings as marginally independent. Correct L1 ($G = \mathrm{AND}(C, G_{\text{sub}})$ where $G_{\text{sub}} = \mathrm{OR}(A_1, A_2)$) gives the exact answer because A_1 and A_2 are conditionally independent given C *and because* $\theta_{i|\neg C} = 0$*, so the AND-prerequisite is the correct encoding*. The sector condition is verified (Prop B.6 in N) with $\alpha_\Sigma = \min(1/(n_C+1),\ \theta_C(1-\varepsilon)/(n_{A_1}+1),\ \theta_C\varepsilon/(n_{A_2}+1))$ — combining evidence-starvation and exploration-gating effects. B.5b transfers losslessly.

All L0 formal results transfer to correctly constructed L1 DAGs *for strict-prerequisite common causes* because the augmented DAG is a standard AND/OR DAG that satisfies causal sufficiency. For L1' (mixture form) with **observable common cause**, the formal results transfer through Prop B.7 (N) — five-way gating with $\alpha_{L1'} = \min(1/(n_C + 1), \theta_C\pi_{j|C}/(n_{j|C}+1), (1-\theta_C)\pi_{j|\neg C}/(n_{j|\neg C}+1))$. For L1' with **unobservable common cause**, the per-conditional decomposition is identifiability-obstructed by the Cramér-Rao floor — the per-trial Fisher matrix is rank 1 rather than rank $2K+1$, so no unbiased online estimator on the joint conditional vector admits a sector parameter $\alpha > 0$. This is not "open" but structurally refuted; the agent must augment C-observability, run multi-child joint observations, or fall back to L0-on-marginals (losing per-conditional diagnostics). The practical challenges are: (1) identifying which common causes matter enough to model explicitly, (2) classifying each identified common cause as strict or soft, (3) verifying C-observability for soft cases, (4) restructuring the DAG correctly for each. All four are modeling judgments, not mechanical procedures.

L2 (Full correlation) is a mathematical ideal. Specifying the full joint failure distribution over m edges requires $O(2^m)$ parameters, which violates the bounded-cognition constraint that motivates the DAG representation in the first place. L2 is useful as a reference point for characterizing the L0/L1 approximation error, not as a practical representation.

Choosing among L1, L1', and L2 requires classifying each common cause and verifying observability. $\theta_{\text{child}|\neg C} \approx 0$ → L1 (factor above the correlation, B.6). $\theta_{\text{child}|\neg C} > 0$ with C **observable** → L1' (B.7, five-way gating). $\theta_{\text{child}|\neg C} > 0$ with C unobservable → L1' is identifiability-obstructed by the Cramér-Rao floor; either augment C-observability (preferred), use L2 explicit conditioning, or fall back to L0-on-marginals (losing per-conditional diagnostics). Mixed-classification environments (some strict, some soft, varying observability) combine L1 factoring for the strict and L1' mixtures for the soft observable. Treating all common causes as strict prerequisites under L1 alone systematically *understates* success on soft-facilitator branches (the symmetric failure mode to L0's overestimation); treating L0 as the default leads to systematic overconfidence — the agent believes it has more redundancy than it actually does (OR-nodes) or more fragility than it actually does (AND-nodes). L0 remains appropriate for domains with genuinely independent risks (independent parallel experiments, diversified portfolios with low correlation) and as a computational stepping stone during strategy construction.

Detecting latent common causes. An agent at L0 can detect causal insufficiency *under joint sibling observability generated by its own interventions* — not from on-policy traces alone. On-policy execution alone cannot distinguish the with-and-without-latent-common-cause worlds (the no-go in 8.2); the unique broadly-available escape is joint sibling observability under exploration, where the agent's interventions (6.3) generate the joint outcome data the test requires. Persistent overestimation of plan success after edge credences have converged is the *signal* of insufficiency (4.5 applied to the strategy layer); pairwise covariance among observed siblings is the *test* (positive covariance rejects independence and identifies where to add L1 nodes). Full treatment: 8.2; numerical instantiation: #example-L1. Summary references in dependent segments that elide the on-policy / interventional distinction should be treated with caution — the load-bearing scope is "joint sibling observability under exploration," not "from observational data alone."

Scope of the terminal construction. Terminal conditions as Boolean predicates with AND/OR aggregation work naturally for threshold, constraint, and composite objectives. For continuous-valued objectives without natural thresholds, the agent must set an operational threshold — introducing a discretization that is a practical proxy, not a lossless encoding of V_O. The primary $O_t \leftrightarrow$ theory interface remains V_O through the value object (5.6); terminal conditions are Σ_t's internal operational encoding.

Single-parameter edges. Each edge carries one number (p_{ij}), not two. An earlier formalism attempt used (p_{ij}, θ_{ij}) where θ was "contribution magnitude." This was dropped because the AND/OR combination rules at nodes absorb θ's role — the complexity budget goes to one bit per node (γ) instead of one float per edge.

Edge-credence presentation coordinates. Each edge's single-number credence has two equivalent presentations: the *probability-space* coordinate $p_{ij} \in [0, 1]$ (convenient for AND/OR propagation and interpretation) and the *log-odds* coordinate $\lambda_{ij} = \log(p_{ij}/(1 - p_{ij})) \in \mathbb{R}$ (the unique additive-evidence coordinate forced by the evidential-additivity axiom; see P). The two coordinates are related by the sigmoid $p_{ij} = \sigma(\lambda_{ij})$. AAT's AND/OR status propagation and Correlation Hierarchy algebra operate in probability space; the continuous-gradient update machinery in 8.6 operates natively in log-odds and projects back to $[0, 1]$ via sigmoid at the readout interface. Props B.1–B.7 of N are stated in moment-parameter (probability-space) form; their sector-parameter content is Fisher-equivalent in either coordinate.

Acyclicity is Derived.

Each node in Σ_t represents a future event or state with temporal position $\tau_i > t$. An edge $X_i \to X_j$ requires $\tau_i < \tau_j$ (1.8: causes precede effects). A cycle $X_i \to X_j \to \cdots \to X_i$ would require $\tau_i < \tau_j < \cdots < \tau_i$, which is impossible for a real-valued time index.

Derived (from causal-structure + finite planning horizon).

Strategies involving iteration ("try A, if fail try B, if fail try A again") are acyclic when time-indexed. The sequence unfolds as:

$$A_1 \to \text{check}_1 \to B_1 \to \text{check}_2 \to A_2 \to \dots \tag{7.13}$$

Each attempt is a distinct node at a distinct time. The apparent cycle is a linear chain in the unrolled view.

Formally: a finite set with a strict partial order (future events ordered by time) is representable as a DAG. This is a standard result in order theory.

Scope of the acyclicity result. This applies to Σ_t (the agent's strategy over the future), not to M_t's model of the environment, which may include cyclic causal processes (feedback loops in the physical world, market dynamics, ecosystem interactions). The acyclicity is specific to the purposeful substate.

Epistemic Status.

Conditional on the 7.2 restriction. The DAG structure itself follows from operational postulates: temporal ordering (acyclicity — exact), probabilistic uncertainty (Cox's theorem — exact; Cox 1946, "Probability, Frequency and Reasonable Expectation," *American Journal of Physics* 14(1):1–13; Jaynes 2003, *Probability Theory: The Logic of Science*, Cambridge University Press, Chapter 2), and the Causal Markov Condition theorem under causal sufficiency (Markov factorization — proved conditional). The full argument is in M: P1 + P2 + causal sufficiency → CMC → DAG with Markov property. The result is *conditional on causal sufficiency* of the strategy (no latent common causes among strategy nodes). Causal sufficiency is a modeling ideal — the agent designed the graph, so *intended* causal relationships are explicit, but environmental common causes routinely go unmodeled in complex domains. When causal sufficiency fails, the Markov factorization breaks down, edge outcomes become correlated, and $\hat{P}_\Sigma$ overestimates success. This is model inadequacy (4.5), repairable by adding the missing common-cause nodes (L1 augmentation). See the Correlation Hierarchy above for the practical framework.

Causal sufficiency and edge independence. The AND/OR status propagation is correct when the DAG is causally sufficient (all common causes represented as nodes) — this follows from the Causal Markov Condition theorem (M). In complex real-world systems, causal sufficiency is systematically violated: shared infrastructure, common-mode risks, supply chain dependencies, and correlated adversary actions introduce latent common causes. Correlated failure is the dominant case, not the exception. The Correlation Hierarchy (above) characterizes three levels: independence (L0, tractable baseline), augmented DAG (L1, practical sweet spot), and full correlation (L2, mathematical ideal). AAT's formal strategy-layer results are proved under L0 but transfer to L1 (which is just a larger L0-compliant DAG). The independence model remains the tractable foundation for formal analysis; L1 augmentation is the recommended practice for deployment.

The AND/OR parameterization is a parsimony-motivated formulation choice within the strongly motivated graphical structure, not a derived necessity (7.2). The single-parameter edge convention is similarly a formulation choice motivated by convergence across three independent attempts.

Discussion.

The graph structure is sufficient under causal sufficiency; the parameterization is chosen. The DAG structure follows from temporal ordering (acyclicity — proved) and the Causal Markov Condition theorem under causal sufficiency (Markov factorization — proved conditional; see M). This is a theorem-backed result, not a sketch: the CMC (Spirtes et al. 2000, Pearl 2009) proves that any causally sufficient causal DAG with independent exogenous noise satisfies the Markov factorization. The status is therefore: DAG-with-Markov-property is *sufficient* given operational postulates plus causal sufficiency — the desiderata are satisfied by this representation, but the necessity direction (whether some non-DAG structure could also satisfy them) is an open stronger result (M). AAT uses AND/OR parameterization within this strongly motivated structure because (a) AND/OR is the most parsimonious complete basis for binary combination, (b) the representation converged across three independent formalism attempts. Alternative parameterizations within the strongly motivated graphical structure are legitimate research directions.

Combination assignment is principled but fallible. The question "if I remove one parent, can v still be achieved?" is derivable from M_t's causal model — it's a principled assignment, not arbitrary. But the assignment can be wrong (false AND = pessimistic over-investment; false OR = optimistic under-investment), and should be updateable when evidence reveals a different structural relationship.

Connection to Pearl's framework. Under causal sufficiency, the strategy DAG is a causal Bayesian network in the sense of structural causal models, not merely a structural analog: M proves that the operational postulates (P1, P2, P4) plus causal sufficiency yield the Markov factorization via the Causal Markov Condition theorem (Spirtes et al. 2000, Pearl 2009). Pearl's do-calculus therefore applies to status propagation and plan evaluation, with its causal content scoped by the identification regime of the data feeding the edge credences (8.5). In Regime A domains, where the agent performs genuine interventions and observes isolated outcomes, the DAG's edge credences approximate interventional probabilities and do-calculus yields clean causal estimates. In Regimes B and C, the edge credences are the agent's working causal beliefs, updated from data of weaker identification strength. The DAG remains useful for planning — the agent must act on *some* causal model — but the strategy-plan-confidence score $\hat{P}_\Sigma$ inherits the identification weaknesses of its constituent edges. An agent operating primarily in Regime C should treat $\hat{P}_\Sigma$ as a rough heuristic, not a calibrated probability. When causal sufficiency itself fails (latent common causes among strategy nodes), the Markov factorization is violated and the DAG is the agent's *intended* but uncalibrated causal model; the L1/L1' augmentation in the Correlation Hierarchy is the principled repair.

Depth penalties on calibration. Beyond the confidence decay that deeper DAGs suffer (7.1), the two-edge strategic dynamics analysis (N (Props B.2-B.3)) shows that deeper edges are also harder to calibrate. Edge k in a chain is tested only when all upstream edges succeed, so its effective correction rate is attenuated by $\prod_{j<k} \theta_j$ (the evidence-starvation effect). Deeper DAGs therefore face a double penalty: lower aggregate confidence AND slower convergence of edge credences toward truth. This reinforces the structural pressure toward shallow, observable strategies — deep plans require both high per-edge reliability and sustained observability at every intermediate level to remain calibratable.

Edge independence is a consequence of causal sufficiency, not a separate assumption. The Correlation Hierarchy (Formal Expression, above) establishes this precisely: the AND/OR propagation's correctness is not a matter of "assuming independence" — it is a consequence of whether the DAG is causally sufficient. The CMC theorem (M) proves that causal sufficiency → exogenous independence → Markov factorization → correct AND/OR propagation. The assumption is causal sufficiency; independence is the consequence.

What L0 buys: Tractable $O(|V| + |E|)$ status propagation; single-parameter edges; clean persistence proofs (the sector condition transfers from per-edge credence to plan value via the Jacobian — Prop B.5 in N).

What L0 costs when the DAG is causally insufficient: $\hat{P}_\Sigma$ systematically overestimates success. The strategy-plan-confidence error $\delta_s = \hat{P}_\Sigma - \Phi$ proved persistent by B.5 tracks calibration *within the L0 model* — Φ is the AND/OR formula at true edge rates, not actual plan success (8.1). Gradient-based credit assignment (8.6) inherits the same bias: the residual $(y_G - \hat{P}_\Sigma)$ conflates per-edge miscalibration with omitted correlation structure.

The L1 remedy: Add common-cause nodes and restructure the DAG so each common cause is factored above the correlation it creates (see Correlation Hierarchy above). The AND/OR propagation then applies correctly to the restructured DAG, and all L0 formal results transfer because the restructured DAG is itself L0-compliant. The key engineering challenge is twofold: identifying which common causes matter, and positioning them correctly in the graph topology.

Relationship to Moore machines / finite-state automata — behavioral surface vs epistemic interior. The strategy DAG and the Moore machine $(S, A, \tilde{A}, \delta, \lambda, s_0)$ formalize different aspects of what "strategy" means; they are not competing representations of the same object but representations of *different* objects that both get called "strategy" in different literatures. The Moore machine encodes a *reactive policy* — given the current state and the other agent's last action, produce an output and transition; it captures a complete input-output mapping over an indefinite horizon. The strategy DAG encodes a *causal plan* — a theory of which conditions must hold (AND/OR) and with what credence for the objective to be achieved; it captures the agent's beliefs about the causal structure of its problem. **The Moore machine is the behavioral surface; the strategy DAG is the epistemic interior.**

Moore machine to DAG (partial embedding). A Moore machine with n states unrolls over a finite horizon H into a tree of depth H with branching factor $|\tilde{A}|$. Each path is a sequence of (state, output) pairs — a scenario. The unrolled tree is a DAG (acyclic by construction, per `#deriv-graph-structure-uniqueness`), but it lacks credence weights and AND/OR semantics. To produce a strategy DAG, two pieces of external information must be supplied: (a) probabilities over opponent actions at each branch (from M_t); (b) AND/OR combination rules reflecting which branches the agent treats as jointly required vs. alternatively sufficient. The Moore machine does not contain this information — it is agnostic about which opponent responses are likely or which paths matter. The Moore-to-DAG map is *injective on skeletons* but requires external information to complete.

DAG to Moore machine (lossy compilation). A strategy DAG can be compiled into a reactive policy by enumerating leaves in topological order and defining a Moore machine state per action leaf that transitions on observed success/failure. This compilation *discards* three pieces of structure: (a) the credence weights (the compiled machine is deterministic); (b) the AND/OR structure (the machine commits to a fixed execution order); (c) the causal semantics (the machine has no representation of *why* actions are ordered this way). The compiled machine executes the plan but cannot reason about it — it cannot recompute plan-confidence when a leaf credence changes, which is exactly what status propagation (`#disc-credit-assignment-boundary`) provides for the strategy DAG.

Neither representation is strictly more general; the two are *orthogonal*. Moore machines can represent reactive strategies with no causal model (tit-for-tat, grim trigger) that have no natural DAG form — there is no plan to encode, just a response rule. Strategy DAGs can represent plans with rich conditional structure and uncertainty that would require exponentially many Moore-machine states (because the Moore machine must enumerate every possible observation sequence, while the DAG factors through conditional independence). The orthogonality is structural: behavioral surface vs epistemic interior. The DAG's acyclicity is not a restriction — it is a consequence of temporal indexing per `#deriv-graph-structure-uniqueness`; what looks like a Moore-machine cycle (revisiting state "cooperate" at $t = 1$ and $t = 5$) is two distinct nodes $v_{t=1}, v_{t=5}$ in the time-unrolled DAG. For finite horizons the two have equivalent expressive power over behavioral sequences, with the Moore machine exponentially more compact for repetitive strategies and the DAG exponentially more compact for plans factoring through conditional independence.

Composition behaviour. Miller's product-automaton construction gives *exact* composition: two Moore machines interacting produce a meta-machine with state space $S_1 \times S_2$ and deterministic transitions. AAT's composition closure (`#form-composition-closure`) uses an *approximate* dynamical homomorphism with closure defect ε^*. These address different questions. The product automaton asks "what behaviour does the pair produce?" — answer: another automaton, exactly. AAT asks "can this pair be described as a single AAT agent?" — answer: approximately, with bounded error. Even if strategies were Moore machines, composition of *behaviour* would become exact (product automaton) but composition of *agent descriptions* (M_t, O_t, Σ_t) would still require the approximate framework — the closure defect comes from projecting the joint internal state, not from composing the policy.

Why AAT uses the DAG rather than the Moore machine. The defining feature of an adaptive agent (`#scope-adaptive-system`) is that the agent revises its strategy when M_t changes. Strategy revision requires the agent to know *why* it is doing something, not just *what*. A Moore machine specifies what to do but contains no theory of why this is the right thing to do — when M_t changes, the Moore machine has no internal handle to revise from. The strategy DAG carries the causal semantics that strategy revision operates on (per `#disc-credit-assignment-boundary` and the gradient-based candidate signal); compiling to a Moore machine would discard the very structure adaptation requires.

Working Notes.

- The Correlation Hierarchy (L0/L1/L2) is now first-class in the Formal Expression. The relationship between causal sufficiency, edge independence, and $\hat{P}_\Sigma$ accuracy is grounded by the CMC theorem. The main open question is practical: what heuristics help agents identify which common causes are worth modeling at L1? This is a domain-specific engineering question, not a theoretical gap.
- The graph-sufficiency argument (P1, P2, P4 + causal sufficiency $\Longrightarrow$ DAG with Markov property via CMC) is proved, not sketched. See M for the full derivation and the parallel to Cox's theorem (Cox is necessary-and-sufficient; this argument is sufficient only). The DAG-with-Markov-property is sufficient given the postulates plus causal sufficiency; the AND/OR parameterization remains a formulation choice. strategy-dag stays typed as Definition because the *parameterization* is chosen, and the necessity of the *graphical structure* itself is an open stronger result.
- Health metrics (groundedness, observability coverage, weighted redundancy, bottleneck scores) are scaffold — engineering quantities for monitoring DAG health, not principled derivations. They may be useful for implementation but should not enter the theory's formal chain.
- **Satisfaction criterion.** $V_{O_t}^{\min}$ is now introduced in 5.4 as a parameter of the objective — the minimum acceptable trajectory value. The well-formedness constraint references it from there, not from 7.4. The satisfaction gap diagnostic builds on $V_{O_t}^{\min}$ but does not define it.
- **Terminal alignment error.** When the agent achieves its terminal conditions but evaluates $V_{O_t}(\tau) < V_{O_t}^{\min}$ on the actual trajectory, the well-formedness belief was wrong — the operational success criteria didn't capture what the objective actually required. This is detectable only through experience (achieve the terminals, evaluate V_{O_t}), not through a priori analysis. It triggers terminal reassessment — a structural change in Σ_t driven by the $O_t \leftrightarrow$ terminal mismatch. Whether this should be formalized as a named diagnostic signal (δ_{align}) alongside δ_{sat}, δ_{regret}, and $\delta_{\text{strategic}}$ is open.

7.4 *Definition:* Satisfaction Gap

The satisfaction gap measures the distance between what the objective requires and what the best available one-step policy improvement can deliver, under the current model and horizon. Under the canonical continuation convention (5.6), this is a *local* diagnostic — it answers "can I improve toward the goal from here?" not "is the goal globally feasible?" A multi-step recoverable objective may show positive δ_{sat} because continuation is frozen at π_{current}. Different continuation conventions yield different gap values; see Epistemic Status.

Formal Expression.

Definition (objective-attainability).

$$A_O(M_t;\, \Pi, N_h) = \sup_{\pi\in\Pi} V_O(M_t, \pi;\, N_h) \tag{7.14}$$

The **objective attainability** — the best achievable value given current beliefs M_t, available policy class Π, and horizon N_h.

Definition (satisfaction-gap).

$$\delta_{\text{sat}} = V_{O_t}^{\min} - A_O(M_t;\, \Pi, N_h) \tag{7.15}$$

where $V_{O_t}^{\min}$ is the minimum trajectory value that counts as "objective met" — a threshold set by the objective itself (for constraints: all satisfied; for utility: a minimum acceptable level).

- $\delta_{\text{sat}} > 0$: The objective is **unmet** under the best available policy, current model, and horizon.
- $\delta_{\text{sat}} \le 0$: The objective is **attainable** in principle.

Disambiguation. $\delta_{\text{sat}} > 0$ does NOT automatically mean the goal is wrong. It means the goal is unmet given (M_t, Π, N_h). The positive signal has multiple possible causes:

Table 7.5

Cause	*Fix*	*How to distinguish*
Goal is genuinely infeasible	Revise O_t	Persists across M_t, Π, N_h improvements
Policy class too narrow	Expand Π (structural adaptation of Σ_t)	δ_{sat} decreases when richer policies are tried
Horizon too short	Extend N_h	δ_{sat} decreases with longer planning horizon
Model is wrong about feasibility	Improve M_t (reduce $\delta_{\text{epistemic}}$)	δ_{sat} changes when M_t is corrected
Objectives jointly infeasible	Revise O_t or relax constraints	Individual terminal satisfaction gaps are zero but AND-node fails; cross-terminal tradeoff is required

Objective revision is the **last resort**, not the first response to unmet goals. The orient cascade (9.1) formalizes this ordering.

Epistemic Status.

Exact as a definition — convention-relative as a diagnostic. The satisfaction gap is a mathematical definition — the difference between a threshold and a supremum over a function class. The definition is precise; the *computation* of A_O is generally intractable (it requires optimization over a policy class), but the quantity is well-defined. However, the *value* of δ_{sat} depends on three external parameters: the continuation convention (π_{cont}), the horizon (N_h), and the scalarization of the objective (V_{O_t}). These are not derived by AAT — they are choices the analyst makes. The satisfaction gap is therefore an intrinsic architectural diagnostic *given* a measurement convention, not an absolute property of the agent.

Convention dependence and the hierarchy. A_O inherits the continuation convention from the value object (5.6), which defines three named conventions forming a monotonicity hierarchy:

- **C1** (one-step improvement, canonical): $\delta_{\text{sat}}^{(1)} = V_{O_t}^{\min} - A_O^{(1)}$. Tests whether the agent can improve toward the goal in one step. Most conservative — a multi-step recoverable objective may show $\delta_{\text{sat}}^{(1)} > 0$ because continuation is frozen at π_{current}.
- **C2** (receding-horizon): $\delta_{\text{sat}}^{\text{RH}} = V_{O_t}^{\min} - A_O^{\text{RH}}$. Tests whether the agent can reach the goal with N_r-step replanning. Captures recovery paths invisible to C1.
- **C3** (Bellman): $\delta_{\text{sat}}^{\text{B}} = V_{O_t}^{\min} - A_O^{\text{B}}$. Tests whether the goal is genuinely infeasible given (M_t, Π, N_h). Strongest diagnostic — $\delta_{\text{sat}}^{\text{B}} > 0$ means no policy in Π can achieve the objective.

The monotonicity result (5.6): $\delta_{\text{sat}}^{\text{B}} \leq \delta_{\text{sat}}^{\text{RH}} \leq \delta_{\text{sat}}^{(1)}$. C1 gives the most false "unattainable" diagnoses; C3 gives none (modulo model error in M_t). Analyses using different conventions should not be compared directly — the convention is part of the measurement.

Discussion.

Why two gap measures, not one. An earlier formulation used a single $\delta_{\text{objective}}$ for goal-related mismatch. This conflates two distinct situations: "the goal is too hard" and "the strategy is too weak." When the agent is optimally pursuing an infeasible goal, $\delta_{\text{objective}}$ is large but there's no strategy to improve — the problem is the goal, not the plan. The satisfaction gap (δ_{sat}) and control regret (7.5) separate these cases, enabling the right corrective action.

The disambiguation table is load-bearing. Without it, an agent facing $\delta_{\text{sat}} > 0$ might immediately revise its objective when the real problem is an inadequate model or a too-narrow policy class. The table encodes the diagnostic procedure: check M_t adequacy first (maybe the goal IS feasible but the model doesn't know it), then check Π and N_h, and only then consider revising O_t.

Dependence on M_t. A_O is computed from M_t, not from the true environment state Ω_t. The agent's assessment of attainability could be wrong — an achievable goal might look unachievable with a bad model, or vice versa. Improving M_t (reducing $\delta_{\text{epistemic}}$) brings the agent's attainability assessment closer to reality. This is why the orient cascade puts epistemic update before attainability evaluation.

Diagnostic content vs. AI's expected-free-energy decomposition. Active inference's expected free energy (EFE) decomposes into *pragmatic value* (how preferred are the outcomes the policy expects?) and *epistemic value* (how much does the policy reduce uncertainty?) (Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017, "Active inference: a process theory," *Neural Computation* 29; Da Costa, Parr, Sajid, Veselic, Neacsu & Friston 2020, "Active inference on discrete state-spaces," *J. Math. Psych.* 99 §2.4; Sajid, Ball, Parr & Friston 2021, "Active inference: demystified and compared," *Neural Computation* 33). The decomposition supports policy ranking but does not separate two distinct diagnoses that AAT's apparatus does separate: "the goal is unattainable from here" ($\delta_{\text{sat}} > 0$, this segment) versus "the current policy is not the best available" ($\delta_{\text{regret}} > 0$ in 7.5). Both increase EFE without distinguishing the cause. The 2×2 cell map in the disambiguation table above gives the four diagnoses the orient cascade (9.1) acts on differently — strategy revision, objective revision, action vs. learning. AI's pragmatic-epistemic split does not produce this disambiguation.

The diagnostic structure depends on V_{O_t} being a *value functional* on trajectories (5.4) and A_O being an *attainability supremum*, not on outcomes encoded as log-priors — AI's preferences-as-priors form ($C(o) = \log P_{\text{pref}}(o)$) collapses the diagnostic by making "wanting o" and "expecting o" formally the same operation (the dark-room critique, Sun & Firestone 2020, "The dark room problem," *Trends Cog. Sci.* 24). Sun & Firestone diagnose the preferences-as-priors collapse; AAT's value-functional reformulation is AAT's own downstream architectural response to that diagnosis, not a move Sun & Firestone themselves propose. The conservative reading of the citation: AAT takes their critique as motivation for abandoning the preferences-as-priors form in favor of value functionals that separate outcome value from outcome prediction, enabling the diagnostic structure the 2×2 cell map requires. This is a deliberate divergence from AI, not an oversight.

Working Notes.

- $V_{O_t}^{\min}$ (the satisfaction threshold) is a property of the objective, not of the agent. For constraint-satisfaction objectives, it's natural (all constraints met = satisfied). For utility-maximizing objectives, it's less obvious — what counts as "good enough"? This threshold may need to be explicitly modeled as part of O_t.
- The supremum in A_O is over Π, the available policy class. For agents with explicit Σ_t, Π corresponds to the set of strategies representable in the agent's DAG formalism. Expanding Π (structural adaptation of Σ_t — adding nodes, edges, or changing γ assignments) is the purposeful analog of 4.5.
- **Cross-reference to NeurIPS Paper 2.** The satisfaction-gap / control-regret two-gap diagnostic is **Component 1** of NeurIPS 2026 Paper 2's composition theorem ("A Unified Convergence Theory for Non-Stationary Reinforcement Learning", `~/src/neurips/02-unified-convergence-rl/`, §4 Theorem 4.1). The diagnostic separates "you're not doing it well enough" (control regret) from "the world doesn't permit it" (satisfaction gap), preventing dark-room collapse in the composed regret bound. The paper's hypothesis (A1) carries the extended-real reading required for the two-gap structure to hold even when the satisfaction frontier is empty. See `msc/neurips-back-integration-2026-05-08.md` §1 Paper 2.

7.5 *Definition:* Control Regret

Control regret measures the gap between the best available one-step policy improvement and the agent's current policy, under the current model and horizon. Under the canonical continuation convention (5.6), this is a *local* diagnostic — it answers "could I do better right now?" not "is my overall strategy globally suboptimal?" A revisable policy may show low δ_regret simply because continuation is frozen. This is the signal for strategy revision, with the caveat that the signal's scope matches the continuation convention's scope.

Formal Expression.

Definition (control-regret).

$$\delta_{\text{regret}} = A_O(M_t;\, \Pi, N_h) - V_O(M_t, \pi_{\text{current}};\, N_h) \geq 0 \tag{7.16}$$

Always non-negative: the current policy cannot outperform the best in its class.

- $\delta_{\text{regret}} \approx 0$: The agent is doing the best it can within current (Π, N_h, M_t). If $\delta_{\text{sat}} > 0$ simultaneously, the problem is not the current strategy — it's either the goal, the capability (Π, N_h), or the model (M_t). See 7.4's disambiguation.
- $\delta_{\text{regret}} \gg 0$: There's room for improvement without changing O_t. → Revise Σ_t.

Epistemic Status.

Exact as a definition — convention-relative as a diagnostic. Like the satisfaction gap, this is a mathematical definition — a difference between two values of the same functional. The quantity is well-defined; computing it requires evaluating A_O (generally intractable) and V_O under the current policy (tractable in simulation, approximate in practice).

Convention hierarchy. δ_{regret} inherits the continuation convention from 5.6. Under the monotonicity result: $\delta^{(1)}_{\text{regret}} \leq \delta^{\text{RH}}_{\text{regret}} \leq \delta^{\text{B}}_{\text{regret}}$. C1 (one-step) reveals only the gap between the current first action and the best one-step deviation — a policy that is "locally near-optimal" under C1 may be globally suboptimal. C3 (Bellman) reveals the full gap to the globally optimal policy. C2 (receding-horizon) interpolates: it captures regret from suboptimal first actions that become visible with N_r-step lookahead. For strategy revision, C2 is often the most useful convention: it reveals recoverable suboptimality without requiring the full Bellman solution.

Discussion.

The diagnostic power of the two-gap system. The satisfaction gap and control regret together encode a 2×2 diagnostic:

Table 7.6

	$\delta_{sat} \leq 0$ *(attainable)*	$\delta_{sat} > 0$ *(unmet)*
$\delta_{\text{regret}} \approx 0$ (near-optimal)	**Success**: goal achievable, policy good	**Capability limit**: optimally pursuing an unmet goal → check M_t, Π, N_h, then consider revising O_t
$\delta_{\text{regret}} \gg 0$ (suboptimal)	**Strategy problem**: goal achievable, policy poor → revise Σ_t	**Both**: goal hard AND strategy weak → revise Σ_t first, then reassess δ_{sat}

This diagnostic is what makes the orient cascade (9.1) actionable: each cell prescribes a different corrective action.

Control regret as the signal for Σ_t revision. When δ_{regret} is high, the agent knows it could do better with a different strategy. The *specific* corrections — which edges to revise, which branches to prune, which alternatives to add — come from the strategic calibration residual (8.1), which localizes the regret to specific parts of Σ_t.

Regret approaching zero when optimally failing. This is the key insight motivating the two-gap split. A single $\delta_{\text{objective}}$ would show "large gap" for both "bad strategy, achievable goal" and "good strategy, impossible goal." The first warrants strategy revision; the second warrants goal revision (after ruling out $M_t/\Pi/N_h$ inadequacy). Without the split, the agent cannot distinguish these cases and may waste effort optimizing a strategy that's already near-optimal for an infeasible goal.

Diagnostic content vs. AI's expected-free-energy decomposition. The 2×2 disambiguation depends on the satisfaction gap / control regret split being orthogonal — distinguishing "goal too hard" from "strategy too weak." Active inference's expected free energy decomposition (pragmatic value + epistemic value; Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017, "Active inference: a process theory," *Neural Computation* 29) supports policy ranking but does not separate these diagnoses — both increase EFE without distinguishing cause. See 7.4 for the full analysis of why the diagnostic structure depends on V_{O_t} being a value functional rather than log-priors over outcomes (Sun & Firestone 2020, "The dark room problem," *Trends Cog. Sci.* 24).

Working Notes.

- **Cross-reference to NeurIPS Paper 2.** Together with `#def-satisfaction-gap`, this segment is **Component 1** of NeurIPS 2026 Paper 2's composition theorem ("A Unified Convergence Theory for Non-Stationary Reinforcement Learning", `~/src/neurips/02-unified-convergence-rl/`, §4 Theorem 4.1). The two-gap orthogonality is what gives the composed regret bound a principled coordinate for "is the binding pressure on goal-feasibility or on policy-quality?" Sibling segment `#def-satisfaction-gap` carries the matching cross-reference. See `msc/neurips-back-integration-2026-05-08.md` §1 Paper 2.

7.6 *Discussion:* Additional Implications & Discussion

The chapter delivered three things that work together at the chapter-end better than they do separately at chapter-middle: a formal strategy representation whose load-bearing structure is theorem-backed under stated conditions and whose parameterization is principled but chosen; an orthogonal diagnostic split that distinguishes goal-attainability from policy-quality and whose meaning is convention-indexed rather than absolute; and an honest scope statement about correlated failure that turns out to be the framework's first explicit instance of *naming an information-theoretic floor and the unique broadly-available escape via AAT machinery*. The chain-confidence-decay identity at the head of the chapter is doing work three layers deeper than its elementary form suggests. None of these are findings on their own that the intro doesn't preview; what the chapter-end can do that the intro can't is read them together, point at the bridges to the rest of the framework, and surface the methodological commitments they encode.

Theorem-backed structure, convergent-choice parameterization.

The chapter is careful about a distinction that downstream segments rely on without restating: what falls out of theorems, and what is a parameterization choice within the theorem-backed structure. Acyclicity of Σ_t is proved — directed temporal order over a finite horizon forces it (1.8 + standard order theory). The Markov factorization is proved under causal sufficiency via the Causal Markov Condition theorem (Spirtes-Glymour-Scheines 2000, Pearl 2009) — `#deriv-graph-structure-uniqueness` does the work. Together they constitute a sufficient condition for the DAG-with-Markov-property representation. The argument parallels Cox's theorem for probability in form (operational desiderata force a measure) but not yet in strength (Cox is necessary-and-sufficient; this is sufficient only — whether some non-DAG structure could also satisfy the postulates is an open stronger result).

What remains a *choice* within the theorem-backed structure is the parameterization: AND/OR combination semantics at nodes, single-parameter credences on edges. These are not derived — but the chapter is also careful to note they are not arbitrary. The catalog draft names a third epistemic category worth surfacing as a methodological finding in its own right: *convergent representational choices*, where every investigated alternative either failed or recovered the same structure. AND/OR converged across three independent formalism attempts; alternatives (general CPTs, noisy-OR everywhere, weighted combinations) were rejected for non-identifiability or parameter explosion. Single-parameter edges replaced an earlier (p_{ij}, θ_{ij}) two-parameter form when the AND/OR combination rules at nodes turned out to absorb θ's role. The P3→Markov route from local revisability — superseded now by the CMC derivation — was a convergent choice before its derivation, and the convergence pattern is what motivated looking for a derivation. The category between *derived from first principles* and *arbitrary framework decision* is itself a research-trail-honesty artifact: a catalog that tells the reader "we didn't just pick this; we tried everything else and it didn't work." Future promotions from convergent-choice to derived (the open parsimony theorem for AND/OR, the open necessity direction in `#deriv-graph-structure-uniqueness`) close the gap one item at a time.

For practitioners, the implication worth carrying is methodological: when AAT's strategy machinery shows up downstream in regret bounds, persistence schemas, or composition arguments, the load-bearing structural commitments are theorem-backed under causal sufficiency, not modeling conveniences. The conditioning on causal sufficiency is the right level of conditionality — it is a property of *strategy quality* (did the agent represent the relevant common causes as nodes?), not of the proof. It is testable in principle (correlated residuals after convergence indicate missing common causes) and repairable in practice (add the common-cause node, restoring causal sufficiency). The Markov-factorization-as-theorem position is what makes the strategy-layer results inherit the strength of Pearl-Bareinboim causal machinery rather than reading as agent-theoretic analogies of it.

The diagnostic split, lifted by the convention hierarchy.

The chapter's orthogonal diagnostic — satisfaction gap ($\delta_{\text{sat}} = V_{O_t}^{\min} - A_O$) and control regret ($\delta_{\text{regret}} = A_O - V_O(\pi_{\text{current}})$) — separates "the world doesn't permit it" from "you're not doing it well enough." The 2×2 cell map (goal × policy) routes four regimes to four corrective actions, and the orient cascade in Part II Ch.5 will use this routing to direct revision pressure to the right substate. This is the implications-side framing of the diagnostic: it is not a measurement but a *routing* device, and getting the routing right is what makes Section II's strategy revision actionable.

The lift the chapter delivers but does not entirely surface is that the diagnostic is *convention-indexed*, not absolute. The value object's three named conventions — C1 (one-step improvement, canonical), C2 (receding-horizon), C3 (Bellman) — form a monotonicity hierarchy on the satisfaction gap: $\delta_{\text{sat}}^{\text{B}} \le \delta_{\text{sat}}^{\text{RH}} \le \delta_{\text{sat}}^{(1)}$. C1 gives the most false "unattainable" diagnoses; C3 gives none modulo model error. The 2×2 cell map is therefore not a single object but a *convention-indexed family*, and a practitioner using satisfaction-gap diagnostics under C1 cannot conclude what a C3 practitioner would.

The monotonicity result tells the practitioner exactly what additional inferential force the higher convention buys for what additional computational cost. Most decision-theory frameworks treat convention choice as a purely computational tradeoff; AAT adds the inferential-force semantics that makes the choice diagnostic-meaningful rather than merely budget-meaningful.

The split also has a positive content visible only against active-inference's preferences-as-priors form, where outcome value and outcome probability are collapsed into a single quantity. Friston et al.'s expected free energy decomposes into pragmatic and epistemic value — useful for policy ranking, but the two halves don't separate the two cases the 2×2 cell map separates. "The goal is unattainable from here" ($\delta_{\text{sat}} > 0$) and "the current policy is not the best available" ($\delta_{\text{regret}} > 0$) both increase EFE without distinguishing cause, and the orient cascade acts on the two differently. The diagnostic structure depends on V_{O_t} being a value functional on trajectories rather than a log-prior on outcomes — that is what makes "wanting o" and "expecting o" formally distinguishable rather than collapsed (Sun & Firestone 2020 diagnose the collapse; AAT's value-functional reformulation is the downstream architectural response). This is a deliberate divergence from active inference, not an oversight.

The Correlation Hierarchy and identifiability-floor Instance 2.

The chapter's `#def-strategy-dag` carries an honest scope statement about correlated failure that is more consequential than it reads on first encounter. The Correlation Hierarchy (L0 / L1 / L1' / L2) names what AAT's formal strategy-layer results actually claim and where they degrade. L0 (independence — the tractable baseline under causal sufficiency) is where the chapter's persistence schema, gradient-based credit assignment, and sector-condition transfer are all proved. L1 (augmented DAG with strict-prerequisite common-cause nodes restructured so the common cause is factored above the correlation it creates) is where the formal results transfer exactly. L1' (mixture form for soft facilitators with $\theta_{\text{child}|\neg C} > 0$ when C is observable per trial) transfers with five-way gating via Prop B.7 in `#deriv-edge-credence-dynamics`. L2 (full joint distribution) is mathematically ideal but exponential in parameter count and incompatible with the bounded-cognition constraint that motivates the DAG in the first place.

The substantive finding lives in the boundary between L1' (observable C) and L1' (unobservable C). When the common cause is unobservable, the per-conditional decomposition is not "open" — it is *structurally refuted* by the Cramér-Rao floor (Fisher rank-1 per-trial information matrix), and no unbiased online estimator on the joint conditional vector admits a sector parameter $\alpha > 0$. This is the framework's second formally-named instance of the identifiability-floor pattern (Instance 2 in `#disc-identifiability-floor`), and the agent's three escape routes — augment C-observability via an instrument variable, run multi-child joint observations under exploration, or fall back to L0-on-marginals (losing per-conditional diagnostics) — are the operational instantiation of *naming the floor and naming the unique broadly-available escape via AAT machinery*. The framework's recurring posture is in view here: each identifiability floor is a *constructive use of impossibility*, where the no-go elevates the load-bearing role of the specific AAT machinery that supplies the escape.

The chapter is also where the first instance of the same pattern (Instance 1) becomes statable: the on-policy L0-insufficiency-detection no-go. An agent at L0 cannot detect its own causal insufficiency from on-policy traces alone — the worlds with and without a latent common cause are observationally indistinguishable under policy-perfect execution (the Bareinboim Causal Hierarchy Theorem applied at the L0/L1 distinction). The unique broadly-available escape is joint sibling observability under exploration, with the agent's own interventions (6.3) supplying the joint outcome data the test requires. The full formal treatment lives in `#der-causal-insufficiency-detection` in Part II Ch.4; the chapter's Correlation Hierarchy is where the scope condition is named and where the no-go becomes anticipatable. Read Ch.3's L0/L1' boundary and Ch.4's on-policy detection no-go together and the identifiability-floor pattern's first two instances come into view as duals: Instance 2 says identification fails inside a single agent under unobservable common causes; Instance 1 says self-diagnosis fails under policy-perfect execution. The single escape AAT names — interventional access supplying the joint outcome data — applies to both.

Chain-confidence decay, coordinate forcing, and the bridge into Strategy Dynamics.

The chapter opens with `#der-chain-confidence-decay` — the chain rule of probability lifted to log-space, $\log P(\text{chain}) = \sum_i \log P(E_i \mid E_{<i})$. The identity is elementary; what the chapter intro flags but the implications can develop is that it is the *first instance* of a meta-pattern AAT will deploy at three more representational layers. AAT forces a coordinate at each of chain, divergence, update, and metric layers via uniqueness theorems on AAT-internally-motivated additivity or invariance axioms: log-probability at chain (Cauchy-FE on the probability chain rule); reverse-KL at divergence (Cauchy-FE via Hobson 1969 / Csiszár 1991 / Shore-Johnson 1980 on chain-rule additivity over conditional factorizations); log-odds at update (Cauchy-FE via Aczél 1966 on evidential additivity); Fisher information metric at metric (Čencov 1982 invariance under (PI) parameterization-invariance commitment). The four are interlocked by cross-cites; the convergence across independent axioms onto a single exponential-family Legendre-Fenchel geometry is the substance — see `#disc-additive-coordinate-forcing` for the catalog.

The methodological implication worth surfacing: AAT's coordinates are not chosen by convention. They are forced by AAT-internal axioms via classical uniqueness theorems on additivity or invariance, and they all live on the same geometric object. Lyapunov quadratic (the chapter's persistence machinery) and the IB Lagrangian (the modeling-class machinery in Part I) are adjacent family members — Bregman divergence on Euclidean potential, and adopted-rather-than-derived respectively. The framework's posture is that where a coordinate matters, the framework can usually point at the axiom that forces it; where the axiom isn't AAT-internal, the framework can say what was adopted from outside. The chapter is where this posture lives its first explicit instance.

The downstream bite is wide. Each of the four coordinate-forcing instances backs at least one of the chapter's catalog-grade findings elsewhere: log-probability at chain anchors the depth-fragility result whose engineering reading is "deep plans are structurally fragile, with the rate of fragility growing exponentially in depth"; reverse-KL anchors the Bretagnolle-Huber regret bound in Part II Ch.4; log-odds anchors the detection-latency no-go (S) and the forgetting prerequisite (`#schema-strategy-persistence`); Fisher metric anchors the Class 3 (Coupled) bias bound (Track 2 with $C_{FR} = \sqrt{2}$ universal under (PI) + Čencov, dimension-free) and the survival-LMI directional exploration bonus. The chapter's elementary identity at chain layer is the canonical example of a load-bearing AAT claim that looks like algebra and turns out to be the entry point to a four-layer architectural commitment.

The widest downstream bite is in Part II Ch.4 (Strategy Dynamics), where the strategy-edge persistence schema instantiates the chapter's DAG structure with Beta-Bernoulli update dynamics, and where `#der-causal-insufficiency-detection`'s on-policy no-go (this chapter's Instance 1 scope-named) becomes the formal statement. The Ch.4 chapter-end implications will read against this one — the framework's identifiability-floor pattern delivers its first formal instance there, with the Correlation Hierarchy from this chapter as the scope statement and the loop-interventional access from Part II Ch.2 as the escape mechanism.

What Part II Ch.3 closes on, then, is the formal substrate Strategy Dynamics will operate on: a theorem-backed graphical structure with a principled parameterization, a convention-indexed diagnostic that routes revision pressure to the right substate, a scope-honest treatment of correlated failure with one Cramér-Rao floor already named and one Bareinboim-CHT no-go anticipated, and the first instance of a four-layer coordinate-forcing meta-pattern whose remaining three layers reach across the rest of the framework. The chapter is small in segment count and wide in downstream reach.

Working Notes.

- This segment is a chapter-end discussion grouping findings from `#def-strategy-dag`, `#def-satisfaction-gap`, `#def-control-regret`, `#der-chain-confidence-decay`, `#deriv-graph-structure-uniqueness`, `#scope-and-or`, and `#def-value-object` (the convention-hierarchy host, formally in Part II Ch.1 but consequential for the diagnostic split here). The cross-pattern reading in §3 cross-references `#disc-identifiability-floor`'s meta-segment, and §4 cross-references `#disc-additive-coordinate-forcing`. Each finding's load-bearing math lives in its home segment or meta-segment; this segment surfaces what the chapter delivers as a chapter.

- **Cross-reference vs canonical-home policy.** None of this chapter's source segments carries a `## Findings` section. The chapter intro `#strategy-structure-intro` is unusually rich — it already develops the derived-vs-chosen distinction, the Correlation Hierarchy and the on-policy no-go, the diagnostic split with the 2×2 cell map, and the chain-confidence-decay anchor-of-three-downstream-layers framing. This segment intentionally does not restate that content; the implications-flavored payoff worth pulling out beyond the intro is (a) the convention-hierarchy lift of the diagnostic split (the intro covers the 2×2 cell map but not the C1/C2/C3 indexing of it), (b) the explicit positioning of the Correlation Hierarchy's L1'-unobservable boundary as Instance 2 of the identifiability-floor pattern with the cross-instance dual to Instance 1, (c) the methodological framing of the *convergent representational choices* category as a research-trail-honesty artifact (#60 in the source catalog), and (d) the framework's posture on coordinate-forcing as a methodological commitment with four-layer reach.
- **Connection to NeurIPS submissions.** The satisfaction-gap / control-regret diagnostic is Component 1 of NeurIPS 2026 Paper 2's composition theorem (`~/src/neurips/02-unified-convergence-rl/` §4 Theorem 4.1); the paper's (A1) hypothesis carries the extended-real reading required for the two-gap structure to hold when the satisfaction frontier is empty. The convention-hierarchy monotonicity result is a property of `#def-value-object`; whether it is theorem-grade in any NeurIPS submission is not flagged in current cross-references.
- **Open: a "convergent choice" register in segment frontmatter?** The convergent-representational-choices framing names a third epistemic category between *derived* and *chosen*. The current type / status / stage frontmatter does not carry a field for this. A `convergent-choice:` boolean (or a `status: convergent-choice` extension) would let `bin/extract-findings` or a sibling extractor surface the category systematically. Open for a future FORMAT.md cycle; the meta-architectural pattern's current home is the `#disc-*` meta-segments and the chapter-end implications series.

Chapter 8

Strategy Dynamics

[**Gap**] Discussion

8.1 *Definition:* Strategic Calibration

The strategic calibration residual measures whether the strategy's causal model is correct: are the edges in Σ_t accurate predictors of how much value each step actually produces? This is the fine-grained diagnostic that localizes control regret to specific parts of the strategy.

Formal Expression.

For each edge (i, j) in Σ_t with credence p_{ij}, the **edge residual**:

Definition (strategic-calibration).

$$r_{ij} = \mathbb{E}[\Delta V_O \mid \text{edge } (i, j) \text{ traversed}, \ M_t] - \Delta V_O^{\text{observed}} \tag{8.1}$$

where ΔV_O is the change in $V_O(M_t, \pi; N_h)$ attributable to completing step j — as predicted by Σ_t versus as observed.

The **strategic calibration residual** aggregates across active edges:

$$\delta_{\text{strategic}} = \left(\sum_{(i,j) \in \text{active}} w_{ij} \cdot r_{ij}^2 \right)^{1/2} \tag{8.2}$$

where w_{ij} weights edges by importance (e.g., criticality to the current plan's critical path).

Conditioning. The edge residual r_{ij} is meaningful only when: - The edge was actually traversed (the agent attempted the step) - M_t is adequate (so the observed ΔV_O is meaningful, not noise) - The agent followed Σ_t's prescription for step j (execution fidelity — otherwise the residual conflates bad plan with bad execution)

Without the execution fidelity condition, a positive residual could mean "the plan is wrong" or "the agent didn't follow the plan." These require different corrections (Σ_t revision vs. execution improvement).

Epistemic Status.

Discussion-grade. The edge residual concept is well-motivated: each edge predicts a value increment, and comparing prediction to observation is standard calibration. But the specific aggregation (L^2 norm with importance weights) is a reasonable first pass, not a derived result. The weighting scheme (w_{ij} by criticality) is sensible but ungrounded. The conditioning requirements (especially execution fidelity) make this quantity hard to estimate in practice — the agent must know whether it followed its own plan, which requires a level of self-monitoring that many agents lack.

This is a **second-order inference** — it requires accumulating evidence over multiple edge traversals. It is inherently slower to evaluate than $\delta_{\text{epistemic}}$ (which updates on every observation) or δ_{sat} (which can be evaluated from M_t and Σ_t alone).

Discussion.

Connection to 8.10. There are two distinct strategic mismatch quantities:

1. **Strategy-strategy-plan-confidence error** $\delta_s = \hat{P}_\Sigma - \Phi$ — the scalar gap between the agent's strategy-plan-confidence score and the independence-model plan value at true edge parameters. Computable from status propagation alone, without credit assignment. The sector condition transfers to δ_s (Prop B.5 in N). Note: Φ is the AND/OR propagation formula evaluated at true edge rates — it equals actual plan success probability only when the DAG is causally sufficient (L0 of the Correlation Hierarchy in 7.3). Under correlated failure (causally insufficient DAG, the dominant real-world case), Φ overestimates actual success. δ_s tracks calibration *within the independence model*, not calibration to strategic reality. For L1 (augmented DAGs with explicit common-cause nodes), δ_s of the augmented graph tracks calibration within the augmented model, which is more accurate.
2. **Strategic calibration residual** $\delta_{\text{strategic}}$ (this segment) — an L^2 aggregation of per-edge value-increment residuals requiring credit assignment to compute.

These are related (both measure strategy-reality mismatch) but not interchangeable. δ_s is the proven persistence target; $\delta_{\text{strategic}}$ provides finer-grained diagnostics but its persistence properties remain open, pending the credit-assignment machinery in 8.6. The correction function for both is edge-credence revision (8.4); the disturbance is environmental changes that alter edge-traversal outcomes.

Typing as value-increment residuals. Each edge predicts a scalar (value increment), not a full state transition. This is the most tractable typing because it connects directly to the value object V_O and allows aggregation across heterogeneous step types (a military advance and a logistics delivery produce different state changes but both produce value increments measurable on the same scale).

Credit-assignment problem. The edge residual r_{ij} subtracts "predicted value increment" from "observed value change." These are different types: the prediction comes from Σ_t's internal causal model, while the observation comes from empirical measurement of ΔV_O. The subtraction is meaningful only when the observed value change can be *attributed to the specific edge* — a credit-assignment problem the current formulation does not address. For edges with a single parent, attribution is straightforward. For multi-parent AND/OR nodes, the observed ΔV_O at the child reflects the combined effect of all parent edges, and decomposing it into per-edge contributions requires additional structure (e.g., Shapley-value decomposition, or sequential observation of parent completions). In the absence of clean attribution, r_{ij} measures "how well did the overall prediction work?" rather than "how well did this specific edge predict?" — still useful for aggregate calibration, but not sufficient for targeted edge revision.

Working Notes.

- The aggregation into a single $\delta_{\text{strategic}}$ may lose important structure. A per-edge or per-path profile of residuals would be more informative for diagnosis: which parts of the strategy are well-calibrated and which are not? The scalar summary is useful for the persistence condition (which needs a single mismatch magnitude) but not for strategy revision (which needs to know WHERE the problem is).

- Alternative aggregation: maximum edge residual (worst-calibrated edge), or weighted by information value (edges the agent is most uncertain about). The right aggregation depends on the use case.
- Execution fidelity monitoring is a genuine challenge for agents that don't have a clear execution trace. For software agents operating through tool calls, execution fidelity is relatively easy to assess (did the agent issue the right commands?). For organizational agents, it's much harder (did the subordinate actually follow the directive, or reinterpret it?).

8.2 *Derived:* Causal Insufficiency Detection

An agent operating at L0 of the Correlation Hierarchy (7.3) faces a structural impossibility: under purely on-policy execution, no detection mechanism can distinguish an L0-insufficient world (latent common causes present) from an L0-sufficient world matched to the on-policy regime conditionals. This is a consequence of the causal hierarchy theorem (6.1, 6.2) — observational data does not in general identify interventional structure. Detection is therefore *only* possible by capabilities that violate the "purely on-policy" condition: joint sibling observability under exploration (the canonical AAT route, exploiting 6.3), intermediate-state observability, structural priors, or direct intervention on the candidate latent. The pairwise sibling covariance test is the AAT-canonical detector; the L0 plan-level residual is a degenerate special case of the no-go.

Formal Expression.

The No-Go Theorem: Purely On-Policy Detection Is Impossible.

Let $\mathcal{M}_{L0}$ be the agent's L0 strategy model with sequential short-circuit AND/OR execution policy π_{L0}. Let $\mathcal{W}_{L1}$ be a world with a latent common cause C acting on multiple sibling action propositions, and $\mathcal{W}^*_{L0}$ be an L0 world with edge probabilities $\{\theta^*_j\}$ matched to the on-policy regime conditionals of $\mathcal{W}_{L1}$. Let $\mathbb{P}^{\text{obs}}_{\pi_{L0}}[\cdot]$ denote the joint distribution over the agent's on-policy observable events under π_{L0}.

Derived (no-go-on-policy, from causal hierarchy theorem + observational equivalence under sequential short-circuit), conditional on (S1)–(S5) below.

Observational equivalence. $\mathbb{P}^{\text{obs}}_{\pi_{L0}}[\mathcal{W}_{L1}] = \mathbb{P}^{\text{obs}}_{\pi_{L0}}[\mathcal{W}^*_{L0}]$.

No-go conclusion. Any function of the agent's on-policy observable history alone cannot distinguish $\mathcal{W}_{L1}$ from $\mathcal{W}^*_{L0}$. Therefore no purely on-policy detection mechanism — no test, statistic, or Bayesian comparison taking only the on-policy distribution as input — can detect L0 causal insufficiency.

Scope conditions (S1)–(S5).

- (S1) Pure on-policy execution; no off-policy sampling.
- (S2) Sequential short-circuit AND/OR evaluation.
- (S3) Censored sibling observation: short-circuited siblings are not observed.
- (S4) No interventional access to candidate latents.
- (S5) No structural priors positing specific common causes.

Tier. *Exact* for shallow strict-prerequisite cases (2-sibling OR or AND with binary common cause and $\theta_{j|\neg C} = 0$ — see #example-L1). *Robust qualitative* for general DAG topology, soft facilitators, and deeper structures: the structural argument transfers, but explicit $\mathcal{W}_{L0}^*$ construction has been carried out only for shallow cases.

Construction of $\mathcal{W}_{L0}^*$. For a 2-sibling OR with strict-prerequisite latent C, $P(C) = \theta_C$, conditional success rates $\theta_{j|C}$:

$$\theta_1^* = \theta_C \cdot \theta_{1|C}, \qquad \theta_2^* = p_2^c = \frac{\theta_C\,(1-\theta_{1|C})\,\theta_{2|C}}{1-\theta_C\,\theta_{1|C}} \tag{8.3}$$

Direct verification (see #example-L1) shows $\mathbb{P}^{\text{obs}}_{\pi_{L0}}[\mathcal{W}_{L1}] = \mathbb{P}^{\text{obs}}_{\pi_{L0}}[\mathcal{W}_{L0}^*]$ on the three on-policy observable events. The Bareinboim, Correa, Ibeling & Icard (2022) Causal Hierarchy Theorem then gives the no-go: any two SCMs that agree on Level 1 (associational) data cannot in general be distinguished on Level 2 (interventional) questions, and the L0/L1 distinction — whether siblings share a common cause — is precisely a Level 2 question about $P(A_2 \mid do(\neg A_1))$ versus $P(A_2 \mid \neg A_1)$.

Why this matters. The no-go is the structural reason the prior aggregate-residual mechanism collapses on-policy: the residual is a function of $\mathbb{P}^{\text{obs}}_{\pi_{L0}}$, which is identical between the two worlds, so the residual is identically zero under both. The collapse is not a quirk of the residual statistic — it is a special case of the no-go applied to that specific function. No replacement aggregate statistic can do better.

The Detection Routes: What Circumvents the No-Go.

The no-go's scope conditions (S1)–(S5) define "purely on-policy." Each condition's violation corresponds to an AAT capability that admits (partial) detection:

Derived (boundary-routes, from no-go scope conditions).

Table 8.1

Route	*Scope violated*	*AAT capability*	*Detection strength*
(a) ε-exploration	(S1)	SA3 exploration (N Prop B.4)	Partial, scales with ε
(b) Joint sibling observability	(S3)	Covariance test under SA3 + 6.3	Strong
(c) Intermediate observability	(S3) at finer grain	Observability investment (8.3)	Very strong when available
(d) Structural priors	(S5)	Hypothesized common-cause nodes in DAG construction	Prior-quality-dependent
(e) Direct intervention on latent	(S4)	Domain-specific latent control	Strongest when available

The covariance test (route (b)) is the AAT-canonical detector: it uses only machinery the theory already requires (exploration via SA3, interventional data via the loop) and is available in the broadest range of domains. The remaining sections operationalize this primary mechanism.

Primary Detection Mechanism: Pairwise Sibling Covariance Under Intervention.

Under L0 (the independence model in 7.3's Correlation Hierarchy), sibling outcomes under a common parent are uncorrelated:

Derived (from loop-interventional-access + independence test, conditional on SA3 exploration providing joint observability).

$$H_0 : \ \mathrm{Cov}(Y_{A_i}, Y_{A_j}) = 0 \quad \forall\, i \neq j \text{ siblings under the same parent} \tag{8.4}$$

Under causal insufficiency (latent common cause C acting on multiple siblings), sibling outcomes are positively correlated:

$$H_1 : \ \exists\, i \neq j \text{ with } \mathrm{Cov}(Y_{A_i}, Y_{A_j}) > 0 \tag{8.5}$$

The agent generates test data through the standard exploration mechanism (SA3 — ε-greedy or similar). On trials where both siblings are observable — the agent tries one and can also observe the other's outcome, or tries them in rapid succession before the environment state changes — it accumulates the empirical covariance:

$$\hat{\rho}_{ij} = \frac{1}{N} \sum_t (Y_{A_i,t} - \bar{Y}_{A_i})(Y_{A_j,t} - \bar{Y}_{A_j}) \tag{8.6}$$

A significantly positive $\hat{\rho}_{ij}$ rejects the L0 independence hypothesis. Joint observability (6.3 supplies the interventional character; SA3 supplies the joint sampling) is precisely the violation of scope condition (S3) that admits the test under the no-go.

Detection criterion. A statistically significant positive $\hat{\rho}_{ij}$ at sample size N sufficient for the desired test power, after per-edge credences have stabilized:

$$\hat{\rho}_{ij} > z_{1-\alpha}\, \hat{\sigma}_{\rho_{ij}} / \sqrt{N} \qquad \Longrightarrow \qquad \text{DAG is causally insufficient between siblings } i, j \tag{8.7}$$

(Standard hypothesis-testing form; threshold and test power depend on application.)

Preconditions for the covariance test.

1. **Joint observability.** The agent can occasionally observe (Y_{A_i}, Y_{A_j}) pairs in the same environment state. Pure short-circuit execution censors one of each pair; SA3 exploration or simultaneous-attempt regimes provide uncensored pairs.
2. **Per-edge credence stabilization.** Edge credences $\hat{p}_i, \hat{p}_j$ have stopped drifting at the timescale of the covariance accumulation, so $\bar{Y}_{A_i}, \bar{Y}_{A_j}$ are well-defined empirical means.
3. **Approximate stationarity over the test window.** The latent common cause's frequency and the conditional success rates are not drifting faster than the test's accumulation timescale.

When these preconditions hold, $\hat{\rho}_{ij} > 0$ is diagnostic of a missing common cause acting on (A_i, A_j). When they do not, the signal is ambiguous.

The Aggregate Residual as a Degenerate Special Case of the No-Go.

A historically prominent diagnostic uses the L0 plan-level residual $\Phi^{L0}(\hat{p}) - \bar{y}_G$ as a detection signal. The no-go theorem subsumes this as a special case: under pure on-policy execution, the residual is *identically zero* in both $\mathcal{W}_{L1}$ and $\mathcal{W}_{L0}^*$.

Derived (residual-degeneracy, as instance of no-go theorem).

Direct verification. Under sequential short-circuit, the agent's empirical credences converge to the on-policy regime conditionals: $\hat{p}_j \to p_j^c$. Plugging these into the L0 arithmetic recovers the chain rule of probability (e.g., for OR: $1 - (1 - p_1^c)(1 - p_2^c) = 1 - P(\neg A_1, \neg A_2) = P(A_1 \cup A_2)$, which equals $\bar{y}_G$ under the executed policy). The residual is zero by algebraic identity.

This is *not* a separate finding from the no-go: it is the no-go's prediction for the specific aggregate-residual statistic. The no-go forbids *any* on-policy statistic from distinguishing $\mathcal{W}_{L1}$ from $\mathcal{W}_{L0}^*$; the residual evaluates to the same value (zero) in both, as expected.

Off-policy boundary. Under ε-exploration (route (a)), the residual scales as $O(\varepsilon)$ to leading order with sign matching the dominant node-type bias (+ for OR-heavy, − for AND-heavy):

$$\Phi^{L0}(\hat{p}) - \bar{y}_G = \varepsilon \cdot R + O(\varepsilon^2), \quad \text{sign}(R) = \text{sign}(\rho) \tag{8.8}$$

where R is structure-dependent and recovers the marginal-limit ρ at $\varepsilon = 1$. *[Heuristic]* The qualitative form is robust; the exact coefficient depends on the gap between conditional and marginal credences. The widely-quoted "$\varepsilon \cdot \rho$" scaling is correct as an order-of-magnitude statement. For a 2-sibling OR with conditional credences p_j^c, the exact two-OR formula is $\varepsilon R_1 - \varepsilon^2 R_2$ with $R_1 - R_2 = \rho$; the leading-order coefficient R_1 is structure-dependent and equals ρ only at $\varepsilon = 1$.

The residual is therefore a *confirmatory* signal under route (a): when the agent has material off-policy exploration and the covariance test (route (b)) has localized a candidate latent, the residual sign confirms the bias direction. It is not a primary detector and cannot replace the covariance test.

From Detection to L1 Construction.

Once the agent detects $\hat{\rho}_{ij} > 0$ between siblings A_i and A_j, it knows a latent common cause exists but not its identity. The construction process:

Derived (from positive covariance signal + L1 construction principle in #def-strategy-dag).

1. **Hypothesize** a common-cause node C that explains the correlation.
2. **Estimate** θ_C from the pattern of joint outcomes. The joint failure rate $P(A_i \text{ fails}, A_j \text{ fails})$ exceeds $(1-\theta_i)(1-\theta_j)$ by $\hat{\rho}_{ij}$; the excess localizes the common cause's frequency.
3. **Restructure** the DAG: factor C above the correlated siblings (7.3, L1 construction principle: factor the common cause above the correlation it creates).
4. **Re-estimate** conditional edge credences $\theta_{k|C}$ from the data, conditioned on the inferred C state.

This is structural adaptation (4.5) at the strategy level: the agent changes its model class from L0 to L1, adding representational capacity for a pattern the L0 model cannot express. The cost is the standard cost of structural change: temporary performance degradation while the new credences converge, and increased graph complexity. (Soft-facilitator common causes require L1' rather than L1 — see 7.3 and #example-L1 for the strict-prerequisite vs soft-facilitator distinction.)

Diagnostic CIY.

Which actions are most informative for detecting latent common causes? Under the no-go, only actions that violate one of (S1)–(S5) yield detection signal. The explore-exploit tradeoff extends with a third axis tied to the boundary characterization:

Discussion (diagnostic-ciy).

- **Exploit**: pursue the current best plan (no scope violation; no detection signal).
- **Explore**: test unknown edges for individual success rates (route (a); partial detection).
- **Diagnose**: test known edges for joint correlation structure (route (b); strong detection).

Diagnosis is a form of internal exploration — the agent probes its own model's structural assumptions by violating (S3) deliberately, generating joint sibling outcomes that the no-go forbids the agent to obtain on-policy. The information value of diagnostic actions is highest when:

- Edge credences have converged (the agent has good marginals/conditionals but unknown joint structure).
- Joint outcomes for sibling pairs are observable in the same environment state (the covariance test has data — route (b) is operational).
- The agent has sufficient off-policy budget that the secondary residual signal corroborates (route (a) is also operational).

Epistemic Status.

Conditional on the no-go's scope conditions (S1)–(S5) and on strategy-layer instantiation of 4.5. The **no-go theorem** is *exact* for shallow strict-prerequisite cases (2-sibling OR or AND, single binary common cause) by direct construction; *robust qualitative* for general DAG topology, soft facilitators, and deeper structures. The structural argument (observational equivalence of regime-conditional L0 and latent-cause L1) transfers to the general case; explicit construction of $\mathcal{W}^*_{L0}$ has been carried out only for shallow cases.

The **boundary characterization** (routes (a)–(e)) is *robust qualitative*: each route maps to a specific scope-condition violation and to existing AAT machinery, but the precise detection power of each route depends on domain particulars. Routes (a) and (b) have explicit AAT scaffolding (N, 6.3); routes (c)–(e) depend on domain capability.

The **primary detection mechanism** (pairwise sibling covariance) is *robust qualitative*: standard hypothesis testing applied to interventional data from the feedback loop, with explicit preconditions. Its sensitivity depends on how cleanly the agent can separate sibling-covariance signal from edge-credence noise at convergence; in adversarial or fast-drifting environments the test's effective sample size shrinks.

The **aggregate residual** as a confirmatory signal is *exact* for the on-policy collapse (no-go's prediction is direct); the off-policy mixed-regime scaling is *heuristic* (linear-in-ε with structure-dependent coefficient).

The **detection-to-construction pipeline** is *discussion-grade*: the trigger is the (statistically rigorous) covariance signal, but the specific procedures for estimating θ_C and $\theta_{k|C}$ from correlated outcome data are domain engineering.

What Cannot Be Detected. By the no-go and its boundary characterization, several latent structures remain undetectable by *any* AAT route:

- **Latents with no joint-observability route.** If the latent affects siblings that cannot be jointly observed (mutually exclusive with long horizons, no cause-indicator availability, no intervention capability, no informative prior), the no-go applies in full strength and detection is impossible.
- **Latents affecting only one edge.** By definition not common causes; appear as noise in individual edge credences.
- **Latents too rare to produce observable joint outcomes.** Even with route (b) operational, a latent with $\theta_C \approx 1$ rarely reveals itself — the agent needs enough $C = 0$ events to estimate the covariance.
- **Negatively-correlating latents.** The formulation assumes positive correlation from shared enabling factors. Negative correlation (competing for a shared resource) produces the opposite bias pattern and requires a different model.

These limitations parallel the information-theoretic underdetermination in 8.6: detection requires data with the right structure, and the no-go specifies precisely what "right structure" means.

Discussion.

Why the no-go is a strengthening, not a softening. The prior framing ("the residual mechanism collapses on-policy") was a local observation about one statistic. The no-go is the structural reason: any on-policy statistic must collapse, because the on-policy distribution is identical between L0 and L1 worlds matched on regime conditionals. The covariance test is not just *a* working detector — it is the unique broadly-available violation of the no-go's scope. This sharpens the load-bearing of `#der-loop-interventional-access`: without the loop's interventional data, the no-go forbids detection entirely; with it, route (b) is operational.

Connection to Pearl's hierarchy. The L0/L1 distinction is a Level 2 distinction in Pearl's framework — it concerns whether $P(A_2 \mid do(\neg A_1)) = P(A_2 \mid \neg A_1)$. The Causal Hierarchy Theorem (Bareinboim, Correa, Ibeling & Icard 2022, Theorem 1) proves that Level 2 distinctions are not in general identifiable from Level 1 (associational) data. The no-go is the AAT-specific instantiation: on-policy data is Level 1; the L0/L1 question is Level 2; therefore detection requires more than on-policy data. The five circumvention routes are all ways the agent obtains supra-Level-1 information.

The censoring mechanism is the structural source. Sequential short-circuit evaluation is what makes on-policy data Level 1 only — it censors the joint outcomes that would constitute Level 2 evidence. An agent that *did not* short-circuit would obtain joint sibling outcomes naturally, and the no-go would not apply. But short-circuit is forced by efficiency: testing A_2 when A_1 has already succeeded is wasted action. The no-go is therefore a tradeoff between execution efficiency (favoring short-circuit) and structural diagnosis (favoring joint observation). SA3 ε-exploration is the AAT compromise: short-circuit by default, occasional non-short-circuit excursions that pay the efficiency cost to maintain detection capability.

Connection to the orient cascade. The detection signal enters the orient cascade (9.1) at step 4c (causal-sufficiency check). Step 4c's reference to "pairwise sibling covariance under an augmented test" aligns with the primary detection mechanism here. The no-go strengthens the cascade's load-bearing: step 4c is not "one possible diagnostic" but "the unique broadly-available diagnostic given the structural impossibility of purely on-policy detection."

Connection to the broader identifiability-floor pattern. The no-go is one of an emerging class of structural impossibility results in AAT — limits on what can be inferred from limited information, derived from external information-theoretic theorems. See X for the meta-pattern collecting this result alongside the L1' mixture-identifiability obstruction (N Prop B.7) and the open causal-IB extension for interventional relevance variables.

Domain instantiations. The covariance test (route (b)) applies concretely in: - **Software deployment**: two services sharing infrastructure fail together more often than independent failure rates predict → add infrastructure-health node. -

Military operations: two concurrent operations fail together under adverse weather → add weather-condition node. - **Investment**: two positions lose value together during market stress → add market-regime node. - **Organizational strategy**: two initiatives stall together during leadership transitions → add organizational-stability node.

In each, what makes detection feasible is the agent's ability to occasionally observe *both* sibling outcomes — the route (b) capability. Pure short-circuit ("only run service B if A is down") suppresses the joint-observation events the test relies on; some routine joint exposure is necessary. When joint observation is impossible (routes (b) and (c) both unavailable) and intervention on the candidate latent is impossible (route (e) unavailable), the agent must rely on structural priors (route (d)) — domain knowledge positing the common cause. This is the regime in which intuition-driven causal modeling is the only tractable approach.

Findings.

On-Policy L0 Insufficiency Is Structurally Undetectable. **Brief:** An agent operating with a strategy model that assumes its action propositions are causally independent (an L0 model) faces a structural impossibility when its world contains a latent common cause acting on multiple of those actions: under purely on-policy execution, no test, statistic, or Bayesian comparison built from the agent's observable history can distinguish the latent-cause world from a no-latent-cause world whose edge probabilities are matched to the on-policy regime conditionals. The two worlds emit identical on-policy distributions. This is not a quirk of any particular diagnostic — it is the agent-theoretic instance of the Causal Hierarchy Theorem applied to the L0/L1 distinction, which is a Level-2 question being asked of Level-1 data. The agent that wants to discover its own model's structural insufficiency must source data the on-policy regime structurally cannot produce.

Impact: Reframes exploration from a discretionary diagnostic activity into a structural prerequisite for self-correction at the strategy layer. Each scope-condition violation (S1)–(S5) maps to a specific AAT capability that admits partial detection — most importantly, joint sibling observability under exploration, which lifts `#der-loop-interventional-access` from "useful machinery" to "the unique broadly-available violation of the no-go." Downstream, this hardens the orient cascade's causal-sufficiency check from "one possible diagnostic" to "the only widely-applicable diagnostic"; it explains why the prior aggregate-residual mechanism collapses on-policy (the residual is one statistic the no-go forbids, and *every* on-policy statistic is forbidden); and it gives a precise account of when L0→L1 escalation can and cannot be triggered from the agent's own data.

Novelty Claim: *Claim differentiation* on the framing of why structure-aware exploration is required. Causal bandit and causal MDP work under hidden confounding establishes that observational and interventional data are non-interchangeable and that intervention can be necessary for low-regret learning; this finding sharpens that line into a no-go for *self-diagnosis* under policy-perfect execution, tied to latent strategic correlation structure rather than to regret minimization, and characterizes five boundary routes by which AAT machinery escapes it.

Related Work:

Table 8.2

ASF concern	*Prior-art language*	*Relationship / Positioning*
Observational vs experimental data are non-equivalent under hidden confounding	Bareinboim, Forney & Pearl 2015, "Bandits with Unobserved Confounders" *NeurIPS* (published 2015, found 2026-04 via Undermind report)	*conceptual precursor* — the underlying observational/interventional asymmetry is shared; this finding recasts it as a no-go on agent self-diagnosis rather than a regret bound on action selection

ASF concern	*Prior-art language*	*Relationship / Positioning*
Sequential extension to MDPs	Zhang & Bareinboim 2016, "Markov Decision Processes with Unobserved Confounders" (published 2016, found 2026-04)	*conceptual precursor* — sharpens the bandit asymmetry to sequential control; the present finding's L0/L1 distinction is the strategy-DAG analog at the structural-detection layer rather than the policy-optimality layer
Naive randomization can be insufficient under confounding	Forney, Pearl & Bareinboim 2017, "Counterfactual Data-Fusion for Online Reinforcement Learners" *ICML* (published 2017, found 2026-04)	*conceptual precursor* — narrows the gap between generic exploration and structure-aware experimentation; the present finding's covariance test under joint sibling observability is one such structure-aware mechanism, derived as the unique broadly-available violation of the no-go's scope conditions
Not all interventions are useful	Lee & Bareinboim 2018, "Structural Causal Bandits" *NeurIPS*; Lee & Bareinboim 2020, "Characterizing Optimal Mixed Policies" *NeurIPS* (published 2018/2020, found 2026-04)	*conceptual precursor* — closest formal neighborhood for "exploration design"; the present finding addresses a different question (when *any* on-policy diagnosis is impossible) but inherits the principle that not all interventional data carries equal structural-detection value
Exploration as epistemic-value drive	Friston, Rigoli, Ognibene, Mathys, FitzGerald & Pezzulo 2015, "Active inference and epistemic value" *Cognitive Neuroscience* 6:187–214 (published 2015, found 2026-04)	*adjacent literature* — weaker threat because EFE-based exploration reduces uncertainty under the generative model rather than Pearl-style observational equivalence under hidden common causes; the structural-undetectability claim does not arise in the EFE framework, which presumes a generative model expressive enough to represent the latent
Theorem the no-go imports	Causal Hierarchy Theorem: SCMs agreeing on Level 1 may disagree on Level 2 (Bareinboim, Correa, Ibeling & Icard 2022, in *Probabilistic and Causal Inference: The Works of Judea Pearl*; published 2022, found 2025)	*formal antecedent* — the no-go is the AAT-specific instantiation; the L0/L1 distinction is precisely a Level-2 question (concerns $P(A_2 \mid do(\neg A_1))$ vs $P(A_2 \mid \neg A_1)$) being asked of on-policy Level-1 data

Search Log:

- 2026-04 (*nominally comprehensive*, via `ref/Novelty_defense_and_integration.md` Pillar 1): Undermind report on the causal-bandits-and-MDPs-under-hidden-confounding literature, plus active-inference epistemic-value exploration, established the prior-art landscape; verdict was *Conceptual Precursor* (High confidence). The closest formal neighborhood is the Bareinboim/Zhang/Forney/Lee line; no paper surveyed states the claim in the same form (no-go for self-diagnosis under policy-perfect execution tied to latent strategic correlation). The defense strategy positions ASF's increment narrowly: a no-go theorem on structural detection rather than a regret bound on action selection, with five explicit boundary routes mapped to existing AAT machinery.
- 2025 (*targeted*): Bareinboim, Correa, Ibeling & Icard 2022 identified as the formal antecedent for the Causal Hierarchy Theorem invocation; the segment's invocation of the CHT was already grounded in this source before the comprehensive Pillar-1 defense.

Working Notes.

- The general-topology construction is a structural argument; the explicit $\mathcal{W}^*_{L0}$ for arbitrary AND/OR DAGs with mixed common-cause patterns has not been carried out. For load-bearing application of the no-go to specific complex topologies, the construction should be specialized. Currently the load-bearing applications (orient-cascade step 4c, strategy-dag's L0/L1 escalation principle) reference shallow strict-prerequisite cases for which the no-go is exact.
- The boundary characterization's routes (c) and (e) depend on domain capability and are not formalized in AAT beyond cross-references to `#der-observability-dominance` and `#der-loop-interventional-access`. A future refinement could quantify "detection power" per route as a function of domain parameters (e.g., observability cost, intervention availability, prior strength).

- The no-go is asymmetric: it forbids on-policy *detection* of L1 from L0, but it does *not* forbid on-policy *parameter learning within L0*. The agent can learn its L0 conditionals to arbitrary precision on-policy; it just cannot determine whether those conditionals hide a latent. This distinction sharpens the diagnosis-vs-calibration split that 4.5 makes at the parametric/structural boundary.
- The CHT (Bareinboim et al. 2022) is invoked as an external theorem. AAT inherits its conditions (well-defined SCMs over compatible variable sets); these are satisfied for the strategy-DAG setting by construction.

8.3 *Derived:* Observability Dominance

Unobservable strategy edges cannot be updated — the gain principle drives their update rate to zero. This means the agent's effective strategy is limited to the parts it can observe, regardless of the nominal confidence in unobservable paths. Observability dominates nominal confidence in determining which strategies are epistemically alive.

Formal Expression.

For a path P through Σ_t, the **path observability**:

Derived (observability-dominance, from update-gain + strategy-dag).

$$\mathrm{obs}(P) = \min_{v \in P} \sigma_v \tag{8.9}$$

where σ_v is the observability of node v — how well the agent can determine whether v has been achieved. The weakest link determines the path's observability.

Observability-adjusted confidence:

$$\mathrm{conf}_{\mathrm{obs}}(P) = \mathrm{conf}(P) \cdot \mathrm{obs}(P) \tag{8.10}$$

When $\sigma_v \approx 0$ for any node v on the path: by 3.6, $\eta_{\mathrm{edge}} = U_{\mathrm{edge}}/(U_{\mathrm{edge}} + U_{\mathrm{obs}}) \to 0$ as $U_{\mathrm{obs}} \to \infty$. The edges connecting to v are **frozen at their prior** — the agent cannot update them regardless of what happens. The path is epistemically dead.

Epistemic Status.

Robust qualitative. The mechanism (high observation noise → low gain → frozen edges) is a direct consequence of 3.6's uncertainty ratio. The specific functional form ($\mathrm{conf}_{\mathrm{obs}} = \mathrm{conf} \cdot \mathrm{obs}$) is a first-order approximation — the actual relationship between observability and effective confidence is more complex (it depends on how many observations are accumulated, the prior strength, and the noise structure). The qualitative prediction (low observability → frozen beliefs → ineffective strategy) is robust.

Discussion.

Observability as the gateway to learning. An agent choosing between a strong-but-blind path (high conf(P), low obs(P)) and a weak-but-visible path (lower conf(P), high obs(P)) should prefer the visible one. After one attempt: the visible path yields large η_{edge} — the agent quickly learns whether it works and can redirect. The blind path yields tiny η_{edge} — the agent is still guessing after n attempts. Observability enables learning; opacity prevents it.

Unobservable regions are absorbing. Once significant strategy investment operates through unobservable nodes: frozen beliefs → no mismatch signal → no reason to revise → the agent cannot learn and cannot recognize that it cannot learn. Escape requires external shock, proactive observability investment (instrumenting previously unmonitored nodes), or another agent whose observations cover the blind spot (12.4).

Connection to `#der-code-quality-as-observation-infrastructure` (cross-component reference — see `02-tst-core/`). In the software domain, code quality directly determines σ_v — well-structured code with good tests makes strategy steps (features, refactors, deployments) observable. Poor code quality reduces observability, freezing the developer's causal beliefs about what changes will accomplish. This makes code quality a strategic concern, not just an aesthetic one.

Optimal decomposition depth. Finer decomposition (more intermediate nodes) provides earlier failure detection — detect a problem at step k rather than discovering it at step n. But finer decomposition also increases the number of uncertain edges (7.1). The optimal decomposition depth balances incremental confirmation against compound decay: decompose as finely as observation channels allow, but no finer.

Quantitative content from the two-edge case. The analysis in N (Props B.2-B.3) instantiates this result for the minimal multi-edge chain $A \to B \to G$, giving the qualitative claims above precise mathematical form.

Observable intermediate B. When the agent can observe whether B was achieved, each edge gets independent Bayesian updates. The per-edge sector parameters are $\alpha_1 = 1/(n_1 + 1)$ and $\alpha_2 = \theta_1/(n_2 + 1)$, where θ_1 is the true success probability of the first edge. The overall sector parameter is $\alpha_\Sigma = \min(\alpha_1, \alpha_2)$ — a weakest-link result. The θ_1 factor in α_2 is the **evidence-starvation effect**: downstream edge 2 can only be tested when upstream edge 1 succeeds (with probability θ_1), so its effective correction rate is attenuated by θ_1. For a depth-d chain, this generalizes to $\alpha_k = \prod_{j<k} \theta_j/(n_k + 1)$ — the deepest edge faces exponential attenuation. This is derived for the two-edge case and conjectured (with clear inductive mechanism) for depth-d.

Unobservable intermediate B. When the agent observes only the terminal outcome y_G, per-edge identification fails entirely. The marginal Bayesian update has a systematic bias: the zero-correction-at-truth property (A1) is violated with bias $O(1/n)$. The agent's point-estimate updates for each edge are downward-biased because success always credits both edges fully ($\alpha_k \to \alpha_k + 1$), but failure distributes blame fractionally via proportional attribution. The proportional-blame update turns out to be exactly the marginal Bayesian point estimate — not a heuristic — but it discards the posterior correlation that failure introduces between the edge beliefs.

The consequence is stronger than "frozen edges": unobservable intermediates don't just prevent updating — they make per-edge identification impossible, forcing the agent to plan-level aggregation. If the agent tracks $\hat{\Phi} = p_1 p_2$ as a single Beta on the plan's overall success probability, the single-edge sector condition applies with $\alpha_{\Sigma,\text{plan}} = 1/(n_\Phi + 1)$. But diagnostic resolution is lost: the agent knows the plan is failing but cannot localize which step needs revision.

The observability investment tradeoff. Making B observable yields a quantifiable improvement in α_Σ: from the plan-level rate $1/(n_\Phi + 1)$ to the per-edge weakest-link rate $\min(1/(n_1 + 1), \theta_1/(n_2 + 1))$. The improvement is positive whenever $\theta_1 > 1/2$ and experience is distributed similarly across edges. This gives the observability-dominance principle concrete economic content: the value of instrumenting an intermediate node is the difference in sector parameters, which translates directly to persistence margin.

Working Notes.

- The absorbing-state property of unobservable regions is a strong prediction. In organizational settings, it predicts that departments with poor measurement (R&D, strategy groups, some management functions) will develop persistent, untested beliefs about their own effectiveness. The theory predicts this is structural (frozen η_{edge}), not motivational.
- Observability is not binary — it's a spectrum. Partial observability (noisy observations of intermediate results) gives partial gain, which gives slow but nonzero learning. The diagnostic question is whether the learning rate is fast enough to maintain strategy persistence given the environment's rate of change (ρ).

8.4 *Hypothesis:* Edge Update via Gain

The uncertainty-ratio gain principle (3.6) extends from epistemic updates to strategy-edge updates: edge credences revise in proportion to the ratio of edge uncertainty to observation noise, modulated by the identifiability of the causal link (8.5). This gives a principled, conservative update rule that avoids overreacting to single observations and degrades gracefully when causal identification is weak.

Formal Expression.

Edge credences update via:

Hypothesis (edge-update-via-gain).

$$p_{ij}^{\text{new}} = p_{ij}^{\text{old}} + \eta_{\text{edge}} \cdot \left(\text{signal}(o_t, i, j) - p_{ij}^{\text{old}}\right) \tag{8.11}$$

where: - $\text{signal}(o_t, i, j) \in [0, 1]$: evidential content of observation o_t about the causal link $i \to j$ - $\eta_{\text{edge}} = U_{\text{edge}}/(U_{\text{edge}} + U_{\text{obs}})$: update gain

with: - U_{edge}: uncertainty about this specific causal link. If $p_{ij} \sim \text{Beta}(\alpha_{ij}, \beta_{ij})$: $U_{\text{edge}} = \text{Var}[\text{Beta}] = \alpha\beta/((\alpha + \beta)^2(\alpha + \beta + 1))$ - U_{obs}: observation noise on the channel confirming this link. $U_{\text{obs}} \propto 1/\sigma_j$ (inverse of observability of node j)

Beta-Bernoulli instantiation. For binary observations (success/failure of step j): - Observe success: $\alpha_{ij} \to \alpha_{ij} + 1$ - Observe failure: $\beta_{ij} \to \beta_{ij} + 1$ - Point estimate: $p_{ij} = \alpha/(\alpha + \beta)$ - Effective gain: $\eta_{\text{edge}} = 1/(n + 1)$ where $n = \alpha + \beta$

This is the standard Bayesian conjugate update, yielding $\Delta\hat{p} = (y - \hat{p})/(n+1)$. The gain $1/(n+1)$ is the exact conjugate update rate — not a literal substitution into $\eta = U_{\text{edge}}/(U_{\text{edge}} + U_{\text{obs}})$, which is a structural principle (conservative updating proportional to relative uncertainty), not a universal algebraic formula. The Beta-Bernoulli gain satisfies the same principle: it decreases as posterior certainty increases, trusting accumulated evidence over individual observations. But the algebraic derivation is conjugate analysis, not Kalman-style variance ratios, because the Bernoulli observation model has different noise structure than the Gaussian case where the variance-ratio formula is exact. The sector-condition analysis in N (Props B.1-B.4) uses the exact Beta-Bernoulli gain $1/(n + 1)$ directly.

Parallel log-odds presentation. The same update is natively additive in the log-odds coordinate $\lambda_{ij} = \log(p_{ij}/(1 - p_{ij}))$:

$$\lambda_{ij}^{\text{new}} = \lambda_{ij}^{\text{old}} + \ell(y) \tag{8.12}$$

where $\ell(y) = \log[P(y \mid \text{edge true})/P(y \mid \text{edge false})]$ is the per-observation log-likelihood ratio. The log-odds coordinate is the unique (up to positive affine transformation) parameterization on which Bayesian independent-evidence accumulation is additive — forced by the evidential-additivity axiom derived in P, which is the update-level analog of 7.1's chain-level additive log-confidence decomposition. The moment-parameter (probability-space) form

$\Delta\hat{p} = (y - \hat{p})/(n+1)$ is the projected image of the log-odds additive update under the sigmoid readout $p = \sigma(\lambda)$; the two coordinates carry equivalent content.

The log-odds presentation matters for 8.6's default signal function, where continuous-gradient updates can break the $[0, 1]$ domain in probability-space presentation but are well-posed globally on $\lambda \in \mathbb{R}$. For the per-edge Beta-Bernoulli derivation of Props B.1–B.4 in N, the moment-parameter form is retained because algebraic tightness (the gain $1/(n+1)$ is exact in probability space) is pedagogically cleaner; the sector-parameter content is Fisher-equivalent in either coordinate.

Epistemic Status.

Hypothesis. The extension of the gain principle from M_t to Σ_t edges is structurally motivated: the same uncertainty-ratio logic applies wherever an agent updates a belief in response to noisy evidence. The update *form* is validated for Beta-Bernoulli edges across four topologies (N, Props B.1-B.4). The signal function — how to compute signal(o_t, i, j) — is now characterized at the theory level:

- **Theory-level resolution.** 8.6 shows that persistence does not require per-edge credit assignment (Prop B.5), that any signal function with directional fidelity suffices for persistence (Level 1), and that exact attribution is #P-hard in general (Level 3). The gradient-based candidate signal$_k = p_k + J_k \cdot (y_G - \hat{P}_\Sigma)/\|\mathbf{J}\|^2$ satisfies directional fidelity for monotone DAGs and is computable in $O(|V| + |E|)$.
- **What remains domain-specific.** The choice among signal function implementations (gradient attribution, proportional blame, belief propagation, exact Bayesian) depends on the domain's observability structure and computational budget. This is engineering, analogous to how the gain *principle* ($\eta^* = U_M/(U_M + U_o)$) is theory while the gain *estimator* (Kalman, TD-learning, intuition) is engineering.
- **Remaining open questions.** (1) Continuous outcomes with multi-parent nodes — signal function candidates exist but are not verified. (2) Interaction with M_t updates — edge updates use observations already processed for M_t; whether this creates statistical dependencies that bias edge updates is not analyzed. (3) The gradient candidate inherits $\hat{P}_\Sigma$'s overestimation bias under correlated failures (7.3) — see 8.6 Working Notes.

Discussion.

Connection to 8.3. When $\sigma_j \approx 0$: $U_{\text{obs}} \to \infty, \eta_{\text{edge}} \to 0$. The edge is frozen at its prior. Unobservable links cannot be updated — this is the mechanism that makes unobservable paths epistemically dead.

Connection to 8.1. The edge update rule is the correction function that the strategic calibration residual calls for: persistently positive r_{ij} (predicted value increment exceeds observed) should reduce p_{ij}; persistently negative r_{ij} should increase it. The gain principle provides the update magnitude — how much to adjust given the evidence strength.

Conservative by design. The uncertainty ratio ensures the agent doesn't overreact to single observations: a well-established edge ($\alpha + \beta$ large, U_{edge} small) requires strong evidence to revise; a newly hypothesized edge ($\alpha + \beta$ small, U_{edge} large) is easily moved by evidence. This mirrors the epistemic gain's behavior for M_t — the agent trusts accumulated experience over individual observations.

Working Notes.

- The signal function is characterized at the theory level (8.6) but remains domain-specific in implementation. The gradient-based candidate is the theory's default Level 1 scheme; domains with richer observability can do better (Level 2-3). For continuous outcomes with multiple contributing edges, even the gradient candidate requires further validation.
- **Partial progress on the signal function** (from N (Props B.2-B.3)). For observable binary outcomes in a chain $A \to B \to G$: when B is observable, the signal function is trivial — signal$(o_t, A, B) = y_B$ and signal$(o_t, B, G) = y_G$ (given $y_B = 1$), with no signal when the upstream edge fails. When B is unobservable, the proportional-blame signal $q_k = P(\text{edge } k \text{ caused failure} \mid y_G = 0)$ turns out to be exactly the marginal Bayesian point estimate — not a heuristic. This is a genuine result: the "obvious"

blame-assignment heuristic is the optimal marginal update for the posterior mean. However, the marginal point-estimate update is biased at truth (violates A1 with $O(1/n)$ bias), so the signal function is correct as a Bayesian computation but produces a biased correction when used in the factored point-estimate representation. The signal function question remains open for continuous outcomes, multi-parent nodes, and general DAG topologies.

- The Beta-Bernoulli model assumes edge outcomes are i.i.d. Bernoulli draws. This is adequate for many settings but misses: (a) time-varying edge reliability (p_{ij} drifting), (b) context-dependent reliability (p_{ij} varies with environmental conditions), (c) correlated edge outcomes. Extending to time-varying priors (discounting old evidence) would connect to the mismatch dynamics framework.
- M_t/**edge evidence double-counting** analyzed in `spikes/spike-edge-semantics-resolution.md` §7. Verdict: mostly unfounded — the two updates extract different information from the same observation (M_t update: "what does o_t say about the world?"; edge update: "what does o_t say about causal link $i \to j$?"). A bounded effect exists when M_t updates change leaf credences that then also influence edge updates via the same observation. The cascade ordering (process M_t first) is correct; the effect is self-correcting under the gain mechanism.
- **Dependency removed 2026-05-11.** `deriv-edge-update-natural-parameter` was previously listed in `depends:` and is now removed. The core hypothesis (moment-parameter form $p^{\text{new}} = p^{\text{old}} + \eta_{\text{edge}} \cdot (\text{signal} - p^{\text{old}})$) is logically self-contained — the log-odds derivation is referenced only via the "Parallel log-odds presentation" paragraph in Formal Expression, where it grounds the log-odds *as an alternative coordinate*, not as a logical prerequisite for the hypothesis itself. The reference there ("forced by the evidential-additivity axiom derived in P") is a forward pointer in the FORMAT.md §"Cross-references" sense: a linear reader can encounter the hypothesis here and proceed without having read the derivation; the moment-parameter form is fully specified by the time the log-odds paragraph appears. Removing this dep also breaks the inner 2-cycle with `deriv-edge-update-natural-parameter` and four detected cycles in the edge-update subgraph (verified 2026-05-11). The reverse direction (`deriv-edge-update-natural-parameter` → `hyp-edge-update-via-gain`) is suspect on the same Gate 1 "genuine use vs. mentioned" grounds and likely deserves removal too, but was left alone — surgical move only, the disc-separability-pattern subtree is deferred from the foundation paper's critical path.

8.5 *Scope:* Causal Validity of Edge Updates

The gain-based edge update (8.4) revises edge credences p_{ij} — causal efficacy estimates whose identification strength varies with the data regime (7.3). This segment scopes where the update yields credences that approximate the interventional quantity $P(j \mid do(i), M_t)$, where it yields partially identified estimates, and where it yields associational proxies.

Formal Expression.

Scope Condition (edge-update-causal-validity).

Where the agent has direct intervention. By 7.3, leaf action nodes are propositions about the agent's own actions: "action a succeeds at τ_v." When the agent executes an action leaf, it performs a genuine $do(\cdot)$ operation. The edge from that leaf to its child carries credence $p_{ij} = \text{Cr}(j \text{ advances} \mid do(i), M_t)$, and the execution-observation pair $(do(i), o_j)$ is interventional data for that edge.

However, interventional data does not automatically yield clean causal identification. By 6.3, the loop provides data *generated under intervention* — but between the intervention and a usable causal estimate stand coverage, within-step confounding, delay, and partial observability. The following conditions determine when the interventional data is strong enough for valid edge revision:

(C1) The parent is an action leaf under the agent's control. The agent directly executed the action. This makes the data interventional in character — the agent chose the action, the environment responded. For condition leaves (observable states the agent doesn't control) and internal nodes (propositional combinations achieved indirectly), $do(i)$ is not directly available. See "Indirect edges" below for the weaker identification available at those positions.

(C2) The outcome is attributable. The agent can distinguish whether j advanced specifically because of $do(i)$, or for other concurrent reasons. This is the credit-assignment problem identified in 8.1. It is trivially satisfied for single-parent nodes (one possible cause) and for well-isolated interventions. It is violated when multiple parent edges of j fire concurrently.

(C3) Execution conditions vary. The agent does not systematically execute i only under conditions that independently favor j's success. If it does, the observed success rate carries selection bias: $P(j \mid \text{chose to execute } i) \neq P(j \mid do(i))$ because the decision to execute correlates with favorable conditions. This is mitigated when the agent varies execution contexts across episodes, or when external factors (CI pipelines, scheduled operations) force execution regardless of conditions.

C1 establishes that the data is interventional. C2 and C3 determine whether the interventional signal can be cleanly extracted. All three are satisfied simultaneously in Regime A domains; they degrade together in Regime B and C domains.

Three regimes. These conditions partition domains into admissibility regimes, paralleling 6.4:

Table 8.3

Regime	*C1*	*C2*	*C3*	*Causal validity of leaf-edge updates*
A: Intervention-rich	Agent controls leaf actions	Good isolation (one action at a time)	Conditions vary (CI, diverse contexts)	**Strong.** Updates approximate interventional.
B: Partial intervention	Agent acts but with coordination constraints	Concurrent actions blur attribution	Self-selection likely	**Moderate.** Updates carry optimistic bias.
C: Observation-only	Agent did not act (condition leaves, passive monitoring)	Attribution impossible	Confounding dominant	**Weak.** Updates reflect association, not causation.

Indirect edges. For edges between non-leaf nodes, or edges whose parent is a condition node, the agent does not directly intervene on the parent. Instead, the agent's interventions at the leaves propagate upward through the DAG. The edge (i, j) where i is an internal node receives *indirect* interventional evidence: the agent intervened on leaves below i, observed that i was (or wasn't) achieved, and then observed whether j advanced.

This indirect evidence is weaker for two reasons: 1. **Compounding attribution**: the agent must attribute j's outcome to edge (i, j) after already attributing i's achievement to the leaf-level interventions. Each attribution step introduces uncertainty. 2. **Confounding from below**: i's achievement depends on multiple leaf actions and condition states. Even if each leaf intervention is clean, their combined effect on i may be confounded.

The identification strength for indirect edges decreases with depth in the DAG — deeper edges are farther from the agent's direct interventions and have more confounding pathways.

Identifiability-adjusted gain.

The update gain should be adjusted by the agent's confidence in causal attribution:

Hypothesis (identifiability-coefficient).

$$\eta_{\text{edge}}^{\text{adj}} = \eta_{\text{edge}} \cdot \iota_{ij} \tag{8.13}$$

where $\iota_{ij} \in [0, 1]$ is the **identifiability coefficient** — the agent's estimate of how cleanly the observed outcome can be attributed to edge (i, j) specifically.

- $\iota_{ij} = 1$: clean attribution (leaf-originating edge, single parent, isolated execution in Regime A).
- $\iota_{ij} \approx 0$: no attribution possible (deep internal edge, many concurrent causes, Regime C).

For leaf-originating edges in Regime A: $\iota_{ij} \approx 1$. For internal edges at depth d: ι_{ij} decreases with d (each level of indirect inference degrades attribution). The precise functional form is domain-dependent.

This unifies two sources of frozen edges: 1. Low **observability** ($\sigma_v \approx 0$ from 8.3): the node's outcome is hard to *measure*. 2. Low **identifiability** ($\iota_{ij} \approx 0$): the outcome is measurable but can't be *attributed* to this edge.

Both drive $\eta_{\text{edge}} \to 0$ and produce the same effect: the edge is frozen at its prior.

Epistemic Status.

Conditional on the three conditions C1–C3 and the DAG position of the edge. Max attainable: conditional (the conditions are genuine restrictions, not removable assumptions).

The restriction to leaf-originating edges for strong identification: **derived** from 7.3's node definitions. Only action-leaf nodes correspond to operations the agent directly performs. This is a structural property of the DAG, not an empirical claim.

The three-regime classification: **robust qualitative**. The regimes parallel the CIY admissibility regimes in 6.4. The underlying logic is Pearl's interventional/observational distinction applied to strategy-edge updates.

The identifiability coefficient ι_{ij}: **hypothesis**. The concept is sound (discounting by attribution confidence is standard in causal inference), but the specific form — a scalar multiplier on the gain — is a first-order approximation.

The depth-dependent degradation for indirect edges: **robust qualitative**. Each level of indirect inference adds an attribution step, and uncertainty compounds. The specific degradation rate is domain-dependent.

Discussion.

Why this gap matters. Without causal validity conditions, 8.4's status is ambiguous: it might be updating credences that claim to be interventional using evidence that is merely associational. The conditions make explicit what's needed for the update to preserve the interventional semantics of 7.3's edge credences.

Connection to 8.3. Observability gates whether the agent can *detect* an outcome. Identifiability gates whether the agent can *attribute* an outcome to a cause. Both are prerequisites for learning. The combined effective gain:

$$\eta_{\text{eff}} = \frac{U_{\text{edge}}}{U_{\text{edge}} + U_{\text{obs}}} \cdot \iota_{ij} \tag{8.14}$$

captures both gates in a single quantity. When either gate closes ($U_{\text{obs}} \to \infty$ or $\iota_{ij} \to 0$), effective gain goes to zero.

Connection to the signal function. The identifiability coefficient is one component of the unspecified signal function flagged in 8.4's working notes. The full signal function signal(o_t, i, j) decomposes into: (a) what outcome was observed (o_t), (b) how attributable is it to edge (i, j) (ι_{ij}), and (c) what does the attributed outcome imply about the edge's causal strength. This segment addresses (b); (a) and (c) remain open.

Software as Regime A. In software development, the agent runs a specific test (leaf action) and observes the result. C1 is satisfied (the agent ran the test), C2 is satisfied (the test targets one behavior), C3 is satisfied (CI runs regardless of conditions). Leaf-originating edges in software strategies have $\iota \approx 1$. This is the maximally favorable domain for causal edge updates.

Optimal decomposition depth revisited. 8.3 notes that finer decomposition (more intermediate nodes) provides earlier failure detection but adds uncertain edges via 7.1. This segment adds another cost of depth: deeper edges have lower ι_{ij}, making them harder to learn causally. The optimal decomposition depth balances three factors: (a) observability of intermediate nodes, (b) confidence decay through chains, and (c) identifiability degradation with depth.

Working Notes.

- The interaction between M_t updates and edge updates flagged in 8.4's working notes deserves attention. The orient cascade processes M_t first, then edge updates. Both use the same observation o_t. There may be statistical dependencies (double-counting of evidence) that bias the edge update.
- Can ι_{ij} be estimated online? In software (Regime A), it's nearly 1 by construction for leaf edges. In organizations (Regime B), a simple heuristic: $\iota_{ij} \approx 1/|\mathrm{pa}(j)|$ when parent edges fire concurrently (maximum-entropy attribution). This is crude but principled.
- **Regime C edges should be labeled.** An agent operating in Regime C should tag its edge credences as "observational" rather than "interventional." Observational credences should be trusted less in high-stakes decisions, and the agent should actively seek probe actions (high CIY) to promote edges from observational to interventional status.
- The depth-dependent degradation for indirect edges may make very deep strategies epistemically unlearnable: the leaves are learnable, edges one level up are partially learnable, but edges near the root may be effectively frozen regardless of observability. This would constrain the useful depth of strategy DAGs from the identification side, complementing the chain-confidence-decay constraint from the propagation side.

8.6 *Discussion:* Credit Assignment Boundary

The strategy-revision loop requires assigning credit for observed outcomes to specific edges in the strategy DAG — decomposing "the plan partially worked" into "step 3 failed, step 5 was irrelevant, step 1 succeeded." This is AAT's version of RL's temporal credit assignment problem. This segment characterizes the boundary between tractable and intractable cases, states what the theory requires of any credit-assignment scheme, and identifies what the theory can guarantee without solving the problem at all.

Formal Expression.

The Credit Assignment Problem for Strategy DAGs.

Given strategy DAG $\Sigma_t = (V, E, p, \gamma)$, an observed outcome at the root (and possibly at some intermediate nodes), produce per-edge signals signal(o_t, i, j) for each edge $(i, j) \in E$ such that the edge update

Discussion (credit-assignment-problem).

$$p_{ij}^{\text{new}} = p_{ij} + \eta_{\text{edge}} \cdot (\text{signal}(o_t, i, j) - p_{ij}) \tag{8.15}$$

drives credences toward truth. The problem is trivial when all intermediates are observable (each edge updates from its own observation) and genuinely hard when intermediates are unobservable and the DAG has shared structure.

Default Signal Function (Gradient-Based Attribution, Regime-Aware).

Any edge-update signal function decomposes along three independent axes: *what happened* at the child node, *whether that outcome is attributable* to the specific edge being updated, and *how causal* the evidence is:

Formulation (gradient-signal-function).

$$\text{signal}(o_t, i, j) = f\big(\text{outcome}(o_t, j),\ \text{attribution}(o_t, i, j),\ \text{regime}(i, j)\big) \tag{8.16}$$

The outcome component answers "what was observed at j?" The attribution component answers "can we credit that outcome to edge (i, j) specifically, or did other parents contribute?" The regime component answers "how causal is the evidence — is it interventional (Regime A), partially identified (Regime B), or observational (Regime C)?" This decomposition is derived from the regime-indexed edge semantics of 7.3 and 8.5: the same signal pipeline must carry the regime distinction through to the update.

AAT's default implementation, analogous to $\eta^* = U_M/(U_M + U_o)$ being the default gain. The update is stated in the log-odds coordinate $\lambda_k = \log(p_k/(1-p_k))$ — the unique additive-evidence coordinate forced by the evidential-additivity axiom (P). The probability-space form is the projected image via $p_k = \sigma(\lambda_k)$ at the readout interface:

$$\lambda_k^{\text{new}} = \lambda_k + \eta_{\text{edge}} \cdot \iota_k \cdot \frac{J_k \cdot (y_G - \hat{P}_\Sigma)}{\|\mathbf{J}\|^2}, \qquad p_k^{\text{new}} = \sigma(\lambda_k^{\text{new}}) \tag{8.17}$$

where: - $\mathbf{J} = \nabla_{\mathbf{p}} P_\Sigma$ is the plan-value gradient (computable from status propagation in $O(|V| + |E|)$) — supplying the *attribution* component - $(y_G - \hat{P}_\Sigma)$ is the plan-level residual — the *outcome* component - $\iota_k \in [0, 1]$ is the edge's identifiability coefficient (8.5) — the *regime* modulation, scaling the correction by the fraction of evidence that genuinely identifies this edge's causal effect - η_{edge} is the edge-level update gain (8.4); for Beta-Bernoulli the moment-parameter form recovers $\eta_{\text{edge}} = 1/(n_k + 1)$ under the sufficient-statistic correspondence

For Regime-A edges ($\iota_k \approx 1$), the log-odds update recovers the pure gradient-based form $\Delta\lambda_k = \eta_{\text{edge}} \cdot J_k(y_G - \hat{P}_\Sigma)/\|\mathbf{J}\|^2$. For Regime-C edges ($\iota_k \approx 0$), the update is essentially zero — no meaningful update is made because no meaningful causal evidence is available. Regime-B edges receive proportionally reduced updates that honestly reflect the weaker identification.

Why log-odds rather than probability-space. In probability space, the same update written as $\text{signal}_k - p_k$ can push p_k^{new} outside $[0, 1]$ when $\|\mathbf{J}\|^2$ is small — a mechanical break of the $[0, 1]$ probability domain. The log-odds coordinate has domain $\mathbb{R}$, so additive updates never escape the domain; the sigmoid projection at the readout interface guarantees $p_k^{\text{new}} \in (0, 1)$ by construction. The log-odds coordinate is the unique (up to positive affine transformation) parameterization on which Bayesian independent-evidence accumulation is additive (P); this makes it the natural presentation for any continuous-gradient update rule that aims to preserve Bayesian coherence. The probability-space presentation remains useful for interpretation but is the projected image, not the native update coordinate.

Properties: - **Domain closure (no mechanical break):** Updates live on $\lambda_k \in \mathbb{R}$, so no update magnitude can push credence outside the valid probability domain. The sigmoid projection $p_k = \sigma(\lambda_k) \in (0, 1)$ at the readout interface guarantees $[0, 1]$ boundedness by construction — not by clipping. The historical probability-space presentation required a normalization constant and could diverge when $\|\mathbf{J}\|^2 \to 0$; the log-odds presentation eliminates this failure mode. - **Directional fidelity (B1):** For $\iota_k > 0$, satisfies $\mathbb{E}[\Delta\lambda_k] \propto \iota_k \cdot J_k(\Phi - \hat{P}_\Sigma) \propto \iota_k \cdot J_k \cdot \delta_s$. Since $J_k \geq 0$ for monotone AND/OR DAGs and $\iota_k \geq 0$ by definition, the expected log-odds update pushes each edge's credence toward truth whenever evidence is available to push it. The probability-space image inherits directional fidelity through the monotonic sigmoid. - **Sector parameter:** $\alpha_s = \iota_k \cdot \eta_{\text{edge}}$ for componentwise corrections (regime-adjusted Prop B.5b); $\alpha_s = \iota_k \cdot \eta_{\text{edge}} / \kappa(\mathbf{J})^2$ for coupled corrections. Regime-C edges have $\alpha_s \approx 0$, making them effectively frozen — consistent with 8.3's treatment of unobservable edges. The sector parameter is Fisher-equivalent across coordinates (see Epistemic Status); the probability-space and log-odds-space statements of Props B.1–B.7 carry the same content. - **Computational cost:** $O(|V| + |E|)$ — the same forward pass that computes $\hat{P}_\Sigma$ also yields $\mathbf{J}$. The ι factors are per-edge domain parameters, not computed from the DAG structure. The sigmoid projection is $O(|E|)$ per update step. - **Relationship to RL:** This is the AAT analog of REINFORCE with a causal-identification weighting — the Jacobian $\mathbf{J}$ is the score function, $(y_G - \hat{P}_\Sigma)$ is the advantage, and ι_k is the causal-validity discount on each edge's update.

Correlated-failure interaction (L0 vs L1). The gradient signal operates at L0 of the Correlation Hierarchy (7.3). When the DAG is causally insufficient (the dominant real-world case), the residual $(y_G - \hat{P}_\Sigma)$ decomposes into per-edge miscalibration *plus* omitted correlation structure. $\hat{P}_\Sigma$ systematically overestimates success, making the residual systematically negative on failure. The gradient signal then attributes to individual edges what is actually causal insufficiency (missing common-cause nodes). The signal retains directional fidelity *on average* (it pushes edges downward when the plan is overconfident, which is the correct direction), but the per-edge attribution is contaminated. The principled fix is L1: add common-cause nodes to restore causal sufficiency, then apply gradient attribution to the augmented DAG. In the augmented DAG, the residual correctly decomposes into per-edge miscalibration because the correlation structure is explicitly represented.

Domains with richer observation structure can do better (Thompson sampling, full belief propagation, domain-specific attribution). The gradient-based signal is the *concrete Level 1 default* — the minimum viable credit-assignment scheme that satisfies the theory's requirements.

What the Theory Can Guarantee Without Solving Credit Assignment. Three results hold independently of any specific credit-assignment scheme:

1. Persistence is credit-assignment-free. Proposition B.5 in N shows that the sector condition transfers from per-edge credence space to **strategy-plan-confidence error** $\delta_s = \hat{P}_\Sigma - \Phi$ via the Jacobian $\mathbf{J} = \nabla_{\mathbf{p}} P_\Sigma$. The Jacobian is computable from status propagation in $O(|V| + |E|)$ — no outcome decomposition required. The persistence guarantee (whether the strategy's plan-level self-assessment can be maintained) does not depend on the agent's ability to attribute outcomes to edges. **Note:** this proves persistence of δ_s (strategy-plan-confidence error), not of $\delta_{\text{strategic}}$ (the per-edge calibration residual from 8.1). Extending persistence to $\delta_{\text{strategic}}$ requires solving the credit-assignment problem — that is the gap this segment characterizes.

2. The diagnostic framework is plan-level. The satisfaction gap (7.4), control regret (7.5), and the orient cascade ordering (9.1) operate on aggregate value, not per-edge quantities. They tell the agent *whether* the strategy is failing (and whether the failure is feasibility vs. optimality vs. calibration), without requiring per-edge attribution.

3. Observability-dominance identifies the tractable edges. 8.3 determines which edges have nonzero observability — only these can receive informative signals. Edges with zero observability are frozen regardless of the credit-assignment scheme. The tractable boundary is the observable subgraph of Σ_t.

The Tractable Cases.

Credit assignment is solved (exact, polynomial-time) when:

Discussion (tractable-credit-assignment).

Table 8.4

Condition	*Why tractable*	*Update rule*
All intermediates observable	Each edge has its own observation; updates decouple	Beta-Bernoulli per edge (Prop B.2)
Binary outcomes, independent edges, linear chain	Marginal Bayesian update = proportional blame	Prop B.3 (with plan-level fallback for unobservable)
Tree DAG, observable leaves	No shared descendants; message passing is exact	Belief propagation (standard)

The Intractable Cases: Three Independent Barriers.

Exact per-edge attribution in general AND/OR DAGs with partial observability faces three independent barriers:

Discussion (intractable-credit-assignment).

1. Computational intractability (#P-hardness). The "contribution of edge k to the observed outcome" has the form of a Shapley value over a cooperative game defined by the AND/OR propagation. Since AND/OR DAGs can represent any monotone Boolean function (including weighted threshold functions), and Shapley value computation for weighted voting games is #P-complete (Deng and Papadimitriou, 1994), exact attribution is #P-hard. *Caveat:* the reduction is to *exact* computation; approximate Shapley values are computable in polynomial time with sampling.

2. Information-theoretic underdetermination. When intermediates are unobservable, per-edge attribution is *under-determined*, not just hard. The identifiable subspace has dimension bounded by the number of observable nodes:

$$\dim(\mathcal{I}(\mathcal{V}_{\text{obs}})) \leq |\mathcal{V}_{\text{obs}}| \tag{8.18}$$

When $|\mathcal{V}_{\text{obs}}| < |E|$ (fewer observable nodes than edges), some directions in θ-space are fundamentally unresolvable from the available data. Any attribution in the unidentifiable directions relies on prior beliefs, not evidence.

3. The posterior correlation barrier. Even for approximately identifiable cases, any factored representation (independent Beta posteriors per edge) necessarily discards the correlation introduced by failure at multi-parent nodes. The exact posterior complexity grows exponentially with the number of observed failures. The factored representation is an approximation by construction — coupled corrections are inherent to the problem, not an artifact of a bad algorithm.

The Design Requirement.

The theory does not prescribe a specific credit-assignment scheme. It states what any scheme must satisfy for the persistence guarantees to hold:

Discussion (credit-assignment-design-requirement).

Minimal requirement (from 4.2): The per-edge signal function must have **directional fidelity** — the expected update for each edge must point toward the true credence:

$$\mathbb{E}[(\text{signal}(o_t, i, j) - p_{ij}) \cdot (p_{ij} - \theta_{ij})] \leq 0 \tag{8.19}$$

(the expected correction is non-positively correlated with the current error). This is the per-component version of condition B1 from the bridge theorem. Any signal function satisfying this produces sector-satisfying corrections that transfer losslessly to value space (Prop B.5b, componentwise case).

Sufficient condition for persistence: Per-component directional fidelity + bounded gain ($\eta_{\text{edge}} > 0$). The theory guarantees persistence when these hold, regardless of how the signals are computed.

What's NOT required: Exact attribution, unbiased estimation, minimum-variance estimation, or optimality of any kind. The persistence guarantee is robust to approximation — a sloppy but directionally correct signal function still produces bounded strategic mismatch. The *quality* of the approximation affects the *tightness* of the persistence bound (how close R^*_Σ is to zero), not whether persistence holds at all.

The Hierarchy of Credit Assignment Quality.

Table 8.5

Level	*Requirement*	*What it buys*	*Cost*
0 (none)	Plan-level tracking only	Persistence guarantee (Prop B.5)	No per-edge diagnostics
1 (directional)	Directional fidelity per edge	Persistence + rough per-edge diagnostics	Gradient computation $O(\lvert V\rvert + \lvert E\rvert)$
2 (approximate)	Proportional blame / expectation propagation	Persistence + per-edge diagnostics (with bias)	Factor-graph inference
3 (exact)	Full Bayesian posterior	Persistence + optimal per-edge calibration	#P-hard (general case)

AAT's formal guarantees require only Level 0. Practical agents need at least Level 1 for adaptive behavior — and the default signal function (above) provides a concrete Level 1 scheme. Level 2 is the sweet spot for most applications. Useful Level 2 factor-graph approximations include: exact Belief Propagation (BP) on tree or polytree cases, loopy BP or max-sum for MAP-style diagnosis, Expectation Propagation (EP) for approximate marginals, and structured variational methods only where common-cause structure is explicitly modeled. Level 3 is a mathematical ideal that is computationally unattainable in the general case.

Epistemic Status.

Mixed. The default signal function (gradient-based attribution, stated in log-odds) is a *formulation* — a concrete, well-motivated representational choice analogous to the gain principle's η^* formula. It satisfies directional fidelity for monotone AND/OR DAGs (derivable from Jacobian non-negativity). The log-odds presentation is *canonically selected* by the evidential-additivity axiom — the unique additive-evidence parameterization up to positive affine transformation (P). The boundary characterization (tractable cases, intractability barriers, design requirement) is *discussion-grade*, with the intractability argument at sketch level and the design requirement derived from the bridge theorem (4.2, N Prop B.5).

Historical note on the presentation choice. An earlier probability-space presentation of the default signal function, written as

$$\text{signal}_k = p_k + \iota_k \cdot J_k(y_G - \hat{P}_\Sigma)/\|\mathbf{J}\|^2 \tag{8.20}$$

with the update $p_k^{\text{new}} = p_k + \eta(\text{signal}_k - p_k)$, exhibited a mechanical break: when $\|\mathbf{J}\|^2 \to 0$ the signal magnitude could push p_k^{new} outside $[0, 1]$. The log-odds presentation above eliminates this failure mode by construction. Props B.1–B.7 of N remain as stated — their sector-parameter content is Fisher-equivalent across parameterizations, and the moment-parameter form is retained for algebraic clarity. The current segment writes the *default* signal in log-odds (native coordinate), with the moment-parameter form understood as the projected image via sigmoid.

Max attainable: *conditional* — with a formal intractability reduction, the boundary characterization could be promoted. The design requirement is already exact (it follows from the bridge theorem). The log-odds presentation is *derived-conditional* (on the evidential-additivity axiom) via P, structurally parallel to the reverse-KL uniqueness result under the chain-rule axiom in O §6.1.

Discussion.

AAT characterizes the structure, not the algorithm. The theory's contribution to credit assignment is not a solution but a characterization: when it's trivial (observable intermediates), when it's hard (general partial observability), what any solution must satisfy (directional fidelity), and what guarantees hold without it (persistence, plan-level diagnostics). This is the right level of ambition for a theory of adaptive systems — the specific algorithm is domain-specific engineering; the structural characterization is universal.

The analogy to the gain principle. 3.6 characterizes the optimal gain structure ($\eta^* = U_M/(U_M + U_o)$) without prescribing how U_M and U_o are estimated. A Kalman filter computes them exactly; an RL agent approximates them; a human intuits them. The gain *principle* is theory; the gain *estimator* is engineering. Credit assignment has the same structure: the *requirement* (directional fidelity) is theory; the *implementation* (proportional blame, gradient attribution, belief propagation) is engineering.

The observability lever. The most powerful insight from this analysis is that credit assignment is primarily an *observability design problem*, not an algorithm design problem. An agent that designs its strategy with observable intermediates (instrumented plans, staged rollouts, checkpoints) sidesteps the intractability entirely. This connects to 8.3's practical guidance: invest in making intermediate states observable, because unobservable regions are both epistemically dead (frozen credences) and computationally intractable (no efficient attribution).

OKRs as observability-by-design. The Objectives and Key Results framework is a direct organizational instantiation of this principle. The OKR discipline converts a deep, partially-unobservable strategy DAG (objective → vague initiatives → daily actions) into one with explicitly observable intermediate nodes (objective → measurable Key Results → tracked initiatives). In AAT terms: Key Results are intermediate nodes with $\sigma_v \approx 1$ by construction, making credit assignment between actions and objectives componentwise (Prop B.2) rather than intractable.

OKR failure modes map to AAT predictions:

Table 8.6

OKR Failure	*AAT Analog*	*Formal Quantity*	*Consequence*
Vanity metrics (measurable but irrelevant)	Observable node not causally connected to objective	High σ_v, low p_{ij}	Edge updates, but the edge doesn't lead where the agent thinks — correction effort is wasted
Too many Key Results	Wide OR-node exploration-gating	$\alpha_\Sigma \propto 1/k$ (Prop B.4)	Correction capacity diluted across alternatives; persistence threshold harder to meet
Lagging indicators	Evidence starvation by delay	$\nu_{\text{obs}} \ll \rho$	By the time the KR is measured, the correction window has passed; mismatch accumulated beyond R_Σ
Goodhart's Law (metric becomes the goal)	Terminal-condition misalignment with O_t	$V_{O_t}(\tau) < V_{O_t}^{\min}$ despite terminals achieved	Well-formedness constraint (7.3) violated; agent optimizes the intermediate rather than the objective

This is not an analogy — it is a domain instantiation. The same formal machinery that predicts when strategic persistence holds also predicts when OKRs work: when Key Results are genuinely causally connected to the Objective (p_{ij} is high and calibrated), few enough to monitor effectively (k is small), and measured on a timescale that permits correction (observation rate exceeds environment drift rate). The TST bridge to software team dynamics runs through exactly this connection.

Working Notes.

- A formal reduction from AND/OR credit assignment to Shapley value computation would promote the intractability claim from sketch to derived. The key step: mapping the AND/OR propagation to a weighted threshold game.
- The proportional-blame heuristic (attribute in proportion to prior credence) satisfies directional fidelity for independent edges. Does it satisfy directional fidelity in general? The two-edge analysis (B.3) shows it introduces O(1/n) bias — which means it has directional fidelity in expectation but not per-step. Whether the expected-value directional fidelity is sufficient for the sector condition (it should be, since the sector condition is also stated in expectation) is worth verifying formally.
- The connection between AAT's credit assignment and RL's temporal credit assignment (TD learning, eligibility traces) deserves formal treatment. Both face the same structural problem (partial observability of the causal chain) and both use similar heuristics (discounted attribution to recent actions). AAT's DAG structure is richer than RL's linear temporal chain, which may make the problem harder but also provides more structural information.
- The log-odds presentation of the default signal function (this segment) closes the Finding 2 mechanical break by construction (`audits/pending-findings-2026-04-22.md` §Finding 2, `spikes/spike-gbp1-logit-scoping.md`). Props B.1–B.7 remain in moment-parameter form because Fisher-equivalence keeps their sector-parameter content parameterization-agnostic; if a future pass moves the entire strategy layer to log-odds as its primary coordinate (full G-BP1 execution, likely paired with G-BP3 Fisher unification), the Prop B.1–B.7 restatements should follow. The scoping found the current narrow fix sufficient.

8.7 *Formulation:* Structural Change as Parametric Limit

In the probabilistic DAG, "structural" changes to Σ_t are continuous operations on edge weights and node sets — not a separate mechanism. Pruning is a credence dropping below threshold; **strategic grafting** is a new causal hypothesis initialized at a prior. This dissolves the sharp line between parametric update (adjusting weights) and structural change (adding/removing edges). *(The unqualified term "grafting" — familiar from horticulture, graph rewriting, and decision-tree learning — is sanctioned in-segment shorthand for "strategic grafting" once the canonical compound has been introduced.)*

Formal Expression.

The six operations on Σ_t, ordered from most to least frequent:

Formulation (structural-change-as-parametric-limit).

Table 8.7

Operation	*What changes*	*Trigger*
Reweighting	Edge credence p_{ij}	New observation about the link (8.4)
γ reclassification	Node combination type AND↔OR	Strong structural evidence that combination semantics changed
Pruning	Remove failed branch ($p_{ij} \to\approx 0$)	Credence drops below viability threshold
Strategic grafting	Add new branch ($0 \to p_{ij}$)	Discovery of a new possible path (initialized at prior)
Objective revision	Change terminal nodes	Feasibility failure or opportunity (7.4)
Full restructure	Replace entire Σ_t	Catastrophic failure (4.5)

A healthy agent does continuous strategic maintenance (reweight, occasionally prune and graft) and rarely reaches catastrophic restructuring. Full restructure is the strategic analog of 4.5's model-class change — the rare, expensive event when the entire representational structure must be replaced.

Epistemic Status.

Robust qualitative. The continuity claim (structural change as parametric limit) is a property of the probabilistic representation: in a space where edges carry real-valued credences, adding or removing an edge is a boundary event, not a discontinuity. The frequency ordering of operations is an empirical pattern, not a derived result — it's consistent with the observation that deeper changes (restructuring > pruning > reweighting) require more evidence and are more costly.

Discussion.

Connection to structural-adaptation necessity. 4.5 describes the rare case when the entire model class must be replaced — the agent's representational framework is fundamentally inadequate. In the strategy DAG, this corresponds to full restructure: the causal theory encoded in Σ_t is so wrong that incremental revision (reweighting, pruning) cannot fix it. The DAG must be rebuilt from a different starting point.

The key insight is that this extreme case is the *limit* of the continuous process, not a separate mechanism. An agent that maintains healthy strategic hygiene (regular reweighting, timely pruning of failing branches, proactive grafting of alternatives) will rarely need full restructure. An agent that neglects maintenance — letting failing branches persist, ignoring negative evidence — accumulates strategic debt until catastrophic restructuring becomes unavoidable.

Neutral variation as the constructive bridge. This segment claims that radical restructuring is the *limit* of continuous operations, but the mechanism by which small changes accumulate into radical transformation was unspecified. Miller (2022, *Ex Machina*) provides a constructive existence proof for exactly this bridge in finite-state automata. In Moore machines, each state has accessible and inaccessible regions. A mutation that alters a transition into an inaccessible region is *neutral* — the machine's observable behavior is unchanged because those states are never visited. But such mutations alter the machine's *latent structure*: they change what the machine *would do* if novel inputs were encountered. A sequence of neutral mutations can transform a machine from one behavioral equivalence class to a structurally different one without ever changing its observable behavior — until a single additional mutation opens a transition to the restructured region, causing a radical behavioral change. In Miller's terminology, this is the *extreme transition motif*: neutral invasion → neutral drift → niche creation → cascading displacement → consolidation. The automaton setting makes the mechanism precise: "inaccessible states" are the formal representation of latent structural capacity, "neutral mutations" are the incremental changes, and "a transition opening to a previously inaccessible region" is the parametric-limit event where continuous change produces discontinuous behavioral effect. The strategy-DAG analog: edges with near-zero credence ($p_{ij} \approx 0$) are the latent structure. They consume minimal cognitive cost (8.9) but represent alternative causal hypotheses that become load-bearing if circumstances change. An agent that prunes all low-credence edges for efficiency loses this latent diversity and becomes brittle to regime change.

Pruning and grafting thresholds. When should the agent prune (remove an edge with very low credence) rather than just keeping it at low p_{ij}? The answer involves the cognitive cost of maintaining edges in Σ_t — each edge consumes representational capacity and evaluation time. For agents with bounded representational capacity (LLMs with finite context windows), pruning is necessary to keep Σ_t within capacity. The threshold is domain-dependent and connects to the open question of cognitive cost of Σ_t (5.7 Working Notes).

Cross-agent grafting: symbiogenic composition. The grafting operation above is within-agent — the agent incorporates a new causal hypothesis into its own Σ_t, with the source (internal model, external information, exploration) being a stimulus rather than an integrating party. When the grafted structure originates in *another* agent and that agent's objective is also absorbed in the process, the mechanism is symbiogenic composition (10.3) — a bilateral cross-agent process rather than a unilateral within-agent operation. Biological example: mitochondria integrating into a host cell transfers gene-based structure (grafting on the host side) *and* subsumes the endosymbiont's independent objective (outside the scope of this segment's operations). Social analog: firm acquisitions integrate processes and decision-making authority simultaneously. The within-agent grafting operation is the host-side component of symbiogenesis, but symbiogenesis as a whole is governed by additional dynamics captured in 10.3.

Working Notes.

- The timescale ordering (reweight ≫ reclassify ≫ prune/graft ≫ revise O_t ≫ full restructure) is an empirical observation that should be testable. In software development: edge reweighting ≈ updating confidence after a test passes/fails; γ reclassification ≈ realizing two tasks are both required (not alternatives); pruning ≈ abandoning an approach that isn't working; grafting ≈ discovering a new approach; objective revision ≈ changing the feature scope; full restructure ≈ starting the project over.
- Grafting (adding new edges) is qualitatively different from other operations because it requires the agent to hypothesize a causal relationship that isn't in its current Σ_t. Where do new causal hypotheses come from? From M_t (the model suggests a possible path), from external sources (12.4 — another agent suggests an approach), or from exploration (3.7 — the agent discovers an unexpected effect). This is the creative step in strategic revision.

8.8 *Definition:* Strategic Tempo

The effective rate at which an agent acquires useful revisions to its strategy Σ_t — the sum of per-edge correction capacities across the strategy DAG, weighted by each edge's causal identifiability.

Formal Expression.

Definition (strategic-tempo).

$$\mathcal{T}_\Sigma = \sum_{(i,j)\in E} \nu_{ij} \cdot \eta_{\mathrm{edge},ij} \cdot \iota_{ij} \tag{8.21}$$

where: - (i, j) indexes the edges of the strategy DAG (7.3) - ν_{ij} is the effective observation rate for edge (i, j) — how often the agent obtains evidence about the causal link $i \to j$ - $\eta_{\mathrm{edge},ij}$ is the per-edge update gain (8.4) - $\iota_{ij} \in [0, 1]$ is the identifiability coefficient (8.5): the fraction of the evidence stream that genuinely identifies the edge's causal effect

Regime contributions. The identifiability coefficient captures the regime distinction from 8.5:

Table 8.8

Regime	ι_{ij}	*Contribution to* $\mathcal{T}_\Sigma$	*Example domain*
A Intervention-rich	≈ 1	Full: $\nu_{ij} \cdot \eta_{\mathrm{edge},ij}$	Software tests, laboratory experiments
B Partial intervention	$\in (0, 1)$	Reduced proportionally	Organizational actions with concurrent effects
C Observation-only	≈ 0	Near-zero: edges contribute negligibly	Passive monitoring, intelligence analysis

An agent cannot improve the parts of its strategy that it cannot test interventionally. This is the operational content of the ι factor: Regime-C edges contribute essentially nothing to $\mathcal{T}_\Sigma$ regardless of how fast the agent acts or how many observations it makes. Regime-A edges yield full strategic tempo at their observation rate. The ι factor is where edge-causal-validity (8.5) enters the operational machinery of strategy revision.

Parallel with epistemic tempo. The definition mirrors 3.8's $\mathcal{T} = \sum_k \nu^{(k)} \cdot \eta^{(k)*}$, replacing observation channels with strategy edges and adding the ι modulation for causal identifiability. The structural parallel is exact for Regime-A edges (where $\iota = 1$ recovers the direct rate-times-gain form); in mixed regimes, ι carries the additional content distinguishing interventional evidence from associational proxy.

Key difference: endogenous edge rates. Epistemic tempo's channel rates $\nu^{(k)}$ are largely exogenous — the environment generates observations at its own pace. Strategic tempo's edge rates ν_{ij} are *endogenous*: they depend on the agent's action policy (which edges get tested) and on upstream success (downstream edges are tested only when upstream edges fire). This endogeneity is the source of the structural differences between epistemic and strategic persistence.

Consistency verification. The definition is consistent with the four verified topologies from N:

Case B.1 (single edge $A \to G$). One edge, $\nu = \nu_{AG}, \eta_{\text{edge}} = 1/(n+1)$. $\mathcal{T}_\Sigma = \nu_{AG}/(n+1)$. The sector parameter $\alpha_\Sigma = 1/(n+1)$ is the per-observation correction quality; $\mathcal{T}_\Sigma = \nu \cdot \alpha_\Sigma$, matching the epistemic tempo pattern exactly.

Case B.2 (two-edge AND chain, $A \to B \to G$, B observable). Two edges. Edge 1 is tested at rate $\nu_1 = \nu$ (every execution). Edge 2 is tested only when edge 1 succeeds: $\nu_2 = \nu \cdot \theta_1$. The bottleneck edge has $\alpha_\Sigma = \min(1/(n_1 + 1),\ \theta_1/(n_2+1))$. $\mathcal{T}_\Sigma = \nu/(n_1+1) + \nu\theta_1/(n_2+1)$, consistent with depth-gated attenuation.

Case B.3 (two-edge AND chain, B unobservable). Per-edge tempo is ill-defined (the marginal point estimate is biased). Plan-level tempo is well-defined: $\mathcal{T}_{\Sigma,\text{plan}} = \nu/(n_\Phi + 1)$, treating $\hat{\Phi} = p_1 p_2$ as a single tracked quantity.

Case B.4 (two-arm OR node, ε-greedy). Edge 1 tested at rate $\nu_1 = \nu(1-\varepsilon)$, edge 2 at rate $\nu_2 = \nu\varepsilon$. $\mathcal{T}_\Sigma = \nu(1-\varepsilon)/(n_1+1) + \nu\varepsilon/(n_2+1)$. Action selection directly controls the rate allocation — exploration-gated, not depth-gated.

Structural decomposition. **AND-chains: depth-gated (geometric attenuation).** In a chain of depth d with edge success probabilities θ_k, the effective observation rate for edge k is:

Derived (Conditional on independent edges).

$$\nu_k = \nu \cdot \prod_{j<k} \theta_j \tag{8.22}$$

Each additional depth level attenuates by a factor $\theta_k < 1$. For a uniform chain ($\theta_k = \theta, n_k = n$ for all k):

$$\mathcal{T}_\Sigma = \frac{\nu}{n+1} \sum_{k=1}^{d} \theta^{k-1} = \frac{\nu}{n+1} \cdot \frac{1-\theta^d}{1-\theta} \tag{8.23}$$

This converges to $\nu/((n+1)(1-\theta))$ as $d \to \infty$ — total strategic tempo is bounded even for arbitrarily deep chains. The marginal tempo contribution of edge k decays as θ^{k-1}, falling below any fixed threshold at depth d^* (8.9). Deep AND-chains have low $\mathcal{T}_\Sigma$ at their leaves regardless of how fast the agent acts — the evidence-starvation effect identified in N.

OR-nodes: exploration-gated. At an OR-node with m alternatives under ε-exploration, the rate allocated to alternative l is:

Definition (OR-node rate allocation).

$$\nu_l = \begin{cases} \nu(1 - \varepsilon + \varepsilon/m) & l = l^* \text{ (greedy arm)} \\ \nu \cdot \varepsilon/m & l \neq l^* \text{ (exploratory arms)} \end{cases} \tag{8.24}$$

The bottleneck is the least-explored alternative. Pure greedy ($\varepsilon = 0$) gives $\nu_l = 0$ for non-greedy arms, making those edges permanently uncorrectable.

Per-edge persistence.

For Σ_t to persist, every edge must maintain bounded mismatch. The bottleneck condition is:

Derived (from persistence-condition applied per edge).

$$\forall (i,j) \in E: \quad \nu_{ij} \cdot \iota_{ij} \cdot \eta_{\text{edge},ij} > \frac{\rho_{\Sigma,ij}}{R_{\Sigma,ij}} \tag{8.25}$$

This is the per-edge analog of 14.5's per-dimension condition for M_t. The aggregate relationship between $\mathcal{T}_\Sigma$ and the average correction rate α_Σ is:

$$\alpha_\Sigma \leq \frac{\mathcal{T}_\Sigma}{|E|} \leq \mathcal{T}_\Sigma \tag{8.26}$$

(minimum $\leq$ average $\leq$ sum). Consequently, $\mathcal{T}_\Sigma > |E| \cdot \rho_\Sigma / R_\Sigma$ is *necessary* for persistence but not sufficient — the persistence condition is bottleneck-limited by the weakest edge, not governed by the aggregate.

Epistemic Status.

The definition itself is axiomatic — it names a quantity by analogy with 3.8. The consistency verification with the four cases from N is *derived* (conditional on Beta-Bernoulli dynamics). The AND-chain geometric attenuation is *derived* (conditional on independent edge outcomes). The OR-node exploration gating and identifiability adjustment are *hypotheses* in the general DAG case, though verified for the specific topologies above. The bottleneck-limited persistence observation is *derived* from 14.5's result applied to the strategy substate.

Max attainable: conditional. Currently conditional because the general DAG case (mixed AND/OR topologies, correlated edges) has not been verified.

Discussion.

Connection to 14.5. The per-edge persistence condition inherits the same structure as the per-dimension epistemic result: the weak edge is the bottleneck. Scalar $\mathcal{T}_\Sigma$ overestimates effective strategic adaptation for the same reason scalar $\mathcal{T}$ overestimates epistemic adaptation — it averages over heterogeneous correction capacities.

Three-way tradeoff. Strategic tempo competes with both epistemic tempo and exploitation for the agent's finite action capacity. Each action that tests a strategy edge (improving $\mathcal{T}_\Sigma$) is an action not spent gathering epistemic information (improving $\mathcal{T}$) or pursuing the current best action (exploitation). The allocation is addressed in 9.2 — the extended deliberation threshold is derived, but the broader three-way framing is discussion-grade. Deliberation (internal computation) supplements $\mathcal{T}_\Sigma$ via an internal channel distinct from action-generated evidence.

Software as Regime A. In software development, tests are genuine interventions with attributable outcomes ($\iota_{ij} \approx 1$). This makes software agents naturally high-$\mathcal{T}_\Sigma$ — a structural advantage identified but not yet formalized in the TST domain (`02-tst-core/`).

Working Notes.

- **Mixed AND/OR DAGs.** The structural decomposition treats AND-chains and OR-nodes separately. Real strategy DAGs interleave both. How the geometric attenuation (AND) and exploration gating (OR) interact in mixed topologies is unverified.

The per-edge persistence condition sidesteps this by treating each edge independently, but the *aggregate* behavior of $\mathcal{T}_\Sigma$ in mixed DAGs may exhibit interference effects (e.g., evidence starvation in an AND-chain feeding into an OR-node's greedy arm).

- **Optimal topology.** Given a fixed action budget ν, what DAG topology maximizes $\mathcal{T}_\Sigma$? Shallow OR-heavy structures maximize tempo (more edges are directly observable, no attenuation). Deep AND-chains minimize tempo. This may yield a principled argument for preferring flat, option-rich strategies over deep sequential plans — complementing the confidence-decay argument in 7.1.
- **Dynamic complexity.** As n_{ij} grows (more observations per edge), $\eta_{\text{edge},ij}$ shrinks (diminishing returns). Strategic tempo declines over time even in a static environment — the agent's corrections become smaller as edges converge. This is the correct qualitative behavior (converged edges need less correction) but means $\mathcal{T}_\Sigma$ is not a fixed property of the agent-environment pair.
- **Cross-reference to NeurIPS Paper 2.** The strategic-tempo aggregator is **bottleneck-not-sum** in NeurIPS 2026 Paper 2 ("A Unified Convergence Theory for Non-Stationary Reinforcement Learning", `~/src/neurips/02-unified-convergence-rl/`, §5 Lemma 4.2 / `#lem-forgetting`): $\mathcal{T}_\Sigma^{\text{bn,ss}} := \min_{(i,j)\in E} \nu_{ij} \cdot \iota_{ij} \cdot (1-\lambda_{ij})$. Adversarial disturbance concentrates on the slowest-forgetting / weakest-attribution / slowest-revisited element — the same weakest-link pattern as `#result-per-dimension-persistence` at strategic-tempo level. This segment's $\mathcal{T}_\Sigma = \sum_{(i,j)} \nu_{ij} \cdot \eta_{\text{edge},ij}$ is the throughput aggregation; the bottleneck form is the persistence-relevant aggregation. The two are compatible: throughput-sum bounds the budget of corrective work the agent can allocate; bottleneck-min bounds the rate at which the worst element survives adversarial concentration. See `msc/neurips-back-integration-2026-05-08.md` §1 Paper 2 entry 8.

8.9 *Formulation:* Cognitive Cost of Strategy

The complexity cost of maintaining an explicit strategy Σ_t, formulated via minimum description length and the information bottleneck principle — connecting DAG structure to the maintenance term C_{maintain} in the explicit strategy condition (6.6).

Formal Expression.

Strategy description length.

The minimum description length of a strategy DAG $\Sigma_t = (V, E, p, \gamma)$ (7.3) decomposes as:

Formulation (strategy-description-length).

$$\mathrm{DL}(\Sigma_t) = \mathrm{DL}_{\text{struct}}(G) + \mathrm{DL}_{\text{param}}(p \mid G) \tag{8.27}$$

where: - $\mathrm{DL}_{\text{struct}}(G)$: bits to encode the DAG topology — node identities, edge connectivity, AND/OR labels γ. Scales as $O(|E| \log|V|)$ for sparse DAGs. - $\mathrm{DL}_{\text{param}}(p \mid G)$: bits to encode the edge credences given the topology. For Beta-distributed credences, each edge requires $O(\log n_{ij})$ bits where $n_{ij} = \alpha_{ij} + \beta_{ij}$ is the effective sample size.

The total scales as $O(|E| \log|V|)$ for moderate-precision credences, growing linearly in the number of edges and logarithmically in the number of nodes.

Strategy IB objective.

The optimal strategy complexity balances parsimony against decision-relevance. Σ_t is the IB-compression of the interaction history $\mathcal{C}_t$ *for guidance*, parallel to M_t as the IB-compression of $\mathcal{C}_t$ *for prediction* (V for the shared IB shape across AAT's compression operations, and for the relationship between the theoretical $I(\mathcal{C}_t; \Sigma_t)$ compression cost and the operational DL-based minimization below):

Formulation (strategy-IB-objective; KL-direction strengthened by regret bound — see Epistemic Status).

Theoretical form (variational). Σ_t is a tractable variational approximation of the optimal-policy posterior $Q^*(\pi \mid M_t)$. The strategy-cost objective:

$$\Sigma_t^* = \arg\min_{\Sigma_t} \Big[I(\mathcal{C}_t;\, \Sigma_t) \;+\; \beta_\Sigma \cdot D_{\mathrm{KL}}\big(\pi^*(\cdot \mid M_t) \,\big\|\, Q_{\Sigma_t}(\pi \mid M_t)\big) \Big] \tag{8.28}$$

where $Q_{\Sigma_t}(\pi \mid M_t)$ is the action distribution induced by the strategy DAG given the current model state, and $\pi^*(\cdot \mid M_t)$ is the optimal-policy reference. The KL direction — π^*-first — is forced by the regret-bound derivation (next paragraph); the opposite direction is vacuous under deterministic π^*.

Regret-bound derivation of KL direction. Under AAT's canonical scope, $\pi^* = \delta_{a^*}$ is deterministic (5.6). Define the strategy-induced regret against π^* as $R(Q_{\Sigma_t}) := V(a^*) - \mathbb{E}_{a \sim Q_{\Sigma_t}}[V(a)]$, where $V(a) = Q_O(M_t, a; \pi_{\mathrm{cont}}, N_h)$ is the action-value (5.6, O_t induces V via 5.4). Under bounded value range $V_{\max} := \max_a V(a) - \min_a V(a)$:

$$R(Q_{\Sigma_t}) \;\le\; V_{\max} \cdot (1 - Q_{\Sigma_t}(a^*)) \;=\; V_{\max} \cdot \mathrm{TV}(\pi^*, Q_{\Sigma_t}) \tag{8.29}$$

Applying Pinsker's inequality ($\mathrm{TV}(P, Q) \le \sqrt{\tfrac{1}{2} D_{\mathrm{KL}}(P \| Q)}$) with $P = \pi^*$, $Q = Q_{\Sigma_t}$:

$$R(Q_{\Sigma_t}) \;\le\; V_{\max} \cdot \sqrt{\frac{1}{2}\, D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})} \tag{8.30}$$

Under deterministic π^*, $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t}) = -\log Q_{\Sigma_t}(a^*)$ — finite and graded whenever $Q_{\Sigma_t}(a^*) > 0$. The opposite-direction $D_{\mathrm{KL}}(Q_{\Sigma_t} \| \pi^*)$ equals $+\infty$ whenever Q_{Σ_t} places any mass off a^*, giving a vacuous bound. The regret-bound argument therefore **forces the KL direction** with π^* first. Within the direction-forced f-divergence family, reverse-KL is *uniquely* selected under the chain-rule additivity axiom (Hobson 1969; Csiszár 1991 Theorem 3 corollary and Theorem 5; standard functional-equation derivation per Aczél & Daróczy 1975), which is AAT-internally motivated as the divergence-level analog of additive log-confidence decay (7.1). See O §6.1 for the uniqueness theorem, §6.2 for secondary supporting characterizations (gradient-tractability, VI-alignment, MDL), and §7 for the linear-vs-square-root β_Σ trade-off.

The variational form is the strategy-layer analog of variational free energy minimization in active inference (Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017, "Active inference: a process theory," *Neural Computation* 29; Da Costa, Parr, Sajid, Veselic, Neacsu & Friston 2020, "Active inference on discrete state-spaces," *J. Math. Psych.* 99; Parr & Pezzulo 2022, *Active Inference*, MIT Press). AAT borrows the variational form as the appropriate generalization of the Shannon-MI relevance term and now derives the direction of KL from an internal regret-bound argument — without committing to AI's preferences-as-priors encoding ($C(o) = \log P_{\mathrm{pref}}(o)$; AAT's O_t remains a value functional on trajectories, 5.4) or to expected free energy as master objective (AAT's CIY-unified objective is a related but distinct decomposition; 6.5).

Operational form. Since $I(\mathcal{C}_t; \Sigma_t)$ is not computable in closed form for general DAG encodings, the operational minimization replaces the information cost with a description-length surrogate and the KL term with a sample-based estimate (a per-edge calibration discrepancy weighted by decision-relevance — see 8.6 for the gradient form):

$$\Sigma_t^* \approx \arg\min_{\Sigma_t} \left[\mathrm{DL}(\Sigma_t) + \beta_\Sigma \cdot \widehat{D_{\mathrm{KL}}}(\pi^* \parallel Q_{\Sigma_t}) \right] \tag{8.31}$$

where: - $\mathrm{DL}(\Sigma_t)$: description length (coding-cost upper bound on $I(\mathcal{C}_t; \Sigma_t)$ for the given DAG encoding scheme — see §2.2 below) - $\widehat{D_{\mathrm{KL}}}(\pi^* \parallel Q_{\Sigma_t})$: sample-based estimate of the KL divergence from the optimal-policy reference to the strategy-induced policy - $\beta_\Sigma > 0$: trade-off parameter — cognitive cost per decision-relevant bit (the Σ_t instance of the shared β framework in V); under the regret-bound derivation, β_Σ has a *local* interpretation as $V_{\max}/(2\sqrt{2D_{\mathrm{KL}}})$ via the Pinsker form (the linear-KL form is the IB-shape instance; the square-root form is the tighter regret-scale form — see Epistemic Status and the spike for the trade-off)

The two forms agree in the limit where the DAG encoding is rate-distortion optimal and the policy posterior is sample-recoverable; the operational form is the one an agent actually runs. The theoretical form places the objective on the same variational frontier as M_t, shared intent, and composition projection, with the π^*-first KL-form relevance term resolving the Shannon-zero degeneracy under deterministic π^* *and* the forward-KL infinity degeneracy that the opposite direction would introduce.

When β_Σ is low (high maintenance cost relative to decision value), the agent prefers simple strategies. When β_Σ is high (strategy is cheap to maintain relative to its decision value), the agent can afford complex plans. The explicit strategy condition (6.6) is the binary threshold: β_Σ large enough that *any* Σ_t is worth maintaining.

Maximum useful chain depth.

From 8.8's per-edge persistence condition, an AND-chain of depth d with per-edge observation rate ν, true success probability θ per edge, and effective sample size n per edge persists only if the deepest edge satisfies:

Derived (Conditional on Beta-Bernoulli, per-edge persistence).

$$\nu \cdot \theta^{d-1} \cdot \frac{1}{n+1} > \frac{\rho_\Sigma}{R_\Sigma} \tag{8.32}$$

Solving for the maximum depth at which persistence is achievable:

$$d^* = 1 + \left\lfloor \frac{\log\left(\frac{\nu}{(n+1)\rho_\Sigma/R_\Sigma}\right)}{\log(1/\theta)} \right\rfloor \tag{8.33}$$

When $\nu/((n+1)\rho_\Sigma/R_\Sigma) \leq 1$, even $d = 1$ fails — the agent cannot maintain a single edge under these conditions.

Interpretation. Beyond depth d^*, evidence starvation makes edges uncorrectable faster than the environment invalidates them. The agent accumulates strategic mismatch on deep edges regardless of how fast it acts at the top of the chain.

Quantitative illustration ($\theta = 0.8, \nu = 1$):

Table 8.9

n	ρ_Σ/R_Σ	d^*
10	0.01	10
10	0.1	0
100	0.01	5
100	0.1	0

High evidence requirements (n large) and volatile environments (ρ_Σ / R_Σ large) severely limit useful chain depth.

Triple depth penalty. Deep AND-chains suffer three independent penalties that compound:

1. **Confidence decay** (7.1): aggregate confidence $\prod p_k$ decays geometrically with depth. The plan is *less likely to succeed*.
2. **Evidence starvation** (N): effective observation rate $\nu_k = \nu \cdot \prod_{j<k} \theta_j$ decays geometrically. The plan is *harder to calibrate*.
3. **Cognitive cost** (this segment): each additional depth level adds $O(\log|V|)$ bits to description length. The plan is *more expensive to maintain*.

All three are multiplicative in depth, making deep sequential strategies exponentially costly along three independent dimensions.

Enriched explicit strategy condition.

The maintenance cost C_{maintain} from 6.6 decomposes as:

Formulation (enriched-strategy-condition).

$$C_{\text{maintain}} = C_{\text{represent}} + C_{\text{revise}} + C_{\text{monitor}} \tag{8.34}$$

where: - $C_{\text{represent}} \propto \text{DL}(\Sigma_t)$: cognitive cost of holding the strategy in working memory (proportional to description length) - $C_{\text{revise}} \propto \sum_{(i,j)} \nu_{ij} \cdot c_{\text{update}}$: cost of processing edge updates (proportional to strategic tempo $\mathcal{T}_\Sigma$ times per-update cost) - $C_{\text{monitor}} \propto |\{(i,j) : \iota_{ij} < 1\}|$: cost of monitoring edges with partial identifiability (the agent must do extra causal reasoning for non-trivial edges)

This decomposition makes the 6.6's maintenance term concrete: each component maps to a quantity defined elsewhere in the theory.

Complexity compression operations.

The IB objective suggests three compression operations, corresponding to structural changes from 8.7:

Discussion (complexity-compression).

1. **Edge pruning** (operation 3 in 8.7): remove edges with $\eta_{\text{edge},ij} \cdot I_{\text{edge},ij} < c_{\text{bit}}$, where $I_{\text{edge},ij}$ is the decision-relevance of that edge and c_{bit} is the per-bit maintenance cost. Edges that contribute less decision value than their representational cost are candidates for removal.
2. **Node merging** (reducing $|V|$): collapse intermediate nodes that serve no decision-distinguishing function. This reduces $\text{DL}_{\text{struct}}$ by a factor proportional to the reduction in $|V|$.
3. **Depth truncation** at d^*: prune all edges beyond the maximum useful depth. This is not optimization but necessity — edges beyond d^* cannot maintain bounded mismatch.

Epistemic Status.

The description length formulation is a *formulation* — it applies standard MDL to the strategy DAG, which is a representational choice not a derived necessity. The IB objective is *formulation (strengthened by regret bound)* in its variational form (above): the specific KL *direction* (π^*-first, i.e., reverse-KL in the variational-inference vocabulary) is *derived* as an upper regret bound via Pinsker's inequality under bounded value range and deterministic π^* (see Regret-bound derivation paragraph above; full derivation in appendix O). The choice of reverse-KL *within* the direction-forced family upgrades from canonical-formulation to **derived (conditional on chain-rule additivity axiom)** — the chain-rule axiom (Hobson 1969; Csiszár 1991 Theorem 3 corollary and Theorem 5; standard functional-equation derivation per Aczél & Daróczy 1975) picks reverse-KL uniquely among f-divergences, and the axiom is AAT-internally motivated as the divergence-level analog of 7.1 (see O §6.1). Secondary properties (gradient-tractability, variational-inference alignment with Friston et al. 2017, Da Costa et al. 2020, Parr & Pezzulo 2022; MDL coding; compatibility with Amari & Nagaoka 2000 Fisher geometry) are convergent evidence rather than independent uniqueness grounds — in particular, Fisher-metric-at-second-order is *not* distinguishing within f-divergences (Eguchi 1983, *Ann. Statist.* 11(3):793–803). The regret-bound derivation closes the direction ambiguity that the earlier V-medium move (commit `a14682e`) left open: the initial V-medium form used $D_{\mathrm{KL}}(Q_{\Sigma_t} \| \pi^*)$ (forward-KL), which is $+\infty$ under deterministic π^* whenever Q_{Σ_t} has any off-optimum mass — a different-valued but structurally identical degeneracy to the Shannon-MI zero it replaced. The π^*-first reverse-KL direction escapes both degeneracies by construction.

The variational form replaces an earlier Shannon-MI form $-\beta_\Sigma \cdot I(\Sigma_t;\, \pi^* \mid M_t)$ which had a Shannon-zero degeneracy: when π^* is a deterministic function of M_t (the standard scope), Shannon mutual information to a constant vanishes identically, collapsing the objective to arg min $\mathrm{DL}(\Sigma_t)$. The π^*-first KL form does not have this degeneracy and is graded whenever Q_{Σ_t} places any mass on a^*. The maximum useful depth d^* is *derived* conditional on Beta-Bernoulli dynamics and the per-edge persistence condition from 8.8. The triple depth penalty is an *observation* combining results from three independent segments. The enriched maintenance decomposition is *formulation*. The compression operations are *discussion-grade*.

On β_Σ interpretation. Under the Pinsker-reverse-KL bound $R(Q_{\Sigma_t}) \le V_{\max}\sqrt{\frac{1}{2} D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})}$, a square-root-in-KL trade-off would naturalize β_Σ globally as $\beta_\Sigma \propto V_{\max}$. The segment retains the linear-in-KL form (to preserve the rate-distortion-Lagrangian IB shape shared with V); under the linear form, β_Σ has only a *local* regret-bound interpretation ($\partial R / \partial D_{\mathrm{KL}}$ at the operating point). This is a trade-off between IB-shape alignment and regret-scale naturalization. Under AAT's canonical deterministic-π^* scope, the sharper Bretagnolle-Huber identity $R \le V_{\max}(1 - e^{-D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})})$ holds as the primary form (O §4); Pinsker is retained here for IB-shape alignment and as the loose general form correct for stochastic-π^* extensions.

Assumption explicitly stated: bounded value range. The regret-bound derivation requires $V_{\max} := \max_a V(a) - \min_a V(a) < \infty$ over $\mathcal{A}$ at fixed M_t. This is mild but not automatic — 5.4 specifies V_{O_t} : trajectories $\to \mathbb{R}$, and bounded range at fixed state is an additional assumption stated here.

Max attainable: *robust-qualitative* for the IB objective with the direction-forced derivation; conditional for the specific functional form (linear vs. square-root in KL). The DL formulation is standard; the depth bound could reach exact status for specific edge models. The regret-bound derivation does not extend to stochastic π^* (outside AAT canonical scope); see 5.6 continuation conventions for scope.

Discussion.

Connection to 6.6. The enriched maintenance decomposition gives the cost inequality quantitative content: an agent can now *compute* whether a proposed strategy is worth maintaining by estimating its description length, revision cost, and monitoring burden, and comparing against the exploration/repair cost of operating without it.

LLM context windows as DL constraint. For language-constituted agents (`03-llm-core/`), the strategy must fit in the context window. A context window of W tokens imposes $\mathrm{DL}(\Sigma_t) \le W \cdot \log_2(|\mathrm{vocab}|)$ as a hard constraint.

This makes the IB trade-off non-optional: the agent *must* compress its strategy, and the depth bound d^* becomes a context-window-limited quantity. A 128K-token context window may support a 500-edge DAG encoded in natural language; a 4K-token window may support only a 15-edge sketch.

Computational compression from interaction horizon (Miller 2022). The maximum useful depth d^* derived above constrains complexity from the *maintenance* side (evidence starvation). Miller (2022, *Ex Machina*, Table 12.2) provides a complementary constraint from the *interaction* side: the number of interaction rounds compresses the space of behaviorally distinguishable strategies regardless of the agent's internal complexity. For Moore machines with binary actions:

Table 8.10

Agent states	*Unique computations*	*After 1 round*	*After 2 rounds*	*After 4 rounds*
1	2	2	2	2
2	26	2	8	26
3	1,054	2	8	690
4	57,068	2	8	5,936

The pattern: effective complexity $=$ min(agent complexity, interaction-horizon complexity). With only one round, even a four-state machine (57,068 unique computations) reduces to two distinguishable behaviors — equivalent to a one-state machine. With two rounds, all machines with two or more states reduce to eight distinguishable behaviors. The interaction horizon compresses the strategy space from above, just as the maintenance cost and evidence starvation compress it from below. For AAT's strategy DAG, the analog: a complex Σ_t whose edges are only tested over a short horizon gains nothing from its depth — the untested structure is indistinguishable from a simpler strategy. This reinforces the d^* bound and provides empirical grounding beyond the Beta-Bernoulli derivation.

Strategy simplification pressure. The triple depth penalty creates systematic pressure toward shallow, wide (OR-heavy) strategies over deep, sequential (AND-heavy) ones. This aligns with the structural pressure identified in 7.1's Discussion, now grounded in three independent mechanisms rather than one.

Working Notes.

- **Mixed topologies.** The depth bound d^* assumes uniform AND-chains. Mixed AND/OR DAGs have heterogeneous depth penalties: OR-nodes reset the evidence-starvation clock (each alternative is tested independently), while AND-nodes compound it. The effective depth for computing d^* may be the longest AND-chain in the DAG, not the total graph depth.
- **Optimal topology.** Given a fixed DL budget and action rate ν, what DAG topology maximizes decision-relevant information $I(\Sigma_t; \pi^* \mid M_t)$? This is a combinatorial optimization over graph structures — likely NP-hard in general but potentially tractable for specific graph families (trees, bounded treewidth).
- **Dynamic complexity.** As edges converge (high n_{ij}), their per-edge $\eta_{\text{edge},ij}$ shrinks, but their description length DL_{param} grows (more bits to encode the precise credence). The IB objective would favor *dropping* converged edges (they contribute little decision-relevant information since the agent already acts correctly on them), replacing them with a default "high confidence" summary. This is a principled version of "stop tracking what you already know."
- **Stochastic $\mathcal{T}_\Sigma$.** If strategic disturbance follows Model S (stochastic) rather than Model D (deterministic), the steady-state mismatch scales as $\rho_\Sigma/\sqrt{\mathcal{T}_\Sigma}$ rather than $\rho_\Sigma/\mathcal{T}_\Sigma$. The depth bound d^* would change accordingly. Not yet derived.

8.10 *Proposed Schema:* Proposed-schema: Strategy Persistence Schema

The sector-persistence template (C) proves bounded state for any system with a state variable, a correction function satisfying the sector condition, and bounded disturbance. The template is domain-agnostic — it applies to any state variable meeting its preconditions (T1)–(T3). This schema is the strategic-layer instantiation: if strategic update dynamics satisfy the template's preconditions, strategy persistence follows as a direct instance. A key additional requirement — absent from the epistemic instantiation but load-bearing here — is **experience discounting**, because the strategic sector parameter α_Σ decays monotonically with experience and requires an explicit forgetting mechanism to remain bounded below.

Formal Expression.

If strategic update dynamics satisfy the template preconditions (T1)–(T3) of C for $\xi = \delta_\Sigma$ (a strategic mismatch state), together with:

Proposed Schema (strategy-persistence-schema, from sector-persistence-template).

- **(SA1)** Zero correction at zero strategic mismatch (the template's (T1)): when the mismatch state is zero, no revision occurs
- **(SA2')** Local sector condition on strategic correction (the template's (T2) for $\xi = \delta_\Sigma$): the correction function points inward with baseline efficiency α_Σ within a strategic reserve R_Σ
- **(SA3)** Sufficient exploration (OR-nodes only): the action selection policy allocates correction capacity to all OR alternatives at a rate exceeding the strategic disturbance-to-reserve ratio
- Bounded strategic disturbance at rate ρ_Σ (the template's (T3)): the rate at which the environment invalidates causal links is bounded

Then Σ_t persists iff:

$$\alpha_\Sigma > \frac{\rho_\Sigma}{R_\Sigma} \tag{8.35}$$

directly by the template's Model D result. Here α_Σ is the strategic correction rate, ρ_Σ is the strategic disturbance rate, and R_Σ is the strategic reserve (tolerance for strategic mismatch before performance degrades catastrophically). The Model S instantiation replaces ρ_Σ with σ_Σ and gives $\alpha_\Sigma > n\sigma_\Sigma^2/(2R_\Sigma^2)$ under the same template.

Forgetting as Prerequisite.

The schema form above is an **instantaneous persistence check at the current experience level**, not a trajectory guarantee. For Beta-Bernoulli edge updates (the canonical verified case; see N Props B.1–B.6), the sector parameter has the form:

Formulation (forgetting-prerequisite).

$$\alpha_\Sigma = \frac{1}{n+1} \tag{8.36}$$

where n is the edge's accumulated experience (pseudo-count). Without a forgetting mechanism, n grows monotonically with each observation, so $\alpha_\Sigma \to 0$ for every edge asymptotically. For any fixed (ρ_Σ, R_Σ) with $\rho_\Sigma > 0$, every agent eventually violates the threshold. The structural identity with 4.4 — where α can be stationary — holds for the strategic case only under an explicit forgetting mechanism.

Exponential forgetting. Replace the raw Beta-Bernoulli update with a discounted update: at each step, shrink the pseudo-counts by a factor $\lambda \in (0, 1)$:

$$\alpha_k \mapsto \lambda\,\alpha_k + y_k, \qquad \beta_k \mapsto \lambda\,\beta_k + (1 - y_k) \tag{8.37}$$

The effective sample size stabilizes at $n_{\text{eff}} = 1/(1 - \lambda)$, and substituting into Prop B.1's $\alpha_\Sigma = 1/(n + 1)$ gives the exact steady-state sector parameter:

$$\alpha_\Sigma^{\text{ss}} = \frac{1}{n_{\text{eff}} + 1} = \frac{1 - \lambda}{2 - \lambda} \tag{8.38}$$

For slow forgetting ($\lambda \to 1$, the regime where the prerequisite is most likely to bind), the leading-order expansion gives the simpler form $\alpha_\Sigma^{\text{ss}} \approx 1 - \lambda$ — which agrees with the forgetting rate itself. Outside the high-λ regime the approximation deteriorates: at $\lambda = 0.5$, the exact form gives $\alpha_\Sigma^{\text{ss}} = 1/3$ while the linear approximation gives $1/2$ ($\approx$ 50% overestimate); at $\lambda = 0.9$, exact $\alpha_\Sigma^{\text{ss}} = 1/11$ versus approximation $1/10$ ($\approx$ 10% overestimate).

The forgetting prerequisite. Combining with the schema's persistence form $\alpha_\Sigma > \rho_\Sigma / R_\Sigma$:

$$\frac{1 - \lambda}{2 - \lambda} \;>\; \frac{\rho_\Sigma}{R_\Sigma} \quad\iff\quad \lambda \;<\; \frac{R_\Sigma - 2\rho_\Sigma}{R_\Sigma - \rho_\Sigma} \tag{8.39}$$

(valid when $\rho_\Sigma < R_\Sigma/2$; for $\rho_\Sigma \geq R_\Sigma/2$ no λ satisfies the prerequisite and the schema's trajectory guarantee fails for any forgetting rate).

This is a **prerequisite of the schema's trajectory guarantee, not a tunable heuristic**. An agent without forgetting has no long-run strategic persistence regardless of its initial α_Σ. The steady-state sector parameter must exceed the disturbance-to-reserve ratio, or the instantaneous persistence check — no matter how comfortably it holds at any given time — eventually fails as experience accumulates.

In the slow-forgetting regime the threshold simplifies to the linear-form analog of 4.4:

$$(1 - \lambda) \;>\; \frac{\rho_\Sigma}{R_\Sigma} \qquad \text{(slow-forgetting limit, } \lambda \to 1\text{)} \tag{8.40}$$

playing the role that $\mathcal{T} > \rho / \|\delta_{\text{critical}}\|$ plays for the epistemic case. The forgetting rate $(1 - \lambda)$ is the strategic analog of adaptive tempo: faster forgetting means faster tracking but noisier estimates; slower forgetting means stable estimates but slower tracking. The optimal λ balances bias and variance for the specific ρ_Σ the environment presents. Outside the slow-forgetting regime, the exact form $(1 - \lambda)/(2 - \lambda) > \rho_\Sigma / R_\Sigma$ is the operating threshold and the simpler linear form is unsafe (it permits λ values that violate the actual prerequisite).

Which mismatch state? The schema applies to any mismatch state for which conditions (SA1)-(SA3) can be verified. Two candidates exist:

- **Strategy-plan-confidence error $\delta_s = \hat{P}_\Sigma - \Phi$**: the scalar difference between the agent's strategy-plan-confidence score and the independence-model plan value at true edge parameters. This is the mismatch for which persistence IS proved (Props B.1-B.5 in N). It is computable from status propagation without credit assignment. **Scope:** δ_s operates at L0 of the Correlation Hierarchy (7.3) — it tracks calibration within the independence model. For L1 (augmented DAG), the same persistence result applies to the augmented graph's $\hat{P}_\Sigma$. The gap between L0's Φ and actual plan success under correlated failure is a model-class limitation, not an estimation error.

- **Strategic-calibration residual** $\delta_{\text{strategic}}$: the per-edge value-increment residual aggregation defined in 8.1. This is the mismatch the orient cascade (9.1) uses for edge-level revision. Persistence of $\delta_{\text{strategic}}$ remains **open** and requires the credit-assignment machinery in 8.6.

The verified instances below all use per-edge credence error $\delta_c = (\hat{p}_k - \theta_k)$ or the plan-level surrogate δ_s. They do not verify the schema for $\delta_{\text{strategic}}$ directly.

Epistemic Status.

Sketch, with verified instances. This is a **result schema**, not a proven result in the general case. The mathematical template (sector conditions → bounded mismatch) is derived (A). What was missing was instantiation — showing that specific strategic update dynamics satisfy the template's preconditions. Four cases have now been verified (full derivations in N):

1. **Single edge, Beta-Bernoulli** ($A \to G$): Sector condition satisfied globally with $\alpha_\Sigma = 1/(n+1)$. The bound is tight (expected correction is exactly linear). (A1) satisfied. Persistence condition: $1/(n+1) > \rho_\Sigma / R_\Sigma$.
2. **Two-edge chain, observable intermediate** ($A \to B \to G$, B observable): Sector condition satisfied globally with $\alpha_\Sigma = \min(1/(n_1+1), \theta_1/(n_2+1))$ — a weakest-link result. Correction function is diagonal (no cross-edge coupling). (A1) satisfied. The θ_1 factor in edge 2's rate is the evidence-starvation effect.
3. **Two-edge chain, unobservable intermediate** ($A \to B \to G$, B not observable): Per-edge sector condition **fails** — the marginal Bayesian update violates (A1) with bias $O(1/n)$. But plan-level tracking (treating $\hat{\Phi} = p_1 p_2$ as a single Beta) recovers the sector condition with $\alpha_{\Sigma,\text{plan}} = 1/(n_\Phi + 1)$, at the cost of per-edge diagnostic resolution.
4. **Two-arm OR-node, ε-greedy** ($A_1, A_2 \to G$, G is OR): Sector condition satisfied with $\alpha_\Sigma = \min((1-\varepsilon)/(n_1 + 1),\ \varepsilon/(n_2+1))$ — an **exploration-gated** weakest-link, not depth-gated as in AND chains. (SA3) required: minimum exploration rate $\varepsilon > \rho_\Sigma (n_{\max} + 1)/R_\Sigma$. Pure greedy ($\varepsilon = 0$) **fails** the sector condition. With optimal equal-rate allocation, $\alpha_\Sigma = 1/(n_1 + n_2 + 2)$ — the correction capacity is split across alternatives.

The schema is no longer purely hypothetical. The sector parameter for strategic dynamics is the edge update gain η_{edge} — the same quantity that governs epistemic persistence. The structural parallel between epistemic and strategic persistence is not an analogy but a mathematical identity at the sector-framework level.

Discussion.

What's needed to promote this from schema to result.

1. **Strategic mismatch state**: partially resolved. Prop B.5 in N shows the sector condition transfers from per-edge credence error to **strategy-plan-confidence error** $\delta_s = \hat{P}_\Sigma - \Phi$ — the scalar difference between the agent's strategy-plan-confidence score and the independence-model plan value at true edge parameters (note: Φ is NOT actual plan success probability under correlated failure — see 7.3 edge-independence caveat). For linear correction (Beta-Bernoulli), the transfer is exact ($\alpha_s = \alpha_c$); for nonlinear correction, $\alpha_s \geq \alpha_c / \kappa(\mathbf{J})^2$. **However**, δ_s is distinct from the **strategic-calibration residual** $\delta_{\text{strategic}}$ defined in 8.1, which is an L^2 aggregation of per-edge value-increment residuals requiring credit assignment to compute. Persistence of δ_s (plan-level tracking) is proved; persistence of $\delta_{\text{strategic}}$ (per-edge diagnostics) remains open and requires the credit-assignment machinery in 8.6.
2. ~~**Strategic correction function**: needs to satisfy the sector condition.~~ **Resolved** for Beta-Bernoulli edges. Props B.1-B.4 in N verify the sector condition for four topologies (single edge, two-edge AND observable/unobservable, two-arm OR).
3. **Strategic disturbance**: The rate at which the environment invalidates causal links in Σ_t. **Still open** as a formalized quantity — currently a domain parameter (ρ_Σ), analogous to how ρ for epistemic disturbance is a domain parameter in 4.4.

4. ~~**Sector condition verification**: the critical mathematical step.~~ **Resolved** for four topologies. See N.
5. ~~**Credit assignment / signal function**: needed for edge updates.~~ **Characterized at the theory level.** 8.6 shows persistence does not require credit assignment (Prop B.5), establishes directional fidelity as the minimal requirement, and provides a gradient-based default signal function. The specific update algorithm is domain engineering, not theory — the same way the gain *estimator* is domain engineering while the gain *principle* ($\eta^* = U_M/(U_M + U_o)$) is theory. Caveat: the default gradient signal inherits $\hat{P}_\Sigma$'s overestimation bias under correlated failures (7.3, 8.6).
6. **Time-varying α_Σ**: this is where the strategic case genuinely differs from the epistemic one. For Beta-Bernoulli edges, $\alpha_\Sigma = 1/(n+1)$ decays monotonically with experience, so the sector-persistence template's constant-α precondition cannot hold asymptotically under any raw Bayesian update. The resolution is the **forgetting prerequisite** promoted into the Formal Expression above: exponential forgetting at rate λ stabilizes α_Σ at the exact value $(1-\lambda)/(2-\lambda)$, with the simpler linear form $1-\lambda$ as the slow-forgetting asymptote. The exact threshold $(1-\lambda)/(2-\lambda) > \rho_\Sigma/R_\Sigma$ converts the schema's instantaneous check into a trajectory guarantee. This is structural, not heuristic — without forgetting, the schema's form holds only until the agent accumulates enough experience to cross below threshold; with insufficient forgetting (linear-form approximation that treats ρ_Σ as small relative to R_Σ when in fact $\rho_\Sigma \geq R_\Sigma/2$), the prerequisite is unsatisfiable and the schema's trajectory guarantee fails.

The structural parallel is genuine, but conditional on forgetting. This schema extends *structural persistence* (see Persistence in `LEXICON.md`) from the epistemic substate to the strategy substate — asking whether the strategy correction machinery can outpace the rate at which the environment invalidates the agent's causal theory. It inherits the same limitation: structural persistence of Σ_t does not address operational persistence (how close $\|\delta_{\text{strategic}}\|$ is to R_Σ) or continuity persistence (whether the agent's strategic identity coheres through time). The persistence condition for M_t (4.4) says: adaptive tempo must exceed the ratio of disturbance to critical mismatch. If the same mathematics applies to Σ_t, then strategy persistence requires strategic tempo to exceed the ratio of strategic disturbance to critical strategic mismatch. The strategic analog of "the environment changes faster than the agent can learn" is "the world invalidates plans faster than the agent can revise them." Both lead to the same catastrophic outcome: the system cannot maintain bounded mismatch and begins to degrade.

What this would buy the theory. If promoted to a result, strategy persistence would: - Provide a formal criterion for "when does a strategy remain viable?" - Connect strategic failure modes to the same mathematical framework as epistemic failure modes - Enable quantitative comparison: is the bottleneck epistemic persistence (model can't keep up with reality changes) or strategic persistence (plans can't keep up with requirement changes)? - Ground the organizational intuition that plans need to be revised faster than the situation changes

Findings.

The Forgetting Prerequisite for Strategic Persistence. **Brief:** With infinite memory, standard Bayesian updating guarantees eventual strategic failure. The sector parameter for Beta-Bernoulli edge updates is $\alpha_\Sigma = 1/(n+1)$, which decays monotonically as experience n accumulates. For any positive disturbance rate, the persistence threshold $\alpha_\Sigma > \rho_\Sigma/R_\Sigma$ is eventually violated regardless of the agent's initial calibration. Exponential forgetting with discount factor λ stabilizes the effective sample size at $1/(1-\lambda)$, giving exact steady-state $\alpha_\Sigma^{ss} = (1-\lambda)/(2-\lambda)$ — with the simpler linear form $1-\lambda$ as the slow-forgetting asymptote ($\lambda \to 1$). The forgetting prerequisite — $(1-\lambda)/(2-\lambda) > \rho_\Sigma/R_\Sigma$ — converts the schema's instantaneous persistence check into a trajectory guarantee. Forgetting is therefore a structural prerequisite, not a tunable heuristic: the rate at which the agent discounts old evidence must satisfy the threshold. The threshold has a hard ceiling at $\rho_\Sigma \geq R_\Sigma/2$: no λ satisfies the prerequisite when disturbance reaches half the strategic reserve, regardless of forgetting design.

Impact: Translates an organizational platitude ("stay adaptive") into a quantitative survival inequality with explicit failure mode. Identifies a structural calcification process common to all long-running adaptive systems whose update mechanism accumulates evidence: institutional rigidity, RL value-function staleness, scientific-paradigm lock-in, and the loss-of-edge phenomenon in incumbent firms all instantiate the same dynamic. The threshold is sharp — no amount of prior learning protects against an environment that invalidates plans faster than the agent forgets, and the disturbance-to-reserve ratio has a hard ceiling at 1/2 above which no forgetting rate suffices. For long-lived AI agents this mandates explicit memory-pruning mechanisms; the design choice is not whether to forget but at what rate, and how to keep the strategic disturbance well below half the reserve. The structural identity with `#result-persistence-condition` (where α can be stationary) is recovered for the strategic case *only* under explicit forgetting, which is itself a non-obvious finding about the asymmetry between epistemic and strategic dynamics.

Novelty Claim: *Claim differentiation* on Bayesian update dynamics with experience discounting. The Beta-Bernoulli decay $\alpha = 1/(n+1)$ is elementary and the forgetting-factor remedy is standard adaptive-control / online-learning practice. The AAT-distinctive contribution is the connection from these standard mechanics to a *survival inequality with environment-side parameters* (disturbance rate, reserve), making forgetting a structural prerequisite of strategic persistence rather than a hyperparameter. The exact threshold $(1-\lambda)/(2-\lambda) > \rho_\Sigma/R_\Sigma$ — with the slow-forgetting linear form $(1-\lambda) > \rho_\Sigma/R_\Sigma$ as its asymptotic — is the strategic analog of the linear operational form of the persistence condition, with the forgetting rate $(1-\lambda)$ playing the role of adaptive tempo and the $1/(2-\lambda)$ damping factor capturing the integration of finite-horizon evidence.

Related Work:

Table 8.11

ASF Concern	*Prior-art Language*	*Relationship / Positioning*
Exponential forgetting / discounted least squares	Ljung 1987 *System Identification: Theory for the User*, MIT Press (published 1987, found pre-2026)	*formal antecedent* — supplies the discounted-update mechanism; AAT adopts it directly and connects it to a survival threshold
Plasticity/stability tradeoff in continual learning	Kirkpatrick et al. 2017 *Overcoming Catastrophic Forgetting in Neural Networks*, PNAS 114(13) (published 2017, found pre-2026)	*conceptual precursor* — frames the same tradeoff at the network-weight level; does not connect to a Lyapunov-style survival inequality with environment parameters
Drift detection in streaming learning	Gama et al. 2014 *A Survey on Concept Drift Adaptation*, ACM Computing Surveys 46(4) (published 2014, found pre-2026)	*conceptual precursor* — recognizes the need to track changing distributions; does not formalize "forgetting rate must exceed disturbance-to-reserve ratio" as a structural prerequisite
Requisite variety as cybernetic principle	Ashby 1956 *An Introduction to Cybernetics* §11; Conant & Ashby 1970 (published 1956/1970, found pre-2026)	*conceptual precursor* — the regulator's variety must match the disturbance's variety; AAT's forgetting prerequisite is a quantitative time-domain analog applied to the strategic substate
Organizational "core rigidities" / competency traps	Leonard-Barton 1992 *Strategic Management Journal* 13:111; Levitt & March 1988 *Annual Review of Sociology* 14:319 (published 1988/1992, found pre-2026)	*conceptual precursor* — qualitative recognition that accumulated competence becomes liability under environmental change; AAT supplies the structural threshold

Search Log: - 2026-04 (*intuition-only* on the survival-inequality framing): the Beta-Bernoulli $1/(n+1)$ decay is elementary, the forgetting-factor remedy is standard, and the qualitative organizational-calcification intuition is well-established. The unsearched claim is whether the framing as a *structural survival prerequisite* — with environment-side ρ_Σ/R_Σ on the right-hand side — has been formalized elsewhere as a threshold rather than a hyperparameter. Targeted future search candidates: bounded-rationality with memory cost (Genewein, Leibfried, Grau-Moya, Braun 2015 *Frontiers in Robotics and AI*), regret analysis of forgetting-factor estimators in adaptive control (Anderson-Moore line), organizational-learning literature on competency traps and "core rigidities" connected to environmental volatility metrics. Pre-search expectation: the constituent moves are individually well-precedented; the integrated framing as a Lyapunov-style survival prerequisite for the *strategic* substate, with the structural identity to the epistemic persistence condition under forgetting, is plausibly AAT-distinctive but not yet verified under nominally-comprehensive search.

Working Notes.

- **Done.** Five cases verified: single-edge AND, two-edge AND (observable and unobservable intermediate), two-arm OR (ε-greedy), and mixed AND/OR with common-cause node (L1 augmented DAG). Full derivations in N (Props B.1–B.6). Key findings: AND-node persistence is depth-gated (evidence starvation); OR-node persistence is exploration-gated (action selection policy); L1 augmented DAGs exhibit three-way gating (condition testing × starvation × exploration). All satisfy the schema's form ($\alpha_\Sigma > \rho_\Sigma / R_\Sigma$). B.6 is the first mixed AND/OR case and confirms L1 results transfer from L0 with correct construction. The next step is deeper mixed topologies and multiple common causes.
- **Strategic tempo now formalized.** 8.8 defines $\mathcal{T}_\Sigma = \sum_{(i,j)} \nu_{ij} \cdot \eta_{\text{edge},ij}$ and verifies consistency with all four cases. The relationship to the schema's sector parameter: $\alpha_\Sigma \leq \mathcal{T}_\Sigma / |E|$ (persistence is bottleneck-limited, not throughput-limited). 8.9 provides the IB/MDL framework for strategy compression and derives max useful depth d^*.
- The strategic disturbance ρ_Σ is qualitatively different from epistemic disturbance ρ. Epistemic disturbance is about the environment changing (physical state evolves). Strategic disturbance is about the agent's causal theory becoming invalid (the intervention-outcome mapping shifts). These can be correlated (a changing environment invalidates both model and strategy) but they're not the same quantity.
- The stochastic treatment (from track-b simulations) suggests $\rho_\Sigma / \sqrt{\mathcal{T}_\Sigma}$ rather than $\rho_\Sigma / \mathcal{T}_\Sigma$ for the steady-state strategic mismatch. If this carries over from the epistemic domain, the persistence threshold is different in the stochastic case. Whether strategic disturbance is better modeled as deterministic or stochastic drift is domain-dependent.
- The forgetting prerequisite transforms an organizational platitude ("stay adaptive") into a quantitative constraint: the rate at which an agent discounts old evidence must satisfy the exact threshold $(1-\lambda)/(2-\lambda) > \rho_\Sigma / R_\Sigma$, with the slow-forgetting linear form $(1-\lambda) > \rho_\Sigma / R_\Sigma$ as its asymptote. Institutional examples where this fails — long-running successful firms whose accumulated n suppresses gain below the threshold set by a shifting competitive landscape — are the strategic analog of model-rigidity death spirals.
- **Audit 451729 (D.3) strengthen-first edit, 2026-05-12.** Earlier version of this segment used the linear approximation $\alpha_\Sigma^{ss} \approx 1 - \lambda$ silently throughout (Formal Expression, Discussion, Findings Brief). The exact form derived from $1/(n_{\text{eff}} + 1)$ with $n_{\text{eff}} = 1/(1-\lambda)$ is $(1-\lambda)/(2-\lambda)$; the linear approximation is the slow-forgetting limit, valid asymptotically as $\lambda \to 1$ and degrading rapidly outside that regime (~50% overestimate at $\lambda = 0.5$, ~10% at $\lambda = 0.9$). Per strengthen-first discipline, the exact form is now primary throughout the segment with the linear form retained explicitly as the high-λ asymptote. The hard ceiling at $\rho_\Sigma \geq R_\Sigma / 2$ (no λ satisfies the exact prerequisite) was hidden by the linear approximation — surfaced now in Formal Expression and Findings Brief. Audit report: `audits/audit-451729-FINAL-2026-05-10.md` §D.3.
- **Cross-reference to NeurIPS Paper 2.** The forgetting prerequisite is sharpened in NeurIPS 2026 Paper 2 ("A Unified Convergence Theory for Non-Stationary Reinforcement Learning", `~/src/neurips/02-unified-convergence-rl/`, §App-D / `aux-decay-class`) into a **structural-class theorem on gain-decay updates**: define $\mathcal{A}_{\text{decay}}$ = the class of update mechanisms whose effective gain decays to zero with accumulated experience (count-accumulating Bayesian without forgetting; observation-aggregating without restart; Robbins-Monro / vanishing-step-size gradient methods). Every member of $\mathcal{A}_{\text{decay}}$ universally violates the persistence threshold for any positive disturbance rate. Finite-gain mechanisms (constant-step stochastic approximation, sliding windows, bounded memory, block restart) face *bidirectional* thresholds — both the forgetting prerequisite (lower bound on plasticity) and an upper bound from noise blow-up. This lifts the segment's per-mechanism claim to a class-level no-go: gain-decay mechanisms are structurally disqualified regardless of tuning. See `msc/neurips-back-integration-2026-05-08.md` §1 Paper 2 entry 4.

8.11 *Formulation:* Consolidation Dynamics

Consolidation is a regime of the between-event dynamics g_M of 3.2 in which the agent applies Markov updates driven by replayed or internally-generated pseudo-events, with objective of reducing the rate-distortion gap to the IB-optimal compression $\phi^*(\mathcal{C}_t)$. It is not a new adaptive primitive — the recursive-update form $f(M_{\tau^-}, e_\tau)$ is preserved — but it is a distinct operating regime with its own scope condition, its own objective, and its own failure modes, each of which the theory currently names only implicitly or in a parenthetical. Naming the regime makes the stability-plasticity window visible as an AAT-expressible failure boundary and supplies the architectural primitive that logogenic agents (`03-llm-core/`) require under context-turnover.

Formal Expression.

Regime definition.

Let the agent's between-event dynamics be $g_M(M_\tau)$ per H (Corollary: Between-events dynamics, $dM/d\tau = g(M_\tau)$). **Consolidation** is the regime of g_M in which the agent applies updates $M_{\tau^+} = f(M_{\tau^-}, e_\tau^{\text{replay}})$ where e_τ^{replay} is a pseudo-event synthesized from M_{τ^-} itself — a sample drawn from the agent's retained trace (replay buffer, hippocampal reinstatement, a remembered episode, an earlier paragraph re-read). The recursive-update form is preserved; what distinguishes consolidation is the *objective* these updates optimize:

Formulation (consolidation-regime, specializes #der-recursive-update).

$$\text{consolidation objective:}\ \min_{M_\tau} \mathcal{J}_{\text{IB}}(M_\tau) \ := \ I(M_\tau; \mathcal{C}_t) - \beta I(M_\tau; o_{t+1:\infty} \mid a_{t:\infty}) \tag{8.41}$$

— the 2.2 Lagrangian evaluated against the agent's accumulated chronica $\mathcal{C}_t$. By contrast, online update's objective (per 3.6) is one-step predictive mismatch minimization at the current event; it has no representation of $\mathcal{J}_{\text{IB}}$.

Under C3 (state completeness, per H), $\mathcal{I}(e_\tau^{\text{replay}} \mid M_{\tau^-}) = 0$: the pseudo-event carries no new external information. Yet the update still does work — it *redistributes* existing information across the factorization structure of M_τ. The distinguishing content is not the information brought in (zero, by construction) but the rate-distortion gap closed (nonzero, when the agent has not yet reached ϕ^*).

Scope condition — timescale separation.

Let ν_{online} be the rate of external events (3.1) and ν_{consol} the rate of consolidation updates. The consolidation regime is well-defined only when

Scope (timescale-separation).

$$\nu_{\text{consol}} \ll \nu_{\text{online}} \tag{8.42}$$

— the convergence constraint of 4.6 applied to an additional intermediate timescale between parametric update (fast) and structural adaptation (slow). Violating this constraint makes consolidation act on online transients rather than settled state, producing the same oscillation failures 4.6 warns about.

Necessity condition.

Consolidation is necessary — online-only cannot reach the IB optimum — when *both* of the following hold:

Derived (consolidation-necessity, conditional).

(N1) Sub-state factorization. M_t factors into sub-states M_t^{fast} and M_t^{slow} with divergent compression-prediction trade-offs. M_t^{fast} favors high-capacity sparse representation; M_t^{slow} favors distributed compressed representation. The two sub-states capture cross-episode regularities versus verbatim traces respectively — the Complementary Learning Systems factorization (McClelland, McNaughton & O'Reilly 1995; Kumaran, Hassabis & McClelland 2016).

(N2) Bounded per-event budget. The per-event processing budget B_{online} is strictly less than the integration cost $B_{\text{consol-needed}}$ for updating M_t^{slow} against cross-episode regularities. Online updates can move at most B_{online} bits of model-state change per event; updating M_t^{slow} to represent a cross-episode pattern requires comparison against a distribution of prior episodes, which exceeds B_{online}.

When (N1) *or* (N2) fails, consolidation is a *luxury*: online update with sufficient per-event budget or without sub-state factorization can reach ϕ^* in the limit. Kalman filters with persistent covariance, conjugate-Bayesian agents with full posterior, and linear-Gaussian systems with online Riccati updates all satisfy neither (N1) nor (N2) and have no consolidation need. When (N1) *and* (N2) both hold, consolidation is a *necessity*: no online-only policy reaches ϕ^* under the joint constraint.

Stability-plasticity feasibility window.

8.10 derives the *plasticity lower bound* on forgetting rate λ:

Derived (stability-plasticity-window, conditional).

$$(1-\lambda) \;>\; \rho_\Sigma / R_\Sigma \qquad \text{(plasticity lower bound, from \#schema-strategy-persistence)} \tag{8.43}$$

— forgetting fast enough to track non-stationarity. This segment's complement is a *stability upper bound*:

$$(1-\lambda) \;<\; \phi(\nu_{\text{consol}}, \text{consolidation-budget}) \qquad \text{(stability upper bound, see Working Notes for derivation sketch)} \tag{8.44}$$

— forgetting slow enough to let consolidation integrate cross-episode patterns before they are discarded. Between these bounds is the **feasibility window** for λ. Empty window — rapid non-stationarity with slow consolidation cadence — is the catastrophic-forgetting regime (French 1999; Kirkpatrick et al. 2017): no λ satisfies both constraints and the agent's long-run IB objective is strictly worse than a slower-environment or faster-consolidation counterpart.

The upper bound's exact form depends on the consolidation mechanism and is not derived here — it is a candidate derivation flagged in Working Notes. What is derived: the window's *existence* as a structural object (plasticity must satisfy both bounds) and the catastrophic-forgetting regime as its empty-window limit.

Structural-adaptation enablement.

Under (N1)+(N2), structural adaptation (per 4.5) cannot be executed online. Parametric update has a hard timescale constraint — mismatch decays at rate $\mathcal{T}$; delayed updates accumulate mismatch at rate ρ. Structural adaptation tolerates much larger delay because the slow process operates on what R_Σ or R are *measuring tolerance against* — the model class itself. Consolidation provides the operating regime where structural operations (decomposition-and-recombination, expansion, compression, grafting per 8.7) become executable: the per-event budget is irrelevant when updates are offline, and interleaved replay supports stability-preserving structural change.

Derived (structural-adaptation-requires-consolidation, conditional).

Pure online structural adaptation is the luxury case where per-event budget equals or exceeds integration cost for structural-class operations — this is where Bayesian nonparametric agents with unlimited compute sit. All finite-budget agents require consolidation for quality-preserving structural change.

What Is Derived vs. What Is Chosen.

Table 8.12

Property	*Source*	*Strength*
Consolidation as regime of g_M with replayed/pseudo-event driver	Specialization of 3.2's Discussion (consolidation listed as an example)	Formulation choice (the "regime" framing; could alternatively be presented as a distinct adaptive primitive with its own update form)
IB-gap reduction as consolidation's objective	Identification with 2.2's Lagrangian against $\mathcal{C}_t$	Derived (given IB acceptance; the alternative — online-update-only optimization — leaves the gap un-reducible under (N1)+(N2))
Scope condition $\nu_{\text{consol}} \ll \nu_{\text{online}}$	Direct application of 4.6 convergence constraint	Derived
Necessity condition (N1)+(N2)	Structural argument: under (N1), cross-episode information cannot enter one event; under (N2), online budget insufficient for cross-episode integration	Derived (qualitatively; the quantitative version requires specifying B_{online} and $B_{\text{consol-needed}}$ per architecture)
Stability-plasticity window existence	8.10's lower bound + this segment's upper bound (form open)	Derived (existence); upper-bound functional form open
Catastrophic-forgetting regime = empty window	Direct from both-bounds-unsatisfiable	Derived
Structural adaptation requires consolidation under (N1)+(N2)	4.5's per-step timescale + consolidation's offline budget	Derived (qualitatively)
CLS factorization (hippocampal fast / neocortical slow) as canonical (N1) instance	McClelland-McNaughton-O'Reilly 1995; Kumaran-Hassabis-McClelland 2016	External theorem (CLS literature)
Quantitative online-only no-go under (N1)+(N2)	Rate-distortion argument sketched but not rigorously derived here	Sketch (candidate X Instance 3)

Epistemic Status.

Robust qualitative. Max attainable: *robust qualitative* for the regime characterization and the necessity condition; *conditional* for the feasibility-window claim pending upper-bound derivation; *sketch* for the IB-optimum no-go claim.

The regime characterization is a formulation choice — consolidation can always be re-described as a regime of 3.2's g_M with appropriate pseudo-events. The distinguishing objective (IB-gap reduction via replayed pseudo-events) is well-defined and distinct from online update's one-step mismatch objective; this is the cleanest formal separation the spike analysis uncovered.

The necessity condition (N1)+(N2) is *qualitatively derived* — the argument is structural: (N1) implies cross-episode information is not in any single event, (N2) implies online updates cannot cross-compare. Together they force consolidation. The quantitative version — the precise bit-budget boundary below which online fails and above which it succeeds — depends on the architecture's sub-state structure and is not derived here. Nor are (N1) and (N2) individually necessary: an agent may need consolidation for reasons outside this segment's scope (e.g., structural adaptation at pure cost-of-delay, unrelated to cross-episode regularities).

The stability-plasticity window's *existence* follows from 8.10's lower bound plus any monotone upper bound on $(1-\lambda)$ from consolidation-cadence considerations. The specific functional form of the upper bound is open work; candidate form in Working Notes. The catastrophic-forgetting regime as empty-window is then an immediate structural consequence, not a new claim.

The online-only no-go claim (that under (N1)+(N2), no online-only policy reaches ϕ^* in steady state) is *sketch-level*. The argument reduces to a rate-distortion inequality close to known results in continual-learning theory and rate-distortion with side information, but the rigorous version requires specifying the budget geometry carefully. It is a candidate Instance 3 for X (an external information-theoretic obstruction with AAT machinery — the consolidation regime in g_M — as the unique escape).

What this segment does not claim. It does not introduce a new adaptive primitive — the recursive-update form $f(M_{\tau^-}, e_\tau)$ is preserved, with e_τ^{replay} playing the role of e_τ. It does not derive the quantitative feasibility-window upper bound. It does not resolve the (N1) factorization question for specific architectures (e.g., whether transformer attention heads satisfy (N1) is a logogenic-agents question, not an AAT-core one).

Discussion.

Current AAT surface — why naming is warranted. Consolidation is visible in the theory today as:

- A parenthetical example in 3.2 Discussion ("includes prediction generation, uncertainty growth, and internal reorganization (consolidation, abstraction)").
- The implicit slow timescale in 4.6.
- The plasticity lower bound in 8.10 — with no stability upper bound.
- A compression-by-convergence Working Note in 8.9 ("as edges converge, drop them").
- The PULSUS MEMORATA / VERA / AXIOMATA cadences in `ref/agentic-tft/agentic-tft-cognitive-loop-spec.md` for logogenic agents — where consolidation is *already* a first-class architectural commitment.

Naming the regime explicitly promotes what the theory implicitly depends on. The asymmetric treatment in 8.10 (plasticity lower bound only, no stability upper bound) predicts faster forgetting is always better — empirically false whenever the slow sub-state matters (Complementary Learning Systems literature; continual-learning benchmarks; organizational memory research). The feasibility-window framing closes this asymmetry.

Distinguishing axes examined. Four candidate axes could distinguish consolidation from online update. Only one yields a clean formal distinction:

- *Timescale*: consolidation is slower. But 4.6 already admits arbitrary timescale separation, and any periodic process can be modeled as a channel with clock-driven $\nu^{(k)}$. Timescale is a *scope condition*, not a distinguishing feature.
- *Information source*: consolidation operates on replayed data. This is a real difference but lives inside g_M without new primitive structure — the recursive-update form is preserved with e_τ^{replay} in place of external e_τ.
- *Objective*: **clean formal distinction.** Online = one-step predictive-mismatch minimization; consolidation = IB-gap reduction. This is what this segment adopts as the defining axis.
- *Scope of change*: under bounded per-event budget, structural adaptation is temporally decouplable from online but parametric update is not. Structural-adaptation operations naturally live offline — true under (N2), but this is a consequence of the necessity condition, not an independent axis.

Relation to Complementary Learning Systems (CLS) theory. The (N1) factorization — fast sparse-conjunctive sub-state + slow distributed-overlapping sub-state — is the CLS architecture (McClelland-McNaughton-O'Reilly 1995). CLS's core claim is that hippocampal replay during sleep supports interleaved neocortical learning that cross-episode structure requires. In AAT vocabulary, this is: online-only on the slow sub-state catastrophically forgets (French 1999); replay-based offline updates (experience replay: Mnih et al. 2015; prioritized replay: Schaul et al. 2016) or regularization-based approaches (EWC: Kirkpatrick et al. 2017) provide the escape.

The AAT reading of EWC is worth noting: EWC adds a stability-weighted update gain (per-parameter Fisher-information weighting) — a *tensor-valued* generalization of 3.6's scalar η^*. This is a different direction from consolidation (it keeps updates online but weights them by prior-task importance). Both escape the catastrophic-forgetting regime but via different mechanisms; consolidation reuses 3.2's structure, EWC requires the new tensor-valued gain.

Logogenic implications. Consolidation is a primitive in a stronger sense for logogenic agents (`03-llm-core/`) than for AAT-core for three composing reasons.

First, **context-turnover.** Logogenic agents have near-100% reset of the fast sub-state (context window) per session. The only continuity is the slow sub-state (persistent memory, weights, external files). The between-session interval is a *forced* consolidation window — the agent must transfer signal from the about-to-be-lost fast state to the persistent slow state, or it is lost. This is qualitatively different from non-logogenic agents where the fast sub-state persists across events.

Second, **linguistic medium of reflection.** The PULSUS MEMORATA / VERA / AXIOMATA cadences in `ref/agentic-tft/agentic-tft-cognitive-loop-spec.md` are scheduled consolidation processes with different cadences and different target representations. Each is a linguistic operation ("What from recent experience should be compressed into lasting memory?", "Are my beliefs still justified?", "Who am I becoming?") — using language to reorganize language-structured state. This is consolidation operating as the primary unit of cross-session cognition.

Third, **pre-consolidated embedding space.** Per `ref/agentic-tft/agentic-tft-narrative-as-implementation.md`, pre-trained language embeddings encode structured epistemic geometry at training time. The logogenic agent doing linguistic reflection is operating in a representational space *already* consolidated into a high-structure form, with access to cross-episode generalization that sub-linguistic agents would have to build online. This is a load-bearing asymmetry in `03-llm-core/`.

Luxury vs necessity mapping. When is consolidation a luxury (subsumed by online update)? When at least one of (N1)/(N2) fails:

- *Rich-state luxury*: Kalman filter with persistent covariance, conjugate-Bayesian agent with full posterior over parameters. The posterior *is* the consolidated representation.
- *Large-budget luxury*: per-event budget ≥ integration cost allows online slow-track update per event.
- *Stationary-environment luxury*: online update converges to ϕ^* in the limit if the environment holds still long enough.

When is consolidation a necessity? When both (N1) and (N2) hold, which is the empirically-ubiquitous case: CLS-architected agents, bounded-budget deep RL agents (hence experience replay's empirical necessity in DQN), organizations with event-arrival rates exceeding real-time cognitive bandwidth, logogenic agents under context-turnover.

Predictive statement. *The depth of consolidation machinery an agent needs scales with (a) the factorization depth of its representational structure, (b) the gap between its per-event processing budget and the integration cost of its slowest sub-state, and (c) the rate of cross-episode structural regularities in its environment relative to its event arrival rate.* This is a scope-indexed claim — testable, domain-general, and makes AAT's position on continual learning explicit.

Connection to AAT's meta-architecture. - *Y* (positive half): the regime is the repair machinery between separable-core and structured-repair ladders along the representation-factorization axis. Where the (N1)+(N2) necessity conditions hold, the repair is consolidation; where either fails, the online regime suffices. - *X* (negative half): candidate Instance 3 — the under-bounded-budget + no-reach-of-IB-optimum no-go, with consolidation as the unique escape. - *Z* (constructive half): the IB Lagrangian is an adjacent family member (adopted as applied external theorem, not re-derived from an AAT-internal additivity axiom), which the consolidation objective directly uses.

Working Notes.

- **Stability upper bound derivation (open).** The claim $(1-\lambda) < \phi(\nu_{\text{consol}}, \text{budget})$ is stated but the functional form of ϕ is not derived. Candidate form: ϕ is the minimum forgetting rate that leaves enough retained signal in M_t^{fast} for the next consolidation cycle to integrate cross-episode patterns — a function of replay-buffer size, consolidation-budget B_{offline}, and event-arrival density. Rigorous derivation would connect to continual-learning theory (Parisi et al. 2019 survey) and is a natural follow-up spike.
- **Online-only no-go rigorization (open).** The sketch argument reduces to a rate-distortion inequality: online update's effective rate is $B_{\text{online}} \cdot \nu$ bits/time; rate required to drive M_t toward ϕ^* while also integrating new events is strictly larger when the environment has cross-episode structural regularities. Making this rigorous (ideally via rate-distortion with side information) would promote the no-go to derived-exact and establish this segment as Instance 3 of X. Adjacent candidate for X §"Adjacent Floors" open extensions.
- **EWC formulation as stability-weighted gain.** An alternative escape from catastrophic forgetting is Elastic Weight Consolidation (Kirkpatrick et al. 2017) — a stability-weighted per-parameter update that penalizes changes to parameters important for prior tasks. In AAT, this would be a tensor-valued generalization of 3.6's scalar η^*, with the stability weighting coming from per-parameter Fisher information. Not pursued here; naming it distinguishes the two escapes.
- **Relationship to 8.9 compression-by-convergence.** Working Notes there observe that as edges converge (high n_{ij}), the IB objective favors dropping them — compression-by-convergence. This is a consolidation operation in the edge-credence sub-state: once an edge's credence has concentrated, the consolidation regime can compress the representation by pruning. Worth cross-referencing when both segments stabilize.
- **Quantitative CLS instantiation.** A focused spike could work out the quantitative form of (N1)+(N2) for a specific CLS-like architecture (sparse-conjunctive + distributed-overlapping with specific capacity ratios) and derive the online-only no-go as a rate-distortion bound. This would also give a quantitative stability-upper-bound for the feasibility window.
- **Logogenic primitive status.** `03-llm-core/` should declare consolidation as an architectural primitive (not a regime of a more basic primitive) — the context-turnover forcing makes it non-optional. The relationship between that declaration and this AAT-core formulation is: this segment gives the formal shape; the logogenic treatment adds a context-turnover-specific scope condition and the three PULSUS cadences as instantiations. Cross-reference from `#obs-context-turnover` (in the logogenic component) back to this segment is expected.
- **Is consolidation one segment or two?** A plausible split: this segment on the formulation + necessity, a separate segment on the stability-plasticity window as a derivation-type once the upper bound is derived. Recommended to land as one segment at `robust-qualitative` now; if the upper-bound derivation lands, split off into its own segment at `conditional` or `derived`.

8.12 *Discussion:* Additional Implications & Discussion

The chapter does more work than its segment count suggests. Strategy edges acquire dynamics; the structural commitments from Ch.3 acquire formal failure modes; and the framework's signature pattern — *naming an information-theoretic floor and the unique broadly-available escape via AAT machinery* — delivers its first formal instance as a derived no-go. Several of the chapter's findings compose with each other and with findings from Part I Ch.4 and Part II Ch.2/Ch.3 into cross-segment results that are not visible at any single segment: a triple-pressure picture of long-lived adaptive failure, a unified RL convergence story under non-stationarity, and a sub-scope refinement of Part I's persistence template that explains a decade of empirical RL folklore on optimizer choice and approximate inference. None of these is a single derivation; each is what becomes visible when the chapter's segments are read against the rest of the framework.

Edge dynamics: detection latency, forgetting, and the stability-plasticity window.

The chapter's strategic-edge persistence schema (8.10) carries the forgetting prerequisite in its own Findings section: forgetting at rate $(1-\lambda) > \rho_\Sigma / R_\Sigma$ is structurally required for asymptotic strategic persistence, not a tunable heuristic. With infinite memory, standard Bayesian updating guarantees eventual strategic failure because the Beta-Bernoulli sector parameter $\alpha_\Sigma = 1/(n+1)$ decays monotonically with accumulated experience. The implications-side payoff is that this prerequisite is one of three structural pressures on plasticity in long-lived agents, and the chapter is where the other two become statable.

The first complementary pressure is detection latency. `#deriv-update-detection-latency` derives that for a Beta-Bernoulli strategy-edge agent without forgetting, the expected cycles to detect a within-class regime change of footprint ε scales as $\Omega((n_{\min}+1)/\varepsilon)$. The result has a structural rather than parametric character — the $1/(n+1)$ rate is *forced*, not chosen, through the composition of two AAT-internal moves. The log-odds coordinate is uniquely forced by the evidential-additivity axiom via Aczél 1966's Cauchy-functional-equation uniqueness theorem (P); in that forced coordinate, Beta-Bernoulli accumulation gives $1/(n+1)$. No reparameterization escapes the rate without abandoning evidential additivity, which would invalidate the update rule on AAT-internal grounds. The chapter therefore sharpens the forgetting prerequisite from "required for asymptotic persistence" to "required for *bounded detection latency at every operating point*" — long-lived agents face detection latency that grows linearly with accumulated experience unless forgetting bounds the effective sample size at $1/(1-\lambda)$. The framework's name for the empirical pattern is *stability-induced myopia*: successful agents that have accumulated experience on their working plans systematically take longer to notice regime changes than the same agents earlier in their lives, with a precise rate of slowdown.

The second complementary pressure is consolidation cadence. `#form-consolidation-dynamics` adds a *stability upper bound* on plasticity: forgetting slow enough that consolidation can integrate cross-episode patterns before they are discarded. Between the plasticity lower bound (forgetting prerequisite) and the stability upper bound (consolidation cadence) lies the *feasibility window* for λ. An empty window — rapid non-stationarity with slow consolidation cadence — is the *catastrophic-forgetting regime*: no λ satisfies both constraints and the agent's long-run IB objective is strictly worse than a slower-environment or faster-consolidation counterpart. Existence is derived under (N1) sub-state factorization (the Complementary Learning Systems factorization — McClelland-McNaughton-O'Reilly 1995's hippocampus-like fast traces + neocortex-like slow integrals) and (N2) bounded per-event budget. The window's upper-bound functional form is candidate-derivation in `#form-consolidation-dynamics`'s Working Notes; *window existence* is what's derived here.

The three pressures compose into a triple-pressure picture that no single segment carries: long-lived agents face plasticity bounded above and below, with detection latency growing inside the viable window. The prescriptive content — which intervention works depends on which bound is binding — is the cross-segment finding worth surfacing as its own object. A *short-attention* agent (forgetting-bound binding from below) needs retention. A *calcified* agent (latency-bound binding inside the window) needs targeted intervention or external Pearl-do (causal-IB exploration drive from `#deriv-causal-ib-exploration`). A *fragmented* agent (consolidation-bound binding from above) needs cadence engineering. The same agent in different operating regimes faces different binding pressures and benefits from different interventions; treating any one mechanism as universal misses the routing. Continual-learning machinery — experience replay (Mnih et al. 2015 DQN), prioritized replay (Schaul et al. 2016), Elastic Weight Consolidation (Kirkpatrick et al. 2017) — are all window-restoration mechanisms; the framework's contribution is naming the window as the structural object and the routing as the choice. For logogenic agents under near-100% context turnover the picture is sharpest: (N1) holds by construction (the context window is the fast sub-state, persistent memory the slow sub-state), (N2) holds by token-budget, and the consolidation regime is forced rather than optional.

On-policy detection structurally fails — identifiability-floor Instance 1.

The chapter's `#der-causal-insufficiency-detection` is the first formal instance of the framework's identifiability-floor pattern, and its Findings section carries the result well: an agent at L0 cannot distinguish a world with a latent common cause from a no-latent-cause world matched in on-policy regime conditionals. The two worlds emit identical on-policy distributions; no test, statistic, or Bayesian comparison built from the agent's observable history can tell them apart under policy-perfect execution. The result is the agent-theoretic instance of Bareinboim's Causal Hierarchy Theorem (Bareinboim-Correa-Ibeling-Icard 2022) applied at the L0/L1 distinction of `#def-strategy-dag`'s Correlation Hierarchy — a Level-2 question (does the latent common cause exist?) being asked of Level-1 data (on-policy traces).

The implication worth surfacing beyond the segment's Findings is the *constructive* use of the no-go. The pattern's signature posture is to name the floor, then name the unique broadly-available escape via AAT machinery, and the chapter delivers this signature for the first time formally. Five boundary routes (S1–S5) characterize what scope condition the agent must violate to escape the no-go: (S1) joint sibling observability under exploration, (S2) instrumental variation, (S3) cross-episode covariance under structural breaks, (S4) off-policy intervention, (S5) external structural knowledge. Joint sibling observability under exploration is the *unique broadly-available* violation — every AAT agent within agency scope has it (via `#der-loop-interventional-access`'s loop-Level-2 access); the others require additional structure not provided by the cycle alone. The chapter therefore lifts `#der-loop-interventional-access` (which Part II Ch.2 introduced as a useful enabling result) from "useful machinery" to *the unique broadly-available violation of the no-go*. This is the move that makes the framework's identifiability-floor pattern load-bearing rather than discouraging — each floor names a specific machinery's load-bearing role with maximum sharpness.

The chapter also makes precise why the no-go is asymmetric in a way the prior aggregate-residual mechanism could not surface. Persistent overestimation of plan success after edge credences have converged is the *signal* of insufficiency, not the *test*; the test is pairwise covariance among observed siblings, which the on-policy regime structurally cannot produce. The aggregate-residual approach collapsed on-policy precisely because the residual is one of the statistics the no-go forbids — *every* on-policy statistic is forbidden. The framework's reformulation in terms of the no-go-and-escape pattern is what gives the joint-sibling-covariance test its load-bearing role rather than treating it as one diagnostic among many. The connection to Ch.3 closes the loop: Ch.3's Correlation Hierarchy is the scope statement that makes the no-go formal here; this chapter's no-go is the dual to Ch.3's L1' Cramér-Rao floor (Instance 2). Together they cover both agent-internal (Instance 2 — single agent, unobservable common cause) and self-diagnostic (Instance 1 — policy-perfect execution) failures of identification, with `#der-loop-interventional-access` as the shared escape mechanism.

Observability dominates, and the depth penalty compounds.

`#der-observability-dominance` carries an underrated structural claim: unobservable strategy edges *freeze* — the gain principle drives their update rate to zero. Paths through unobservable nodes become epistemically dead, an absorbing state the agent cannot escape without external shock, proactive observability investment, or communication-gain machinery (`#hyp-communication-gain`). This is the chapter's strong-prediction result that organizational settings with poor measurement (R&D, strategy groups, some management functions) develop persistent, untested beliefs about their own effectiveness — *structurally* (frozen η_{edge}), not motivationally. The same mechanism predicts that LLM agents operating with high-uncertainty intermediate outputs cannot update their strategy through those nodes, regardless of nominal confidence.

The observability dominance compounds with the chapter's other depth-related results into a triple penalty on planning horizons. Chain-confidence decay from Ch.3 (probabilities multiply, exponential in depth, log-additive coordinate forced by the chain rule) is one penalty. Evidence starvation is another: in a d-depth chain, edge k is tested only when all upstream edges succeed, so its effective correction rate is attenuated by $\prod_{j<k} \theta_j$ — the deepest edge faces *exponential* attenuation regardless of how reliable each upstream edge is. The IB complexity cost (`#form-strategy-complexity-cost`) is the third: maintaining Σ_t in working memory has an information-theoretic cost that compounds with depth, eventually exceeding the agent's cognitive budget. Deep plans are therefore structurally fragile along three independent axes — fragility, calibratability, and tractability — and the structural pressure toward shallow, parallelizable strategies is grounded in three independent mechanisms rather than one. For LLM-based agents working through context-window-bounded planning, the triple penalty is the framework's account of why deep chains of LLM reasoning tend to fail in ways that look "creative" but are actually structural.

The chapter's quantification of the observability-investment tradeoff (from `#der-observability-dominance`'s two-edge analysis and `#deriv-edge-credence-dynamics`'s Props B.2–B.3) gives the principle concrete economic content. Making an intermediate node observable improves α_Σ from the plan-level rate $1/(n_\Phi + 1)$ to the per-edge weakest-link rate $\min(1/(n_1 + 1),\ \theta_1/(n_2 + 1))$; the improvement is positive whenever $\theta_1 > 1/2$ and experience is distributed similarly across edges. This translates directly to persistence margin in Part I Ch.4's vocabulary — instrumenting an intermediate node buys *measured* persistence-margin gain. The TST connection is concrete and stated in `#der-observability-dominance`'s Discussion: code quality directly determines σ_v for software-domain strategy steps; poor code quality reduces observability, freezing the developer's causal beliefs about what changes will accomplish, and the strategic concern is structural rather than aesthetic.

Adaptive gain, variational extensions, and sub-scope refinement of the persistence template.

The chapter's appendix-side derivations refine Part I Ch.4's sector-persistence template into sub-scope α / α_2 / β partitions that explain several decades of empirical RL and approximate-inference folklore on principled grounds. `#deriv-adaptive-gain-dynamics` derives the A2'-split: sub-scope α_1 (fixed-gain Kalman / exponential family / strongly-convex gradient) where α is derived from the update rule's structure under B1 directional fidelity; sub-scope α_2 (adaptive-gain under named meta-gain conditions MG-1 through MG-4) where two-timescale Lyapunov composition applies; sub-scope β (PID / rule-based / human-judgment) where A2' is *assumed* per-instance rather than derived. The partition is operationally consequential: *AMSGrad is a meta-gain repair that restores (MG-1) by construction*, which is why it outperforms Adam on ill-conditioned problems — the empirical superiority is now derived, not folklore. *MAML's outer loop sits in β*, which is why Fallah et al. 2020's non-convexity is structurally forced rather than a specific algorithmic accident. *IMM regime transitions are in β across transition*, which is why adaptive Kalman in regime-switching needs explicit dwell-time machinery. The Mehra non-identifiability result is flagged as a candidate fifth instance of the identifiability-floor pattern (Instance 5) — an additional opening for the meta-pattern.

`#deriv-variational-sector-condition` extends the persistence template to variational-inference agents under a controlled-KL budget. Under $\mathrm{KL}(q\|p) \leq \varepsilon$, an ε-fidelity directional-fidelity condition transfers at $O(\sqrt{\varepsilon})$ rate via Pinsker, with the sector parameter degrading proportionally. The sub-scope α' tier covers controlled-KL variational agents as a workhorse case; *natural-gradient variational inference recovers the full* α (no degradation) because the geometry matches the Fisher metric; *mean-field VI suffers the $O(\sqrt{\varepsilon})$ degradation* because it does not. The implication-worth-surfacing as its own object: *mean-field VI agents are structurally suboptimal as adaptive systems by a factor proportional to $\sqrt{\text{KL-budget}}$, regardless of compute or training data*. The decade-plus Bayesian-deep-learning debate about MF-VI versus natural-gradient on heuristic grounds has a principled answer: if persistence-optimality matters, there is no choice. The geometry of the variational family is load-bearing for the persistence machinery, not merely an aesthetic consideration.

Both refinements compose with Part I Ch.4's bandwidth result via the persistence-information-rate floor: each sub-scope inherits the $\dot{R} \geq n\alpha/2$ bound from `#deriv-persistence-cost`, with its own α playing the role. An agent that adopts a coarser sub-scope (moving from α_1 to α_2 or β, or from natural-gradient to mean-field VI) pays in the persistence-bandwidth requirement: the same target distortion costs more information per unit time because the effective sector constant is smaller. The framework's posture across these refinements is that the choice of agent class is not free —

it has structural consequences for how much observation bandwidth the agent must consume to hold its persistence guarantees.

The Bretagnolle-Huber regret bound, reverse-KL coordinate forcing, and the unified RL convergence theory.

`#deriv-strategy-cost-regret-bound` carries one of the chapter's sharpest mathematical results: under deterministic optimal policy π^* — AAT's canonical scope — the Bretagnolle-Huber identity $D_{\mathrm{KL}}(\pi^* \| Q) = -\log(1 - \mathrm{TV}(\pi^*, Q))$ holds *exactly* (not as inequality), and the strategy-cost regret obeys the tight two-sided bound $\Delta_{\min}(1 - e^{-D_{\mathrm{KL}}}) \le R \le V_{\max}(1 - e^{-D_{\mathrm{KL}}})$. *Strictly sharper than Pinsker* by a factor of 2 in canonical regimes — a quantitative improvement that matters for practical RL bound-tightness on the common deterministic-optimum case. The reverse-KL direction in AAT's strategy-cost objective (π^*-first) is *forced* by the regret-bound derivation: forward-KL is vacuous under deterministic π^* because it cannot bound regret against a point mass.

The result is also the second AAT-internal instance of the additive-coordinate-forcing meta-pattern Ch.3 anchored — reverse-KL at the divergence layer (Cauchy-FE on chain-rule additivity over conditional factorizations, via Hobson 1969 / Csiszár 1991 / Shore-Johnson 1980), following log-probability at the chain layer. The same exponential-family Legendre-Fenchel geometry both coordinates live on is where `#deriv-edge-update-natural-parameter`'s log-odds (instance three, update layer) and `#scope-agent-identity`'s (PI)+Čencov Fisher metric (instance four, metric layer) also live. The framework's posture is unified: coordinates are forced by AAT-internal axioms via classical uniqueness theorems, and the four-layer architecture is what makes derived results at any one layer compose cleanly with results at the others.

The widest payoff of the chapter is what becomes the *unified RL convergence theory under non-stationarity* once the four contributions are composed. Reinforcement learning has many regret-bound results (UCB, Thompson sampling, RLSVI) but typically (a) assumes stationarity, (b) lacks explicit metric structure on policy space, (c) lacks connection to a satisfaction-gap diagnostic that prevents dark-room collapse. AAT's RL convergence story has all three by composing four findings that are derived independently and turn out to be mutually consistent for principled reasons. The two-gap diagnostic from Ch.3 (7.4 + 7.5) supplies the local separation between "you're not doing it well enough" and "the world doesn't permit it." The Bretagnolle-Huber identity here supplies the tight two-sided regret bound with PAC-Bayes generalization (Rubin 2012). The strategic-tempo + sector-persistence schema supplies the rate of Σ_t revision that must outpace strategic disturbance — verified across four canonical topologies (linear chain, balanced tree, unbalanced tree, full DAG with feedback) by `#def-strategic-tempo` + `#deriv-edge-credence-dynamics`. The loop-Level-2 result from Ch.2 makes the regret bound *learnable* — interventional data is automatic rather than earned. The four together give a quantitative RL convergence theory under non-stationarity with three properties no existing RL framework has all of. The composition is NeurIPS Paper 2's structural argument ("A Unified Convergence Theory for Non-Stationary Reinforcement Learning", `~/src/neurips/02-unified-convergence-rl/` §4 Theorem 4.1); each piece is derived elsewhere, and the synthesis is what becomes visible only at the chapter level.

What Part II Ch.4 closes on, then, is the substrate Strategy Dynamics needed: edge dynamics with derived rates and bounded operating-point trajectories, an identifiability-floor instance with a uniquely-named escape, a sub-scope partition that explains optimizer and inference-method folklore on principled grounds, a tight reverse-KL regret bound that anchors the second instance of the coordinate-forcing pattern, and a cross-segment composition (unified RL convergence under non-stationarity) that resolves what RL theory has had open for a long time. Chapter 5's orient cascade is what AAT does with strategy dynamics; this chapter is where the dynamics gets its formal architecture.

Working Notes.

- This segment is a chapter-end discussion grouping findings from `#schema-strategy-persistence`, `#form-consolidation-dynamics`, `#deriv-update-detection-latency`, `#der-causal-insufficiency-detection`, `#der-observability-dominance`, `#deriv-edge-credence-dynamics`, `#def-strategic-tempo`, `#form-strategy-complexity-cost`, `#deriv-strategy-cost-regret-bound`, `#deriv-adaptive-gain-dynamics`, and `#deriv-variational-sector-condition`. Cross-pattern reading across §§2–4 leans on `#disc-identifiability-`

`floor` and `#disc-additive-coordinate-forcing` meta-segments; the cross-segment compositions (triple-pressure routing in §1, unified RL convergence theory in §5) are findings whose substance lives between segments rather than within any one.

- **Cross-reference vs canonical-home policy.** Two source segments in this chapter carry `## Findings` sections — `#schema-strategy-persistence` (the forgetting prerequisite) and `#der-causal-insufficiency-detection` (the on-policy no-go). Both are well-written and the implications-side payoff cross-references them rather than restating. The remaining findings (detection latency, stability-plasticity window, observability dominance, depth penalty, adaptive-gain refinement, variational sector, BH regret bound, strategic tempo, unified RL convergence, triple-pressure routing) are treated more substantively here because their home segments do not currently carry Findings sections — the implications segment is their canonical home for catalog-grade surfacing.
- **Cross-segment findings — the heaviest content in this chapter.** Two cross-segment findings live here as canonical home: CS2 *Compounding Stability-Induced Myopia* (triple-pressure picture composing forgetting prerequisite + detection latency + stability-plasticity window with prescriptive routing in §1) and CS1 *Unified RL Convergence Theory Under Non-Stationarity* (composing two-gap diagnostic from Ch.3 + BH identity here + strategic-tempo machinery here + loop-Level-2 from Ch.2, in §5). Both are findings whose substance does not live in any single segment; the chapter-end is the natural home because the chapter is where the last prerequisite for each lands.
- **Identifiability-floor pattern coverage so far.** The pattern's first two formal instances are now in view: Instance 1 (on-policy L0-insufficiency detection, this chapter) and Instance 2 (L1' mixture identifiability under unobservable common cause, Ch.3's Correlation Hierarchy). Instance 4 (universal information-to-distance constant under heteroscedastic-Gaussian counterexample, in `#deriv-observation-ambiguity-bias-bound`) lives downstream in Part II Ch.5. Instance 3 (composite contraction certification from component data, via Liberzon common-Lyapunov-nonexistence) lives in Part III. Instance 5 (Mehra non-identifiability in adaptive Kalman regime-switching, flagged in `#deriv-adaptive-gain-dynamics`) is a candidate awaiting formal promotion. The chapter-by-chapter implications work surfaces the pattern instance-by-instance as the formal prerequisites land.
- **Connection to NeurIPS submissions.** CS1's composition is the structural argument behind NeurIPS Paper 2's Theorem 4.1; the chapter's BH identity result is §6.2 of that paper. Paper 3's hallucination bias bound depends on the (PI)+Čencov axiom from `#scope-agent-identity` and lifts the chapter's variational sector framing to Class 3 (Coupled) agents — that bridge is the subject of Part II Ch.5's implications. AMSGrad / MAML / IMM connections (via `#deriv-adaptive-gain-dynamics`) are not currently formalized in any submission and remain in the framework's empirical-RL-explanation register.
- **Density flag and check-in.** This is the densest chapter-end implications segment in the planned series — five findings synthesized via two cross-segment compositions, drawing on Part I Ch.4, Part II Ch.2, and Ch.3 in addition to this chapter's own. The chapter-by-chapter pace through Part II Ch.5 and Part III Ch.2–5 will run lighter on per-segment density now that the heaviest cross-segment composition has landed.

Chapter 9

The Orient Cascade

[**Gap**] Discussion

9.1 *Derived:* Orient Cascade

For actuated agents, epistrophe (the corrective phase of the cycle) expands into a multi-step cascade. The resolution order is forced by information dependency: epistemic update first, then attainability assessment, then strategy evaluation, then (if needed) objective revision. Each step's input depends on the output of prior steps. The ordering is not a design choice — it's a consequence of which quantities require which others.

Formal Expression.

1. **Reduce $\delta_{\text{epistemic}}$** — understand reality. Update M_t via 3.4 and 3.6. Prerequisite for all purposeful evaluation, because M_t appears in every subsequent formula.

   Derived (orient-cascade, from information dependency between mismatch types).

2. **Evaluate δ_{sat}** — is the goal achievable? Compute $A_O(M_t; \Pi, N_h)$ using the updated M_t. Requires adequate M_t to assess attainability (7.4).

3. **Evaluate δ_{regret}** — is the policy suboptimal? Compare A_O to $V_O(M_t, \pi_{\text{current}}; N_h)$ (7.5). This step applies regardless of δ_{sat}'s sign — the 2×2 diagnostic (7.5) requires both quantities to distinguish four cases:

   - $\delta_{\text{sat}} \leq 0$, $\delta_{\text{regret}} \approx 0$: **success** — goal attainable, policy near-optimal.
   - $\delta_{\text{sat}} \leq 0$, $\delta_{\text{regret}} \gg 0$: **strategy problem** — goal attainable, policy poor → revise Σ_t.
   - $\delta_{\text{sat}} > 0$, $\delta_{\text{regret}} \gg 0$: **both** — goal hard AND strategy weak → revise Σ_t first (cheaper than revising O_t), then reassess δ_{sat}.
   - $\delta_{\text{sat}} > 0$, $\delta_{\text{regret}} \approx 0$: **capability limit** — already doing the best available; proceed to step 5.

4. **If δ_{regret} high, evaluate strategy calibration** — is the plan's causal model wrong?

   (4a) Plan-level calibration (default within-L0). Evaluate strategy-plan-confidence error $\delta_s = \hat{P}_\Sigma - \Phi$ — the gap between the agent's strategy-plan-confidence score and the independence-model plan value at true edge parameters. δ_s is credit-assignment-free (requires only status propagation), and its persistence is proved (8.10, Prop B.5 in N). This is the cheapest operational signal, but its persistence guarantee is within the **independence**

model (L0) of the Correlation Hierarchy (7.3): $\delta_s \to 0$ means $\hat{P}_\Sigma$ has converged to Φ, not to actual plan success probability. When the DAG is causally insufficient, Φ itself is a biased target. Step 4c checks whether this is the binding regime.

(4b) Edge-level localization (when credit assignment is available). When sufficient observability and attribution quality exist (Level 1+ per 8.6), the agent can compute per-edge residuals $\delta_{\text{strategic}}$ (8.1) to localize which edges need revision. $\delta_{\text{strategic}}$ provides finer-grained diagnostics but its persistence is open and it requires the credit-assignment machinery that δ_s avoids. Step 4b is optional — it improves diagnostic resolution but is not required for the cascade's corrective function.

(4c) Causal-sufficiency check (L0→L1 escalation). If persistent $\delta_s \approx 0$ coincides with persistent negative plan-outcome residuals ($y_G < \hat{P}_\Sigma$ on average, after edge credences have converged), this is evidence that the DAG is causally insufficient and L0 calibration is converging to a biased target. The diagnostic is pairwise sibling covariance under an augmented test (8.2): positive covariance among sibling edges, at timescales where each edge's individual credence has stabilized, localizes where a latent common cause is missing. When the signal for L1 augmentation is present, step 4c directs the agent to add common-cause nodes (7.3 Correlation Hierarchy) *before* escalating via 5a–5d. Running the cascade's exploitation recommendations under L0 when the signal for L1 is present compounds miscalibration — the agent acts confidently on a model whose own residual structure is telling it the model is wrong.

Practical sensitivity. Step 4c is the unique broadly-available L0→L1 diagnostic (8.2 no-go), but its effective power depends on signal-to-noise: small samples, weak common-cause effects, or noisy residuals can produce false negatives, and edge-credence drift can mimic sibling covariance. The convergence framing — "after edge credences have converged" — is a precondition, not a guarantee: in non-stationary environments where per-edge credences keep drifting, the trigger may never cleanly fire and L1 augmentation should be considered the default rather than gated on 4c's signal (7.3 Correlation Hierarchy). Formal preconditions (joint observability, per-edge credence stabilization, approximate stationarity over the test window) and detection scope live in 8.2; consult them before treating a 4c null as evidence of L0 sufficiency.

5. **If $\delta_{\text{sat}} > 0$ persists** — escalate before revising O_t.

   Under C1 (the canonical default), $\delta_{\text{sat}} > 0$ means *locally stuck*, not *globally infeasible* (5.6). Before concluding the objective is wrong, the agent should check whether the gap reflects a limitation of the current analysis rather than genuine infeasibility:

   (5a) Check whether M_t correction changes the feasibility assessment — a wrong model may make an achievable goal appear unattainable.

   (5b) Check whether a richer policy class Π would close the gap — structural Σ_t adaptation (expanding the strategy space, not just revising edge credences). This includes L1 augmentation of the strategy DAG (7.3 Correlation Hierarchy): if step 4c detected causal insufficiency, adding common-cause nodes here is the structural repair.

   (5c) Check whether convention escalation reveals recovery paths — evaluating under C2 (receding-horizon) may show $\delta_{\text{sat}}^{\text{RH}} \leq 0$ for a goal that appeared unattainable under C1.

   (5d) If $\delta_{\text{sat}} > 0$ persists across M_t correction, Π expansion (including L1 augmentation), and convention escalation — **revise O_t**.

   The cascade's ordering ensures objective revision is the last resort, not the first response to unmet goals. The agent reaches step 5d only after exhausting the alternatives that the satisfaction-gap disambiguation table (7.4) identifies: wrong M_t, narrow Π, short N_h, and only then genuinely infeasible goal.

Derivation. Each step's input depends on prior steps' outputs: - You cannot evaluate strategy quality with a broken reality model (step 3 requires step 1) - You cannot distinguish "locally bad strategy" from "locally unattainable goal" without both δ_{sat} and δ_{regret} (step 3 requires step 2) - You cannot localize strategy failures (4b) without first detecting plan-level miscalibration (4a) - You cannot diagnose causal insufficiency (4c) until after edge credences have had time to converge (4a), because the diagnostic signal — persistent negative plan-outcome residuals — requires $\delta_s \approx 0$ to be separable from ordinary calibration error - You should not revise the objective until you've verified that improving M_t, Π (including L1 augmentation when 4c signals it), and N_h cannot close the gap (step 5 requires steps 3–4 and the escalation substeps)

The ordering is forced by information dependency. The split of step 4 into 4a/4b/4c reflects three distinct diagnostic levels: 4a gives a within-L0 persistence signal (plan-level tracking via δ_s), 4b gives within-L0 edge-level localization when credit assignment is available (Level 1+), and 4c exits L0 entirely when the independence model is the binding constraint. The escalation substeps in step 5 reflect the satisfaction-gap disambiguation (7.4): multiple causes of $\delta_{\text{sat}} > 0$ must be ruled out before the agent concludes the goal itself is wrong. L1 augmentation (7.3 Correlation Hierarchy) enters 5b as a structural Σ_t adaptation when 4c's signal is present.

Convention hierarchy and diagnostic power. The 2×2 diagnostic and the inferences drawn from it are relative to the continuation convention in the value object (5.6), which defines a hierarchy of three conventions with a proved monotonicity result.

Under **C1** (one-step improvement, canonical default), δ_{sat} and δ_{regret} are one-step-improvement quantities. A multi-step recoverable objective may appear locally unattainable ($\delta_{\text{sat}} > 0$) because continuation is frozen at π_{current}. The "capability limit" quadrant ($\delta_{\text{sat}} > 0, \delta_{\text{regret}} \approx 0$) means *locally stuck* — the agent may be globally recoverable but cannot see the recovery path from one-step analysis.

Under **C2** (receding-horizon, N_r-step replanning), the diagnostics capture multi-step recovery potential. An objective that appeared unattainable under C1 may show $\delta^{\text{RH}}_{\text{sat}} \leq 0$ because replanning reveals a viable path. The "capability limit" quadrant means *stuck with N_r-step replanning* — stronger evidence of genuine difficulty.

Under **C3** (Bellman), the diagnostics are global. $\delta^{\text{B}}_{\text{sat}} > 0$ means the goal is genuinely infeasible given (M_t, Π, N_h) — no policy can achieve it. The "capability limit" quadrant is a definitive diagnosis: the agent is optimally pursuing an infeasible goal (modulo model error in M_t). This is the inference the cascade was designed to support; C1 and C2 provide progressively weaker versions of it.

The monotonicity (5.6): $\delta^{\text{B}}_{\text{sat}} \leq \delta^{\text{RH}}_{\text{sat}} \leq \delta^{(1)}_{\text{sat}}$. Every "unattainable" diagnosis under C3 persists under C1 and C2. A C1 "unattainable" diagnosis may be overturned by C2 or C3 (the goal is reachable with replanning or globally optimal play). The cascade's *ordering* is convention-independent (forced by information dependency). The *inferential force* at each step scales with the convention: C1 gives local heuristics, C2 gives moderate-horizon diagnostics, C3 gives global conclusions.

Epistemic Status.

The cascade **ordering** is *exact*: it is a logical consequence of which quantities appear in which formulas. Steps 1-2 (epistemic update, attainability assessment) rest on well-typed quantities (3.4, 7.4) and exact derivation. Step 3 (control regret) is exact (7.5). Step 4a (plan-level calibration via δ_s) is grounded in a proved quantity — the sector condition transfers to δ_s (Prop B.5 in N) — but the formal guarantee is *within the L0 independence model*: $\delta_s \to 0$ means convergence to Φ, which equals actual plan success only when the DAG is causally sufficient. Step 4b (per-edge localization via $\delta_{\text{strategic}}$) inherits strategic-calibration's discussion-grade status — the credit-assignment problem and execution-fidelity requirement are acknowledged but unresolved (8.1, Epistemic Status). Step 4c (causal-sufficiency check) is the mechanism for exiting L0 when L1 is the binding regime; it is *robust-qualitative* — the diagnostic logic is sound (8.2), but sensitivity depends on how cleanly the agent can separate sibling-covariance signal from edge-credence noise at convergence. Step 5's escalation substeps (5a-5c) are derived from the satisfaction-gap disambiguation table (7.4); step 5d (objective revision) is the residual case after alternatives are exhausted. The ordering of all steps is forced by information dependency (each step's input depends on prior steps' output). What is NOT derived is the *timing* — how long the agent should spend on each step before proceeding, and how long $\delta_s \approx 0$ must persist before 4c's signal is trusted.

The **convention hierarchy** (5.6) is *exact*: the three conventions (C1, C2, C3) are definitions, and the monotonicity result is a direct consequence of "better continuation policy yields higher expected value." The diagnostic implications table states what each convention's quantities mean by construction. The cascade's inferential force at steps 2-5 scales with the convention but the ordering is convention-independent.

Discussion.

G_t complexity bounded by M_t capacity. Σ_t's evaluable complexity is bounded by M_t's ability to observe which strategy edges are intact. An agent with poor model sufficiency ($S(M_t) \ll 1$) cannot meaningfully evaluate a complex Σ_t — the strategic calibration residual requires adequate M_t to distinguish "edge prediction wrong" from "observation too noisy to tell."

This creates a **virtuous cycle**: better M_t → richer evaluable Σ_t → better-directed action → faster M_t improvement. And a **vicious one**: degraded M_t → forced strategy simplification → cruder action → further M_t degradation. The vicious cycle is the strategic analog of the death spiral described in the persistence condition (4.4) — the agent loses the capacity to maintain complex plans, which reduces the quality of its actions, which further degrades its model.

Connection to Boyd's OODA. The orient cascade is the formal analog of Boyd's "Orient" — not just model updating, but the structured interaction between reality-understanding and strategy. Boyd's insight was that Orient is the critical step, not Decide. The cascade provides a mathematical mechanism consistent with this insight: Orient resolves the information dependencies that make Decide meaningful. An agent that skips to Decide (strategy revision) without adequate Orient (model update + attainability assessment) will revise its strategy based on stale or incorrect beliefs. Whether the dependency structure in the cascade captures the actual cognitive process Boyd described is an empirical question.

Timescale structure. The cascade implies a natural timescale ordering for the different update types:

$$\nu_{\text{epistemic}} \gg \nu_{\text{edge-update}} \gg \nu_{\gamma\text{-reclassify}} \gg \nu_{\text{prune/graft}} \gg \nu_{O\text{-revision}} \tag{9.1}$$

Weight updates are frequent (every observation). Combination-type reclassification is rare (needs strong structural evidence). Pruning/grafting is rarer still (abandon or create causal hypotheses). Objective revision is rarest (change what you want, not how you get it). This ordering is an empirical observation for many agent populations, consistent with the cascade but not derived from it.

Working Notes.

- The cascade as stated is sequential. In practice, agents may run steps in partial overlap — beginning to assess δ_{sat} before M_t is fully updated, or revising edges while still processing observations. The cascade describes the *logical* dependency, not the *temporal* scheduling. An agent that parallelizes steps must still respect the dependencies (don't finalize strategy revision using a stale M_t).
- The resource allocation question (how much of the agent's tempo budget to spend on each step) is open and may be domain-dependent. In fast-changing environments, the agent may need to truncate early steps to keep up. In stable environments, the agent can spend more time on deep strategic evaluation.
- The virtuous/vicious cycle between M_t quality and Σ_t evaluable complexity is structurally motivated but not formally derived. Formalizing it would require a coupled dynamics model — possibly an extension of the persistence framework to the (M_t, Σ_t) pair.
- **Strategy-maintenance status (updated).** The cascade's ORDERING is exact — forced by information dependency. The cascade's CONTENT for steps 3-5 has progressed: 8.6 characterizes the tractable/intractable boundary and establishes that persistence does not require credit assignment (Prop B.5); N verifies sector conditions for four topologies; 8.4 has a default signal function candidate (gradient-based, satisfying directional fidelity). What remains domain-specific: the choice of signal

function implementation, execution-fidelity monitoring, and handling of correlated failures (where $\hat{P}_\Sigma$ overestimates — 7.3). The theory now provides the structural requirements and a default scheme; specific implementations are engineering, parallel to how the gain principle provides η^* while Kalman/TD-learning/etc. are implementations.

9.2 *Discussion:* Three-Way Resource Allocation

At each decision point, an actuated agent with explicit strategy Σ_t faces a three-way allocation of its finite cycle budget across exploit (take the currently-best action), explore (take an information-gathering action), and deliberate (pause acting to revise the model or strategy internally).

Formal Expression.

The three activities.

An actuated agent's cycle budget decomposes into three activities:

Definition (three-activity-decomposition).

1. **Exploit**: select and execute the action maximizing expected value under the current model and strategy. Earns immediate value $Q_O(M_t, a; \pi_{\text{cont}}, N_h)$.
2. **Explore**: select and execute an action maximizing causal information yield. Earns future model improvement via $\text{CIY}(a; M_t)$.
3. **Deliberate**: pause external action to run the orient cascade (9.1) — internal model refinement ($\Delta\eta^*$), strategy revision ($\Delta\alpha_\Sigma$), or objective reassessment. Earns improved future action quality.

The key distinction: Exploit and explore both involve external actions with environmental consequences. Deliberation is *internal exploration* — simulation, counterfactual reasoning, cross-domain synthesis, and contingency planning conducted in model-space rather than environment-space. It acquires no new environmental data, but it can generate genuinely new information by combining, recombining, and extrapolating from what the agent already knows — including knowledge from entirely different domains. A chess engine deliberating explores millions of future board states; a commander war-gaming evaluates counterfactual scenarios; a developer doing architecture review synthesizes cross-domain expertise. Deliberation's fidelity ceiling is set by model quality, not by data quantity.

Extended deliberation threshold.

Extending 4.1 to incorporate strategic deliberation, the deliberation threshold becomes:

Derived (Conditional on GA-4, deliberation-drift assumption from #der-deliberation-cost).

$$\Delta\tau^* = \arg\max_{\Delta\tau \geq 0} \left[\underbrace{\Delta\eta^*(\Delta\tau) \cdot \|\delta_{\text{post}}\|}_{\text{epistemic benefit}} + \underbrace{\Delta V_\Sigma(\Delta\tau)}_{\text{strategic benefit}} - \underbrace{\rho_{\text{delib}} \cdot \Delta\tau}_{\text{drift cost}} \right] \tag{9.2}$$

where: - $\Delta\eta^*(\Delta\tau)$ is the improvement in model update gain from $\Delta\tau$ of internal simulation (4.1) - $\|\delta_{\text{post}}\|$ is the post-deliberation mismatch magnitude - $\Delta V_\Sigma(\Delta\tau)$ is the improvement in strategy value from deliberation: the expected increase in V_O from revising Σ_t during the pause - ρ_{delib} is the local mismatch drift rate during inaction (GA-4)

First-order condition:

$$\frac{\partial \Delta\eta^*}{\partial \Delta\tau} \cdot \|\delta_{\text{post}}\| + \frac{\partial \Delta V_\Sigma}{\partial \Delta\tau} = \rho_{\text{delib}} \tag{9.3}$$

Stop deliberating when the marginal joint improvement rate (epistemic plus strategic) drops below the mismatch drift rate.

Reduction to existing results. When $\Delta V_\Sigma = 0$ (no strategy revision benefit), this reduces to the deliberation threshold in 4.1. When $\Delta\tau^* = 0$ (deliberation never beneficial), the agent acts immediately.

On the structure of the allocation.

Two-stage decomposition. The three-way allocation can be decomposed into two nested decisions: (1) how long to deliberate before acting, (2) conditional on acting, which action to take (solved by 6.5). This decomposition is natural because exploit and explore are both external actions ($\in \mathcal{A}$), while deliberation is internal processing. However, this decomposition is a *modeling convenience*, not a structural necessity. A unified objective over $\mathcal{A} \cup \{\text{deliberate}\}$ with temporal discount γ is equally valid:

Discussion (allocation-structure).

$$\bar{\pi}^* = \arg\max_{\bar{a} \in \mathcal{A} \cup \{\text{deliberate}\}} \bar{Q}(\bar{a}; M_t, \gamma) \tag{9.4}$$

where $\bar{Q}(\text{deliberate}) = \gamma \cdot \mathbb{E}[V_{\text{act}}(f_{\text{delib}}(M_t))]$ and $\bar{Q}(a) = r(a) + \gamma \cdot \mathbb{E}[V_{\text{act}}(M_{t+1})]$.

The two-stage decomposition is useful because it allows reusing the CIY-unified objective for the action-selection step, but it is not the unique or forced structure.

Additive benefit decomposition. The objective function above decomposes deliberation benefit into additive epistemic and strategic terms. This assumes the two benefits are approximately independent. Under directed separation (5.3), M_t updates independently of G_t, which makes the decomposition structurally motivated for Class 1 (Separated) agents. However, strategic benefit depends on the improved model ($\Delta V_\Sigma(\Delta\tau, M_t + \Delta M_t(\Delta\tau))$), creating an interaction that the additive form ignores. The additive form is a linearization that holds when ΔM_t is small relative to M_t. For Class 3 (Coupled) agents, epistemic and strategic deliberation are coupled, and the additive decomposition is a convenience.

Control regret as deliberation ceiling.

When $\delta_{\text{regret}} \approx 0$ (7.5), strategic deliberation has no value: the agent is already executing the best available strategy. This is the formal reason why low control regret suppresses deliberation. The ceiling on strategic deliberation value is δ_{regret} — the most the agent could gain by finding a better strategy.

Discussion (deliberation-ceiling).

However, this ceiling applies only to the *strategy-revision* channel of deliberation. Deliberation may also: - Reveal that the *objective* should change (step 5 of the orient cascade), producing value not bounded by δ_{regret} - Reduce *uncertainty about* strategy quality, enabling more confident exploitation even when the strategy doesn't change

Connection to active inference.

The exploit/explore decomposition maps cleanly onto the expected free energy (EFE) decomposition into *pragmatic value* (preferences-aligned outcomes) and *epistemic value* (expected information gain about hidden states) in active inference (Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017, "Active inference: a process theory," *Neural Computation* 29; Da Costa, Parr, Sajid, Veselic, Neacsu & Friston 2020, "Active inference on discrete state-spaces," *J. Math. Psych.* 99 §2.4; Sajid, Ball, Parr & Friston 2021, "Active inference: demystified and compared," *Neural Computation* 33). Under the mapping, AAT's exploit term Q_O corresponds to EFE's pragmatic value when the value functional V_{O_t} is read as expected log-preferences, and AAT's CIY corresponds to EFE's epistemic value when the relevance variable is read as the hidden-state posterior. The structural correspondence is at the shared-shape level: both decomposed into value-and-information terms, but AAT does *not* commit to AI's preferences-as-priors form (preserving the satisfaction-gap diagnostic in 7.4, which the priors-as-preferences collapse would erase).

Discussion (ai-connection).

The deliberation axis — internal exploration in model-space rather than environment-space — is structurally adjacent to "sophisticated active inference" (Friston, Da Costa, Hafner, Hesp & Parr 2021, "Sophisticated inference," *Neural Computation* 33), which handles bounded computation via recursive expected free energy with depth-limited belief-about-belief reasoning. AAT's machinery for deliberation cost (4.1) and the extended deliberation threshold above derives the same trade-off via per-edge persistence and evidence starvation, with the structural advantage that the depth bound d^* is causally derived from the strategy DAG rather than from belief-recursion depth. The two frameworks address the same problem with different machinery; AAT can credit the AI lineage as a co-developed alternative.

Exploit-regret and the strategy-cost objective. The exploit term's control regret $\delta_{\text{regret}} = A_O - V_O(\pi_{\text{current}})$ (7.5) is the same decision-theoretic regret that underwrites the KL-direction derivation in 8.9. There, strategy-induced regret $R(Q_{\Sigma_t}) = V(a^*) - \mathbb{E}_{a \sim Q_{\Sigma_t}}[V(a)]$ is bounded above by $V_{\max}\sqrt{\frac{1}{2} D_{\text{KL}}(\pi^* \| Q_{\Sigma_t})}$ (Pinsker), forcing the KL direction with π^* first. The exploit term thus shares structure with the strategy-cost relevance term: both are regret-aligned, with the action-selection objective operating pointwise and the strategy-cost objective operating over the strategy's induced distribution. This is the structural cleanup that "value and information term" shares with EFE's pragmatic-epistemic decomposition at the shape level, without the preferences-as-priors commitment — the regret framing is the AAT-internal derivation of the direction that the variational-inference literature arrives at from free-energy-gradient arguments.

Qualitative regime descriptions.

The three activities have natural dominance conditions that follow from the structure of the problem:

Discussion (qualitative-regimes).

Exploit dominates when the strategy is near-optimal ($\delta_{\text{regret}} \approx 0$), model uncertainty is low ($\lambda(M_t) \approx 0$), and the environment penalizes pauses (ρ_{delib} high). The agent knows what to do and must act now. This is Boyd's implicit guidance and control.

Explore dominates when model uncertainty is high (U_M large), and the model needs correction via external evidence that internal simulation cannot provide. Exploration acquires new data; deliberation only processes existing data.

Deliberate dominates when the strategy needs revision ($\delta_{\text{regret}} \gg 0$ or $\delta_{\text{strategic}} \gg 0$), the environment is stable during pauses (ρ_{delib} low), and the agent has unprocessed information or structural hypotheses it can evaluate internally. This regime requires that internal computation is cheaper or more efficient than external probing — otherwise exploration dominates deliberation.

Boundary conditions. As $\rho_{\text{delib}} \to \infty$: $\Delta\tau^* \to 0$, and the allocation collapses to the binary exploit/explore tradeoff. As $\lambda \to 0$ and $\Delta V_\Sigma \to 0$: the allocation collapses to pure exploitation. As $\delta_{\text{regret}} \to 0$: strategic deliberation value vanishes, and the allocation reduces to 4.1's epistemic think/act threshold combined with 6.5's exploit/explore balance.

These regime descriptions are qualitative observations about the problem structure, not derived predictions. They restate what the constituent objectives already imply (high uncertainty favors information acquisition, low cost favors deliberation) without adding quantitative content. The theory does not provide the thresholds at which one regime transitions to another — these are domain-specific.

Epistemic Status.

Discussion-grade. The extended deliberation threshold is *derived* (conditional on GA-4 and the deliberation-drift assumption from 4.1) and is the one genuinely formal contribution of this segment. The regime descriptions are *discussion-grade* — qualitatively correct observations that restate the problem structure. The two-stage decomposition is a *formulation choice*, not a derived result. The additive benefit decomposition is a *linearization* that is structurally motivated under directed separation but not derived.

Simulation check (drifting multi-armed bandit, `spikes/sim-three-way-tradeoff.py`): The three-way allocation with perfect foresight outperformed binary exploit/explore by 2–6% across five configurations. The oracle almost never chose deliberation (0–8 out of 200 steps). Exploration earned 3.5x more information per step than deliberation. A unified objective (single argmax with temporal discount) outperformed two-stage UCB by 8–13%. The simulation is UNFAVORABLE to deliberation (no strategy structure, no irreversible actions, small action space), so these results are a lower bound on deliberation's value in complex domains. The simulation confirms that deliberation adds little in simple settings and that the two-stage decomposition is not necessary.

Max attainable: *discussion-grade* for the overall framing. The extended deliberation threshold could reach *conditional* (it inherits its status from 4.1). The regime descriptions are unlikely to advance beyond discussion-grade without domain-specific instantiation that provides quantitative thresholds.

Discussion.

Why this is Section II content, not Section I. Section I's deliberation-cost handles the binary think/act threshold using only M_t and η^*. The three-way allocation is inherently a Section II result because it requires: - Σ_t to exist (the strategy that deliberation can improve) - δ_{regret} to be defined (the signal that motivates strategic deliberation) - The orient cascade (the internal structure of what "deliberation" does for an actuated agent)

Without G_t, there is no strategic deliberation mode — the agent can only improve M_t or act, which is the Section I binary.

Deliberation as internal exploration. The deepest characterization of deliberation is not "computation on existing data" but *lower-cost, more efficient, usually lower-fidelity exploration in model-space rather than environment-space*. All three activities are forms of information acquisition; they differ in source, cost, fidelity, and scope:

- **Exploitation** acquires value directly from the environment.
- **Exploration** acquires information from the environment via novel observations — high fidelity, high cost (real actions have real consequences).
- **Deliberation** acquires information from the agent's internal model via simulation, synthesis, and counterfactual reasoning — lower fidelity (model may be wrong), lower cost (no environmental consequences), but potentially unlimited scope.

A chess engine deliberating is not "reprocessing existing data" — it is exploring a vast space of futures that no amount of external probing could cover at the same rate. A military commander war-gaming is performing Level 3 counterfactual reasoning (6.1): "what *would have happened* if the enemy had moved north?" A developer doing architecture review is synthesizing patterns from entirely different domains — cross-domain expertise that no local exploration of the current codebase could surface. An AI agent planning its approach is simulating the universe it inhabits and weighing future possibilities.

What deliberation can do that external exploration cannot: - **Counterfactual reasoning** (Level 3): evaluate actions never taken, in situations never encountered - **Cross-domain synthesis**: bring expertise and experience from other contexts to bear on the current problem - **Combinatorial search**: systematically evaluate a space of futures too large to probe externally (chess, Go, multi-step planning) - **Contingency planning**: prepare responses to events that haven't happened yet - **Risk assessment**: evaluate irreversible actions before committing to them

Why deliberation has diminishing returns: The agent's internal model has finite fidelity. Deeper simulation eventually hits the model's accuracy boundary — further deliberation generates predictions the model cannot reliably distinguish. This explains why deliberation is most valuable early (when many model-derivable insights remain unextracted) and why high-fidelity internal models increase deliberation's value. But unlike "computation on existing data," this ceiling is about model fidelity, not data quantity — an agent with a perfect model and finite data can still generate unbounded value from deliberation (this is exactly what MCTS does).

The computational-extraction-gap framing. A sharper structural characterization than "third activity on par with exploit and explore" reads deliberation's value as the *gap between information in the existing data and information already extracted into* M_t. The three activities differ in which bit-channel they operate on: exploitation earns value-bits (direct reward, no new information); exploration earns environment-bits (reduces $H(\Omega_t \mid M_t)$ via new observations); deliberation earns inference-bits (reduces $H(\theta \mid M_t, \mathcal{C}_t)$ via additional computation on existing data). The information ceiling for deliberation is

$$I_{\text{delib}}(\Delta\tau) \;\leq\; I_{\mathcal{C}_t} \;-\; I_{\text{already-extracted}} \tag{9.5}$$

For an agent that processes its observation history optimally (full Bayesian update, exact inference), this gap is zero — deliberation has no value. The gap exists when the agent uses point estimates instead of full posteriors, when inference is intractable and approximate methods (MCMC, variational inference) have not converged, or when the agent has structural hypotheses it has not yet tested against existing data (cross-arm correlations in bandits; common-cause structure at L1; etc.). Three structural consequences follow:

1. *Agent-architecture-dependent*: a hypothetical perfect-Bayesian has zero deliberation value; real bounded-rational agents have non-zero deliberation value calibrated by their inference budget.
2. *Diminishing within a cycle*: each deliberation step closes part of the computation gap, with the next step's marginal value bounded by the remaining gap.
3. *Not independent of exploration*: new data changes what's available to extract; an exploration step expands $I_{\mathcal{C}_t}$ and therefore can *raise* the ceiling on subsequent deliberation value.

This places deliberation as a different *kind* of thing than exploit/explore — it operates on existing data rather than acquiring new data or value — even though the three-way framing in the Formal Expression treats them as comparable allocation targets. Both readings are valid: the allocation framing is useful for engineering decisions about $\Delta\tau$; the gap-closing framing is useful for understanding *why* deliberation has the structural properties it does.

Deliberation's comparative advantage is greatest when: - Internal simulation is cheaper than external probing (bits per unit cost, not bits per timestep) - The action space is too large for systematic exploration (planning/search) - Actions are irreversible and error costs are high (deliberation as risk management) - The agent has cross-domain knowledge exploitable by synthesis - The model supports counterfactual reasoning (Level 3 access)

Connection to 8.9. Deliberation duration required to revise Σ_t grows with strategy complexity. The description length $\mathrm{DL}(\Sigma_t)$ from 8.9 provides a lower bound on the deliberation time needed for meaningful revision.

Connection to 8.8. Strategic tempo $\mathcal{T}_\Sigma$ is the rate of useful strategy revision from *external* evidence. Deliberation provides an *internal* channel for strategy revision that supplements $\mathcal{T}_\Sigma$.

Domain instantiations:

Table 9.1

Domain	*Exploit*	*Explore*	*Deliberate*
Military (Boyd)	Execute the mission	Reconnaissance, probing attacks	Staff planning, war-gaming
RL agent	Greedy action	ε-exploration, UCB	MCTS planning, model-based rollouts
Software developer	Write code along current plan	Try unfamiliar approach, spike	Architecture review, read docs
Organization	Execute strategy	A/B test, pilot program	Strategic planning offsite
AI coding agent	Execute planned edit	Read unfamiliar code, run tests	Plan approach, analyze requirements

Working Notes.

- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. Removed at `candidate` stage per FORMAT.md Gate 4.
- **Open: conditions for deliberation dominance.** The simulation showed deliberation rarely dominates in bandit settings (no strategy structure, small action space, reversible actions). The open question is: what formal conditions make deliberation strictly dominate external exploration? Likely candidates: large action spaces requiring combinatorial search, irreversible actions requiring risk assessment, cross-domain knowledge available for synthesis, and high model fidelity enabling accurate counterfactual reasoning. The bandit simulation was unfavorable to deliberation by design (no structure to exploit internally); a planning-heavy domain (chess, logistics, architecture design) would likely show the opposite.
- **Open: the additive decomposition.** Under what conditions is the additive separation of epistemic and strategic benefit a good approximation? The interaction term (ΔV_Σ depends on ΔM_t) may be quantitatively significant when deliberation produces a model insight that transforms the strategy evaluation. Formalizing this requires a coupled model of epistemic and strategic improvement.
- **The ΔV_Σ approximation misses two channels of deliberation value.** The strategic-deliberation ceiling $\Delta V_\Sigma \approx \delta_{\text{regret}} \cdot$ Pr[revision succeeds] captures only the *strategy-revision* channel — deliberation that finds a better Σ_t within the current objective O_t. Two other channels of value are not in the approximation: (a) the *objective-revision* channel, where deliberation reveals that O_t should change (step 5 of the orient cascade); the value is then $V_{O'}(M_t, \pi') - V_O(M_t, \pi)$ where O' is the revised objective, a quantity with no necessary relationship to δ_{regret} (which measures regret *within* the current objective); (b) the *uncertainty-reduction* channel, where deliberation confirms that the current strategy is good and shifts the agent from cautious mixed-mode to confident exploitation; the value of *certainty about your strategy* is a behavioral effect not captured by δ_{regret}. The approximation is best read as a *strategy-revision ceiling* rather than a full deliberation-value ceiling; the other two channels operate on different quantities and should be tracked separately if they matter for the use case.
- **Landing context.** The adversarial analysis behind the Epistemic Status downgrade (formulation-choice + linearization + discussion-grade tags rather than "derived" headlines) landed in the 2026-05-12 (late) Group-I surgical-promotion sweep; see CHANGELOG 2026-05-13. Its four attacks (two-stage decomposition not forced; additive form unjustified; ΔV_Σ approximation inadequate; dominance regimes trivially correct) are absorbed into the Epistemic Status block, the §Discussion "Computational-extraction-gap framing" sub-section, and the missing-channels Working Note above. Originating spike is absorbed archaeology, not a live reference.
- **Open: the unified vs two-stage question.** Simulation suggests the unified approach may be strictly better. If so, the segment should present the unified formulation as primary and the two-stage decomposition as a computational simplification.
- **Relationship to working memory.** Deliberation requires maintaining internal state. For LLM agents, this manifests as context-window competition: tokens spent "thinking" are tokens unavailable for representing the problem state. This architectural constraint on deliberation is real but domain-specific.

9.3 *Discussion:* Additional Implications & Discussion

The chapter closes Part II by tying together what the cascade does, what forces it to take the shape it does, and what the cascade looks like at two structural extremes — when the agent is *too confident* in a drifting world (where the cascade has a Lyapunov-forced exploration drive its information-value cousins cannot supply) and when the agent is *coupled-architecture* by construction (where the cascade cannot be produced in cascade order by the model itself and must be recovered at the loop level). The chapter is small in segment count but its results connect Part II to Part III via the directed-separation architectural classification, to `03-llm-core/` via the Class-3-Coupled scaffolding requirement, and back to Part I Ch.4 via the bandwidth floor that deliberation does not escape.

The cascade is forced by information dependency, not designed.

`#der-orient-cascade` carries the chapter's central architectural move: the Orient phase of OODA expands into a multi-step cascade — reduce $\delta_{\text{epistemic}}$ → assess attainability → evaluate strategy → (if needed) revise objective — whose *ordering is not a design choice but a consequence of which quantities require which others*. Each step's input depends on the prior step's output. M_t enters every subsequent formula, so it must be updated first; attainability A_O is computed against M_t, so it precedes strategy evaluation; strategy revision needs A_O to know whether the problem is the strategy or the goal; objective revision is last because it requires both the model and the strategy assessment to know that no policy in Π can deliver the objective. The cascade is *derived* from information dependency, not engineered — and that is the framework's load-bearing position against alternative cascade orderings (objective-first, strategy-first, or all-at-once) that have appeared in adjacent literatures.

The downstream consequence is that Section II's diagnostic core — the satisfaction-gap and control-regret split from Ch.3 — has its corrective machinery here. The 2×2 cell map routes to four different revision targets via the cascade: M_t first under epistemic mismatch; Σ_t under positive control regret; Π expansion under control regret persisting after strategy revision; and O_t only when satisfaction gap persists after all earlier revisions. Objective revision is the *last resort*, not the first response to unmet goals. This is the chapter's implications-side payoff: the diagnostic from Ch.3 is what tells the agent *what* to revise; the cascade from this chapter is what tells the agent *in what order*; and the orient cascade's forced ordering is what makes the diagnostic actionable rather than merely informative. Section II would have been a collection of diagnostics without revision machinery without it.

The cascade also depends on the directed-separation scope condition from `#der-directed-separation` (Part II Ch.1) for the *sequential resolution* to be meaningful. In a Class 1 (Separated) architecture the cascade runs sequentially because epistemic processing has no causal path from G_t; in a Class 3 (Coupled) architecture the cascade lives in one merged computation and the resolution order cannot be produced by the model itself. The framework's prescription for Class 3 — that scaffolded loops around coupled components recover the cascade ordering at the wrapper level — depends on this chapter's derivation: what scaffolding *is* structurally doing is reproducing the orient cascade externally when the component cannot reproduce it internally. The implication is concrete and significant for practitioners: agentic systems (loops around LLMs) are not engineering convenience; they are the structural mechanism by which Section II's persistence guarantees become available to Class 3 (Coupled) substrates.

Exploration at two ends of the uncertainty spectrum.

The chapter's appendix-side derivations carry one of AAT's load-bearing methodological positions: exploration is forced by Lyapunov persistence collapse, not by epistemic value-of-information. `#deriv-causal-ib-exploration`'s Findings cover this in full — the maximum tolerable observation-noise expectation $U_o^{\max} = U_M(Rc_{\min}/\rho^{\text{eff}} - 1)$ scales with U_M, so as the agent becomes confident ($U_M \to 0$), the tolerable noise tightens, and the agent is forced to actively seek pristine, low-noise observations to penetrate its own stubbornness. The framework therefore carries *two parallel exploration drives at opposite ends of the uncertainty spectrum*: the standard epistemic drive $\lambda_{\text{info}} \propto U_M$ (explore when uncertain — dominant when U_M is high) and a Lyapunov-survival drive $\lambda_{\text{surv}} \propto 1/U_M$ (explore when confident in a drifting world, because the agent's own update gain has dropped below the rate at which reality is shifting — dominant when U_M is low).

The implications-side payoff that the segment's Findings flags but is worth repeating is that this *sidesteps active-inference's dark-room objection at structural rather than parametric level*: an EFE agent with strong priors can stop exploring, an AAT agent in a drifting world physically cannot, because exploration is forced by the same Lyapunov machinery that grounds the rest of Section II.

`#deriv-causal-ib-lmi` carries the matrix-form sharpening. The scalar form of the survival imperative admits a "blank wall" attack — an action that selects a low-noise channel orthogonal to the drift direction satisfies the scalar constraint while the agent's mismatch in the drifting subspace grows unbounded. The LMI lift to a Linear Matrix Inequality on the Fisher Information Matrix, with a positive-semidefinite matrix Lagrange multiplier $\Lambda \succeq 0$ as directional shadow price, resolves the attack by complementary slackness on direction — the exploration bonus for blank-wall actions is mathematically zeroed. The result also makes precise that the Fisher Information Matrix is doing *three structurally distinct* loads in the framework: credit assignment via `#deriv-fisher-whitened-update-rule`, coordinate-forcing via `#disc-additive-coordinate-forcing`'s fourth instance, and now survival via the LMI here. The same geometric object recurs because the framework's coordinates are forced rather than chosen; the LMI form is the first explicit cross-instance composition where two of those loads — coordinate-forcing and survival — interact within the same machinery.

The composition with Part I Ch.4's persistence-information-rate floor is worth surfacing. The persistence-bandwidth-floor $C \geq \mathcal{T}/2$ is what a confident agent in a drifting world is structurally required to *meet*; the survival-imperative LMI is what tells the agent *in what directions* to spend the bandwidth. Without the LMI, the bandwidth could in principle be supplied in irrelevant directions; with the LMI, complementary slackness routes the bandwidth toward the drifting modes. The two results compose into a directional version of Part I Ch.4's central inequality: *bandwidth-per-direction is the binding constraint*, and the LMI is what makes the directionality structurally enforceable rather than implicit.

Coupled architecture, the bias bound, and the scaffolding requirement.

The chapter is also where the framework's treatment of Class 3 (Coupled) agents lands its most consequential structural commitment. `#der-directed-separation`'s Findings section covers the architectural classification in full — Class 1 (Separated), Class 2 (Partial), Class 3 (Coupled) — with the explicit Class 3 scope exit that hands the agents off to `03-llm-core/` rather than treating directed separation as an unenforced approximation. `#deriv-observation-ambiguity-bias-bound`'s Findings cover the universal constant under the (PI) parameterization-invariance axiom: Track 1 (transport-inequality, linear in I) gives $C^2_{W_2} = 2L^2_{\text{post}}/\rho_{\text{LSI}}$ under log-Sobolev + Lipschitz-posterior; Track 2 (Fisher-Rao, $\sqrt{I}$ scaling) gives universal dimension-free $C_{FR} = \sqrt{2}$ under (PI)+Čencov + small-information regime. The no-go counterexample (heteroscedastic-Gaussian, demonstrating that no universal C exists under Euclidean-parameter norm) is what forces (PI) from convergent representational choice to load-bearing for theorem-level status — and what adds the fourth instance to `#disc-additive-coordinate-forcing`'s catalog and the fourth identifiability-floor Instance 4 to `#disc-identifiability-floor`.

What the implications segment can develop beyond the source-segment Findings is the *coupled-diagnostic framework* that the bias bound enables, and which does not currently have its own segment home. For Class 2 (Partial) and Class 3 (Coupled) agents — LLMs, transformer-based agents with attention processing goals and observations together — the orient cascade's information-dependency-forced ordering cannot be produced in cascade order by the model itself; the cascade lives in one merged computation. But the Section II diagnostic quantities (δ_{sat}, δ_{regret}) remain *defined* on post-update state, and the Lipschitz bound $|\delta^{(\text{coupled})}_{\text{sat}} - \delta^{(\text{clean})}_{\text{sat}}| \leq L_A \cdot \|\Delta M_{\text{bias}}\|$ shows the diagnostic error is bounded by the bias bound, which is itself bounded by $\kappa \cdot \mathcal{A}$ (coupling × ambiguity). The structural conclusion is the load-bearing implication: **agentic systems — scaffolded loops around coupled components — are not just engineering convenience but a structural requirement to recover Section II's persistence guarantees for Class 3 (Coupled) substrates**. The cascade ordering happens at the loop level when the model cannot produce it; the scaffolding is doing real structural work rather than aesthetic work. For a production LLM-agent practitioner running structured-reasoning templates, multi-step prompts, external monitors, or retrieval-augmented calls, the framework's claim is that what is being built is *necessary* (not contingent) for AAT-grade behavior in a Class 2 / Class 3 substrate. The cross-domain transfer to human bicameral / dual-process models (Kahneman 2011) and to organizations where executive scaffolding recovers cascade ordering across coupled individuals follows the same structural mechanism.

The engineering anchor worth carrying is the $\varkappa \cdot \mathcal{A}$ decomposition. The architectural coupling $\varkappa$ is *fixed* for a Class 3 substrate (you cannot un-couple an LLM); the ambiguity $\mathcal{A}$ is *variable* (you can engineer the prompt and environment). Prompt engineering, in the framework's vocabulary, is literally $\mathcal{A}$-reduction — and the bias bound says reducing $\mathcal{A}$ tightens the bound proportionally, with the constant C now bounded by Track 1 or Track 2. The framework predicts (and the empirical pattern supports) that LLMs excel at low-$\mathcal{A}$ tasks (coding, structured reasoning, well-defined transformations) and struggle with high-$\mathcal{A}$ tasks (open interpretation, ambiguous goals, contested values). The framework's prescription is concrete: where $\varkappa$ is structurally large, design $\mathcal{A}$ structurally small.

Deliberation is internal exploration that does not relax the bandwidth floor.

`#disc-exploit-explore-deliberate` extends the standard exploit / explore tradeoff into a three-way framing by surfacing deliberation as *internal exploration in model space rather than environment space*. The implications-side payoff is structural: deliberation is Pearl-do on a simulated trajectory rather than on the agent's singular causal trajectory, and the chapter is where this characterization first becomes formally well-posed — Part II Ch.2's loop-Level-2 result is what makes the deliberation-as-internal-exploration framing distinct from "the agent thinks more" rather than a verbal reframing.

The composition with Part I Ch.4's bandwidth floor is the finding that has no other natural home: *deliberation does not relax the persistence-information-rate floor; it only changes channel allocation*. The agent that deliberates is internally rerouting its information-processing budget from environmental observation channels to internal-simulation channels, but the total Shannon information rate required to maintain bounded mismatch ($\dot{R} \geq n\alpha/2$ nats/time per dimension) is unchanged. An agent that deliberates indefinitely is *not* exempt from receiving observations at the floor rate — environmental drift continues during deliberation, and the post-deliberation mismatch grows at rate $\rho_{\text{delib}} \cdot \Delta\tau$ per `#der-deliberation-cost`'s drift term. The extended deliberation threshold from `#disc-exploit-explore-deliberate` formalizes this: the agent should deliberate until the marginal benefit (epistemic gain + strategic gain) equals the marginal drift cost. The framework's posture is that deliberation is *temporary reallocation*, not substitute for external bandwidth — and the cross-domain transfer follows naturally to chess engines (search depth and observation freshness must both be funded), military command (war-gaming and reconnaissance must both be funded), and science (theory-building and experiment-running must both be funded).

The connection to Part II Ch.4's strategy-complexity-cost (8.9) is also worth surfacing here. Part of the cost of deliberation is the cognitive cost of holding Σ_t in the agent's representational capacity while operating on it. For an LLM agent, this means fitting the strategy in the context window — and the framework's claim is that this is the same constraint that bounds online operation, not a separate deliberation-mode constraint. Deliberation buys epistemic and strategic improvement at the cost of cycle budget *and* representational capacity simultaneously; both are bandwidth-flavored constraints, and the persistence-bandwidth-floor framework subsumes them. This is one of the chapter's quieter findings — the framework's bandwidth vocabulary unifies environmental-observation bandwidth, strategy-representation capacity, and deliberation budget under a single Shannon-rate floor.

What Part II closes on, and the bridge into Part III.

The chapter is the structural close of Part II, and what Section II has delivered as a whole is now worth naming. The diagnostic core (satisfaction-gap and control-regret split, convention-indexed by the C1/C2/C3 hierarchy) gives the framework two orthogonal signals that distinguish "the world doesn't permit it" from "you're not doing it well enough." The strategy-DAG commitment with Markov factorization under causal sufficiency gives the formal representation the diagnostic operates on. The Pearl-Level-2 access from the loop, combined with the Correlation Hierarchy's scope statement and the chapter's directed-separation architectural classification, gives the agent the data substrate and the architectural class it sits in. The bias bound for Class 3 (Coupled) agents, with its conditional-theorem constant under named sub-scopes, gives the framework's quantitative reach into LLM-class substrates without overclaim. The orient cascade gives the corrective machinery; the survival-imperative drive gives the structural exploration mandate; the coupled-diagnostic framework gives the scaffolding-as-structural-requirement claim.

The result of all this is what `#result-section-ii-survival` formalizes: Section II's definitional and structural architecture transfers exactly to fully-coupled (Class 3) agents in 16 of 24 cases, with the remaining results either approximating with bounded error (5/24) or naming the precise modification required (2/24 plus 1 boundary-defining failure). The bias-bound theorem under named sub-scopes is what lifts the approximate-survival category from heuristic to conditional theorem. Section II's reach is wider than the most-restrictive scope condition suggests; this chapter is where the chapter-end can name that reach without overclaiming.

The bridge to Part III is direct. The directed-separation classification opens to *composite-level class inheritance*: Class-1 sub-agents with partially-opposing objectives produce Class-2 (or Class-3) composites — the architectural classification is not preserved under composition with goal divergence. The third instance of the identifiability-floor pattern (Instance 3) lives in Part III: composite contraction certification from component-marginal data is structurally impossible (Liberzon 2003 common-Lyapunov-nonexistence), and the unique broadly-available escape is observer-on-sub-agent interventional access (Mode 2 of the loop-Level-2 pattern from Ch.2). Part III is where AAT's machinery operates at the composite scope, with the chapter-end here as the last single-agent prerequisite to land.

What this chapter closes on, then, is the structural completion of Section II: the cascade that makes the diagnostics actionable, the dual-drive exploration that prevents dark-room collapse, the bias bound that quantifies Class 3 (Coupled) reach with conditional-theorem precision, the scaffolding requirement that makes agentic systems structurally non-optional rather than aesthetic, and the deliberation framing that grounds the three-way resource allocation in the same Shannon-rate floor as everything else. Part III's machinery operates on what Section II built; the implications work here is the last single-agent-scope synthesis before the composite chapters take over.

Working Notes.

- This segment is a chapter-end discussion grouping findings from `#der-orient-cascade`, `#disc-exploit-explore-deliberate`, `#der-directed-separation` (the cascade depends on it for the sequential-resolution scope), `#deriv-observation-ambiguity-bias-bound` (Appendix A, the bias-bound constant), `#deriv-causal-ib-exploration` and `#deriv-causal-ib-lmi` (Appendix A, the survival-imperative drive and its matrix lift), and `#der-deliberation-cost` (Part I Ch.4, the deliberation threshold this chapter extends). The closing-Part-II reading in §5 leans on `#result-section-ii-survival` which is the formal classification of which Section II results transfer to Class 3 (Coupled) agents at exact / approximate / modified status.
- **Cross-reference vs canonical-home policy.** Four source segments in this chapter's scope carry well-written `## Findings` sections — `#der-directed-separation` (architectural classification), `#deriv-observation-ambiguity-bias-bound` (universal constant under (PI)), `#deriv-causal-ib-exploration` (survival-imperative drive), and `#deriv-causal-ib-lmi` (matrix-form lift). All four are cross-referenced rather than restated. The findings whose canonical home is this implications segment are: `#20` (Orient Cascade as derived from information dependency), `#13` (Coupled Diagnostic Framework — scaffolded loops recover the cascade for Class 3 substrates; no segment currently carries this finding directly), and `#35` (Internal Deliberation Pays Bandwidth Floor — compositional finding with no segment home).
- **`#13` (Coupled Diagnostic Framework) — segment exists in `03-llm-core/`.** Corrected from earlier draft: `#result-coupled-diagnostic-framework` exists as a segment in `03-llm-core/src/` (where the Class 3 (Coupled) treatment lives), not in `01-aat-core/`. The segment does not currently carry a `## Findings` section, so the implications-side surfacing here covers what AAT Ch.5 unlocks (cascade recovery via scaffolding at the loop level, bias bound via $\kappa \times \mathcal{A}$); the Logogenic chapter-end implications carry the finding from the logogenic-substrate-specific side. A future cycle may add a `## Findings` section to the existing segment in `03-llm-core/`; until then, the chapter-end implications carry the catalog-grade content.
- **Connection to NeurIPS submissions.** The bias-bound result is NeurIPS Paper 3 (`~/src/neurips/03-llm-hallucinate-bound/`) in formal theorem form; the (PI) axiom from `#scope-agent-identity` is load-bearing for the universal-constant route under the no-go in §4 of Paper 3. The cascade-recovery-via-scaffolding argument is a structural claim AAT has been carrying since the Section II reorganization; whether it lifts to theorem grade in any current submission is not flagged. The

survival-imperative drive composed with the LMI matrix lift is the formal apparatus behind NeurIPS Paper 1 ("Tragedy of the Confident Agent", ~`/src/neurips/01-tragedy-confident-agent/`).

- **Identifiability-floor pattern coverage update.** With this chapter's content the pattern's four formal instances are now all in view: Instance 1 (on-policy L0-insufficiency, Part II Ch.4), Instance 2 (L1' mixture under unobservable common cause, Part II Ch.3), Instance 4 (universal constant under heteroscedastic-Gaussian no-go, this chapter / Appendix A). Instance 3 (composite contraction certification from component-marginals via Liberzon common-Lyapunov-nonexistence) lives in Part III. Instance 5 (Mehra non-identifiability in adaptive Kalman) is candidate. The chapter-end pattern coverage is now five-sixths complete on the planned AAT chapter-end work; the Instance 3 surfacing in Part III Ch.5 completes it.

Part III

Agentic Composites

Chapter 10

Scope and Composition Formation

10.1 *Scope:* Multi-Agent Scope

Section III applies wherever multiple agents satisfying the scope condition interact through a shared environment. Each agent observes, acts, and faces uncertainty; their actions affect each other's environments. This is the general case for organizations, teams, ecosystems, and adversarial encounters. Independence (agents whose actions don't affect each other) is the special case requiring justification.

Formal Expression.

A multi-agent system consists of N agents $\{A_1, \ldots, A_n\}$, each satisfying the scope condition (1.6), interacting through a shared environment with state $\Omega_t \in \mathcal{S}_{env}$:

Scope (multi-agent-scope).

- Each agent A_i has state $X_t^{(i)} = (M_t^{(i)}, G_t^{(i)})$
- Each agent observes: $o_t^{(i)} = h^{(i)}(\Omega_t, a_t^{(\neg i)}, \xi_t^{(i)})$ — observations may depend on other agents' actions
- Each agent acts: $a_t^{(i)} = \pi^{(i)}(X_t^{(i)})$
- The environment evolves: $\Omega_{t+1} = T(\Omega_t, a_t^{(1)}, \ldots, a_t^{(n)}, \omega_t)$

The coupling is through the environment: agent i's actions enter agent j's observation function and the shared environment transition. Agents may also communicate directly (a special case of action-observation coupling with a dedicated channel).

Observation decomposition and routing.

Each agent's observation decomposes into environmental and inter-agent components:

Definition (observation-decomposition).

$$o_t^{(i)} = \left(o_{\text{env},t}^{(i)},\ \{m_{ji,t}\}_{j \in \mathcal{N}_t(i)}\right) \tag{10.1}$$

where: - $o^{(i)}_{\text{env},t} = h^{(i)}_{\text{env}}(\Omega_t, \xi^{(i)}_t)$: direct environmental observation (no inter-agent content) - $\mathcal{N}_t(i) \subseteq \{1, \dots, N\} \setminus \{i\}$: the **communication neighborhood** — which agents send messages to i at time t - $m_{ji,t} = c^{(j\to i)}_t(X^{(j)}_t)$: message from j to i, determined by the sender's full state and the communication protocol

The **multi-agent routing structure** $R_t = (\mathcal{N}_t, \{c^{(j\to i)}_t\})$ specifies: - The **topology** $\mathcal{N}_t$: who communicates with whom - The **protocol** $c^{(j\to i)}_t$: the rule governing what class of information flows from j to i

Definition (multi-agent-routing-structure).

Note: the protocol $c^{(j\to i)}_t$ is a *rule* specifying the channel, not the specific content of any message. Individual messages reflect the sender's state $X^{(j)}_t$ — including their individual goals — through the sender's policy. What the routing structure governs is the *infrastructure*: which channels exist and what kind of information they carry. *Bare-prose shorthand: the term "routing structure" is sanctioned within this segment after the first compound-form introduction; cross-segment citation should use the full "multi-agent routing structure" form.*

Routing is **goal-blind** when neither the topology nor the protocol depends on the composite's goal state:

Definition (goal-blind-routing).

$$\mathcal{N}_t \perp G^c_t \qquad \text{and} \qquad c^{(j\to i)}_t \perp G^c_t \quad \forall\, j, i \tag{10.2}$$

This means the communication infrastructure does not change based on what the composite is trying to achieve. Individual messages naturally reflect individual agents' goals through their policies — this is action, not routing. The routing condition is about the *structure* of information flow, not the *content* of individual messages.

Goal-dependent routing occurs when either the topology or the protocol varies with G^c_t. Examples: activating crisis-specific communication channels, changing intelligence-sharing protocols based on the current mission, reassigning reporting chains based on the operational objective.

Epistemic Status.

Axiomatic. This is a scope definition — it describes the class of systems Section III addresses. The only substantive choice is that coupling goes through the shared environment rather than through direct state modification. This follows from the agent boundary assumption (1.1): agents affect each other by affecting the environment, not by directly altering each other's internal states.

Discussion.

Correlated observations as default. When agents share an environment, their observations are generically correlated — they see aspects of the same reality. Independence (uncorrelated observations) requires the agents to observe non-overlapping aspects of the environment, which is the special case. Most multi-agent settings of interest involve substantial observation correlation.

The adversarial case sits along a spectrum — partly inside, partly outside the composite scope. Agents whose objectives conflict are multi-agent systems with negative teleological unity (12). Whether they form a composite depends on whether their strategic interaction admits an equilibrium structure: equilibrium-convergent adversarial pairs (potential or monotone games per Monderer-Shapley / Rosen) satisfy 10.2 via route (C-iv) as **strategic composites** with equilibrium-based macro-state — see 14.1 for the equilibrium-theoretic machinery and the (SC-1)–(SC-3) decomposition. Cyclic or non-convergent adversarial pairs satisfy none of (C-i)–(C-iv) and remain within 10.1 only. The asymmetric attacker / target case (one agent treated as an exogenous parameter rather than a sub-agent running its full AAT loop) is also a 10.1 phenomenon, handled by 13.2.

Two distinct classes of machinery apply across this spectrum:

- **Agent-level machinery** (individual persistence, agent tempo, per-agent mismatch, sector-condition stability) applies to *every* agent in *every* multi-agent configuration — cooperative, adversarial, or indifferent — because each agent individually satisfies 1.6 and Section I/II results apply directly. The adversarial tempo advantage (13.4) and adversarial destabilization (13.2) results are applications of this agent-level machinery to the case where one agent's actions are a disturbance source for another.

- **Composite-level machinery** applies *only* when 10.2 is satisfied via at least one of (C-i)–(C-iv). The (C-i)–(C-iii) alignment routes give composites with a shared objective O_c and admit closure-defect / team-persistence / composite-tempo / unity-closure machinery via 11.1. The (C-iv) strategic route gives composites with an equilibrium-based macro-state and admits the equilibrium-convergence / sector-template / closure-defect machinery in 14.1; the alignment-route and strategic-route forms are structurally distinct (an equilibrium-convergent adversarial composite has no shared O_c, only a shared $\mathcal{E}$).

The adversarial regime spans the negative-U_O end of the unity spectrum. Equilibrium-convergent adversarial pairs are captured by (C-iv) strategic-composite machinery; cyclic / non-convergent adversarial dynamics and asymmetric attacker / target configurations are captured by agent-level machinery applied across the pair. The existing adversarial segments (13.2, 13.4) operate at the agent-level for cases that fall outside (C-iv); 14.1 operates at the composite-level for cases that fall inside it.

Inter-agent communication as a special observation channel. Messages from j to i are actions by j and observations by i. The routing structure formalizes the infrastructure: who talks to whom ($\mathcal{N}_t$) and under what protocol ($c_t^{(j\to i)}$). The sender controls the content (unlike passive environmental observation), which introduces strategic manipulation (12.4). The gain from inter-agent communication enters the distributed tempo (12.4, Working Notes).

The routing/content distinction matters for directed separation. Individual messages reflect senders' goals — that's just action through policy. The directed-separation question at the composite level (11.3) is about the *routing structure*: does the infrastructure change based on the composite's goals? Goal-blind routing preserves directed separation; goal-dependent routing breaks it.

10.2 *Scope:* Composite Agent

A set of purposeful sub-agents, each satisfying 1.6, constitutes a *composite agent* only when their objectives exhibit sufficient teleological alignment to define a coherent composite purpose. Without this condition, the sub-agents form a multi-agent system (10.1) that may still be analyzed — but the machinery of composition (closure defect, team persistence, composite tempo) does not apply because there is no composite agent whose quantities to compute. This parallels the single-agent 1.6: before asking how an agent adapts, one must check that an agent is present.

Formal Expression.

Given sub-agents $\{A_1, \dots, A_N\}$ each satisfying 1.6, they constitute a *composite agent* iff there exists a common objective structure under at least one of the following routes:

Scope (scope-composite-agent).

(C-i) Shared composite objective. There exists O_c such that each sub-agent's effective policy is ϵ-compatible with O_c-optimal:

$$\exists\, O_c \,:\, \forall i,\; D\big(\pi_i,\; \pi_i^{O_c}\big) \leq \epsilon \tag{10.3}$$

where $\pi_i^{O_c}$ is sub-agent i's optimal policy under the shared objective and D is an appropriate policy divergence. Strongest route; the sub-agents are all optimizing for the same thing, possibly with local decompositions.

(C-ii) Hierarchical derivation. There exists a parent objective O_c from which sub-objectives $\{O_i\}$ are derivable by decomposition consistent with 1.7:

$$\{O_i\} = \mathcal{D}(O_c) \tag{10.4}$$

where $\mathcal{D}$ is a structure-preserving decomposition. Each sub-agent optimizes its own O_i, but all O_i trace back to O_c. Military chain of command; corporate department structure. Sub-agents may not individually know O_c.

(C-iii) Mutual-benefit alignment. There exists a relevance variable Y such that the sub-agents' joint actions raise $\mathbb{E}[Y]$ above the non-cooperation baseline for each sub-agent:

$$\exists\, Y \,:\, \forall i,\; \mathbb{E}[Y \mid \text{joint}] > \mathbb{E}[Y \mid \text{non-coop}] \tag{10.5}$$

Weakest route. No explicit common objective, but interactions are positive-sum in some dimension. Symbiotic coexistence; commensal ecologies; trading partners who share no goals beyond mutual benefit.

(C-iv) Equilibrium-convergent strategic interaction. There exists an equilibrium concept $\mathcal{E}$ (Nash, correlated, or coarse correlated) such that coupled best-response dynamics of $\{A_i\}$ converge to (or cycle within the support of) $\mathcal{E}$:

$$\exists\, \mathcal{E} \,:\; \text{coupled best-response dynamics converge to the support of } \mathcal{E}. \tag{10.6}$$

Qualitatively distinct from (C-i)–(C-iii): requires neither shared objectives, nor hierarchical derivation, nor mutual benefit. Requires only **structural convergence** of the strategic interaction in the game-theoretic sense. The sub-agents' objectives may be partially opposing, but the interaction admits a stable joint behaviour. Composites satisfying (C-iv) are **strategic composites**, distinguished from alignment composites (C-i, C-ii) and mutual-benefit composites (C-iii). The composite's macro-state is defined relative to the equilibrium structure $\mathcal{E}$ rather than relative to a shared target state. See `#deriv-strategic-composition` for the A2'-analog transfer of `#result-sector-persistence-template` under potential-game (Monderer-Shapley 1996) or monotone-game (Rosen 1965) conditions, the (SC-1)–(SC-3) existence/stability/convergence decomposition, and the honest sub-scope β' scope exit for non-potential non-monotone games (VI existence + regret-minimization CCE set-convergence only).

Disjunctive form.

The scope condition is satisfied when **any** of (C-i), (C-ii), (C-iii), or (C-iv) applies. (C-i)–(C-iii) are progressively weaker qualitative requirements for what "teleological alignment sufficient to define a coherent composite purpose" means; (C-iv) covers strategic interaction with partially-opposing objectives via equilibrium convergence rather than alignment. The routes are *not* shown to reduce to a common scalar threshold. Each route carries its own operationalization and its own N-agent aggregation:

Scope (scope-composite-agent, disjunctive).

- (C-i) uses a value-function divergence $D(\pi_i, \pi_i^{O_c})$ aggregated across sub-agents.
- (C-ii) uses decomposition consistency of a parent objective O_c — a structural check, not a scalar.
- (C-iii) uses the existence of a relevance variable on which each sub-agent's marginal contribution is positive — a per-pair existential check, not a magnitude.
- (C-iv) uses existence of an equilibrium structure $\mathcal{E}$ under coupled best-response dynamics — a fixed-point check (Nash / VI / regret-minimization CCE), structurally distinct from the alignment checks in (C-i)–(C-iii).

The teleological unity measure U_O from 12 (pairwise value-correlation aggregated to the group) tracks one projection of alignment — primarily route (C-i) — but is not a reduction of all three routes to a single scalar. Downstream segments (12.1, 10.3, 13.1) describe quality *conditional on scope-satisfaction* without assuming a common threshold: they presume the scope condition holds via at least one route, then analyze composite quantities within that regime.

(C-i) gives the strongest alignment; (C-iii) gives the weakest that still qualifies. A composite may satisfy multiple routes simultaneously; only one is required for scope.

What fails the scope condition: sub-agents with orthogonal objectives that also fail to admit equilibrium convergence — no shared or derivable O_c, no relevance variable providing mutual benefit, and no equilibrium concept (pure Nash, mixed Nash, or CCE) whose support the coupled best-response or no-regret dynamics reach — or unclassifiable objective-structure coupling. Such systems remain within 10.1 but not 10.2. Adversarial pairs that admit equilibrium convergence via (C-iv) — whether pure-strategy Nash under α' (potential / monotone games), mixed Nash under β' (Nash 1950 existence for finite games), or CCE in distribution under β' (Hart-Mas-Colell 2000) — DO satisfy composition-scope-condition as strategic composites. Cyclic games (rock-paper-scissors, matching pennies) lack a pure-strategy Nash but admit mixed Nash and CCE convergence; they fall within β' and satisfy (C-iv). The narrow category that fails (C-iv) is games with no equilibrium concept whose support is reachable by any admissible dynamic — a genuinely small class within the standard game-theoretic landscape.

Epistemic Status.

Robust-qualitative. Max attainable: *robust-qualitative*. The composite-agent scope is a structural proposal paralleling the single-agent scopes (1.5 and 1.6, both axiomatic — the minimal conditions for adaptive and purposeful machinery to be non-vacuous). Unlike them, 10.2 is a theoretical choice: we could admit teleologically misaligned groups as composites (via the closure-defect framework alone) and separately track whether their composite objective is meaningful. The proposal here is that this separate tracking is better done as a scope restriction: composition quantities are defined for systems satisfying at least one of (C-i)–(C-iv), not for systems satisfying none.

The three alternative routes are not exhaustive but cover the well-understood cases. The relationship between the routes (whether they partition or overlap) and whether there is a single underlying scalar that reduces them all is open; the disjunctive form above is chosen because no such reduction has been demonstrated.

Discussion.

Why this is a scope condition, not merely a quality metric. The existing 11.1 framework treats any group admitting a low-ε^* projection as a composite. This is mechanically defensible but runs into a category problem: a composite's purposeful substate $G_c = (O_c, \Sigma_c)$ requires a well-defined composite objective O_c. If the sub-agents' objectives do not align under any common O_c, then G_c is ill-defined and the composite is a fiction. Closure defect can be computed (as a projection property), but calling the result a "composite agent" strains the term. Making this a scope condition resolves the category issue: composition applies where G_c is well-defined.

Relationship to the single-agent scopes. The parallel is tight. The single-agent scopes (1.5 + 1.6) together say: before asking whether an agent persists, check that it is an agent (it observes under residual uncertainty, and at least one action has causal effect). 10.2 says: before asking whether a composite persists, check that it is a composite (teleological alignment sufficient to define O_c). In both cases, the scope precedes the quantitative machinery.

Relationship to 12. U_O in 12 has a dual role: (i) scope tracking — primarily the (C-i) route, where value-correlation is the operationalization; (ii) quality parameter for action-closure ε_a, via 12.1. The two roles are not in tension, but they are also not identical across routes: scope-satisfaction via (C-ii) or (C-iii) does not directly correspond to a U_O value, so U_O is best read as a quality metric *conditional on scope*, not as the defining scope variable.

Four routes: three alignment routes plus one strategic-equilibrium route. Routes (C-i), (C-ii), (C-iii) — the **alignment routes** — differ in how directly the sub-agents share purpose. They are progressively weaker qualitative requirements — (C-i) strongest, (C-iii) weakest — but they are not shown to reduce to a single scalar. (C-i) uses value-correlation directly; (C-ii) uses hierarchical decomposition consistency; (C-iii) uses marginal benefit existence. A composite can satisfy multiple alignment routes simultaneously (a well-aligned team satisfies both (C-i) and (C-ii)). Route (C-iv) — the **strategic-equilibrium route** — is qualitatively distinct: it admits composites with partially-opposing objectives whose strategic interaction is equilibrium-convergent (potential or monotone games per Monderer-Shapley / Rosen; see `#deriv-strategic-composition`), and its composite macro-state is defined relative to the equilibrium structure $\mathcal{E}$ rather than a shared target. What matters for scope is that at least one of the four routes holds.

Miller's IAM definition. Miller (2022, *Ex Machina*, Ch 1) proposes that social behavior requires Interaction, Agency, and Mutual benefit (IAM). The mutual-benefit requirement in (C-iii) is the AAT analog of Miller's third element. Miller treats IAM as the definition of social behavior; the proposal here treats alignment as the scope condition for composite-agent analysis. The two align substantively: Miller's mutual benefit is a minimum teleological unity.

Multi-agent systems with and without composite status. Adversarial pairs partition along a finer line than alignment / non-alignment. Adversarial pairs that admit equilibrium convergence — via pure-strategy Nash under α' (potential or monotone strategic games per `#deriv-strategic-composition`), mixed Nash under β' (Nash 1950 existence for finite games), or CCE in distribution under β' (Hart-Mas-Colell 2000 under no-regret dynamics) — satisfy the scope condition via (C-iv) as strategic composites. Cyclic games (rock-paper-scissors, matching pennies) sit inside β': they lack pure-strategy Nash but admit mixed Nash and CCE convergence, so they satisfy (C-iv). The narrow category that genuinely fails (C-iv) is games with no equilibrium concept whose support is reachable by any admissible dynamic — a small class within the standard game-theoretic landscape. The asymmetric case — one agent as exogenous attacker, one as target — is a `#scope-multi-agent` phenomenon handled by `#der-adversarial-destabilization`; it does not form a composite in either (C-i)–(C-iii) or (C-iv) senses because the attacker is treated as a parameter rather than a sub-agent running its full AAT loop. Symmetric strategic composition under partially-opposing objectives is covered by (C-iv) and forms a strategic composite with equilibrium-based macro-state.

Load-bearing enabling role under `#disc-identifiability-floor` Instance 3. The composition-layer identifiability-floor instance establishes a no-go: composite contraction $\kappa_c > 0$ is not in general certifiable from component-level data alone — the coupling-sign bit distinguishing cooperative from adversarial regimes is unidentifiable from component marginals. Four structural escapes are available (observable coupling topology; matched Tier at composite level; passivity certificate; common contraction metric). Scope-satisfaction via at least one of (C-i)–(C-iv) in this segment is the **enabling condition** for any of these escapes: without scope-satisfaction, the "composite" is ill-defined (O_c does not exist under (C-i)–(C-iii); equilibrium structure $\mathcal{E}$ does not exist under (C-iv); R_c is ill-typed), and the escapes have no coherent target to certify. The scope condition is therefore load-bearing apparatus that the composition-layer floor depends on — scope-satisfaction is what distinguishes a composite whose contraction or equilibrium convergence can be certified (under some escape) from a multi-agent system whose "composite contraction" is not a well-defined object.

How composites come into being: symbiogenesis. The formal mechanism by which a group passes from failing all three routes to satisfying at least one is described in 10.3. Briefly: in biological and social symbiogenesis, a host integrates an endosymbiont; the endosymbiont's objective is subsumed into the host's; the combined entity's objective O_c emerges where two independent objectives stood. After the transition, (C-ii) applies: the endosymbiont's objective is a sub-objective of O_c. This is the dynamical process of composite-agent identity creation; the scope condition here describes its result.

Working Notes.

- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. Removed at `candidate` stage per FORMAT.md Gate 4.
- **Whether a common scalar exists across the alignment routes.** The current form treats the three alignment routes (C-i)–(C-iii) as a disjunction of qualitatively distinct conditions rather than as instances of a unified threshold; the strategic-equilibrium route (C-iv) is structurally distinct from all three and would need to be handled separately under any aggregation. Whether there is a single scalar that reduces (C-i)–(C-iii) — and whether such a scalar would have a phase-transition structure (a sharp threshold beyond which alignment-composite analysis becomes well-posed) — is open. If such a scalar exists, U_O is one candidate but covers only (C-i) directly; (C-ii) and (C-iii) would require separate operationalization and then an aggregation across routes. (C-iv)'s fixed-point existence check is not a scalar threshold on alignment and is not unified with the alignment-side aggregation.
- **Asymmetric unity.** Routes (C-i) and (C-ii) implicitly assume symmetric alignment (sub-agents equally represented in O_c). Symbiogenesis, firm acquisitions, and hierarchical delegation all feature *asymmetric* unity (host dominates; endosymbiont subordinate). The scope condition should accommodate asymmetric alignment; the current (C-ii) hierarchical form partially covers it, but an explicit treatment of role-asymmetric composites is open.
- **Interaction with Class 3 (Coupled) architectures.** 5.3's architectural classification (Class 1 Separated, Class 2 Partial, Class 3 Coupled) applies to individual agents. Composites of Class 3 (Coupled) agents — e.g., multi-LLM systems — may satisfy this scope condition via shared training rather than shared explicit objective. Whether (C-i)-(C-iii) cover this adequately, or whether a fourth route for implicit alignment is needed, is open.
- **Transitivity.** If A_1, A_2 align and A_2, A_3 align, do A_1, A_3? Probably yes via (C-ii) if there is a common parent; not necessarily via (C-i) without a shared objective. Transitivity would make composite formation more robust; non-transitivity would explain why coalition-building requires explicit alignment work.
- **Promotion dependencies.** Before `claims-verified`: settle whether the three alignment routes (C-i)–(C-iii) reduce to a common scalar or remain an honest disjunction (first bullet); audit whether any existing Section III segment implicitly assumes this scope without naming it (likely yes for team-persistence, tempo-composition, closure-based results) and update each to reference 10.2 for scope-satisfaction explicitly, including whether they apply to (C-i)–(C-iii), to (C-iv), or to both.

10.3 *Hypothesis:* Symbiogenic Composition

Symbiogenesis is an asymmetric composition mechanism in which one agent (the *host*) integrates another (the *endosymbiont*) as a specialized sub-component, with the endosymbiont's objective gradually subsumed into the host's. It is distinct from peer coupling (11.1) and from population-level restructuring (the extreme transition motif drawn from Miller 2022, discussed in 4.5): symbiogenesis is how composite agents *come into existence* by crossing the 10.2 from below. The mechanism is well-attested empirically (eukaryote formation, firm mergers, legal-precedent adoption, language families) but formally underspecified within AAT. This segment captures the phenomenon and flags the specific formalization gaps.

Formal Expression.

Given two purposeful agents A_h (host) and A_e (endosymbiont), each satisfying 1.6, symbiogenic composition is a process on the joint state space with three coupled dynamics:

Hypothesis (symbiogenic-composition).

(S-1) Objective absorption. The endosymbiont's objective O_e transforms toward alignment with or derivation from the host's objective O_h:

$$O_e(\tau) \xrightarrow{\tau \to \tau_{\text{consolidated}}} \mathcal{D}_e(O_h) \tag{10.7}$$

where $\mathcal{D}_e$ is a derivation functional (in the sense of route (C-ii) in 10.2): O_e becomes a sub-objective derived from O_h. Before: O_h and O_e are independent objectives, no route of 10.2 applies, and the pair is a multi-agent system (10.1) rather than a composite. After: O_e is a role within O_h; route (C-ii) applies; the composite (A_h, A_e) satisfies the composition scope condition.

(S-2) Function transfer. Structural content from the endosymbiont's state (elements of M_e or Σ_e) transfers to or becomes accessible by the host:

$$\{M_h, \Sigma_h\}(\tau) \xrightarrow{\tau \to \tau_{\text{consolidated}}} \{M_h, \Sigma_h\} \cup \mathcal{F}(M_e, \Sigma_e) \tag{10.8}$$

where $\mathcal{F}$ is a transfer mapping (structure-preserving integration of endosymbiont functions into host state). In biological symbiogenesis: gene transfer. In organizational symbiogenesis: acquired firm's processes, patents, know-how integrated into acquirer's operations. This is the grafting operation of 8.7 in its cross-agent form — the host grafts structure originating in the endosymbiont.

(S-3) Autonomy reduction. The endosymbiont's effective action space contracts; many of its choices become fixed by the host's coordination:

$$\mathcal{A}_e^{\text{effective}}(\tau) \xrightarrow{\tau \to \tau_{\text{consolidated}}} \mathcal{A}_e^{\text{restricted}} \subsetneq \mathcal{A}_e^{\text{initial}} \tag{10.9}$$

The endosymbiont retains enough autonomy to avoid catastrophic transfers (e.g., mitochondria retain some genome to handle local fast-timescale responses that would be hazardous to route through the host nucleus) but loses most independent decision-making.

Integrated transition. At consolidation, the joint system is a single composite agent A_c whose substate contains the integrated structure:

$$X_c = \big(M_c, G_c\big) = \big(M_h \cup \mathcal{F}(M_e, \Sigma_e),\ (O_c, \Sigma_c)\big) \quad \text{with } O_c \approx O_h \tag{10.10}$$

The endosymbiont persists as a specialized sub-component of the host, not as an independent agent. The 10.2 is now satisfied; the peer-coupling machinery of 11.1 applies to the resulting composite.

Epistemic Status.

Robust qualitative. Max attainable: *robust qualitative* — the phenomenon is well-attested empirically across biological and social domains, but a general mathematical formalization within AAT is open.

What is well-established (externally):

- The existence of symbiogenesis as a distinct evolutionary mechanism (Mereschkowsky 1905, 1910; Sagan 1967; Margulis & Sagan 1997). Mitochondria and chloroplasts are the paradigm cases.
- The social analog in firms, technology, language, legal systems, religions (Miller 2022, Appendix B).
- "Innovation by parts" as qualitatively different from "innovation by sparks" (gradual mutation) — Miller's framing.

What is *not* derived within AAT:

- A formal model of the objective-transfer dynamics (S-1). What evolutionary or optimization process drives $O_e \to \mathcal{D}_e(O_h)$?
- A formal specification of the transfer functional $\mathcal{F}$ in (S-2). What structure is preserved, what is lost, what is transformed?
- A precise characterization of autonomy reduction (S-3). Why does the endosymbiont retain some autonomy rather than becoming fully deterministic?
- Quantitative predictions — e.g., when symbiogenesis is favored over peer coupling, what governs the timescale of consolidation, under what conditions it reverses.

The three dynamics (S-1), (S-2), (S-3) are proposed schemas, not results. A follow-up development of an AAT-specific dynamical model is the natural next step.

Discussion.

The role of this mechanism in Section III. Three distinct composition mechanisms are now in scope:

1. **Peer coupling** (11.1, 13.1, 11.2) — sub-agents interact through shared environment; closure defect measures faithfulness of projection. Presumes scope-satisfaction via at least one route of 10.2 (not a scalar U_O threshold).
2. **Extreme transition motif** (Miller 2022; introduced in 4.5; pending dedicated segments for composition-transition dynamics, latent structural diversity, and endogenous coupling) — population-level restructuring via neutral drift / niche creation / cascading displacement. U_O shifts across a population as agent types replace one another.
3. **Symbiogenesis** (this segment) — hierarchical absorption. U_O crosses the composition scope condition from below, creating a composite that did not previously exist.

Before symbiogenesis, the sub-agents were a multi-agent system (10.1) but not a composite. After, the resulting composite is subject to all of AAT's composition machinery. The symbiogenic transition is the specific dynamical process of composite-agent identity creation.

Why this cannot be modeled as peer coupling. Peer coupling assumes pre-existing sub-agents being projected into a macro-description. The closure-defect framework presupposes the composite exists; it measures how faithfully the macro tracks the micro. Symbiogenesis is about a composite *coming into being* from two previously-independent agents. No projection Λ of the pre-symbiogenic system yields the post-symbiogenic composite, because the endosymbiont's objective *changes* during the process — its objective is different before and after. The transformation is intrinsic to the sub-agents' state, not an external projection choice.

Why this cannot be modeled as extreme transition. The extreme transition motif operates at the population level with many agents, neutral drift of types, and niche-construction dynamics. Symbiogenesis is typically between two specific agents (or a small number) and proceeds through a specific asymmetric integration rather than through statistical population dynamics. The mechanisms overlap — symbiogenesis often occurs as part of a larger transition — but the core mechanism of symbiogenesis (bilateral asymmetric integration) is distinct from the population-level dynamics of extreme transitions.

Examples across domains.

Table 10.1

Domain	*Host*	*Endosymbiont*	*Integrated composite*
Biology	Archaeal host cell	α-proteobacterium	Eukaryotic cell (mitochondrion persists as organelle)
Biology	Eukaryotic cell	Cyanobacterium	Plant cell (chloroplast persists as organelle)
Commerce	Acquiring firm	Acquired firm	Merged firm (acquired operates as division)
Technology	Base platform	Integrated component	Composite product (component operates within host system)
Linguistics	Host language	Adopted vocabulary/grammar	Creolized / evolved language
Law	Legal system	Adopted precedent	Evolved jurisprudence (precedent operates as doctrine)
Religion	Host tradition	Absorbed elements	Syncretic practice

In each case: asymmetric integration, autonomy reduction of the absorbed entity, gradual objective subsumption, functional specialization.

Connection to 8.7. The single-agent "grafting" operation in 8.7 is within-agent — an agent incorporates external structure into its own Σ_t. Symbiogenesis is cross-agent — the grafted structure originates in another agent, and the integration is accompanied by that other agent's objective being absorbed. These are related but distinct: grafting is the structural-change mechanism on the host side; symbiogenesis is the bilateral process that includes grafting plus objective-absorption plus autonomy reduction.

Rate-distortion interpretation (connecting to 12.1). Under the Information Bottleneck conjecture in 12.1, peer coupling is IB compression with the relevance variable defined by a shared composite objective. Symbiogenesis is the process by which the relevance variable itself shifts: from two separate IB problems (each sub-agent's own survival objective) to a single IB problem (the composite's survival objective). The symbiogenic transition creates the shared relevance variable, which in turn makes the IB frontier well-defined for the composite. This is a structural shift in the IB problem, not a compression along a fixed IB frontier.

Working Notes.

- **Objective-transfer dynamics (S-1).** The most load-bearing open formalization. What process drives $O_e \to \mathcal{D}_e(O_h)$? Candidates: evolutionary selection (endosymbionts whose objectives align with host survival are selected for, since the alternative is extinction); bounded-rationality constraint (coordinating two divergent objectives exceeds the endosymbiont's capacity, forcing simplification); explicit design (firm mergers where acquired objectives are deliberately restructured). Each gives a different dynamical equation.
- **Function transfer $\mathcal{F}$ (S-2).** Needs to respect the structure of the host's M_h and Σ_h. In biology, gene transfer preserves molecular functions but changes regulatory context. In social analogs, the analog is: functions are preserved, but their triggers and dependencies change. A general specification is open.
- **Autonomy reduction (S-3).** Why not complete? The endosymbiont retains some autonomy because complete integration would eliminate the fast local response capacity that made symbiogenesis advantageous in the first place. A cost-benefit analysis on autonomy retention (in the style of 8.9) would make this quantitative.
- **(S-3) as weighted-Lyapunov limit (sketch-level).** F's asymmetric limit $\alpha_2 \to 0$ under weighted Lyapunov $V_\mu(\xi) = \frac{1}{2}(\|\delta_1\|^2 + \mu\|\delta_2\|^2)$ with $\mu \to 0$ formalizes (S-3): the endosymbiont's autonomous correction dynamics are weighted out of the joint stability argument, leaving the host's sector condition as the composite's persistence condition. This is a smooth deformation of the peer-coupling (CM4) inequality, not a discontinuous regime change — symbiogenesis and peer coupling are parameter-limits of the same weighted-Lyapunov analysis. The sketch is promotable to derived once (S-2) function transfer is formalized in this segment (the weighted Lyapunov limit does not address what happens to M_h when structure from M_e is inherited).

- **Reverse symbiogenesis.** Endosymbionts occasionally regain autonomy (biological examples: some organelle-hosted genes return to the nucleus; organizational examples: acquired divisions spun off). Theoretically: the scope condition can be crossed in either direction. A composite that loses U_O dissolves back into a multi-agent system. The triggering conditions and typical dynamics are open.

- **Interaction with logogenic agents.** In LLM-based agent architectures, multiple models can compose through shared training or through interface-specified protocols. Whether this constitutes symbiogenesis (with one model dominating) or peer coupling depends on whether the component models retain independent objectives. Worth investigating in `03-llm-core/`.

- **Quantitative predictions.** When is symbiogenesis favored over peer coupling? Hypothesis: when the coordination overhead C_{coord} between would-be peer-coupled agents exceeds the integration cost of symbiogenesis. Transaction-cost theory (Coase / Williamson) is the economic analog. The AAT version would connect C_{coord} (11.2) to the energetic or informational cost of maintaining separate objectives, with symbiogenesis favored when the latter exceeds the former.

- **Timescale of consolidation.** In biology, symbiogenesis takes evolutionary time (millions of years). In firms, months to years. In software/ideas, potentially much faster. The consolidation timescale $\tau_{\text{consolidated}}$ is domain-dependent; a general characterization is open.

- **Saddle-node bifurcation analysis — quantitative threshold form, conditional on a nonlinear coordination penalty.** Under the *hypothesis* that aggregate multi-agent mismatch δ obeys $\dot{\delta} = \rho_{\text{env}} - \alpha_{\text{auto}}\delta + k\delta^2$ with a nonlinear coordination penalty $+k\delta^2$ ($k > 0$) capturing compounding coordination failure between misaligned autonomous sub-agents, the steady-state fixed points $\delta^* = (\alpha_{\text{auto}} \pm \sqrt{\alpha_{\text{auto}}^2 - 4k\rho_{\text{env}}})/(2k)$ exist only for $\rho_{\text{env}} \leq \rho_c := \alpha_{\text{auto}}^2/(4k)$. Above ρ_c the two fixed points collide in a *saddle-node bifurcation* and disappear: $\dot{\delta} > 0$ everywhere, and the autonomous multi-agent system has no stable equilibrium. The symbiogenic escape route is structural: the composite merges sub-agent objectives and state ($\mu \to 0$), eliminating the coordination-penalty term and recovering linear dynamics on the merged state. The bifurcation analysis therefore predicts symbiogenesis as a *mathematically forced phase transition* at critical environmental volatility ρ_c rather than as a contingent organizational choice. **Status: conditional on derivation of the $+k\delta^2$ coordination penalty from `#def-shared-intent`.** The $+k\delta^2$ form is currently stipulated rather than derived; making it rigorous requires showing that compounding coordination failure between agents with mismatch δ_A, δ_B produces an aggregate-mismatch dynamics with this specific quadratic-in-δ structure under named conditions on the shared-intent quantity. Without that derivation, the threshold $\rho_c = \alpha_{\text{auto}}^2/(4k)$ is a *formulation*, not a theorem; with it, the threshold form is exact under the named hypothesis. The closed-negative result here — that the bifurcation derivation is conditional on a derivation step not yet attempted — is itself load-bearing: any future strengthening of `#hyp-symbiogenic-composition` from hypothesis-tier to derived-result-tier must produce the missing derivation of the coordination penalty from a more fundamental AAT construct (the natural candidate is shared-intent mutual information between sub-agent models $M_t^{(A)}$ and $M_t^{(B)}$ under coupled-evidence regimes).

Chapter 11

Composition Machinery

[**Gap**] Discussion

11.1 *Formulation:* Composition Closure Criterion

We define a group of interacting agents as a valid composite macro-agent when its closed-loop dynamics approximately commute with coarse-graining — that is, when projecting micro-states to macro-states and then running macro-dynamics yields approximately the same result as running micro-dynamics and then projecting.

Formal Expression.

Let a system consist of N sub-agents interacting in a shared environment with state space $\mathcal{S}_{\text{env}}$. The micro-state, micro-observations, and micro-actions are:

$$X_{\text{micro},t} = \{(M_{i,t}, G_{i,t})\}_{i=1}^{N} \in \mathcal{X}_{\text{micro}} \tag{11.1}$$

$$o_{\text{micro},t} = \{o_{i,t}\}_{i=1}^{N} \in \mathcal{O}_{\text{micro}} \tag{11.2}$$

$$a_{\text{micro},t} = \{a_{i,t}\}_{i=1}^{N} \in \mathcal{A}_{\text{micro}} \tag{11.3}$$

The coupled micro-dynamics form an action-observation loop:

$$X_{\text{micro},t} \xrightarrow{\pi_{\text{micro}}} a_{\text{micro},t} \xrightarrow{E} (\Omega_{t+1}, o_{\text{micro},t+1}) \xrightarrow{f_{\text{micro}}} X_{\text{micro},t+1} \tag{11.4}$$

We constrain our search to an admissible class of projections $\Lambda \in \mathcal{P}_{\text{adm}}$ mapping micro to macro, an admissible class of macro-dynamics $(\pi_c, E_c, f_c) \in \mathcal{M}_{\text{adm}}$, and a **timescale ratio** $K_c \geq 1$: the number of micro-timesteps per macro-step. Micro-time is indexed by $t \in \{0, \dots, H\}$; macro-time by $m \in \{0, \dots, \lfloor H/K_c \rfloor\}$ with macro-step m corresponding to micro-timestep $t = mK_c$. The projection components then have type signatures:

- $\Lambda_x : \mathcal{X}_{\text{micro}} \to \mathcal{X}_c = (M_c, G_c)$ — pointwise, evaluated at macro-boundaries.
- $\Lambda_\Omega : \mathcal{S}_{\text{env}} \to \mathcal{S}_{\text{env},c}$ — pointwise, evaluated at macro-boundaries.

- $\Lambda_o : \mathcal{O}_{\text{micro}}^{K_c} \to \mathcal{O}_c$ — aggregates the window of K_c micro-observations between successive macro-boundaries.
- $\Lambda_a : \mathcal{A}_{\text{micro}}^{K_c} \to \mathcal{A}_c$ — aggregates the window of K_c micro-actions similarly (used on the observation-comparison side; the macro-policy π_c emits one macro-action per macro-step).

Aggregation is part of the projection specification — common choices are mean/sum (linear systems), first/last (event-rate projections), or task-specific sufficient statistics. When $K_c = 1$ the windows collapse, Λ_o, Λ_a reduce to pointwise maps, and micro- and macro-tempos coincide (synchronous-update composites). When $K_c \gg 1$ the composite operates on the quasi-steady-state output of its sub-agents, the regime that 4.6 asserts as the natural one for composition. K_c is part of the problem specification; it does not appear in (A1)-(A4) or (P1)-(P3) below, but it determines at what granularity closure is measured.

Let $\mathcal{D}_{\text{micro}}$ be the distribution of reachable trajectories generated entirely by the true micro-system over horizon H, and let $o_{\text{micro},(m-1)K_c:mK_c}$ denote the window of K_c micro-observations $(o_{\text{micro},(m-1)K_c+1}, \dots, o_{\text{micro},mK_c})$ between macro-boundaries $m-1$ and m (similarly for actions).

[Definition (Composition Closure)] We define the minimal achievable closure defect ε^* over the admissible classes as:

$$\varepsilon^* = \inf_{\Lambda \in \mathcal{P}_{\text{adm}},\, (\pi_c, E_c, f_c) \in \mathcal{M}_{\text{adm}}} \left\| (\varepsilon_x, \varepsilon_a, \varepsilon_o) \right\| \tag{11.5}$$

where the expected component errors evaluated over true micro-trajectories $\tau \sim \mathcal{D}_{\text{micro}}$, measured **per macro-step**, are:

- $\varepsilon_x = \mathbb{E}_\tau\Big[\frac{1}{M_H}\sum_{m=1}^{M_H} \Big\|\Lambda_x(X_{\text{micro},\,mK_c}) - f_c\big(\Lambda_x(X_{\text{micro},\,(m-1)K_c}),\ \Lambda_o(o_{\text{micro},\,(m-1)K_c:mK_c})\big)\Big\|_{\mathcal{X}}\Big]$
- $\varepsilon_a = \mathbb{E}_\tau\Big[\frac{1}{M_H}\sum_{m=1}^{M_H} \Big\|\Lambda_a(a_{\text{micro},\,(m-1)K_c:mK_c}) - \pi_c\big(\Lambda_x(X_{\text{micro},\,(m-1)K_c})\big)\Big\|_{\mathcal{A}}\Big]$
- $\varepsilon_o = \mathbb{E}_\tau\Big[\frac{1}{M_H}\sum_{m=1}^{M_H} \Big\|\Lambda_o\big(E_{\text{obs}}(\Omega,\ a_{\text{micro}})\big|_{(m-1)K_c:mK_c}\big) - E_{c,\text{obs}}\big(\Lambda_\Omega(\Omega_{(m-1)K_c}),\ \pi_c(\Lambda_x(X_{\text{micro},\,(m-1)K_c}))\big)\Big\|_{\mathcal{O}}\Big]$

where $M_H = \lfloor H/K_c \rfloor$ is the number of macro-steps in horizon H, and $E_{\text{obs}}(\Omega, a_{\text{micro}})\big|_{(m-1)K_c:mK_c}$ denotes the window of micro-observations the environment produces across the macro-step.

Units. $\varepsilon_x, \varepsilon_a, \varepsilon_o$ carry units of distance — per-macro-step norm errors. Multiplied by the macro-update rate ν_c (units [time^{-1}]), they yield the disturbance-rate units [distance] $\cdot$ [time^{-1}] demanded by the sector-persistence template's ρ_ξ (C) and by the dimensional accounting in 11.2. The previous per-micro-step formulation conflated ε^*'s granularity with ν_c's — the macro-step formulation resolves the unit inconsistency.

$K_c = 1$ **special case.** When micro and macro update in lockstep, $M_H = H$, the observation/action windows collapse, and the formulas reduce to the synchronous form. This is the regime of tightly-coupled groups (distributed algorithms running in phase, ensemble filters where every member steps per tick) and is where the two-Kalman worked example below operates.

$K_c \gg 1$ **case.** When the composite lives at a strictly slower timescale than its sub-agents (the regime 4.6 asserts as natural), only micro-states at macro-boundaries enter ε_x, and macro-observations aggregate K_c micro-observations. The closure defect then measures how well the macro-dynamics predicts the *next* macro-state from the *previous* macro-state and the aggregated micro-observation window — not any intermediate micro-state. Timescale abstraction is recovered: the composite's description need not — and under $K_c \gg 1$ should not — track the micro-trajectory between macro-boundaries.

The closure-defect framework applies to sets that satisfy 10.2 — i.e., that form composites via at least one of the three alignment routes (shared objective, hierarchical derivation, mutual benefit). Given scope-satisfaction, a set forms a *meaningful* composite agent — distinguished from a multi-agent system whose low-ε^* projection is a happenstance of the environment rather than a reflection of composite coherence — when $\varepsilon^* \leq \varepsilon_{\max}$. Without scope-satisfaction, the ε^*

infimum is still well-defined as a projection property, but the resulting "composite" has ill-defined purposeful substate $G_c = (O_c, \Sigma_c)$ and composition-level machinery (team persistence, composite tempo) does not apply.

Admissibility constraints on macro-dynamics.

$\mathcal{M}_{\text{adm}}$ is the class of macro-dynamics (π_c, E_c, f_c) satisfying:

Formulation (macro-dynamics-admissibility).

(A1) AAT agent structure. The macro-state decomposes as $X_c = (M_c, G_c)$. The macro-update is recursive at macro-time m:

$$X_{c,m+1} = f_c(X_{c,m}, o_{c,m+1}) \tag{11.6}$$

The macro-policy is state-dependent: $a_{c,m} = \pi_c(X_{c,m})$. This ensures the macro-agent has the same structural components as a single AAT agent (1.1, 3.2, 3.3).

(A2) Macro-mismatch. A mismatch signal is well-defined:

$$\delta_{c,m} = o_{c,m} - \hat{o}_{c,m}(M_c, a_{c,m-1}) \tag{11.7}$$

where $\hat{o}_{c,m}$ is the macro-model's prediction of the next macro-observation. This ensures the macro-agent can detect prediction errors — the foundation of adaptation (3.4).

(A3) Macro-tempo. The macro-update has well-defined adaptive tempo:

$$\mathcal{T}_c = \sum_k \nu_c^{(k)} \cdot \eta_c^{(k)*} \tag{11.8}$$

where k indexes the macro-agent's observation channels, $\nu_c^{(k)}$ is the event rate, and $\eta_c^{(k)*}$ is the optimal gain (3.8, 3.6). This ensures the persistence condition is formulable at the macro level.

(A4) Bounded macro-correction. The macro-correction function satisfies the sector condition (4.3):

$$\delta_c^T F_c(\mathcal{T}_c, \delta_c) \geq \alpha_c \|\delta_c\|^2 \quad \text{for } \|\delta_c\| \leq R_c \tag{11.9}$$

with $\alpha_c > 0$ (positive correction rate) and $R_c > 0$ (finite reserve). This ensures the macro-agent's corrections work — they reduce mismatch rather than amplifying it, within the macro-model's capacity.

What (A1)-(A4) prevent. Without these constraints, the infimum over $\mathcal{M}_{\text{adm}}$ is trivially zero (any dynamical system can curve-fit micro-trajectories over a finite horizon). The constraints force the macro-dynamics to be a genuine AAT agent — with decomposed state, prediction errors, adaptive tempo, and stable correction — not an arbitrary function approximator. ε^* then measures the irreducible cost of representing multiple agents as one AAT agent.

What (A1)-(A4) do NOT require. Directed separation (5.3) is not part of the admissibility constraints. It is an additional structural property that some composites have and others don't (11.3). Results that depend on directed separation declare it as an additional assumption. Similarly, strategy structure ($G_c = (O_c, \Sigma_c)$ with a DAG) is not required — simpler goal representations are admissible.

Admissibility constraints on projections.

The admissible class of projections $\mathcal{P}_{\text{adm}}(\epsilon_I, L)$ consists of projections $\Lambda = (\Lambda_x, \Lambda_o, \Lambda_a, \Lambda_\Omega)$ satisfying three conditions:

Definition (projection-admissibility, proposed from spike).

(P1) Information preservation. The projection retains at least a fraction $(1 - \epsilon_I)$ of the predictive mutual information, evaluated across a macro-step:

$$I\big(\Lambda_x(X_{\text{micro},(m-1)K_c});\ \Lambda_o(o_{\text{micro},(m-1)K_c:mK_c}) \mid \Lambda_a(a_{\text{micro},(m-1)K_c:mK_c})\big) \geq (1-\epsilon_I)\cdot I\big(X_{\text{micro},(m-1)K_c};\ o_{\text{micro},(m-1)K_c:mK_c} \mid a_{\text{micro},(n} \tag{11.10}$$

The left side is the predictive information the macro-state has about the next macro-observation (aggregated over the K_c-window), conditioned on the macro-action. The right side is the micro-state's predictive information about the *same* macro-observation, conditioned on the same aggregated actions — both sides share a common target so the ratio is well-defined. The parameter $\epsilon_I \in [0, 1)$ controls how much predictive power may be sacrificed by projection. This is the information bottleneck (Tishby et al. 1999) applied to the projection map. When $K_c = 1$, the windows collapse to singletons and (P1) reduces to the per-micro-step form previously stated.

(P2) Lipschitz continuity. Each component of Λ is Lipschitz-continuous with constant L:

$$\|\Lambda_x(X) - \Lambda_x(X')\|_{\mathcal{X}_c} \leq L \cdot \|X - X'\|_{\mathcal{X}_{\text{micro}}} \quad \forall X, X' \in \mathcal{X}_{\text{micro}} \tag{11.11}$$

(and analogously for $\Lambda_o, \Lambda_a, \Lambda_\Omega$, each with its own Lipschitz constant). Without Lipschitz regularity, bounded closure defect does not imply bounded trajectory error — the bridge lemma requires this or something equivalent. When the projection is L-Lipschitz, the trajectory error bound becomes $L \cdot \varepsilon^*/\alpha_c$.

(P3) Dimensionality reduction. The macro-state space has strictly lower dimension than the micro-state space:

$$\dim(\mathcal{X}_c) < \dim(\mathcal{X}_{\text{micro}}) \tag{11.12}$$

This prevents the trivial identity projection, which achieves $\varepsilon^* = 0$ but is not a genuine abstraction.

Why three conditions. No single condition suffices alone. (P1) prevents degenerate projections (a constant map $\Lambda_x = c$ has zero mutual information). (P2) prevents pathological projections (discontinuous maps that amplify micro-errors into unbounded macro-errors). (P3) prevents trivial projections (the identity map satisfies (P1) and (P2) but achieves nothing). Together they constrain $\mathcal{P}_{\text{adm}}$ to projections that are informative, regular, and genuinely reductive.

The (ϵ_I, L) parameters are part of the problem specification, not derived quantities. The natural choice for many applications is $L = 1$ (non-expansive projection) and ϵ_I chosen as the minimum information loss compatible with genuine dimensionality reduction.

Independence from (A1)-(A4). The macro-dynamics admissibility (A1)-(A4) partially constrains the projection — (A1) requires Λ_x to preserve the M_c/G_c decomposition, and (A4) implicitly requires regularity through the sector condition. But (A1)-(A4) do not specify how much predictive information the projection must retain. A macro-agent with a very coarse projection can satisfy (A1)-(A4) while being a poor representation of the micro-system. The information-preservation condition (P1) fills this gap. See `spikes/spike-projection-admissibility.md` §5 for the full independence analysis.

Bridge lemma: closure defect to trajectory error.

The bridge lemma is the sector-persistence template (C) applied with state variable $\xi = e_m = X_{c,m} - \Lambda_x(X_{\text{micro},\,mK_c})$ (trajectory error at macro-boundaries, between macro-evolved state and projected micro-state) and effective disturbance rate $\rho_\xi = \varepsilon^* \nu_c$ (closure defect per macro-step, multiplied by macro-update rate). The units are consistent by construction of the macro-step formulation above: ε^* is per-macro-step (distance) and ν_c is macro-step rate ([time^{-1}]), so ρ_ξ has the disturbance-rate units the template requires. The template yields directly:

Derived (bridge-lemma, from sector-persistence-template + A4 + contraction assumption).

$$\limsup_{m\to\infty} \|e_m\| \leq \frac{\varepsilon^* \nu_c}{\alpha_c} \tag{11.13}$$

This bound fits within the composite's sector-condition region iff $\varepsilon^* < \alpha_c R_c / \nu_c$, which is the persistence condition applied to the closure-error rate: if it fails, the macro-description diverges from micro-reality.

What the bridge lemma adds beyond the template. The template's precondition (T2) is the one-point sector bound. For the trajectory-error instantiation, the bound propagation requires a *strictly stronger* condition: the macro-update map $f_c(\cdot, o)$ must be contracting in its state argument (**incremental sector bound**, DA2'-inc — strong monotonicity of F_c), not merely one-point sector-bounded at each state. The per-step contraction factor is $\lambda = 1 - \alpha_c/\nu_c < 1$ (automatic when $\alpha_c < \nu_c$, true for any realistic agent that doesn't fully correct in a single step). This stronger condition is where the tier structure enters; see Epistemic Status for the Tier 1/2/3 taxonomy.

The sector condition (A4) does double duty — conditionally. (A4) ensures the macro-agent corrects external mismatch (single-agent persistence) AND — under the incremental sector bound — that the macro-description tracks micro-reality (this bridge). Both are instances of the same template applied to different state variables with different effective disturbance rates.

What Is Derived vs. What Is Chosen.

Table 11.1

Property	*Source*	*Strength*
Closure-defect quantity ε^* as infimum over admissible classes	Definition built from (A1)-(A4), (P1)-(P3), and K_c	Formulation choice
Three-component decomposition into $\varepsilon_x, \varepsilon_a, \varepsilon_o$	Match to the three arrows of the action-observation loop (state, action, observation)	Formulation choice
Per-macro-step formulation; ε^* has units of distance-per-macro-step	2026-04-22 temporal coarse-graining repair (Finding A); forced by dimensional consistency with $\rho_\xi = \varepsilon^* \nu_c$ in C and 11.2	Derived (dimensional consistency)
Timescale ratio $K_c \geq 1$ as independent problem-specification parameter	Application-specific; $K_c = 1$ (synchronous) and $K_c \gg 1$ (4.6 regime) both admissible	Formulation choice
(A1) Macro AAT structure $X_c = (M_c, G_c)$ with recursive update	Import from 1.1 + 3.2 + 3.3	Formulation choice (requirement)
(A2) Well-defined macro-mismatch	Import from 3.4	Formulation choice (requirement)
(A3) Well-defined macro-tempo	Import from 3.8 + 3.6	Formulation choice (requirement)
(A4) Sector-bounded macro-correction	Import from 4.3	Formulation choice (requirement)
(A1)-(A4) as a requirement *set* render the infimum non-trivial (exclude curve-fitting macro-dynamics)	Consequence of restricting $\mathcal{M}_{\text{adm}}$ to genuine AAT agents	Derived (under the chosen requirement set)

Property	*Source*	*Strength*
(P1) Information-preservation constraint	IB Lagrangian-dual with source X_{micro}, relevance $o_{\text{micro},t+1}$; formally connects to V	Formulation choice (requirement)
(P2) Lipschitz continuity of Λ	Required for bridge-lemma trajectory-error bound (without it, bounded ε^* does not imply bounded trajectory error)	Formulation choice (requirement)
(P3) Strict dimensionality reduction	Rules out identity projection (which trivially achieves $\varepsilon^* = 0$ but is not abstraction)	Formulation choice (requirement)
(P1)-(P3) independent of (A1)-(A4)	Spike analysis (`spikes/spike-projection-admissibility.md` §5) — (A1) and (A4) constrain Λ_x partially but do not fix information-preservation level	Derived
Bridge lemma: $\limsup\|e_m\| \le \varepsilon^* \nu_c/\alpha_c$	Sector-persistence template (C) applied with state $\xi = e_m$, disturbance rate $\rho_\xi = \varepsilon^* \nu_c$	Derived (conditional on incremental sector bound DA2'a-inc)
Persistence-as-closure-boundedness condition $\varepsilon^* < \alpha_c R_c/\nu_c$	Sector-condition region fits the asymptotic error ball	Derived (under A4 + DA2'a-inc)
Incremental sector bound (DA2'a-inc) is strictly stronger than (A4)'s one-point sector bound	Counterexample exhibited (oscillatory globally-inward, locally non-monotone corrections); `spikes/spike-bridge-lemma-contraction.md` §4.1	Proved
Three-tier agent classification (Tier 1 / 2 / 3) for bridge-lemma applicability	Taxonomy induced by whether DA2'a-inc holds globally (T1), locally (T2), or must be verified per-domain (T3)	Formulation choice (classification)
Tier 1: bridge lemma derived for Bayesian updaters on exponential families, gradient descent on strongly convex losses, linear correction with positive-definite gain-observation product	Standard monotone-operator / Kalman analysis; `spikes/spike-bridge-lemma-contraction.md`	Derived (conditional on DA2'a-inc + linear prediction, i.e., C2)
Tier 2: local contraction with factor degraded by $\kappa(D\hat{o})^2$	Extended-Kalman / locally-convex analysis	Derived (local)
Tier 3: contraction must be verified per-domain	No general result available for non-convex / discontinuous / non-mismatch-driven components	Discussion-grade
Composite (A4) from sub-agent properties: $\alpha_c \ge \min_i(\alpha_i - \Delta\mathcal{T}_i^{\text{cost}}), R_c \le \min_i R_i$	Weakest-link bound; `msc/working-composition-admissibility.md` §6.2	Derived (conditional on bounded coordination cost per 13.1)
Composite critical-mass inequality $(\alpha - C)R > \rho + \gamma\mathcal{T}$ (symmetric-matched-Tier-1) — sign-sensitive refinement of the weakest-link bound; recovers 13.1 (cooperative) and 13.2 (adversarial) as signed special cases	F	Derived (conditional on Tier 1 + matched-symmetric + Model D + coupling model C1/C2)
Composite persistence as (contraction) $\wedge$ (scope-satisfaction): $\kappa_c(U_O) > 0 \wedge$ 10.2	F (CM4); U_O enters multiplicatively on γ plus scope-gate, not additively	Derived (with (UO-mult) discussion-grade)
Asymmetric-limit connection to 10.3 (S-3) autonomy reduction	F §asymmetric-limit; weighted Lyapunov V_μ with $\mu \to 0$	Sketch (S-2 function transfer not yet formalized; promotable when 10.3 closes (S-2))
Meta-machine exact composition ($\varepsilon^* = 0$) for finite-state Moore machines	Miller 2022 §3.3; product automaton is the micro-dynamics by construction	Derived (external theorem)
Two-Kalman instantiation: $\varepsilon^* = 0$ at all ρ_{corr} for the means-only projection at steady state	Analytic steady-state calculation; `spikes/spike-composition-correlated-kalman.md`	Proved (closed form)
ε_x depends on both sub-agent unity *and* update-rule heterogeneity (ΔK)	Heterogeneous-gain Kalman case; 12.1	Derived (specific closed form for the two-Kalman case)

Property	*Source*	*Strength*
Mahalanobis norm as natural choice for estimation-type agents	Two-Kalman verification; Kalman gain minimizes expected squared Mahalanobis distance	Formulation choice (canonical for estimation; general-domain norm specification open)
N-agent scaling of ε^* (polynomial vs. exponential in team size)	Coupling-structure dependent; `spikes/spike-composition-scaling-N.md`	Open
Computability of (P1) for nonlinear / non-Gaussian systems	Requires Monte Carlo or variational bounds; closed-form only for linear-Gaussian	Open
Principled setting of the ϵ_I threshold	No formal criterion currently; candidate tying to team size and coupling structure	Open
Strategy-DAG projection under Λ_x	Domain-specific; not resolved by (P1)-(P3)	Open
Mori-Zwanzig zero-lag bound $\varepsilon^* \geq \|Q_\Lambda U P_\Lambda\|_{\text{op}}$	Koopman/MZ projection (stationary coordinate-compatible case)	Derived (conditional on stationarity)
Full-kernel MZ bound on ε^*	Type mismatch (trajectory-accumulation vs. per-step)	Refuted at the ε^*-level; natural home is the bridge-lemma trajectory-error bound

The dividing line: (A1)-(A4) and (P1)-(P3) are *formulation choices of the requirement set* — alternative requirement sets are possible (e.g., entropy-preservation instead of MI-preservation; contraction bound in place of Lipschitz), and this one is chosen for parsimony, direct import from prior AAT segments, and downstream tractability. The *consequences* that follow under this requirement set — the non-triviality of the infimum, the bridge-lemma bound, the weakest-link composite-(A4) derivation, the two-Kalman $\varepsilon^* = 0$ result — are Derived, with bridge-lemma-level results carrying the additional DA2'a-inc condition (which is proved strictly stronger than (A4) alone). The Tier 1/2/3 classification is itself a formulation, but the per-tier results are derived under tier-specific conditions. Two classes of genuinely open questions remain: the computability / principled-setting questions for (P1) and ϵ_I, and the N-agent scaling of ε^* — the latter decides whether very large teams are representable as single AAT agents at all.

Epistemic Status.

Conditional. Max attainable: conditional (formulation choice).

The closure defect ε^* is well-defined given (A1)-(A4), (P1)-(P3), the timescale ratio K_c, and specified norms. The macro-dynamics admissibility (A1)-(A4) and the projection admissibility (P1)-(P3) are independent formulation choices — (A1)-(A4) specify what "AAT-shaped macro-dynamics" means; (P1)-(P3) specify what "informative, regular, genuinely reductive projections" means. Both are needed for ε^* to be an infimum over a non-trivial, non-degenerate class. The timescale ratio $K_c \geq 1$ is a third, independent piece of problem specification: it sets the granularity at which closure is measured and the type signatures of Λ_o, Λ_a. K_c does not constrain the macro-dynamics class and is not derived — it is chosen from the timescale separation of the application (per 4.6). Both $K_c = 1$ (tightly-coupled composites) and $K_c \gg 1$ (hierarchically composed composites on their own timescale) are admissible. The projection admissibility conditions have a proposed definition with one exact instantiation (two-Kalman case at $K_c = 1$, `spikes/spike-projection-admissibility.md`); general-case computability of (P1) and the dependence of ε^* on K_c for a fixed application remain open.

The bridge lemma is *conditional* — not fully derived from (A4) alone. The precise additional condition is the **incremental sector bound** (DA2'a-inc): the correction function F_d must be *strongly monotone* ($\langle \delta - \delta', F_d(\delta) - F_d(\delta') \rangle \geq c_{\min} \|\delta - \delta'\|^2$ for all pairs), not just satisfy the one-point sector bound. This is strictly stronger than (A4) — a counterexample exists: oscillatory corrections that are globally inward-pointing but locally non-monotone (`spikes/spike-bridge-lemma-contraction.md`, §4.1). Three tiers of agents result:

- **Tier 1 (contraction proved):** Kalman filters, exponential-family Bayesian updaters, gradient descent on strongly convex losses, all linear corrections with positive definite gain-observation product. For these agents the bridge lemma is *derived* from (A4) + DA2'a-inc + linear prediction (C2), with contraction factor λ_{eff} from K.
- **Tier 2 (local contraction):** Extended Kalman, locally convex gradients, nonlinear prediction models. Contraction holds locally with factor degraded by $\kappa(D\hat{o})^2$.
- **Tier 3 (independent verification):** Non-convex optimization, discontinuous/rule-based corrections, agents with non-mismatch-driven state components. Contraction must be verified per-domain.

The discrete-time recurrence argument itself is standard (Elaydi 2005); the contribution is identifying the incremental sector bound as the precise condition that bridges sector-bounded correction to full-update-map contraction. Full derivation: `spikes/spike-bridge-lemma-contraction.md`.

The norm choices ($\|\cdot\|_{\mathcal{X}}$, $\|\cdot\|_{\mathcal{A}}$, $\|\cdot\|_{\mathcal{O}}$, and the combination norm) are load-bearing. For estimation-type agents, the Mahalanobis norm (weighted by inverse prediction-error covariance) is the natural choice — verified exactly in the two-Kalman case. For general domains, norm specification remains open.

Discussion.

This criterion replaces intuitive questions about "where the boundary of an agent is" with a functional test: does a macroscopic AAT description preserve the underlying micro-dynamics well enough to remain predictive and capable? The core requirement is an **approximate dynamical homomorphism** — the macro-dynamics approximately commute with the projection.

Relationship to 1.7. The Section I postulate requires that AAT's machinery be scale-invariant — predictions at different levels of description must be compatible. This segment operationalizes "compatible" as "bounded closure defect under admissible coarse-graining." The admissibility constraints ensure the macro-description is genuinely AAT-shaped, so the same persistence condition, the same tempo framework, and the same mismatch dynamics apply at the macro level with macro-level parameters.

Projection-behaviour facet of the stability certificate (W). Coarse-graining is a non-invertible projection of the certificate (D): the certificate-as-metric survives (the Schur complement of a positive-definite form is positive-definite) but the *dynamic* guarantee does not — the closure defect ε^* is the certificate's projection-residue, equal to the Mori–Zwanzig zero-lag memory-commitator norm (derivation table above), zero exactly when the resolved subspace is invariant. Read through the spine, this is the question "does the agent's measuring-stick survive being viewed coarsely": its shape survives, its guarantee leaks by exactly the memory term. The composition-level identifiability floor (no common certificate across sub-agents; Liberzon, X Instance 3) is this same residue read as a boundary event — a *distinct* obstruction from the rank-collapse floors (a non-invertible projection, not a congruence), which is why the cross-sectional structure is several meta-patterns and not one.

The Jacobian-level observation. The incremental sector bound (DA2'-inc) used to prove the bridge lemma is mathematically equivalent to the contraction-metric condition (CT2) evaluated at the identity metric $M = I$, for continuously differentiable (C^1) correction functions F on convex domains. This equivalence (cf. Rockafellar & Wets 1998) means that AAT's bridge lemma is a specialized application of generalized contraction theory. See `#result-contraction-template` for the full lifting of this condition to arbitrary Riemannian metrics.

Topology-indexed composition closures (via). `#result-contraction-template` lifts the bridge-lemma's DA2'-inc condition into the contraction-metric framework of Lohmiller & Slotine 1998. Under its (CT1)–(CT3) preconditions, three topology-indexed closure results apply:

- **Parallel composition** preserves contraction under blockdiag metric at rate $\min(\lambda_1, \lambda_2)$ — recovers this segment's weakest-link bound.

- **Cascade composition** ($\dot{x}_1 = f_1(x_1)$, $\dot{x}_2 = f_2(x_1, x_2)$) preserves contraction under lower-triangular weighted metric at rate bounded by $\min(\lambda_1, \lambda_2)$ up to coupling-gain adjustment (Slotine 2003 Thm 2). Handles hierarchical agent structures.
- **Negative feedback with bounded loop gain** (Slotine 2003 Thm 3) gives the heterogeneous critical-mass inequality $(\lambda_1 - C_1)(\lambda_2 - C_2) > k_{12}k_{21}/4$ via (CM2-M) in — extending F's matched-symmetric (CM2) to heterogeneous sub-agents with different architectures and coupling strengths.

The Tier 1/2/3 bridge-lemma taxonomy of this segment maps cleanly to Tier 1M/2M/3M under (globally metric-contracting / locally metric-contracting / no globally valid metric). The observation that (CT2) at $M = I$ is *equivalent* to DA2'-inc means this segment has been carrying the Jacobian-level Euclidean contraction condition at the composite level all along; surfaces this at the single-agent level and adds the compositional theorems that generalize this segment's closure content beyond the matched-symmetric case.

DA2'-inc ≡ (CT2) at $M = I$ — structural-transparency equivalence. *[Derived, status: exact.]* For C^1 correction functions F on convex domains, the following three conditions are equivalent (standard monotone-operator theory; Rockafellar & Wets 1998 *Variational Analysis* Corollary 12.4; Nesterov 2004 *Introductory Lectures on Convex Optimization* §2.1.3 Theorem 2.1.9 — strong monotonicity ⇔ symmetric-part of Jacobian uniformly positive-definite):

(a) DA2'-inc (strong monotonicity): $(\delta - \delta')^\top (F(\delta) - F(\delta')) \geq c\|\delta - \delta'\|^2$ for all δ, δ'.

(b) Jacobian-symmetric-part PD: $(\partial F/\partial \delta)_{\text{sym}} \succeq cI$ pointwise on the domain.

(c) (CT2) at $M = I$ with $\lambda = c$: $-\partial F/\partial\delta - (\partial F/\partial\delta)^\top \preceq -2\lambda I$.

Derivation. (a) → (b): take $\delta' \to \delta$ along direction v, divide by $\|\delta - \delta'\|^2$, take the limit. (b) → (a): integrate $v^\top(\partial F/\partial\delta)_{\text{sym}} v \geq c\|v\|^2$ along the segment from δ' to δ. (b) ↔ (c): direct algebraic identity.

Implication for AAT-internal derivability. Under this equivalence, the Euclidean sub-scope metric-α_1 cases of `#result-contraction-template` (Kalman-scalar, Euclidean strongly-convex-gradient, L2-regularized, linear-PD-symmetric) are AAT-internally derived from the DA2'-inc commitment already carried at the composite level — no new axiom required. What the (CT2)-at-$M = I$ framing adds is *visibility*: the Jacobian-level condition was implicit under the DA2'-inc name; surfacing the equivalence converts the Euclidean sub-scope from "theorem-imported from Lohmiller-Slotine" to "AAT-internally derived via a standard monotone-operator identity." This is structural transparency, not new content. Non-Euclidean metric-α_2 cases (Fisher, Hessian, Lyapunov-metric) remain separately treated in `#result-contraction-template` — the equivalence is specific to $M = I$.

The sector condition does double duty — conditionally. (A4) ensures the macro-agent's corrections work (structural persistence), AND — given the additional contraction assumption (see Epistemic Status) — it ensures the macro-description tracks micro-reality (bridge lemma). Both uses address *structural persistence* in the sense of `LEXICON.md` — the capacity of the correction machinery, not the current operating point or the composite's identity through time. The central insight is that structural persistence of the composite and faithfulness to its constituents are related through the same mechanism — contracting correction dynamics under bounded perturbation — but the bridge from sector-bounded correction to full-update-map contraction requires an assumption beyond (A4) that has been verified for estimation-type agents (Kalman) and remains open for general agents. When this assumption holds, the composite that persists in its own right also persists as a faithful representation of its constituents.

Deriving composite (A4) from sub-agent properties. If each sub-agent satisfies the sector condition with parameters (α_i, R_i), and coordination costs are bounded by $\Delta\mathcal{T}_i^{\text{cost}}$ per agent (13.1), then the composite satisfies (A4) with:

$$\alpha_c \geq \min_i(\alpha_i - \Delta\mathcal{T}_i^{\text{cost}}) \tag{11.14}$$

$$R_c \leq \min_i R_i \tag{11.15}$$

This is a weakest-link bound (conservative but clean). Cooperative coupling (13.1) can improve α_c beyond this bound by reducing effective disturbance. The key implication: (A4) is *verifiable* from micro-level properties — compute each sub-agent's sector-condition parameters, estimate coordination costs, and check whether the composite has positive correction rate. No need to compute f_c directly. See `msc/working-composition-admissibility.md` §6.2 for the derivation.

Sign-sensitive refinement (derived). The sign-blind weakest-link bound is subsumed by F's closed-form inequality $(\alpha - C)R > \rho + \gamma\mathcal{T}$ (CM2) in the matched-symmetric-Tier-1 case, where the right-hand side makes the effective disturbance explicit and the sign of the coupling γ (cooperative $\gamma < 0$ vs adversarial $\gamma > 0$) enters directly. (CM2) can yield composite persistence even when the weakest-link bound alone fails — when cooperative coupling reduces $\rho + \gamma\mathcal{T}$ below what the raw $\alpha - C$ margin would permit. 13.1 (cooperative) and 13.2 (adversarial) are recovered as signed special cases. The honest composite-persistence statement is the conjunction of contraction (CM3) **and** scope-satisfaction (10.2) — (CM4) — making explicit that composite persistence can fail in two qualitatively different ways: the composite fails to contract, or the composite was never a composite. The asymmetric-parameter limit $\alpha_2 \to 0$ via weighted Lyapunov V_μ formalizes 10.3's (S-3) autonomy-reduction mechanism as a smooth deformation of (CM4), not a discontinuous regime change.

Connection to team-persistence. 13.1 derives persistence conditions for sub-agents in a cooperative-adversarial network. This segment provides the macro-level complement: the conditions under which the composite itself is a valid AAT agent. Together they close the loop: sub-agents persist individually (team-persistence) AND the composite is a meaningful macro-agent (composition closure with admissibility).

Constructive (A1)–(A4) via wrapping. The wrapping construction in `#der-class-coercion-via-wrapping` is a special-case instantiation where (A1)–(A4) hold *by construction* through the wrapper's type signatures rather than by post-hoc verification of an admissible projection Λ. The wrapper builds explicit $X_W = (M_W, G_W)$ with belief-update and strategy-update maps that have specified type signatures; (A1) is automatic; (A2)–(A4) follow from wrapper-design constraints (commit to a prediction map; choose f_M from the Tier-1 belief-update class). The closure-defect ε^* in this special case has a different reading than the standard fidelity reading — the wrapper *deliberately* changes the underlying component's behavior to enforce structural separation. Two distinct ε^* quantities appear in wrapper analyses: $\varepsilon^*_{\text{track}}$ (the standard fidelity quantity defined here, used in the bridge lemma) and $\varepsilon^*_{\text{coerce}}$ (the wrapper-vs-component behavioral divergence, distinct from $\varepsilon^*_{\text{track}}$ and not bounded by the bridge lemma). See `#der-class-coercion-via-wrapping` Discussion for the disambiguation.

Meta-machine: exact composition for finite automata (Miller 2022). When agents are finite-state automata (Moore machines), composition is *exact*: two machines with state sets S_1, S_2 interacting through their output/input channels form a **meta-machine** (Miller 2022, *Ex Machina*, §3.3) — itself a Moore machine with state set $S_1 \times S_2$, deterministic transitions (each machine's output determines the other's input), and a single starting state from the constituent starting states. The closure defect $\varepsilon^* = 0$ trivially, because the product automaton *is* the micro-dynamics — there is no approximation. The meta-machine always falls into a cycle (finite states with deterministic transitions must eventually revisit a state), so two interacting automata produce an eventually-periodic joint behavior. The composition-closure framework becomes interesting when we ask for a *compressed* macro-description: can a smaller automaton (fewer than $|S_1| \times |S_2|$ states) approximate the meta-machine? This is where $\varepsilon^* > 0$ and the admissibility constraints become non-trivial — (P3) requires genuine dimensionality reduction, and (P1) asks how much predictive information the compression preserves. Machine minimization (the theorem that every automaton has a unique minimized equivalent; Hopcroft et al. 2006) is a natural candidate for the optimal projection: the minimized meta-machine has the fewest states that reproduce the full joint behavior, achieving $\varepsilon^* = 0$ with (P3) satisfied whenever the minimized machine is smaller than the product. See (planned) for the full instantiation.

Two-Kalman instantiation. The simplest nontrivial worked case: two Kalman filters tracking correlated scalar random walks with correlation ρ_{corr}, no communication. This case sits at $K_c = 1$ — both sub-agents and the composite step together — so the macro-step formulation reduces to the pointwise form; the Λ_o aggregation is the identity. The natural projection keeps the state estimates and discards the covariance states (means-only projection, dimension 2 from micro-dimension 4). At steady state, the closure defect is exactly $\varepsilon^* = 0$ for **all** values of ρ_{corr} — the means-only projection perfectly represents the micro-dynamics because the Kalman gains converge to constants and the discarded covariance state carries no information. The "cost of independence" (the estimation performance lost by not exploiting cross-correlations) is a **performance gap** $\Delta_{\text{perf}} \approx 2\rho_{\text{corr}}^2 q^2 r/(S^*)^2$ (quadratic in ρ_{corr} for small correlations), not a closure defect — it measures suboptimality relative to a joint filter, not failure to track the micro-dynamics. The composition-closure framework diagnoses representability ("is this group a coherent AAT agent?"), not optimality ("is this group as good as a centralized agent?"). See `spikes/spike-composition-correlated-kalman.md` for the full derivation, (A1)-(A4) verification, and the first genuine $\varepsilon^* > 0$ case (purposeful agents with Beta-Bernoulli strategy updates). All three (P1)-(P3) conditions are verified exactly: $\epsilon_I = 0$ (no information loss at steady state, since the discarded covariance state is constant), $L = 1$ (the projection is a coordinate map), and $\dim(\mathcal{X}_c) = 2 < 4 = \dim(\mathcal{X}_{\text{micro}})$.

Norm specification for estimation-type agents. The two-Kalman case identifies the Mahalanobis norm as the natural choice for agents whose primary function is state estimation. The state norm weights by the inverse prediction-error covariance: $\|X_c - X_c'\|^2 = (\hat{\omega}_c - \hat{\omega}_c')^T (P_{\text{pred}}^*)^{-1} (\hat{\omega}_c - \hat{\omega}_c')$. The observation norm weights by the inverse innovation covariance $S^{-1} = (P_{\text{pred}}^* + R)^{-1}$. The general principle: norms should weight by inverse uncertainty, so that differences in well-estimated components count more than differences in poorly-estimated ones. This is the norm the Kalman filter implicitly uses — the Kalman gain minimizes the expected squared Mahalanobis distance from truth. For domains without a natural covariance structure (discrete states, non-Gaussian models), the Euclidean norm remains the default. See `spikes/spike-projection-admissibility.md` §4 for derivation and §4.5 for the general-case pattern.

Findings.

Composition-Closure Defect and Bridge Lemma. **Brief:** When does a group of interacting agents legitimately count as one larger agent? The closure-defect ε^* is the minimum residual prediction error from collapsing the micro-system into an AAT-shaped macro-description, taken over admissible coarse-grainings. The bridge lemma converts this prediction-level error into a trajectory-level guarantee: under sector-bounded macro-correction plus strong monotonicity of the macro-update, the macro-description tracks micro-reality with bounded asymptotic error $\varepsilon^* \nu_c / \alpha_c$. Composition is therefore not asserted by interface or by intuition — it is certified by an inequality, with explicit failure modes on both sides (the macro-dynamics fail to be AAT-shaped; the projection fails to be informative-and-regular; the macro-correction is not strongly monotone).

Impact: Closes the gap that Markov-blanket and active-inference accounts of nested agency leave open: those accounts give a statistical / thermodynamic boundary criterion but no control-theoretic loss bound on treating a collection as one agent. The closure-defect framework supplies the bound, attaches it to AAT's existing sector-condition machinery, and routes Section I and Section II results across composite boundaries on Tier-1-classified sub-agents. Downstream this lets `#deriv-critical-mass-composition` derive the composite sector constant from sub-agent parameters in closed form, lets `#der-tempo-composition` give Brooks's Law a formal expression in tempo units, and supplies the escape route (b) for `#disc-identifiability-floor` Instance 3. The criterion also draws an honest scope line: $\varepsilon^* = 0$ does not mean optimality (two non-communicating Kalman filters achieve $\varepsilon^* = 0$ while being suboptimal versus a joint filter); the framework diagnoses representability, not optimality.

Novelty Claim: *Claim differentiation* on bounded-loss composition as agent-boundary criterion. The bridge-lemma form is mathematically continuous with the approximate-information-state and influence-based-abstraction lines (Subramanian 2020; Congeduti 2020; Abel 2016; Taylor 2008), which prove approximation-loss results for compressed control under coarse-graining or aggregation. The differentiation is the *use* of such a bound as a criterion for when a multi-agent system can be re-described as a single agent — not as a control-policy compression result on a fixed agent — combined with explicit admissibility conditions (A1)-(A4) on the macro-dynamics that force the macro-description to itself be AAT-shaped, ruling out trivial curve-fits.

Related Work:

Table 11.2

ASF concern	*Prior-art language*	*Relationship / Positioning*
Statistical / thermodynamic boundary of a nested agent	Parr, da Costa & Friston 2019, “Markov blankets, information geometry and stochastic thermodynamics” *Phil. Trans. R. Soc. A* 378 (published 2019, found via Undermind 2026-04 Pillar 2)	*conceptual precursor* — supplies the boundary-and-nesting intuition; does not provide a coarse-graining error bound, a coordination-overhead theorem, or a strong-monotonicity condition for valid macro-agent treatment
Hierarchical-blanket nested agency with temporal depth	Kirchhoff, Parr, Palacios, Friston & Kiverstein 2018, “The Markov blankets of life” *J. R. Soc. Interface* 15 (published 2018, found 2026-04)	*conceptual precursor* — invokes synergetic ordering and self-organization at the compositional step but remains conceptual; no theorem bounding predictive loss of coarse-graining
Bound on control loss from compressed history	Subramanian, Sinha, Seraj & Mahajan 2020, “Approximate information state for approximate planning and reinforcement learning in partially observed systems” *arXiv:2010.08843* (published 2020, found 2026-04)	*closest mathematical neighbor* — proves a bound from predictive-compression error to control error in the same shape as the bridge lemma. ASF reuses the predictive-compression-to-control-loss mechanism and applies it at the coarser granularity of *agent boundary formation*, not single-agent state compression
Loss bound from approximate influence-based abstraction	Congeduti, Mey & Oliehoek 2020, “Loss Bounds for Approximate Influence-Based Abstraction” *arXiv:2011.01788* (published 2020, found 2026-04)	*closest mathematical neighbor* — closer than the Markov-blanket line to coordination-overhead mathematics. ASF differs in framing: Congeduti bounds policy-value loss under approximate cross-agent influence; the bridge lemma bounds *macro-state trajectory error* under coarse-graining
Near-optimality under state aggregation	Abel, Hershkowitz & Littman 2016, “Near Optimal Behavior via Approximate State Abstraction” *ICML*; Taylor, Precup & Panangaden 2008, “Bounding Performance Loss in Approximate MDP Homomorphisms” *NeurIPS* (published 2016/2008, found 2026-04)	*adjacent literature* — single-agent MDP framing with aggregation bounds; same loss-from-coarse-graining mechanism, not framed as composite agency or organizational closure
Exact reduction of decentralized control via common information	Nayyar, Mahajan & Teneketzis 2012, “Decentralized Stochastic Control with Partial History Sharing” *IEEE TAC* 58 (published 2012, found 2026-04)	*adjacent literature* — names the formal space where a lossy composition theorem would live but proves an exact reduction, not a closure-defect bound
Approximate dynamical homomorphism / model reduction	Mori-Zwanzig projection (1965); balanced truncation (Moore 1981); Koopman-operator analysis (Koopman 1931; modern review Mezić 2005)	*formal antecedent* — supplies the approximate-dynamical-homomorphism framing of the closure defect. The MZ zero-lag bound surfaces in this segment’s derivation table under stationarity. Adopted as standard machinery
Information bottleneck for the projection	Tishby, Pereira & Bialek 1999, “The information bottleneck method” *Allerton*	*formal antecedent* — admissibility condition (P1) is the IB Lagrangian-dual with source X_{micro} and relevance “next macro-observation given action”

Search Log:

- 2026-04 (*nominally comprehensive*, via `ref/Novelty_defense_and_integration.md` Pillar 2): Two-search Undermind pass over composite agency, approximate information states, Markov-blanket nested agency, decentralized control, state abstraction, and influence-based abstraction. Verdict: *Conceptual Precursor* (High confidence). Markov-blanket line (Parr 2019; Kirchhoff 2018) positioned as conceptual precursor on nested agency without bridge-lemma mathematics; approximate-information-states / state-abstraction / influence-based-abstraction line (Sub20; Con20; Abe16; Tay08; Nay12) positioned as closest mathematical neighbors with shared predictive-loss-to-control-error mechanism but applied to single-agent or already-decomposed control problems rather than to agent boundary formation.

- 2026-04 (*intuition-only*, prior to comprehensive search): adjacent literatures expected to host prior art were dynamical-systems model reduction (Mori-Zwanzig, balanced truncation), active-inference Markov-blanket work, and PAC-MDP state-abstraction. The comprehensive search confirmed all three families and added the influence-based-abstraction and decentralized-control-via-common-information lines that intuition did not surface.

Working Notes.

- **Resolved: $\mathcal{P}_{\mathrm{adm}}$ now has a proposed definition.** (P1)-(P3) above. Confirmed independent of (A1)-(A4) — the macro-dynamics admissibility partially constrains projections but does not specify information-preservation level. See `spikes/spike-projection-admissibility.md` §5 for the analysis.
- **Resolved (2026-04-22): temporal coarse-graining gap.** The previous formulation summed $\varepsilon_x, \varepsilon_a, \varepsilon_o$ over micro-timesteps ($t = 1, \ldots, H$), forcing synchronous micro-and-macro cadence. This contradicted 4.6 (composites naturally update on slower timescales, $\nu_{\text{level } n+1} \ll \nu_{\text{level } n}$) and conflicted with 11.2's dimensional accounting, which treats ε^* as a per-macro-step quantity and ν_c as the macro-update rate. Introducing the timescale ratio $K_c \geq 1$ and reformulating the defects per macro-step fixes both issues: Λ_o, Λ_a become window-aware (aggregating K_c micro-observations/actions), the sum runs over macro-steps m, and $\varepsilon^* \nu_c$ has consistent rate units by construction. $K_c = 1$ recovers the previous formula; $K_c \gg 1$ enables the timescale abstraction that was always part of the composition theory's intent. The bridge lemma and the sector-persistence-template instantiation are unchanged in substance — the template's state variable is now e_m at macro-boundaries rather than e_t at micro-timesteps, but the Lyapunov argument is identical. Option 3 of `audits/pending-findings-2026-04-21.md` Finding A; Option 2 (full Mori-Zwanzig equilibrium-residual form) remains a possible future refinement if one wants an explicit singular-perturbation framing.
- **Resolved: norm choices for estimation-type agents.** The Mahalanobis norm (inverse-covariance-weighted) is the natural choice for Kalman-type agents, verified exactly in the two-Kalman case. The general principle (weight by inverse uncertainty) extends to other estimation frameworks. For non-estimation agents (discrete states, non-Gaussian), norms remain domain-specific.
- **Open: computing (P1) for nonlinear/non-Gaussian systems.** The information-preservation condition requires conditional mutual information over the joint distribution of micro-states, observations, and actions. Tractable for linear-Gaussian systems (closed-form); requires Monte Carlo estimation or variational bounds for general systems.
- **Open: the right value of ϵ_I.** The information-preservation threshold is a free parameter. Too small ($\epsilon_I \to 0$) excludes useful projections; too large ($\epsilon_I \to 1$) admits degenerate ones. A natural candidate: ϵ_I comparable to the fractional information loss from adding one agent to the composite — tying it to team size and coupling structure. Formalizing this is open.
- **Open: N-agent scaling of ε^*.** Whether the closure defect scales polynomially or exponentially with N depends on coupling structure. Tree-structured coupling (hierarchical organizations) may allow efficient dimensionality reduction; fully-connected coupling may not. This is the formal analog of the claim that very large teams cannot be treated as single agents.
- **Partial: Mori-Zwanzig connection.** Under stationarity and coordinate-compatibility assumptions on the micro-dynamics, the Koopman-operator framing lifts Λ_x to a Hilbert-space projection P_Λ with $Q_\Lambda = I - P_\Lambda$. The MZ-optimal Markovian macro-dynamics is $f_c^{\mathrm{MZ}} = P_\Lambda U P_\Lambda$ (the conditional expectation given the macro-state), providing a concrete benchmark against which admissibility constraints can be measured. When $f_c^{\mathrm{MZ}} \notin \mathcal{M}_{\mathrm{adm}}$, the per-step bound $\varepsilon^* \geq \|Q_\Lambda U P_\Lambda\|_{\mathrm{op}}$ holds — a zero-lag memory-kernel bound. The bridge-lemma contraction on f_c in state space corresponds to spectral gap of $Q_\Lambda U$ in observable space (distinct quantities but related). *What does not close*: the full-kernel bound $\varepsilon^* \geq C \cdot \|K\|_{\ell^1}$ (sum-of-lags) is a type mismatch — ε^* is per-step; $\|K\|_{\ell^1}$ is trajectory-accumulation. Natural home for the full-kernel norm is the bridge lemma's trajectory-error bound, not ε^* directly. *Hard obstruction*: MZ's stationarity assumption fails for purposeful agents

with non-stationary auxiliary state (Beta-Bernoulli with diverging n) — exactly the cases where $\varepsilon^* > 0$ genuinely. Extending to these requires innovation-frame reformulation, not yet developed.

- **(P1) as IB Lagrangian-dual — resolved.** (P1) is the Lagrangian-dual of a standard IB constraint with source X_{micro}, compressed representation $\Lambda_x(X_{\text{micro}})$, relevance variable $o_{\text{micro},t+1} \mid a_{\text{micro},t}$, and $\beta(\varepsilon_I)$ the rate-distortion multiplier. See V for the derivation and for the shared IB shape across AAT's four compression operations (M_t, Σ_t, shared intent, Λ). (P2) Lipschitz continuity remains a separate analytic admissibility condition. (P3) dimensional reduction remains separate in the Gaussian case (the IB-optimal T at any finite β typically uses full support; categorical dimensional reduction is harder than any rate constraint). The "all admissibility is IB" slogan overclaims; the accurate slogan is "(P1) is IB; (P2) and (P3) compose with it as separate conditions." Cross-instance unification is shape-sharing (U-medium), not a reduction to a single master problem (U-strong) — cross-instance theorems do not follow from the shared IB shape alone.
- **Two independent drivers of ε_x.** The non-degenerate Kalman case (heterogeneous gains $K_1^* \neq K_2^*$ with projection $\Lambda_x(\hat{\omega}_1, \hat{\omega}_2) = (\hat{\omega}_1 + \hat{\omega}_2)/\sqrt{2}$) shows ε_x depends on *both* sub-agent unity (process correlation, captured by U_M) *and* update-rule heterogeneity ($\Delta K = K_1^* - K_2^*$). $\Delta K = 0$ gives $\varepsilon_x = 0$ at every correlation; $\Delta K \neq 0$ gives $\varepsilon_x > 0$ even at perfect correlation. The closed form is $\varepsilon_x^2 = (\Delta K/2)^2[S_- - C_{+-}^2/S_+]$ where $S_\pm$ are innovation variances in the $\pm$ rotation and C_{+-} their cross-covariance. Heterogeneity is not captured by the four unity dimensions in 12 as currently defined; this is a gap in that segment, not in this one — the closure defect correctly registers both drivers. See 12.1 for the full analysis.
- **Open: strategy DAG projection.** How individual strategy DAGs compose under Λ_x (union, abstracted DAG, or no macro-strategy) is deeply domain-specific and not resolved by (P1)-(P3). The information-preservation condition provides a functional test but not a specific mechanism.
- ~~The bridge lemma's contraction assumption — proving it from (A4) alone.~~ **CHARACTERIZED 2026-04-06** (`spikes/spike-bridge-lemma-contraction.md`). The contraction assumption cannot be proved from (A4) alone. The precise additional condition is the **incremental sector bound** (DA2'a-inc): the correction function F_d must be *strongly monotone* ($\langle \delta - \delta', F_d(\delta) - F_d(\delta') \rangle \geq c_{\min} \|\delta - \delta'\|^2$), not just satisfy the one-point sector bound. Strong monotonicity is strictly stronger than the one-point sector bound (counterexample: oscillatory corrections that are globally inward-pointing but locally non-monotone). Three agent tiers emerge: **Tier 1** (contraction proved — all Bayesian updaters on exponential families, all gradient agents on strongly convex losses, all linear corrections with positive definite gain-observation product); **Tier 2** (local contraction — nonlinear prediction models, with factor degraded by $\kappa(D\hat{o})^2$); **Tier 3** (independent verification required — non-convex optimization, discontinuous rules, agents with non-mismatch-driven state components). For Tier 1 agents, the bridge lemma is promoted from "conditional" to "derived (conditional on DA2'-inc + linear prediction)." The contraction factor equals λ_{eff} from K.
- The weakest-link structure ($\alpha_c = \min_i \alpha_i^{\text{eff}}$) is conservative. In practice, strong sub-agents may compensate for weak ones through cooperative coupling. A tighter bound would account for cross-agent compensation, likely through the cooperative disturbance reduction terms in 13.1.
- **A richer toy case** is needed: two purposeful agents (Section II) with strategy DAGs, cooperative communication, and a shared objective. This would exercise (A1) fully (including the G_c component) and test whether the admissibility constraints are tight enough to be useful without being so tight they exclude interesting composites.
- The approach is an approximate dynamical homomorphism condition, a standard tool in dynamical systems and model reduction (cf. Mori-Zwanzig projection, balanced truncation). The specific contribution is applying it to AAT's closed-loop agent structure with sector-condition-based stability guarantees.
- **Mori-Zwanzig memory-kernel relationship — upper-bound direction closes; named lower bound does not.** The closure defect ε^* has a tractable upper-bound relationship to a Mori-Zwanzig (MZ) memory kernel norm $\|K\|$ when AAT's state-space projection Λ is composed with an MZ Hilbert-space projection P on a stationary observable measure π: under (i) stationarity of π under micro-dynamics, (ii) compatibility of Λ with P's range (the projected observables include the coarse-grained state coordinates AAT's Λ retains), and (iii) bounded operator-norm of K, the upper bound $\varepsilon^* \leq C \cdot \|K\|$ holds for a constant C depending on α_c, ν_c. The *named target* lower bound $\varepsilon^* \geq C' \cdot \|K\|$ — which would make slow-decaying memory kernels an *obstruction* to small closure defect — does NOT close under the current AAT formulation without additional structural hypotheses. The obstruction is that Λ (state-space coarse-graining) and P (Hilbert-space projection on observables) are *different objects*, and a generic AAT coarse-graining does not give an orthogonal MZ-style projection on the observable

Hilbert space; the lower bound would require either a stronger compatibility assumption between Λ and P or a separate orthogonality-of-residual argument that has not been worked out. The honest structural takeaway: MZ memory-kernel decay is a *sufficient* condition for small ε^* (via the upper bound direction) but is not *necessary*; AAT's closure defect can be small for reasons that don't reduce to MZ memory-kernel norms. Future work would either close the lower-bound direction under specific Hilbert-space hypotheses on the agent's observable algebra, or settle the asymmetry as a structural feature distinguishing AAT's coarse-graining from MZ projection-operator formalism.

11.2 *Sketch:* Derived: Composite Tempo Inequality

The adaptive tempo of a composite agent is bounded from above by the sum of its sub-agents' tempos. The gap between aggregate potential and realized composite tempo is the coordination overhead — tempo consumed by internal reconciliation rather than external mismatch correction.

Formal Expression.

This segment instantiates the sector-persistence template (C) at the composite level with state variable $\xi = \delta_c$ (composite mismatch) and a decomposed effective disturbance $\rho_{\text{eff}} = \rho_{\text{ext}} + \varepsilon^* \nu_c$ — external environment plus closure-defect contribution from 11.1's bridge lemma. The template supplies the Lyapunov machinery; this segment's distinctive content is the tempo-equivalent form of the coordination overhead lower bound $C_{\text{coord}} \geq \varepsilon^* \nu_c$.

Let $\mathcal{T}_i$ be the adaptive tempo of sub-agent i within a composite group of N agents. Let $\mathcal{T}_c$ be the adaptive tempo of the composite macro-agent A_c defined by an admissible coarse-graining Λ.

[Derived (Sub-additive Tempo, sketch)] For any composite agent A_c with minimal closure defect $\varepsilon^* \geq 0$, the composite tempo is bounded by the sum of individual tempos: $\mathcal{T}_c \leq \sum_{i=1}^{N} \mathcal{T}_i$

[Definition (Coordination Overhead)] We define the **coordination overhead penalty** C_{coord} as the difference between aggregate potential and realized macro-tempo: $C_{\text{coord}} := \left(\sum_{i=1}^{N} \mathcal{T}_i \right) - \mathcal{T}_c$

Epistemic Status.

Sketch. Max attainable: exact (with a complete proof). The inequality is structurally motivated: composition cannot create corrective capacity out of nothing, and internal reconciliation consumes tempo that doesn't contribute to external mismatch correction. The $\varepsilon^* \to C_{\text{coord}}$ connection is sketched: closure defect as internal disturbance rate $\varepsilon^* \nu_c$ (units [distance] $\cdot$ [time^{-1}]) converts to a tempo-equivalent penalty $C_{\text{coord}} \geq \varepsilon^* \nu_c / \|\delta_{\text{critical}}\|$ (units [time^{-1}]) through the same distance normalization used by the persistence condition. The inequality $\mathcal{T}_c \leq \sum \mathcal{T}_i$ follows. **Caveat:** this connection depends on the bridge lemma in 11.1, which requires a contraction assumption on the macro-update map — an assumption that is structurally motivated by (A4) but not formally derived from it (see 11.1, Epistemic Status). The "follows directly" language below should be read as "follows from the bridge lemma under its contraction assumption." The remaining gap: whether the lower bound is tight or whether additional coordination costs contribute beyond the closure-defect mechanism.

Intuition. Tempo is a rate of mismatch correction. When $\varepsilon^* = 0$ (exact closure), every sub-agent correction cycle contributes directly to the macro AAT loop without friction, and $\mathcal{T}_c = \sum \mathcal{T}_i$. When $\varepsilon^* > 0$, sub-agents spend part of their tempo correcting *internal* mismatches generated by other sub-agents (Alice changes an API; Bob must update his M_t) or taking actions that counteract each other. That fraction is C_{coord} and is unavailable for external mismatch correction.

The closure-defect to coordination-overhead connection.

The closure defect ε^* introduces internal mismatch at rate $\varepsilon^* \nu_c$ — units of [distance] $\cdot$ [time^{-1}], a disturbance rate matching the units of environmental ρ. This internal mismatch must be corrected by the macro-agent's correction capacity, and the correction consumes tempo that would otherwise address external mismatch.

Derived (coordination-overhead-bound, from composition-closure bridge lemma).

Dimensional accounting. Three quantities appear in the composite persistence analysis, each with its own units:

Table 11.3

Quantity	*Units*	*Role*
Tempo $\mathcal{T}, \mathcal{T}_c, \mathcal{T}_c^{\text{ext}}, C_{\text{coord}}$	[time^{-1}]	Rate of mismatch correction
Disturbance rate ρ, ρ_{eff}	[distance] $\cdot$ [time^{-1}]	Rate of mismatch injection
Closure defect ε^*	[distance]	Per-step norm error
Critical scale $\|\delta_{\text{critical}}\|$	[distance]	Task-adequacy boundary

The conversion between a disturbance rate and a tempo-equivalent requires division by a distance scale, mirroring the persistence condition form $\mathcal{T} > \rho / \|\delta_{\text{critical}}\|$.

Effective disturbance. The macro-agent faces total effective disturbance combining external and closure-defect contributions:

$$\rho_{\text{eff}} = \rho_{\text{ext}} + \varepsilon^* \nu_c \quad (\text{units: [distance]} \cdot [\text{time}^{-1}]) \tag{11.16}$$

Coordination overhead (tempo-equivalent). The fraction of macro-tempo consumed by closure correction, expressed as a tempo:

Derived (coordination-overhead-tempo-form).

$$C_{\text{coord}} \;\geq\; \frac{\varepsilon^* \nu_c}{\|\delta_{\text{critical}}\|} \quad (\text{units: } [\text{time}^{-1}]) \tag{11.17}$$

The closure error rate is converted to a tempo penalty by normalizing against the task-relevant distance scale. The inequality is a lower bound because additional coordination costs (negotiation, synchronization, conflict resolution — see Working Notes) contribute beyond the closure-defect mechanism. When the macro-agent's correction of internal mismatch is less efficient than its correction of external mismatch (because internal mismatches may be harder to diagnose — they look like environmental change from the macro perspective), the actual overhead exceeds this bound.

Consequences (dimensionally correct):

- The realized external tempo: $\mathcal{T}_c^{\text{ext}} = \mathcal{T}_c - C_{\text{coord}} \;\leq\; \sum \mathcal{T}_i - \frac{\varepsilon^* \nu_c}{\|\delta_{\text{critical}}\|}$
- **Brooks's Law.** Adding an agent increases $\sum \mathcal{T}_i$ but may also increase ε^*. The turning point — where adding a member lowers realized external tempo — occurs when: $\frac{\Delta\varepsilon^* \nu_c}{\|\delta_{\text{critical}}\|} > \Delta \mathcal{T}_i$ The closure-defect increase (expressed as tempo penalty) exceeds the new member's tempo contribution.

- **Composite persistence condition.** The composite persists iff $\mathcal{T}_c > \rho_{\text{eff}}/\|\delta_{\text{critical}}\|$, equivalently: $\sum \mathcal{T}_i > \frac{\rho_{\text{ext}}+\varepsilon^* \nu_c}{\|\delta_{\text{critical}}\|}$ which explicitly separates internal-coordination overhead from external challenge while keeping all quantities in tempo units on the LHS and disturbance-rate units on the RHS.

Discussion.

Equality conditions. Strict equality ($\mathcal{T}_c = \sum \mathcal{T}_i$, i.e. $C_{\text{coord}} = 0$) requires: 1. **Orthogonal Routing:** The information dependency DAG exactly matches organizational boundaries (no costly cross-talk required). 2. **Perfect Shared Intent:** No tempo is lost to internal negotiation or conflicting pursuit of the macro-objective. 3. **No Net Macro-Information Loss:** Observations may be redundant, but they do not result in wasted tempo after fusion, nor is critical macro-information lost during coordination.

These are stated as sufficient conditions; whether they are also necessary is open.

Brooks's Law. Adding more agents (increasing $\sum \mathcal{T}_i$) only increases the composite tempo $\mathcal{T}_c$ if the corresponding increase in closure defect ε^* doesn't let C_{coord} dominate. The model provides a formal analog of Brooks's Law: if communication overhead (C_{coord}) grows faster than aggregate capability ($\sum \mathcal{T}_i$) when adding agents, then adding people to a late project makes it later. Whether this specific mechanism (coordination overhead consuming tempo) is the dominant cause in practice is an empirical question.

Connection to 13.1. The persistence condition for the composite agent requires $\mathcal{T}_c > \rho_{\text{ext}}/\|\delta_{\text{critical}}\|$. Since $\mathcal{T}_c = \sum \mathcal{T}_i - C_{\text{coord}}$, high coordination overhead can push the composite below the persistence threshold, causing it to disintegrate as a coherent entity — even though each sub-agent individually persists.

Wrapping construction as a Brooks's-Law instance. Class coercion of a Class 2 (Partial) or Class 3 (Coupled) component via wrapping (`#der-class-coercion-via-wrapping`) is a concrete instance of the same coordination-overhead form. A wrapper makes $K \geq 2$ component calls per macro-step (one for each goal-blind belief-update query, one for the goal-conditioned strategy-update query, more in richer wrapper designs). The wrapper-level macro-update rate is $\nu_W = \nu_A/K$ where ν_A is the underlying component's nominal call rate, so the wrapper-level tempo is $\mathcal{T}_W \leq \mathcal{T}_A^{\text{nominal}} - C_{\text{coord}}^{\text{wrap}}$, with the coordination overhead $C_{\text{coord}}^{\text{wrap}}$ scaling with K. The cost of class coercion — the structural enforcement of directed separation at the wrapper level when the underlying component does not provide it — is paid in macro-tempo. Strict-wrapping (W_1) regimes have $K \geq 2$ minimum; partial-wrapping (W_2) regimes have $K = 1$ for the LLM call plus some parsing overhead, but the structural-separation guarantee is correspondingly weaker (see `#der-class-coercion-via-wrapping`).

Working Notes.

- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. Removed at `candidate` stage per FORMAT.md Gate 4.
- The lower bound $C_{\text{coord}} \geq \varepsilon^* \nu_c/\|\delta_{\text{critical}}\|$ captures the minimum tempo-equivalent overhead from closure defect. Additional coordination costs likely contribute beyond this: negotiation costs (reaching shared decisions), synchronization costs (waiting for slower agents), and conflict resolution costs (undoing contradictory actions). These are not captured by ε^* because they arise from the coordination PROCESS, not from the closure DEFECT. A fuller model would add: $C_{\text{coord}} = \varepsilon^* \nu_c/\|\delta_{\text{critical}}\| + C_{\text{negotiation}} + C_{\text{sync}} + C_{\text{conflict}}$, with all terms in tempo units.
- **Heterogeneity drives closure defect.** For agents with equal correction rates ($\alpha_1 = \alpha_2$), $\varepsilon^* = 0$ and $C_{\text{coord}} = 0$. For agents with different correction rates, $\varepsilon^* \propto |\alpha_1 - \alpha_2|$ — heterogeneity drives closure defect, confirmed analytically in the 2-agent toy case.
- The internal/external mismatch distinction is formalized through the bridge lemma: trajectory error e_t is the internal mismatch (divergence between macro-prediction and micro-reality, units of distance), and the tempo spent reducing e_t

is the coordination overhead. The steady-state internal mismatch is $\varepsilon^* \nu_c / \alpha_c$ (a distance, since α_c has units [time^{-1}]), and maintaining this at the critical-distance scale requires ongoing tempo expenditure of $\varepsilon^* \nu_c / \|\delta_{\text{critical}}\|$.

- **Norm specification note.** The closure defect ε^* depends on the norm choices in 11.1. For estimation-type agents, the natural norm is Mahalanobis (weighted by inverse prediction-error covariance). In these norms, ε^* measures the ratio of missed cross-correction to expected estimation uncertainty — a quantity with direct tempo interpretation: it's the fraction of macro-tempo consumed by correcting errors that optimal cross-exploitation would eliminate.
- **Channel-independence assumption.** The sub-additive bound $\mathcal{T}_c \leq \sum \mathcal{T}_i$ uses scalar tempos that each inherit the channel-independence assumption from 3.8. When sub-agents' observation channels overlap, the individual $\mathcal{T}_i$ are already overcounted, making the upper bound looser than it appears. The coordination overhead C_{coord} is defined relative to the (possibly inflated) sum, so the *fraction* of tempo consumed by coordination may be underestimated.
- **Open: does ε^* scale with N?** For orthogonal agents: no (each agent adds independent channels, no cross-talk). For coupled agents: likely yes (more agents → more cross-interactions → harder to approximate as one macro-agent). The scaling determines whether Brooks's Law is linear or superlinear in team size.

11.3 *Hypothesis:* Directed Separation Under Composition

When individual agents satisfy directed separation (5.3), does the composite macro-agent also satisfy it? The answer depends on whether the composite's internal information routing — which observations reach which sub-agents — is itself goal-dependent. Two cases arise, corresponding to the first two classes in 5.3's architectural classification.

Formal Expression.

Consider N agents $A_1, \ldots, A_N$, each satisfying directed separation individually: $M_{i,\tau^+} = f_M^{(i)}(M_{i,\tau^-}, e_{i,\tau})$ with no $G_t^{(i)}$ argument.

Hypothesis (directed-separation-under-composition, extending directed-separation to composites).

The question. Directed separation is about **processing**, not **selection** (5.3, scope condition). A single agent's goals affect which events it seeks (through π), but f_M processes whatever event arrives without reference to G_t. At the composite level, the analogous question is: does the composite's routing structure R_t (10.1) depend on the composite's goals G_t^c?

Note: sub-agents' goal-driven actions shape the environment, which other sub-agents observe. This is NOT a violation of directed separation — it is the same mechanism directed separation explicitly allows at the single-agent level: goals influence events through action, but processing of realized events is goal-blind. Agent B observing environmental changes caused by A's goal-driven behavior is B processing a realized event goal-blindly. The observations carry information about A's goals, but that is a property of the event's content, not a failure of goal-blind processing. Similarly, individual messages reflecting individual agents' goals is action through policy, not a routing-structure dependence on G_t^c.

Case 1: Goal-blind routing.

If the routing structure satisfies $R_t \perp G_t^c$ (10.1, goal-blind routing) — neither the communication topology $\mathcal{N}_t$ nor the protocol $c_t^{(j\to i)}$ depends on the composite's goals — then:

Hypothesis (Case 1).

- Each sub-agent processes observations goal-blindly (individual directed separation)
- The routing is goal-blind (by construction)
- Therefore f_M^c is G_t^c-independent

Directed separation **survives** at the composite level. Examples: military command structures with doctrinal communication protocols, software teams with defined code-review processes, multi-agent AI systems with protocol-specified message passing.

Case 2: Goal-dependent routing.

If R_t depends on G_t^c — either the topology $\mathcal{N}_t$ changes (different reporting chains activated depending on the mission) or the protocol $c_t^{(j\to i)}$ changes (different intelligence products shared depending on the objective) — then the composite's effective observation function has a goal argument:

Hypothesis (Case 2).

$$o_c = h^c(\Omega, a_{\text{micro}}, G_t^c, \xi) \tag{11.18}$$

Even if each sub-agent's $f_M^{(i)}$ processes o_i goal-blindly, the **set of observations reaching each sub-agent** depends on G_t^c through the routing function. Directed separation **fails** at the composite level. The composite is analogous to a Class 3 (Coupled) architecture: goal content shapes the information pathway, not through individual interpretation but through collective routing.

Epistemic Status.

Conditional on the routing structure of the composite. Max attainable: conditional (the two cases are genuine architectural alternatives). Previously discussion-grade; upgraded after routing formalization in 10.1 and architectural classification promotion in 5.3 (2026-04-01).

Foundation status.

1. The **architectural classification** (Class 1/2/3) is in 5.3's Formal Expression with formal operationalization ($\kappa_{\text{processing}}$). Status: robust qualitative.
2. The **routing structure** R_t is defined in 10.1 with a formal goal-independence condition ($R_t \perp G_t^c$). The definition decomposes into topology independence and protocol independence. Whether this captures all relevant ways that composite information flow can depend on goals is an open question.
3. The **admissible coarse-graining** Λ from 11.1 now has specified admissibility constraints: (A1)-(A4) for macro-dynamics and (P1)-(P3) for projections. The remaining gaps are validation (exercising the machinery on a purposeful multi-agent example) and computability (P1 requires conditional mutual information, tractable only for linear-Gaussian), not missing definitions.

The logic of each case is sound given the routing definition. Case 1: goal-blind routing + goal-blind processing = goal-blind composite. Case 2: goal-dependent routing means the composite's observation function depends on G_t^c, regardless of individual processing. Both follow from directed separation's scope condition.

The claim that the architectural classification lifts to composition is **structurally motivated** by 1.7 (the theory must give consistent answers at every level of description). This is an argument from theoretical coherence, not a derivation.

Discussion.

What this is and what it isn't. This segment provides a two-case taxonomy for directed separation under composition. It is not yet a derivation — the remaining caveat (admissibility placeholders) prevents that. It IS a useful classification that identifies which composites fall within Section III's scope (Case 1) and which require a coupled formulation (Case 2).

Constructive special case: wrapper-around-component. For the special case of a single-component composite — one underlying primitive component embedded inside an external scaffold — the question of when directed separation holds is *derived* by `#der-class-coercion-via-wrapping`. Under explicit conditions on the component (admissibility of goal-blind queries, stationary conditional, no implicit goal-inference) and on the wrapper's update maps (type signatures that exclude G_W from the belief-update path), directed separation holds at the wrapper level by structural commitment. The construction promotes the wrapper-around-component case from hypothesis to derived; the general N-agent composition case (Case 1 vs. Case 2 above) remains a hypothesis pending further work.

Most composites of interest are Case 1. Fixed communication structures are the norm in designed multi-agent systems: military doctrine specifies information flow regardless of the current mission; software development processes define code review, CI, and deployment pipelines independent of the feature being built; multi-agent AI protocols specify message formats and channels. Goal-dependent routing (Case 2) is rarer: crisis management (where communication structure changes with threat type), ad hoc teams assembled for specific objectives, and attention-based multi-agent architectures where query routing depends on the shared goal.

Connection to logogenic agents. LLM-based agents are individually Class 3 (Coupled; goal-conditioned attention). Whether composites of LLM agents are Case 1 or Case 2 depends on whether the inter-agent communication protocol is fixed or goal-dependent. A multi-agent LLM system with fixed API contracts between agents is Case 1 at the composite level even though each agent is individually Class 3 (Coupled). The internal architecture of each agent and the composition architecture are separate questions.

Adversarial implications. Goal-dependent routing (Case 2) makes the composite's objective partially observable through its communication patterns. An adversary who can observe which information is being routed where can infer the composite's current goal. This creates a meta-strategic incentive for fixed routing: it preserves not just epistemic hygiene but operational security.

Working Notes.

- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. Removed at `candidate` stage per FORMAT.md Gate 4.
- **Goal-information leakage is a separate phenomenon.** In any composite, sub-agents' goal-driven actions shape the environment, so observations carry statistical information about goals. This is real and may matter for adversarial dynamics, trust calibration, and OPSEC — but it is NOT a directed-separation issue. It's the normal action-environment coupling that directed separation explicitly allows. If this phenomenon deserves formalization (and it may — the mutual information $I(o_c; G_t^c \mid \Omega_t)$ is well-defined and operationally relevant), it should be its own segment about goal-information leakage, not a case within directed-separation analysis. An earlier draft of this segment conflated the two; this was caught by external review.
- **Partial goal-dependence.** Real composites may have mostly-fixed routing with occasional goal-dependent exceptions (a military unit that follows doctrinal comms but occasionally switches to crisis-specific channels). The error from treating such composites as Case 1 depends on the frequency and magnitude of the goal-dependent exceptions. A formal treatment would need to quantify this, possibly through a "routing independence fraction."
- **Testable prediction.** Organizations that restructure communication channels based on the current strategic objective should exhibit lower prediction accuracy at the organizational level (the shared model becomes goal-contaminated through routing, not just through environmental coupling). Compare prediction accuracy before and after mission-dependent communication restructuring.

11.4 *Derived:* Class Coercion via Wrapping

A Class 2 (Partial) or Class 3 (Coupled) component (one whose forward pass entangles belief-update and goal-conditioning) can be embedded inside an external scaffold whose state $X_W = (M_W, G_W)$ is updated by structurally distinct query channels: **goal-blind queries** to the component update M_W; **goal-conditioned queries** update G_W. Under stated conditions on the component, directed separation holds at the wrapper level by construction, and the composite system is Class 1 (Separated) — even though the underlying component is not. This is the constructive direction of `#hyp-directed-separation-under-composition` for the wrapper-around-component special case: a procedure for *making* directed separation hold when the underlying component does not provide it.

This segment establishes the directed-separation claim. The companion segment `#der-class-coercion-in-composition` establishes that the wrapped system is also a valid AAT composite agent (satisfying (A1)–(A4) of `#form-composition-closure`) and inherits the sector-persistence and tempo-composition machinery at the wrapper level.

Formal Expression.

Setup. Let $A : \mathcal{I}_A \to \mathcal{O}_A$ be a primitive component, treated by the wrapper as a black-box oracle: the wrapper issues queries (inputs) and consumes responses (outputs), without access to A's internal state. $\mathcal{Q}_A \subseteq \mathcal{I}_A$ is the set of admissible queries.

A **wrapper** W over A has state $X_W = (M_W, G_W) \in \mathcal{X}_M \times \mathcal{X}_G$ with $\mathcal{X}_G = \mathcal{X}_O \times \mathcal{X}_\Sigma$ per `#def-strategy-dimension`. The wrapper interacts with an environment via observations $o_W \in \mathcal{O}_W$ and actions $a_W \in \mathcal{A}_W$.

[Definition (wrapper-update-maps)] The wrapper's update at macro-step m uses four type-signed components:

- **Belief-side query selector:** $q_M : \mathcal{X}_M \times \mathcal{O}_W \to \mathcal{Q}_A$. The wrapper chooses the query for M_W updates from belief and observation only — *no G_W argument*.
- **Strategy-side query selector:** $q_G : \mathcal{X}_M \times \mathcal{X}_G \to \mathcal{Q}_A$. May depend on G_W.
- **Belief-update map:** $f_M : \mathcal{X}_M \times \mathcal{O}_W \times \mathcal{Q}_A \times \mathcal{O}_A \to \mathcal{X}_M$. Updates M_W from prior belief, observation, the query made, and the component's response. *No G_W argument.*
- **Strategy-update map:** $f_G : \mathcal{X}_G \times \mathcal{X}_M \times \mathcal{Q}_A \times \mathcal{O}_A \to \mathcal{X}_G$. May depend on G_W.

The external policy $\pi_W : \mathcal{X}_W \to \mathcal{A}_W$ selects the wrapper's external action.

A macro-step proceeds: construct $q_M(M_W, o_W)$ → query A → apply f_M; construct $q_G(M'_W, G_W)$ → query A → apply f_G; emit $\pi_W(X'_W)$. The wrapper makes $K \geq 2$ component calls per macro-step in this minimal form (more in richer wrapper designs).

Conditions. *[Conditions (component-admissibility)]* The theorem applies under three conditions on the component A:

(C1) Goal-blind admissibility. $\mathcal{Q}_A$ contains queries whose specification can be constructed from (M_W, o_W) alone — i.e., a non-trivial q_M exists. Components partition into three classes: - **Class A (goal-blind by design).** A's interface is goal-blind by construction — POMDP belief-state filters, world models, sensory pipelines, retrieval systems, calculators. (C1) holds trivially. - **Class B (admit a goal-blind query mode).** A supports goal-conditioned queries but also goal-blind ones. Large language models in summarization or fact-extraction modes; hybrid RL agents with separable value/policy; multi-modal models. (C1) holds operationally — the wrapper *chooses* to use the goal-blind mode. - **Class C (fundamentally goal-conditioned).** A's only operating mode requires goal-conditioning. Pure end-to-end goal-conditioned policy networks. (C1) fails; the construction does not apply.

(C2) Stationary component conditional. A's output distribution conditional on input is fixed during the wrapper's operation: $P(A(\cdot) \mid q)$ does not depend on prior queries or on side information beyond q. Adaptation-during-deployment systems are out of scope.

(C3) No implicit goal-inference. A's response to a goal-blind query does not depend on G_W via inference from query patterns:

$$P(A(q_M) \mid q_M, G_W) = P(A(q_M) \mid q_M) \quad \forall\, q_M, G_W \tag{11.19}$$

For pretrained components (notably LLMs), (C3) holds *exactly* only when query content is statistically independent of G_W in the pretraining distribution. The approximate form weakens (C3) to a leakage bound (Theorem 2 below).

Theorem 1: Directed separation at the wrapper level (exact form).

Under (C1)–(C3), directed separation holds *exactly* at the wrapper level:

Derived (directed-separation-at-wrapper-exact, from C1+C2+C3).

$$P(M_{W,m+1} \mid M_{W,m},\ o_{W,m+1},\ G_{W,m}) = P(M_{W,m+1} \mid M_{W,m},\ o_{W,m+1}) \tag{11.20}$$

Therefore W is a Class 1 (Separated) architecture per `#der-directed-separation`.

Proof. Identify all paths from $G_{W,m}$ to $M_{W,m+1}$ given $(M_{W,m}, o_{W,m+1})$. The update is

$$M_{W,m+1} = f_M\big(M_{W,m},\ o_{W,m+1},\ q_M(M_{W,m}, o_{W,m+1}),\ A(q_M(M_{W,m}, o_{W,m+1}))\big) \tag{11.21}$$

f_M has no G_W argument by type signature (D-pathway-1 closed). q_M has no G_W argument by type signature (D-pathway-2 closed). The remaining pathway is $A(q_M)$ depending on G_W given q_M. Under (C3), $P(A(q_M) \mid q_M, G_W) = P(A(q_M) \mid q_M)$ — the response is conditionally independent of G_W given q_M. Since q_M is itself a deterministic function of $(M_{W,m}, o_{W,m+1})$, conditioning on $(M_{W,m}, o_{W,m+1})$ determines q_M, and the integrand $P(M_{W,m+1} \mid M_{W,m}, o_{W,m+1}, q_M, A(q_M)) \cdot P(A(q_M) \mid q_M, G_W)$ no longer depends on G_W. The conditional distribution of $M_{W,m+1}$ given $(M_{W,m}, o_{W,m+1}, G_{W,m})$ equals that given $(M_{W,m}, o_{W,m+1})$. ∎

Theorem 2: Directed separation (approximate form, C3 weakened to leakage bound).

If (C3) is replaced by a KL-leakage bound

Derived (directed-separation-at-wrapper-approximate, from C1+C2+leakage-bound).

$$D_{\mathrm{KL}}\big(P(A(q_M) \mid q_M, G_W) \,\big\|\, P(A(q_M) \mid q_M)\big) \le \kappa \quad \forall\, q_M, G_W \tag{11.22}$$

then the wrapper-level KL-divergence on M_W updates is bounded by the same κ:

$$D_{\mathrm{KL}}\big(P(M_{W,m+1} \mid M_{W,m}, o_{W,m+1}, G_{W,m}) \,\big\|\, P(M_{W,m+1} \mid M_{W,m}, o_{W,m+1})\big) \le \kappa \tag{11.23}$$

The wrapper is *almost-Class-1 (Separated)* with leakage rate $\le \kappa$. *Proof.* The wrapper-level M_W update is a deterministic function of the component response given the wrapper's other inputs; the data-processing inequality propagates the KL bound from response distribution to wrapper-state distribution. ∎

Wrapping regime hierarchy. The construction supports three regimes, distinguished by where structural separation lives:

Table 11.4

Regime	*Construction*	*Leakage bound*	*Leakage source*
$\mathbf{W_0}$ (no wrapping)	Raw Class 2 (Partial) or Class 3 (Coupled) component	κ_{W_0} at the component's maximum goal-conditioning sensitivity	No constraint
$\mathbf{W_2}$ (partial wrapping)	One goal-conditioned call per macro-step; structurally typed parsed response routes updates to M_W vs. G_W slots	κ_{W_2} bounded *behaviorally* — by the component's compliance with the prompted instruction-to-separate; **no structural bound**	Component's instruction-following fidelity
$\mathbf{W_1}$ (strict wrapping)	Theorem 1 / 2 — separate q_M and q_G calls per macro-step	$\kappa_{W_1} \leq I(A(q_M); G_W \mid q_M)$ — bounded *structurally* by mutual information in the pretraining distribution	Pretraining-induced query-content / goal-content correlation

W_1 admits a structural bound from (C3) or its weakening; W_2 admits only a behavioral bound from the component's compliance fidelity. The two are different in kind — structural bounds are derivable from query content; behavioral bounds depend on the component's training and prompt-following. The same KL-form bound of Theorem 2 covers both regimes; what changes is *what determines* κ.

The W_0 / W_2 / W_1 distinction refines the Class 1 (Separated) cell of `#der-directed-separation`: within Class 1 (Separated), **Class-1-by-structure** (natively goal-blind components, or W_1 wrapping) has a structurally derivable directed-separation guarantee; **Class-1-by-behavior** (W_2 wrapping) has only an empirically estimable guarantee that depends on the component's instruction-following.

Epistemic Status.

Conditional on (C1), (C2), and (C3) (or its weakening to a leakage bound). The proofs are short conditional-independence reasoning (Theorem 1) and a single application of the data-processing inequality (Theorem 2); both are standard.

Max attainable: derived under stated conditions. (C3)'s exact form is a structural ideal that pretrained components (notably LLMs with goal-rich training data) generally satisfy only approximately; the realistic regime is Theorem 2 with κ characterized empirically.

The wrapping regime hierarchy (W_0/W_2/W_1) is a *formulation* — the partition is made by the structural choice of where to place the separation commitment. The leakage bounds within each regime are derived once the regime is fixed.

Discussion.

Quality–separation tradeoff inside Class B. For Class-B components (admitting both goal-blind and goal-conditioned modes), the wrapper has a design choice: how aggressively to restrict q_M to goal-blind content, vs. how much context to allow that may carry goal-correlated information. Maximally goal-blind queries (only the current observation, no context, no history) reduce the pretraining-induced leakage that bounds κ_{W_1}, but may produce information-poor responses that hurt f_M's update quality. Maximally informed queries (full history, retrieved context) produce richer responses but increase the mutual information $I(q_M; G_W)$ and therefore the upper bound on κ_{W_1}. The tradeoff is real and resolved per application.

Component-admissibility partition. Class A components (goal-blind by design) satisfy (C1) trivially and don't need wrapping in the substantive sense — wrapping for Class A is organizational rather than structural. Class B components (LLMs, hybrid RL with separable value/policy, multi-modal models) are the substantive wrapping case — the wrapper *chooses* to use the goal-blind mode. Class C components (pure end-to-end goal-conditioned policy networks) fail (C1) and are scope-out for the basic theorem. Salvage paths for Class C — null-goal queries, goal-uniform averaging, auxiliary distilled goal-blind heads — exist but cost something (information loss, computation, training).

Resolution of the LLM scope question. The "Class 3 (Coupled) exit" framing — *directed separation violated by goal-conditioned agents (LLMs); handled as architectural scope, not approximation* — is refined by this segment from a scope exit to a constructive route through. Class 3 (Coupled) LLMs are scope-in *for the wrapper construction* (under Class-B admissibility). The cost is paid in residual leakage rate κ_{W_1} bounded by pretraining-distribution mutual information; the tempo cost is established separately in `#der-class-coercion-in-composition`. Whether this construction yields an operationally useful agent depends on how favorable the pretraining-distribution-induced bounds are for the application.

Relationship to `#hyp-directed-separation-under-composition`. The hypothesis is descriptive — when does directed separation hold under composition? This segment provides the constructive answer for the wrapper-around-component special case: directed separation holds whenever the wrapper's type signatures are respected and (C1)–(C3) hold (or their weakenings). The general N-agent composition question remains a hypothesis; the wrapper-around-component case is now derived.

Wrapping as a truthification mechanism. The wrapping construction is the *rigorous formal version* of what `#disc-adversarial-coupling-pressure` §"Defensive scaffolding as composition" gestures at informally — peer review, prediction registers, double-entry bookkeeping, adversarial procedure, structured red-teaming. Those external scaffolds are operational mechanisms for *increasing* the modularity of a composite agent in the face of forces that would couple it; the wrapping construction is the structural version of the same operation applied internally rather than externally. Both share the discipline: a goal-blind belief-update query path is structurally enforced (W_1 strict) or behaviorally bounded (W_2), at a definite cost (extra component calls per macro-step plus residual leakage rate). The $W_0/W_2/W_1$ regime hierarchy is the *graded* characterization of how thoroughly the truthification has been applied — W_0 is the un-truthified base state, W_2 behavioral truthification, W_1 structural truthification. Forward-reference: `#disc-modularity-state-dynamics` (queued; scoped in `msc/modularity-cycle-plan-2026-05-09.md`) is the meta-segment in which the truthification operation sits as one of three operations on the modularity state — alongside strategic self-coupling (self-driven-decreasing) and adversarial coupling pressure (externally-driven-decreasing). When that meta-segment lands, this segment becomes the canonical *formal* instance of the truthification operation, paired with the *informal* defensive-scaffolding instance from the adversarial-pressure segment. Until then, the connection is named here and in `#impl-composition-machinery` §"Class-coercion as truthification mechanism."

Findings.

Constructive Directed Separation via Wrapping. **Brief:** When you have a component (like an LLM) whose belief-update and goal-conditioning are entangled in a single forward pass, you can build a scaffold around it that maintains explicit, separate stores for what the system believes and what it wants. The structural rule is that belief updates only see queries to the component that don't include the goal as input. Under reasonable conditions on the component, the wrapped system is goal-blind in its belief updates *by construction* — even though the underlying component isn't. The cost shows up as a residual leakage from the component's pretraining (the component might still infer the goal from query content, even when the goal isn't explicit in the input). Two practical regimes appear: strict wrapping with separate goal-blind and goal-conditioned calls (theoretically clean, with a structural leakage bound), and partial wrapping with one goal-conditioned call whose response is parsed into separate update fields (operationally common, with only a behavioral leakage bound — depending on the component's instruction-following fidelity rather than its query structure).

Impact: Promotes `#hyp-directed-separation-under-composition` to derived (in the wrapper-around-component special case). Refines the Class 1 (Separated) cell of `#der-directed-separation` with a structural-vs-behavioral sub-distinction (W_1 vs. W_2). Resolves the LLM scope question — Class 3 (Coupled) components are scope-in for the wrapper construction at a measurable cost, not scope-out. The composition-level consequences (wrapper as valid AAT composite agent, persistence-template inheritance, tempo cost) are derived in the companion segment `#der-class-coercion-in-composition`.

Novelty Claim: *Claim integration* of POMDP / cognitive-architecture prior art with the AAT Class 1/2/3 (Separated/Partial/Coupled) directed-separation taxonomy, plus the $W_0/W_2/W_1$ regime hierarchy that surfaces the structural-vs-behavioral leakage distinction and the LLM-specific (C1)–(C3) admissibility/leakage conditions. The wrapping move itself is rediscovery of patterns established in POMDP theory (Bayesian belief-update is goal-blind by construction) and cognitive architectures (modular agent design with separated belief/goal/action state, four decades). AAT's contribution is the structural-leakage analysis at the directed-separation level and the regime hierarchy that names where the separation guarantee lives.

Related Work:

Table 11.5

ASF concern	*Prior-art language*	*Relationship / Positioning*
Goal-blind belief-update by construction	Astrom 1965, "Optimal control of Markov processes with incomplete state information," *J. Math. Anal. Appl.* 10; Kaelbling, Littman, Cassandra 1998, "Planning and acting in partially observable stochastic domains," *Artificial Intelligence* 101	*formal antecedent* — POMDP belief-state filters are goal-blind by construction; the wrapping move recapitulates this in the AAT vocabulary. The closest formal prior art for the directed-separation guarantee.
Modular agent design with separated belief/goal/action	Newell 1990, *Unified Theories of Cognition*; Laird 2012, *The Soar Cognitive Architecture*; Anderson 2007, *How Can the Human Mind Occur in the Physical Universe?*; Sun 2016 *Anatomy of the Mind* (CLARION); Baars 1988 *A Cognitive Theory of Consciousness* / Dehaene 2014 *Consciousness and the Brain* (Global Workspace)	*formal antecedent* — cognitive architectures have done modular agent design with separated belief/goal/action state for 40+ years. The W_1 wrapping move is essentially the per-cycle commitment that cognitive architectures make at the system level.
Tool-using language-model agent frameworks	Yao et al. 2022, "ReAct: Synergizing reasoning and acting in language models"; Shinn et al. 2023, "Reflexion: language agents with verbal reinforcement learning"; Park et al. 2023, "Generative Agents: Interactive Simulacra of Human Behavior"; Packer et al. 2023, "MemGPT: Towards LLMs as operating systems"; Schick et al. 2023, "Toolformer: language models can teach themselves to use tools"	*empirical instantiation* — practical wrappers around language-model substrates. Most fall in W_2 (partial wrapping / output-structuring); Generative Agents' observation→memory step is the closest empirical instance of W_1. AAT's regime hierarchy gives these constructions a structural reading.

Search Log:

- 2026-05-09 (*targeted*): Web + training-data search across POMDP / cognitive-architecture / scaffolded-LLM threads. Verdict: **substantial overlap** with the POMDP and cognitive-architecture lines as the closest formal prior art. AAT's contribution is the structural-leakage analysis and regime hierarchy rather than novelty in the wrapping move itself.
- 2026-05-09 (*intuition-only*, prior to the targeted search): adjacent literatures expected to host prior art were active inference (Markov blankets), control theory (approximate simulation), and scaffolded-LLM frameworks. The targeted search confirmed all three and added the POMDP and cognitive-architecture lines as the formal precedents.

Working Notes.

- **Empirical $\varkappa$ measurement.** $\varkappa_{W_1} \leq I(A(q_M); G_W \mid q_M)$ is computable for any component with stochastic outputs by sampling responses under multiple goal-conditioning histories and estimating the divergence. For a fixed component, this bound is a property of the wrapper's choice of q_M — narrower queries reduce the bound, richer queries increase it. The empirical instantiation is open follow-on.
- **Compositional wrapping (wrapper-of-wrapper).** How leakage rates compose under iterated wrapping is open. Conjecture: additive in KL ($\varkappa_{\text{outer}} \leq \varkappa_{\text{inner}} + \varkappa_{\text{outer-shell}}$) by data-processing inequality applied at each level, but tightness is unclear.
- **Behavioral compliance axiom for W_2.** $\varkappa_{W_2}$ has no structural bound; it depends on the component's instruction-following fidelity. Whether a behavioral-compliance axiom (assuming the component honestly attempts to follow structural-separation instructions) yields a bound is an open hypothesis. If so, it would be hypothesis-grade rather than derived.
- **Identifying the regime in the wild.** Practical scaffolded-LLM frameworks (ReAct, Reflexion, MemGPT, etc.) almost universally implement W_2. Distinguishing W_2 from W_1 in a deployed system requires inspection of the per-cycle query structure — does f_M's update path receive a query that contains G_W or not? This is the diagnostic question.
- **Segment split provenance (2026-05-11).** This segment was bifurcated from a combined "class coercion" derivation. Claim A (directed separation at the wrapper level) lives here; Claim B (wrapper as valid AAT composite agent — (A1)–(A4) verification, persistence-template inheritance, Brooks's-Law tempo cost) lives in `#der-class-coercion-in-composition` (which declares this segment as prerequisite). The split reflects FORMAT.md Gate 1 discipline: this segment's depends list (`der-directed-separation`, `def-agent-environment`) reflects exactly what the directed-separation theorem actually requires. The composition-level dependencies (`form-composition-closure`, `deriv-sector-condition`, `result-sector-persistence-template`, `der-tempo-composition`) are Claim B's load and now live with Claim B.
- Reasoning-trail provenance: spike directories at `spikes/class-coercion-wrapping/` and `spikes/temporal-nesting-rg/` carry the working-out of these results.
- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. Three independent axes in this segment: (a) GUC Class 1/2/3 — renamed and swapped; (b) W_0/W_1/W_2 wrapping regimes — UNCHANGED; (c) Class A/B/C component-admissibility partition — UNCHANGED.

11.5 *Derived:* Class Coercion in Composition

Under the wrapper construction of `#der-class-coercion-via-wrapping`, the wrapped system W is a valid AAT composite agent: it satisfies (A1)–(A4) of `#form-composition-closure`, inherits the sector-persistence template from `#result-sector-persistence-template`, and incurs a tempo cost in the Brooks's-Law form of `#der-tempo-composition`. This segment establishes those composition-level consequences. The directed-separation guarantee that motivates the construction is established in the prerequisite segment.

Formal Expression.

Setup (inherited). The wrapper structure, the four type-signed update components (q_M, q_G, f_M, f_G), the macro-step schedule ($K \geq 2$ component calls per macro-step), and the admissibility conditions (C1)–(C3) on the component A are defined in `#der-class-coercion-via-wrapping`. This segment uses that setup without redefinition.

Wrapper-design constraints. For (A2)–(A4) of `#form-composition-closure` to hold at the wrapper level, the wrapper's update structure must satisfy three additional constraints:

(D-A2) The wrapper commits to a prediction map $\hat{o}_W : \mathcal{X}_M \times \mathcal{A}_W \to \mathcal{O}_W$ so that macro-mismatch $\delta_W = o_W - \hat{o}_W$ is well-defined.

Conditions (wrapper-design).

(D-A3) f_M supports a gain interpretation per `#def-adaptive-tempo`. Holds for Tier-1 belief-update maps — Bayesian on exponential families, gradient on strongly convex losses, linear-PD with bounded gain. Tier-2/3 cases inherit the corresponding tier-restricted scope from `#deriv-sector-condition`.

(D-A4) f_M satisfies the sector condition with positive correction rate. Automatic for Tier-1 belief-update maps via `#deriv-sector-condition` Prop A.1 and `#der-gain-sector-bridge`.

(A1) of `#form-composition-closure` holds by construction — $X_W = (M_W, G_W)$ has the AAT form because the wrapper *builds it in*.

Theorem: Wrapper as valid AAT composite agent.

Under (D-A2)–(D-A4) and the directed-separation conditions of `#der-class-coercion-via-wrapping`, the wrapper W satisfies (A1)–(A4) of `#form-composition-closure` and therefore qualifies as an AAT composite agent.

Derived (wrapper-as-composite, from (D-A2)+(D-A3)+(D-A4)).

Proof. (A1) by construction of the type signatures; the wrapper's state is $X_W = (M_W, G_W)$ in the AAT shape. (A2) under (D-A2): the prediction map closes the mismatch definition. (A3) under (D-A3): Tier-1 belief-update maps inherit gain interpretation via `#deriv-sector-condition` Prop A.1, with the Tier-2/3 lifts deferring to the tier-restricted scope from `#form-composition-closure`'s bridge-lemma classification. (A4) under (D-A4): the sector condition transfers from the (Tier-1) belief-update map to the wrapper level via `#der-gain-sector-bridge`. ∎

Inheritance of the persistence template. Under (D-A4), the wrapper inherits `#result-sector-persistence-template` at the wrapper level: persistence holds when $\alpha_W R_W > \rho_W$. The wrapper-level effective disturbance has two contributions: external environmental disturbance ρ_{ext} acting through o_W, and internal disturbance from the component's response variance, ρ_{int}, bounded by the variance of A's responses to goal-blind queries. Total: $\rho_W = \rho_{\text{ext}} + \rho_{\text{int}}$. Persistence at the wrapper level requires $\alpha_W R_W > \rho_{\text{ext}} + \rho_{\text{int}}$.

Tempo cost — Brooks's-Law instance. The wrapper makes $K \geq 2$ component calls per macro-step (more in richer wrapper designs). If the component's nominal call rate is ν_A, the wrapper-level macro-update rate is $\nu_W = \nu_A / K$. By `#der-tempo-composition`,

$$\mathcal{T}_W \leq \mathcal{T}_A^{\text{nominal}} - C_{\text{coord}}^{\text{wrap}} \tag{11.24}$$

where $C_{\text{coord}}^{\text{wrap}}$ is the coordination overhead specific to the wrapping construction — the tempo consumed by maintaining the wrapper's (M_W, G_W) state separately from the component's internal state. This is the cost of class coercion paid in tempo: the same Brooks's-Law form whose general statement is in `#der-tempo-composition`. Adding state-management infrastructure reduces realized external tempo even when the underlying component's compute rate is unchanged.

Epistemic Status.

Conditional on the wrapper-design constraints (D-A2), (D-A3), (D-A4), and on the prerequisite directed-separation result of `#der-class-coercion-via-wrapping`. The proof is direct verification using inheritance from `#deriv-sector-condition` and `#def-adaptive-tempo`; both are standard. (D-A4) is automatic for Tier-1 belief-update maps; for Tier-2/3 belief-update maps the wrapper inherits the tier-restricted scope from `#deriv-sector-condition` and `#form-composition-closure`'s bridge-lemma classification.

Max attainable: derived under stated conditions. The composition-level claim is parametrized by the belief-update map's tier; Tier-1 cases inherit cleanly, Tier-2/3 cases inherit under additional restrictions.

Discussion.

Coercion-distance vs. tracking-distance. The closure-defect ε^* in `#form-composition-closure` is read as a *fidelity* measure: how well the macro-system tracks the projected micro-system. In the wrapping construction, the wrapper deliberately changes the underlying component's behavior to enforce separation — the wrapper *should* differ from the unwrapped component in the goal-conditioning direction. Two distinct quantities are involved:

- $\varepsilon^*_{\text{track}}$ — the standard fidelity quantity from `#form-composition-closure`. Used in the bridge lemma to bound trajectory error.
- $\varepsilon^*_{\text{coerce}}$ — the wrapping-specific quantity measuring behavioral divergence between the wrapped system and the unwrapped component. Higher means more aggressive coercion.

For wrapper analyses: the persistence-template inheritance (above) uses $\varepsilon^*_{\text{track}}$. Cost-of-class-coercion analyses use $\varepsilon^*_{\text{coerce}}$. The leakage analysis from `#der-class-coercion-via-wrapping` uses κ — a KL-divergence on response distributions, distinct from both trajectory-error quantities. Conflating them propagates downstream confusion.

Form-preservation reading. Read in form-preservation language, (A1)–(A4) of `#form-composition-closure` are the requirement that AAT's form survive coarse-graining: macro must itself be AAT. The wrapping construction is the *constructive* version — when the underlying component doesn't preserve form on its own (Class 2 (Partial) or Class 3 (Coupled)), the wrapper enforces form-preservation at the macro level by structural commitment of the type signatures (from `#der-class-coercion-via-wrapping`) plus the wrapper-design constraints here. The closure-defect $\varepsilon^*_{\text{track}}$ then plays the role of distance from the form-preserved fixed point in the constructive setup.

Findings.

Wrapper as Valid AAT Composite Agent with Brooks's-Law Tempo Cost. **Brief:** Once you've wrapped a coupled component to get directed separation (the structural rule from `#der-class-coercion-via-wrapping`), is the wrapped system actually a valid AAT agent that AAT's machinery applies to? The answer is yes, under reasonable design constraints on the wrapper's prediction map, gain interpretation, and sector behavior. This means the wrapped system inherits the persistence-condition machinery — the wrapper's own version of "can the agent keep up with disturbance" applies, with disturbance now decomposing into the external environment's drift plus the underlying component's response variance. There's a cost, and it's Brooks's Law in tempo coordinates: every additional component call per cycle (needed to maintain separate belief and goal stores) divides the macro-update rate. The wrapped system runs slower than the component would running alone, by a structurally determined factor.

Impact: Closes the loop on whether the wrapping construction yields a *usable* AAT agent: yes, the wrapper inherits `#result-sector-persistence-template` and `#der-tempo-composition` at the wrapper level. Makes the tempo cost of class coercion explicit and quantifiable — it's a Brooks's-Law instance, the same general form that governs all AAT compositions, not a special-case tax. Provides the basis for analyzing logogenic-substrate wrappers (e.g., PROPRIUM's auxilia hierarchy, agentic-tft's CONTEXTUALIZE-CHOOSE phase separation) as valid composite agents with computable persistence and tempo characteristics.

Novelty Claim: *Claim integration* of the AAT sector-Lyapunov persistence template, Brooks's-Law tempo accounting, and the form-composition-closure (A1)–(A4) discipline, applied to the wrapper-around-component construction. The composition-level inheritance is straightforward given the persistence-template and tempo-composition machinery; the contribution is establishing that the wrapped system is in the right shape for those results to apply, and naming the Brooks's-Law form of the coercion cost.

Related Work:

Table 11.6

ASF concern	*Prior-art language*	*Relationship / Positioning*
Predictive-loss bounds under coarse-graining of MDPs	Ravindran-Barto 2004, "An algebraic approach to abstraction in reinforcement learning"; Taylor, Precup, Panangaden 2008, "Bounding performance loss in approximate MDP homomorphisms," NeurIPS; Abel, Hershkowitz, Littman 2016, "Near optimal behavior via approximate state abstraction," ICML; Subramanian, Sinha, Seraj, Mahajan 2020, "Approximate information state for approximate planning," arXiv:2010.08843; Congeduti, Mey, Oliehoek 2020, "Loss bounds for approximate influence-based abstraction," arXiv:2011.01788	*adjacent literature* — control-theoretic predictive-loss bounds. AAT connects to this cluster via `#form-composition-closure`'s bridge lemma at the wrapper level.
Compositional structure for nested agents	Smithe 2024, "Structured Active Inference," arXiv:2406.07577; Capucci, Gavranović, Hedges, Rischel 2022, "Towards foundations of categorical cybernetics," ACT 2021	*adjacent literature* — categorical / lens framing of compositional agents. The wrapper construction is consistent with the lens framing; an alternative reading in categorical-systems-theory terms is available.
Form-preservation under coarse-graining	Friston 2019 *J. Theor. Biol.*, "On Markov blankets and hierarchical self-organisation"; Friston, Heins, Verbelen, Da Costa et al. 2025 *Front. Network Physiology*, "From pixels to planning: scale-free active inference"	*adjacent literature* — Friston's framing of the renormalization group as form-conservation under coarse-graining. The wrapping construction reads as a constructive form-preservation move.
IB-Lagrangian semigroup composition	Mehta, Schwab 2014, "An exact mapping between the Variational Renormalization Group and Deep Learning," arXiv:1410.3831; Kline, Palmer 2022, "Gaussian Information Bottleneck and the Non-Perturbative Renormalization Group," PMC8967309	*adjacent literature* — IB-as-RG. The wrapper's (P1) information-preservation condition (from `#form-composition-closure`) is IB-shaped; the semigroup composition rule is the closest analog of "AAT form preserved under iterated wrapping."
Singular-perturbation as unified RG	Chen, Goldenfeld, Oono 1996, "Renormalization group and singular perturbations," *Phys. Rev. E* 54:376; Chiba 2009, SIAM J. Appl. Dyn. Syst. 8:1066	*formal antecedent* — RG framework subsumes singular perturbation theory. The $K_c \gg 1$ timescale-separation regime in `#form-composition-closure` invokes this for free.

Search Log:

- 2026-05-09 (*targeted*): Search across MDP-homomorphism / hierarchical-control / categorical-systems-theory / FEP-RG / IB-RG / singular-perturbation-RG threads. Verdict: **substantial overlap** with the MDP-homomorphism and categorical-cybernetics lines as the closest formal prior art for composition under coarse-graining. AAT's contribution is the specific Brooks's-Law tempo form and the persistence-template inheritance pattern.

Working Notes.

- **Detailed tempo accounting for canonical wrapper architectures.** $C_{\text{coord}}^{\text{wrap}}$ for ReAct-shape, Reflexion-shape, PROPRIUM-shape wrappers is an empirically-relevant computation deferred from this segment's promotion. The general bound (Brooks's-Law form via `#der-tempo-composition`) is established here; specific architectural breakdowns are follow-on.
- **Connection to ELI-specific structure in `04-eli-core/`.** Most ELI-specific content (sovereignty axes, accountability infrastructure, identity factors, substrate-independence) is *added structure* beyond what the composition-level inheritance provides — this segment establishes the wrapper as a valid composite agent; ELI work is what happens on top of that substrate.
- **Segment split provenance (2026-05-11).** This segment was bifurcated from a combined "class coercion" derivation. The prerequisite segment `#der-class-coercion-via-wrapping` carries the directed-separation claim (Claim A: wrapper is Class 1 (Separated) by construction); this segment carries the composition-level claim (Claim B: wrapper is a valid AAT composite agent with persistence-template inheritance and Brooks's-Law tempo cost). The split reflects FORMAT.md Gate 1 discipline: each segment's depends list reflects exactly what its claim actually requires.

11.6 *Discussion:* Additional Implications & Discussion

The chapter is where AAT's per-agent machinery lifts to the composite level. Closure defect supplies the criterion for when a group of agents legitimately counts as one larger agent; the bridge lemma converts predictive defect into trajectory error; the wrapping construction supplies the constructive route from Class 3 (Coupled) components to Class 1 (Separated) composites; and the closed-form critical-mass derivation unifies four previously-separate persistence inequalities as signed special cases of a single composite-sector condition. The chapter's two segments with `## Findings` carry the load-bearing technical content well; the implications-side payoff is the cross-segment composition — how the four pieces produce composite-level results that no single segment carries — and what those results buy for `03-llm-core/`, AI-safety architecture, and the framework's identifiability-floor pattern at the composition layer.

Closure defect, bridge lemma, and the meaning of composition.

`#form-composition-closure`'s Findings cover the closure-defect / bridge-lemma machinery in full: when does a group of interacting agents legitimately count as one larger agent? — answered by an inequality, with explicit failure modes on both sides. The closure-defect ε^* is the minimum residual prediction error from collapsing the micro-system into an AAT-shaped macro-description; the bridge lemma converts that prediction-level error into a trajectory-level guarantee bounded by $\varepsilon^* \nu_c / \alpha_c$ under sector-bounded macro-correction plus strong monotonicity of the macro-update. The implication worth surfacing beyond the segment's own statement is *what* the framework is offering against adjacent positions, and *what* it is honest about not offering.

Against Markov-blanket and active-inference accounts of nested agency, AAT's bridge-lemma form supplies a *control-theoretic loss bound* on treating a collection as one agent — something the Markov-blanket apparatus does not produce, even when the boundary criterion is in hand. The closure-defect framework is consistent with the approximate-information-state and influence-based-abstraction lines from PAC-MDP work (Subramanian 2020, Congeduti 2020, Abel 2016, Taylor 2008) but applies the same predictive-loss-to-control-error machinery at the *agent boundary formation* scope rather than at the single-agent compression scope. The framework's distinctive content is the *use* of bounded-loss composition as an agent-boundary criterion, together with the admissibility conditions (A1)–(A4) on the macro-dynamics that force the macro-description to itself be AAT-shaped (ruling out trivial curve-fits where any composite could be called an "agent" by re-defining state).

The honest scope statement worth keeping in view is that $\varepsilon^* = 0$ does not mean *optimality* — two non-communicating Kalman filters achieve $\varepsilon^* = 0$ while being suboptimal versus a joint filter, because the joint filter could exploit cross-correlation the non-communicating pair cannot. The framework diagnoses *representability* of a composite as an AAT agent, not optimality of that composite against alternative architectures. Strong monotonicity is the hinge — sub-agents that are merely Lyapunov-stable do not compose into a coherent macro-agent, and this is one of the chapter's quieter surprises. Three agent tiers fall out of the strong-monotonicity requirement: Tier 1 where contraction is proved for the full class (Bayesian on exponential families, gradient descent on strongly convex losses, linear correctors with positive-definite gain); Tier 2 where it holds locally; Tier 3 where it must be verified per-domain.

The critical-mass closed form unifies four signed special cases.

The chapter's deepest cross-segment finding is what `#deriv-critical-mass-composition` produces by composing the bridge lemma with `#der-team-persistence`'s signed-coupling decomposition. For the symmetric-matched-Tier-1 dyad, the composite sector constant is now derived in closed form:

$$(\alpha - C)R > \rho + \gamma\mathcal{T} \tag{11.25}$$

with signed γ (negative cooperative, positive adversarial) and C the coordination-overhead cost. The single inequality recovers *four prior results as signed special cases*: single-agent persistence ($\gamma = 0, C = 0$, reducing to Part I Ch.4's central inequality); team-persistence with cooperative coupling ($\gamma < 0$, where the negative term enters as effective-disturbance reduction); adversarial destabilization ($\gamma > 0$, where the positive term amplifies disturbance); and coordination-dominated failure ($C > \alpha$, where the composite's effective correction rate goes negative regardless of individual quality — *Brooks's Law as a derived inequality* rather than a folkloric observation). The teleological unity U_O enters multiplicatively on γ plus scope-gate, not additively — a distinction with downstream implications for Ch.3's unity-closure-mapping result.

The implications-side payoff that the segment's Working Notes flag but is worth surfacing: this is the framework's *cleanest internal-economy demonstration*. One inequality, with signed parameters indexing the regime, recovers what previously read as four separate results. The result composes with the sector-persistence template economy (Part I Ch.4 implications, §4): the template absorbed the Lyapunov boilerplate; the critical-mass derivation specializes the template's persistence inequality with a signed-coupling effective-disturbance decomposition that handles single, cooperative, adversarial, and coordination-failure regimes simultaneously. For a practitioner, the engineering anchor is that Brooks's Law (adding people to a late project makes it later) has a *physics-level* derivation here: a development team's macro-correction rate is $\alpha - C$ where C is coordination overhead; once C exceeds α, adding people degrades persistence regardless of individual quality. The threshold is sharp, not gradual, and the inequality predicts exactly where it bites.

The closed form also produces `#disc-identifiability-floor`'s Instance 3 in its fullest form. Composite contraction certification from component-marginal data is structurally impossible — the coupling-sign bit ($\gamma < 0$ vs $\gamma > 0$) is unidentifiable from how each component looks in isolation (Liberzon 2003's common-Lyapunov-nonexistence is the external no-go). The five boundary routes named in `#disc-identifiability-floor` for Instance 3 escape this: under matched-Tier conditions the closed form supplies the coupling sign by interventional access at the composite scope, via Mode 2 of the loop-Level-2 pattern (observer-on-sub-agent, from Part II Ch.2 implications). The chapter is therefore where the *third* formal instance of the identifiability-floor pattern becomes statable, completing the four-instance pattern with Instance 1 (Part II Ch.4), Instance 2 (Part II Ch.3), Instance 3 (here), and Instance 4 (Part II Ch.5).

Class coercion via wrapping — what the construction unlocks.

`#der-class-coercion-via-wrapping`'s Findings cover the constructive directed-separation result well: when a component (like an LLM) has belief-update and goal-conditioning entangled in a single forward pass, a scaffold around it can maintain explicit separate stores for belief and goal, with belief updates only seeing queries that don't include the goal as input. Under named admissibility conditions on the component, the wrapped system is goal-blind in its belief updates *by construction* — even though the underlying component is not. The $W_0/W_2/W_1$ regime hierarchy names the practical landscape: W_1 (strict wrapping, structural leakage bound, theoretically clean), W_2 (partial wrapping with goal-conditioned call and parsed response, behavioral leakage bound, operationally common), W_0 (no wrapping — the unconverted Class 3 case).

The implications-side payoff is what the construction unlocks beyond the segment's statement: it is the *constructive bridge* that resolves how AAT's Section II results — derived for Class 1 (Separated) agents — apply to Class 3 (Coupled) substrates. Without the wrapping construction, Class 3 components would sit outside Section II's persistence machinery as a scope exit; with the wrapping construction, the wrapped system at the *wrapper level* is Class 1 (Separated) by construction, inherits the sector-persistence template, and operates under the standard AAT results. The cost is paid in two places: a Brooks's-Law tempo overhead (more component calls per macro-step), and a residual leakage rate bounded structurally (W_1, by the wrapper's query design) or behaviorally (W_2, by the component's instruction-following fidelity).

The composition-level consequences are derived in `#der-class-coercion-in-composition`: the wrapped system is a valid AAT composite agent, satisfying admissibility conditions (A1)–(A4) of `#form-composition-closure` under wrapper-design constraints (D-A2)–(D-A4). It inherits the sector-persistence template at the wrapper level; its tempo cost is the Brooks's-Law composition cost from `#der-tempo-composition`. This is the chapter's quiet load-bearing claim for `03-llm-core/`: language-model substrates are *constructively* in AAT scope, at a measurable cost, rather than scope-out. The implications for AI-safety practice are concrete: scaffolded loops around LLMs are not just engineering convenience — they are the structural mechanism that converts a Class 3 (Coupled) substrate into a Class 1 (Separated) composite, and the leakage bounds tell the practitioner what residual coupling remains. The Generative Agents observation→memory step (Park et al. 2023) is the canonical empirical W_1 instance; most production tool-using LLM frameworks (ReAct, Reflexion, Toolformer, MemGPT) sit in W_2. The framework's regime hierarchy gives all of these constructions a *structural* reading rather than an engineering-pattern reading.

The cross-domain transfer follows naturally. Human bicameral / dual-process models (Kahneman 2011) are W_1-style internal scaffolding when the deliberative system constrains the intuitive system through structural commitment to goal-blind belief-update queries; organizations where executive scaffolding (boards, committees, decision rights) recovers cascade ordering across coupled individuals are W_1/W_2 at the organizational scale. The framework's structural claim is the same in all three cases: *separation at the wrapper level is what gives AAT-grade behavior; the wrapping construction is structurally non-optional for Class 3 (Coupled) substrates*.

Tempo composition, Brooks's Law, and the bandwidth connection.

`#der-tempo-composition` carries the sub-additive tempo inequality: $\mathcal{T}_c \leq \sum_i \mathcal{T}_i$, with the gap bounded below by $C_{\text{coord}} \geq \varepsilon^* \nu_c$. The implication is that *coordination overhead acts as effective disturbance at the composite level* — closure defect $\varepsilon^* \nu_c$ is what the template's instantiation at the composite identifies as the internal contribution to effective disturbance, alongside the external environmental rate ρ_{ext}. The composite persists when $\alpha_c R_c > \rho_{\text{ext}} + \varepsilon^* \nu_c$, and Brooks's Law follows: adding agents increases $\sum_i \mathcal{T}_i$ but may increase $\varepsilon^* \nu_c$ faster, pushing the composite's effective disturbance past its persistence threshold even as nominal capacity grows.

The composition with Part I Ch.4's information-rate floor is worth surfacing as a cross-segment finding. Each agent in the composite pays its own $\dot{R} \geq n\alpha/2$ bandwidth cost; the composite as a whole pays the sum *minus what coordinated sub-agents save on shared observation*. But there is a complementary cost: the closure defect $\varepsilon^* \nu_c$ has its own information-rate signature, because maintaining the bridge-lemma's strong-monotonicity condition at the macro-update requires the composite to acquire information about how its sub-agents are interacting. The framework's prescription is that *composite persistence pays for both the sub-agents' individual bandwidth and the coordination-overhead bandwidth*, and the trade-off between adding agents (more $\sum_i \mathcal{T}_i$, more individual bandwidth) and increasing coordination overhead (more $\varepsilon^* \nu_c$, more coordination bandwidth) has a precise inflection point at $C_{\text{coord}} = \alpha_c$. Past the inflection point, additional sub-agents *cost more bandwidth than they buy*, and the composite's persistence-information-rate floor moves up faster than its persistence threshold. This is the implication of the chapter's machinery for distributed-systems design: the sweet spot for sub-agent count is bounded by the coordination-overhead growth rate, not by individual sub-agent capacity.

The chapter is also where Mode 2 of the loop-Level-2 deployment pattern (observer-on-sub-agent, from Part II Ch.2 implications) becomes structurally load-bearing. The Liberzon-anchored identifiability-floor Instance 3 says the coupling sign cannot be recovered from component marginals; the unique broadly-available escape is interventional access at the composite scope. Operationally, this means an external observer (or the composite agent itself acting on its sub-agents) can recover the sign by performing do-interventions on one sub-agent and observing the response of another. The closed-form critical-mass derivation gives the mathematical form of the recovered coupling; the loop-Level-2 access at the composite level gives the data substrate that recovery requires.

Closing notes and the bridge into Unity, Communication, and Shared Intent.

What this chapter closes on is the composite-agent foundation Part III needs: a representability criterion via closure defect, a trajectory-level loss bound via the bridge lemma, a closed-form persistence inequality that recovers single-agent, team, adversarial, and Brooks's-Law regimes as signed special cases, a constructive route from Class 3 (Coupled) components to Class 1 (Separated) composites with measured leakage bounds, and a sub-additive tempo inequality whose coordination-overhead term enters the persistence template as effective disturbance at the composite level. The chapter's two segment-level `## Findings` sections carry the load-bearing technical content; this implications segment surfaces the cross-segment composition and the engineering anchors that the segments individually cannot reach.

The bridge into Part III Ch.3 (Unity, Communication, and Shared Intent) is the closure-defect ε^* itself. The dimensions of unity — content axes (U_M, U_O, U_Σ, U_{obs}) and structural axis (U_f) — parametrize *what sub-agents can share*; Ch.3's `#result-unity-closure-mapping` shows that unity parametrizes a rate-distortion curve for closure defect. The chapter here gives the closure defect a precise form; Ch.3 gives unity its quantitative content; together they say *which dimensions of unity matter how much for which kinds of composite-agent persistence*. The cooperative-vs-adversarial machinery in Ch.4 then operates on what Ch.2 and Ch.3 jointly build: cooperative coupling reduces γ (lowers effective disturbance); adversarial coupling raises γ (amplifies it); the signed-coupling structure is the same inequality viewed from opposite directions. Ch.5's strategic composition lifts the symmetric partially-opposing case from scalar coupling to equilibrium machinery (potential / monotone games), and the identifiability-floor pattern's Instance 3 instance becomes operational there in its strongest form.

Working Notes.

- This segment is a chapter-end discussion grouping findings from `#form-composition-closure`, `#der-tempo-composition`, `#hyp-directed-separation-under-composition`, `#der-class-coercion-via-wrapping`, `#der-class-coercion-in-composition`, `#deriv-critical-mass-composition` (Appendix A), `#result-sector-persistence-template` (Appendix A), and `#der-directed-separation` (Part II Ch.1, the architectural classification the wrapping construction lifts to composite status). Cross-pattern reading leans on `#disc-identifiability-floor`'s meta-segment for the Instance 3 surfacing and `#disc-modularity-state-dynamics` for the truthification-mechanism reading of the wrapping construction.
- **Cross-reference vs canonical-home policy.** Two source segments in this chapter carry well-written `## Findings` sections — `#form-composition-closure` (closure-defect and bridge lemma) and `#der-class-coercion-via-wrapping` (constructive directed separation). Both are cross-referenced rather than restated. The findings whose canonical home is this implications segment are the *cross-segment compositions*: the critical-mass closed form as recovery of four signed special cases (composing `#deriv-critical-mass-composition` with `#der-team-persistence` and Part I Ch.4's persistence condition), the Brooks's-Law derivation as inequality (composing `#der-tempo-composition` with the template), the composition-level identifiability-floor Instance 3 (composing `#deriv-critical-mass-composition` with `#disc-identifiability-floor`'s pattern and Part II Ch.2's

loop-Level-2 access), and the wrapping-construction-as-truthification (composing `#der-class-coercion-via-wrapping` with `#disc-modularity-state-dynamics`'s three-operation pattern).

- **Class-coercion as truthification mechanism — connection to `#disc-modularity-state-dynamics`.** The M4 meta-segment positions modularity as a *contested property* under three operations: truthification (self-driven-increasing), strategic self-coupling (self-driven-decreasing), and adversarial coupling pressure (externally-driven-decreasing). Class coercion via wrapping is one of two operational mechanisms for truthification — alongside defensive scaffolding (peer review, prediction registers, etc., from `#disc-adversarial-coupling-pressure`). The chapter is therefore where modularity-as-state lands its first construction: wrapping is a structural commitment that *moves* a composite agent's modularity state from W_0 (no wrapping) toward W_1 (strict wrapping) under explicit cost, and the cost is the framework's quantitative content for the M4 truthification operation.
- **Identifiability-floor pattern coverage now four-of-four.** With this chapter's content, the pattern's four formal instances are all in view: Instance 1 (Part II Ch.4 — on-policy L0-insufficiency), Instance 2 (Part II Ch.3 — L1' mixture under unobservable common cause), Instance 3 (here — composite contraction certification from component marginals), Instance 4 (Part II Ch.5 — universal information-to-distance constant). Instance 5 (Mehra non-identifiability in adaptive Kalman) remains candidate. The implications-side payoff is the cross-instance dual structure: Instance 1 and Instance 2 cover agent-internal failures of identification (single agent); Instance 3 covers composition-level failure; Instance 4 covers the universal-constant-no-go that forces the (PI) commitment. The unique broadly-available escapes share Pearl-do structure (loop-interventional access) but differ in deployment mode (agent-self-intervention for Instance 1, instrument variable for Instance 2, observer-on-sub-agent for Instance 3, parameterization commitment for Instance 4).
- **No NeurIPS submission directly formalizes this chapter's machinery.** The closure-defect framework, the bridge lemma, and the critical-mass closed form are AAT-internal; their cross-references in Paper 2 (RL convergence) and Paper 3 (LLM hallucinate bound) are scope and structural-argument links rather than theorem-grade adoption. A composition-themed submission would naturally land here in a future cycle.

Chapter 12

Unity, Communication, and Shared Intent

[**Gap**] Discussion

The quality of a composite agent's composition — *conditional on 10.2 being satisfied via at least one of its four routes (three alignment routes + the strategic-equilibrium route (C-iv))* — is parametrized along **two architecturally distinct axes**:

- **Content axis (four unity dimensions).** What the sub-agents *share*: epistemic (U_M, shared model), teleological (U_O, shared objective), strategic (U_Σ, coordinated action), and perceptual (U_{obs}, shared observations).
- **Structural axis (update-rule homogeneity, U_f).** Whether sub-agents implement the *same* correction rule: how similar their f_M updates are across the population.

Together, the two axes parametrize the rate-distortion curves for the component closure defects (11.1, 12.1). Higher unity along either axis permits more aggressive compression at lower closure defect; neither axis alone is sufficient. In pure Section I composition (passive estimators, no G_t), agents with identical content can still produce non-zero ε_x if their update rules differ — the content axis cannot detect this, which is why the structural axis is required. Unity (in either sense) does not directly predict closure-defect magnitude; it controls the compressibility of the corresponding state, observation, or action component under projection.

Scope. The decomposition applies to composites that satisfy 10.2. U_O plays a role in the (C-i) route of the scope condition via value-correlation, but the scope condition is a disjunction of four routes — three alignment routes (shared objective, hierarchical derivation, mutual benefit) plus the strategic-equilibrium route (C-iv) — not a scalar threshold on U_O. Below scope-satisfaction (no route applies), the sub-agents are a multi-agent system per 10.1 and composition-level quantities are not well-defined.

For a composite agent A_c composed of sub-agents $\{A_1, \ldots, A_n\}$, the unity profile consists of *four content dimensions* (this section, below) and *one structural dimension* (U_f, defined after the content dimensions).

Definition (definition-unity-dimensions).

Epistemic unity U_M — how much of the reality model is shared:

$$U_M = \frac{I(M_t^{(1)}; \ldots ; M_t^{(n)})}{H(M_t^{(1)}, \ldots , M_t^{(n)})} \tag{12.1}$$

The fraction of total model information that is shared (multi-information / total-correlation ratio). $U_M = 1$ for identical models; $U_M = 0$ for independent models.

Teleological unity U_O — how aligned are the objectives:

$$U_O^{(i,j)} = \text{corr}\Big(V_{O_t^{(i)}}(\tau),\ V_{O_t^{(j)}}(\tau)\Big) \tag{12.2}$$

over trajectories the composite encounters. $+1$ for identical objectives; -1 for perfectly opposed; 0 for orthogonal. The composite teleological unity is an aggregation over all pairs. The scalar ranges from fully cooperative to fully adversarial per objective dimension.

[Scope note] U_O plays a role in 10.2 — primarily along route (C-i), where value-correlation is the operationalization of teleological alignment. The scope condition itself is disjunctive: it is satisfied when *any* of the alignment routes (C-i), (C-ii), (C-iii) applies, or when the strategic-equilibrium route (C-iv) applies; not by U_O alone crossing a common scalar threshold. The quality-metric role of U_O captured in this segment presumes scope-satisfaction via *some* route; on the (C-iv) side it tracks at best the alignment projection of a strategic composite whose macro-state is defined by equilibrium structure rather than by a shared target. When the sub-agents fail all four routes, they form a multi-agent system (10.1) but not a composite, and composition-level quantities (closure defect, composite tempo, team persistence) are not well-defined.

Strategic unity U_Σ — how coordinated is the joint policy:

Discussion.

$$U_\Sigma = 1 - \frac{D_{\text{KL}}(\pi^c_{\text{actual}} \| \pi^c_{\text{optimal}})}{D_{\text{KL}}(\pi^c_{\text{independent}} \| \pi^c_{\text{optimal}})} \tag{12.3}$$

where π^c_{optimal} is the jointly optimal policy. $U_\Sigma = 1$ when actual matches optimal; $U_\Sigma = 0$ when actual matches independent (no coordination). Requires knowing the jointly optimal policy, which is itself a strong assumption.

Perceptual unity U_{obs} — how much of the observation stream is shared:

The fraction of total observation information that reaches all sub-agents. Full perceptual unity means all agents observe the same signals; zero means private observations only. Enables epistemic convergence without explicit model-sharing.

Update-rule homogeneity U_f — how similar the sub-agent update rules are:

Definition.

$$U_f = 1 - d\Big(f_M^{(1)}, \dots, f_M^{(n)}\Big) \tag{12.4}$$

where d is a distance over the space of update operators $f_M : (M, o, a) \mapsto M'$, normalized so that $U_f = 1$ when all sub-agents implement the same correction rule and $U_f = 0$ at maximal heterogeneity. The choice of d is case-specific — for parametric Kalman-like updates, $d \propto |\Delta K|/K_{\max}$ on the gain parameter; for Bayesian updates with shared structural form but different priors, d tracks divergence between the induced kernels; for arbitrary f_M in function space, candidates include operator-norm distance, Fisher-information-weighted distance, or IB-style comparison.

Where the four content unities measure shared *information* across sub-agents (state, objective, policy, observation), U_f measures shared *structure* — whether the agents instantiate the same update law. The two axes are independent: agents can share a model ($U_M = 1$) while updating it differently ($U_f < 1$), and conversely. In purposeful settings (G_t present), U_Σ partially absorbs structural variation in the policy half of the cycle, but the model-update half remains uncovered without U_f. In pure Section I composition (passive estimators, no G_t), U_f is the only handle on structural homogeneity. The closed-form linear-Gaussian instance — heterogeneous Kalman gains $\Delta K = K_1^* - K_2^*$ producing $\varepsilon_x \propto |\Delta K|$ — is derived in 12.1 §Two-axis structure.

Discussion.

The achievable component closure defect $\varepsilon_d^{\min}(k_d)$ under a projection of macro-dimension k_d is a function of *both* axes — the relevant content unity U_d and the structural unity U_f — together with the projection-dimension parameter:

$$\varepsilon_d^{\min}(k_d) = f_d\big(k_d;\ U_d,\ U_f\big) \tag{12.5}$$

monotone decreasing in each unity argument and monotone increasing in compression aggressiveness (smaller k_d). The form is derived in 12.1; in linear-Gaussian scalar cases it admits closed-form expressions for $d \in \{x, o, a\}$. U_O and U_Σ enter ε_a jointly rather than separately.

Discussion-grade. Max attainable: empirical. The four content dimensions are qualitatively motivated by correspondence with the four components of agent state (M_t, O_t, Σ_t, and the observation channel); the structural dimension U_f is forced by the linear-Gaussian two-Kalman case (12.1 §Two-axis structure), where heterogeneous gains produce non-zero ε_x that no content dimension can register. The two-axis architecture (content × structure) is therefore a definitional commitment of this segment, not a heuristic.

The specific metrics are sketches. The information-theoretic formulations (U_M, U_Σ) are well-defined in principle but require specifying distributions and distance measures for practical computation. U_f is even less prescriptive — the choice of operator distance d is case-specific, and a general theory of structural-variation measures across arbitrary f_M classes is open.

The claim that the dimensions are *substantially independent* is a hypothesis, not derived. Epistemic unity may enable strategic unity (shared models allow coordination without explicit planning); content unity along U_M does not constrain U_f (agents can share a posterior while updating it differently). Independence holds approximately, with documented joint dependencies: (U_O, U_Σ) jointly control ε_a and cannot be separated; U_f enters all three closure components $(\varepsilon_x, \varepsilon_o, \varepsilon_a)$ as a structural multiplier on what content unity alone would predict.

Clausewitz's three gaps. These dimensions map to the gaps identified by Clausewitz (systematized by Bungay in *The Art of Action*):

Table 12.1

Clausewitz Gap	*Unity Dimension*	*Formal Quantity*
Knowledge gap	Epistemic unity (U_M)	$1 - U_M$: fraction of model not shared
Alignment gap	Teleological unity (U_O)	$1 - U_O$: objective misalignment
Effects gap	Strategic + Perceptual unity	$1 - U_\Sigma$ + observation routing costs

The mapping is not perfect — Clausewitz's "effects gap" blends action coordination with observation feedback — but it provides 200+ years of organizational evidence for the qualitative decomposition.

Connection to closure defect. The unity dimensions parametrize a rate-distortion relation with the component closure errors in 11.1, not a direct correspondence. The formal statement is in 12.1: the achievable closure-defect component $\varepsilon_d(k_d)$ under a projection of macro-dimension k_d decreases monotonically in both the relevant content unity U_d and the structural unity U_f, with closed-form expressions in the linear-Gaussian case. Qualitative direction along the content axis: U_M governs the compressibility of state information (ε_x), (U_O, U_Σ) jointly govern action compressibility (ε_a), U_{obs} governs observation compressibility (ε_o). The naive reading "high U_d predicts low ε_d" fails: for non-compressing projections (e.g., the means-only projection in the two-Kalman case) $\varepsilon_x = 0$ regardless of U_M, while for heterogeneous-gain projections $\varepsilon_x > 0$ even at perfect content correlation. Both observations point at the same correction — closure defect lives on a rate-distortion surface with two unity arguments, not a single one.

What each dimension's absence costs.

- Low U_M: prediction conflicts → uncoordinated actions based on contradictory beliefs. Internal mismatch component from model disagreement.
- Low U_O: strategic friction → sub-agents pursue conflicting sub-goals. Effort wasted or counterproductive.
- Low U_Σ: redundancy and gaps → two agents fix the same bug while a critical one goes unnoticed.
- Low U_{obs}: information silos → critical signals observed by one agent but not actionable by the composite.
- Low U_f: structural drift → even agents sharing identical models, objectives, observations, and policy targets produce divergent macro-trajectories under aggressive projection, because their corrections respond to the same evidence with different gains. The composite cannot be summarized at low macro-dimension without residual error scaling with the gain mismatch.

- The independence of unity dimensions needs careful examination. High epistemic unity likely enables (but does not guarantee) high strategic unity — if agents share models, they can coordinate implicitly. The dimensions may be better described as "substantially independent inputs to a joint prediction of ε^*" rather than "independent properties." Independence between content axis and structural axis ($U_d \perp U_f$) is cleaner — content sharing and update-rule similarity are categorically distinct properties — but a formal proof is open.
- The specific metric formulations need testing on concrete cases (software team, military unit) to determine if they discriminate meaningfully between well-composed and poorly-composed groups.
- The teleological unity scalar per objective dimension ($+1$ to -1) captures mixed cooperative-competitive situations: a company can be cooperative on product quality and competitive on internal resource allocation simultaneously.
- **U_f operator-distance choice is open.** The definition leaves d case-specific. A general theory of structural-variation measures across arbitrary f_M classes (operator-norm distance, Fisher-information-weighted distance, IB-style comparison) is unsettled. The linear-Gaussian Kalman case ($d \propto |\Delta K|/K_{\max}$) is the only worked closed form; non-Gaussian and non-linear cases are open follow-up work tracked in 12.1 Working Notes.
- **Joint $(U_O, U_\Sigma) \to \varepsilon_a$ dependence.** State error tracks U_M; action error tracks *both* U_O (target alignment) *and* U_Σ (policy alignment); observation error tracks U_{obs}. The two dimensions jointly controlling action error are physically distinct: U_O is about evaluation/preference agreement; U_Σ is about execution-path agreement. Agents with identical objectives but different execution plans have high U_O, low U_Σ; agents coordinating on arbitrary shared routines have high U_Σ, low U_O. See 12.1 for the quantitative relationship.
- **U_O as scope vs. quality.** The scope role is in 10.2 as a disjunction of four routes (three alignment + one strategic-equilibrium) — U_O is the value-correlation operationalization of route (C-i), not a universal scope variable. The quality-metric role remains here, presumed conditional on scope-satisfaction via at least one route. Open: whether the three alignment routes (C-i)–(C-iii) reduce to a single scalar is not established, and the strategic-equilibrium route (C-iv) is structurally distinct from any alignment-side aggregation, so "scope-satisfaction" and "U_O value" should be treated as distinct concerns in downstream uses.

12.1 *Result:* Unity-to-Closure Rate-Distortion Mapping

Unity dimensions parametrize rate-distortion curves for closure-defect components, not point-valued predictors. The achievable closure-defect component ε_d under projection of macro-dimension k_d is monotone decreasing in both the relevant content unity U_d and the structural unity U_f (update-rule homogeneity), with higher unity along either axis lowering achievable defect at a given compression. Closed forms hold in the linear-Gaussian case; structural monotonicity survives more broadly. The two-axis structure (content × structure) is forced by the heterogeneous-Kalman case below and is reflected definitionally in 12.

Formal Expression.

Rate-distortion framing (general).

Fix a composite agent satisfying the admissibility conditions (A1)-(A4) in 11.1. For each content unity dimension U_d (with $d \in \{M, \Sigma, \text{obs}\}$, and U_O contributing jointly with U_Σ — see below) and the structural unity U_f (update-rule homogeneity, defined in 12), the achievable component closure defect under a projection whose corresponding macro-dimension is k_d satisfies:

Formulation (unity-rate-distortion).

$$\varepsilon_d^{\min}(k_d) = f_d\big(k_d;\ U_d,\ U_f\big) \tag{12.6}$$

where f_d is monotone decreasing in both unity arguments, monotone increasing in aggressiveness of compression (smaller k_d). The mapping from unity to closure-defect *magnitude* is via the shape of this rate-distortion surface; unity does not directly predict closure-defect value. In the linear-Gaussian Kalman case the structural argument reduces to $1 - U_f \propto |\Delta K|/K_{\max}$ on the gain mismatch.

Linear-Gaussian closed forms (two-agent scalar case).

For two agents with scalar observations correlated at $\rho_{o,\text{eff}}$ (combining ρ_{env} and ρ_{obs}), under 1D principal-component projection of observations, the minimum achievable observation closure defect is:

Derived (obs-closure-linear-Gaussian, from unity-dimensions, composition-closure).

$$\varepsilon_o^2(k_o = 1) = \sigma_o^2 \cdot \frac{1 - \rho_{o,\text{eff}}}{2} \ \propto\ 1 - U_{\text{obs}} \tag{12.7}$$

Higher perceptual unity → observations are more redundant → 1D summary suffices → ε_o small. Exact in the linear-Gaussian scalar case.

For two agents with scalar quadratic objectives, independent LQR policies ($\rho_\Sigma = 0$), scalar targets r_1, r_2 with correlation ρ_O, under 1D state projection, the minimum achievable action closure defect is:

Derived (action-closure-independent-policies, from unity-dimensions, composition-closure).

$$\varepsilon_a^2 \propto \kappa^2 \cdot (1 - \rho_O) \ \propto\ \kappa^2 \cdot (1 - U_O) \tag{12.8}$$

where κ is the scalar LQR gain. Adding policy coordination ($\rho_\Sigma > 0$) further reduces ε_a through a multiplicative factor. The joint (U_O, U_Σ) dependence takes the form:

$$\varepsilon_a^2 \propto (1 - U_O) \cdot f_1(U_\Sigma) + g(U_\Sigma) \tag{12.9}$$

with f_1 decreasing in U_Σ and g capturing residual strategic-misalignment error even when targets coincide.

State closure in linear-Gaussian.

For linear-Gaussian micro-dynamics with consistent linear projections Λ_x and Λ_o (the macro observation projection is the same linear combination as the macro state projection), the state closure defect vanishes:

Derived (state-closure-linear-Gaussian-trivial, from composition-closure).

$$\varepsilon_x = 0 \tag{12.10}$$

regardless of U_M or compression dimension. Linear projections of linear dynamics are exact *when the range of* Λ_x *is invariant under the micro-dynamics matrix*. ε_x becomes non-trivial when:

- the projection's range is non-invariant under the dynamics matrix — even with linear dynamics, consistent projections, and homogeneous updates, cross-coordinate coupling or anisotropic noise scales that mix macro-subspace components with their orthogonal complement give $\varepsilon_x > 0$ (the Mori-Zwanzig zero-lag bound $\varepsilon^* \geq \|Q_\Lambda U P_\Lambda\|_{\mathrm{op}}$ in 11.1 is the general expression of this obstruction),
- the projection is inconsistent (macro state and macro observation projections disagree),
- the micro-dynamics are nonlinear, or
- sub-agent update rules are heterogeneous (see Two-axis structure below).

Two-axis structure (update heterogeneity).

In the non-degenerate linear-Gaussian case with heterogeneous sub-agent update rules — e.g., two Kalman filters with different gains $K_1^* \neq K_2^*$ tracking correlated processes, projected to the 1D sum $\hat{\omega}_+ = (\hat{\omega}_1 + \hat{\omega}_2)/\sqrt{2}$ — the state closure defect has the closed form:

Derived (two-axis-structure, from composition-closure, linear-Gaussian case).

$$\varepsilon_x^2 = (\Delta K/2)^2 \left[S_- - C_{+-}^2/S_+\right] \tag{12.11}$$

where $\Delta K = K_1^* - K_2^*$, $S_\pm$ are the innovation variances in the $\pm$ directions, and C_{+-} is their cross-covariance.

This exhibits two independent drivers of ε_x, one along each unity axis of 12:

1. **Content unity** (U_M, via process correlation ρ): higher correlation → lower ε_x.
2. **Structural unity** (U_f, via gain mismatch ΔK): when $\Delta K = 0$ (i.e., $U_f = 1$), $\varepsilon_x = 0$ at every ρ; when $\Delta K \neq 0$, $\varepsilon_x > 0$ even at perfect content correlation.

The four content unities measure shared information (goals, policies, observations, model state); U_f measures whether sub-agents implement the same correction rule. The two axes contribute to the closure-defect rate-distortion surface independently — content unity controls compressibility of what the agents agree on; structural unity controls whether projection induces memory by mixing the discarded subspace into the retained one.

Epistemic Status.

Conditional. Max attainable: *exact* (linear-Gaussian scalar cases) to *robust qualitative* (general).

- The observation and action closed forms are *exact* in the linear-Gaussian scalar case with stated projection choices.
- The state closure form $\varepsilon_x^2 = (\Delta K/2)^2[S_- - C_{+-}^2/S_+]$ is *exact* in the two-Kalman heterogeneous case.
- The rate-distortion framing (unity as compressibility parameter rather than direct predictor) is *robust qualitative* — it survives beyond linear-Gaussian, but concrete rate-distortion curves require case-by-case derivation.
- The joint $(U_O, U_\Sigma) \to \varepsilon_a$ formula is a *sketch* — the leading structure is derived; the precise forms of f_1 and g are mechanical extensions not fully computed here.

Ceiling-limiting factors: non-Gaussian cases require information-theoretic bounds (Gaussian IB is fully tractable; general IB is not), and the structural-unity axis U_f has a worked closed form only in the linear-Gaussian Kalman gain-mismatch case — a general theory of f_M structural variation across arbitrary update operators is open.

Discussion.

Why a one-axis reading fails. A "high U_M predicts low ε_x" reading is wrong in the two-Kalman case with the standard means-only projection: $\varepsilon_x \equiv 0$ for every correlation value, irrespective of U_M. The closure-defect surface depends on the projection choice, on the content-unity axis, and on the structural-unity axis U_f — high content unity with mismatched update rules still produces $\varepsilon_x > 0$, while low content unity under a non-compressing projection still produces $\varepsilon_x = 0$. The rate-distortion framing is what makes the multi-parameter dependence explicit.

Connection to the Information Bottleneck (2.2). The rate-distortion shape is not coincidental. Projection admissibility condition (P1) in 11.1 is the Lagrangian-dual of the IB constraint: the projection sits on or above the IB frontier at rate $I(X;T) \le I_{\max}(\varepsilon_I)$ for the relevance variable "next observation given action" (V supplies the derivation). Unity dimensions — measured as mutual-information-like quantities between sub-agent state components — parametrize the frontier's shape. The four AAT compression operations (M_t, Σ_t, shared intent, Λ) share IB shape but are not shown to reduce to a single master problem (U-medium, per V); cross-instance theorems do not follow from shared shape alone. (P2) Lipschitz continuity is not naturally IB and remains a separate admissibility condition; (P3) dimensional reduction remains separate in the Gaussian case. The Gaussian-IB closed form applies to linear-Gaussian composition setups; beyond them, the IB frontier is definitional but requires variational or numerical approximation.

Two-axis structure. The unity profile in 12 decomposes into a content axis (four dimensions: U_M, U_O, U_Σ, U_{obs}) measuring shared information, and a structural axis (U_f) measuring shared correction rules. In purposeful-agent settings (G_t present), U_Σ already absorbs structural variation in the policy half of the cycle — agents with different action laws have different effective policies — but the model-update half remains uncovered without U_f. In pure Section I composition (passive estimators, no G_t), U_f is the only handle on structural homogeneity, and the heterogeneous-Kalman case in this segment is the canonical instance where it bites.

Interpretation of "low closure defect." Unity controls the rate-distortion curve; low closure defect is achievable with aggressive compression when unity is high. But closure defect alone does not measure composite *optimality* (see 11.1 §5.1): two independent Kalman filters can have $\varepsilon^* = 0$ (perfectly representable) while failing to exploit cross-correlations (suboptimal relative to a joint filter). The rate-distortion mapping is about representability, not optimality.

Working Notes.

- **Extension to nonlinear cases.** The framing is linear-Gaussian because that's where rate-distortion has closed forms. Extension to nonlinear micro-dynamics would likely show $\varepsilon_x > 0$ even with consistent projections (the identity-propagation argument in Formal Expression relies on linearity). Worth a follow-up spike.

- **Structural-unity formalization.** A quantitative measure U_f across arbitrary f_M functions (beyond the linear-Gaussian gain-mismatch closed form) is open. Candidates for the underlying operator distance: operator-norm distance in function space, Fisher-information-weighted distance, or IB-style comparison. See 12 Working Notes.
- **Joint (U_O, U_Σ) derivation.** The exact f_1 and g functional forms require a full joint-LQR vs independent-LQR comparison. Mechanical but deferred.
- U_O → **sector-constant pathway (partial via F).** The LQR-compatibility sketch $\gamma(U_O) = -\gamma_{\max} U_O$ in F §5.2 (flagged discussion-grade) is a structural complement to this segment's rate-distortion framing: it channels U_O into the composite sector-constant κ_c through the signed coupling γ rather than through the closure defect ε. Upgrading (UO-mult) from discussion-grade to derived requires the action-space inner-product analysis natural to this segment: define the environment's action-coupling operator, show that LQR-linear policies produce cross-actions with inner product proportional to target correlation, and pin $\gamma_{\max}$ in terms of the quadratic objective's Hessian and the environment's coupling gain. Natural extension to the linear-Gaussian closed-form section above.
- **Mori-Zwanzig cross-check.** Under a stationary-measure setting, the Koopman-operator formulation of the projection-induced dynamics identifies the non-degenerate Kalman case as exercising the zero-lag memory kernel K_0 non-trivially, with $\|K_0\|$ scaling with $|\Delta K|$. This is consistent with the two-axis finding here — a Mori-Zwanzig lower bound on ε^* via the zero-lag kernel and the rate-distortion bound via IB should coincide at the same linear-algebraic quantity (the L^2 residual of projecting off an eigenspace of the micro-propagator). Formal equivalence not yet established. The MZ connection is developed further in 11.1 Epistemic Status.
- **Relationship to 10.2.** This segment describes quality *conditional* on composition existing — i.e., on 10.2 being satisfied via at least one of its three disjunctive routes (shared objective, hierarchical derivation, mutual benefit), *not* via a scalar U_O threshold. The rate-distortion curves parametrize quality given scope-satisfaction; they do not address whether a composite exists at all. For multi-agent systems where no scope route applies, closure-defect quality talk is a category error — there is no composite whose closure defect to measure.

12.2 *Definition:* Shared Intent

When sub-agents within a composite must coordinate, they face a communication problem: transmitting the full objective O_t and strategy Σ_t is expensive (high bandwidth, high latency), but acting without any shared purpose wastes coordination potential. The Information Bottleneck (2.2) applied to inter-agent communication predicts an optimal compression: transmit enough of G_t to align behavior, not more.

Formal Expression.

Let G_t^{full} be the source agent's complete purposeful state (O_t, Σ_t). Let G_t^{shared} be the compressed representation communicated to partners. The shared intent is the IB-optimal compression:

Definition (shared-intent).

$$G_t^{\text{shared}} = \arg\min_{G_s} \left[I(G_t^{\text{full}}; G_s) - \beta \cdot I(G_s; a_t^{\text{coordinated}}) \right] \tag{12.12}$$

where $a_t^{\text{coordinated}}$ is the jointly optimal action and β controls the complexity-relevance tradeoff. At high β, the agent communicates more detail (approaching full model sharing). At low β, communication is minimal (approaching independent action).

The shared intent is the *minimal sufficient statistic* of the sender's purposeful state for predicting the jointly optimal coordination behavior.

Epistemic Status.

Discussion-grade. Max attainable: conditional (conditional on the IB framework being appropriate for inter-agent communication). The application of IB to inter-agent communication is structurally motivated — IB compresses optimally given a relevance criterion, and coordination relevance is the natural criterion — but the specific formulation assumes: (1) the sender knows the jointly optimal action (which requires knowing other agents' states), (2) the compression is lossless in the IB sense (real communication introduces noise, delay, and misinterpretation), (3) the β parameter is fixed rather than dynamically adjusted. These are strong assumptions. The qualitative prediction (communicate purpose before plans before models) is more robust than the specific IB formulation.

Discussion.

What gets compressed out. The IB compression preferentially preserves: 1. Terminal objectives (what the agent is trying to achieve) — these are compact and change slowly 2. High-level strategy (which approach, not which specific steps) — moderate size, moderate change rate 3. Strategic details (specific edge credences in Σ_t) — large, change fast, low coordination value

Connection to cognitive cost of Σ_t. For agents with bounded communication capacity (bandwidth-limited channels, finite context windows), the DAG must be summarized for transmission. A 500-node strategy DAG cannot be shared in full; the IB compression identifies which structural features of the DAG matter for coordination.

Organizational communication patterns. Commander's intent in military doctrine is an empirical instantiation: the commander communicates *what* to achieve and *why*, not *how*. This is IB-optimal if objectives change slowly (low ν_O) and strategies change fast (high ν_Σ) — communicating objectives gives a long shelf life per bit transmitted.

Working Notes.

- The IB formulation assumes a single relevance variable ($a_t^{\text{coordinated}}$). In practice, coordination relevance is multi-dimensional: shared intent needs to support action coordination, conflict resolution, resource allocation, and adaptive replanning. A richer relevance variable might be needed.
- How does shared intent interact with 100% context turnover? An AI agent starting a new session needs to reconstruct G_t^{shared} from persistent storage. The compression from full G_t to shared intent is also useful for M_t preservation (`#disc-m-preservation`) — store the compressed version, not the full state.

12.3 *Hypothesis:* Auftragstaktik Principle

For a composite agent with limited communication bandwidth, the optimal allocation prioritizes sharing objectives over strategies over models. This captures the structural insight of Auftragstaktik (mission-type tactics): investing communication bandwidth in shared purpose (teleological unity) while accepting lower epistemic and strategic unity, granting sub-agents autonomy to adapt locally. The model predicts the same priority ordering that military doctrine discovered empirically; whether the mechanism (IB-optimal bandwidth allocation) is the actual reason Auftragstaktik works is an open question.

Formal Expression.

Let a composite agent's total inter-agent communication bandwidth be $B = B_O + B_\Sigma + B_M$, allocated across objective sharing (B_O), strategy coordination (B_Σ), and model synchronization (B_M).

Hypothesis (auftragstaktik-principle).

The hypothesis: the allocation that maximizes composite tempo $\mathcal{T}_c$ (or equivalently, minimizes coordination overhead C_{coord}) prioritizes:

$$B_O > B_\Sigma > B_M \tag{12.13}$$

when: - Objectives change slowly relative to strategies: $\nu_O \ll \nu_\Sigma$ - Strategies change slowly relative to models: $\nu_\Sigma \ll \nu_M$ - Sub-agents have sufficient local adaptive capacity: each $\mathcal{T}_i > \rho_i^{\text{local}} / \|\delta^i_{\text{critical}}\|$

The priority ordering follows from the IB framework (12.2): the bits with the longest shelf life and highest coordination value per bit should be transmitted first. Objectives change slowly and enable autonomous coordination (sub-agents who share objectives can independently choose compatible strategies). Models change fast and provide diminishing coordination value (two agents with the same model but different objectives still conflict).

Epistemic Status.

Discussion-grade. Max attainable: empirical. The priority ordering is a qualitative prediction grounded in the IB framework and supported by extensive military-organizational evidence (Bungay's analysis of Clausewitz, Wehrmacht doctrine, modern mission command). But it is not derived — the IB optimization would need to be solved explicitly with realistic cost functions to confirm the ordering, and the conditions under which the ordering reverses (e.g., when model synchronization is critical because the situation is genuinely ambiguous) are not characterized. The empirical evidence is strong but comes primarily from one domain (military command); generalization to software teams, AI agent swarms, and other settings is plausible but unverified.

Discussion.

When the ordering reverses. The prioritization $B_O > B_\Sigma > B_M$ assumes sub-agents can independently construct adequate local models. When the environment is genuinely ambiguous and local observations are insufficient (fog of war, novel codebase, unprecedented market conditions), model synchronization may be worth more than objective sharing — sub-agents who share the same wrong model at least err consistently, which is sometimes better than each having a different wrong model.

Bungay's evidence. In *The Art of Action*, Bungay documents that organizations consistently fail by inverting this priority: they over-invest in controlling *how* subordinates act (strategy sharing, B_Σ) rather than ensuring subordinates understand *why* (objective sharing, B_O). The result: subordinates who follow instructions precisely but cannot adapt when conditions change, because they lack the teleological context to improvise intelligently.

The software team instantiation. A well-functioning development team has: - High B_O: clear product goals, understood by all (sprint goals, feature objectives) - Moderate B_Σ: architectural decisions shared, implementation details autonomous - Low B_M: each developer builds their own mental model of the code they touch; full codebase understanding is neither expected nor efficient

When this inverts (micromanagement of implementation details, unclear product goals), team tempo drops — consistent with the Auftragstaktik prediction.

Connection to Conway's Law. Conway's Law (system structure mirrors communication structure) is a consequence: when B_Σ is low and B_O is high, sub-agents coordinate through shared objectives rather than explicit action coordination, producing systems whose boundaries reflect objective decomposition rather than communication channels.

Working Notes.

- The formal IB derivation of the priority ordering would need: (1) a model of how each unity dimension contributes to composite tempo, (2) the rate of change of each shared quantity (ν_O, ν_Σ, ν_M), (3) the communication cost per bit for each type. The qualitative argument is that objectives are compact and slow-changing (high bits-per-cost, long shelf life), while models are large and fast-changing (low bits-per-cost, short shelf life). Formalizing this is tractable but has not been done.
- The principle may need qualification for AI agent teams where model synchronization is cheap (shared vector databases, persistent memory) but objective alignment is hard (prompt engineering, RLHF). The cost structure differs from human organizations.

12.4 *Hypothesis:* Communication Gain

When an agent incorporates information from another agent (rather than from direct observation), the optimal update gain extends the uncertainty ratio with additional terms for source quality and teleological-unity uncertainty.

Formal Expression.

Hypothesis (communication-gain).

$$\eta_{ji}^* = \frac{U_{M_i}}{U_{M_i} + U_{o,ji} + U_{\text{src},j} + U_{\text{align},ji}} \tag{12.14}$$

where: - U_{M_i}: agent i's model uncertainty (same as 3.6) - $U_{o,ji}$: **communication channel noise** — latency, ambiguity, compression loss, bandwidth limitations of the channel between j and i - $U_{\text{src},j}$: **source quality uncertainty** — i's uncertainty about j's model calibration and domain competence - $U_{\text{align},ji}$: **teleological-unity uncertainty** — i's uncertainty about whether j's communications serve i's interests or j's potentially conflicting objectives

When all additional terms are zero (perfect channel, calibrated and aligned source): $\eta_{ji}^* \to 1$ (full trust). When any term is large: $\eta_{ji}^* \to 0$ (ignore the signal).

Connection to single-agent case. When j is the environment (direct observation): $U_{\text{src}} = U_{\text{align}} = 0$, recovering 3.6's standard form $\eta^* = U_M/(U_M + U_o)$.

Epistemic Status.

Hypothesis. The additive denominator treats all uncertainty sources as independent, zero-mean noise — a structural heuristic, not a strict variance derivation. This is appropriate for $U_{o,ji}$ (channel noise) and $U_{\text{src},j}$ (miscalibration), which are typically unstructured. For $U_{\text{align},ji}$ (deception), additivity is conservative: it correctly drives η_{ji}^* toward zero when teleological-unity uncertainty is high, but misses the adversary's *actual* strategy — presenting as trustworthy to exploit high η_{ji}^*. The additive model captures the *defender's* response to detected misalignment; it does not model the *attacker's* optimization over the defender's trust dynamics.

All four uncertainty terms must be expressed in a **common predictive-dispersion scale** before summation — the same units as U_{M_i} (variance of the predictive distribution over the observed quantity). When hard to estimate directly, a conservative approximation: set $U_{\text{src}} + U_{\text{align}}$ to the empirical variance of j's past prediction residuals as observed by i, minus the known channel noise $U_{o,ji}$.

Discussion.

The denominator terms have different natures. $U_{o,ji}$ is a property of the *channel* — improvable by infrastructure. $U_{\text{src},j}$ is a property of the *source* — improvable by j improving its model, or estimable by i through calibration tracking. $U_{\text{align},ji}$ is a property of the *relationship* — the game-theoretic variable.

Trust calibration as a meta-model. Agent i's estimates of $U_{\text{src},j}$ and $U_{\text{align},ji}$ constitute a **trust meta-model** — a model of models. This meta-model is itself subject to AAT's full apparatus: it has mismatch (trust prediction errors), should be updated with appropriate gain (not overreacting to single disagreements), and can be structurally inadequate (4.5 — the agent's trust model class may not capture the actual reliability structure of its sources).

Risk-asymmetric trust. The Bayesian posterior on source reliability gives the best *estimate*, but the *decision* about how much to trust should be risk-weighted. Trusting a deceptive agent (HILP — high impact, low probability) can cause catastrophic model corruption (13.2, effects spiral). Mild miscalibration toward a reliable source (LIHP) causes small ongoing inefficiency. For high-stakes interactions, use a conservative quantile of the trust posterior rather than the mean — require more evidence before granting high trust.

Trust transitivity. When agent i has no direct experience with agent k, but trusted intermediary j provides an assessment, the transitive trust question arises. A Bayesian mixture model discounts the recommendation by the intermediary's own reliability:

$$P_i(\theta_k \mid s_j) \propto \left[r_{ji} \cdot P(s_j \mid \theta_k) + (1 - r_{ji}) \cdot P_0(s_j)\right] \cdot P_i(\theta_k) \tag{12.15}$$

where r_{ji} is i's reliability estimate of j and θ_k is k's true alignment. When $r_{ji} \to 0$, the posterior collapses to the prior (no update); when $r_{ji} \to 1$, the full informative likelihood applies. This model gives a principled three-phase trust formation: prior → transitive update → direct experience (which eventually swamps the prior).

Working Notes.

- The communication gain enters the distributed tempo: $\mathcal{T}_i = \sum_k \nu_i^{(k)} \eta_i^{(k)*} + \sum_j \nu_{ji}^{\text{comm}} \eta_{ji}^*$. This is the formal basis for 13.1 — teams persist where individuals cannot because cooperative communication adds to each agent's effective tempo.
- Coordination overhead limits team size: adding members increases communication tempo with diminishing returns while coordination costs grow. The optimal team size occurs where marginal communication tempo equals marginal coordination cost. This connects to organizational theory (span of control, communication overhead).
- The adversary's strategy (making U_{align} *appear* low) creates a meta-game on trust estimation. This is where game theory enters — the trust calibration itself is strategic. AAT provides the state variables (mismatch, gain, tempo, reserve); game theory provides the equilibrium analysis.
- Open: when multiple intermediaries provide corroborating recommendations about k, correlation matters. If all got their information from the same source, corroboration is illusory.
- Consider eventually **splitting $U_{\text{src},j}$ from $U_{\text{align},ji}$** into separate treatment tracks, not just separate denominator terms. Source calibration uncertainty is an *estimation* problem (estimable, improvable, converges with data). Alignment uncertainty is a *strategic* problem (the adversary optimizes *over* the defender's trust policy, not independently of it). The additive heuristic correctly drives η_{ji}^* toward zero in both cases, but a richer model would separate the estimation problem (how good is this source?) from the trust-policy problem (how much should I trust, given that the source knows my trust policy?). The latter requires game-theoretic treatment — AAT provides the state variables; the equilibrium analysis is external.

12.5 *Discussion:* Additional Implications & Discussion

The chapter sits between Ch.2's composition machinery (closure defect, bridge lemma, wrapping construction) and Ch.4's cooperative-vs-adversarial coupling, and its job is to give *unity* a quantitative content rather than leaving it as a verbal property of composites that work. Unity decomposes into named axes that parametrize a rate-distortion curve for closure defect; shared intent decomposes into an IB-optimal compression of the sender's purposeful state; communication gain decomposes the Bayesian update for inter-agent channels into source-noise, source-competence, and source-alignment terms. None of the three segments carries a `## Findings` section, so the implications segment is their canonical catalog home — but the work is mostly synthesis: surfacing what the chapter delivers as a chapter, what's hypothesis-grade rather than derived, and what bridges into Ch.4 and the rest of Part III.

Unity dimensions and the rate-distortion content of "well-coordinated".

`#def-unity-dimensions` decomposes unity into a *content axis* (four dimensions — U_M, U_O, U_Σ, U_{obs} — measuring sub-agent agreement on models, objectives, strategies, and observations respectively) and a *structural axis* (U_f, update-rule homogeneity). The implications-side payoff comes from `#result-unity-closure-mapping`: unity dimensions are not point-valued predictors of composite quality; they parametrize *rate-distortion curves* for closure-defect components. The achievable closure-defect component ε_d under projection of macro-dimension k_d is monotone decreasing in both the relevant content unity U_d and the structural unity U_f — higher unity along either axis lowers achievable closure defect at a given compression. Closed forms hold in the linear-Gaussian case; structural monotonicity survives more broadly.

The two-axis structure is not aesthetic — it is *forced* by the heterogeneous-Kalman case. Two agents can have identical content unity (same models, same objectives, same observations) and still produce composite closure defect if their update *machinery* differs ($K_1 \neq K_2$ in the Kalman case, or $\eta_1^* \neq \eta_2^*$ in the gain-principle case). The structural-unity term picks this up; the content axes alone would miss it. The framework's prediction is concrete: composite agents with high content unity but low structural unity will show closure-defect signatures the content-only analysis cannot diagnose, and the remedy is *update-rule homogenization* (synchronizing how sub-agents update, not just what they believe). For multi-agent ML systems, this is the underlying mechanism behind the empirical pattern that an ensemble of identically-trained models behaves differently from an ensemble whose members were trained with different optimizers or schedules — same content, different structural unity, different composite behavior.

The composition with Ch.2's closure-defect framework is the implication that anchors Part III's coordination story. Unity's rate-distortion content is *what makes the bridge lemma actionable*: the lemma converts predictive closure defect into trajectory error, and unity tells the composite-design practitioner *which dimensions to invest in* to reduce ε^*. Investing in U_O (objective alignment) reduces ε_a multiplicatively; investing in U_Σ (strategy coordination) reduces it further through f_1; investing in U_f (update homogeneity) reduces all components simultaneously. The closed-form predictions are linear-Gaussian; the structural pattern (unity dimensions monotone in closure defect, two-axis structure load-bearing) survives more broadly as robust-qualitative.

Shared intent and the Auftragstaktik bandwidth allocation.

`#def-shared-intent` formalizes commander's-intent / mission-command doctrine as an Information Bottleneck compression: the optimal inter-agent communication transmits enough of the sender's purposeful state $G_t = (O_t, \Sigma_t)$ to align coordination behavior, not more. The minimal sufficient statistic for predicting jointly optimal coordination is what gets transmitted; everything below that bar is discarded as IB-suboptimal at a given complexity-relevance tradeoff parameter β.

The implications-side payoff is what `#hyp-auftragstaktik-principle` extracts from this: under named regime conditions (objectives change slowly relative to strategies which change slowly relative to models; sub-agents have sufficient local adaptive capacity), the IB-optimal bandwidth allocation prioritizes $B_O > B_\Sigma > B_M$ — invest in objective sharing, then strategy coordination, then model synchronization, in that order. The framework's prediction is the priority ordering

Auftragstaktik discovered empirically in 19th-century Prussian military doctrine (Moltke, Clausewitz) and that Bungay 2011 documents organizations consistently *fail* by inverting (micromanaging *how* subordinates act rather than ensuring they understand *why*). The mechanism the framework proposes — bits with longer shelf life and higher coordination value per bit transmitted first — is hypothesis-grade rather than derived, and the segment carries that honestly. What the IB framework predicts and what military doctrine empirically converged on agree at the *qualitative* level; whether the underlying mechanism is in fact IB-optimal bandwidth allocation is open.

What this buys for AAT's composition machinery is a quantitative reading of the cost in Ch.2's tempo-composition inequality. The closure defect $\varepsilon^* \nu_c$ that enters the composite's effective disturbance has a *bandwidth signature*: maintaining ε^* small requires the composite to spend inter-agent bandwidth on the dimensions that close the closure defect. The Auftragstaktik prediction is that this spending is *front-loaded* in the priority ordering — most of the closure-defect reduction comes from a small bandwidth investment in objective sharing, with strategy and model investment delivering diminishing returns. For software teams the picture is concrete: well-functioning teams have high B_O (clear product goals understood by all), moderate B_Σ (architectural decisions shared, implementation autonomous), low B_M (each developer builds their own mental model of the code they touch). When this inverts — micromanagement of implementation details, unclear product goals — team tempo drops, consistent with the framework's prediction.

The cross-domain transfer follows naturally and the segment surfaces it: Conway's Law (system structure mirrors communication structure) is a consequence rather than a separate observation. When B_Σ is low and B_O is high, sub-agents coordinate through shared objectives rather than explicit action coordination, producing systems whose boundaries reflect objective decomposition rather than communication channels. The framework's contribution is the structural argument — Conway's Law follows from IB-optimal bandwidth allocation under the regime conditions — rather than the empirical observation, which long predates AAT.

The honest scope statement worth surfacing: when the regime conditions fail, the priority ordering reverses. The prioritization assumes sub-agents can independently construct adequate local models; when the environment is genuinely ambiguous and local observations are insufficient (fog of war, novel codebase, unprecedented market conditions), model synchronization may be worth more than objective sharing — sub-agents who share the same wrong model at least err consistently, which is sometimes better than each having a different wrong model. The framework's prediction is that high-ambiguity regimes shift the IB-optimal allocation toward B_M, and the priority inversion is observable.

Trust as inverse effective noise — the communication-gain decomposition.

`#hyp-communication-gain` extends the single-agent uncertainty ratio from `#emp-update-gain` ($\eta^* = U_M/(U_M + U_o)$) to inter-agent channels by adding two new noise sources to the denominator: $U_{\text{src},j}$ (source-competence uncertainty, *i*'s uncertainty about *j*'s model calibration) and $U_{\text{align},ji}$ (teleological-unity uncertainty, *i*'s uncertainty about whether *j*'s communications serve *i*'s interests or *j*'s potentially conflicting objectives). The implication is that *trust formalizes as the inverse of effective noise on the inter-agent channel*: $\eta^*_{ji} \to 1$ when source-noise, source-competence, and source-alignment uncertainties are all small; $\eta^*_{ji} \to 0$ when any is large. The single-agent case is recovered when j is the environment: $U_{\text{src}} = U_{\text{align}} = 0$, and the standard η^* form returns.

The structural payoff is that *trust calibration is itself an AAT process*. Agent *i*'s estimates of $U_{\text{src},j}$ and $U_{\text{align},ji}$ constitute a *trust meta-model* — a model of models — which is subject to the framework's full apparatus: mismatch (trust prediction errors), gain (not overreacting to single disagreements), and structural inadequacy (`#result-structural-adaptation-necessity` applied at the meta-level — the agent's trust-model class may not capture the actual reliability structure of its sources). This is the segment's quietest claim and one of the framework's more reach-y consequences: relationships between agents are AAT processes operating on agent-pair-indexed quantities, with structure mirroring the agent-environment loop one level up. For AI-agent practitioners this prescribes a specific architectural pattern: trust assessments should be Bayesian-update-style with explicit calibration tracking, not threshold-based or static.

The decomposition also names a sharp distinction between the three uncertainty terms that downstream segments rely on. $U_{o,ji}$ is a property of the *channel* — improvable by infrastructure. $U_{\text{src},j}$ is a property of the *source* — improvable by

j improving its model, or estimable by i through calibration tracking. $U_{\text{align},ji}$ is a property of the *relationship* — the game-theoretic variable, and the one that becomes load-bearing in Ch.4's cooperative-vs-adversarial coupling and Ch.5's strategic composition. The signed-coupling structure of those chapters is what determines $U_{\text{align},ji}$: cooperative coupling ($\gamma < 0$, from `#deriv-critical-mass-composition`) corresponds to $U_{\text{align},ji} \to 0$ (aligned objectives, trust calibrates to high η^*_{ji}); adversarial coupling ($\gamma > 0$) corresponds to $U_{\text{align},ji}$ large (misaligned objectives, trust calibrates to low η^*_{ji}). The communication-gain decomposition here is what makes the signed-coupling structure of Ch.4 *operational* at the trust-meta-model level.

The risk-asymmetric trust discussion in the segment is worth surfacing as an implications-flavored point in its own right: the Bayesian posterior on source reliability gives the best *estimate*, but the *decision* about how much to trust should be risk-weighted. Trusting a deceptive agent (high impact, low probability) can cause catastrophic model corruption via `#der-adversarial-destabilization`'s effects spiral; mild miscalibration toward a reliable source (low impact, high probability) causes small ongoing inefficiency. For high-stakes interactions, the framework prescribes a conservative quantile of the trust posterior rather than the mean — require more evidence before granting high trust. This is the AAT-internal grounding for the empirical pattern that high-trust relationships build slowly and break quickly: the structural reason is that the decision-side risk-asymmetry favors slow accumulation and fast erosion.

Composition with the framework's other machinery.

The chapter's three findings compose with the rest of Part III in a way worth surfacing as a unifying point. Ch.2's closure defect ε^* acquires its rate-distortion content from this chapter's `#result-unity-closure-mapping`; Ch.2's coordination overhead $C_{\text{coord}} \geq \varepsilon^* \nu_c$ in `#der-tempo-composition`'s tempo composition acquires its bandwidth signature from this chapter's Auftragstaktik prediction. Ch.4's signed coupling γ acquires its trust-meta-model interpretation from this chapter's $U_{\text{align},ji}$; Ch.5's strategic composition under partially-opposing objectives acquires its $U_{\text{align},ji}$ dynamics from how `#hyp-communication-gain`'s trust calibration responds to observed coupling sign. The chapter is the *bandwidth and trust layer* between Ch.2's composition machinery and Ch.4/Ch.5's coupling dynamics — and the framework's posture is that all four chapters are operating on the same composite-agent persistence machinery, with each chapter supplying a different parametric content (machinery, bandwidth/trust, coupling sign, equilibrium).

Two cross-domain transfers worth noting before leaving the chapter. *Multi-agent AI safety*: the trust-calibration apparatus prescribes that agent-to-agent communication should be filtered through Bayesian-update-style assessment with explicit calibration tracking. Multi-agent systems that rely on flat trust models (one threshold, applied uniformly to all interlocutors) systematically miscalibrate the way `#hyp-communication-gain` predicts — they trust unreliable sources too much in low-stakes regimes and too little in high-stakes regimes, because the structural decomposition into channel / source / alignment noise is collapsed. *Human-AI teaming*: the same apparatus predicts which dimensions of teaming matter. Shared intent first (high-bandwidth objective alignment); strategy coordination next (architectural decisions shared, implementation autonomous); model synchronization last (the AI agent doesn't need the human's full mental model and vice versa). The framework's contribution against the empirical-design literature on human-AI teaming is that the priority ordering is structural, not contingent.

Working Notes.

- This segment is a chapter-end discussion grouping findings from `#def-unity-dimensions`, `#result-unity-closure-mapping`, `#def-shared-intent`, `#hyp-auftragstaktik-principle`, and `#hyp-communication-gain`. The cross-segment composition with Ch.2's machinery in §4 leans on `#form-composition-closure` and `#der-tempo-composition`; the forward bridges into Ch.4 lean on `#der-team-persistence`, `#der-adversarial-destabilization`, and `#deriv-strategic-composition`.
- **Cross-reference vs canonical-home policy.** None of this chapter's five source segments carry `## Findings` sections. This implications segment is the canonical catalog home for #22 (Shared Intent / Auftragstaktik via IB), #47 (Rate-Distortion Mapping of Team Unity), and #51 (Trust Formalization via Communication Gain), and treats each in full. The treatment is

honest about the hypothesis-grade nature of #22 and #51 — both are structurally motivated but not derived — and about the linear-Gaussian-specific closed forms for #47 with structural monotonicity surviving more broadly.

- **Cross-segment finding — bandwidth-and-trust as the layer between Ch.2 and Ch.4.** The unifying point in §4 is itself a cross-segment finding without a single segment home: the framework's composition story is operating on a single composite-persistence machinery across Ch.2 (machinery), Ch.3 (bandwidth and trust), Ch.4 (coupling sign), and Ch.5 (equilibrium). Each chapter supplies a different parametric content. This synthesis is what the implications-segment series at chapter-end can do that the individual segments cannot.
- **Hypothesis-grade content honest scoping.** Three of the chapter's findings sit at hypothesis-grade epistemic strength — #22 (Auftragstaktik priority ordering is qualitatively predicted but the IB-optimal-allocation mechanism is not derived in closed form), #51 (the additive denominator for communication gain treats uncertainty sources as independent, which is structural rather than derived), and parts of #47 (the rate-distortion mapping has closed-form predictions only in the linear-Gaussian case; structural monotonicity survives more broadly as robust-qualitative). The implications segment reflects this honestly rather than over-claiming theorem-grade status.
- **No identifiability-floor instance lands in this chapter.** The chapter's findings are about *machinery* (unity dimensions, shared intent compression, trust calibration) rather than about identifiability boundaries. The four formal identifiability-floor instances are now all surfaced in the implications series (Instance 1 in Ch.4-strategy-dynamics, Instance 2 in Ch.3-strategy-structure, Instance 3 in Ch.2-composition-machinery, Instance 4 in Ch.5-orient-cascade). Instance 5 (Mehra non-identifiability) remains candidate.
- **Connection to NeurIPS submissions.** The chapter's findings are AAT-internal; their cross-references in current submissions are scope and structural-argument links rather than theorem-grade adoption. The Auftragstaktik prediction could land in a future multi-agent-AI-safety submission if the regime conditions can be operationalized as testable conditions on a real multi-agent system.

Chapter 13

Cooperative and Adversarial Coupling

Once composites exist, the question becomes how their members interact. This chapter develops the dynamics — cooperative coupling that helps allies, adversarial coupling that hurts opponents, and the four-regime classification of how an arriving event lands on its recipient — under a single signed-coupling structure. Cooperation and adversarial dynamics are not two theories; they are the same machinery with opposite signs.

The framing comes from the disturbance decomposition. The disturbance rate an agent faces decomposes into environment, adversarial, and cooperative components,

$$\rho_i = \rho_{i,\mathrm{env}} + \sum_j \gamma_{j\to i}^{\mathrm{adv}} \mathcal{T}_j - \sum_j \gamma_{j\to i}^{\mathrm{coop}} \mathcal{T}_j \tag{13.1}$$

with $\gamma > 0$ in both cases, but opposite signs in the sum. Cooperative allies reduce agent i's effective disturbance by acting in the shared environment — stabilizing resources, preempting threats, absorbing perturbations. Adversaries amplify it by acting against. The machinery is identical; the sign is what differs. This is the unification that "adversarial dynamics as a separate theory" misses, and it is one of the clearer cases of AAT's framework character: not many results, one principle applied with a sign flip.

Two physically distinct cooperative mechanisms compose under this structure, and the chapter is careful about distinguishing them. *Communication tempo*: allies *tell* agent i things that improve its correction — the communication-tempo term that adds to $\mathcal{T}_i$ via 12.4. *Cooperative action*: allies *do* things that reduce the disturbance i faces — the negative term in ρ_i above. The two enter the persistence inequality at different points (one raising $\mathcal{T}$, the other lowering ρ) and warrant separate accounting. Counting a single ally-event in both would double-count its benefit. There is a useful concrete analogy: this is the mathematical difference between a consultant and an employee. A consultant boosts your tempo by telling you what's happening; if your α is too low to act on the information, the consultant doesn't save you — you'll just understand your failure more clearly. An employee lowers your effective disturbance by intercepting the volatility before it reaches your state. Communication and cooperative action are not interchangeable, and structurally failing agents need the second more than the first.

The adversarial side is where the famous results live. Under coupling-dominant deterministic disturbance — the canonical adversarial regime — the steady-state mismatch ratio between two agents scales as the *squared* tempo ratio:

$$\frac{\|\delta_B\|_{ss}}{\|\delta_A\|_{ss}} = \left(\frac{\mathcal{T}_A}{\mathcal{T}_B}\right)^2 \tag{13.2}$$

The faster agent both corrects its own mismatch faster *and* generates disturbance for the slower agent faster. Speed advantage is multiplicative, not additive: a 2:1 tempo ratio yields a 4:1 mismatch ratio; a 3:1 ratio yields 9:1; a 10:1

ratio yields 100:1. This is the formal analog of Boyd's "getting inside the opponent's OODA loop" — it turns military intuition into physics. The math itself is not mysterious; it is just the steady-state formula applied to both sides and ratioed. But the consequence is sharp. The squared scaling means that a sufficiently large tempo asymmetry in a coupling-dominant regime is not a quantitative disadvantage but a regime change: at 100:1 mismatch ratio, the slower agent is almost certainly past its operating reserve R, which means structural collapse rather than degraded performance. Speed advantage in a coupled regime *compounds*, and the consequence is qualitatively different from what additive-intuition predicts. Under stochastic adversarial coupling the exponent drops from 2 to 3/2; under non-coupling-dominant base disturbance it approaches 1. The exponent depends on the regime — the chapter names them explicitly — but the regime where it is sharpest is also where it matters most.

There is a caveat in the other direction: speed without coupling is useless. The destabilization threshold from 13.2 makes this explicit — if A's coupling to B is weak ($\gamma_A \to 0$), A can have infinite tempo and still fail to destabilize B. Being fast in a vacuum doesn't accomplish anything; you have to be fast *and* coupled. The two are jointly required and neither suffices.

The effects spiral — the corollary in 13.2 where B's degrading model causes B's actions to become erratic in ways that increase A's coupling effectiveness — is the mathematics of panic. When an agent is pushed past its adaptive reserve, its corrections become unreliable, which makes it more legible to the adversary, which increases the disturbance, which pushes the agent further past reserve. The spiral terminates only when the agent undergoes structural adaptation (a different model class, a different mode of operation) or ceases to function as an adaptive agent entirely. The structural implication for any agent under sustained adversarial pressure is that resilience is not just high α or high R — it is also having a mechanism to break the spiral before structural collapse, which usually means external intervention from outside the agent.

The other interesting move is the recipient-side decomposition. From the emitter's perspective, coupling is a single number — $\gamma_A \cdot \mathcal{T}_A$ — applied to the target's disturbance budget. From the recipient's perspective, that scalar collapses real structure. An arriving event from A lands on B as one of four qualitatively different things, determined by three independent boundary conditions in B's existing AAT quantities. It might be a *Regime I informative update* (small enough to be absorbed, large enough to register, within B's model class — the good case). It might be a *Regime II-a magnitude shock* (large enough to exit B's sector region — the destabilizing-fast-events case). It might be a *Regime II-b structural shock* (within sector but outside B's model class — the wrong-kind-of-event case, repairable only by structural adaptation rather than by faster correction). Or it might be a *Regime III ambient erosion* (below the observability floor — death by a thousand memos, where every event is small enough to be processed but the cognitive overhead of processing accumulates until the agent has no tempo left for real work; this is the formal version of why DDoS-of-low-priority-alerts is a real attack and not just inconvenience). Different regimes call for different repairs. Magnitude shocks call for more bandwidth; structural shocks call for a different model class; ambient erosion calls for better filtering, often at the infrastructure layer before signals reach the agent. Conflating them, which the scalar emitter view does, confuses diagnosis.

This is one of the chapter's load-bearing pedagogical moves. Most multi-agent frameworks treat "coupling" as a single coefficient. The recipient-side classification says: the coefficient is what the emitter cares about; the regime is what the recipient cares about; and they are not interchangeable for purposes of figuring out what to do.

The flow of the chapter: team persistence and the signed disturbance decomposition (13.1) → adversarial destabilization and the effects spiral (13.2) → the recipient-side four-regime classification (13.3) → the superlinear tempo advantage and its regime-dependent exponents (13.4). Chapter 5 extends to symmetric strategic composition (equilibrium-convergence framing for partially-opposing objectives), agent opacity as the formal dual of observation quality, and the qualifications that gate when the superlinear advantage actually obtains.

- This is a chapter-introduction segment; it bridges Chapter 3's quality/communication machinery to Chapter 4's dynamics under signed coupling. It carries no formal claim of its own.

- The "consultant vs employee" analogy comes from Gemini's first-encounter reaction to 13.1 — it captures the communication-vs-cooperative-action distinction at the level of intuition where the math lands. The intro adopts it because the original abstract phrasing didn't convey why the distinction is load-bearing.
- The "regime change rather than quantitative disadvantage" framing of the $b = 2$ scaling is the consequential corollary worth surfacing in the intro — most readers will hit the squared scaling and think "an order of magnitude" rather than "structural collapse." The intro pulls the corollary forward so it lands at the same time as the result.
- The "mathematics of panic" framing for the effects spiral is another Gemini observation worth keeping — it does in one phrase what a paragraph of formal commentary does less effectively.
- The "death by a thousand memos" framing for Regime III is concrete and load-bearing: the regime exists not because the events are individually significant but because processing them consumes tempo that should be going to actual work. This connects directly to logogenic-agent infrastructure design (PROPRIUM ASM, attention-management) where Regime III filtering at the infrastructure layer is operational.

13.1 *Derived:* Team Persistence

Teams persist where individuals cannot through two physically distinct cooperative mechanisms: communication (allies share observations that improve correction) and action (allies act in the shared environment to reduce disturbance at its source). These mechanisms enter the persistence condition at different points — tempo and disturbance respectively — and a given cooperative interaction contributes through one mechanism or the other, not both.

Formal Expression.

This segment instantiates the sector-persistence template (C) at the multi-agent level with state variable $\xi = \delta_i$ (sub-agent i's mismatch) and a decomposed effective disturbance $\rho_i^{\text{eff}} = \rho_{i,\text{env}} + \sum_j \gamma_{j\to i}^{\text{adv}} \mathcal{T}_j - \sum_j \gamma_{j\to i}^{\text{coop}} \mathcal{T}_j$ that accounts for adversarial and cooperative coupling. The template supplies the Lyapunov machinery; this segment's distinctive content is the disturbance decomposition and the corresponding tempo extension.

Distributed Tempo.

Agent i's effective tempo includes contributions from both direct observation and communication from allies:

Definition (distributed-tempo).

$$\mathcal{T}_i = \underbrace{\sum_k \nu_i^{(k)} \eta_i^{(k)*}}_{\text{direct observation tempo}} + \underbrace{\sum_{j \in \mathcal{N}(i)} \nu_{ji}^{\text{comm}}\, \eta_{ji}^{*}}_{\text{communication tempo}} \tag{13.3}$$

where ν_{ji}^{comm} is the rate of communication events from agent j to agent i, and η_{ji}^{*} is the communication gain (12.4). Faster team adaptation comes not only from faster individual sensing but from faster, more reliable knowledge transfer.

Channel independence caveat. Both sums are additive, inheriting the channel-independence assumption from 3.8: each channel and each communication source contributes non-redundant correction capacity. When allies report correlated information (overlapping observations, shared intelligence sources, redundant status reports), the communication tempo overcounts. The additive formula is an upper bound; the redundancy penalty depends on the mutual information between communication sources. See 3.8 for the single-agent version of this caveat.

Cooperative-Adversarial Disturbance Decomposition.

The disturbance rate experienced by agent i decomposes into environment, adversarial, and cooperative components:

Formulation (disturbance-decomposition).

$$\rho_i = \rho_{i,\text{env}} + \sum_{j \in \mathcal{A}_i} \gamma_{j\to i}^{\text{adv}} \mathcal{T}_j - \sum_{j \in \mathcal{C}_i} \gamma_{j\to i}^{\text{coop}} \mathcal{T}_j \tag{13.4}$$

where $\mathcal{A}_i$ is the set of agents adversarially coupled to i, $\mathcal{C}_i$ is the set cooperatively coupled, and the γ coefficients capture coupling effectiveness (as in 13.2).

The cooperative term is negative — but through action, not communication. Allies reduce agent i's effective disturbance by *acting in the shared environment* to prevent or mitigate disturbance at its source. Examples: an ally stabilizes a shared resource, neutralizes a threat before it reaches i, or absorbs environmental variation through their own actions. The mechanism is causal coupling through the shared environment, not information transfer.

Separation from communication tempo. The communication tempo term (above) captures allies *telling* agent i things that improve its correction. The cooperative disturbance term captures allies *doing* things that reduce the disturbance i faces. These are physically distinct: communication improves i's correction function (better α_i, higher $\mathcal{T}_i$); cooperative action reduces the disturbance ρ_i that i must correct against. A single cooperative event contributes through one channel or the other. An ally's message about a threat enters through communication tempo; an ally's action that eliminates the threat enters through disturbance reduction. Counting a single event in both terms would double-count the benefit and make the persistence threshold systematically optimistic.

Effective disturbance rate. The decomposition can yield $\rho_i < 0$ when cooperative coupling dominates both environment disturbance and adversarial coupling. The sector-condition analysis (4.3) assumes non-negative disturbance (GA-2). Define:

Definition (effective-disturbance).

$$\rho_i^{\text{eff}} = \max(\rho_i,\, 0) \tag{13.5}$$

When $\rho_i^{\text{eff}} = 0$, the agent's cooperative network fully absorbs all disturbance — the persistence condition is trivially satisfied and mismatch decays to zero. This is an idealized limit; in practice, $\rho_i^{\text{eff}} > 0$ because cooperative coupling is imperfect and environment disturbance is never fully preempted. All downstream uses of ρ_i in the persistence and reserve conditions should be read as ρ_i^{eff}.

Team Persistence Condition.

Applying the sector-condition framework (4.3) with ρ_i^{eff}, agent i persists iff:

Derived (team-persistence, from sector-condition-stability, persistence-condition).

$$\frac{\rho_i^{\text{eff}}}{\alpha_i} < R_i \tag{13.6}$$

Substituting the decomposition (the $\max(\cdot, 0)$ in ρ_i^{eff} is omitted to expose the three levers; the condition is trivially satisfied when the numerator is non-positive):

$$\frac{\rho_{i,\text{env}} + \sum_j \gamma_{j\to i}^{\text{adv}} \mathcal{T}_j - \sum_j \gamma_{j\to i}^{\text{coop}} \mathcal{T}_j}{\alpha_i} < R_i \tag{13.7}$$

This reveals three distinct levers for team persistence:

1. **Increase** α_i (individual correction efficiency) — better models, better gain calibration, including communication-improved tempo from allies (12.4)
2. **Increase cooperative disturbance reduction** ($\gamma^{\text{coop}}\mathcal{T}_j$) — more effective allied action in the shared environment: stabilizing shared resources, preempting threats, absorbing environmental variation. This is the action-based mechanism distinguished above, not the communication channel.
3. **Reduce adversarial coupling** ($\gamma^{\text{adv}}\mathcal{T}_j$) — better deception detection, reduced exposure to adversarial actions

Coordination Overhead Threshold.

Communication channels have costs: time to compose and parse messages, bandwidth limitations, synchronization requirements. These costs reduce the agent's effective tempo by diverting capacity from direct adaptation. Let $\Delta\mathcal{T}_i^{\text{cost}}(j)$ represent the tempo-equivalent coordination cost of maintaining the channel with j — the reduction in i's direct observation tempo caused by the overhead, in units of $[t^{-1}]$.

Discussion — Coordination Threshold.

The net benefit of adding agent j to i's communication network is positive only when:

$$\nu_{ji}^{\text{comm}}\,\eta_{ji}^{*} > \Delta\mathcal{T}_i^{\text{cost}}(j) \tag{13.8}$$

Both sides have units $[t^{-1}]$: the LHS is communication tempo gained, the RHS is direct-adaptation tempo lost to coordination overhead. This implies a natural team-size limit: adding members increases communication tempo with diminishing returns (as U_{src} and U_o accumulate across diverse sources) while coordination costs grow, potentially superlinearly. The optimal team size occurs where the marginal communication tempo equals the marginal coordination cost.

Epistemic Status.

Conditional on the communication-gain hypothesis (12.4). The distributed tempo definition is a *formulation* — a representational choice extending 3.8 to the multi-agent case. The disturbance decomposition is a *formulation* — the additive structure and the sign convention are modeling choices, not derivations. The persistence condition is *derived* from the sector-condition framework (4.3) given the decomposition: the derivation is exact under the same assumptions (GA-2, GA-3 applied to ρ_i^{eff}). The coordination overhead threshold is *discussion-grade* — qualitatively clear but the claim about diminishing returns and superlinear costs is asserted, not derived.

Max attainable: *robust-qualitative* for the persistence condition (it inherits the sector-condition's robustness but the decomposition is a modeling choice). The coordination threshold could reach *conditional* with a concrete cost model.

Discussion.

Compositional analog of 4.4. Like the single-agent persistence condition, this segment addresses *structural persistence* (see Persistence in `LEXICON.md`) — whether the composite correction machinery can outpace the effective disturbance rate. It does not address operational persistence (whether any particular sub-agent is near its boundary) or continuity persistence (whether the team maintains coherent identity through personnel changes). A team can be structurally persistent as a composite while individual members are operationally fragile or while the team's continuity is interrupted by turnover. The single-agent persistence condition says an agent persists when $\mathcal{T} > \rho/\|\delta_{\text{critical}}\|$. This segment extends that condition to agents embedded in a cooperative-adversarial network. The formal structure is identical — the sector-condition machinery applies unchanged — but the *inputs* ($\mathcal{T}_i$ and ρ_i) now include inter-agent terms. This is consistent with 1.7: the same dynamical laws apply at every level of description; what changes between levels is which channels contribute to tempo and which sources contribute to disturbance.

Why teams can persist where individuals cannot. Two distinct mechanisms combine. First, communication tempo raises $\mathcal{T}_i$ — allies provide observations that improve correction. Second, cooperative action lowers ρ_i — allies act in the environment to reduce disturbance at its source. An individual agent facing $\rho_{i,\text{env}} > \alpha_i R_i$ fails the persistence condition. Adding cooperative allies can either raise the numerator's denominator (tempo) or lower the numerator directly (disturbance), or both — through physically distinct mechanisms.

Timescale separation and 1.7. The distributed tempo definition presumes that communication events and direct observation events are comparable — they enter additively into $\mathcal{T}_i$. This requires that the communication timescale is not so slow relative to the environment dynamics that communicated information is stale on arrival. When communication latency approaches $1/\rho_{i,\text{env}}$, the effective η^*_{ji} degrades (the observation uncertainty $U_{o,ji}$ increases with staleness), naturally suppressing the communication tempo contribution.

Complement to 13.2. That segment characterizes when an adversary can push an agent past its stability boundary. This segment characterizes the cooperative counterpart: when allies can pull an agent back from instability. The γ coefficients have the same structure — coupling effectiveness — but opposite sign in the disturbance decomposition.

Composite-level complement: F. This segment gives the *per-sub-agent* persistence condition within a team. F supplies the *composite-level* analog: a closed-form critical-mass inequality in the matched-symmetric-Tier-1 two-agent case, with the same signed-γ coupling structure used here but applied to the joint Lyapunov on the concatenated mismatch state. The two are complementary: the team persists at the sub-agent level when each i satisfies this segment's condition; the team persists at the composite level when F's (CM4) holds. Cooperative coupling ($\gamma < 0$) reduces ρ_i^{eff} here and reduces $\rho + \gamma\mathcal{T}$ in (CM2) there — the same mechanism viewed at two scales.

Working Notes.

- The topology-dependent analysis (F.4 in the source material — peer networks, ensemble architectures, hierarchical structures) and game-theoretic integration (F.5) are related but separate concerns, not covered here. They may warrant their own segments.
- The coordination cost model $\Delta\mathcal{T}_i^{\text{cost}}(j)$ needs further specification to be useful. In software systems, coordination cost is empirically measurable (meeting time, code review latency, merge conflict rates). In military contexts, it maps to C2 overhead. The question is whether there is a useful *general* cost model or whether it is always domain-specific.
- The disturbance decomposition treats cooperative and adversarial coupling as additive and independent. In practice, the same agent j might be cooperatively coupled on some dimensions and adversarially coupled on others (e.g., a competitor who shares some information). The per-dimension persistence condition (4.4's per-dimension extension) may be relevant here.

13.2 *Derived:* Adversarial Destabilization

When two agents are coupled such that one's praxis contributes to the other's disturbance rate, the faster agent can generate aporia in the target faster than the target's epistrophe can resolve it — driving the target outside its invariant region and causing the correction mechanism to break down entirely.

Formal Expression.

This segment is the sector-persistence template (C) applied with coupling-amplified disturbance: $\rho_B = \rho_{B,\text{base}} + \gamma_A \mathcal{T}_A$ (Model D) or $\sigma_B = \sigma_{B,\text{base}} + \gamma_A \mathcal{T}_A$ (Model S). The destabilization threshold is the **negation** of the template's persistence condition for agent B: destabilization occurs precisely when the coupling-amplified disturbance violates $\alpha_B R_B > \rho_B$. Persistence and destabilization are the same inequality viewed in opposite directions. The superlinear adversarial scaling (13.4) follows from the template's $1/\alpha$ (Model D) versus $1/\sqrt{\alpha}$ (Model S) scaling, not from separate derivation.

Setup. Both agents satisfy the single-agent sector-persistence template (C) with parameters (α_A, R_A) and (α_B, R_B). Coupling amplifies B's effective disturbance rate by $\gamma_A \cdot \mathcal{T}_A$; destabilization is the negation of the template's persistence condition $\alpha_B R_B > \rho_B^{\text{eff}}$ for B. See 14.3 for regime taxonomy.

Derived (adversarial-destabilization, from sector-persistence-template).

Model D: deterministic drift coupling. *[Assumption (Coupling Model D)]* $\rho_B = \rho_{B,\text{base}} + \gamma_A \cdot \mathcal{T}_A$. The template's Model D conclusion $R_B^* = \rho_B/\alpha_B$ applied with the coupling model yields B's destabilization threshold $R_B^* > R_B$:

$$\boxed{\mathcal{T}_A \; > \; \frac{\alpha_B R_B - \rho_{B,\text{base}}}{\gamma_A}} \quad \text{(Model D)} \tag{13.9}$$

Denote $\Delta\rho_B^* = \alpha_B R_B - \rho_{B,\text{base}}$, B's adaptive reserve — the template's reserve quantity applied with the baseline disturbance. □

Model S: stochastic noise coupling. *[Assumption (Coupling Model S)]* $\sigma_B = \sigma_{B,\text{base}} + \gamma_A \cdot \mathcal{T}_A$ — the adversary's tempo increases unpredictability, not systematic direction. The template's Model S conclusion $R_B^* = \sigma_B\sqrt{n/(2\alpha_B)}$ (scalar $n = 1$) applied with the coupling yields the destabilization threshold:

$$\boxed{\mathcal{T}_A \; > \; \frac{R_B\sqrt{2\alpha_B} - \sigma_{B,\text{base}}}{\gamma_A}} \quad \text{(Model S)} \tag{13.10}$$

Scaling difference. The Model D threshold is linear in α_B; the Model S threshold is linear in $\sqrt{\alpha_B}$ — the same $1/\alpha$ versus $1/\sqrt{\alpha}$ split the template gives for the two disturbance models, propagated through the destabilization negation. This is the direct origin of the $b = 2$ versus $b = 3/2$ exponent distinction in 14.3, not a separate derivation. □

Unified view. Symmetrically, B destabilizes A when the analogous threshold on $\mathcal{T}_B$ is exceeded, using whichever model describes A's disturbance. The adversarial outcome depends on whether either agent can push the other past its stability limit.

Regime selection in practice. Model D fits situations where adversarial action produces persistent positional shifts (military maneuvering, API changes propagating through dependents, doctrinal initiative). Model S fits situations where adversarial action produces unpredictable perturbations around a stationary level (feints, randomized probing, market volatility). Mixed cases are handled by decomposing the disturbance into drift and noise components and applying both bounds additively.

Interpretation. "Getting inside the opponent's OODA loop" has a precise Lyapunov characterization: Agent A destabilizes Agent B when A's praxis, multiplied by coupling effectiveness, generates aporia in B faster than B's epistrophe can resolve it — specifically, when A's tempo times coupling exceeds B's adaptive reserve $\Delta\rho_B^*$. This captures:

- **Asymmetric coupling** ($\gamma_A \neq \gamma_B$): an agent with lower tempo but higher coupling effectiveness can still win.
- **Finite reserves**: an agent with very high $\mathcal{T}$ but operating near its model-class limit ($\Delta\rho^*$ small) is vulnerable despite high tempo.
- **Structural collapse**: when $R_B^* > R_B$, the failure mode is not merely "large mismatch" but "correction mechanism breakdown" — connecting to 4.5.

Corollary: The Effects Spiral. When Agent B is driven past its stability boundary ($R_B^* > R_B$), and B's degrading model causes B's actions to become erratic in a way that increases A's coupling effectiveness (γ_A increases with $\|\delta_B\|$), the result is a positive-feedback Lyapunov instability:

Discussion — Mechanism Schematic.

$$\|\delta_B\| \uparrow \Rightarrow \ B\text{'s actions become erratic} \ \Rightarrow \ \gamma_A \uparrow \Rightarrow \ \rho_B \uparrow \Rightarrow \ \|\delta_B\| \uparrow \tag{13.11}$$

With γ_A now an increasing function of $\|\delta_B\|$, the disturbance term in B's dynamics grows superlinearly. $\dot{V}_B > 0$ and increasing — mismatch accelerates away from the stability region. The spiral terminates only when B undergoes structural adaptation (4.5 — changing the model class) or ceases to function as an adaptive agent entirely.

Epistemic Status.

Both Model D and Model S destabilization thresholds are *exact* under their respective coupling assumptions (which treat $\mathcal{T}_A$ as exogenous). The Model D threshold follows from the deterministic sector-condition steady state $R^* = \rho/\alpha$ (Prop A.1); the Model S threshold follows from the stochastic sector-condition steady state $R_S^* = \sigma\sqrt{n/(2\alpha)}$ (Prop A.1S). Both coupling models (additive to ρ in Model D, additive to σ in Model S) are *assumptions* — they decouple the agents rather than modeling the fully coupled dynamical system where both agents' mismatch states co-evolve. The analysis therefore characterizes the *destabilization threshold* (the conditions under which A *can* push B past its stability boundary) rather than the full transient dynamics. This is a worst-case bound, treating A as operating at its steady-state tempo.

The effects spiral (corollary) is *discussion-grade* — the positive-feedback mechanism is qualitatively clear, but formalizing the $\gamma_A(\|\delta_B\|)$ functional form and proving instability under it requires specifying how an agent's degrading model affects its action quality, which the theory does not yet formalize.

A full coupled Lyapunov analysis with a joint function $V(\delta_A, \delta_B)$ would capture mutual feedback effects but requires specifying how each agent's mismatch state affects the other's disturbance in real time — an open extension.

Discussion.

Destabilization vs. steady-state ratio. The destabilization threshold is a failure of *structural persistence* (see Persistence in `LEXICON.md`) — the point where the correction machinery can no longer outpace the adversarially amplified disturbance. An adversary does not need to attack operational persistence (pushing the target near its boundary) or continuity persistence (disrupting identity) directly; destroying structural persistence is sufficient, because without it, operational persistence degrades to zero and the agent ceases to function regardless of its continuity stance. The linear analysis in 3.9 gives the steady-state mismatch ratio under coupling: a quantitative result about how much worse B does. This segment gives the qualitative result: under what conditions does B *fail entirely*, not merely fall behind. The linear analysis tells you the score; the Lyapunov analysis tells you when the game is over.

Connection to 13.4. The simulation results show the tempo advantage is superlinear (exponent ≈ 2 in pure adversarial regimes). This Lyapunov result explains WHY: the destabilization threshold creates a phase transition — below it, B persists (possibly with degraded performance); above it, B's correction mechanism collapses entirely, and the effects spiral accelerates the collapse.

Recipient-side refinement. This segment's $\gamma_A \mathcal{T}_A$ scalar increment compresses a richer structure. Per 13.3, events arriving at B fall into four regimes with three independent boundaries (sector-region / model-class / observability). This segment's destabilization story is the Regime II integration — specifically, the magnitude-shock sub-regime (II-a) where $\|e\|_B > R_B$. The structural-shock sub-regime (II-b), where the signal exceeds B's *model-class capacity* regardless of magnitude, produces destabilization via a different mechanism (per 4.5's trigger condition at truth) and admits a different repair (structural adaptation, not more tempo). Both sub-regimes manifest as "adaptive reserve exceeded" in the scalar view, but the distinction is load-bearing for diagnosis and repair. The 13.3 segment also surfaces an adversarial move this segment's formulation cannot express: *Regime-I-with-adversarial-content* — exploiting B's openness to informative updates by injecting misinformation with adversarially-chosen sign on the log-odds signal. See 13.3 Discussion for the full decomposition.

Agent opacity and coupling effectiveness. γ_A is stated as a parameter but its determinants are not decomposed. One key factor is how *legible* or *opaque* the target agent B is to the adversary A. We adopt from Hafez et al. (2026) the backward predictive uncertainty $H_b = H(S, A \mid S')$ as a measure of agent opacity — how many distinct (state, action) pairs produce indistinguishable environment transitions. High H_b means the agent is opaque; low H_b means it is legible. In adversarial settings, B's opacity directly affects A's ability to model B's correction function: low $H_b^{(B)}$ (transparent target) enables targeted disruption that maximizes γ_A, while high $H_b^{(B)}$ forces A to act against an uncertain model, reducing effective γ_A. Symmetrically, high $H_b^{(A)}$ (opaque adversary) degrades B's ability to anticipate A's disruptions — increasing the effective observation uncertainty U_o in B's model of A. The coupling effectiveness is thus modulated by opacity in both directions: $\gamma_A \propto 1/H_b^{(A)} \cdot 1/H_b^{(B)}$ is a qualitative relationship (precise functional form is open). H_b is the formal dual of observation quality U_o: where U_o characterizes how well the agent sees the world, H_b characterizes how well the world sees the agent. See `#der-agent-opacity` for the formal definition, the sign-flip derivation via signed coupling, the emitter-side four-regime classification, and the 16-cell emitter-recipient composition that operationalizes adversarial-edge-targeting.

Connection to extreme transition dynamics (Miller 2022). The effects spiral has a constructive counterpart: the same self-reinforcing coupling mechanism that drives destabilization here can drive *regime transitions* rather than collapse when the coupling is constructive rather than destructive. An environmentally neutral variant accumulates through drift, creating a niche that a mutant in the opposing population exploits — a positive-feedback cascade that rapidly transforms both populations. The Lyapunov coupling model applies to both signs: destructive coupling (this segment) increases ρ; constructive coupling (13.1) decreases it. The difference is sign, not structure. The endogenous emergence of coupling — where γ changes as population composition shifts — is the critical extension needed to formalize the full transition motif; it is flagged as a gap in Section III's dynamics (see 4.5 for the single-agent analog and the dynamics-level gaps enumerated in the OUTLINE).

Working Notes.

- The decoupled analysis (treating $\mathcal{T}_A$ as exogenous) is conservative — it's the best case for A. In a fully coupled system, A's actions against B may divert adaptive capacity from A's own mismatch correction, creating a self-limiting effect. The coupled analysis for symmetric adversarial composition is not a Lyapunov problem: it is a fixed-point / equilibrium problem on the joint best-response dynamics, and its formal home is `#deriv-strategic-composition`. The effects spiral in this segment's Corollary becomes a joint-Jacobian eigenvalue condition there. **Scope boundary:** `#result-contraction-template` (the contraction-metric generalization of `#result-sector-persistence-template`) covers the cooperative half of Section III composition. Adversarial / strategic composition lies structurally outside the contraction-metric framework (saddle-point equilibria are not attracting fixed points; Slotine compositional theorems do not apply); this segment's `#result-sector-persistence-template` instantiation with coupling-amplified disturbance is the correct tool for the asymmetric-adversarial regime.
- γ_A is the product of coupling strength, observability, and action impact — it captures the full spectrum from tightly coupled (direct disruption) to loosely coupled (indirect environmental effects). In the software domain, coupling is precisely measurable from the dependency graph (`#def-system-coupling`).
- The effects spiral is the formal analog of Boyd's cascading disorientation of the slower adversary — the same structural pattern (tempo advantage → destabilization → accelerating breakdown) appears in the Lyapunov analysis. The model captures the pattern; whether it captures the actual mechanisms of human disorientation is an empirical question, not a mathematical one. Future work should formalize the $\gamma_A(\|\delta_B\|)$ relationship to make the spiral a result rather than a discussion-grade observation.

13.3 *Derived:* Interaction-Channel Classification (Recipient-Side)

The same signal from agent A lands on recipient B as one of four qualitatively different things — informative update, magnitude-shock, structural-shock, or ambient noise — determined by three independent boundary conditions stated entirely in B's existing AAT quantities. The emitter-side collapse of this variation into a scalar $\gamma_A \mathcal{T}_A$ loses information that is load-bearing: the recipient's repair path depends on which regime the event falls into, and "more tempo" vs "different model class" address structurally different failure modes.

Formal Expression.

Setup and Notation. Two purposeful agents A and B coupled through a shared environment. A's praxis produces an event e_τ^A that enters B's observation channel. On B's side the event is processed by the standard AAT machinery: h_B maps the A-induced environment state to observation o_τ^B (1.3); mismatch is $\delta_\tau^B = o_\tau^B - \hat{o}_\tau^B$; update absorbs δ_τ^B with gain $\eta_B^* = U_{M,B}/(U_{M,B} + U_{o,B})$ (3.6).

Two event-level quantities enter the classification and must not be conflated:

- $\|e_\tau^A\|_B$ — the **magnitude** of the event in B's observation space (how large a perturbation it produces in δ_τ^B on arrival).
- $\mathcal{I}(e_\tau^A)$ — the **information content** of the event conditional on B's prior, formally $I(e_\tau^A; \Omega \mid M_{B,\tau^-})$ per NOTATION.md's event-information quantity. A large-magnitude already-predicted event has large $\|e\|$ but small $\mathcal{I}$; a tiny-magnitude structurally novel event has small $\|e\|$ but large $\mathcal{I}$.

Let $\mathcal{F}(\mathcal{M}_B)$ denote B's model-class fitness (2.4), and $\mathcal{I}_{\max}(\mathcal{M}_B)$ the maximum per-event information content representable within the class (see Working Notes for the cleaner sufficient-statistics-span formulation).

Classification Boundaries.

Event e_τ^A arriving at B falls into one of four regimes, determined by three independent boundary conditions:

Definition (regime-boundaries).

Regime I (Informative update) when all three hold:

$$\text{(I-a)} \quad \|e_\tau^A\|_B \leq R_B \qquad \text{(within sector-condition region)} \tag{13.12}$$

$$\text{(I-b)} \quad \mathcal{I}(e_\tau^A) \leq \mathcal{F}(\mathcal{M}_B) \cdot \mathcal{I}_{\max}(\mathcal{M}_B) \qquad \text{(representable within model class)} \tag{13.13}$$

$$\text{(I-c)} \quad \mathcal{I}(e_\tau^A) \cdot \nu^{(k)} \geq U_{o,B}^{(k)} \cdot c_{\text{floor}} \qquad \text{(above observability floor)} \tag{13.14}$$

where k is the arrival channel, $\nu^{(k)}$ its event rate, and c_{floor} a detection-theory constant controlling the false-alarm tolerance.

Regime II-a (Magnitude-shock destabilization) when (I-a) fails:

$$\|e_\tau^A\|_B > R_B \tag{13.15}$$

The event exits B's sector-condition region on arrival. B's correction function does not point inward strongly enough to discharge the mismatch before the next event; under sustained rate $\nu > rsim\alpha_B$, destabilization proceeds per 13.2.

Regime II-b (Structural-shock destabilization) when (I-a) holds but (I-b) fails:

$$\|e_\tau^A\|_B \le R_B, \qquad \mathcal{I}(e_\tau^A) > \mathcal{F}(\mathcal{M}_B) \cdot \mathcal{I}_{\max}(\mathcal{M}_B) \tag{13.16}$$

The event's information content exceeds what B's model class can represent. By 4.5, parametric update within $\mathcal{M}_B$ cannot close the mismatch; residuals retain systematic structure. Repair requires structural adaptation (a different model class), not more bandwidth.

Regime III (Ambient noise / slow erosion) when (I-a) and (I-b) hold but (I-c) fails:

$$\|e_\tau^A\|_B \le R_B, \qquad \mathcal{I}(e_\tau^A) \le \mathcal{F}(\mathcal{M}_B) \cdot \mathcal{I}_{\max}(\mathcal{M}_B), \qquad \mathcal{I}(e_\tau^A) \cdot \nu^{(k)} < U_{o,B}^{(k)} \cdot c_{\text{floor}} \tag{13.17}$$

The event is representable and within capacity but its information content sits below the observability floor. It contributes to δ_B's variance (enters Model S as part of $\sigma_{w,B}^2$) without triggering a usable update; B's adaptive reserve $\Delta\rho_B^*$ slowly drains.

Three Independent Boundaries. The three boundary conditions are structurally independent, each stated in quantities AAT already carries:

Table 13.1

Boundary	*AAT quantities*	*Failure mode*
(I-a) / (II-a): sector-region	$\|e\|_B$, R_B (from 2.4 / C)	*magnitude* — more capacity cures
(I-b) / (II-b): model-class	$\mathcal{I}(e), \mathcal{F}(\mathcal{M}_B), \mathcal{I}_{\max}(\mathcal{M}_B)$	*class* — structural adaptation cures
(I-c) / (III): observability	$\mathcal{I}(e), \nu_B^{(k)}, U_{o,B}^{(k)}$ (from 14.4)	*rate* — lower observation noise or higher event rate cures

No new ad-hoc thresholds are introduced. $\mathcal{I}_{\max}(\mathcal{M}_B)$ is the only new symbol; see Working Notes for its cleaner sufficient-statistics-span formulation.

Regime-Typed Disturbance Decomposition.

Under a stream of events $\{e_\tau^A\}$, B's **regime-typed effective disturbance** rate decomposes into regime-typed contributions:

Derived (regime-typed-rho-eff, from regime-boundaries + sector-persistence-template).

$$\rho_B^{\text{eff}} = \underbrace{\sum_{e\in\text{II-a}} \|e\|_B \cdot \nu_e}_{\text{magnitude disturbance}} + \underbrace{\text{floor}(\mathcal{M}_B) \cdot \sum_{e\in\text{II-b}} \nu_e}_{\text{structural mismatch floor}} + \underbrace{\sum_{e\in\text{III}} \sigma_e^2 \cdot \nu_e}_{\text{ambient variance}} - \underbrace{\sum_{e\in\text{I}} \iota_B(e)\, \mathcal{I}(e) \cdot \nu_e}_{\text{informative correction}} \tag{13.18}$$

The Regime-I term is **negative**: informative events reduce B's effective disturbance rate, not increase it. This generalizes 13.1's cooperative-action term $-\gamma^{\text{coop}}\mathcal{J}_j$: a cooperative event is precisely a Regime-I event from an aligned emitter, and the sign flip in the emitter-side decomposition falls out of the regime assignment on the recipient side. Adversarial events land in Regimes II-a/II-b; ambient-noise events in Regime III.

The emitter-side formulation $\gamma_A \mathcal{J}_A \to \rho_B^{\text{eff}}$ compresses the regime-typed sum into a single scalar, losing (i) the sign of cooperative coupling, (ii) the magnitude vs structural distinction in destabilization, and (iii) the observability-floor loss to Regime III.

Structured Derivation — Kalman-over-Kalman.

For a concrete check, take B as a Kalman filter on a scalar linear-Gaussian state with model class $\mathcal{M}_B = \{\theta \in [\theta_{\min}, \theta_{\max}]\}$, process noise q, observation noise r, sector parameter $\alpha_B = \eta_B^* = P_{\text{pred}}/(P_{\text{pred}} + r)$. B's sector-region radius is $R_B = \sqrt{q/(1-\theta_{\max}^2)}$ (stationary standard deviation at the class edge).

Derivation (Kalman-over-Kalman, from regime-boundaries + update-gain).

A's emitted perturbation ξ_A enters B's innovation as $\delta_\tau^B = \xi_A + \varepsilon_\tau + (\omega_\tau - \hat{\omega}_\tau)$. Four canonical distributions for ξ_A partition the classification:

Case 1 — Small-variance Gaussian $\xi_A \sim \mathcal{N}(0, s^2)$, $s^2 \ll r$ (expected Regime III). $\mathcal{I}(\xi_A) \approx s^2/(2r \ln 2)$ nats — small. (I-a) holds ($s \ll R_B$); (I-b) holds (Gaussian-within-Gaussian); (I-c) fails because $\mathcal{I}(\xi_A) < U_{o,B} \cdot c_{\text{floor}}$. Result: Regime III. Derived consequence — the contribution to ρ_B^{eff} is through $\sigma_{w,B}^2$; adaptive reserve drains by $\sum_e \eta_B^{*2} s_e^2 \cdot \nu_e$.

Case 2 — Moderate Gaussian $\xi_A \sim \mathcal{N}(\mu, s^2)$, $\mu \ll R_B$, $s^2 \sim r$ (expected Regime I). $\mathcal{I}(\xi_A) = \frac{1}{2}\log(1 + (s^2+\mu^2)/r)$ — substantial. All three (I-a)/(I-b)/(I-c) hold. Result: Regime I. Standard Kalman update; M_B refines; Regime-I term contributes negatively to ρ_B^{eff}.

Case 3 — Binary kick $\xi_A \in \{\pm\Delta\}$ with $\Delta > R_B$ (expected Regime II-a). (I-a) fails by construction. The Kalman update $\hat{x}^+ = \hat{x}^- + \eta^*\Delta$ undershoots by $(1-\eta^*)\Delta$ per event. If events arrive at rate $\nu > rsim\alpha_B$, lag accumulates and $\alpha_B R_B > \rho_B^{\text{eff}}$ is violated. Result: Regime II-a — destabilization per 13.2. Notice: the signal is *within* the model class (Gaussian handles $\pm\Delta$ mathematically), but correction cannot discharge it fast enough.

Case 4 — Heavy-tailed ξ_A with $\mathbb{E}[\xi_A^2] \sim r$ but kurtosis $\kappa \to \infty$ (expected Regime II-b). Mean contribution is fine; the problem is the distribution shape. The Kalman filter — Gaussian-optimal — mis-gains: too aggressive for small events, too conservative for genuine large ones. The per-event KL gap $D_{\text{KL}}(P_{\text{true}} \| P_{\mathcal{M}_B}) > 0$ for any heavy-tailed P_{true} against Gaussian. By 2.4, $\mathcal{F}(\mathcal{M}_B) < 1 - \varepsilon$ with ε lower-bounded by the KL gap; by 4.5, no parametric update within $\mathcal{M}_B$ closes the mismatch. Result: Regime II-b — residuals retain non-Gaussian structure (visible in kurtosis tests); repair requires expanding the model class (e.g., Student-t observation model), not more Kalman tuning.

Each case lands where the classification predicts. The derivation transfers to any recipient architecture in which the underlying AAT quantities are well-defined — this is the scope inherited from C + 2.4 + 3.8.

Recovery of Emitter-Side Results.

Each existing emitter-side result is a restriction of the four-regime decomposition:

Derived (emitter-side-recovery).

- **13.2** is Regime II-a integrated over a tempo-proportional event stream. The magnitude-shock sub-regime corresponds directly; the structural-shock II-b subcase is implicit in that segment, collapsed into "adaptive reserve exceeded" but here made explicit.
- **13.4** — superlinear tempo scaling follows from the sector-persistence template's $1/\alpha$ (Model D) vs $1/\sqrt{\alpha}$ (Model S) applied to Regime II events. The b-exponent drops toward zero in the high-$U_{o,B}$ limit because the fraction of A's events landing in Regime II drops (more fall into Regime III).
- **14.4** is the recipient-side expression of boundary (I-c): high $U_{o,B}$ pushes events into Regime III where they add to variance without contributing to destabilization.
- **10.3** corresponds to asymmetric classification: host's signals to endosymbiont contain high-$\mathcal{I}$ structure the endosymbiont's class cannot initially represent (Regime II-b for the endosymbiont, forcing structural adaptation toward the host's class); endosymbiont's signals to the host land in Regime I (host absorbs endosymbiont's accumulated structure). Consolidation is the fixed-point where both streams are Regime I.

- **Cooperative signaling** (in 13.1) — Regime-I events from aligned emitters contribute negatively to ρ_B^{eff} via the cooperative-action term. The communication-tempo $\nu_{ji}^{\text{comm}} \cdot \eta_{ji}^{*}$ is the rate of Regime-I events times the recipient's informative gain.

What Is Derived vs. What Is Chosen.

Table 13.2

Property	*Source*	*Strength*
Three independent boundaries (sector-region / model-class / observability)	Import from 2.4 + C + 14.4	Formulation choice (three-way partition; coarser / finer alternatives possible)
Four-regime partition (I / II-a / II-b / III)	The three boundaries yield four boundary-state combinations; only four are non-degenerate	Derived from the boundary structure
(I-a) / (II-a) boundary at R_B	C's sector-region radius	Derived
(I-b) / (II-b) boundary at $\mathcal{F}(\mathcal{M}_B) \cdot \mathcal{I}_{\max}$	2.4 + class-capacity normalization	Formulation (the $\mathcal{I}_{\max}$ normalization is heuristic; sufficient-statistics-span form is cleaner — see Working Notes)
(I-c) / (III) boundary at $U_{o,B} \cdot c_{\text{floor}}$	14.4 + detection-theory threshold	Formulation (c_{floor} is a detection-power parameter)
Regime-typed ρ_B^{eff} decomposition with negative Regime-I term	Aggregation over event stream using regime classification	Derived (the sign of the Regime-I term is structural, not a choice)
Kalman-over-Kalman four-case derivation	Direct application of Kalman gain + sector + KL-gap + SNR analyses	Proved (for the stated case)
Recovery of 13.2, 10.3, 14.4, 13.1 as restrictions	Each emitter-side result is exhibited as a per-regime special case	Derived
Non-Gaussian Case 4 derivation (heavy-tailed → II-b via KL gap)	Informal argument grounded in robust-filtering literature (Huber, Masreliez)	Discussion-grade (rigorous version requires per-family KL computation)
Class 2 (Partial) approximation with $\kappa_{\text{processing}}$ degradation	Transfer from Class 1 (Separated) with goal-blind-update failure	Formulation (qualitative); exact form requires spelling out the goal-contamination coupling

Epistemic Status.

Conditional. Max attainable: *derived* for the classification structure in Class 1 (Separated) architectures; *exact* for the Kalman-over-Kalman worked case in sub-scope α; *robust qualitative* for general sub-scope β recipients.

The three boundaries are structurally independent and each is stated in an existing AAT quantity, so the four-regime partition is not ad-hoc; it is forced by the structure of the quantities already in the theory. Within Class 1 (Separated; goal-blind epistemic update per 5.3) and sub-scope α recipients (Kalman / conjugate-Bayesian / exponential-family / strongly-convex-gradient / linear-PD), the Kalman-over-Kalman derivation transfers with A2' derived per 4.2, and each case yields the predicted regime by direct substitution. In sub-scope β (PID / rule-based / human-judgment), the classification's form transfers but boundary (I-a) requires per-instantiation verification of the sector bound, inheriting from C's sub-scope β caveat.

Scope limits. Class 3 (Coupled) recipients are out of formal scope: the coupled epistemic-purposeful update violates 5.3, so the regime assignment cannot be cleanly computed against M_B alone. This is `03-llm-core/` territory. Class 2 (Partial) recipients inherit with degradation proportional to $\kappa_{\text{processing}}$ (see `spikes/spike-kappa-hb-operationalization.md`): the Regime-I update may be goal-contaminated, and Regime II-b may be misdiagnosed as II-a or III under goal-blind-update failure. Per-event classification is singular-trajectory-indexed by 4.7; aggregation over event streams is deferred to C.

What the classification does not claim. (i) It does not determine *semantic content*: knowing an event is Regime I tells you B will update, not *what* B will believe afterward — that lives in 8.6's default signal function. (ii) It does not make the four regimes sharp in practice: real recipients' sector regions and class capacities have estimation uncertainty, so borderline events oscillate across boundaries under parameter noise. (iii) It does not replace C's temporal aggregation; it provides the per-event regime label that feeds into the aggregate disturbance bound.

Discussion.

Complement, not replacement, of the emitter-side segments. 13.2, 13.4, 14.4, 10.3, and 13.1 each describe what happens when A's praxis couples into B. They collapse the variation into a scalar increment on B's effective disturbance rate. The recipient-side classification decomposes that increment into regime-typed contributions with (importantly) a **signed** Regime-I component. The five emitter-side segments are recovered as restrictions — the classification provides the pattern-recognizer to match the emitter-side optimizer.

Why II-a vs II-b matters. Both destabilization regimes manifest as "adaptive reserve exceeded" in the emitter-side view. But the repairs are different: magnitude-shock (II-a) calls for more bandwidth — larger R_B, higher α_B, faster tempo; structural-shock (II-b) calls for a different model class — an expansion, grafting, or compression in the sense of 4.5. An organization running rapid-response incident drills against a bureaucratic-process adversary faces II-a and responds with more capacity; the same organization hit by a technology-regime change faces II-b and responds with restructuring. The two produce similar pain signals but admit opposite cures. Collapsing them confuses diagnosis.

Pairing with `#adversarial-edge-targeting` (Section III GAP). The classification provides the recipient-side pattern-recognizer; the emitter-side question — which of B's strategy edges are most valuable for A to attack — provides the adversary-optimization complement. The two compose: the emitter chooses which edge to target; the classification determines what happens at that edge. A particularly sharp move the classification makes visible but the emitter-side formulation hides: the **Regime-I-with-adversarial-content attack** — an emitter who can control the content y_G fed into B's default log-odds signal function (per 8.6) can choose the sign of $(y_G - \hat{P}_\Sigma)$ to push λ_k in the direction that degrades Σ_B most. This is not a destabilizing attack (which lands in Regime II); it is an *informational* attack that exploits B's openness to Regime-I updates to inject misinformation. The emitter-side formulation, which represents coupling as a scalar γ_A, cannot express the sign lever; the recipient-side decomposition does. The value is maximized at edges with large plan-sensitivity J_k and moderate credence p_k — exactly the edges load-bearing for Σ_B.

Agent opacity and the emitter-side regime distribution. Hafez et al. (2026)'s agent opacity $H_b^B = H(S, A \mid S')$ is primarily *emitter-side*: it gates A's ability to model which of B's regimes a given signal will land in. As $H_b^B \to 0$ the emitter can target specific regimes adversarially; as $H_b^B \to \infty$ the emitter's signals distribute across regimes and the targeting advantage collapses. Formally, $\gamma_A^{\text{effective}} = \gamma_A^{\max} \cdot f(H_b^B)$ with f monotonically decreasing. This is the dual of 14.4's story: where observation-gates-advantage says the emitter's advantage degrades when its observations of the *environment* are noisy, opacity says the advantage degrades when its observations of B are noisy. Cooperative signaling is the mirror: low H_b^B (legibility) enables aligned emitters to target signals into B's Regime I. This is 12.3's recipient-side translation: shared objectives give A a better model of B's prior and model class, which lets A produce signals that land in B's informative regime.

Organizational reception intuition. The four regimes correspond to a familiar organizational diagnostic: (Regime I) *heard* — the organization updated on the relevant edges of its Σ_{org}; (II-a) *shattered by shock* — signal was too large/fast, the organization's correction bandwidth was exceeded; (II-b) *couldn't hear* — signal required reorganization beyond the organization's structural capacity, matching Cohen & Levinthal's (1990) absorptive-capacity framing; (III) *absorbed as fatigue* — signal drained $\Delta\rho^*$ without producing structural change, the "death by a thousand memos" pattern. The cross-domain instantiation is cleanest where $\mathcal{F}(\mathcal{M}_B)$ maps to absorptive capacity and $U_{o,B}$ to communication-channel quality.

Future meta-segment candidate. The recipient-side classification is the *signal-reception* complement to C's *persistence-under-disturbance* structure. If a future pattern emerges elsewhere (e.g., consolidation dynamics classifying inputs, or a more general theory of inter-agent coupling) a meta-segment `#signal-reception-pattern` could name the recipient-classification shape. Premature at this stage; log as speculation.

Working Notes.

- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. Scope-limits paragraph updated (Class 2 fully-merged → Class 3 Coupled; Class 3 partially-modular → Class 2 Partial). What-Is-Derived table updated (Class 3 approximation → Class 2 Partial approximation). The four-regime classification (I / II-a / II-b / III) is recipient-side signal-processing vocabulary — unaffected by the GUC rename. Removed at `candidate` stage per FORMAT.md Gate 4.
- **$\mathcal{I}_{\max}(\mathcal{M}_B)$ — replace with sufficient-statistics-span.** The boundary (I-b) uses $\mathcal{F}(\mathcal{M}_B) \cdot \mathcal{I}_{\max}(\mathcal{M}_B)$ as the class's representational ceiling on per-event information. The cleaner formulation: (I-b) holds iff the event's sufficient statistics for prediction lie in the span of $\mathcal{M}_B$'s sufficient statistics. For parametric families (exponential families, Gaussian) the explicit form is routine; for non-parametric classes it requires the projection-to-class formalism of 2.2. Worth refining at the next candidacy review.
- **Four regimes vs finer splits.** The partition is the *coarsest* useful split where each regime has a distinct AAT-machinery response. Regime I could split into within-observability / within-calibration (point update vs structural refinement) as a fifth-regime refinement; Regime II-a could split by transient vs sustained dynamics. These duplicate the structural-adaptation hierarchy and are not pursued here.
- **Heavy-tailed derivation rigorization.** The Case-4 argument is informal; a rigorous version would compute the effective Kalman gain under heavy-tailed observation misspecification and show residuals retain signal. This is the robust-filtering territory (Huber, Masreliez) and is standard; the classification does not depend on a specific non-Gaussian family, just on the KL gap being nonzero.
- **Regime II-b as candidate X Instance 3.** Under a sustained Regime II-b stream, B's ρ_B^{eff} degrades at a rate lower-bounded by the KL gap between $\mathcal{M}_B$ and the true event distribution — a quantitative misspecification-cost floor. This is adjacent to X's "Misspecification-cost quantification" open extension. Worth a focused spike to formalize.
- **Connection to active inference.** Regime II-b corresponds to high-surprise events that exceed the agent's generative model — what AI calls "novel generative context." The classification avoids AI's variational-free-energy-as-master commitment (per `spikes/spike-active-inference-vs-aad.md`); it uses only the information-theoretic content of $\mathcal{I}(e)$, not the variational machinery.
- **Next spike candidates.** (1) Formalize the $f(H_b^B)$ emitter-side-effect function — tighten the qualitative $\gamma_A \propto f(H_b^B)$ into a derived form. (2) Formalize Regime II-b misspecification-cost as a structural result (X Instance 3 candidate). (3) Solve the `#adversarial-edge-targeting` optimization formally — the arg-max over edges given B's plan Jacobian and edge observabilities pairs with this segment as emitter-side counterpart.

13.4 *Result:* Adversarial Tempo Advantage

Under adversarial coupling where one agent's actions contribute to the other's disturbance rate, the steady-state mismatch ratio scales superlinearly with the tempo ratio.

Formal Expression.

Setup. Two agents A, B with adaptive tempos $\mathcal{T}_A, \mathcal{T}_B$, each instantiating C with linear correction ($\alpha = \mathcal{T}$). The adversarial coupling of 13.2 enters each agent's effective disturbance:

Derived (adversarial-tempo-advantage, from sector-persistence-template + adversarial-destabilization coupling model).

$$\rho_A^{\text{eff}} = \rho_{\text{base}} + \gamma_B \cdot \mathcal{T}_B, \qquad \rho_B^{\text{eff}} = \rho_{\text{base}} + \gamma_A \cdot \mathcal{T}_A \tag{13.19}$$

with $\gamma_A, \gamma_B > 0$ coupling effectivenesses and ρ_{base} the shared background rate (asymmetric ρ_{base} generalizes straightforwardly).

Model D: Deterministic coupling, $b = 2$.

The template's Model D conclusion $\|\delta\|_{ss} = \rho^{\text{eff}}/\mathcal{T}$ (linear case, $\alpha = \mathcal{T}$) applied to both agents and ratioed:

Result (adversarial-tempo-advantage, Model D).

$$\frac{\|\delta_B\|_{ss}}{\|\delta_A\|_{ss}} = \frac{(\rho_{\text{base}} + \gamma_A \mathcal{T}_A)\,\mathcal{T}_A}{(\rho_{\text{base}} + \gamma_B \mathcal{T}_B)\,\mathcal{T}_B} \tag{13.20}$$

In the coupling-dominant limit ($\gamma\mathcal{T} \gg \rho_{\text{base}}$) with symmetric coupling ($\gamma_A = \gamma_B$):

$$\frac{\|\delta_B\|_{ss}}{\|\delta_A\|_{ss}} \to \left(\frac{\mathcal{T}_A}{\mathcal{T}_B}\right)^2 \tag{13.21}$$

The exponent is $b = 2$: a **squared** tempo advantage. A 2:1 tempo ratio yields a 4:1 mismatch ratio. The faster agent both (a) corrects its own mismatch faster and (b) generates disturbance for the opponent faster — the two effects compound rather than add. □

Model S: Stochastic coupling, $b = 3/2$.

Under Model S the coupling enters the noise scale: $\sigma_B^{\text{eff}} = \sigma_{\text{base}} + \gamma_A \mathcal{T}_A$. Adversary tempo increases unpredictability, not systematic direction. The template's Model S steady state $\|\delta\|_{\text{rms}} = \sigma/\sqrt{2\mathcal{T}}$ (linear $\alpha = \mathcal{T}$, scalar $n = 1$) applied to both agents:

Derived (stochastic-tempo-advantage, from sector-persistence-template Model S + coupling).

$$\frac{\|\delta_B\|_{\text{rms}}}{\|\delta_A\|_{\text{rms}}} = \frac{(\sigma_{\text{base}} + \gamma_A \mathcal{T}_A)\sqrt{\mathcal{T}_A}}{(\sigma_{\text{base}} + \gamma_B \mathcal{T}_B)\sqrt{\mathcal{T}_B}} \tag{13.22}$$

In the coupling-dominant, symmetric limit:

$$\frac{\|\delta_B\|_{\text{rms}}}{\|\delta_A\|_{\text{rms}}} \to \left(\frac{\mathcal{T}_A}{\mathcal{T}_B}\right)^{3/2} \tag{13.23}$$

The exponent is $b = 3/2$. □

Why 3/2, not 2. The half-power difference between the template's Model D ($1/\alpha$) and Model S ($1/\sqrt{\alpha}$) scalings propagates through the ratio. Numerator contributes $\mathcal{T}_A^1$ from the coupling; denominator contributes $\mathcal{T}_B^{1/2}$ from noise averaging; combined with the A-side $1/\mathcal{T}_A^{1/2}$ gives $\mathcal{T}_A^{3/2}/\mathcal{T}_B^{3/2}$.

Summary of Regime-Dependent Exponents.

Table 13.3

Regime	*Coupling type*	*Dominance*	*Exponent b*	*Source*
1	Deterministic drift (Model D)	Coupling-dominant	2	Derived above
2	Stochastic noise (Model S)	Coupling-dominant	3/2	Derived above
3	Either	Non-coupling-dominant	$\to$ 1 (det.) or $\to$ 1/2 (stoch.)	Asymptotic limit

Regime 3 (non-coupling-dominant). When $\rho_{\text{base}} > rsim\gamma \cdot \mathcal{T}$ (or $\sigma_{\text{base}} > rsim\gamma \cdot \mathcal{T}$), the base disturbance dominates and the coupling terms become a perturbation. The mismatch ratio degrades toward $\mathcal{T}_A/\mathcal{T}_B$ (linear, $b = 1$) for Model D, or toward $(\mathcal{T}_A/\mathcal{T}_B)^{1/2}$ for Model S.

The simulation validation across all three regimes is in 14.3.

Epistemic Status.

Both coupling-dominant exponents are *exact* conditional on their respective disturbance models. The squared law ($b = 2$) is *exact* under Model D (deterministic bounded disturbance, GA-2) with coupling-dominant conditions. The 3/2 law ($b = 3/2$) is *exact* under Model S (stochastic disturbance, GA-2S) with coupling-dominant conditions, derived from the $1/\sqrt{\alpha}$ steady-state scaling (Prop A.1S in A). Both derivations are straightforward algebra from the respective steady-state formulas and the coupling model. The coupling model itself is an *assumption* — the same one used in 13.2.

The non-coupling-dominant limits ($b \to 1$ for Model D, $b \to 1/2$ for Model S) are derived asymptotically. The smooth transition between regimes is confirmed by simulation (14.3) but the interpolation formula is empirical. The transition between regimes is smooth, not sharp.

Max attainable: exact conditional on the disturbance model and coupling model. The result is as strong as its assumptions; no additional work changes the epistemic status without changing the dynamical model.

Discussion.

Superlinearity is the key result. The naive expectation — twice as fast yields twice the advantage — is wrong under adversarial coupling. The mechanism is that the faster agent both (a) corrects its own mismatch faster and (b) generates disturbance for the opponent faster. These two effects multiply, producing the squared exponent. Speed advantage is not additive; it compounds.

Relationship to 13.2. The steady-state mismatch ratio quantifies how much worse the slower agent does *while both agents persist*. The destabilization threshold (13.2) marks where the slower agent fails entirely — its correction mechanism breaks down. Below the threshold, this segment's mismatch ratio applies. Above it, 13.2's Lyapunov divergence takes over. The two results are complementary: this one gives the score; that one gives the game-ending condition.

Regime dependence is operationally significant. Whether an adversary's tempo increase produces systematic drift (positional maneuvering, API changes, doctrinal initiative) or unpredictable noise (feints, randomized attacks, market volatility) determines the scaling law. The distinction is not academic — $b = 2$ vs. $b = 3/2$ means a 3:1 tempo ratio yields 9:1 vs. 5.2:1 mismatch ratio. The model predicts that consistent, directional pressure is more effective per unit of tempo than unpredictable disruption.

Formal analog of OODA-loop observations. The squared scaling is consistent with Boyd's observation that getting inside the opponent's decision cycle has disproportionate effects. The theory identifies a specific mechanism (multiplicative interaction of correction speed and disturbance generation) and a specific condition (coupling-dominant regime) under which this disproportionality holds. Whether this mechanism is the dominant one in actual adversarial interactions is an empirical question, not a mathematical one.

Working Notes.

- **Channel-independence assumption.** The tempo ratio $\mathcal{T}_A/\mathcal{T}_B$ uses scalar tempo, which inherits the channel-independence assumption from 3.8. When either agent's observation channels are correlated, the additive formula overcounts their tempo, inflating or deflating the ratio and the derived mismatch advantage. The superlinear exponents ($b = 2$, $b = 3/2$) are exact given the scalar tempos; the caveat concerns whether the scalar tempos themselves are accurate.
- The analysis treats each agent's tempo as exogenous — $\mathcal{T}_A$ does not change in response to B's actions and vice versa. A fully coupled analysis where both agents' mismatch states co-evolve simultaneously (joint Lyapunov function over (δ_A, δ_B)) is the open extension. The decoupled result is a worst-case bound for the slower agent: in practice, the faster agent may divert adaptive capacity to generating disturbance rather than correcting its own mismatch, creating a self-limiting effect.
- The stochastic exponent ($b = 3/2$) is now derived from both the AR(1) stationary variance (discrete) and the Itô-Lyapunov analysis (continuous, Prop A.1S). The continuous-time analog (Ornstein-Uhlenbeck) gives the same scaling, confirming the asymptotic-scaling claim is the fluid-limit value. The 0.019 gap between the simulation $b = 1.481$ and the asymptotic $b = 3/2$ is *not pure numerical noise*: it is consistent with a derivable finite-ν correction factor (proportional to $\sqrt{(2c_{\min} - \eta_A^* c_{\max}^2)/(2c_{\min} - \eta_B^* c_{\max}^2)}$ when $\eta_A^* > \eta_B^*$) that arises because the discrete steady-state variance carries the $O(\eta^* c_{\max}^2/c_{\min}^2) = O(c_{\max}^2/(c_{\min}^2 \nu))$ gap from K. In the fluid limit ($\nu \to \infty$, $\eta^* \to 0$ at fixed $\mathcal{T}$), the correction factor approaches 1 and the asymptotic $b = 3/2$ is recovered exactly. The two models (D and S) are unified by the common sector-condition framework with different disturbance assumptions (GA-2 vs. GA-2S).
- Asymmetric coupling ($\gamma_A \neq \gamma_B$) appears as a multiplicative prefactor γ_A/γ_B that shifts the mismatch ratio without changing the exponent. An agent with lower tempo but higher coupling effectiveness (γ) can partially compensate — but the squared dependence on tempo dominates for large tempo ratios.

13.5 *Discussion:* Additional Implications & Discussion

The chapter is where AAT's signed-coupling structure first comes into formal view. Team persistence and adversarial destabilization turn out to be the same inequality viewed in opposite directions — both fall out of the sector-persistence template with effective disturbance decomposed into environmental and coupling-induced terms — and the chapter's headline result (superlinear adversarial tempo advantage) follows from the template's scaling without separate derivation. The recipient-side four-regime classification gives the chapter its distinctive contribution: what the emitter-side scalar $\gamma_A \mathcal{T}_A$ collapses, the recipient-side decomposition recovers, with three independent boundary conditions in B's existing AAT quantities and operational consequence for what kind of repair the receiving agent needs. None of the chapter's segments carries a `## Findings` section, so the implications segment is their canonical catalog home; the work is mostly synthesis: surfacing how the chapter's machinery composes with Ch.2's closure-defect framework and what the equilibrium hand-off to Ch.5 looks like.

Cooperative and adversarial are one signed coupling, not two theories.

The chapter intro `#cooperative-adversarial-intro` flags it as a chapter-level architectural commitment, and the chapter's segments deliver on it: cooperative coupling and adversarial coupling are not separate dynamics with their own machinery. They are the *same* signed-coupling structure with γ of opposite sign. `#der-team-persistence`'s composite-persistence condition decomposes effective disturbance for sub-agent i as $\rho_i^{\text{eff}} = \rho_{i,\text{env}} + \sum_j \gamma_{j\to i}^{\text{adv}} \mathcal{T}_j - \sum_j \gamma_{j\to i}^{\text{coop}} \mathcal{T}_j$ — adversarial contributions enter as positive terms (effective disturbance amplification), cooperative contributions as negative terms (effective disturbance reduction). The signed-coupling form is what makes cooperative coupling *quantitatively comparable* to adversarial coupling rather than addressing them in separate vocabularies, and it is what lets the framework's persistence template (Part I Ch.4 implications, §4) absorb both regimes as instantiations of a single Lyapunov argument.

The implications-side payoff is the duality between persistence and destabilization. `#der-adversarial-destabilization` makes this precise: B destabilizes iff $\rho_B^{\text{eff}} > \alpha_B R_B$, where ρ_B^{eff} includes the coupling-amplified term $\gamma_A \mathcal{T}_A$. *This is the* *negation of B's persistence condition under the same coupling model*. The two are the same inequality, read in opposite directions: persistence asks whether the inequality holds, destabilization asks whether it fails, and the coupling parameter $\gamma_A \mathcal{T}_A$ is what determines which side of the threshold the composite sits on. The framework's posture is therefore that adversarial attacks are *not* a separate kind of phenomenon requiring separate machinery — they are the same machinery, with the attacker's contribution entering the target's effective disturbance via the signed-coupling structure.

The closed-form unification this enables, surfaced in Part III Ch.2 implications, deserves to be repeated as it bites here: the four prior persistence results — single-agent ($\gamma = 0$), team-persistence ($\gamma < 0$), adversarial-destabilization ($\gamma > 0$), coordination-failure ($C > \alpha$, Brooks's Law) — are signed special cases of one inequality. Cooperative and adversarial coupling sit at opposite ends of the same γ-axis; the framework's machinery does not pick a side, and the practitioner who understands one regime understands the other by sign change.

Universal taxonomy of inter-agent events — what the recipient sees.

`#der-interaction-channel-classification` is the chapter's distinctive AAT-internal contribution: the same signal from emitter A lands on recipient B as one of *four* qualitatively different things — *informative update*, *magnitude shock*, *structural shock*, or *ambient noise* — and which one is determined by three independent boundary conditions stated entirely in B's existing AAT quantities. The emitter-side scalar $\gamma_A \mathcal{T}_A$ that previous segments have used as a stand-in for "how strong the coupling is" loses this distinction; the recipient-side classification recovers it. The implication for the framework's repair machinery is concrete: the four regimes route to four different corrective paths, and "more tempo" works for some regimes but not for others.

The three boundary conditions deserve to be named even at the implications-segment grain. The first separates informative updates from magnitude shocks: whether the residual δ_τ^B is within the current model's prediction-confidence interval or outside it. Inside the interval, the event is informative — B's standard gain-principle update absorbs it with full effect. Outside the interval, the event is a magnitude shock — the gain principle still applies, but the residual is large enough to suggest model-class strain rather than parameter drift. The second separates magnitude shocks from structural shocks: whether the model class $\mathcal{M}$ can absorb the event at all. When the residual reflects a regime the model class cannot represent (`#result-structural-adaptation-necessity` applied locally), the event triggers structural adaptation rather than parametric update. The third separates ambient noise from informative updates: whether the event's expected magnitude clears the agent's signal-detection threshold; below it, the event is filtered as noise and produces no update.

The framework's prescription is that *repair path depends on regime*, not on event magnitude or coupling strength alone. Informative updates respond to tempo investment (more $\mathcal{T}_B$ closes the residual faster). Magnitude shocks respond to gain investment (lower $U_{o,B}$ — better observation quality — closes more of each shock per cycle). Structural shocks respond to model-class expansion (`#result-structural-adaptation-necessity`'s structural change) — no amount of tempo or gain suffices. Ambient noise does not call for response at all; investing in detecting it is wasted bandwidth that should go to the other three regimes. The implication for adversarial robustness: an attacker who knows the target's regime boundaries can choose events that *miss* the target's repair vocabulary — structural shocks against a tempo-investment defender, magnitude shocks against a model-class-expansion defender. The framework's design prescription is that defenders should *cover all four repair paths*, not optimize for one.

The composite-level payoff is the regime-typed effective disturbance decomposition. The chapter's ρ_B^{eff} now decomposes per regime: informative updates contribute to ordinary tempo demand; magnitude shocks contribute to peak-load demand; structural shocks contribute to structural-adaptation cadence; ambient noise contributes to zero. The aggregate effective-disturbance number from `#der-team-persistence` is the *sum* across regimes; the per-regime decomposition is what tells the practitioner *where the disturbance is coming from*. For composite-agent design under uncertainty about the disturbance regime, the framework's prescription is to instrument the four regime detectors independently — knowing ρ_B^{eff} is small is less useful than knowing which regime is contributing what.

Adversarial tempo advantage as superlinear scaling — and the disturbance-model split.

`#result-adversarial-tempo-advantage` carries the chapter's headline result, and the implication worth surfacing is *why* the scaling is superlinear. The template's Model D conclusion gives $\|\delta\|_{ss} = \rho^{\text{eff}}/\mathcal{T}$ in the linear case; substituting the coupling-amplified disturbance gives the steady-state mismatch ratio scaling as $(\mathcal{T}_A/\mathcal{T}_B)^2$ — superlinear by exponent $b = 2$ under deterministic drift coupling. Under Model S (stochastic) the template's $1/\sqrt{\alpha}$ scaling produces $b = 3/2$. The two exponents are not arbitrary parameters: they fall out of Part I Ch.4's persistence-template structure, $1/\alpha$ versus $1/\sqrt{\alpha}$, applied to the coupling model. The framework's posture is that *the same template that explains why adaptive systems persist also predicts the precise rate at which they lose to a faster attacker* — the dual is structural.

The exponent splits further in Ch.5 — `#result-adversarial-exponent-regimes` gives three regime-typed values ($b = 2, 3/2, \sim 1$), with the coupling-non-dominant case dropping to $b \to 1$. The non-superlinear regime is operationally important: when observation noise is high enough that the attacker's tempo advantage doesn't penetrate the defender's observation channel cleanly, the superlinear scaling collapses to roughly linear, and the framework predicts that *cheap noise injection* into the defender's observation channel actually *helps* the defender against a high-tempo attacker by reducing the exponent. This is the structural reason the empirical pattern that highly-noisy environments produce more cooperative-game-like dynamics than expected — high observation noise gates the adversarial tempo advantage to near-linear scaling, where the attacker cannot exploit superlinear scaling to drive the defender past the destabilization threshold. The full regime taxonomy lives in Ch.5; the chapter here establishes that superlinear *is* the dominant regime under coupling-dominant noise structures.

The operational anchor worth carrying is Boyd's OODA-loop aphorism, which the framework now gives precise mathematical content. *Inside the opponent's loop* means $\mathcal{T}_A > \mathcal{T}_B/k$ for some regime-dependent k, and what the inside-the-loop position buys the attacker is superlinear destabilization rate rather than mere tempo advantage. The chapter's intro framing — "superlinear tempo advantage as regime change rather than quantitative gap" — is the implications-flavored point: at the destabilization threshold, the target does not gradually degrade. It loses bounded behavior, the way a balance held just barely beneath a tipping point is qualitatively different from one well above it. The threshold is the same one Part I Ch.4's central inequality names; the chapter's contribution is identifying the coupling-induced disturbance term that pushes the target past it.

Three adversarial obstructions to contraction — the equilibrium hand-off.

`#result-contraction-template` (Appendix A, depended on by Ch.4 via the persistence template) names the chapter's most consequential structural recognition: contraction-metric machinery *cannot* handle strategic / adversarial regimes for three convergent reasons that are individually classical results from different mathematical literatures. *Slotine 2003 compositional applicability fails*: saddle-point equilibria break the attracting-fixed-point assumption that contraction analysis requires. *Passivity universality fails*: adversarial inputs drive any storage function, so passivity-based stability does not give bounds. *Daskalakis et al. 2018 last-iterate non-convergence*: for generic non-zero-sum games, learning dynamics fail to converge to equilibrium in the last-iterate sense — only in time-average sense.

The implications-side payoff is structural rather than parametric: three *unrelated* mathematical obstructions converge on the same conclusion. Contraction analysis works for cooperative / coordination regimes (where there is an attracting fixed point) and fails for strategic / adversarial regimes (where equilibrium machinery is required). The convergence of three obstructions makes the limit *structural* rather than framework-specific — it is not that AAT's contraction machinery is too weak; it is that the regime is outside the scope condition contraction analysis depends on across multiple formal traditions. The framework's posture is to recognize the limit precisely and hand off cleanly to equilibrium-theoretic methods at the boundary.

The hand-off lands in Ch.5: `#deriv-strategic-composition` is where equilibrium machinery (potential games via Monderer-Shapley 1996, monotone games via Rosen 1965) takes over from contraction analysis for the symmetric partially-opposing case. The chapter's contribution is naming the limit; Ch.5's contribution is supplying the replacement machinery. The framework's posture across the boundary is that *scope-honesty is load-bearing architecture*: knowing where each tool applies cleanly is more useful than trying to extend one tool across all regimes. The three convergent obstructions are why this matters operationally — an AAT-style sector-condition argument applied to a strategic regime would not just produce wrong predictions, it would produce *systematically* wrong predictions in each of the three directions the obstructions name.

The chapter also surfaces a quieter consequence worth carrying: cooperative coupling that is *too strong* can produce equilibrium-stability failures even without adversarial components. When sub-agents' cooperative coupling drives them toward a saddle-point equilibrium (rather than a stable attractor), the framework's prediction is the same as the strategic case — contraction analysis fails, and equilibrium machinery is required. This is empirically the situation in tightly-coupled multi-agent ML systems where reward-shaping creates implicit competition among cooperators; the framework predicts the failure mode is structurally identical to the adversarial case, with the unique broadly-available remedy being explicit equilibrium-theoretic analysis rather than tempo or gain investment.

Closing the chapter and bridging to Strategic Composition.

The chapter delivers the signed-coupling structure that the rest of Part III operates on. Cooperative coupling reduces effective disturbance; adversarial coupling amplifies it; the same template handles both via the sign of γ. The four-regime taxonomy says what the recipient actually sees, and the framework's repair vocabulary diversifies accordingly. The superlinear adversarial tempo advantage falls out of the template's $1/\alpha$ versus $1/\sqrt{\alpha}$ scaling; OODA's "inside the opponent's loop" acquires mathematical content. The three convergent obstructions to contraction analysis name precisely where the chapter's machinery hands off to equilibrium-theoretic tools.

What the chapter does *not* deliver is the equilibrium machinery itself, the agent-opacity formalism, the recipient-side classification's emitter-side dual, or the per-dimension persistence treatment that adversarial targeting depends on. Those all live in Ch.5, where the framework's machinery operates at full Part III scope: strategic composition with partially-opposing objectives, agent opacity as the formal dual of observation quality, the 16-cell emitter-recipient adversarial-targeting decomposition, the regime-typed exponent decomposition (b = 2, 3/2, ~1), and the matrix-Loewner persistence formulation that adversarial concentration on weak dimensions requires. The chapter-end here is the last single-layer-of-coupling synthesis before Ch.5's strategic-equilibrium machinery takes over.

The bridge into Ch.5 runs through three threads. First, the contraction-template obstructions hand off to equilibrium machinery (potential / monotone games). Second, the recipient-side classification will compose with the emitter-side opacity classification in Ch.5 into a 16-cell joint composition with closed-form arg-max for adversarial-targeting decisions — the chapter here supplies one half, Ch.5 supplies the other. Third, the signed coupling γ structure becomes the *trust-meta-model signal* via Ch.3's `#hyp-communication-gain`: observed cooperative coupling drives $U_{\text{align},ji} \to 0$ (high trust calibration); observed adversarial coupling drives $U_{\text{align},ji}$ large (low trust calibration). The implications-segment series continues to weave back across earlier chapters, with each chapter's machinery operating on what previous chapters have built.

Working Notes.

- This segment is a chapter-end discussion grouping findings from `#der-team-persistence`, `#der-adversarial-destabilization`, `#der-interaction-channel-classification`, `#result-adversarial-tempo-advantage`, `#cooperative-adversarial-intro`, `#result-sector-persistence-template` (Appendix A, the template these segments instantiate), `#result-contraction-template` (Appendix A, the contraction-obstructions surfacing), and `#deriv-strategic-composition` (Ch.5, the forward-pointer for equilibrium machinery).
- **Cross-reference vs canonical-home policy.** None of this chapter's source segments carry `## Findings` sections. The implications segment is the canonical catalog home for: #21 (Universal Taxonomy of Inter-Agent Events — `#der-interaction-channel-classification` content), the cooperative-vs-adversarial duality (the signed-coupling structure as one architectural commitment), the headline adversarial tempo advantage result (#44 partial — the $b = 2$ and $b = 3/2$ exponents from

this chapter; the regime-decomposition refinement lives in Ch.5), and #24 (Three Adversarial Obstructions to Contraction Analysis — `#result-contraction-template` content surfaced at this chapter's hand-off boundary).

- **Forward-pointers into Ch.5 are heavy.** The chapter sits structurally between cooperative-coupling machinery (here) and strategic-composition machinery (Ch.5). Several findings are partially treated here with completions in Ch.5: regime-typed exponent decomposition, observation-noise gating of advantage, agent-opacity formalism as dual to observation quality, 16-cell emitter-recipient adversarial-targeting decomposition, and the matrix-Loewner persistence formulation that adversarial concentration depends on. Reading this implications segment together with Ch.5's gives the full coupling picture; reading either alone gives the half-picture.
- **Identifiability-floor pattern coverage update.** No new instances surface in this chapter — the four formal instances (Instances 1–4) were all in view by Part II Ch.5's implications, and Ch.5 here will give Instance 3 (composite contraction certification from component marginals via Liberzon common-Lyapunov-nonexistence) its operational form in the strategic-composition setting. The contraction-obstructions in `#result-contraction-template` are *adjacent* to but not the same as the identifiability-floor pattern — they are obstructions to a specific *analysis method* rather than no-go theorems about *identification*. The distinction is worth carrying: identifiability-floor instances forbid identifying a quantity from data; contraction-obstructions name where a method fails. Both are scope-honesty-load-bearing for AAT, but they sit in different meta-architectural slots.
- **Connection to NeurIPS submissions.** The superlinear tempo advantage result has not been formalized to theorem grade in current submissions; the signed-coupling structure underlies Paper 2's multi-agent extension (which is a candidate for a future revision rather than current scope). The chapter's content stays in the AAT-internal register.

Chapter 14

Strategic Composition and Channel Effects

[**Gap**] Discussion

14.1 *Derivation:* Strategic Composition via Equilibrium Convergence

When two or more AAT agents interact through a shared environment with **partially-opposing objectives** $\{O_t^{(i)}\}$, the composition-level question is not "does the trajectory contract to zero closure-defect?" but "does the coupled best-response dynamics admit an equilibrium and converge to it?" Contraction to shared truth is a $U_O = 1$ special case; strategic composition is the $U_O < 1$ companion regime in which the correct primitive is **fixed-point existence and stability**, not Lyapunov contraction on a shared state. Under potential-game or monotone-game conditions (sub-scope α'), the sector-persistence template transfers to the gradient of the joint potential (resp. to a weighted-norm variational inequality), recovering AAT's persistence machinery at the equilibrium layer. Outside sub-scope α', only set-convergence to coarse correlated equilibria is available (sub-scope β'). This segment establishes the framing, derives the α'-transfer, documents the β' scope limits honestly, and relates to `#der-adversarial-destabilization` (asymmetric adversarial) and `#deriv-critical-mass-composition` (aligned composition with shared target) as siblings on the composition axis.

Formal Expression.

The framing move.

For N purposeful sub-agents with partially-opposing $\{O_t^{(i)}\}$ running coupled AAT loops through a shared environment, the composition-level question is:

Formulation (strategic-composition-framing).

(SC-1) Existence of equilibrium. Does there exist a joint state $(X_{c,1}^*, \dots, X_{c,N}^*)$ — equivalently a joint policy profile $(\pi_1^*, \dots, \pi_N^*)$ — such that no sub-agent has a unilateral mismatch reduction available?

(SC-2) Stability of equilibrium. Do coupled best-response dynamics, initialised near $(X_{c,i}^*)$, remain there or return to it?

(SC-3) Convergence from interior. Do coupled dynamics, initialised away from any equilibrium, converge to the equilibrium set?

(SC-1)–(SC-3) are a fixed-point-existence question, a local-stability question, and a reachability question respectively. None is a Lyapunov contraction question on a shared state variable. The contraction framing in `#form-composition-closure` / `#deriv-critical-mass-composition` is recovered as the $U_O = 1$ special case (unique Nash equilibrium at the shared-objective optimum; best-response dynamics collapse to single-objective optimization).

Sub-scope α': potential and monotone games (A2'-analog derived).

A strategic interaction $(\{A_i\}, \{O_t^{(i)}\})$ is a **potential game** (Monderer & Shapley 1996, *Games and Economic Behavior* 14) when there exists a scalar potential Φ such that each sub-agent's unilateral improvement matches the potential's unilateral improvement:

Derived (A2'-analog-potential, from Monderer-Shapley 1996).

$$O_t^{(i)}(\pi_i', \pi_{-i}) - O_t^{(i)}(\pi_i, \pi_{-i}) = \Phi(\pi_i', \pi_{-i}) - \Phi(\pi_i, \pi_{-i}) \quad \forall i. \tag{14.1}$$

Under this condition plus each sub-agent's B1 directional fidelity (4.2) to $\nabla_{\pi_i} O_t^{(i)}$, the joint best-response dynamics satisfy

$$\frac{d\Phi(\pi)}{dt} = \sum_i \langle \nabla_{\pi_i}\Phi,\ \dot{\pi}_i \rangle \geq \alpha_{\text{joint}} \|\nabla\Phi\|^2 \quad \text{for some } \alpha_{\text{joint}} > 0 \tag{14.2}$$

whenever the joint configuration is not at a stationary point of Φ. This is (T2) transcribed to state variable $\xi =$ gradient-of-potential, correction function $F =$ joint best-response velocity field; the quadratic Lyapunov structure is the same. **The sector-persistence template transfers** with $\xi = \pi - \pi^*$ (deviation from Nash), $\alpha = \alpha_{\text{joint}}$, $R =$ basin-of-attraction radius, and ρ_ξ the exogenous disturbance rate on strategy space. Equilibrium existence follows from potential-function compactness on compact strategy spaces; equilibrium stability follows from Φ's role as a joint Lyapunov function.

Weaker than potential: a game is **diagonally strictly concave** (Rosen 1965, *Econometrica* 33) when the Jacobian of the joint gradient field is negative-definite on the joint strategy space. Under this condition, there exists a unique Nash equilibrium and the joint gradient dynamics converge to it exponentially. The convergence rate is bounded below by the smallest eigenvalue of the symmetric part of the joint Jacobian, playing the role of α_{joint}. No scalar potential need exist; a *weighted-norm* Lyapunov argument on the joint Jacobian's symmetric part substitutes.

Derived (A2'-analog-monotone, from Rosen 1965).

Sub-scope α' comprises: potential games (Monderer-Shapley), monotone games (Rosen), strongly-monotone games, and exponential-family dual-averaging under concave objectives. For these classes, the sector-persistence template extends to equilibrium convergence with composite sector constant α_{joint} inheriting from the joint-gradient-field structure.

Sub-scope β': non-potential non-monotone games.

Every strategic interaction with continuous strategy spaces and regular payoffs can be reformulated as a **variational inequality**: find $\pi^* \in \mathcal{K}$ such that $\langle F(\pi^*), \pi - \pi^* \rangle \geq 0$ for all $\pi \in \mathcal{K}$, where F is the joint pseudo-gradient field. When $\mathcal{K}$ is compact-convex and F is continuous, a solution exists (Hartman-Stampacchia theorem). **Pure-strategy Nash equilibrium existence is therefore guaranteed** for continuous-strategy games with compact convex strategy sets and continuous payoffs. But the VI framework gives *existence* only, not *convergence of any specific dynamic to the solution*; solutions may be non-unique.

Derived (equilibrium-existence-via-VI, from Facchinei-Pang 2003).

Under no-regret learning (e.g., Hedge / multiplicative weights, Freund-Schapire 1997), the empirical joint distribution converges to the set of **coarse correlated equilibria** (CCE) at rate $O(1/\sqrt{T})$. This requires no structure on the game beyond each sub-agent computing its own regret.

Derived (regret-minimization-convergence-to-CCE, from Hart-Mas-Colell 2000).

Under β', the macro-state of a strategic composite is a *distribution* on the joint strategy space — the empirical joint play whose support is the CCE — rather than a state-space point. This is the structural shape of "convergence" in the β' regime: distributional convergence, not pointwise. Pure-strategy Nash may or may not exist (cyclic games — rock-paper-scissors, matching pennies — lack pure Nash but retain mixed Nash via Nash 1950 and CCE convergence via Hart-Mas-Colell 2000); the β' machinery's guarantees are at the distributional layer regardless.

Sub-scope β' scope-honesty: AAT can predict that long-run joint play lies in the CCE support; it cannot predict short-run trajectory, per-sub-agent mismatch convergence, or selection among multiple equilibria. The sector-persistence template does *not* apply in β'. This is a genuine scope limit shared with game theory as a whole, not a defect of AAT.

Zero-sum scalar worked example.

Two agents A, B with scalar actions $a_i \in [-1, 1]$ and state $s_{t+1} = s_t + a_A - a_B + w_t$, $w_t \sim \mathcal{N}(0, \sigma^2)$. Objectives $O_t^{(A)}(s) = s$ (maximize s), $O_t^{(B)}(s) = -s$ (minimize s); zero-sum at the state-dependent payoff level.

Derived (zero-sum-scalar-instantiation).

Action-coefficient analysis. The agents' state-preferences are opposite, but the action-coefficients in s_{t+1} are also opposite: $\partial s/\partial a_A = +1$ and $\partial s/\partial a_B = -1$. Composing with the objectives' state-dependence:

$$\frac{\partial O^{(A)}}{\partial a_A} = \frac{\partial O^{(A)}}{\partial s} \cdot \frac{\partial s}{\partial a_A} = (+1)(+1) = +1 \tag{14.3}$$

$$\frac{\partial O^{(B)}}{\partial a_B} = \frac{\partial O^{(B)}}{\partial s} \cdot \frac{\partial s}{\partial a_B} = (-1)(-1) = +1 \tag{14.4}$$

Both agents marginal-prefer *increasing* their own action — the opposing state-preferences and opposing action-coefficients compose to aligned action-preferences.

Potential function. For Monderer-Shapley, $\partial\Phi/\partial a_i = \partial O^{(i)}/\partial a_i$ for each i. With both partial derivatives equal to $+1$, the potential is $\Phi(a_A, a_B) = a_A + a_B$ (modulo additive constant).

Nash equilibrium. Each agent's best response on $[-1, 1]$ is to push its own action to $+1$; the unique Nash equilibrium is $(a_A^*, a_B^*) = (1, 1)$. At equilibrium, $\Delta s = a_A^* - a_B^* = 0$: the substantive zero-sum property is that equal opposing action-coefficients cancel under the agents' aligned action-preferences, so joint action contributes no net displacement to s (state drift comes only from w_t). This is qualitatively different from the naive picture in which the agents push the state in opposite directions.

Sector-persistence template — interior-NE instantiation via Cournot-style payoffs. The unregularized example has a *linear* potential Φ and a *corner* equilibrium at $(1, 1)$; the unconstrained gradient field $\nabla\Phi = (1, 1)$ is constant in $\xi = \pi - \pi^*$, so there is no interior linear restoring force around the equilibrium and the template's (T2) sector lower bound $\xi^T F(\xi) \geq \alpha\|\xi\|^2$ does not hold without modification — the saturation, not a sector contraction, is what enforces equilibrium. The aligned-action-preferences finding is the conceptual payload of the unregularized form; the sector-template transfer requires a payoff structure whose curvature is genuinely quadratic, not merely projected onto a corner.

A Cournot-style duopoly (Cournot 1838; Monderer-Shapley 1996 §3 example) supplies that structure naturally. Two firms produce quantities $q_i \in [0, Q_{\max}]$; market price is $P(q_A, q_B) = a_0 - b(q_A + q_B)$; per-firm cost is linear at rate κ; per-firm profit is

$$O^{(i)}(q_A, q_B) = q_i\,(a_0 - b(q_A + q_B) - \kappa). \tag{14.5}$$

This is partially-opposing in the policy-conflict sense — each firm's marginal payoff is decreased by the other's production through the shared price — without being strictly zero-sum. It is a potential game with quadratic potential.

Derived (Cournot-as-potential, from Monderer-Shapley 1996 §3).

$$\Phi(q_A, q_B) = (a_0 - \kappa)(q_A + q_B) - b(q_A^2 + q_B^2) - b\,q_A q_B, \tag{14.6}$$

verified by $\partial\Phi/\partial q_i = (a_0 - \kappa) - 2bq_i - bq_{-i} = \partial O^{(i)}/\partial q_i$. The symmetric interior Nash equilibrium is $q^* = (a_0 - \kappa)/(3b)$, located strictly inside the action box whenever $0 < q^* < Q_{\max}$.

Under continuous-time best-response $\dot{q}_i = \partial O^{(i)}/\partial q_i$, writing $\xi = q - q^*\mathbf{1}$, the joint dynamics are $\dot{\xi} = -bM\xi$ with $M = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$. Setting $F(\xi) = bM\xi$ (correction-strength sign convention; $\dot{\xi} = -F(\xi)$):

- (T1) $F(0) = 0$. ✓
- (T2) $\xi^T F(\xi) = b(2\xi_A^2 + 2\xi_B^2 + 2\xi_A\xi_B) \geq b\|\xi\|^2$, since M's eigenvalues are $\{1, 3\}$. So $\alpha_{\text{joint}} = b$ — the smallest eigenvalue of the symmetric part of the joint Jacobian, exactly the quantity Rosen 1965 identifies as the convergence rate. ✓
- (T2-upper) $\|F(\xi)\| \leq 3b\|\xi\|$ from M's largest eigenvalue.
- (T3) $R = \min(q^*, Q_{\max} - q^*)$ — the radius of the largest origin-centered ball in ξ-space contained in the feasible region; the closer of the two active constraints (zero-production and capacity) determines the basin.

The template's preconditions hold with α_{joint} inherited from the demand-side curvature parameter b rather than from any individual sub-agent's α. This is the substantive transfer: the composite-level sector constant lives at the *joint potential's curvature*, with economic interpretation (market-saturation slope) rather than ad-hoc parametrization. Other linear-quadratic strategic-substitutes games (LQR with coupled state, network-coordination with quadratic disagreement cost, public-goods with quadratic externalities) produce structurally equivalent template instantiations, with α_{joint} tracking the symmetric part of their respective joint Jacobians.

The unregularized scalar zero-sum and the Cournot instantiation are complementary: the former illustrates the framing's surprise — opposing state-preferences combined with opposing action-coefficients yield aligned action-preferences — while the latter exhibits the template transfer with genuine quadratic curvature and interior NE.

New scope route (C-iv) in `#scope-composite-agent`.

Strategic composition with partially-opposing $\{O_t^{(i)}\}$ admits a joint equilibrium structure $\mathcal{E}$ (Nash, correlated, or coarse correlated) such that coupled best-response dynamics converge to the support of $\mathcal{E}$. The composite exists as an AAT agent with macro-state defined relative to $\mathcal{E}$ rather than relative to a shared target state.

Proposed Scope (composition-scope-condition, route C-iv).

(C-iv) is qualitatively distinct from (C-i)–(C-iii): it does *not* require shared objectives, hierarchical derivation, or mutual benefit. It requires only structural convergence of the strategic interaction in the game-theoretic sense. Composites satisfying (C-iv) are **strategic composites**, distinguished from alignment composites (C-i, C-ii) and mutual-benefit composites (C-iii).

What Is Derived vs. What Is Chosen.

Table 14.1

Property	*Source*	*Strength*
Framing move (contraction → equilibrium convergence)	Standard game-theoretic positioning	Formulation choice
(SC-1) / (SC-2) / (SC-3) three-question decomposition	Parallel to fixed-point / stability / reachability in dynamical systems	Formulation choice
A2'-analog under potential-game condition	Monderer-Shapley 1996 transcribed into AAT notation with B1 directional fidelity	Derived (exact under potential-game + B1)
A2'-analog under monotone-game condition	Rosen 1965 transcribed with weighted-norm Lyapunov	Derived (exact under diagonal strict concavity)
Equilibrium existence via VI	Facchinei-Pang 2003 (Hartman-Stampacchia)	Derived (external theorem, applied to strategic composition setting)
Regret-minimization CCE convergence	Hart-Mas-Colell 2000	Derived (external theorem applied)
Sub-scope α' / β' partition	Parallel to `#deriv-sector-condition` α/β	Formulation choice
Zero-sum scalar instantiation (corner-NE conceptual lesson)	Direct substitution into potential-game framework	Exact (within stated setup)
Cournot-style sector-template instantiation (interior-NE quadratic)	Monderer-Shapley 1996 §3 + sector-persistence-template; demand-side curvature b supplies α_{joint}	Exact (within stated setup)
(C-iv) scope route	Extension to `#scope-composite-agent` disjunction	Formulation choice (scope extension)
Effects-spiral eigenvalue condition	Joint-Jacobian $\mathrm{Re}(\lambda_{\max}) > 0$ at candidate equilibria — formalizes `#der-adversarial-destabilization`'s discussion-grade effects spiral	Sketch (specific AAT instantiations open)
Strategic composition produces Class 2 (Partial) composites from Class 1 (Separated) sub-agents	Across-agent coupling through environment + cross-agent observation; preserves within-agent modularity	Derived (scope-structural)
Mechanism-design impossibility as candidate adjacent-floor instance in `#disc-identifiability-floor`	Gibbard-Satterthwaite 1973-75, Arrow 1951, Myerson-Satterthwaite 1983	Flagged; not derived in this segment
Potential function Φ as additive-coordinate-forcing instance	Φ's additivity is definitional consequence of being a potential game, not forced by AAT-internal axiom; adjacent family member	Discussion-grade
Bridge from strategic composition to `#form-composition-closure` ε^*	Macro-description is an equilibrium statistic, not an equilibrium state	Open
General equilibrium-selection under multiple Nash	Existence theorems do not pin down which equilibrium; selection (risk-dominance, payoff-dominance, Pareto) is partial	Open
Mean-field-game limit ($N \to \infty$)	Lasry-Lions 2007; Huang-Malhamé-Caines 2006; requires population scope condition	Open (pending Section III population gaps)

Epistemic Status.

Conditional. Max attainable: *exact* under potential-game + B1 (sub-scope α'); *derived* under monotone-game + diagonal strict concavity; *discussion-grade* for the framing and sub-scope β' set-convergence-only claim.

The A2'-analog results under Monderer-Shapley 1996 and Rosen 1965 are transcriptions of established game-theoretic theorems into AAT notation. AAT's contribution is not the mathematics but the recognition that the sector-persistence template transfers cleanly when these conditions hold, and that the composite-level sector constant α_{joint} lives at the *joint potential gradient* or *joint Jacobian's symmetric part*, not at any individual sub-agent's α. The framing move (contraction → equilibrium convergence) is a structural positioning move.

Sub-scope β' gives AAT substantially weaker predictive power than sub-scope α': *set-convergence to CCE only*, not trajectory convergence or per-agent mismatch convergence. This is an honest scope limit, not a defect — it mirrors game theory's own scope at the no-potential-no-monotonicity regime. AAT does not claim to predict equilibrium selection under multiple Nash, short-run dynamics in cyclic games (rock-paper-scissors), or convergence rates better than $O(1/\sqrt{T})$ in β'.

What this segment does not establish:

- General convergence proof for non-potential games under AAT-native update rules (imports Monderer-Shapley + Rosen as scope statements, doesn't re-prove them).
- Explicit joint-Jacobian analysis for specific AAT agent architectures (e.g., two Beta-Bernoulli agents on a shared DAG). Effects-spiral formalization is sketched, not derived per-instance.
- Bridge from strategic composition to `#form-composition-closure`'s closure-defect ε^* — the macro-description of a strategic composite is a distributional equilibrium statistic rather than a state, so the closure-defect formulation needs re-specification. Open.
- Full mechanism-design derivation (VCG, Bayesian-Nash) for specific AAT-relevant settings.

Honest Limits.

- **No pure-strategy Nash equilibrium** in cyclic games (rock-paper-scissors; matching pennies). Mixed-Nash equilibrium exists universally for finite games (Nash 1950) but is a saddle point of best-response dynamics rather than a basin attractor; fictitious play orbits the mixed-Nash without converging pointwise to it. No-regret dynamics still drive the *empirical joint distribution* to the CCE set at rate $O(1/\sqrt{T})$ (Hart-Mas-Colell 2000), so β' machinery applies; the macro-state of a strategic composite in this regime is a distribution over the strategy space, not a state-space point.

  Scope honesty — strategic-composition-failures.

- **Multiple equilibria with ambiguous selection** (coordination games; drive-left vs drive-right). Convergence to *some* equilibrium; AAT does not predict which.
- **Slow mixing / finite-sample failure** under regret-minimization: $O(1/\sqrt{T})$ cumulative regret implies finite-horizon play may be far from CCE support.
- **Non-compact strategy spaces** (unbounded prices, resource allocations). No equilibrium without compactification; compactification may be artificial.
- **Bayesian games with strategic + epistemic uncertainty compounding.** Existence and uniqueness become pathological under certain information structures.
- **Mean-field games** ($N \to \infty$) require population-scope machinery not covered here.

Discussion.

Sibling to `#der-adversarial-destabilization`. `#der-adversarial-destabilization` handles the *asymmetric* adversarial case: one agent is target; attacker's tempo is exogenous parameter; sector-persistence template applies to target's mismatch with coupling-amplified disturbance. This segment handles the *symmetric* strategic case: both agents running full AAT loops; state variable is joint deviation from equilibrium; correction function is joint best-response field. The two are siblings. `#der-adversarial-destabilization`'s Working-Notes-flagged "coupled Lyapunov analysis is the open problem" has its formal home here — but the analysis is *not* a Lyapunov problem; it is a fixed-point problem, and the correct primitive is equilibrium convergence rather than Lyapunov descent. `#der-team-persistence` is recovered as the cooperative limit (all γ^{coop} dominate; joint dynamics reduce to parallel single-agent sector-persistence with reduced effective disturbance); `#deriv-critical-mass-composition` sits in the middle with signed γ on a matched-symmetric-Tier-1 dyad under shared target.

The effects spiral's formal home. `#der-adversarial-destabilization`'s effects spiral — B's degrading model causes B's actions to destabilize A further — is a **joint-Jacobian eigenvalue condition** in the strategic-composition framing: the spiral exists iff $\max_{\text{candidate equilibria }\pi^*} \text{Re}(\lambda_{\max}(\nabla F(\pi^*))) > 0$, where F is the joint best-response field. This condition specializes monotone-game failure (the Jacobian's symmetric part fails to be negative-definite at equilibrium). The asymmetric formulation in `#der-adversarial-destabilization` cannot express this condition because it treats one agent's tempo as exogenous; the symmetric coupled formulation here does.

Class-2-(Partial)-composite-from-Class-1-(Separated)-sub-agents. `#der-directed-separation` classifies agents (Class 1 Separated, Class 3 Coupled, Class 2 Partial) based on within-agent coupling. Strategic composition introduces a distinct form of coupling: *across-agent* coupling through the shared environment and cross-agent observation — each sub-agent's $M_t^{(i)}$ contains a model of other sub-agents' policies, but its own $f_M^{(i)}$ remains goal-blind. Strategic composites of Class 1 (Separated) sub-agents are therefore **Class 2 (Partial) composites**: within-agent processing is Separated, but composite-level directed separation fails because composite (M_c, G_c) acquires intrinsic coupling through the sub-agents' observations of each other. This is a bounded coupling — within-agent separation is preserved — placing the composite in Partial (Class 2) rather than Coupled (Class 3). This refines `#der-directed-separation`'s classification: class-type is a property of composites, not just of individual agents; strategic composition is the canonical Class 1 (Separated) sub-agents → Class 2 (Partial) composite case.

Mechanism-design impossibility. If an outside designer can shape $\{O_t^{(i)}\}$, strategic composition becomes a **mechanism-design problem**: choose objectives so that the induced equilibrium *is* contraction to a designed state. Impossibility results — Gibbard-Satterthwaite 1973–75 (no dominant-strategy non-dictatorial Pareto-efficient voting mechanism for ≥3 alternatives); Myerson-Satterthwaite 1983 (no efficient, individually-rational, incentive-compatible bilateral-trade mechanism without subsidies); Arrow 1951 (no social welfare function satisfying unrestricted-domain, Pareto-efficient, IIA, non-dictatorial) — are **candidate adjacent-floor instances of `#disc-identifiability-floor`**: external theorems forbidding design-of-alignment under stated constraints, with AAT machinery (Bayes-Nash relaxation, randomized allocations, subsidy injection) as structural escapes. Flagged for future follow-up spike; not derived in this segment.

Meta-pattern positioning.

- `#disc-separability-pattern`*:* sub-scope α' (potential / monotone — template transfers) is separable-core; sub-scope β' (VI / regret-minimization — set-convergence only) is structured-repair; cyclic / non-convergent / multi-equilibrium-selection-ambiguous is general-open. Candidate additional ladder (strategic-interaction regime); decide whether to surface as 8th ladder or merge into an existing ladder.
- `#disc-identifiability-floor`*:* mechanism-design impossibility is a candidate fourth instance (flagged above).
- `#disc-additive-coordinate-forcing`*:* the potential function Φ plays an additive-coordinate role at the strategic layer, but its additivity is a *definitional consequence* of being a potential game (Monderer-Shapley require the additivity property by definition) rather than forced by a uniqueness theorem on an AAT-internal axiom. This

positions Φ as an adjacent family member, parallel to Lyapunov quadratic and IB Lagrangian, not a primary instance.

Active-inference sharpening. Sun-Firestone 2020's dark-room argument observes that preferences-as-priors collapse under mutual prediction. Strategic composition gives this argument a formal substrate: two agents mutually predicting each other become a **fixed-point problem**, not a Lyapunov descent. The active-inference attempt to unify goal-seeking and prediction fails here in a derivable way — the fixed point is not at the prediction-optimum; it is at the strategic-equilibrium, which is structurally different. This strengthens `#def-satisfaction-gap`'s positioning against preferences-as-priors without requiring additional refutation machinery.

Working Notes.

- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. The canonical strategic-composition pattern "Class-1-sub-agents → Class-3-composite" (old vocab; Class 3 = partially modular) is now "Class 1 (Separated) sub-agents → Class 2 (Partial) composite." Removed at `candidate` stage per FORMAT.md Gate 4.
- **No γ' sub-scope.** A γ' sub-scope for cyclic-distributional equilibria (games where no pure-strategy Nash exists but mixed Nash and/or CCE do exist, with macro-state as equilibrium distribution rather than equilibrium point) was considered and collapsed into β': cyclic games are paradigmatic β' instances, with the same machinery (regret-minimization), the same convergence guarantee (CCE set-convergence), and the same rate ($O(1/\sqrt{T})$). The α'/β' decomposition is the right granularity at the regret-minimization layer; the type-of-object distinction (distribution vs state-space point) is surfaced within β' framing rather than as a third sub-scope.
- **Effects-spiral eigenvalue condition formalization.** The condition $\max_{\pi^*} \mathrm{Re}(\lambda_{\max}(\nabla F(\pi^*))) > 0$ is sketched in §Discussion. Deriving it for specific AAT agent classes (two Beta-Bernoulli agents on a shared DAG; two Kalman agents with coupled observations) would upgrade the spiral from discussion-grade in `#der-adversarial-destabilization` to derived here. Follow-on spike candidate.
- **Mechanism-design impossibility as `#disc-identifiability-floor` Instance 4.** Would require working out the external-theorem anchoring, the boundary characterization (Bayes-Nash / randomization / subsidies as structural escapes), and the strengthened consequence (which AAT machinery the escapes operationalize). Follow-on spike.
- **Composite-class-inheritance refinement.** The Class 1 (Separated) sub-agents → Class 2 (Partial) composite argument needs cross-checking against `#hyp-directed-separation-under-composition` for consistency. The refinement is structurally sound but deserves dedicated cross-reference work.
- **Bridge from strategic composition to `#form-composition-closure` ε^*.** The macro-description of a strategic composite is an equilibrium *statistic* (joint play distribution) rather than an equilibrium *state*. The closure-defect formulation needs re-specification for this case. Open.
- **Replicator / evolutionary dynamics.** Replicator-based strategic dynamics (Sandholm 2010, *Population Games and Evolutionary Dynamics*) converge to ESS (evolutionarily stable strategies), a subset of Nash. Under which AAT-native update rules does the induced strategic dynamic match replicator? Multi-armed-bandit with softmax choice approximates replicator; gradient-descent on log-likelihood of mixed strategy is replicator exactly. Worth follow-on spike on AAT → evolutionary-dynamics correspondence.
- **Mean-field extension.** $N \to \infty$ limit with each agent interacting with a population *distribution* rather than named others. Natural extension for market / population strategic composition. Lasry-Lions 2007; Huang-Malhamé-Caines 2006. Requires population-scope condition of Section III that is currently marked GAP.
- **Worked-example structure — Cournot substitution for sector-template instantiation.** The unregularized scalar zero-sum case carries a non-trivial conceptual lesson: with $\Phi = a_A + a_B$, both agents' marginal preferences align (+1 each) and the unique Nash equilibrium is the *corner* $(1, 1)$, where joint action contributes no net displacement to the shared state via cancellation of opposing action-coefficients rather than via opposing action-directions. This is preserved as the framing illustration. Sector-template instantiation is then carried by a Cournot-style duopoly (Cournot 1838; Monderer-Shapley

1996 §3) whose interior NE arises from genuine demand-side curvature rather than from an ad-hoc per-agent quadratic action cost. The reasoning trail — including a discarded $-\frac{c}{2}a_i^2$ regularization that placed an interior NE at $(1/c, 1/c)$ but introduced a parameter without economic interpretation, and an algebra error in that regularization's R (stated as $(1-1/c)\sqrt{2}$ where the inscribed-ball radius of the action box around the interior NE is $1-1/c$, the corner-distance over-counting by $\sqrt{2}$) — is recorded in `spikes/spike-strategic-composition.md` §5.2–5.4 along with the original sign error in the potential function. Brainstormed alternatives (LQR with coupled state; network-coordination with quadratic disagreement cost; public-goods with quadratic externalities) all give structurally equivalent template instantiations; Cournot was chosen for textbook authority and for the strategic-substitutes interpretation that retains the partially-opposing-objectives flavor without requiring strict zero-sum.

14.2 *Derived:* Agent Opacity (H_b)

Alongside AAT's heavily formalized *forward* observation quality (how well the agent sees the world — observation ambiguity, model-class fitness, identifiability floor on what the agent can infer), AAT carries a **dual quantity** measuring how well the world sees the agent: **backward predictive uncertainty** H_b, an observer-indexed, horizon-indexed, trajectory-indexed entropy of the agent's future actions given another agent's filtration. Adopted from Hafez et al. 2026 as a first-class multi-agent quantity. H_b is the dual of observation quality U_o: where U_o characterizes how well the agent sees the world, H_b characterizes how well the world sees the agent. It is **sign-flipped in value across regimes**: low H_b (legibility) enables cooperative coordination (13.1); high H_b (opacity) enables adversarial effectiveness (13.2, 13.4). The sign-flip is a direct consequence of AAT's existing signed-coupling structure rather than a separate posit. This segment's emitter-side four-regime classification is the dual of `#der-interaction-channel-classification`'s recipient-side theory; together they close `#adversarial-edge-targeting` as emitter-optimizer paired with recipient-classifier.

Formal Expression.

For agent A on singular trajectory $\mathcal{C}_A$ and observer agent B with filtration $\mathcal{F}_B^t$ (per-trajectory observable history per `#scope-agent-identity`'s token-level commitment):

Definition (agent-opacity-Hb).

$$H_b^{A|B}(t, \tau) := H(a_{A,t+\tau} \mid \mathcal{F}_B^t) \tag{14.7}$$

the entropy of agent A's action at horizon τ conditional on observer B's filtration at time t. **Four indexing arguments:** observer B, time t, horizon τ, trajectory $\mathcal{C}_A$. Each is load-bearing:

- **Observer-indexed.** Different observers (allies with shared infrastructure; adversaries with limited instrumentation; environment itself) have different filtrations $\mathcal{F}_B^t$; H_b varies accordingly.
- **Horizon-indexed.** Immediate-next-action opacity ($\tau = 1$) and long-horizon-plan opacity ($\tau \gg 1$) decouple: an agent may be predictable at immediate action but unpredictable at plan level, or vice versa.
- **Trajectory-indexed.** Per `#scope-agent-identity`, AAT applies to agents on singular trajectories. $H_b^{A|B}$ is the opacity of *this* trajectory's continuation, not a type-level claim.
- **Time-indexed.** Opacity may drift with learning (as B's model of A improves, $H_b^{A|B}(t)$ decreases); steady-state values exist for ergodic regimes.

Under the IDT-observer specialization — B operates as Hafez's Information Digital Twin monitoring (S_A, a_A, S'_A) from outside A's processing — and under ergodicity, $H_b^{A|B}(t, \tau) \to H(S, A \mid S')$ as defined in Hafez et al. 2026. AAT's added features (observer-indexing, horizon-indexing, trajectory-indexing) are the distinctive extensions.

Sign-flip via signed coupling.

The value of H_b^A *to A* depends on the sign of A's coupling to other agents — the same signed-coupling structure that organizes `#der-team-persistence`, `#der-adversarial-destabilization`, and `#deriv-critical-mass-composition`'s (CM2) γ parameter.

Derived (sign-flip-from-signed-coupling).

- **Cooperative coupling ($\gamma^{\text{coop}} > 0$, reducing allies' disturbance).** For B to treat A's action as cooperation rather than disturbance, B must predict A's action well enough to preempt or complement it. Under `#der-interaction-channel-classification`'s recipient-side decomposition, unpredictable ally actions fall into Regime II (magnitude/structural shock) rather than Regime I (informative update). Therefore cooperative coupling effectiveness $\gamma_{A\to B}^{\text{coop}}$ is *increasing in legibility*, equivalently decreasing in $H_b^{A|B}$. Under sub-scope α Gaussian coupling: $\gamma^{\text{coop, effective}} \propto (1 - H_b^{A|B}/H_b^{\max})$.
- **Adversarial coupling ($\gamma^{\text{adv}} > 0$, amplifying target's disturbance).** Predicted attacks are neutralized; unpredicted attacks deliver effective disturbance. Adversarial coupling effectiveness is *increasing in opacity* — the mechanism of adversarial advantage (per `#result-adversarial-tempo-advantage`) operates *through* B's failure to predict A. Under the same sub-scope α setup: $\gamma^{\text{adv, effective}} \propto H_b^{A|B}/H_b^{\max}$.

The sign-flip on H_b's value-to-A lives in the sign of γ itself, not in a different sign on H_b. Cooperative regime ($\gamma^{\text{coop}} > 0$) rewards low H_b; adversarial regime ($\gamma^{\text{adv}} > 0$) rewards high H_b. The same H_b quantity, the same monotone dependence; opposite value-to-A because the signs of the coupling terms differ.

Emitter-side four-regime classification.

Parallel to `#der-interaction-channel-classification`'s recipient-side four regimes, the emitter A sends events that fall into four emitter-side regimes based on A's opacity signal structure and self-model quality:

Formulation (emitter-regimes, dual to #der-interaction-channel-classification).

- **E-I Broadcast.** A emits actions transparently; $H_b^{A|B}$ is low for any observer B with standard instrumentation. Examples: public announcements, published decisions, legible industrial controllers.
- **E-II Selective-signal.** A is transparent to some observers and opaque to others (e.g., shared allied infrastructure gives allies lower H_b than adversaries without that infrastructure). Boundary: differential instrumentation in $\mathcal{F}_B^t$ across observers.
- **E-III Information-hide.** A is uniformly opaque to observers; actions are randomized, encrypted, or routed through dead-drops. $H_b^{A|B}$ near $H_b^{\max}$ for all observers lacking the key / pattern / channel.
- **E-IV Active-deceive.** A emits actions that mispredict — the observer's model of A converges to a *wrong* prediction that differs from the actual action by a larger margin than the same observer's model of the environment would accommodate. Boundary: A's self-model quality (for active-deceive, A must model the observer's model of A well enough to choose actions that exploit it).

The 16-cell emitter-recipient composition (four emitter regimes × four recipient regimes) gives a closed-form *adversarial-targeting arg-max* under `#adversarial-edge-targeting`: the most-valuable-to-attack edge is the one where the product of emitter's opacity-to-target and target's vulnerability-to-shock is maximized. This closes the Section III gap that `#adversarial-edge-targeting` (previously GAP) was reserved for; the segment is now operationalized with targeting-fidelity factor $(1 - H_b^{B|A}/H_b^{\max})$ from A's self-model quality plus the four-regime recipient classification from `#der-interaction-channel-classification`.

Tempo amplification by opacity.

`#result-adversarial-tempo-advantage`'s tempo-multiplier $\gamma_A \mathcal{T}_A$ in `#der-adversarial-destabilization` decomposes into a tempo term and an opacity term:

Derived (tempo-amplification-by-opacity).

$$\mathcal{T}_A^{\text{effective}} = \mathcal{T}_A \cdot \frac{H_b^{A|B}}{H_b^{\max}} \quad \text{(Model D adversarial coupling)} \tag{14.8}$$

The superlinear formula $(\mathcal{T}_A/\mathcal{T}_B)^2$ becomes $(\mathcal{T}_A/\mathcal{T}_B)^2 \cdot (H_b^{A|B}/H_b^{B|A})^2$ under bilateral opacity — a higher-order tensor product with the same exponent $b = 2$ (Model D) or $b = 3/2$ (Model S) from `#result-adversarial-exponent-regimes`. Whether b itself is reshaped under bilateral opacity is open; the leading-order scaling is the tempo-opacity product.

What Is Derived vs. What Is Chosen.

Table 14.2

Property	*Source*	*Strength*
$H_b^{A\|B}(t, \tau)$ definition	Adopted from Hafez et al. 2026; extended with observer / horizon / trajectory indexing per `#scope-agent-identity`	Formulation choice (adoption + AAT-extension)
Reduction to Hafez's $H(S, A \mid S')$ under IDT-observer + ergodic regime	Direct substitution	Derived (exact under IDT + ergodicity)
Sign-flip via signed coupling	Cooperative coupling requires predictability (allies preempt); adversarial coupling operates via disturbance-injection (predicted attack is neutralized)	Derived (from existing `#der-team-persistence` + `#der-adversarial-destabilization` signed-γ structure)
Emitter-side four-regime classification	Dual construction to `#der-interaction-channel-classification`'s recipient-side four regimes	Formulation choice
16-cell emitter-recipient composition closes `#adversarial-edge-targeting`	Product of emitter opacity × recipient vulnerability-to-shock over four × four cells	Derived (arg-max construction)
Tempo-amplification leading-order: $\mathcal{T}^{\text{eff}} = \mathcal{T} \cdot H_b/H_b^{\max}$	First-order substitution into `#result-adversarial-tempo-advantage`'s tempo-multiplier under Model D	Derived (conditional on Gaussian-coupling sub-scope α)
Parameterization-invariance of H_b	H_b is an action-marginal entropy; action space is coordinate-free per `#scope-agent-identity`	Derived
Candidate 4th `#disc-identifiability-floor` instance (generic observer-side form)	H_b's formal structure — "observer cannot predict agent's future action better than $H_b^{A\|B}$" — is a CHT-style no-go at the observer-side-inference task	Discussion-grade (framing; precise external theorem not yet identified)
Candidate opacity ladder for `#disc-separability-pattern`	Transparent-core / partial-transparency / full-opacity across observer filtrations	Formulation choice (ladder proposal)
Effects-spiral opacity amplification (higher H_b → higher γ_A → larger $\dot{V}_B$ → B's actions become more erratic → observer's model of B degrades → higher $H_b^{B\|A}$)	Composition of sign-flip derivation with `#der-adversarial-destabilization`'s effects spiral	Sketch (discussion-grade; specific functional form open)
Dual-filtration apparatus (each agent's M_t carries an other-filtration as feature)	Would unify observer-indexing with `#scope-agent-identity`'s single-trajectory formalism more tightly	Open extension (mild architectural, orthogonal to derivations)

Property	*Source*	*Strength*
Sharp functional form for $\gamma^{\text{adv}}_{\text{effective}} = f(H_b)$	Leading-order: $\gamma \propto H_b$. Exact function depends on sub-scope — Gaussian-coupling linear; sigmoid-coupling saturating	Open per sub-scope

Epistemic Status.

Conditional. Max attainable: *exact* for the Hafez-reduction under IDT + ergodicity; *derived* for the sign-flip via signed-coupling; *formulation choice* for the emitter-regime classification structure; *conditional* for the tempo-amplification formula; *discussion-grade* for the meta-pattern candidate status.

Load-bearing: - The sign-flip derivation from existing signed-γ structure is the segment's core structural contribution — the adversarial/cooperative opacity duality is not a separate posit; it falls out of AAT's existing signed-coupling apparatus. - The emitter-side four-regime classification is a clean dual to `#der-interaction-channel-classification`; its derivation is parallel (boundaries in AAT-native quantities — emitter opacity signal structure, self-model quality, coupling regime). - The closure of `#adversarial-edge-targeting` via the 16-cell composition is a derived arg-max; the previously-stated GAP is filled.

Not established: - Sharp functional forms for $\gamma^{\text{adv}}_{\text{effective}}(H_b)$ outside Gaussian-coupling sub-scope α. - Whether b (the adversarial exponent from `#result-adversarial-exponent-regimes`) is reshaped under bilateral opacity. - Formal fourth `#disc-identifiability-floor` instance (requires external-theorem anchoring not yet identified); discussion-grade framing only. - The effects-spiral's opacity amplification (composition with `#der-adversarial-destabilization`'s spiral) is sketch-level.

Honest Limits.

- **Observer-indexing under complex information structures.** $\mathcal{F}_B^t$ may include shared memory, cryptographic keys, insider knowledge, or partial access to *A*'s internal state. Enumerating all relevant observer-filtration structures for a specific application is task-specific; the segment provides the formal framework, not the per-instance enumeration.
- **Active-deceive (E-IV) requires *A* to model *B*'s model of *A*.** Mutual-modeling regress under partially-opposing objectives connects to `#deriv-strategic-composition`'s joint-Jacobian analysis. Active-deceive is formally reachable only when *A*'s GUC architecture admits modeling another agent's model — typically Class 3 (Coupled; LLM-style) or Class 2 (Partial) architectures.
- **Type-level vs token-level opacity.** H_b is trajectory-indexed per `#scope-agent-identity`. Statements about "the opacity of model X" (aggregated across deployments) are type-level claims outside AAT's formal scope; they require additional machinery (e.g., population-level dynamics per Section III gaps).

Discussion.

Dual to observation quality. U_o characterizes how well the agent sees the world: observation noise, ambiguity, model-class fitness. H_b characterizes how well the world sees the agent: predictability to observers. The duality is structural — both quantify information flow through the agent-environment boundary, in opposite directions. High U_o agent (observes the world well) and low H_b agent (is observed well) are independent properties; an agent can have one without the other.

Closing `#adversarial-edge-targeting`. The segment provides the emitter-side arg-max structure missing from the Section III adversarial machinery. Paired with `#der-interaction-channel-classification`'s recipient-side four-regime decomposition, the full adversarial-targeting problem has a closed-form: choose edges where emitter's H_b (to target) and target's recipient-side vulnerability (Regime II magnitude/structural shock) are jointly maximized. This operationalizes what "inside the opponent's loop" means at the targeting layer — Boyd's aphorism becomes an explicit optimization over the 16-cell emitter-recipient product.

Meta-pattern positioning. - `#disc-identifiability-floor`*:* H_b's structure suggests a generic observer-side floor — "the observer cannot predict the agent's action better than H_b" — that Instances 1/2/3 specialize on specific variables (causal structure / mixture parameters / coupling sign). The generic framing is candidate-status: it lacks a single external-theorem anchor clear enough to match F1's CHT or F13's Cramér-Rao, but H_b appears naturally in Instance 3's coupling-sign unidentifiability and in Instance 1's on-policy detection no-go (an observer watching the agent's on-policy play has non-zero H_b on the agent's interventional regime). - `#disc-separability-pattern`*:* candidate opacity ladder — transparent-core (E-I Broadcast; allies / public interfaces) / structured-repair (E-II Selective-signal; trust-weighted partial instrumentation) / general-open (E-III, E-IV; uniformly opaque or active-deceive). Adds to the ladder count if adopted. - `#disc-additive-coordinate-forcing`*:* H_b's logarithmic form is adopted from Shannon via Khinchin-Aczél axiomatics, imported as an applied external theorem rather than re-forced under an AAT-internal additivity axiom. Cross-agent additivity fails under the coupling regimes AAT cares about (correlated opacity structures break independence). Adjacent family member, parallel to the IB Lagrangian's position — not a primary instance.

Parameterization-invariance composes cleanly. H_b is an action-marginal entropy. Under `#scope-agent-identity`'s (PI) axiom, the action space is coordinate-free; H_b is invariant under change of the agent's internal-state parameterization. This composes with the (PI)/Čencov fourth primary instance of `#disc-additive-coordinate-forcing` without adding a new axiom.

Relation to `#der-directed-separation`. Class 3 (Coupled) agents have high structural opacity to any observer without internal access — their f_M and G_t are entangled, so predicting the next action requires joint state modelling. Class 1 (Separated) agents are more transparent at the interface level because the decomposed update admits separate modelling of epistemic vs. purposeful components. H_b therefore tends to be *architecturally higher* for Coupled (Class 3) agents — a structural consequence of architecture rather than choice, beyond what E-III Information-hide captures.

Hafez integration note. The IDT pattern (Hafez et al. 2026) uses bi-predictability P (how well a sidecar observer can predict the agent's next state-action) and entropy change ΔH as diagnostics. The IDT's reported 89% perturbation-detection accuracy (vs. 44% for reward-based monitoring) operates on the Level-2 structure per `#der-loop-interventional-access`. In AAT terms, the IDT is a low-H_b-preserving observation channel — its presence as a modular sidecar reduces H_b for the operator without increasing the agent's internal complexity. For `03-llm-core/`, this validates that modular monitoring of internally-merged agents is feasible and effective even when the agent itself is architecturally Class 3 (Coupled).

Findings.

Agent Opacity (H_b) as Dual to Observation Quality (U_o). **Brief:** How well the agent sees the world (observation quality U_o) and how well the world sees the agent (backward predictive uncertainty H_b) are formal duals — both quantify information flow through the agent-environment boundary, in opposite directions. H_b is observer-, horizon-, and trajectory-indexed, extending Hafez's $H(S, A \mid S')$ with these load-bearing indices. The same H_b quantity has opposite value-to-the-agent depending on coupling sign: cooperative coupling rewards low H_b (allies must predict to coordinate); adversarial coupling rewards high H_b (adversaries cannot neutralize what they cannot predict). The sign-flip is not a separate posit — it falls out of AAT's existing signed-coupling structure ($\gamma^{\text{coop}} > 0$ for ally-disturbance reduction; $\gamma^{\text{adv}} > 0$ for target-disturbance amplification). Adversarial tempo advantage decomposes into a tempo term and an opacity term: $\mathcal{T}^{\text{eff}} = \mathcal{T} \cdot H_b / H_b^{\max}$, so the superlinear $(\mathcal{T}_A / \mathcal{T}_B)^2$ advantage from `#result-adversarial-tempo-advantage` scales with the bilateral opacity ratio $(H_b^{A|B} / H_b^{B|A})^2$ under Model D.

Impact: Operationalizes Boyd's OODA-loop aphorism ("inside the opponent's loop") as a precise multiplicative relationship between tempo and opacity, replacing a metaphor with a coupled differential structure. Closes the previously-reserved `#adversarial-edge-targeting` gap as a 16-cell emitter-recipient composition (4 emitter regimes × 4 recipient regimes from the dual `#der-interaction-channel-classification`), giving a closed-form arg-max for "where to attack."

The duality with U_o lets the same machinery cover opposite information regimes — high-U_o low-H_b agents (sees well, is seen well) versus low-U_o high-H_b agents (sees poorly, is hidden) — without parallel theorems for each. Direct relevance to multi-agent safety (legibility for cooperation), adversarial robustness (opacity as targeting-vulnerability shaping), and IDT-style monitoring of opaque AI systems (the IDT pattern is a low-H_b-preserving observation channel).

Novelty Claim: *Claim differentiation* on Hafez's H_b. The base quantity is *adopted* from Hafez et al. 2026 with citation and original name; the AAT-distinctive contributions are (i) the observer/horizon/trajectory indexing per `#scope-agent-identity`, (ii) the sign-flip derivation from existing signed-coupling structure rather than as a separate posit, (iii) the emitter-side four-regime classification dual to AAT's recipient-side classification, closing the adversarial-edge-targeting arg-max via the 16-cell composition, and (iv) the tempo-opacity decomposition of `#result-adversarial-tempo-advantage`. The $U_o \leftrightarrow H_b$ duality framing is the integrative move; predictability has been studied in games and information-theoretic security, but the formal-dual treatment with shared signed-coupling structure appears AAT-distinctive.

Related Work:

Table 14.3

ASF Concern	*Prior-art Language*	*Relationship / Positioning*
The base H_b quantity (entropy of agent's action conditional on observer filtration)	Hafez, Khan & Iqbal 2026 *A Mathematical Theory of Agency and Intelligence* — $H(S, A \mid S')$ under IDT-observer (published 2026, found 2026-03)	*formal antecedent* — adopted directly; AAT reduces to Hafez's form under IDT + ergodicity. The four-index extension is the AAT-distinctive layering
OODA loop and tempo-opacity advantage	Boyd 1986–1995 briefings: *Patterns of Conflict*, *The Essence of Winning and Losing* (delivered 1986–1995, found pre-2026)	*conceptual precursor* — supplies the "inside the opponent's loop" intuition for tempo and opacity advantage; the AAT result quantifies the coupling and decomposes the multiplier
Predictability as strategic resource in games	Camerer 2003 *Behavioral Game Theory*; Aumann & Maschler 1995 *Repeated Games with Incomplete Information* (published 1995/2003, found pre-2026)	*conceptual precursor* — predictability and information advantage are well-studied in game theory; the AAT treatment as a formal dual to sensor noise via signed coupling is distinctive
Information-theoretic security and channel opacity	Shannon 1949 *Communication Theory of Secrecy Systems*, Bell System Technical Journal 28(4); modern steganography literature (Cachin 1998) (published 1949/1998, found pre-2026)	*conceptual precursor* — opacity as a quantifiable information-channel property; AAT applies the same machinery to action-prediction rather than message-transmission
Causal Hierarchy Theorem and observer-side identifiability	Bareinboim, Correa, Ibeling & Icard 2022 (published 2022, found 2025)	*adjacent* — observer-side identifiability floor for causal structure; H_b plays an analogous role for action-prediction. Candidate fourth instance of `#disc-identifiability-floor` (discussion-grade)
Bi-predictability P as substrate-independent diagnostic	Hafez et al. 2026 (cited above)	*formal antecedent* — IDT-based bi-predictability complements H_b as the practitioner-facing diagnostic; AAT's observer-indexing makes the dependence on filtration explicit

Search Log: - 2026-04 (*intuition-only* on the U_o-H_b duality framing): no targeted Undermind-grade search has been conducted on whether information-theoretic agent frameworks elsewhere treat predictability-of-agent and observability-of-world as formal duals via shared signed-coupling structure. Hafez 2026's H_b is the explicit formal antecedent for the base quantity; the AAT-distinctive layering (observer/horizon/trajectory indexing + sign-flip derivation + emitter-regime dual) is the segment's contribution. Targeted future search candidates: epistemic game theory (Aumann's tradition); inverse reinforcement learning where the observer infers agent policy and predictability bounds the inference task (Ng & Russell 2000; Ziebart et al. 2008); legibility / readability literature in human-robot interaction (Dragan & Srinivasa 2013); multi-agent identifiability and observability literature. Pre-search expectation: the constituent moves are well-precedented; the integration as formal dual via shared coupling structure within an agent theory may be novel under nominally-comprehensive search. - 2026-03 (*targeted*): Hafez et al. 2026 confirmed as formal antecedent for the base H_b quantity and bi-predictability diagnostic; the IDT-as-low-H_b-channel framing follows directly.

Working Notes.

- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. Three occurrences updated: (1) Honest Limits "Class 2 (LLM-style)" → "Class 3 (Coupled; LLM-style)"; (2) Discussion "Class 2 (fully merged) / Class 1 (modular)" → "Class 3 (Coupled) / Class 1 (Separated)"; (3) Hafez integration note "architecturally Class 2" → "Class 3 (Coupled)". H_b symbol and the E-I/E-II/E-III/E-IV emitter-regime classification are independent of the GUC rename — untouched. Removed at `candidate` stage per FORMAT.md Gate 4.
- The (C-iv) scope route of `#scope-composite-agent` accommodates adversarial composition via equilibrium convergence; the effects-spiral joint-Jacobian eigenvalue condition of `#deriv-strategic-composition` composes with this segment's opacity-amplification story to give a fully-coupled picture of symmetric adversarial dynamics. Full composition is open work.
- Candidate fourth-instance formalization for `#disc-identifiability-floor`: the most natural external-theorem anchor is Fano's inequality (relating H_b to error-probability lower bounds) applied to the observer-side prediction task. Open; not pursued here.
- The 16-cell emitter-recipient composition admits closed-form arg-max only under sub-scope α coupling; general non-convex coupling requires per-case optimization.
- Dual-filtration apparatus (M_A carries $\mathcal{F}_B^t$ as feature, M_B carries $\mathcal{F}_A^t$ as feature) would tighten the formalism by unifying observer-indexing with the single-trajectory scope of `#scope-agent-identity`. Architecturally clean; not needed for the derivations here.

14.3 *Observation:* Result: Adversarial Exponent Regimes

The adversarial tempo advantage exponent — the power b in $\|\delta_B\|/\|\delta_A\| \sim (\mathcal{T}_A/\mathcal{T}_B)^b$ — is not a single number. It depends on two structural features of the disturbance: whether the adversarial coupling enters as deterministic drift (Model D) or stochastic noise (Model S), and whether the coupling dominates the base disturbance rate. Three regimes, with the coupling-dominant exponents now derived analytically from the respective disturbance models.

Formal Expression.

Regime 1: Model D (deterministic drift), coupling-dominant. When adversarial coupling enters as a persistent directional disturbance ($\rho_B = \rho_{\text{base}} + \gamma \cdot \mathcal{T}_A$, GA-2) and coupling dominates ($\gamma \cdot \mathcal{T}_B \gg \rho_{\text{base}}$):

Derived (adversarial-exponent-regimes, from Model D/S steady states + coupling model; validated by simulation).

$$b = 2 \qquad \text{(simulation: 1.999)} \tag{14.9}$$

Derived from the Model D steady state $\|\delta\|_{ss} = \rho/\mathcal{T}$ (Prop A.1). See 13.4.

Regime 2: Model S (stochastic noise), coupling-dominant. When adversarial coupling enters through the noise scale of zero-mean perturbations ($\sigma_B = \sigma_{\text{base}} + \gamma \cdot \mathcal{T}_A$, GA-2S) and coupling dominates:

$$b = \frac{3}{2} \qquad \text{(simulation: 1.481)} \tag{14.10}$$

Derived from the Model S steady state $\|\delta\|_{\text{rms}} = \sigma_w/\sqrt{2\mathcal{T}}$ (Prop A.1S). The $1/\sqrt{\mathcal{T}}$ scaling (vs. $1/\mathcal{T}$ for Model D) removes one half-power from the denominator, reducing the exponent from 2 to 3/2. See 13.4.

Regime 3: Non-coupling-dominant. When base disturbance is comparable to or exceeds the adversarial coupling ($\rho_{\text{base}} > rsim\gamma \cdot \mathcal{T}_B$):

$$b \to 1.0\,\text{(Model D)} \quad \text{or} \quad b \to 0.5\,\text{(Model S)} \tag{14.11}$$

The exponent degrades smoothly as the base-to-coupling ratio increases. The asymptotic limits are derived (they reflect the $1/\mathcal{T}$ or $1/\sqrt{\mathcal{T}}$ scaling without the coupling numerator); the smooth interpolation is empirical.

Table 14.4

$\rho_{base}/(\gamma \cdot \mathcal{T}_B)$	*Exponent (deterministic)*	*Exponent (stochastic)*
0.002	1.999	1.481
0.20	1.877	1.101
2.0	1.445	0.791
6.3	1.213	0.577

Epistemic Status.

Exact conditional on disturbance model. The coupling-dominant exponents are derived, not empirical: $b = 2$ follows from the Model D steady state (Prop A.1) and the coupling model; $b = 3/2$ follows from the Model S steady state (Prop A.1S) and the coupling model. The simulation results (6 variants, multiple parameter sweeps) now serve as validation of the derived exponents, not as their epistemic foundation. The non-coupling-dominant limits ($b \to 1, b \to 1/2$) are derived asymptotically; the smooth interpolation between coupling-dominant and non-coupling-dominant is empirical. What remains empirical is whether a given real adversarial interaction is better modeled as Model D or Model S — that is a domain question, not a theory question.

Discussion.

The disturbance model determines the exponent. The mismatch dynamics (3.9) now distinguish two disturbance models: Model D (bounded deterministic, GA-2) with steady-state $\rho/\mathcal{T}$, and Model S (stochastic zero-mean, GA-2S) with steady-state $\sigma_w/\sqrt{2\mathcal{T}}$. The different steady-state scaling is the root cause of the different exponents. This resolves the ambiguity that previously existed in the single-ρ formulation.

Why the squared law held for the coupling-dominance sweep. In Variant A, the coupling enters as deterministic drift: $\rho_B = \rho_{\text{base}} + \gamma \cdot \mathcal{T}_A$, and the steady state is $\|\delta_B\| = \rho_B/\mathcal{T}_B$. The ratio $\|\delta_B\|/\|\delta_A\|$ in the coupling-dominant limit gives $(\mathcal{T}_A/\mathcal{T}_B)^2$ directly.

Nonlinear correction creates thresholds, not lower exponents. For saturating, sigmoid, and breakdown correction functions under deterministic drift, the issue is not a reduced exponent but a catastrophic divergence when ρ exceeds the correction capacity ($\rho > \mathcal{T} \cdot R$). This is exactly the persistence threshold failure (4.4), observed directly in simulation.

Domain interpretation. Whether a given opponent's tempo increase causes deterministic drift or stochastic noise depends on the domain: - Military: an opponent who maneuvers faster creates systematic positional change (drift, $b \approx 2$) - Market: a competitor who acts unpredictably creates noise in signals ($b \approx 1.5$) - Software: a fast-changing API creates systematic drift in the codebase state (drift) - Adversarial ML: an opponent who varies attack vectors increases observation noise ($b \approx 1.5$)

Working Notes.

- The interpolation between drift and noise regimes (Variant B) shows smooth transition, not a sharp boundary. At mixed drift-noise coupling, the exponent lies between the two asymptotes. The drift fraction $f = \mu/(\mu + \sigma)$ continuously parameterizes the transition.
- The exponent of 1.05 from the original sim2 was not a falsification of Corollary 11.2 — it reflected a stochastic model (noise-variance coupling) tested in a non-coupling-dominant regime. The original simulation was testing the wrong regime for the ODE's prediction.
- Simulation code: `../../spikes/track-b-nonlinear-sims/variants/variant_ab_drift.py`, `variant_cd_regimes.py`. Results: `variant_ab_results.md`, `variant_cd_results.md`.

14.4 *Observation:* Gated Tempo Advantage

Observation noise collapses the adversarial tempo advantage. When agents observe their mismatch through a noisy channel, the faster agent's additional corrections become noisy, partially offsetting its tempo advantage. The optimal gain (3.6) partially restores the advantage but cannot fully recover it.

Formal Expression.

In a two-agent adversarial system with observation noise σ_{obs} added to each agent's mismatch signal:

Observation (obs-gated-tempo-advantage, from track-b Variant E).

Table 14.5

σ_{obs}	*Exponent (fixed η)*	*Exponent (optimal η^*)*
0.00	1.04	1.04
0.10	1.00	0.97
0.20	0.92	0.94
0.50	0.60	0.63
1.00	0.18	0.40

At $\sigma_{\text{obs}} = 1.0$ (10x the process noise), the fixed-gain adversarial exponent drops from ~ 1.0 to ~ 0.2 — tempo advantage nearly vanishes. The Riccati-optimal gain restores it to ~ 0.4, more than doubling the advantage but not recovering the noise-free level.

The mechanism. When observation noise is high, each correction step adds noise to the mismatch estimate. The faster agent makes more corrections per unit time, each noisy, partially offsetting the benefit of higher tempo. The optimal gain mitigates this by reducing η to match the noise level — correcting less aggressively but more accurately.

Epistemic Status.

Empirical. Max attainable: derived (the mechanism is analytically tractable via Riccati analysis of noisy AR(1) processes). The observation that noise degrades advantage is confirmed by simulation. The optimal gain's partial restoration is consistent with the uncertainty ratio principle (3.6: $\eta^* = U_M/(U_M + U_o)$). The quantitative degradation curve (b vs. σ_{obs}) is empirical at these parameters; a general analytical expression would require solving the coupled noisy-AR(1) system.

Discussion.

Observation quality gates tempo advantage. Boyd insisted that the quality of Orient (observation processing) matters more than raw OODA speed. The simulation results show a formal analog of this pattern: faster tempo with noisy observations (σ_{obs} high) gives nearly zero advantage over a slower agent with equally noisy observations. The tempo advantage is gated by observation quality — consistent with Boyd's emphasis, though the model captures a specific mechanism (noisy correction steps) rather than the full richness of Orient processing.

The optimal gain helps most in the moderate-noise regime. At $\sigma_{\text{obs}} = 0.05$ (observation noise half of process noise), the optimal gain cuts steady-state mismatch by 52% compared to fixed gain. At very high noise, the improvement is less dramatic in absolute terms but more important relatively (0.40 vs. 0.18 exponent).

Practical implication. An agent facing an adversary with superior tempo should invest in degrading the adversary's observation quality rather than trying to match their speed. Conversely, an agent with superior tempo should protect its observation channels — the tempo advantage is only as good as the observation quality that supports it.

Connection to code quality. In the software domain (`#der-code-quality-as-observation-infrastructure` — cross-component reference, see `02-tst-core/`), code quality IS observation infrastructure. A well-structured codebase provides low-noise observations (clear tests, readable code, explicit interfaces). A poorly structured codebase adds observation noise to every development cycle, degrading the developer's effective tempo regardless of how fast they work.

Recipient-side mechanism. High $U_{o,B}$ pushes adversarial events below the observability floor (boundary (I-c) in 13.3). Events that would otherwise land in Regime II (destabilizing) instead fall into Regime III (ambient noise): they contribute to $\sigma^2_{w,B}$ without producing destabilizing mismatch. The tempo-advantage exponent drops because the *fraction* of *A*'s events landing in Regime II shrinks — *A*'s tempo still matters, but more of it is dissipated into the noise floor. This is the recipient-side expression of the rate boundary.

Working Notes.

- The finding that fixed $\eta = 0.1$ is "remarkably robust" to observation noise (42% degradation at $\sigma_{\text{obs}} = 10 \times q_{\text{env}}$) suggests that conservative gains are a reasonable default for environments with unknown noise levels. The cost of being slightly below optimal is much less than the cost of being above optimal (overcorrection amplifies noise).
- The interaction between observation noise and adversarial exponent regime (drift vs. stochastic) has not been tested. The Variant E results use stochastic coupling only. Whether observation noise degrades the deterministic-drift exponent ($b = 2$) by the same proportion is an open question.
- Simulation code: `../../spikes/track-b-nonlinear-sims/variants/variant_ef_extensions.py`. Results: `variant_ef_results.md`.

14.5 *Result:* Per-Dimension Persistence

The scalar persistence condition overestimates adaptive capacity when the agent's correction gain varies across dimensions. The weak dimension is the bottleneck — it dominates the aggregate mismatch regardless of performance on strong dimensions. The correct condition is per-dimension, with the form depending on whether the disturbance is deterministic (Model D) or stochastic (Model S).

Formal Expression.

For an agent with d-dimensional mismatch $\delta_t \in \mathbb{R}^d$, diagonal correction gain $\eta = \text{diag}(\eta_1, \ldots, \eta_d)$, and per-dimension disturbance:

Result (per-dimension-persistence).

Model D: Deterministic Per-Dimension Threshold. Under bounded disturbance $|w_k(t)| \leq \rho_k$ (GA-2, per dimension), the per-dimension steady-state mismatch is:

$$|\delta_k|_{ss} = \frac{\rho_k}{\alpha_k} \tag{14.12}$$

Persistence requires $\alpha_k > \rho_k / R_k$ for each dimension, or in linear operational form:

$$\mathcal{T}_k > \frac{\rho_k}{\delta_{\text{critical},k}} \quad \text{for each dimension } k \tag{14.13}$$

This is the deterministic worst-case bound — exact under bounded disturbance by the same Lyapunov argument as Prop A.1, applied per dimension.

Model S: Stochastic Per-Dimension Steady State. Under stochastic disturbance $w_{k,t} \sim N(0, \rho_k^2)$ (GA-2S, per dimension), the discrete AR(1) process $\delta_{k,t+1} = (1 - \eta_k)\delta_{k,t} + w_{k,t}$ has stationary distribution:

$$\delta_k \sim N\left(0,\ \frac{\rho_k^2}{2\eta_k - \eta_k^2}\right) \tag{14.14}$$

The stationary distribution supplies three task-adequacy criteria, each with its own threshold. The choice of criterion is an engineering decision; the three are related by exact constants for Gaussian δ_k.

(a) RMS bound (mean-square, matches the scalar form in 4.4):

$$\sqrt{E[\delta_k^2]} = \frac{\rho_k}{\sqrt{2\eta_k - \eta_k^2}} \tag{14.15}$$

Requiring $\sqrt{E[\delta_k^2]} < \delta_{\text{critical},k}$ and using the small-η_k approximation $2\eta_k - \eta_k^2 \approx 2\eta_k$:

$$\boxed{\eta_k > \frac{\rho_k^2}{2\,\delta_{\text{critical},k}^2}} \quad \text{(RMS criterion)} \tag{14.16}$$

This is the scalar Model S threshold in 4.4 ($\alpha > n\sigma_w^2/(2R^2)$) applied per dimension.

(b) MAE bound (mean absolute error; bounds the expected deviation rather than its square):

$$E[|\delta_k|] = \sqrt{E[\delta_k^2]} \cdot \sqrt{\frac{2}{\pi}} = \frac{\rho_k}{\sqrt{2\eta_k - \eta_k^2}} \cdot \sqrt{\frac{2}{\pi}} \tag{14.17}$$

Requiring $E[|\delta_k|] < \delta_{\text{critical},k}$:

$$\eta_k > \frac{\rho_k^2}{\pi\, \delta_{\text{critical},k}^2} \quad \text{(MAE criterion)} \tag{14.18}$$

MAE is smaller than RMS by the factor $\sqrt{2/\pi} \approx 0.798$, so the MAE threshold is $2/\pi \approx 0.637$ times the RMS threshold. The criteria differ by a constant but bound different quantities; applying the same numerical $\delta_{\text{critical},k}$ under both does not mean the same thing.

(c) Probability bound (tail-risk criterion, for applications where occasional excursions matter):

$$P\big(|\delta_k| > \delta_{\text{critical},k}\big) < \epsilon \iff \eta_k > \frac{\rho_k^2 \cdot z_{1-\epsilon/2}^2}{2\, \delta_{\text{critical},k}^2} \tag{14.19}$$

where $z_{1-\epsilon/2}$ is the two-sided Gaussian quantile. The probability bound at $\epsilon = 0.05$ (two-sided $z \approx 1.96$) is about $1.96^2 \approx 3.84$ times the RMS threshold — stricter because it bounds tail excursions rather than typical magnitudes.

Recommended primary form. The RMS criterion (a) is the canonical form for Model S persistence, matching the scalar treatment in 4.4 and the Lyapunov-based derivation in A (Prop A.1S). The MAE and probability-bound variants are provided for applications where those are the natural task-adequacy measures. All three thresholds are quadratic in $\rho_k/\delta_{\text{critical},k}$ (not linear as in Model D), reflecting the $1/\sqrt{\alpha}$ scaling of the Model S stationary variance.

Common Structure. The aggregate L_2 mismatch $\|\delta\| = \sqrt{\sum_k \delta_k^2}$ is dominated by the dimension with the largest ρ_k/η_k ratio (Model S) or ρ_k/α_k ratio (Model D). The qualitative conclusion — the weak dimension is the bottleneck — holds for both models.

Epistemic Status.

Exact conditional on disturbance model. Both per-dimension forms are now derived from their respective disturbance models:

1. **Model D threshold** ($\mathcal{T}_k > \rho_k/\delta_{\text{critical},k}$) follows from Prop A.1 applied per dimension — the same Lyapunov argument with bounded disturbance, restricted to each coordinate. This is exact under GA-2 + GA-3.

2. **Model S steady state** follows from the AR(1) stationary distribution under Gaussian disturbance (GA-2S): $\delta_k \sim N(0, \rho_k^2/(2\eta_k - \eta_k^2))$. The RMS, MAE, and probability-bound thresholds are all exact under this distribution, differing by the constants 1, $\sqrt{2/\pi}$, and $z_{1-\epsilon/2}$ respectively. The RMS form $\eta_k > \rho_k^2/(2\delta_{\text{critical},k}^2)$ matches the scalar Model S treatment in 4.4 applied per dimension. The 4-significant-figure simulation match validates Model S, not Model D.

The previously noted "regime mixing" is resolved: the two threshold forms belong to different disturbance models. The Model D threshold is linear in ρ_k; the Model S threshold is quadratic. The 72% scalar overestimate and weak-dimension bottleneck are structural properties that hold under both models.

Discussion.

Scalar tempo overestimates. In a 3D system with gains $\eta = (0.15, 0.03, 0.03)$ and disturbances $\rho = (0.20, 0.20, 0.02)$:

Table 14.6

Dimension	η_k	ρ_k	ρ_k/η_k	$E[\|\delta_k\|]$
1 (well-tracked)	0.15	0.20	1.33	0.303
2 (weak)	0.03	0.20	6.67	0.656
3 (unimportant)	0.03	0.02	0.67	0.066

Scalar prediction: $\rho/\mathcal{T} = 0.284/0.21 = 1.35$. Actual $\|\delta\|_{L_2} = 0.785$. Overestimate: 72%. Dimension 2 alone accounts for 84% of the L_2 mismatch.

Isotropic allocation dominates. Equalizing the same total gain budget ($\eta = 0.07$ per dimension) reduces $\|\delta\|_{L_2}$ from 0.785 to 0.685 — a 13% improvement — because it reduces the bottleneck effect on the weak dimension.

Adversarial exploitation. An adversary who identifies the target's weak dimension can concentrate disturbance there. Targeted attack (80% on the weak dimension) amplifies the mismatch ratio by 17% (from 2.70 to 3.15). The real danger is structural: if the weak dimension's mismatch exceeds its critical threshold ($R_{\max}$), correction fails on that dimension while the aggregate $\|\delta\|_{L_2}$ may still look manageable. Per-dimension monitoring is essential.

Implications for the persistence condition. Like the scalar result, per-dimension persistence addresses *structural persistence* (see Persistence in `LEXICON.md`) — whether the correction machinery on each dimension can outpace that dimension's disturbance rate. An agent can be structurally persistent on every dimension while still being operationally fragile on one (near its per-dimension R_k boundary). The scalar persistence condition (4.4) remains correct as a *necessary* condition: if the aggregate tempo is insufficient, the agent fails. But it is not *sufficient* — an agent can satisfy the scalar condition while failing on a single dimension. The per-dimension condition has two forms: Model D ($\mathcal{T}_k > \rho_k/\delta_{\text{critical},k}$, exact under bounded disturbance) and Model S ($\eta_k > \rho_k^2/(\pi \cdot \delta_{\text{critical},k}^2)$, exact under Gaussian disturbance). Both predict per-dimension failure correctly; the choice depends on the disturbance character in the domain.

Connection to multi-agent systems. The per-dimension result has a direct multi-agent analog: in a composite agent, each sub-agent's contribution to composite tempo may be strong in some dimensions and weak in others. The composite's persistence requires coverage across all relevant dimensions — a team of specialists who each handle one dimension well composes better than a team of generalists who are mediocre at everything, provided the dimension assignment matches.

Findings.

The Weakest-Link Dimensional Persistence Law. **Brief:** When mismatch is multi-dimensional, persistence is governed by the worst-served dimension, not by aggregate or average performance. The scalar persistence condition (`#result-persistence-condition`) overestimates adaptive capacity whenever per-dimension correction gains differ from per-dimension disturbance rates. The correct condition is *per-dimension*: $\alpha_k > \rho_k/R_k$ under bounded disturbance (Model D, linear in ρ_k), or $\eta_k > \rho_k^2/(2\,\delta^2_{\text{critical},k})$ under Gaussian disturbance (Model S, quadratic in ρ_k). The aggregate L^2 mismatch is dominated by the dimension with the largest ρ_k/η_k ratio. A simulated 3D system shows the scalar form overestimating adaptive capacity by 72% with a single dimension accounting for 84% of the L^2 mismatch; equalizing the gain budget across dimensions (isotropic allocation) reduces aggregate mismatch by 13% by raising the bottleneck dimension's gain.

Impact: Establishes that survival in any multi-attribute environment is a *min* operation, not a sum or average — a structural critique of scalar capability metrics in adversarial and high-stakes settings. Adversarial implication is sharp: an opponent who identifies the target's weak dimension can concentrate disturbance there, amplifying the mismatch ratio asymmetrically while the aggregate may still appear acceptable. Per-dimension monitoring is therefore not optimization polish but a structural requirement for any agent operating under directed pressure. Carries through composition: a team of specialists each handling one dimension well composes better than a team of generalists mediocre at everything — provided the dimension assignment matches the actual disturbance structure. Calibrates how much the scalar form misleads (Model D linear, Model S quadratic in ρ_k), so the cost of conflating the two is itself quantified rather than asserted.

Novelty Claim: *Claim differentiation* on per-dimension Lyapunov stability. The per-coordinate Lyapunov argument is standard (control theory routinely does diagonal stability analysis); weakest-link arguments appear throughout reliability theory; multi-attribute critiques of scalar evaluation are familiar in decision theory. The AAT-distinctive contributions are (i) the explicit Model D / Model S decomposition with the corresponding linear vs quadratic threshold scaling in ρ_k, (ii) the quantitative overestimate calibration (~72% in a simulated 3D system) showing the scalar form is not just incomplete but materially misleading, and (iii) the connection to adversarial *concentration* — an opponent's optimal targeting strategy is to maximize ρ_k at the weakest η_k, not to spread effort uniformly.

Related Work:

Table 14.7

ASF Concern	*Prior-art Language*	*Relationship / Positioning*
Per-coordinate Lyapunov stability and ultimate boundedness	Khalil 2002 *Nonlinear Systems* (3rd ed.), Prentice Hall, chapters 4 and 9 (published 2002, found pre-2026)	*formal antecedent* — supplies the per-coordinate Lyapunov machinery; the per-dimension result is its application to AAT's correction structure under both bounded and Gaussian disturbance
AR(1) stationary distribution under Gaussian forcing (Ornstein-Uhlenbeck stationary)	Uhlenbeck & Ornstein 1930 *Physical Review* 36:823; Karatzas & Shreve 1991 *Brownian Motion and Stochastic Calculus* (published 1930/1991, found pre-2026)	*formal antecedent* — supplies the AR(1) stationary form used in the Model S threshold derivation
Weakest-link reliability / serial-system reliability	Gnedenko, Belyayev & Solovyev 1969 *Mathematical Methods of Reliability Theory*; Barlow & Proschan 1975 *Statistical Theory of Reliability* (published 1969/1975, found pre-2026)	*conceptual precursor* — the min-operation intuition for serially-dependent failures; AAT instantiates this for adaptive correction rather than for component failure
Multi-attribute decision theory and aggregate-vs-attribute critique	Keeney & Raiffa 1976 *Decisions with Multiple Objectives*, Wiley (published 1976, found pre-2026)	*conceptual precursor* — recognizes that aggregate scoring obscures per-attribute deficits; AAT supplies the dynamics-side analog with explicit failure thresholds

ASF Concern	*Prior-art Language*	*Relationship / Positioning*
Adversarial robustness and per-feature attack budgets	Szegedy et al. 2014 *Intriguing Properties of Neural Networks*, ICLR; Madry et al. 2018 *Towards Deep Learning Models Resistant to Adversarial Attacks*, ICLR (published 2014/2018, found pre-2026)	*adjacent* — adversarial-ML literature on per-feature perturbation budgets; AAT's concentration analysis (ρ_k at weakest η_k) gives an adaptive-system framing of the same vulnerability
Critique of scalar AI capability metrics	Various AI safety / evaluation literature (Hendrycks et al. 2021 benchmarks; capability evaluation methodology) (published 2021–, found 2026)	*adjacent* — the AAT result supplies a dynamics-grounded structural argument for why aggregate capability scores under-predict adversarial failure

Search Log: - 2026-04 (*intuition-only* on the integrated Model D / Model S framing): per-coordinate Lyapunov stability is standard control theory; weakest-link reliability and multi-attribute critiques of aggregate scoring are well-precedented in their own literatures. The AAT-distinctive contributions are (i) the side-by-side Model D / Model S threshold scaling difference (linear vs quadratic in ρ_k), (ii) the calibrated simulation overestimate as quantitative evidence the scalar form is misleading rather than merely incomplete, and (iii) the adversarial-concentration framing connecting weakest-dimension targeting to the per-dimension threshold. Targeted future search candidates: robustness in adversarial ML (per-feature adversarial attack literature, Goodfellow-Madry line); capability-evaluation methodology in AI safety (the critique-of-scalar-metrics framing has natural relevance there); portfolio-theory analogs of weakest-dimension exposure (Markowitz tradition with downside-risk concentration); reliability-theory weakest-link dynamics with stochastic forcing. - 2025 (*targeted*): Khalil 2002 confirmed as formal antecedent for per-coordinate Lyapunov machinery; the segment cites it inline.

Working Notes.

- The diagonal-correction assumption: closed. The matrix-Loewner persistence condition `#deriv-matrix-persistence-condition` lifts the per-coordinate form to general (not-necessarily-diagonal, not-necessarily-axis-aligned-with-D_δ) matrix tempo $\mathcal{T}$. Cross-dimensional correction produces off-diagonal entries in $\mathcal{T}$; under those off-diagonals, per-coordinate is *unsafe* — there exist regimes where per-coordinate declares persistence on a system whose stationary mismatch will exceed task-adequacy along the diagonal direction (the §4 counterexample of `#deriv-matrix-persistence-condition` is the constructive demonstration: $\mathcal{T} = \begin{pmatrix} 1 & -0.9 \\ -0.9 & 1 \end{pmatrix}$, $\Sigma_w = I$, $\delta_{\text{critical}} = (1.7, 1.7)$ — per-coordinate says PASS, matrix-Loewner correctly says FAIL with the bad direction $(1,1)/\sqrt{2}$). The matrix-Loewner form is the canonical anisotropic persistence condition; per-coordinate is its diagonal-$\mathcal{T}$-axis-aligned-D_δ special case, sharp where the coordinate basis happens to be the correction-machinery's eigenbasis and unsafe outside it. The weak-direction bottleneck argument of this segment generalizes from "the weak coordinate dimension" to "the weak eigendirection of Σ_∞ relative to D_δ" in the matrix form.
- The tensor formulation of tempo (tracking per-dimension adaptive capacity) is now in `#def-adaptive-tempo`'s Tensor extension sub-block: $\mathcal{T} = \sum_k \nu^{(k)} \cdot K^{(k)}$ with $K^{(k)} = (H_M + H_L^{(k)})^{-1} H_L^{(k)}$ the matrix gain operator derived in `#deriv-fisher-local-update-gain`. The per-dimension persistence condition $\mathcal{T}_k > \rho_k / \delta_{\text{critical},k}$ is the diagonal projection of the tensor form along each natural-gradient direction; the matrix gain $K^{(k)}$ is the per-coordinate primitive this result invokes. Open: tightening this result's Formal Expression to invoke the tensor form directly rather than per-coordinate scalars would make the connection structural rather than discursive.
- Simulation code: `../../spikes/track-b-nonlinear-sims/variants/variant_ef_extensions.py`. Results: `variant_ef_results.md`.

14.6 *Discussion:* Additional Implications & Discussion

The chapter is where Part III closes, and several distinct strands converge here that have been building across the preceding chapters. The strategic-composition derivation lifts the symmetric partially-opposing case from Ch.4's scalar-coupling framing to equilibrium machinery, completing the contraction-to-equilibrium hand-off that Ch.4 surfaced as a boundary. Agent opacity formalizes the dual of observation quality — `#der-agent-opacity` is the chapter's strongest single segment, with rich `## Findings` content — and the 16-cell adversarial-targeting decomposition closes a previously-reserved gap. The adversarial-exponent regime taxonomy ($b = 2, 3/2, \sim 1$) gives the superlinear tempo advantage its disturbance-model decomposition; observation-noise gating gives the structural condition under which the superlinear advantage collapses to roughly linear. Per-dimension persistence — formally located here but content-wise belonging with Part I Ch.4 — gives adversarial concentration its mathematical form. The chapter delivers the framework's first operational instance of identifiability-floor pattern Instance 3 (composite-contraction certification from component-marginal data), and the closing notes hand off cleanly to TST and logogenic-agents work.

Strategic composition lifts contraction to equilibrium.

`#deriv-strategic-composition` is the chapter's central machinery move. When sub-agents are individually Class 1 (Separated — directed separation holds by construction) but pursue *partially-opposing* objectives, *the composite is necessarily Class 3 (Coupled)*. The architectural classification from Part II Ch.1 is *not preserved under composition with goal divergence*. The framework's contribution is making this precise via equilibrium-convergence framing: under sub-scope α' (potential games — Monderer-Shapley 1996; monotone games — Rosen 1965 diagonally-strictly-concave), AAT's persistence machinery transfers cleanly with A2'-analog conditions derived; under sub-scope β', the framework reaches only CCE set-convergence rather than last-iterate convergence (the honest scope limit). The segment also provides the formal home for *effects spiral* via the joint-Jacobian eigenvalue condition — what previously read as a metaphor for cascading misalignment becomes a derivable structural property of the composite.

The implications-side payoff has three threads worth surfacing. First, the result *closes the framework's contraction-to-equilibrium hand-off* that Ch.4's `#result-contraction-template` named as a boundary. Saddle-point equilibria break attracting-fixed-point under Slotine 2003; passivity universality fails under adversarial inputs; Daskalakis et al. 2018's last-iterate non-convergence completes the trio of obstructions. The strategic-composition derivation supplies the *replacement* machinery for those obstructions: equilibrium theory applies in the partially-opposing case where contraction analysis fails. The framework's posture is that *scope-honesty is load-bearing architecture* — knowing where each tool applies cleanly is more useful than trying to extend one tool across all regimes, and the chapter delivers the equilibrium-theoretic tool that handles where contraction stops.

Second, the Class-1-subs → Class-3-composite result composes with `#der-directed-separation`'s architectural classification into a sharper structural commitment: *the framework cannot avoid Class 3 (Coupled) treatment by limiting attention to Class 1 (Separated) sub-agents*. As soon as sub-agents have partially-opposing objectives — which is the empirical norm in multi-agent settings, including most human organizations and many ML systems — the composite is Class 3 (Coupled) regardless of the sub-agents' architectural class. The wrapping construction from Ch.2 (`#der-class-coercion-via-wrapping`) gives the constructive route in the other direction (Class 3 component → Class 1 composite via external scaffolding); the strategic-composition derivation gives the failure direction (Class 1 sub-agents → Class 3 composite under goal divergence). The framework's posture is that the architectural classification is a *property of the dynamics*, not of the components alone, and goal divergence is one of two operations (alongside adversarial coupling pressure from `#disc-adversarial-coupling-pressure`) that move modularity in the wrong direction.

Third, the chapter formalizes mechanism-design impossibility (Gibbard-Satterthwaite / Myerson-Satterthwaite / Arrow) as a *candidate fifth instance* of the identifiability-floor pattern (Instance 5). The structural shape is the same: a no-go theorem from a classical literature (mechanism design) maps onto a specific AAT setting (multi-agent composition with strategic incentives) and identifies the unique broadly-available escape via AAT machinery (the equilibrium-convergence machinery in this segment). The promotion of Instance 5 to formal status awaits a focused derivation cycle; the chapter flags the candidate and the structural shape.

Symbiogenic composition and the asymmetric absorption mechanism.

`#hyp-symbiogenic-composition` (formally in Ch.1) gives the chapter its account of *how composite agents form* from sub-agents that were previously not composed: asymmetric absorption, where a host integrates an endosymbiont and the resulting composite's teleological unity U_O crosses the scope threshold from below. The implication is concrete: the framework recovers the empirical pattern of biological symbiogenesis (mitochondria, chloroplasts, gut microbiota-host integration) and organizational acquisition (objective absorption, function transfer, autonomy reduction) as the *asymmetric-parameter limit* of `#deriv-critical-mass-composition`'s closed form. Specifically, when sub-agent parameters $(\alpha_A, R_A, \gamma_A)$ and $(\alpha_B, R_B, \gamma_B)$ are *strongly asymmetric* (one sub-agent dominates), the composite sector constant α_c collapses toward the dominant sub-agent's α, and the absorbed sub-agent's objectives are progressively folded into the dominant sub-agent's via the U_O multiplier on γ. The framework's posture is that symbiogenesis is *not a separate kind of composition* — it is the asymmetric-parameter limit of the same composition machinery that handles the matched-Tier dyad and the heterogeneous-Tier cases.

The cross-domain transfer is concrete enough to ground in specific examples. The eukaryotic-cell formation (Margulis's serial endosymbiosis) is the canonical biological instance: an archaean host absorbing a bacterial endosymbiont produces a composite with the host's broader objective surface and the endosymbiont's specialized metabolic function. Corporate acquisition produces the same dynamics at a different scale: the acquirer's strategic surface absorbs the target's operational function, with the target's objectives folded in through the U_O alignment process. The framework's contribution is the *mathematical form* — the closed-form critical-mass derivation specialized to the asymmetric-parameter limit — that makes the symbiogenic process predictable rather than purely empirical.

Modular safety architectures fail under goal divergence.

The most consequential implication of strategic composition for AI-safety practice is what `#28` in the catalog names but does not yet have a segment home: modular safety architectures of the form "compose modular safety modules with a central planner" are *structurally guaranteed to fail under goal divergence*. The argument composes `#deriv-strategic-composition` (Class-1 sub-agents with partially-opposing objectives → Class-3 composite) directly: when safety modules and the central planner have non-aligned objectives — exactly when the safety modules are *needed* (to constrain the planner against its own objectives that may produce undesired behavior) — the composite is structurally Class 3 (Coupled), and the modular guarantees of the individual safety modules do not transfer to the composite.

The framework's reading of several empirical AI-safety patterns follows directly. Red-team penetrations of constitutional AI exploit the structural mismatch between the constitutional constraints (one set of objectives) and the model's training objectives (another set); under partial opposition the composite has Class 3 (Coupled) dynamics regardless of each component's nominal modularity. Mesa-optimizer formation within nominally-modular systems (Hubinger et al. 2019) instantiates the same structural failure: a sub-component developing its own objective surface that opposes the outer system's objective produces partial opposition, and the composite's modular guarantees collapse. The recurring difficulty of "scalable oversight" without per-module identifiability is the chapter's identifiability-floor Instance 3 instance manifesting: composite contraction certification from component-marginal data is structurally impossible (Liberzon 2003), and oversight regimes that rely on component-only inspection cannot, in general, certify the composite's safety properties.

The framework's prescription is not that modular safety architectures cannot work — it is that they require either *goal-alignment commitment by design* (sub-agents constructed to share objectives, removing the partial-opposition condition) *or Class 3 (Coupled) treatment from the start* (acknowledging the composite is structurally Coupled and applying AAT's wrapping construction at the composite scope, with the resulting Brooks's-Law tempo cost and structural-leakage bound). The implications-side payoff for AI-safety methodology is concrete: any safety architecture that *assumes* modularity preserves at composition is making a structural commitment that holds only under goal alignment. Where goal alignment cannot be designed-in by construction (which is the empirical case for most current deployment settings), the framework prescribes the wrapping construction with explicit leakage accounting.

Agent opacity, the $U_o \leftrightarrow H_b$ duality, and 16-cell adversarial targeting.

`#der-agent-opacity` carries the chapter's strongest single-segment finding, well-developed in its `## Findings` section. *How well the agent sees the world* (U_o) and *how well the world sees the agent* (backward predictive uncertainty H_b) are formal duals. Both quantify information flow through the agent-environment boundary, in opposite directions; both are observer/horizon/trajectory-indexed; and the same H_b quantity has *opposite value to the agent* depending on coupling sign — cooperative coupling rewards low H_b (allies must predict to coordinate), adversarial coupling rewards high H_b (adversaries cannot neutralize what they cannot predict). The sign-flip is not a separate posit; it falls out of the signed-coupling structure from Ch.4.

The implications-side payoff that the segment's Findings flag but is worth elaborating is the *16-cell joint composition* with the recipient-side classification from Ch.4. Four emitter regimes (high/low H_b, cooperative/adversarial) × four recipient regimes (informative / magnitude-shock / structural-shock / ambient-noise) produce 16 cells, each with a different optimization signature for "where to attack" or "where to coordinate." The closed-form arg-max for adversarial targeting routes to the cell where the attacker's opacity is high AND the recipient's regime detector treats the event as structural-shock (which the recipient cannot absorb without architectural change). The framework therefore replaces the OODA metaphor "inside the opponent's loop" with a precise bilinear optimization problem — the attacker chooses the cell that maximizes the recipient-side regime-induced repair cost, given the attacker's opacity budget.

The decomposition $\mathcal{T}^{\text{eff}} = \mathcal{T} \cdot H_b / H_b^{\max}$ gives the framework's quantitative reading of the OODA-loop tempo-opacity multiplier. The superlinear $(\mathcal{T}_A/\mathcal{T}_B)^2$ adversarial scaling from Ch.4 acquires a second multiplier: the *bilateral opacity ratio* $(H_b^{A|B}/H_b^{B|A})^2$ under Model D, with the corresponding 3/2 exponent under Model S. The framework predicts that high-opacity attackers (those who can act without being predicted by the target) get *quadratic* opacity-leverage on top of *quadratic* tempo-leverage — total quartic scaling in the opacity-times-tempo regime. This is the formal grounding for what Boyd's framing called "operating inside the OODA loop": tempo and opacity multiply, and the multiplication is quadratic-in-quadratic at the structural extreme.

The duality also unifies what would otherwise be parallel theorems. High-U_o low-H_b agents (sees well, is seen well) sit at one corner of the duality; low-U_o high-H_b agents (sees poorly, is hidden) sit at the opposite corner. The same machinery covers both via the signed-coupling structure rather than separate analyses, and the cross-domain transfer to multi-agent safety (legibility for cooperation), adversarial robustness (opacity as targeting-vulnerability shaping), and IDT-style monitoring of opaque AI systems all follow from the same dual.

Adversarial scaling regimes and the OODA self-bound.

`#result-adversarial-exponent-regimes` and `#obs-gated-tempo-advantage` together decompose the superlinear-tempo-advantage from Ch.4 into a regime-typed taxonomy: $b = 2$ under deterministic drift (coupling-dominant); $b = 3/2$ under stochastic noise (coupling-dominant); $b \to 1$ in the coupling-non-dominant regime where observation noise is high enough to gate the tempo advantage to near-linear scaling. Both segments are AAT-internal derivations; neither carries a `## Findings` section, so the catalog-grade treatment lives in this implications segment.

The implication worth surfacing is that the *non-superlinear regime* is operationally important and predictable from AAT's existing machinery. When observation noise $U_{o,B}$ is high enough that the attacker's tempo advantage doesn't penetrate the defender's observation channel cleanly, the gain-principle update $\eta_B^* = U_{M,B}/(U_{M,B} + U_{o,B})$ collapses toward zero, and the defender's effective correction rate α_B shrinks. But — counterintuitively — this *helps* the defender at the system level by gating the exponent. The framework predicts that *cheap noise injection* into the defender's observation channel can be a defensive strategy against a high-tempo attacker, by reducing the exponent from superlinear to linear, even at the cost of slowing the defender's individual updates. This is the structural reason the empirical pattern that highly-noisy environments produce more cooperative-game-like dynamics than expected — observation noise gates the adversarial advantage to near-linear scaling, where the attacker cannot exploit superlinear scaling to drive the defender past the destabilization threshold.

The composition with detection latency from `#deriv-update-detection-latency` produces a cross-segment finding (`#40` in the catalog — *OODA Tempo + Detection Latency Self-Bounds Adversarial Targeting*) that has no other natural home. The 16-cell adversarial-targeting matrix is *self-bounded*: the most damaging cell to attack is the one the defender's *detection latency* makes hardest to recognize. The composition is $\mathcal{T}^{\text{eff}} = \mathcal{T} \cdot H_b/H_b^{\max}$ (from `#der-agent-opacity`) and $\mathbb{E}[T_{\text{detect}}] = \Omega((n+1)/\varepsilon)$ (from `#deriv-update-detection-latency`): the cell with maximal damage is the one with maximal opacity and minimal ε given the defender's $n_{\min}$. Adversarial targeting therefore concentrates *where defender's response is slowest*, and the framework predicts the empirical pattern that cybersecurity incidents cluster on long-trusted dependencies (longest accumulated $n_{\min}$) and that adversarial-ML examples cluster on high-confidence model regions. Both observed empirically; the framework supplies the unifying mechanism.

The cross-segment finding's prescription is concrete: defenders should *short-circuit accumulated trust* on critical dependencies (forced forgetting to reset $n_{\min}$) and *probe high-confidence model regions* with explicit interventional checks. The detection-latency self-bound says these are not optional optimizations — they are the structural countermeasures the adversary's optimal-targeting argument forces the defender to deploy. The framework's contribution against the empirical adversarial-robustness literature is that the targeting concentration has a structural derivation, not merely a folkloric observation, and the countermeasures follow directly.

Closing Part III.

Per-dimension persistence (`#result-per-dimension-persistence`) is formally in this chapter though its content belongs with Part I Ch.4. The implications-segment treatment landed at Part I Ch.4; here it appears as the adversarial concentration mechanism: an opponent who identifies the target's weak dimension can concentrate disturbance there, amplifying the mismatch ratio asymmetrically while the aggregate L^2 metric still looks acceptable. The framework's prescription — *per-dimension monitoring is a structural requirement for any agent operating under directed pressure* — bites hardest here, where strategic composition delivers attackers who can locate the weak direction and adversarial targeting decomposes the attack into recipient-side regimes that exploit the dimension-asymmetry.

The chapter is also where Part III's open population-dynamics questions surface as recognized gaps: latent structural diversity (variation invisible to current performance but consequential under regime change), endogenous coupling (γ as function of population composition rather than exogenous parameter), composition-transition dynamics (Miller 2022's extreme transition motif), and computational thresholds for social behavior (Miller's ICE framework). These four gaps are the bridge to *population-level* dynamics that the current pairwise/strategic-equilibrium scope does not cover. When fleshed out they would likely form their own chapter on population dynamics, possibly relocating the per-dimension result alongside. The implications-segment treatment flags this rather than over-claiming present coverage.

What Part III closes on is the composite-agent machinery in its operational form: closure-defect / bridge-lemma representability criterion (Ch.2), unity dimensions and Auftragstaktik bandwidth allocation (Ch.3), signed-coupling structure with cooperative-vs-adversarial as one inequality and the recipient-side four-regime taxonomy (Ch.4), strategic composition under equilibrium machinery and agent opacity with 16-cell adversarial targeting (Ch.5). The four chapters compose into AAT's full composite-agent treatment, with the identifiability-floor pattern's four formal instances (Instances 1–4) covering the framework's scope-honest commitments at the boundaries.

The bridge into TST (`02-tst-core/`) and the logogenic-agent stages (`03-llm-core/`, `04-eli-core/`) is direct from here. TST instantiates the AAT machinery in the software domain, with code quality as observation infrastructure, Lindy-Effect as the temporal-optimality grounding, and software development as the *privileged calibration laboratory* (high P1–P6 identification properties). Logogenic agents — Class 3 (Coupled) by construction — inherit the wrapping construction from Ch.2 and the bias bound from Part II Ch.5, with the framework's reach into LLM-class substrates running through those two segments. The orient-cascade-recovery-via-scaffolding argument from Part II Ch.5 implications is the structural claim that connects AAT's persistence machinery to production agentic systems.

The implications-segment series for AAT chapter-ends is complete with this segment. The TST and logogenic chapter-ends follow the same pattern; that work picks up under task 11 after Joseph's check-in.

Working Notes.

- This segment is a chapter-end discussion grouping findings from `#deriv-strategic-composition`, `#der-agent-opacity`, `#result-adversarial-exponent-regimes`, `#obs-gated-tempo-advantage`, `#result-per-dimension-persistence` (content cross-referenced from Part I Ch.4 implications), `#hyp-symbiogenic-composition` (formally in Ch.1, content surfacing here), and `#der-interaction-channel-classification` (Ch.4, the recipient-side dual to this chapter's emitter-side classification). Cross-pattern reading leans on `#disc-identifiability-floor` for Instance 3's operational form, `#deriv-update-detection-latency` for the OODA self-bound composition (`#40`), and `#disc-modularity-state-dynamics` for the goal-divergence-as-modularity-decreasing operation.
- **Cross-reference vs canonical-home policy.** Two source segments in this chapter carry well-written `## Findings` sections — `#der-agent-opacity` and `#result-per-dimension-persistence`. Both are cross-referenced rather than restated. The findings whose canonical home is this implications segment are: `#23` (Class-1-sub-agents → Class-3-composite under partially-opposing objectives), `#28` (modular safety architectures fail under goal divergence — no segment home), `#44` partial (the exponent-regime decomposition $b = 2, 3/2, \sim 1$), `#54` (observation noise gates advantage), `#52` (symbiogenic composition dynamics, asymmetric-limit reading), `#40` (OODA tempo + detection latency self-bounds — cross-segment finding without home), and `#50` (strategic composition game-theory bridge — companion to `#23`).
- **`#28` Modular Safety as segment-promotion candidate.** The catalog draft entry `#28` references a finding that has no home segment. The implications segment is its canonical home for now; a future cycle may promote a dedicated `#disc-modular-safety-under-goal-divergence` segment under Appendix A with `## Findings`, at which point this segment downgrades to cross-reference.
- **Cross-segment finding without home — `#40` OODA + Detection Latency Self-Bound.** Composes `#der-agent-opacity` (16-cell), `#result-adversarial-tempo-advantage` (superlinear), and `#deriv-update-detection-latency` ($\Omega((n+1)/\varepsilon)$) into the self-bounded adversarial-targeting prediction. No source segment carries this composition; the implications segment is its canonical home. Empirical signature (cybersecurity incident concentration on long-trusted dependencies, adversarial-ML clustering on high-confidence regions) is observed but not pulled together; the framework's contribution is the unifying mechanism.
- **Identifiability-floor pattern's full four-instance treatment.** The pattern is now fully surfaced in the implications-segment series: Instance 1 (Part II Ch.4-strategy-dynamics — on-policy L0-insufficiency detection), Instance 2 (Part II Ch.3-strategy-structure — L1' mixture identifiability under unobservable common cause), Instance 3 (Part III Ch.2-composition-machinery, operationalized here — composite contraction certification from component marginals), Instance 4 (Part II Ch.5-orient-cascade — universal information-to-distance constant under heteroscedastic-Gaussian no-go). Instance 5 (Mehra non-identifiability in adaptive Kalman) is candidate; mechanism-design impossibility (Gibbard-Satterthwaite / Myerson-Satterthwaite / Arrow) is candidate-fifth via the strategic-composition derivation in this chapter. The pattern's load-bearing role across AAT's scope is now visible end-to-end.
- **Connection to NeurIPS submissions.** The strategic-composition machinery is closest to a future multi-agent extension of Paper 2's convergence theory; the modular-safety implication is closest to a future AI-safety-track submission. Current cross-references in Paper 2 / Paper 3 are scope and structural-argument links rather than theorem-grade adoption from this chapter.
- **AAT chapter-end implications series complete.** With this segment, the AAT-core chapter-end implications work covers Part I Ch.4 + Part II Ch.2/3/4/5 + Part III Ch.2/3/4/5 — seven chapter-end segments plus the pilot. Each cross-references existing `## Findings` where they exist and treats canonical-home findings in full where they do not. Cross-segment findings (CS1 unified RL convergence, CS2 compounding stability-induced myopia, CS3 three matrix-form bounds on Fisher backbone, `#40` OODA self-bound) land in the chapter whose last prerequisite completes. The optional Appendix-A end segment (CS3 + cross-references to M1/M2/M3/M4) and the TST / logogenic-agents chapter-ends are remaining work flagged in tasks 10 and 11.

  [Gap] Latent structural diversity: variation in correction architectures invisible to persistence analysis, consequential under regime change

[Gap] Endogenous coupling: γ as function of population composition, not exogenous parameter; coupling emergence threshold

[Gap] Composition transition dynamics: epochal stability → latent diversification → niche emergence → cascading restructuring → re-equilibration (adopts Miller 2022's extreme transition motif)

[Gap] Computational thresholds for social behavior: minimum agent complexity and interaction depth for composition dynamics (adopts Miller 2022's ICE framework; grounds 8.9)

Part IV

Appendices: Details

Appendix A

Derivation: Sector Condition Stability — Lyapunov Derivation

Complete Lyapunov derivations of bounded mismatch and adaptive reserve for the sector-condition results stated in 4.3.

Motivation.

3.9 hypothesizes the linear ODE $d\|\delta\|/dt = -\mathcal{T}\|\delta\| + \rho$ as a first-order approximation. The linear form yields clean closed-form results but commits to a specific functional relationship between mismatch magnitude and correction rate. True correction dynamics are almost certainly nonlinear — exhibiting saturation at large mismatch, threshold effects near zero, and structural breakdown when the model class is exhausted.

A Lyapunov approach proves persistence and stability under much weaker assumptions: any correction dynamics satisfying qualitative monotonicity properties (the sector condition). The results below are strictly more general — the linear case is recovered where the sector bounds coincide.

Setup.

Let $\delta(t) \in \mathbb{R}^n$ be the mismatch vector — the difference between the model's predictions and reality across n observable dimensions. The vector treatment connects to per-dimension tempo analysis (14.5).

The mismatch dynamics are:

Definition (Dynamics Setup).

$$\frac{d\delta}{dt} = -F(\mathcal{T}, \delta) + w(t) \tag{A.1}$$

where:

- $F(\mathcal{T}, \delta) : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^n$ is the **correction function** — how the agent's adaptive process reduces mismatch. It maps to the same space as δ (so that the inner product $\delta^T F$ in the sector condition is well-defined). This subsumes the update gain η^* (3.6), event rate ν, and the structure of the update rule.
- $w(t)$ is the **disturbance** — new mismatch introduced by environmental change, with $\|w(t)\| \leq \rho$ (bounded disturbance rate).

The linear case from 3.9 has $F(\mathcal{T}, \delta) = \mathcal{T} \cdot \delta$.

Assumptions on F**.**

We require only qualitative properties, not a specific functional form:

(A1) Zero Correction at Zero Mismatch.

Assumption A1.

$$F(\mathcal{T}, 0) = 0 \tag{A.2}$$

No correction is applied when the model perfectly matches reality.

(A2') Local Sector Condition. There exists a region $\mathcal{B}_R = \{\delta : \|\delta\| \leq R\}$ and $\alpha > 0$ such that (following the sector-condition framework of Lur'e[1]):

Assumption A2' (sector-condition) — derived in sub-scope α, assumed in sub-scope β (see below).

$$\delta^T F(\mathcal{T}, \delta) \geq \alpha \|\delta\|^2 \quad \forall \delta \in \mathcal{B}_R \tag{A.3}$$

The correction function always points "inward" (reducing mismatch), and its magnitude is bounded below relative to $\|\delta\|^2$. The linear case has $\alpha = \mathcal{T}$. A saturating correction has α decreasing for large $\|\delta\|$. A threshold correction has $\alpha = 0$ for small $\|\delta\|$.

The local form allows the correction to break down outside $\mathcal{B}_R$ — the structural adaptation regime of 4.5.

Grounding of GA-3 — sub-scope α (A2' derived). For a characterized class of AAT-in-scope agents, A2' is a *derived* consequence of the update rule, not a primitive assumption. 4.2 (Prop B.3) shows that the gain-based update $M_t = M_{t-1} + \eta^* g(\delta_t)$ (3.6) induces a correction function satisfying A2' whenever the update rule has **directional fidelity (B1)** — $\delta^T H g(\delta) \geq c_{\min} \|\delta\|^2$ on $\mathcal{B}_R$. Sub-scope α — the agent classes where B1 holds structurally — includes:

- *Optimal Bayesian updates* (Kalman, conjugate families): B1 holds by Bayes-risk minimization. $\alpha = \eta^* \cdot c_{\min}$, reducing to $\alpha = \eta^*$ in the scalar case.
- *Exponential families in natural parameters*, on a bounded interior scope $\Theta_0 \subset \mathrm{int}(\Theta)$: the Hessian is the Fisher information matrix, PD on the interior. $\alpha = \eta \cdot \mu_0$ where $\mu_0 = \inf_{\theta \in \Theta_0} \lambda_{\min}(\mathbf{I}(\theta)) > 0$. Pointwise PD does not imply a uniform global lower bound: $\inf_{\theta \in \Theta} \lambda_{\min}(\mathbf{I}(\theta))$ can be zero (Poisson: $\mathbf{I}(\theta) = e^\theta$, $\inf_{\theta \in \mathbb{R}} e^\theta = 0$), so Θ_0 supplies the local-region scope R that A2' requires. Families with a uniform Fisher floor on Θ (Gaussian-mean, Beta-Bernoulli) extend to global α.
- *Gradient descent on locally strongly convex losses* (Prop B.4): B1 is *equivalent* to strong convexity via the gradient-monotonicity characterization (Nesterov 2004[2], Thm 2.1.10). $\alpha = \eta \cdot \mu$ where μ is the strong convexity modulus.
- *L2-regularized convex losses*: regularization provides a floor $\mu \geq \lambda$, so $\alpha \geq \eta\lambda$ globally.
- *Linear corrections with positive-definite gain–observation product*: $\alpha = \lambda^+_{\min}(KH)$ (matrix Kalman, restricted to the observable subspace).

Within sub-scope α, A2' is written down by inspection of the update rule; no independent posit is required.

Grounding of GA-3 — sub-scope β (A2' assumed as empirical claim). For the remaining AAT-in-scope agents, A2' stands as a well-scoped empirical claim about the agent's correction geometry. Sub-scope β includes:

[1]Lur'e, A. I. (1957). *Some Nonlinear Problems in the Theory of Automatic Control*.
[2]Nesterov, Y. (2004). *Introductory Lectures on Convex Optimization*. Springer. Theorem 2.1.10.

- *PID controllers with fixed gains* — no gradient / optimality structure; B1 is a tuning question, not a structural consequence.
- *Rule-based systems* — no continuous update rule; A2' is domain-specific.
- *Human judgment / organizational learning* — structural analogy in 3.6; no formal B1 guarantee.
- *Severely misspecified agents* (FM-5 in 4.2) — proper-gradient rules can aim at the wrong target.
- *Variational / approximate posteriors* — B1 not guaranteed by optimality because approximation error can rotate the correction.
- *Non-convex gradient agents beyond the basin* — A2' fails at basin boundary; this IS the 4.5 trigger.
- *Stochastic gradients (per-step)* — A2' holds in expectation; per-step noise enters as effective disturbance under Prop A.1S.

Within sub-scope β, A2' must be verified per-system — the claim is stronger than what AAT's postulates + gain structure alone can force (an agent with $g(\delta) = R_{90°}\delta$ satisfies every AAT postulate but violates B1 trivially; see 4.2 FM-1). The Lyapunov proofs below are unchanged — they operate downstream of A2' regardless of whether it is assumed or derived.

A2' / DA2' α/β as operator-family classification (external mathematical lineage). The α/β partition can be restated precisely in the operator-theoretic language of Rockafellar 1970 (*Convex Analysis*, §24) and Bauschke & Combettes 2017 (*Convex Analysis and Monotone Operator Theory in Hilbert Spaces*, 2nd ed., §§22–28). Casting the discrete update map as $T_d(\delta) = \delta - \eta^* F_d(\delta)$ with $T_d(0) = 0$, A2' at $\delta^* = 0$ is exactly a **one-point strong monotonicity condition** on $A = I - T_d$, and DA2'-inc (the incremental strengthening required by `#form-composition-closure`'s bridge lemma) is **full two-point strong monotonicity** in the Rockafellar sense. Sub-scope α maps onto established operator families:

- **Optimal Bayesian updates** (Kalman, conjugate, exponential family) ≡ proximal / firmly nonexpansive operators (Bauschke-Combettes Prop 23.7). Firmly nonexpansive is equivalent to $\frac{1}{2}$-averaged; operator-sector condition holds with the gain as the sector constant.
- **Gradient operators on strongly convex functions** ≡ cocoercive operators via the Baillon-Haddad theorem; strong monotonicity with modulus $\eta\mu$.
- **Exponential families in natural parameters** ≡ natural-gradient operators in the Fisher-weighted Hilbert space (Amari 1998; cf. 4.2 "Fisher-metric cases under parameterization-invariance" for the AAT-internal forcing via (PI)/Čencov named in Z).
- **L2-regularized convex gradients** ≡ shifted monotone-gradient operators with regularization-provided floor.
- **Linear corrections with PD gain–observation product** ≡ linear contraction in the $(P^-)^{-1}$-weighted inner product.

Sub-scope β maps onto "operator families that are not cocoercive in any natural inner product": PID (tuning-dependent; not cocoercive generically); rule-based / symbolic (no inner-product structure); variational / amortized (approximation-gap rotation); non-convex-gradient beyond basin (non-monotone on the full domain); per-step SGD (noise-dominated); human judgment (no formal rule). The α/β epistemic labeling in AAT tracks whether the operator family admits operator-sector structurally (derived from the class definition) or only under empirical per-instance verification — a scope-honesty move, not a mathematical reclassification.

This recognition positions AAT's sector-condition framework as a specialization of monotone-operator theory (Minty 1962; Browder 1968; Rockafellar 1970; Bauschke-Combettes 2017) to one-point-anchored strong monotonicity under a specific inner product structure. AAT's distinctive content — one-point anchoring at the equilibrium (strictly weaker than full two-point strong monotonicity, matched to fixed-point-at-target semantics); Model D / Model S disturbance decomposition; composition with the identifiability-floor meta-pattern; composition-consistency postulate; the α/β epistemic labeling itself — sits as specialization + repurposing rather than strict generalization. This acknowledgment is load-bearing for scope honesty: the mathematical machinery is established; AAT's value lies in its AAT-internal architecture (organization-of-scope under a singular-trajectory agent, signed-coupling structure, forced coordinates via uniqueness theorems, three meta-patterns) rather than in novel monotone-operator mathematics. The unification has honest limits: the coarse-graining projection Λ (`#form-composition-closure`) does not fit the operator-sector primitive; three of five metric-α_2 cases in `#result-contraction-template` remain theorem-imported; the identifiability-floor axis is orthogonal.

Sub-scope β "rule-based / discontinuous" — structural Lipschitz floor. *[Derived, status: exact scope-exit statement.]* The rule-based / state-machine / threshold-triggered / discontinuous entry in sub-scope β is not merely "empirical verification required"; it is a **structural scope-exit for contraction-based bridge-lemma analysis**. For correction functions F that are not locally Lipschitz, no scalar sector bound $\delta^\top F(\delta) \geq \alpha\|\delta\|^2$ implies the full-update-map contraction required by `#form-composition-closure`'s bridge lemma — the update map $f_c(X, o) = X - \eta^* F(X)$ can exhibit $\Omega(1)$ jumps between arbitrarily close X, X' at rule-firing boundaries, so no $\lambda < 1$ Lipschitz constant exists. Concrete counterexample: $F(\delta) = \alpha\delta$ for $|\delta| < 1$ and $F(\delta) = \alpha\delta + \text{sign}(\delta)$ for $|\delta| \geq 1$ — the sector bound holds with α, but f_c jumps by η^* at $|X| = 1$, violating contraction. The appropriate external apparatus for this class is the **hybrid-dissipative framework** (van der Schaft & Schumacher 2000, *An Introduction to Hybrid Dynamical Systems*, Springer; Di Bernardo, Liuzza & Russo 2014, *SIAM J. Control Optim.* 52) — distinct machinery (dwell-time, Filippov solutions, impulsive-dissipative inequalities) operating on a different regularity class. AAT's Lyapunov machinery in this segment and the contraction machinery in `#result-contraction-template` both require C^1 or better regularity; the scope-exit to hybrid-dissipative analysis is honest and structural, not a deficiency to be repaired within AAT's current apparatus. Under `#disc-identifiability-floor`'s Instance 2 pattern (Cramér-Rao floor under unobservable common cause), rule-based agents whose rule firing depends on regime structure — e.g., threshold-triggered state-machines with regime-dependent thresholds — additionally suffer regime-C identifiability collapse when the regime variable is unobservable; the non-contractibility and the non-identifiability compose in that case.

(A3) Tempo Monotonicity. For fixed δ, $\delta^T F(\mathcal{T}, \delta)$ is monotone increasing in $\mathcal{T}$. Higher tempo means faster correction.

Connection to AAT parameters. The sector parameter α is determined by the adaptive tempo $\mathcal{T}$ (3.8) and the structure of the correction function. In the linear case, $\alpha = \mathcal{T} = \sum_k \nu^{(k)} \cdot \eta^{(k)*}$. In nonlinear cases, α represents the *worst-case* correction efficiency within the valid region — the minimum ratio of correction power to mismatch magnitude. The radius R represents the model class capacity: how large a mismatch can grow before the correction mechanism fails (i.e., before the sector condition ceases to hold), at which point structural adaptation (4.5) becomes necessary.

Candidate Lyapunov Function.

Definition (Lyapunov Candidate).

$$V(\delta) = \frac{1}{2}\|\delta\|^2 \tag{A.4}$$

Positive definite, radially unbounded, continuously differentiable. Its level sets $V = c$ are spheres of radius $\sqrt{2c}$ in mismatch space.

Proposition A.1: Bounded Mismatch (Single Agent).

Statement. Under (A1), (A2'), (A3), with bounded disturbance $\|w(t)\| \leq \rho$, the mismatch $\delta(t)$ is **ultimately bounded**: there exists $R^* > 0$ such that $\|\delta(t)\| \leq R^*$ for all sufficiently large t, provided $R^* < R$ (the ultimately bounded region fits within the sector-condition region).

Proof.

Compute $\dot{V}$ along trajectories:

Derived (Proof Step).

$$\dot{V} = \delta^T \dot{\delta} = \delta^T[-F(\mathcal{T}, \delta) + w(t)] \tag{A.5}$$

Derived (Proof Step).

$$= -\delta^T F(\mathcal{T}, \delta) + \delta^T w(t) \tag{A.6}$$

By (A2'): $\delta^T F(\mathcal{T}, \delta) \geq \alpha\|\delta\|^2$

By Cauchy-Schwarz: $\delta^T w(t) \leq \|\delta\| \cdot \|w(t)\| \leq \rho\|\delta\|$

Therefore:

Derived (Proof Step).

$$\dot{V} \leq -\alpha\|\delta\|^2 + \rho\|\delta\| = -\|\delta\|(\alpha\|\delta\| - \rho) \tag{A.7}$$

$\dot{V} < 0$ whenever $\|\delta\| > \rho/\alpha$ **and** $\|\delta\| \leq R$ (where A2' holds).

Define $R^* = \rho/\alpha$.

Invariance of $\mathcal{B}_R$. When $R^* < R$ (the persistence condition), the ball $\mathcal{B}_R$ is *positively invariant*: any trajectory starting in $\mathcal{B}_R$ remains in $\mathcal{B}_R$ for all future time. At the boundary $\|\delta\| = R$, the sector condition holds and $\dot{V} = -\|\delta\|(\alpha\|\delta\| - \rho) = -R(\alpha R - \rho) < 0$ (since $\alpha R > \rho$ by the persistence condition). Trajectories at the boundary point inward. Therefore $\mathcal{B}_R$ is invariant, and the sector condition applies to the entire future trajectory of any trajectory starting inside $\mathcal{B}_R$.

Ultimate boundedness. Within $\mathcal{B}_R$, the Lyapunov function is strictly decreasing for $\|\delta\| > R^*$ and may increase for $\|\delta\| < R^*$. All trajectories starting in $\mathcal{B}_R$ are ultimately bounded by R^* — they are driven into $\mathcal{B}_{R^*}$ and remain in a neighborhood of it (with possible oscillation at the boundary due to the disturbance).

Initial condition requirement. The result requires the trajectory to start inside $\mathcal{B}_R$ (or to be brought inside by some external mechanism). A trajectory starting outside $\mathcal{B}_R$ is not covered — the sector condition does not hold there, and the correction function may fail to reduce mismatch. This is precisely the structural adaptation regime of 4.5: an agent whose mismatch exceeds R has exhausted its model class capacity and needs structural change to re-enter the region where parametric correction works.

The agent persists (from within $\mathcal{B}_R$) iff $R^* < R$, i.e., iff $\alpha > \rho/R$. □

Interpretation. The ultimately bounded region has radius $R^* = \rho/\alpha$. In the linear case, $\alpha = \mathcal{T}$, recovering 4.4's steady-state result $R^* = \rho/\mathcal{T}$ exactly. But Proposition A.1 holds for *any* correction function satisfying the sector condition, not just the linear one.

The persistence threshold, generalized. The agent persists (mismatch remains bounded within the model class capacity) iff $\rho/\alpha < R$. If the correction function breaks down (A2' fails) before R^* is reached, the agent may diverge. This IS 4.5's trigger: when $\rho/\alpha > R$ (the environment demands more correction than the model class can provide), parametric adaptation fails and structural change is required.

Proposition A.2: Stability Margin (Adaptive Reserve).

Statement. Under the conditions of A.1, the agent can tolerate a sudden increase in disturbance rate of:

Derived (adaptive-reserve).

$$\Delta\rho^* = \alpha R - \rho \tag{A.8}$$

without mismatch diverging (where R is the radius of the sector-condition region from A2'). Beyond this, R^* exceeds R and the correction function may fail.

Proof. After a shock, the new disturbance rate is $\rho + \Delta\rho$. The new ultimately bounded radius is $(\rho + \Delta\rho)/\alpha$. This remains within the valid region iff $(\rho + \Delta\rho)/\alpha \le R$, i.e., $\Delta\rho \le \alpha R - \rho$. □

Interpretation. $\Delta\rho^*$ is the agent's **adaptive reserve** — how much additional environmental volatility it can absorb before its model breaks down. This is a single number characterizing an agent's robustness to shock:

- An agent operating well below capacity ($\rho \ll \alpha R$) has a large reserve — it is **robust**.
- An agent near its limit ($\rho \approx \alpha R$) has a small reserve — it is **fragile**.

Table A.1

Domain	*Large $\Delta\rho^*$ (robust)*	*Small $\Delta\rho^*$ (fragile)*
Control	Kalman filter on slow target	Same filter on erratic target
Biology	Organism in stable niche	Same organism under climate change
Organization	Well-capitalized firm, stable market	Startup in volatile market
Military	Force with operational depth	Force at culmination point

Proposition A.1S: Bounded Mismatch Under Stochastic Disturbance.

Statement (region-aware form). Under (A1), (A2') on $\mathcal{B}_R$, (A3), with stochastic disturbance (GA-2S: $w(t)$ is zero-mean with $\mathbb{E}[\|w(t)\|^2] = \sigma_w^2$), let $\tau_R = \inf\{t : \|\delta(t)\| > R\}$ be the first-exit time from the sector-condition region. Then:

(i) *Stopped bound* — the stopped process satisfies for all $t \ge 0$:

Derived (stochastic-bounded-mismatch, stopping-time localization, Khasminskii 2012 ch. 5).

$$\mathbb{E}[\|\delta(t \wedge \tau_R)\|^2] \le \|\delta(0)\|^2 e^{-2\alpha t} + \frac{n\sigma_w^2}{2\alpha} \tag{A.9}$$

(ii) *Mean-square persistence condition* — when $R_S^* := \sigma_w\sqrt{n/(2\alpha)} < R$, equivalently

$$\alpha > \frac{n\sigma_w^2}{2R^2} \tag{A.10}$$

the ultimately-bounded RMS radius fits inside the sector-condition region.

(iii′) *Fixed-time tail* — under the mean-square persistence condition (ii), Markov's inequality on the stopped second moment gives, for each fixed t (sharpest in the stationary $t \to \infty$ limit),

$$P\big(\|\delta(t)\| > R\big) \ \le\ \frac{\mathbb{E}[\|\delta(t \wedge \tau_R)\|^2]}{R^2} \ \xrightarrow[t\to\infty]{}\ \frac{n\sigma_w^2}{2\alpha R^2}. \tag{A.11}$$

This controls the mismatch at any single time. It is *not* an infinite-horizon containment statement: under additive Brownian forcing the diffusion is recurrent on $\mathbb{R}^n$ (the additive-noise generator has no bounded non-constant harmonic function), so $P(\tau_R < \infty) = 1$ over an unbounded horizon — there is no $P(\tau_R < \infty) < 1$ bound, and none is claimed. Pathwise containment of $\mathcal{B}_R$ is a Model-D guarantee only (Prop A.1's positive invariance); Model S controls the *typical scale* (ii) and the *fixed-time tail* (iii′), not the sample path over an unbounded horizon. The kind-of-guarantee dichotomy this exposes is itself a result — see the Discussion.

(iv) *Finite-horizon sample-path bound* — when a sup-over-an-interval statement is wanted, the sound one is the additive first-exit bound

$$P\Big(\sup_{0\le s\le T} \|\delta(s)\| > R\Big) \ \le\ \frac{\|\delta(0)\|^2 + n\sigma_w^2\, T}{R^2}, \tag{A.12}$$

rigorous under (A2') alone (Khasminskii 2012 ch. 5). It controls the whole path on $[0, T]$ — stronger than (iii′) within its regime — but grows linearly in T and is vacuous for $T > rsimR^2/(n\sigma_w^2)$, consistent with $P(\tau_R < \infty) = 1$.

The proof below establishes the Grönwall-type bound first, then the stopping-time localization at the end.

Proof.

The SDE form of the mismatch dynamics under stochastic disturbance is:

$$d\delta = -F(\mathcal{T}, \delta)\, dt + \sigma_w\, dW_t \tag{A.13}$$

where W_t is a standard n-dimensional Wiener process.

Apply Itô's formula to $V(\delta) = \frac{1}{2}\|\delta\|^2$:

Derived (Proof Step).

$$dV = \delta^T(-F)\, dt + \delta^T\sigma_w\, dW_t + \frac{1}{2}\sigma_w^2 n\, dt \tag{A.14}$$

The last term is the Itô correction: $\frac{1}{2}\mathrm{tr}(\sigma_w^2 I_n) = \frac{n}{2}\sigma_w^2$.

Taking expectations (the Itô integral $\delta^T\sigma_w\, dW_t$ has zero expectation):

Derived (Proof Step).

$$\frac{d}{dt}\mathbb{E}[V] = \mathbb{E}[\delta^T(-F)] + \frac{n}{2}\sigma_w^2 \tag{A.15}$$

By (A2'): $\delta^T F \geq \alpha \|\delta\|^2 = 2\alpha V$. Therefore:

Derived (Proof Step).

$$\frac{d}{dt}\mathbb{E}[V] \leq -2\alpha\,\mathbb{E}[V] + \frac{n}{2}\sigma_w^2 \tag{A.16}$$

This is a linear ODE in $\mathbb{E}[V]$ with solution:

$$\mathbb{E}[V(t)] \leq V(0)\,e^{-2\alpha t} + \frac{n\sigma_w^2}{4\alpha}(1 - e^{-2\alpha t}) \tag{A.17}$$

Since $V = \frac{1}{2}\|\delta\|^2$, the steady-state mean-square mismatch is:

$$\mathbb{E}[\|\delta\|^2]_{ss} = \frac{n\sigma_w^2}{2\alpha} \tag{A.18}$$

The RMS steady-state mismatch is:

Derived (stochastic-steady-state).

$$R_S^* \;=\; \|\delta\|_{\text{rms}} = \sigma_w\sqrt{\frac{n}{2\alpha}} \tag{A.19}$$

Persistence requires $R_S^* < R$, giving $\alpha > n\sigma_w^2/(2R^2)$.

Stopping-time localization. The Itô-Lyapunov step above used A2' ($\delta^T F \geq \alpha\|\delta\|^2$) on the entire trajectory, but A2' is posited only on $\mathcal{B}_R$. Replacing t with the stopped time $t \wedge \tau_R$ makes the argument valid: on $[0, t \wedge \tau_R]$, $\delta(s) \in \mathcal{B}_R$ almost surely, so A2' applies. The Itô integral $\int_0^{t\wedge\tau_R} \delta^T \sigma_w\, dW_s$ remains a martingale with zero expectation (optional stopping for L^2-bounded martingales). The stopped Grönwall bound (i) follows. For (iii′), Markov's inequality applied to the stopped second moment (i) gives the fixed-time tail directly. The *infinite-horizon* ever-exit probability is 1 under additive Brownian forcing, and no nonnegative supermartingale certifies any $P(\tau_R < \infty) < 1$ bound — the natural Ville/Doob maximal-inequality route provably cannot exist (demonstrated in `#deriv-stochastic-non-exit`). (iv)'s finite-horizon sup-bound follows from the same Itô-Lyapunov step without the decay term. □

Interpretation. Model D (Prop A.1) gives $R^* = \rho/\alpha$, scaling as $1/\alpha$. Model S gives $R_S^* = \sigma_w\sqrt{n/(2\alpha)}$, scaling as $1/\sqrt{\alpha}$. Doubling the correction efficiency halves the deterministic steady-state mismatch but only reduces the stochastic steady-state by a factor of $\sqrt{2} \approx 1.41$. Correction is less effective against noise than against drift. This difference in scaling propagates into the adversarial exponent regimes (14.3): $b = 2$ under Model D, $b = 3/2$ under Model S.

Tail bound. At steady state, Markov's inequality gives $P(\|\delta\| > R) \leq n\sigma_w^2/(2\alpha R^2)$. The agent stays within R with probability $\geq 1 - \epsilon$ provided $\alpha \geq n\sigma_w^2/(2\epsilon R^2)$. For the linear case (Ornstein-Uhlenbeck), the stationary distribution is Gaussian and exact tail probabilities are available. The Markov bound is the general result holding under (A2') alone.

Corollary A.1S.1: Disturbance-Model Containment Dichotomy.

For a trajectory started in $\mathcal{B}_R$ under (A1), (A2') on $\mathcal{B}_R$, (A3), the persistence-region first-exit probability is **categorical**:

Derived.

$$P(\tau_R < \infty) \;=\; \begin{cases} 0 & \text{Model D (bounded } w,\ \|w\| \le \rho) \text{ under } \alpha R > \rho, \\ 1 & \text{Model S (additive stochastic, } d\delta = -F\,dt + \sigma_w\,dW_t). \end{cases} \tag{A.20}$$

The Model-D value is the deterministic positive invariance of Prop A.1 (boundary-inward: at $\|\delta\| = R$, $\dot{V} < 0$ when $\alpha R > \rho$). The Model-S value holds for **every** $\alpha > 0$, **every** $\sigma_w > 0$, and **every** correction function F satisfying (A2'): a non-degenerate diffusion (additive forcing $\sigma_w\,dW_t$, $\sigma_w > 0$) exits any bounded region in finite time almost surely, for any locally bounded drift — no finite inward correction can defeat the Brownian crossing near $\partial\mathcal{B}_R$. The Ornstein–Uhlenbeck benchmark makes the mechanism explicit (scale function $s'(u) \propto e^{\alpha u^2/\sigma_w^2}$ unbounded, hence recurrent; Khasminskii 2012[3] ch. 3–4), but the conclusion needs only non-degeneracy of the additive noise on a bounded region, not the linear structure. That the natural maximal-inequality route to a $P(\tau_R < \infty) < 1$ bound *cannot* exist — the question "are you sure you can't just Doob/Ville this?" — is demonstrated in `#deriv-stochastic-non-exit`.

The achievable first-exit probability is therefore exactly the two-point set $\{0, 1\}$, and **which point obtains is fixed by the disturbance model's support structure (bounded vs. unbounded), not by the correction strength** α. Increasing α tightens the typical scale ($R_S^* \propto 1/\sqrt{\alpha}$) and the fixed-time tail (iii′); it cannot move $P(\tau_R < \infty)$ off 1 under Model S, and is unnecessary for the 0 under Model D. Pathwise containment of $\mathcal{B}_R$ is categorically a bounded-disturbance property: additive stochastic forcing removes the *kind* of guarantee available, not merely its rate. This sharpens the hand-off into 4.5 — in any genuinely stochastic environment region-exit is a certain eventual event, so the structural-adaptation trigger is *generic, not exceptional*, for a sufficiently long-lived agent.

Summary of Results.

Table A.2

Result	*What it proves*	*Assumptions*	*Linear case recovery*
A.1 (Bounded Mismatch)	$R^* = \rho/\alpha$	(A1), (A2'), bounded ρ (GA-2)	$\alpha = \mathcal{T}$ gives $R^* = \rho/\mathcal{T}$
A.1S (Stochastic Bounded Mismatch, region-aware)	$R_S^* = \sigma_w\sqrt{n/(2\alpha)}$ (stopped bound); fixed-time tail $P(\|\delta(t)\| > R) \le n\sigma_w^2/(2\alpha R^2)$ (stationary-sharp); no infinite-horizon non-exit bound — pathwise containment is Model-D-only	(A1), (A2') on $\mathcal{B}_R$, stochastic w (GA-2S)	$\alpha = \mathcal{T}$ gives $R_S^* = \sigma_w\sqrt{n/(2\mathcal{T})}$
Cor A.1S.1 (Disturbance-Model Containment Dichotomy)	$P(\tau_R < \infty)$ is exactly $\{0, 1\}$ — 0 under Model D (positive invariance), 1 under Model S (a.s. exit of a non-degenerate diffusion); α-invariant, correction strength cannot interpolate	(A1), (A2') on $\mathcal{B}_R$; Model D needs $\alpha R > \rho$	Model D → Prop A.1 invariance at $\alpha = \mathcal{T}$; Model S → OU recurrence is the explicit instance — exact generally (any F under A2'), not linear-scoped
A.2 (Stability Margin)	$\Delta\rho^* = \alpha R - \rho$	Same as A.1	$R \to \infty$ for linear (always stable if $\mathcal{T} > 0$)

[3] Khasminskii, R. (2012). *Stochastic Stability of Differential Equations* (2nd ed.). Springer. Chapter 5: Lyapunov functions and stochastic stability; stopping-time localization for diffusion processes.

What Is Derived vs. What Is Chosen.

Table A.3

Property	*Source*	*Strength*
Dynamics setup $\dot{\delta} = -F(\mathcal{T}, \delta) + w(t)$ with F as correction function and w as disturbance	Definitional scope of the appendix; generalizes the linear hypothesis of 3.9	Definition
(A1) $F(\mathcal{T}, 0) = 0$	Qualitative property of any correction process by construction	Assumption (uncontroversial)
(A2') Local sector condition $\delta^T F \geq \alpha \|\delta\|^2$ on $\mathcal{B}_R$	Sector-condition framework (Lur'e 1957); sub-scope α (Kalman/conjugate Bayesian, exponential family in natural params, gradient descent on strongly convex losses, L2-regularized convex, linear with PD KH) derived via 4.2 Prop B.3 under B1 directional fidelity (then $\alpha = \eta^* \cdot c_{\min}$); sub-scope β (PID, rule-based, human judgment, severely misspecified, variational approximations, non-convex beyond basin, per-step SGD) requires independent verification	**Derived in sub-scope α; Assumption in sub-scope β** (sub-scopes named explicitly in the Grounding paragraphs)
(A3) Tempo-monotonicity of $\delta^T F$	Qualitative requirement tying $\mathcal{T}$ to correction power	Assumption
Quadratic Lyapunov candidate $V = \frac{1}{2}\|\delta\|^2$	Canonical choice for norm-bounded stability in Euclidean state spaces (Khalil 2002 ch. 4)	Formulation choice
Prop A.1: ultimate bound $R^* = \rho/\alpha$	$\dot{V} \leq -\|\delta\|(\alpha\|\delta\| - \rho)$ via A2' + Cauchy-Schwarz; standard Lyapunov ultimate-boundedness theorem (Khalil 2002 Thm 4.18)	**Proved (conditional on** $R^* < R$**)**
Positive invariance of $\mathcal{B}_R$ under the persistence condition	Boundary-inward argument: at $\|\delta\| = R$, $\dot{V} < 0$ when $\alpha R > \rho$	Proved
Initial-condition scope (trajectory must start inside $\mathcal{B}_R$)	Direct consequence of A2' being local; trajectories outside are not covered	Derived (scope statement)
Identification of structural-adaptation trigger ($\rho/\alpha > R$ forces exit from parametric regime)	Connects A.1's persistence threshold to 4.5's regime boundary	Derived
Prop A.2: adaptive reserve $\Delta\rho^* = \alpha R - \rho$	Algebraic corollary of Prop A.1 under shock $\rho \to \rho + \Delta\rho$; requires R fixed under shock	Proved (corollary of A.1)
Prop A.1S (region-aware): stopped bound $\mathbb{E}[\|\delta(t \wedge \tau_R)\|^2] \leq \|\delta(0)\|^2 e^{-2\alpha t} + n\sigma_w^2/(2\alpha)$ **+ fixed-time tail** $P(\|\delta(t)\| > R) \leq n\sigma_w^2/(2\alpha R^2)$ **(stationary-sharp) + finite-horizon sup-bound (iv)**	Itô's formula on V + A2' on $\mathcal{B}_R$ + Grönwall + stopping-time localization at τ_R (Khasminskii 2012 ch. 5; Khalil 2002 ch. 9)	**Proved**; the infinite-horizon non-exit probability is structurally 1 for additive Model S (no-go, Khasminskii ch. 3–4) — pathwise containment is Model-D-only
Cor A.1S.1: containment dichotomy $P(\tau_R < \infty) \in \{0, 1\}$, **$\alpha$-invariant**	Prop A.1 positive invariance (Model D $= 0$) + a.s. finite exit of a non-degenerate diffusion from a bounded region for any F under A2' (Model S $= 1$); Khasminskii 2012 ch. 3–4	**Proved** — new exact result
Stochastic steady-state RMS $R_S^* = \sigma_w\sqrt{n/(2\alpha)}$	Steady state of the Grönwall bound; scales as $1/\sqrt{\alpha}$ versus $1/\alpha$ deterministic	Derived
Tail bound $P(\|\delta\| > R) \leq n\sigma_w^2/(2\alpha R^2)$	Markov's inequality applied to steady-state second moment	Proved
Linear-case recovery ($\alpha = \mathcal{T}$ reproduces 4.4)	Substitution $F = \mathcal{T}\delta$ into A2'	Derived
Adversarial-exponent scaling $b = 2$ (Model D) vs $b = 3/2$ (Model S) into 14.3	Direct transfer of the $1/\alpha$ vs $1/\sqrt{\alpha}$ scalings	Derived (transfer claim)
Bounded-disturbance model (GA-2) vs stochastic-disturbance model (GA-2S) as distinct environmental regimes, not approximations to each other	Discussion-section positioning; neither regime handles heavy tails	Discussion-grade

Property	*Source*	*Strength*
Global sector condition would give $\Delta\rho^* = \infty$	Limit analysis of A.2 as $R \to \infty$; rejected as unrealistic for finite model classes	Derived (scope-limit observation)

The dividing line: the two persistence results (ultimate bound, adaptive reserve) and the stochastic mean-square bound are **proved** within standard Lyapunov / Itô-Lyapunov theory — the appendix's contribution is the application to AAT's correction-function object, not the mathematics. What is **chosen** is the quadratic Lyapunov candidate $V = \frac{1}{2}\|\delta\|^2$ (canonical for norm-bounded stability but not uniquely forced; see Path 3 note below) and the bounded-vs-stochastic disturbance partition (two structurally distinct empirical regimes rather than a hierarchy). The **sector condition itself (A2')** now carries its sub-scope explicitly: **derived in sub-scope** α (optimal Bayesian / exponential-family / strongly-convex-gradient / L2-regularized / linear-PD corrections, via 4.2 Prop B.3 under B1), **assumed in sub-scope** β (PID / rule-based / human-judgment / severely-misspecified / variational / non-convex-beyond-basin / per-step SGD). Prop A.1S's region condition is no longer deferred to Epistemic Status — the proposition statement is region-aware via a standard stopping-time localization (Khasminskii 2012 ch. 5); the fixed-time tail (iii′) quantifies the typical-scale cost of Wiener excursions, and the *absence* of any infinite-horizon non-exit bound (a structural no-go for additive Model S) is itself the result — pathwise containment is a Model-D guarantee only.

Epistemic Status.

The setup and assumptions are *definitions* — they specify what we mean by "correction function" and "disturbance." Propositions A.1 and A.2 are *exact* — they follow from the assumptions via standard Lyapunov theory (Khalil 2002[4], Chapters 4 and 9). Proposition A.1S is *exact* in its region-aware form: the stopped Grönwall bound (i), the mean-square persistence condition (ii), the fixed-time tail (iii′), and the finite-horizon sup-bound (iv) are each exact (stopping-time localization at τ_R, Khasminskii 2012[5], ch. 5). **Corollary A.1S.1 is itself an exact result** — and a new one: the first-exit probability is *exactly* the two-point set $\{0, 1\}$ (0 under bounded Model D, 1 under additive stochastic Model S), categorical and α-invariant, the value selected by the disturbance model's support structure rather than by correction strength. Both halves are exact (Model-D: deterministic positive invariance, Prop A.1; Model-S: almost-sure finite exit of a non-degenerate diffusion from a bounded region, for any F satisfying A2'). No implicit strengthening of A2' is required.

Sub-scope α / β status of A2'. The sector condition A2' is *derived* within sub-scope α (the five agent classes named in the Grounding paragraph — Bayesian / exponential family / strongly-convex gradient / L2-regularized / linear-PD corrections) via 4.2 Prop B.3 under B1 directional fidelity. Within sub-scope α the magnitude α is computable from the update-rule parameters. Within sub-scope β (PID, rule-based, human judgment, severely misspecified, variational approximations, non-convex beyond basin, per-step SGD), A2' stands as an empirical claim that must be verified per-agent. The Lyapunov proofs above apply to both sub-scopes — they operate downstream of A2' regardless of how it is established. This is *scope narrowing*, not scope retreat.

Why Euclidean A2' specifically. The A2' form $\delta^T F \geq \alpha\|\delta\|^2$ is the sector condition *matched to the quadratic Lyapunov candidate* $V = \frac{1}{2}\|\delta\|^2$. A converse-Lyapunov argument (Khalil 2002, Thm 4.17) gives: if persistence holds under the dynamics $\dot{\delta} = -F(\delta)$ on $\mathcal{B}_R$, then there exists a quadratic-equivalent Lyapunov function $V_*(\delta)$ with $c_1\|\delta\|^2 \leq V_* \leq c_2\|\delta\|^2$ — but V_* may not be the Euclidean norm itself. Under a weighted Lyapunov candidate $V(\delta) = \frac{1}{2}\delta^T P\delta$, the natural sector condition is $\delta^T P F(\delta) \geq \alpha\, \delta^T P\delta$; the matrix-Kalman case of 4.2 is exactly this in the $(P^-)^{-1}$-weighted inner product, with a norm-equivalence transfer to Euclidean A2' degraded by the condition

[4]Khalil, H. K. (2002). *Nonlinear Systems* (3rd ed.). Prentice Hall.

[5]Khasminskii, R. (2012). *Stochastic Stability of Differential Equations* (2nd ed.). Springer. Chapter 5: Lyapunov functions and stochastic stability; stopping-time localization for diffusion processes.

number $\kappa(P^{-})$. Euclidean A2' is therefore not the unique sector form — it is the canonical one matched to the canonical V. An agent that persists under a non-Euclidean metric satisfies a weighted-sector A2' that transfers to Euclidean A2' only up to norm equivalence. Z classifies the Lyapunov case as an *adjacent family member* to AAT's three primary additive-coordinate-forcing instances (chain / divergence / update, where a logarithmic coordinate is uniquely forced via Cauchy's functional equation under an AAT-internal additivity axiom): here the quadratic coordinate is matched to the sector form rather than forced by one.

The assumptions themselves (sector condition on a region, bounded disturbance) are *empirical claims* about the qualitative behavior of real correction dynamics. The sector-condition framework originates with Lur'e (1957); the Lyapunov stability results are standard. The application to adaptive agents under the AAT framework is new but the mathematics is not.

Discussion.

Key value. The persistence threshold and adaptive reserve are no longer contingent on the linear hypothesis in 3.9. They hold for any correction dynamics satisfying the sector condition — a mild qualitative assumption that says "correction points inward with at least baseline efficiency α." This is a significant epistemic upgrade: from *hypothesis-dependent* to *robust under qualitative assumptions*.

What the proofs do NOT illuminate. (1) Quantitative steady-state values — Lyapunov gives *bounds*, not exact values; the linear analysis remains necessary for quantitative predictions. (2) Convergence rates — standard Lyapunov tells you stable/unstable, not how fast. (3) Optimal gain structure — 3.6 comes from estimation theory, not stability theory. (4) Model sufficiency — the 2.2 framework is information-theoretic, complementary to but independent of stability analysis.

Two disturbance models. Proposition A.1 assumes bounded disturbance (GA-2: $\|w(t)\| \leq \rho$); Proposition A.1S assumes stochastic disturbance (GA-2S: $\mathbb{E}[\|w(t)\|^2] = \sigma_w^2$). These are not approximations to each other — they capture structurally different environments. Model D (bounded) covers persistent directional change: an adversary who maneuvers, an API that drifts, a climate that shifts. Model S (stochastic) covers unpredictable fluctuations around a stable mean: market noise, sensor noise, random perturbations. The choice of model is an empirical question for each domain; the theory provides both tools. Neither model handles heavy-tailed environmental shocks (financial crises, ecological catastrophes, strategic surprise) where $\|w(t)\|$ can exceed any finite bound with non-negligible probability. Extreme tail events are better understood as triggers for structural adaptation (4.5) rather than disturbances to be absorbed parametrically.

Kind of guarantee, not just rate. The two models differ not only in how *tightly* the mismatch is held but in *what kind* of guarantee is available at all. Corollary A.1S.1 makes this exact and categorical: $P(\tau_R < \infty) = 0$ under Model D (positive invariance) versus 1 under Model S (recurrent additive-noise diffusion), with no intermediate value reachable by any correction strength. What Model S controls instead is the *typical scale* (the $1/\sqrt{\alpha}$ RMS radius, ii) and the *fixed-time tail* (iii′) — distributional, instantaneous guarantees, not a sample-path-forever one. Increasing α shrinks the typical scatter; it does not convert a distributional guarantee into a pathwise one. This is the deeper content of "these are not approximations to each other": additive stochastic forcing removes the *kind* of containment available, not merely its rate. It sharpens — rather than caveats — the hand-off into 4.5: in a genuinely stochastic environment, region-exit is not a measure-zero pathology to be assumed away but a *certain eventual event*, so the trigger for structural (non-parametric) adaptation is **generic, not exceptional**, for any sufficiently long-lived agent.

Findings.

The disturbance-model containment dichotomy: $P(\tau_R < \infty)$ is exactly $\{0, 1\}$, α-invariant. **Result (exact, new).** Corollary A.1S.1: under the sector-condition correction dynamics, the persistence-region first-exit probability is *categorical* — exactly 0 under bounded disturbance (Model D, positive invariance) and exactly 1 under additive stochastic disturbance (Model S, almost-sure exit of a non-degenerate diffusion from a bounded region). The achievable value is the two-point set $\{0, 1\}$; which point obtains is fixed by the disturbance model's support structure, **not** by correction strength α.

Brief: Against a *bounded* disturbance — a steady wind you can lean into — a sector-stable corrector holds the mismatch inside a fixed region *forever and with certainty*: once in, never out. Against *stochastic* disturbance — random gusts — no correction strength buys that. The agent stays near target on average and almost all of the time, but over an unbounded horizon some fluctuation eventually pushes it past any fixed boundary, with probability one. Stronger correction tightens the *typical* scatter (the RMS radius shrinks as $1/\sqrt{\alpha}$); it does not erect a wall. So moving from bounded to stochastic environments is not "the same guarantee with a weaker constant" — it is a change in the *kind* of guarantee available: pathwise-and-forever (Model D) versus distributional-and-fixed-time (Model S). Practical consequence: in any genuinely stochastic environment, leaving the parametric-correction region is not a rare pathology to assume away — it is a certain eventual event, so the trigger for *structural* adaptation is generic, not exceptional.

Impact: Sharpens the Model-D/Model-S architecture and the hand-off into 4.5 (structural adaptation is a generic eventual necessity for long-lived agents in stochastic environments, not an edge case). The companion controls under Model S are the fixed-time tail (iii′, the Markov bound $P(\|\delta(t)\| > R) \leq n\sigma_w^2/(2\alpha R^2)$, stationary-sharp) and the finite-horizon sample-path bound (iv); pathwise containment is categorically Model-D-only. The Model-S half is the load-bearing proof step demonstrated at length in `#deriv-stochastic-non-exit`, which also packages the reusable no-go signature (*unbounded scale function ⇒ no non-constant bounded harmonic function ⇒ no horizon-independent non-exit certificate*) that future stochastic-containment proposals can be settled against rather than re-attempted.

Novelty Claim: *Synthesis* — an exact result built from classical components. The component facts are textbook: deterministic positive invariance under bounded disturbance (Khalil 2002 ch. 4) and almost-sure finite exit of a non-degenerate diffusion from a bounded region (Khasminskii 2012 ch. 3–4; the OU scale-function computation is the explicit instance). The new exact result is their synthesis into a *categorical, α-invariant containment dichotomy* stated as a structural property of AAT's own Model-D/Model-S architecture: correction strength cannot interpolate $P(\tau_R < \infty)$ between the two disturbance regimes — the value is exactly $\{0, 1\}$, selected by disturbance-support structure. This exact dichotomy is new to the framework; it sharpens the structural-adaptation hand-off (4.5). No novelty is claimed on the SDE mathematics — the novelty is the exact dichotomy as a framework theorem and its α-invariance.

Related Work:

Table A.4

ASF concern	*Prior-art language*	*Relationship / Positioning*
No infinite-horizon non-exit bound for additive-noise diffusions	Khasminskii 2012, *Stochastic Stability of Differential Equations* (2nd ed.) ch. 3–4 (recurrence; scale function; no bounded non-constant harmonic function)	*formal antecedent* — supplies the classical recurrence fact; AAT recognizes it as the structural reason Model S admits no pathwise-forever containment in contrast to Model D
Fixed-time second-moment tail	Markov's inequality on the stopped Itô-Lyapunov second moment (Khasminskii 2012 ch. 5)	*standard machinery* — (iii′) is Markov on (i); the AAT-specific content is the Model-D/S kind-of-guarantee contrast it forces
Pathwise positive invariance under bounded disturbance	Khalil 2002 ch. 4 (ultimate boundedness; boundary-inward invariance)	*formal antecedent* — Model D's pathwise-forever guarantee (Prop A.1) is the contrast term that makes the Model-S no-go a *kind* distinction, not a rate one

Search Log:

- 2026-05-16 (*targeted, derivation-driven*): the question was internal — does Prop A.1S(iii)'s infinite-horizon non-exit bound hold? The strengthening attempt (Doob/Ville maximal inequality on the Itô-Lyapunov supermartingale) was worked in full and shown to fail structurally; the classical recurrence fact (Khasminskii ch. 3–4) is the obstruction and is well-known. No prior-art search beyond the standard SDE references was needed — no novelty is claimed on the mathematics, only on the framework-level kind-of-guarantee recognition. Reasoning trail: `spikes/spike-stochastic-non-exit-strengthening-2026-05-16.md`.

Working Notes.

- The adversarial extension (Prop A.3, coupled agents) and effects spiral (Cor A.3.1) are in 13.2. The multi-timescale sketch (A.4) is in I.
- The vector treatment of $\delta(t) \in \mathbb{R}^n$ connects directly to per-dimension tempo analysis (14.5). Each dimension can have different effective α_k values, and the weakest dimension determines overall persistence — a tensor generalization of the scalar results here.
- A global sector condition (A2 without the local restriction to $\mathcal{B}_R$) would give global stability, making $\Delta\rho^*$ infinite — the agent could absorb any finite disturbance shock. But this requires the correction function to work perfectly at arbitrary mismatch magnitudes, which is unrealistic for any finite model class. The local form (A2') is the honest one.
- Landing-context provenance: the sub-scope α/β partition reflects the strengthening trail recorded in `spikes/spike-a2-prime-strengthening.md` — the analysis that ruled out a universal A2' derivation and identified the five operator families where the bridge is structural.
- Landing-context provenance (iii′)/(iv)/dichotomy: the prior (iii) asserted an infinite-horizon $P(\tau_R < \infty) \le n\sigma_w^2/(2\alpha R^2)$ via a "Markov tail on the supermartingale," conflating a fixed-time second-moment bound with the ever-exit probability. Audit findings 742613-SUPPLEMENT §2 and 613842-F2 flagged it and recommended a *soften* (restate as fixed-time). Per strengthen-before-soften the strengthening (Doob/Ville maximal inequality, the route the 628401 adjudication predicted would succeed) was attempted in full first and **fails structurally** — documented dead-end in `spikes/spike-stochastic-non-exit-strengthening-2026-05-16.md` (no nonnegative supermartingale dominates V; additive-noise generator has no bounded non-constant harmonic function; recurrent OU exits a.s., EM-corroborated). Completion-state (3): the failure *is* the result — the (iii′)+(iv)+kind-of-guarantee-dichotomy package is strictly more honest and more informative than the false (iii). The 628401 prediction that the strengthening would succeed is recorded as disconfirmed so it is not re-attempted.
- *Low-confidence ideation (flagged per "working notes FTW" — glimpses worth recording, not claims):*
  - The dichotomy is now stated as the labeled exact result **Corollary A.1S.1**; the general-F concern is resolved *within* it (the Model-S half needs only non-degeneracy of the additive noise on a bounded region — no linear structure, any F under A2'). The open glimpse is whether the *dichotomy itself* recurs at the corpus's other bounded/additive-stochastic pairs — `#deriv-discrete-sector-condition`, `#deriv-matrix-persistence-condition`, `#deriv-adaptive-gain-dynamics`, possibly `#result-per-dimension-persistence`. If Cor-A.1S.1-shaped dichotomies appear at ≥3 sites, that is a candidate instance-family for the theorem-import-architecture meta-segment (PROPOSALS SP-23) or a sibling of `#disc-identifiability-floor` (the infinite-horizon non-exit object is *structurally absent*, not merely hard to bound — an identifiability-floor-shaped no-go, now with an exact dichotomy as its positive-content companion).
  - Sharper consequence for `03-llm-core/04-eli-core`: ELI/LLM operating environments are Model-S-like (stochastic), so the corrected (iii′)+dichotomy says structural adaptation is *certain over an unbounded horizon* for long-lived language-constituted agents — not contingent. That tightens the persistence ↔ continuity / Three-Deaths line: continuity *requires* recurring structural (not just parametric) adaptation, and the Model-S no-go makes that requirement inevitable rather than a risk. Worth a deliberate look when the parts III/IV persistence segments are next touched (low confidence on exact form; high confidence the connection is real).

- The no-go's signature — "compensating to restore the supermartingale destroys nonnegativity / generator has no bounded non-constant harmonic function" — is a reusable *diagnostic*: a fast structural test for whether any proposed "stays-in-region-forever w.h.p." claim elsewhere is impossible vs. merely unproven. Candidate content for whichever meta-segment catalogs the framework's no-go boundary results.
- Downstream worth a check (not chased here): does the $b = 3/2$ Model-S adversarial-exponent scaling in `#result-adversarial-exponent-regimes` implicitly lean on the old infinite-horizon framing anywhere? Expected not (it transfers the $1/\sqrt{\alpha}$ RMS scaling, which is unchanged), but flagged so a future pass confirms rather than assumes.

Appendix B

Derivation: No Horizon-Independent Non-Exit Bound Under Additive Stochastic Forcing (the Model-S No-Go)

Under additive stochastic forcing there is no finite bound on the infinite-horizon first-exit probability — $P(\tau_R < \infty) = 1$ for every correction strength — and the natural maximal-inequality route to one provably cannot exist; this is the load-bearing proof step behind Prop A.1S(iii′)/(iv) and the Model-S half of Corollary A.1S.1.

Formal Expression.

Setup. Mismatch dynamics under additive stochastic disturbance, started inside the sector-condition region $\mathcal{B}_R$:

$$d\delta = -F(\mathcal{T}, \delta)\, dt + \sigma_w\, dW_t, \qquad \delta(0) \in \mathcal{B}_R, \quad \sigma_w > 0, \tag{B.1}$$

with W_t a standard n-dimensional Wiener process, (A2') $\delta^\top F \geq \alpha\|\delta\|^2$ on $\mathcal{B}_R$, and $\tau_R = \inf\{t : \|\delta(t)\| > R\}$ the first-exit time. Let $V = \frac{1}{2}\|\delta\|^2$.

Theorem (Model-S no-go). *[Derived]* For every $\alpha > 0$, every $\sigma_w > 0$, and every correction function F satisfying (A2'),

$$P(\tau_R < \infty) = 1, \tag{B.2}$$

and there exists no nonnegative supermartingale dominating V that certifies a horizon-independent bound $P(\tau_R < \infty) \leq c < 1$. Pathwise containment of $\mathcal{B}_R$ is unattainable under additive stochastic forcing at any correction strength.

Why the natural route cannot work. The route a careful reader reaches for first is a time-uniform maximal inequality (Ville / Doob) on an Itô–Lyapunov supermartingale — the same instinct that makes the fixed-time mean-square bound (Prop A.1S(i)) succeed. It fails, and the failure is structural, not a matter of a missing trick.

Define $G(t) = e^{2\alpha t} V(\delta(t))$. On $[0, \tau_R]$, by Itô and (A2'),

$$dG = e^{2\alpha t}\Big[\underbrace{(2\alpha V - \delta^\top F)}_{\leq 0 \text{ on } \mathcal{B}_R}\, dt + \tfrac{n}{2}\sigma_w^2\, dt + \delta^\top \sigma_w\, dW_t\Big] \;\leq\; e^{2\alpha t}\tfrac{n}{2}\sigma_w^2\, dt + e^{2\alpha t}\delta^\top \sigma_w\, dW_t. \tag{B.3}$$

G is **not** a supermartingale: the $+\,e^{2\alpha t}\frac{n}{2}\sigma_w^2\,dt$ term has strictly positive drift growing exponentially. Removing it by compensation gives

$$S(t) = e^{2\alpha(t\wedge\tau_R)}V(\delta_{t\wedge\tau_R}) - \frac{n\sigma_w^2}{4\alpha}\big(e^{2\alpha(t\wedge\tau_R)} - 1\big), \tag{B.4}$$

and $dS \le e^{2\alpha t}\delta^\top\sigma_w\,dW_t$ on $[0,\tau_R]$ — so S *is* a supermartingale. **But S is not nonnegative.** The subtracted $\frac{n\sigma_w^2}{4\alpha}(e^{2\alpha t} - 1)$ dominates $e^{2\alpha t}V$ whenever $V(\delta(t)) < \frac{n\sigma_w^2}{4\alpha}$ — i.e. exactly inside the persistence basin, which under the mean-square persistence condition is *most of the time* (that condition places the RMS radius $R_S^* = \sigma_w\sqrt{n/2\alpha}$ well inside $\mathcal{B}_R$, so $V \ll \frac{n\sigma_w^2}{4\alpha}$ typically). Ville's inequality requires a nonnegative supermartingale; Doob's maximal inequality a nonnegative sub/supermartingale. Both are inapplicable to a sign-indefinite S, and the obstruction is not removable: for additive non-degenerate Brownian forcing the diffusion's scale function is unbounded (the OU scale density $\propto e^{\alpha u^2/\sigma_w^2}$), so the only bounded harmonic functions of the generator are constants — the gambler's-ruin / Lyapunov-exit machinery cannot certify "stays inside forever with positive probability." There is no nonnegative supermartingale dominating V with finite expected initial value that yields a horizon-independent exit bound.

Why $P(\tau_R < \infty) = 1$, generally. The conclusion does not depend on the linear structure. A non-degenerate diffusion (additive forcing $\sigma_w\,dW_t$, $\sigma_w > 0$) exits any bounded region in finite time almost surely, for *any* locally bounded drift: near $\partial\mathcal{B}_R$ the Brownian increment has positive probability of crossing in any time interval, and no finite inward correction satisfying (A2') can suppress this (A2' bounds $\delta^\top F$ from below by $\alpha\|\delta\|^2$, a finite inward push, not an impassable wall). Hence $\tau_R < \infty$ a.s. for every F under (A2'), every α, every σ_w. The Ornstein–Uhlenbeck case is the explicit instance (positively recurrent on $\mathbb{R}^n$, unbounded stationary support, exits any finite ball a.s.), not the basis.

Epistemic Status.

Exact. Both the impossibility of the maximal-inequality certificate (sign-indefiniteness of the only candidate compensated supermartingale; unbounded scale function ⇒ no non-constant bounded harmonic function) and the positive fact $P(\tau_R < \infty) = 1$ (almost-sure finite exit of a non-degenerate diffusion from a bounded region under any locally bounded drift) are classical (Khasminskii 2012[1] ch. 3–4). The contribution here is not the SDE mathematics but the demonstration that the route which *succeeds* for the fixed-time second moment (Prop A.1S(i)) structurally *cannot* deliver an infinite-horizon containment statement — the answer to "are you sure you can't just bound $\sup_t$ with Doob/Ville?" is *yes, sure, and here is precisely the step that breaks.* "Exact" is claimed in the framework's defeasible sense (valid under stated assumptions, subject to a found error), not as a claim beyond a found-mistake.

This derivation is load-bearing for the critical path: it is the proof step that forces Prop A.1S's corrected (iii′) fixed-time tail and (iv) finite-horizon sup-bound (in `#deriv-sector-condition`) and grounds the Model-S half of Corollary A.1S.1. It is a **reusable no-go signature**: wherever a "stays-in-region-forever with positive probability" claim is proposed under additive non-degenerate forcing, the unbounded-scale-function / no-non-constant-bounded-harmonic-function obstruction settles it negatively without a fresh attempt.

[1] Khasminskii, R. (2012). *Stochastic Stability of Differential Equations* (2nd ed.). Springer. Chapters 3–4 (recurrence, scale function, bounded harmonic functions); Chapter 5 (Lyapunov functions, stopping-time localization).

Discussion.

The structural reading: additive stochastic forcing does not weaken the *rate* of containment, it removes a *kind* of guarantee. Bounded disturbance (Model D) gives deterministic positive invariance of $\mathcal{B}_R$ ($P(\tau_R < \infty) = 0$); additive stochastic disturbance gives almost-sure eventual exit ($P(\tau_R < \infty) = 1$); the achievable value is the two-point set $\{0, 1\}$, selected by the disturbance's support structure, not by correction strength — Corollary A.1S.1 in `#deriv-sector-condition`. What survives under Model S is distributional and instantaneous: the fixed-time / stationary Markov tail (Prop A.1S(iii′)) and the finite-horizon sample-path bound (Prop A.1S(iv), which grows linearly in the horizon and is vacuous for $T > rsimR^2/(n\sigma_w^2)$).

Direct simulation confirms the obstruction is real rather than an artifact of the proof technique. Exact 1-D OU benchmark $dX = -\alpha X\,dt + \sigma\,dW$ (the segment's own linear case; (A2') global and sharp), Monte-Carlo started *at the origin* (the most favorable initial condition), under the mean-square persistence condition (ii) holding:

Table B.1

α	σ	R	R_S^*	$P(\sup_{[0,50]} > R)$	T=200	T=800	T=3200	*seg. const* $\frac{n\sigma^2}{2\alpha R^2}$
1.0	0.3	1.0	0.21	0.002	0.010	0.030	**0.114**	0.045
0.5	0.2	0.8	0.20	0.020	0.074	0.276	**0.720**	0.063
1.0	0.5	1.0	0.35	0.781	0.998	1.000	**1.000**	0.125

The ever-exit probability climbs monotonically toward 1 with the horizon even started at the origin and even with (ii) comfortably satisfied — the fixed-time constant $n\sigma^2/(2\alpha R^2)$ is *not* an upper bound on the sup-over-horizon probability, exactly as the no-go predicts. This sharpens the hand-off into `#result-structural-adaptation-necessity`: under any genuinely stochastic environment region-exit is a certain eventual event, so the trigger for structural (non-parametric) adaptation is generic, not exceptional, for a sufficiently long-lived agent.

Findings.

The natural maximal-inequality route to infinite-horizon containment provably cannot exist under additive stochastic forcing. **Brief:** If a system is held near a target by inward correction but also kicked by persistent random noise, you can bound how far off it is at any *given* moment, and you can bound the worst excursion over any *fixed window* — but you can never bound "it stays inside the safe region *forever*," because given unlimited time the noise will, with certainty, eventually push it out, no matter how strong the correction. The tempting fix — find a single "energy" quantity that only ever drifts downward, so a classic gambler's-ruin argument caps the lifetime escape probability — fails for a precise reason: the only candidate energy that has the right drift is negative exactly where the system spends most of its time, and the gambler's-ruin machinery requires a non-negative quantity. This is not a missing trick; it is structural, and it is the same obstruction every time, so it can be invoked rather than re-derived.

Impact: Supplies the load-bearing proof step for Prop A.1S(iii′)/(iv) and Corollary A.1S.1 in `#deriv-sector-condition` (the Model-S half of the containment dichotomy). Provides a reusable no-go signature — *unbounded scale function* $\Rightarrow$ *no non-constant bounded harmonic function* $\Rightarrow$ *no horizon-independent non-exit certificate* — that future stochastic-containment proposals in the corpus can be settled against without re-attempting the maximal-inequality route.

Novelty Claim: *Recognition*. The component facts (non-degenerate-diffusion exit from a bounded region; unbounded scale function of additive-noise generators; Ville/Doob preconditions) are classical (Khasminskii 2012 ch. 3–4). The contribution is the recognition, made explicit and citable, that the maximal-inequality route which succeeds for the fixed-time second moment structurally cannot deliver the infinite-horizon statement — i.e. *where exactly* the natural attempt breaks. No novelty is claimed on the SDE mathematics.

Related Work:

Table B.2

ASF concern	*Prior-art language*	*Relationship / Positioning*
No horizon-independent non-exit certificate for additive-noise diffusions	Khasminskii 2012, *Stochastic Stability of Differential Equations* (2nd ed.) ch. 3–4 (scale function; recurrence; bounded harmonic functions)	*formal antecedent* — supplies the recurrence / unbounded-scale-function fact; this derivation recognizes it as the structural reason the Ville/Doob route fails
Maximal-inequality preconditions	Ville's inequality; Doob's maximal inequality (nonnegative sub/supermartingale required)	*standard machinery* — the demonstration is that the only compensated candidate is sign-indefinite exactly inside the persistence basin

Search Log:

- 2026-05-16 (*targeted, derivation-driven*): internal question — can Prop A.1S's infinite-horizon non-exit claim be certified by Ville/Doob on an Itô–Lyapunov supermartingale? Worked in full; the candidate compensated supermartingale is sign-indefinite, and the classical unbounded-scale-function fact (Khasminskii ch. 3–4) closes it. No prior-art search beyond standard SDE references — no novelty claimed on the mathematics. Reasoning trail: `spikes/spike-stochastic-non-exit-strengthening-2026-05-16.md`.

Working Notes.

- Provenance: the prior Prop A.1S(iii) asserted an infinite-horizon $P(\tau_R < \infty) \le n\sigma_w^2/(2\alpha R^2)$ via a fixed-time "Markov tail on the supermartingale." Two audit records (742613-SUPPLEMENT §2, 613842-F2) flagged it and recommended a soften; per strengthen-before-soften the strengthening (the Ville/Doob route, which a peer adjudication confidently predicted would succeed — "standard textbook, should not fail") was worked in full *first* and fails structurally, as demonstrated above. Completion-state (3): the failure is the result. The peer prediction that the strengthening would succeed is recorded here and in `spikes/spike-stochastic-non-exit-strengthening-2026-05-16.md` as disconfirmed, so it is not re-attempted. 742613-SUPPLEMENT §2 + 613842-F2 are resolved by strengthening-then-no-go (state 3), not by soften.
- The generality (any F under A2', not just OU) rests on the elementary non-degenerate-diffusion-exits-bounded-region fact, not on the OU scale-function computation; the latter is the explicit instance. No further strengthening of this half is open.
- Low-confidence glimpse (per "working notes FTW"): the same obstruction shape should recur wherever the corpus pairs a bounded-disturbance result with an additive-stochastic counterpart — `#deriv-discrete-sector-condition`, `#deriv-matrix-persistence-condition`, `#deriv-adaptive-gain-dynamics`, possibly `#result-per-dimension-persistence`. If a Corollary-A.1S.1-shaped dichotomy appears at ≥3 sites, that is a candidate instance-family for the theorem-import-architecture meta-segment (PROPOSALS SP-23) or a sibling of `#disc-identifiability-floor` (the infinite-horizon non-exit object is *structurally absent*, not merely hard to bound).

Appendix C

Result: Sector-Persistence Template

Any state variable evolving under bounded-correction dynamics with bounded disturbance admits the same Lyapunov persistence argument. AAT's persistence-flavored results — epistemic, strategic, team, composite closure, composite tempo, adversarial destabilization — are instances of a single template. This segment states the template once in parameter-free form so that each instantiation can cite it and specify only what varies: its state variable, correction function, effective disturbance rate, and reserve.

Formal Expression.

Let $\xi(t) \in \mathbb{R}^n$ be a state variable evolving under

Template preconditions (sector-persistence-template).

$$\frac{d\xi}{dt} = -F(\xi) + w(t) \tag{C.1}$$

where F is a correction function and $w(t)$ is a disturbance. The template applies when:

(T1) Zero correction at zero state. $F(0) = 0$ — no correction is applied when the state is at its target.

(T2) Local sector condition. There exist $\alpha > 0$ and $R > 0$ such that

$$\xi^T F(\xi) \geq \alpha \|\xi\|^2 \quad \text{for } \|\xi\| \leq R. \tag{C.2}$$

The correction points inward with at least baseline efficiency α throughout the region of radius R.

(T3) Bounded disturbance. Either:

- *Model D (deterministic bound):* $\|w(t)\| \leq \rho_\xi$, or
- *Model S (stochastic zero-mean):* $\mathbb{E}[\|w(t)\|^2] = \sigma_\xi^2$ with $w(t)$ a Wiener-process increment.

Under (T1)–(T3), with $V(\xi) = \frac{1}{2}\|\xi\|^2$ as Lyapunov function:

Template result (from sector-condition-derivation Props A.1, A.1S, A.2).

Model D. The state is ultimately bounded by $R^* = \rho_\xi/\alpha$. Structural persistence (the ultimate bound fits within the sector-condition region) requires

$$\alpha > \frac{\rho_\xi}{R}. \tag{C.3}$$

The adaptive reserve — the additional disturbance the system can absorb before persistence fails — is $\Delta\rho_\xi^* = \alpha R - \rho_\xi$.

Model S. The state satisfies $\mathbb{E}[\|\xi(t)\|^2] \to n\sigma_\xi^2/(2\alpha)$ in mean square, giving RMS bound $R_S^* = \sigma_\xi\sqrt{n/(2\alpha)}$. Structural persistence in the mean-square sense requires

$$\alpha > \frac{n\sigma_\xi^2}{2R^2}. \tag{C.4}$$

The Model D result scales as $1/\alpha$; the Model S result scales as $1/\sqrt{\alpha}$ — correction is less effective against noise than against drift. This scaling difference propagates into the adversarial exponent regimes (14.3): $b = 2$ under Model D, $b = 3/2$ under Model S.

Instantiations in AAT. The template is invoked across six segments. Each specifies its own (ξ, F, ρ_ξ, R) and verifies (T1)–(T3) locally:

Table C.1

Segment	ξ	*Effective* ρ_ξ	R	*Locally-verified precondition*
4.4	δ_t (epistemic mismatch)	ρ (environmental disturbance rate)	model-class capacity, or task-adequacy threshold $\|\delta_{\text{critical}}\|$ if stricter	(T2) via 4.2
8.10	δ_Σ (strategic mismatch)	ρ_Σ (rate of edge invalidation)	R_Σ (strategic reserve)	(T2) via Beta-Bernoulli edge updates (N Props B.1–B.6); constant-α requires experience discounting
13.1	δ_i (sub-agent mismatch)	$\rho_i^{\text{eff}} = \rho_{i,\text{env}} + \sum_j \gamma_{j\to i}^{\text{adv}} \mathcal{T}_j - \sum_j \gamma_{j\to i}^{\text{coop}} \mathcal{T}_j$	R_i	Cooperative coupling can drive ρ_i^{eff} below the single-agent ρ
11.1 (bridge lemma)	e_m (trajectory error at macro-boundaries m)	$\varepsilon^* \nu_c$ (closure-defect per macro-step $\times$ macro-update rate)	the composite's R_c	Tier-specific contraction stronger than (T2): incremental sector bound (DA2'-inc). Tier 1 proved; Tier 2 local; Tier 3 domain-specific
11.2	δ_c (composite mismatch)	$\rho_{\text{ext}} + \varepsilon^* \nu_c$ (external + internal)	R_c	(T3) requires the bridge-lemma contraction; $C_{\text{coord}} \geq \varepsilon^* \nu_c$ is the tempo-equivalent of the internal disturbance
13.2	δ_B (target-agent mismatch)	$\rho_{B,\text{base}} + \gamma_A \mathcal{T}_A$ (Model D) or $\sigma_{B,\text{base}} + \gamma_A \mathcal{T}_A$ (Model S)	R_B	Destabilization is the negation of persistence: (T3)'s coupling-amplified disturbance violates $\alpha R > \rho_\xi$

The same Lyapunov argument applies in every row. What varies is how each instantiation defines its state variable, its correction function, and — most importantly — what counts as "effective disturbance" for its context (environmental noise, adversarial coupling, closure defect, or a decomposition of these).

External mathematical lineage: monotone-operator theory. The sector-Lyapunov apparatus this segment factors out has a well-established mathematical home: AAT's sector condition (T2) is a **one-point strong monotonicity** condition in the sense of Rockafellar 1970 (*Convex Analysis*, §24) anchored at the equilibrium $\xi^* = 0$, and the incremental strengthening DA2'-inc required by `#form-composition-closure`'s bridge lemma is **full two-point strong monotonicity** in the Bauschke-Combettes 2017 (*Convex Analysis and Monotone Operator Theory in Hilbert Spaces*, 2nd ed., §§22–28) monotone-operator framework. The Lyapunov-plus-Grönwall argument proving the Model D ultimate bound and the Itô-Lyapunov argument proving Prop A.1S are standard specializations of the **monotone-operator perturbation theorem** (Bauschke-Combettes Thm 5.14–5.16; Parikh & Boyd 2014, *Proximal Algorithms*, §2). Banach-Picard contraction under bounded perturbation, monotone-operator convergence under square-summable noise, operator-splitting for composite systems (Douglas-Rachford, ADMM) — the full supporting apparatus is available in that literature.

AAT's relationship to monotone-operator theory is *specialization + repurposing*, not generalization. AAT-distinctive content: (i) **one-point anchoring at the equilibrium** — strictly weaker than full two-point strong monotonicity, matched to fixed-point-at-target semantics, admitting agent classes (PID-bounded-plant, variational-approximate) where full monotonicity fails but persistence-at-the-target is available. (ii) The **Model D / Model S disturbance decomposition** with distinct $1/\alpha$ vs $1/\sqrt{\alpha}$ scaling — monotone-operator theory has perturbation theorems but no systematic bounded-adversarial vs stochastic-zero-mean split propagating to adversarial-exponent regimes at $b = 2$ vs $b = 3/2$. (iii) Composition with the `#disc-identifiability-floor` meta-pattern supplies a second (information-theoretic) axis orthogonal to the operator-theoretic machinery. (iv) The `#post-composition-consistency` postulate operates at AAT's level of description (agent / composite / macro-agent) rather than at the abstract operator level. (v) The sub-scope α/β epistemic labeling is scope-honesty, not a mathematical partition — it tracks which agent classes give the monotone-operator structure *by construction* versus *by per-instance verification*.

This acknowledgment is load-bearing for scope honesty: the mathematical machinery is established external to AAT; AAT's distinctive content is the agent-architecture specialization (singular-trajectory identity per `#scope-agent-identity`; signed-coupling composition; coordinate-forcing via uniqueness theorems per `#disc-additive-coordinate-forcing`; three meta-patterns) rather than novel monotone-operator mathematics. `#deriv-sector-condition`'s Grounding paragraphs name the specific operator-family correspondence (proximal / firmly-nonexpansive / cocoercive / strongly-monotone-gradient / linear-PD) for the five sub-scope-α agent classes. `#result-contraction-template` extends to non-Euclidean metrics via Lohmiller-Slotine differential-contraction (also within the broader monotone-operator lineage). The honest limits of the unification: the coarse-graining projection Λ (11.1) does not fit the operator-sector primitive (heterogeneous spaces, three independent admissibility conditions); three of five metric-α_2 cases in `#result-contraction-template` remain theorem-imported rather than AAT-internally derived; the identifiability-floor axis is orthogonal to the operator-sector axis.

Comparison with the FEP-flow stability argument. Active inference's stability arguments come from the geometry of the variational free-energy landscape — agents are argued to flow toward the minimum of variational free energy on a non-equilibrium-steady-state (NESS) density. The primary source for the NESS-density framing is Friston 2019, "A free energy principle for a particular physics," arXiv:1906.10184; the path-integral / particular-kinds methodological extension is Friston, Da Costa, Sakthivadivel, Heins, Pavliotis, Ramstead & Parr 2023, "Path integrals, particular kinds, and strange things," *Phys. Life Rev.* 47 (which rewrites the FEP-flow argument in path-integral language rather than proving new stability bounds). Aguilera, Millidge, Tschantz & Buckley (2022, "How particular is the physics of the free energy principle?", *Phys. Life Rev.* 40:24–50) showed that the FEP-flow argument's mathematical validity is narrow: the NESS-density framing holds only in a small parameter regime for non-equilibrium linear stochastic systems, and natural extensions (nonlinear, non-Gaussian, non-equilibrium) often fall outside the proven regime.

The AAT persistence template is structurally different: it is a Lyapunov-based argument requiring only (T1) zero-correction-at-zero-state, (T2) local sector condition (correction points inward), and (T3) bounded disturbance — all of which are checked locally for each instantiation (A Props A.1, A.1S, A.2). The template applies to bounded and to mean-square-stochastic disturbance, gives explicit ultimate-bound and adaptive-reserve formulas, and does not depend on NESS structure or on a free-energy gradient.

The breadth difference is not rhetorical: where the FEP-flow argument's parameter regime is debated in the AI literature itself, the sector-Lyapunov apparatus is the standard machinery of nonlinear control theory (Khalil 2002, *Nonlinear Systems*, 3rd ed., Prentice Hall, ch. 4) and applies wherever (T1)–(T3) hold. This is one of AAT's stronger structural positions and is worth making explicit when comparing AAT to active inference: AAT does the persistence work AI tries to do, with broader validity and explicit ultimate-bound formulas.

Epistemic Status.

Exact. The template is the abstract form of the Lyapunov result proved in A (Props A.1, A.1S, A.2). The proofs transfer without modification whenever (T1)–(T3) hold; the template's contribution is the recognition that AAT's persistence-flavored results are instances of a single pattern and the enumeration of what each instantiation must verify.

On (T2) and A2' sub-scoping. (T2) is the local sector condition A2' transcribed to a generic state variable ξ. Within the state-variable spaces relevant to AAT's instantiations (epistemic mismatch δ, strategic mismatch δ_Σ, sub-agent mismatch δ_i, composite trajectory error e_m, composite mismatch δ_c, target-agent mismatch δ_B), the update rules are overwhelmingly sub-scope α in the A / 4.2 sense — Bayesian, exponential-family, strongly-convex-gradient, or linear-PD. For these, (T2) is *derived* from the update-rule structure under B1 directional fidelity. **Important, and easy to miss: this is not what carries forward universally.** For sub-scope β instantiations (e.g. a team with a rule-based sub-agent) (T2) is an *empirical precondition, verified per-instantiation*, not derived. That distinction scopes which instantiations satisfy (T2) by which route; it does not touch the template's exactness, which is the conditional "(T1)–(T3) $\Rightarrow$ persistence" — anything satisfying (T2) by either route inherits the exact result.

On Prop A.1S region-awareness. The Model S case uses the region-aware form of Prop A.1S (stopped second-moment bound + the fixed-time tail $P(\|\xi(t)\| > R) \leq$ const under the mean-square persistence condition). Instantiations that rely on the Model S result automatically inherit the stopping-time localization — no extra work is required at the template level. As at A, Model S provides a distributional / fixed-time guarantee, not pathwise-forever containment (which is a Model-D-only guarantee); template instantiations inherit that kind-of-guarantee distinction, not an infinite-horizon non-exit bound.

Max attainable: *exact*. The result is as strong as Lyapunov stability theory. Additional work could extend the template (state-dependent noise, time-varying α, non-quadratic Lyapunov functions) but would not strengthen it within its stated scope.

What the template does not establish:

- *Quantitative convergence rates.* Lyapunov gives stability and ultimate bounds, not convergence speed. Specific rates require instantiation-level analysis.
- *Behavior outside the sector-condition region.* (T2) holds only for $\|\xi\| \leq R$. Beyond R, correction may break down — the structural-adaptation regime of 4.5.
- *Time-varying parameters.* The template assumes constant α and R. Time-varying cases require additional machinery. 8.10 documents the most important AAT example: $\alpha_\Sigma = 1/(n+1)$ decays with experience, so strategic persistence requires experience discounting at rate $(1-\lambda) > \rho_\Sigma / R_\Sigma$ to recover constant-α.
- *Heavy-tailed disturbances.* Neither (T3-D) nor (T3-S) covers disturbances with unbounded moments. Extreme tail events are better handled as triggers for structural adaptation than as disturbances the template can absorb.

Discussion.

Why factor this out. Every persistence-flavored result in AAT takes the form "correction rate exceeds effective disturbance rate (relative to reserve)" and is proved by the same Lyapunov function $V(\xi) = \frac{1}{2}\|\xi\|^2$. Stating the argument once, parameter-free, makes visible that the theory is *one result about bounded-correction dynamics applied to several state spaces*, not six separately-proved results that happen to look alike.

Each instantiation's distinctive content is now sharply visible: *what is the state variable, and what counts as its effective disturbance?* The Lyapunov machinery is shared; the characterization of ρ_ξ is where the domain-specific insight lives. Adversarial destabilization's content is the coupling term $\gamma_A \mathcal{T}_A$; team persistence's content is the cooperative-minus-adversarial decomposition; composition closure's content is the closure-defect rate $\varepsilon^* \nu_c$. In each case, the template absorbs the Lyapunov boilerplate and lets the distinctive claim stand without it.

This template is the *interior facet* of the stability certificate (W): the "correction rate exceeds effective disturbance rate" condition is the certificate being positive-definite on the scope ball, and the existence of such a certificate is equivalent to exponential stability (D). The shared Lyapunov function $V(\xi) = \frac{1}{2}\|\xi\|^2$ is the certificate in the Euclidean metric; carries the non-Euclidean-metric interior.

What each instantiation must still verify. Invoking the template is not trivial. Each instantiation must establish (T1)–(T3) for its specific (ξ, F, ρ_ξ, R), and the non-trivial verifications differ substantively:

- *Strategic persistence* (8.10): (T2) is verified for Beta-Bernoulli edge updates across five DAG topologies (N Props B.1–B.6). But $\alpha_\Sigma = 1/(n+1)$ is time-varying: it decays monotonically with experience. Constant-α — and therefore the template's trajectory guarantee — requires experience discounting as a prerequisite, not a heuristic.
- *Composition closure* (11.1): (T2) as stated is insufficient. The bridge lemma requires the *incremental sector bound* (DA2'-inc) — strongly monotone F across the whole state space, strictly stronger than the one-point sector bound (T2). Three agent tiers result: Tier 1 where contraction is proved for the full class (Bayesian on exponential families, gradient descent on strongly convex losses, linear correctors with positive-definite gain), Tier 2 where it holds locally, Tier 3 where it must be verified per-domain.
- *Team persistence* (13.1): (T3) is the decomposition $\rho_i^{\text{eff}} = \rho_{i,\text{env}} + \sum_j \gamma_{j\to i}^{\text{adv}} \mathcal{T}_j - \sum_j \gamma_{j\to i}^{\text{coop}} \mathcal{T}_j$. Cooperative coupling enters through a negative term, which can drive ρ_i^{eff} below the single-agent ρ — this is formally how teams persist where individuals cannot.

Persistence and destabilization as one result. Adversarial destabilization (13.2) is the template applied with coupling-amplified disturbance: $\rho_B = \rho_{B,\text{base}} + \gamma_A \mathcal{T}_A$. The "destabilization threshold" — the condition under which agent A pushes agent B past its stability boundary — is precisely the *negation* of the template's persistence condition for B's instantiation. B persists iff $\alpha_B R_B > \rho_B$; B destabilizes iff $\rho_B > \alpha_B R_B$. These are the same inequality viewed in opposite directions, not independent results. The superlinear adversarial scaling ($b = 2$ under Model D, $b = 3/2$ under Model S) follows from the template's $1/\alpha$ vs $1/\sqrt{\alpha}$ scaling without further derivation.

Signed-coupling pattern across instantiations. The six instantiations above — plus F, which derives a closed-form composite (T2) for the matched-symmetric-Tier-1 dyad — share a signed-coupling structure: each effective ρ_ξ is a sum of environmental disturbance plus a cross-agent contribution whose sign encodes cooperative vs adversarial coupling. Team persistence, composition closure (via bridge-lemma), tempo composition, adversarial destabilization, and composite critical-mass are instances of one inequality at different state-variable levels. The template is where this pattern lives at the meta level; the per-segment instantiations supply the domain-specific (ξ, F, ρ_ξ, R) quadruples.

Cost complement: the template's information-rate floor. Alongside the threshold condition, each instantiation inherits an information-rate lower bound from E: under Model S with Gaussian-OU-shaped disturbance statistics on ξ, the sustained information rate required to maintain $\mathbb{E}[\|\xi\|^2]_{ss}$ at the ultimate bound satisfies $\dot{R} \geq n\alpha/2$ nats/time with the instantiation's own $(\alpha, n, \sigma_\xi^2)$. For the epistemic-mismatch instance, this translates to the channel-capacity floor $C \geq \mathcal{T}/2$ — a first-class persistence prerequisite that the threshold condition alone does not surface. Each instantiation can derive its own cost bound by direct substitution into the main theorem of E; the consolidation of this into a parametric template-cost subsection is flagged in that segment's Working Notes.

The coordination overhead C_{coord} is effective disturbance at the composite level. The tempo-composition inequality $\mathcal{T}_c \leq \sum_i \mathcal{T}_i$ has a lower bound on the gap: $C_{\text{coord}} \geq \varepsilon^* \nu_c$ (11.2). This is the template's instantiation with $\xi = \delta_c$ and effective disturbance $\rho_{\text{ext}} + \varepsilon^* \nu_c$ — the composite's internal closure error acts as an additional disturbance at the macro level, absorbing correction capacity that would otherwise address the external environment. Brooks's Law follows as an instance: adding agents increases $\sum_i \mathcal{T}_i$ but may increase $\varepsilon^* \nu_c$ faster, pushing ρ_c^{eff} above $\alpha_c R_c$.

Relationship to 4.4. 4.4 is the canonical single-agent instantiation. It carries additional content beyond the template — *task adequacy*, a domain-specific constraint $R^* < \|\delta_{\text{critical}}\|$ that further restricts the usable region — and it instantiates the template in the linear special case where $F = \mathcal{T}\delta$, $\alpha = \mathcal{T}$, and (T2) holds globally. Task adequacy is not part of the template because it is not part of the Lyapunov argument; it is an application-level constraint the agent's domain imposes.

Relationship to 1.7. The composition-consistency postulate requires that AAT's predictions be compatible across levels of description. The template is how this compatibility cashes out operationally: the same persistence argument applies at every level where a state variable with a sector-bounded correction function is present. Section III segments that invoke the template (team persistence, composition closure, tempo composition) are applying AAT's single persistence argument at the composite level, with effective disturbance decompositions that capture what is distinctive about the composite scope.

Relationship to (metric-formulation generalization). The Euclidean sector inequality (T2) is the $M = I$ specialization of a broader contraction-metric condition (CT2) under a Riemannian metric M (Lohmiller & Slotine 1998). For several AAT-relevant agent classes, the natural Lyapunov lives in a non-Euclidean metric — Fisher for statistical-manifold learning (matrix Kalman in information metric; exponential family in natural parameters), Hessian-induced for ill-conditioned strongly-convex optimization, Lyapunov-equation-determined for asymmetric-stable linear systems, or Lyapunov-metric for PID-with-bounded-plant-nonlinearity. states the generalization once with (CT1)–(CT3) preconditions matching (T1)–(T3), adds compositional theorems (parallel / cascade / negative-feedback with small-gain) that extend `#deriv-critical-mass-composition` (CM2) to heterogeneous sub-agents via (CM2-M), and fills `#disc-separability-pattern`'s seventh ladder (A2'-scope into metric-α_1 / metric-α_2 / metric-β). The Euclidean formulation stated in this segment remains the default for Euclidean-natural instances; is invoked when the natural coordinate is non-Euclidean. Structural consequence worth noting: the (CT2) condition at $M = I$ is equivalent to `#form-composition-closure`'s DA2'-inc (incremental sector bound), so AAT has been carrying the Jacobian-level Euclidean contraction condition at the composite level all along; makes this explicit at the single-agent level.

Working Notes.

- The template is type `result` because it states a result abstractly; the proof lives in A (the canonical instance for δ), and the transfer to other state variables is routine once (T1)–(T3) are verified. Treating it as a `formulation` would misrepresent its status — the Lyapunov argument does not leave room for alternative formulations within its stated scope.
- Candidate extensions that would strengthen the template: (i) time-varying $\alpha(t)$ with lower bound, for the strategic case; (ii) state-dependent noise intensity, for cases where the disturbance magnitude depends on the current mismatch; (iii) non-quadratic Lyapunov functions, for richer stability regions. Each is a real extension of Lyapunov theory, not specific to AAT.
- The template's role in 1.7's "applies at every level" claim is now explicit: the claim holds at every level where (T1)–(T3) can be verified for the level's state variable. This sharpens the composition postulate from "applies at every level" to "applies at every level the template applies to," which is a testable condition rather than a universal assertion.

Appendix D

Result: Certificate Existence — Operator-Sector in Some Metric Is Exponential Stability

A stability certificate exists for an agent exactly when the agent is exponentially stable about its target; the organizing slogan *an adaptive system is an operator whose contraction rate exceeds its target's drift rate* is not a heuristic but this equivalence, with the certificate as its witness.

Formal Expression.

The object.

For an agent with error dynamics $\dot{e} = -F(e)$ about an equilibrium e^* ($F(e^*) = 0$, $F \in C^1$ near e^*, Jacobian $J := DF(e^*)$), a **stability certificate** is a symmetric positive-definite $\mathcal{M}$ for which the one-point sector condition holds in the $\mathcal{M}$-inner-product on a ball $\mathcal{B}_R(e^*)$:

Definition (stability-certificate).

$$\langle F(e),\, e - e^* \rangle_{\mathcal{M}} \;\geq\; \kappa \, \|e - e^*\|^2_{\mathcal{M}}, \qquad \kappa > 0. \tag{C}$$

The certificate is not unique: it is whatever positive-definite form makes the dynamics contract. In the recurring sub-cases it specializes — to the Fisher information for Bayesian agents, to $(P^-)^{-1}$ for Kalman agents, to the loss Hessian for gradient agents, and to a plant-selected Lyapunov metric for linear-Hurwitz or PID agents. These are not four separate stories; they are one object under four certificates.

The equivalence.

At the linearized level (C) reads $\mathcal{M}J + J^\top \mathcal{M} \succeq 2\kappa\mathcal{M} \succ 0$, a strict Lyapunov inequality for the system matrix $A = -J$. The following are equivalent:

Result (certificate-existence), exact at the linearized level.

1. $A = -J$ is Hurwitz — the linearized error dynamics $\dot{e} = -Je$ is exponentially stable;

2. there exist $\mathcal{M} \succ 0$ and $\kappa > 0$ with $\mathcal{M}J + J^\top \mathcal{M} \succeq 2\kappa\mathcal{M}$ — a one-point sector condition (a stability certificate) in *some* inner product;
3. for every $Q \succ 0$ there exists a unique $\mathcal{M} \succ 0$ with $\mathcal{M}J + J^\top \mathcal{M} = Q$.

So "operator-sector in *some* inner product" and "the equilibrium is exponentially stable" are **the same statement**, with the certificate $\mathcal{M}$ as the converse-Lyapunov witness — an equivalence, not an analogy.

The certificate-strength ladder.

The certificate admits three strictly-ordered strengths, all on the one object:

Derived (ordering of conditions; exact).

Table D.1

Rung	*Condition*	*Equivalent to*	*Certificate is*
R0	one-point (C), some $\mathcal{M}$, local	$A = -J$ Hurwitz + remainder dominated	converse-Lyapunov $\mathcal{M}$ (exists; generally not forced)
R1	incremental (two-point) $\mathcal{M}$-strong-monotonicity on $\mathcal{B}_R$	global $\mathcal{M}$-strong-monotone (cocoercive class)	curvature-like $\mathcal{M}$ (potential sub-case)
R2	R1 with $\mathcal{M}$ *forced* by a uniqueness theorem on an AAT-internal axiom	natural-gradient in the Čencov-unique Fisher metric	Fisher metric (Čencov-forced)

R0 $\Longleftarrow$ R1 $\Longleftarrow$ R2 strictly. R0 is the *widest* rung — it reaches the plant-Lyapunov cases (linear-Hurwitz-non-symmetric, PID) where no potential exists; R1 is the cocoercive/proximal class where a variational structure is available; R2 is the uniqueness-theorem-forced statistical case. The widest rung is not a weakness: it is exactly the reach the narrower rungs cannot give. (R2's forcing is established in Z; R1's cocoercive class in .)

Derivation.

(1 ⟹ 3, 2). $A = -J$ Hurwitz. By the Lyapunov theorem (Lyapunov 1892; Khalil, *Nonlinear Systems* 3rd ed., Thm 4.6), for every $Q \succ 0$ the equation $A^\top \mathcal{M} + \mathcal{M}A = -Q$ has a unique solution $\mathcal{M} \succ 0$. Substituting $A = -J$ gives $\mathcal{M}J + J^\top \mathcal{M} = Q \succ 0$ (clause 3). For the rate: $Q \succeq \lambda_{\min}(Q)\, I \succeq \dfrac{\lambda_{\min}(Q)}{\lambda_{\max}(\mathcal{M})}\,\mathcal{M}$, so (C) holds at the linear level with $\kappa = \dfrac{\lambda_{\min}(Q)}{2\,\lambda_{\max}(\mathcal{M})} > 0$ (clause 2).

(2 ⟹ 1). Suppose $\mathcal{M} \succ 0$, $\kappa > 0$, $\mathcal{M}J + J^\top \mathcal{M} \succeq 2\kappa\mathcal{M}$. Take $V(e) = e^\top \mathcal{M} e$ (valid Lyapunov candidate, $\mathcal{M} \succ 0$). Along $\dot{e} = -Je$: $\dot{V} = -e^\top(\mathcal{M}J + J^\top \mathcal{M})e \le -2\kappa\, e^\top \mathcal{M} e = -2\kappa V < 0$ for $e \neq 0$. Hence $\|e(t)\|_{\mathcal{M}} \le e^{-\kappa t}\|e(0)\|_{\mathcal{M}}$ and $A = -J$ is Hurwitz.

(3 ⟺ 2) is the rate-extraction argument (3 gives 2 with the displayed κ; 2 is the special case $Q = \mathcal{M}J + J^\top \mathcal{M}$).

Local nonlinear extension. With $F(e) = Je + r(e)$, $\|r(e)\| = o(\|e\|)$, (C) holds on $\mathcal{B}_R(e^*)$ iff the linearization is exponentially stable *and* the second-order remainder is dominated on the ball ($\|r(e)\|_{\mathcal{M}} \le c\|e\|^2_{\mathcal{M}}$ with $cR < \kappa$). This is the standard Lyapunov indirect method (Khalil Thm 4.7); the remainder-domination radius is exactly the contraction-template Tier-2 degradation radius (), not new machinery.

Epistemic Status.

Exact at the linearized level (the equivalence is the standard Lyapunov theorem, recognized and applied — not a fresh derivation), and *exact with the standard second-order remainder condition* locally. The constituent theorem is classical; what is contributed here is the recognition that AAT's one-point sector condition (A2'/(T2)) in a free choice of inner product is *exactly* the converse-Lyapunov certificate, which makes the framework's organizing slogan a theorem rather than a heuristic.

Scope honesty. Linearized/local — this is the level at which AAT's persistence results already operate (sector conditions, contraction templates, the bridge lemma all linearize about the equilibrium), so it is not a weakening relative to the rest of the theory, but it is a genuine scope statement and is not papered over: there is **no** claim that one-point operator-sector is equivalent to *global* exponential stability (it is not, in general — the global statement requires the incremental rung R1, the cocoercive class). The disturbance-to-ultimate-bound step that turns "contracts" into "stays within a bounded ball under disturbance" is proved separately in A and C; this result supplies the contraction half of the slogan, those supply the exceeds-the-drift half.

Max attainable: *exact* (it is the Lyapunov theorem, in scope). Novelty posture is *recognition*, not new mathematics.

Discussion.

This is the segment-level home of the contraction-over-drift organizing principle. The slogan *an adaptive system is an operator whose contraction rate exceeds its target's drift rate* had, until now, no segment to point to — it was carried as a framing heuristic. The equivalence above is its formal content: the "contraction" half is clause 2 $\Leftrightarrow$ clause 1 (a certificate exists iff the dynamics is exponentially stable); the "exceeds the drift" half is the ultimate-bound argument of C, which takes the certificate's κ and the disturbance rate and returns the bounded-error guarantee. A reader can now cite the principle to a result, not to a slogan.

Why the equivalence matters beyond tidiness. It is what licenses the cross-sectional reading in W: because "a certificate exists" is the same statement as "the agent can keep up," every structural question about AAT's reach becomes a question about this one object — does a certificate exist here (scope), is it forced to a unique form (forced identity), where does it degenerate (the boundary), does it survive coarse-graining (composition). Without the equivalence those would be four loosely-related observations; with it they are four questions about one mathematical object.

The certificate is not the Lyapunov function. $V(e) = e^\top \mathcal{M} e$ is the Lyapunov function; $\mathcal{M}$ is the certificate. The distinction is load-bearing downstream: the framework's representational freedom (choice of coordinate / metric) acts on $\mathcal{M}$, and what that freedom can and cannot do to $\mathcal{M}$ — change its conditioning but not its inertia — is the content of the boundary facet (X, the Sylvester mechanism). Naming $\mathcal{M}$ as the object, rather than V, is what makes that downstream statement expressible.

Findings.

The Contraction-Over-Drift Principle, Grounded. **Brief:** Picture an agent trying to stay on a target that keeps drifting. Ask one question: is there a way of measuring "how far off am I" — a measuring-stick — such that *every* correction the agent makes provably shrinks that measure? This result says: such a measuring-stick exists exactly when the agent is, in the ordinary control-theory sense, exponentially stable. The two are not analogous, they are the *same fact* (the Lyapunov theorem) — which is why the long-standing slogan "an adaptive system is one whose contraction outpaces its target's drift" is a theorem and not a vibe. A thoughtful non-specialist can carry the whole thing away from the picture: *the measuring-stick exists = the agent can keep up*; everything else AAT says about reach, blind spots, and composition is downstream questions about that one stick.

Impact: Discharges the long-standing "organizing slogan not yet surfaced at segment level" status by giving the contraction-over-drift principle an exact, citable home. Supplies the anchor that licenses the certificate-cone cross-sectional reading (W): the equivalence is what turns "four loosely-related structural observations" into "four questions about one object." Cleanly separates the exact mathematical core (this result) from the discussion-grade organizing recognition (the spine), so each is housed at its true epistemic tier rather than the exact equivalence being buried as a derived-tag inside a discussion-grade segment. Names the certificate (not the Lyapunov function) as the object, which is the precondition for the boundary-facet's Sylvester statement being expressible downstream.

Novelty Claim: *Claim recognition* that AAT's one-point sector condition under a free choice of inner product is exactly the converse-Lyapunov certificate, making the framework's contraction-over-drift organizing principle the Lyapunov-theorem equivalence rather than a heuristic. The theorem is classical (Lyapunov 1892); the contribution is the identification that the framework's A2'/(T2) machinery *is* this object, and the consequent segment-level grounding of the organizing slogan.

Related Work: - Lyapunov, A. M. (1892), *The General Problem of the Stability of Motion* (Engl. transl. 1992, Taylor & Francis); Khalil, H. K. (2002), *Nonlinear Systems* 3rd ed., Thm 4.6 (Lyapunov equation), Thm 4.7 (indirect method) (found 2026-05-14) — *formal antecedent* — the equivalence and the local extension are this theorem applied; the contribution is the recognition, not the proof. - A, C (adjacent) — supply the disturbance-to-ultimate-bound half of the slogan; this result supplies the contraction half.

Search Log: - 2026-05-14 (*targeted*): The equivalence is the textbook Lyapunov theorem; no search needed for the theorem itself. The search target was whether the *recognition* "AAT's one-point sector condition in a free inner product = the converse-Lyapunov certificate, hence the contraction-over-drift slogan is this equivalence" appears articulated elsewhere as a framework-grounding move. Not found at this depth; expected to remain *recognition*-tier (the theorem is universal; the framework-grounding identification is the contribution).

Working Notes.

- **Provenance.** Split out of W 2026-05-14 (it had been a `*[Derived]*`-tagged block) so the exact equivalence is housed at its true tier (`status: exact`) rather than inside a `discussion-grade` segment. Landed in the 2026-05-14 operator-family-unification cycle; see CHANGELOG 2026-05-14. The full proof and the R0/R1/R2 ladder are in the Derivation above; the originating spike is absorbed archaeology, not a live reference.
- **Provisional slug.** `result-certificate-existence`. Subject-noun = "certificate existence" (the result establishes when a certificate exists). Route through the naming pipeline if a better name surfaces.
- **Organizing-principle linkage.** This segment is the intended segment-level home for the PROPOSALS Bundle-1 organizing-principle slogan ("contraction-over-drift"); the PROPOSALS entry should point here once Joseph confirms the propagation step. Bundle-1 not auto-rewritten.

Appendix E

Derivation: Persistence Cost — Information Rate to Maintain Bounded Mismatch

AAT's persistence machinery establishes that under the sector condition, mismatch stays bounded. It does not quantify the *sustained rate of effort* an agent must expend to hold that bound. Two agents with identical persistence guarantees can face wildly different demands — a Kalman filter tracking a stationary process vs one tracking a rapidly non-stationary process are both persistent; one is dormant, the other running hot. Under Model S with Gaussian-OU signal, the sustained Shannon information rate the agent must acquire from observations to maintain the sector-persistence ultimate bound is $\dot{R}_{\min} \geq n\alpha/2$ nats per unit time — a Landauer-analog lower bound that depends only on the signal's second-order statistics and the sector constant α, and that Kalman-Bucy saturates in steady state. The bound promotes channel capacity $C \geq \mathcal{T}/2$ into a first-class persistence prerequisite that the current theory does not name.

Formal Expression.

Setup. The agent is in scope of 4.3 Model S (GA-2S, stochastic disturbance). Per Prop A.1S the state satisfies $\mathbb{E}[\|\delta\|^2]_{ss} = n\sigma_w^2/(2\alpha)$. The RMS bound is $R_S^* = \sigma_w\sqrt{n/(2\alpha)}$. The environmental signal is n-dimensional independent-component Ornstein-Uhlenbeck with per-component intrinsic drift λ_s and diffusion coefficient σ_w^2. The mean-square persistence condition $\alpha > n\sigma_w^2/(2R^2)$ holds. Sector-persistence is achieved at the tight bound $D^2 = R_S^{*2}$.

Persistence Information Rate (main theorem).

Proposition. Any adaptive process achieving the tight Model-S ultimate bound $\mathbb{E}[\|\delta\|^2]_{ss} = n\sigma_w^2/(2\alpha)$ under the stated setup must acquire information from observations at sustained rate

Proved (persistence-information-rate, from Shannon RDF + Prop A.1S).

$$\boxed{\dot{R} \;\geq\; \dot{R}_{\min} \;=\; \frac{n\alpha}{2}\text{ nats per unit time}} \tag{E.1}$$

Derivation. Shannon's rate-distortion theorem (Shannon 1948; Berger 1971; Cover & Thomas 2006 Theorem 10.2.1) states that for any source-code achieving mean-square distortion D^2, the coding rate satisfies $\dot{R} \;\geq\; R(D^2)$ where $R(\cdot)$ is the rate-distortion function. For n-dimensional independent-component OU in the high-resolution regime ($D^2 \ll \sigma_x^2 = \sigma_w^2/(2\lambda_s)$), the RDF per unit time is (Ihara 1993 Theorem 4.6.4; Gray 1972 Theorem 2):

$$\dot{R}(D^2) = \frac{n\sigma_w^2}{4D^2} \tag{E.2}$$

Substituting $D^2 = R_S^{*2} = n\sigma_w^2/(2\alpha)$:

$$\dot{R}_{\min} = \frac{n\sigma_w^2}{4 \cdot n\sigma_w^2/(2\alpha)} = \frac{\alpha}{2} \cdot 1 = \frac{n\alpha}{2} \text{ nats per unit time} \tag{E.3}$$

(the calculation gives $\alpha/2$ per dimension and $n\alpha/2$ total for n independent OU components). The agent's observation-channel information rate must meet this bound regardless of the specific correction function, filter structure, or implementation. □

Kalman-Bucy Saturates the Bound.

The Kalman-Bucy filter attains the bound exactly in steady state. Mitter & Newton (2005, *J. Stat. Phys.* 118:145–176) derive the filter's rate of information supply as

Derived (Kalman-tight, Mitter-Newton 2005).

$$\dot{I} = \tfrac{1}{2}\,\mathrm{tr}(H^T \Sigma_o^{-1} H P_{ss}) \tag{E.4}$$

Scalar case ($H = 1, \Sigma_o = \sigma_o^2$) with steady-state covariance $P_{ss} = \sigma_o\sigma_w$ in the drift-dominated limit and Kalman gain $K_{ss} = P_{ss}/\sigma_o^2 = \sigma_w/\sigma_o$:

$$\dot{I} = \frac{P_{ss}}{2\sigma_o^2} = \frac{K_{ss}}{2} = \frac{\alpha}{2} \tag{E.5}$$

Under the linear-correction identification $\alpha = K_{ss}$ (from 4.2, scalar Kalman case), the Kalman filter's information supply rate equals $\alpha/2$ exactly. This matches the RDF lower bound, confirming tightness. **The bound is not merely a lower bound — it is achieved by the Kalman-optimal filter.**

Channel-Capacity Prerequisite.

By Shannon's channel coding theorem (Cover & Thomas 2006 §7.7), an observation channel of Shannon capacity C_{channel} can support any information rate up to C_{channel} and no higher. Combining with the persistence information rate:

Derived (channel-capacity-floor, from main theorem + Shannon capacity).

$$C_{\text{channel}} \;\geq\; \dot{R}_{\min} \;=\; \frac{n\alpha}{2} \tag{E.6}$$

Under the linear-correction identification $\alpha = \mathcal{T}$ (from 3.8 + 4.2 scalar Kalman):

$$\boxed{C_{\text{channel}} \;\geq\; \mathcal{T}/2 \text{ nats/time per dimension}} \tag{E.7}$$

Persistence demands observation-channel capacity at least half the adaptive tempo. This is a *new first-class persistence diagnostic* not present in the current theory. Its binding matters most in capacity-constrained settings — bandwidth-limited distributed systems, biological neurons, context-window-limited LLMs — where the tempo framework alone underestimates the difficulty of maintaining bounded mismatch.

Rejected Candidate Cost Metrics. The information-rate bound is not the only candidate for a cost-of-persistence quantity. Three alternatives fail structurally. Recording them here keeps the scope-honesty visible.

[Observation (gain-magnitude-tautological)] $\mathbb{E}[\|K(t)\|]$ as a cost metric: in the linear Kalman case $K_{ss} = \alpha$ (sub-scope α per 4.2), so "gain magnitude" equals the sector constant itself. Any bound of shape $\mathbb{E}[\|K\|] \geq f(\alpha, \ldots)$ becomes tautological. Rejected as fundamental cost metric — it recapitulates α.

[Observation (control-effort-filter-specific)] $\mathbb{E}[\|u(t)\|^2]$ (per-unit-time control-effort integral): filter-specific. Different filters achieving the same steady-state variance P_{ss} have different control-effort integrals. The Kalman filter is minimum-effort among linear filters (optimal-control interpretation of the Riccati equation); nonlinear filters can trade effort vs variance differently. A filter-agnostic bound cannot be stated in this quantity — it is not invariant under the equivalence class of filters meeting the persistence condition.

$\mathbb{E}[\alpha\|\delta\|^2]_{ss}$ (Lyapunov dissipation rate) at steady state equals $n\sigma_w^2/2$ regardless of α — a non-equilibrium-steady-state conservation law (dissipation balances disturbance-power injection). The quantity is *structurally invariant*: it does not depend on the quality of adaptation, only on the disturbance statistics. This is what makes the RDF bound tight at the Model-S ultimate bound (the steady state is active, not slack), but it cannot itself serve as a cost metric because it does not distinguish well-adapted from poorly-adapted agents at a given α.

Observation (Lyapunov-dissipation-conservation).

Candidate 4 — the Shannon information rate above — is the one that closes. The structural reason: information rate is filter-agnostic (depends only on signal second-order statistics and target distortion), universal (any filter implementation is lower-bounded), and has a clean thermodynamic interpretation (Still et al. 2012, "Thermodynamics of Prediction", *Phys. Rev. Lett.* 109:120604: nonpredictive information retained equals dissipation during interaction).

What Is Derived vs. What Is Chosen.

Table E.1

Property	*Source*	*Strength*
Persistence information rate $\dot{R}_{\min} \geq n\alpha/2$	Shannon RDF (Berger 1971; Gray 1972; Ihara 1993 Thm 4.6.4) composed with Prop A.1S	Derived (conditional on Model S + Gaussian-OU + high-resolution regime)
Kalman-Bucy saturates bound	Mitter-Newton 2005 Theorem (information-supply identity); linear-correction identification $\alpha = K_{ss}$ per 4.2	Derived (linear-Gaussian; exact when A2' is derived in sub-scope α)
Channel-capacity prerequisite $C \geq \mathcal{T}/2$	Main theorem + Shannon channel-capacity theorem (Cover-Thomas 2006 §7.7) + $\alpha = \mathcal{T}$ identification	Derived
Gain-magnitude as cost metric	$\mathbb{E}[\|K\|] \approx \alpha$ in sub-scope α	Rejected as fundamental (tautological)
Control-effort integral as cost metric	Filter-specific; depends on optimality class	Rejected as universal (filter-dependent)
Lyapunov dissipation rate	Conservation law $= n\sigma_w^2/2$ at steady state	Not a cost metric — structural observation enabling tightness of main theorem
Non-Gaussian signal extension	Different RDF form (Berger 1971 Ch. 4)	Open — qualitative scaling $\dot{R} \propto$ innovation-rate$/D^2$ likely preserved; exact prefactor changes
Model D (bounded disturbance) analog	Requires adversarial / minimax information argument (Csiszár-Körner 2011 Ch. 11)	Open
Transient-regime rate	Bound is steady-state; transient rates during structural adaptation much higher	Open
Sub-scope β transfer	RDF bound holds as inequality (data-processing); tightness not guaranteed	Conditional (holds as lower bound; Kalman-Bucy saturation requires sub-scope α)

Epistemic Status.

Conditional. Max attainable: *exact* in the linear-Gaussian case; *robust qualitative* for general stationary Gaussian signals; *heuristic* for non-Gaussian.

The theorem rests on three named conditions: (i) Model S stochastic disturbance per 4.3 (GA-2S); (ii) n-dimensional independent-component Ornstein-Uhlenbeck signal structure; (iii) high-resolution regime $D^2 \ll \sigma_x^2$. Within this scope the result is as tight as the Shannon RDF allows — Kalman-Bucy exactly saturates per Mitter-Newton 2005. For non-Gaussian signals the RDF has a different form (Berger 1971 Ch. 4) but the scaling $\dot{R}_{\min} \propto \sigma_w^2/D^2$ is expected to persist qualitatively; exact prefactors change per signal class. Under Model D (bounded disturbance) the argument requires worst-case channel-capacity machinery (Csiszár-Körner 2011 Ch. 11) and gives a different expression.

Sub-scope transfer. Sub-scope α agents (Kalman / conjugate-Bayesian / exponential-family / strongly-convex-gradient / linear-PD) have α derived from the update rule's structure per A + 4.2, so the $n\alpha/2$ bound holds with known α and is tight in the linear-Gaussian case. Sub-scope β agents (PID / rule-based / human-judgment) assume A2' rather than deriving it — the RDF bound still holds as an inequality (by the data-processing inequality applied to any filter mapping observation to update), but tightness is not guaranteed and the assumed α enters the bound.

Confidence in correctness. The Shannon RDF for Gaussian-OU is a classical result (Gray 1972; Ihara 1993). The sector-persistence bound is already in AAT (Prop A.1S of A). The theorem's mathematical core is a two-line composition of both; a careful reader can verify. The contribution is not the mathematics but the AAT-framing — reading $\alpha/2$ off the sector constant as the fundamental information-rate cost of persistence — and the channel-capacity-as-first-class-quantity opening.

What this does not establish.

- Upper bounds on persistence cost (how much rate a specific filter consumes; the bound is a lower limit).
- Optimal filter design for minimum information throughput.
- Non-Gaussian signal cost bounds (qualitative scaling expected; quantitative forms open).
- Cost under Model D adversarial disturbance.
- Cost for composite agents beyond simple additivity.
- The stability upper bound on plasticity that 8.11 references — that involves the consolidation cadence, not the online information rate.

Discussion.

The $\alpha/2$ per-dimension Landauer analog. The theorem has a clean thermodynamic reading per Still et al. 2012: each nat of information about the signal costs at least $k_B T$ of dissipation (Landauer 1961). Combined, persistence at sector constant α in n dimensions costs at least $n\alpha/2$ nats/time of information acquisition and at least $n\alpha k_B T/(2\ln 2) \approx 0.35 n\alpha k_B T$ of thermodynamic dissipation per unit time in any physical substrate. AAT does not commit to a specific substrate, but the bound is substrate-agnostic: it constrains any filter implementation via the RDF (information-theoretic) and, when physical, any computational realization via Landauer (thermodynamic).

Why channel capacity matters as a first-class quantity. AAT's 4.4 and 3.8 currently frame persistence as a correction-rate vs disturbance-rate inequality. This theorem adds a *lower* constraint: observation channels must jointly supply Shannon capacity $\geq \mathcal{T}/2$ nats/time per dimension, else persistence fails regardless of correction-function design. The constraint is binding in any setting where observation bandwidth is non-abundant — most real systems. Three domains where the capacity floor is more binding than the tempo bound:

- *Biological systems*: neural channel capacity is finite; this bound gives a quantitative minimum on sensory bandwidth for a given adaptive rate.

- *Bandwidth-constrained distributed systems*: agents operating over noisy or low-bandwidth links face channel capacity directly as a hard constraint.
- *Context-window-limited LLMs*: the effective information rate per unit of cognition is bounded by context size and token-throughput; this theorem predicts a minimum tempo achievable at a given context budget.

In each, knowing that $\mathcal{T}/2$ is the floor converts an opaque "just needs more capacity" observation into a specific dimensional requirement.

Connection to AAT's meta-architecture. The result composes with AAT's three meta-segments.

- *Y (positive half)*: the bound sits in structured-repair along the identification-regime ladder — derived in sub-scope α via composition of two external theorems; holds as inequality (not tightness) in sub-scope β.
- *X (negative half)*: this is the positive-dual of the floor pattern. Where the floor says "AAT cannot distinguish X without information augmentation" (external no-go + AAT escape), this result says "AAT requires at least $n\alpha/2$ nats/time of information supply to operate" (external lower bound + AAT bridge through sector-persistence template). The two patterns are duals: external-theorem-forbids vs external-theorem-lower-bounds.
- *Z (constructive half)*: the bound is linear in α, not logarithmic. It sits *outside* the three-layer logarithmic-coordinate family (chain / divergence / update). No Cauchy-FE argument forces the coordinate here — the result is a direct substitution from classical information theory. This is a useful non-example: AAT's additive-coordinate-forcing pattern is not universal; specific results live on different coordinates.

Relationship to C. The template enumerates six instantiations (epistemic mismatch; strategic mismatch; sub-agent mismatch; composite trajectory error; composite mismatch; target-agent mismatch). Each has its own state variable ξ, sector constant α, and disturbance statistics. The persistence cost bound *specializes* for each instantiation: under Model S with Gaussian-OU-shaped disturbance on the state variable, the information-rate cost is $n\alpha/2$ with the template's α and n. A compact template-cost bound — parametric in $(\xi, \alpha, n, \sigma_\xi)$ — could land as a subsection of C so all six instances inherit the cost bound by substitution. That move is flagged in Working Notes as an optional consolidation.

The relationship to Mitter-Newton's thermodynamic reading. Mitter & Newton (2005) showed the Kalman-Bucy filter operates as a Maxwellian demon: it returns signal energy to the heat bath without entropy increase, but only because new information is continually supplied at rate $\dot{I}$. This supply rate *is* what persistence costs — the filter maintains bounded mismatch by paying, information-theoretically, for new observations at a rate matched to the environment's innovation rate. Our theorem makes this quantitative as a function of the sector constant: at a given α, the matching rate is $\alpha/2$. Faster correction requires more information supply; the bound is tight.

Working Notes.

- **Template-cost subsection.** A compact parametric statement — "under (T1)–(T3) with Model S and Gaussian-OU disturbance at per-component parameters $(\lambda_\xi, \sigma_\xi^2)$, sustained information rate to maintain steady-state mean-square ξ at ultimate bound $D^2 = n\sigma_\xi^2/(2\alpha)$ satisfies $\dot{R} \geq n\alpha/2$ nats/time" — would let each of C's six instances inherit the cost bound by substitution. Worth considering on next template revision; would consolidate this segment's core result into the template.
- **Model D adversarial analog.** Rate-distortion is inherently stochastic; Model D's bounded-disturbance version of the theorem requires minimax / worst-case channel capacity (Csiszár-Körner 2011 Ch. 11). Candidate follow-up spike. Expected form: similar $\propto \alpha$ scaling, different prefactor.
- **Non-Gaussian signals.** For heavy-tailed or power-law signals the RDF has a different exact form (Berger 1971 Ch. 4). The qualitative scaling $\dot{R}_{\min} \propto \sigma_w^2/D^2$ is expected to persist; quantitative extensions are per-family.
- **Misspecification cost.** When $\mathcal{F}(\mathcal{M}) < 1 - \varepsilon$ per 2.4, achievable distortion is bounded away from zero by an additional floor. Natural extension has the sustained information rate lower-bounded by $n\sigma_w^2$ divided by 4 times the larger of the Model-S ultimate bound and D^2_{floor}. Connects to X's "Misspecification-cost quantification" open extension.

- **Composite persistence cost.** For a composite agent, the information-rate bound's scaling under composition is not trivial. Candidate: $\dot{R}_{c,\min} \leq \sum_i \dot{R}_{i,\min}$ due to coordination overhead eating capacity. Cost-analog of 11.2's sub-additivity + 13.1's cooperative-coupling reduction. Open.
- **Observation-channel capacity as notation.** Currently AAT uses U_o (observation uncertainty) as a noise parameter. Lifting Shannon channel capacity $C^{(k)}$ of channel k into NOTATION.md and relating it to U_o (via the channel-capacity-from-noise standard transform) would make the capacity-floor condition a first-class persistence diagnostic. This is the biggest architectural opening from this theorem and worth a follow-up scoping decision.
- **Connection to 8.10.** The strategy-edge persistence condition has its own disturbance statistics ρ_Σ and sector constant α_Σ. Specializing this theorem gives the information rate required to track strategy-edge invalidation: $\dot{R}_{\Sigma,\min} = n_\Sigma \alpha_\Sigma / 2$. If $\alpha_\Sigma = 1/(n+1)$ with experience discounting, the rate decays with accumulated n — which connects to the stability-induced myopia result in the pending spike G.
- **Not forced by a Cauchy-FE axiom.** Unlike the three primary instances of Z (chain, divergence, update), the linear-in-α coordinate of this bound is not forced by an AAT-internal additivity axiom. It is a direct consequence of Gaussian-RDF's specific functional form. This places the result *outside* the three-layer family — a useful non-example that shows the additive-coordinate-forcing pattern is not universal.

Appendix F

Derivation: Critical-Mass Composition

The composite sector constant α_c is derived — not merely bounded from below — for the symmetric-matched-Tier-1 two-agent case, yielding a closed-form critical-mass inequality in which the sign of the inter-agent coupling γ and the teleological unity U_O enter explicitly. The result subsumes the weakest-link bound, recovers 13.1 (cooperative) and 13.2 (adversarial) as signed special cases, formalizes 10.3's autonomy-reduction mechanism as an asymmetric Lyapunov-weight limit, and makes the scope-gate from 10.2 explicit as the second conjunct of composite persistence.

Formal Expression.

Setup. Two sub-agents A_1, A_2, each a **Tier 1 agent** in the sense of 11.1's bridge-lemma taxonomy — mismatch-driven update, linear prediction, incremental sector-Lipschitz correction (Kalman, exponential-family Bayesian, gradient-on-strongly-convex, linear-with-PD-KH). **Matched architectures**: f_1, f_2 are structurally the same function, with $\alpha_1 = \alpha_2 = \alpha, R_1 = R_2 = R$. Disturbance statistics shared: each sees bounded $w_i(t)$ with $\|w_i\| \leq \rho$ (Model D, per C).

Inter-agent coupling enters additively to the disturbance at rate $\gamma \mathcal{T}_j$:

Formulation (coupling-model-C1, from #der-team-persistence + #der-adversarial-destabilization).

$$\rho_i^{\text{eff}} = \rho + \gamma \mathcal{T}_j \tag{C1}$$

with sign convention $\gamma < 0$ cooperative (ally's tempo-contribution reduces disturbance, recovering 13.1's $-\gamma^{\text{coop}} \mathcal{T}_j$ term), $\gamma > 0$ adversarial (ally's tempo-contribution amplifies disturbance, recovering 13.2's $+\gamma_A \mathcal{T}_A$ term). Symmetric case: $\gamma_{1\to 2} = \gamma_{2\to 1} = \gamma$.

Coordination overhead reduces each agent's effective correction rate symmetrically:

Formulation (coordination-cost-C2).

$$\alpha_i^{\text{eff}} = \alpha - C \tag{C2}$$

with $C \geq 0$ the $\Delta \mathcal{T}_i^{\text{cost}}$ from 13.1's coordination-overhead threshold.

Critical-mass inequality (symmetric-matched-Tier-1 case).

Let $\xi = (\delta_1, \delta_2)^T$ and take the joint quadratic Lyapunov candidate $V(\xi) = \frac{1}{2}(\|\delta_1\|^2 + \|\delta_2\|^2)$. Under the block-diagonal correction structure with cross-coupling absorbed into ρ_i^{eff} via (C1), and using $\|\delta_1\| + \|\delta_2\| \leq \sqrt{2(\|\delta_1\|^2 + \|\delta_2\|^2)}$ (Cauchy–Schwarz):

Derived (critical-mass-symmetric, from #result-sector-persistence-template + C1 + C2).

$$\dot{V} \leq -(\alpha - C)(\|\delta_1\|^2 + \|\delta_2\|^2) + (\rho + \gamma\mathcal{T})\sqrt{2(\|\delta_1\|^2 + \|\delta_2\|^2)}. \tag{F.1}$$

Setting $\dot{V} = 0$ gives the ultimate bound on $\|\xi\|$ and, projecting to the macro-state $\delta_c = (\delta_1 + \delta_2)/\sqrt{2}$, the ultimate composite mismatch

$$R_c^* \leq \frac{\rho + \gamma\mathcal{T}}{\alpha - C}. \tag{L4}$$

The composite persists iff $R_c^* < R_c$. Inheriting $R_c = R$ from the symmetric-matched averaging projection:

$$\boxed{(\alpha - C)R \;>\; \rho + \gamma\mathcal{T}} \tag{CM2}$$

Rearranging into the composite contraction-rate form:

$$\kappa_c \;:=\; (\alpha - C) \;-\; \frac{\rho + \gamma\mathcal{T}}{R}, \qquad \text{composite persists iff } \kappa_c > 0. \tag{KC}$$

Specialization checks. Under matched symmetry, (CM2) reduces correctly in four limits:

Table F.1

Limit	*Setting*	*(CM2) reduces to*	*Recovers*
No coupling	$\gamma = 0$, $C = 0$	$\alpha R > \rho$	Single-agent 4.4
Cooperative-symmetric	$\gamma < 0$, $C = 0$	$\alpha R > \rho + \gamma\mathcal{T}$ (easier than individual)	13.1's "teams persist where individuals can't"
Adversarial-symmetric	$\gamma > 0$, $C = 0$	Fails when $\gamma\mathcal{T} > \alpha R - \rho$	13.2 threshold (symmetric)
Coordination-dominated	$C > \alpha$, $\gamma = 0$	LHS < 0; composite fails	Brooks's Law

Subsumption of the weakest-link bound.

The weakest-link bound $\alpha_c \geq \min_i(\alpha_i - \Delta\mathcal{J}_i^{\text{cost}})$ from 11.1's derivation table specializes under matched symmetry to $\alpha_c \geq \alpha - C$. (KC) refines this by making the composite's effective disturbance explicit as $\rho + \gamma\mathcal{T}$, turning a correction-rate bound into a full persistence inequality. Critically, (KC) can yield $\kappa_c > 0$ even when the weakest-link bound alone fails — when cooperative coupling ($\gamma\mathcal{T} < 0$) reduces the effective disturbance below what the raw $\alpha - C$ margin would permit. The weakest-link bound cannot see this because it does not account for γ's sign.

Derived (weakest-link-subsumption).

U_O entry: multiplicative-on-γ plus scope-gate.

In a purposeful-agent setting where each sub-agent optimizes a quadratic objective $L_i(\omega) = \frac{1}{2}(\omega - r_i)^T Q(\omega - r_i)$ with target r_i, and $U_O := \text{corr}(r_1, r_2)$ is the target correlation per 12' U_O, the cross-coupling in the joint dynamics has sign and magnitude controlled by U_O:

Derived (unity-multiplicative-modulator, conditional on LQR-compatible action structure).

$$\gamma(U_O) \;=\; -\gamma_{\max} U_O, \qquad \gamma_{\max} > 0, \tag{UO-mult}$$

via aligned targets → aligned action directions in the shared environment → constructive (cooperative) cross-contribution in the symmetric eigendirection. Substituting into (KC):

$$\kappa_c(U_O) \;=\; (\alpha - C)R \;-\; \rho \;+\; \gamma_{\max} U_O \mathcal{T}. \tag{CM3}$$

(CM3) is necessary but not sufficient for composite existence. Under 10.2, a composite exists as an AAT agent only when one of the three disjunctive alignment routes (shared objective, hierarchical derivation, mutual benefit) is satisfied. Below this threshold, no coherent O_c is definable and composite-level quantities — including R_c on the right of (CM2) — are ill-typed. The honest statement of composite persistence is therefore the conjunction:

Scope (scope-gate-from-composition-scope-condition).

$$\boxed{\kappa_c(U_O) > 0 \;\wedge\; \text{\#scope-composite-agent satisfied} \;\Leftrightarrow\; \text{composite persists as AAT agent}} \tag{CM4}$$

U_O enters (CM4) in two independent ways: multiplicatively within (CM3), and as scope-gate via 10.2. It does **not** enter purely additively as a separate reserve term — there is no free-floating "U_O contribution" detached from the coupling it modulates.

Asymmetric limit and symbiogenic composition.

Drop the matched-symmetric assumption. Let $\alpha_1 \gg \alpha_2$ with $\alpha_2 \to 0$. The unweighted joint Lyapunov $V = \frac{1}{2}(\|\delta_1\|^2 + \|\delta_2\|^2)$ fails (the weakest-link ultimate bound diverges as $\alpha_2 \to 0$). A **weighted** Lyapunov $V_\mu(\xi) = \frac{1}{2}(\|\delta_1\|^2 + \mu\|\delta_2\|^2)$ with $\mu \to 0$ yields

Sketch (asymmetric-limit-symbiogenesis, from weighted Lyapunov).

$$\dot{V}_\mu \leq -\alpha_1\|\delta_1\|^2 + \rho_1\|\delta_1\| \;+\; O(\mu), \tag{F.2}$$

so in the limit the composite's stability is controlled **entirely by agent 1**; agent 2's autonomous correction dynamics are weighted out of the stability accounting.

This provides a Lyapunov-weighted formalization of 10.3's **(S-3) autonomy reduction**: the endosymbiont's effective action space contracts ($\mathcal{A}_e^{\text{effective}} \to \mathcal{A}_e^{\text{restricted}}$) and its autonomous dynamics fall out of the joint Lyapunov argument. The asymmetric limit is a smooth deformation of (CM4), not a discontinuous regime change — symbiogenesis and peer coupling are parameter-limits of the same weighted-Lyapunov analysis. The result does **not** close 10.3's (S-1) objective absorption or (S-2) function transfer: what happens to agent 1's state space when it inherits structure from agent 2 is a separate question the weighting argument does not address.

What Is Derived vs. What Is Chosen.

Table F.2

Property	*Source*	*Strength*
Coupling model (C1): $\rho_i^{\text{eff}} = \rho + \gamma\mathcal{T}_j$	Import from 13.1 and 13.2	Formulation choice (requirement for the derivation)
Coordination-cost model (C2): $\alpha_i^{\text{eff}} = \alpha - C$	Import from 13.1's coordination-overhead threshold	Formulation choice
Joint quadratic Lyapunov $V = \frac{1}{2}(\lVert\delta_1\rVert^2 + \lVert\delta_2\rVert^2)$	Standard vector-Lyapunov construction (Matrosov 1962; Bellman 1962)	Formulation choice (canonical for matched-symmetric dyads)
Ultimate bound $R_c^* \leq (\rho + \gamma\mathcal{T})/(\alpha - C)$	Lyapunov dissipation + Cauchy–Schwarz	Derived
Critical-mass inequality (CM2): $(\alpha - C)R > \rho + \gamma\mathcal{T}$	(L4) + sector-region fit $R_c^* < R_c = R$	Derived (conditional on Tier 1 + matched-symmetric + Model D)
Four specialization checks (no-coupling / cooperative / adversarial / coordination-dominated)	Direct substitution into (CM2)	Proved (within stated scope)
Subsumption of weakest-link bound	(CM2) sign-sensitive; weakest-link is sign-blind	Proved
(UO-mult): $\gamma(U_O) = -\gamma_{\max} U_O$	LQR-compatibility sketch; aligned targets → aligned actions → constructive cross-contribution	Discussion-grade
Composite persistence as (CM3) ∧ scope-satisfaction: (CM4)	(KC) with (UO-mult) + 10.2	Derived (conditional)
Asymmetric limit → 10.3 (S-3) via weighted Lyapunov	Matrosov-style weighting; $\mu \to 0$ limit	Sketch (the weighting is standard; the identification with (S-3) is structurally motivated but not a theorem)
(S-1) objective absorption and (S-2) function transfer formalizations	Not addressed by this derivation	Open (in 10.3 Working Notes)
Heterogeneous-architecture case (A_1 Tier 1, A_2 Tier 2/3)	Requires per-sub-agent tiering per 11.1	Open
Heterogeneous-metric Tier-1M dyad ($\lambda_1 \neq \lambda_2$, $C_1 \neq C_2, k_{12} \neq k_{21}$)	(CM2-M) via Slotine 2003 negative-feedback small-gain: $(\lambda_1 - C_1)(\lambda_2 - C_2) > k_{12}k_{21}/4$	Derived (conditional on (CT2) preconditions + Slotine 2003)
Nonlinear coupling $\gamma = \gamma(\delta_j)$	Requires full joint-Lyapunov machinery from 13.2 (effects-spiral corollary)	Open
Dynamic coordination cost $C = C_0 + C_1\lVert\delta_j\rVert$	Quadratic inequality; admits closed form, loses interpretive cleanliness	Open
Fully-coupled tempo dynamics ($\mathcal{T}_i$ responsive to δ_j)	Requires joint tempo analysis from 13.2 Working Notes	Open
$N > 2$ scaling of (CM4)	Conjunction over pairwise terms generalizes but loses closed form; see `spikes/spike-composition-scaling-N.md`	Open

The dividing line: (C1), (C2), and the quadratic Lyapunov candidate are **formulation choices** imported from adjacent segments or from standard Lyapunov practice. The *consequences* under these choices — (L4), (CM2), (KC), the specialization checks, the weakest-link subsumption, and (CM4) with its scope-gate conjunct — are **derived**. The U_O-multiplicative modulator (UO-mult) is discussion-grade: it uses an LQR-compatibility argument whose rigor depends on an action-space inner-product analysis deferred to 12.1. The asymmetric-limit identification with 10.3 (S-3) is sketch-level — the weighted-Lyapunov argument is standard but the semantic identification with autonomy reduction is structural, not proved.

Epistemic Status.

Conditional. Max attainable: *exact* (within the matched-symmetric-Tier-1 scope); *conditional* beyond.

(CM2) and (KC) are **proved** under Tier 1 architecture + matched-symmetric parameters + Model D disturbance + the (C1) disturbance-coupling model + the (C2) coordination-cost model. These conditions cover Kalman filters, exponential-family Bayesian updaters, gradient-on-strongly-convex agents, and linear-PD correctors — the same architecture class for which 11.1's bridge lemma is promoted from "conditional" to "derived." Within this scope the result is as strong as the standard Lyapunov argument allows.

(CM3) inherits the conditional status of (UO-mult), which is **discussion-grade**: the LQR-compatibility sketch is qualitatively clear, but a rigorous action-space inner-product derivation that pins down $\gamma_{\max}$ has not been produced. (CM4) is therefore conditional on both (UO-mult)'s rigor and on 10.2's scope-gate being independently verified for the given composite candidate.

The asymmetric limit (§asymmetric-limit) is **sketch-level**. The weighted-Lyapunov argument is textbook (Matrosov 1962); the identification of the $\mu \to 0$ limit with 10.3's (S-3) autonomy-reduction mechanism is structurally motivated but not a theorem, because (S-2) function transfer is not formalized in 10.3. When (S-2) lands, the symbiogenic-limit result can be promoted to derived.

What this segment does **not** establish:

- Composite **incremental** sector bound (DA2'-inc) — still the domain of 11.1's bridge lemma and `spikes/spike-bridge-lemma-contraction.md`. (CM4) gives composite (T2) at the macro level; the bridge lemma's contraction is a separate, stronger condition.

- Heterogeneous-architecture composites (Kalman + PID, Tier 1 + Tier 3): the joint Lyapunov construction requires agent-by-agent tiering; closed form is not available.

- Nonlinear or state-dependent coupling: (C1) assumes γ independent of δ. State-dependent γ produces a nonlinear inequality in δ; this is the effects-spiral corollary territory of 13.2, still open.

- $N > 2$ scaling: the matched-symmetric pairwise result generalizes by conjunction, but the Cauchy–Schwarz step degrades with team size; the closed form does not survive cleanly. See `spikes/spike-composition-scaling-N.md`.

On (T2) and sub-scoping. The joint quadratic Lyapunov candidate presumes each sub-agent's correction is in sub-scope α of A (Bayesian / exponential-family / strongly-convex-gradient / linear-PD) under directional fidelity per 4.2. Composites with sub-scope β sub-agents (PID, rule-based, human-judgment) require (T2) verification per sub-agent at the composite level — the template's A2'-sub-scope label is inherited pairwise.

Discussion.

Relationship to the bridge lemma. 11.1's bridge lemma establishes when the macro-update map f_c is **incrementally** contracting, at the trajectory-error level — a condition strictly stronger than the one-point sector bound. This segment is the complement at the macro-state level: given the bridge lemma's admissibility, (CM4) derives composite (T2), the one-point sector bound that C's instantiation for the composite requires. Both are necessary for the composite to be a stable macro-agent with bounded mismatch: bridge lemma says the macro-description tracks micro-reality (trajectory error stays bounded); (CM4) says the composite's own corrections drive composite mismatch back inside the sector region. Together they close the composite-persistence argument at both layers.

Pattern across the signed-coupling instances. (CM4) has the same shape as several persistence-flavored results already in AAT:

- 13.1: per-sub-agent inequality $\alpha_i R_i > \rho_{i,\text{env}} + \sum_j \gamma_{j\to i}^{\text{adv}} \mathcal{T}_j - \sum_j \gamma_{j\to i}^{\text{coop}} \mathcal{T}_j$
- 11.2: composite inequality with effective disturbance $\rho_{\text{ext}} + \varepsilon^* \nu_c$
- 13.2: failure condition $\gamma_A \mathcal{T}_A > \alpha_B R_B - \rho_B$ (negation of persistence)
- this segment: matched-symmetric dyad with signed γ and scope-gate

All four are instances of C's pattern with signed coupling controlling the sign of the cross-agent contribution to ρ_ξ. The template already names this pattern at the meta level; the present segment is the dyadic closed-form instance that the other three segments reference pairwise. A dedicated meta-segment for "signed-coupling critical-mass" was considered during spike work and judged redundant with C — the cross-instance structure is already visible in that segment's instantiation table.

Potential-game generalization (via 14.1). (CM4) applies to composites with shared target (scope routes C-i / C-ii / C-iii from `#scope-composite-agent`) where contraction-to-shared-truth is the correct primitive. `#deriv-strategic-composition` carries the sibling result for composites with partially-opposing objectives (scope route C-iv — strategic composites): under the potential-game condition (Monderer-Shapley 1996), the sector-persistence template transfers to the gradient of the joint potential Φ, with α_{joint} playing the role of the composite sector constant and $\xi = \pi - \pi^*$ (deviation from Nash) playing the role of the composite state. For matched-symmetric potential games, the structural form of (CM2) survives with $(\alpha, R, \rho, \gamma, C)$ replaced by $(\alpha_{\text{joint}}, R_{\text{Nash-basin}}, \rho_\xi, \gamma_{\text{strategic}}, C_{\text{strategic}})$; the specific mapping is instance-dependent and typically not closed-form for non-zero-sum games. The joint-quadratic-Lyapunov machinery of this segment and the potential-function-Lyapunov machinery of `#deriv-strategic-composition` are both instances of `#result-sector-persistence-template` at the composite-state-variable level; what varies is whether the composite state is *mismatch-to-shared-target* (this segment) or *deviation-from-Nash* (`#deriv-strategic-composition`).

What (CM4) contributes beyond existing segments.

1. **Derivation, not assumption.** The composite sector constant appearing in 11.1's (A4) was previously bounded by the weakest-link formula (a derived lower bound) but treated as an assumption for the bridge lemma. (CM4) *derives* α_c as a closed-form function of sub-agent parameters + coupling + unity + coordination overhead, in the matched-symmetric-Tier-1 case.
2. **Sign-sensitive.** The weakest-link bound is sign-blind — it gives the same $\alpha_c \geq \alpha - C$ regardless of whether the coupling is cooperative or adversarial. (CM4) makes the sign explicit through the $\rho + \gamma\mathcal{T}$ right-hand side and turns a correction-rate bound into a full persistence inequality.
3. **Unifies cooperative and adversarial regimes.** Teams persisting where individuals can't and adversarial destabilization are the same inequality viewed from opposite signs of γ, not independent results.

4. **Scope-gate made explicit.** Composite persistence requires both contraction (CM3) **and** scope-satisfaction (10.2). (CM4) states this conjunction honestly; the absence of one or the other is a different failure mode (composite fails to contract vs. composite was never a composite).

5. **Symbiogenic and peer regimes connected.** The asymmetric limit shows symbiogenesis is a parameter-limit of peer coupling under a Lyapunov-weight deformation, not a discontinuous regime change requiring separate machinery.

Why the matched-symmetric restriction is load-bearing. Relaxing matched architectures requires per-sub-agent tiering in the sense of 11.1 (Tier 1 / 2 / 3). Heterogeneous composites can still satisfy (T2) at the composite level, but the joint Lyapunov construction must be weighted per sub-agent's contraction capacity — and the weighted form loses the clean closed-form (CM2). Relaxing symmetric coupling ($\gamma_{1\to 2} \neq \gamma_{2\to 1}$) produces a non-symmetric matrix whose smallest eigenvalue must be computed explicitly per instance. The matched-symmetric case is the one where both sub-agent tiering and coupling symmetry collapse the Lyapunov construction to a scalar inequality — which is why closed form is available there and not elsewhere.

Load-bearing role under `#disc-identifiability-floor` Instance 3. The composition-layer identifiability floor (Instance 3 of `#disc-identifiability-floor`) establishes a no-go theorem: there exist pairs of coupled systems with identical marginal component-level observation distributions but opposite composite-contraction signs, so composite contraction is not in general identifiable from component data alone. Four structural escapes are named there; escape (b) — matched Tier at the composite level — is operationalized by the closed-form (CM2) in this segment. Under the floor, (CM2) is not just "a closed-form result in a special case" but **the unique broadly-available composition-contraction certificate** among the four escape routes listed. Without (CM2) or its metric-formulation generalization (CM2-M) via `#result-contraction-template`, the weakest-link bound (WL) is sign-blind and cannot distinguish the cooperative-contracting from the adversarial-destabilizing composite. This load-bearing status positions this segment as "the machinery that escapes the composition-layer floor" rather than as a closed-form curiosity in the matched-symmetric special case.

Adjacent literature. The joint Lyapunov construction is an instance of the classical **vector Lyapunov function** method (Matrosov 1962; Bellman 1962). The critical-mass inequality is the AAT sector-bounded analog of the **small-gain theorem for ISS systems** (Jiang–Teel–Praly 1994; Sontag 1989): in the small-gain framework, composition of two ISS systems is ISS iff the product of their gains is less than one; in AAT's sector-bounded framework, the composite persists iff the *sum* of parent reserve-rates exceeds the coupling-amplified disturbance (additive rather than multiplicative because the averaging projection averages rather than multiplies gains). In the linear-diagonal-coupling case, the matrix $A = \alpha I - \beta(I - J)$ where J is the averaging operator is the graph-Laplacian-shifted form governing consensus convergence (Olfati-Saber & Murray 2004); (CM2) is consensus convergence rewritten as a persistence inequality. Relative to active inference's FEP-flow stability arguments (Friston 2019; narrowed by Aguilera et al. 2022 to small parameter regimes in NESS-density models), the present result inherits C's broader validity: it applies wherever (T1)–(T3) hold pairwise + (C1)/(C2) + matched symmetry, without requiring free-energy landscapes or NESS structure.

Findings.

Strong Monotonicity as the Hinge for Legitimate Macro-Agent Coarse-Graining. **Brief:** Two coupled adaptive systems can have identical marginal observation distributions while one composite contracts and the other diverges — the difference is a single bit (the sign of the cross-coupling) that is invisible from component data alone. The matched-symmetric-Tier-1 critical-mass inequality $(\alpha - C)R > \rho + \gamma\mathcal{T}$ exposes this bit explicitly: the same inequality recovers cooperative team-persistence (where coupling helps), adversarial destabilization (where coupling hurts), and Brooks's-Law coordination collapse (where overhead C exceeds correction rate α) as signed special cases of one closed-form result. The strong-monotonicity / DA2'-inc condition picks out precisely the agent classes (Kalman, exp-family Bayes, gradient on strongly-convex losses, linear with positive-definite gain-observation product) where the inequality is exact rather than heuristic.

Impact: Promotes the composite sector constant in `#form-composition-closure`'s admissibility condition (A4) from a weakest-link lower bound (sign-blind, treats cooperative and adversarial coupling identically) to a derived closed form that depends explicitly on coupling sign, target-correlation U_O, and coordination overhead C. Operationalizes Instance 3 of `#disc-identifiability-floor` (composition-layer common-Lyapunov no-go via Liberzon 2003) through escape route (b): under matched Tier at the composite level, the joint Lyapunov collapses to a scalar inequality and the otherwise-unidentifiable sign bit becomes a structural input to (CM2). Within the four `#disc-identifiability-floor` Instance-3 escapes, (CM2) and its metric generalization (CM2-M) in `#result-contraction-template` are the unique broadly-available coupling-sign certificate. The asymmetric-parameter limit gives `#hyp-symbiogenic-composition`'s (S-3) autonomy-reduction a Lyapunov-weighted formulation as a smooth deformation of (CM4) rather than a discontinuous regime change.

Novelty Claim: *Claim novelty* on strong monotonicity as the criterion separating legitimate macro-agent coarse-graining from coexistence-only multi-agent description. The closest mathematical neighbors (Subramanian 2020; Congeduti 2020) prove bounded-loss composition under predictive compression but do not isolate a single condition whose failure flips composite contraction; the Markov-blanket line (Parr 2019; Kirchhoff 2018) discusses nested agency without a control-theoretic monotonicity criterion; the small-gain-theorem line (Jiang-Teel-Praly 1994; Sontag 1989) provides a multiplicative composition criterion under input-to-state stability rather than the additive sign-sensitive form (CM2) takes here.

Related Work:

Table F.3

ASF concern	*Prior-art language*	*Relationship / Positioning*
Strong monotonicity as composition-validity hinge	Subramanian, Sinha, Seraj & Mahajan 2020, "Approximate information state for approximate planning and reinforcement learning in partially observed systems" *arXiv:2010.08843* (published 2020, found 2026-04)	*closest mathematical neighbor* — proves a bridge-shape bound but does not foreground a single monotonicity-grade condition that determines whether composite description is legitimate. Per the Pillar-2 defense strategy, foregrounding strong monotonicity (DA2'-inc; equivalently (CT2)-at-$M = I$ in `#result-contraction-template`) as the hinge between stable coexistence and valid macro-agent coarse-graining is the increment ASF supplies on top of this neighbor
Coordination-overhead / influence loss bound	Congeduti, Mey & Oliehoek 2020, "Loss Bounds for Approximate Influence-Based Abstraction" *arXiv:2011.01788* (published 2020, found 2026-04)	*closest mathematical neighbor* — explicit bounds on value loss from approximate influence in multi-agent settings. (CM2)'s sign-sensitive RHS $\rho + \gamma\mathcal{T}$ encodes the same loss-vs-cooperation tradeoff that Congeduti's influence-approximation bound encodes for value, applied at the persistence-inequality layer rather than the policy-value layer
Composite-Lyapunov nonexistence under component-marginal data	Liberzon 2003, *Switching in Systems and Control* Theorem 2.1; Dayawansa & Martin 1999, *IEEE TAC* 44; Shorten et al. 2007 *SIAM Review* 49 (published 2003/1999/2007, found 2025-04)	*formal antecedent* — supplies the no-go theorem `#disc-identifiability-floor` Instance 3 invokes. (CM2) is the AAT escape route (b) under matched Tier at the composite level
Vector-Lyapunov composition	Matrosov 1962; Bellman 1962	*formal antecedent* — joint quadratic Lyapunov $V = \frac{1}{2}(\|\delta_1\|^2 + \|\delta_2\|^2)$ is a vector-Lyapunov construction; standard machinery

ASF concern	*Prior-art language*	*Relationship / Positioning*
Multiplicative composition criterion via small-gain	Jiang, Teel & Praly 1994, "Small-gain theorem for ISS systems and applications" *Math. Control Signals Syst.* 7; Sontag 1989	*partial anticipation* — same compositional-validity question, multiplicative form (gain product < 1) under input-to-state stability. (CM2)'s additive form $(\alpha - C)R > \rho + \gamma\mathcal{T}$ comes from sector-bounded correction rather than ISS; the two are different scaling regimes of the same composition-validity question, with (CM2) sharper in AAT's setting because the averaging projection averages rather than multiplies sub-agent gains
Negative-feedback heterogeneous extension	Slotine 2003, "Modular stability tools for distributed computation and control" *Int. J. Adapt. Control Signal Process.* 17:397–416 — Theorem 3	*formal antecedent* — supplies (CC-feedback) used in `#result-contraction-template` to lift (CM2) to heterogeneous Tier-1M sub-agents via (CM2-M)
Consensus dynamics graph-Laplacian shift	Olfati-Saber & Murray 2004, "Consensus problems in networks of agents with switching topology and time-delays" *IEEE TAC* 49:1520	*adjacent literature* — in the linear-diagonal-coupling case, (CM2) is consensus convergence rewritten as a persistence inequality; different domain framing, same mathematical object
Markov-blanket nested-agency at composition	Parr, da Costa & Friston 2019; Kirchhoff et al. 2018 (published 2019/2018, found 2026-04)	*conceptual precursor* — provides nested-agency framing without a control-theoretic monotonicity criterion that separates valid macro-agent coarse-graining from coexistence

Search Log:

- 2026-04 (*nominally comprehensive*, via `ref/Novelty_defense_and_integration.md` Pillar 2): Per Pillar-2 defense strategy, the strong-monotonicity hinge is foregrounded as the criterion not visible in retrieved prior art. Subramanian / Congeduti provide the closest math (bounded-loss composition) without isolating a single condition whose failure flips composite legitimacy; Markov-blanket line provides nested-agency intuition without monotonicity machinery.
- 2026-04 (*targeted*, prior to comprehensive search): the small-gain-theorem line (Jiang-Teel-Praly 1994; Sontag 1989) and consensus-dynamics line (Olfati-Saber & Murray 2004) were known and cited in the segment's existing Discussion as adjacent literature. The comprehensive search added the Sub20 / Con20 closest-neighbor positioning and confirmed no surfaced paper foregrounds the strong-monotonicity hinge.
- 2026-04 (*intuition-only*, prior): expected adjacent literature was switched-systems stability (Liberzon 2003) and game-theoretic equilibrium analysis. Switched-systems hit confirmed via `#disc-identifiability-floor` Instance 3; game-theoretic side is properly the territory of `#deriv-strategic-composition`, scope-separated.

Working Notes.

- **Partial-derivation: heterogeneous-architecture dyad.** The clean closed form (CM2) relies on matched sub-agent architectures collapsing the joint Lyapunov candidate to a scalar inequality. A natural next move is the (A_1, A_2) = (Tier 1, Tier 2) case: weighted Lyapunov $V_\mu = \frac{1}{2}(\|\delta_1\|^2 + \mu(\delta_2)\|\delta_2\|^2)$ with the weight a function of agent 2's local contraction modulus. Likely produces a range-valued (not closed-form) critical-mass inequality. Defer until a specific Tier-mismatch composite motivates it.
- **Sharpen (UO-mult).** Upgrading $\gamma(U_O) = -\gamma_{\max} U_O$ from discussion-grade to derived requires an action-space inner-product analysis: define the environment's action-coupling operator, show that LQR-linear policies produce cross-actions

with inner product proportional to target correlation, and pin down $\gamma_{\max}$ in terms of the quadratic objective's Hessian and the environment's coupling gain. This is mechanical but non-trivial; natural home is an extension to 12.1's linear-Gaussian closed-form section.

- **Close (S-2) function transfer in 10.3.** The asymmetric-limit result here identifies the Lyapunov-weight limit as (S-3) autonomy reduction but leaves (S-2) function transfer unformalized. A complete symbiogenic-limit theorem requires specifying what happens to agent 1's state space when it inherits structure from agent 2 — how $\mathcal{F}(M_e, \Sigma_e)$ lands in M_h. This is flagged in 10.3's Working Notes and is a prerequisite for promoting the limit result to derived.
- **Connection to Miller-2022 extreme-transition motif.** The asymmetric-limit smooth-deformation result contrasts with extreme-transition dynamics (pending dedicated Section III segments) where population-level niche-replacement proceeds discontinuously. The two mechanisms co-exist and are not reducible to one another; the weighted-Lyapunov analysis is specific to bilateral asymmetric integration.
- ***N*-agent scaling.** The conjunction generalization "composite persists iff (CM4) holds pairwise for all i, j" is probably too strong — pairwise cooperative effects can compose super-additively in particular topologies (ensemble filters, committee agents). A sharper N-agent theorem requires a graph-structured Lyapunov construction. See `spikes/spike-composition-scaling-N.md` for a framing of the question.

Appendix G

Derivation: Gain-Sector Bridge — Proofs and Verification

Complete derivations for the results stated in 4.2: gain-based updating produces correction functions satisfying the sector condition (GA-3) whenever the update rule has directional fidelity. Covers the Kalman case (scalar and matrix), the general bridge theorem, the gradient-descent equivalence, and numerical verification.

Motivation.

A proves persistence and stability downstream of the sector condition (GA-3): $\delta^T F(\mathcal{T}, \delta) \geq \alpha\|\delta\|^2$. The sector parameter α governs the persistence bound ($R^* = \rho/\alpha$), the adaptive reserve ($\Delta\rho^* = \alpha R - \rho$), and the adversarial scaling exponent. But whether the correction function F induced by the gain principle (3.6) actually satisfies GA-3 was not derived — GA-3 was the theory's softest structural joint.

This appendix closes the gap. For optimal Bayesian updates (Kalman, conjugate families), the bridge is exact: the sector parameter equals the gain. For gradient-based agents, GA-3 is equivalent to local strong convexity of the loss function. The load-bearing assumption shifts from "the correction function satisfies the sector condition" (opaque) to "the update rule has directional fidelity" (transparent, checkable).

Proposition B.1: Scalar Kalman Sector Condition.

Statement. For a scalar linear-Gaussian system ($n = m = 1, H = 1$) with Kalman filter gain $K = P^-/(P^- + R_{\text{obs}})$, the correction function $F(e) = K \cdot e$ satisfies the sector condition with:

Derived (scalar-Kalman-sector, from Kalman update).

$$\alpha = K = \frac{P^-}{P^- + R_{\text{obs}}} = \eta^* \tag{G.1}$$

The sector parameter equals the Kalman gain, which equals the uncertainty-ratio gain from 3.6. The bound is tight (the correction is linear).

Proof. The scalar Kalman update for state estimate $\hat{x}$ given innovation $\delta = o - \hat{x}$ is $\hat{x}_{\text{post}} = \hat{x}_{\text{prior}} + K\delta$, where the gain is $K = P^-/(P^- + R_{\text{obs}})$ with P^- the prior variance and R_{obs} the observation noise variance.

The state-space estimation error $e = x - \hat{x}$ evolves under the correction step as:

$$e_{\text{post}} = e_{\text{prior}} - K(He_{\text{prior}} + \varepsilon) = (1 - K)e_{\text{prior}} - K\varepsilon \tag{G.2}$$

With $H = 1$, the expected correction function is $F(e) = K \cdot e$. The sector product is:

$$e \cdot F(e) = K \cdot e^2 \tag{G.3}$$

Since $P^- > 0$ and $R_{\text{obs}} > 0$, we have $K \in (0, 1)$. The sector condition $e \cdot F(e) \geq \alpha \cdot e^2$ holds with $\alpha = K$. The bound is tight because the correction is exactly linear. □

Steady-state sector parameter. At steady state (random walk dynamics, $A = 1$), the prior variance satisfies the algebraic Riccati equation. Solving $K^2 R_{\text{obs}} + QK - Q = 0$ (where Q is the process noise variance) gives:

Derived (steady-state scalar sector parameter).

$$\alpha_{ss} = K_{ss} = \frac{-Q + \sqrt{Q^2 + 4QR_{\text{obs}}}}{2R_{\text{obs}}} \tag{G.4}$$

Limiting behavior: when $Q \gg R_{\text{obs}}$ (fast dynamics, clean observations), $K_{ss} \to 1$; when $Q \ll R_{\text{obs}}$ (slow dynamics, noisy observations), $K_{ss} \approx \sqrt{Q/R_{\text{obs}}}$.

Connection to AAT quantities. The adaptive tempo for a single observation channel at rate ν is $\mathcal{T} = \nu \cdot K_{ss}$, and the sector parameter in the continuous-time framework is $\alpha = \mathcal{T}$. The bridge is trivial in the scalar case: the gain IS the sector parameter.

Proposition B.2: Matrix Kalman Sector Condition.

Statement. For a linear-Gaussian system with state $x \in \mathbb{R}^n$, observations $o \in \mathbb{R}^m$, observation matrix H, and Kalman gain $K = P^- H^T (HP^- H^T + R_{\text{obs}})^{-1}$, the correction function $F(e) = KH \cdot e$ satisfies the sector condition in the $(P^-)^{-1}$-weighted inner product, restricted to the observable subspace $\mathcal{O} = \text{range}(H^T)$:

Derived (matrix-Kalman-sector, from Kalman covariance reduction).

$$e^T (P^-)^{-1} KH e = e^T H^T S^{-1} H e \geq \alpha \cdot e^T (P^-)^{-1} e \tag{G.5}$$

where $S = HP^- H^T + R_{\text{obs}}$ is the innovation covariance, and:

$$\alpha = 1 - \lambda_{\max}(P_{t|t}\, P_{t|t-1}^{-1}) \tag{G.6}$$

restricted to observable directions. In unobservable directions ($e \in \ker(H)$), $\alpha = 0$.

Proof. Define the $(P^-)^{-1}$-weighted inner product: $\langle u, v \rangle_{P^-} = u^T (P^-)^{-1} v$. In this inner product the sector product becomes:

$$\langle e, KH e \rangle_{P^-} = e^T (P^-)^{-1} KH e \tag{G.7}$$

Substituting $K = P^- H^T S^{-1}$:

$$e^T (P^-)^{-1} P^- H^T S^{-1} H e = e^T H^T S^{-1} H e \tag{G.8}$$

Since $S \succ 0$, the matrix $H^T S^{-1} H$ is symmetric positive semidefinite, with null space $\ker(H)$. So $\langle e, KH\,e\rangle_{P^-} \geq 0$ for all e, with equality iff $He = 0$.

For the sector bound, we need $e^T H^T S^{-1} H\,e \geq \alpha \cdot e^T (P^-)^{-1} e$. From the Kalman covariance identity $KH = I - P(P^-)^{-1}$ (where $P = P_{t|t}$ is the posterior covariance), the eigenvalues of KH in the $(P^-)^{-1}$-weighted sense are $1 - \lambda_i(P(P^-)^{-1})$, where λ_i denotes generalized eigenvalues.

Since $P \preceq P^-$ (the posterior is no less certain than the prior), $\lambda_i(P(P^-)^{-1}) \leq 1$, so all eigenvalues of KH are non-negative. In observable directions, $P \prec P^-$ strictly, giving $\alpha > 0$. In unobservable directions, $P = P^-$ (no information gained), so $\alpha = 0$. □

Fully observable diagonal case. When $H = I, A = I, Q = \mathrm{diag}(q_1, \ldots, q_n), R_{\mathrm{obs}} = \mathrm{diag}(r_1, \ldots, r_n)$, the problem decouples into n scalar problems (each governed by Prop B.1), and:

Derived (diagonal-Kalman-sector).

$$\alpha = \min_{i=1}^{n} K_i = \min_{i=1}^{n} \frac{-q_i + \sqrt{q_i^2 + 4q_i r_i}}{2r_i} \tag{G.9}$$

The bottleneck is the dimension with the worst signal-to-noise ratio, consistent with 14.5.

The weighted-norm subtlety. The Lyapunov proofs in A use the Euclidean norm ($V = \frac{1}{2}\|e\|^2$), while the matrix Kalman sector condition holds in the $(P^-)^{-1}$-weighted norm. For fully observable systems with bounded condition number $\kappa(P^-)$, the norms are equivalent:

$$\alpha_{\mathrm{Euclidean}} \geq \alpha_{\mathrm{weighted}} / \kappa(P^-) \tag{G.10}$$

The Lyapunov results remain valid with this quantitative adjustment. The weighted Lyapunov function $V(e) = \frac{1}{2} e^T (P^-)^{-1} e$ yields the tighter bound directly.

Proposition B.3: Bridge Theorem.

Statement. Given the gain-based update $M_t = M_{t-1} + \eta^* \cdot \mathrm{g}(\delta_t)$ (3.6), with H mapping state-space corrections to observation-space mismatch reduction, the induced correction function $F(\delta) = \eta^* \cdot H\,\mathrm{g}(\delta)$ satisfies the sector condition (GA-3) with parameter $\alpha > 0$ whenever:

(B1) Directional fidelity. The composite mismatch transform preserves the mismatch-reducing direction:

Derived (bridge-theorem, from update-gain + B1).

$$\delta^T H\,\mathrm{g}(\delta) \geq c\|\delta\|^2 \quad \text{for } \|\delta\| \leq R \tag{G.11}$$

for some $c > 0$. The sector parameter is:

$$\alpha = \eta^* \cdot c_{\min}, \qquad c_{\min} = \inf_{\|\delta\| \leq R} \frac{\delta^T H\,\mathrm{g}(\delta)}{\|\delta\|^2} \tag{G.12}$$

Proof. The gain-based update changes the model state by $\Delta M = \eta^* \cdot \mathrm{g}(\delta)$. The induced change in predicted observation is $H \cdot \Delta M = \eta^* \cdot H\,\mathrm{g}(\delta)$, so the expected correction function (the mismatch reduction per step) is:

$$F(\delta) = \eta^* \cdot H\, g(\delta) \tag{G.13}$$

The sector product is:

$$\delta^T F(\delta) = \eta^* \cdot \delta^T H\, g(\delta) \tag{G.14}$$

By B1: $\delta^T H\, g(\delta) \geq c_{\min}\|\delta\|^2$. Since $\eta^* > 0$ (the agent updates), we have:

$$\delta^T F(\delta) \geq \eta^* \cdot c_{\min}\|\delta\|^2 = \alpha\|\delta\|^2 \qquad \square \tag{G.15}$$

When B1 holds by construction. For optimal Bayesian updates (Kalman, conjugate families, exponential families), the posterior update minimizes expected loss, ensuring the correction aligns with the mismatch. B1 holds by optimality, not by assumption. The gain principle $\eta^* = U_M/(U_M + U_o)$ determines only the magnitude; the direction is guaranteed by the update rule's optimality. For gradient descent on a convex loss, B1 follows from gradient monotonicity (the defining property of convexity).

Proposition B.4: Gradient Equivalence.

Statement. For any agent updating via gradient descent on a differentiable loss L with learning rate η and minimizer M^* (where $\nabla L(M^*) = 0$), there are two distinct results corresponding to the one-point and two-point sector conditions:

(B.4-i) One-point sector $\Leftarrow$ strong convexity (one direction only). The one-point sector condition at M^* — equivalent to AAT's GA-3 / A2' as stated in A — is

$$\delta^T F(\delta) \geq \alpha\|\delta\|^2 \quad \text{for } \|\delta\| \leq R \tag{G.16}$$

evaluated at fixed equilibrium M^*. Local (α/η)-strong convexity of L on $\mathcal{B}_R(M^*)$ implies this one-point sector condition with $\alpha = \eta\mu$, but the converse fails: there exist losses satisfying the one-point sector condition that are *not* strongly convex on any neighborhood of M^* (counterexample below).

(B.4-ii) Two-point sector $\Leftrightarrow$ strong convexity (full equivalence). The two-point / incremental sector condition — DA2'-inc in K and the bridge-lemma precondition in 11.1 — is

Derived (sector-convexity-equivalence, two-point form).

$$(F(\delta_1) - F(\delta_2))^T(\delta_1 - \delta_2) \geq \alpha\|\delta_1 - \delta_2\|^2 \quad \text{for } \delta_1, \delta_2 \in \mathcal{B}_R(M^*) \tag{G.17}$$

Under this strengthened condition the iff holds:

$$\text{Two-point sector with } (\alpha, R) \iff L \text{ is } (\alpha/\eta)\text{-strongly convex on } \mathcal{B}_R(M^*) \tag{G.18}$$

with $\alpha = \eta\mu$ and $\mu = \inf_{\delta\in\mathcal{B}_R(M^*)} \lambda_{\min}(\nabla^2 L(M^* + \delta))$. The basin radius R is the largest ball around M^* where $\nabla^2 L$ remains positive definite.

Proof of (B.4-ii). The correction function for gradient descent is $F(\delta) = \eta \cdot \nabla L(M^* + \delta)$ (continuous-time form absorbing event rate ν into $\mathcal{T} = \nu \cdot \eta$). The two-point sector condition becomes:

$$\eta \cdot (\nabla L(M^* + \delta_1) - \nabla L(M^* + \delta_2))^T(\delta_1 - \delta_2) \geq \alpha\|\delta_1 - \delta_2\|^2 \tag{G.19}$$

Dividing by $\eta > 0$ gives gradient monotonicity with modulus $\mu = \alpha/\eta$. By Nesterov 2004, Theorem 2.1.10, L is μ-strongly convex on $\mathcal{B}_R(M^*)$ iff this gradient monotonicity holds for all $x, y \in \mathcal{B}_R(M^*)$. The equivalence is bidirectional:

($\Rightarrow$) Two-point sector with (α, R) yields gradient monotonicity with $\mu = \alpha/\eta$, hence strong convexity.

($\Leftarrow$) μ-strong convexity gives gradient monotonicity, multiplying by η recovers the two-point sector condition with $\alpha = \eta\mu$. □

Proof of (B.4-i) one direction. Setting $\delta_1 = \delta, \delta_2 = 0$ in the two-point condition and using $\nabla L(M^*) = 0$ recovers the one-point form $\delta^T F(\delta) \geq \alpha\|\delta\|^2$. So strong convexity (which is equivalent to the two-point sector by (B.4-ii)) implies the one-point sector. The reverse direction does not hold; see the counterexample below. □

Counterexample showing one-point $\not\Rightarrow$ strong convexity. Let $L : \mathbb{R} \to \mathbb{R}$ have $L'(x) = x(1 + \frac{1}{2}\sin(10x))$ with $L'(0) = 0$ (so $x^* = 0$). The one-point sector at x^* is

$$x \cdot L'(x) = x^2\left(1 + \tfrac{1}{2}\sin(10x)\right) \geq \tfrac{1}{2}x^2, \tag{G.20}$$

so the one-point sector holds globally with $\mu = \frac{1}{2}$. But $L''(x) = 1 + \frac{1}{2}\sin(10x) + 5x\cos(10x)$ takes values such as $L''(\pi/10) = 1 - \pi/2 \approx -0.57 < 0$, so L is not convex on any neighborhood of x^* that contains $\pi/10$. The one-point sector at the equilibrium is genuinely weaker than full local strong convexity — a structural distinction, not a smoothness artifact. The full two-point sector fails on this L for any pair (δ_1, δ_2) straddling a region where $L'' < 0$ (gradient monotonicity fails there), so (B.4-ii) correctly identifies this L as a non-instance.

The α factorization (under (B.4-ii)). The decomposition $\alpha = \eta \cdot \mu$ separates agent design (η, responsiveness) from environment structure (μ, loss curvature). The adaptive tempo maps as $\alpha \propto \mathcal{T}$: higher gain or steeper curvature means faster mismatch correction.

Where each direction lands in AAT. The one-point form is what `#deriv-sector-condition`'s A2' ((A2') applied at $\delta^* = 0$) requires for Lyapunov persistence — and what AAT's six persistence-flavored results inherit through `#result-sector-persistence-template` (T2). The two-point / incremental form is what `#form-composition-closure`'s bridge lemma requires (DA2'-inc) for full-update-map contraction at the composite level. Strong convexity sits at the strict end of this scale: it implies both. The one-point $\not\Rightarrow$ two-point counterexample above is exactly why AAT distinguishes the two — Lyapunov persistence is available for some agent classes (variational, PID-bounded-plant) where bridge-lemma contraction is not.

Loss Function Classification.

The gradient equivalence (Prop B.4) classifies which loss functions satisfy GA-3, and with what parameters:

Table G.1

Loss class	*Strong convexity*	α	*Basin R*	*Status*
Quadratic (Kalman, linear regression)	Global: $\mu = \lambda_{\min}(H)$	$\eta \cdot \lambda_{\min}(H)$	∞	Exact
Exponential family (natural params), bounded scope	Pointwise on the interior; uniform on any compact $\Theta_0 \subset \mathrm{int}(\Theta)$: $\mu_0 = \inf_{\theta\in\Theta_0} \lambda_{\min}(\mathbf{I}(\theta)) > 0$	$\eta \cdot \mu_0$	Θ_0 (compact / interior-bounded)	Exact within Θ_0
L2-regularized convex loss	Global: $\mu \geq \lambda$	$\eta \cdot \lambda$ (floor)	∞	Exact
Unregularized logistic	Global convex, not strongly	$\eta \cdot \mu(R)$, decaying	∞ (but $\alpha \to 0$)	Local

Loss class	*Strong convexity*	α	*Basin R*	*Status*
Quasi-convex	$\delta^T \nabla L \geq 0$ but ratio $\to 0$	$\eta \cdot \mu(R) \to 0$	Finite effective	Local
Non-convex within basin	Local: $\mu > 0$ in basin	$\eta \cdot \mu_{\text{local}}$	Distance to nearest saddle	Local
Non-convex beyond basin	Fails: $\lambda_{\min}(\nabla^2 L) < 0$	N/A	N/A	Fails

Key observations:

1. *Exponential family models in natural parameter form satisfy GA-3 pointwise on the interior of the natural-parameter space, and uniformly on any compact subset of the interior.* The Hessian is the Fisher information matrix $\mathbf{I}(\theta)$, which is positive definite in $\text{int}(\Theta)$ — but pointwise positive definiteness does not imply a uniform global lower bound on $\lambda_{\min}(\mathbf{I}(\theta))$. The Poisson natural parameter is the canonical counterexample: $\mathbf{I}(\theta) = e^{\theta}$ on $\Theta = \mathbb{R}$, so $\inf_{\theta \in \mathbb{R}} \mathbf{I}(\theta) = 0$ and no global sector constant exists. A uniform $\mu_0 > 0$ — and hence a global $\alpha = \eta\mu_0$ — requires either (i) restricting the operating region to a compact $\Theta_0 \subset \text{int}(\Theta)$ on which $\mu_0 := \inf_{\theta \in \Theta_0} \lambda_{\min}(\mathbf{I}(\theta)) > 0$, or (ii) verifying a uniform Fisher lower bound for the family in question (Gaussian-mean and Beta-Bernoulli, whose Fisher information is bounded below on the interior of Θ, qualify; Poisson and Gamma-shape do not on the full natural-parameter line). The compact-scope condition is verifiable per-application from the agent's prior support and corresponds to taking R as the diameter of Θ_0 in the natural-parameter coordinate, matching A2's local-region structure (A).

2. *L2 regularization transforms locally strongly convex losses into globally strongly convex ones.* The regularization parameter λ provides a floor: $\alpha \geq \eta\lambda$ everywhere.

3. *Non-convex losses satisfy GA-3 within each basin of attraction.* The basin radius R is the distance from the local minimum to the nearest inflection surface (where $\lambda_{\min}(\nabla^2 L) = 0$). Beyond the basin, the sector condition fails — this IS the structural-adaptation trigger from 4.5.

Failure Modes.

The bridge fails in precisely five cases. These characterize the exact conditions under which the gain-based update does NOT produce a correction function satisfying the sector condition.

FM-1: Directional Infidelity. The mismatch transform g rotates the correction away from the mismatch direction, so $\delta^T H g(\delta) \leq 0$.

Example. If $\delta \in \mathbb{R}^2$ and $g(\delta) = R_{90}\delta$ (a 90-degree rotation), then $\delta^T g(\delta) = 0$ for all δ. The correction is perpendicular to the mismatch and never reduces it.

When it occurs. Pathological parameterizations where the model-space update direction is misaligned with observation-space mismatch. For optimal Bayesian updates, B1 holds by construction — the posterior minimizes expected loss.

FM-2: Gain Collapse. $\eta^* \to 0$ while $\rho > 0$, so $\alpha \to 0$ and the persistence condition $\alpha > \rho/R$ eventually fails. An agent accumulating experience in a non-stationary environment will eventually have gain too small to track changes. This is not a bridge failure but a persistence-condition failure — the bridge holds ($\alpha = \eta^* \cdot c > 0$ for finite experience), but α falls below ρ/R. See 3.6.

FM-3: Nonlinear Saturation. The correction function g saturates at large $\|\delta\|$, so the sector ratio $\delta^T g(\delta)/\|\delta\|^2$ decays. For a saturating function such as $g(\delta) = \tanh(\delta)$, the local sector parameter is:

$$\alpha(R) = \eta^* \cdot \frac{\tanh(R)}{R} \tag{G.21}$$

For small R, $\alpha \approx \eta^*$ (linear regime). For large R, $\alpha \approx \eta^*/R$ (saturated regime). The sector condition holds locally — the valid region shrinks as the nonlinearity strengthens. This is exactly what the local form A2' in A anticipates.

FM-4: Unobservable Directions. When $\ker(H) \neq \{0\}$ (fewer observations than state dimensions), the correction has no effect on mismatch components in $\ker(H)$. The sector condition holds only in the observable subspace $\mathcal{O} = \mathrm{range}(H^T)$. This is a structural limitation: the agent cannot correct errors it cannot observe. See 8.3.

For persistence in the full state space, the system must be *detectable*: all unstable modes must be observable. Stable unobservable modes decay naturally without correction.

FM-5: Model Misspecification. The model class does not contain the truth, so the gradient direction is systematically wrong. B1 fails because the correction aims at the wrong target. The sector parameter degrades proportionally to the misspecification. This is the 4.5 trigger: the correction function fails not because the gain is wrong but because the correction *direction* is wrong.

Simulation Results Summary.

Six numerical experiments validate the gradient equivalence (Prop B.4) and the loss-function classification.

Experiment 1: Quadratic Loss (Global Strong Convexity). $L(w) = \frac{1}{2}(w - w^*)^T H(w - w^*)$ with $w \in \mathbb{R}^5$, Hessian eigenvalues $\{0.5, 1.0, 2.0, 3.0, 5.0\}$, $\eta = 0.1$. The sector ratio $\delta^T F/\|\delta\|^2$ at every gradient-descent step fell within $[\eta\mu, \eta L] = [0.05, 0.50]$, confirming Prop B.4 exactly. No step violated the bound.

Experiment 2: Logistic Regression (Position-Dependent Convexity). Logistic regression with $d = 2, N = 5000$ samples. Unregularized: sector ratio positive along the entire trajectory, with local μ ranging from 0.076 (at MLE) to higher values far from it. With L2 regularization ($\lambda = 0.1$): sector ratio bounded below by $\eta\lambda = 0.03$ at every point, confirming the global floor.

Experiment 3: Gaussian Mixture (Non-Convex, Basin Structure). Two-component Gaussian mixture. Within basin ($r < 3.1$): Hessian positive definite, sector ratio positive, gradient descent converges to correct optimum. At basin boundary ($r \approx 3.1$): Hessian eigenvalue crosses zero. Beyond basin ($r > 3.5$): sector ratio negative, gradient descent converges to wrong optimum. Basin boundary confirmed as sharp transition, matching the 4.5 trigger.

Experiment 4: Local Sector Parameter Measurement. Same Gaussian mixture. Sector parameter $\alpha(r)$ decreases monotonically from 181.4 (at $r = 0.5$) to 125.8 (at $r = 3.0$), then drops to zero at $r \approx 3.1$. The decay is moderate within the basin (69% retention at the boundary); the transition at the boundary is sharp.

Experiment 5: Exponential Family (Poisson, Natural Parameters). Poisson in natural parameter space. Sector ratio positive at every step, ranging from 0.00177 to 0.00240 (with $\eta = 0.001$). Confirms global strong convexity of exponential family losses in natural parameterization.

Experiment 6: Quasi-Convex and Multi-Modal Losses. *Quasi-convex* ($L(w) = -1/(1+w^2)$): sector ratio always positive but decays as $O(1/w^4)$. No uniform lower bound exists — $\alpha(R) \to 0$, making persistence bounds vacuous for large R.

Multi-modal ($L(w) = \sin(w) + 0.01w^2$): sector condition holds within each basin ($|\delta| < 3.0$), fails at basin boundary (gradient reversal at local maximum). Basin radius ≈ distance from local minimum to nearest local maximum.

Epistemic Status.

Propositions B.1 and B.2 are *exact* — they follow from standard Kalman filter algebra. The only subtlety is the weighted norm in the matrix case (B.2), which is a quantitative refinement, not a qualitative gap. Both are standard results in estimation theory reframed in sector-condition language.

Proposition B.3 (Bridge Theorem) is a *conditional derivation*: exact under B1 (directional fidelity). B1 holds by construction for optimal Bayesian updates (the posterior minimizes expected loss, ensuring the correction aligns with the mismatch). For approximate update rules, B1 is a design condition, not a global assumption. The condition is transparent and checkable for specific systems.

Proposition B.4 (Gradient Equivalence) splits along the one-point / two-point distinction. (B.4-ii) — the iff between strong convexity and the *two-point / incremental* sector condition (DA2'-inc) — is an *exact mathematical equivalence*: bidirectional via Nesterov 2.1.10. (B.4-i) — the relation to AAT's one-point sector A2' as stated in A — is *one-directional only*: strong convexity implies the one-point sector at M^*, but not conversely. The Codex-style counterexample $L'(x) = x(1 + \frac{1}{2}\sin(10x))$ satisfies the one-point sector globally yet has L'' negative on intervals, ruling out a converse. The loss-function classification table is read accordingly: rows that supply strong convexity satisfy both forms; rows that supply only one-point sector (none in the current table — strong convexity is the universal column) would not transfer to two-point. The simulation experiments confirm the theoretical predictions across all six loss classes tested; no violations were observed where the theory predicts the sector condition holds.

Max attainable: *conditional* for the bridge (B1 is inherent — pathological update rules exist), *exact* for the gradient equivalence in its two-point form (B.4-ii) and *exact one-direction* for its one-point form (B.4-i). The condition cannot be removed: there exist correction functions that violate the sector condition (FM-1 through FM-5), and there exist losses that satisfy the one-point sector at M^* without being strongly convex (the Codex-style counterexample), which is why the one-point $\not\Rightarrow$ two-point gap is structural rather than a smoothness artifact.

What remains open. (1) Non-gradient agents (PID controllers, rule-based systems, human judgment): GA-3 remains an empirical claim for these agent classes. (2) Time-varying α: when the gain adapts (Adam, RMSprop, Kalman transient), $\alpha(t)$ varies; the Lyapunov analysis extends to time-varying $\alpha(t) \geq \underline{\alpha} > 0$ via standard results (Khalil 2002, Chapter 8). (3) Stochastic gradients: SGD satisfies the sector condition *in expectation*; the per-step noise enters as effective disturbance in the Prop A.1S framework.

Discussion.

The formal chain is closed for well-designed agents. The prediction chain becomes:

$$\text{gain principle+B1 (directional fidelity)} \xrightarrow{\text{Prop B.3 (exact)}} \text{sector condition (GA-3)} \xrightarrow{\text{Props A.1, A.1S, A.2 (exact)}} \text{persistence, reserve, scaling} \tag{G.22}$$

For optimal Bayesian updates, B1 holds by construction, making the full chain a derivation. For gradient agents, B1 is equivalent to local strong convexity — a well-characterized property.

GA-3 is grounded, not floating. The assumption load shifts from "the correction function satisfies the sector condition" (what correction function?) to "the update rule has directional fidelity" (the correction at least points toward reality).

For the important agent classes — Kalman filters, conjugate posteriors, exponential families, gradient descent on convex losses — this is automatic.

Basin boundary = structural adaptation trigger. For gradient agents with non-convex losses, the basin radius R is the convexity radius of the loss landscape. When mismatch exceeds R, the correction function reverses direction. This is the 4.5 trigger with a precise geometric characterization: structural adaptation is needed when the agent crosses an inflection surface of its loss landscape.

Working Notes.

- The fluid-limit gap: the bridge analysis works in expected value. The actual per-step correction is stochastic; observation noise ε contributes $K\varepsilon$ to each update, acting as effective disturbance of magnitude $\|K\|\sigma_\varepsilon$ per step. This maps to the Prop A.1S (stochastic) framework — the sector condition on the expected correction plus stochastic noise.
- For variational inference and other approximate methods, B1 is not guaranteed by optimality and must be verified per approximation scheme.
- The adaptive-reserve factorization for gradient agents is $\Delta\rho^* = \eta\mu R - \rho$, with three controllable factors: gain η, curvature μ, and basin width R. An agent is robust when $\eta\mu R \gg \rho$.
- Full simulation code, raw traces, and intermediate analysis live in `spikes/spike-gain-sector-bridge-nonlinear.md`; the segment carries the parameters and outcomes that establish the result. Landing-context provenance: this segment consolidates `spikes/spike-gain-sector-bridge.md` (Kalman / bridge theorem) and `spikes/spike-gain-sector-bridge-nonlinear.md` (gradient equivalence and the six experiments).

Appendix H

Derivation: Recursive Update — Uniqueness Derivation

Derivation showing that $M_{\tau^+} = f(M_{\tau^-}, e_\tau)$ is the *unique* update form consistent with directed time, partial observability, and state completeness. Not merely one option, but the only one.

Setup.

We work within AAT's scope (1.5): an agent coupled to an environment Ω through observation and action channels, with residual uncertainty.

Universe of information at event time τ. The following information exists (in the broadest ontological sense) at the moment event e_τ occurs:

Table H.1

Information	*Description*
Ω_τ	The environment state
$\mathcal{C}_{\tau^-}$	The complete interaction history (1.4) up to (but not including) e_τ
$\{M_{\tau'}\}_{\tau' \leq \tau^-}$	The agent's prior internal states, culminating in M_{τ^-}
e_τ	The current event (observation arriving or action completing)
$\{e_{\tau'}\}_{\tau' > \tau}$	Future events (not yet occurred)

The question: of these, which can the update M_{τ^+} depend on?

The Three Constraints.

Constraint 1 — Arrow of time (1.8 postulate). Events are temporally ordered and this ordering is irreversible. An update occurring at time τ cannot depend on events that have not yet occurred:

$$M_{\tau^+} \text{ cannot depend on } \{e_{\tau'}\}_{\tau' > \tau} \tag{H.1}$$

This is a physical constraint — the most primitive one. In a classical universe, information from the future is simply not available. Even if the agent can *predict* future events, those predictions are part of M_{τ^-} (they are internal computations, not future information).

Constraint 2 — Partial observability (1.5). The agent cannot access Ω_τ directly. Its only interface with the environment is through the event e_τ, which is a lossy function of Ω_τ (via 1.3):

$$M_{\tau^+} \text{ cannot depend on } \Omega_\tau \text{ except through } e_\tau \tag{H.2}$$

This is a scope constraint. If the agent could access Ω directly, the residual uncertainty condition in 1.5 would be trivially violable.

Constraint 3 — State completeness (2.1). M_{τ^-} is the agent's *complete* internal state just before event e_τ. There is no information about the agent's past that is available to the update mechanism but not encoded in M_{τ^-}:

$$M_{\tau^+} \text{ cannot depend on } \mathcal{C}_{\tau^-} \text{ or } \{M_{\tau'}\}_{\tau'<\tau^-} \text{ except through } M_{\tau^-} \tag{H.3}$$

This constraint does the most interesting work and deserves careful examination (see Discussion below).

The Derivation.

Result (Recursive Update Uniqueness). Under Constraints 1–3, the model update at event time τ must have the form

$$M_{\tau^+} = f(M_{\tau^-}, e_\tau) \tag{H.4}$$

for some (possibly stochastic) function $f : \mathcal{M} \times \mathcal{E} \to \mathcal{M}$. No other update form is consistent with the three constraints.

Derivation. Consider the most general possible update. The updated state M_{τ^+} is a function of *all accessible information*:

$$M_{\tau^+} = F(\text{accessible information at } \tau) \tag{H.5}$$

We characterize the accessible information by eliminating what is not accessible.

(i) Eliminate future events. By C1 (arrow of time), $\{e_{\tau'}\}_{\tau'>\tau}$ is not accessible.

After this elimination, the candidate dependency set is: $\{\Omega_\tau,\ \mathcal{C}_{\tau^-},\ \{M_{\tau'}\}_{\tau'\leq\tau^-},\ e_\tau\}$

(ii) Eliminate direct environment access. By C2 (partial observability), the agent cannot access Ω_τ except through the event e_τ. Any information from Ω_τ that reaches the agent does so through e_τ — already in the dependency set.

After this elimination: $\{\mathcal{C}_{\tau^-},\ \{M_{\tau'}\}_{\tau'\leq\tau^-},\ e_\tau\}$

(iii) Reduce past information to M_{τ^-}. By C3 (state completeness), M_{τ^-} is the agent's complete internal state. Every element of $\mathcal{C}_{\tau^-}$ and every prior model state $M_{\tau'}$ $(\tau' < \tau^-)$ that could influence the update can do so *only through* its effect on M_{τ^-}. The agent's internal state evolves through a sequence of updates; the cumulative effect of all prior events is exactly M_{τ^-}. The raw events that produced this state are no longer separately available — they were "consumed" by the update mechanism and their information (to the extent it was retained) is now encoded in M_{τ^-}.

Could the agent maintain a separate log of raw events outside of M? It could — but that log *is part of* M. Whatever information the agent retains in any form — model parameters, cached data, raw event buffers, metadata — is by definition part of its complete internal state M_{τ^-}. If something is available to the update mechanism and not in M_{τ^-}, then M_{τ^-} was not the complete state — contradicting C3.

After this elimination: $\{M_{\tau^-},\ e_\tau\}$

Therefore: $M_{\tau^+} = F(M_{\tau^-}, e_\tau) \equiv f(M_{\tau^-}, e_\tau)$

This is the unique form: no information beyond (M_{τ^-}, e_τ) is accessible under the three constraints, so no update form depending on anything else is realizable. □

Corollary (Between-events dynamics). Between events, no new event e arrives. The same argument applies with e_τ removed from the accessible set:

$$\frac{dM}{d\tau} = g(M_\tau) \tag{H.6}$$

The agent's internal evolution between events (prediction, decay, internal simulation) depends only on the current state. □

Corollary (Serial special case). When observations and actions alternate at a uniform rate on a single channel, each event e_t is the pair (o_t, a_{t-1}). The update becomes:

$$M_t = f(M_{t-1}, o_t, a_{t-1}) \tag{H.7}$$

This is the familiar discrete-time form. □

Information-Set Formalization.

For readers who prefer a measure-theoretic framing:

The agent's **information set** at time τ is the sigma-algebra $\mathcal{I}_\tau^{agent}$ — the collection of events (in the probability-theoretic sense) about which the agent can condition its update.

- **C1** restricts $\mathcal{I}_\tau^{agent} \subseteq \sigma(\{e_{\tau'} : \tau' \leq \tau\} \cup \{\Omega_\tau\} \cup \{M_{\tau'} : \tau' \leq \tau^-\})$ — no future information.
- **C2** further restricts: $\sigma(\Omega_\tau) \setminus \sigma(e_\tau)$ is not in $\mathcal{I}_\tau^{agent}$ — the agent cannot condition on aspects of Ω_τ not captured by e_τ.
- **C3** further restricts: $\sigma(\{e_{\tau'} : \tau' < \tau\} \cup \{M_{\tau'} : \tau' < \tau^-\}) \subseteq \sigma(M_{\tau^-})$ from the agent's perspective.

After all three restrictions: $\mathcal{I}_\tau^{agent} = \sigma(M_{\tau^-}, e_\tau)$.

By the Doob–Dynkin lemma[1], any $\sigma(M_{\tau^-}, e_\tau)$-measurable random variable is a (Borel) function of (M_{τ^-}, e_τ). Therefore $M_{\tau^+} = f(M_{\tau^-}, e_\tau)$ for some measurable f. □

Attempts to Break the Result.

Before trusting the proof, seven counterexample attacks:

Attack 1: Simultaneous events. Two events arrive at exactly the same time: $e_\tau^{(1)}$ and $e_\tau^{(2)}$. The update has three arguments: $f(M_{\tau^-}, e_\tau^{(1)}, e_\tau^{(2)})$.

Verdict: Not deep — 3.1 defines events as atomic. If we allow bundled events, the form holds with e_τ as a set. Reveals that "event" needs careful definition, but the form is preserved.

[1] Kallenberg, O. (2002). *Foundations of Modern Probability* (2nd ed.). Springer. §1.2 (measurability and the Doob–Dynkin lemma).

Attack 2: Continuous environmental influence. An agent embedded in a physical system experiences continuous forces (gravity, temperature, electromagnetic fields). These aren't "events" in 3.1's sense; they're continuous signals. The true dynamics would be $dM/d\tau = g(M_\tau, o(\tau))$ where $o(\tau)$ is a continuous observation stream.

Verdict: Genuine limitation of the event-driven formulation. The between-events corollary $dM/d\tau = g(M_\tau)$ holds only when the agent is truly isolated between events. For continuous coupling, the analogous result is the general state-space representation $\dot{M} = g(M, u)$ from control theory — arrived at by the same three constraints. The event-driven version is a special case for digital/sampled systems.

Attack 3: The C3 circularity. C3 defines M as the agent's complete internal state. Any apparent counterexample is dissolved by expanding M. Consider: an agent has a "model" (neural net weights) and a "replay buffer" (stored raw events). C3 says $M =$ (weights, buffer). The model space is just larger than you thought.

Verdict: The deepest objection. The proof essentially: (1) Define M to be everything the agent has. (2) Observe the update can only use what the agent has. (3) Therefore $f(M_{\tau^-}, e_\tau)$. The real content is the *analytical commitment*: by defining M as complete, we commit to Markovian analysis, which then makes 2.3 the right quality metric. See Epistemic Status below.

Attack 4: Shared state between agents. Agents A and B share a common memory bank (shared database). The clean resolution is the multi-agent framework: the shared memory is part of the *composite* system's state, and each agent's interaction with it is mediated by events (reads and writes). Not a true counterexample but highlights that C3 requires careful delineation of agent boundaries.

Attack 5: External randomness not in e_τ. Hardware thermal noise used in the update. The stochastic case $M_{\tau^+} \sim P(\cdot \mid M_{\tau^-}, e_\tau)$ is a special case of f where f is a randomized function. The *form* — dependence on exactly (M_{τ^-}, e_τ) — is preserved. The result statement should explicitly allow stochastic f.

Attack 6: Time-dependent updates. Could f depend on the timestamp τ itself? Yes — consistently. The event e_τ in 3.1 carries a timestamp: $e_\tau =$ (type, channel, payload, τ). So time-dependence enters through e_τ. Alternatively, the agent may maintain an internal clock as part of M_{τ^-}. Either way, $f(M_{\tau^-}, e_\tau)$ accommodates time-dependence.

Attack 7: Agents that store full history. An agent with $M_{\tau^-} \supseteq \mathcal{C}_{\tau^-}$ is entirely consistent. The model space $\mathcal{M}$ is simply large enough to include the raw history. The 2.2 argues compression is *wise* — but the recursive update form holds regardless of compression level.

What Is Derived vs. What Is Chosen.

Table H.2

Property	*Source*	*Strength*
Constraint C1 (arrow of time: update depends on τ^-, not future events)	Physical law — not a formulation choice	Postulate (physical)
Constraint C2 (partial observability: update depends on e_τ, not raw Ω_τ)	Scope definition of AAT	Postulate (scope-defining)
Constraint C3 (state completeness: M_{τ^-} summarizes the agent's relevant past)	Analytical commitment — the definition of M as complete	Definition
Recursive form $M_\tau = f(M_{\tau^-}, e_\tau)$	C1 + C2 + C3	Proved (unique form compatible with the three constraints)
Future-dependent updates eliminated	C1 alone	Derived (direct consequence)
Ω_τ-dependent updates eliminated	C2 alone	Derived (direct consequence)

Property	*Source*	*Strength*
Full-history-dependent updates reducible to recursive form	C3 + any choice of $M \supseteq \mathcal{C}_{\tau^-}$	Proved (compatibility, not elimination)
Markov property of the update	C3 (completeness) + recursive form	Proved (follows from C3 definition)
Seven attack counterexamples (simultaneous events, continuous coupling, C3 circularity, shared state, external randomness, time-dependence, full history)	Case-by-case reduction to the recursive form	Proved (each)
C3 is definitional, not eliminative	Analysis of what C3 asserts vs. what it rules out	Discussion-grade (clarifying observation)

The dividing line: C1 and C2 do genuine *eliminative* work — they rule out physically or scope-excluded update forms. C3 is a *definitional commitment* that forces the Markov structure by making M complete by construction; it cannot be "violated" because any apparent violation means M was misspecified. The recursive form's uniqueness is therefore conditional on the three-constraint set being accepted, not on the constraints being independently inescapable — C3 in particular could be refused (yielding non-Markovian analysis), at the cost of leaving AAT's scope.

Epistemic Status.

The result is correct but partly definitional. The three constraints have different epistemic characters:

Table H.3

Constraint	*Character*	*Can it be violated?*
C1 (arrow of time)	Physical law	Not in a classical universe
C2 (partial observability)	Scope definition	Only by leaving AAT's scope
C3 (state completeness)	Analytical commitment	Not without redefining M

C1 and C2 do genuine eliminative work — they rule out update forms that depend on future events or on raw Ω. These are non-trivial constraints.

C3 is a definitional commitment that produces the Markov structure. It cannot be "violated" because any violation is absorbed by expanding M. This is not a weakness — it's the nature of the claim. The result says: *the Markovian analysis is the only one consistent with C1 + C2 + the definition of M as complete*. The alternative — an update that depends on something outside M — is not "wrong" but rather means M was misspecified.

What the result says: C1 eliminates a physically impossible class of updates (future-dependent). C2 eliminates a scope-excluded class (Ω-dependent). After (1) and (2), the *only remaining question* is how the past enters: through the full history $\mathcal{C}_{\tau^-}$ or through a compressed state M_{τ^-}. C3 says the agent *has* a complete state, and whatever that state is, it's all the agent has. The Markov form follows.

What the result does NOT say: That M must be a lossy compression (the agent could store full history). That the Markov property is "natural" or "optimal" (it's a consequence of how M is defined). That continuous-coupling systems are event-driven (the event framework is one abstraction; $\dot{M} = g(M, u)$ is the more general one, arrived at by the same three constraints).

Discussion.

Recursion as a consequence of completeness. The recursive form is not an assumption bolted on — it follows from the definition of M_t as complete. The sufficiency of the recursive form is precisely what 2.3 measures: when $S(M_t) = 1$, the recursive update loses nothing.

What this opens. The proof yields the *form*. It immediately invites the follow-up questions that the rest of the theory addresses: What should f preserve? → 2.2 and 2.3. How should f weight new information? → 3.6. When is $\mathcal{M}$ itself inadequate? → 4.5.

Working Notes.

- C3's definitional character is a feature, not a bug — but it must be stated honestly. The result is not "the update must be Markovian" but rather "the Markovian analysis is the *only* consistent one, given the modeling commitment of 2.1." These sound the same but have different epistemic status.
- The continuous-coupling generalization (Attack 2) deserves a proper note somewhere: $\dot{M} = g(M, u)$ is the more general form, with event-driven updates as a special case. The three constraints produce the same argument structure in both cases.
- The information-set formalization (Doob-Dynkin) provides the cleanest technical proof. It should probably be considered the primary proof path, with the elimination argument as the more intuitive exposition.

Appendix I

Sketch: Multi-Timescale Stability

When adaptive processes operate at N nested timescales, composite stability requires each level to be stable given its slower levels, with sufficient timescale separation between adjacent pairs.

Formal Expression.

Formulation (multi-timescale-stability sketch).

The General *N*-Timescale System. The temporal nesting in 4.6 creates a coupled multi-timescale system with N levels. Singular perturbation theory provides tools to analyze such systems. Define a hierarchy of state variables:

Definition (State Hierarchy).

$$x^{(1)},\ x^{(2)},\ \ldots,\ x^{(N)} \tag{I.1}$$

where $x^{(1)}$ is the fastest (e.g., mismatch at the reactive/parametric level) and $x^{(N)}$ is the slowest (e.g., architectural or meta-structural state). The coupled dynamics:

Formulation (N-Timescale Dynamics).

$$\dot{x}^{(k)} = \frac{1}{\epsilon_k}\, G^{(k)}\big(x^{(1)}, \ldots, x^{(N)}\big) + w^{(k)}(t) \tag{I.2}$$

where $\epsilon_1 \ll \epsilon_2 \ll \cdots \ll \epsilon_N$ encode the timescale separation and each $G^{(k)}$ may depend on the states at all levels.

The Two-Timescale Special Case. The simplest nontrivial instance has $N = 2$:

- Fast state $x^{(1)} = \delta$ (mismatch under parametric adaptation)
- Slow state $x^{(2)} = \mathcal{M}$ (model class, changing on a structural timescale)

$$\dot{x}^{(1)} = -F(\mathcal{T}, x^{(1)}; x^{(2)}) + w(t) \quad \text{(fast: parametric adaptation)} \tag{I.3}$$

$$\dot{x}^{(2)} = \epsilon\, G(x^{(1)}, x^{(2)}) \quad \text{(slow: structural adaptation)} \tag{I.4}$$

where $\epsilon \ll 1$ reflects the timescale separation and F depends on $x^{(2)}$ (the correction function is determined by the current model class).

Sketch of Approach (General Case). The standard singular perturbation result (Tikhonov 1952, "Systems of differential equations containing a small parameter multiplying the derivative," *Matematicheskii Sbornik* 31(3):575–586; generalized N-level form per Khalil 2002, *Nonlinear Systems* (3rd ed.), Prentice Hall, Chapter 11) applies layer by layer: if level k is stable for each fixed configuration of the slower levels $k+1, \ldots, N$ (each level has a stable attractor given the levels above it), and each successive slow manifold is itself stable, then the composite N-level system is stable.

4.6's convergence constraint $\nu_{n+1} \ll \nu_n$ is the condition ensuring sufficient timescale separation at each boundary — i.e., $\epsilon_k/\epsilon_{k+1} \ll 1$ for each k. When this separation is violated between any adjacent pair, the faster level's transients contaminate the slower level's dynamics, potentially destabilizing the composite system.

Epistemic Status.

This is a *sketch*, not a complete result. The framework and approach are presented as a guide for future development. The claim that timescale separation ensures composite stability is a standard result in singular perturbation theory; the application to AAT's nested adaptive levels is new but follows the standard pattern.

Making it rigorous requires specifying the dynamics $G^{(k)}$ for levels deeper than parametric adaptation. 4.5 gives the *trigger condition* for structural change but not the *dynamics* of how change at deeper levels proceeds. Specifying these would require theories of how agents search over model classes, modify their own architecture, or restructure their adaptive mechanisms — open problems in RL (architecture search, meta-learning), biology (evolutionary dynamics), and organizational theory (institutional change).

Discussion.

The convergence constraint as stability condition. The sketch suggests that 4.6's convergence constraint is not merely a heuristic but a formal condition for composite stability across arbitrarily many timescales. This connects the empirical observation (don't let deeper-level changes happen too fast) to a stability-theoretic foundation.

Applicability to LLM systems. LLMs involve many parallel adaptive processes — pretraining (slowest), fine-tuning, LoRA-style adaptation, in-context learning, retrieval/RAG updates, tool-use feedback, and within-generation attention dynamics — without clean boundaries between "parametric" and "structural." The N-timescale framework accommodates this naturally: each mechanism operates at its characteristic rate, and the stability analysis requires only that adjacent timescales be sufficiently separated, regardless of how many levels exist or how they are labeled.

Working Notes.

- The key open problem: formalizing $G^{(k)}$ for structural adaptation levels. The two-timescale case (parametric + structural) is the tractable starting point.
- The connection to 8.10 is direct: strategy operates at its own timescale, and strategy persistence requires timescale separation from the faster epistemic updates and the slower objective revisions.

- When timescale separation breaks down between organizational levels, the result is "micromanagement" — the organizational analog of control-theoretic instability from gain mismatch. This observation connects to the hierarchical topology analysis in the multi-agent coupling material.

Appendix J

Sketch: Does the Model-S Containment Dichotomy License Structural-Adaptation Genericity?

The Model-S segments claim Corollary A.1S.1 "sharpens the hand-off into `#result-structural-adaptation-necessity` — the structural-adaptation trigger is *generic, not exceptional*, for a sufficiently long-lived agent." This sketch isolates that hand-off as an open question: as stated it is plausibly a mild overclaim, and the honest move is to *derive the bridge* rather than soften the prose.

The Open Question.

`#deriv-sector-condition` Corollary A.1S.1 establishes (exactly) that under additive stochastic forcing the persistence-region first-exit probability is $P(\tau_R < \infty) = 1$ — a long-lived agent leaves the sector-condition region $\mathcal{B}_R$ with certainty. The Model-S segments and `#deriv-stochastic-non-exit` then assert this "sharpens the hand-off" so that structural adaptation is *generic, not exceptional*.

Discussion.

But `#result-structural-adaptation-necessity` derives necessity from a **different trigger**: model-class inadequacy ($\mathcal{F}(\mathcal{M}) < 1 - \varepsilon$ — no model in the current class can represent reality). It does not reference Model-S, region-exit, or Corollary A.1S.1; its mechanism is fitness-driven, not noise-driven. And the corollary's own explicit instance (the OU benchmark) is **positively recurrent**: the process exits any bounded region a.s. *and returns* a.s. — exit-and-return, not escape.

So there is a gap between what Cor A.1S.1 proves and what the hand-off prose claims:

- **What the corollary licenses (honest narrow claim):** the *pathwise-forever parametric-containment guarantee* is structurally unavailable under stochastic forcing; a long-lived agent generically enters the regime where the parametric persistence guarantee is no longer *proven* (the out-of-region regime `#result-structural-adaptation-necessity` addresses).
- **What it does not license without further argument:** that *model-class* structural adaptation is generically *necessary*. Recurrent region-exit (exit-and-return) is not model-class inadequacy; a transient noise excursion beyond $\mathcal{B}_R$ that the (un-posited-but-possibly-still-inward) correction pulls back is not a structural-adaptation trigger in the `#result-structural-adaptation-necessity` sense.

Proposed Direction (rigor pending).

Per strengthen-before-soften, the preferred resolution is **derive the bridge**, not tighten the prose. Candidate conditions under which certain region-exit *would* imply structural-adaptation necessity, to be attempted:

Discussion.

1. **A2' genuinely fails outside $\mathcal{B}_R$** (not merely "not posited"): if the correction function provably ceases to point inward beyond R for the agent's model class, then a.s. exit composes with non-recovery, and recurrent exit *does* force the structural regime. This connects to the sub-scope β / model-class-capacity reading of R in `#deriv-sector-condition`.
2. **Excursion statistics interact with $\mathcal{F}(\mathcal{M})$:** if recurrent large excursions degrade *effective* model-class fitness (the model fitted within $\mathcal{M}$ becomes systematically wrong during excursions faster than it re-converges between them), the noise-driven and fitness-driven triggers couple, and genericity follows from the coupling rather than from exit alone.
3. **Timescale-separation argument:** if structural adaptation operates so much slower than parametric correction that the *fraction of time* spent out-of-region (positive under recurrence, however small) accumulates an unbounded structural-adaptation debt over an unbounded horizon.

Each is a derivation target; any one that closes turns the hand-off from asserted to proven and is a genuine strengthening (a new conditional result, not a softening). If all provably fail, the honest outcome is the no-go protocol applied to the hand-off claim (per `doc/audit-routing-instructions.md` §4) and the Model-S Discussion/Findings prose tightened to the narrow licensed claim above — but only *after* the strengthening attempt is exhausted and recorded.

Epistemic Status.

Sketch. Direction identified, formalization not attempted. This segment makes **no** claim that the bridge holds or fails; it records that the hand-off currently asserted in `#deriv-sector-condition` / `#deriv-stochastic-non-exit` Discussion/Findings is **plausibly a mild overclaim at the Discussion/Findings tier** (a Gate-2 class concern: a claim that *sounds* like it follows from Cor A.1S.1 but is not derived from it), and that the resolution path is to attempt the bridge derivation. The narrow licensed claim (pathwise-parametric-guarantee unavailability ⇒ generic entry into the out-of-region *regime*) is sound and uncontroversial; the disputed increment is "⇒ model-class structural adaptation is generically *necessary*." Ceiling: a closed bridge would be `derived (conditional)` on the named condition; a clean impossibility would be a no-go result. Until attempted, this stays `sketch`.

Discussion.

Why this matters enough to surface rather than leave as a soft ledger row: the genericity claim is load-bearing downstream — it is the form in which the Model-S no-go's consequence reaches the ELI / long-lived-agent persistence framing ("in any genuinely stochastic environment, structural adaptation is not an edge case"). If the bridge holds under a named condition, that is a sharp, citable strengthening of exactly the kind the framework's no-go signature tends to produce. If it does not, the framework's honesty requires the hand-off prose to state only the narrow licensed claim. Either way the question should be visible in the argument's structure, not invisible in a tracking file — which is the point of carrying it as an `exploratory` OUTLINE node.

Working Notes.

- Provenance: parent-generated 2026-05-16 during the audit-routing cycle, from a first-hand deep-read of `#result-structural-adaptation-necessity` against the Model-S segments (not a cluster audit finding). Originally filed as consolidated-ledger research-seed **S30**; Joseph adjudicated it up to a first-class `exploratory` OUTLINE node ("it will get its day in the light") rather than a soft ledger row. The ledger row is correspondingly demoted to a pointer here.

- Segment-vs-spike judgment (recorded so it is not re-litigated): this is recognized open *territory* between two landed results with **no attempt yet** — segment-shaped (visible structural node), not spike-shaped (a spike is the reasoning trail of an *attempt*). The spike is the eventual *work*: when this is picked up, a strengthening spike attempts directions 1–3; math lands back here (or in a new derivation/result segment), per `math-lives-in-segments`.
- Related: the `#disc-identifiability-floor` family — if the bridge is a no-go, the "no horizon-independent non-exit certificate" structural absence may itself compose with a structural-adaptation-necessity floor; worth checking whether this is an identifiability-floor-shaped instance.

Appendix K

Derivation: Discrete-Time Sector Condition

Discrete-time analogs of Props A.1, A.1S, and A.2 via contraction mapping, closing the fluid-limit gap (GA-5) between the event-driven dynamics (3.1) and the continuous-time Lyapunov results in A.

Formal Expression.

Setup. The discrete mismatch dynamics at event step k are:

Definition (Discrete Dynamics).

$$\delta_{k+1} = \delta_k - \eta^* F_d(\delta_k) + w_k \tag{K.1}$$

where η^* is the update gain (3.6), F_d is the discrete correction direction, and w_k is the per-step disturbance. The continuous correction function $F(\mathcal{T}, \delta)$ from A decomposes as $F = \nu \cdot \eta^* \cdot F_d$ at event rate ν.

(DA2') Discrete Sector-Lipschitz Condition.

There exist constants $c_{\min} > 0$ and $c_{\max} < 2/\eta^*$ such that for all $\|\delta\| \leq R$:

Assumption DA2' (discrete-sector-condition).

(DA2rsquoa) Lower sector bound (directional fidelity):

$$\delta^T F_d(\delta) \geq c_{\min} \|\delta\|^2 \tag{K.2}$$

(DA2rsquob) Lipschitz bound (bounded correction magnitude):

$$\|F_d(\delta)\| \leq c_{\max}\|\delta\| \tag{K.3}$$

The **lower bound** (DA2'a) is directional fidelity — the correction points inward, identical to the continuous sector condition (A2') from A via 4.2.

The **Lipschitz bound** (DA2'b) controls the *magnitude* of the correction, not merely its projection onto the mismatch direction. The combined constraint $c_{\max} < 2/\eta^*$ is the **no-overshoot condition**: each correction step must not reverse the mismatch. This is the classical step-size condition $\eta^* < 2/L$ for gradient descent (where L is the Lipschitz constant of the gradient). For Bayesian updates, this is satisfied by construction — the posterior lies between prior and data.

Why DA2'b is stronger than an inner-product upper bound. A two-sided inner-product condition $\delta^T F_d(\delta) \leq c_{\max}\|\delta\|^2$ constrains only the projection of F_d onto δ. By Cauchy-Schwarz, the Lipschitz bound (DA2'b) implies the inner-product upper bound: $\delta^T F_d(\delta) \leq \|\delta\| \cdot \|F_d(\delta)\| \leq c_{\max}\|\delta\|^2$. But the converse fails — a correction function with a large transverse component (orthogonal to δ) can satisfy the inner-product bound while violating the norm bound. The proofs below (especially DA.1S) require the norm bound $\|F_d(\delta)\|^2 \leq c_{\max}^2\|\delta\|^2$, which follows from DA2'b but not from an inner-product condition alone.

Scalar case. In one dimension, DA2'a and DA2'b together reduce to the classical sector condition $c_{\min} \leq F_d(\delta)/\delta \leq c_{\max}$, since norm and inner product coincide. No generality is lost for scalar systems.

Relationship to the continuous-time condition. The continuous sector condition (A2'/GA-3) is a one-sided inner-product bound $\delta^T F \geq \alpha\|\delta\|^2$ — this suffices for continuous-time Lyapunov analysis because $\dot{V}$ involves only $\delta^T F$, not $\|F\|$. Discretization introduces the quadratic term $(\eta^*)^2\|F_d\|^2$ (see DA.1S proof), which requires the Lipschitz bound. This is the standard sector-vs-Lipschitz distinction in nonlinear systems theory.

Contraction factor. Under DA2', the per-step Lyapunov function $V_k = \frac{1}{2}\|\delta_k\|^2$ satisfies (in the zero-disturbance case $w_k = 0$):

Derived (contraction, from DA2').

$$\|\delta_{k+1}\|^2 = \|\delta_k - \eta^* F_d(\delta_k)\|^2 = \|\delta_k\|^2 - 2\eta^*\delta_k^T F_d(\delta_k) + (\eta^*)^2\|F_d(\delta_k)\|^2 \tag{K.4}$$

Applying DA2'a (lower sector bound on $\delta^T F_d$) and DA2'b (Lipschitz bound on $\|F_d\|$):

$$\|\delta_{k+1}\|^2 \leq (1 - 2\eta^* c_{\min} + (\eta^*)^2 c_{\max}^2)\|\delta_k\|^2 = \lambda_{\text{eff}}^2\|\delta_k\|^2 \tag{K.5}$$

where:

$$\lambda_{\text{eff}}^2 = 1 - 2\eta^* c_{\min} + (\eta^*)^2 c_{\max}^2 \tag{K.6}$$

Stability condition. $\lambda_{\text{eff}}^2 < 1$ requires $2\eta^* c_{\min} > (\eta^*)^2 c_{\max}^2$, i.e.:

$$\eta^* < \frac{2c_{\min}}{c_{\max}^2} \tag{K.7}$$

This is automatically satisfied when $c_{\min} \approx c_{\max}$ (well-conditioned correction), recovering the standard step-size condition $\eta^* < 2/c_{\max}$. For ill-conditioned corrections ($c_{\min} \ll c_{\max}$), the constraint is tighter. For Bayesian updates with bounded condition number, both conditions are satisfied.

Scalar (colinear) specialization. When $F_d(\delta) \parallel \delta$ (scalar system or colinear correction), $\|F_d(\delta)\| = |F_d(\delta)/\delta| \cdot \|\delta\|$ and the contraction factor simplifies to $\lambda = \max(|1 - \eta^* c_{\min}|, |1 - \eta^* c_{\max}|)$, the classical form. The general vector formula λ_{eff}^2 reduces to λ^2 in this case.

With disturbance $w_k \neq 0$:

$$\|\delta_{k+1}\|^2 \leq \lambda_{\text{eff}}^2 \|\delta_k\|^2 + 2\|\delta_k\|\|w_k\| + \|w_k\|^2 \tag{K.8}$$

Proposition DA.1: Bounded Mismatch (Deterministic). **Statement.** Under DA2' with $\eta^* < 2c_{\min}/c_{\max}^2$ and bounded per-step disturbance $\|w_k\| \leq \rho_{\text{step}}$, the mismatch is ultimately bounded:

Derived (DA.1, discrete bounded mismatch).

$$R_D^* = \frac{\rho_{\text{step}}}{1 - \lambda_{\text{eff}}} \tag{K.9}$$

Proof. By the triangle inequality: $\|\delta_{k+1}\| = \|(\delta_k - \eta^* F_d(\delta_k)) + w_k\| \leq \|\delta_k - \eta^* F_d(\delta_k)\| + \|w_k\|$. The contraction bound gives $\|\delta_k - \eta^* F_d(\delta_k)\| \leq \lambda_{\text{eff}} \|\delta_k\|$. Therefore:

$$\|\delta_{k+1}\| \leq \lambda_{\text{eff}} \|\delta_k\| + \rho_{\text{step}} \tag{K.10}$$

This is an affine contraction with $\lambda_{\text{eff}} < 1$. By the Banach fixed-point theorem, all trajectories starting in $\mathcal{B}_R$ converge to the ball of radius $R_D^* = \rho_{\text{step}}/(1 - \lambda_{\text{eff}})$, provided $R_D^* < R$. $\square$

Recovery of continuous result. In the fluid limit ($\eta^* \to 0, \nu \to \infty, \nu\eta^* = \mathcal{T}$ fixed), $\lambda_{\text{eff}}^2 = 1 - 2\eta^* c_{\min} + O((\eta^*)^2)$, so $\lambda_{\text{eff}} \to 1 - \eta^* c_{\min}$ and $\rho_{\text{step}} \to \rho/\nu$. Then:

$$R_D^* = \frac{\rho/\nu}{\eta^* c_{\min}} = \frac{\rho}{\nu\eta^* c_{\min}} = \frac{\rho}{\alpha} \tag{K.11}$$

recovering Prop A.1 exactly. The discrete-to-continuous gap for Model D steady state is **zero**.

Proposition DA.2: Adaptive Reserve (Discrete). **Statement.** Under the conditions of DA.1, the agent can absorb an additional per-step disturbance of:

Derived (DA.2, discrete adaptive reserve).

$$\Delta\rho_{\text{step}}^* = (1 - \lambda_{\text{eff}})R - \rho_{\text{step}} \tag{K.12}$$

Proof. Identical to Prop A.2: the new $R_D^* = (\rho_{\text{step}} + \Delta\rho)/(1 - \lambda_{\text{eff}})$ must satisfy $R_D^* \leq R$. $\square$

The structure is identical to the continuous adaptive reserve $\Delta\rho^* = \alpha R - \rho$. The per-event contraction rate is $(1 - \lambda_{\text{eff}})$; the per-unit-time rate is $\nu(1 - \lambda_{\text{eff}})$. In the fluid limit, $\nu(1 - \lambda_{\text{eff}}) \to \nu \cdot \eta^* c_{\min} = \alpha$, recovering the continuous result. (Note: the per-event rate $(1 - \lambda_{\text{eff}})$ converges to $\eta^* c_{\min}$, not to α directly — the factor of ν converts between per-event and per-unit-time.)

Proposition DA.1S: Stochastic Bounded Mismatch (Discrete). Statement. Under DA2' with $\eta^* < 2c_{\min}/c_{\max}^2$ and i.i.d. zero-mean disturbance $\mathbb{E}[w_k] = 0$, $\mathbb{E}[\|w_k\|^2] = \sigma_{\text{step}}^2$, the mismatch satisfies:

Derived (DA.1S, discrete stochastic bounded mismatch).

$$\mathbb{E}[\|\delta_k\|^2] \le \lambda_{\text{eff}}^{2k}\|\delta_0\|^2 + \frac{\sigma_{\text{step}}^2}{1-\lambda_{\text{eff}}^2} \tag{K.13}$$

where $\lambda_{\text{eff}}^2 = 1 - 2\eta^* c_{\min} + (\eta^*)^2 c_{\max}^2$.

Proof. Define $V_k = \|\delta_k\|^2$. Taking expectations of the squared update:

$$\mathbb{E}[V_{k+1} \mid \delta_k] = \|\delta_k - \eta^* F_d(\delta_k)\|^2 + \sigma_{\text{step}}^2 \tag{K.14}$$

The first term expands as:

$$\|\delta_k - \eta^* F_d(\delta_k)\|^2 = V_k - 2\eta^* \delta_k^T F_d(\delta_k) + (\eta^*)^2 \|F_d(\delta_k)\|^2 \tag{K.15}$$

By DA2'a (lower sector bound): $\delta_k^T F_d(\delta_k) \ge c_{\min} V_k$.

By DA2'b (Lipschitz bound): $\|F_d(\delta_k)\|^2 \le c_{\max}^2 V_k$.

Note that the second step requires the *norm* bound DA2'b, not merely an inner-product upper bound — this is where the Lipschitz condition is essential. Combining:

$$\mathbb{E}[V_{k+1} \mid \delta_k] \le \lambda_{\text{eff}}^2 V_k + \sigma_{\text{step}}^2 \tag{K.16}$$

The condition $\eta^* < 2c_{\min}/c_{\max}^2$ ensures $\lambda_{\text{eff}}^2 < 1$. This is a supermartingale (when V_k is large enough). Iterating:

$$\mathbb{E}[V_k] \le \lambda_{\text{eff}}^{2k} V_0 + \frac{\sigma_{\text{step}}^2}{1-\lambda_{\text{eff}}^2} \tag{K.17}$$

The steady-state mean-square mismatch is $\sigma_{\text{step}}^2/(1-\lambda_{\text{eff}}^2)$. □

Recovery of continuous result. In the fluid limit: $\sigma_{\text{step}}^2 \to n\sigma_w^2/\nu$, $\lambda_{\text{eff}}^2 \to 1 - 2\eta^* c_{\min}$, and $(1-\lambda_{\text{eff}}^2) \to 2\eta^* c_{\min}$. The steady-state becomes $n\sigma_w^2/(2\nu\eta^* c_{\min}) = n\sigma_w^2/(2\alpha)$, recovering Prop A.1S exactly.

The discrete-to-continuous gap for Model S variance is $O(\eta^* c_{\max}^2/c_{\min}) = O(c_{\max}^2/(c_{\min}^2\,\nu))$, dominated by the conditioning ratio $c_{\max}^2/c_{\min}^2$. Substituting $\sigma_{\text{step}}^2 = n\sigma_w^2/\nu$ and $\eta^* = \mathcal{T}/\nu$ into $V_{ss} = \sigma_{\text{step}}^2/(1-\lambda_{\text{eff}}^2)$ and Taylor-expanding at small η^*:

$$\frac{V_{ss}}{V_c} = \frac{1}{1-\eta^* c_{\max}^2/(2c_{\min})} = 1 + \frac{\eta^* c_{\max}^2}{2c_{\min}} + O((\eta^*)^2) \tag{K.18}$$

so

$$V_{ss} - V_c \approx \frac{n\sigma_w^2\, c_{\max}^2}{4c_{\min}^2\,\nu} \tag{K.19}$$

— a leading correction that scales as $1/\nu$. The $(\eta^*)^2\|F_d\|^2$ term in the per-step recurrence enters $1-\lambda_{\text{eff}}^2$ at order $(\eta^*)^2$, but the steady-state ratio $\sigma_{\text{step}}^2/(1-\lambda_{\text{eff}}^2)$ inverts the leading $2\eta^* c_{\min}$ contraction, yielding $O(\eta^*)$ asymptotic gap rather than $O((\eta^*)^2)$. This quantifies the error introduced by GA-5 and confirms it is small whenever $c_{\max}^2/(c_{\min}^2\nu) \ll 1$.

Fluid Limit Theorem.

Statement. Let F_d be Lipschitz continuous with constant L_F on $\mathcal{B}_R$. Let $\delta^{(\nu)}(t)$ denote the piecewise-constant interpolation of the discrete trajectory at event rate ν, and $\delta(t)$ the solution of the continuous ODE $d\delta/dt = -F(\mathcal{T},\delta) + w(t)$ with $F = \nu\eta^* F_d$. Then:

Derived (Conditional on Lipschitz regularity).

$$\sup_{t\in[0,T]} \|\delta^{(\nu)}(t) - \delta(t)\| \le C \cdot \frac{\eta^* c_{\max}}{\nu^{1/2}} \tag{K.20}$$

for a constant C depending on T, L_F, and R.

Sketch. This follows from the standard ODE-approximation theory for Euler schemes (Kushner & Yin, 2003, Ch. 5). The discrete update $\delta_{k+1} = \delta_k - \eta^* F_d(\delta_k) + w_k$ is a forward Euler step for the ODE with step size $1/\nu$. The Lipschitz condition on F_d ensures the local truncation error is $O(1/\nu^2)$. Summing over $O(\nu T)$ steps and applying the Gronwall inequality gives the $O(1/\nu^{1/2})$ bound.

For Model D (deterministic): the steady-state gap is exactly zero (both discrete and continuous converge to the same fixed point). The fluid-limit error affects only transients.

For Model S (stochastic): the steady-state variance gap is $O(\eta^* c_{\max}^2/c_{\min}^2) = O(c_{\max}^2/(c_{\min}^2\,\nu))$, dominated by the conditioning ratio $c_{\max}^2/c_{\min}^2$. The $(\eta^*)^2\|F_d\|^2$ term in the per-step recurrence enters $1-\lambda_{\text{eff}}^2$ at order $(\eta^*)^2$, but inverting the leading $2\eta^* c_{\min}$ in $V_{ss} = \sigma_{\text{step}}^2/(1-\lambda_{\text{eff}}^2)$ produces an $O(\eta^*)$ asymptotic gap, not $O((\eta^*)^2)$ — see the recovery calculation in Prop DA.1S above.

Epistemic Status.

DA.1, DA.1S, and DA.2 are *conditional* on DA2', which now includes both a sector lower bound (DA2'a) and a Lipschitz upper bound (DA2'b). The Lipschitz condition is strictly stronger than an inner-product upper bound — this is a real assumption, not a technicality. For Bayesian updates and gradient descent on smooth losses, DA2'b is satisfied (the correction magnitude is controlled by the loss curvature). For agents with non-smooth or pathological correction functions, DA2'b must be verified independently.

The proofs themselves are standard contraction-mapping (DA.1/DA.2) and supermartingale (DA.1S) arguments. The step-size constraint $\eta^* < 2c_{\min}/c_{\max}^2$ is essential and is stated explicitly in each proposition.

Fluid limit theorem is *conditional* on Lipschitz regularity of F_d — a standard regularity condition satisfied by all correction functions in the verified instances table (4.2). The convergence rate follows from classical ODE approximation theory (Kushner & Yin, 2003); the application to the AAT mismatch dynamics is new but the mathematics is not.

Max attainable: conditional — the Lipschitz bound DA2'b is inherent to the discrete-time treatment and cannot be replaced by a weaker inner-product condition without losing the norm bound used in the proofs.

Discussion.

GA-5 is closed. The fluid-limit bridging assumption is no longer required as an ungrounded assumption. For Model D, the discrete and continuous steady states are identical — the gap is zero. For Model S, the gap is $O(\eta^* c_{\max}^2/c_{\min}^2) = O(c_{\max}^2/(c_{\min}^2\,\nu))$ in variance, quantitatively bounded and small in the regime where $c_{\max}^2/(c_{\min}^2\,\nu) \ll 1$. The continuous-time results in A are formally justified as the fluid limit of the discrete results here.

DA2' is a sector-Lipschitz condition, not a pure sector condition. The continuous-time treatment (A) requires only the one-sided sector bound $\delta^T F \geq \alpha\|\delta\|^2$ because the Lyapunov derivative $\dot{V}$ depends only on $\delta^T F$. Discretization introduces a quadratic term $(\eta^*)^2\|F_d\|^2$ that requires the Lipschitz bound $\|F_d\| \leq c_{\max}\|\delta\|$. This is the standard distinction between sector conditions and Lipschitz conditions in nonlinear systems theory. For all agents in the verified instances table (4.2), the Lipschitz bound holds alongside the sector bound, so DA2' imposes no additional restriction in practice.

The step-size constraint has two forms. For well-conditioned corrections ($c_{\min} \approx c_{\max}$), the constraint reduces to $\eta^* < 2/c_{\max}$ — the classical form. For ill-conditioned corrections ($c_{\min} \ll c_{\max}$), the tighter constraint $\eta^* < 2c_{\min}/c_{\max}^2$ applies. This distinction is invisible in scalar systems but matters for high-dimensional agents with anisotropic correction.

No downstream results change qualitatively. The persistence condition, adaptive reserve, and adversarial scaling laws derived in A and 13.2 hold as stated. The discrete framework provides sharper constants (replacing α with $\nu(1-\lambda_{\text{eff}})$) and a quantitative transient error bound, but the qualitative structure is unchanged.

Section I's formal chain is now complete. The prediction chain:

$$\text{gain principle} + \text{B1} \xrightarrow{\text{derived}} \text{sector condition} \xrightarrow{\text{Lyapunov/contraction}} \text{persistence, reserve, scaling} \tag{K.21}$$

holds in both continuous time (via A) and discrete time (via this segment), with a quantitative bridge between them (the fluid limit theorem).

Working Notes.

- Non-stationary gain: when η^* varies over time (as in Kalman filtering with growing P^-), the contraction factor λ changes per step. The ultimate bound still holds if $\sup_k|\lambda_k| < 1$, but the transient analysis requires time-varying contraction arguments.
- Heavy-tailed disturbances: DA.1S assumes finite second moment. Sub-Weibull or heavy-tailed w_k would need truncation arguments or alternative Lyapunov functions. These extreme cases are better modeled as triggers for structural adaptation (4.5).
- Non-i.i.d. disturbances: correlated w_k (e.g., Markov environment noise) weakens the supermartingale argument. The contraction still holds per-step, but the steady-state bound requires mixing conditions. This connects to the adversarial coupling analysis in 13.2.

Appendix L

Detail: Linear ODE Approximation

The full linear treatment of mismatch dynamics: scalar and vector forms, steady-state solutions under both disturbance models, transient behavior, convergence rate, validity conditions, breakdown modes, discrete-time connection, and adversarial coupling. This appendix collects the linear-case results that 3.9 states as a hypothesis, derives them explicitly, and delineates exactly where the linear approximation is valid and where it fails.

Formal Expression.

1. The scalar and vector forms. The mismatch vector $\delta(t) \in \mathbb{R}^n$ evolves under the general dynamics from A:

Definition (vector dynamics).

$$\frac{d\delta}{dt} = -F(\mathcal{T}, \delta) + w(t) \tag{L.1}$$

The **linear approximation** sets $F(\mathcal{T}, \delta) = \mathcal{T} \cdot \delta$, giving:

Hypothesis (linear-vector-ode).

$$\frac{d\delta}{dt} = -\mathcal{T} \cdot \delta + w(t) \tag{L.2}$$

This is exact when the correction function is linear in δ (Kalman filter, Beta-Bernoulli conjugate update). It is an approximation otherwise.

Scalar form (equality). For a scalar mismatch $\delta(t) \in \mathbb{R}$:

Derived (scalar-ode, from linear-vector-ode with $n = 1$).

$$\frac{d\delta}{dt} = -\mathcal{T} \cdot \delta + w(t) \tag{L.3}$$

This is an equality: the scalar ODE is exactly the $n = 1$ case of the vector ODE.

Norm form (inequality). For vector $\delta \in \mathbb{R}^n$, take the time derivative of $\|\delta\| = (\delta^T \delta)^{1/2}$:

$$\frac{d\|\delta\|}{dt} = \frac{\delta^T\dot{\delta}}{\|\delta\|} = \frac{\delta^T(-\mathcal{T}\delta + w)}{\|\delta\|} = -\mathcal{T}\|\delta\| + \frac{\delta^T w}{\|\delta\|} \tag{L.4}$$

By Cauchy-Schwarz, $\delta^T w/\|\delta\| \le \|w\|$. Under Model D (GA-2: $\|w(t)\| \le \rho$):

Derived (norm-inequality, from Cauchy-Schwarz).

$$\frac{d\|\delta\|}{dt} \le -\mathcal{T} \cdot \|\delta\| + \rho \tag{L.5}$$

This is an inequality, not an equality. The Cauchy-Schwarz step is tight only when $w(t)$ is parallel to $\delta(t)$ (worst-case disturbance). The norm form stated in 3.9 is this upper bound. For the scalar case ($n = 1$), Cauchy-Schwarz is automatically tight and the inequality becomes an equality.

2. Steady-state solutions.

Model D (deterministic bounded disturbance, GA-2: $\|w(t)\| \le \rho$). Setting $d\|\delta\|/dt = 0$ in the norm inequality gives the worst-case steady state:

Derived (model-d-steady-state, from norm inequality).

$$\|\delta\|_{ss} = \frac{\rho}{\mathcal{T}} \tag{L.6}$$

This is the tight upper bound: steady-state mismatch equals the ratio of disturbance to correction. In the scalar case ($n = 1$), this is exact (the Ornstein-Uhlenbeck process with deterministic forcing converges to $\rho/\mathcal{T}$). In the vector case, it is the worst case over all disturbance directions.

Model S (stochastic zero-mean disturbance, GA-2S: $d\delta = -\mathcal{T}\delta\, dt + \sigma_w\, dW_t$). The Ornstein-Uhlenbeck process has stationary variance derived via Ito-Lyapunov analysis (Prop A.1S in A, specialized to $\alpha = \mathcal{T}$):

Derived (model-s-steady-state, from Ito-Lyapunov with linear correction).

$$\mathbb{E}[\|\delta\|^2]_{ss} = \frac{n\sigma_w^2}{2\mathcal{T}} \tag{L.7}$$

The RMS steady-state mismatch is:

$$\|\delta\|_{\mathrm{rms}} = \sigma_w\sqrt{\frac{n}{2\mathcal{T}}} \tag{L.8}$$

For the scalar case ($n = 1$): $\|\delta\|_{\mathrm{rms}} = \sigma_w/\sqrt{2\mathcal{T}}$.

Scaling difference. Model D gives $\|\delta\|_{ss} \propto 1/\mathcal{T}$; Model S gives $\|\delta\|_{\mathrm{rms}} \propto 1/\sqrt{\mathcal{T}}$. Doubling the correction rate halves the deterministic steady-state mismatch but only reduces the stochastic steady-state by a factor of $\sqrt{2} \approx 1.41$. Correction is less effective against noise than against drift.

3. Transient solution. **Model D, constant ρ.** The linear ODE $d\|\delta\|/dt = -\mathcal{T}\|\delta\| + \rho$ with initial condition $\|\delta(0)\| = \|\delta_0\|$ has the standard first-order linear solution:

Derived (transient-model-d).

$$\|\delta(t)\| = \|\delta_0\|\, e^{-\mathcal{T}t} + \frac{\rho}{\mathcal{T}}(1 - e^{-\mathcal{T}t}) \tag{L.9}$$

Mismatch decays exponentially from initial conditions toward the steady state $\rho/\mathcal{T}$ with time constant $\tau = 1/\mathcal{T}$. After k time constants, the transient has decayed by a factor of e^{-k}: 63% after one time constant, 95% after three, 99.3% after five.

Model S, Ornstein-Uhlenbeck. The mean-square mismatch evolves as (from Prop A.1S with $\alpha = \mathcal{T}$):

Derived (transient-model-s).

$$\mathbb{E}[\|\delta(t)\|^2] = \|\delta_0\|^2\, e^{-2\mathcal{T}t} + \frac{n\sigma_w^2}{2\mathcal{T}}(1 - e^{-2\mathcal{T}t}) \tag{L.10}$$

The variance converges twice as fast as the mean (rate $2\mathcal{T}$ vs. $\mathcal{T}$) because the Lyapunov function $V = \frac{1}{2}\|\delta\|^2$ is quadratic — its dynamics have doubled exponential rate.

4. When the linear approximation is valid. The linear case $F(\mathcal{T}, \delta) = \mathcal{T} \cdot \delta$ is a special case of the sector condition (GA-3) from 4.3 with:

Derived (linear-sector-parameters).

$$\alpha = \mathcal{T}, \qquad R \to \infty \tag{L.11}$$

That is, the sector condition holds globally with lower bound α equal to the tempo, and the sector-condition region has infinite radius. In the notation of 4.2:

- **Directional fidelity parameter:** $c_{\min} = 1$ (the correction points exactly at the mismatch).
- **Sector parameter:** $\alpha = \eta^* \cdot c_{\min} = \eta^* = \mathcal{T}/\nu$.
- **No upper saturation:** $c_{\max} = c_{\min} = 1$ (the sector bounds coincide).

Consequence for persistence. With $R \to \infty$, structural persistence ($\alpha > \rho/R$) is trivially satisfied for any $\mathcal{T} > 0$. The binding constraint is task adequacy alone: $\mathcal{T} > \rho/\|\delta_{\text{critical}}\|$ (Model D) or $\mathcal{T} > n\sigma_w^2/(2\|\delta_{\text{critical}}\|^2)$ (Model S). This is why the linear operational forms in 4.4 exist: in the linear world, structural persistence is free.

The linear approximation is exact when:

1. **Kalman filter** (scalar or matrix): the gain K_t produces $F(\delta) = K_t H\delta$, which is linear. The sector parameter equals the Kalman gain: $\alpha = K$ (scalar) or $\alpha = \lambda^+_{\min}(KH)$ (matrix). See 4.2.
2. **Beta-Bernoulli conjugate update:** the correction $F(\delta) = \delta/(n+1)$ is linear with $\alpha = 1/(n+1) = \eta_{\text{edge}}$.
3. **Exponential family with natural parameters:** the natural-parameter update is linear in the sufficient statistic, giving $\alpha = \eta \cdot \lambda_{\min}(\text{Fisher})$.
4. **Gradient descent on quadratic loss:** $F(\delta) = \eta\nabla^2 L \cdot \delta$ is linear with $\alpha = \eta \cdot \lambda_{\min}(\nabla^2 L)$.

5. When the linear approximation breaks down. The linear form $F = \mathcal{T} \cdot \delta$ fails when the true correction function deviates from linearity. Three failure modes, each corresponding to a specific nonlinearity:

Saturation at large $\|\delta\|$. The correction mechanism is overwhelmed by large mismatch. The true correction satisfies $F(\delta) < \mathcal{T}\|\delta\|$ for large $\|\delta\|$ — correction is slower than the linear prediction. Simulation confirms: for a saturating function $g(\delta) = \delta/(1 + |\delta|/R)$, the sector parameter at the capacity boundary is $\alpha \approx \mathcal{T}/2$ (half the linear value). The linear approximation overstates persistence margins. The sector-condition framework (4.3) handles this via finite R and reduced α.

Threshold effects at small $\|\delta\|$. Below a detection threshold ε, small mismatches go uncorrected: $F(\delta) \approx 0$ for $\|\delta\| < \varepsilon$. This creates a dead zone where the model drifts. The linear approximation (which predicts correction at all scales) misses this. The sector condition fails locally at small $\|\delta\|$ (the ratio $\delta^T F/\|\delta\|^2 \to 0$), but the dead zone is bounded and does not affect large-scale stability.

Structural breakdown. Beyond some critical mismatch, the correction rate drops to zero because the model class is no longer appropriate: the mismatch exceeds the model's representational capacity. $F(\delta) \approx 0$ for $\|\delta\| > R$. This is the structural adaptation trigger (4.5). The linear approximation, which predicts correction growing without bound, misses this entirely. The sector-condition framework captures it via the finite radius R of the sector-condition region.

6. Discrete-time connection. The continuous ODE is a fluid-limit approximation of the discrete event-driven dynamics (3.1). The discrete mismatch dynamics are:

$$\delta_{k+1} = \delta_k - \eta^* F_d(\delta_k) + w_k \tag{L.12}$$

In the linear case ($F_d(\delta) = \delta$), this becomes:

$$\delta_{k+1} = (1 - \eta^*)\delta_k + w_k \tag{L.13}$$

which is an AR(1) process with coefficient $\lambda = 1 - \eta^*$.

K proves the formal connection:

Model D. The discrete and continuous steady states are *identical* — the fluid-limit gap is exactly zero. Both give $R^* = \rho/\alpha$. The discrete form has $\alpha = (1-|\lambda|)/\eta^* = (1-|1-\eta^*|)/\eta^*$, which equals 1 (hence $\alpha_{\text{continuous}} = \nu \cdot 1 = \mathcal{T}$) when $0 < \eta^* < 1$.

Model S. The discrete steady-state variance is $\sigma^2_{\text{step}}/(1 - \lambda^2_{\text{eff}})$, which recovers the continuous result $n\sigma^2_w/(2\mathcal{T})$ in the fluid limit ($\eta^* \to 0, \nu \to \infty, \nu\eta^* = \mathcal{T}$ fixed). The variance gap is $O(\eta^* c^2_{\max}/c^2_{\min}) = O(c^2_{\max}/(c^2_{\min}\,\nu))$ — quantitatively small whenever $c^2_{\max}/(c^2_{\min}\,\nu) \ll 1$. See K for the leading-order Taylor expansion that gives this scaling.

Fluid-limit validity. The ODE approximation is formally justified by the fluid limit theorem in K (conditional on Lipschitz regularity of F_d). The approximation is most accurate when $\eta^* \ll 1$ (the small-gain regime). It is least accurate during initial transients when η^* is large, but this phase is short-lived and the transient error is bounded by $O(\eta^* c_{\max}/\nu^{1/2})$.

7. Adversarial coupling in the linear case. When two agents are coupled, A's praxis contributes to B's disturbance:

Hypothesis (Linear Coupling Model).

$$\rho_B = \rho_{B,\text{base}} + \gamma_A \cdot \mathcal{T}_A \tag{L.14}$$

Under the linear approximation, the steady-state mismatches (Model D, coupling-dominant: $\gamma_A \mathcal{T}_A \gg \rho_{B,\text{base}}$) are:

Derived (linear-adversarial-steady-state, from Model D steady state + coupling model).

$$\|\delta_B\|_{ss} \approx \frac{\gamma_A \mathcal{T}_A}{\mathcal{T}_B}, \qquad \|\delta_A\|_{ss} \approx \frac{\gamma_B \mathcal{T}_B}{\mathcal{T}_A} \tag{L.15}$$

The ratio:

Derived (squared-tempo-law, from linear steady states).

$$\frac{\|\delta_B\|_{ss}}{\|\delta_A\|_{ss}} = \frac{\gamma_A}{\gamma_B} \left(\frac{\mathcal{T}_A}{\mathcal{T}_B}\right)^2 \tag{L.16}$$

Under symmetric coupling ($\gamma_A \approx \gamma_B$), tempo advantage squares: a 2:1 tempo ratio yields a 4:1 mismatch ratio; a 3:1 ratio yields 9:1.

Why the squared law. The exponent 2 arises because the $1/\mathcal{T}$ steady-state scaling appears in both numerator (faster agent generates more disturbance) and denominator (faster agent corrects better): one power from the coupling, one from the steady-state. Under Model S (stochastic coupling, $1/\sqrt{\mathcal{T}}$ scaling), the exponent reduces to 3/2. See 14.3 for the full regime analysis.

Simulation validation. Variant A confirmed $b = 1.999$ (Model D, coupling-dominant). Variants C-D confirmed $b = 1.481$ (Model S, coupling-dominant). Both within numerical precision of the derived values. See .

Epistemic Status.

Exact for the linear case. The linear ODE is a standard first-order linear system; the solutions (transient and steady-state, both Model D and Model S) are textbook results for the Ornstein-Uhlenbeck process and its deterministic analog. The sector-condition parameters ($\alpha = \mathcal{T}, R \to \infty$) are exact when the correction function is linear. The discrete-time connection is exact (zero gap for Model D, $O(\eta^*)$ gap for Model S — see K for the leading-order Taylor expansion). The adversarial scaling law ($b = 2$ for Model D, $b = 3/2$ for Model S) is derived and simulation-validated.

What is exact vs. what is an approximation. The linear ODE is exact for systems with linear correction dynamics (Kalman, Beta-Bernoulli, quadratic loss). For nonlinear systems, the linear ODE is a local Taylor approximation valid near the steady state. The sector-condition framework (4.3, A) is the general treatment; this appendix derives what falls out when the sector bounds coincide.

The norm form is an upper bound for $n > 1$. The inequality introduced by Cauchy-Schwarz at the norm step is tight only in the worst case. For typical disturbances that are not persistently aligned with the mismatch direction, the actual norm trajectory lies below the upper bound. The scalar case ($n = 1$) is an equality. This distinction is load-bearing: downstream results that use $d\|\delta\|/dt = -\mathcal{T}\|\delta\| + \rho$ as an equality (e.g., the exact transient solution, the exact steady-state ratio) are strictly correct only for $n = 1$ or for worst-case disturbance. The Lyapunov analysis in A avoids this issue by working with $V = \frac{1}{2}\|\delta\|^2$ directly, without passing through the norm derivative.

Max attainable: exact (the linear case is its own exact analysis; there is nothing to tighten).

Discussion.

Why this appendix exists. The linear ODE in 3.9 is labeled as a hypothesis and described as a "first-order approximation." That is correct for the general case. But for agents with linear correction dynamics, the ODE is not an approximation at all – it is exact. This appendix serves two roles: (1) providing the complete derivation that 3.9 points to rather than carrying inline, and (2) making precise when the "approximation" is exact and when it genuinely introduces error.

Pedagogical value. The linear case is the entry point for understanding AAT's mismatch dynamics. The closed-form solutions (exponential convergence, ratio steady state, squared scaling law) build intuition that the general sector-condition results (4.3) cannot provide – Lyapunov gives bounds and existence, not quantitative trajectories. The linear treatment also connects directly to the AR(1) simulation framework () and classical control theory (PI controllers, Kalman filters).

Speed-quality substitutability. From $\mathcal{T} = \nu \cdot \eta^*$ (single channel): doubling event rate ν has the same effect on $\|\delta\|_{ss}$ as doubling update quality η^*. They are multiplicative when both improve: 50% improvement in each yields $1.5 \times 1.5 = 2.25\times$, not $3\times$. This is the formal analog of Boyd's insight that Orient quality often matters more than raw OODA speed.

The persistence threshold in the linear world. From the steady state: $\|\delta\|_{ss} < \|\delta_{\text{critical}}\|$ iff $\mathcal{T} > \rho/\|\delta_{\text{critical}}\|$ (Model D). This is task adequacy only – structural persistence is trivially satisfied because $R \to \infty$. The linear world makes persistence look easier than it is. For nonlinear agents, both structural and task-adequacy conditions must hold (4.4).

Working Notes.

- The vector ODE $d\delta/dt = -\mathcal{T}\delta + w$ assumes isotropic correction (same rate in all directions). For anisotropic agents (tempo tensor $\mathbf{T}$), the linear ODE generalizes to $d\delta/dt = -\mathbf{T}\delta + w$ with matrix exponential solutions and per-eigenvalue convergence rates. The per-dimension persistence condition ($\mathcal{T}_k > \rho_k/\delta_{\text{critical},k}$) from 14.5 is the diagonal special case.
- The norm-inequality subtlety (Section 1 above) propagates into the adversarial scaling law: the squared law ($b = 2$) is exact for $n = 1$ and a worst-case upper bound for $n > 1$. Whether this matters quantitatively for multi-dimensional adversarial dynamics has not been tested in simulation.
- Model S's Ito-Lyapunov derivation assumes σ_w is constant. When the noise scale depends on state ($\sigma_w(\delta)$, as in multiplicative noise models), the Ornstein-Uhlenbeck analysis no longer applies and the steady-state formula changes. The sector-condition results in A handle additive noise only.

Appendix M

Derivation: Graph Structure Uniqueness

Operational requirements on the agent's representation — directed temporal ordering, probabilistic uncertainty, and the ability to test strategy components — are *sufficient* for the strategy to be a directed acyclic graph with the Markov factorization property. The argument parallels Cox's theorem for probability in *form* but not yet in *strength*: Cox's theorem is necessary-and-sufficient (the only measure satisfying the desiderata is probability); this result is sufficient only (the desiderata guarantee DAG+Markov, but no one has shown a non-DAG structure cannot satisfy them). Acyclicity and directed edges are *proved* from temporal ordering over a finite horizon; the Markov factorization is *proved under causal sufficiency* via the Causal Markov Condition theorem (Spirtes–Glymour–Scheines, Pearl). A fourth postulate (observable intermediates) is required for localized strategic diagnosis but not for the representation or persistence results themselves.

How far the Cox parallel goes. Cox's theorem starts from desiderata on how a rational agent should quantify uncertainty (consistency, universality, continuous functional composition) and proves that the *only* measure satisfying them is probability — necessity. This segment starts from desiderata on how a bounded agent can represent its strategy under causal action (directed temporal order, probabilistic edge uncertainty, causal sufficiency of the chosen nodes) and proves that DAG+Markov *suffices* to satisfy them — sufficiency. The necessity direction — no non-DAG structure (factor graphs, junction trees with cyclic message schedules, chain graphs) can satisfy P1–P4 plus causal sufficiency — is not established here. In practical terms this gap is unimportant because the proved sufficiency gives a rigorous grounding for the DAG structure; claims that AAT's strategy *must* be a DAG should be read as "must-if-sufficient-via-this-route," not as a proved necessity. A stronger Cox-style result is open. The placement of the DAG structure on a footing comparable to probability — a consequence of operational requirements rather than a modeling convenience — holds in the sufficiency direction; the full parallel with Cox awaits a uniqueness argument.

Formal Expression.

The Postulates. Four properties that a strategy representation must satisfy. Each is independently motivated from the adaptive-systems foundation.

P1: Directed Temporal Ordering.

If component A of the strategy causally produces component B, then A temporally precedes B. The strategy representation must respect this directionality — edges point from causes to effects, from actions to outcomes, from prerequisites to goals.

Derived (from #post-causal-structure).

This is a consequence of the temporal postulate (1.8): the arrow of time is constitutive, not incidental. Reversing a causal edge would mean effects precede causes, which is physically impossible.

P2: Probabilistic Uncertainty.

The agent's uncertainty about whether each step of the strategy will succeed must be quantified by a measure satisfying Cox's axioms (consistency, universality, non-negativity). The unique such measure is probability (Cox 1946, "Probability, Frequency and Reasonable Expectation," *American Journal of Physics* 14(1):1–13; modern exposition in Jaynes 2003, *Probability Theory: The Logic of Science*, Cambridge University Press, Chapter 2). The agent may use other representations internally (confidence scores, fuzzy logic), but these must be mappable to probability to be consistent.

Derived (from Cox's theorem).

P3: State-Local Revisability.

When the agent observes evidence about one component of its strategy (e.g., "step 3 succeeded" or "prerequisite 2 is blocked"), it must be able to update its beliefs about that component and its consequences without recomputing the entire strategy from scratch.

Derived (from #der-chain-confidence-decay + bounded computation).

Why this is forced, not chosen:

From fragility. Additive log-confidence (7.1) means longer chains are exponentially less reliable. The agent will frequently encounter partial failures. Each partial failure requires re-evaluation of the affected portion of the strategy. If each re-evaluation requires full recomputation, the agent's planning tempo $\mathcal{T}_\Sigma$ is catastrophically slow — potentially violating the strategy persistence condition.

From bounded computation. The agent has finite computational resources (the IB constraint applies to planning as well as model maintenance). Full recomputation of a strategy with N components costs $O(N)$ or worse. Local revision costs $O(|\text{affected}|)$, which can be much smaller.

From the persistence condition. Strategy must be revised faster than the environment invalidates it. Local revision directly increases $\mathcal{T}_\Sigma$ by reducing the per-update cost. An agent that must recompute everything on each update has lower $\mathcal{T}_\Sigma$ and is more likely to fall below the persistence threshold.

P4: Observable Intermediates.

To support **localized strategic diagnosis and revision**, the strategy representation benefits from internal checkpoints — observable states between the initial action and the final goal — that the agent can monitor to detect partial failure.

Derived (from #der-chain-confidence-decay + monitoring requirement).

Without intermediates, the agent cannot detect chain failure until the final outcome. By the time the final outcome reveals failure, all intermediate actions are wasted. With intermediates, the agent can detect failure at step k and revise, saving the cost of steps $k + 1$ through n. The value of early detection grows with chain length, because longer chains fail more often (P2 + 7.1).

Observable intermediates are not required for strategy representation or persistence. When intermediates are unobservable, plan-level tracking (8.10, Case 3) preserves the sector condition at the cost of per-edge diagnostic resolution — the agent knows the plan is failing but cannot localize which step needs revision (8.3). P4 is therefore a requirement for *strong diagnostics*, not for strategy representation per se. The observability investment tradeoff (8.3) quantifies the payoff: making an intermediate observable improves the sector parameter from $1/(n_\Phi + 1)$ (plan-level) to $\min(1/(n_1 + 1),\ \theta_1/(n_2 + 1))$ (per-edge weakest-link).

The Derivation.

Step 1: P1 implies directed edges. Each component X_i of the strategy has a set of direct causes $\mathrm{Pa}(X_i)$ — the components whose outcomes directly influence X_i's outcome. P1 requires that these causal relationships are directed: $\mathrm{Pa}(X_i)$ temporally precedes X_i. This gives directed edges $\mathrm{Pa}(X_i) \to X_i$.

Step 2: P2 implies probability distributions on edges. Each edge carries uncertainty: $P(X_i \mid \mathrm{Pa}(X_i))$. By P2 (Cox), this is a probability distribution. The joint distribution over all strategy components is some $P(X_1, \ldots, X_n)$.

Step 3: Causal sufficiency implies the Markov condition (proved).

Claim. For a causally sufficient strategy DAG, the Markov factorization property is a theorem — a consequence of the Causal Markov Condition (CMC).

Derived (Conditional on causal sufficiency of Σ_t).

The Markov factorization property. Each variable X_i is conditionally independent of its non-descendants given its parents:

$$X_i \perp \mathrm{NonDesc}(X_i) \mid \mathrm{Pa}(X_i) \tag{M.1}$$

Equivalently, the joint distribution factorizes as:

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid \mathrm{Pa}(X_i)) \tag{M.2}$$

(The equivalence holds for positive distributions — Lauritzen 1996, Theorem 3.27.)

The argument has five parts:

(a) The DAG is a causal model. P1 establishes that edges represent causal relationships: completing a parent step causally advances the child step. P2 establishes probabilistic uncertainty over outcomes. Together: Σ_t is a causal DAG in the sense of structural causal models (Pearl 2009, Definition 7.1.1) — each node's outcome is determined by its parents' outcomes (through the causal mechanism encoded in the edge credences) plus exogenous uncertainty specific to that step. Formally, each node admits a structural equation:

$$X_i = f_i(\mathrm{Pa}(X_i), \varepsilon_i) \tag{M.3}$$

where f_i is the local causal mechanism and ε_i is the exogenous noise (the residual uncertainty at step i not determined by its parents).

(b) Causal sufficiency implies exogenous independence. The exogenous terms ε_i are mutually independent if and only if no unmodeled common cause affects two or more nodes in the graph. This is precisely the **causal sufficiency** assumption: every variable that is a direct common cause of two or more nodes in Σ_t is itself a node in Σ_t.

For agent-constructed strategies, causal sufficiency is a **modeling ideal, not a typical condition**. The agent designed the graph, so all *intended* causal relationships are explicit — but environmental common causes (shared infrastructure, weather, market shifts, correlated adversary actions) routinely affect multiple strategy steps without appearing as nodes. In complex, multi-stakeholder, or adversarial environments, causal insufficiency is the dominant case (7.3, Correlation Hierarchy). When an environmental factor is omitted, the exogenous terms become correlated and the Markov condition fails. This is model inadequacy (4.5), and the remedy is to add the missing common-cause node — but identifying which common causes matter is a modeling judgment, not a mechanical procedure (7.3, L1 construction principle). The proof's conditional on causal sufficiency is therefore a condition on model quality: the result holds exactly when the DAG is well-constructed, approximately when it is close, and fails when major common causes are missing. The Correlation Hierarchy in 7.3 provides the practical framework: L0 (independence, this proof's assumption) gives tractable results; L1 (augmented DAG with explicit common-cause nodes) is the practical default in complex domains; L0 formal results transfer to correctly constructed L1 DAGs.

(c) The Causal Markov Condition theorem. For a DAG G over variables $V = \{X_1, \dots, X_n\}$ with structural equations $X_i = f_i(\text{Pa}(X_i), \varepsilon_i)$ where the ε_i are mutually independent:

$$P(X_1, \dots, X_n) = \prod_{i=1}^{n} P(X_i \mid \text{Pa}(X_i)) \tag{M.4}$$

This is the **Causal Markov Condition** — a proved theorem, not a modeling assumption. The standard references are Spirtes, Glymour, and Scheines (2000, Theorem 3.4) and Pearl (2009, §1.4.1, Theorem 1.4.1). The proof applies the chain rule in topological order: $P(X_1, \dots, X_n) = \prod_i P(X_i \mid X_1, \dots, X_{i-1})$, then uses the independence of ε_i to show that conditioning on all predecessors reduces to conditioning on parents only. Each non-parent predecessor's influence on X_i is fully mediated through the parents — its direct contribution enters through the causal mechanism f_i, not through ε_i.

(d) P3 as consequence. State-local revisability (P3) was originally stated as an independent postulate. The CMC reveals it is a *consequence* of the causal structure under causal sufficiency: since $X_i \perp \text{NonDesc}(X_i) \mid \text{Pa}(X_i)$, updating beliefs about X_i requires only $\text{Pa}(X_i)$ — local revision is automatically correct. No information from the rest of the graph changes the conditional distribution of X_i given its parents. P3 was motivated as an operational requirement (agents *need* local revision for computational tractability, and the persistence condition demands it). The CMC shows the requirement is automatically satisfied by any causally sufficient causal DAG. The two arguments converge from different directions: P3 says local revision is *needed*; the CMC says it is *guaranteed* (under causal sufficiency).

(e) Connection to edge independence. The CMC's exogenous independence condition (ε_i mutually independent) is precisely the **edge-independence assumption** in the AND/OR status propagation (7.3). When exogenous noise terms are independent, edge outcomes are conditionally independent given parents, and the AND/OR formulas compute correct probabilities. When they are correlated (causal insufficiency — latent common causes), the AND/OR propagation systematically overestimates success because it treats joint failure probability as the product of marginals. The validity of the Markov factorization and the validity of the independence model are the *same condition*: causal sufficiency of Σ_t. See 7.3 for the full treatment of correlated failure as the primary case.

Assembling (a)-(e). P1-P2 establish that Σ_t is a causal DAG with probabilistic uncertainty. Under causal sufficiency (exogenous independence), the CMC theorem proves the Markov factorization. P3 (local revisability) follows as a validated consequence. The Markov property is both operationally required (P3) and structurally guaranteed (CMC). When causal sufficiency fails, the Markov factorization is still the agent's *intended* factorization — the one its DAG represents — but it is wrong about the world. The gap between intended and actual factorization manifests as correlated failure and $\hat{P}_\Sigma$ overestimation, and the fix is structural: add the missing common-cause nodes to restore causal sufficiency.

Step 4: P1 + finite horizon implies acyclicity (proved). This is the strongest piece of the argument. See the dedicated section below.

Step 5: Assembly. P1 (directed edges + causal interpretation) + P2 (probabilistic) + causal sufficiency (CMC → Markov factorization) + P4 (internal nodes) + finite horizon (acyclicity):

The strategy representation must be representable as a directed acyclic graph with probability distributions at each node conditioned on its parents — a Bayesian network. P3 (local revisability) is validated as a consequence of this structure, not required as a premise.

Acyclicity Derivation.

This resolves a former known fragility in the theory. Acyclicity of Σ_t is derived, not assumed.

Derived (from #post-causal-structure + finite planning horizon).

Result. For a strategy representation over a finite future horizon, temporal ordering forces acyclicity.

Derivation.

1. Each node X_i in Σ_t represents a future event or state with temporal position $\tau_i > t$ (the future time at which the step occurs or the state is evaluated).
2. Each edge $X_i \to X_j$ requires $\tau_i < \tau_j$ (P1: causes temporally precede effects).
3. A cycle $X_i \to X_j \to \cdots \to X_i$ would require $\tau_i < \tau_j < \cdots < \tau_i$, which is impossible for a real-valued time index.
4. Therefore the graph is acyclic. □

Formally: a finite set with a strict partial order (future events ordered by time) is representable as a DAG. This is a standard result in order theory — every finite partial order has a Hasse diagram, which is a DAG.

The iteration objection resolved. A strategy that says "try A, if fail try B, if fail try A again" appears cyclic.

In the time-indexed representation:

$$A_1 \to \text{check}_1 \to B_1 \to \text{check}_2 \to A_2 \to \ldots \tag{M.5}$$

Each attempt is a distinct node at a distinct time. The apparent cycle is a linear chain in the unrolled view. Iteration "terminates" when either a node succeeds (remaining retry nodes become probability-zero), the agent exhausts its resource budget (a constraint truncating the chain), or the horizon ends. Any finite-horizon strategy, including those with "loops" in the informal sense, is acyclic when time-indexed.

Scope. This applies to Σ_t (the agent's strategy over the future), not to M_t's model of the environment. M_t may include cyclic causal processes — feedback loops in the physical world, market dynamics, ecosystem interactions. The acyclicity is specific to the purposeful substate because Σ_t represents planned future actions and the future is partially ordered by time. M_t's model of environmental dynamics may need to represent cycles (via time-unrolled DBNs or other cyclic structures).

Connection to Pearl. Pearl's do-calculus is defined on DAGs. Extensions to cyclic structures exist (cyclic SCMs, equilibrium models) but are substantially more complex and lose some of do-calculus's clean properties. The temporal argument here shows that for strategy representations (future-looking plans), acyclicity is not a convenience restriction on Pearl's framework — it is a consequence of the temporal structure of planning.

What Is Derived vs. What Is Chosen.

Table M.1

Property	*Motivating postulate*	*Strength*
Directed edges	Temporal ordering (P1, 1.8)	Proved
Probabilistic uncertainty	Cox's theorem (P2)	Proved
Acyclicity	Temporal ordering + finite horizon (P1)	Proved
Internal structure	Fragility + monitoring (P4, 7.1)	Derived

Property	*Motivating postulate*	*Strength*
Markov factorization	Causal Markov Condition theorem (P1 causal interpretation + P2 probability + causal sufficiency)	Proved under causal sufficiency (CMC theorem)
DAG with Markov property	**P1 + P2 + causal sufficiency (CMC) + P4**	**Conditional on causal sufficiency — which is testable and repairable**
AND/OR parameterization	Boolean completeness + parsimony	Hypothesis (binary outcomes only)
Single-parameter edges	Parsimony / IB	Formulation choice
Specific node ontology	—	Formulation choice

The dividing line: acyclicity and directed edges are proved; the full DAG-with-Markov-property is conditional on causal sufficiency. The parameterization (AND/OR, CPT form, edge semantics) is a formulation choice within the strongly motivated structure, motivated by parsimony and domain fit but not by mathematical necessity.

Equivalence Class. **Within the DAG class.** Multiple DAGs can encode the same conditional independence relations, forming a Markov equivalence class identified by a CPDAG (completed partially directed acyclic graph). Two DAGs in the same equivalence class make identical probabilistic predictions but may differ in causal interpretation.

Across representation types. Factor graphs, junction trees, influence diagrams, and chain graphs are NOT simple presentational variants of DAGs:

- **Factor graphs** and **junction trees** preserve factorization and inference structure without necessarily preserving directed causal semantics.
- **Influence diagrams** add decision and utility nodes — a richer object, not an equivalent one.
- **Chain graphs** can express independence models that are not representable as DAGs at all.
- **Markov equivalence** is a statement within DAG classes, not across all graphical model types.

The correct claim is narrow: for a given factorized distribution, DAG and factor-graph representations can compute the same marginals. But causal semantics (do-calculus) are DAG-specific and do not transfer to undirected or mixed representations without additional structure.

AAT's choice. AAT uses DAG + AND/OR because: (a) AND/OR is the most parsimonious complete basis for binary combination (7.2), (b) the DAG naturally supports causal/interventional reasoning (Pearl's do-calculus), and (c) the representation converged across three independent formalism attempts.

Epistemic Status.

The acyclicity derivation is *exact* — it follows from temporal ordering over a finite horizon via standard order theory. The individual postulates P1, P2, and P4 are each well-grounded (temporal structure, Cox's theorem, and chain fragility respectively).

The Markov property is now *proved under causal sufficiency* via the Causal Markov Condition theorem (Spirtes, Glymour & Scheines 2000, Theorem 3.4; Pearl 2009, Theorem 1.4.1). The previous sketch argument (P3 requires locality → P1 identifies parents → therefore Markov) has been replaced by a rigorous chain: P1-P2 establish the causal DAG structure, causal sufficiency guarantees exogenous independence, and the CMC theorem proves the factorization. P3 (local revisability) is now a *consequence* of the Markov property, not a premise — the CMC shows that local revision is automatically correct for causally sufficient DAGs, validating P3's operational requirement.

The conditioning on causal sufficiency is the right level of conditionality: it is a property of *strategy quality* (did the agent include all relevant common causes as nodes?), not of agent architecture. It is testable in principle (correlated residuals after convergence indicate missing common causes) and repairable in practice (add the common-cause node). When causal sufficiency fails, the Markov factorization fails, and this manifests as correlated failure and $\hat{P}_\Sigma$ overestimation (7.3). The edge-independence assumption in AND/OR propagation and the causal sufficiency condition for the Markov property are the same condition viewed from different angles.

Max attainable: *exact* for acyclicity (already there). *Derived conditional* for the full DAG-with-Markov-property — the derivation is rigorous (invokes a proved theorem), and the remaining condition (causal sufficiency) is about model quality, not proof quality.

The AND/OR restriction is a *hypothesis* for binary outcomes, grounded in Boolean completeness and parsimony. For non-binary outcomes, it does not apply and richer parameterizations within the derived graphical structure are needed.

The parallel to Cox's theorem is now tighter than previously stated: Cox's theorem proves that consistency axioms force probability; the CMC theorem proves that causal structure under sufficiency forces the Markov factorization. Both are formal results, not analogies. The remaining gap: Cox's axioms are necessary and sufficient for probability; AAT's postulates are sufficient for DAG+Markov structure, but the necessity direction (could a non-DAG structure satisfy P1-P4?) is not established. For practical purposes this gap is unimportant — the proved sufficiency gives a rigorous foundation for the strategy representation.

Discussion.

The contribution of this analysis. The dividing line between derived/conditional structure and chosen parameterization is the key result. The agent community (and future researchers) know exactly what is proved (acyclicity, directed edges), what is conditional (Markov property, contingent on causal sufficiency), and what is a formulation choice (the parameterization within the graph). Alternative parameterizations can be explored without abandoning the theoretical foundation.

Acyclicity deserves emphasis. The existing theory previously flagged "DAG acyclicity is an assumption, not forced" as a known fragility. The derivation above resolves this: acyclicity of Σ_t follows from temporal ordering over a finite planning horizon. This is specific to the strategy — M_t's environment model is not restricted to acyclic structures.

The causal sufficiency assumption. The Markov condition assumes causal sufficiency — no hidden common causes within the strategy. Since the agent constructs Σ_t, there are no "hidden" variables internal to the strategy (the agent chose all the nodes). But environmental common causes that affect multiple strategy steps may violate the Markov condition within Σ_t. If so, latent variable models or causal graphs with hidden nodes would be needed. This connects to the edge identifiability problem noted in 7.3.

Working Notes.

- **P3→Markov: resolved.** The previous gap (P3 could be satisfied by non-Markov sparse factorizations) is resolved by grounding the Markov property in the CMC theorem rather than in P3. The CMC proves the factorization from the causal structure (P1) and causal sufficiency (exogenous independence), making P3 a consequence rather than a premise. The alternative-factorization concern (factor graphs, junction trees) is now moot: the Markov property follows from the causal semantics, not from locality. P3 remains valuable as an *operational requirement* that the CMC-guaranteed factorization satisfies.
- **Parsimony theorem for AND/OR.** Is there a formal result that AND/OR (noisy-AND + noisy-OR) is the unique $O(k)$-parameter complete basis for binary nodes? This would strengthen 7.2 from formulation choice to derived. The Boolean completeness half is standard; the uniqueness-under-parsimony half is not established.
- **Non-binary outcome analogs.** For continuous or multi-valued outcomes, the natural analogs of AND/OR might be min/max or additive/multiplicative combination. Is there a completeness result for these? The current argument applies cleanly only to binary outcomes.

- **Environmental common causes and the CMC.** The CMC proof makes the failure mode precise: when unmodeled environmental dependencies (weather, shared infrastructure, market conditions) affect multiple strategy steps, the exogenous noise terms ε_i become correlated, causal sufficiency fails, and the Markov factorization is violated. The consequence is exactly the correlated-failure phenomenon in 7.3: $\hat{P}_\Sigma$ overestimates success because it treats joint failure probability as the product of marginals. The fix is structural: add the common-cause as a condition node in Σ_t, restoring causal sufficiency and the Markov property for the augmented DAG. This connects the graph-theoretic result (Markov property) to the strategy-layer result (independence model validity) through a single condition: causal sufficiency.
- **Promotion potential.** With the CMC-based proof, 7.3's DAG structure claim is now strongly grounded: acyclicity is proved, the Markov property is proved under causal sufficiency. The remaining "definition" character of strategy-dag is about the *parameterization* (AND/OR, single-parameter edges), which is a formulation choice within the proved graphical structure.

Appendix N

Derivation: Sector Condition Verification for Edge-Credence Dynamics

Complete verification that specific strategy update dynamics satisfy the sector condition from A, backing the schema proposed in 8.10. Four cases are treated: a single-edge strategy, a two-edge AND-chain with observable intermediate, the same chain with unobservable intermediate, and a two-arm OR-node under exploratory action selection.

Motivation.

8.10 proposes that the sector-condition mathematics (Props A.1, A.1S, A.2 in A) extends from epistemic to strategic mismatch. The schema identifies three structural conditions (SA1, SA2', SA3) and conjectures the persistence form $\alpha_\Sigma > \rho_\Sigma / R_\Sigma$. This appendix instantiates the schema for concrete Beta-Bernoulli edge dynamics, verifying the conditions case by case. Each proposition establishes the sector parameter α_Σ for a specific DAG topology; the persistence condition then follows by direct substitution into Proposition A.1 of A.

Common Setup.

Throughout, edge credences follow Beta-Bernoulli dynamics: edge k has true success probability $\theta_k \in (0, 1)$, agent belief $p_k \sim \text{Beta}(\alpha_k, \beta_k)$ with point estimate $\hat{p}_k = \alpha_k / n_k$ where $n_k = \alpha_k + \beta_k$, and conjugate update on observing $y_k \in \{0, 1\}$:

Definition (Beta-Bernoulli edge dynamics).

$$\Delta \hat{p}_k = \frac{y_k - \hat{p}_k}{n_k + 1}, \qquad \eta_k = \frac{1}{n_k + 1} \tag{N.1}$$

The single-step update matches the gain-based form from 8.4 with $\eta_{\text{edge}} = 1/(n_k + 1)$.

Parameterization note. Propositions B.1–B.7 below are stated in the *moment-parameter* (probability-space) coordinate $\hat{p}_k \in [0, 1]$, consistent with the Beta-Bernoulli conjugate presentation. The log-odds coordinate $\lambda_k = \log(\hat{p}_k / (1 - \hat{p}_k)) \in \mathbb{R}$ is the unique additive-evidence presentation forced by the evidential-additivity axiom (P); the sector-parameter content of each proposition is Fisher-equivalent in either coordinate, so the derivations are retained in moment-parameter form for algebraic tightness ($\eta_k = 1/(n_k + 1)$ is exact in probability space). The log-odds coordinate becomes load-bearing in 8.6 for the continuous-gradient signal function, where the probability-space presentation would exhibit a mechanical domain break.

Strategic mismatch per edge is $\delta_k = \hat{p}_k - \theta_k$. The sector-condition framework (A) requires: (SA1) $F(\mathbf{0}) = \mathbf{0}$, (SA2') $\delta^T \mathbf{F}(\delta) \geq \alpha_\Sigma \|\delta\|^2$ within a region $\mathcal{B}_R$.

Regime-adjustment convention. The propositions below state sector parameters α_Σ for the **canonical Regime-A case**, in which every edge's evidence is fully interventional and attributable (identifiability coefficient $\iota_{ij} = 1$ per 8.5). For edges with $\iota_{ij} < 1$ (Regime B or C), each sector parameter carries an implicit ι_{ij} factor:

$$\alpha_\Sigma^{\text{eff}} = \iota_{ij} \cdot \alpha_\Sigma^{\text{stated}} \tag{N.2}$$

This commutes through every diagonal sector product in Props B.1–B.4 and B.6 because each edge's contribution to $\delta_\Sigma^T \mathbf{F}$ is scaled by ι_{ij} independently: Regime-C edges ($\iota \approx 0$) contribute essentially nothing to the weakest-link minimum, and Regime-A edges ($\iota \approx 1$) recover the stated formulas exactly. The persistence condition $\alpha_\Sigma > \rho_\Sigma / R_\Sigma$ becomes $\iota_{ij} \cdot \alpha_\Sigma^{\text{stated}} > \rho_{\Sigma,ij}/R_{\Sigma,ij}$ per edge. The propositions below state everything in Regime-A form for clarity; the ι-adjustment is uniform across cases and does not alter the structural results (evidence starvation, exploration gating, three-way gating) — only their quantitative thresholds per edge.

For Prop B.3 (unobservable intermediate), the regime adjustment does not change the A1 violation result: the bias is structural to the marginal Bayesian update, not to the identifiability quality. The plan-level fallback still recovers the sector condition with an ι-adjusted $\alpha_{\Sigma,\text{plan}}$ governed by the aggregate's identifiability.

Proposition B.1: Single Edge ($A \to G$).

Setup. One action A, one goal G, one edge with credence $\hat{p}$ and true probability θ. Mismatch: $\delta_\Sigma = \hat{p} - \theta$ (scalar).

Statement. Under Beta-Bernoulli updating, the expected correction function satisfies the sector condition globally (all $|\delta_\Sigma| \leq 1$) with:

Derived (Conditional on Beta-Bernoulli model).

$$\alpha_\Sigma = \frac{1}{n+1} \tag{N.3}$$

The bound is tight.

Proof.

The expected correction. Since $y \sim \text{Bernoulli}(\theta)$:

Derived (Proof Step).

$$\mathbb{E}[\Delta \hat{p}] = \frac{\theta - \hat{p}}{n+1} = -\frac{\delta_\Sigma}{n+1} \tag{N.4}$$

Identifying $F_\Sigma(\delta_\Sigma) = -\mathbb{E}[\Delta \delta_\Sigma] = \delta_\Sigma/(n+1)$:

Derived (Proof Step).

$$\delta_\Sigma \cdot F_\Sigma(\delta_\Sigma) = \frac{\delta_\Sigma^2}{n+1} = \frac{1}{n+1} \|\delta_\Sigma\|^2 \tag{N.5}$$

So $\alpha_\Sigma = 1/(n+1)$. The correction is exactly linear, so the sector bound is tight (not merely a lower bound).

Verification of (SA1). At $\delta_\Sigma = 0$: $F_\Sigma(0) = 0$. Satisfied.

Stochastic ultimate bound. Decomposing $\delta_{\Sigma,t+1} = \delta_\Sigma \cdot n/(n+1) + (y-\theta)/(n+1)$ and computing $\mathbb{E}[\delta^2_{\Sigma,t+1}]$:

Derived (stochastic ultimate bound, single edge).

$$\mathbb{E}[\delta^2_{\Sigma,t+1}] = \delta^2_\Sigma \cdot \frac{n^2}{(n+1)^2} + \frac{\theta(1-\theta)}{(n+1)^2} \tag{N.6}$$

The expected Lyapunov derivative $\mathbb{E}[\Delta V] < 0$ whenever:

$$|\delta_\Sigma| > \sqrt{\frac{\theta(1-\theta)}{2n+1}} \tag{N.7}$$

The stochastic noise floor decreases as $O(1/\sqrt{n})$ — the standard Bayesian posterior concentration rate. □

Proposition B.2: Two-Edge AND-Chain, Observable Intermediate ($A \to B \to G$, B ***observed).***

Setup. Action A leads to intermediate B, then to goal G. Both y_B and y_G are observed. Edge credences $\hat{p}_1, \hat{p}_2$ with true probabilities θ_1, θ_2. Plan confidence: $\hat{P}_\Sigma = \hat{p}_1 \hat{p}_2$ (AND propagation from 7.2). Mismatch vector: $\delta_\Sigma = (\delta_1, \delta_2)^T$.

The generative model:

$$y_B \sim \text{Bernoulli}(\theta_1), \qquad y_G \mid y_B = 1 \sim \text{Bernoulli}(\theta_2), \qquad y_G \mid y_B = 0 = 0 \tag{N.8}$$

Statement. Under Beta-Bernoulli updating with observable intermediate, the expected correction function is diagonal and satisfies the sector condition globally with:

Derived (Conditional on Beta-Bernoulli model, observable intermediate).

$$\alpha_\Sigma = \min\left(\frac{1}{n_1+1}, \frac{\theta_1}{n_2+1}\right) \tag{N.9}$$

Proof.

Edge 1 is tested every trial: $\Delta\hat{p}_1 = (y_B - \hat{p}_1)/(n_1+1)$, giving $\mathbb{E}[\Delta\hat{p}_1] = -\delta_1/(n_1+1)$.

Edge 2 is tested only when $y_B = 1$ (probability θ_1). When $y_B = 0$, edge 2 receives no information:

Derived (Proof Step).

$$\mathbb{E}[\Delta\hat{p}_2] = \theta_1 \cdot \frac{\theta_2 - \hat{p}_2}{n_2+1} + (1-\theta_1)\cdot 0 = -\frac{\theta_1\delta_2}{n_2+1} \tag{N.10}$$

The expected correction function is diagonal:

$$\mathbf{F}(\delta_\Sigma) = \begin{pmatrix} \delta_1/(n_1+1) \\ \theta_1\delta_2/(n_2+1) \end{pmatrix} \tag{N.11}$$

Verification of (SA1). At $\delta_\Sigma = \mathbf{0}$: $\mathbf{F}(\mathbf{0}) = \mathbf{0}$. Satisfied.

The sector product:

Derived (Proof Step).

$$\delta_\Sigma^T \mathbf{F}(\delta_\Sigma) = \frac{\delta_1^2}{n_1+1} + \frac{\theta_1 \delta_2^2}{n_2+1} \geq \min\left(\frac{1}{n_1+1}, \frac{\theta_1}{n_2+1}\right)(\delta_1^2 + \delta_2^2) \tag{N.12}$$

The minimum is achieved by whichever edge has the lower effective correction rate. □

The evidence-starvation effect. Edge 2's effective correction rate $\theta_1/(n_2+1)$ is attenuated by the upstream success probability θ_1. When θ_1 is small, edge 2 is starved of evidence — it is tested rarely, learns slowly, and becomes the weakest link. This gives a precise mechanism for 7.1: downstream edges in AND-chains are harder to calibrate, with correction rate proportional to the product of all upstream success probabilities.

Generalization to depth d. For a chain of depth d with all intermediates observable, edge k's effective correction rate is $\alpha_k = \prod_{j=1}^{k-1} \theta_j/(n_k+1)$, giving:

Derived (depth-d chain sector parameter).

$$\alpha_\Sigma = \min_{k=1}^{d} \frac{\prod_{j=1}^{k-1} \theta_j}{n_k+1} \tag{N.13}$$

Proposition B.3: Two-Edge AND-Chain, Unobservable Intermediate ($A \to B \to G$, B not observed).

Setup. Same DAG as Proposition B.2, but only $y_G \in \{0,1\}$ is observed. The agent cannot distinguish "$A \to B$ failed" from "$B \to G$ failed" when $y_G = 0$.

Statement. (a) The per-edge sector condition fails: the marginal Bayesian update violates (SA1) with systematic bias $O(1/n)$. (b) Plan-level tracking (treating $\hat{\Phi} = \hat{p}_1 \hat{p}_2$ as a single Beta over the composite probability $\Phi = \theta_1\theta_2$) recovers the sector condition with:

Derived (Conditional on Beta-Bernoulli model, unobservable intermediate).

$$\alpha_{\Sigma,\text{plan}} = \frac{1}{n_\Phi + 1} \tag{N.14}$$

Proof of (a): Per-edge (SA1) violation.

The joint posterior after success ($y_G = 1$) factors cleanly: $\alpha_k \to \alpha_k + 1$ for both edges. The joint posterior after failure ($y_G = 0$) includes the factor $(1 - \theta_1\theta_2)$, which does not decompose into separate functions of θ_1 and θ_2. Failure introduces posterior correlation.

The exact marginal Bayesian update for edge 1 on failure is:

Derived (Proof Step).

$$\Delta \hat{p}_1^{(\text{fail})} = -\frac{\hat{p}_1 \, q_1}{n_1 + 1} \tag{N.15}$$

where $q_1 = (1 - \hat{p}_1)/(1 - \hat{p}_1\hat{p}_2)$ is the posterior probability that edge 1 caused the failure (the proportional-blame weight). The expected update, combining success (probability $\theta_1\theta_2$) and failure (probability $1 - \theta_1\theta_2$):

Derived (Proof Step).

$$\mathbb{E}[\Delta \hat{p}_1] = \theta_1\theta_2 \cdot \frac{1-\hat{p}_1}{n_1+1} - (1-\theta_1\theta_2) \cdot \frac{\hat{p}_1\, q_1}{n_1+1} \tag{N.16}$$

Evaluating at truth ($\hat{p}_k = \theta_k$):

Derived (SA1 violation magnitude).

$$\mathbb{E}[\Delta \hat{p}_1]\big|_{\hat{p}=\theta} = -\frac{\theta_1(1-\theta_1)(1-\theta_2)}{n_1+1} \neq 0 \tag{N.17}$$

The bias is negative (pushes credence below truth), of magnitude $O(1/n)$, and arises because success always increments α_1 by a full unit while failure increments β_1 by the fractional weight $q_1 < 1$. The success case over-credits edge 1 because the agent cannot separate edge 1's contribution from edge 2's. By symmetry, $\hat{p}_2$ is also biased downward. □

Proof of (b): Plan-level recovery.

If the agent tracks $\hat{\Phi}$ directly as a Beta posterior over $\Phi = \theta_1\theta_2$ using $y_G \sim$ Bernoulli(Φ), the setup reduces to a single-edge problem (Proposition B.1) with mismatch $\delta_\Phi = \hat{\Phi} - \Phi$ and sector parameter $\alpha_{\Sigma,\text{plan}} = 1/(n_\Phi + 1)$. All single-edge results apply. □

Tradeoff. Per-edge tracking provides diagnostic resolution (which edge is failing?) but yields biased, non-sector-conforming updates when the intermediate is unobservable. Plan-level tracking sacrifices diagnostic resolution but provides clean sector-condition satisfaction. This is a concrete instance of 8.3: unobservable nodes freeze per-edge learning, making the aggregate the correct unit of analysis.

Non-identifiability. From y_G observations alone, only $\Phi = \theta_1\theta_2$ is identifiable. The per-edge decomposition (θ_1, θ_2) is underdetermined — any pair with the same product produces identical observations. Per-edge identification requires per-edge observability (or structural constraints that break the symmetry).

Proposition B.4: Two-Arm OR-Node ($A_1, A_2 \to G$, ε-greedy).

Setup. Goal G reachable via two alternative actions A_1, A_2 (an OR-node from 7.2). True probabilities θ_1, θ_2. Agent selects one action per trial: with probability $1-\varepsilon$ the greedy choice (higher $\hat{p}_k$), with probability ε the other. Mismatch vector: $\delta_\Sigma = (\delta_1, \delta_2)^T$.

Statement. (a) Under a greedy policy ($\varepsilon = 0$), the sector condition fails for the full mismatch vector. (b) Under ε-greedy action selection with $\varepsilon > 0$, the sector condition is satisfied with:

Derived (Conditional on Beta-Bernoulli model, ε-greedy OR-node).

$$\alpha_\Sigma = \min\!\left(\frac{1-\varepsilon}{n_1+1},\ \frac{\varepsilon}{n_2+1}\right) \tag{N.18}$$

where arm 1 is the greedy choice.

Proof of (a): Greedy failure.

Under greedy selection, only the chosen arm updates. Assuming $\hat{p}_1 > \hat{p}_2$ (arm 1 is greedy):

$$\mathbf{F}(\delta_\Sigma) = \begin{pmatrix} \delta_1/(n_1+1) \\ 0 \end{pmatrix} \tag{N.19}$$

The sector product $\delta_\Sigma^T \mathbf{F} = \delta_1^2/(n_1+1)$. For the sector condition to hold, we need $\delta_1^2/(n_1+1) \geq \alpha_\Sigma(\delta_1^2+\delta_2^2)$. Setting $\delta_1 = 0$: the LHS is zero but $\alpha_\Sigma \delta_2^2 > 0$ for $\delta_2 \neq 0$. No $\alpha_\Sigma > 0$ works. □

Proof of (b): ε-greedy recovery.

The expected correction (arm 1 greedy):

Derived (Proof Step).

$$\mathbb{E}[\mathbf{F}(\delta_\Sigma)] = \begin{pmatrix} (1-\varepsilon)\,\delta_1/(n_1+1) \\ \varepsilon\,\delta_2/(n_2+1) \end{pmatrix} \tag{N.20}$$

The sector product:

Derived (Proof Step).

$$\delta_\Sigma^T \mathbf{F} = \frac{(1-\varepsilon)\,\delta_1^2}{n_1+1} + \frac{\varepsilon\,\delta_2^2}{n_2+1} \geq \min\!\left(\frac{1-\varepsilon}{n_1+1},\ \frac{\varepsilon}{n_2+1}\right)(\delta_1^2+\delta_2^2) \tag{N.21}$$

Verification of (SA1). At $\delta_\Sigma = \mathbf{0}$: $\mathbf{F}(\mathbf{0}) = \mathbf{0}$. Satisfied. □

Optimal exploration rate. Maximizing α_Σ over ε by equalizing the two terms:

Derived (optimal exploration rate).

$$\varepsilon^* = \frac{n_2+1}{n_1+n_2+2}, \qquad \alpha_\Sigma^* = \frac{1}{n_1+n_2+2} \tag{N.22}$$

For equal experience ($n_1 = n_2$): $\varepsilon^* = 1/2$ (equal allocation). For unequal experience, ε^* allocates more trials to the arm with higher n (lower gain) to equalize correction rates.

The k-arm generalization. For k OR-alternatives with ε-uniform exploration (probability $\varepsilon/(k-1)$ per non-greedy arm):

Derived (k-arm OR-node sector parameter).

$$\alpha_\Sigma = \min\!\left(\frac{1-\varepsilon}{n_{\text{greedy}}+1},\ \min_{j\neq\text{greedy}} \frac{\varepsilon/(k-1)}{n_j+1}\right) \tag{N.23}$$

With optimal equal-rate exploration and equal experience: $\alpha_\Sigma = 1/(k(n+1))$. The sector parameter scales as $1/k$ — each additional alternative dilutes correction capacity.

Condition (SA3). For the sector condition to be satisfiable at all, the exploration rate must satisfy:

Derived (minimum exploration rate for persistence).

$$\varepsilon > \frac{\rho_\Sigma(n_{\max}+1)}{R_\Sigma} \tag{N.24}$$

A purely greedy agent ($\varepsilon = 0$) violates the sector condition because the unchosen alternative's mismatch grows without correction. This is the structural justification for (SA3) in 8.10: OR-nodes require deliberate exploration to maintain strategic persistence.

Persistence Conditions.

Substituting each proposition's α_Σ into the persistence threshold $\alpha_\Sigma > \rho_\Sigma / R_\Sigma$ from Proposition A.1 of A:

Single edge (B.1):

Derived (single-edge persistence threshold).

$$\frac{1}{n+1} > \frac{\rho_\Sigma}{R_\Sigma} \quad \iff \quad n < \frac{R_\Sigma}{\rho_\Sigma} - 1 \tag{N.25}$$

The critical experience level $n^* = R_\Sigma/\rho_\Sigma - 1$ is the point at which gain collapse occurs: the posterior has concentrated so much that the agent can no longer track environmental drift.

Two-edge observable (B.2):

Derived (two-edge persistence threshold, observable).

$$\min\left(\frac{1}{n_1+1}, \frac{\theta_1}{n_2+1}\right) > \frac{\rho_\Sigma}{R_\Sigma} \tag{N.26}$$

This decomposes into two independent conditions: $n_1 < R_\Sigma/\rho_\Sigma - 1$ and $n_2 < \theta_1 R_\Sigma/\rho_\Sigma - 1$. The downstream edge has a stricter bound, tightened by θ_1. Gain collapse hits downstream edges first.

Two-edge unobservable (B.3):

$$\frac{1}{n_\Phi+1} > \frac{\rho_\Sigma}{R_\Sigma} \quad \iff \quad n_\Phi < \frac{R_\Sigma}{\rho_\Sigma} - 1 \tag{N.27}$$

Same form as single-edge, applied to the plan-level aggregate.

Two-arm OR, ε-greedy (B.4):

Derived (OR-node persistence threshold).

$$\min\left(\frac{1-\varepsilon}{n_1+1}, \frac{\varepsilon}{n_2+1}\right) > \frac{\rho_\Sigma}{R_\Sigma} \tag{N.28}$$

With optimal exploration and equal experience ($n_1 = n_2 = n$):

$$n < \frac{R_\Sigma}{2\rho_\Sigma} - 1 \tag{N.29}$$

The critical experience is halved relative to the single-edge case. Each additional OR-alternative further reduces the critical level by $1/k$.

Proposition B.6: Mixed AND/OR with Common-Cause Node (L1 Augmented DAG).

Setup. Goal G reachable via two alternative actions A_1, A_2 (an OR-node), but both actions require a shared condition C (a common-cause node). The L1 DAG factors the common cause above the OR-structure: $G = \mathrm{AND}(C, G_{\mathrm{sub}})$ where $G_{\mathrm{sub}} = \mathrm{OR}(A_1, A_2)$. Three leaves: condition C with true probability θ_C, actions A_1, A_2 with true conditional probabilities $\theta_{1|C}, \theta_{2|C}$. Agent selects ε-greedy among A_1, A_2 (only testable when $C = 1$). Mismatch vector: $\delta_\Sigma = (\delta_C, \delta_{A_1}, \delta_{A_2})^T$.

Plan confidence: $\hat{P}_\Sigma = \hat{p}_C \cdot [1 - (1 - \hat{p}_{A_1})(1 - \hat{p}_{A_2})]$. Reference value: $\Phi = \theta_C \cdot [1 - (1 - \theta_{1|C})(1 - \theta_{2|C})]$. At truth, $\hat{P}_\Sigma = \Phi =$ actual plan success probability (L1 is unbiased).

Statement. Under Beta-Bernoulli updating with observable condition C and ε-greedy path selection, the expected correction function satisfies the sector condition globally with:

Derived (Conditional on Beta-Bernoulli model, L1 augmented DAG, observable common cause C, ε-greedy action selection).

$$\alpha_\Sigma = \min\left(\frac{1}{n_C + 1}, \frac{\theta_C(1 - \varepsilon)}{n_{A_1} + 1}, \frac{\theta_C \varepsilon}{n_{A_2} + 1}\right) \tag{N.30}$$

where A_1 is the greedy choice.

Proof.

Edge C (condition leaf, observed every trial):

$$\mathbb{E}[\Delta \hat{p}_C] = \frac{\theta_C - \hat{p}_C}{n_C + 1} = -\frac{\delta_C}{n_C + 1} \tag{N.31}$$

Edge A_1 (action leaf, tested when $C = 1$ and path 1 selected — probability $\theta_C(1 - \varepsilon)$):

$$\mathbb{E}[\Delta \hat{p}_{A_1}] = -\frac{\theta_C(1 - \varepsilon)\, \delta_{A_1}}{n_{A_1} + 1} \tag{N.32}$$

Edge A_2 (action leaf, tested when $C = 1$ and path 2 selected — probability $\theta_C \varepsilon$):

$$\mathbb{E}[\Delta \hat{p}_{A_2}] = -\frac{\theta_C \varepsilon\, \delta_{A_2}}{n_{A_2} + 1} \tag{N.33}$$

The expected correction function is diagonal:

$$\mathbf{F}(\delta_\Sigma) = \begin{pmatrix} \delta_C/(n_C + 1) \\ \theta_C(1 - \varepsilon)\, \delta_{A_1}/(n_{A_1} + 1) \\ \theta_C \varepsilon\, \delta_{A_2}/(n_{A_2} + 1) \end{pmatrix} \tag{N.34}$$

Verification of (SA1). $\mathbf{F}(\mathbf{0}) = \mathbf{0}$. ✓

Sector product:

$$\delta_\Sigma^T \mathbf{F} = \frac{\delta_C^2}{n_C + 1} + \frac{\theta_C(1-\varepsilon)\,\delta_{A_1}^2}{n_{A_1} + 1} + \frac{\theta_C\varepsilon\,\delta_{A_2}^2}{n_{A_2} + 1} \;\geq\; \alpha_\Sigma \|\delta_\Sigma\|^2 \quad \square \tag{N.35}$$

Three-way gating. This is the first mixed AND/OR case. The sector parameter α_Σ is determined by three independent gating mechanisms: 1. **Condition testing** $(1/(n_C + 1))$: the common cause is the easiest component to calibrate — tested every trial, same as B.1. 2. **Evidence starvation** (θ_C factor on action terms): both action edges are gated by the condition probability, same mechanism as B.2. 3. **Exploration gating** ($(1-\varepsilon)$ and ε factors): the two OR-alternatives compete for test opportunities, same mechanism as B.4.

The bottleneck is typically the explore action behind the condition: $\theta_C\varepsilon/(n_{A_2} + 1)$.

B.5b applies. The Jacobian $\mathbf{J} = (s_{G_{\text{sub}}},\ \hat{p}_C(1-\hat{p}_{A_2}),\ \hat{p}_C(1-\hat{p}_{A_1}))^T$ is non-negative (monotone AND/OR). Corrections are componentwise (each leaf updates independently). Therefore $\alpha_s = \alpha_c = \alpha_\Sigma$ — the strategy-plan-confidence error inherits the sector condition with no penalty.

Optimal exploration. Equalizing the action terms: $\varepsilon^* = (n_{A_2} + 1)/(n_{A_1} + n_{A_2} + 2)$, giving $\alpha_\Sigma^* = \min(1/(n_C + 1),\ \theta_C/(n_{A_1} + n_{A_2} + 2))$.

L0 comparison. An L0 model of the same scenario (no C node, marginal probabilities $\theta_k = \theta_C \cdot \theta_{k|C}$) has $\alpha_\Sigma^{L0} = \min((1-\varepsilon)/(n_1 + 1),\ \varepsilon/(n_2 + 1))$ and reference value $\Phi^{L0} = 1 - (1-\theta_1)(1-\theta_2)$. The L0 sector parameter is *higher* (no θ_C attenuation) but Φ^{L0} systematically overestimates actual plan success. L1 persistence is harder (lower α_Σ) but calibration is honest (Φ^{L1} matches reality). The tradeoff: L0 is easier to maintain but calibrates to a biased target; L1 is harder to maintain but calibrates to truth.

The L1 construction principle. The common-cause node C must be factored *above* the OR-structure whose children it correlates. A naive construction (adding C as a parent of both A_1 and A_2 while keeping them as OR-siblings) does not fix the overestimation because the OR-propagation formula treats siblings as marginally independent. Correct L1 construction makes C an AND-prerequisite above the OR-structure, so the OR-children are conditionally independent given C. Full worked example with quantitative L0/L1 comparison: #example-L1.

Summary of Results.

Table N.1

Case	*Topology*	α_Σ	*SA1*	*SA3*	*Weakest link*
B.1 Single edge	$A \to G$	$1/(n+1)$	Yes	N/A	—
B.2 Two-edge, B observable	$A \to B \to G$ (AND)	$\min(1/(n_1{+}1),\ \theta_1/(n_2{+}1))$	Yes	N/A	Depth-gated (evidence starvation)
B.3 Two-edge, B unob-servable	$A \to B \to G$ (AND)	$1/(n_\Phi{+}1)$ (plan-level)	Per-edge: No ($O(1/n)$ bias); Plan: Yes	N/A	Credit-assignment collapse

Case	*Topology*	α_Σ	*SA1*	*SA3*	*Weakest link*
B.4 Two-arm OR, ε-greedy	$A_1, A_2 \to G$ (OR)	$\min((1-\varepsilon)/(n_1+1),\ \varepsilon/(n_2+1))$	Yes	Required	Exploration-gated
B.5a Credence→value (linear)	Any	$\alpha_s = \alpha_c$ (Jacobian cancels)	—	—	None (exact transfer)
B.5b Credence→value (nonlinear, componentwise)	Any	$\alpha_s = \alpha_c$ ($J_k \geq 0$ preserves bound)	—	—	None (exact transfer)
B.5b Credence→value (coupled)	Any	$\alpha_s \geq \alpha_c/\kappa(\mathbf{J})^2$	—	—	Inter-edge coupling
B.6 L1 augmented, mixed AND/OR	$G = \mathrm{AND}(C, \mathrm{OR}(A_1, A_2))$	$\min(1/(n_C+1),\ \theta_C(1-\varepsilon)/(n_{A_1}+1),\ \theta_C\varepsilon/(n_{A_2}+1))$	Yes	Required	Three-way gating
B.7 L1' mixture form, observable C	$\hat{P}_\Sigma^{L1'} = \hat{\theta}_C P_\Sigma(\hat{\mathbf{p}}_{\mid C}) + (1-\hat{\theta}_C)P_\Sigma(\hat{\mathbf{p}}_{\mid\neg C})$	$\min(1/(n_C+1),\ \min_j \theta_C\pi_{j\mid C}/(n_{j\mid C}+1),\ \min_j(1-\theta_C)\pi_{j\mid\neg C}/(n_{j\mid\neg C}+1))$	Yes	Required *and* facilitator monotonicity	Five-way gating; **refuted under unobservable** C (Cramér-Rao floor)

Structural results across cases:

- **Gain as sector parameter.** In every case, α_Σ is determined by the edge update gain $\eta_k = 1/(n_k+1)$, modulated by observability (θ_1 attenuation in B.2), credit-assignment (plan-level collapse in B.3), action-selection policy (ε in B.4), or a combination of these (B.6). The structural parallel between epistemic and strategic persistence is not an analogy — it is a mathematical identity at the sector-framework level.
- **Evidence starvation (AND-chains).** Downstream edges in AND-chains have correction rates attenuated by the product of upstream success probabilities. Depth increases fragility: α_Σ decays exponentially with chain depth.
- **Exploration gating (OR-nodes).** Unchosen OR-alternatives receive zero correction. The sector condition requires deliberate exploration (SA3) at a rate exceeding $\rho_\Sigma(n_{\max}+1)/R_\Sigma$. Pure greedy policies fail.
- **Three-way gating (L1 augmented).** When a common cause gates OR-alternatives, calibration faces condition testing, evidence starvation, and exploration gating simultaneously (B.6). The bottleneck is the least-explored action behind the rarest condition.
- **Five-way gating (L1' mixture, observable C).** When the common cause is a *soft facilitator* ($\theta_{j\mid\neg C} > 0$) and is observable per trial, calibration faces five gating mechanisms simultaneously (B.7): condition testing, evidence starvation on each conditional branch ($C = 1$ and $C = 0$), and exploration gating on each branch. The bottleneck is whichever branch is rarer, modulated by exploration on that branch. **Identifiability obstruction under unobservable C:** without direct C-observation, the mixture parameters are non-identifiable from a single observation channel — the per-trial Fisher information matrix is rank 1 rather than rank $2K+1$, so by the Cramér-Rao bound (Cramér 1946, *Mathematical Methods of Statistics*, Princeton University Press) no unbiased online estimator on the joint conditional vector admits $\alpha > 0$. The agent must augment C-observability, run multi-child joint observations, or fall back to plan-level (L0-on-marginals) tracking.

- **Gain-collapse threshold.** In all cases, increasing experience eventually violates persistence when the environment drifts. The critical experience level n^* is inversely proportional to the drift rate ρ_Σ.
- **L1 tradeoff.** L1 augmented DAGs have lower α_Σ (harder to persist) but honest calibration (Φ^{L1} matches reality). L0 is easier to maintain but calibrates to a biased target. The choice depends on whether accuracy or trackability is the binding constraint.

Proposition B.7: L1' Mixture-Form Sector Transfer (Observable Common Cause).

Where Prop B.6 handles a strict-prerequisite common cause via an AND-prerequisite construction (L1), this proposition handles a *soft-facilitator* common cause via the mixture form L1' (7.3, Correlation Hierarchy). It is the L1' analog of B.6, generalizing three-way gating to five-way gating across two parallel conditional branches.

Setup. L1' (mixture-form) DAG with binary common cause C, true prior $\theta_C \in (0,1)$, gating a sub-plan with conditional structures $G_{|C}$ (when $C = 1$) and $G_{|\neg C}$ (when $C = 0$). Each affected child j carries two conditional credences $\hat{p}_{j|C}$ and $\hat{p}_{j|\neg C}$. Plan confidence:

$$\hat{P}_\Sigma^{L1'} = \hat{\theta}_C \cdot P_\Sigma(\hat{\mathbf{p}}_{|C}) + (1 - \hat{\theta}_C) \cdot P_\Sigma(\hat{\mathbf{p}}_{|\neg C}) \tag{N.36}$$

The agent observes C-state directly each trial (i.e., C is treated as an observable condition leaf, in parallel with B.6). Per trial, the agent (i) updates $\hat{\theta}_C$ from observed C via Beta-Bernoulli; (ii) on $C = 1$ trials, attempts a child j via the active selection policy $\pi_{|C}$ and updates $\hat{p}_{j|C}$ from y_j; (iii) on $C = 0$ trials, attempts a child j via $\pi_{|\neg C}$ and updates $\hat{p}_{j|\neg C}$ from y_j.

Mismatch state: $\xi = (\xi_C, \{\xi_{j|C}\}, \{\xi_{j|\neg C}\})$ with $\xi_C = \theta_C - \hat{\theta}_C$, $\xi_{j|s} = \theta_{j|s} - \hat{p}_{j|s}$.

Statement.

Under Beta-Bernoulli updating with observable common cause, componentwise updates per conditional branch, and *facilitator monotonicity* ($P_\Sigma(G_{|C}) \geq P_\Sigma(G_{|\neg C})$), the expected correction function on the joint state satisfies the sector condition globally with:

Derived (Conditional on Beta-Bernoulli model, observable common cause C, componentwise conditional-branch update, facilitator monotonicity).

$$\alpha_{L1'} = \min\left(\frac{1}{n_C + 1},\ \min_{j \in \mathcal{J}_C} \frac{\theta_C \pi_{j|C}}{n_{j|C} + 1},\ \min_{j \in \mathcal{J}_{\neg C}} \frac{(1 - \theta_C)\pi_{j|\neg C}}{n_{j|\neg C} + 1}\right) \tag{N.37}$$

where $\mathcal{J}_s$ is the set of children tested on the $C = s$ branch and $\pi_{j|s}$ is the action-selection probability for child j on that branch.

Proof. *Edge C (condition leaf, observable every trial).* Standard Beta-Bernoulli on direct C-observation:

$$\mathbb{E}[\Delta\hat{\theta}_C] = \frac{\theta_C - \hat{\theta}_C}{n_C + 1} = -\frac{\xi_C}{n_C + 1} \tag{N.38}$$

so $F_C(\xi) = \xi_C/(n_C + 1)$.

Edge $p_{j|C}$ (conditional credence, $C = 1$ branch). Tested when $C = 1$ (probability θ_C) and the agent's $C = 1$-conditional policy selects j (probability $\pi_{j|C}$):

$$\mathbb{E}[\Delta \hat{p}_{j|C}] = \theta_C \pi_{j|C} \cdot \frac{\theta_{j|C} - \hat{p}_{j|C}}{n_{j|C} + 1} = -\frac{\theta_C \pi_{j|C} \xi_{j|C}}{n_{j|C} + 1} \tag{N.39}$$

so $F_{j|C}(\xi) = \theta_C \pi_{j|C} \xi_{j|C} / (n_{j|C} + 1)$.

Edge $p_{j|\neg C}$ (conditional credence, $C = 0$ branch). Symmetric: tested with probability $(1 - \theta_C)\pi_{j|\neg C}$:

$$F_{j|\neg C}(\xi) = \frac{(1 - \theta_C)\pi_{j|\neg C} \xi_{j|\neg C}}{n_{j|\neg C} + 1} \tag{N.40}$$

Sector product. The correction function is diagonal in ξ with positive entries:

$$\xi^T \mathbf{F}(\xi) = \frac{\xi_C^2}{n_C + 1} + \sum_{j \in \mathcal{J}_C} \frac{\theta_C \pi_{j|C} \xi_{j|C}^2}{n_{j|C} + 1} + \sum_{j \in \mathcal{J}_{\neg C}} \frac{(1 - \theta_C)\pi_{j|\neg C} \xi_{j|\neg C}^2}{n_{j|\neg C} + 1} \ \geq \ \alpha_{L1'} \cdot \|\xi\|^2 \tag{N.41}$$

with $\alpha_{L1'}$ as stated.

SA1 check. $\mathbf{F}(\mathbf{0}) = \mathbf{0}$ at truth ($\xi = 0$). ✓

Globality. The Beta-Bernoulli updates are linear in ξ on $[-1, 1]^{2K+1}$, so the diagonal Jacobian computed at truth is the exact $\mathbf{J}_\mathbf{F}$ throughout. The sector inequality holds globally with the same $\alpha_{L1'}$.

B.5b bridge to plan value. Jacobian of $\hat{P}_\Sigma^{L1'}$ with respect to the joint state:

$$\mathbf{J}_{P_\Sigma} = \begin{pmatrix} P_\Sigma(\hat{\mathbf{p}}_{|C}) - P_\Sigma(\hat{\mathbf{p}}_{|\neg C}) \\ \hat{\theta}_C \cdot \nabla_{\hat{\mathbf{p}}_{|C}} P_\Sigma(\hat{\mathbf{p}}_{|C}) \\ (1 - \hat{\theta}_C) \cdot \nabla_{\hat{\mathbf{p}}_{|\neg C}} P_\Sigma(\hat{\mathbf{p}}_{|\neg C}) \end{pmatrix} \tag{N.42}$$

By facilitator monotonicity, the first entry is ≥ 0. The other entries are ≥ 0 by AND/OR monotonicity on each conditional sub-DAG. So $\mathbf{J}_{P_\Sigma} \geq 0$ componentwise. By the componentwise nonlinear case of B.5b, the sector condition transfers losslessly: $\alpha_s = \alpha_c = \alpha_{L1'}$. □

Five-Way Gating. This generalizes B.6's three-way gating to five gating mechanisms, one per term in the min:

1. **Condition testing** ($1/(n_C + 1)$): the common cause is the easiest component to calibrate — directly observed every trial.
2. **Evidence starvation, $C = 1$ branch** (θ_C factor): conditional edges on the $C = 1$ branch are gated by the prior frequency of $C = 1$.
3. **Exploration gating, $C = 1$ branch** ($\pi_{j|C}$ factor): OR-alternatives within the $C = 1$ branch compete for test opportunities.
4. **Evidence starvation, $C = 0$ branch** ($1 - \theta_C$ factor): conditional edges on the $C = 0$ branch are gated by the prior frequency of $C = 0$.
5. **Exploration gating, $C = 0$ branch** ($\pi_{j|\neg C}$ factor): OR-alternatives within the $C = 0$ branch.

The bottleneck is typically whichever of θ_C and $1 - \theta_C$ is smaller (the rare branch sees fewer trials), modulated by exploration on that branch.

Reduction to B.6 (Strict-Prerequisite Limit). Setting $\theta_{j|\neg C} \to 0$ collapses the $\neg C$ branch (the agent learns the $\neg C$ conditionals are zero with no remaining variance). The per-edge state for the $\neg C$ branch drops out of the sector inequality, and the formula reduces to B.6's three-way gating with $\alpha_\Sigma = \min(1/(n_C + 1), \theta_C(1-\varepsilon)/(n_{A_1} + 1), \theta_C\varepsilon/(n_{A_2} + 1))$. ✓

Refuted Under Unobservable C (Mixture Identifiability Obstruction).

When C is not observable, the agent can only update the joint state via online soft EM on the marginal y_j. Computing the Fisher information of the mixture model $\mu_j = \theta_C\theta_{j|C} + (1-\theta_C)\theta_{j|\neg C}$ at truth, the score vector for parameters $\phi = (\theta_C, p_{j|C}, p_{j|\neg C})$ admits the rank-1 factorization:

Derived (cramer-rao-floor, from Fisher information of mixture model + Cramér-Rao bound).

$$\mathcal{F}(\phi) = \frac{1}{\mu_j(1-\mu_j)}\, uu^T, \qquad u = (\Delta_j,\ \theta_C,\ 1-\theta_C) \tag{N.43}$$

where $\Delta_j = p_{j|C} - p_{j|\neg C}$ is the *separability gap*. The matrix is rank 1 for any non-degenerate mixture; the two-dimensional null space corresponds to perturbations along the indeterminacy manifold $\{\hat{\phi} : \hat{\theta}_C\hat{p}_{|C} + (1-\hat{\theta}_C)\hat{p}_{|\neg C} = \mu_j\}$ — directions unobservable from a single binary signal.

Since the soft-EM step at truth equals (up to the $1/(n+1)$ scaling) the natural-gradient ascent on the expected log-likelihood, the Jacobian of the correction function equals $\mathcal{F}/(n+1)$, also rank 1. Therefore $\lambda_{\min}(\mathbf{J}_F) = 0$, and **no SA1-preserving update on the joint conditional vector admits a sector parameter $\alpha > 0$ under SUB-B.** This is a Cramér-Rao bound, not a defect of any particular update rule: any unbiased online estimator must respect the bound, which is infinite in the unidentifiable directions.

Repair routes. When C is unobservable, the agent must:

(i) **Augment C-observability** (recover B.7 above) — instrument secondary signals that identify C per trial, transforming the problem from refuted to globally derived. (ii) **Run $K \geq 2$ children jointly under the same C-realization** — when the joint Fisher matrix reaches rank $2K + 1$, the mixture is identifiable; local sector condition holds with $\alpha \geq \sigma_{\min}(\mathcal{F}_{\text{joint}})/\max_k(n_k+1)$. Strong structural requirement (joint observation typically unavailable in standard sequential strategy execution). (iii) **Fall back to plan-level tracking on the marginal $\hat{\mu}_j$** — recovers B.1's $\alpha = 1/(n_\mu + 1)$ on the scalar marginal but loses the per-conditional decomposition (equivalent to L0-on-marginals).

Connection to the Identifiability-Floor Pattern. The B.7-vs-refuted-unobservable-C structure is one of two no-go theorems in AAT's strategy layer. The other is the on-policy detection no-go in 8.2, which prohibits L0-vs-L1 distinction from purely on-policy data via the causal hierarchy theorem. Both are structural impossibility results derived from external information-theoretic theorems (Bareinboim et al. 2022 Causal Hierarchy Theorem; Cramér-Rao bound on the Fisher information). Both strengthen related machinery — the on-policy no-go strengthens 6.3; the mixture no-go strengthens the case for *observability-as-information-augmentation*. See X for the meta-pattern.

Proposition B.5: Bridge from Credence Error to Value Residuals.

The propositions above verify the sector condition for the credence-error mismatch state $\delta_c = (\hat{p}_k - \theta_k)$. This proposition extends the result to **strategy-plan-confidence error** — the scalar difference between the agent's strategy-plan-confidence score and the independence-model plan value evaluated at true edge parameters. This is a natural plan-level mismatch that is distinct from (but related to) the per-edge value-increment residuals $\delta_{\text{strategic}}$ defined in 8.1.

Relationship to $\delta_{\text{strategic}}$. 8.1 defines $\delta_{\text{strategic}}$ as an L^2 aggregation of per-edge value-increment residuals — a quantity that requires credit assignment to compute. Plan-confidence error δ_s defined here is a *different* quantity: it is the error in the DAG's aggregate self-assessment, computable from status propagation without credit assignment. The two are related (both measure strategy-reality mismatch) but not identical. This proposition shows the sector condition holds for strategy-plan-confidence error; extending it to $\delta_{\text{strategic}}$ directly would require the credit-assignment machinery discussed in 8.6.

Setup. The plan value $\hat{P}_\Sigma = P_\Sigma(\mathbf{p})$ is a function of the edge credence vector $\mathbf{p} = (p_1, \dots, p_m)$. The **independence-model reference value** is $\Phi = P_\Sigma(\theta)$ — the AND/OR propagation formula evaluated at the true edge parameters θ. Note: Φ is NOT the actual probability of plan success when edge outcomes are correlated (7.3, edge-independence caveat). It is the best the independence model can do with perfect edge knowledge. Under correlated failure, the actual plan success probability may be lower than Φ; the gap is a model-class limitation of the AND/OR DAG representation, not an estimation error. What δ_s tracks is calibration *within the independence model*, not calibration to reality. Define:

- **Credence-error mismatch:** $\delta_c = \mathbf{p} - \theta \in \mathbb{R}^m$
- **Plan-confidence error:** $\delta_s = \hat{P}_\Sigma - \Phi = P_\Sigma(\mathbf{p}) - P_\Sigma(\theta)$ (scalar)

The gradient of plan value with respect to credences is $\mathbf{J} = \nabla_{\mathbf{p}} P_\Sigma \in \mathbb{R}^m$ (a vector, since P_Σ is scalar). By first-order Taylor expansion:

$$\delta_s \approx \mathbf{J}^T \delta_c \tag{N.44}$$

B.5a: Linear Correction (Beta-Bernoulli).

When the correction in credence space is linear — $\mathbf{F}_c(\delta_c) = \eta \odot \delta_c$ where $\eta_k = 1/(n_k + 1)$ and $\odot$ is elementwise product — consider the sector ratio directly. Since $\delta_s = \mathbf{J}^T \delta_c$ and $F_s = \mathbf{J}^T \mathbf{F}_c$:

Derived (value-residual sector condition, linear case).

$$\delta_s \cdot F_s = (\mathbf{J}^T \delta_c)^T (\mathbf{J}^T \mathbf{F}_c) = \delta_c^T \mathbf{J}\mathbf{J}^T \mathbf{F}_c \tag{N.45}$$

For the linear case $\mathbf{F}_c = \eta_{\min} \delta_c$ (using the worst-case uniform gain for the bound):

$$\delta_s \cdot F_s = \eta_{\min} \cdot \delta_c^T \mathbf{J}\mathbf{J}^T \delta_c = \eta_{\min} \cdot \|\mathbf{J}^T \delta_c\|^2 = \eta_{\min} \cdot \delta_s^2 \tag{N.46}$$

Therefore:

$$\frac{\delta_s \cdot F_s}{\delta_s^2} = \eta_{\min} \tag{N.47}$$

The Jacobian cancels. The sector parameter in value-residual space equals the sector parameter in credence space, regardless of DAG structure:

$$\alpha_s = \alpha_c = \eta_{\min} = \min_k \frac{1}{n_k + 1} \tag{N.48}$$

This holds because $\mathbf{J}\mathbf{J}^T$ appears in both numerator and denominator of the sector ratio, and the linear correction commutes with the Jacobian mapping.

Interpretation. The Jacobian $\mathbf{J}$ determines *which* credence errors matter most for plan value (high-sensitivity edges contribute more to value residuals). But it does not change the *rate* at which those errors are corrected, because the correction is proportional to the error regardless of the error's plan-value impact.

B.5b: Nonlinear Componentwise Correction.

When the correction is nonlinear but **componentwise** — each edge corrects independently, so $(\mathbf{F}_c)_k$ depends only on $(\delta_c)_k$ — and the plan-value Jacobian is **non-negative** ($J_k \geq 0$ for all k), the transfer is lossless regardless of nonlinearity.

Derived (from per-component sector condition + Jacobian non-negativity).

The per-component sector condition (from Props B.1-B.4): for each edge k,

$$(\delta_c)_k \cdot (\mathbf{F}_c)_k \geq \alpha_c \cdot (\delta_c)_k^2 \tag{N.49}$$

which gives $(\mathbf{F}_c)_k/(\delta_c)_k \geq \alpha_c$ (the correction-to-mismatch ratio exceeds α_c for each edge independently).

Jacobian non-negativity holds for all well-formed AND/OR DAGs: increasing any edge credence p_k never decreases plan value P_Σ (monotonicity of AND/OR propagation). Therefore $J_k = \partial P_\Sigma/\partial p_k \geq 0$ for all k.

The transfer. Since $J_k \geq 0$ preserves the inequality direction:

$$J_k \cdot (\mathbf{F}_c)_k \geq J_k \cdot \alpha_c \cdot (\delta_c)_k \quad \text{for each } k \tag{N.50}$$

Summing:

$$\mathbf{J}^T\mathbf{F}_c = \sum_k J_k(\mathbf{F}_c)_k \geq \alpha_c \sum_k J_k(\delta_c)_k = \alpha_c \cdot \mathbf{J}^T\delta_c \tag{N.51}$$

Therefore $F_s \geq \alpha_c \cdot \delta_s$, giving:

$$\alpha_s = \alpha_c \tag{N.52}$$

No condition-number penalty. The Jacobian's anisotropy is irrelevant when the correction is componentwise and the Jacobian is non-negative. The non-negativity ensures that every edge's correction contributes in the right direction to the value correction, regardless of how unevenly the Jacobian weights different edges.

When does componentwise correction hold? For all cases in Props B.1, B.2, and B.4 — the standard Beta-Bernoulli update on each edge, including the stochastic (nonlinear) single-step realization. The correction IS nonlinear (the actual per-step update $(y-\hat{p})/(n+1)$ is random, not proportional to δ_k), but it is componentwise (each edge updates from its own observation independently).

When does it fail? When edge corrections are coupled — correcting one edge changes another's correction direction. This occurs for: - **Unobservable intermediates (B.3) under marginal Bayesian updates:** The marginal Bayesian update couples edge posteriors. The correction for edge 2 depends on the joint posterior over both edges, not just on $(\delta_c)_2$ alone. *And — critically — Prop B.3(a) showed that this scheme violates SA1 with $O(1/n)$ bias: there is no valid α_c for the marginal Bayesian update on coupled edges in the first place.* The per-edge bound must therefore be re-established before the Jacobian transfer can be applied. - **Shared observations:** If testing one arm gives partial information about another (correlated arms), corrections couple across edges — with an SA1 status that depends on the specific attribution scheme used to project the shared observation onto per-edge updates.

The coupled-case Cauchy-Schwarz bound presupposes a valid α_c. For the bound $\alpha_s \geq \alpha_c/\kappa(\mathbf{J})^2$ to apply meaningfully, the per-edge sector condition must hold — i.e., the attribution scheme must satisfy SA1. Marginal Bayesian updates on coupled edges do *not* satisfy this precondition (Prop B.3(a)). The resolution is to replace marginal Bayesian with **gradient-based attribution** (Prop B.5d below), which recovers SA1 and yields a valid per-edge sector parameter that admits the Jacobian transfer with the κ^2 penalty.

Scope (coupled-case-applicability).

Per-edge vs. plan-level routes through coupling. Prop B.3(b) already established an alternative route for the two-edge unobservable case: tracking at the plan level ($\hat{\Phi} = \hat{p}_1\hat{p}_2$) with a single Beta posterior recovers SA1 directly, yielding $\alpha_\Sigma = 1/(n_\Phi + 1)$. The plan-level route works *without* needing the Jacobian bridge because it sidesteps per-edge attribution entirely. The gradient-based per-edge route works *through* the Jacobian bridge, preserving per-edge credences at the cost of the κ^2 penalty. Both routes give valid sector parameters; the choice depends on whether per-edge diagnostics or aggregate efficiency matters more.

The refined picture: nonlinearity is not the problem; inter-edge coupling combined with the attribution scheme is. A nonlinear but componentwise correction transfers losslessly through any non-negative Jacobian. A coupled correction under a scheme that preserves SA1 (gradient-based) incurs the condition-number penalty because the coupling can redirect correction away from the value-relevant direction. A coupled correction under a scheme that violates SA1 (marginal Bayesian) has no valid α_c to transfer at all and must be replaced before B.5 applies.

Proposition B.5d: Gradient-Based Attribution Restores SA1 on Coupled Edges. **Setup.** Same two-edge AND topology as B.3: edges with true probabilities θ_1, θ_2, credences $\hat{p}_1, \hat{p}_2$, plan probability $\hat{P}_\Sigma = \hat{p}_1\hat{p}_2$, observable plan outcome $y_G \in \{0, 1\}$ with $\mathbb{E}[y_G] = \Phi = \theta_1\theta_2$. Intermediate is unobservable.

Gradient-based per-edge update. Distribute plan-level surprise proportionally to the Jacobian component:

Formulation (gradient-attribution).

$$\Delta\hat{p}_k \;=\; \frac{1}{n_k + 1} \cdot J_k \cdot (y_G - \hat{P}_\Sigma), \qquad J_k = \frac{\partial \hat{P}_\Sigma}{\partial \hat{p}_k} \tag{N.53}$$

For the two-edge AND case, $J_1 = \hat{p}_2, J_2 = \hat{p}_1$.

Verification of SA1. At truth ($\hat{p}_k = \theta_k$):

$$\mathbb{E}[\Delta\hat{p}_k]\big|_{\hat{\mathbf{p}}=\theta} \;=\; \frac{J_k}{n_k + 1} \cdot \mathbb{E}[y_G - \hat{P}_\Sigma]\big|_{\hat{\mathbf{p}}=\theta} \tag{N.54}$$

The bracket evaluates as: $\mathbb{E}[y_G] = \Phi$ by definition of y_G as a Bernoulli outcome with mean Φ; and $\hat{P}_\Sigma\big|_{\hat{\mathbf{p}}=\theta} = \theta_1\theta_2 = \Phi$. Therefore $\mathbb{E}[y_G - \hat{P}_\Sigma]\big|_{\hat{\mathbf{p}}=\theta} = 0$, and:

$$\mathbb{E}[\Delta\hat{p}_k]\big|_{\hat{\mathbf{p}}=\theta} \;=\; 0 \quad \text{for all } k. \qquad \text{(SA1 satisfied)} \tag{N.55}$$

The contrast with marginal Bayesian is precise: the marginal scheme suffers from a Φ-vs-θ_k mismatch (the update treats each edge's marginal likelihood rather than the joint), producing the $O(1/n)$ bias derived in Prop B.3(a). The gradient scheme distributes surprise via the Jacobian, which aligns the per-edge update direction with the plan-level gradient — and this gradient is zero at truth by construction.

Sector parameter (SA2'). With the gradient-based scheme, the per-edge correction is:

$$F_k(\delta) \;=\; \frac{J_k}{n_k + 1} \cdot (\Phi - \hat{P}_\Sigma) \;\approx\; \frac{J_k}{n_k + 1} \cdot \mathbf{J}^T\delta \tag{N.56}$$

where the approximation uses the first-order expansion $\hat{P}_\Sigma - \Phi \approx \mathbf{J}^T\delta$ valid near truth. The per-edge sector bound is:

$$\delta^T \mathbf{F}(\delta) \;\approx\; \sum_k \frac{J_k^2}{n_k+1} \cdot (\mathbf{J}^T\delta)^2/\|\delta\| \;\geq\; \frac{\sigma_{\min}(\mathbf{J})^2}{\max_k(n_k+1)}\|\delta\|^2 \tag{N.57}$$

giving:

$$\alpha_c^{\text{grad}} \;\geq\; \frac{\sigma_{\min}(\mathbf{J})^2}{\max_k(n_k+1)} \tag{N.58}$$

Transfer through B.5c. Applying the coupled-case Cauchy-Schwarz bound:

$$\alpha_s \;\geq\; \frac{\alpha_c^{\text{grad}}}{\kappa(\mathbf{J})^2} \;=\; \frac{\sigma_{\min}(\mathbf{J})^2/\max_k(n_k+1)}{\sigma_{\max}(\mathbf{J})^2/\sigma_{\min}(\mathbf{J})^2} \;=\; \frac{\sigma_{\min}(\mathbf{J})^4}{\sigma_{\max}(\mathbf{J})^2 \cdot \max_k(n_k+1)} \tag{N.59}$$

The $\sigma_{\min}(\mathbf{J})^4/\sigma_{\max}(\mathbf{J})^2$ factor is the full cost of coupled per-edge attribution — a steep penalty compared to the componentwise case. The plan-level route (B.3(b)) avoids this penalty entirely by tracking Φ as a single quantity with $\alpha_\Sigma = 1/(n_\Phi + 1)$, at the cost of losing per-edge diagnostics.

Why not proportional-blame or other schemes? Proportional-blame distributes surprise proportionally to current credences: $\Delta\hat{p}_k \propto \hat{p}_k \cdot (y_G - \hat{P}_\Sigma)/\sum_j \hat{p}_j$. This violates SA1 because the proportionality factor $\hat{p}_k/\sum_j \hat{p}_j$ is not the gradient direction, so the distribution over edges does not align with the plan-level surprise's actual attribution. Only schemes whose per-edge update aligns with the Jacobian direction preserve SA1 for coupled edges; gradient-based is the canonical such scheme.

Claim. For coupled edges with unobservable intermediates, gradient-based attribution is the minimal scheme that (a) satisfies SA1, (b) yields a well-defined per-edge sector parameter, (c) admits the Cauchy-Schwarz Jacobian transfer, and (d) permits per-edge diagnostics. Its $\sigma^4_{\min}/\sigma^2_{\max}$ scaling is inherent to the coupled case, not an artifact of the scheme. □

B.5c: Implications. Four regimes for the credence-to-value bridge, indexed by correction type and attribution scheme:

Table N.2

Correction type	*Edge coupling*	*Attribution*	*SA1*	*Transfer*	α_s
Linear	Any	—	Yes	Exact (Jacobian cancels)	α_c
Nonlinear, componentwise	Independent edges	Per-edge Bayesian	Yes	Exact ($J_k \geq 0$ preserves bound)	α_c
Nonlinear, coupled	Coupled edges	Gradient-based (Jacobian distribution)	Yes	$\kappa(\mathbf{J})^2$ penalty	$\alpha_c^{\text{grad}}/\kappa^2$
Nonlinear, coupled	Coupled edges	Marginal Bayesian	**No** (O(1/n) bias)	— (Jacobian bridge does not apply)	plan-level alternative (7.3)

Componentwise correction closes the gap for all verified cases. Props B.1, B.2, and B.4 all use componentwise edge updates with non-negative Jacobian. The operational diagnostic $\delta_{\text{strategic}}$ inherits the sector condition from credence error with no penalty — even for the nonlinear (stochastic) single-step realization. The derivation validates the operational mismatch, not merely a surrogate.

Inter-edge coupling — not nonlinearity — is the fragility. The κ^2 penalty arises only when correcting one edge changes another's correction direction (B.3's unobservable case, correlated observations). Deep or unbalanced DAGs have high $\kappa(\mathbf{J})$ when corrections couple, but the coupling, not the depth or imbalance per se, is what degrades the persistence guarantee.

Connection to credit assignment. The Jacobian $\mathbf{J} = \nabla_{\mathbf{p}} P_\Sigma$ is computable from the DAG's status propagation formulas — it does not require solving the credit-assignment problem. The credit-assignment problem is about *decomposing observed value changes into per-edge contributions* (needed for the update rule). The Jacobian bridge is about *transferring a per-edge sector condition to the value-residual space* (needed for the persistence guarantee). These are different problems: the bridge works even when credit assignment is unsolved, because it only requires the sensitivity structure, not the causal attribution.

What Is Derived vs. What Is Chosen.

The segment carries derivations at three distinct strengths: (i) exact sector-parameter algebra for concrete Beta-Bernoulli topologies (B.1, B.2, B.4, B.6, B.7 under observable C + facilitator monotonicity), (ii) two structural refutations that are *themselves* derivations of impossibility (B.3(a) SA1 bias; B.7 Cramér-Rao floor under unobservable C), and (iii) formulation choices (Beta-Bernoulli dynamics, componentwise conditional-branch updates, the particular ε-greedy policy family). The dividing line runs between the Beta-Bernoulli *model* (chosen) and the sector-parameter *algebra conditional on that model* (proved). B.7's positive transfer requires two load-bearing conditions — observability of C *and* facilitator monotonicity $P_\Sigma(G \mid C) \geq P_\Sigma(G \mid \neg C)$ — either of which failing disqualifies the transfer.

Table N.3

Property	*Source*	*Strength*
Beta-Bernoulli edge dynamics ($\Delta\hat{p}_k = (y_k - \hat{p}_k)/(n_k+1)$)	Conjugate-prior model; matches 8.4	Formulation choice
Regime-A adjustment: $\alpha_\Sigma^{\text{eff}} = \iota_{ij} \cdot \alpha_\Sigma^{\text{stated}}$	Diagonal commutation across Props B.1–B.4, B.6; 8.5	Derived
B.1 Single-edge sector parameter $\alpha_\Sigma = 1/(n+1)$	Expected Beta-Bernoulli update + sector product algebra	Proved (tight, conditional on Beta-Bernoulli)
B.1 Stochastic ultimate bound $\lvert\delta\rvert > \sqrt{\theta(1-\theta)/(2n+1)}$	Second-moment recursion + expected-Lyapunov decrement	Derived
B.2 Observable AND-chain sector parameter $\min(1/(n_1+1),\ \theta_1/(n_2+1))$	Diagonal correction function; sector product algebra	Proved (conditional on Beta-Bernoulli + observable intermediate)
B.2 Depth-d generalization $\alpha_\Sigma = \min_k \prod_{j<k} \theta_j/(n_k+1)$	Iteration of B.2 pattern under observable intermediates	Derived
B.3(a) Per-edge SA1 violation under marginal Bayesian attribution ($O(1/n)$ bias)	Exact marginal update + evaluation at truth	Derived as refutation (no valid α_c exists)
B.3(b) Plan-level recovery $\alpha_{\Sigma,\text{plan}} = 1/(n_\Phi+1)$	Reduction of aggregate $\hat{\Phi}$ tracking to B.1	Derived (reduction)
B.3 Non-identifiability of (θ_1, θ_2) from y_G alone	Product $\Phi = \theta_1\theta_2$ symmetry in observation channel	Proved
B.4(a) Pure-greedy OR ($\varepsilon = 0$) violates sector condition	Counterexample at $\delta_1 = 0,\ \delta_2 \neq 0$	Derived as refutation
B.4(b) ε-greedy sector parameter $\min((1-\varepsilon)/(n_1+1),\ \varepsilon/(n_2+1))$	Policy-weighted correction; sector product algebra	Proved (conditional on Beta-Bernoulli + ε-greedy)
B.4 Optimal exploration $\varepsilon^* = (n_2+1)/(n_1+n_2+2)$	Term-equalization in min	Derived
B.4 Minimum exploration rate (SA3 requirement) $\varepsilon > \rho_\Sigma(n_{\max}+1)/R_\Sigma$	Substitution into persistence threshold $\alpha_\Sigma > \rho_\Sigma/R_\Sigma$	Derived

Property	*Source*	*Strength*
B.5a Linear credence-to-value transfer $\alpha_s = \alpha_c$ (Jacobian cancels)	Algebraic cancellation of $\mathbf{J}\mathbf{J}^T$	Proved (exact, DAG-structure independent)
B.5b Componentwise nonlinear transfer $\alpha_s = \alpha_c$	Per-component sector bound + monotone AND/OR ($\mathbf{J} \geq 0$)	Proved (conditional on componentwise update + non-negative Jacobian)
B.5b Coupled-edge Cauchy-Schwarz bound $\alpha_s \geq \alpha_c/\kappa(\mathbf{J})^2$	Cauchy-Schwarz on $\mathbf{J}^T\mathbf{F}_c$ and $\mathbf{J}^T\delta_c$	Derived (conditional on Jacobian regularity *and* an SA1-preserving attribution scheme)
B.5c Four-regime classification of credence-to-value transfer	Combination of linear/nonlinear × componentwise/coupled × attribution type	Derived
B.5d Gradient-based attribution restores SA1 on coupled edges	Jacobian-weighted surprise distribution; $\mathbb{E}[y_G - \hat{P}_\Sigma]\|_\theta = 0$	Derived
B.5d Sector parameter under gradient attribution $\alpha_c^{\text{grad}} \geq \sigma_{\min}(\mathbf{J})^2/\max_k(n_k+1)$	First-order expansion near truth + singular-value analysis	Derived
B.5d Gradient-based as *minimal* SA1-preserving scheme for coupled edges	Argument excluding proportional-blame and other schemes	Derived qualitatively (not a uniqueness theorem)
B.6 L1 augmented-DAG sector parameter (three-way gating: condition / starvation / exploration)	Diagonal correction function over $(\delta_C, \delta_{A_1}, \delta_{A_2})$; sector product	Proved (conditional on Beta-Bernoulli + observable C + ε-greedy)
B.6 L0-vs-L1 comparison: L1 is lower α_Σ but honestly calibrated	Direct Φ^{L0} vs Φ^{L1} computation at truth	Derived
B.6 L1-construction principle (common cause factored *above* OR-structure)	Requirement for conditional independence of OR-siblings given C	Derived
B.7 L1' mixture-form sector parameter (five-way gating)	Diagonal correction function over $(\xi_C, \{\xi_{j\|C}\}, \{\xi_{j\|\neg C}\})$; sector product	Derived (conditional on Beta-Bernoulli + observable C + componentwise conditional-branch update + *facilitator monotonicity* $P_\Sigma(G \mid C) \geq P_\Sigma(G \mid \neg C)$)
B.7 Componentwise update per conditional branch	Matches Beta-Bernoulli on the branch actually executed each trial	Formulation choice
B.7 Facilitator-monotonicity condition $P_\Sigma(G \mid C) \geq P_\Sigma(G \mid \neg C)$	Required so the first Jacobian entry of $\hat{P}_\Sigma^{L1'}$ is non-negative, enabling the lossless B.5b transfer	Load-bearing scope condition
B.7 Reduction to B.6 under strict-prerequisite limit ($\theta_{j\|\neg C} \to 0$)	Collapse of $\neg C$-branch terms in the sector inequality	Verification (algebraic consistency)
B.7 Refutation under unobservable C (single-channel Fisher rank 1)	Cramér-Rao bound applied to rank-1 Fisher $\mathcal{F}(\phi) = uu^T/[\mu_j(1-\mu_j)]$	Proved by external theorem (Cramér-Rao bound) — no unbiased online estimator admits $\alpha > 0$
B.7 Repair route (i): augment C-observability	Recovers the positive B.7 derivation	Formulation / scope choice (maps to 6.3)
B.7 Repair route (ii): $K \geq 2$ joint children under same C-realization	Fisher matrix rank reaches $2K+1$; local sector condition	Derived (sketch; applicability narrow)
B.7 Repair route (iii): plan-level fallback on marginal $\hat{\mu}_j$	Reduction to B.1 on scalar marginal	Derived (at cost of per-conditional decomposition)
Persistence thresholds (substitution of each α_Σ into $\alpha_\Sigma > \rho_\Sigma/R_\Sigma$)	Proposition A.1 of A	Derived (each)
Gain-collapse threshold $n^* \propto R_\Sigma/\rho_\Sigma$ common across cases	Direct inversion of persistence condition	Derived
Expected-value (rather than almost-sure) sector analysis	Pedagogical simplification; a.s. convergence via standard Bayesian consistency	Formulation choice (scope)
Constant-α assumption with experience-discounting (λ) stabilization	Instantaneous persistence check + forgetting-factor analysis	Derived (instantaneous only)

Epistemic Status.

Conditional on the Beta-Bernoulli model. Propositions B.1, B.2, B.4, and B.6 are *derived*: the sector-condition verification is exact algebra under the stated generative model. The persistence conditions follow by direct application of Proposition A.1 (A), which is independently established. Proposition B.3(a) (the SA1 violation) is also derived; B.3(b) reduces to B.1 by construction. Proposition B.5a (the credence-to-value bridge for linear correction) is *exact* — the Jacobian cancellation is algebraic, independent of DAG structure. Proposition B.5b (the componentwise nonlinear transfer) is *exact under non-negative Jacobian* — algebraic, no condition-number penalty. The coupled-edge case of B.5b is *conditional on (i) Jacobian regularity* — the condition-number bound requires $\sigma_{\min}(\mathbf{J}) > 0$, which holds for non-degenerate DAGs — *and (ii) an SA1-preserving attribution scheme* (gradient-based, Prop B.5d); marginal Bayesian attribution on coupled edges does not admit the Cauchy-Schwarz bound because SA1 fails upstream. Proposition B.5d (gradient-based attribution) is *derived* — SA1 verification is algebraic and the sector-parameter bound follows from standard singular-value analysis of the Jacobian.

All results use the *expected-value* sector condition. A full stochastic treatment (Foster-Lyapunov or supermartingale convergence) would give probability bounds rather than expected-value bounds. The expected-value analysis gives the correct asymptotic behavior (posterior concentration at $O(1/\sqrt{n})$) but does not prove almost-sure convergence. For the stationary Beta-Bernoulli case, almost-sure convergence is guaranteed by the standard Bayesian consistency theorem independently of this derivation.

The time-varying α_Σ issue remains: since n_k increases with each observation, α_Σ decreases over time. The sector-condition framework assumes constant α. The results here give an *instantaneous* persistence check at the current experience level, not a trajectory guarantee. Experience discounting (exponential forgetting with factor λ) stabilizes α_Σ at the exact value $(1-\lambda)/(2-\lambda)$ from $1/(n_{\text{eff}}+1)$ with $n_{\text{eff}} = 1/(1-\lambda)$ — with the simpler $\approx 1-\lambda$ as the slow-forgetting asymptote ($\lambda \to 1$) — yielding the persistence requirement $(1-\lambda)/(2-\lambda) > \rho_\Sigma/R_\Sigma$ (full treatment in 8.10).

What remains open:

1. ~~*General DAG topology.*~~ **Partially resolved.** Proposition B.6 verifies the first mixed AND/OR case (L1 augmented DAG with common-cause node). The three-way gating structure (condition testing × evidence starvation × exploration gating) appears to be the general pattern. Remaining: arbitrary mixed AND/OR DAGs with multiple common causes and deeper nesting.

2. *Continuous outcomes.* The Beta-Bernoulli model gives conjugate, closed-form updates. Non-conjugate cases (continuous signals, partial observability) require approximate inference, and the sector condition must be verified for the approximation.

3. ~~*Modified sector condition for biased correction.*~~ **Resolved via Prop B.5d (gradient-based attribution).** The $O(1/n)$ bias in the unobservable case (B.3a) under marginal Bayesian updates is eliminated by switching to gradient-based attribution, which satisfies SA1 exactly at truth. Per-edge results are recovered under the gradient scheme at the $\sigma_{\min}^4/\sigma_{\max}^2$ scaling. Alternative route (plan-level tracking, B.3(b)) remains available without the condition-number penalty.

4. ~~*Correlated edges (L1/L2 scope).*~~ **Resolved.** Proposition B.6 verifies the sector condition for L1 (strict-prerequisite augmented DAG); Proposition B.7 verifies it for L1' (soft-facilitator mixture form, observable common cause) with five-way gating. The unobservable-C single-channel case is *refuted* by the Cramér-Rao floor (mixture identifiability obstruction; see B.7 §"Refuted Under Unobservable C") — not merely "open" but structurally impossible without additional information augmentation. Practical paths under unobservable C: augment C-observability, run multi-child joint observations (sketch only, narrow applicability), or fall back to plan-level tracking on the marginal. Strategies requiring full L2 conditioning over k unobservable common causes still cost $O(2^k)$.

5. *Adaptive exploration.* Proposition B.4 uses fixed ε. Adaptive strategies (UCB, Thompson sampling) allocate exploration based on current uncertainty and should yield tighter sector bounds.

Discussion.

The sector parameter is the edge update gain. Across all four cases, α_Σ is a function of $\eta_k = 1/(n_k + 1)$ — the same quantity that governs epistemic persistence (3.8, 3.6). Strategic persistence and epistemic persistence are governed by the same mathematical machinery, not merely structurally parallel. This validates the schema proposed in 8.10 at the level of concrete dynamics.

AND vs. OR: structural vs. behavioral fragility. The evidence-starvation effect (B.2) is structural — it depends on the DAG topology and upstream reliability, not on the agent's policy. The exploration-gating effect (B.4) is behavioral — it depends on the action-selection policy. AND-node persistence is improved by increasing upstream reliability (making early steps more likely to succeed). OR-node persistence is improved by increasing exploration (testing alternatives more frequently). The distinction maps to the familiar depth-vs-breadth tradeoff in planning.

Gain collapse as the dominant failure mode. In every drifting-environment case, the persistence threshold takes the form $n < c/\rho_\Sigma$ for some constant c. Accumulated experience suppresses the update gain, eventually making the agent unable to track environmental change. This gives the gain-collapse phenomenon (3.6) a precise quantitative threshold for each topology.

The credence-to-value bridge decouples persistence from credit assignment. Proposition B.5 shows that strategic persistence (in value-residual space) can be established from per-edge sector conditions (in credence space) without solving the credit-assignment problem. The bridge requires only the value-sensitivity Jacobian $\mathbf{J} = \nabla_{\mathbf{p}} P_\Sigma$, which is computable from the DAG's status propagation formulas in $O(|V| + |E|)$ time. Credit assignment — attributing observed value changes to specific edges — is needed for the *update rule* (computing the signal function in 8.4), not for the *persistence guarantee*. This means the persistence analysis can proceed even while the update rule remains incompletely specified.

Inter-edge coupling is the true fragility, not nonlinearity or DAG depth. Proposition B.5b shows that nonlinear componentwise corrections transfer losslessly through any non-negative Jacobian — the $\kappa(\mathbf{J})^2$ penalty arises only when corrections are *coupled* across edges. This refines the earlier qualitative argument from 7.1: the structural pressure toward shallow, balanced strategies is not about depth per se but about the coupling that deep or complex structures tend to introduce. A deep AND-chain with fully observable intermediates (B.2) has componentwise corrections and transfers losslessly despite high depth. The same chain with unobservable intermediates (B.3) has coupled corrections and incurs the penalty. The distinction is observability-mediated coupling, not topological complexity.

Appendix O

Derivation: Regret-Bound Derivation of the Strategy-Cost KL Direction

The variational form of the strategy-cost objective (8.9) carries a $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})$ relevance term. This appendix derives that **the π^*-first KL direction is forced** as the non-vacuous regret-bound form, and proves (conditional on a chain-rule additivity axiom motivated by 7.1) that **reverse-KL is uniquely forced within the direction-forced f-divergence family**.

Formal Expression.

§1 — Setup. Fix O_t, M_t, continuation policy π_{cont}, and horizon N_h. The action-value is the objective-functional evaluation at a: $V(a) := Q_O(M_t, a; \pi_{\mathrm{cont}}, N_h)$ (5.6; V derives from O_t per 5.4). The optimal action is $a^* := \arg\max_a V(a)$; under AAT's canonical scope (5.6), the optimal policy $\pi^*(\cdot \mid M_t) = \delta_{a^*}$ is the point mass on a^* (tied-optimum extensions §8). Write $V_{\max} := \max_a V(a) - \min_a V(a)$ for the value range at fixed M_t; this is finite whenever $\mathcal{A}$ is a bounded action set under V.

$Q_{\Sigma_t}(\cdot \mid M_t)$ is the action distribution induced by the strategy DAG. In general Q_{Σ_t} is stochastic — the DAG's edge-credence uncertainty induces policy stochasticity downstream of propagation.

§2 — Regret against the optimal policy.

The strategy-induced regret of Q_{Σ_t} against π^* is:

Definition (strategy-induced-regret).

$$R(Q_{\Sigma_t}) \;:=\; V(a^*) \;-\; \mathbb{E}_{a \sim Q_{\Sigma_t}}[V(a)] \;=\; \sum_a Q_{\Sigma_t}(a) \cdot \left[V(a^*) - V(a)\right] \tag{O.1}$$

Writing $\Delta(a) := V(a^*) - V(a) \in [0, V_{\max}]$ for the per-action regret gap. Three regret forms in the literature:

$$\mathbb{E}_{\pi^*}[V] - \mathbb{E}_{Q_{\Sigma_t}}[V], \qquad V(a^*) - \mathbb{E}_{Q_{\Sigma_t}}[V], \qquad \mathbb{E}_{\pi^*}[V - V_{Q_{\Sigma_t}}] \tag{O.2}$$

coincide identically under deterministic π^*.

§3 — Total-variation bound (tight).

Derived (tv-regret-bound, from bounded V + deterministic π^*).

$$R(Q_{\Sigma_t}) \;=\; \sum_a Q_{\Sigma_t}(a)\,\Delta(a) \;\le\; V_{\max} \cdot \sum_{a \neq a^*} Q_{\Sigma_t}(a) \;=\; V_{\max} \cdot (1 - Q_{\Sigma_t}(a^*)) \tag{O.3}$$

For deterministic $\pi^* = \delta_{a^*}$:

$$\mathrm{TV}(\pi^*, Q_{\Sigma_t}) \;=\; \tfrac{1}{2} \sum_a |\pi^*(a) - Q_{\Sigma_t}(a)| \;=\; 1 - Q_{\Sigma_t}(a^*) \tag{O.4}$$

Therefore:

$$\boxed{R(Q_{\Sigma_t}) \;\le\; V_{\max} \cdot \mathrm{TV}(\pi^*, Q_{\Sigma_t})} \qquad \text{(tight)} \tag{O.5}$$

Tightness. Equality holds exactly when $\Delta(a) = V_{\max}$ for all $a \neq a^*$ (degenerate value landscape: every suboptimal action incurs the full value range). For typical landscapes the bound is loose by a factor $\mathbb{E}_{Q_{\Sigma_t}}[\Delta \mid a \neq a^*]/V_{\max} \in (0, 1]$.

§4 — Exact regret-reverse-KL identity under deterministic π^*. Under AAT's canonical scope (deterministic $\pi^* = \delta_{a^*}$), reverse-KL and total variation are related by an *identity*, not merely an inequality. This gives a tight two-sided regret bound that the Pinsker inequality strictly weakens.

Identity. For deterministic $\pi^* = \delta_{a^*}$ and any Q with $Q(a^*) > 0$:

Derived (bh-identity-deterministic, under deterministic π^*).

$$\boxed{D_{\mathrm{KL}}(\pi^* \| Q) \;=\; -\log Q(a^*) \;=\; -\log\bigl(1 - \mathrm{TV}(\pi^*, Q)\bigr)} \tag{O.6}$$

Derivation. $D_{\mathrm{KL}}(\delta_{a^*} \| Q) = \sum_a \delta_{a^*}(a) \log(\delta_{a^*}(a)/Q(a)) = -\log Q(a^*)$ (only the $a = a^*$ term contributes; $0 \log 0$ terms are conventionally 0). And $\mathrm{TV}(\delta_{a^*}, Q) = 1 - Q(a^*)$ (direct from §3). Substituting gives the identity. □

This is the Bretagnolle-Huber inequality (Bretagnolle & Huber 1978, "Estimation des densités," *Séminaire de probabilités XII*, Springer LNM 649; Tsybakov 2009 §2.4; Sason & Verdú 2016) specialized to the deterministic-P case, where the general inequality $\mathrm{TV} \le \sqrt{1 - e^{-D_{\mathrm{KL}}}}$ becomes an equality.

Combining with the tight TV-regret bound of §3:

Derived (exact-regret-reverse-kl, under deterministic π^*).

$$\boxed{R(Q_{\Sigma_t}) \;\le\; V_{\max} \cdot \bigl(1 - e^{-D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})}\bigr)} \tag{O.7}$$

Matching lower bound. Define the action-gap $\Delta_{\min} := \min_{a \neq a^*} \Delta(a) > 0$ (standard in RL theory; well-defined whenever the optimum is isolated over finite $\mathcal{A}$ with bounded V). Then:

$$R(Q_{\Sigma_t}) \;=\; \sum_{a \neq a^*} Q_{\Sigma_t}(a)\,\Delta(a) \;\ge\; \Delta_{\min} \cdot \sum_{a \neq a^*} Q_{\Sigma_t}(a) \;=\; \Delta_{\min} \cdot \mathrm{TV}(\pi^*, Q_{\Sigma_t}) \tag{O.8}$$

Under the exact BH identity:

$$R(Q_{\Sigma_t}) \;\geq\; \Delta_{\min} \cdot \left(1 - e^{-D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})}\right) \tag{O.9}$$

Lipschitz-equivalence corollary. Regret and $(1 - e^{-D_{\mathrm{KL}}})$ are Lipschitz-equivalent with constants $\Delta_{\min}$ (below) and $V_{\max}$ (above):

Derived (matched-lower-bound, under deterministic π^* + isolated optimum).

$$\frac{\Delta_{\min}}{V_{\max}} \;\leq\; \frac{R(Q_{\Sigma_t})}{V_{\max}(1 - e^{-D_{\mathrm{KL}}})} \;\leq\; 1 \tag{O.10}$$

The upper bound is tight when the value landscape is extremal ($\Delta(a) = V_{\max}$ for all $a \neq a^*$); the lower bound is tight when sub-optimal actions are uniformly bad ($\Delta_{\min} = \max_{a \neq a^*} \Delta(a)$).

$D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t}) \in [0, +\infty)$: finite and graded whenever $Q_{\Sigma_t}(a^*) > 0$; diverges only in the structural-failure limit where the strategy places zero mass on the optimum.

§4.1 — Pinsker bound as loose general form. The Pinsker inequality $\mathrm{TV}(P, Q) \leq \sqrt{\frac{1}{2} D_{\mathrm{KL}}(P \| Q)}$ (Tsybakov 2009 §2.4; Cover & Thomas 2006 §11.6) yields a regret bound that does not assume deterministic π^*:

$$R(Q_{\Sigma_t}) \;\leq\; V_{\max} \cdot \sqrt{\frac{1}{2}\, D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})} \tag{O.11}$$

Under AAT's canonical scope, this is strictly weaker than the BH-identity form of §4. For any $D_{\mathrm{KL}} > 0$: $1 - e^{-D_{\mathrm{KL}}} < \sqrt{D_{\mathrm{KL}}/2}$ for small D_{KL} (the BH form is linear in D_{KL} while Pinsker is $\sqrt{D_{\mathrm{KL}}}$), and Pinsker's $V_{\max}\sqrt{D_{\mathrm{KL}}/2}$ exceeds the trivial $V_{\max}$ bound once $D_{\mathrm{KL}} > 2$, giving vacuous content there. The BH identity is informative uniformly in D_{KL}.

Pinsker remains the correct tool for stochastic-π^* extensions (§9) where the deterministic-π^* scope is relaxed and the BH identity degrades back to inequality.

§5 — Direction-forcing claim (the load-bearing result).

Claim. The KL direction with π^* *first* is forced as the non-vacuous regret-bound form. Forward-KL $D_{\mathrm{KL}}(Q_{\Sigma_t} \| \pi^*)$ is *not* a non-vacuous bound on $R(Q_{\Sigma_t})$ under deterministic π^*.

Derived (kl-direction-forced, from deterministic π^*).

Derivation. Expanding:

$$D_{\mathrm{KL}}(Q_{\Sigma_t} \| \pi^*) \;=\; \sum_a Q_{\Sigma_t}(a) \log\!\left(Q_{\Sigma_t}(a)/\pi^*(a)\right) \tag{O.12}$$

For any $a \neq a^*$: $\pi^*(a) = 0$ (point mass assumption). The summand $Q_{\Sigma_t}(a) \log(Q_{\Sigma_t}(a)/0) = +\infty$ unless $Q_{\Sigma_t}(a) = 0$. Therefore $D_{\mathrm{KL}}(Q_{\Sigma_t} \| \pi^*) = +\infty$ whenever Q_{Σ_t} places any mass off a^* — for *all but a measure-zero subset* of stochastic strategies. A bound "$R \leq +\infty$" is vacuous.

This is the same shape of degeneracy as the original Shannon-MI form (Gemini Finding 2) that the V-medium move was introduced to escape: same degeneracy-when-π^*-is-deterministic, different value (0 vs $+\infty$). Only the *reverse* direction escapes both.

Alignment with the structural problem. Forward-KL is natural for *mode-covering* inference (where Q is asked to cover P's full support) and variational inference when Q is learned to match a full distribution. Reverse-KL is natural for *mode-seeking* (concentrate Q on P's mode). The AAT strategy problem is mode-seeking by construction: the strategy should concentrate on a^*. The direction alignment with the structural problem is not accidental.

Asymmetry is forced by regret's one-sidedness. A second, independent argument for asymmetry — one that does not rely on the chain-rule axiom of §6.1. Under deterministic π^*, regret is a *one-sided* quantity: $R(Q_{\Sigma_t}) = \sum_{a \neq a^*} Q_{\Sigma_t}(a)\Delta(a)$ contributes only from Q_{Σ_t}'s off-optimum mass; π^*'s "mass off Q_{Σ_t}" is vacuous since π^* has no support off a^*. Any divergence whose role is to bound this quantity should therefore be asymmetric — it should penalize Q_{Σ_t}'s deviation from π^* without symmetrically penalizing the (trivially zero) deviation of π^* from Q_{Σ_t}. Symmetric divergences (squared Hellinger, Jensen-Shannon, symmetrized KL) treat the two deviations interchangeably; under the one-sided regret quantity this is operationally wrong — it introduces a term with no semantic role. The asymmetry requirement is thus *structural*, emerging directly from what regret is as a functional, and is established independently of the chain-rule axiom that picks the specific asymmetric form in §6.1. The two arguments compose: *direction-forcing* (vacuity under deterministic π^*) + *asymmetry-forcing* (regret is one-sided) ⇒ the admissible bounding divergence is π^*-first and asymmetric; chain-rule additivity (§6.1) then picks reverse-KL uniquely within that family.

Terminology note (Bishop-vs-AAT "reverse-KL"). AAT's canonical reverse-KL is $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})$ — optimum-first, agent-second. Under Bishop 2006's convention ($D_{\mathrm{KL}}(q\|p)$ with q the approximation is called "reverse-KL"), AAT's form would be labeled "forward-KL." The two naming conventions disagree in surface labeling but agree in the operational property — both pick out the *mode-seeking* direction (concentrate the approximation on the target's mode). Literature that uses the Bishop convention (variational inference, generative modeling, Levine 2018) and literature that uses the π^*-first / target-first convention (decision theory, regret analysis, Rubin-Shamir-Tishby 2012) describe the same operational quantity under deterministic π^*. When this segment writes "reverse-KL," the π^*-first / AAT convention is meant; under softened targets the two conventions describe different objects and must be disambiguated explicitly.

§6 — Admissible-divergence family and uniqueness of reverse-KL.

The regret-bound argument forces the *direction* (the reference distribution π^* appears first) but admits a family of divergences. Each member yields a valid regret bound; they differ in tightness and operational properties:

Discussion (admissible-regret-divergences).

Table O.1

Divergence	*Bound on R*	*Tightness*	*Finite under det. π^*?*	*Gradient-tractable?*
$\mathrm{TV}(\pi^*, Q_{\Sigma_t})$	$V_{\max} \cdot \mathrm{TV}$	**Tight** (extremal V)	Yes	No (non-differentiable)
$D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})$ via Pinsker	$V_{\max}\sqrt{\frac{1}{2}D_{\mathrm{KL}}}$	Loose by $\sqrt{\cdot}$	Yes	Yes
$D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})$ via Bretagnolle-Huber	$V_{\max}\sqrt{1 - e^{-D_{\mathrm{KL}}}}$	Tighter than Pinsker for large D_{KL}	Yes	Yes
$\chi^2(\pi^* \| Q_{\Sigma_t})$ (Le Cam)	$V_{\max} \cdot \frac{1}{2}\sqrt{\chi^2}$	Typically looser than Pinsker	Yes: $\chi^2 = 1/Q_{\Sigma_t}(a^*) - 1$	Yes
$D_\alpha(\pi^* \| Q_{\Sigma_t})$ (Rényi, $\alpha \geq 1$)	Various	Interpolates KL ($\alpha \to 1$) and χ^2 ($\alpha = 2$)	Yes for $\alpha \geq 1$	Yes
$D_{\mathrm{KL}}(Q_{\Sigma_t} \| \pi^*)$ (forward-KL)	$+\infty$	Vacuous	**No**	—

What is uniquely forced by the regret-bound argument alone. The direction: the reference π^* is first. This is a real derivation outcome, not a selection. Within the π^*-first family, multiple f-divergences each give valid bounds; an additional structural axiom is required to pick one uniquely.

§6.1 — Chain-rule uniqueness theorem.

Theorem (chain-rule / additivity uniqueness of KL among f-divergences; folk theorem, standard functional-equation derivation). *Let* $D_f(P\|Q) = \sum_x Q(x) f(P(x)/Q(x))$ *be a smooth f-divergence with* f *convex and* $f(1) = 0$. *The chain rule*

Derived (reverse-kl-uniqueness, Conditional on chain-rule axiom).

$$D_f(P_{XY}\|Q_{XY}) \;=\; D_f(P_X\|Q_X) \;+\; \mathbb{E}_{P_X}\big[D_f(P_{Y|X}\|Q_{Y|X})\big] \tag{O.13}$$

holds for all joint distributions (X, Y) *if and only if* $f(t) = c \cdot t \log t$ *for some* $c > 0$ *— i.e.,* D_f *is reverse-KL up to positive scaling.*

References. The theorem is a classical folk result obtainable by direct functional-equation argument (sketched below). The canonical published axiomatic characterizations equivalent to the chain-rule statement are: **Hobson 1969**, "A new theorem of information theory," *J. Stat. Phys.* 1(3):383–391 — uniqueness of the Kullback expression via a composition/additivity axiom; **Csiszár 1991**, "Why least squares and maximum entropy? An axiomatic approach to inference for linear inverse problems," *Annals of Statistics* 19(4):2032–2066 — Theorem 3 corollary: the only transitive statistical projection rule is the I-divergence projection rule; Theorem 5: product-consistency characterizes I-divergence uniquely. See also **Shore & Johnson 1980**, "Axiomatic derivation of the principle of maximum entropy and the principle of minimum cross-entropy," *IEEE Trans. Info. Theory* 26(1):26–37 (system-independence axiom); **Sanov 1957**, "On the probability of large deviations of random variables," *Mat. Sb.* 42(84):11–44 (large-deviation rate function for empirical-distribution concentration); **Aczél & Daróczy 1975**, *On Measures of Information and Their Characterizations* (Academic Press) for the general functional-equation machinery.

These are not independent uniqueness routes. Hobson's composition axiom, Csiszár's transitive-projection axiom, Shore-Johnson's system-independence axiom, and Sanov's sampling-consistency condition are *structurally-equivalent re-formulations* of the same underlying requirement — all factor through independence-of-sub-problems (joint inference on independent subsystems decomposes additively over factorizations). The Cauchy functional equation that each reduces to is the common structural content beneath varied surface formulations. This matters for `#disc-additive-coordinate-forcing`: the chain-rule axiom's load-bearing role is not weakened by noting multiple canonical references; conversely, the multiple references do not provide multiple independent uniqueness arguments. No known uniqueness route outside the independence-on-sub-problems family exists.

Proof sketch. Write $r_x = P(x)/Q(x)$, $s_{y|x} = P(y \mid x)/Q(y \mid x)$. The chain-rule identity must hold for all joint decompositions, which forces the functional equation $f(rs) = f(r) + r f(s) + g(r)$ for all $r, s > 0$. Combined with $f(1) = 0$ and convexity, the unique solution is $f(t) = c \cdot t \log t$ for $c > 0$ (Aczél & Daróczy 1975 §4).

Why other members of the family fail the chain rule. Concrete counterexample for χ^2: take Q_X uniform on $\{x_1, x_2\}$, $P_X = (3/4, 1/4)$, $Q(y \mid x)$ uniform, $P(y \mid x) = (3/4, 1/4)$ for both x. Direct calculation gives $\chi^2(P_{XY}\|Q_{XY}) = 9/16$, while $\chi^2(P_X\|Q_X) + \mathbb{E}_{P_X}[\chi^2(P_{Y|X}\|Q_{Y|X})] = 1/4 + 1/4 = 8/16$. Non-additive: $9/16 \neq 8/16$. Rényi-α for $\alpha \neq 1$ fails analogously; Bretagnolle-Huber is a monotone transform of reverse-KL (not an independent f-divergence); Hellinger-squared ($\alpha = 0$) likewise fails.

AAT-internal motivation for the chain-rule axiom. The chain rule is the *divergence-level analog* of the additive log-confidence decay in `#der-chain-confidence-decay`. AAT has already committed to additive-mismatch-decomposition-along-causal-chains in that segment; the divergence-level parallel is:

$$D(\pi^*\|Q_{\Sigma_t}) \;=\; \sum_{t=1}^{T} \mathbb{E}_{\pi^*}\Big[D\big(\pi^*(\cdot \mid a^*_{<t}) \,\Big\|\, Q(\cdot \mid a^*_{<t})\big)\Big] \tag{O.14}$$

— total mismatch between π^* and Q_{Σ_t} decomposes additively over the DAG's causal layers along the optimal trajectory. Non-chain-rule divergences (e.g., χ^2) give super-additive decompositions in which layer-mismatches amplify multiplicatively — structurally discordant with the DAG factorization (7.3). Adopting the chain-rule axiom is therefore not arbitrary; it is the divergence-level version of a decomposition principle AAT already relies on.

Corollary (within the chain-rule axiom). Under deterministic π^* + bounded value + chain-rule additivity, reverse-KL is the unique smooth f-divergence in the direction-forced admissible family. The "Reverse-KL canonical among smooth divergences" status upgrades from formulation choice (under the pre-axiom reading) to *Derived (conditional on the chain-rule axiom)*.

What the uniqueness theorem does not do. It fixes the divergence within the f-divergence family; it does not fix the bounding function g around the divergence (Pinsker vs Bretagnolle-Huber vs Le Cam-on-χ^2 are *different bounds using different divergences*, so only Pinsker and BH-on-KL remain in scope after uniqueness — and these differ only in the g-envelope on top of reverse-KL, not in the divergence itself). It does not resolve the linear-vs-square-root form question of §7 (that is a Lagrangian-shape choice orthogonal to the divergence choice). Monotone transforms of reverse-KL (e.g., $1 - e^{-D_{\mathrm{KL}}}$) are equivalence-class members for gradient-based optimization and are not ruled out.

Scope of the axiom. The chain rule is stated for arbitrary joint decompositions. AAT's strategy spaces are discrete at the proposition level; the theorem applies directly. For continuous strategy spaces (e.g., low-level control), the chain rule extends by replacing sums with integrals against the dominating measure; the functional-equation derivation is identical in the continuous case (see Liese & Vajda 1987, *Convex Statistical Distances*, Teubner, for the standard measure-theoretic treatment of f-divergences and their decomposition properties).

§6.2 — Secondary characterizations (now supporting rather than load-bearing). Four additional properties further motivate reverse-KL as the canonical choice; under §6.1 they are no longer load-bearing for uniqueness but remain informative about why reverse-KL is a comfortable fit with the rest of the AAT framework:

1. **Gradient-tractability.** TV is non-differentiable at mass boundaries; reverse-KL is smooth. Operational minimization via gradient methods rules out TV despite its tightness — independent of the uniqueness argument.

2. **Variational-inference alignment.** Reverse-KL is the standard divergence in the variational-inference lineage (ELBO derivation uses reverse-KL; Friston et al. 2017; Da Costa et al. 2020). The V-medium move's rhetorical payoff — shared vocabulary with active inference — lives on reverse-KL specifically. This is *convergent evidence* that reverse-KL is natural in multiple frames, not an independent uniqueness argument (Path D in `spikes/spike-reverse-kl-uniqueness.md` §6 shows the ELBO decomposition itself uses the chain rule as a sub-step; logical priority is chain rule → reverse-KL → ELBO).

3. **Fisher geometry.** Reverse-KL's second-order expansion gives the Fisher information metric (Amari & Nagaoka 2000). *But by the differential-geometric framework in Eguchi 1983* ("Second order efficiency of minimum contrast estimators in a curved exponential family," *Annals of Statistics* 11(3):793–803, §2 contrast-function development; modern restatement at Amari & Cichocki 2010 Theorem 5 eq. (126)), every smooth f-divergence with $f''(1) > 0$ induces the Fisher metric at second order up to the scalar $f''(1)$. Fisher-metric-at-second-order is *not* a distinguishing axiom within the f-divergence family — it is satisfied by every member. Noted here because it is a commonly-invoked distinguishing property; it does not distinguish. (Eguchi's Theorem 3 itself is about estimator efficiency via Γ^1-transversality; the f-divergence/Fisher-metric result is a consequence of the §2 contrast-function machinery that supports Theorem 3, not the theorem statement.)

4. **Information-budget / MDL interpretation.** $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t}) = -\log Q_{\Sigma_t}(a^*)$ is the expected extra bits needed to code samples from π^* under Q_{Σ_t}'s code. The coding interpretation aligns with the segment's MDL framing. MDL is itself a chain-rule-respecting coding scheme (the bits-to-code-a-joint decompose additively over factorizations); consistency with MDL is convergent with §6.1.

Čencov/Morozova-Chentsov background (Čencov 1982 *Statistical Decision Rules and Optimal Inference*, AMS Translations of Mathematical Monographs 53; Morozova & Chentsov 1991, "Markov invariant geometry on state manifolds," *Itogi Nauki i Tekhniki*, translated in *J. Sov. Math.* 56(5):2648–2669; Ay, Jost, Lê & Schwachhöfer 2017, *Information Geometry*, Springer). The f-divergence family itself is characterized by Markov-morphism invariance (sufficient-statistic coarse-graining invariance) — this is what makes "f-divergences" the right background family to search within. Amari 2009 shows the alpha-divergences are the sub-family that are simultaneously f-divergences and Bregman divergences. Reverse-KL sits at $\alpha = 1$ within the alpha-family; the chain-rule axiom of §6.1 is what picks $\alpha = 1$ out of the one-parameter alpha-family.

§6.3 — Bregman-Fenchel identification: reverse-KL and log-odds as dual coordinates.

On the probability simplex Δ^{n-1}, the **negative-entropy potential** $\phi(Q) = \sum_a Q_a \log Q_a$ is strictly convex, Legendre, and essentially smooth on the relative interior (Rockafellar 1970 *Convex Analysis* §26). Its Fenchel conjugate is the **log-partition function** $\phi^*(\eta) = \log \sum_a e^{\eta_a}$ on the natural-parameter space $\mathbb{R}^n$ modulo the affine-gauge direction. The primal-dual correspondence is softmax: $Q = \nabla\phi^*(\eta)$ gives $Q_a = e^{\eta_a} / \sum_{a'} e^{\eta_{a'}}$, and $\eta = \nabla\phi(Q) = \log Q + \mathbf{1}$ (up to the constraint-normal direction) so $\eta_a - \eta_b = \log(Q_a/Q_b)$ is the **log-odds ratio**.

Derived (bregman-fenchel-dual-pair, exact; standard Legendre-Fenchel).

The Bregman divergence induced by ϕ on the primal simplex is **reverse-KL** exactly:

$$B_\phi(P, Q) \ := \ \phi(P) - \phi(Q) - \langle \nabla\phi(Q), P - Q \rangle \ = \ \sum_a P_a \log \frac{P_a}{Q_a} \ = \ D_{\mathrm{KL}}(P \| Q) \tag{O.15}$$

Derivation. Expand: $\phi(P) - \phi(Q) = \sum_a (P_a \log P_a - Q_a \log Q_a)$; the inner-product term using $\nabla\phi(Q)_a = \log Q_a + 1$ gives $\sum_a (\log Q_a + 1)(P_a - Q_a) = \sum_a (\log Q_a)(P_a - Q_a)$ since $\sum_a (P_a - Q_a) = 0$. Substituting and simplifying yields $\sum_a P_a \log(P_a/Q_a) = D_{\mathrm{KL}}(P \| Q)$. □

Identification with `#deriv-edge-update-natural-parameter`. The log-odds coordinate derived under evidential-additivity in `#deriv-edge-update-natural-parameter` is the *Fenchel-dual natural parameter* of this Legendre-Fenchel pair. The divergence-layer coordinate of this segment (reverse-KL on Δ^{n-1}) and the update-layer coordinate of `#deriv-edge-update-natural-parameter` (log-odds on $\mathbb{R}^n/\mathbf{1}$) are two sides of one exponential-family geometric structure, viewed through primal and dual coordinates respectively.

What this adds. The two segments independently derive their coordinates via Cauchy-FE on logically-independent AAT-internal axioms (chain-rule additivity here; evidential additivity in `#deriv-edge-update-natural-parameter`). The Fenchel-Bregman correspondence shows that *the coordinates the two axioms pick out are related by Legendre duality* — the axioms are not redundant (they have different logical forms and constrain different objects; see §6.1 vs. `#deriv-edge-update-natural-parameter`'s axiom) but the forced coordinates *coincide on one geometric object*. This is a Discussion-grade observation about the relationship between two primary instances of `#disc-additive-coordinate-forcing`; it is textbook Legendre-Fenchel (Amari & Nagaoka 2000 §3.5; Bregman 1967; Beck & Teboulle 2003 *Mirror descent and nonlinear projected subgradient methods* — relevant for the operational consequence that mirror-descent on the policy simplex with negative-entropy regularizer is equivalent to exponentiated-gradient descent on log-odds coordinates). The full geometric-unification reading sits in `#disc-additive-coordinate-forcing`'s meta-treatment.

§6.4 — Information-theoretic-MDP lineage and AAT's direction choice.

AAT's reverse-KL strategy-cost form sits within an established research lineage — the information-theoretic treatment of decision problems initiated by **Tishby & Polani 2011** ("The Information Theory of Decision and Action," in *Perception-Action Cycle: Models, Architectures, and Hardware*, Springer, pp. 601–636; Eq. 15 p. 19 introduces **Information-to-Go** $\mathfrak{I}^\pi(s_t, a_t)$, the conditional multi-information of the entire future state-action trajectory; Eq. 17 p. 21 the trade-off objective $\arg\min_\pi[\mathfrak{I}^\pi - \beta Q^\pi]$) and developed by **Rubin, Shamir & Tishby 2012** ("Trading value and information in MDPs," in *Decision Making with Imperfect Decision Makers*, Springer, pp. 57–74; §2.2 Eq. 3 p. 4 **control information**

$\Delta I(s) = \sum_a \pi_s(a) \log(\pi_s(a)/\rho_s(a))$; §3.1 Theorem 1 p. 5 Bellman recursion for the free-energy $F_\pi = I_\pi - \beta V_\pi$). The control-as-inference lineage via **Levine 2018** ("Reinforcement Learning and Control as Probabilistic Inference," arXiv:1805.00909; §3.1 Eq. 11 and §5.2 p. 15) develops the max-entropy-RL framework using $D_{\mathrm{KL}}(p_\theta \| p_{\mathrm{tgt}})$ throughout. The shared framework — value and information as joint first-class quantities in MDP optimization, with a KL divergence measuring the information cost of a policy against a reference — is the context in which AAT's strategy-cost objective naturally lives.

Discussion (information-theoretic-mdp-lineage), positioning rather than derivation.

AAT's direction is distinctive within this lineage. Tishby-Polani 2011 uses the stationary marginal $\pi(a)$ of the policy itself as reference (Eq. 16 p. 20, Eq. 22 p. 22) — a direction-neutral multi-information rather than a KL-to-external-reference. Rubin et al. 2012 uses a default policy $\rho_s(a)$ (typically uniform) with direction $D_{\mathrm{KL}}(\pi_s \| \rho_s)$ — **agent-first, default-second**, a regularization/complexity cost. Levine 2018 uses $D_{\mathrm{KL}}(p_\theta \| p_{\mathrm{tgt}})$ — **proposal-first, target-second**, a variational ELBO form with softened exponentiated-reward target. **AAT's** $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})$ — **optimum-first, agent-second** — is the *opposite* direction to all three.

The direction is forced, not chosen. AAT's direction is not a convention adopted from the surrounding literature; it is forced by two independent structural requirements of this segment's derivation: (i) the regret-bound argument of §5 requires the optimum as the reference for the bound to be tight (mode-seeking toward a^*); (ii) the Bretagnolle-Huber identity of §4 is *exact* in the π^*-first direction under deterministic π^* and degenerate in the flipped direction. The flipped direction (Rubin 2012's / Levine 2018's form with $\rho = \pi^*$) is vacuous here for the same reason forward-KL is vacuous in §5: π^* has no support off a^*. AAT *owns* the direction-distinctiveness rather than inheriting it.

Rubin 2012 Theorem 3 PAC-Bayesian motivation. A fourth independent operational property of the π^*-first KL form comes from Rubin-Shamir-Tishby 2012 Theorem 3 (p. 10): under a-priori stochastic default policy and empirical reward estimate, the control information I_π bounds the generalization gap of the value function:

$$\tilde{D}_{\mathrm{KL}}\big(\hat{V}_\pi(s_0) \,\big\|\, V_\pi(s_0)\big) \;\le\; \frac{I_\pi(s_0) + \log(2m/\delta)}{m-1} \tag{O.16}$$

This is a PAC-Bayesian concentration bound. In AAT terms: policies with lower strategy-cost have tighter value-function generalization under finite empirical data. Together with (a) the regret-bound direction-forcing of §5, (b) the chain-rule uniqueness of §6.1, and (c) the Bregman-Fenchel dual identification of §6.3, this is a fourth independent motivation for the KL-to-reference form — operational rather than axiomatic, and independent of the chain-rule axiom. The direction differs (Rubin uses agent-first; AAT uses optimum-first), but the generalization-bound structure transfers by the same PAC-Bayesian machinery: I_π as regularizer against a reference.

Relation to `#form-information-bottleneck`. The canonical IB objective is $I(X;T) - \beta I(T;Y)$ with Shannon mutual information on both terms. AAT's strategy-cost relevance term is $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})$ — a KL divergence to a target, not a mutual information. The two are *sibling lineages*, both descended from Shannon rate-distortion theory (the IB instance uses MI-to-relevant-label as the fidelity term; the information-theoretic-MDP instance uses KL-to-reference-policy); neither reduces to the other without a change of relevance variable. AAT's compression-operations framework (`#disc-compression-operations`) uses the IB form for M_t, G_t^{shared}, and Λ compressions; the strategy-cost compression uses the information-theoretic-MDP form. Both are valid Lagrangian relaxations of an underlying rate-distortion problem; the choice of fidelity term depends on whether the compressed variable is meant to preserve information about an observable (IB) or to match a reference policy (information-theoretic-MDP). The "abandoned IB purity" concern raised in external reviews (Gemini 2026-04-24) is defused at this level: AAT has not abandoned IB; its strategy-cost compression uses a different relevance quantity than its model-compression, and both are principled under the rate-distortion umbrella.

§7 — β_Σ interpretation under the regret bound.

The Pinsker bound gives $R(Q_{\Sigma_t}) \leq V_{\max}\sqrt{\frac{1}{2}D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})}$. Two scale forms for the segment's trade-off parameter β_Σ arise:

Discussion (β_Σ-local-vs-global).

(a) Square-root-KL form (tight regret-bound scale):

$$\mathcal{L}(\Sigma_t) \;=\; \mathcal{R}(\Sigma_t) \;+\; \beta_\Sigma \cdot \sqrt{D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})}, \qquad \beta_\Sigma = V_{\max}/\sqrt{2} \tag{O.17}$$

β_Σ is *globally naturalized* as a constant scale proportional to the value range at fixed M_t. Cost: the form departs from the linear-Lagrangian IB shape that V uses across the four compression operations.

(b) Linear-KL form (IB-shape instance, preserved in the segment):

$$\mathcal{L}(\Sigma_t) \;=\; \mathcal{R}(\Sigma_t) \;+\; \beta_\Sigma \cdot D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t}) \tag{O.18}$$

Under this form, β_Σ's regret-bound interpretation is *local*: differentiating the Pinsker bound, $\partial R/\partial D_{\mathrm{KL}} = V_{\max}/(2\sqrt{2D_{\mathrm{KL}}})$, so β_Σ represents the local cost-per-bit at the operating point but varies with D_{KL}. Upside: consistency with the IB linear-Lagrangian shape (V). Downside: no uniform global β_Σ scale.

Choice made in 8.9. Keep the linear form (IB-shape alignment); note the square-root form in the Epistemic Status as the "source" derivation. The direction-forcing claim is the load-bearing strengthening; the linear-vs-square-root choice is a second-order trade-off that preserves architectural consistency at the cost of a fully naturalized β_Σ.

§8 — Bound tightness and scope limits. **Vacuity regimes** (where the regret bound fails to be informative):

1. $V_{\max} = \infty$ **(unbounded value).** Then all bounds are ∞. AAT's O_t is $\mathbb{R}$-valued (5.4); boundedness of V over $\mathcal{A}$ at fixed M_t is an additional assumption. Stated here as a scope condition of the derivation.
2. $Q_{\Sigma_t}(a^*) = 0$ **(strategy cannot express the optimum).** Then $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t}) = +\infty$. This is structural failure — the strategy's DAG cannot produce the optimal action — and the infinite KL correctly flags it. *Informative degeneracy*, not pathology: operational minimization rejects such Σ_t.
3. π^* **not deterministic (outside canonical scope).** For stochastic π^* with larger support than Q_{Σ_t}, reverse-KL may still be infinite. §9 addresses tied-optimum extensions; stochastic π^* under softmax-smoothing is future work.

Tightness comparison. Under deterministic π^* (AAT's canonical scope), the Bretagnolle-Huber **identity** $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t}) = -\log(1 - \mathrm{TV}(\pi^*, Q_{\Sigma_t}))$ makes the KL and TV bounds *operationally equivalent in reverse-KL coordinates*: the KL-based bound $R \leq V_{\max}(1 - e^{-D_{\mathrm{KL}}})$ of §4 is exactly the TV-based bound of §3 re-expressed in the KL coordinate, with matching lower bound $R \geq \Delta_{\min}(1 - e^{-D_{\mathrm{KL}}})$ (§4). Pinsker is strictly weaker under this scope (see §4.1). Outside the deterministic-π^* scope, the BH identity degrades back to the inequality $\mathrm{TV} \leq \sqrt{1 - e^{-D_{\mathrm{KL}}}}$ (Bretagnolle-Huber 1978; tighter than Pinsker for large D_{KL} but no longer an identity); Pinsker is the textbook fallback for general distributions. The net: reverse-KL is not merely "the tightest smooth divergence" — in AAT's canonical scope it carries the *exact* regret-bound relationship via BH, with Pinsker as a loose alternative retained only for stochastic-π^* extensions.

§9 — Extensions. **Tied-optimum case.** If π^* has support on a tied-optimum set $\mathcal{A}^* = \{a : V(a) = V(a^*)\}$ with uniform mass, reverse-KL is finite whenever Q_{Σ_t} covers $\mathcal{A}^*$. The regret-bound argument extends directly: $R(Q_{\Sigma_t}) = \sum_a Q_{\Sigma_t}(a)\Delta(a) \le V_{\max} \cdot \mathbb{P}_{Q_{\Sigma_t}}(a \notin \mathcal{A}^*)$ and Pinsker applies unchanged.

Softmax-smoothed π^* (stochastic π^* for non-degeneracy reasons). A regret bound of the form

$$\mathbb{E}_{\pi^*}[V] - \mathbb{E}_{Q_{\Sigma_t}}[V] \tag{O.19}$$

with softmax-weighted π^* also admits the Pinsker-KL step, with reverse-KL again the non-vacuous direction (forward-KL remains vacuous whenever π^* has wider support than Q_{Σ_t}, and vice versa). Deferred for explicit treatment.

What Is Derived vs. What Is Chosen.

Table O.2

Property	*Source*	*Strength*
Regret definition $R(Q_{\Sigma_t}) := V(a^*) - \mathbb{E}_{Q_{\Sigma_t}}[V]$	Definitional; collapses to the three literature forms under deterministic π^*	Definition
TV-regret bound $R \le V_{\max} \cdot \mathrm{TV}(\pi^*, Q_{\Sigma_t})$	Bounded value range + deterministic π^*	Proved (tight)
BH identity $D_{\mathrm{KL}}(\pi^* \Vert Q_{\Sigma_t}) = -\log(1 - \mathrm{TV}(\pi^*, Q_{\Sigma_t}))$ under deterministic π^*	Direct calculation (Bretagnolle-Huber 1978 specialized to $P = \delta_{a^*}$)	**Proved (exact identity)**
Regret-reverse-KL bound $R \le V_{\max}(1 - e^{-D_{\mathrm{KL}}(\pi^* \Vert Q_{\Sigma_t})})$	TV-regret bound + BH identity, under deterministic π^*	**Proved (tight in upper direction)**
Matching lower bound $R \ge \Delta_{\min}(1 - e^{-D_{\mathrm{KL}}(\pi^* \Vert Q_{\Sigma_t})})$	Isolated-optimum action-gap $\Delta_{\min} > 0$ + BH identity	**Proved (Lipschitz-equivalence with $V_{\max}$ / $\Delta_{\min}$ constants)**
Pinsker-KL bound $R \le V_{\max}\sqrt{\frac{1}{2} D_{\mathrm{KL}}(\pi^* \Vert Q_{\Sigma_t})}$	Pinsker's inequality applied to TV bound	Proved (strictly weaker than BH identity under deterministic π^*; correct form for stochastic-π^* extensions)
KL direction forced (reverse-KL, π^*-first)	Forward-KL is vacuous ($+\infty$) under deterministic π^* whenever Q_{Σ_t} has off-optimum mass	**Proved (direction uniquely forced)**
Asymmetry forced by regret's one-sidedness	Regret contributes only from Q_{Σ_t}'s off-optimum mass; π^*'s off-Q deviation is vacuous; symmetric divergences penalize interchangeably, which is operationally wrong	Proved (asymmetry forced; independent of chain-rule axiom)
Reverse-KL uniquely forced within direction-forced f-divergences	Chain-rule additivity axiom (Hobson 1969; Csiszár 1991 Theorem 3 corollary and Theorem 5; standard functional-equation derivation per Aczél & Daróczy 1975); AAT-internally motivated as divergence-level analog of 7.1 (§6.1)	**Derived (conditional on chain-rule axiom)**
Secondary characterizations (gradient-tractability, VI-alignment, MDL coding)	Independent operational properties; each compatible with reverse-KL, none individually distinguishing	Formulation support (non-load-bearing under §6.1)
Fisher-metric-at-second-order does not distinguish reverse-KL	Eguchi 1983 §2 contrast-function framework (*Ann. Statist.* 11(3):793–803): every smooth f-divergence with $f''(1) > 0$ induces the Fisher metric up to scalar $f''(1)$	Proved (no-go for Path A as uniqueness axiom)

Property	*Source*	*Strength*
Admissible family members (TV, Bretagnolle-Huber, χ^2, Rényi-α)	Each yields a valid regret bound; survey §6	Derived (each)
Under deterministic π^*: TV and KL-coordinate bounds are equivalent via BH identity; Pinsker strictly weaker	§4 BH identity + §4.1 Pinsker comparison	Proved (scope-dependent tightness ordering)
Reverse-KL is the Bregman divergence of negative-entropy on Δ^{n-1}; log-odds is the Fenchel-dual natural coordinate (Legendre-Fenchel pair)	Standard Legendre-Fenchel (Rockafellar 1970 §26; Amari & Nagaoka 2000 §3.5; Bregman 1967) applied to $\phi(Q) = \sum_a Q_a \log Q_a$	Proved (textbook identification; §6.3)
AAT's π^*-first KL direction is distinctive within the information-theoretic-MDP lineage (TP2011 / Rubin 2012 / Levine 2018 all put agent-first) and is *forced* (not inherited) by the regret-bound + BH-identity derivations	Literature positioning (§6.4) + derivation structure (§§4–5)	Discussion (positioning; direction forced by §§4–5 derivations)
PAC-Bayesian generalization bound: $\tilde{D}_{\text{KL}}(\hat{V}_\pi \| V_\pi) \leq (I_\pi + \log(2m/\delta))/(m-1)$	Rubin-Shamir-Tishby 2012 Theorem 3	Derived (external theorem applied; independent operational motivation alongside regret-bound + chain-rule + Fenchel-dual)
$\beta_\Sigma \propto V_{\max}$ naturalization (square-root-KL form)	Direct identification from Pinsker bound	Derived (under square-root form)
β_Σ local interpretation (linear-KL form, segment-retained)	Differentiation of Pinsker bound at operating point	Derived (local only)
Linear-KL form retained over square-root form	IB-shape alignment with V	Formulation choice
Vacuity regimes ($V_{\max} = \infty$, $Q_{\Sigma_t}(a^*) = 0$, stochastic π^*)	Direct analysis of the bound	Proved (boundary)
Tied-optimum extension	π^* uniform on $\mathcal{A}^*$; bound adapts directly	Derived
Softmax-smoothed π^* extension	Sketched in §9	Hypothesis (deferred)
Uniqueness under chain-rule axiom (§6.1)	Hobson 1969 / Csiszár 1991 Theorem 3 corollary and Theorem 5 / standard functional-equation argument per Aczél & Daróczy 1975, applied to the direction-forced family; axiom AAT-internally motivated (§6.1)	Derived (conditional)

The dividing line: the KL **direction** is forced by the regret-bound derivation (strong result). The specific reverse-KL **form** within the direction-forced f-divergence family is *derived under the chain-rule additivity axiom* (§6.1) — conditional on an axiom that AAT independently motivates as the divergence-level version of the additive-decomposition principle already in 7.1. The β_Σ naturalization is partial — available globally only under the square-root form, locally under the linear form retained for IB-shape alignment.

Epistemic Status.

Robust qualitative overall, with several specific claims at *exact* tier. The direction-forcing claim (§5) is derived under standard assumptions (bounded value, deterministic π^* on the canonical AAT scope). The **Bretagnolle-Huber identity under deterministic** π^* (§4, $D_{\text{KL}}(\pi^* \| Q) = -\log(1 - \text{TV})$) is *exact* by direct calculation, and yields the tight regret bound $R \leq V_{\max}(1 - e^{-D_{\text{KL}}})$ with matching lower bound $\Delta_{\min}(1 - e^{-D_{\text{KL}}})$ — the segment's primary bound form. Pinsker (§4.1) is retained as the strictly-weaker general form, correct for stochastic-π^* extensions. The TV bound (§3) is textbook. The admissible-divergence family characterization (§6) is a truthful survey. The **chain-rule uniqueness theorem (§6.1)** upgrades the reverse-KL selection to "uniquely forced conditional on the chain-rule axiom"; the axiom is AAT-internally motivated as the divergence-level analog of `#der-chain-confidence-decay`. **Asymmetry-from-one-sidedness (§5)** provides a second independent argument for the direction-forcing side, not requiring the chain-rule axiom.

Bregman-Fenchel identification (§6.3) is textbook Legendre-Fenchel; ties the segment's divergence coordinate to `#deriv-edge-update-natural-parameter`'s update coordinate as Fenchel-dual aspects of one exponential-family structure. Fisher-metric-at-second-order is explicitly identified as *not* a distinguishing axiom within the f-divergence family (Eguchi's theorem), closing the most-obvious alternative candidate uniqueness route as a no-go. **Information-theoretic-MDP lineage positioning (§6.4)** is *Discussion-grade* (positioning in the literature, not a derivation); its role is to place AAT's direction-distinctive form honestly within the TP2011 / Rubin 2012 / Levine 2018 lineage rather than overclaim equivalence with any of them. The Rubin 2012 Theorem 3 PAC-Bayesian bound (§6.4) is a fourth independent operational motivation for the KL-to-reference form. The β_Σ naturalization (§7) is a trade-off analysis, not a derivation. The extensions (§9) are sketches, not derivations.

Max attainable: *exact* for (a) "forward-KL is a vacuous bound on R under deterministic π^*" (direct calculation; §5); (b) "under deterministic π^*, $D_{\mathrm{KL}}(\pi^* \| Q) = -\log(1 - \mathrm{TV}(\pi^*, Q))$" (BH identity; §4); (c) "regret is Lipschitz-equivalent to $(1 - e^{-D_{\mathrm{KL}}})$ with constants $\Delta_{\min}$ / $V_{\max}$" (§4 matched lower bound); (d) "reverse-KL is the Bregman divergence of negative-entropy; log-odds is its Fenchel-dual natural coordinate" (standard Legendre-Fenchel; §6.3); (e) "reverse-KL is the unique chain-rule-respecting f-divergence in the direction-forced family" (Hobson 1969 / Csiszár 1991; §6.1); (f) "asymmetry is forced by regret's one-sidedness" (direct; §5). *Robust qualitative* for the overall strengthened characterization — the chain-rule uniqueness is conditional on the chain-rule axiom, which is a principled axiom choice (AAT-internally motivated) but remains an axiom rather than a consequence of prior AAT commitments. The information-theoretic-MDP lineage positioning (§6.4) is *Discussion-grade* and honest about where AAT sits in the broader lineage rather than claiming formal equivalence.

Discussion.

The convergence with active inference's reverse-KL. Active inference's expected free energy (Friston et al. 2017; Parr & Pezzulo 2022) also uses reverse-KL against a preferred distribution, with the direction arising from the variational-free-energy-gradient lineage. AAT derives the direction from a regret-bound argument internal to the decision-theoretic scope. The convergence (two distinct derivations reach the same direction) is worth noting but is not a uniqueness argument — it shows reverse-KL is *natural* in multiple frames. The lineage difference preserves AAT's distinctive content: AAT's derivation does not require VFE as master objective or preferences-as-priors (6.5, 7.4 for the contrast).

Why the regret bound is the right derivation, not Donsker-Varadhan. The Donsker-Varadhan variational representation

$$\log \mathbb{E}_Q[e^f] \geq \mathbb{E}_P[f] - D_{\mathrm{KL}}(P \| Q) \tag{O.20}$$

gives reverse-KL bounds for exponential-moment functionals, but requires V to have bounded exponential moments (a stronger-than-bounded assumption for unbounded V). The Pinsker-TV route used here requires only bounded value range and is closer to the AAT scope's natural assumptions. Donsker-Varadhan becomes relevant if unbounded-V extensions are attempted.

Relationship to standard policy-gradient regret bounds. The argument here is structurally identical to standard policy-gradient / online MDP regret bounds (Agarwal, Kakade, Lee, Mahajan 2021 "On the theory of policy gradient methods" Lemma 3.2; Even-Dar, Kakade, Mansour 2009 "Online Markov decision processes"). AAT's contribution is *re-using the standard bound as the derivation of the KL direction in the strategy-cost objective*, not inventing the bound. This is consistent with AAT's integration-over-invention character.

Position in the additive-coordinate-forcing pattern. The chain-rule uniqueness theorem of §6.1 is the *divergence-level* instance of an AAT-wide pattern in which logarithmic coordinates are uniquely forced via Cauchy's functional equation on an AAT-internally-motivated additivity axiom. The pattern has two other primary instances: the chain-layer identity (7.1, the motivating anchor) and the update-layer log-odds uniqueness (P). Z catalogs the pattern, states the 1-anchor-plus-2-theorem structure precisely, and documents Lyapunov and IB Lagrangian as adjacent family members that share the additive-decomposition shape without the AAT-internal forcing structure.

Working Notes.

- **Linear vs. square-root form — revisit with downstream evidence.** The segment keeps the linear-KL form for IB-shape alignment. If downstream work shows the IB-shape-across-four-operations claim in V is strained and the four βs want separate scales anyway, switching 8.9 to the square-root form (naturalizing β_Σ) becomes attractive. Revisit with the O-BP2 (four compressions as one hierarchy) work if pursued.

- **Stochastic-π^* extension.** Flagged in §9 but not derived. The tied-optimum case is immediate; softmax-smoothed π^* needs a careful treatment of what regret quantity is being bounded (expected-vs-max).

- **Family characterization strengthening.** *Resolved 2026-04-22* — see §6.1. The chain-rule additivity axiom (Hobson 1969; Csiszár 1991 Theorem 3 corollary and Theorem 5; standard functional-equation derivation per Aczél & Daróczy 1975) picks reverse-KL uniquely within the direction-forced f-divergence family. The Fisher-metric candidate was ruled out by Eguchi's theorem (every smooth f-divergence induces the Fisher metric at second order up to scale; not distinguishing). Full reasoning trail in `spikes/spike-reverse-kl-uniqueness.md`, including the citation audit that corrected the initial draft's misattributions (Csiszár 1972, Amari 2009 Theorem 1, and Amari & Cichocki 2010 Prop 3.2 do not contain the chain-rule uniqueness theorem; the correct attributions are above).

- **BH identity under deterministic π^* — elementary fact the earlier presentation underplayed.** *Resolved 2026-04-24.* §4 now presents the BH identity as primary, with Pinsker demoted to §4.1 as the loose general form. The earlier framing "non-Pinsker tighter bounds… BH flagged for operational work where the tightness difference matters" under-sold the result: under AAT's canonical deterministic-π^* scope, the BH inequality specializes to an *identity* (not merely a tighter inequality), making the KL-coordinate bound exact rather than approximate. The segment now carries the exact two-sided relation with matching lower bound via $\Delta_{\min}$.

- **Citation verification (Path 1 content, §6.4).** The information-theoretic-MDP lineage content in §6.4 was PDF-verified 2026-04-24 against TP2011 / Rubin-Shamir-Tishby 2012 / Levine 2018. The verification process found a specific attribution error in an earlier spike draft — the objective form $\max_\pi \mathbb{E}[R] - \beta^{-1} I(\Pi; X)$ with KL-to-reference reference is **Rubin 2012's**, not TP2011's (TP2011's actual quantity is Information-to-Go multi-information $\mathfrak{I}^\pi$, a different object). The segment reflects the corrected attributions with explicit equation/page references. Full trail in `spikes/spike-ib-purity-strategy-cost-strengthening-2026-04-24.md` §12 "Citation-verification addendum."

- **AAT-outlier-on-KL-direction — distinctive feature, not a departure.** *Documented 2026-04-24.* The literature uses agent-first or proposal-first directions; AAT uses optimum-first, forced by the regret-bound derivation and required for the BH identity to be exact. Owned rather than inherited; see §6.4.

- **Cross-reference to NeurIPS Paper 2.** The deterministic-π^* corner identity is sharpened in NeurIPS 2026 Paper 2 ("A Unified Convergence Theory for Non-Stationary Reinforcement Learning", `~/src/neurips/02-unified-convergence-rl/`, §5 Key Lemma 1 / `#lem-pointmass-identity`): under deterministic optimum, $\mathrm{TV}(\delta_{a^*}, Q) = 1 - e^{-D_{\mathrm{KL}}(\delta_{a^*} \| Q)}$ holds as an *exact algebraic identity*, lying strictly below the Pinsker and Bretagnolle-Huber envelopes at this corner (numerical comparison in Paper §App-E). The identity is **coordinate-optimal among bounds depending only on TV** — Paper 2 uses it as Component 2 of the composition theorem and as the structural reason why the segment's optimum-first reverse-KL direction is the right coordinate. The classical BH bound is recovered as the loose general form away from the deterministic-π^* corner. See `msc/neurips-back-integration-2026-05-08.md` §1 Paper 2 entry 1; standalone-segment proposal under back-integration §3 (proposed slug `#deriv-bh-pointmass-identity` or `#result-pointmass-tv-kl-identity`).

- **Fenchel-Bregman connection to `#deriv-edge-update-natural-parameter`.** *Documented 2026-04-24.* §6.3 identifies reverse-KL and log-odds as Fenchel-dual coordinates of one exponential-family Legendre-Fenchel structure. The full geometric-unification reading (covering all four layers of `#disc-additive-coordinate-forcing`) is a meta-segment-level observation; trail in `spikes/spike-fenchel-bregman-reframe-additive-coordinate-forcing-2026-04-24.md`.

Appendix P

Derivation: Log-Odds as the Unique Additive-Evidence Parameterization for Edge Credences

The log-odds coordinate $\lambda_{ij} = \log(p_{ij}/(1 - p_{ij}))$ is the unique parameterization (up to positive affine transformation) on which independent Bernoulli evidence updates edge credences additively, under an evidential-additivity axiom motivated as the update-level analog of 7.1's chain-level additive log-confidence decomposition. This segment states the uniqueness theorem, derives it, and explains how it positions log-odds as the natural parameterization for AAT's continuous-gradient edge-update machinery.

Formal Expression.

Setup. Let $p \in (0, 1)$ denote a scalar Bernoulli credence (the probability that a proposition is true) and let $\psi : (0, 1) \to \mathbb{R}$ be a smooth, strictly monotone reparameterization. Consider a sequence of independent Bernoulli observations $y_1, \ldots, y_n \in \{0, 1\}$ drawn from a channel with likelihood ratio $P(y \mid H_1)/P(y \mid H_0)$.

Evidential-additivity axiom. The posterior update, applied to a single observation y, takes the form

Assumption (evidential-additivity axiom).

$$\psi(p_{\text{post}}) = \psi(p_{\text{prior}}) + g(y) \tag{P.1}$$

for some function $g : \{0, 1\} \to \mathbb{R}$ that depends only on the observation y — not on p_{prior} nor on observation history.

Theorem.

Theorem. The functional equation above admits solutions if and only if

Derived (evidential-additivity uniqueness of log-odds, conditional on the axiom above).

$$\psi(p) = c \cdot \log \frac{p}{1 - p} + d \tag{P.2}$$

for constants $c > 0$ and $d \in \mathbb{R}$, with $g(y) = c \cdot \ell(y)$ where $\ell(y) = \log[P(y \mid H_1)/P(y \mid H_0)]$ is the log-likelihood ratio.

Derivation.

By Bayes' theorem applied to binary hypotheses,

Derived (Proof Step: Bayesian form of the update).

$$\frac{p_{\text{post}}}{1 - p_{\text{post}}} = \frac{p_{\text{prior}}}{1 - p_{\text{prior}}} \cdot \frac{P(y \mid H_1)}{P(y \mid H_0)} \tag{P.3}$$

Taking the logarithm of both sides and writing $h(p) := \log(p/(1-p))$,

$$h(p_{\text{post}}) = h(p_{\text{prior}}) + \ell(y) \tag{P.4}$$

So $\psi = h$ trivially satisfies the axiom with $g = \ell$.

Suppose ψ is any smooth, strictly monotone reparameterization satisfying the axiom. Since the Bayesian mapping $p_{\text{prior}} \mapsto p_{\text{post}}$ is fully determined by y through the likelihood ratio, the difference $\psi(p_{\text{post}}) - \psi(p_{\text{prior}})$ depends only on y, and by the axiom must equal $g(y)$.

Derived (Proof Step: uniqueness by Cauchy functional equation).

Change variables via $\lambda = h(p) = \log(p/(1-p))$ and define $\Psi(\lambda) := \psi(\sigma(\lambda))$ where $\sigma(\lambda) = 1/(1+e^{-\lambda})$ is the logistic sigmoid. The axiom becomes

$$\Psi(\lambda + \ell(y)) - \Psi(\lambda) = g(y) \quad \text{for all } \lambda \in \mathbb{R},\ y \in \{0, 1\} \tag{P.5}$$

Extending to continuous-valued evidence (or considering mixtures of Bernoulli channels with varying likelihood ratios, which span all of $\mathbb{R}$ in the ℓ-value space), the identity

$$\Psi(\lambda + \ell) - \Psi(\lambda) = G(\ell) \quad \text{for all } \lambda, \ell \in \mathbb{R} \tag{P.6}$$

holds for a function G independent of λ. This is the Cauchy functional equation (translation-additivity). Combined with the smoothness assumption on ψ, the unique solution class is $\Psi(\lambda) = c \cdot \lambda + d$ for constants c and d (Aczél 1966, *Lectures on Functional Equations and Their Applications*, §2.1).

Strict monotonicity of ψ forces $c \neq 0$. Taking ψ to have the same monotonicity sense as $p \mapsto p$ (credence increasing with ψ), we need $c > 0$. Thus $\psi(p) = c \cdot h(p) + d = c \cdot \log(p/(1-p)) + d$, and $g(y) = c \cdot \ell(y)$.

Derived (Proof Step: determining the constants).

This completes the proof. □

Three-Layer Parallel.

The evidential-additivity axiom is the update-level instance of an additive-decomposition principle that AAT has already committed to at two prior layers:

Discussion (three-layer additive decomposition).

Table P.1

Layer	*Quantity decomposed*	*Decomposition form*	*Source*
Chain level	Confidence along a causal chain	$\log P(\text{chain}) = \sum_i \log P(E_i \mid E_{<i})$	7.1
Divergence level	Mismatch between optimal and strategy policies	$D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})$ decomposes additively across DAG layers along the optimal trajectory	O §6.1
Update level	Credence evolution under independent evidence	$\psi(p_{\text{post}}) - \psi(p_{\text{prior}}) = g(y_1) + \cdots + g(y_n)$ for n observations, with ψ = log-odds	This segment

Each layer forces a logarithmic coordinate through essentially the same structural move: products of independent factors become sums on a log scale. At the chain level, $p^n \to n \log p$; at the divergence level, $\prod Q \to \sum \log Q$; at the update level, $\prod \mathrm{LR} \to \sum \log \mathrm{LR}$. The three are the same transform applied to different quantities.

Interpretation for the Edge-Update Machinery.

For edge credence p_{ij} with log-odds $\lambda_{ij} = \log(p_{ij}/(1 - p_{ij}))$, the Bayesian update under independent Bernoulli evidence is

Discussion (operational consequence).

$$\lambda_{ij}^{\text{post}} = \lambda_{ij}^{\text{prior}} + \ell(y) \tag{P.7}$$

where $\ell(y)$ is the per-observation log-likelihood ratio.

Two operational consequences that follow from the uniqueness theorem:

1. **Domain unboundedness.** The log-odds coordinate has domain $\mathbb{R}$, not $[0, 1]$. Additive updates cannot escape the domain, regardless of update magnitude. The probability-space presentation $p_{ij} \in [0, 1]$ is the projected image of the log-odds coordinate, obtained via $p_{ij} = \sigma(\lambda_{ij})$ at the readout interface.
2. **Invariance under the chain of causal reasoning.** Because the log-odds coordinate is the unique additive evidence coordinate, evidence accumulated along one edge in a strategy DAG can be composed with evidence accumulated along another edge by addition in the log-odds vector space, provided the evidence is conditionally independent. The Beta-Bernoulli moment-parameter form $\hat{p} = \alpha/(\alpha + \beta)$ is the projected image, where α, β are the cumulative sufficient statistics in exponential-family form.

These consequences are why the continuous-gradient edge-update machinery in 8.6 is well-posed globally in log-odds but exhibits the Finding 2 mechanical break (unbounded updates pushing credences outside $[0, 1]$) when stated directly in probability space.

Scope Condition.

The evidential-additivity axiom applies to agent classes that treat observations as independent Bernoulli likelihood evidence — the Bayesian-coherent sub-scope of AAT. Non-Bayesian agents (PID controllers, rule-based systems, human judgment per 3.6) do not invoke likelihood-ratio accumulation and are outside the axiom's scope. This matches the sub-scope α / sub-scope β partition in 4.2 (see also `spikes/spike-a2-prime-strengthening.md`): the uniqueness applies within sub-scope α, where B1 (directional fidelity) is already derived from Bayesian coherence.

Scope (evidential-additivity scope).

For multinomial / categorical edge credences with $K > 2$ outcomes, the analog is softmax / canonical exponential-family parameters: the softmax natural parameters $\eta_k = \log \pi_k$ (up to a reference-class shift) satisfy the same evidential-additivity axiom. The Bernoulli case ($K = 2$) collapses to log-odds.

Epistemic Status.

Derived (conditional on the evidential-additivity axiom). The uniqueness theorem is a standard functional-equation result (Cauchy functional equation with smoothness; Aczél 1966 §2.1). The AAT-internal motivation of the axiom is structural: it is the update-level analog of 7.1's chain-level additive log-confidence decomposition and O §6.1's divergence-level chain-rule additivity. Without the axiom, the selection of log-odds weakens to canonical-not-unique on convergent grounds (exponential-family naturalness, Fisher-information / natural-gradient canonicity, domain-well-posedness of the continuous gradient).

Max attainable: *exact* for the Cauchy functional equation step (standard); *derived-conditional* for the overall uniqueness claim because the axiom, though AAT-internally motivated, is itself a commitment rather than a consequence of prior AAT commitments. The axiom's status is parallel to the chain-rule additivity axiom in O §6.1 — both are AAT-internally motivated as the "right level" instances of a single additive-decomposition principle.

Not uniqueness in the unconditional sense. The theorem states that *if* credence updates live on a single additive coordinate whose increment depends only on the observation, *then* that coordinate is log-odds up to positive affine. It does not rule out non-additive update schemes, nor schemes that live on multiple coordinates (e.g., Beta-Bernoulli with sufficient statistics (α, β) rather than a single credence coordinate). The Beta-Bernoulli presentation is *equivalent* in content to the log-odds presentation under a change of coordinates; the uniqueness is about what single-coordinate additive form is possible.

Discussion.

Why this matters for G-BP1. The scoping spike `spikes/spike-gbp1-logit-scoping.md` examined whether log-odds reparameterization (G-BP1 in `msc/architectural-proposals-2026-04-22.md`) is *uniquely correct* or merely canonical. Paths A (exponential-family canonical), C (Fisher / natural-gradient), and D (sector-condition preservation) supply convergent grounds but no uniqueness theorem. Path B (this segment) gives the uniqueness result conditional on the evidential-additivity axiom. The axiom is AAT-internally motivated, so the result is genuine strengthening — not "log-odds happens to work" but "log-odds is forced by the update-level analog of a principle AAT already relies on at chain and divergence levels."

Why this matters for Finding 2. The mechanical break in 8.6's default signal function (unbounded gradient updates pushing credences outside $[0, 1]$ when $\|\mathbf{J}\|^2 \to 0$; see `audits/pending-findings-2026-04-22.md` §Finding 2) is a presentation artifact of the probability-space coordinate. In log-odds, the domain is $\mathbb{R}$ and the update is well-posed globally. The fix is not "clip the update" but "present the update in its native additive coordinate."

Relationship to Bayesian conjugate analysis. The Beta-Bernoulli conjugate update produces posterior hyperparameters $(\alpha + y, \beta + 1 - y)$ that are additive in the sufficient statistics, giving the point estimate $\hat{p} = \alpha/(\alpha + \beta)$ that updates *non-additively* in probability space. The content is the same — the log-odds coordinate makes the additivity manifest at the level of credence itself, while Beta-Bernoulli shows the same additivity at the level of sufficient statistics. The two are dual presentations of the same exponential-family structure.

Non-uniqueness outside the Bayesian sub-scope. For non-Bayesian agents in sub-scope β of 4.2 (PID controllers, rule-based systems, human judgment), the evidential-additivity axiom does not apply. Such agents may update credences via coordinate-free heuristics (proportional blame, threshold rules) that do not decompose into likelihood-ratio addition. The uniqueness of log-odds is therefore scope-conditional, not universal.

Path B's role in the overall pattern. The segment makes visible that AAT is committed to a family of consistent coordinate-forcing moves at chain, divergence, update, and metric layers, each forcing a specific coordinate via a uniqueness theorem on an AAT-internally-motivated axiom. Future coordinate-forcing structures in AAT (e.g., any novel decomposition along trajectories, across channels, or between agents) should be examined for whether Cauchy-FE on an additivity axiom or Čencov-invariance on a parameterization-invariance axiom applies. If yes, log-odds / natural-parameter / reverse-KL / Fisher-metric-style results compose cleanly. If no, the new layer is a genuine departure requiring its own analysis. The pattern is catalogued at the meta-pattern level in Z, including the 1-anchor-plus-3-theorem characterization (the chain layer is a mathematical identity; divergence, update, and metric are theorems conditional on AAT-internally-motivated axioms) and the adjacent-family classification (Lyapunov quadratic and IB Lagrangian share the additive shape but not the AAT-internal forcing structure).

Block-structured evidential additivity under L1' correlated evidence. When sibling edges share a latent common cause C (L1' correlated evidence per `#def-strategy-dag`'s Correlation Hierarchy), the independent-evidence axiom grounding Path B is literally false. Under *observable* C, the likelihood factorizes per-cluster as $P(\mathbf{y}, c) = P(c) \prod_k P(y_k \mid c)$, and the per-factor Aczél-1966 uniqueness argument applies independently to each factor. The forced coordinate becomes a $(2K + 1)$-dimensional vector log-odds (one coordinate for θ_C, two per child edge for $p_{j|C}$ and $p_{j|\neg C}$), reducing to the scalar log-odds when the cluster is a singleton and no latent is present. This is a **generalization-in-scope** of the theorem above, not a new primary instance of `#disc-additive-coordinate-forcing`: the forcing machinery is unchanged (Cauchy-FE on per-factor additivity); the scope widens to block-factorized likelihoods under observable latents. Under *unobservable* C, the soft-EM responsibility-reweighted posterior depends nonlinearly on the prior, so no smooth ψ satisfies the block-additivity axiom — the axiom is structurally inconsistent with Bayesian mixture updates. This is the same scope boundary that `#disc-identifiability-floor` Instance 2 names via the Cramér-Rao Fisher-rank-1 obstruction; two independent analytical routes (Cauchy-FE failure at update layer; Cramér-Rao rank deficiency at observation layer) converge on the same unobservable-C structural floor, strengthening the "structural, not analytical-artifact" reading of Instance 2.

Findings.

Log-Odds as Uniquely-Forced Edge-Update Coordinate. **Brief:** When updating beliefs about whether something is true, there's a question of "what's the natural number to update?" — the probability itself, or some transformation of it. This result identifies a particular transformation (the log-odds, also used in logistic regression and Bayesian inference broadly) as the *uniquely* natural one for agents updating on independent evidence, in the precise sense that any equivalent representation must be log-odds up to scale. The choice isn't aesthetic; it's forced by what "independent evidence should add up" means mathematically. The uniqueness follows from Cauchy's functional equation operating on an evidential-additivity axiom motivated as the update-level analog of the chain-layer log-additive identity in `#der-chain-confidence-decay`.

Impact: Promotes a representational choice that previously had only convergent grounds (exponential-family canonicity, Fisher-information naturalness, sector-condition preservation) to a uniqueness theorem under an explicit AAT-internal axiom, joining the additive-coordinate-forcing meta-pattern (`#disc-additive-coordinate-forcing`) as one of its three theorem-grade instances. The result resolves a mechanical issue in the credit-assignment default signal function (`#disc-credit-assignment-boundary`), where unbounded gradient updates pushing credences outside $[0,1]$ in the probability-space presentation become globally well-posed in the log-odds presentation (domain $\mathbb{R}$). It also positions log-odds as the coordinate against which downstream gain-and-update machinery should be analyzed for sub-scope α (Bayesian-coherent) agents.

Novelty Claim: *Claim differentiation* on an already-canonical representational choice (log-odds as the natural Bayesian-update coordinate, well-known from logistic regression / exponential-family / information-geometry traditions) by deriving its uniqueness under an AAT-internally-motivated evidential-additivity axiom. The Cauchy-FE machinery is classical; the AAT-internal axiomatization and the meta-pattern membership are the contribution.

Related Work:

- Aczél 1966, *Lectures on Functional Equations and Their Applications* §2.1 — *formal antecedent* — Cauchy-FE uniqueness machinery directly adopted in the derivation; cited inline.
- Cox 1946, "Probability, frequency and reasonable expectation" *Am. J. Phys.* 14:1–13; Jaynes 2003, *Probability Theory: The Logic of Science* — *conceptual precursor* — log-odds and additive-evidence accumulation as the natural form for Bayesian update; folk knowledge in this tradition.
- Amari & Nagaoka 2000, *Methods of Information Geometry* §2 — *formal antecedent* — log-odds as the natural parameter in the Bernoulli exponential family; the canonical-natural-parameter framing.
- Bishop 2006, *Pattern Recognition and Machine Learning* §4.2; Murphy 2012, *Machine Learning: A Probabilistic Perspective* — *adjacent literature* — standard statistical-learning treatments of log-odds and logistic regression; representational standard.

Search Log: - 2026-04 (*intuition-only* on AAT-internal axiomatization angle): no targeted search for prior derivations of log-odds via the specific evidential-additivity axiom (update-level analog of chain-rule additivity) has been conducted. Pre-search expectation: the Cauchy-FE move is classical (Aczél 1966); the log-odds canonical parameter is exponential-family standard; the *AAT-internal axiomatization* (motivating the additivity axiom as the update-level analog of `#der-chain-confidence-decay`) is the contribution being claimed. A targeted search would query the Bayesian-coherence and information-geometry literatures specifically for prior axiomatic derivations of log-odds via Cauchy-FE in agent-update settings. - 2025 (*targeted*): Aczél 1966 confirmed as the formal antecedent for the Cauchy-FE machinery; cited inline.

Appendix Q

Derivation: Adaptive-Gain Dynamics — A2' Under a Learning Gain

AAT's gain structure (3.6, 4.2) derives the optimal gain η^* per regime — the gain is a function of the noise model, chosen to minimize one-step mismatch variance. Real adaptive agents often *learn the noise model itself* (adaptive Kalman), *switch regimes* (IMM), *adapt step-size online* (RMSProp / Adam), or *optimize gain across tasks* (MAML). The gain becomes a state variable with its own update dynamics. The question is whether the sector-persistence machinery extends to this case, and where inside the A2' sub-scope partition adaptive gain sits. The result: sub-scope α splits into α_1 (fixed-gain per Prop B.3) and α_2 (adaptive-gain under four derivable conditions named (MG-1)–(MG-4)), with sub-scope β catching the rest. The two-timescale argument is an augmented-state Lyapunov composition (standard Khalil Thm 4.18), not a Tikhonov reduction — the primary and meta-gain sector conditions compose rather than one being eliminated.

Formal Expression.

Augmented-state setup. Treat the gain K_t as state. Define $\tilde{K}_t = K_t - K^*$ (error relative to a target optimal gain, specified per case: Riccati-steady-state for adaptive Kalman, EMA-fixed-point for RMSProp, etc.). The augmented state is $z_t = (\delta_t, \tilde{K}_t)$. Primary and meta-gain dynamics:

$$\dot{\delta} = -F(\delta; K^* + \tilde{K}) + w(t), \qquad \dot{\tilde{K}} = -\Phi(\tilde{K}, \delta) + v(t) \tag{Q.1}$$

where F is the primary correction function (depending on the current gain via its argument), Φ is the gain-update contraction, w is primary-channel disturbance, v is gain-channel disturbance (estimator noise, innovation variability, etc.).

Meta-gain sector conditions (MG-1)–(MG-4).

(MG-1) Primary sector floor under bounded gain error. There exist $\underline{\alpha} > 0$ and $r_K > 0$ such that for all $\|\tilde{K}\| \leq r_K$ and $\|\delta\| \leq R$:

Formulation (meta-gain-conditions, extend A2' to adaptive-gain setting).

$$\delta^T F(\delta; K^* + \tilde{K}) \geq \underline{\alpha}\|\delta\|^2 \tag{Q.2}$$

— A2' uniform in the gain-error ball. The sector floor is preserved across the gain-state range the meta-learner visits.

(MG-2) Meta-gain sector condition. The gain-update map Φ satisfies (T1) (zero at $\tilde{K} = 0$) and a local sector bound:

$$\tilde{K}^T \Phi(\tilde{K}, \delta) \geq \alpha_K \|\tilde{K}\|^2 \quad \text{for } \|\tilde{K}\| \leq r_K, \text{ uniformly in } \|\delta\| \leq R \tag{Q.3}$$

with $\alpha_K > 0$. This is a sector condition in the gain-error state — the adaptive-gain analog of A2' itself.

(MG-3) Timescale separation. $\alpha_K \ll \underline{\alpha}$. The gain adapts slower than the primary state contracts. This is 4.6's convergence constraint transcribed onto Lyapunov decay rates instead of event rates.

(MG-4) Coupling boundedness. The gain-channel disturbance $\upsilon(t)$ has bounded contribution from the primary state:

$$\mathbb{E}[\|\upsilon(t)\|^2 \mid \delta] \leq \sigma_{K,0}^2 + c_\upsilon \|\delta\|^2 \tag{Q.4}$$

for some $c_\upsilon \geq 0$. (MG-4) with $c_\upsilon = 0$ is clean two-timescale decoupling; $c_\upsilon > 0$ is δ-coupled meta-gain disturbance (RMSProp near minimizer), requiring fixed-point closure.

Composed persistence result.

Under (MG-1)–(MG-4), the augmented state $z = (\delta, \tilde{K})$ is ultimately bounded in mean square. The Lyapunov candidate $V(z) = \frac{1}{2}\|\delta\|^2 + \frac{c}{2}\|\tilde{K}\|^2$ for appropriate weight c satisfies, along trajectories:

Derived (augmented-state-persistence, from sector-persistence-template applied twice with coupling).

$$\dot{V} \leq -\underline{\alpha}\|\delta\|^2 - c\alpha_K\|\tilde{K}\|^2 + \rho\|\delta\| + c(\sigma_{K,0} + \sqrt{c_\upsilon}\|\delta\|)\|\tilde{K}\| \tag{Q.5}$$

Complete-the-square on the cross term (requires $c\sqrt{c_\upsilon}$ small compared to $\underline{\alpha} \cdot c\alpha_K$, i.e., (MG-3) timescale separation plus (MG-4) coupling smallness):

$$\dot{V} \leq -\frac{\underline{\alpha}}{2}\|\delta\|^2 - \frac{c\alpha_K}{2}\|\tilde{K}\|^2 + \frac{\rho^2}{2\underline{\alpha}} + \frac{c\sigma_{K,0}^2}{2\alpha_K} \tag{Q.6}$$

Standard Lyapunov ultimate-boundedness (Khalil 2002 Thm 4.18) applied to the augmented state gives the composed persistence bound. Both δ and $\tilde{K}$ are ultimately bounded with explicit bounds in $(\underline{\alpha}, \alpha_K, \rho, \sigma_{K,0}, c_\upsilon)$. □

A2' sub-scope partition: α_1 / α_2 / β.

The adaptive-gain analysis refines A's A2' sub-scope partition into three tiers:

Formulation (sub-scope-refinement).

Sub-scope α_1 — fixed-gain, A2' derived. 4.2 Prop B.3's current scope: the gain K is treated as a static function of fixed noise model parameters. A2' is derived from B1 directional fidelity. Covers Kalman with known (Q, R), conjugate-Bayesian updates, exponential-family MLE, linear correction with PD KH, strongly-convex-gradient fixed-step-size.

Sub-scope α_2 — adaptive-gain, A2' derived through augmented-state Lyapunov. When (MG-1)–(MG-4) hold with all four conditions derivable from the update-rule structure:

- Adaptive Kalman with Mehra-type innovation-based (Q, R) estimator under timescale separation
- RMSProp/Adam with strongly-convex loss, large β (slow EMA), and coupling-smallness; AMSGrad fix (Reddi et al. 2018) structurally a meta-gain repair preserving (MG-1)

- Any adaptive-gain scheme admitting a clean (MG-1)–(MG-4) derivation from the update rule

For these, A2' is derived at the augmented-state level, not merely assumed. Setting $\tilde{K} \equiv 0$ recovers sub-scope α_1 cleanly.

Sub-scope β — A2' assumed (possibly under scope narrowing). When any of (MG-1)–(MG-4) must be assumed per-agent rather than derived. Concrete instances:

- MAML outer loop: meta-loss non-convex even under per-task convexity (Fallah et al. 2020), so (MG-2) cannot be derived from per-task structure. Inner loop is α_1; outer loop is β.
- IMM regime transitions: (MG-1) fails uniformly in time during posterior-reconcentration windows of duration τ_{IMM}. Between-transition regime is α_2; across-transition window is β (scope narrowing via dwell-time + impulsive-disturbance absorption).
- Adam without AMSGrad on ill-conditioned problems: (MG-1) fails under aggressive-β + small-gradient-noise (Reddi et al. 2018 counterexample).
- Rule-based / PID / human-judgment adaptive gains: no structural argument for either (MG-1) or (MG-2); both must be assumed.

This refinement preserves the existing A2' sub-scope α as a specialization (α_1) and identifies a derivable adaptive-gain extension (α_2) with honest fallback to β when the derivability conditions fail.

Structured cases.

Case A — Adaptive Kalman with Mehra estimator. Scalar linear-Gaussian setting with unknown (Q^*, R^*). The innovation-based Mehra estimator (Mehra 1970, 1972; Dunik et al. 2021 for identifiability) yields $\hat{Q}_t, \hat{R}_t$ from a sliding-window autocorrelation of the innovation sequence. For window length N, the gain-update map to first order in $\tilde{K}$ is a scalar Ornstein-Uhlenbeck process:

Derivation (case-adaptive-kalman-alpha2).

$$\tilde{K}_{t+1} = (1 - \lambda_N)\tilde{K}_t + \lambda_N \eta_t^{\mathrm{inn}} \tag{Q.7}$$

with $\lambda_N \asymp 1/N$ (contraction rate set by window length) and η^{inn} zero-mean innovation noise with variance $O(1/N)$. This is itself an instance of C with:

- (T1): $\Phi_K(0) = 0$ — when gain is optimal, estimator returns optimal in expectation.
- (T2): $\tilde{K} \cdot \Phi_K(\tilde{K}) = \lambda_N \tilde{K}^2$, so $\alpha_K = \lambda_N$.
- (T3-S): bounded stochastic disturbance on the gain channel.

Prop A.1S applied to the meta-gain channel gives $R^*_{S,K} \asymp 1/\sqrt{N}$ — the classical Mehra asymptotic rate, now derived from (MG-2). Primary sector floor is preserved: $\underline{\alpha} = K^* - |\tilde{K}|_{\max}$ via triangle. Composed persistence via augmented-state Lyapunov gives $O(1/\sqrt{N})$ degradation from the fixed-gain case. **This case is derived at sub-scope α_2.**

Under Mehra non-identifiability (rank-deficient transform matrix; see Zagrobelny-Rawlings 2015, Dunik et al. 2021), (MG-2) fails structurally — an instance of the X pattern on the meta-gain channel.

Case B — RMSProp on strongly-convex loss. The per-step effective step is $\eta_t^{\mathrm{eff}} = \eta_t/(\sqrt{v_t} + \varepsilon)$ where $v_t = \beta v_{t-1} + (1-\beta)\hat{g}_t^2$ tracks the second moment. Near the minimizer, $\mathbb{E}[\hat{g}_t^2] \to \|\nabla L\|^2 + \sigma_g^2$. Writing $\tilde{v}_t = v_t - \mathbb{E}[\hat{g}_t^2]$:

Derivation (case-rmsprop-alpha2-conditional).

$$\tilde{v}_{t+1} = \beta\tilde{v}_t + (1-\beta)(\hat{g}_t^2 - \mathbb{E}[\hat{g}_t^2]) + \beta(\mathbb{E}[\hat{g}_{t-1}^2] - \mathbb{E}[\hat{g}_t^2]) \tag{Q.8}$$

The first two terms give (MG-2) with $\alpha_v = 1-\beta$ and $\sigma_v^2 \asymp (1-\beta)^2\mathrm{Var}(\hat{g}_t^2)$. The third is δ-coupled: $\mathbb{E}[\hat{g}_t^2]$ depends on δ through $\|\nabla L\|^2$, giving $c_v > 0$ in (MG-4).

Composed persistence under design conditions β close to 1 (slow EMA) and $\lambda_{\max}(H)\cdot R_S^* \ll \sqrt{\sigma_g^2}$ (coupling smallness): fixed-point closure between primary R_S^* and meta-gain R_v yields existence of a stable equilibrium. Sub-scope α_2 under design conditions.

Outside those conditions (aggressive β + ill-conditioning + small gradient noise — Reddi et al. 2018's Adam counterexample), fixed-point iteration diverges. **AMSGrad's monotonicity on v_t is structurally a meta-gain repair that restores (MG-1) by construction** — preserving sub-scope α_2 by forcing a condition (MG-1) the vanilla algorithm would violate.

Case C — IMM / regime-switching Kalman. Mixture of M Kalman filters with Markov-transition posterior over regimes. Between regime transitions, the posterior concentrates on the true regime and the effective gain approaches the regime-conditional optimum: sub-scope α_2 in the steady portion with Mehra-style derivation. Across regime transitions, posterior re-concentration takes $\tau_{\mathrm{IMM}} \asymp 1/(1-p)$ observations (self-loop probability p); during this window (MG-1) fails uniformly in time — the gain can be aligned with the wrong regime. Scope narrowing via dwell-time + impulsive-disturbance absorption: regime stable for $T_{\mathrm{dwell}} \gg \tau_{\mathrm{IMM}}$, transient mismatch bounded by counting-argument. Sub-scope α_2 between-transition; β across-transition.

Sketch (case-imm-alpha2-plus-dwelltime).

Case D — MAML inner-outer structure. Inner loop (per-task adaptation with k gradient steps) — sub-scope α_1 from Prop B.4 under per-task convexity. Outer loop (meta-parameter update via gradient on meta-loss $\sum_i L_i(\theta_i'(\theta))$) — Fallah et al. 2020's convergence analysis shows the meta-loss is non-convex even under per-task convexity because of the non-linearity of inner-loop updates in θ. (MG-2) is not derivable from per-task structure. Outer loop is β — A2' assumed per basin.

Classification (case-maml-mixed).

What Is Derived vs. What Is Chosen.

Table Q.1

Property	*Source*	*Strength*
Augmented-state setup $z = (\delta, \tilde{K})$ with coupled sector dynamics	Definitional reformulation extending 3.2 + 3.6	Formulation choice
(MG-1) primary sector floor under bounded gain error	Derived in Cases A and B via triangle / ε-floor arguments	Derived (per case, conditional on regularity)
(MG-2) meta-gain sector condition	Derived in Cases A (Mehra OU) and B (EMA second-moment) from estimator structure	Derived (per named case)
(MG-3) timescale separation $\alpha_K \ll \underline{\alpha}$	Design condition on estimator window / EMA rate; Lyapunov decay-rate transcription of 4.6	Formulation (design condition; violations dissolve composition)
(MG-4) coupling boundedness of gain-channel disturbance	Decoupled ($c_v = 0$) in Case A; δ-coupled ($c_v > 0$) in Case B with fixed-point closure	Derived (per case)
Composed augmented-state persistence under (MG-1)–(MG-4)	Quadratic Lyapunov on z + Khalil 2002 Thm 4.18 ultimate-boundedness	Derived (conditional on MG-1 through MG-4)

Property	*Source*	*Strength*
Sub-scope refinement α_1 / α_2 / β	Extension of current A α/β partition	Formulation (classification)
α_2 reduces to α_1 under $\tilde{K} \equiv 0$	Direct substitution into augmented-state setup	Proved
Case A (adaptive Kalman): sub-scope α_2 under identifiability + window-length timescale separation	Mehra OU-form of estimator + Prop A.1S	Derived
Case B (RMSProp): sub-scope α_2 under design conditions; β otherwise; AMSGrad as meta-gain repair	EMA second-moment derivation + fixed-point closure + Reddi et al. 2018 counterexample	Derived (conditional); AMSGrad framing is discussion
Case C (IMM): sub-scope α_2 between-transition + β across-transition via dwell-time	Posterior re-concentration + impulsive-disturbance absorption	Sketch
Case D (MAML): inner-loop α_1 + outer-loop β	Fallah et al. 2020 meta-loss non-convexity	Classification (not derivation)
Mehra non-identifiability as meta-gain X instance	Rank-deficient transform matrix blocks (MG-2) derivation	Discussion (candidate floor instance)

Epistemic Status.

Conditional. Max attainable: *derived* for the composed persistence result under (MG-1)–(MG-4); *exact* for Case A (adaptive Kalman) under Mehra identifiability; *conditional* for Case B (RMSProp) under design conditions; *sketch / classification* for Cases C (IMM) and D (MAML).

The core claim — that (MG-1)–(MG-4) together give an augmented-state sector-persistence result via standard Khalil Thm 4.18 — is a clean application of two-timescale Lyapunov analysis to the gain-as-state formulation. The mathematical core is textbook (vector Lyapunov, augmented-state dissipation, completing-the-square on cross terms). The contribution is the AAT-framing: (i) naming the four conditions that make adaptive gain composable with sector-persistence, (ii) identifying which adaptive-gain schemes satisfy all four from update-rule structure (Cases A, B-design) and which require assumption (Cases C-transition, D-outer, rule-based), and (iii) refining the A2' sub-scope partition into $\alpha_1/\alpha_2/\beta$ as a principled extension rather than overhaul.

Imported machinery, acknowledged. The underlying techniques — two-timescale stochastic approximation (Borkar 1997, 2008), adaptive filtering (Mehra 1970–72; Mohamed-Schwarz 1999; Dunik et al. 2021), MRAC stability theory (Narendra-Annaswamy 1989; Ioannou-Sun 1996), non-convex meta-learning convergence (Fallah et al. 2020; Ji-Yang-Liang 2022) — are standard in control theory and ML. AAT's value-add here is the integration: connecting these techniques to C via the augmented-state Lyapunov composition, and producing a sub-scope partition that matches AAT's scope-honesty architecture rather than importing the techniques' native terminology. The existing sub-scope α / β partition from A becomes α_1 / α_2 / β, with α_2 labeling the newly-derivable adaptive-gain region.

What this does not establish:

- Convergence rates beyond ultimate-boundedness (the bound gives stability, not speed).
- Behavior under (MG-3) violation — fast-adapting gain on a slow primary state — which is structurally different (the composite system becomes singular-perturbation pathological, not sector-persistent).
- Rigorous fixed-point closure for the Case-B δ-coupled case; the qualitative argument is clear but quantitative bounds on when the closure exists are per-problem.
- Composition at the multi-agent level — the augmented-state Lyapunov is for a single adaptive agent; team adaptive-gain (simultaneous meta-learning across agents) is beyond scope.

Discussion.

Resolving Epistemic Opacity. In `#def-observation-function`, the agent is structurally forbidden from knowing the true noise distribution ε_t. Yet the optimal gain `#emp-update-gain` requires knowing U_o. The adaptive Kalman filter with an innovation-based estimator (e.g., Mehra 1970) resolves this. The agent computes the autocorrelation of its observable mismatch sequence (innovations): $\iota_t = o_t - \hat{o}_t$. Because the innovation statistics are observable, the agent can estimate U_o (and U_M) purely from its own history $\mathcal{C}_t$, satisfying both the opacity axiom and the optimality requirement. The error in this estimation ($\tilde{K}$) is exactly what the meta-gain sector condition (MG-2) bounds.

Why this is not a singular-perturbation reduction. Singular perturbation theory (Tikhonov 1952; Khalil 2002 Ch. 11) replaces fast dynamics with their quasi-steady-state manifold — the fast variable is *eliminated* in the slow reduction. The augmented-state analysis here does something different: it *keeps both states* and bounds their joint persistence via a weighted Lyapunov. Tikhonov gives "the fast variable tracks its slow manifold up to $O(\epsilon)$"; template-composition gives "each level's state is ultimately bounded with coupling-amplified disturbance" — the persistence-flavored statement aligned with AAT's other persistence results. They answer different questions; both are useful; neither replaces the other. I's current sketch leans toward the Tikhonov framing; this segment adds the template-composition framing as an alternative.

Three timescale patterns in AAT. I distinguishes model-class (slow) and parametric (fast) timescales. This segment introduces a *third* pattern: gain-parameter timescale, sitting between parametric-correction (fastest) and model-class (slowest). Adaptive gain is a different two-timescale pattern from structural adaptation — the gain is a parameter of the update rule, not of the model class. The convergence constraint $\nu_{n+1} \ll \nu_n$ from 4.6 translates in this segment to $\alpha_K \ll \underline{\alpha}$ on Lyapunov decay rates, which is the (MG-3) condition.

Meta-gain scope-honesty failure modes. - *Identifiability failures* on the meta-gain channel (Mehra non-identifiability) are instances of the X pattern: a structural no-go (non-identifiability) on the meta-gain channel blocks the sub-scope α_2 derivation. AMSGrad as a meta-gain repair is the constructive complement — an added structural condition (monotonicity on v_t) that restores (MG-1) and keeps the adaptive scheme in α_2. - *Non-convex meta-losses* (MAML outer loop) put the outer loop in sub-scope β: A2' must be assumed within a basin, not derived from the meta-loss's global structure. This is the honest scope-narrowing posture — different from saying "it works" or "it doesn't" without qualification. - *Regime transitions* (IMM across switches) violate (MG-1) uniformly in time during posterior re-concentration windows. Repair via dwell-time is clean but significant scope narrowing.

Connection to 8.10's $\alpha_\Sigma = 1/(n+1)$. The strategy-edge persistence condition has α_Σ decaying with experience — this is adaptive gain of a specific form, where the gain (effective update rate per edge) is a function of accumulated pseudo-count. Without forgetting, $\alpha_\Sigma \to 0$ and (MG-1) fails asymptotically. 8.10's forgetting prerequisite $(1-\lambda) > \rho_\Sigma / R_\Sigma$ is precisely the condition that keeps strategy-edge adaptation in sub-scope α_2 rather than letting it fall to β as n grows.

Relation to catastrophic forgetting via EWC. Elastic Weight Consolidation (Kirkpatrick et al. 2017) introduces a per-parameter stability-weighted update: gain is scaled by inverse Fisher information of prior tasks. In AAT vocabulary, EWC is a tensor-valued adaptive gain with stability weighting, distinct from the scalar adaptive-gain case treated here but fitting the same augmented-state framework with (MG-1)–(MG-4) reinterpreted per-parameter. Adapting the present derivation to EWC would tensorize the Lyapunov argument — a natural but not-yet-executed extension.

Working Notes.

- **Identifiability as meta-gain obstruction.** Mehra non-identifiability for certain system structures (Dunik et al. 2021) is an instance of X on the meta-gain channel — a structural no-go blocking (MG-2) derivation. Candidate for explicit cross-reference in X's §"Adjacent Floors" as a fourth derived instance (after F1 CHT no-go, F13 Cramér-Rao mixture-identifiability, and Regime II-b misspecification-cost from 13.3).

- **Rate-specific results.** The composed persistence result is a bound, not a rate. Can the $1/\sqrt{N}$ rate of Mehra-adaptive-Kalman be derived rigorously from the augmented-state Lyapunov? Classical Mehra asymptotics give it directly; deriving it from (MG-1)–(MG-4) would confirm the framework's completeness.
- **Adversarial adaptive-gain.** If the environment adversarially varies K^* (hostile regime-switching cadence), dwell-time repair of Case C fails. Adversarial-tempo-advantage analog at the meta-gain level: faster meta-gain tracking beats faster environmental regime switching iff $\alpha_K T_{\text{dwell}}$ exceeds the transition rate. Not derived here; adjacent spike.
- **Discrete-time form.** Everything above is continuous-time. Discrete-time version involves the Lipschitz bound of K on both F and Φ. Extension is mechanical but deferred.
- **Relationship to 8.10 time-varying α.** 8.10's $\alpha_\Sigma = 1/(n+1)$ decay is a concrete instance of adaptive-gain dynamics where (MG-1) fails asymptotically without forgetting. Worth re-reading that segment through the α_2 lens; the forgetting prerequisite is the specific move that keeps strategic adaptation in α_2 as n grows.
- **Template-composition as general technique.** The augmented-state weighted-Lyapunov argument is a general two-timescale composition technique for C. It composes two template instances with a cross-coupling bound. A generalization to three or more timescales (inner loop + outer loop + meta-meta-loop, or fast + gain + slow + structural) would give a chain-composition form; not derived here but structurally available.
- **AMSGrad as α_2-preserving meta-gain repair.** Characterizing AMSGrad as "enforce (MG-1) by construction" is a clean AAT-framing of a pragmatically-motivated algorithm. Worth noting as an example of adaptive-gain schemes being AAT-classifiable by which of (MG-1)–(MG-4) they structurally ensure vs leave to luck.
- **Deterministic-meta-gain special cases.** When the meta-gain is a deterministic function of the primary state rather than an independently learned variable — $K_t = \mathbf{I}^{-1}(\lambda_t)$ for Fisher whitening at the edge-update layer (), or $K = (H_M + H_L)^{-1} H_L$ for the Fisher-local model-parameter update () — all four meta-gain conditions are satisfied trivially under the regularity conditions of those segments. The non-degenerate adaptive-Kalman-with-Mehra case is this segment's primary instance and adds a Mehra-style meta-channel with its own (MG-1)–(MG-4); the Fisher-local cases are degenerate special cases where the meta-gain is read off the primary state directly.

Appendix R

Derivation: Internal-External Decomposition of Agent Viability

The persistence condition ($\alpha R > \rho$) provides a binary survival threshold. However, diagnosing *why* an agent persists requires decomposing its margin of survival (viability) into internal agent capacity versus external environmental affordance.

Log-Viability.

We define log-viability $\mathcal{V}$ as the margin by which the agent's steady-state mismatch R^* stays below the critical task boundary $\|\delta_{\text{critical}}\|$:

$$\mathcal{V} = \log \frac{\|\delta_{\text{critical}}\|}{R^*} = \log\|\delta_{\text{critical}}\| - \log R^* \tag{R.1}$$

Persistence holds exactly when $\mathcal{V} > 0$. Because $R^* = \rho/\alpha$ (under linear Model D), we can additively decompose viability:

$$\mathcal{V} = \log\|\delta_{\text{critical}}\| - \log \rho + \log \alpha \tag{R.2}$$

The Full Decomposition ($\mathcal{V}_E + \mathcal{V}_I$).

By expanding α (tempo $\times$ gain $\times$ fidelity), ρ (volatility $\times$ policy benignity $\times$ model expressiveness), and ν (cap vs chosen usage), we split the margin into two additive components:

$\mathcal{V} = \mathcal{V}_E + \mathcal{V}_I$

Environmental Affordance ($\mathcal{V}_E$). Properties purely dictated by the environment: $\mathcal{V}_E = \log\|\delta_{\text{critical}}^{\text{domain}}\| - \log \rho_{\text{external}} + \log \nu_{\text{external-cap}}$

Internal-Operational Health ($\mathcal{V}_I$). Properties controlled by the agent's architecture, policy, and goals: $\mathcal{V}_I = \log h(\mathcal{O}_t) - \log f(\mathcal{M}) - \log g(\pi) + \log \frac{\nu_{\text{chosen}}}{\nu_{\text{external-cap}}} + \log \eta^* + \log c_{\min}$

Confounding and Identifiability.

In a feedback system, high internal capacity often creates high environmental affordance (e.g., a good team refactors the codebase, lowering future U_0 and ρ_{external}). Separating these terms using purely observational data is formally impossible (identifiability floor). Separating them requires Level 2 interventional data: rotating agents across different environments (Regime A) to isolate $\Delta\mathcal{V}_E$ from $\Delta\mathcal{V}_I$.

Appendix S

Derivation: Update Detection Latency Under Accumulated Experience

For a Beta-Bernoulli strategy-edge agent without forgetting, the expected number of cycles required to detect a within-class regime change of observable footprint ε scales as $\Omega((n_{\min}+1)/\varepsilon)$ with $n_{\min}$ the minimum accumulated pseudo-count on load-bearing edges. The $1/(n+1)$ rate is **structurally forced** — it is the log-odds update magnitude per cycle under P's Aczél-Cauchy-FE uniqueness theorem, and no choice of coordinate escapes it. The result sharpens 8.10's forgetting prerequisite from "required for asymptotic persistence" to "required for detection-latency bounded independently of operating point." The broader myopia observation — that successful, high-capability organizations systematically underinvest in detecting regime changes that would require structural adaptation — admits a complementary decision-theoretic account via 9.2's oracle analysis; this segment's contribution is the signal-side lower bound, not the decision-side account.

Formal Expression.

Setup. An agent with a Beta-Bernoulli strategy DAG per 8.10 / N, credit assignment via the log-odds signal of 8.6 (forced by P), and no forgetting ($\lambda = 1$, pseudo-counts n_k accumulate monotonically). Let E_{load} denote the load-bearing edges on the current active plan; $n_{\min} = \min_{k \in E_{\text{load}}} n_k$. The agent has been operating with model-class fitness $\mathcal{F}(\mathcal{M})$ near 1, adaptive reserve $\Delta\rho^* > 0$, control regret δ_{regret} small — a high-operating-point configuration.

At cycle t_0, a regime change occurs within the current model class (a true edge probability shifts by ε; the L0 graph remains correct; the agent's model class still suffices). This is case R1 in the spike taxonomy — a within-class drift change. Other regime-change cases (R2 model-class inadequacy; R3 L0→L1 structural transition) are deferred to Working Notes and X respectively.

Detection-latency theorem.

Proposition. Under the setup, the expected number of cycles T_{detect} required for the log-odds coordinate on any load-bearing edge $k \in E_{\text{load}}$ to cross a fixed detection threshold $\Delta\lambda_{\text{detect}}$ in response to the regime change satisfies

Derived (detection-latency-R1, from edge-update-natural-parameter + deriv-edge-credence-dynamics).

$$\boxed{\mathbb{E}[T_{\text{detect}}] \;=\; \Omega(\Delta\lambda_{\text{detect}} \cdot (n_{\min} + 1)/\varepsilon)} \tag{S.1}$$

Derivation. The default signal function of 8.6 (under the log-odds coordinate forced by P) updates an edge's log-odds credence by

$$\lambda_k^{\text{new}} = \lambda_k + \eta_{\text{edge}} \cdot \iota_k \cdot J_k \cdot (y_G - \hat{P}_\Sigma)/\|\mathbf{J}\|^2 \tag{S.2}$$

with $\eta_{\text{edge}} = 1/(n_k + 1)$ for Beta-Bernoulli (Prop B.4 of N). Per-cycle the expected log-odds update magnitude is bounded:

$$\mathbb{E}|\Delta\lambda_k| \leq \frac{|J_k| \cdot \mathbb{E}|y_G - \hat{P}_\Sigma|}{\|\mathbf{J}\|^2 \cdot (n_k + 1)} \tag{S.3}$$

Under regime change R1 with observable footprint ε, the expected systematic residual $\mathbb{E}|y_G - \hat{P}_\Sigma| = \Theta(\varepsilon)$ as $\varepsilon \to 0$ (the misspecified-edge residual is proportional to the edge's probability shift, via the linearization of the Bernoulli likelihood in a neighborhood of the pre-change parameter). Combining: the per-cycle expected log-odds increment on load-bearing edges is $O(\varepsilon/(n_{\min} + 1))$.

For the agent to cross $\Delta\lambda_{\text{detect}}$, expected cycles required is at least $\Delta\lambda_{\text{detect}}/(O(\varepsilon/(n_{\min}+1))) = \Omega(\Delta\lambda_{\text{detect}} \cdot (n_{\min}+1)/\varepsilon)$. □

The rate is structurally forced.

The $1/(n + 1)$ scaling is not a property of the specific update rule choice. Per P, the log-odds coordinate is the *unique* additive-evidence coordinate satisfying the evidential-additivity axiom — the Aczél 1966 Cauchy-functional-equation uniqueness theorem forces it up to positive affine transformation. In this forced coordinate, the per-cycle increment for Beta-Bernoulli edges is forced to be $O(1/(n + 1))$: the Fisher-equivalent statement holds in any sensible coordinate. Rearranging the update to a different scale does not change the rate — it just changes the units in which the rate is measured.

Observation (rate-forced-by-aczel).

The forcing composition:

1. P forces the log-odds coordinate via evidential additivity.
2. In that coordinate, Beta-Bernoulli accumulation gives $\eta_{\text{edge}} = 1/(n + 1)$.
3. Therefore the per-cycle update magnitude is structurally forced at $O(1/(n + 1))$.

This is the specific link between AAT's constructive meta-pattern (Z, via P's theorem) and a downstream detection-latency consequence. The rate cannot be escaped without abandoning evidential additivity — which would invalidate the update rule on AAT-internal grounds, not merely operational ones.

Sharpening the forgetting prerequisite.

8.10 derives the forgetting prerequisite $(1 - \lambda) > \rho_\Sigma/R_\Sigma$ as required for *asymptotic persistence* — without forgetting, $\alpha_\Sigma = 1/(n + 1) \to 0$ and persistence eventually fails. The detection-latency theorem sharpens this to a **load-bearing claim about detection latency on the way to asymptotic failure**:

Corollary (forgetting-as-latency-bound, sharpens #schema-strategy-persistence).

Forgetting is required not only for asymptotic persistence, but also for detection latency to be bounded independently of operating point. Without forgetting, $n_{\min}$ grows monotonically, and $\mathbb{E}[T_{\text{detect}}]$ grows linearly with $n_{\min}$. With forgetting at rate $\lambda < 1$, the effective pseudo-count $n_{\text{eff}} = 1/(1 - \lambda)$ is bounded, and the detection latency caps at $\Omega(1/((1 - \lambda)\varepsilon))$ regardless of how long the agent has been operating.

This dualizes 8.10's asymptotic claim: forgetting is operationally load-bearing at every step, not only in the limit. An agent with the right λ has bounded detection latency throughout its lifetime; an agent with $\lambda = 1$ has detection latency that grows unboundedly with experience, producing the phenomenon of stability-induced myopia in practice.

What Is Derived vs. What Is Chosen.

Table S.1

Property	*Source*	*Strength*
R1 detection-latency theorem $\mathbb{E}[T_{\text{detect}}] = \Omega((n_{\min} + 1)/\varepsilon)$	P log-odds coordinate + N Prop B.4 $\eta_{\text{edge}} = 1/(n+1)$ + Pinsker-type linearization	Derived (conditional on Beta-Bernoulli + log-odds + no forgetting)
$1/(n+1)$ rate structurally forced	Composition of P's Aczél-Cauchy-FE theorem with Beta-Bernoulli accumulation	Proved (conditional on evidential-additivity axiom)
Sharpening of forgetting prerequisite from asymptotic persistence to bounded detection latency	8.10 + this theorem	Derived
R2 model-class inadequacy sub-case (C1-diagnostic blindness under misspecification)	Common-mode-bias argument on $A_O^{(1)}$ vs $V_O(\pi_{\text{current}})$	Discussion (sketch; see Working Notes)
R3 L0→L1 structural transition sub-case	Direct application of X Instance 1 — already derived there	Reference (not new content)
Probability-of-deferral bound	Derivable from the expected-time bound via Markov inequality	Discussion
Connection to institutional inertia / Christensen / competency traps	Stability-induced-myopia as a mechanism for the empirical pattern	Discussion
Connection to IDT sidecar monitoring (Hafez et al. 2026 89% vs 44%)	On-policy reward-based monitoring is the log-odds accumulator; IDT bi-predictability is a change-detector that bypasses the $1/(n+1)$ rate	Discussion

Epistemic Status.

Conditional. Max attainable: *exact* for R1 under stated scope (Beta-Bernoulli + log-odds forced + no forgetting + linearized residual near pre-change parameter); *robust qualitative* for general exponential-family edge with similar sufficient-statistic accumulation; *discussion-grade* for extensions beyond the Beta-Bernoulli / log-odds specialization.

The R1 theorem is a direct consequence of two derived AAT results (P's forced log-odds coordinate + N's Beta-Bernoulli η_{edge}) composed with a standard residual-magnitude argument (Pinsker-type inequality applied to the linearized Bernoulli likelihood). The mathematical core is not novel — what is novel is the AAT-framing: reading the $1/(n+1)$ scaling as *structurally forced* through composition of existing AAT theorems, not contingent on update-rule choice. The rate is unescapable without abandoning evidential additivity.

The forgetting-prerequisite sharpening follows by direct substitution — replace unbounded $n_{\min}$ with bounded $n_{\text{eff}} = 1/(1-\lambda)$ and the latency bound inherits the cap. This is where the segment's most concrete contribution lies: the existing forgetting prerequisite's justification expands from "asymptotic persistence fails without it" to "detection latency is unbounded in operating point without it." Two load-bearing consequences instead of one, from the same $\lambda < 1$ condition.

What this does not establish:

- R2 (model-class inadequacy) as a full derivation. The sketch argument — that $A_O^{(1)}$ and $V_O(\pi_{\text{current}})$ share common-mode bias under misspecification, so the C1 regret diagnostic under-reports — is qualitatively clear but quantitatively informal. Making it rigorous requires a model of the bias's functional form.
- R3 (L0→L1 transition) as new content. This is X Instance 1 — a structural no-go, not a latency bound. This segment references it but does not extend it.
- Decision-theoretic consequences. The theorem is about signal latency, not about whether the signal once received yields the right decision. 9.2's oracle analysis (which shows deliberation rarely chosen) operates at the decision-theoretic level; the myopia theorem operates at the signal-quality level. An agent with degraded signals + perfect decision rule still defers; this is complementary to the oracle's degraded-decision-with-perfect-signals result.
- Extension beyond Beta-Bernoulli. Gaussian edges admit $1/(n+c)$ scaling with analogous behavior; general exponential-family edges admit scaling following the sufficient-statistic accumulation. The qualitative form — expected detection time monotone in accumulated sufficient statistic — is more general than Beta-Bernoulli; the quantitative $1/(n+1)$ is specific.

Discussion.

The forcing composition is where the meta-pattern interacts with downstream results. Z is an AAT meta-segment naming the Cauchy-FE structural move at three layers (chain / divergence / update). This segment exhibits the specific case where that move's downstream consequences bind: the log-odds forcing at the update layer implies — through Beta-Bernoulli accumulation — a specific detection rate that cannot be escaped via re-parameterization. The meta-pattern's constructive power is visible here as a *negative* downstream consequence (bounded detection rate), which then becomes the positive motivator for forgetting. This is an example of AAT's meta-architecture composing constructively: the forcing move (positive) produces a bound (negative) that sharpens an existing requirement (positive).

Distinction from adjacent phenomena:

- 4.5 characterizes WHEN restructuring is needed at truth (model-class inadequacy). This segment characterizes WHEN (in terms of cycles) the agent's detection signal crosses threshold. The two together say: the agent will eventually detect a structural problem, but the latency can be large at high operating point.
- 9.2 gives a decision rule over exploit / explore / deliberate given signals. This segment is about signal quality — specifically, a predictable bias in signals as a function of operating point. The two compose: even an oracle's decision rule applied to myopia-biased signals produces deferred action.
- 8.10's forgetting prerequisite is a condition for asymptotic persistence. The corollary here is a condition for bounded detection latency — the same $\lambda < 1$ buys both.
- X Instance 1 (on-policy L0-insufficiency no-go) is a structural unboundedness result specific to L0/L1 transitions. The R1 detection-latency theorem is a quantitative bound on a broader class of regime changes (within-class drift, not just structural transitions).

Connection to cross-domain empirical patterns. Christensen's *Innovator's Dilemma*, Levitt & March competency traps, Hannan & Freeman structural inertia, March exploration/exploitation, and Eldredge-Gould punctuated equilibria all identify empirical patterns of stability-induced myopia across domains. The common story is that successful, mature, capable systems systematically underinvest in detecting regime changes that would require restructuring. The theorem supplies a mechanism: as experience accumulates ($n_{\min}$ grows), per-observation learning rate decays as $1/(n+1)$, expected detection time grows linearly, and the structural-adaptation trigger signal is systematically delayed. The phenomenon is not only decision-theoretic (incentive mis-alignment, short-horizon myopia) — it has a signal-side epistemic component forced by the log-odds coordinate. The signal-side and decision-side stories compose; neither subsumes the other.

Connection to IDT (Hafez et al. 2026) empirical result. Hafez et al. 2026 observed 89% vs 44% detection accuracy for Information Digital Twin (IDT) sidecar monitoring vs reward-based monitoring. The AAT-native explanation: reward-based monitoring is the log-odds accumulator — it inherits the $1/(n+1)$ detection rate and is subject to the theorem's latency bound. IDT's bi-predictability and entropy-change measures are structural-information-theoretic tests that bypass the log-odds coordinate entirely; they are change-detectors in the CUSUM / GLR tradition (Page 1954; generalized likelihood ratio tests) rather than Bayesian credence accumulators. The predicted gap between the two monitoring styles grows with accumulated experience — a testable refinement of the 89%/44% observation. A sidecar change-detector is always going to beat the log-odds accumulator on detection latency for within-class drift; the AAT theorem explains why without invoking the specific IDT mechanism.

The signal side and decision side as complementary. The fact that the oracle in 9.2 rarely chose deliberation is usually read as "agents correctly defer when deliberation is expensive and information is limited." This segment's theorem adds: "and the information is limited in a specific quantitative way that scales with operating point." Both are true; neither is contradicted. The detection-latency result does not overturn the oracle analysis; it provides the signal-quality context in which that decision analysis operates.

Working Notes.

- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. Removed at `candidate` stage per FORMAT.md Gate 4.
- **R2 sharpening (model-class inadequacy).** The argument: under C1 convention, $\delta^{(1)}_{\text{regret}} = A^{(1)}_O - V_O(\pi_{\text{current}})$ where both terms are computed under the current (now misspecified) M_t. Both terms bias in the same direction when the model class is structurally wrong; the gap stays small under common-mode bias. A clean version would compute the common-mode bias as a function of 3.5's model-error component, show it is $O(1)$ (not $O(\varepsilon)$) when the misspecification is directional, and derive a convention-dependent detection latency. Candidate one-evening spike.
- **R3 reference, not extension.** The L0→L1 structural-transition case is X Instance 1 — on-policy detection is forbidden by Bareinboim CHT, not merely slow. The detection-latency theorem here is about within-class drift (R1), which does not hit that no-go. When the regime change *does* involve an L0→L1 transition, the theorem does not apply and the no-go does; the two coexist at different regime-change types.
- **Extension beyond Beta-Bernoulli.** Gaussian edges: $\eta \propto 1/(n+c)$ for the Gaussian-mean update. Exponential-family edges in natural-parameter form: η follows sufficient-statistic accumulation. The *qualitative* form (detection time monotone in accumulated sufficient statistic) is general; the *quantitative* $1/(n+1)$ is specific to Beta-Bernoulli. A parametric version of the theorem covering general exponential-family edges would be a clean follow-up.
- **Probability bound.** The expected-time bound implies a probability bound via Markov: Pr[detect within T] $\leq T/\mathbb{E}[T_{\text{detect}}]$. Tighter concentration requires specifying the residual distribution's shape; candidate one-paragraph extension.
- **Testable prediction.** In agents with controlled accumulated experience n, detection latency for regime change with fixed observable footprint ε should scale linearly in n. Directly testable in multi-arm bandits with drifting reward distributions, in simulated organizations with commit-log-mediated task environments, in human-subjects studies on noisy-signal regime detection. Each would confirm or refute the $1/(n+1)$ scaling in its native setting.
- **Multi-timescale connection.** The theorem adds empirical content to 4.6: the slow (structural-adaptation) timescale slows with accumulated fast-timescale state — the two timescales are not independent. The fast one's accumulated state feeds back into the slow one's responsiveness. This is the formal version of the institutional-inertia observation and is adjacent to, but not subsumed by, I's current formulation.

- **Section III analog.** Do larger teams / organizations face proportionally longer detection latency via analogous accumulation? Candidate: team effective detection latency is bounded below by the slowest member's latency (the weakest-dimension version of 14.5). Connects to 13.1. Not derived here.
- **Class 3 (Coupled) architectures.** For fully-coupled agents (LLMs), the Beta-Bernoulli accumulator is absorbed into the single merged computation — there is no literal edge-credence accumulator. But there *is* an analog at pretraining: token-statistic accumulation. Do LLMs exhibit a myopia analog? This is in the domain of `03-llm-core/`, potentially connecting to `#obs-context-turnover` and the frozen-weights framing.

Appendix T

Discussion: Independence Audit

AAT's results depend on a recurring modeling move: treat some quantity as independent of another to obtain tractable mathematics, then identify the failure regime where independence breaks and specify the repair. This segment enumerates the independence assumptions used across the theory, their failure regimes, their diagnostic signals, and the repair operations AAT provides. The enumeration makes visible what is *not* an independence assumption — acyclicity of Σ_t, Cox-derived probability, the Lyapunov machinery of C — and therefore what survives when a particular independence assumption fails.

Formal Expression.

Six load-bearing independence assumptions in AAT, each paired with its failure regime and repair:

Discussion (independence-audit).

1. Directed separation: M_t update independent of G_t. **Statement:** $f_M(M_{\tau^-}, e_\tau)$ has no G_t argument — the epistemic update is goal-blind.

Where it appears: 5.3, the structural backbone of Section II. Feeds the orient cascade's sequential resolution (9.1), the causal validity of Q_O (5.6), and the scope of all Section II results to Class 1 (Separated) agents.

Failure regime: Class 3 (Coupled) architectures — transformer LLMs where attention processes goals and observations together. Motivated reasoning, confirmation bias, prompt-conditioned perception. Partially also Class 2 (Partial) agents.

Diagnostic signal: $\kappa_{\text{processing}} = I(G_t; M_{\tau^+} \mid e_\tau, M_{\tau^-}) / H(G_t \mid e_\tau, M_{\tau^-})$. Zero for Separated agents; near one for Coupled; intermediate for Partial.

Repair operation: Class 2 (Partial) approximation quality scales with $\kappa_{\text{processing}}$. Class 3 (Coupled) agents require the coupled formulation $X_{\tau^+} = f_X(X_{\tau^-}, e_\tau)$ without (M_t, G_t) decomposition — the scope of `03-llm-core/`. At the system level, Class 3 (Coupled) components can be wrapped in modular topology (separate observation processing, external monitoring — see `#der-directed-separation` Working Notes on the IDT pattern).

2. Causal sufficiency: no latent common causes among strategy nodes. **Statement:** Every common cause of two or more nodes in Σ_t is itself a node in Σ_t.

Where it appears: M (precondition for the CMC-based Markov proof); 7.3 edge-independence in status propagation; all of Section II's strategy-layer formal results (Props B.1–B.6, the sector condition transfer of B.5, persistence of δ_s).

Failure regime: The dominant real-world case in complex, multi-stakeholder, or adversarial environments — shared infrastructure, market conditions, correlated adversary actions, common-mode risks, supply-chain dependencies.

Diagnostic signal: Pairwise covariance among sibling edges after edge credences have converged. Positive covariance rejects the independence hypothesis and localizes where a common cause is missing. See 8.2.

Repair operation: L1 augmentation — add common-cause nodes and restructure the DAG so each common cause is factored *above* the correlation it creates (7.3 Correlation Hierarchy; N Prop B.6). L0 formal results transfer exactly to correctly constructed L1 DAGs because L1 restores causal sufficiency by construction. The orient cascade's step 4c triggers this escalation.

3. Channel independence: observation channels contribute non-redundant correction. **Statement:** $\mathcal{T} = \sum_k \nu^{(k)} \cdot \eta^{(k)*}$ with each channel contributing independently to effective tempo.

Where it appears: 3.8 (core definition); inherited by 4.4, 13.4, 13.1 (communication tempo), 11.2, and any result using scalar $\mathcal{T}$.

Failure regime: Any system with redundant or correlated information sources — overlapping sensors, correlated teammate reports, redundant telemetry. In multi-agent settings, allies reporting the same intelligence source.

Diagnostic signal: $I(e^{(1)}; e^{(2)} \mid M_{\tau^-})$ — pairwise mutual information between event streams conditioned on prior model state. Non-zero mutual information signals redundancy.

Repair operation: Under correlation, $\mathcal{T}$ satisfies a strict inequality: $\mathcal{T} \leq \sum_k \nu^{(k)}\eta^{(k)*}$, with equality iff channels are informationally independent. The additive formula is an upper bound. For precise tempo, the effective capacity must account for mutual information between channels. Propagates as a redundancy penalty into every segment using scalar tempo.

4. Unity-dimension independence: $(U_M, U_O, U_\Sigma, U_{\mathbf{obs}})$ substantially independent. **Statement:** The four unity dimensions of 12 parametrize composite quality along substantially independent axes.

Where it appears: 12 framing; 12.1's rate-distortion family indexed by each unity separately.

Failure regime: Shared models enable implicit strategic coordination (high $U_M \to$ high U_Σ without explicit policy sharing); aligned objectives often induce strategic coordination. Also *update-rule heterogeneity* (an axis not covered by any unity dimension) contributes to closure defect independently — the two-Kalman non-degenerate case shows $\varepsilon_x > 0$ from differing Kalman gains at perfect U_M.

Diagnostic signal: No clean one. The hypothesis is empirical and the theory currently logs it as working-position rather than resolved (see 12 Working Notes: update-heterogeneity gap).

Repair operation: Two options explored in the theory: (a) accept a two-axis structure (unity × homogeneity) for closure defect, (b) add a fifth update-homogeneity dimension. The current working position is (a); formal resolution open. Downstream uses of unity dimensions should not treat them as cleanly orthogonal.

5. Independent edge outcomes in AND/OR propagation. **Statement:** In $\hat{P}_\Sigma$ status propagation, sibling edge outcomes are independent given their parents.

Where it appears: 7.3 status propagation formula; 7.2; every downstream quantity computed from $\hat{P}_\Sigma$ (satisfaction gap, control regret via A_O, strategy-plan-confidence error δ_s).

Failure regime: Same as causal sufficiency (item 2) — the CMC theorem makes them the same condition: causal sufficiency $\Leftrightarrow$ exogenous noise independence $\Leftrightarrow$ edge-outcome independence. When one fails, they all fail.

Diagnostic signal: Same as item 2.

Repair operation: Same as item 2 (L1 augmentation). The assumption is not a separate modeling choice from causal sufficiency; the theory treats it as the *consequence* of causal sufficiency (via CMC), not an independent axiom.

6. Scalar tempo / isotropic correction. **Statement:** $\mathcal{T}$ as a scalar captures the agent's correction capacity.

Where it appears: 4.4 linear operational form; most Section I results stated in scalar ($\mathcal{T}, \rho, \|\delta_{\text{critical}}\|$) form; 8.8 aggregate.

Failure regime: Any real multi-dimensional system with non-uniform correction capacity across dimensions. Simulation confirms scalar $\rho/\mathcal{T}$ overestimates by up to 72% in anisotropic systems, with the weak dimension accounting for 84% of total mismatch.

Diagnostic signal: Gain variation across dimensions (easily computable when gains are explicit — Kalman, gradient descent with diagonal preconditioning).

Repair operation: Per-dimension persistence: $\mathcal{T}_k > \rho_k/\|\delta_{\text{critical},k}\|$ for each dimension. See 14.5. The weakest dimension is the bottleneck. Same structural pattern propagates to strategic tempo (8.8 Per-edge persistence) — the bottleneck edge determines persistence, not the aggregate.

What is* not *an independence assumption.

Several results that might appear to depend on independence actually do not:

- **Acyclicity of Σ_t** is proved from temporal ordering over a finite horizon (M). Independent of any independence assumption.
- **Cox-derived probability structure** at edges is independent of edge-independence — Cox forces the *measure* to be probability, not the joint structure.
- **Sector-persistence template machinery** (C) depends only on the sector condition (T2) and bounded disturbance (T3); it does not require any independence assumption on the state variable or correction function.
- **The DAG structure derivation** (directed edges, acyclicity, representability as a Bayesian network) does not require causal sufficiency — only the *Markov factorization* within the derived structure does.

This is the boundary between AAT's robust results and its conditional ones: robust results survive when the independence assumptions fail; conditional results break and require their specific repair operations.

Epistemic Status.

Robust-qualitative. Max attainable: *robust-qualitative*. The enumeration is definitional — each independence assumption is stated in a named segment, and the repair operations are either formally specified (causal sufficiency ↔ L1 augmentation) or named open problems (unity-dimension independence). What this segment adds is the cross-cutting view: the six assumptions are the recurring modeling move that distinguishes AAT's exact-tractable results from its conditional results, and the repairs are (mostly) already present in the theory but scattered across segments.

The enumeration is not exhaustive. Other independence-flavored assumptions exist in specific instantiations (e.g., independent Beta priors across edges in the credit-assignment analysis of 8.6). The six listed are the load-bearing ones — those whose failure regime the theory has actually characterized and provided a repair for.

Discussion.

Why enumerate these in one place. AAT's theoretical contribution is often described as "integration": bringing control theory, causal inference, information theory, and agent architecture under one framework. Integration *is* achieved largely through the independence assumptions listed above: each is a bridge that lets a module from one discipline plug into a module from another. Directed separation lets control-theoretic Lyapunov analysis operate on the epistemic substate without goal-entanglement. Causal sufficiency lets Pearl's DAG machinery apply to strategy without requiring the agent to model every environmental common cause. Channel independence lets the tempo framework sum over heterogeneous observation modalities.

When these bridges fail, the integration does not fail catastrophically — each failure has a characterized repair. But the theory's "exact" claims depend on the bridges holding, and recognizing this is the honest way to present the framework's scope. A reader who understands the six assumptions and their repairs understands where AAT's results apply exactly, where they apply approximately, and where they require structural revision.

Connection to U. The L0/L1 distinction for causal sufficiency (items 2 and 5) is one instance of a meta-pattern: AAT's results are parameterized by approximation level, with proved monotonicity between levels. See U for the full treatment — the pattern also applies to the C1/C2/C3 value-convention hierarchy and the Tier 1/2/3 bridge-lemma contraction structure.

What this does NOT say. The enumeration does not claim the theory's results are fragile or routinely fail. Most independence assumptions hold approximately in most well-behaved domains, and the theory's quantitative predictions are typically within the regime where assumptions apply. The audit is a scope clarification, not a criticism.

Working Notes.

- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. Failure-regime references under Assumption 1 updated accordingly. The C1/C2/C3 mention at line 120 is the value-convention hierarchy (a different ladder), not the GUC axis — left unchanged. Removed at `candidate` stage per FORMAT.md Gate 4.
- **Interaction between assumptions.** The six assumptions are not fully orthogonal. Causal sufficiency failure and edge-independence failure are the same failure (item 5 is a consequence of item 2 via CMC). Directed separation failure can propagate through the orient cascade into the strategy-layer results (which inherit their structure from the cascade). A fuller treatment would map the dependency graph among assumptions and characterize cascading failure modes.
- **Quantitative failure modes.** For each independence assumption, the theory has or could have a quantitative failure bound — how much the dependent result degrades as a function of the independence-violation magnitude ($\varkappa_{\text{processing}}$, the covariance magnitude, the mutual information between channels). These bounds are partially specified in segment-level caveats; collecting them in one place would strengthen the audit.

- **Empirical audit.** A concrete agent could be scored against the six assumptions to produce an "independence profile." For software agents: directed separation ~holds (Separated), causal sufficiency ~holds (test-based Regime A), channel independence ~holds (tests target disjoint behaviors), scalar tempo may fail (component-level gain variation). For LLM agents: directed separation fails structurally (Class 3 Coupled), causal sufficiency varies by task, channel independence varies. The profile would be a diagnostic for where the theory's exact claims apply to the agent and where they degrade.

Appendix U

Discussion: Approximation Tiering

AAT uses a recurring meta-pattern for handling intractability: when a problem admits no tractable exact treatment in general, introduce a tiered hierarchy of approximations with proved monotonicity between tiers and a diagnostic for ascending when needed. Three such hierarchies exist in the theory — the Correlation Hierarchy (L0/L1/L2) in 7.3, the Convention Hierarchy (C1/C2/C3) in 5.6, and the Tier 1/2/3 contraction taxonomy in 11.1. This segment articulates the pattern explicitly, identifies what makes a successful approximation tiering, and notes where other scattered simplifications might fit the same shape.

Formal Expression.

A successful approximation tiering has four components:

Discussion (approximation-tiering).

(AT1) A parameter indexing tractability. Some quantity (correlation modeling depth; continuation-policy fidelity; operator regularity) that admits a natural ordering from least to most demanding computation.

(AT2) Monotonicity. A proved ordering of the results produced at each level — each higher tier dominates the lower in the direction of interest (calibration accuracy, diagnostic force, guarantee strength).

(AT3) Graceful degradation. Lower tiers produce usable (not vacuous) results, with the specific form of the degradation characterized. An agent stuck at the lowest tier knows what it is missing, not just that it is incomplete.

(AT4) Ascension diagnostic. A signal the agent can detect at a lower tier that indicates escalation to a higher tier is warranted. Without this, the tiering is descriptive but not operational.

The three AAT tierings.

Table U.1

Hierarchy	*Parameter*	*Monotonicity*	*Lowest tier*	*Highest tier*	*Ascension diagnostic*
Correlation (7.3)	Modeling depth of inter-edge dependence	L0 is conservative; L1 is unbiased on augmented graph; L2 is the full joint	L0 (independence model, $O(\|V\|+\|E\|)$ propagation)	L2 (full joint, $O(2^m)$ in general)	Sibling-edge covariance after credence convergence (8.2)

Hierarchy	*Parameter*	*Monotonicity*	*Lowest tier*	*Highest tier*	*Ascension diagnostic*
Convention (5.6)	Continuation-policy fidelity	$A_O^{(1)} \leq A_O^{\mathrm{RH}} \leq A_O^{\mathrm{B}}$ (proved monotonicity)	C1 (one-step improvement)	C3 (Bellman optimal)	Persistent $\delta_{\mathrm{sat}} > 0$ under C1 that C2 replanning may resolve
Contraction (11.1)	Operator regularity for bridge lemma	Tier 1 ⊃ Tier 2 ⊃ Tier 3 in terms of proved contraction strength	Tier 3 (domain-specific verification required)	Tier 1 (contraction proved for full class)	Structural test: is the correction strongly monotone, locally convex, or neither?

Each row satisfies (AT1)–(AT4) fully — these are mature tierings in the theory. Each tiering is independently motivated by a different source of intractability, and each has its own ascension mechanism. Their structural similarity is not designed-in; it is the consequence of facing three separate intractable problems with the same modeling strategy.

Candidate future tierings. Several other scattered simplifications in the theory have the right shape to become tierings but are not currently organized that way:

- **Scalar vs. per-dimension tempo** (around 3.8, 14.5). Tier 0 (scalar isotropic), Tier 1 (per-dimension), Tier 2 (full tensor with off-diagonal coupling). Monotonicity: scalar overestimates margin, per-dimension is exact for diagonal-gain systems, full tensor handles off-diagonal gain coupling. Ascension diagnostic: gain variance across dimensions.
- **AND/OR parameterization** (7.2). Tier 0 (AND/OR, $O(k)$ parameters per node), Tier 1 (weighted/threshold, $O(k)$ additional parameters), Tier 2 (full CPT, $O(2^k)$ parameters). Monotonicity: expressiveness. Ascension diagnostic: systematic miscalibration at AND/OR nodes that survives L1 augmentation.
- **Identifiability regimes** (8.5). Regime A/B/C is already a tiering of causal identification quality, with ι_{ij} as the indexing parameter. Monotonicity: interventional ≥ partial ≥ observational. Ascension diagnostic: available intervention opportunities in the domain.

Promoting these to named tierings with explicit (AT1)–(AT4) structure would make AAT's scope parameterization more uniform. This is not pursued here — it is noted as a direction for editorial consolidation.

Epistemic Status.

Robust-qualitative. Max attainable: *robust-qualitative*. The meta-pattern is an editorial observation: three existing hierarchies in the theory share the same four-component structure, and the shared structure is worth naming because it makes the theory's scope-parameterization move visible rather than implicit. The pattern itself is not a theorem — there is no result that says "every intractable AAT problem admits a (AT1)–(AT4) tiering." The candidate future tierings are conjectures about where the pattern could be extended; each would require its own monotonicity proof and ascension diagnostic.

The three existing tierings are individually grounded: the L0→L1 monotonicity holds under the CMC theorem and augmentation construction; the C1→C2→C3 monotonicity is proved in 5.6; the Tier 1/2/3 taxonomy is grounded in operator theory (strong monotonicity, local convexity, arbitrary).

Discussion.

Why this matters. AAT's results are often presented as "exact under assumption X." A reader can misread this as "the theory works when X holds and breaks otherwise." The tiering pattern reveals a different structure: when X fails, a named lower tier takes over with a characterized weaker result, and the theory provides a diagnostic for when to escalate to reinstating X by structural repair. This is graceful degradation — a property that a theory wanting to apply across a range of agents and domains needs, and that AAT achieves in three places independently.

Reading AAT through the tiering lens: the theory's "exact" core is what it guarantees at the highest tier of each hierarchy. Its "conditional" periphery is where specific tiers are in force. The claim is not that agents should always operate at the highest tier — that would often be intractable — but that the theory tells the agent which tier it is in and how to get to a higher one when the binding constraint changes.

Connection to T. The two meta-segments are complementary. T enumerates the independence assumptions whose failure drops a result from exact to conditional. This segment enumerates the tierings that provide the structured *recovery* from those drops. Together they characterize AAT's scope parameterization: the independence audit says where the boundaries are; the tiering pattern says how to navigate within them.

What a full tiering promotion would look like. If the scattered simplifications listed above were promoted to named tierings, the theory's scope-parameterization would be dramatically more uniform: every conditional result would come with its own tiering and ascension diagnostic. This is aspirational. The three existing tierings took substantial segment-work to formalize (a full Appendix derivation for the CMC, a monotonicity proof for the convention hierarchy, a spike for the incremental sector bound). Extending the pattern to five or six hierarchies would be correspondingly substantial.

The pattern is not unique to AAT. Approximation tiering is a common move in applied mathematics — numerical methods (low-order vs. high-order integrators with error bounds), information theory (typical vs. exact coding with rate-distortion monotonicity), statistical inference (maximum-likelihood vs. Bayesian vs. nonparametric). AAT's contribution is not the tiering pattern itself but its deployment across the specific intractable problems that adaptive-agent theory faces.

Working Notes.

- **Formal characterization of the tiering pattern.** Is there a mathematical structure that unifies approximation tiering across domains? Candidates: lattices of model classes ordered by inclusion, rate-distortion curves with explicit corners, hierarchies of Galerkin approximations. None of these is an exact match; the AAT tierings are closer to "ordered families of sufficient conditions" than to any standard mathematical hierarchy. Worth investigating whether a cleaner abstract pattern exists.
- **Interaction between tierings.** An agent operating at L1 correlation, C2 convention, and Tier 2 contraction is in a specific combined regime. The cross-hierarchy interactions are not worked out — is there cross-hierarchy monotonicity? (E.g., does L1 change anything about the convention hierarchy's guarantees?) Each tiering is independently-grounded but the joint structure is not yet mapped.
- **Diagnostic costs.** Each ascension diagnostic has a cost: detecting sibling covariance requires observing convergence; detecting C1-inadequacy requires comparing replanning values; detecting Tier 1 violations requires verifying strong monotonicity. A unified treatment of "when is the diagnostic itself worth running?" would connect this segment to 4.1 and the allocation analysis in 9.2.

Appendix V

Discussion: Compression Operations in AAT

AAT contains four compression operations — the epistemic model M_t, the strategy DAG Σ_t, shared intent G_t^{shared}, and the composition projection Λ — each formulated in its own segment with its own objective. Three of the four are written in Information Bottleneck (IB) form already; the fourth is stated as an IB constraint. This segment makes the shared shape explicit, promotes one underspecified source (the ontologically ambiguous "true causal structure" for Σ_t) to a cleaner formulation parallel to M_t, and establishes that composition admissibility (P1) is the Lagrangian-dual of a standard IB objective. It does *not* claim the four operations reduce to a single optimization problem — cross-instance theorems do not follow from the shared shape alone, and several conditions (Lipschitz regularity (P2), dimensional reduction (P3) in the Gaussian case, interventional relevance for Level-2 edges) remain outside the IB frame.

Formal Expression.

The shared IB shape.

Every compression operation in AAT has an objective or constraint of the form:

Discussion (ib-shape).

$$T^* = \arg\min_{T|X} \left[\, I(X;T) \;-\; \beta \cdot I(T;Y) \,\right] \tag{V.1}$$

with the Markov chain $Y - X - T$. The four AAT instances specialize this with different bindings:

Table V.1

Instance	*X (source)*	*T (compressed)*	*Y (relevance variable)*	*β (trade-off)*
Model compression (2.2)	$\mathcal{C}_t$	M_t	$o_{t+1:\infty} \mid a_{t:\infty}$	$\beta(\rho, \pi)$ — volatility and policy
Strategy compression (8.9)	$\mathcal{C}_t$	Σ_t	$\pi^* \mid M_t$	β_Σ — cognitive cost per decision-bit
Shared intent (12.2)	$G_t^{\text{full}} = (O_t, \Sigma_t)$	G_t^{shared}	$a_t^{\text{coordinated}}$	bandwidth per coordination-bit
Composition projection (11.1 P1)	$X_{\text{micro},t}$	$\Lambda_X(X_{\text{micro},t})$	$o_{\text{micro},t+1} \mid a_{\text{micro},t}$	$\beta(\epsilon_I)$ — rate-distortion Lagrange multiplier

What the four instances share: *shape* (the objective structure), *variational calculus* (minimization over stochastic compressors), and *rate-distortion interpretation* (each trade-off parameter indexes a point on the frontier). What they do not share: source type (history vs. structured state), relevance-variable availability (observed vs. latent), computability (Gaussian closed forms vs. variational approximation), or a single joint optimization across the four. The level of unification is *medium*: shared shape and vocabulary, not a shared master problem.

Strategy compression: source reformulation.

The current statement in 8.9 has the Σ_t IB objective as:

Formulation (strategy-compression-source).

$$\Sigma_t^* = \arg\min_{\Sigma_t} \big[\, \mathrm{DL}(\Sigma_t) \;-\; \beta_\Sigma \cdot I(\Sigma_t;\; \pi^* \mid M_t) \,\big] \tag{V.2}$$

Two issues with this as currently written:

1. **The compression cost is description length, not mutual information.** DL and $I(X;T)$ are related through coding-theoretic equivalences but coincide only under specific coding schemes. Using DL in the complexity term blocks the identification of this objective as an IB instance directly.
2. **The "source" is not an AAT object.** To fit the IB shape, the objective implicitly treats Σ_t as a compression of "the true causal structure." That structure is not part of AAT's ontology — the agent never has access to it; it is only ever implicit.

Reformulation. Treat Σ_t as a compression of $\mathcal{C}_t$ (the interaction history — the agent's only evidence) *for decision-relevance*, parallel to M_t which is a compression of $\mathcal{C}_t$ *for prediction-relevance*:

$$\Sigma_t^* = \arg\min_{\Sigma_t} \big[\, I(\mathcal{C}_t;\Sigma_t) \;-\; \beta_\Sigma \cdot I(\Sigma_t;\; \pi^* \mid M_t) \,\big] \tag{V.3}$$

Under this reformulation:

- The source $\mathcal{C}_t$ is a well-defined AAT object, shared with the M_t instance.
- The two instances differ cleanly in relevance variable: M_t is compressed for prediction ($Y = o_{t+1:\infty} \mid a$); Σ_t is compressed for guidance ($Y = \pi^* \mid M_t$). Prediction is about *what will happen*; guidance is about *what to do*. Both are computed from the same history, with different targets.
- The information cost $I(\mathcal{C}_t;\Sigma_t)$ replaces $\mathrm{DL}(\Sigma_t)$ as the theory-level compression term. The DL formulation remains useful as an *operational* cost measure for specific DAG encodings; it is not the theoretical quantity the IB objective minimizes.

Relationship between the two cost measures. Under MDL with a specific encoding scheme for DAGs (the one in 8.9), $\mathrm{DL}(\Sigma_t)$ is an upper bound on $I(\mathcal{C}_t;\Sigma_t)$ for DAGs produced by the given encoder — coding cost dominates distinguishability cost. In practice, DL is computable and I is not, so the operational minimization uses DL as a proxy; the theoretical minimization uses I. The IB objective above is the theoretical statement; the DL-based minimization in 8.9 remains the practical one.

Variational form. The Shannon mutual information $I(\Sigma_t;\pi^* \mid M_t)$ in the relevance term collapses to zero when π^* is deterministic-from-M_t. The variational form (cf. 8.9) replaces the relevance term with the KL-divergence $D_{\mathrm{KL}}(\pi^*(\cdot \mid M_t) \,\|\, Q_{\Sigma_t}(\pi \mid M_t))$ — note the π^*-first direction, which is *forced* by a regret-bound derivation (full derivation and admissible-divergence family analysis in O). Under bounded value range and deterministic π^*, Pinsker's inequality gives $R(Q_{\Sigma_t}) \le V_{\max}\sqrt{\frac{1}{2}D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})}$ where R is the strategy-induced regret against π^*; the opposite KL direction is vacuous ($+\infty$ under deterministic π^* whenever Q_{Σ_t} has off-optimum mass). The π^*-first KL is well-defined and graded under deterministic π^*. Under the variational reading, the AAT Σ_t is a tractable approximation of the policy-relevant posterior, and the KL term measures approximation quality — aligning the strategy compression with the variational free energy decomposition $-F = \text{accuracy} - \text{complexity}$ in active inference (Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017; Da Costa, Parr, Sajid, Veselic, Neacsu & Friston 2020). The direction alignment is convergent: both AAT's regret-bound and active inference's variational-free-energy derivations pick π^*-first KL (reverse-KL in the variational-inference vocabulary), with AAT's additional interpretation being an upper-regret-bound rather than free-energy-gradient. The shared-IB-shape framing of the four AAT compression operations is the rate-distortion specialization of this variational picture; AAT's commitment is to the rate-distortion form (which gives the (P1) Lagrangian-dual derivation and the four-instance unification at U-medium), not to the full variational free-energy interpretation.

Composition admissibility (P1) as IB Lagrangian-dual.

11.1 condition (P1) is currently stated as a lower-bound constraint:

Derived (p1-ib-dual, from composition-closure + rate-distortion duality).

$$I\big(\Lambda_x(X_{\text{micro},t});\ \Lambda_o(o_{\text{micro},t+1}) \mid \Lambda_a(a_{\text{micro},t})\big)\ \geq\ (1-\epsilon_I)\cdot I\big(X_{\text{micro},t};\ o_{\text{micro},t+1} \mid a_{\text{micro},t}\big) \tag{V.4}$$

This is the *constraint form* of an IB problem. In Lagrangian form:

$$\Lambda^*\ \in\ \arg\min_{\Lambda\in\mathcal{P}} \big[\, I(X_{\text{micro}};\Lambda_x(X_{\text{micro}}))\ -\ \beta(\epsilon_I)\cdot I(\Lambda_x(X_{\text{micro}});\ Y_{\text{rel}})\,\big] \tag{V.5}$$

where $Y_{\text{rel}} = (o_{\text{micro},t+1} \mid a_{\text{micro},t})$ and $\beta(\epsilon_I)$ is the Lagrange multiplier corresponding to the relevance-preservation tolerance ϵ_I. The correspondence $\epsilon_I \leftrightarrow \beta$ is the standard rate-distortion duality: smaller ϵ_I (more relevance preserved) corresponds to larger β (less aggressive compression).

Consequence: admissible projections are those that sit *on or above* the IB frontier at rate $I(X;T) \leq I_{\max}(\epsilon_I)$. The information-theoretic content of (P1) is exactly "project onto the IB frontier with a tolerance ϵ_I." This formalizes the connection previously logged in 11.1's Working Notes and in 12.1's §Connection to the Information Bottleneck.

What this resolves. The 11.1 Working Note "Open: Information Bottleneck unification" is now resolved for (P1): it is the Lagrangian-dual of the IB constraint at $\beta(\epsilon_I)$. The corresponding Working Note in 12.1 §6 moves from conjecture to derived result. Nothing else about (P1) changes — the condition continues to define admissible projections; only its information-theoretic reading is now explicit.

What stays separate from the IB frame. Three admissibility and structural conditions do *not* reduce to IB:

- **(P2) Lipschitz continuity.** Not an IB constraint. The bridge lemma in 11.1 requires (P2) for analytic reasons (propagating bounded closure defect into bounded trajectory error); IB does not impose any continuity condition on compressors. (P2) remains a separate admissibility condition.
- **(P3) Dimensional reduction.** In the Gaussian-IB case relevant to composition, the IB-optimal T at any finite β typically uses full support of $\mathbb{R}^{\dim X}$; the categorical dimensionality reduction $\dim \mathcal{X}_c < \dim \mathcal{X}_{\text{micro}}$ is a *harder* condition than any rate constraint. (P3) remains separate. (In discrete cases it may be rate-implied, but the composition instance is Gaussian.)

- **Interventional relevance (Level 2).** The relevance variable in all four instances is associational (Y in a joint distribution with X and T). Strategy edges in the regime-indexed interpretation (8.5) want interventional relevance for Regime A: "what edge (i, j) predicts *under* $do(i)$." This is a strictly stronger requirement than IB provides. Adapting IB to interventional relevance (causal IB, Wieczorek & Roth 2017 and follow-ups) is an extension direction, not a specialization of the master IB.

The honest slogan is therefore "the (P1)-analog in each compression operation is IB; regularity, dimensionality, and interventional relevance are separate conditions that compose with it."

Epistemic Status.

Robust-qualitative. The claim that the four compression operations share IB shape is *discussion-grade* — it is a presentational observation supported by examining each segment's formulation and noting structural alignment. The Σ_t source reformulation is a *formulation choice* replacing one underspecified source with a cleaner one; no derivation is needed because the new formulation and the old are not the same claim. The (P1) as Lagrangian-dual of IB is *derived* — rate-distortion duality is standard (see §I.12–13 of Cover & Thomas) and the constraint-form ↔ Lagrangian-form equivalence is mechanical.

Max attainable: *robust-qualitative* for the shared-shape claim; *derived conditional on rate-distortion duality* for (P1) as IB Lagrangian-dual; *formulation* for the Σ_t source reformulation. The strongest version of the unification claim — "U-strong," that the four operations are the same optimization problem with different bindings — is *not* established and is unlikely to be, for reasons in the Discussion.

This segment does not promote any of the four instance segments. Each remains the primary site for its own instance. What this segment adds is the shared-shape framing, the Σ_t source fix, and the (P1) derivation.

Discussion.

Why not go further. Two distinct claims could be made: (a) U-strong — the four operations are the *same* optimization problem with different bindings, so a single derivation specializes to each instance; (b) U-medium — the four share *shape*, vocabulary, and variational structure but differ in substantive ways (source type, relevance-variable availability, computability) that prevent a single derivation from covering all four. U-strong requires cross-instance theorems, results of the form "because these are all IB, property X holds across all four." No such theorems are identified. The shared IB shape is legibility-enhancing but does not produce deductions for free. This segment therefore states U-medium (shared shape) honestly rather than overclaiming U-strong.

Honest credit to the hierarchical-generative-model lineage. AAT's four compression operations are a structurally narrower family than the operations a hierarchical generative model in the predictive-coding lineage (Friston 2008, "Hierarchical models in the brain," *PLoS Comp. Biol.* 4; Friston 2010, "The free-energy principle: a unified brain theory?" *Nat. Rev. Neurosci.* 11; Clark 2013, "Whatever next?", *Behav. Brain Sci.* 36; Hohwy 2013, *The Predictive Mind*, OUP) could express. A hierarchical generative model layers compressions of compressions, with each layer producing a representation tuned to the next layer's prediction target — and the AAT compressions (M_t for prediction, Σ_t for guidance, shared intent for coordination, Λ for level-bridging) are all expressible within that frame as specific layer-bindings. What AAT adds is structural: (a) the *relevance variables* Y are made first-class with explicit per-instance bindings (see the table in §"The shared IB shape"); (b) the (P1)–(P3) admissibility conditions for composition give a measurable closure-defect bound that hierarchical-generative-model layering does not natively produce (11.1); (c) regime-indexed edges (8.5) introduce Pearl-Level-2 relevance for Regime A, which standard hierarchical generative models do not address. The shared family is real; AAT's additions are also real. The honest framing is "AAT's compressions are a structured subset of the hierarchical-generative-model family with additional structure load-bearing for AAT's specific results."

Why this is not the "biggest available unification." It is better described as the *most legible* unification. The pattern is already half-named at each instance site (three of four segments already write their objectives in IB form); the segment-level contribution is sharpening the naming and fixing one formulation issue (Σ_t source), not discovering a new structural result. The biggest leverage concision move in the theory — the sector-Lyapunov template in C — is substantively different: it removes repeated content and absorbs proofs that previously appeared in multiple segments. This segment adds legibility without removing much.

What each β means. The four trade-off parameters are not unified in value, only in role:

- $\beta(\rho, \pi)$ for M_t depends on environmental volatility and policy; fast-changing environments or high-value policies both warrant higher β.
- β_Σ for Σ_t is the cognitive cost per decision-relevant bit retained; bounded-context agents (LLMs with finite windows) have low β_Σ.
- The bandwidth β for shared intent is the communication-capacity-per-coordination-bit; low-bandwidth teams have low β.
- The $\beta(\epsilon_I)$ for (P1) is the rate-distortion Lagrange multiplier at tolerance ϵ_I.

These are four calibration problems, not one. A theory of *resource allocation across the four compression operations* — where a single cognitive budget is split across modeling, strategy, communication, and composition — is not currently in AAT, and the unification does not produce it. If such a theory is ever needed, it would tie the four βs together through an outer optimization; the spike flags this as an open direction.

Relationship to 12 and 12.1. The unity dimensions parametrize rate-distortion curves for closure defects (12.1). Under the present segment's framing, unity dimensions are *curve parameters for the* Λ *instance of the IB schema*. This tightens the existing 12.1 Discussion, which already presented the rate-distortion reading as the correct interpretation; it is now stated uniformly across the four compression operations.

Relationship to M. The DAG-uniqueness result derives the *shape* of Σ_t from operational postulates (M, Cox-analog). The present segment characterizes the *compression level* of Σ_t. These operate at different levels: uniqueness supplies the representational type (the compressed T is a DAG), IB supplies the rate-distortion trade-off within that type (how richly the DAG is populated). The two arguments are independent and compatible — IB-compressed Σ_t is a DAG-with-fewer-nodes, not a non-DAG.

Position relative to the additive-coordinate-forcing pattern. The IB Lagrangian's $I(X;T) - \beta \cdot I(T;Y)$ is an additive decomposition on a log-scale coordinate (mutual information is logarithmic by construction) — the same shape that Z catalogs at the chain / divergence / update layers. The IB case is classified as an *adjacent family member* rather than a fourth primary instance because 2.2 adopts the Lagrangian form from Tishby, Pereira & Bialek 1999 as an applied external theorem rather than re-deriving it under an AAT-internally-motivated additivity axiom. The Shore-Johnson 1980 system-independence axioms that axiomatize IB's additivity are cited in O §6.1; if a future framing pass promotes Shore-Johnson to an explicit AAT axiom, IB would move from adjacent member to fourth primary instance.

Working Notes.

- **The Σ_t source reformulation changes the theoretical quantity without changing the operational computation.** 8.9's DL-based minimization remains the practical procedure; what changes is the theoretical claim about what that procedure approximates. Depending on promotion-workflow preferences, the reformulation may warrant a small note in 8.9 referencing this segment.
- **(P1) as IB Lagrangian-dual does not close 11.1's other Working Notes.** Computability of (P1) for nonlinear/non-Gaussian systems, choice of ϵ_I, and N-agent scaling remain open. The unification resolves the *framing* question but not the *computation* questions.

- **Synthesis segment vs. full unification.** This segment is a targeted synthesis: it fixes the Σ_t source, derives (P1) as IB Lagrangian-dual, and tables the four instances. It does *not* rewrite the four instance segments as specializations of a master schema. If future work reveals that the four instance segments are pulling in inconsistent directions (e.g., if the Σ_t reformulation creates friction with the DL-based minimization in 8.9), that is the signal to escalate to a full rewrite in which the instance segments become thin specializations. Short of that signal, this targeted synthesis is the stopping point.
- **The Gaussian IB frontier derivation is deferred.** Completing the derivation that the means-sum projection attains the Gaussian IB frontier in the symmetric two-Kalman case would promote 12.1's rate-distortion claim from discussion-grade to derived. The obstruction is mechanical (specializing Chechik, Globerson, Tishby & Weiss 2005, "Information Bottleneck for Gaussian Variables," *J. Machine Learning Research* 6:165–188, Theorem 3.1 to the symmetric-gain setup) rather than structural — feasible as a follow-up derivation, not required for this segment.

Appendix W

Discussion: The Stability Certificate — One Object Behind the Cross-Sectional Meta-Patterns

AAT's cross-sectional structure is the geometry of a single object — the **equilibrium stability certificate**, the positive-definite form whose existence certifies that an agent can correct itself faster than its world drifts; operator-sector is that object's interior, the separability pattern its scope of existence, additive-coordinate-forcing its forced identity, and the identifiability floor its boundary, with composition the question of whether it survives projection.

The three meta-segments Y, X, and Z read, separately, as three independent organizing insights that happen to recur. This segment names the object they are facets of. The relationship is the same one Z already runs at smaller scale ("layer-specific manifestations of a single geometric object"), raised to the framework: not a fourth meta-pattern *alongside* the three, but the spine the three are projections of.

Formal Expression.

The object.

For an agent with error dynamics $\dot{e} = -F(e)$ about an equilibrium e^* ($F(e^*) = 0$, $F \in C^1$ near e^*, Jacobian $J := DF(e^*)$), a **stability certificate** is a symmetric positive-definite $\mathcal{M}$ for which the one-point sector condition holds in the $\mathcal{M}$-inner-product on a ball $\mathcal{B}_R(e^*)$:

Definition (stability-certificate).

$$\langle F(e),\, e - e^* \rangle_{\mathcal{M}} \;\geq\; \kappa \, \| e - e^* \|^2_{\mathcal{M}}, \qquad \kappa > 0. \tag{C}$$

The certificate is not unique: it is whatever positive-definite form makes the dynamics contract. In the recurring sub-cases it specializes — to the Fisher information for Bayesian agents, to $(P^-)^{-1}$ for Kalman agents, to the loss Hessian for gradient agents, and to a plant-selected Lyapunov metric for linear-Hurwitz or PID agents. These are not four separate stories; they are one object under four certificates.

The anchor.

The object is load-bearing only because its existence is not a definition but an equivalence: **a stability certificate exists iff the agent is exponentially stable about its target** — operator-sector in *some* inner product and exponential stability are the same statement, with the certificate as the converse-Lyapunov witness, and the certificate admits a strict strength ladder R0 ⟸ R1 ⟸ R2 (widest one-point/local; cocoercive; Čencov-forced). This is the segment-level form of the contraction-over-drift organizing principle. The equivalence, the ladder, and the proof are stated and derived exactly in D; this spine cites that result and builds the cross-sectional reading on it rather than re-deriving it.

Result (cited: #result-certificate-existence).

The four facets.

The certificate is one object; the cross-sectional meta-patterns are its facets on the positive-semidefinite cone $\mathbb{S}^n_{\geq 0}$:

Discussion.

Table W.1

Facet	*Meta-segment*	*What the facet is*	*Canonical home*
Interior	C,	$\mathcal{M} \succ 0$ on the scope ball: the contraction holds	the template segments
Scope of existence	Y	the region where a certificate exists at all (separable core / structured repair / general open)	M2
Forced identity	Z	*which* certificate: Čencov forces $\mathcal{M} =$ Fisher uniquely in statistical scope; matched (existence-only) elsewhere	M3
Boundary	X	$\mathcal{M}$ drops rank ($\partial\mathbb{S}^n_{\geq 0}$): the inferential task is structurally impossible	M1
Projection behaviour	11.1	whether a *common* certificate survives coarse-graining; the closure defect ε^* is the certificate's projection-residue	composition-closure

Each meta-segment retains its own canonical home and per-instance derivations; this segment claims only the recognition that they are facets of one object, and what that buys (Discussion below).

The three obstructions are distinct — the plurality is the content.

A tempting reading is that the certificate's failures are one obstruction seen three ways (a single "failure of integrability"). They are not. The three failure modes are irreducibly distinct, each invariant under the others' degrees of freedom:

Discussion.

- **Forced-identity failure — Helmholtz–Hodge.** J non-symmetric ⟹ the field is not a gradient ⟹ no potential ⟹ the certificate is *matched* (converse-Lyapunov existence), not *forced* (Čencov). A non-symmetric Hurwitz J still has a certificate (it is *not* on the boundary), so this is an M3 failure, not an M1 one. Invariant: symmetry of J.
- **Existence failure — Sylvester's law of inertia.** The certificate drops rank. Every coordinate/metric change acts on the certificate by congruence; congruence preserves inertia; so a rank-deficient certificate is rank-deficient in *every* coordinate. The boundary is invariant under the agent's entire representational freedom (that freedom *is* the congruence orbit); the only escape is rank-augmentation — genuinely new information, not a re-mapping. (Detailed in X's Sylvester finding.) Invariant: inertia under congruence.

- **Projection failure — Mori–Zwanzig / Schur.** Coarse-graining is a non-invertible projection. The certificate-as-metric survives (the Schur complement of a positive-definite form is positive-definite) but the *dynamic* guarantee does not: the closure defect ε^* equals the norm of the Mori–Zwanzig memory commutator, zero exactly when the resolved subspace is J-invariant. Invariant: J-invariance of the resolved subspace.

Each obstruction is untouched by the others' freedoms: a metric change does not fix non-invariance; projection does not fix non-symmetry; rank-augmentation does not fix a memory kernel. That mutual invariance is the reason the cross-sectional structure is *several* meta-patterns and not one — stated as a structural fact rather than left as "they are different concerns."

Epistemic Status.

Discussion-grade at the organizing-principle level. This segment names a single object behind separately-derived results; it is not itself a new theorem. What is derivative here is the recognition that the cross-sectional meta-patterns are facets of one certificate — the recognition, not a fresh derivation, is the content.

Constituent results retain their own, higher status. The certificate-existence equivalence (Lyapunov theorem; Formal Expression "anchor") is *exact* at the linearized level and *exact-with-standard-remainder* locally. The certificate-strength ladder R0/R1/R2 is *exact* as an ordering of conditions. The boundary-irreducibility mechanism (Sylvester's law) is *exact*; its full statement and per-instance verification live in X. The projection-residue identification ($\varepsilon^* =$ memory-commutator norm) is *robust-qualitative*, with the per-case content in 11.1. The forced-vs-matched distinction (Čencov statistical-scope-only) is established in Z.

Scope honesty. The anchor equivalence is linearized/local — this is the level at which AAT's persistence results already operate (sector conditions, contraction templates, the bridge lemma all linearize about the equilibrium), so it is not a weakening relative to the rest of the theory, but it is a genuine scope statement and is not papered over: there is no claim that one-point operator-sector is equivalent to *global* exponential stability (it is not, in general). The synthesis claim — that AAT's *entire* cross-sectional structure is this one cone — is *robust-qualitative*: each per-facet identification is exact or cited, and the "all of it" is as strong as those identifications jointly, no stronger. Whether exactly three obstructions exhaust the failure modes is not proved; three are established and each is exact, but exhaustiveness is open.

Max attainable: *discussion-grade* for the organizing claim (it is a presentational spine, not a derivation). The anchor equivalence and the Sylvester mechanism retain their *exact* status at their own canonical homes.

Discussion.

What the spine buys. Three things. (i) It converts "AAT has several recurring organizing insights" into "AAT's cross-section is one object's facets," which is the difference between a catalog and a structure — a reader who holds the certificate picture can predict where the meta-patterns will bite (interior / scope / forced-identity / boundary / projection) rather than meeting them as separate surprises. (ii) It grounds the long-standing organizing slogan — *an adaptive system is an operator whose contraction rate exceeds its disturbance rate* — at segment level, as the anchor equivalence rather than as a heuristic. (iii) It makes the framework's scope-honesty sharper: the prior positioning that the identifiability floor is "orthogonal" to the contraction machinery is replaced by the exact geometric statement that the floor is the *boundary* of the very cone whose interior the contraction machinery is, with the boundary held invariant by Sylvester's law against the framework's only representational freedom.

Relationship to the facet segments. This segment cross-references; it does not restate. X remains the canonical home of the floor instances and the Sylvester mechanism; Y of the separable-core / structured-repair / general-open ladders; Z of the Čencov/Cauchy-FE forcing; C and of the contraction machinery; 11.1 of the closure defect. The spine's contribution is the cross-segment recognition and the anchor equivalence, not any per-facet content.

Why the plurality of obstructions is a feature. Had the three failure modes collapsed to one, the spine would be a single clean theorem — elegant but false on the evidence (the integrability collapse fails: a non-symmetric Hurwitz field has a certificate yet no potential; congruence is invertible while projection is not). The honest structure is more useful: it tells a future agent precisely *not* to seek a single mechanism unifying the floor, the closure defect, and the forcing failure — they are Sylvester, Mori–Zwanzig, and Helmholtz respectively, and the proof of their distinctness is their mutual invariance. The unification is real at the object level and the no-go is real and plural at the failure level; both are load-bearing.

Complementarity with the existing meta-segment framing. Y names AAT's positive half and X its negative half; Z names the constructive half. The spine names what all three are halves *of*. The three remain the right reading lenses for any individual segment; the spine is the reason those lenses compose rather than merely coexist.

Findings.

The Cross-Sectional Meta-Patterns Are Facets of One Stability Certificate. **Brief:** Think of an adaptive agent as trying to stay on a moving target, and ask one question: is there a way of measuring "how far off am I?" such that every correction the agent makes provably shrinks that measure faster than the world pushes it back out? That measuring-stick is the stability certificate. The whole cross-sectional skeleton of AAT turns out to be facts about *this one stick*: the agent can keep up exactly when the stick exists and is positive (operator-sector — the interior); the framework's reach is exactly the region where some such stick exists (the separability pattern); when the stick is pinned down uniquely it is pinned by one classical invariance theorem and only in the statistical case (additive-coordinate-forcing); and the agent provably *cannot* keep up exactly when the stick goes flat in some direction (the identifiability floor — the boundary), with a second classical theorem (Sylvester's law of inertia) proving no change of measuring units ever un-flattens it — only genuinely new information does. A thoughtful non-specialist can carry the whole structure away from the one picture: existence of the stick = can adapt; flatness of the stick = a blind spot no re-measurement fixes; uniqueness of the stick = a special (statistical) privilege, not the general case; and looking at the stick through a coarse lens (composition) keeps its shape but loses its guarantee by exactly a memory term.

Impact: Reorganizes AAT's self-description from three independent meta-patterns plus a contraction mechanism into one object with four facets and one anchor equivalence, so the framework's cross-section can be read predictively rather than as a catalog of separate recurrences. Grounds the organizing slogan (contraction-rate-exceeds-drift) at segment level via the Lyapunov-theorem equivalence, discharging the long-standing "not yet surfaced at segment level" status. Sharpens the scope-honesty posture: "the floor is orthogonal to operator-sector" becomes "the floor is the boundary of the cone whose interior is operator-sector, held invariant by Sylvester's law against the framework's only representational freedom." Bounds its own claim honestly: the unification is at the object level; the three failure obstructions are provably *distinct* (Helmholtz / Sylvester / Mori–Zwanzig), and that plurality is precisely why AAT carries multiple cross-sectional meta-patterns rather than one — now a stated structural fact instead of an intuition. Gives every future organizing-pattern candidate a test: is it a new facet of the certificate, or a genuinely new object?

Novelty Claim: *Claim recognition* that AAT's cross-sectional meta-patterns (separability, identifiability-floor, additive-coordinate-forcing) and its contraction machinery are facets — interior, scope-of-existence, forced-identity, boundary, projection-behaviour — of a single object, the equilibrium stability certificate; together with *claim synthesis* binding the Lyapunov-theorem certificate-existence equivalence, the Sylvester-law boundary-irreducibility, and the Mori–Zwanzig projection-residue into one cross-sectional structure. The constituent theorems are classical; the contribution is the recognition that AAT's separately-derived meta-patterns are one object's facets and that their failure modes are provably plural.

Related Work: - Lyapunov, A. M. (1892), *The General Problem of the Stability of Motion*; Khalil, H. K. (2002), *Nonlinear Systems* 3rd ed., Thm 4.6 (found 2026-05-14) — *formal antecedent* — the certificate-existence equivalence (operator-sector in some metric ⟺ Hurwitz). - Sylvester, J. J. (1852), *Phil. Mag.* 4(23):138–142; Horn & Johnson (2013), *Matrix Analysis* 2nd ed., Thm 4.5.8 (found 2026-05-14) — *formal antecedent* — the boundary-irreducibility mechanism; full treatment in X. - Mori (1965) / Zwanzig (1961); Chorin, Hald & Kupferman (2002), *Physica D* 166:239 (found 2026-05-14) — *formal antecedent* — the projection-residue (memory kernel) underlying the composition facet; per-case content in 11.1. - The facet segments X, Y, Z — *adjacent* — each carries its own per-instance prior-art landscape; the spine adds the cross-segment object, not the per-facet priors.

Search Log: - 2026-05-14 (*targeted*): The recognition was assembled from classical pieces (Lyapunov / Sylvester / Mori–Zwanzig) plus the framework's own meta-segments. The search target was whether "an integrated agent theory's cross-sectional meta-patterns are facets of a single stability-certificate cone" appears as an articulated structure elsewhere. Not found at this depth; the constituent theorems are textbook and the per-facet identifications are individually well-precedented, but the cross-segment unification as a framework spine is a fresh presentational recognition. Expected to remain *recognition*/*synthesis*-tier under deeper search — the pieces are classical; the assembly is the contribution. Per-facet comprehensiveness is inherited from the facet segments, not from a fresh cross-facet search.

Working Notes.

- **Provenance.** The certificate-spine recognition, the anchor equivalence, the Sylvester boundary mechanism, and the broken-integrability-triad result were worked out in the 2026-05-14 operator-family-unification cycle (the deep push of the predecessor C1 "2-instance-plus-1-consequence" question and the prior co-owner "do not elevate unless O-BP10 surfaces at segment level" gate, now met by the anchor equivalence). See CHANGELOG 2026-05-14 for the cycle narrative. The originating spike is absorbed archaeology, not a live reference; the load-bearing content is in this segment, D, and the X Sylvester finding.
- **Dependency rationale (for Gate-1 audit).** `depends:` lists `result-certificate-existence` (the anchor equivalence this spine builds the cross-sectional reading on — a genuine dependency, consumed not merely recognized) plus `result-sector-persistence-template` and `deriv-sector-condition` (the persistence machinery the anchor's drift half rests on). X, Y, Z, and 11.1 are **cross-referenced as facets, not depended on**: per FORMAT.md Gate 1, a dependency is genuine only when the segment uses the referenced segment's definitions/results, not when it is recognized or related in Discussion. The spine *recognizes* the meta-segments as facets of one object; it does not consume their definitions to make its claim. This is why the spine is correctly placed *before the three meta-segments* in OUTLINE (it is the object they are facets of, so it reads first) without an ordering violation — the facet relationship is lateral recognition, not a dependency edge — while being placed *after* `#result-certificate-existence` (a genuine dependency, which therefore precedes it). Treat the facet `#…` references as expected forward/lateral cross-refs (FORMAT.md §Cross-References).
- **Provisional slug.** `disc-stability-certificate`. Alternative considered: `disc-certificate-cone` (names the geometry rather than the object). Subject-noun discipline favours the object ("the stability certificate"); the cone/interior/boundary is what the segment *says about* it. Route through the naming pipeline if a better name surfaces; the spike verdict floated both.
- **Provisional OUTLINE position.** Placed in `## *Appendices* Details` immediately before X, as the lead of the four-row meta-segment cluster (spine → M1 → M2 → M3). Provisional because: (a) the meta-segments may eventually warrant their own chapter rather than Appendix-A residence; (b) if the OUTLINE preamble is reframed to lead with the spine, the cluster likely moves to a more prominent position. Both are propagation steps below, not landed here.
- **Propagation plan (ordered by commitment; steps 6–7 are framework-voice keystones gated on Joseph, not auto-executed):**
  1. X — cross-ref line in its "complementarity" Discussion paragraph naming the spine as the object whose boundary it is. (Sylvester finding + Working Note already point here.) Cross-ref only.
  2. Y — parallel line: it is the *scope-of-existence* facet (where the certificate $\succ 0$). Cross-ref only.
  3. Z — frame its (PI)/Čencov content as the *forced-identity* facet. Cross-ref only; Čencov machinery's canonical home stays M3.
  4. C / — one Discussion sentence: the template condition is the certificate interior; cross-ref the spine for the cone reading. No formal change.
  5. 11.1 — Discussion line framing ε^* as the certificate's projection-residue and Liberzon as "no common certificate"; cross-ref the spine. (Its Mori–Zwanzig Working Note already exists.)

6. **O-BP10 keystone (Joseph's call).** Recommend this segment *is* O-BP10's segment-level home: the slogan is its one-sentence summary, the equivalence is its anchor. The PROPOSALS Bundle-1 O-BP10 entry then points here. Do not auto-rewrite Bundle 1.
7. **OUTLINE preamble reframe (highest commitment; Joseph's call).** OUTLINE.md line 17 currently opens "Three meta-segments form AAT's cross-sectional structure: Y … X … Z …". Proposed replacement, for Joseph's confirmation before it goes live in the auditor-visible preamble: *"AAT's cross-sectional structure is one object — the equilibrium stability certificate (W). Its interior is operator-sector (the contraction machinery); its scope of existence is the separability pattern (Y, positive half); its forced identity is additive-coordinate-forcing (Z, constructive half, Čencov-forced in statistical scope only); its boundary is the identifiability floor (X, negative half); and its behaviour under coarse-graining is the composition-closure defect. Reading any segment through the certificate and its facets surfaces what makes it load-bearing: whether a certificate exists for it, which one is forced, what boundary it abuts, and whether it survives projection."* Execute only on Joseph's confirmation, having seen the segment land first.

- **Open edges (from `99-verdict.md`).** Anchor equivalence is linearized/local (stated in Epistemic Status). "Exactly three obstructions" is robust-qualitative, not proved exhaustive — a fourth (e.g. non-autonomous certificate drift for time-varying systems) is not searched. Sylvester is proved finite-dimensional; the function-space (M_t for logogenic agents) extension is unchecked and flagged for any future logogenic application, not load-bearing for the AAT-core claim.

Appendix X

Discussion: The Identifiability Floor — A Class of Structural No-Go Results

AAT has derived a class of structural impossibility results — *floors below which* identification or detection is impossible from limited information. Each floor arises by applying an external information-theoretic theorem (the Pearl/Bareinboim causal hierarchy; the Cramér-Rao bound on Fisher information) to a specific AAT setting. The floors are negative results in form but positive in consequence: they precisely characterize what additional structure (loop-interventional access, multi-channel observability, observable latents) is required to escape the floor, and thereby strengthen the load-bearing role of the AAT machinery that supplies it.

This segment names the meta-pattern, collects the current instances, and identifies adjacent floors that are open research directions.

The Pattern.

Each instance of the identifiability floor has the form:

1. **Setting.** An AAT inferential task — detect a structural property, identify a parameter, distinguish two model classes — under a specific information regime (purely observational data, single observation channel, observation of marginals only, etc.).

2. **External theorem.** An information-theoretic limit independent of AAT: the causal hierarchy theorem (Bareinboim, Correa, Ibeling & Icard 2022) for distinguishing observational and interventional content; the Cramér-Rao bound for unbiased estimation under finite Fisher information.

3. **No-go.** The external theorem is invoked to prove that the inferential task is impossible under the regime: no statistic, no Bayesian comparison, no online estimator can succeed using only the available information.

4. **Boundary characterization.** The conditions under which the floor's regime fails — i.e., the agent has *more* information than the regime allows — admit (partial) identification. Each boundary route maps onto specific AAT machinery already required by the theory.

5. **Strengthened consequence.** The floor strengthens the load-bearing role of whichever AAT machinery is the unique broadly-available violation of the regime. Often this elevates a piece of machinery from "useful" to "structurally required by the theory."

The pattern is *not* a negative posture. AAT does not say "many things are impossible." It says: "here is precisely what cannot be inferred from limited data; here is exactly which additional capability is required to recover identification; the AAT machinery that supplies that capability is therefore load-bearing in the strongest possible sense — without it, the no-go forbids the inference entirely."

Current Instances.

Instance 1 — On-Policy L0 Insufficiency Detection (8.2). Setting. Detect whether an L0 strategy DAG is causally insufficient (a latent common cause is acting on multiple sibling action propositions) using only the agent's on-policy observation history under sequential short-circuit AND/OR execution.

External theorem. Bareinboim, Correa, Ibeling & Icard (2022) Causal Hierarchy Theorem: there exist SCMs that agree on Level 1 (associational) data but disagree on Level 2 (interventional) questions. Therefore Level 2 distinctions are not in general identifiable from Level 1 data.

No-go. For any L1 world $\mathcal{W}_{L1}$ with latent common cause C, there exists an L0 world $\mathcal{W}_{L0}^*$ with edge probabilities matched to the on-policy regime conditionals such that the on-policy observation distribution is *identical*: $\mathbb{P}^{\text{obs}}_{\pi_{L0}}[\mathcal{W}_{L1}] = \mathbb{P}^{\text{obs}}_{\pi_{L0}}[\mathcal{W}_{L0}^*]$. The L0/L1 distinction is a Level 2 question (it concerns $P(A_2 \mid do(\neg A_1))$ vs $P(A_2 \mid \neg A_1)$); on-policy data is Level 1; the CHT forbids distinguishing them.

Boundary characterization. Five routes (cf. 8.2): - (a) ε-exploration violates pure on-policy execution → partial detection at scale $O(\varepsilon)$. - (b) Joint sibling observability under exploration violates short-circuit censoring → strong detection via the pairwise covariance test under 6.3. - (c) Intermediate-state observability at finer grain → very strong detection where available. - (d) Structural priors positing common causes → prior-quality-dependent. - (e) Direct intervention on the candidate latent → strongest where available.

Strengthened consequence. The covariance test under joint observability (route (b)) becomes the unique broadly-available detection mechanism. This sharpens 6.3 from "useful machinery" to "structurally required to escape the no-go." The orient cascade's step 4c (causal-sufficiency check) is no longer "one possible diagnostic" but "the unique broadly-available diagnostic given the structural impossibility of purely on-policy detection."

Tier. *Exact* for shallow strict-prerequisite cases (2-sibling OR or AND with binary common cause). *Robust qualitative* for general DAG topology, soft facilitators, and deeper structures.

Instance 2 — L1' Mixture Identifiability from Single-Channel Observations (N Prop B.7). Setting. Identify the mixture parameters $(\theta_C, p_{j|C}, p_{j|\neg C})$ of a soft-facilitator L1' DAG using single-channel observations y_j of one child at a time, with C unobservable.

External theorem. The Cramér-Rao bound (Cramér 1946, *Mathematical Methods of Statistics*, Princeton University Press): the variance of any unbiased estimator is at least the inverse of the Fisher information matrix; if the Fisher matrix is rank-deficient, the bound is infinite in the null directions and no unbiased estimator achieves finite variance there.

No-go. Computing the Fisher information of the mixture model $\mu_j = \theta_C\theta_{j|C} + (1-\theta_C)\theta_{j|\neg C}$ at truth, the matrix admits the rank-1 factorization $\mathcal{F} = uu^T/(\mu_j(1-\mu_j))$ with $u = (\Delta_j, \theta_C, 1-\theta_C)$ and $\Delta_j = p_{j|C} - p_{j|\neg C}$ the separability gap. The two-dimensional null space corresponds to perturbations along the indeterminacy manifold $\{\hat{\phi} : \hat{\theta}_C\hat{p}_{|C} + (1-\hat{\theta}_C)\hat{p}_{|\neg C} = \mu_j\}$ — directions unobservable from a single binary signal. The smallest eigenvalue of the soft-EM update Jacobian is therefore zero; no SA1-preserving update on the joint conditional vector admits a sector parameter $\alpha > 0$.

Boundary characterization. Three repair routes: - (i) Augment C-observability — instrument secondary signals identifying C per trial. Recovers Prop B.7 globally with the five-way-gating $\alpha_{L1'}$. - (ii) Joint multi-child observation — when $K \geq 2$ children share C with linearly independent conditional profiles AND are observed jointly under the same C-realization, the joint Fisher matrix can reach rank $2K+1$. Strong structural requirement (not satisfied by typical sequential strategy execution). - (iii) Plan-level fallback — track the marginal $\hat{\mu}_j$ scalar (which is identifiable; recovers B.1's $\alpha = 1/(n_\mu + 1)$) at the cost of losing the per-conditional decomposition, equivalent to L0-on-marginals.

Strengthened consequence. *Observability-as-information-augmentation* becomes load-bearing: when the agent can treat C as an observable feature of the environment (e.g., a regime indicator the environment broadcasts — common in software/operational settings: build state, deployment regime, user tier), the problem transforms from refuted to globally derived. This elevates the engineering choice "instrument the latent" from a convenience to a theoretical prerequisite for L1' identifiability.

Tier. *Exact* (Cramér-Rao bound is exact for unbiased online estimators).

Instance 3 — Composite Contraction Certification from Component Data (F,). **Setting.** Certify $\kappa_c > 0$ (composite contracting in a combined metric) for N sub-agents each verified at its own level (individual sector conditions with parameters (α_i, R_i), individual Tier 1 classifications per `spikes/spike-bridge-lemma-contraction.md`, individual modularity per `#der-directed-separation`), using only component-level data: per-sub-agent trajectories, mismatch observations, update rules — no observation of the coupling topology (sign pattern of cross-agent influence), no common contraction metric chosen across sub-agents, no passivity certificate on the coupling channels, no shared Lyapunov function.

External theorem. Common-Lyapunov nonexistence for switched linear systems — Liberzon 2003, *Switching in Systems and Control*, Theorem 2.1; explicit 2×2 counterexample in Dayawansa & Martin 1999, "A converse Lyapunov theorem for a class of dynamical systems which undergo switching," *IEEE Trans. Automat. Control* 44:751; systematic review in Shorten, Wirth, Mason, Wulff & King 2007, "Stability criteria for switched and hybrid systems," *SIAM Review* 49:545. Complementary anchor: small-gain contrapositive — Jiang, Teel & Praly 1994, "Small-gain theorem for ISS systems and applications," *Math. Control Signals Syst.* 7:95; if interconnection gain is unbounded or not observable from component data, no composite-ISS certificate from that data. The Liberzon/Shorten common-Lyapunov result is the sharper anchor for AAT's setting.

No-go. There exist pairs of coupled systems (Σ_1, Σ_2) and (Σ_1, Σ_2') with **identical marginal component-level observation distributions** but opposite composite-contraction signs ($\kappa_c > 0$ in the cooperative regime per `#deriv-critical-mass-composition` (CM2) with $\gamma < 0$; $\kappa_c < 0$ in the adversarial regime per `#der-adversarial-destabilization` with $\gamma > 0$ past the destabilization threshold). Concretely, take two symmetric-matched-Tier-1 scalar agents with coupling term $\gamma\mathcal{T} \cdot \text{sign}(\delta_{\bar{\imath}})$. Each sub-agent in isolation sees $\dot{\delta}_i = -\alpha\delta_i + w_i^{\text{total}}$ with total disturbance bound $\rho + |\gamma|\mathcal{T}$ — consistent with both $\gamma = +\gamma_0$ (adversarial) and $\gamma = -\gamma_0$ (cooperative) since the cross-term is absorbed into bounded-disturbance regardless of sign. Only observation of the *joint* dynamics (or structural knowledge of the coupling sign) distinguishes them. **The single bit of coupling-sign distinguishing cooperative from adversarial regimes is unidentifiable from component marginals, and that bit is exactly what flips composite persistence.**

Boundary characterization. Four structural escapes:

- (a) **Observable coupling topology** via composite-extended `#der-loop-interventional-access` — interventions on sub-agent A_j reveal A_i's cross-coupling response, which is a $do(\cdot)$-data distinction between the two coupled constructions.
- (b) **Matched Tier at the composite level** — shared architecture (matched Tier 1, same norm/metric) admits a joint quadratic Lyapunov $V = \sum V_i$, yielding `#deriv-critical-mass-composition`'s (CM2) closed form. Under `#result-contraction-template`'s topology-indexed closure results, this extends to heterogeneous composites via (CM2-M) for matched contraction-metric structure across agents.
- (c) **Passivity / storage-function certificate** on the coupling channel (Willems 1972 *Arch. Ration. Mech. Anal.* 45:321). Adjacent machinery not currently in an AAT segment.

- (d) **Common contraction metric** (Lohmiller & Slotine 1998). Operationalized in `#result-contraction-template`: composite metric M_c constructed compositionally from sub-agent metrics per topology (parallel / cascade / negative-feedback), with rate λ_c from Slotine 2003.

Strengthened consequence. The no-go elevates three pieces of AAT machinery from "useful" to "structurally required":

1. `#deriv-critical-mass-composition` moves from "closed-form result in a special case" to **the unique broadly-available composition-contraction certificate under the matched-Tier structural escape (b)**; without (CM2) or its contraction-metric generalization (CM2-M) in `#result-contraction-template`, the weakest-link bound (WL) in `#form-composition-closure` cannot see coupling sign and so cannot distinguish the two coupled constructions.
2. **Composite-extended `#der-loop-interventional-access`** becomes the unique coupling-sign identifier under heterogeneous Tier structures — the composite-layer analog of the single-agent interventional-access-escape for Instance 1.
3. `#scope-composite-agent` acquires **load-bearing enabling status**: scope-satisfaction (one of C-i, C-ii, C-iii) is what positions the composite within a regime where one of (a)–(d) can operate. Without scope-satisfaction, the composite might not be a composite, and the escapes have no coherent target.

Tier. *Exact* for the symmetric-matched-Tier-1-scalar construction exhibited above. *Robust qualitative* for general heterogeneous composites, inheriting the common-Lyapunov-nonexistence structure from Liberzon / Shorten / Dayawansa-Martin without a closed-form AAT-level counterexample.

Instance 4 — Universal Information-to-Distance Constant under Non-(PI) Norms ( ). **Setting.** Derive a universal constant C that bounds the parameter-space displacement $\|\Delta M_{\text{bias}}\|$ caused by observation ambiguity, scaling predictably with the mutual information $I(G; \Omega \mid e, M)$.

External theorem. The Cramér-Rao lower bound, or equivalent metric equivalences linking Kullback-Leibler divergence to local distance metrics.

No-go. Without a canonical metric on the model space $\mathcal{M}$, no universal information-to-distance constant exists. If the parameter space uses an arbitrary Euclidean norm, the relationship between KL divergence (an information-theoretic quantity) and parameter displacement is unbounded and coordinate-dependent. An explicit counterexample exists in heteroscedastic-normal families (see `#deriv-observation-ambiguity-bias-bound`, Attempt E) where C can be made arbitrarily large by reparameterization.

Boundary characterization. The unique escape is to adopt the Parameterization Invariance (PI) axiom. Under (PI), Čencov's theorem forces the geometry to be the Fisher-Rao metric. In this canonical geometry, the relationship between mutual information and parameter displacement is universally bounded (e.g., $C_{FR} = \sqrt{2}$ in the small-information regime).

Strengthened consequence. This elevates the (PI) axiom from a "nice-to-have" geometric property to a **load-bearing requirement** for the Class 3 (Coupled) bias bound in `#result-section-ii-survival` and `#scope-observation-ambiguity-modulation`. Without (PI), the bias bound is merely an order-of-magnitude heuristic; with (PI), it is a rigorous theorem.

Tier. *Exact (counter-example-grade).* The non-existence of a universal C in arbitrary Euclidean norms is proven by explicit construction.

Adjacent Floors (Open Research Directions).

Causal-IB Extension for Interventional Relevance Variables. The standard Information Bottleneck (2.2) and the four AAT compression operations (V) work with associational relevance variables — Y in a joint distribution with X and T. Strategy edges in the regime-indexed interpretation (8.5) want *interventional* relevance for Regime A: "what edge (i, j) predicts under $do(i)$, not under observation of i." Standard IB cannot supply this; a causal-IB variant (Wieczorek & Roth 2017 and follow-ups) is the natural framework. The expected floor: under purely associational data, the interventional relevance content is bounded; the gap is recoverable via do-data from the loop. Open: formalize the gap as an identifiability floor parallel to Instances 1 and 2.

Misspecification-Cost Quantification. Structural adaptation (4.5) names *when* to switch model classes. It does not quantify the *continuous degradation* from a mildly misspecified model under a finite information budget. Adjacent to `#scope-observation-ambiguity-modulation`'s $\kappa \cdot \mathcal{A}$ bound but not covered by it. The expected floor: under fixed information budget, the degradation rate from misspecification is bounded below by an information-theoretic quantity (likely related to the KL gap between true and assumed model classes). Open.

Tier-Switching Policy Cost. Approximation tiering (U) enumerates AAT's tiered approximations (L0/L1/L1'/L2 in correlation, C1/C2/C3 in convention, Tier 1/2/3 in contraction). The cost of switching tiers — when should the agent move from L0 to L1, or from C1 to C2 — is itself a deliberation-cost problem. Under finite computation budget, the optimal switching policy faces an identifiability floor on its own switching diagnostics. Open.

Mechanism-Design Impossibility (candidate 4th instance from 14.1). Under the mechanism-design framing of strategic composition (14.1 §Discussion), an outside designer may be able to shape sub-agents' objectives $\{O_t^{(i)}\}$ so that the induced strategic equilibrium coincides with a desired joint state. Impossibility results from social-choice theory — **Gibbard-Satterthwaite** 1973-75 (no dominant-strategy non-dictatorial Pareto-efficient voting mechanism for ≥3 alternatives); **Myerson-Satterthwaite** 1983 (no efficient, individually-rational, incentive-compatible bilateral-trade mechanism without subsidies); **Arrow** 1951 (no social welfare function satisfying unrestricted-domain + Pareto-efficient + IIA + non-dictatorial simultaneously) — establish that certain mechanism-design goals are **structurally unachievable** under stated constraints. This matches the meta-pattern shape: setting (composite-design task under specific-constraint regime) → external theorem (social-choice impossibility) → no-go → boundary characterization (relaxation of constraints: Bayes-Nash in place of dominant-strategy; randomized allocations; subsidy injection; strategy-space restriction) → strengthened consequence (the AAT machinery of `#deriv-strategic-composition`'s sub-scope α' potential-game conditions becomes a load-bearing target for mechanism design). Candidate fourth instance; would require a dedicated formalization of the AAT-machinery escape route specifically rather than the general social-choice escape. Open.

Why This Pattern Matters.

Strengthens the case for AAT's foundational machinery. Each floor identifies a piece of AAT machinery as load-bearing in the strongest possible sense — without it, the corresponding inferential task is *impossible*, not merely *harder*. The loop's interventional access is not just useful; it is the unique broadly-available violation of the on-policy detection no-go. Observability of latents is not just convenient; it is the unique route from refuted-by-Cramér-Rao to globally-derived for L1' transfer.

Maps the limits of AAT's machinery. The floors precisely characterize what AAT cannot do without additional capability. This is honest scope-marking — at each floor, the theory states "here is what is impossible; here is what you need to escape the impossibility; here is what AAT supplies (or doesn't supply, as the case may be)."

Provides a unifying framework for future no-go results. Adjacent floors (causal-IB extension, misspecification cost, tier-switching cost) are open research directions that would extend the pattern. Each, if formalized, would have the same shape: setting → external theorem → no-go → boundary characterization → strengthened machinery.

Connects AAT to information-theoretic foundations. The external theorems (Pearl/Bareinboim hierarchy; Cramér-Rao bound; rate-distortion / IB) are the mature literature AAT inherits. Each instance positions AAT as a domain-specific application of an established theorem — not a re-derivation, but a consequential application that shapes downstream segment structure.

Epistemic Status.

Discussion-grade at the meta-pattern level. The segment is a presentational organizing principle — it names a shared shape across separately-derived results, not a theorem in its own right. What is derivative here is the recognition that two independent AAT findings share the pattern (setting → external theorem → no-go → boundary characterization → strengthened consequence); the pattern itself is not derived and has no identification claim of its own.

Individual instances retain their own, higher, epistemic status. Instance 1 (on-policy L0 insufficiency detection, via the Causal Hierarchy Theorem) is *exact* for shallow strict-prerequisite cases and *robust qualitative* for general DAG topology, derived in 8.2. Instance 2 (L1' under unobservable common cause, via the Cramér-Rao bound on Fisher information) is *exact* under the Fisher rank-1 calculation in N Prop B.7 refutation. Readers citing this segment for a specific no-go should cite the instance's own derivation, not the meta-pattern.

The segment makes one additional claim: that AAT's machinery (loop-interventional access, observability-as-information-augmentation) acquires sharper load-bearing roles when read through the floors that motivate them. This is a *discussion-grade* observation about the theory's architecture — it is visible once the instances are assembled, but is not itself a theorem.

Whether the pattern is a generative principle — whether future AAT work will systematically encounter and derive more instances — is a *hypothesis* that the adjacent open floors test.

Max attainable: *discussion-grade* for the meta-pattern (it is a presentational organizing principle, not a derivation). The individual instances retain their own epistemic status as derived above.

Discussion.

The pattern is asymmetric. Each floor forbids inference *from* limited data; it does not forbid inference *with* the augmenting capability. The asymmetry is the source of the pattern's positive content — it tells the reader exactly what to instrument, observe, or intervene upon to escape the floor.

The asymmetry has a named mechanism for the rank-collapse instances: Sylvester's law of inertia. For the instances whose floor is a rank-deficiency of an information operator — Instance 2 (Fisher rank-1 under an unobservable common cause) cleanly, and Instance 1 (the causal-parameter block carries no score under a purely observational regime) structurally — the reason no change of coordinates escapes the floor is one theorem. The agent's only representational freedom is the choice of coordinate / metric, and every such change acts on the information operator by *congruence* ($\mathcal{G} \mapsto S^\top \mathcal{G} S$ for invertible S — the standard Fisher-information reparameterization law). Sylvester's law of inertia states that congruence preserves inertia: the number of zero eigenvalues is invariant under every invertible reparameterization. So a rank-deficient information operator is rank-deficient in *every* coordinate; the floor is the boundary of the positive-definite cone, and the entire reparameterization freedom is a congruence orbit that cannot cross that boundary. This is why the only escape is *rank-augmentation* — adding a genuinely new score component (interventional data via the loop, a side channel, a witness), which is not a coordinate change — and never reweighting. The per-instance "no reparameterization removes the degeneracy" computations (the Fisher rank-1 calculation in N Prop B.7; the on-policy no-detection argument in 8.2) are special cases of this single law. **The mechanism is specific to the rank-collapse subclass, not to all floors.** Instance 3 (composite contraction certification) is a *different* obstruction: composition is a non-invertible projection, not a congruence, so its no-go is a Schur-complement / memory-kernel statement (the certificate's metric survives projection but its dynamic guarantee does not), not an inertia statement. That the floors do not share one mechanism — Sylvester for rank-collapse, a projection/closure obstruction for composition — is itself load-bearing: it is why the floor pattern is a presentational family rather than a single theorem.

The pattern composes with AAT's scope honesty. Directed separation (5.3) classifies architectures by where Section II's exact results apply (Class 1 Separated; Class 3 Coupled needs coupled formulation). The identifiability floors are a different kind of scope claim: they specify what the theory's machinery *cannot do* under specific information regimes, with explicit characterization of the regime escapes. Together, the architectural classification and the identifiability floors mark AAT's scope at two levels — what kinds of agents the theory applies to, and what those agents can and cannot infer from given data.

Complementarity with the separability pattern (Y). This segment names the *negative half* of AAT's scope; the companion meta-segment Y names the *positive half* — separable-core / structured-repair / general-open across six ladders (correlation, convention, architecture, contraction, identification, scope). Each identifiability-floor instance here has a positive counterpart there: Instance 1's on-policy detection no-go matches the observable-sibling-covariance structured-repair in the correlation ladder; Instance 2's unobservable-C L1' refutation matches the observable-C / facilitator-monotonicity structured-repair in the same ladder. The two halves together characterize AAT's scope at both extremes — what succeeds and under what machinery, and what structurally cannot succeed without specific information augmentation.

This segment is the *boundary facet* of the stability certificate (W). The rank-collapse floors are the certificate dropping rank — the boundary of the positive-definite cone whose interior is operator-sector — with the Sylvester-law irreducibility (above) the statement that the framework's representational freedom is a congruence orbit that cannot cross that boundary. Read through the spine, this segment answers "where does the agent's measuring-stick go flat, and why does no re-graduation un-flatten it"; Y answers "where does a stick exist at all" (scope-of-existence facet) and Z "which stick is forced" (forced-identity facet).

The pattern is conservative in style. Each floor invokes a published external theorem (Bareinboim et al. 2022; standard Cramér-Rao bound) rather than deriving a new impossibility result. AAT's contribution is the *application* — recognizing the AAT setting falls within the theorem's scope, characterizing the boundary conditions, and identifying which AAT machinery is the unique broadly-available escape. This style aligns with the broader posture of AAT as an integrating framework that connects established results across control theory, causal inference, and information theory under a common formalism.

Findings.

The Identifiability Floor as Cross-Cutting Meta-Pattern. **Brief:** Across four independently-derived results, AAT has converged on a recurring shape: an inferential task (detecting a structural property, identifying a parameter, distinguishing two model classes) is shown structurally impossible under a specific information regime, by importing an external information-theoretic theorem (Pearl/Bareinboim's causal hierarchy, the Cramér-Rao bound, Liberzon's common-Lyapunov-nonexistence, Čencov's invariance theorem); the conditions under which the regime fails are characterized as boundary routes, each of which maps onto specific AAT machinery the theory already requires; and the floor strengthens the load-bearing role of that machinery by elevating it from "useful" to "the unique broadly-available escape." The pattern is a presentational organizing principle, not a theorem of its own — but the recognition that four distinct AAT results share this shape is itself an architectural finding worth surfacing externally, because it tells the reader where to look for AAT's distinctive scope-honesty moves and which adjacent floors are open research.

Impact: Surfaces a unifying frame for results that would otherwise read as disparate impossibility theorems. Each instance gains interpretive context (it is one of an emerging class) and the meta-segment provides a consistent template for evaluating candidate future floors (causal-IB extension, misspecification cost, tier-switching cost, mechanism-design impossibility). The pattern's positive content — that each floor names exactly which AAT machinery the theory requires to escape it — converts a sequence of negative results into a structural argument for the load-bearing status of the loop's interventional access (Instances 1 and 3, two semantically distinct deployment modes), of observability-as-information-augmentation (Instance 2), and of the (PI) parameterization-invariance axiom (Instance 4). The complementary

`#disc-separability-pattern` carries the positive half (separable-core / structured-repair / general-open across seven ladders); together the two meta-segments mark AAT's scope honestly at both extremes.

Novelty Claim: *Claim recognition* of structural pattern across four AAT results that import external information-theoretic theorems to derive impossibility statements with mapped boundary-route escapes; the meta-pattern is an organizing principle rather than a theorem, and the per-instance prior-art positioning lives in the instance segments (`#der-causal-insufficiency-detection`, `#deriv-edge-credence-dynamics`, `#deriv-critical-mass-composition` / `#result-contraction-template`, `#deriv-observation-ambiguity-bias-bound`).

Related Work:

The four instances import distinct external theorems; per-instance prior-art landscapes live in the instance segments. The meta-pattern itself has no direct anticipation in the AAT-adjacent literature surveyed; the closest cousins are scope-honesty moves in formal-method literatures (Liberzon's switched-systems analysis, which itself instances the pattern; Cramér-Rao tradition in identifiability theory) and the broader posture of conservative-form causal inference (Pearl 2009, *Causality*; Bareinboim et al. 2022). What the meta-pattern adds is the cross-instance observation: AAT repeatedly recognizes that its own setting falls within a published external impossibility theorem, characterizes the boundary-route escape via existing AAT machinery, and uses the floor to elevate that machinery to load-bearing status.

Table X.1

Pattern element	*Where the move is established in the prior literature*	*Relationship / Positioning*
Importing CHT to derive on-policy detection no-go (Instance 1)	Bareinboim, Correa, Ibeling & Icard 2022 (published 2022, found 2025)	*formal antecedent* — see `#der-causal-insufficiency-detection` Findings for the full Pillar-1 prior-art table; the meta-pattern subsumes this instance
Importing Cramér-Rao to derive mixture-identifiability no-go (Instance 2)	Cramér 1946, *Mathematical Methods of Statistics* (published 1946, found 2025-04)	*formal antecedent* — Fisher rank-deficiency forces the floor; see `#deriv-edge-credence-dynamics` Prop B.7 for the explicit calculation
Importing common-Lyapunov-nonexistence to derive composite-contraction no-go (Instance 3)	Liberzon 2003, *Switching in Systems and Control* §2.1; Dayawansa & Martin 1999 *IEEE TAC* 44:751; Shorten et al. 2007 *SIAM Review* 49:545 (found 2025-04)	*formal antecedent* — the symmetric-matched-Tier-1-scalar counterexample uses the standard switched-systems machinery; see `#deriv-critical-mass-composition` and `#result-contraction-template` for the AAT-internal closures
Importing (PI)/Čencov to derive universal-constant no-go (Instance 4)	Čencov 1982, *Statistical Decision Rules and Optimal Inference* (published 1982, found 2024); explicit heteroscedastic-normal counterexample derived AAT-internally	*formal antecedent for (PI) escape, AAT-internal for the no-go construction* — see `#deriv-observation-ambiguity-bias-bound` Attempt E
Cross-instance pattern recognition	No direct anticipation surfaced at search depth	*claim novelty* under cursory search — the meta-pattern as an articulated framework has not been found in the surveyed literatures; the constituent moves (scope-honesty, importing external no-go theorems, boundary characterization) are individually well-precedented but their cross-instance unification within an integrated agent-theoretic framework is a presentational contribution rather than a borrowed schema

Search Log:

- 2026-04 (*nominally comprehensive at the per-instance level*, via `ref/Novelty_defense_and_integration.md` Pillar 1): The Undermind defense covered Instance 1's prior-art landscape (causal bandits / causal MDPs under hidden confounding); the meta-pattern itself was not the search target. Instances 2–4 inherit per-instance comprehensiveness from their constituent segments rather than from a cross-instance search.
- 2026-04 (*intuition-only* on the meta-pattern as an articulated framework): The cross-instance recognition that four AAT results share the *setting → external theorem → no-go → boundary characterization → strengthened-consequence* shape has not been searched for as a unified concept. Targeted future search candidates: scope-honesty traditions in formal verification (refinement-mapping work; Lamport's TLA; Abadi-Lamport refinement); identifiability-theory meta-frameworks; the broader "no-go theorems" literature in physics and economics. Expected outcome: the constituent moves are well-precedented; the unified framework as applied to an integrated agent theory may be novel under cursory search but is unlikely to be novel under comprehensive search of methodological-essay literature.

The Rank-Collapse Floor's Irreducibility Is Sylvester's Law of Inertia. **Brief:** Picture the agent trying to pin down a parameter as localizing a point in a landscape whose curvature is the information it has. A "floor" is a *flat direction* in that landscape — a direction the data cannot resolve (rank-deficient Fisher information). The agent's only freedom is to re-draw its map: choose different coordinates, a different metric. Sylvester's law of inertia says that re-drawing the map with any invertible change of coordinates can bend, rotate, and rescale the contours but can never change *how many genuinely flat directions* exist at a point — a flat valley stays flat in every coordinate system. So when the information operator is rank-deficient, no choice of representation recovers the lost direction; the only thing that ever helps is *taking a new measurement* — adding genuinely new information (an intervention, a side channel, a witness), which is not a re-drawing of the map. A thoughtful non-specialist can re-derive the qualitative claim from the analog alone: you cannot survey your way out of a blind spot by changing the units on the ruler; you have to look from a new vantage point. This is the exact mechanism behind every rank-collapse identifiability floor, and it explains in one sentence why those floors are escapable only by capability, never by cleverness of representation.

Impact: Converts a sequence of per-instance "we checked, no reparameterization removes the degeneracy" computations (Fisher rank-1 in N Prop B.7; on-policy no-detection in 8.2) into one named classical theorem (Sylvester 1852), sharpening the meta-pattern's load-bearing claim. It converts the prior honest-but-soft positioning "the identifiability floor is *orthogonal* to the contraction/operator-sector machinery" into the sharp geometric statement: operator-sector is the *interior* of the positive-definite (information / stability-certificate) cone; the rank-collapse floor is its *boundary*; and the boundary is invariant under the agent's entire representational freedom because that freedom is exactly the congruence orbit Sylvester's law fixes. The escape routes named per instance (loop-interventional access; observability of latents) are unified as *rank-augmentation* of the information operator — the unique category of move that is not a congruence. The result also bounds its own scope honestly: it is the mechanism for the rank-collapse subclass only. Instance 3 (composite contraction certification) has a structurally different obstruction — composition is a non-invertible projection, so its no-go is a Schur-complement / memory-kernel statement, not an inertia statement. That the floors are *plural in mechanism* (Sylvester for rank-collapse; a projection-closure obstruction for composition) is itself the reason the floor pattern is a presentational family and not a single theorem — this Finding makes that plurality precise rather than leaving it as "they are different concerns."

Novelty Claim: *Claim recognition* that the irreducibility of AAT's rank-collapse identifiability floors is a single named classical theorem — Sylvester's law of inertia applied to the Fisher-information reparameterization law — rather than a coincidence of per-instance computations; and *claim differentiation* that this mechanism is specific to the rank-collapse subclass and provably distinct from the composition floor's projection-closure obstruction. Sylvester's law itself is classical (1852); the contribution is the recognition that AAT's representational freedom is exactly its congruence orbit, which makes the floor's irreducibility a corollary rather than a checked property.

Related Work: - Sylvester, J. J. (1852), "A demonstration of the theorem that every homogeneous quadratic polynomial is reducible by real orthogonal substitutions to the form of a sum of positive and negative squares," *Phil. Mag.* 4(23):138–142 (found 2026-05-14) — *formal antecedent* — the inertia-invariance theorem; applied here to the Fisher-information congruence. - Horn, R. A., & Johnson, C. R. (2013), *Matrix Analysis* (2nd ed.), Thm 4.5.8 (found 2026-05-14) — *formal antecedent* — standard modern statement of the law of inertia used in the argument. - Lehmann, E. L., & Casella, G. (1998), *Theory of Point Estimation* (2nd ed.), §2.5 (found 2026-05-14) — *formal antecedent* — the Fisher-information reparameterization law $\mathcal{G}_\varphi = S^\top \mathcal{G}_\theta S$ that makes every coordinate change a congruence. - Per-instance external theorems (Bareinboim et al. 2022; Cramér 1946) — *adjacent* — established in the instance segments' own Findings; this Finding adds the cross-instance mechanism, not the per-instance prior art.

Search Log: - 2026-05-14 (*targeted*): The recognition arose in the operator-family-unification spike from the certificate-cone framing. Sylvester's law and the Fisher reparameterization law are textbook; the search target was whether the *recognition* "AAT's identifiability-floor irreducibility = Sylvester's law via the Fisher congruence" appears as an articulated statement in the identifiability-theory or causal-inference methodological literature. Not found at this depth; the constituent facts are universally known but the cross-instance unification as the named irreducibility mechanism for an integrated agent theory's floor pattern appears to be a fresh presentational recognition. Expected to remain *recognition*-tier under deeper search (the pieces are classical; the assembly is the contribution).

Working Notes.

- **Sylvester-recognition provenance.** The rank-collapse-floor-as-Sylvester finding and the certificate-cone framing it sits in were worked out in the 2026-05-14 operator-family-unification cycle; see CHANGELOG 2026-05-14. The broader spine question (reorganizing M1/M2/M3 as facets of one certificate-cone object) is *resolved* — landed as W + D, with this segment as the boundary facet (see the Discussion cross-ref above). Originating spike is absorbed archaeology, not a live reference.
- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. Removed at `candidate` stage per FORMAT.md Gate 4.
- **Naming convention.** "Identifiability floor" frames the pattern positively: the floor is what the agent cannot go below given limited information, but the boundary characterization tells the agent exactly how to climb above it. An alternative name "no-go theorems" would emphasize the negative form. Recommend retaining "floor" — it captures the asymmetry.
- **Instance 3 as a candidate.** The "L1 augmentation when the augmentation graph is itself causally insufficient" question (recurse the no-go: detect when the L1 augmentation is itself missing common causes) is a candidate Instance 3. Likely reduces to Instance 1 applied at the L1 level — the agent at L1 faces the same on-policy detection no-go for L1 → L2 escalation. Worth formalizing if a third instance emerges that does *not* reduce to the existing two.
- **Is the floor pattern unique to causal/identifiability questions?** The two current instances are both about distinguishing two parameter regimes from data. The pattern may also apply to other impossibility-style results in AAT: the inevitability of structural-adaptation thresholds (4.5), the cost of representing high-correlation regimes (L2 exponential blowup), the bandwidth limits in shared intent (12.2). Whether to absorb those into "identifiability floor" or treat them as a separate "structural-cost" pattern is open.
- **Cross-segment integrations.** The meta-pattern surfaces in (at least): 8.2 (Instance 1); N Prop B.7 (Instance 2); 6.3 (load-bearing for Instance 1 escape); 7.3 (the L0/L1/L1'/L2 hierarchy is the regime ladder Instance 2 lives on). Each of these segments cross-references this one for the unifying frame.

Appendix Y

Discussion: The Separability Pattern — Separable Core, Structured Repair, General Open

AAT consistently runs a three-part epistemic posture across state spaces that admit no tractable exact treatment in general: name the **separable core** where identification is clean, name the **structured repair** that recovers identification under explicitly-added machinery, and name the **general open** case where the problem is either intractable or structurally unidentifiable. Six ladders in the current theory share this shape. This segment names the pattern as an organizing principle, catalogs the instances, and makes the complementarity with `#disc-identifiability-floor` (which names the *negative half* of AAT's scope) explicit.

The pattern.

Each instance of the separability pattern has the form:

1. **Separable core.** A sub-class of problem in which identification is tractable by construction — the quantity of interest has a clean estimator that requires only assumptions the agent can verify or control. "Separability" here is broader than statistical independence: it names the regime in which the quantity decomposes along a structural axis (independence, additive decomposition, directed separation, strong contraction, clean intervention, etc.) without interaction terms requiring additional machinery.
2. **Structured repair.** A sub-class where the separability assumption fails but a *specific, named, bounded-cost* additional mechanism recovers identification. The repair is structured: it identifies the failure mode, adds a specific compensating construction, and typically carries a characterized loss in tightness or generality. Repairs are not "apply heuristics"; they are "augment with observability of C and accept $O(1)$ per-node parameter cost," "run receding-horizon replanning at cost $\mathrm{DL}(\Sigma_t)$," or similar.
3. **General open.** The fully-general case where no tractable repair is known. This is where `#disc-identifiability-floor` instances live — structural no-go results that characterize *why* the general case remains open. Naming this boundary positively (as a known frontier) rather than negatively (as a failure) is part of AAT's load-bearing scope honesty.

Current instances — six ladders.

Table Y.1

Ladder	*Separable core*	*Structured repair*	*General open*
Correlation (7.3)	**L0** — independence model, $O(\|V\| + \|E\|)$ propagation	**L1** — strict-prerequisite augmentation with explicit common-cause node; **L1'** — soft-facilitator mixture with five-way gating under observable C + facilitator monotonicity (N Prop B.7)	**L2** — full joint; **L1' unobservable-C** single-channel is *refuted* by Cramér-Rao floor (X Instance 2)
Convention (5.6)	**C1** — one-step improvement	**C2** — receding-horizon replanning	**C3** — Bellman optimal; intractable for large state spaces
Architecture (5.3)	**Class 1 (Separated)** — directed separation holds by construction	**Class 2 (Partial)** — directed separation holds for identified submodules	**Class 3 (Coupled)** — directed separation fails by construction; logogenic territory; coupled formulation required
Contraction (11.1)	**Tier 1** — strong monotonicity; bridge lemma applies with proved ε^*	**Tier 2** — local convexity; bridge applies within a specified basin	**Tier 3** — neither; domain-specific verification required per instance
Identification regime (8.5)	**Regime A** — interventional ($\iota_{ij} = 1$); action is a literal $do(\cdot)$	**Regime B** — partial intervention ($0 < \iota_{ij} < 1$); confounder adjustment, side-channel observation	**Regime C** — observational ($\iota_{ij} \approx 0$); identification depends on external assumptions (e.g., unconfoundedness)
Scope hierarchy (5.1)	**Adaptive** — basic feedback loop; 1.5 satisfied	**Agency** — goal-bearing with $\|\mathcal{A}\| \geq 2$ and at least one action with causal effect; adds O_t and Σ_t machinery (Section II); 1.6 satisfied	**Composite** — multi-agent / team / logogenic; Section III gaps listed at the OUTLINE level (latent structural diversity, endogenous coupling, composition transition dynamics, agent opacity)
A2'-scope ()	**metric-α_1** — Euclidean metric, AAT-internally derived via DA2'-inc ≡ (CT2) at $M = I$ (scalar Kalman, Euclidean strongly-convex, L2-regularized, linear-PD-symmetric)	**metric-α_2** — non-Euclidean metric under explicit conditions; includes five cases (information-metric Kalman, Fisher-metric exp-family, Hessian-metric ill-conditioned, Lyapunov-metric linear-Hurwitz-non-symmetric, Lyapunov-metric PID-bounded-plant). Two of five (Fisher cases) AAT-internally forced under (PI)/Čencov per `#disc-additive-coordinate-forcing`; remaining three theorem-imported from Lohmiller-Slotine 1998	**metric-β** — contraction-metric formulation fails (variational-to-projected-target; rule-based / non-smooth; severely misspecified; per-step SGD / human judgment)

GUC Architecture-row key: Class 1 = Separated (was Modular); Class 2 = Partial (was Class 3 Partially modular); Class 3 = Coupled (was Class 2 Fully merged). See `#der-directed-separation`.

Each row independently satisfies the three-part shape. The shared shape is not designed-in: it arises because AAT faces seven distinct sources of intractability and applies the same posture (name the clean case, name the repair, name what remains open) to each.

Complementarity with the identifiability floor.

The **separability pattern** names the *positive half* of AAT's scope: for each ladder, what succeeds under what conditions. Each separable-core entry is a positive identification claim; each structured-repair entry is a positive identification claim *conditional on* an explicitly-named added mechanism.

This segment is the *scope-of-existence facet* of the stability certificate (W): a separable-core entry is a region where a certificate exists cleanly, a structured-repair entry is a region where one exists under explicitly-added machinery, and a general-open entry is where no certificate is available without further information. Read through the spine, this segment answers "where does the agent's measuring-stick exist at all"; X answers "where does it go flat" (boundary facet) and Z "which stick is forced" (forced-identity facet).

The `#disc-identifiability-floor` names the *negative half*: structural no-go results (impossibility under limited information) that characterize why specific general-open cases remain open. Two floors are currently derived:

- **Instance 1** (on-policy L0-insufficiency detection via Causal Hierarchy Theorem) — forbids L0/L1 distinction from purely on-policy data. The positive counterpart: the **separable core under observable sibling covariance** is the unique broadly-available violation of the no-go's scope, naming what the agent *can* observe to escape the floor.
- **Instance 2** (L1' unobservable-C single-channel via Cramér-Rao) — forbids mixture-parameter identification when C is unobservable. The positive counterpart: the **structured repair under observable C** (Prop B.7 with facilitator monotonicity) is the identification-succeeds-under-augmentation claim matching the no-go's "unless C is observable or multi-child observation is available" scope exit.

Together, the two meta-segments mark AAT's scope at two epistemic layers:

- **Separability pattern** — "here is what succeeds, with clean conditions and explicit repairs."
- **Identifiability floor** — "here is what structurally cannot succeed without specific information augmentation."

Neither alone gives the full picture. A theory stating only the positive half looks unbounded in its claims; a theory stating only the negative half looks conservative in a way that hides its achievements. Together they give scope honesty at both extremes.

Epistemic Status.

Discussion-grade at the meta-pattern level. The segment is a presentational organizing principle. It names a shared shape that AAT already runs across multiple ladders; the pattern itself is not derived and has no theorem of its own. The individual ladder entries retain their own epistemic status — Correlation hierarchy results are at the tier specified in 7.3 and N; Convention hierarchy monotonicity is at the tier specified in 5.6; Contraction tier taxonomy is at the tier specified in 11.1; etc.

Max attainable: *discussion-grade* for the meta-pattern (it is an organizing principle, not a derivation). Individual instances are at their own tiers as above. The six-ladder enumeration could become *robust-qualitative* if a uniqueness argument were derived showing that separable-core / structured-repair / general-open is the unique viable posture across scope-parameterized hierarchies under AAT's scope-honesty architectural principle — but this is not currently in hand, nor clearly attainable.

Discussion.

The pattern is load-bearing for how AAT presents its results. A common failure mode of applied theories is the binary "exact under assumption X, breaks otherwise" presentation. The separability pattern refuses this presentation across six instances: where X fails, a named lower tier takes over with a characterized weaker result, and the theory provides a diagnostic for when to escalate (per U's AT4 component). Readers who learn the pattern once can navigate any instance; this is what makes AAT tractable as an integrating framework rather than a pile of instantiations.

Relationship to U. This segment is complementary to U, not redundant with it. U names the *structural template* (AT1 parameter indexing tractability; AT2 proved monotonicity between tiers; AT3 graceful degradation; AT4 ascension diagnostic) — the how-it-works of each tiering. This segment names the *epistemic posture* (separable core, structured repair, general open) — the what-each-tier-commits-to. Both are needed: U explains what makes a successful tiering; this segment explains the shape of the commitment each tier makes about identification.

Relationship to T. T catalogs the *independence assumptions* whose failure degrades results; this segment catalogs the *ladders of recovery*. Together with U they form a three-part characterization of AAT's scope:

- **T** — where the boundaries are (which assumptions, what breaks if they fail).
- **U** — how to navigate within the boundaries (parameterized hierarchy + monotonicity + ascension).
- **Y** (this) — what each tier positively commits to (separable core / structured repair / general open).

The pattern is distinctive to AAT's integration-over-invention character. Approximation tiering with a separability posture is a common move in applied mathematics — it appears in numerical methods (exact vs. high-order vs. low-order with error bounds), in statistical inference (parametric separable-core / semiparametric repair / nonparametric open), in causal identification (Pearl's three-layer hierarchy is itself an instance). AAT's contribution is not the pattern but its deployment across the six specific intractable problems adaptive-agent theory faces, each with its own positive-half identification claims and negative-half floor instances.

The A2' sub-scope partition is a proper three-part ladder via . The A2' partition carries three tiers: sub-scope α (A2' derived under `#der-gain-sector-bridge` directional fidelity) as the separable core; sub-scope metric-α_2 (non-Euclidean metric under explicit conditions — five cases, with two Fisher-metric cases AAT-internally forced under the (PI)/Čencov fourth primary instance of `#disc-additive-coordinate-forcing`) as the structured-repair middle tier; sub-scope β (A2' assumed) as the general-open tier. A2'-scope therefore qualifies as the seventh ladder of this meta-pattern, as enumerated in the table above. The structured-repair middle tier closes what would otherwise be a binary α/β partition lacking a derivable middle ground.

The software calibration-lab framing (`#obs-software-epistemic-properties`) is a specific instance. C-BP3's calibration-lab reframing (commit `d0373fc`) partitions operational domains into "separable core" (software, where identification conditions P1–P6 are cleanly satisfied) and "structured-repair / general open" (other domains, which inherit under explicitly-named transfer assumptions). This is a domain-axis instance of the same pattern — software is the domain-axis separable core; the transfer-assumption table is the domain-axis structured-repair specification.

Working Notes.

- **Standalone-paper candidacy.** This meta-pattern is a candidate for standalone publication per the *B-N-Sep* portfolio entry. The strategic plan (paper structure, prior-art landscape, cite-and-extend anchors, venue analysis, effort estimate) lives at `msc/separability-standalone-paper-proposal.md`, which subsumes the equivalent section in `~/src/ops/papers/03-asf-tier2-and-cross-segment.md`. **Verified novel** by independent prior-art search: Undermind 31-paper full-text sweep (2026-05-04) returned no exact named cross-domain meta-pattern; closest neighbors are Hintikka 1991 (general theory of identifiability), Pearl/Shpitser ID lineage (most-developed within-domain instance), Bareinboim 2022 (cross-hierarchy meta-discussion at N=2-3 hierarchies), Basse-Bojinov 2020 (modern formal abstraction across fields), Maclaren-Nicholson 2019 (ill-posed inverse problems as cross-field dual structure). Full report at `ref/separability-ladder-prior-art-report.md`.
- **Citations to land on promotion to draft / when paper drafting begins:**
  - **Hintikka, J. (1991).** *"Towards a General Theory of Identifiability."* In *Definitions and Definability: Philosophical Perspectives*, J. H. Fetzer et al. (eds.), Kluwer Academic. DOI: 10.1007/978-94-011-3346-3_7. Locally at `ref/towards-a-general-theory-of-identifiability.pdf`. **Strongest older abstract anchor**: tripartite *definable / identifiable / non-identifiable* trichotomy. Hintikka §1: "P is identifiable on the basis of T[P]" when "the interpretation of P is not determined on the basis of the theory alone, but is determined by the theory together with a number of auxiliary empirical results." The pattern's middle rung in Hintikka's vocabulary; ASF's structured repair generalizes "auxiliary empirical results" to "named bounded-cost structural augmentation" (a real but small extension).

- **Bareinboim, E., Correa, J. D., Ibeling, D. & Icard, T. (2022).** *"On Pearl's Hierarchy and the Foundations of Causal Inference."* In *Probabilistic and Causal Inference: The Works of Judea Pearl*, ACM Books. DOI: 10.1145/3501714.3501743. Treats Pearl's hierarchy as a logical and epistemic hierarchy and compares it to formal-language and complexity hierarchies — closest existing cross-hierarchy meta-discussion at N=2-3, but stops short of the cross-domain meta-pattern this segment names.
- **Robins, J., Richardson, T. & Shpitser, I. (2020).** *"An Interventionist Approach to Mediation Analysis."* In *Probabilistic and Causal Inference*, ACM Books. DOI: 10.1145/3501714.3501754. Cleanest in-domain cite-and-extend anchor: clean separable case + structured-repair via expanded-graph decomposition + recanting-witness no-go.
- **Shpitser, I. & Pearl, J. (2006, 2008).** *"Identification of Joint Interventional Distributions in Recursive Semi-Markovian Causal Models"* (AAAI); *"Complete Identification Methods for the Causal Hierarchy"* (JMLR 9). Foundational ID completeness papers — the cleanest verified no-go-plus-recovery pairing; the ID algorithm IS this template within causal inference.
- **Bareinboim, E. & Pearl, J. (2012, 2014); Lee, S., Correa, J. D. & Bareinboim, E. (2019).** Surrogate experiments (z-identifiability), transportability with limited experiments, general identifiability with arbitrary surrogates. Each is a structured-repair instance with a named bounded augmentation. Several locally at `ref/`.
- **Basse, G. W. & Bojinov, I. (2020).** *"A general theory of identification."* Closest modern formal abstraction across fields (identifiable / partially-identified / strongly-non-identifiable regimes); lacks the bounded-cost-repair operator as middle rung.
- **Maclaren, O. J. & Nicholson, R. (2019).** *"What can be estimated? Identifiability, estimability, causal inference and ill-posed inverse problems."* arXiv 1904.02826. Best dual-structure cross-field neighbor (causal identification ↔ ill-posed inverse problems); centers on stability/regularization rather than structured-repair.
- **Restricted-intervention lineage:** Robins 1986 (treatment-regime semantics; locally at `ref/`); Richardson 2013 SWIGs; Dawid 2000 ("causal inference without counterfactuals"; locally at `ref/`); Dawid 2020 (decision-theoretic foundations); Richardson-Robins 2023 (SWIG/decision-theoretic bridge). Patches the missing-prior-art lineage for restricted/regime-defined intervention semantics.
- **Neighboring recurrence inside causal identification:** Robins-Richardson 2010 (alternative graphical models, expanded graphs); Stensrud et al. 2019 (separable effects in competing events); Díaz 2022 (non-agency interventions); Shpitser-Tchetgen 2014 (intervention hierarchy unification). Pattern recurs in mediation, competing-events, edge-intervention settings within wider causal-identification literature.
- **Adjacent abstractions that stop short:** Iwasaki-Simon 1994 (causality and model abstraction; near-decomposability); Hoover 2012 (causal structure and hierarchies of models; well-made-toaster vs repairman cases). Strong philosophy-and-Simon-line parallels; not a formal tractability/identifiability ladder. Downey-Fellows 1995/1999 (parameterized complexity, FPT/W[1]/paraNP-hard) and Simpson 1999 (reverse mathematics) for hierarchy-of-strength formalisms; verified weak connection to applied-identifiability ladders.

- **Pending family rename and rung-rename decision.** The family name is queued for rename: `discussion-separability-pattern` → `discussion-separability-ladder`, on Round-1 consensus rationale that "ladder" is more evocative for the three-rung shape than "pattern" (which is generic). See `msc/naming/naming-rename-plan.md` §"Deferred to refined Round 1 / Round 2". The **three rung-names** (currently *separable core / structured repair / general open*) are also under consideration for refinement; Hintikka 1991's *definable / identifiable / non-identifiable* trichotomy is a strong candidate for an aligned echo (modulo a small extension on the middle rung from "auxiliary empirical results" to "named bounded-cost structural augmentation"). Decision deferred (2026-05-04). The standalone-paper proposal (above) is ready to draft against either the status-quo names or any of the alternates — the rename can land separately with no impact on paper-drafting tractability.
- **Cross-ladder monotonicity.** The six ladders interact — an agent operating at L0 correlation + C2 convention + Class 1 architecture + Tier 1 contraction + Regime A identification + Agency scope is in a specific combined regime. Does cross-ladder monotonicity hold in any direction? E.g., does moving from L0 to L1 change anything about the Convention hierarchy's guarantees, or do the ladders factor independently? `#disc-approximation-tiering`'s Working Notes flag this as an

open question; the separability pattern's complementarity with the identifiability floor adds a further axis (which combinations are known-unidentifiable per X).

- **Extension candidates beyond the six.** Three candidate future ladders noted in U — scalar-vs-per-dimension tempo, AND/OR parameterization, A/B/C identification (now promoted here) — fit the separability-pattern shape. Promoting them to named ladders with explicit separable-core / structured-repair / general-open entries would extend the pattern to 8–9 ladders. Each promotion requires its own per-instance work.
- **Necessity argument for the pattern itself.** Is there a scope-honesty theorem of the form "any theory that claims exactness under Class-1-style assumptions *and* claims coverage beyond those assumptions must exhibit the separability pattern (or an equivalent three-part decomposition) to avoid latent overclaim"? If so, AAT's deployment of the pattern six times would be a *derived* structural necessity rather than a stylistic consistency. Speculative; not pursued.
- **Does C-BP2 belong in AAT core or in the wider framework narrative?** The segment currently lives in `01-aat-core/` because its instances are AAT segments. If TST adopts the same pattern across its own ladders (e.g., software-calibration-lab as a domain-axis ladder), the segment may want to move up to the root framework level. Deferred; the current placement is defensible while TST adoption is partial.

Table Y.2

Migration note (2026-05-09 GUC rename): Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. Architecture row updated: Class 1	Class 2	Class 3 now reads left-to-right monotonically (cleanest → middle → worst), matching the six other AAT ladders. GUC key added below the row. Removed at `candidate` stage per FORMAT.md Gate 4.

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Appendix Z

Discussion: Additive-Coordinate Forcing

AAT carries a family of structurally connected uniqueness results in which a coordinate is **forced by a uniqueness theorem operating on an independently-motivated AAT-internal axiom**. This meta-pattern previously appeared as a "1-anchor-plus-3-theorem" structure (where the chain-rule anchored three Cauchy-FE and Čencov-invariance results). The deeper structural reality is that **these four layers are layer-specific manifestations of a single geometric object**: the exponential-family Legendre-Fenchel structure.

The pattern is not designed-in: it arises because AAT independently needs to force coordinates at four distinct layers (chain, divergence, update, and metric), and the surrounding apparatus (DAG factorization, Bayesian coherence, regret-bound decision theory, and singular-trajectory scope) has committed to this geometry through multiple independent surface constraints.

This segment names the pattern, defines the underlying geometric object, catalogs the four layer-specific manifestations, and documents adjacent cases that share the shape but not the AAT-internal forcing structure.

The underlying geometric object.

The four forced coordinates all live on the **exponential-family Legendre-Fenchel geometry** (Amari & Nagaoka 2000, *Methods of Information Geometry*, §3.5; Bregman 1967; Rockafellar 1970; Bauschke & Combettes 2017). The structure is characterized by:

1. **A convex potential:** The negative-entropy potential on the probability simplex Δ^{n-1}.
2. **A Fenchel conjugate:** The log-partition function on the dual natural-parameter space.
3. **A primal-dual correspondence:** The softmax / log-odds map translating between probability (primal) and natural parameters (dual).
4. **A Bregman divergence:** The reverse-KL divergence, which measures distance using the primal potential.
5. **A Riemannian metric:** The Fisher information metric, which coincides with the Hessian of the dual potential ($\nabla^2\phi^*$).

When AAT forces an additive or invariant coordinate, it is forcing the theory onto a specific surface of this single geometric object.

The four layer-specific manifestations.

The pattern shares the following structure across four layers: 1. **An AAT-internal axiom (or mathematical identity)** is posited, motivated by the layer's specific needs. 2. **A uniqueness theorem** (Cauchy's functional equation for log-additivity, or Čencov's 1982 invariance theorem) operates on the axiom. 3. **A coordinate is forced**, which corresponds exactly to a feature of the Legendre-Fenchel geometry.

Table Z.1

Layer	*AAT-internal axiom*	*Uniqueness mechanism*	*Forced coordinate*	*Relation to exponential-family geometry*	*Source segment*
Chain	Probability chain rule (mathematical identity, not an AAT axiom)	Cauchy-FE	Log-probability	Log-coordinate of the primal point	7.1
Divergence	Chain-rule additivity over conditional factorizations	Cauchy-FE	Reverse-KL (up to positive scaling)	Bregman divergence generated by the primal potential	O §6.1
Update	Evidential additivity	Cauchy-FE	Log-odds	Natural coordinate on the dual space	P
Metric	Parameterization-invariance (PI) on singular trajectories	Čencov-invariance (Čencov 1982)	Fisher information metric	Hessian of the dual potential	4.2,

The motivational anchor and the structural target. The divergence axiom is stated as "the *divergence-level analog* of the additive log-confidence decay in `#der-chain-confidence-decay`." The update axiom is stated as "the *update-level analog* of `#der-chain-confidence-decay`'s chain-level additive log-confidence decomposition." The (PI) axiom is stated as "the metric-layer analog motivated by the singular-trajectory scope commitment."

The four layers are not independent applications of a generic mathematical idiom; they are interlocked by cross-cites, each positioning itself against an adjacent AAT commitment. **The chain-layer identity is the motivational anchor *and* the exponential-family geometry is the structural target.** The reason the three analog axioms all succeed via uniqueness theorems is that they all independently probe the same underlying geometric object. The convergence across independent axioms is itself the meta-pattern's substance — not a byproduct to be compressed into a single axiom.

The Čencov-Fenchel relationship. On AAT's current Fisher-metric scope (exponential families in natural parameters and matrix-Kalman cases), the Čencov-derived Fisher metric coincides precisely with the Hessian of the dual log-partition potential. This is a scope-dependent equivalence: outside exponential families, Čencov's invariance theorem still forces the Fisher metric, but the Fenchel-Bregman correspondence does not straightforwardly apply. Within AAT's primary operating regime, the two uniqueness machineries (Cauchy-FE and Čencov) resolve to the exact same geometric object.

Downstream consequences of the (PI) commitment. Forcing the Fisher-Rao geometry via (PI) + Čencov is not just an aesthetic unification; it enables specific quantitative theorems downstream. For example, the bias bound derived in `#deriv-observation-ambiguity-bias-bound` (Track 2) relies explicitly on the Fisher-Rao metric to achieve a universal, dimension-free bounding constant ($C_{FR} = \sqrt{2}$) connecting mutual information to parameter displacement. Without the (PI) commitment, attempting this bound in arbitrary Euclidean norms fails completely (yielding arbitrarily large constants).

Adjacent cases that share the shape but not the forcing structure.

Other AAT segments exhibit additive decompositions, but belong to different geometric families or lack the AAT-internal forcing structure.

Lyapunov quadratic (`#deriv-sector-condition`). The sector-Lyapunov argument uses $V(\delta) = \frac{1}{2}\|\delta\|^2$, under which $\dot{V}$ decomposes additively. **Reclassification:** The Lyapunov quadratic is the **Bregman divergence on a Euclidean potential** (squared-norm potential). It belongs to a *distinct* convex potential sub-family, not the negative-entropy exponential family. Furthermore, the coordinate is *chosen* (via a converse-Lyapunov existence argument) to match the sector form; it is not uniquely *forced* by an AAT-internally-motivated additivity axiom. It sits outside this meta-pattern.

Variance-additive cases (AV). Variance-additive statistics (like the Ornstein-Uhlenbeck steady state in `#hyp-mismatch-dynamics`) share Bregman-type additivity but, like Lyapunov, sit on a squared-norm potential rather than the negative-entropy geometry.

IB Lagrangian (`#form-information-bottleneck`, `#disc-compression-operations`). The Information Bottleneck objective $T^* = \arg\min[I(X;T) - \beta \cdot I(T;Y)]$ uses an additive Lagrangian form. **Reclassification:** IB is a direct **application of the negative-entropy / reverse-KL Bregman geometry**. However, AAT *adopts* the IB formulation from external foundations (Tishby, Pereira & Bialek 1999) rather than re-deriving it from an AAT-internally-motivated uniqueness theorem. It is an adjacent family member: it shares the exact geometric object, but its axiomatic provenance is imported.

Using the pattern as a diagnostic filter. This reframe supplies a principled filter for candidate future instances: *does the candidate's coordinate arise from the Legendre-Fenchel / exponential-family geometry?* If yes, check which of the four layer-surfaces it corresponds to. If no, it is either an adjacent family member (different Bregman geometry) or outside the pattern entirely.

Complementarity with X and Y.

The three meta-segments form AAT's scope-and-structure meta-architecture:

- **Z** (this) — the *constructive* half: when AAT forces a coordinate, it forces a surface of the exponential-family geometry (not just a generic logarithmic coordinate, but the specific log-coordinate of the exponential family). Names what the theory *does* at its deepest structural moves.
- **Y** — the *positive scope* half: for each ladder of increasing difficulty, name the separable core, the structured repair, and the general open. Names what the theory *covers* across its instantiations.
- **X** — the *negative scope* half: for each known structural no-go, name the external information-theoretic theorem that enforces it and the unique escape available. Names what the theory *cannot do* without explicit information augmentation.

The three compose. A ladder's separable core typically admits a coordinate-forcing move that makes the clean identification clean; a ladder's general-open case typically sits at an identifiability floor that names the information-theoretic obstruction. Reading any segment through all three meta-lenses surfaces what makes the segment load-bearing.

This segment is the *forced-identity facet* of the stability certificate (W): it answers "*which* certificate" — whether the metric is forced to a unique form by a uniqueness theorem on an AAT-internal axiom (Čencov forces the Fisher metric, statistical scope only) or merely matched. Read through the spine, the three meta-lenses are the certificate's forced-identity (this segment), scope-of-existence (Y), and boundary (X) facets; the certificate exists iff the agent is exponentially stable (D).

Epistemic Status.

Discussion-grade at the meta-pattern level. This segment is a presentational organizing principle: it names a structural shape that AAT runs across four layers. The meta-pattern itself is not derived and carries no theorem of its own.

The individual instances retain their own epistemic statuses: - `#der-chain-confidence-decay` — *exact* (mathematical identity). - `#deriv-strategy-cost-regret-bound` §6.1 — *derived (conditional on chain-rule additivity axiom)*. - `#deriv-edge-update-natural-parameter` — *derived (conditional on evidential-additivity axiom)*. - `#der-gain-sector-bridge` — *derived (conditional on (PI) axiom)*.

Findings.

Cross-Layer Coordinate Forcing on Legendre-Fenchel Geometry. **Brief:** When the framework needs to pick a "natural" way of measuring something (a confidence, a divergence, a parameter), it doesn't choose by convention. It writes down a property the measurement should have (typically: small contributions should add up the way independent evidence should), and a theorem from outside the framework forces the measurement to take a particular form. This pattern shows up at four different places in the framework — chain-rule confidences, divergences between distributions, edge update rules, and the metric on the parameter manifold — and the four places turn out to be different windows into the same underlying mathematical structure (the exponential-family Legendre-Fenchel geometry on probability distributions).

Impact: Names a meta-pattern that is one of AAT's distinctive constructive moves and that organizes a substantial fraction of the theory's load-bearing structural choices. The pattern supplies a principled diagnostic filter for candidate future instances ("does the coordinate arise from the exponential-family geometry?") and explains why AAT's reverse-KL direction in regret bounds, log-odds in edge updates, and Fisher-Rao metric in bias bounds are not arbitrary representational choices but theorems conditional on AAT-internal axioms. The reframe also clarifies which adjacent cases (Lyapunov quadratic, IB Lagrangian) sit on different geometric families or have imported provenance — preventing the meta-pattern from being overclaimed as a single uniform principle. Composes with `#disc-identifiability-floor` (the negative scope half) and `#disc-separability-pattern` (the positive scope half) as the constructive third leg of AAT's three-part meta-architecture.

Novelty Claim: *Claim recognition* of cross-layer pattern across four AAT coordinate-forcing results, with the recognition itself as the contribution rather than any new theorem. The constituent uniqueness machinery (Cauchy's functional equation; Čencov's invariance theorem) is well-precedented; the recognition that four AAT-internal axioms each successfully invoke uniqueness machinery on Legendre-Fenchel-geometry coordinates is the AAT-distinctive observation.

Related Work:

- Aczél 1966, *Lectures on Functional Equations and Their Applications* §2.1 — *formal antecedent* — Cauchy's functional equation; the uniqueness machinery for the chain, divergence, and update layers. Cited inline in each instance segment.
- Čencov 1982, *Statistical Decision Rules and Optimal Inference* — *formal antecedent* — Čencov uniqueness theorem; the uniqueness machinery for the metric layer. Cited inline in `#deriv-observation-ambiguity-bias-bound` and `#deriv-fisher-whitened-update-rule`.
- Amari & Nagaoka 2000, *Methods of Information Geometry* §3.5 — *formal antecedent / synthesis target* — the underlying Legendre-Fenchel geometry on exponential families that all four layers' coordinates resolve to. The recognition that the four forced coordinates are different aspects of one Amari-Nagaoka geometric object is the meta-pattern's payoff.
- Rockafellar 1970, *Convex Analysis* — *formal antecedent* — Legendre-Fenchel duality; the convex-analytic foundation Amari & Nagaoka build on.

- Bregman 1967, "The relaxation method of finding the common point of convex sets" *USSR Comput. Math. Math. Phys.* 7:200–217 — *formal antecedent* — Bregman divergence machinery; the divergence-layer coordinate is the negative-entropy Bregman divergence.

Search Log: - 2026-04 (*intuition-only* on the meta-pattern as a unified framework): the constituent uniqueness machinery (Cauchy-FE, Čencov, Bregman) is well-established, and Amari-Nagaoka information geometry is the standard exponential-family / Legendre-Fenchel framework the four layers resolve to. The recognition that AAT axiomatizes four distinct surfaces of one geometric object via four independently-motivated axioms is the contribution. Whether this cross-layer recognition has been articulated as a unified pattern in any prior framework has not been searched. Targeted future search candidates: information-geometry meta-frameworks (Eguchi 1983 framework discussion; Ay et al. 2017 *Information Geometry* monograph); axiomatic-foundations literature on probability and decision theory (Cox 1946; Jaynes 2003); the broader "uniqueness theorems forcing canonical coordinates" tradition in statistical mechanics. - 2025 (*targeted*): Aczél 1966 / Čencov 1982 / Amari & Nagaoka 2000 / Rockafellar 1970 / Bregman 1967 confirmed as the formal antecedents.

Working Notes.

- **Composition layer admits anchor-style additivity, not theorem-style.** The compositional analog of the four-layer pattern has been explored systematically (log-viability, log-identifiability-shortfall, persistence-cost-rate, log-closure-deficit across a composition tower). None clears the bar set by the three existing instances: each candidate fails to identify an AAT-internally-motivated additivity axiom that forces a logarithmic coordinate via Cauchy FE under smoothness. The strongest candidate — log-closure-deficit along a composition tower — does admit additivity, but as a *mathematical consequence* of operator-norm sub-multiplicativity under the bridge-lemma form, not as a theorem conditional on an AAT-internal axiom independently grounded. That makes it a fourth *anchor* (parallel to `#der-chain-confidence-decay`'s role) rather than a fourth theorem. The structural obstruction is consistent across all candidates: composition-layer quantities do not have a natural chain-rule-like *independence structure* at the agent level because composition is defined by *coupling* (shared environment, shared objective, teleological unity). The very property that makes composition interesting — coupling — is what prevents the "products of independent factors" shape that enables Cauchy-FE forcing. **The honest meta-architectural conclusion is that `#additive-coordinate-forcing` is architecturally a single-agent family, with chain / divergence / update indexing three layers of a *single* agent's internal machinery.** Composition lives in a different structural family — *monotonicity under composition* (the bridge-lemma shape from `#form-composition-closure`) — and the appropriate development for that layer is the monotonicity-under-composition family rather than forcing compositional results into the additive-coordinate-forcing shape. The cross-meta-pattern dependency at `#der-chain-confidence-decay`'s anchor role can be read either as "three single-agent layers indexed by chain/divergence/update + one composition-layer anchor in a different family" or as "a single-agent meta-pattern with explicit scope exit at the composition boundary." Both readings are defensible; the choice is presentational.

Appendix

Derivation: Observation-Ambiguity Bias-Bound Constant C

The observation-ambiguity bias bound carried by Class 3 (Coupled) agents in the logogenic-agents scope — $\|\Delta M_{\text{bias}}\| \leq C \cdot \kappa_{\text{processing}} \cdot I(G; \Omega_\tau \mid e_\tau, M_{\tau^-})$ (`#scope-observation-ambiguity-modulation`, `#result-section-ii-survival`) — previously treated the constant C as "domain-dependent" and the bound as "order-of-magnitude guidance, not a theorem." This appendix derives C under two named sub-scopes, records a no-go showing that C cannot be universal without the (PI) parameterization-invariance axiom of 4.7, and documents two failed derivation routes so future agents do not repeat them.

Formal Expression.

§1 — Setup and the audit before strengthening. Let $M_{\tau^-} \in \mathcal{M}$ be the pre-update epistemic substate. Let e_τ be the event triggering update, G_t the goal substate, and Ω_τ the latent world-state. In Class 1 (Separated) scope, 5.3 guarantees the update $f_M(M_{\tau^-}, e_\tau)$ is goal-blind: $M_{\tau^+}^{\text{decoupled}} = f_M(M_{\tau^-}, e_\tau)$. In Class 3 (Coupled) scope, the coupled update $f_X^M(X_{\tau^-}, e_\tau)$ carries goal-conditional reweighting; the bias is

$$\Delta M_{\text{bias}} := f_X^M(X_{\tau^-}, e_\tau) - f_M(M_{\tau^-}, e_\tau) \tag{.1}$$

Pre-strengthening type audit. For the bound "$\|\Delta M_{\text{bias}}\| \leq C \cdot \kappa \cdot I$" to be well-typed:

1. **Norm on $\mathcal{M}$.** The LHS is a norm on model-space. $\mathcal{M}$ is the model space of 2.1. Three candidate norms: Euclidean on parameters, total variation on induced measures, Fisher-Rao geodesic distance. **Under the (PI) parameterization-invariance axiom in 4.7** (fourth primary instance of Z), Euclidean-on-parameters is a coordinate artifact and Fisher-Rao is the canonical AAT-invariant choice on statistical-manifold sub-cases of $\mathcal{M}$ (Čencov 1982 uniqueness).
2. **Regularity of f_X^M.** For the bound to be a theorem, f_X^M must satisfy some regularity. The Bayesian-posterior model (prior + likelihood reweighting) is the canonical working class; it covers the attention-reweighting mechanism typical of Class 3 (Coupled) architectures to leading order.
3. **Information coordinate.** $I(G; \Omega_\tau \mid e_\tau, M_{\tau^-})$ is in nats (or bits). Converting information to state-distance is exactly what the Pinsker / Bretagnolle-Huber / Otto-Villani / Bakry-Émery machinery does.

The bound is well-typed under (a) named norm, (b) bounded f_X^M regularity, (c) specified information-geometric inequality. Without (PI) adopted, C is norm-dependent; under (PI), Fisher-Rao is canonical and C becomes derivable.

§2 — Track 1: Transport-inequality derivation under log-Sobolev + Lipschitz-posterior.

Named sub-scope (H1-H3).

Derived (w2-transport-track, conditional on H1-H3).

- **(H1) Statistical-manifold sub-case.** Each $M_t \in \mathcal{M}$ corresponds to a probability distribution P_{M_t} over world-states. $\mathcal{M}$ is (locally) a statistical manifold.
- **(H2) Log-Sobolev inequality.** The observation distribution $P_{\Omega|e,M}$ satisfies a log-Sobolev inequality (LSI) with constant $\rho_{\text{LSI}} > 0$. Sufficient condition: $P_{\Omega|e,M}$ is strongly log-concave with constant K (Bakry-Émery 1985 curvature-dimension condition gives $\rho_{\text{LSI}} \geq K$). For Gaussian observation models, $\rho_{\text{LSI}} = 1/\sigma^2$ explicitly.
- **(H3) Lipschitz-posterior stability.** The Bayesian-posterior pushforward from observation to state is L_{post}-Lipschitz in W_2:

$$W_2(P_{M|e,G}, P_{M|e}) \leq L_{\text{post}} \cdot W_2(P_{\Omega|e,G}, P_{\Omega|e}) \tag{.2}$$

  Sufficient condition: well-posed Bayesian inverse problem with bounded log-likelihood Hessian (Stuart 2010 *Inverse problems: a Bayesian perspective*, *Acta Numerica* 19:451–559, Theorem 4.6; Hairer, Stuart & Vollmer 2014 *SIAM J. Math. Anal.* 46(1):415–451 for explicit W_2-bounds in infinite-dimensional settings).

Step 1 — KL from mutual information. By the chain rule of relative entropy (Cover & Thomas 2006 Theorem 2.5.3):

$$\mathbb{E}_G\big[\text{KL}\big(P_{\Omega|e,M,G} \,\|\, P_{\Omega|e,M}\big)\big] = I(G; \Omega_\tau \mid e_\tau, M_{\tau^-}) \tag{.3}$$

Exact identity.

Step 2 — Otto-Villani under LSI. From (H2) and Otto & Villani 2000 *J. Funct. Anal.* 173(2):361–400 Theorem 1:

$$W_2^2\big(P_{\Omega|e,M,G}, P_{\Omega|e,M}\big) \leq \frac{2}{\rho_{\text{LSI}}} \cdot \text{KL}\big(P_{\Omega|e,M,G} \,\|\, P_{\Omega|e,M}\big) \tag{.4}$$

Taking expectation over G and substituting Step 1:

$$\mathbb{E}_G\big[W_2^2(P_{\Omega|e,M,G}, P_{\Omega|e,M})\big] \leq \frac{2}{\rho_{\text{LSI}}} \cdot I(G; \Omega_\tau \mid e_\tau, M_{\tau^-}) \tag{.5}$$

Step 3 — Lipschitz-posterior pushforward. From (H3):

$$\mathbb{E}_G\big[W_2^2(P_{M|e,G}, P_{M|e})\big] \leq L_{\text{post}}^2 \cdot \mathbb{E}_G\big[W_2^2(P_{\Omega|e,G}, P_{\Omega|e})\big] \tag{.6}$$

Step 4 — $\kappa_{\text{processing}}$ factor. The $\kappa_{\text{processing}}$ coefficient from 5.3's Class 1/2/3 (Separated/Partial/Coupled) taxonomy multiplies the goal-conditional reweighting strength by the modularity coefficient. For Class 3 (Coupled), $\kappa_{\text{processing}} \approx 1$; for Class 2 (Partial) with modularity κ, the effective bound carries a factor κ multiplicatively on the information term.

Result (Track 1). *[Derived, conditional on (H1)-(H3)]*

$$\boxed{\mathbb{E}\big[W_2^2(M_{\tau^+}^{\text{coupled}}, M_{\tau^+}^{\text{decoupled}})\big] \;\leq\; \frac{2L_{\text{post}}^2}{\rho_{\text{LSI}}} \cdot \kappa_{\text{processing}} \cdot I(G; \Omega_\tau \mid e_\tau, M_{\tau^-})} \tag{.7}$$

The constant is explicit: $C_{W_2}^2 = 2L_{\text{post}}^2/\rho_{\text{LSI}}$, linear in I, with geometric interpretation:

- L_{post} grows with *prior-likelihood tension* — ill-conditioned inverse problems amplify bias.
- ρ_{LSI} grows with *observation-distribution concentration* — sharper observations yield tighter bounds.
- The ratio $2L_{\text{post}}^2/\rho_{\text{LSI}}$ captures the **geometric stiffness** of the update: small-ρ/large-L = soft, goal-sensitive updates; large-ρ/small-L = stiff updates that resist goal-bias.

§3 — Track 2: Fisher-Rao derivation under (PI) + Čencov + small-information regime.

Named sub-scope (H1 + H4).

Derived (fisher-rao-track, conditional on H1 + H4).

- **(H1) Statistical-manifold sub-case** — as in §2. Under the (PI) parameterization-invariance axiom of 4.7, Čencov's 1982 uniqueness theorem (*Statistical Decision Rules and Optimal Inference*, AMS) forces the Fisher information metric as the canonical Riemannian metric on $\mathcal{M}$ up to global scale (Ay, Jost, Lê & Schwachhöfer 2017 *Information Geometry*, Theorem 5.1).
- **(H4) Small-information regime.** $I(G; \Omega_\tau \mid e_\tau, M_{\tau^-}) \ll 1$ nat. The second-order Taylor expansion of KL at coincident distributions is sharp.

Step 1 — KL-to-Fisher-squared-distance identity. For nearby distributions P, Q on a statistical manifold with Fisher metric $\mathbf{I}$, the KL divergence admits the second-order expansion

$$\mathrm{KL}(P\|Q) \;=\; \frac{1}{2} \cdot d_{FR}^2(P, Q) + O(d_{FR}^3) \tag{.8}$$

where d_{FR} is the Fisher-Rao geodesic distance (Cover & Thomas 2006 §12.5; Amari & Nagaoka 2000 §3.7 Theorem 3.1). This is the infinitesimal form of the Bregman divergence on the exponential family's dual geometry (cf. O §6.3 for the related Fenchel-Bregman identification).

Step 2 — Posterior-displacement in Fisher-Rao. Under (H1)+(H4) + Bayesian-posterior-class f_X^M, the goal-conditional posterior and goal-marginalized posterior are nearby points on the statistical manifold; their Fisher-Rao distance satisfies

$$d_{FR}^2\big(M_{\tau^+}^{\text{coupled}}, M_{\tau^+}^{\text{decoupled}}\big) \;\le\; 2 \cdot \mathrm{KL}\big(P_{M|e,G} \,\|\, P_{M|e}\big) + O(d_{FR}^3) \tag{.9}$$

Using the data-processing inequality $\mathrm{KL}(P_{M|e,G}\|P_{M|e}) \le \mathrm{KL}(P_{\Omega|e,G}\|P_{\Omega|e})$ (Bayesian posterior is a pushforward), and Step 1 of §2 ($\mathbb{E}_G[\mathrm{KL}(P_{\Omega|e,M,G}\|P_{\Omega|e,M})] = I$):

$$\mathbb{E}\big[d_{FR}^2(M_{\tau^+}^{\text{coupled}}, M_{\tau^+}^{\text{decoupled}})\big] \;\le\; 2 \cdot I(G; \Omega_\tau \mid e_\tau, M_{\tau^-}) \cdot (1 + o(1)) \tag{.10}$$

Step 3 — Taking square roots (with κ factor). Under Class 2 (Partial) modularity, $\kappa_{\text{processing}}$ enters multiplicatively on the goal-conditional reweighting strength, hence on the KL and Fisher-Rao distance:

Result (Track 2). *[Derived, conditional on (H1)+(H4), small-information regime]*

$$\boxed{\mathbb{E}\,\|\Delta M_{\text{bias}}\|_{FR} \;\le\; \sqrt{2} \cdot \sqrt{\kappa_{\text{processing}} \cdot I(G; \Omega_\tau \mid e_\tau, M_{\tau^-})} \cdot (1 + o(1))} \tag{.11}$$

The constant is universal and dimension-free: $C_{FR} = \sqrt{2}$ (nats) or $\sqrt{2 \ln 2}$ (bits). **No domain-specific parameters.** The (PI) commitment eliminates the coordinate-dependence that made C ambiguous under Euclidean-parameter norms.

Scaling difference between tracks. Track 1 gives $W_2^2 \propto I$ (linear in information). Track 2 gives $d_{FR}^2 \propto I$, equivalently $d_{FR} \propto \sqrt{I}$ (square-root in information). The two tracks are not contradictory — they are bounds under different assumptions in different metrics. Track 1 is tight in the large-I regime where Otto-Villani's linear form dominates; Track 2 is tight in the small-I regime where the Fisher-metric second-order expansion is sharp. The two compose at intermediate scales by taking the tighter of the two under the local curvature conditions.

§4 — No-go result: universal C under Euclidean-parameter norm fails.

Claim. No universal constant C — independent of $\mathcal{M}$'s geometry, parameterization, and coupled-update structure — exists such that $\|\Delta M_{\text{bias}}\|_{\text{Euclidean}} \leq C \cdot \kappa \cdot I$ under the Euclidean norm on an arbitrary parameter vector.

Proved (no-universal-C-euclidean, counterexample-grade).

Counterexample. Let $\mathcal{M} = \{N(0, \sigma^2) : \sigma > 0\}$ parameterized by σ. The Fisher information in σ scales as $\mathbf{I}(\sigma) = 2/\sigma^2$ — small near large σ, large near small σ. For any $C_0 < \infty$, choose σ_0 large enough that the Fisher-Rao displacement corresponding to $I(G;\Omega) = 1$ nat of goal-conditional reweighting translates to Euclidean-σ displacement $\sigma_0/\sqrt{2} > C_0$. Specifically, by Track 2's $\sqrt{2}$ bound:

$$d_{FR}(M^{\text{coupled}}, M^{\text{decoupled}}) \leq \sqrt{2} \cdot 1 = \sqrt{2} \tag{.12}$$

Under the Fisher metric $\mathbf{I}(\sigma) = 2/\sigma^2$, the Fisher-Rao line element is $ds^2 = 2d\sigma^2/\sigma^2$, so $\Delta\sigma \approx \sigma \cdot d_{FR}/\sqrt{2} = \sigma$ (Euclidean displacement scales linearly in σ). Taking $\sigma \to \infty$ gives arbitrarily large Euclidean-σ displacement for fixed $I = 1$ nat. No universal C in Euclidean-parameter norm exists.

Implication. The no-go strengthens the (PI) commitment rather than weakening the bound: **under (PI), Fisher-Rao is the canonical norm and $C = \sqrt{2}$ is universal; without (PI), C does not exist as a universal constant.** The (PI) axiom is *load-bearing* for this derivation, not coincidental.

Position relative to X. The no-go has the shape of a floor-pattern no-go (external obstruction: Euclidean-parameter norms carry unbounded Fisher condition numbers; escape: (PI) + Fisher-Rao gives universal C), but it does not match the five-element test for a floor instance (see X): it has a *single* escape (the (PI) adoption), not ≥ 2 distinct escapes, and its strengthened consequence is *re-use* of existing (PI)+Čencov machinery (fourth primary instance of Z) rather than elevation of new machinery. The honest position: this no-go is a **downstream theorem of the (PI) commitment**, not a new floor instance. It belongs in this appendix as motivating justification for why (PI) is load-bearing for the derivation, not as a separate meta-segment entry. (Triage detail in `spikes/spike-identifiability-floor-instance-triage-2026-04-24.md`.)

§5 — Failed derivation routes (honest record). Two derivation routes were attempted and failed at a structural level. Documented here so future agents do not re-attempt them without new evidence.

(F1) Cramér-Rao inversion — wrong direction. Attempt: treat G as a parameter estimated from Ω given (e, M); use the Cramér-Rao lower bound $\mathrm{Var}(\hat{G}) \geq \mathbf{I}_G^{-1}$ inverted to bound $\|\Delta M_{\text{bias}}\| \leq L_M \cdot \sqrt{\mathbf{I}_G^{-1}}$ via a posterior-sensitivity factor. Failure: Cramér-Rao bounds estimator error *below*, not above; the bound goes in the opposite direction to what the bias bound needs. Inverting Cramér-Rao for an upper bound on bias propagation requires a fixed-estimator-class assumption that is not available for a Class 3 (Coupled) agent's attention-reweighting mechanism. The transport-inequality route of §2 is the correct machinery.

(F2) Rate-distortion inversion — wrong problem structure. Attempt: apply Shannon's rate-distortion inequality $R(D) \geq h(X) - \frac{1}{2}\log(2\pi e D)$ (Gaussian source), invert to $D \leq \sigma^2 \cdot 2^{-2(R - h(X))}$ with "bits in" = $I(G; \Omega \mid e, M)$. Failure: rate-distortion theory describes the minimum bits-per-symbol required to represent a source within distortion

D; it does **not** describe the maximum distortion induced by injecting side-information into an update. Source-coding theorems are about optimal representations of a source, not spatial displacement induced by side-information injection. The direction is again wrong. Transport inequalities (Attempt B) are the correct machinery.

The pattern shared across (F1) and (F2): **information-theoretic source-coding theorems (Cramér-Rao bounds, rate-distortion lower bounds) are structural lower-bound machinery for an estimator's or encoder's performance; they cannot be inverted to produce upper bounds on displacement induced by side-information.** Upper bounds on displacement require transport-inequality machinery (Pinsker, Otto-Villani, Bakry-Émery, posterior-stability).

§6 — Gaussian worked example.

Setup. Gaussian observation model $P_{\Omega|e,M} = N(\mu_M, \sigma^2 I)$ with fixed σ^2; conjugate-Gaussian prior $P_M = N(0, \tau^2 I)$; goal-conditional likelihood reweighting $P_{\Omega|e,M,G} = N(\mu_M + \beta(G), \sigma^2 I)$ where $\beta(G)$ is a goal-dependent shift with $\|\beta(G)\|^2$ bounded by the information budget $I(G; \Omega_\tau)$.

Illustrative, exact under specified assumptions.

LSI constant. Gaussian $N(\mu, \sigma^2 I)$ satisfies LSI with $\rho_{\text{LSI}} = 1/\sigma^2$ (Bakry-Émery; direct Hessian bound on $-\log p$).

Posterior-Lipschitz constant. Conjugate-Gaussian posterior: $P_{M|\Omega} = N(\mu_{\text{post}}, \Sigma_{\text{post}})$ with $\mu_{\text{post}} = \tau^2/(\tau^2 + \sigma^2) \cdot \Omega$. Lipschitz-in-$\Omega$ constant $L_{\text{post}} = \tau^2/(\tau^2 + \sigma^2) < 1$. Under W_2-Lipschitz (Hairer-Stuart-Vollmer 2014), $L_{\text{post}} \leq 1$ for well-conditioned priors.

Track 1 bound (numerical). $C^2_{W_2} = 2L^2_{\text{post}}/\rho_{\text{LSI}} = 2\sigma^2 \cdot \tau^4/(\tau^2 + \sigma^2)^2$. In the prior-dominant limit ($\tau^2 \ll \sigma^2$): $C^2_{W_2} \approx 2\tau^4/\sigma^2$ (tight prior amplifies bias). In the likelihood-dominant limit ($\sigma^2 \ll \tau^2$): $C^2_{W_2} \approx 2\sigma^2$ (sharp observations limit bias).

Track 2 bound (numerical). Under (PI) + Fisher-Rao on the Gaussian mean-manifold with metric $\mathbf{I}(\mu) = I/\sigma^2$: $d_{FR}(\mu_1, \mu_2) = \|\mu_1 - \mu_2\|/\sigma$. Under small-$I$: $\mathbb{E}\, d_{FR}(\Delta M_{\text{bias}}) \leq \sqrt{2 \cdot \kappa \cdot I}$. In Euclidean-on-$\mu$ norm: $\|\Delta\mu_{\text{bias}}\| \leq \sigma \cdot \sqrt{2\kappa I}$ — the σ prefactor is the Fisher-Rao-to-Euclidean conversion. The Euclidean form is coordinate-dependent (per Attempt E §4); Fisher-Rao form is universal.

Comparison at operating point. For $\tau = \sigma = 1$ (balanced): Track 1 gives $C^2_{W_2} = 2 \cdot (1/4) = 1/2$, so $\mathbb{E}\, W_2 \leq \sqrt{I/2}$. Track 2 gives $\mathbb{E}\, d_{FR} \leq \sqrt{2I}$. The two are within a factor of 2 of each other at balanced scale; they diverge in the ill-conditioned limits where Track 1's L_{post} or ρ_{LSI} blows up while Track 2 remains dimension-free.

What Is Derived vs. What Is Chosen.

Table .1

Property	*Source*	*Strength*
Chain-rule identity $\mathbb{E}_G[\text{KL}(P_{\Omega\|G} \| P_\Omega)] = I(G; \Omega \mid \cdot)$	Cover & Thomas 2006 Theorem 2.5.3	Exact
Pinsker's inequality $\|P - Q\|_{TV} \leq \sqrt{\frac{1}{2}\text{KL}}$	Standard (Csiszár-Körner)	Exact
Otto-Villani $W_2^2 \leq (2/\rho_{\text{LSI}})\text{KL}$ under LSI	Otto & Villani 2000 Theorem 1; Bakry-Émery 1985 curvature-dimension for LSI	Exact
KL-to-Fisher-squared-distance $\text{KL}(P\|Q) = \frac{1}{2}d^2_{FR} + O(d^3_{FR})$	Cover-Thomas 2006 §12.5; Amari-Nagaoka 2000 §3.7 Theorem 3.1	Exact (standard information geometry)
Bayesian-posterior W_2-Lipschitz stability L_{post}	Stuart 2010 Theorem 4.6 under well-posed inverse problem	Conditional on (H3)

Property	*Source*	*Strength*
Track 1 bound $\mathbb{E}[W_2^2(M^{\text{coupled}}, M^{\text{decoupled}})] \le (2L_{\text{post}}^2/\rho_{\text{LSI}}) \cdot \kappa \cdot I$	Composition of §2 Steps 1–4	**Derived (conditional on H1-H3)**
Track 2 bound $\mathbb{E}\,\lVert \Delta M_{\text{bias}} \rVert_{FR} \le \sqrt{2} \cdot \sqrt{\kappa I}(1 + o(1))$	Composition of §3 Steps 1–3 under (PI) + Čencov	**Derived (conditional on H1+H4, small-*I* regime)**
$C_{FR} = \sqrt{2}$ is universal, dimension-free, no domain constants	§3 Result	Exact under (H1)+(H4)
$C_{W_2}^2 = 2L_{\text{post}}^2/\rho_{\text{LSI}}$ with explicit geometric interpretation	§2 Result	Exact under (H1)-(H3)
Attempt E no-go: no universal C in Euclidean-parameter norm	Heteroscedastic-normal counterexample §4	Proved (counterexample-grade)
(F1) Cramér-Rao inversion fails	Direction mismatch (estimator lower bound cannot invert to upper bound)	Recorded failure
(F2) Rate-distortion inversion fails	Problem-structure mismatch (source-coding theorem cannot yield side-information injection bound)	Recorded failure
Gaussian worked example with explicit C_{W_2} and C_{FR}	Conjugate-Gaussian direct computation	Exact under specified assumptions

The dividing line: both tracks are derived theorems under named hypotheses; the hypotheses (LSI, Lipschitz-posterior, small-I, statistical-manifold sub-case, (PI) adoption) are either standard mathematical regularity (LSI, Lipschitz-posterior) or AAT-internal axioms ((PI), already adopted in 4.7 and elevated to fourth primary instance of Z). The no-go §4 justifies the (PI) commitment as load-bearing rather than coincidental. The failed attempts §5 document structural reasons two alternative routes cannot work, preventing future re-attempts.

Epistemic Status.

Conditional — exact under H1-H3 (Track 1) or H1+H4 (Track 2). Both tracks are derived theorems under named hypotheses:

- Track 1 is exact under the three regularity conditions (H1 statistical-manifold sub-case; H2 log-Sobolev on observation distribution; H3 Lipschitz-posterior stability). All three are standard mathematical regularity conditions satisfied for canonical cases (Gaussian observations: (H2) direct; well-posed Bayesian inverse problems: (H3) per Stuart 2010). The bound is linear in I, with explicit geometric-stiffness constants.
- Track 2 is exact under (H1) + (H4) small-information regime, leveraging (PI) + Čencov. The bound is $\sqrt{I}$ scaling with universal dimension-free constant $C_{FR} = \sqrt{2}$. Does not depend on domain-specific L_{post} or ρ_{LSI}; the price is the small-I regime restriction.

The no-go §4 is *proved* via the heteroscedastic-normal family counterexample — universal C in Euclidean-parameter norm does not exist; the (PI) commitment is load-bearing for the derivation. The failed attempts §5 are *recorded* at structural-reason level; they document why Cramér-Rao inversion and rate-distortion inversion cannot produce the needed upper bound, not merely that the attempts failed.

Upgrade over prior formulation. The bias bound in `#result-section-ii-survival` and `#scope-observation-ambiguity-modulation` previously carried the honest admission *"Without the constant, the bound is order-of-magnitude guidance, not a theorem."* This appendix **promotes the bound to theorem-level** under named sub-scopes. The order-of-magnitude admission is superseded by the two explicit tracks with their H-conditions. Updates to the logogenic-agents segments cross-reference this appendix.

Max attainable: *exact* for the composition of the three textbook inequalities (Step 1 chain-rule; Step 2 Otto-Villani; Step 3 Lipschitz-posterior or KL-to-Fisher-Rao identity) and for the no-go counterexample §4. *Conditional* overall —

the theorem statements carry their H-conditions explicitly, and the sub-scope boundary is where "derived" transitions back to "admitted empirical regularity" (cases where H2 or H3 fails require separate treatment, flagged in Working Notes).

Honest failure modes.

The appendix does **not** cover:

1. **Non-statistical-manifold sub-cases of $\mathcal{M}$.** If $\mathcal{M}$ carries structured state-space geometry (mixed continuous/discrete components, graphical-model structure, non-parametric models without sufficient statistics), (H1) is weakened and Čencov's uniqueness does not straightforwardly apply. The Fisher-Rao track degrades; Track 1 may still apply under suitable alternative regularity.
2. **Unbounded KL / structural-failure limit.** If $P_{\Omega|e,M,G}$ has support disjoint from $P_{\Omega|e,M}$ at G-values carrying non-trivial prior probability, the KL (and thus I) diverges. The bound is vacuous in this regime. This is a structural failure of the coupled-update model, not a bound weakness.
3. **Non-Lipschitz pushforward.** If f_X^M is not a well-posed Bayesian posterior (e.g., ill-posed inverse problem, degenerate likelihood, rank-deficient observation channel), (H3) fails and Track 1 does not apply. Track 2's small-I expansion may still apply in the local-linearization sub-region.
4. **Large-information regime beyond the small-I expansion.** Track 2's $\sqrt{I}$ scaling uses the second-order Fisher-metric expansion of KL; the full global relationship is lattice-dependent. A fully-global Fisher-Rao statement requires either compact-manifold assumption (Fisher-Rao distance bounded) or a staged-application argument (many small-I steps composed).
5. **Heavy-tailed observation distributions.** LSI fails for heavy-tailed measures. Track 1 degrades to the Pinsker bound $W_2 \leq \sqrt{\mathrm{KL}/(2\rho_{\mathrm{LSI}})}$ when $\rho_{\mathrm{LSI}} \to 0$; this is the $O(\sqrt{\varepsilon})$ baseline without the LSI tightening.

Discussion.

Position in the Z meta-pattern. Track 2's $C_{FR} = \sqrt{2}$ result is a **downstream theorem of the fourth primary instance** (Čencov-invariance at the metric layer, under (PI)). Once (PI) adoption forces Fisher-Rao as the canonical metric on statistical-manifold sub-cases of $\mathcal{M}$, the KL-to-Fisher-squared-distance identity supplies the universal constant "for free" in the small-I regime. The no-go §4 tightens the connection: the (PI) commitment is not just "useful axiom" but load-bearing for the bound's theorem-level status — without (PI), C does not exist as a universal constant.

Connection to . Both segments use Pinsker's inequality as a first step in propagating a KL bound. After Pinsker, the two segments diverge: this segment's cascade (Pinsker → Otto-Villani → Lipschitz-posterior → W_2 on pushforward) is a KL-to-state-distance machinery; `#deriv-variational-sector-condition`'s use (Pinsker + Cauchy-Schwarz → scalar sector-constant degradation) is a KL-to-scalar-constant machinery. The shared first step is Pinsker; the post-Pinsker cascade is not shared. Candidate shared-lemma extraction is flagged in `spikes/spike-kl-to-state-distance-template-extraction-2026-04-24.md` (Option B, `#posterior-displacement-template`); `#deriv-variational-sector-condition` is positioned there as an adjacent family member (Pinsker shared; cascade not).

Forward-looking clients. Three open extensions in X Working Notes are candidate future primary clients of this appendix's machinery: causal-IB (KL → W_2 on post-intervention pushforward), misspecification-cost (KL → W_2 on posterior under model misspecification), composition-scope-robustness (KL → W_2 on composite-state pushforward). Each would reuse the Track 1 Otto-Villani + Lipschitz-posterior cascade with client-specific (P, Q, f) and $(L_{\mathrm{post}}, \rho_{\mathrm{LSI}})$ instantiations. If two or more of these materialize, factoring the cascade out as a template segment becomes attractive (see the extraction spike).

Why the Fisher-Rao specialization is worth landing even under the small-I restriction. In the AAT-canonical operating regime, Class 3 (Coupled) bias is *moderate* — the goal-conditional reweighting shifts the posterior but does not replace it. Small-I is the honest model of this regime; large-I is the structural-failure limit where the coupled agent's goal-conditioning overwhelms evidence. Track 2's universal $C_{FR} = \sqrt{2}$ is the right bound in the operating regime; Track 1 is the fallback under structural conditions where I is not small or where Fisher-Rao is not available.

Connection to logogenic agents. This appendix's bounds are the quantitative apparatus underlying `#scope-observation-ambiguity-modulation`'s product-form bias bound and `#result-section-ii-survival`'s Class 3 (Coupled) scope-survival analysis. Both segments previously carried the "constant C not computed" caveat; with this appendix landed, C is computed under named conditions and the epistemic tier of the bias bound upgrades to "conditional (exact under H1-H3 or H1+H4)." The logogenic-agents segments retain their client status; the quantitative work moves here.

Findings.

Universal Constant for the Coupled-Agent Bias Bound under Parameterization-Invariance. Brief: When a goal-conditioned agent (like an LLM) processes evidence, its update can be biased by what it's trying to do, not just what it observes. Prior work could only say "this bias is bounded by some constant times the relevant information leakage" without computing the constant. This result computes it under two specific conditions, and shows that at least one of them — committing to a particular geometric structure on the model space (the (PI) parameterization-invariance axiom) — is necessary in a strong sense: without it, no universal constant exists at all. A transport-inequality track gives $C_{W_2}^2 = 2L_{\text{post}}^2/\rho_{\text{LSI}}$ linear in the goal-conditional information leakage, under log-Sobolev concentration of the observation distribution and Lipschitz-stability of the Bayesian posterior. A Fisher-Rao track gives a universal dimension-free $C_{FR} = \sqrt{2}$ in the small-information regime, conditional on the (PI) axiom. Two structurally wrong derivation routes (Cramér-Rao inversion; rate-distortion inversion) are recorded as honest failure cases.

Impact: Promotes the bias bound from "order-of-magnitude guidance, not a theorem" to a conditional theorem under explicitly named hypotheses, closing the quantitative gap that `#scope-observation-ambiguity-modulation` and `#result-section-ii-survival` previously carried as a caveat. The no-go strengthens the (PI) commitment from useful axiom to load-bearing structural choice and adds a fourth primary instance to the `#disc-additive-coordinate-forcing` meta-pattern. The recorded failures prevent future agents from re-attempting Cramér-Rao or rate-distortion inversions for upper bounds on side-information-induced displacement, and identify transport-inequality machinery (Pinsker, Otto-Villani, Bakry-Émery, posterior-stability) as the correct family. Three forward-looking clients in adjacent appendices (causal-IB pushforward, misspecification-cost, composition-scope-robustness) are candidate consumers of the same Track 1 cascade, suggesting eventual extraction of a shared `#posterior-displacement-template` if two or more materialize.

Novelty Claim: *Claim differentiation* on the Lipschitz-posterior + Otto-Villani composition for AAT's coupled-agent bias bound, plus *claim novelty* on the no-go counterexample showing that universal C in Euclidean-parameter norms cannot exist, which jointly elevates the (PI) axiom from convergent representational choice to load-bearing for theorem-level status. The two upstream tracks each compose textbook information-geometric / transport-inequality machinery (Cover-Thomas chain rule; Otto-Villani 2000; Stuart 2010 Bayesian-posterior stability; Amari-Nagaoka 2000 Fisher-metric expansion); the contribution is the composition for the Class 3 (Coupled) agent bias setting and the no-go that justifies the geometric commitment. *(This finding's novelty is the supporting quantitative apparatus; the integrated $\varkappa \times A$ bias law it supports is in* `#scope-observation-ambiguity-modulation`*, where the Pillar-3 verdict applies.)*

Related Work:

Table .2

ASF concern	*Prior-art language*	*Relationship / Positioning*
KL-to-W_2 transport under log-Sobolev	Otto & Villani 2000 *J. Funct. Anal.* 173:361–400 (published 2000); Bakry & Émery 1985 curvature-dimension condition	*formal antecedent* — adopted directly in the Track 1 cascade; standard transport-inequality machinery, cited and used
Lipschitz-stability of Bayesian posteriors	Stuart 2010 *Acta Numerica* 19:451–559 (published 2010); Hairer, Stuart & Vollmer 2014 *SIAM J. Math. Anal.* 46(1):415–451 (published 2014)	*formal antecedent* — adopted directly as (H3); the conjugate-Gaussian case is computed explicitly in §6
Čencov uniqueness of the Fisher metric	Čencov 1982, *Statistical Decision Rules and Optimal Inference* (published 1982); Ay, Jost, Lê & Schwachhöfer 2017 *Information Geometry* Theorem 5.1 (published 2017)	*formal antecedent* — adopted via the (PI) axiom in `#scope-agent-identity` and used in the Track 2 derivation; the no-go in §4 is the AAT-internal demonstration that this adoption is load-bearing for the bound, not just convenient
KL-to-Fisher second-order expansion	Amari & Nagaoka 2000, *Methods of Information Geometry* §3.7 Theorem 3.1; Cover & Thomas 2006 §12.5	*formal antecedent* — supplies the small-I regime sharpness used in Track 2
Information-processing-constraint bounds on decisions	Ortega & Braun 2012 *Proc. R. Soc. A* (published 2012-04, found 2026-04 via Undermind); Genewein et al. 2015 *Front. Robot. AI* 2:27 (published 2015-11, found 2026-04)	*adjacent literature* — addresses a different quantity (decision quality under information constraint rather than belief-update bias under goal-coupling); shares the information-theoretic vocabulary but does not produce a Wasserstein or Fisher-Rao bias bound for coupled-architecture agents
Variational bounds on posterior beliefs in active inference	Friston et al. 2017, "Active inference: a process theory" *Neural Computation* 29:1–49 (published 2017)	*adjacent literature* — uses overlapping transport-inequality machinery in a different framework; consistent with the Track 1 derivation but does not require the variational-free-energy commitment (cf. `#form-information-bottleneck` Discussion)

Search Log:

- 2026-04 (*nominally comprehensive*, via `ref/Novelty_defense_and_integration.md` Pillar 3): Undermind search located bounded-rationality and information-cost literatures (Ort12c, Gen15) as the closest decision-theoretic precursors and confirmed no equivalent of the universal-C derivation under (PI) was surfaced. The Pillar 3 verdict (Wholly Novel, Medium confidence) applies to the integrated $\varkappa \times$ A law downstream in `#scope-observation-ambiguity-modulation`; this segment's contribution is narrower (the constant derivation and the no-go) and inherits the same Medium confidence with a follow-on targeted search recommended on formal posterior-displacement bounds in coupled-architecture agents.
- 2026-04 (*targeted*, AAT-internal trail in `spikes/spike-bias-bound-constant-C-strengthening-2026-04-24.md`): two failed routes (Cramér-Rao inversion, rate-distortion inversion) eliminated at structural-mismatch level; transport-inequality machinery identified as correct; (PI) adoption identified as load-bearing rather than coincidental via §4 counterexample. The shared-cascade extraction question (`#posterior-displacement-template`) is contingent on forward-looking clients materializing.

Working Notes.

- **Trail.** Full derivation history in `spikes/spike-bias-bound-constant-C-strengthening-2026-04-24.md`. The spike's Track 1 and Track 2 structure maps directly to §§2–3 here; the no-go Attempt E maps to §4; the failed attempts (F1 Cramér-Rao, F2 rate-distortion) map to §5. The spike also flags the shared Pinsker / Otto-Villani machinery with `#deriv-variational-sector-condition` as a candidate shared-lemma extraction (`spikes/spike-kl-to-state-distance-template-extraction-2026-04-24.md` Option B); that extraction is contingent on this appendix landing + forward-looking clients materializing.

- **L_{post} for AAT-native cases.** The Lipschitz-posterior stability constant L_{post} is cited from Stuart 2010 / Hairer-Stuart-Vollmer 2014 but not computed for all AAT-relevant cases. The conjugate-Gaussian case of §6 is explicit; general non-parametric Bayesian inverse problems have looser bounds. A future refinement could compute L_{post} for the exponential-family / conjugate-prior cases in 4.2 directly.

- **Large-I Fisher-Rao refinement.** Track 2's $\sqrt{I}$ scaling is the small-I Taylor expansion. Extending to large I requires either compact-manifold assumption or staged composition; flagged for future work.

- **Interaction with the $\mathcal{A}$ factor from `#scope-observation-ambiguity-modulation`.** The product form $\kappa_{\text{eff}} = \kappa \cdot \mathcal{A}$ interacts with the transport-inequality constants in ways not fully worked out here. If $\mathcal{A}$ is itself an information-theoretic ratio, the $\mathcal{A}$ factor may be absorbable into the I term rather than appearing as a separate multiplicative factor. Flagged for a clarification pass when the logogenic-agents segments are next revised.

- **Non-Bayesian pushforwards.** The Track 1 (T3) posterior-Lipschitz step is stated for Bayesian-posterior pushforwards. Non-Bayesian pushforwards (e.g., post-intervention transformations in the causal-IB candidate) would use the general W_2-Lipschitz form; the machinery generalizes, though the concrete constants are client-specific.

- **Relation to active-inference bounds.** Active inference (Friston 2017 *Active inference: a process theory*, *Neural Computation* 29) uses KL bounds on posterior beliefs under the free-energy principle; those bounds are formulated in a different framework but share the underlying transport-inequality machinery. AAT's Track 1 derivation is consistent with the active-inference reading but does not require the variational-free-energy-as-master-objective commitment (see 2.2 Discussion for the lineage distinction).

- **Class 2 (Partial) bias-bound analog — open derivation.** This segment derives the bound C for the Class 3 (Coupled) extreme at $\kappa_{\text{processing}} \to 1$. The corresponding bound for Class 2 (Partial) agents at $\kappa_{\text{processing}} \in (0, 1)$ is open. Both Track 1 (transport-inequality under LSI + Lipschitz-posterior) and Track 2 (Fisher-Rao under (PI) + Čencov + small-I) derive the worst-case envelope; specializing either track to the bounded-coupling regime is the natural starting point. Whether the Class 2 case admits a clean closed-form bound or requires architectural specifics (e.g., per-stage modularity decomposition) is not yet known. Surfaced 2026-05-09 by the GUC rename audit; flagged in `TODO.md` §"Open theory items (MEDIUM)" for cross-referencing.

- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged / fully coupled, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. The segment's canonical claim previously known as the "Class-2 ambiguity bias bound" (fully-coupled case) is now the "Class 3 (Coupled) ambiguity bias bound." Every "Class-2 (fully-coupled)" reference renamed to "Class 3 (Coupled)"; "Class 3 partially-modular" → "Class 2 (Partial)"; "Class-1 (modular)" → "Class 1 (Separated)." Downstream segments citing the old "Class-2 bias bound" name: see external-citation list in the parent Phase 2 commit report — those segments are scope for later Phase 2/3 work. Removed at `candidate` stage per FORMAT.md Gate 4.

- **Cross-reference to NeurIPS Paper 3.** This segment's two-tracks framework is materially sharpened in NeurIPS 2026 Paper 3 ("How Much Can LLMs Hallucinate? An Upper Bound on Goal-Coupling Displacement", `~/src/neurips/03-llm-hallucinate-bound/`). The paper carries: (i) **Two named tracks with explicit hypothesis sets** at §4 Theorem 4.1 / `#thm-umbrella` — Track 1 (transport-inequality, W_2 metric) recovers the canonical Stuart-school cascade form $C_{T_2} \propto L_{\text{post}}^2/\rho_{\text{LSI}}$ under (H2'); Track 2 (Fisher-Rao Čencov uniqueness) gives $C = 2$ **globally on the ambient spherical-arc pseudometric** — the operating constant for adversarial / rare-high-KL prompts including jailbreaks and persona injection (via Rényi-1/2 monotonicity + chord-arc on the unit L^2-sphere; sharp via symmetric N-point witness) — tightening to $C = \sqrt{2}$ **locally** under (R) + (K) + (H4') uniform-locality on goal-conditional slices (first-moment sharp via symmetric two-point witness). (ii) **Chain-rule on post-update law as bridging lemma** at §5 Lemma 5.1 / `#lem-chain-rule`: $\mathbb{E}_G[\mathrm{KL}(P_{M_{\tau+}|e,M_{\tau-},G} \,\|\, P_{M_{\tau+}|e,M_{\tau-}})] =$

$I(G; M_{\tau+}|e_\tau, M_{\tau-})$ — the structural identity connecting Bayesian inverse-problems with architectural classification. (iii) **No-go theorem on Euclidean chart norms** at §4 Theorem 4.2 / `#thm-no-go`: chart-rescaling argument $\phi \mapsto a\phi$ scales chart-Euclidean W_2 linearly while leaving KL/MI/d_{FR}/Hellinger chart-invariant; taking $a \to \infty$ contradicts any candidate fixed $C_0\sqrt{I}$. *Forces (PI) as load-bearing for theorem-level status* and supplies the explicit witness construction for what this segment's Attempt E flagged structurally. The fourth instance of M1 identifiability-floor (named F4 in Paper 3 §4 Lemma 4.2). (iv) **Čencov-uniqueness-and-sharpness** at §4 Theorem 4.3 / `#thm-fr-uniqueness`: under (H1)+(PI)+(R)+(K)+(H4'), $d_{\mathcal{M}}$ is uniquely the ambient Amari-Nagaoka Fisher-Rao spherical-arc distance and $C = \sqrt{2}$ is the unique sharp upper-bound constant. (v) **Class 1 reduction theorem to Stuart-school setup** at §App-D — the Class 1 (Separated) specialization of Track 1's hypothesis space *is* the Stuart-school Lipschitz-posterior cascade (the strict-strengthening from spike A7). The paper's (PI)+(R)+(K) axiom triple — chart-invariance + Riemannian structure + KL second-order matching — is the load-bearing structural commitment for the universal-constant route. See `msc/neurips-back-integration-2026-05-08.md` §1 Paper 3 entries 1, 2, 3, 4, 5, 7. Standalone-segment proposals under back-integration §3: `#deriv-post-update-chain-rule`, `#deriv-chart-rescaling-no-go`.

Appendix

Discussion: Adversarial Coupling Pressure

In adversarial settings, opponents do not merely deceive through the observation channel; they apply strategic pressure that drives a target *across architectural classes* — from Class 1 (Separated) toward Class 3 (Coupled). Coupling is itself an attack surface: a Class 1 (Separated) agent can be deceived only through the observation channel, while a GUC-Coupled agent exposes additional input ports through O_t and Σ_t. The result is a use-case expansion for `03-llm-core/`'s coupled formulation: the population that needs coupled-formulation analysis is broader than the architecturally-coupled. Any agent under sustained adversarial pressure that can drive its $\kappa_{\text{processing}}$ upward sits in the same scope, regardless of native class.

The phenomenon.

A Class 1 (Separated) agent's f_M accepts inputs only through the event stream; its O_t and Σ_t have no legitimate input port from outside that stream. An adversary's only attack channel is engineered observation ambiguity, and the bias they can induce is bounded by `#deriv-observation-ambiguity-bias-bound`'s constant C relative to the ambiguity they can manufacture.

Discussion.

A GUC-Coupled agent — Class 3 (Coupled), or Class 2 (Partial) with active $O_t \to M_t$ pathways — exposes additional channels. Three operational mechanisms recur across the literatures on propaganda, social engineering, marketing, and influence operations:

Table .1

Mechanism	*Channel exploited*	*What the adversary does*
Identity-binding	$O_t \to M_t$	Make the target *want* the false belief to be true (bind belief to identity, group membership, self-image). Revising M_t now requires revising O_t, raising the cost of correction.
Affect / urgency	bypass deliberate f_M	Manufacture time pressure or affective load that prevents the orient cascade from running cleanly; the agent acts before M_t updates resolve.
Sunk-cost engineering	$\Sigma_t \to M_t$	Induce the target to commit publicly or expensively to a strategy; subsequent M_t revisions that would invalidate Σ_t now carry a coupled cost (admit prior commitment was wrong), and the coupling bends M_t revision instead of Σ_t revision.

Each mechanism corresponds to a specific structural pathway through which adversary-controlled signals act on M_t via O_t or Σ_t — not through the legitimate observation channel.

Diagnostic corruption is directional.

The satisfaction-gap / control-regret split is silently corrupted under adversarial coupling pressure, and the corruption is *directional*: a competent adversary biases the diagnostic predictably rather than randomly.

Derived (Conditional on directional adversary objective; from #def-satisfaction-gap, #def-control-regret + adversarial-pressure mechanisms above).

If the adversary's interest is the target's continued effort along an unproductive axis, they bias the reading toward control regret — "the goal is right, try harder." The target compounds effort where it cannot pay, while the adversary extracts whatever value the effort produces (attention, emotional engagement, financial commitment, political mobilization).

If the adversary's interest is the target's *withdrawal* from an axis they could win, they bias the reading toward satisfaction gap — "the goal was wrong, give up." The target abandons achievable wins; the adversary captures the unclaimed ground.

Manufactured-outrage cycles instantiate the first; engineered-helplessness and demoralization campaigns instantiate the second. Both are *diagnostic capture*, not just deception about world-state — the target is reading a corrupted diagnostic and confidently acting on it.

Cascade inversion.

The orient cascade resolves observation → $M_t \to \Sigma_t \to O_t$ in that order, with O_t revising only when forced. Adversarial coupling pressure inverts the cascade: the adversary acts on O_t first (manufacture desire / identity-stake / urgency), which forces Σ_t to align, which then makes legitimate observations flow *against* the gradient and get rejected as inconsistent with the now-committed Σ_t. The cascade still runs, but its information dependency is reversed — and in that reversed form, observation contradicting the pre-loaded objective is the *least* legitimate input the agent will accept.

Derived (Conditional on adversary controls O_t-targeted signal injection; from #der-orient-cascade).

Population expansion for coupled formulation.

The Class 3 (Coupled) formulation handed off to `03-llm-core/` was originally scoped to agents whose native architecture is non-modular: LLMs whose attention processes goals and observations together; biological cognition with motivated-reasoning pathways. Adversarial coupling pressure expands this population: a Class 1 (Separated) organization without internal $O_t \to M_t$ pathways but operating in a hostile information environment that targets its leadership's identity and commitments is, for analytical purposes, in the coupled-agent population during that engagement. The native architecture is irrelevant once the coupling exists in operation.

This is a *use-case expansion*, not a derivation expansion: the coupled-formulation machinery does not need new content for this case. What's new is the recognition that the population is broader than architecture-determined.

Scope: implicit assumptions and the polarity of coupling.

This segment covers one polarity of coupling-as-property — the *vulnerability* polarity, in which decreased modularity is externally driven and produces attack surface. Two implicit assumptions in the framework's machinery back the scope; both are worth surfacing rather than leaving hidden.

Discussion.

Bounded signaling. The framework formally restricts the $G_t \to$ world channel to action selection $a_t = \pi(M_t, G_t)$. A coarse $\mathcal{A}$ implicitly assumes π is sufficiently many-to-one — or the agent's behavioral signature sufficiently bandwidth-restricted — that observers cannot recover G_t from observable behavior alone. In practice the agent's full behavioral signature (micro-expressions, prosody, attention patterns, vocabulary choice, hormonal signals, response latencies) leaks G_t at much higher bandwidth than the discrete-action channel suggests. Against capable adversarial inferrers the bounded-signaling assumption fails, and this is part of *why* the bias bound's C becomes adversarially-tight in the §"Discussion" reframing: sophisticated adversaries effectively read G_t and target identity-bindings precisely. The assumption sits formally in `#der-directed-separation`'s agent-world interface; it is not yet surfaced there as a load-bearing scope statement, and surfacing it back-pressures that segment as a follow-on item.

Fixed action set independent of coupling. The action space $\mathcal{A}$ in the policy $\pi(M_t, G_t) \to a_t$ does not depend on the agent's coupling state. This implicitly assumes the actions available to the agent are independent of $\kappa_{\text{processing}}$. In practice, some actions exist *only* under coupling: Schelling-style commitment devices (where credibility requires foreclosed retreat); the salesperson whose sincere conviction enables persuasion that the modular-but-skeptical salesperson cannot match; the marathoner who can run the race only because retreat options have been burned; religious practice requiring coupled belief-and-action; identity-economics models where coupled identity unlocks coordination outcomes. Coupling is not just a *cost*; it is also *enabling*. Treating $\mathcal{A}$ as fixed therefore mis-specifies the strategic landscape — it omits a class of actions that rational agents may *choose* to acquire by deliberately self-coupling, and it forces the framework to read all coupling as failure mode.

Three operations on modularity state. Together with the segment's core treatment of adversarial coupling pressure, the two assumptions surface a clean three-operation picture of modularity-state dynamics:

Table .2

Operation	*Driver*	*Effect on modularity*	*Effect on agent*
Truthification	self-driven	increases	restores diagnostic; opens Section II results; instantiated by *defensive scaffolding* below
Strategic self-coupling	self-driven	decreases	*enables* actions otherwise unavailable (credibility, conviction, commitment)
Adversarial coupling pressure	externally-driven	decreases	*creates vulnerability* (this segment's core treatment)

This segment's primary subject is the third operation. The first operation is instantiated in §"Defensive scaffolding as composition" below — institutional scaffolding as the truthification leg of the three. The second operation is currently outside the framework's machinery; see Working Notes for the candidate sister segment and the candidate three-operation meta-segment.

Defensive scaffolding as composition.

Restoring modularity under adversarial pressure is structurally a Section III move. Institutional scaffolding — peer review, prediction registers, double-entry bookkeeping, adversarial procedure, prediction markets, source-authentication discipline, structured red-teaming — constructs a composite agent *(individual + scaffolding)* with better Class 1 (Separated) properties than the bare individual, even when the individual's native architecture is Class 3 (Coupled) or driven there by adversarial pressure. The scaffolding externalizes channels of the orient cascade (belief-update checked by parties whose O_t doesn't share the individual's; prediction registers fixing M_t-claims before outcomes resolve; adversarial procedure forcing the feasibility check) so that resolution can run sequentially across the composite even when it cannot run sequentially within any single component.

Discussion.

The defensive composite is a Class 2 (Partial) architecture at the composite level, whose modularity properties exceed those of its sub-components — an inversion of the usual Class 1 (Separated) sub-agents → Class 2 (Partial) composite refinement: here the sub-agents are Class 3 (Coupled) (or under adversarial pressure toward it), and the composite is the modularity-restoring level.

Epistemic Status.

Discussion-grade at the structural-recognition level. The segment articulates a use-case expansion of existing machinery rather than a derivation of new machinery; the substantive claims are inherited from upstream segments at their tiers:

- The Class 1 / 2 / 3 (Separated / Partial / Coupled) partition and $\kappa_{\text{processing}}$ measure are *robust qualitative* per `#der-directed-separation`.
- The bias bound that adversarial pressure attempts to saturate is *conditional* per `#deriv-observation-ambiguity-bias-bound` (under named sub-scopes).
- The diagnostic-corruption directionality is *robust qualitative*: it follows from `#def-satisfaction-gap` and `#def-control-regret` being arithmetic identities once the value objects are fixed, plus the structural observation that adversary preferences are typically directional rather than random; the directional bias is empirically observable across propaganda and influence-operation case studies.
- The orient-cascade inversion claim is *robust qualitative* — it follows from `#der-orient-cascade`'s information dependency under the explicit hypothesis that the adversary controls O_t-targeted signal injection.

Max attainable: *robust qualitative* for the structural pattern; the specific mechanism table is empirical (instantiated by propaganda, security, and influence-operation literatures) and would be promoted to *empirical* with named domain citations once a targeted prior-art search has been conducted.

Discussion.

This is the offensive complement to directed-separation-as-defense. `#der-directed-separation` characterizes modularity as architectural property. This segment names modularity as *contested* property under adversarial pressure: the architecture's modularity is what the adversary is trying to perforate, and what defensive scaffolding is trying to preserve or restore. The two segments together complete the picture — modularity is what the architecture enables (or fails to enable), and modularity is what the adversary attacks (or fails to attack).

Asymmetric advantage compounds in repeated interaction. When two agents face each other, each has structural incentive to (a) reduce its own coupling (defensive scaffolding) and (b) increase the opponent's coupling (offensive pressure). If the asymmetry is sustained, the more-modular agent gets clean observations of the less-modular agent's behavior — informative M_t updates that the bias bound's C does not impose — while the less-modular agent gets contaminated observations of the more-modular agent and bends M_t toward whatever pre-loaded objective the adversary engineered. The more-modular agent compounds informational advantage; the less-modular agent compounds bias. In repeated interaction with sustained asymmetry, this is *structural* rather than tactical advantage — modularity asymmetry produces strategic asymmetry.

Reframing the Class 3 (Coupled) bias bound's operational meaning. `#deriv-observation-ambiguity-bias-bound`'s constant C is presented in that segment as a worst-case bound. Adversarial coupling pressure reframes C as a *target* the adversary actively tries to saturate by engineering observation ambiguity to its maximum admissible value. Under sustained adversarial engineering, the bound is not slack — it is approached. This affects how the bound should be interpreted operationally: in non-adversarial settings, the bound is conservative; in adversarial settings, it is approximately tight under competent adversarial design. Whether the bound *can* be saturated depends on what the adversary controls (the observation channel's ambiguity-engineering bandwidth); under unrestricted adversarial control the saturation is essentially complete.

Connection to the separability pattern. The architectural classification ladder of `#disc-separability-pattern` reads adversarial coupling pressure as an *operational* driver of population movement along the ladder: Class 1 (Separated) agents under sustained adversarial pressure functionally migrate toward the Class 3 (Coupled) / general-open tier for the duration of the engagement, even though their native architecture remains Class 1 (Separated). The structured-repair tier (Class 2, Partial) is then the natural home for *defensive scaffolding* — the explicit added machinery that restores identification under adversarial conditions. The separability pattern's three-tier shape is therefore not just a static catalog of architectural types; it is a phase-space across which adversarial pressure and defensive scaffolding move populations.

Findings.

Adversarial Coupling Pressure as Use-Case Expansion of Coupled Formulation. **Brief:** Adversaries do not just lie; they reshape the *architecture* of the agent they are attacking. A Separated agent (Class 1) — one whose beliefs update from observation alone — can only be deceived through what it sees; the adversary's reach is bounded. But an agent whose beliefs are bent by what it wants to be true exposes new attack surface: every channel through which the adversary can act on what the target wants, or on what plan the target has committed to, becomes a way to reach the target's beliefs without going through legitimate observation. Practical mechanisms — identity-binding, manufactured urgency, sunk-cost engineering — are operational instantiations of this single structural move: drive the target across the architectural boundary so the additional ports open up. The framework's machinery for Coupled agents (Class 3) (originally scoped to architectural cases like LLMs and motivated cognition) therefore covers a broader population than its initial framing suggested: any agent under sustained adversarial pressure, regardless of native architecture. Defensive scaffolding — peer review, prediction registers, double-entry bookkeeping, adversarial procedure — is the structural counter-move: it constructs a composite agent whose modularity properties exceed those of its individual components, restoring at the system level what cannot be sustained at the individual level.

Impact: Names a phenomenon currently treated only obliquely in AAT: the strategic incentive in adversarial interaction to drive opponent coupling. Reframes the Class 3 (Coupled) bias bound (`#deriv-observation-ambiguity-bias-bound`) from passive worst-case to adversarially-saturated target — under competent adversarial design the bound is approximately tight rather than conservative. Connects the orient cascade's clean sequential resolution to its adversarially-inverted form, where observation flows against the information gradient and contradictory inputs are the least legitimate ones the agent will accept. Identifies institutional scaffolding (peer review, prediction markets, double-entry bookkeeping, adversarial procedure) as a Section III composition move that restores Class 1 (Separated) properties at the composite level even when individual sub-components are coupled or driven coupled. Expands the natural population for `03-llm-core/`'s coupled formulation beyond architecture-determined cases to include any agent under sustained adversarial pressure.

Novelty Claim: *Claim recognition* of adversarial coupling pressure as a structural phenomenon in AAT's existing scope architecture — adversaries strategically drive coupling because coupling expands attack surface — combined with *claim differentiation* on the population scope of coupled-formulation analysis: not just architecturally-coupled agents, but any agent under sustained adversarial coupling pressure. The empirical phenomena (manufactured outrage, identity-binding, social engineering, sunk-cost-driven commitment) are well-documented in propaganda, security, and behavioral-economics literatures; the contribution is naming their place in AAT's scope architecture, the corresponding population expansion for coupled formulation, and the asymmetric-advantage-compounding implication for repeated adversarial interaction.

Related Work:

- *Likely adjacent literatures, not yet searched* — propaganda research (Bernays 1928 *Propaganda*; Ellul 1965 *Propagandes*; Stanley 2015 *How Propaganda Works*), social engineering (Mitnick & Simon 2002 *The Art of Deception*; Hadnagy social-engineering corpus), behavioral economics on motivated reasoning (Kunda 1990 *The Case for Motivated Reasoning*; Kahan and the cultural-cognition project), influence operations (Singer & Brooking 2018 *LikeWar*), and Bayesian persuasion / cheap talk (Kamenica-Gentzkow 2011; Crawford-Sobel 1982). Relationship: *adjacent literature, intuition-only* — these literatures document the empirical phenomena instantiating the mechanism table; the contribution this segment claims is structural placement and population expansion, not phenomenon discovery.

Search Log:

- 2026-05-09 (*intuition-only*): Empirical phenomena (manufactured outrage, identity-binding, social engineering, sunk-cost commitment dynamics) are well-documented across propaganda research, social engineering, behavioral economics, and influence operations. The AAT-internal placement (adversarial pressure as driver of $\kappa_{\text{processing}}$ saturation) and the population-expansion claim for coupled formulation have not been searched. Targeted search recommended before promotion to draft, focusing on: (a) whether adversarial-cognitive-architecture framing exists in the active-inference, control-as-inference, or game-theoretic-deception literatures; (b) whether the "adversary drives opponent coupling" framing has been explicitly named in the cognitive-warfare or information-operations academic literatures; (c) whether the asymmetric-advantage-compounding result has been derived in any repeated-game treatment of cognitive influence.

Working Notes.

- **Promotion: requires targeted prior-art search.** The structural-recognition claim is the segment's contribution; the empirical phenomena are not novel. Until a Pillar-style targeted search is conducted, novelty status remains *intuition-only* and the segment should not advance past `draft`.

- **Brief is not yet at Feynman criterion.** The current Brief reaches for the structural recognition rather than an everyday physical analog whose causal structure is isomorphic to coupling-as-attack-surface. A candidate analog: the difference between selling someone a product (must convince them the product is good — bounded by observable quality) and selling someone the *desire* for the product (once they want it, they will not see the flaws — unbounded by observable quality). The salesperson who can drive desire-formation has acquired a channel into belief that does not pass through evidence about the product. Whether this analog carries the load-bearing structure (identity-binding specifically; the cascade inversion; the asymmetric-advantage compounding) needs work; the bathtub-equivalent has not been written.

- **Game-theoretic formalization candidate.** The "modularity asymmetry produces strategic asymmetry" claim in the Discussion is currently presented as a structural intuition rather than a derived result. A formalization candidate: define a two-agent repeated game where each agent's payoff includes a term in the *other agent's* $\kappa_{\text{processing}}$ (offensive value of opponent coupling) and a negative term in *own* $\kappa_{\text{processing}}$ (defensive cost of self coupling). Equilibrium analysis would predict (a) modularity arms races, (b) asymmetric advantage cascades when modularity asymmetry is sustained, and (c) defensive-scaffolding incentives. The information-theoretic structure (the more-modular agent gets cleaner M_t updates from the same observation stream) is what drives the asymmetry. Plausibly tractable; would land as an appendix segment if pursued.

- **Defensive scaffolding deserves its own segment under Section III.** The institutional-scaffolding-as-composite-agent framing in §"Defensive scaffolding as composition" is a specific instantiation of `#post-composition-consistency` with substantive content (peer review, prediction registers, double-entry bookkeeping, adversarial procedure, prediction markets, structured red-teaming all instantiate the same pattern: externalize a channel of the orient cascade so resolution runs sequentially across the composite even when it cannot run sequentially within any single component). Defer until the connection to existing composition results is mapped explicitly; a stub `#deriv-defensive-scaffolding-composition` may be the right home if the formalization tracks.
- **OUTLINE.md placement.** The segment is in `01-aat-core/src/` but is not yet in `01-aat-core/OUTLINE.md`. Natural home: under Section II (it sits at the architecture-classification layer) or as a meta-segment alongside `#disc-separability-pattern` and `#disc-identifiability-floor`. The latter is probably the better fit — it's a meta-pattern that cuts across Section II and Section III rather than a Section II result proper.
- **Sister segment for strategic self-coupling.** The fixed-$\mathcal{A}$ assumption surfaced in §"Scope: implicit assumptions and the polarity of coupling" points at coupling's *enabling* polarity — actions that exist only under coupling (Schelling commitment devices, conviction-enabled persuasion, sunk-cost-foreclosed retreat, identity-coupled coordination outcomes). A sister segment covering this polarity directly is staged at `spikes/spike-strategic-self-coupling.md`. The natural prior-art adoption set is Schelling 1960 *The Strategy of Conflict* (commitment devices), Ainslie 1992 *Picoeconomics* (intertemporal bargaining and willpower), Akerlof-Kranton 2000/2010 (identity economics), Frank 1988 *Passions Within Reason* (emotions as commitment devices). Investigation pending; the spike establishes scope and prior-art landscape.
- **Candidate meta-segment naming the three-operation pattern.** The §"Scope" three-operation table (truthification / strategic self-coupling / adversarial coupling pressure) is structurally a meta-architectural pattern — *modularity-state dynamics* as a structural axis with three distinct operations characterized by driver, polarity-on-modularity, and downstream effect. Could land as `disc-modularity-state-dynamics` alongside `#disc-separability-pattern` and `#disc-identifiability-floor` once both polarity-instantiations have segments (this one for adversarial pressure; the strategic-self-coupling sister for the enabling polarity; truthification is instantiated in §"Defensive scaffolding as composition" but probably wants its own segment if formalized). Defer until at least the strategic-self-coupling segment lands — the three-operation framing is most legible when each operation has a primary segment to point at.
- **Surfacing bounded-signaling in `#der-directed-separation`.** The bounded-signaling assumption surfaced in §"Scope: implicit assumptions" is more naturally located in `#der-directed-separation`'s Discussion as a load-bearing scope statement on the agent-world interface (it's about *that* segment's structure, not specific to adversarial pressure). Back-pressure follow-on: add to that segment a paragraph naming the assumption — that the channel from G_t to the world runs only through π, and that this assumption fails operationally when behavioral leakage is rich relative to action coarseness. Cross-reference back to this segment for the adversarial saturation case.
- **Migration note (2026-05-09 GUC rename):** Class 2 ↔ Class 3 swap. Pre-2026-05-09: Class 2 = fully merged, Class 3 = partially modular. Post: Class 2 = Partial, Class 3 = Coupled. This segment uses "coupling pressure" / "adversarial coupling" as its central concept — that broader vocabulary is unchanged. The specific GUC-class references were renamed: "toward Class 2 (fully merged)" → "toward Class 3 (Coupled)"; "Class-2 coupled formulation" → "Class 3 (Coupled) formulation"; "Class-3 architecture (partially modular)" for the defensive composite → "Class 2 (Partial) architecture"; "Class-2 bias bound" → "Class 3 (Coupled) bias bound"; "Class-1-sub-agents → Class-3-composite refinement" → "Class 1 (Separated) sub-agents → Class 2 (Partial) composite refinement". Removed at `candidate` stage per FORMAT.md Gate 4.

Appendix

Discussion: `#disc-strategic-self-coupling` *(missing)*

Segment `disc-strategic-self-coupling` *not yet written.*

Claim: Self-driven coupling-as-enabling polarity (sister to `#disc-adversarial-coupling-pressure`); coupling-dependent action space $\mathcal{A}(\kappa_t)$; reversibility-cost asymmetry; commitment-device family (Schelling), willpower / intertemporal bargaining (Ainslie), identity economics (Akerlof-Kranton), emotions-as-commitment (Frank). Second leg of the modularity-state-dynamics three-operation pattern

Appendix

Discussion: **#disc-modularity-state-dynamics** *(missing)*

Segment `disc-modularity-state-dynamics` *not yet written.*

Claim: M4 meta-segment alongside M1/M2/M3: modularity as contested property under three operations (truthification self-driven-increasing; strategic self-coupling self-driven-decreasing; adversarial coupling pressure externally-driven-decreasing). Truthification has two operational mechanisms — defensive scaffolding (peer review, prediction registers, etc.) and class-coercion-via-wrapping (`#der-class-coercion-via-wrapping`). Modularity-as-state-not-architecture refines the static GUC Class 1 (Separated)/2 (Partial)/3 (Coupled) partition with operations on the partition

Appendix

Result: Contraction Template

`#result-sector-persistence-template` states AAT's persistence arguments with a Euclidean sector condition (T2) matched to a quadratic Lyapunov in Euclidean norm. Generalizing the sector inequality to a **contraction-metric condition** (Lohmiller & Slotine 1998) preserves the template's ultimate-bound results while extending its coverage in three directions: (a) sub-scope α gains natural non-Euclidean metrics that remove condition-number penalties currently visible in 4.2; (b) two additional sub-scope β items (PID-bounded-plant; non-convex-within-basin) promote to derived sub-scope metric-α_2 under explicit conditions; (c) composition acquires **topology-indexed closure results** (parallel / hierarchical / small-gain-feedback) from Slotine 2003 that generalize `#deriv-critical-mass-composition`'s matched-symmetric-Tier-1 closed form to heterogeneous sub-agents. This segment states the generalization once in parameter-free form so that each lifting citation can reference it and specify only what varies.

`#result-sector-persistence-template` remains as the $M = I$ Euclidean specialization.

Formal Expression.

Preconditions.

Let $\xi(t) \in \mathbb{R}^n$ evolve under $\dot{\xi} = -F(\xi, t) + w(t)$ where F is C^1 in ξ and continuous in t. Let $M : \mathbb{R}^n \times \mathbb{R}_+ \to \mathbb{S}^n_{++}$ be a smooth symmetric positive-definite matrix-valued function with uniform conditioning:

Template preconditions (contraction-template).

$$m_1 I \preceq M(\xi, t) \preceq m_2 I \quad \text{for all } \xi \in \mathcal{B}_R,\ t \geq 0 \tag{M0}$$

with constants $0 < m_1 \leq m_2 < \infty$.

(CT1) Zero correction at zero state. $F(0, t) = 0$ for all t.

(CT2) Local differential-contraction condition. There exist $\lambda > 0$ and $R > 0$ such that for all $\xi \in \mathcal{B}_R$, $t \geq 0$:

$$\dot{M}(\xi, t) + M(\xi, t)\frac{\partial F}{\partial \xi}(\xi, t) + \left(\frac{\partial F}{\partial \xi}(\xi, t)\right)^T M(\xi, t) \succeq 2\lambda M(\xi, t). \tag{CT2}$$

(CT3) Bounded disturbance. Either Model D ($\|w(t)\| \leq \rho_\xi$) or Model S ($w(t)$ Wiener-process increment with $\mathbb{E}[\|w(t)\|^2] = \sigma_\xi^2$).

Ultimate bound — Model D.

Under the preconditions with $V(\xi,t) = \xi^T M(\xi,t)\xi$, the state is ultimately bounded:

Result (contraction-template-D), conditional on (M0), (CT1)–(CT3-D).

$$\limsup_{t\to\infty}\|\xi(t)\| \le \frac{\rho_\xi}{\lambda}\sqrt{\frac{m_2}{m_1}}. \tag{CT-D}$$

Structural persistence (the ultimate bound fits within the contraction region $\mathcal{B}_R$) requires

$$\lambda R\sqrt{m_1/m_2} > \rho_\xi. \tag{CT-D-persist}$$

Proof sketch. Compute $\dot V = \xi^T \dot M\xi - 2\xi^T MF + 2\xi^T Mw$. Integrate (CT2) along the ray $s\xi$, $s \in [0,1]$, using $F(0) = 0$: $2\xi^T MF(\xi,t) \ge 2\lambda\,\xi^T M\xi - \xi^T\dot M\xi$. Substituting: $\dot V \le -2\lambda V + 2\xi^T Mw$. Cauchy-Schwarz in M-inner product + (M0) gives the affine-contraction bound on $W = \sqrt{V}$; conversion to Euclidean norm via (M0) yields (CT-D). □

Ultimate bound — Model S (with Itô correction).

Under stochastic disturbance $d\xi = -F(\xi,t)\,dt + \sigma_\xi\,dW_t$ and a metric M satisfying the Itô-correction bound $\frac{1}{2}\sigma_\xi^2\,\mathrm{tr}(M+$ Hessian correction in drift direction$) \le c_{\text{Itô}}\,m_2$ locally (automatic for state-independent M; bounded by M's Hessian otherwise), the stopped process satisfies

Result (contraction-template-S), conditional on (M0), (CT1)–(CT3-S), Itô-compatible metric.

$$\mathbb{E}[V(\xi(t\wedge\tau_R), t\wedge\tau_R)] \le V(\xi(0),0)\,e^{-2\lambda t} + \frac{n\sigma_\xi^2 c_{\text{Itô}} m_2}{2\lambda}, \tag{.1}$$

where $\tau_R = \inf\{t : \|\xi(t)\| > R\}$. Mean-square structural persistence requires

$$\lambda R^2 > \frac{n\sigma_\xi^2 c_{\text{Itô}}\, m_2^2}{2m_1}. \tag{CT-S-persist}$$

This is the natural extension of `#deriv-sector-condition`'s Prop A.1S region condition (Khasminskii 2012 ch. 5) to weighted metrics. When $M = I$ (Euclidean), $c_{\text{Itô}} = 1$, and the result reduces to Prop A.1S.

Recovery of `#result-sector-persistence-template`. When $M \equiv I$ (Euclidean metric, time-independent), (CT2) reduces to

$$-\frac{\partial F}{\partial\xi} - \Big(\frac{\partial F}{\partial\xi}\Big)^T \preceq -2\lambda I, \tag{.2}$$

which is the **incremental Euclidean sector condition** (strong monotonicity of F in Euclidean norm, equivalently **DA2'-inc** in `#form-composition-closure`'s bridge-lemma language). This implies the one-point sector condition $\xi^T F(\xi) \ge \lambda\|\xi\|^2$ by integration along the ray $s\xi$, so $\alpha = \lambda$ when $M = I$. The template's ultimate bound ρ_ξ/α is reproduced with $m_1 = m_2 = 1$.

The structural observation: (CT2) at $M = I$ is strictly stronger than (T2) — it requires the Jacobian's symmetric part to be uniformly positive definite everywhere, matching DA2'-inc rather than (T2). The contraction formulation therefore **collapses (T2) and DA2'-inc into one condition** at the cost of requiring differential (Jacobian-level) rather than integral (inner-product-at-a-point) information. This is one of the template's structural contributions: the same condition that runs the template runs the bridge-lemma that `#form-composition-closure` depends on, so the gap between single-agent persistence and composite tracking faithfulness closes when (CT2) holds.

Sub-scope metric-α_1 / metric-α_2 / metric-β partition. The contraction formulation refines `#deriv-sector-condition`'s A2' sub-scope partition into three tiers:

Sub-scope metric-α_1 (Euclidean metric, AAT-internally derived via DA2'-inc ≡ (CT2) with M=I). Scalar Kalman, Euclidean strongly-convex gradient, L2-regularized convex, linear-PD-symmetric. These cases carry (CT2) with $M = I$ which is exactly DA2'-inc — a condition `#form-composition-closure` has required at the composite level all along. AAT's existing commitment already forces (CT2)-Euclidean at these cases; the template surfaces it.

Sub-scope metric-α_2 (non-Euclidean metric, derived under explicit conditions). Five cases lift here:

- *Matrix Kalman under information metric* $M = (P^-)^{-1}$: contraction rate $\lambda = \lambda_{\min}(H^T R^{-1} H)/2$ on the observable subspace; Euclidean $\kappa(P^-)$ degradation removed. Under (PI)/Čencov (see `#der-gain-sector-bridge` "Fisher-metric cases under parameterization-invariance"), the information-metric form is *uniquely forced* — AAT-internally derived rather than theorem-imported.
- *Exponential-family natural parameters under Fisher metric* $M = \mathbf{I}(\theta)$: contraction rate $\lambda = \eta$ globally on the interior of the natural-parameter domain (Fisher-conditioning degradation removed). Also AAT-internally forced under (PI)/Čencov.
- *Hessian-metric strongly-convex* $M = \nabla^2 L$: contraction rate $\lambda = \eta$ under bounded metric-derivative in the drift direction. Theorem-imported (Hessian metric chosen to match loss curvature; no AAT-internal axiom forces the specific coordinate).
- *Linear-Hurwitz-non-symmetric under Lyapunov metric*: M solves $MA + A^T M = -Q$ for $Q \succ 0$. Contraction rate $\lambda = \lambda_{\min}(Q)/(2\lambda_{\max}(M))$ in the M-metric. **New coverage:** asymmetric-stable linear corrections where Euclidean A2' fails. Theorem-imported (Lyapunov equation construction is standard; no AAT-internal axiom forces Q).
- *PID with bounded plant nonlinearity under Lyapunov metric*: derived under plant-Lipschitz bound $L_p < \lambda_{\min}(Q)/(2\lambda_{\max}(M))$. **Promotion from sub-scope β.** Theorem-imported for the Lyapunov-metric construction; specializations include SPR-tuned PID (phase margin as sector constant; see `#der-gain-sector-bridge` Verified Instances).

Sub-scope metric-β (contraction-metric formulation fails). Four cases:

- *Variational / amortized / approximate posteriors*: contraction to the *projected* target (best-in-class q^*) is derivable, but the projection error $\mathrm{KL}[q^* \| p_t]$ to the true posterior is a residual disturbance that contraction machinery cannot address.
- *Rule-based / symbolic / discontinuous updates*: non-smooth F; (CT2) requires C^1. Piecewise-smooth extensions (Di Bernardo et al. 2014) cover switched systems but not general rule-based reasoning.
- *Severely misspecified agents*: contraction to a wrong target is still wrong. Metric choice is silent on target validity.
- *Per-step SGD / human judgment*: noise-is-disturbance treatment identical to the Euclidean formulation; no improvement from metric choice.

This is **the seventh ladder** in `#disc-separability-pattern` (A2'-scope): metric-α_1 separable-core / metric-α_2 structured-repair / metric-β general-open.

Compositional contraction.

When sub-agents individually satisfy the preconditions with metrics M_i and rates λ_i, the composite satisfies (CT1)–(CT3) under specific topologies:

Result (compositional-contraction), conditional on Slotine 2003 Thm 1–3 + per-sub-agent (CT1)–(CT3).

(CC-parallel) Parallel composition. $\dot{x} = (\dot{x}_1, \dot{x}_2)^T$ with independent ($\dot{x}_i = -F_i(x_i) + w_i$). The composite contracts under $M_c = \text{blockdiag}(M_1, M_2)$ at rate $\lambda_c = \min(\lambda_1, \lambda_2)$. Recovers `#der-team-persistence`'s weakest-link bound; cooperative coupling improves effective disturbance via the existing signed-coupling mechanism.

(CC-cascade) Hierarchical / lower-triangular cascade. If $\dot{x}_1 = f_1(x_1)$ contracts at λ_1 and $\dot{x}_2 = f_2(x_1, x_2)$ contracts in x_2 at λ_2 uniformly in x_1 with coupling gain $\|\partial f_2/\partial x_1\| \le k$, the cascade contracts. Rate bounded below by $\min(\lambda_1, \lambda_2)$ up to coupling-gain-dependent adjustment.

(CC-feedback) Negative feedback with bounded loop gain (Slotine 2003 Thm 3). Two systems with rates λ_1, λ_2 connected by negative feedback with loop gains k_{12}, k_{21} — the closed loop contracts iff

$$k_{12}k_{21} < 4\lambda_1\lambda_2. \tag{CC-feedback}$$

Heterogeneous critical-mass — (CM2-M).

Specializing (CC-feedback) to the signed-coupling structure of `#deriv-critical-mass-composition` for heterogeneous Tier-1M sub-agents with metric-contraction rates λ_1, λ_2, coordination costs C_1, C_2, and feedback loop gains k_{12}, k_{21}:

Derived (CM2-M), from (CC-feedback) + '#deriv-critical-mass-composition' signed-coupling structure.

$$(\lambda_1 - C_1)(\lambda_2 - C_2) > k_{12}k_{21}/4. \tag{CM2-M}$$

Specializing further to the matched-symmetric case ($\lambda_1 = \lambda_2 = \lambda, C_1 = C_2 = C, k_{12} = k_{21} = k$): $(\lambda - C)^2 > k^2/4 \iff \lambda - C > k/2$, which matches `#deriv-critical-mass-composition`'s (CM2) with $k = 2\gamma\mathcal{T}/R$ (up to normalization). **Heterogeneous-architecture composites now have a closed-form critical-mass inequality** where the matched-symmetric case was the only closed form previously available; `#deriv-critical-mass-composition`'s §6.1 obstruction on heterogeneous composites is thereby closed for the Tier-1M case.

Epistemic Status.

Conditional. Conditional on (M0) + (CT1)–(CT3) and the specific metric M chosen for each sub-scope α entry. The (CT-D) and (CT-S) ultimate-bound results are standard Lohmiller-Slotine theory (Lohmiller & Slotine 1998; Slotine 2003) specialized to AAT's per-instance state-variable choices; the proof technique is integration along the ray $s\xi$ followed by affine contraction. The contraction-metric primitive (Riemannian M) is standard in nonlinear control theory (Lohmiller-Slotine 1998; Forni & Sepulchre 2014 for the Finsler extension); AAT's contribution is the AAT-internal mapping of M-choices to sub-scope α_1/α_2 and the recognition that the compositional theorems (CC-parallel / CC-cascade / CC-feedback / CM2-M) directly generalize Section III's existing closure machinery.

The Jacobian-level observation. `#form-composition-closure`'s DA2'-inc condition is equivalent to (CT2) with $M = I$ for C^1 F on convex domains. AAT has been carrying the Jacobian-level Euclidean contraction condition at the composite level all along under the name DA2'-inc; the contraction template's Euclidean specialization surfaces this at the single-agent level.

The (PI)/Čencov observation. Two metric-α_2 cases (information-metric Kalman, Fisher-metric exp-family) are AAT-internally forced under the (PI) axiom in `#scope-agent-identity` combined with Čencov's 1982 uniqueness theorem

— the fourth primary instance of the meta-pattern in `#disc-additive-coordinate-forcing`. The Euclidean metric-α_1 case is **AAT-internally derived via DA2'-inc equivalence** rather than theorem-imported, structurally connecting it to the bridge lemma in `#form-composition-closure`. The remaining metric-α_2 cases (Hessian, Lyapunov-linear-Hurwitz, Lyapunov-PID) are theorem-imported: the metric choice is problem-specific (matched to loss curvature or plant structure) rather than forced by an AAT-internal axiom.

Max attainable: *conditional*. The condition (the metric specification must be consistent with the system structure; the Slotine 2003 compositional theorems must apply to the topology) is inherent to the contraction-metric framework, not removable. Non-smooth systems (rule-based), strategic/game-theoretic equilibria, and systems without regular metric-state coupling remain honestly outside the template's scope (see §Honest failure modes).

Relationship to monotone-operator theory. The contraction-metric condition (CT2) is a specialization of monotone-operator theory's strong-monotonicity condition in a Hilbert space with a specific inner product structure (Rockafellar 1970; Bauschke & Combettes 2017 §§4, 22–28). AAT's `#deriv-sector-condition` reading of A2' α/β as operator-family classification — proximal / firmly nonexpansive / cocoercive / strongly-monotone-gradient / linear-PD — identifies the same mathematical content in operator-theoretic language; see that segment's Grounding paragraph for the correspondence. The present template does not claim the contraction-metric machinery as distinctively AAT — it claims the specific instantiation to AAT's state-variable choices and the composition with identifiability-floor + composition-scope-condition content as AAT-distinctive.

Honest failure modes.

The contraction-metric formulation does **not** lift AAT's coverage in five cases:

1. **Non-smooth dynamics.** Rule-based agents, state-machine controllers, threshold-triggered switches, and discontinuous policies fail the C^1 requirement. Piecewise-contraction extensions exist (Di Bernardo, Liuzza & Russo 2014, *SIAM J. Control Optim.* 52) but apply to specific switched structures, not general rule-based reasoning. This barrier is structural, not a gap in the metric machinery.

2. **Strategic / adversarial equilibria — three independent convergent obstructions.** The adversarial / strategic half of Section III is structurally outside contraction-metric scope for three distinct reasons that converge on the same conclusion; the convergence matters because it shows the limit is not an accident of one particular framework choice:

   (i) *Compositional-contraction applicability* (Slotine 2003 §III): compositional contraction theorems require each sub-system to be individually contracting *and* for the composition topology to preserve contraction. Two-agent strategic dynamics with opposing objectives yield **saddle-point equilibria of the joint best-response dynamics** — attracting in some directions and repelling in others. Contraction requires attracting fixed points; saddle points break the framework directly.

   (ii) *Passivity universality* (Khalil 2002 ch. 6 Thm 6.4): the dissipation inequality $\dot{S} \leq s(u, y)$ holds for *all* inputs u by the passivity property's universal quantifier. A strategic opponent can select u to maximize the recipient's storage-function growth rather than respecting the port structure. Passivity delivers cooperative/neutral robustness, not adversarial.

   (iii) *Last-iterate convergence failure* (Daskalakis, Ilyas, Syrgkanis & Zeng 2018 "Training GANs with optimism" *ICLR*): adversarial gradient-ascent-descent dynamics (fictitious play, no-regret dynamics) converge to Nash equilibria *in average-iterate* sense (Cesàro-mean) but generically **fail last-iterate convergence** for non-zero-sum games. No contraction metric exists for systems whose last iterates do not converge to an attractor.

   The three obstructions come from different theorem families (compositional-contraction / dissipativity-universality / differential-Lyapunov-attractor-theory) but converge on the same verdict: **contraction metrics cover the cooperative half of Section III** (`#der-team-persistence`, cooperative `#deriv-critical-mass-composition`, cooperative `#form-composition-closure`). The adversarial / strategic half belongs to `#deriv-strategic-composition` (equilibrium convergence via Monderer-Shapley / Rosen) and `#der-adversarial-destabilization` (asymmetric attacker-target dynamics). This is a scope boundary that AAT correctly honors, not a gap to be closed — the apparatus that handles adversarial regimes is equilibrium-theoretic, structurally different from Lyapunov-contraction machinery.

3. **State-dependent metric stochastic coupling.** Pham & Slotine 2007 extends contraction to stochastic systems under small-noise bounds; the Itô-correction $c_{\text{Itô}}$ in (CT-S) is manageable for state-independent or slow-varying metrics. Strongly state-dependent M (adaptive-metric algorithms where M is learned alongside ξ) introduces cross-coupling that can destabilize contraction — the same structural issue `#deriv-adaptive-gain-dynamics` handles with its (MG-4) meta-gain coupling-boundedness condition.

4. **Multi-basin dynamics.** Lohmiller-Slotine contraction is local-on-a-region. Agents with multiple basins of attraction admit only basin-local metrics, giving a basin-chart structure rather than a single global metric. The basin-crossing boundary is exactly the `#result-structural-adaptation-necessity` trigger; the metric formulation makes the basin structure explicit but does not escape it.

5. **Identifiability-floor intersection.** The metric formulation operates downstream of identification. It assumes the correction function F points in a valid direction. Under the floors in `#disc-identifiability-floor` (on-policy L0-insufficiency no-go; L1' unobservable-C Cramér-Rao floor; composition-layer Instance 3 Liberzon no-go), the correction direction is structurally unavailable. Contraction metrics cannot recover identification — the two architectural moves are orthogonal.

Discussion.

Why factor this out. The Euclidean sector inequality in `#result-sector-persistence-template` (T2) matches a quadratic Lyapunov in Euclidean norm. For several AAT-relevant agent classes, the natural Lyapunov lives in a non-Euclidean metric (Fisher for statistical-manifold learning; Mahalanobis for Kalman; Hessian-induced for strongly-convex optimization; Lyapunov-equation-determined for asymmetric-stable linear systems). Stating AAT's persistence machinery once in parameter-free metric-formulation form makes visible that the sector condition and the contraction-metric condition are two states of the same apparatus — Euclidean and non-Euclidean — with the specific metric instantiated per sub-scope entry.

This is the *interior facet* of the stability certificate (W) in the non-Euclidean metric: the contraction-metric M is the certificate, and which metric is available per sub-scope is the certificate-strength ladder (R0 widest / R1 cocoercive / R2 Čencov-forced) of D. The five non-Euclidean metric cases are five certificates of one object, not five separate conditions.

The compositional payoff. The topology-indexed closures (CC-parallel / CC-cascade / CC-feedback) extend to heterogeneous-architecture composites via (CM2-M), closing the §6.1 heterogeneous-architecture obstruction in `#deriv-critical-mass-composition`. This is Section III's single largest coverage expansion: from "closed form for symmetric-matched-Tier-1" to "closed form under any of four topology classes with per-sub-agent (CT1)–(CT3) verification."

Relationship to `#result-sector-persistence-template`. The two segments coexist. `#result-sector-persistence-template`'s Euclidean formulation remains the default for agents whose natural coordinate is already Euclidean; `#result-contraction-template` is invoked when a non-Euclidean metric is natural (matrix Kalman, exp-family, ill-conditioned gradient, asymmetric-stable linear, PID-with-bounded-plant-nonlinearity). Every `#result-sector-persistence-template` invocation can be re-read as a `#result-contraction-template` invocation at $M = I$; the inverse mapping goes from metric to Euclidean with a $\kappa(M)$ degradation. Which direction is natural depends on the instance.

Relationship to `#form-composition-closure`'s bridge lemma. The bridge lemma's DA2'-inc condition is equivalent to (CT2) with $M = I$ on convex C^1 domains. `#result-contraction-template` therefore sharpens the bridge-lemma understanding: Tier 1 (DA2'-inc globally) ≡ metric-contracting-globally at $M = I$; Tier 2 (DA2'-inc locally with $\kappa(D\hat{o})^2$ degradation) ≡ metric-contracting-locally with a specific local metric; Tier 3 (per-domain verification) ≡ no globally valid metric exists. The tier structure of the bridge lemma carries over to `#result-contraction-template` as Tier 1M / 2M / 3M respectively.

Relationship to `#disc-identifiability-floor` Instance 3 (composition-layer no-go). Instance 3's §3.3 construction uses matched-symmetric-Tier-1 scalar systems; the coupling-sign bit is unidentifiable from component marginals. Escape route (E-d) in that Instance names "shared metric structure (common contraction metric; Lohmiller-Slotine 1998)" as an adjacent machinery the no-go motivates; `#result-contraction-template` is the segment that machinery lives in. The Instance 3 escape is therefore operationalized by a `#result-contraction-template` invocation with a composite metric constructed from sub-agent metrics per the topology (parallel / cascade / feedback).

Relationship to meta-patterns.

- `#disc-separability-pattern`*:* metric-α_1 / metric-α_2 / metric-β is the seventh ladder (A2'-scope), closing the "binary special case" gap that segment's Discussion flagged. Separable-core (metric-α_1, Euclidean, AAT-internally derived via DA2'-inc ≡ (CT2)-M=I) / structured-repair (metric-α_2, non-Euclidean, derived under explicit conditions including (PI)/Čencov for Fisher-metric cases) / general-open (metric-β, metric formulation fails).
- `#disc-identifiability-floor`*:* orthogonal axis. Contraction metrics lift the *geometry-of-correction* half of sub-scope α; identifiability-floor operates on the *what-correction-to-make* half. The two architectural moves compose (Instance 3 at composition layer + metric-Tier structure at single-agent layer) but do not interact.
- `#disc-additive-coordinate-forcing`*:* the (PI)/Čencov axiom that grounds two of five metric-α_2 cases is the fourth primary instance of the meta-pattern, adding a Čencov-invariance machinery alongside the three Cauchy-FE instances. See that segment's Four Instances tables.

Findings.

Topology-Indexed Compositional Closures via Contraction-Metric Generalization. **Brief:** AAT's central machinery for proving an adaptive system holds its mismatch bounded was originally written in the simplest geometric setting (the equivalent of measuring distance with a ruler in flat space). Many real systems have a more natural geometry — a Kalman filter "thinks" in terms of an information ellipsoid, a learning agent thinks in Fisher-information geometry, a controller designed for an asymmetric plant thinks in a different metric still. This result restates the persistence machinery in a way that uses each system's natural geometry, lifting five additional agent classes from sub-scope β (assumed) to sub-scope α_1/α_2 (derived under explicit conditions), and inheriting Slotine 2003's topology-indexed compositional closures (parallel / cascade / negative-feedback) at the composition layer.

Impact: Section III's largest single coverage expansion: from "closed-form composite persistence under matched-symmetric Tier-1" to "closed-form composite persistence under any of four topology classes with per-sub-agent contraction verification, including heterogeneous-architecture mixes via (CM2-M)." For Section II, the structural-transparency observation that DA2'-inc ≡ (CT2)-at-$M = I$ shows AAT's existing Euclidean machinery was already carrying the Jacobian-level contraction commitment under another name; what changes is making that commitment legible and lifting non-Euclidean cases (Fisher-metric Kalman, exp-family natural-parameter, ill-conditioned strongly-convex gradient, asymmetric-stable linear, PID with bounded plant) from theorem-imported to AAT-internally derived under named axioms. Three independent obstructions (Slotine 2003 saddle-point; passivity-universality; Daskalakis 2018 last-iterate non-convergence) bound the template's scope honestly — adversarial composition is structurally outside contraction analysis and properly handled by `#deriv-strategic-composition`'s equilibrium-convergence machinery.

Novelty Claim: *Claim synthesis* on contraction-metric machinery + AAT's sub-scope partition + (PI)/Čencov axiom. The contraction-metric primitive itself is standard nonlinear control theory; the topology-indexed closures are Slotine 2003. The synthesis is the AAT-internal mapping of metric choices to per-instance state-variable choices, the recognition that the Euclidean specialization is structurally the bridge-lemma's DA2'-inc, and the (PI)/Čencov axiom forcing two of the metric-α_2 cases to be AAT-internally derived rather than theorem-imported.

Related Work:

Table .1

ASF concern	*Prior-art language*	*Relationship / Positioning*
Differential-contraction primitive on Riemannian metrics	Lohmiller & Slotine 1998, "On Contraction Analysis for Non-linear Systems" *Automatica* 34:683–696 (published 1998, found pre-2026 via `spikes/spike-contraction-metric-generalization.md`)	*formal antecedent* — supplies the (CT1)–(CT3) preconditions and the (CT-D) ultimate-bound argument adopted directly
Compositional contraction under topology	Slotine 2003, "Modular stability tools for distributed computation and control" *Int. J. Adapt. Control Signal Process.* 17:397–416 (published 2003, found pre-2026) — Theorems 1–3 (parallel; cascade; negative-feedback small-gain)	*formal antecedent* — supplies (CC-parallel) / (CC-cascade) / (CC-feedback) and the heterogeneous (CM2-M) extension. AAT adopts these directly, applied to the composition-validity question of `#form-composition-closure`
Finsler (direction-dependent) generalization	Forni & Sepulchre 2014, "A Differential Lyapunov Framework for Contraction Analysis" *IEEE TAC* 59 (published 2014, found 2026-04)	*adjacent literature* — Finsler-metric extension; flagged in Working Notes for highly anisotropic agents; not currently load-bearing
Strong monotonicity / firmly-nonexpansive operators	Rockafellar 1970, "On the maximality of sums of nonlinear monotone operators" *Trans. AMS*; Bauschke & Combettes 2017, *Convex Analysis and Monotone Operator Theory in Hilbert Spaces* §§4, 22–28	*formal antecedent* — (CT2) at $M = I$ is strong monotonicity in Hilbert space; the Rockafellar-Wets / Nesterov equivalence (DA2'-inc ⇔ Jacobian-symmetric-part PD ⇔ (CT2)-at-$M = I$) is the standard monotone-operator identity AAT uses to make the Euclidean metric-α_1 case AAT-internally derived rather than theorem-imported
Fisher-metric / information-geometry uniqueness	Čencov 1982, *Statistical Decision Rules and Optimal Inference* AMS Translations 53	*formal antecedent* — under the (PI) axiom in `#scope-agent-identity`, Čencov's theorem forces the Fisher-metric Kalman and exp-family natural-parameter cases AAT-internally
Bounded-loss composition under predictive compression	Subramanian, Sinha, Seraj & Mahajan 2020, "Approximate information state for approximate planning and reinforcement learning in partially observed systems" *arXiv:2010.08843* (published 2020, found 2026-04)	*adjacent literature* — closest mathematical neighbor to the bridge-lemma side of `#form-composition-closure`. The contraction-metric machinery in this segment supplies the (CT2) condition that, at $M = I$, *is* the bridge-lemma's DA2'-inc; Sub20's predictive-loss-to-control-error result lives in the same shape but at a coarser granularity (single-agent compressed control, not composite agent boundary)
Influence-based abstraction loss bounds	Congeduti, Mey & Oliehoek 2020, "Loss Bounds for Approximate Influence-Based Abstraction" *arXiv:2011.01788* (published 2020, found 2026-04)	*adjacent literature* — bounds value loss from approximate cross-agent influence; the (CC-feedback) topology with bounded loop gain is the contraction-metric counterpart at the dynamics layer
Saddle-point / adversarial limits of contraction	Daskalakis, Ilyas, Syrgkanis & Zeng 2018, "Training GANs with optimism" *ICLR* (published 2018, found 2026-04 via `spikes/spike-contraction-metric-generalization.md`); Khalil 2002 *Nonlinear Systems* 3rd ed. ch. 6 Thm 6.4 (passivity-universality)	*contradicted by this finding* (in the sense of correctly bounding scope) — three independent obstructions to handling adversarial regimes converge: Slotine 2003 §III saddle-point limitation; passivity-universality's universal quantifier; Daskalakis 2018 last-iterate non-convergence. The convergence demonstrates the scope boundary is not an artifact of one framework; adversarial composition is structurally outside contraction analysis

ASF concern	*Prior-art language*	*Relationship / Positioning*
Switched-systems common-Lyapunov nonexistence	Liberzon 2003, *Switching in Systems and Control* Theorem 2.1	*formal antecedent* — supplies `#disc-identifiability-floor` Instance 3's no-go; the contraction-metric construction operationalizes escape route (d) (common contraction metric) for that Instance

Search Log:

- 2026-04 (*nominally comprehensive*, via `ref/Novelty_defense_and_integration.md` Pillar 2): The composition-layer extension of this segment (the (CC-*) closures and (CM2-M)) was reviewed under the same Undermind pass that covered `#form-composition-closure`. Verdict: the contraction-metric machinery itself is standard adopted prior art (formal antecedent); the AAT-internal mapping of metric choices to per-instance sub-scope α_1/α_2 entries and the use of (PI)/Čencov to force two metric-α_2 cases AAT-internally are the synthesis-level contributions. The Sub20 / Con20 positioning at the composition-validity layer is recorded both here and at `#form-composition-closure` since the contraction template is what supplies the formal mechanism the bridge lemma uses.
- 2026-04 (*targeted*, via `spikes/spike-contraction-metric-generalization.md`): Lohmiller-Slotine 1998 / Slotine 2003 / Forni-Sepulchre 2014 known and cited from the spike's literature review. The three adversarial obstructions (Slotine 2003 §III; Khalil 2002 ch. 6; Daskalakis et al. 2018) identified in the spike as convergent scope-boundary evidence.
- 2026-04 (*intuition-only*, prior): expected the contraction-metric primitive to be standard control-theory; expected the composition theorems to be the more proprietary content. Comprehensive search confirmed both: the primitive is well-established, and the composition theorems are Slotine 2003's, with AAT's contribution lying in the application to `#form-composition-closure`'s admissibility-and-bridge-lemma framework.

Working Notes.

- **Landing context.** The segment is the primary result of `spikes/spike-contraction-metric-generalization.md` (2026-04-23 Gap A/B cycle). The Jacobian-level observation (DA2'-inc ≡ (CT2)-M=I) and the (PI)/Čencov observation derive from `spikes/spike-jacobian-b1-strengthening.md` (2026-04-23 follow-up).
- **Heterogeneous general-graph composition.** Slotine 2003 §IV's virtual-system technique covers general directed graphs under a small-gain-along-every-cycle condition. Operationalizing this for `#form-composition-closure`'s general multi-agent setting — including identifying which cycle-gains to measure in practice — is a follow-up derivation. (CM2-M) in this segment handles 2-agent negative-feedback; $N \geq 3$ with general topology remains open but structurally clear.
- **Jacobian-level B1 strengthening is open.** The (CT2) condition on F's Jacobian is a Jacobian-level B1 analog. Whether the integral B1 of `#der-gain-sector-bridge` can be *strengthened* to a Jacobian-level B1* that implies (CT2) AAT-internally (rather than via theorem-import from Lohmiller-Slotine) was attacked in `spikes/spike-jacobian-b1-strengthening.md`. Three of the five metric-α_2 cases (Hessian, Lyapunov-linear-Hurwitz, Lyapunov-PID) remain theorem-imported after that spike; two (Fisher-metric-Kalman, Fisher-metric-exp-family) lift AAT-internally under (PI)/Čencov. The remaining gap is the three theorem-imported cases.
- **Candidate strong-option promotion.** Under a *heredity* axiom (composite admissibility derivable from sub-agent properties — a legitimate AAT-internal strengthening of `#post-composition-consistency`), agent-level B1* is forced, metric-Tier structure collapses, and (CM2-M) promotes from Slotine-imported to AAT-derived. Flagged in the Jacobian-level B1 spike's §11 strong option; promotion is an architectural decision requiring its own scoping.
- **Finsler metrics.** Forni & Sepulchre 2014 extends the differential Lyapunov framework to Finsler (direction-dependent) metrics. Likely useful only for highly anisotropic corrections (coordinate-asymmetric agents); not priority.

- **The (T2) / (CT2) structural observation matters for Section I too.** `#result-persistence-condition`'s linear special case ($F = \mathcal{T}\delta, \alpha = \mathcal{T}$, (T2) global) is trivially contraction-metric with $M = I$ and $\lambda = \alpha$. But non-linear `#result-persistence-condition` instances (nonlinear mismatch dynamics, gradient on strongly convex losses) would benefit from contraction-metric framing to remove the Euclidean-norm-specific condition-number penalty in the persistence constant. A cross-reference from `#result-persistence-condition` back to this segment may be warranted after the next promotion pass.

Appendix

Derivation: Variational Approximate-A2' (ε-Fidelity)

Variational / approximate-posterior agents (VI, amortized VI, active-inference-style variational free energy) currently sit in A2' sub-scope β per `#deriv-sector-condition`: their correction functions target the *best-in-class* variational posterior q^* rather than the true posterior p, and the approximation gap can rotate the correction direction enough to break B1 directional fidelity (4.2). Under a KL bound $\mathrm{KL}(q_\phi \| p) \leq \varepsilon$ on the variational approximation, directional fidelity recovers in a quantifiable form: **ε-fidelity B1**, with sector-constant degradation scaling as $O(\sqrt{\varepsilon})$ (Pinsker-tight). The sector-persistence template applies under a Regime-A / Regime-B decomposition — clean sector bound on an annulus away from the projection-error floor, approximation-dominated on a ball of radius $\delta_0 = O(\sqrt{\varepsilon})$ around the target. This promotes controlled-KL VI from sub-scope β to a new intermediate tier α' within the A2' partition (cf. the α / α_1 / α_2 / β refinements from `#deriv-adaptive-gain-dynamics` and `#deriv-fisher-whitened-update-rule`).

Formal Expression.

ε-fidelity B1.

Let the true posterior be $p(z \mid x)$ and the variational approximation $q_\phi(z \mid x)$ with $\mathrm{KL}(q_\phi \| p) \leq \varepsilon$. Under standard Lipschitz assumptions on the observation model and nested-support on the variational family, the total-variation distance bounds as $\|q_\phi - p\|_{TV} \leq \sqrt{\varepsilon/2}$ (Pinsker's inequality). Propagating this bound through the correction function via Cauchy-Schwarz:

Derived (epsilon-fidelity-B1, from Pinsker + Cauchy-Schwarz).

$$(K\hat{P} - P^*)^T(\hat{P} - P^*) \geq (c_{\min} - C_H\sqrt{2\varepsilon}/\|\delta\|) \cdot \|\delta\|^2 \tag{.1}$$

where C_H is a constant depending on the observation-model Lipschitz constant. The effective sector constant

$$c_\varepsilon(\|\delta\|) = c_{\min} - C_H\sqrt{2\varepsilon}/\|\delta\| \tag{.2}$$

is state-dependent: large-mismatch regions see near-full sector constant; near-target regions see degraded sector constant due to approximation error dominating.

Regime-A / Regime-B decomposition.

Define $\delta_0 = 2C_H\sqrt{2\varepsilon}/c_{\min}$ (the approximation-dominated radius around target). Then:

Formulation (regime-decomposition-variational).

- **Regime A — clean sector bound** ($\|\delta\| > 2\delta_0$). On the annulus $\mathcal{B}_R \setminus \mathcal{B}_{2\delta_0}$, $c_\varepsilon \geq c_{\min}/2$: sector condition holds with constant $c_{\min}/2$. The template's ultimate bound $\rho_\xi/(c_{\min}/2)$ applies.
- **Regime B — approximation-dominated floor** ($\|\delta\| \leq 2\delta_0$). Near target, c_ε can be arbitrarily small; the correction may not contract. The ultimate bound gains an additive $O(\sqrt{\varepsilon})$ term: $R_\varepsilon^* = \rho_\xi/(c_{\min}/2) + \delta_0 = \rho_\xi/(c_{\min}/2) + O(\sqrt{\varepsilon})$.

Sector-persistence template instantiates with: - State variable $\xi = \hat{P} - P^*$ (variational mismatch to target posterior). - Effective ultimate bound R_ε^* on Euclidean norm. - Persistence requires $R_\varepsilon^* < R$, i.e., $\rho_\xi/(c_{\min}/2) + O(\sqrt{\varepsilon}) < R$.

Khasminskii stopping-time localization (same technique as `#deriv-sector-condition` Prop A.1S and the A2' strengthening spike) applies to the annulus Regime A; Regime B is handled by accepting δ_0 as a projection-error floor.

Per-case verdicts.

The α'-membership depends on the specific variational scheme:

Derived (per-variational-case).

Natural-gradient VI with exponential-family q_ϕ. Khan & Lin 2017 (*Conjugate-computation variational inference*) showed natural-gradient VI is equivalent to closed-form conjugate-Bayesian updates for exponential-family variational distributions. This recovers *full* sub-scope α membership (not merely α'), with A2' derived rather than ε-degraded, by converting the variational update into a Bayesian update in a reparameterized family. The $\varepsilon = 0$ limit is exact.

Mean-field VI. When the true posterior is approximately factorized ($p(z) \approx \prod_i p_i(z_i)$), mean-field VI achieves small ε and is the workhorse α' case. Ultimate bound degrades by additive $O(\sqrt{\varepsilon})$; sector persistence holds.

Amortized VI (VAE-style). Amortization adds a second approximation error (the variational network's function-class limit). KL bounds compose *additively*: δ_0 grows as $\sqrt{\varepsilon_{\text{family}} + \varepsilon_{\text{amort}} + \varepsilon_{\text{generalization}}}$. Under controlled-ε amortized VI, α' membership holds with a larger floor.

Diffusion-posterior / energy-based with uncontrolled MCMC. No controlled ε bound; ε grows with mixing time. Stays firmly in sub-scope β.

Active inference (variational free energy). Conditional α' under exponential-family q + natural-gradient; ε-degraded α' otherwise. This does **not** force V-strong G-BP2 (presentation of AAT as control-theoretic specialization of active inference) — V-medium (KL-divergence-based cognitive cost) remains the appropriate scope commitment per `spikes/spike-active-inference-vs-aad.md`.

Sub-scope α' in the A2' partition.

The A2' sub-scope partition is structured as:

Derived (alpha-prime-partition).

- **Sub-scope $\boldsymbol{\alpha}$** (derived under B1 directional fidelity; `#der-gain-sector-bridge`): Kalman, conjugate Bayesian, exponential-family natural parameters, strongly-convex gradient, L2-regularized, linear-PD-symmetric.
- **Sub-scope $\boldsymbol{\alpha_1/\alpha_2}$** (`#result-contraction-template`): metric-formulation generalization of α.

- **Sub-scope α_3** (`#deriv-fisher-whitened-update-rule`): correlated evidence + Fisher-whitened under (PI)/Čencov.
- **Sub-scope α'** (this segment): controlled-KL VI under Pinsker's inequality with $O(\sqrt{\varepsilon})$ sector-constant degradation + Regime-A/B decomposition.
- **Sub-scope β**: uncontrolled-ε agents; non-smooth rule-based; severely misspecified; per-step SGD; human judgment.

This gives the full current picture: α, α_1, α_2, α_3, α', β.

Epistemic Status.

Conditional. Max attainable: *exact* for the Pinsker-bound derivation (standard inequality). The $O(\sqrt{\varepsilon})$ rate is tight in general — improving to $O(\varepsilon)$ requires stronger inequalities (log-Sobolev, Talagrand T2) holding for the specific variational family and target; not chased in this segment but is a potential refinement for concentration-preserving families.

Load-bearing: - Pinsker + Cauchy-Schwarz gives the $O(\sqrt{\varepsilon})$ rate rigorously. - Regime-A annulus sector bound with $c_{\min}/2$ is derived (straightforward bookkeeping on the state-dependent c_ε). - Natural-gradient VI promotion to full α is standard (Khan & Lin 2017). - Mean-field VI as workhorse α' is standard variational analysis.

Not established: - $O(\varepsilon)$ rate (would require stronger inequalities). - Quantitative ε bounds for specific variational families in practice (amortization gaps are typically not estimable tightly). - α' membership under mode-missing VI (multimodal posteriors where q concentrates on a subset of modes). These may require separate analysis.

Honest Failure Modes.

- **Uncontrolled-ε agents**: diffusion posteriors with finite MCMC; VAEs trained without KL control; any VI without a bounded-ε certificate. Stays in β.

  Scope honesty — variational-alpha-prime-limits.

- **Mode-missing on strongly multimodal posteriors**: q misses modes; ε is irreducible regardless of optimization. α' fails structurally.
- **Non-Lipschitz observation models**: the C_H constant diverges; the Pinsker propagation bound doesn't hold. α' fails.
- **Support mismatch**: q's support is not a subset of p's support (or vice versa); KL is $+\infty$. α' fails.
- **Non-exponential-family + non-amortized + uncontrolled + multimodal**: the worst case, strictly β.

Discussion.

Relationship to `#form-strategy-complexity-cost`. The G-BP2 V-medium variational form landed in `#form-strategy-complexity-cost` uses KL divergence in the cognitive-cost term. This segment provides the complementary story on the persistence side: KL-bounded VI has ε-fidelity B1 and α' sector condition. The two together characterize the cognitive/persistence tradeoff for variational agents: ε controls both the cognitive cost (how far from the target) and the persistence degradation (how much sector-constant penalty). Large ε means cheap-but-persistently-weak; small ε means expensive-but-persistently-sharp.

Relationship to `#disc-compression-operations`. Variational compression is the first of the four AAT compression operations; its α' sector structure gives a concrete persistence guarantee for variational representation of M_t. The other three operations (strategy Σ_t, shared-intent, coarse-graining Λ) may admit similar ε-fidelity analyses where a KL-bound on the compression is available.

Meta-pattern positioning. - `#disc-separability-pattern`: α' sits on the structured-repair tier of the A2'-scope ladder (7th ladder from `#result-contraction-template`), alongside metric-α_2 and α_3. Each represents "sector condition recovered under explicit additional structure." - `#disc-additive-coordinate-forcing`: variational persistence sits *outside* this meta-pattern — no logarithmic coordinate is forced; Pinsker is a bound, not a Cauchy-FE argument. α' is an adjacent family member (shape: controlled-approximation-with-quantified-degradation), not a primary instance.

Working Notes.

- Landing context: `spikes/spike-variational-a2prime.md` (2026-04-23 Gap A/B cycle).
- **Open: $O(\varepsilon)$ rate via Talagrand T2.** For concentration-preserving variational families with log-Sobolev or T2-Talagrand inequalities, the Pinsker bound can be sharpened to $O(\varepsilon)$ directly (without square-root). Not chased here; would extend α' to a tighter sub-tier for specific family classes.
- **Amortization gap bounds.** Generalization-gap analysis for VAEs (Rainforth et al. 2018; Nowozin 2018) gives function-class error bounds; composing with variational-family error gives the full δ_0 in amortized VI. Follow-up work.
- **Connection to natural-gradient VI.** Khan & Lin 2017's conjugate-computation framing makes natural-gradient VI full sub-scope α (not just α'). The α' tier is for *non-natural-gradient* VI under KL control.

Appendix

Derivation: L1' Update-Bias Formula

Under L1' correlated-evidence regimes with unobservable common cause, the default log-odds edge-update (P) converges to a **biased fixed point** — edges settle at log-odds values that match the L1' root-probability rather than the marginal-edge truth. This segment derives a closed-form bias formula for the two-edge OR-root case, verifies it via Monte Carlo, and composes it with `#disc-identifiability-floor` Instance 2 and `#schema-strategy-persistence`'s forgetting prerequisite to produce a **dual forgetting-rate requirement** on strategic persistence.

The bias is the quantitative companion to F13's structural-identifiability-floor result: F13 shows identification is *structurally impossible* from single-channel observation; this segment shows the resulting *magnitude drift* is bounded, Lipschitz in correlation strength ρ, and Cramér-Rao-floored under forgetting.

Formal Expression.

Closed-form bias at matched-marginal initial conditions (two-edge OR-root).

Let root node G have two direct causal parents e_1, e_2 via OR-aggregation under L1' with latent common cause C producing correlation ρ between the parent edges. Assume matched-marginal initial conditions: $\hat{\mu}_j^{\text{init}} = \mu_j^*$ for each edge's marginal probability. Under the default log-odds update with per-edge gain $\eta_{\text{edge}} = 1/(n_k + 1)$ and identifiability coefficient ι_k, the per-cycle bias on edge k's log-odds update satisfies

Derived (L1-bias-formula-OR-root).

$$B_k(\rho) = -\frac{\iota_k \cdot (1 - \mu_{\bar{k}}) \cdot \rho}{(n_k + 1) \cdot [(1 - \mu_1)^2 + (1 - \mu_2)^2]} \tag{.1}$$

where $\bar{k} \in \{1, 2\} \setminus \{k\}$ is the sibling edge. The bias is linear in ρ, decays as $1/(n_k + 1)$ with experience, topologically asymmetric via the Jacobian factor $(1 - \mu_{\bar{k}})/\|\mathbf{J}\|^2$.

Admissible region (precise). The denominator is the squared OR-aggregation Jacobian norm $\|\mathbf{J}\|^2 = (1 - \mu_1)^2 + (1 - \mu_2)^2$. The first-order perturbation is well-posed precisely when $\|\mathbf{J}\|$ is bounded below by some fixed $\epsilon_J > 0$ — equivalently, when at least one edge probability is bounded away from 1. This is the **Jacobian-norm admissibility condition** $\|\mathbf{J}\| \geq \epsilon_J$; it is *weaker* than requiring both μ_1, μ_2 to be bounded away from 1, and it is *sharper* than the qualitative "bounded away from determinism" reading: only the joint deterministic-success corner $\mu_1, \mu_2 \to 1$ together is excluded. As either edge alone approaches deterministic success the bias formula remains valid (the surviving sibling edge carries the Jacobian norm); only the diagonal corner where both edges saturate is excluded, and there the per-cycle update itself vanishes (observations carry no residual information), so "no learning, no bias" is the operationally correct behaviour and the formula's apparent singularity is degenerate rather than physical. Inside the admissible region $\|\mathbf{J}\| \geq \epsilon_J$, the closed-form predictions are quantitatively verified by Monte Carlo; at the diagonal corner the segment defers to `#deriv-edge-credence-dynamics`'s degenerate-update analysis. The Jacobian-norm condition is the natural admissibility statement because the bias formula is the projection of the update onto $\mathbf{J}/\|\mathbf{J}\|^2$, so well-posedness requires only that this projection be defined.

For AND-root aggregation, the bias has the same structural form with opposite sign. For mixed OR-AND sub-plans, the bias decomposes sub-plan-wise.

L1'-induced biased fixed point.

Under continued L1' evidence, the log-odds fixed point satisfies

Derived (biased-fixed-point-exists).

$$\lambda_k^{\text{L1'-FP}} = \lambda_k^* + \int_0^\infty e^{-\eta_{\text{edge}}\iota_k t} B_k(\rho)\, dt = \lambda_k^* - \rho \cdot (1-\mu_{\bar{k}}) / \left[(1-\mu_1)^2 + (1-\mu_2)^2\right] \tag{.2}$$

(for constant-ι approximation). Each edge's log-odds converges to a **wrong value** that matches the L1' root probability rather than the marginal truth. **Plan-level correctness is purchased at edge-level miscalibration**: the agent predicts plan outcomes correctly on-policy (the L1' mixture on the sibling matches the L1' marginal), but its per-edge credences are systematically biased.

This is the quantitative companion to F1's on-policy undetectability (`#disc-identifiability-floor` Instance 1) at the marginal level: the bias accumulates at rate $1/(n_k+1)$, converges to a bounded non-zero value, and is *not detectable from on-policy data* because the mixture's marginal is matched to the on-policy observations.

Observable-C zero-bias result.

Under Prop B.7's five-way-gating with C observable, conditioning on C decomposes the L1' mixture into two single-child-per-component problems:

Derived (observable-C-zero-bias).

$$B_k(\rho \mid C = c) = 0 \quad \text{in expectation at conditional truth, for each } c \in \{0,1\}. \tag{.3}$$

Prop B.7's decomposition *exactly eliminates the bias* — the five gating conditions (facilitator monotonicity; C-observability; $\iota > 0$ for common-cause edge; marginal sector condition; identifiability of per-component rates) are precisely the conditions under which conditional-on-C updates restore exactness. This makes Prop B.7's derivation the quantitative escape from the L1' bias.

Unobservable-C Cramér-Rao bias floor under forgetting.

When C is unobservable and the agent uses experience discounting at rate $\lambda \in [0,1)$ to maintain plasticity (`#schema-strategy-persistence`'s forgetting prerequisite), the steady-state bias is

Derived (bias-floor-under-forgetting).

$$B_k^{\text{SS}}(\rho,\lambda) = -\frac{\iota_k \cdot \rho \cdot (1-\mu_{\bar{k}})}{(1-\lambda)\cdot[(1-\mu_1)^2 + (1-\mu_2)^2]}. \tag{.4}$$

The Cramér-Rao floor from `#disc-identifiability-floor` Instance 2's rank-1 Fisher matrix translates directly: no unbiased estimator can reduce the bias below $\rho/(1-\lambda)\cdot$ [constant]. The bias is structurally present in any online estimator operating on single-channel observation of one child edge, not merely the specific default-signal estimator.

Dual forgetting-rate requirement.

`#schema-strategy-persistence` requires forgetting rate $(1-\lambda) > \rho_\Sigma / R_\Sigma$ for asymptotic sector-persistence. The L1' bias floor above imposes a **second** constraint: to keep the bias bounded relative to the sector reserve,

Derived (dual-forgetting-requirement).

$$(1-\lambda) > c_B \cdot \rho / R_\Sigma^{\text{bias}} \tag{.5}$$

where c_B is a topology-dependent constant from the Jacobian factor and R_Σ^{bias} is the tolerable bias-reserve within the sector region. The forgetting rate must satisfy **both**:

$$(1-\lambda) > \max\left(\frac{\rho_\Sigma}{R_\Sigma},\ \frac{c_B \cdot \rho}{R_\Sigma^{\text{bias}}}\right). \tag{.6}$$

When the admissibility window is empty (no forgetting rate satisfies both constraints simultaneously), no standard persistence regime works — the agent must augment (observable-C via instrumented regime indicators; multi-child joint observation; plan-level fallback) or accept biased-fixed-point operation.

Monte Carlo verification.

Numerical simulation (400 trials × 5000 cycles, four scenarios: OR-cooperative, OR-adversarial, AND-cooperative, AND-adversarial) confirms the closed-form predictions: - Sign of bias matches theoretical prediction in all scenarios. - Magnitude matches closed form within $< 5\%$ at $\rho \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. - Vanishing at $\rho = 0$ verified. - Jacobian-induced topological asymmetry verified: OR and AND roots produce opposite-sign biases of matching magnitude. - Initial-cycle rate $dB_k/dt\ |_{t=0}$ matches closed form quantitatively. - Logarithmic cumulative drift matches (biased-fixed-point convergence rate).

Empirical claim (monte-carlo-confirmation).

Full simulation parameters and results in `spikes/spike-l1-update-bias.md` §7.

Epistemic Status.

Conditional. Max attainable: *exact* for the two-edge OR-root / AND-root matched-marginal case. The closed-form bias formula is derived via first-order perturbation of the log-odds update around matched-marginal truth; the Jacobian factor is specific to the DAG topology (OR vs AND vs mixed). *Robust qualitative* for deeper DAG structures where the bias Decompose sub-plan-wise but closed-form coefficients depend on per-sub-plan Jacobians.

What is load-bearing: - The linear-in-ρ scaling is exact (first-order in small correlation). - The $1/(n_k+1)$ experience scaling is exact (follows from `#deriv-edge-update-natural-parameter`'s Beta-Bernoulli gain). - The observable-C zero-bias result under Prop B.7 gating is exact. - The dual forgetting-rate requirement is a first-order composition of two separate constraints.

What is not established: - Closed-form bias for non-matched-marginal initial conditions (transient bias during convergence). Numerical simulation works; analytic form is messier. - Higher-than-second-order DAG structures (L2-general). Scope boundary. - Full coupled bias formula under `#deriv-adaptive-gain-dynamics` meta-gain machinery. Open.

Discussion.

Composition with `#disc-identifiability-floor` Instance 2. F13 establishes the structural no-go for single-channel L1' mixture identification (Cramér-Rao rank-1); this segment gives the **quantitative numerical-floor** companion — the bias is bounded, continuous in ρ, and vanishes at $\rho = 0$. Instance 2 answers "is it identifiable?" (no, structurally); this segment answers "if we update anyway, how wrong is the result?" (bounded drift to a biased fixed point).

Composition with `#deriv-fisher-whitened-update-rule`. The Fisher-whitened update recovers sharp *directional* fidelity but does not reduce the *magnitude* bias — Fisher whitening operates downstream of identification, and Instance 2's obstruction is an identification obstruction. The full picture for the default signal function under L1': **direction preserved (angle ≤ 45° per `#deriv-fisher-whitened-update-rule`), magnitude biased by $B_k(\rho)$ with Cramér-Rao floor under forgetting (this segment), observable-*C* restores exactness (Prop B.7), undetectable on-policy (Instance 1 + Instance 2 composition)**. This is the honest quantitative account of what the default signal function does under Gemini's Gap A (validation under correlated failures).

Relationship to `#schema-strategy-persistence`'s forgetting. The segment's existing forgetting prerequisite (for asymptotic sector-persistence) becomes dual under L1': forgetting must *both* outpace sector disturbance *and* prevent bias accumulation. When the two constraints are compatible (admissibility window non-empty), the existing forgetting-rate requirement suffices with appropriate ρ-dependent tuning. When incompatible, augmentation is required — a structural scope narrowing that elevates observable-*C* instrumentation from convenience to prerequisite.

Connection to `#stability-induced-myopia`'s detection-latency. The detection-latency theorem ($\mathbb{E}[T_{\text{detect}}] = \Omega((n_{\min} + 1)/\varepsilon)$) assumes unbiased marginal tracking; under L1' bias, the pre-accumulated drift degrades effective ε for subsequent regime-change detection. High-operating-point agents with pre-accumulated L1' bias face a compounded myopia floor. Follow-on refinement possible in `#stability-induced-myopia`.

Working Notes.

- Landing context: `spikes/spike-l1-update-bias.md` (2026-04-23 Gap A/B cycle). Closed-form derivation + Monte Carlo verification in the spike.
- **Open: non-matched-marginal transient bias.** Closed-form for arbitrary initial conditions is messier but numerically tractable. A tighter analytic treatment would compose `#deriv-edge-update-natural-parameter`'s β-Bernoulli dynamics with the L1' mixture and likely requires Markov-chain convergence-rate analysis.
- **Open: *N*-edge common-cause extensions.** Three or more children of a single common cause produce a tensor of Jacobian factors. The structural scaling likely remains $O(\rho)$ per edge, but the exact coefficients need working out.

Appendix

Derivation: Fisher-Whitened Edge Update Under Correlated Evidence

Under L1'/L2 correlated-evidence regimes, the default log-odds edge-update (from `#deriv-edge-update-natural-parameter`) retains correct *direction* — the angle between log-odds gradient and natural gradient never exceeds 45° at finite correlation ρ, so B1 directional fidelity (4.2) is never actively violated — but its *magnitude alignment* degrades by a factor $\sqrt{1 - r^2}$ in the sector constant. The Fisher-whitened correction restores sharp B1 on the Fisher-weighted inner product. Under the (PI) parameterization-invariance axiom named in `#scope-agent-identity` and promoted to a primary instance of `#disc-additive-coordinate-forcing` via Čencov 1982, Fisher whitening is **AAT-internally derivable** rather than externally imported. The result adds sub-scope α_3 (correlated evidence + Fisher-whitened update + Bayesian coherence → A2' derived) to the A2' partition and composes cleanly with the meta-gain machinery of `#deriv-adaptive-gain-dynamics` (Fisher whitening is a special case of meta-gain with $K_t = \mathbf{I}^{-1}(\lambda_t)$).

Formal Expression.

Angle characterization under L1' correlation.

For a block-correlated two-edge evidential model (soft-facilitator Prop B.7 parameterization), the Fisher information matrix has off-diagonal entry

Derived (angle-bound-finite-correlation).

$$r = \frac{\theta_C(1 - \theta_C)\Delta_1\Delta_2}{\sigma_1\sigma_2} \in [-1, 1], \tag{.1}$$

where θ_C is the latent-cause probability, $\Delta_j = p_{j|C} - p_{j|\neg C}$ is the separability gap, and σ_j is the marginal-edge standard deviation. The Euclidean angle between the log-odds update direction and the natural-gradient direction (at single-edge plan residual) satisfies

$$\theta_{\text{LO-NG}} = \arccos\frac{1}{\sqrt{1 + r^2}} \leq 45^\circ \quad \text{for all } r \in [-1, 1]. \tag{.2}$$

The log-odds update **never flips sign** under any finite correlation — B1 is never actively violated, only degraded in magnitude. Under the operational claim that "the default signal function needs validation under correlated failures," the finding is: **the default direction is preserved; only magnitude alignment degrades**. The sector constant in B1 degrades by a factor $\sqrt{1 - r^2}$ under Euclidean B1; the Fisher-weighted B1 recovers the unperturbed sector constant.

Fisher-whitened update rule.

In log-odds coordinates $\lambda \in \mathbb{R}^{|E|}$ with Fisher information $\mathbf{I}(\lambda)$, the Fisher-whitened edge update is

Definition (Fisher-whitened-update).

$$T_{\mathrm{FW}}(\lambda) = \lambda - \eta_{\mathrm{edge}} \cdot \mathbf{I}^{-1}(\lambda) \cdot \mathbf{J} \cdot (\hat{P}_{\Sigma}(\sigma(\lambda)) - y_G) \tag{.3}$$

(compared to the Euclidean log-odds update $T_{\mathrm{edge}}(\lambda) = \lambda + \eta_{\mathrm{edge}} \cdot \mathrm{diag}(\iota) \cdot \mathbf{J}(y_G - \hat{P}_{\Sigma})/\|\mathbf{J}\|^2$). The Fisher-weighted inner product $\langle a, b \rangle_{\mathbf{I}} = a^T \mathbf{I}^{-1} b$ makes the update's directional fidelity invariant under reparameterization of the natural-parameter coordinate.

Two AAT-internal axiom paths.

Path A (B1-parameterization-invariance). Require B1 directional fidelity to be *parameterization-invariant* in the sense of `#scope-agent-identity`'s (PI) axiom: the theorems about sub-scope α derivation in `#der-gain-sector-bridge` should not depend on arbitrary coordinate choices for M_t's natural parameters. Under (PI), B1 sub-scope α partition is coordinate-invariant iff the inner product defining the directional-fidelity condition is the Fisher metric (Čencov 1982 uniqueness theorem under (PI); extended by Ay-Jost-Lê-Schwachhöfer 2017). The Fisher-weighted inner product is therefore *forced* by (PI), and the Fisher-whitened update is the AAT-internally derived correction direction for directional-fidelity preservation across parameterizations.

Derived (Fisher-whitening-from-B1-parameterization-invariance).

Path B (Lyapunov-coordinate-matching via adjacent-family classification). In the adjacent-family framing of `#disc-additive-coordinate-forcing`, the Lyapunov coordinate is *matched* (not forced) to the sector condition. For natural-gradient updates, the canonical Lyapunov is Fisher-weighted (Amari 1998, "Natural gradient works efficiently in learning," *Neural Computation* 10); this matches the geometry of the update operator. The two paths converge on the same Fisher-weighted result; Path A forces it via axiomatics, Path B confirms it via adjacent-family coordinate-matching.

Under L0 (no correlation), $r = 0$ and $\mathbf{I}$ is diagonal — Fisher whitening is vacuous (reduces to the existing Euclidean log-odds update). The axioms pick out Fisher whitening *uniquely* only under L1'/L2 (correlated evidence) regimes; they are vacuously satisfied under L0.

New sub-scope α_3.

Correlated-evidence + Fisher-whitened update + Bayesian coherence yields A2' *derived*:

Derived (sub-scope-alpha-3).

$$(T_{\mathrm{FW}}(\lambda) - \lambda^*)^T \mathbf{I}(\lambda^*)^{-1}(\lambda - \lambda^*) \geq \beta \|\lambda - \lambda^*\|_{\mathbf{I}}^2 \quad \text{with } \beta = \eta_{\mathrm{edge}}\mu \tag{.4}$$

on the Fisher-metric sector region around the fixed point $\lambda^* = \mathrm{logit}(\theta^*)$. The directional-fidelity proof of `#der-gain-sector-bridge` carries over with $\kappa(\mathbf{I})$ as Euclidean-transfer penalty — structurally identical to the Kalman case's $(P^-)^{-1}$-norm weighted-norm treatment. The Kalman case in `#der-gain-sector-bridge`'s Verified Instances is a special case of Fisher-weighted sector.

Sub-scope α_3 extends the refinement introduced by `#deriv-adaptive-gain-dynamics`: - **Sub-scope** α_1: fixed-gain, independent evidence (Euclidean B1 applies; the existing A2' partition). - **Sub-scope** α_2: adaptive-gain, independent evidence (meta-gain conditions (MG-1)–(MG-4) per `#deriv-adaptive-gain-dynamics`). - **Sub-scope** α_3: fixed-gain, correlated evidence (Fisher-whitened update; this segment).

Additional composition: adaptive-gain + correlated-evidence (sub-scope α_4) would compose `#deriv-adaptive-gain-dynamics` meta-gain machinery with Fisher whitening at the primary state; open.

Connection to meta-gain as special case.

The Fisher-whitened update is a meta-gain law in the sense of `#deriv-adaptive-gain-dynamics` with $K_t = \mathbf{I}^{-1}(\lambda_t)$ — a *degenerate* special case where the meta-gain is a deterministic function of the primary state rather than an independently learned variable. All four meta-gain conditions (MG-1)–(MG-4) are satisfied trivially: - (MG-1) Meta-gain sector: $\mathbf{I}^{-1}$ is symmetric-positive-definite on the interior of the natural-parameter domain. - (MG-2) Meta-gain bounded: $\|\mathbf{I}^{-1}\|$ bounded on compact natural-parameter subsets. - (MG-3) Smoothness: $\mathbf{I}^{-1}$ depends smoothly on λ for exponential families. - (MG-4) Primary-meta coupling bounded: $\mathbf{I}^{-1}$'s state-derivative in the drift direction is bounded for exponential families.

Derived (Fisher-whitening-as-special-meta-gain).

This hands `#deriv-adaptive-gain-dynamics` a **concrete second instance** of derivable meta-gain alongside adaptive-Kalman-with-Mehra-estimator (its primary instance). The machinery of meta-gain composition via `#result-sector-persistence-template` (augmented-state Lyapunov) applies directly.

L2 regime: candidate third `#disc-identifiability-floor` instance or strengthening of Instance 2.

Under L2-latent regimes (unobservable correlation structure beyond what Fisher information can resolve), Fisher whitening fails: the correlation sub-block of Fisher is rank-deficient (Cramér-Rao floor on unobservable parameters). This parallels `#disc-identifiability-floor` Instance 2's L1'-unobservable-C Fisher-rank-1 obstruction — potentially a new L2 instance, or potentially a generalization of Instance 2 to higher correlation orders. Open whether these are distinct floors or a single unified obstruction.

Hypothesis (L2-latent-floor).

L2-degenerate (perfect correlation, $r \to 1$) is a *structural* collapse — it requires DAG repair (edge merging) rather than update repair. This sits outside the Fisher-whitening framework.

Epistemic Status.

Conditional. Max attainable: *exact* under Path A ((PI) + Čencov combined with B1 directional fidelity) + L1'/L2 regimes + Bayesian coherence. The derivation is textbook information-geometry (Amari 1998; Amari-Nagaoka 2000) plus the AAT-internal (PI) axiom from `#scope-agent-identity`. What is AAT-distinctive is (a) the identification of (PI)/Čencov as the forcing machinery rather than merely a preferred choice; (b) the sub-scope α_3 labeling within the A2' partition; (c) the composition with `#deriv-adaptive-gain-dynamics`' meta-gain framework as a degenerate special case.

The angle bound is ≤ 45° for all finite r. This is a sharp load-bearing observation: AAT's existing default log-odds signal function is *directionally robust* under correlated failures. The bias formula in `#deriv-l1-update-bias` gives the quantitative magnitude-bias; this segment gives the qualitative direction-preservation. Together they characterize the default signal function's behavior under L1' fully: **direction preserved, magnitude biased, observable-C restores exactness (Prop B.7), unobservable-C floor per F13**.

Without (PI) adoption. If (PI) is not adopted as an AAT axiom, Fisher whitening remains a theorem-imported technique from information geometry. The sub-scope α_3 labeling still applies (correlated evidence + Fisher-whitened + Bayesian coherence → A2' derived) but without the AAT-internal forcing. This is the adjacent-family status.

Discussion.

Relationship to the default signal function. `#disc-credit-assignment-boundary`'s default log-odds signal was landed in the 2026-04-22/23 strengthening cycle via the Cauchy-FE uniqueness theorem in `#deriv-edge-update-natural-parameter`. Under L0 evidence independence, the default signal is exact. Under L1' correlated evidence, the default signal's direction is preserved (≤ 45° angle to natural-gradient direction) but magnitude degrades by $\sqrt{1-r^2}$. This segment's Fisher-whitened update is the correction under (PI)/Čencov; it recovers sharp B1 under L1' by lifting the inner product from Euclidean to Fisher.

Relationship to `#disc-identifiability-floor` Instance 2. Instance 2 (L1' unobservable-C Cramér-Rao floor) refutes single-channel identification of mixture parameters. Under observable C, Prop B.7 gives the L1' transfer; under unobservable C, no update — including Fisher-whitened — recovers identification. Fisher whitening operates *downstream* of identification: it corrects the update direction given that the Fisher information is known; it does not manufacture Fisher information that the data cannot yield.

Composition with the (PI)/Čencov 4th primary instance. The Fisher-metric cases in `#der-gain-sector-bridge`'s Verified Instances table (Matrix Kalman, Exponential family in natural parameters) are also AAT-internally forced under (PI)/Čencov. Fisher whitening at the edge-update layer of Section II therefore shares the same AAT-internal axiom as the Kalman / exp-family cases in Section I's update machinery. This is a cross-layer consistency: the (PI) axiom operates uniformly across Sections I and II whenever the natural parameters are parameters on a statistical manifold.

Working Notes.

- **Sub-scope α_4 composition** (adaptive-gain + correlated-evidence). Composing `#deriv-adaptive-gain-dynamics` meta-gain machinery with Fisher whitening gives an augmented-state contraction argument where both the primary state (log-odds) and the meta-state (Fisher information estimate) are updated. Open derivation.
- **L2 regime.** Fisher whitening at L2 (non-tree correlation structure) requires computing higher-order Fisher correlations. For sparse DAG structures with limited correlation order, Fisher-tree approximations may suffice; for dense correlation structures, the computational cost scales exponentially. Scope boundary for Fisher whitening as a practical technique.
- **Relationship to natural-gradient variational inference.** Khan & Lin 2017 (*Conjugate-computation variational inference*) derives natural-gradient VI as a special case of conjugate Bayesian computation. Under (PI)/Čencov, their result becomes AAT-internally derived within sub-scope α_3. This composes with `#deriv-variational-sector-condition` if that segment lands — natural-gradient VI would be the $\varepsilon = 0$ limit of ε-fidelity B1.
- **Landing context.** This segment lands from `spikes/spike-fisher-whitened-update.md` (2026-04-23 Gap A/B cycle). The AAT-internal axiom path via (PI)/Čencov was validated by `spikes/spike-jacobian-b1-strengthening.md` (2026-04-23 follow-up); see that spike for the broader discussion of uniqueness-theorem machinery clearing the `#disc-additive-coordinate-forcing` discipline.
- **Sibling segment at the model-parameter layer.** `#deriv-fisher-local-update-gain` derives update *magnitude* under the Fisher-local invariance regime, sibling to this segment's derivation of update *direction* under correlated evidence. Both share (PI)/Čencov via Path A's axiomatic chain. Together they make the Fisher-local invariance regime AAT-internally complete: direction at the edge-update layer and magnitude at the model-parameter-update layer.

Appendix

Derivation: Fisher-Local Update Gain

Under the **Fisher-local invariance regime** — smooth log-likelihood admitting non-degenerate local quadratic expansion at the current model parameter, with single-step update at first order in the step size — the natural-gradient Bayesian posterior mean shift has the form $\Delta\theta = K \cdot \tilde{\nabla}$ with **gain operator** $K = (H_M + H_L)^{-1} H_L$ and **scalar collapse** $\eta^* = U_M/(U_M + U_o)$ along the natural-gradient direction in the commuting / shared-eigenbasis case (always in 1-D; under (PI)/Čencov in the natural-gradient direction in higher dimensions). The result derives the form `#emp-update-gain` carries as an empirical claim, places it in the Fisher-local regime as exact, and recovers linear-Gaussian (Kalman) and conjugate-Bayesian instances as cases where the local quadratic expansion is globally exact. Companion at the model-parameter-update layer to `#deriv-fisher-whitened-update-rule`'s edge-update derivation: that segment derives update *direction* under correlated evidence; this segment derives update *magnitude* under the Fisher-local regime. Both share the (PI) parameterization-invariance axiom from `#scope-agent-identity` and Čencov 1982 uniqueness as AAT-internal forcing for the Fisher metric.

Formal Expression.

Setup — Fisher-local quadratic regime.

Let $\theta \in \mathbb{R}^d$ parameterize the agent's model and θ_t be the current point estimate (mode of the prior π_0, or center of expansion). Three regime conditions:

Definition (Fisher-local regime, conditions (R1)–(R3)).

- **(R1) Smooth log-likelihood admitting non-degenerate local quadratic expansion.** $\log \pi_0$ and $\log p(o \mid \cdot)$ are C^3 in a neighborhood of θ_t, with prior precision $H_M := -\nabla^2 \log \pi_0(\theta_t)$ and observed information $H_L := -\nabla^2 \log p(o \mid \theta)\big|_{\theta_t}$ satisfying $H_M + H_L \succ 0$ (joint positive-definiteness; see §"Boundary admissibility" for why this is weaker than $H_M \succ 0$ and $H_L \succ 0$ both).
- **(R2) First-order-in-step-size regime.** The posterior update $\Delta\theta = \theta_{t+1} - \theta_t$ is small enough that $O(\|\Delta\theta\|^3)$ terms in the quadratic expansion are negligible compared to the quadratic terms.
- **(R3) Bayesian-coherent update.** θ_{t+1} is taken to be a coordinate of the posterior $p(\theta \mid o) \propto \pi_0(\theta) \cdot p(o \mid \theta)$ — mean, mode, or natural-parameter (they coincide for Gaussian posteriors, which is what the quadratic expansion produces).

Under (R1)–(R3), the local log-posterior is

$$\log p(\theta \mid o) = \text{const} + s^T(\theta - \theta_t) - \frac{1}{2}(\theta - \theta_t)^T (H_M + H_L)(\theta - \theta_t) + O(\|\theta - \theta_t\|^3), \qquad (.1)$$

where $s := \nabla_\theta \log p(o \mid \theta)\big|_{\theta_t}$ is the score. The posterior is approximately $\mathcal{N}(\theta^\star, (H_M + H_L)^{-1})$ with mean shift

Derived (posterior-mean-shift).

$$\theta^\star - \theta_t = (H_M + H_L)^{-1} s. \tag{.2}$$

The AAT-vocabulary correspondences are

$$U_M := H_M^{-1}, \qquad U_o := H_L^{-1} \tag{.3}$$

— model uncertainty is the inverse prior precision, observation uncertainty is the inverse Fisher / inverse observed information (Cramér-Rao floor on what the observation pins down). The matrix forms generalize `#emp-update-gain`'s scalar U_M, U_o in the natural way.

Gain decomposition on the natural-gradient direction.

The **natural gradient** of the log-likelihood at θ_t in the Fisher metric is

Definition (natural gradient at the current point).

$$\tilde{\nabla} \log p(o \mid \theta_t) := \mathcal{I}(\theta_t)^{-1} s. \tag{.4}$$

For a single observation in the Fisher-local regime, $\mathcal{I}(\theta_t) = H_L$ up to the observed-vs-expected information distinction (standard under (R1)+regularity); the derivations below use H_L directly as the Fisher.

Rewriting the posterior mean shift in the natural-gradient direction:

Derived (gain-decomposition).

$$\Delta\theta = (H_M + H_L)^{-1} s = (H_M + H_L)^{-1} H_L \cdot (H_L^{-1} s) = K \cdot \tilde{\nabla}, \tag{.5}$$

with **gain operator**

$$\boxed{K := (H_M + H_L)^{-1} H_L.} \tag{.6}$$

The matrix-valued K is the natural object. The scalar collapse follows in the commuting / shared-eigenbasis case (always in 1-D; under (PI)/Čencov along the natural-gradient direction in higher dimensions): eigenvalues of K are

$$\eta_i^* = \frac{h_{L,i}}{h_{M,i} + h_{L,i}} = \frac{1/u_{o,i}}{1/u_{M,i} + 1/u_{o,i}} = \frac{u_{M,i}}{u_{M,i} + u_{o,i}} \tag{.7}$$

per coordinate, collapsing to the scalar

$$\boxed{\eta^* = \frac{U_M}{U_M + U_o}} \tag{.8}$$

— the form `#emp-update-gain` carries, derived from the Fisher-local quadratic expansion of prior and likelihood and applied to the natural-gradient direction.

Boundary admissibility — improper-prior and degenerate-likelihood directions.

The gain operator $K = (H_M + H_L)^{-1} H_L$ is well-defined whenever $H_M + H_L \succ 0$ — not necessarily when both H_M and H_L are individually positive-definite. This admits **degenerate-likelihood directions** (observation uninformative along some coordinates, H_L rank-deficient): K's eigenvalue along those directions is 0, $u_{o,i} \to \infty$, $\eta_i^* \to 0$, no update along that direction — *trust the prior where the observation says nothing*. It also admits **improper-prior directions** (degenerate prior, $H_M = 0$ along some coordinate) when H_L pins the direction: $\eta_i^* \to 1$ along uninformative-prior directions where the observation is informative — *trust the observation where the prior says nothing*, the maximum-likelihood limit.

Derived (admissibility-of-degenerate-curvature).

The minimal regime condition is $H_M + H_L \succ 0$, with $U_M/(U_M + U_o)$ defined per-eigendirection by the appropriate limit. This corresponds to standard improper-prior admissibility in Bayesian inference and to standard observation-non-identifiability admissibility in maximum-likelihood estimation.

Recovery of linear-Gaussian and conjugate-Bayesian cases.

When the quadratic expansion in (R1) holds *globally* — not just locally — the derivation gives η^* without any expansion error. Two canonical cases:

Derived (global-exactness-special-cases).

- **Linear-Gaussian (Kalman).** Prior $\mathcal{N}(\theta_t, U_M)$, likelihood $\mathcal{N}(o; H\theta, U_o)$ with linear observation operator H: the log-posterior is exactly quadratic. The scalar Kalman gain $K = U_M H^T (H U_M H^T + U_o)^{-1}$ collapses to $\eta^* = U_M/(U_M + U_o)$ in the scalar case (Kalman 1960; Bishop 2006 §3.3). Exact globally.
- **Conjugate Bayesian (Beta-Bernoulli, Dirichlet-multinomial, Gaussian-Gaussian, exponential-family natural-parameter conjugacy).** Natural-gradient VI is the exact posterior update at step size 1 for conjugate families (Khan-Lin 2017 conjugate-computation variational inference). The Fisher-local expansion is globally exact in natural-parameter coordinates; $\eta^* = U_M/(U_M + U_o)$ holds exactly with U_M, U_o read from the conjugate sufficient-statistic precision counts.

For general smooth models, the natural-gradient invariance theorem of Amari 1998 guarantees that the natural gradient is parameterization-invariant; the Fisher-local quadratic expansion is the order at which the natural gradient and the exact posterior agree to first order in step size.

Epistemic Status.

Conditional. Max attainable: *exact* under (R1)–(R3) + (PI) parameterization-invariance from `#scope-agent-identity`. Under (PI) and Čencov 1982 uniqueness, the Fisher metric is the AAT-internally canonical metric on the statistical manifold (the 4th primary instance of `#disc-additive-coordinate-forcing`); the natural-gradient direction is therefore the AAT-internally canonical reference direction along which η^* reads as $U_M/(U_M + U_o)$. The textbook information-geometry result (Amari 1998 natural-gradient invariance theorem) is imported; the (PI)/Čencov forcing in `#deriv-fisher-whitened-update-rule`'s Path A makes the import AAT-internal.

Linear-Gaussian (Kalman) and conjugate-Bayesian instances are cases where the local quadratic expansion is *globally* exact, so $\eta^* = U_M/(U_M + U_o)$ holds without truncation error.

Outside the Fisher-local regime — heavy-tailed posteriors (no second moment), structurally non-smooth likelihoods, multimodal uncertainty where the local quadratic misrepresents global structure — the quantitative form is *not* derived here. The *qualitative* direction (gain rises with model uncertainty, falls with observation noise) is preserved as *robust qualitative* from `#emp-update-gain`'s broader empirical scope; the first-order form is recovered locally; what need not hold is global quantitative fidelity.

This segment depends on (PI) from `#scope-agent-identity` only. The triple (PI)+(R)+(K) of the Markov-morphism layer (carried by the Fisher-Rao bias-bound machinery; the (R) Riemannian-structure and (K) KL-second-order-matching axioms are stronger) is *not* invoked here. If (R) and (K) are asserted at the scope-level in a future cycle, this segment's status remains conditional on (PI) alone unless explicitly extended.

Discussion.

Why read η^* off the natural-gradient direction. The Bayesian-coherent posterior mean shift $\Delta\theta = (H_M + H_L)^{-1}s$ is a fixed vector; its decomposition into "gain times reference direction" depends on the choice of reference direction. Two natural choices give two faces of the same shift:

- **Natural-gradient direction** $\tilde{\nabla} = H_L^{-1}s$. Coefficient: $\eta^*_{\mathrm{NG}} = H_L/(H_M{+}H_L) = U_M/(U_M{+}U_o)$. *AAT-internally canonical* under (PI)/Čencov — the Fisher metric is the unique Markov-invariant metric, so the natural gradient is the unique coordinate-invariant gradient direction (Čencov 1982; extended by Ay-Jost-Lê-Schwachhöfer 2017).
- **Prior-curvature-rescaled direction** $H_M^{-1}s$. Coefficient: $\eta^\star_{\mathrm{prior}} = H_M/(H_M + H_L) = U_o/(U_M + U_o)$. The *complement*; the same algebraic decomposition arises in Kalman literature as the prior weight in the convex combination "posterior = (1-K) prior + K observation."

The two coefficients are duals on the same posterior shift; their product reduces to the posterior covariance $(H_M{+}H_L)^{-1}$. AAT names the natural-gradient coefficient as the canonical η^* because (PI)/Čencov picks out the natural-gradient direction as the unique coordinate-invariant choice.

Three derivation routes converge. The same scalar $\eta^* = U_M/(U_M + U_o)$ falls out of three independent structural derivations:

Table .1

Route	*Lever*	*Surface meaning of η^**
Local-Gaussian Laplace expansion (§Setup–Gain decomposition above)	Algebraic completion of squares on additive log-posterior quadratics	Posterior mean shift coefficient on the natural-gradient direction
Bregman / KL projection onto the local tangent plane	First-order condition on the variational free energy along natural-gradient tilt	Projection coefficient of the posterior onto the local tangent plane (Pythagorean projection in Fisher metric per Amari-Nagaoka 2000 §3.2)
Cramér-Rao / inverse-Fisher	Precision-additive composition $1/U_M + 1/U_o$, inverted	Fractional observation contribution to posterior precision

The agreement is not coincidence: the Fisher-local regime *is* the regime where prior and likelihood compose as a Pythagorean projection in the Fisher metric — the same exponential-family / Bregman / Pythagorean structure underlying `#deriv-strategy-cost-regret-bound`'s ρ-decomposition. The three routes are three faces of one geometric object: precision-additive composition of two Gaussian information sources, read off the natural-gradient direction.

Sibling positioning vs `#deriv-fisher-whitened-update-rule`. That segment lives at the **edge-update layer** of Section II's strategy DAG: it derives the Fisher-whitening correction for the *direction* of edge updates under correlated evidence (L1'/L2). This segment lives at the **model-parameter-update layer** of Section I: it derives the *magnitude* of the natural-gradient Bayesian update via the Fisher-local invariance regime. Both depend on (PI) + Čencov as the AAT-internal axiom forcing the Fisher metric. Together they make the Fisher-local invariance regime AAT-internally complete: direction (whitening) at the edge layer and magnitude (gain) at the model-parameter layer are both derived from the same axiomatic chain.

Special case of `#deriv-adaptive-gain-dynamics`' meta-gain framework. The gain operator $K = (H_M + H_L)^{-1} H_L$ is a *deterministic function of the current state* — it depends on the prior covariance H_M^{-1} (part of the agent's M_t state) and on the observation Fisher H_L (depends on θ_t and o). It is therefore the **deterministic-meta-gain** special case of `#deriv-adaptive-gain-dynamics`' (MG-1)–(MG-4) machinery: the meta-gain is determined by the primary state rather than independently learned. All four meta-gain conditions are satisfied trivially (symmetric-positive-definite K on the interior; smoothness for exponential families; bounded primary-meta coupling). Resolving the epistemic-opacity question that `#def-observation-function` axiomatizes — i.e., the agent does not know U_o a priori and must estimate it from its own mismatch sequence — is one level up: it adds the Mehra-style meta-channel that `#deriv-adaptive-gain-dynamics` treats as its primary instance.

Downstream tempo and persistence machinery. `#emp-update-gain`'s Discussion §"Connection to adaptive tempo" names $\mathcal{T} = \nu \cdot \eta^*$. The derivation here promotes η^*'s tier from `empirical / robust-qualitative` (over the cross-domain validity tail — RL, PID, software-developer) to `derived (conditional on Fisher-local regime)` for the Kalman / conjugate / natural-gradient core. The tempo product $\mathcal{T} = \nu \cdot \eta^*$ inherits the regime distinction directly; the persistence-condition machinery (`#result-persistence-condition`, `#result-sector-condition-stability`) and the adaptive-tempo / aporia diagnostics (`#def-adaptive-tempo`, `#def-aporia`) all gain the tighter η^* exactness statement in their Fisher-local instances. The qualitative direction claim and the failure-mode framing (gain-collapse → epistrophe failure) survive unchanged outside the Fisher-local regime.

Tensor adaptive tempo connection. The matrix gain operator K is the per-coordinate primitive for tensor-valued adaptive tempo: $\mathcal{T} = \nu \cdot K$ as a matrix product, with the existing scalar $\mathcal{T} = \nu \cdot \eta^*$ recovered in the shared-eigenbasis limit. See `#def-adaptive-tempo`'s Tensor extension.

Working Notes.

- **Multi-step extrapolation rate.** The Fisher-local exactness is a one-step statement. Iterating over many observations accumulates higher-order curvature error in the non-Gaussian / non-conjugate case. Bernstein-von Mises asymptotics (van der Vaart 1998 §10.2) give a $1/\sqrt{n}$ posterior concentration rate; whether η^*'s exactness inherits a matching $1/\sqrt{n}$ degradation outside Kalman / conjugate is an open quantitative-rate question. Open spike candidate.
- **Edgeworth higher-order correction.** The $O(\|\Delta\theta\|^3)$ correction to η^* in terms of third / fourth derivatives of the log-likelihood admits a standard Edgeworth-expansion treatment. Worth a follow-up spike if it becomes load-bearing for any AAT use case.
- **Variational-bound extension.** Under variational approximation $\mathrm{KL}(q\|p) \leq \varepsilon$, the gain inherits a Pinsker-tight degradation factor analogous to `#deriv-variational-sector-condition`'s $O(\sqrt{\varepsilon})$ B1 degradation. Composing this with the derivation gives η^* in sub-scope α' under approximate-posterior agents.
- **Multimodal posteriors.** When the prior or posterior is multimodal, the local-quadratic expansion around any one mode misses the global structure. Two natural moves: per-mode $U_M/(U_M + U_o)$ with mode-mixture weights as a separate state; or accept the qualitative direction claim only. Worth a follow-up spike if multimodal posteriors become load-bearing.
- **Consolidation / replay connection.** `#form-consolidation-dynamics`-style between-event consolidation drives U_M up (model uncertainty grows under offline replay via IB-gap reduction). The next online observation gets up-weighted because $U_M/(U_M + U_o)$ rises with U_M. This is the formal connection between consolidation-induced uncertainty growth and increased online responsiveness — worth tightening into a result if `#form-consolidation-dynamics` lands a quantitative form.
- **Honest obstructions.** (O1) The natural-gradient invariance theorem (Amari 1998) is imported, not derived; the (PI)/Čencov forcing in `#deriv-fisher-whitened-update-rule`'s Path A makes the import AAT-internal. Without (PI), the natural-gradient direction is a chosen-not-forced direction and the derivation degrades to "the form on the chosen direction." (O2) Single-observation per-step setup; multi-observation batches generalize mechanically ($H_L = \sum_i H_{L,i}$ for independent observations). (O3) Matrix vs scalar — the matrix K is the natural object; the scalar $U_M/(U_M + U_o)$ is its eigenvalue in the commuting / 1-D / shared-eigenbasis case. (O4) Step-size boundary qualitative; the actual Fisher-local cutoff depends on the third-derivative

norms of $\log \pi_0$ and $\log p(o \mid \cdot)$. (O5) Robust-qualitative downstream survives outside the Fisher-local regime by the general structure of Bayesian updating, not by this derivation.

- **Cross-reference to Paper 3 chart-rescaling no-go.** The (PI) dependence is forced by the chart-rescaling no-go on Euclidean chart norms (NeurIPS 2026 Paper 3, "How Much Can LLMs Hallucinate?", §4 Theorem 4.2 / `#thm-no-go`): outside (PI), no universal-constant claim survives. The natural-gradient direction inherits this forcing; the canonical-direction argument depends on (PI), not on (R) or (K) of the Markov-morphism triple. See `msc/neurips-back-integration-2026-05-08.md` §1 Paper 3 entry #3 and the source `~/src/neurips/03-llm-hallucinate-bound/`.
- **Landing context.** Landed in the 2026-05-12 audit-strengthening cycle (AAT-5); see CHANGELOG 2026-05-12. The load-bearing three-route convergence and boundary-admissibility content is in the Derivation and Discussion above; the originating spike is absorbed archaeology, not a live reference.
- **Downstream consumer at the persistence layer.** The matrix gain operator $K = (H_M + H_L)^{-1} H_L$ derived here is the per-channel primitive that `#def-adaptive-tempo`'s Tensor extension aggregates into $\mathcal{T} = \sum_k \nu^{(k)} K^{(k)}$. The persistence-layer consumer is the matrix-Loewner persistence condition in `#deriv-matrix-persistence-condition`: under matrix $\mathcal{T}$ and matrix disturbance Σ_w, the agent persists iff $\mathcal{T}$ is Hurwitz and the stationary covariance Σ_∞ (solving $\mathcal{T}\Sigma_\infty + \Sigma_\infty \mathcal{T}^T = \Sigma_w$) is strictly Loewner-dominated by $D_\delta = \mathrm{diag}(\delta^2_{\text{critical},k})$. The matrix-Loewner form is *strictly sharper* than per-coordinate: under cross-dimensional correction (off-diagonal $\mathcal{T}$ — the generic Fisher-local case when prior and likelihood do not share the coordinate basis), per-coordinate gives a false-pass while matrix-Loewner reads persistence correctly off the worst direction.

Appendix

Derivation: Matrix-Loewner Persistence Condition

Under matrix adaptive tempo $\mathcal{T}$ from `#def-adaptive-tempo`'s Tensor extension (per-channel primitive $K^{(k)} = (H_M + H_L^{(k)})^{-1} H_L^{(k)}$ from `#deriv-fisher-local-update-gain`) and matrix disturbance covariance Σ_w, the operational persistence condition lifts from scalar to matrix-Loewner form: the agent persists iff $\mathcal{T}$ is Hurwitz (structural persistence) and the stationary covariance Σ_∞ — solving the continuous Lyapunov equation $\mathcal{T}\Sigma_\infty + \Sigma_\infty \mathcal{T}^T = \Sigma_w$ — is strictly Loewner-dominated by the diagonal of squared per-direction critical thresholds $D_\delta := \mathrm{diag}(\delta^2_{\text{critical},k})$. The matrix-Loewner condition $\Sigma_\infty \prec D_\delta$ recovers `#result-persistence-condition`'s scalar Model S form as the isotropic special case and `#result-per-dimension-persistence`'s per-coordinate form as the diagonal special case. Crucially, when $\mathcal{T}$ has off-diagonal entries that misalign with the coordinate basis — the generic situation under cross-dimensional correction — the per-coordinate condition is *unsafe*: it can declare persistence on systems whose stationary mismatch will exceed task-adequacy along the diagonal direction. The matrix-Loewner form is the canonical anisotropic persistence condition; per-coordinate is its diagonal-$\mathcal{T}$-axis-aligned-D_δ special case, sharp where the coordinate basis happens to be the correction-machinery's eigenbasis and unsafe outside it.

Formal Expression.

Setup — Model S with matrix correction.

Consider linear stochastic mismatch dynamics

Definition (matrix-Model-S-mismatch-dynamics).

$$d\delta_t \;=\; -\mathcal{T}\,\delta_t\,dt \;+\; \Sigma_w^{1/2}\,dW_t \tag{.1}$$

with $\delta_t \in \mathbb{R}^d$, matrix adaptive tempo $\mathcal{T} \in \mathbb{R}^{d\times d}$, disturbance covariance $\Sigma_w \succ 0$, and standard d-dimensional Brownian W_t. Per `#def-adaptive-tempo`'s Tensor extension, $\mathcal{T} = \sum_k \nu^{(k)} K^{(k)}$ aggregates the per-channel matrix gain operators from `#deriv-fisher-local-update-gain`.

Stationary covariance — continuous Lyapunov equation.

The stationary distribution of the linear stochastic dynamics is Gaussian $\mathcal{N}(0, \Sigma_\infty)$, with Σ_∞ the unique positive-definite solution of the **continuous Lyapunov equation**

Derived (stationary-covariance, exact under Hurwitz $\mathcal{T}$ and $\Sigma_w \succ 0$).

$$\boxed{\mathcal{T}\Sigma_\infty \;+\; \Sigma_\infty \mathcal{T}^T \;=\; \Sigma_w.} \tag{.2}$$

Existence and uniqueness of $\Sigma_\infty \succ 0$ are equivalent to $\mathcal{T}$ being **Hurwitz** — all eigenvalues having strictly positive real part. The closed-form integral representation is

$$\Sigma_\infty \;=\; \int_0^\infty e^{-\mathcal{T}t}\,\Sigma_w\,e^{-\mathcal{T}^T t}\,dt. \tag{.3}$$

When $\mathcal{T}$ is symmetric and commutes with Σ_w, the solution simplifies to $\Sigma_\infty = \frac{1}{2}\mathcal{T}^{-1}\Sigma_w$. Otherwise the integral form (or any standard Lyapunov-equation solver) gives the stationary covariance.

The matrix-Loewner persistence condition.

The agent persists operationally iff:

Derived (matrix-persistence-loewner-Model-S).

(MP-1) Structural persistence. $\mathcal{T}$ is Hurwitz — equivalently, Σ_∞ exists as the unique positive-definite solution of the Lyapunov equation. This is the matrix analog of `#result-persistence-condition`'s structural condition $\alpha > 0$ (Hurwitz reduces to $\mathcal{T}_0 > 0$ in the scalar case).

(MP-2) Matrix-Loewner task adequacy.

$$\boxed{\Sigma_\infty \;\prec\; D_\delta \;:=\; \mathrm{diag}\big(\delta^2_{\mathrm{critical},k}\big)} \tag{.4}$$

with $\prec$ the strict Loewner (positive-definite) order: $D_\delta - \Sigma_\infty \succ 0$. The stationary mismatch ellipsoid is contained strictly inside the per-direction critical-threshold ellipsoid.

The two conditions are independent in the same sense as `#result-persistence-condition`'s structural-vs-task-adequacy split: (MP-1) is a property of the correction machinery alone; (MP-2) adds the domain-specific bound on per-direction tolerable mismatch.

Three equivalent restatements. The Loewner condition (MP-2) admits three equivalent forms:

(MP-2a) Per-direction. For every unit direction $\hat{v} \in \mathbb{R}^d$:

$$\hat{v}^T\Sigma_\infty\hat{v} \;<\; \hat{v}^T D_\delta \hat{v} \;=\; \sum_k v_k^2\,\delta^2_{\mathrm{critical},k}. \tag{.5}$$

The projected stationary variance along $\hat{v}$ must be less than the direction-projected squared threshold along the same direction.

(MP-2b) Spectral. The generalized eigenvalue problem $\Sigma_\infty x = \lambda D_\delta x$ satisfies $\lambda_{\max} < 1$, equivalently

$$\lambda_{\max}\big(D_\delta^{-1/2}\Sigma_\infty D_\delta^{-1/2}\big) \; < \; 1. \tag{.6}$$

The principal eigenvector identifies the *worst direction*; the eigenvalue's distance from 1 is the worst-direction safety margin.

(MP-2c) Ellipsoid containment. The stationary noise ellipsoid $\{x \,:\, x^T\Sigma_\infty^{-1}x \le 1\}$ is contained strictly inside the threshold ellipsoid $\{x \,:\, x^T D_\delta^{-1} x \le 1\}$.

The three are mathematically equivalent under standard matrix-analysis (Horn & Johnson 2013, §4.6 and §7). (MP-2a) is cleanest for hand reasoning, (MP-2b) for computation (single generalized-eigenvalue solve), (MP-2c) for geometric intuition.

Recovery of existing AAT persistence forms.

Table .1

Special case	*(MP-2) reduces to*	*Matches*
Isotropic ($\mathcal{T} = \mathcal{T}_0 I$, $\Sigma_w = \sigma_w^2 I$, $\delta_{\text{critical}} = \delta_0\mathbf{1}$)	$\sigma_w^2/(2\mathcal{T}_0) < \delta_0^2$, i.e., $\mathcal{T}_0 > \sigma_w^2/(2\delta_0^2)$	Scalar Model S form, `#result-persistence-condition`
Diagonal $\mathcal{T}$, Σ_w in the coordinate basis of D_δ	$\sigma_{w,k}^2/(2\mathcal{T}_{kk}) < \delta_{\text{critical},k}^2$ per coordinate k	Per-coordinate Model S RMS, `#result-per-dimension-persistence`
Symmetric $\mathcal{T}$ commuting with Σ_w, general D_δ	Per-eigendirection inequality in $\mathcal{T}$'s eigenbasis, with direction-projected thresholds	New content recovered from the matrix form — extends per-coordinate to non-axis-aligned $\mathcal{T}$

The third row is the content `#result-per-dimension-persistence` does not currently carry: when $\mathcal{T}$ and Σ_w share an eigenbasis that is *not* the coordinate basis of D_δ, the per-coordinate form misses the right direction-projected thresholds and the matrix-Loewner form handles them correctly.

Where per-coordinate is unsafe — a constructive counterexample. The deeper structural question: is the matrix-Loewner form merely a generalization that agrees with per-coordinate in all real cases, or is it *strictly sharper* — does there exist a regime where per-coordinate says PASS but matrix-Loewner correctly says FAIL? The construction below establishes the latter: the per-coordinate form is **unsafe** under cross-dimensional correction.

Construction. Take $d = 2$, $\Sigma_w = I$, and

$$\mathcal{T} \; = \; \begin{pmatrix} 1 & -0.9 \\ -0.9 & 1 \end{pmatrix}. \tag{.7}$$

This $\mathcal{T}$ is symmetric positive-definite with eigenvalues 1.9 (along $(1,-1)/\sqrt{2}$) and 0.1 (along $(1,1)/\sqrt{2}$) — strongly anisotropic correction, with the $(1,1)$ direction the weak axis. Per-coordinate naive reading of "diagonal of $\mathcal{T}$" sees uniform $\mathcal{T}_{11} = \mathcal{T}_{22} = 1$ and concludes correction is uniform at rate 1 on each coordinate.

Stationary covariance. Since $\mathcal{T}$ is symmetric, $\Sigma_\infty = \frac{1}{2}\mathcal{T}^{-1}\Sigma_w = \frac{1}{2}\mathcal{T}^{-1}$. Direct computation:

$$\Sigma_\infty \; = \; \frac{1}{0.38}\begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix} \; \approx \; \begin{pmatrix} 2.63 & 2.37 \\ 2.37 & 2.63 \end{pmatrix}, \tag{.8}$$

with eigenvalues 5.00 along $(1,1)/\sqrt{2}$ and 0.26 along $(1,-1)/\sqrt{2}$.

Set $\delta_{\mathbf{critical}} = (1.7, 1.7)$, so $D_\delta = 2.89\, I$.

- *Per-coordinate check (* `#result-per-dimension-persistence` *form):* per-coordinate stationary variance is $\Sigma_{\infty,kk} = 2.63 < 2.89 = \delta^2_{\text{critical},k}$ for each k. **Per-coordinate says PASS.** ✓

- *Matrix-Loewner check (MP-2 here):* worst-direction projected variance is $\lambda_{\max}(\Sigma_\infty) = 5.00$ along $(1,1)/\sqrt{2}$. Direction-projected squared threshold along the same direction is $\hat{v}^T D_\delta \hat{v} = 2.89$. The condition $5.00 < 2.89$ fails. Equivalently, $\lambda_{\max}(D_\delta^{-1/2} \Sigma_\infty D_\delta^{-1/2}) = 5.00/2.89 \approx 1.73 > 1$. **Matrix-Loewner says FAIL.** ✗

Conclusion. Per-coordinate gives a false-pass. The agent will routinely produce mismatch vectors with RMS magnitude $\sqrt{5.00} = 2.24$ along the $(1,1)/\sqrt{2}$ direction, exceeding the direction-projected threshold of $\sqrt{2.89} = 1.70$ on that direction. The per-coordinate analysis missed the diagonal direction because each coordinate alone stays within $\delta_{\text{critical},k} = 1.7$ — the failure is *only* visible at the matrix-Loewner level. Cross-dimensional correction (the off-diagonal entries of $\mathcal{T}$) is what creates the diagonal-direction concentration that per-coordinate cannot detect.

Epistemic Status.

Conditional. Max attainable: *exact* under (i) Model S linear stochastic dynamics, (ii) Hurwitz matrix tempo $\mathcal{T}$, (iii) positive-definite disturbance covariance Σ_w, (iv) diagonal critical-threshold structure $D_\delta = \text{diag}(\delta^2_{\text{critical},k})$. The Lyapunov equation, the existence-iff-Hurwitz fact, and the matrix-Loewner equivalence forms are all standard from Bellman 1960 / Khalil 2002 / Horn-Johnson 2013; the AAT-distinctive content is the application to the persistence condition and the recognition that per-coordinate is strictly sharper-or-unsafe rather than uniformly sharper.

What is load-bearing:

- The matrix-Loewner form as the canonical anisotropic persistence condition, with scalar and per-coordinate forms as special cases recovered exactly.

- The strict-sharpness result via the §4 counterexample: per-coordinate gives a false-pass when $\mathcal{T}$ has off-diagonal entries that misalign with the coordinate basis. The per-coordinate form is not merely incomplete in this regime — it is unsafe.

- The three equivalent restatements (per-direction, spectral, ellipsoid-containment) with their different operational readings.

- The structural-vs-task-adequacy decomposition from `#result-persistence-condition` transfers directly: (MP-1) is the matrix structural condition (Hurwitz); (MP-2) is the matrix task adequacy (Loewner $\Sigma_\infty \prec D_\delta$). The two conditions remain independent — Hurwitz $\mathcal{T}$ does not imply $\Sigma_\infty \prec D_\delta$ for any particular D_δ, and conversely.

What is not established here:

- The nonlinear matrix-sector extension. Under sector-bounded $F(\mathcal{T}, \delta)$ with $\delta^T F \succeq \alpha I$ in matrix-Loewner form, the matrix-Lyapunov machinery extends to give an ultimate-bound matrix; the linear case here is the template instantiation. Full matrix-sector treatment is follow-on work in `#deriv-sector-condition`.

- The Model D matrix lift. Under deterministic bounded disturbance $\|w_t\|_M \leq \rho$ in a quadratic norm, the ultimate-bound becomes an ellipsoid containment problem (a linear matrix inequality). Mechanical lift; not derived here.

- The non-Hurwitz boundary behaviour (eigenvalue exactly on the imaginary axis). Outside Hurwitz, the linear analysis fails — no stationary distribution. This matches the scalar case's $\alpha > 0$ requirement.

- The promotion of `#result-adversarial-tempo-advantage` to matrix form. The matrix-Loewner form sharpens the adversarial result via worst-direction targeting (an adversary controlling Σ_w^{adv} maximizes $\lambda_{\max}(D_\delta^{-1/2}\Sigma_w^{\text{adv}}\mathcal{T}_{\text{eff}}^{-1}D_\delta^{-1/2})$), but the full adversarial promotion is a separate cycle.

- The matrix extension of `#deriv-persistence-cost`'s information-rate floor. The natural lift is $\dot{R} \geq \frac{1}{2}\text{Tr}(\mathcal{T})$ — per-eigendirection application of the scalar derivation. Worth a cross-reference Working Note in that segment if it becomes load-bearing; not derived here.

Discussion.

Why the Loewner form is the natural matrix lift. The Lyapunov equation $\mathcal{T}\Sigma_\infty + \Sigma_\infty\mathcal{T}^T = \Sigma_w$ is itself a matrix-Loewner statement: it says Σ_∞ balances correction (via $\mathcal{T}$) against disturbance (via Σ_w) in the PSD ordering. The task-adequacy condition $\Sigma_\infty \prec D_\delta$ reads this balance against the threshold ellipsoid. The whole structure lives in the cone of positive-semidefinite matrices ordered by Loewner dominance; scalar persistence is its 1D shadow, per-coordinate persistence is its axis-aligned-diagonal shadow, and the matrix form is the load-bearing object the shadows project from.

The structural-vs-task-adequacy decomposition transfers. `#result-persistence-condition`'s split — structural persistence (machinery works) vs task adequacy (machinery works well enough) — survives intact in the matrix lift: Hurwitz $\mathcal{T}$ is the matrix structural condition; Loewner $\Sigma_\infty \prec D_\delta$ is the matrix task-adequacy condition. The two are independent; an agent can have Hurwitz $\mathcal{T}$ (correction works in every direction) but fail Loewner on the worst direction (correction is not fast enough on the weak axis for the per-direction threshold). The remedies are different: structural failure requires changing the correction architecture (`#result-structural-adaptation-necessity`); task-adequacy failure requires either increasing $\mathcal{T}$ on the weak direction, decreasing Σ_w on the weak direction, or relaxing the threshold along that direction.

The weak-direction-bottleneck argument transfers and sharpens. `#result-per-dimension-persistence`'s key insight — the weak dimension is the bottleneck — generalizes to the weak *direction* in the matrix lift: the worst eigenvector of Σ_∞ relative to D_δ is what gates persistence. When this eigenvector aligns with a coordinate axis, the per-coordinate analysis catches it. When it points off-axis — the generic situation under cross-dimensional correction — the per-coordinate analysis misses it. The §4 counterexample is the canonical illustration: the worst direction is $(1,1)/\sqrt{2}$, off-axis from both coordinate directions; per-coordinate evaluates the wrong directions; matrix-Loewner evaluates the right ones.

Cross-coordinate correction is the generic case under Fisher-local invariance. The matrix gain operator $K = (H_M + H_L)^{-1}H_L$ from `#deriv-fisher-local-update-gain` is diagonal in the coordinate basis only when prior precision H_M and observation Fisher H_L share an eigenbasis aligned with the coordinate axes. In general — and certainly in the natural-gradient regime where the agent's parameterization is statistical-manifold-natural rather than user-chosen — they do not, and K has off-diagonal entries. The matrix-Loewner condition becomes the canonical form for any Fisher-local-derived persistence analysis; the per-coordinate form is the exceptional case where prior, likelihood, and threshold all happen to share the coordinate axes.

Cross-reference to existing AAT machinery. The matrix-Loewner form composes with:

- `#result-sector-persistence-template` (T1)–(T3) as the linear-case template instantiation; the matrix sector inequality $\delta^T F + F^T\delta \succeq 2\alpha I$ would extend the matrix-Lyapunov result to nonlinear F via the template's (T2) condition lifted to matrix form.

- `#deriv-sector-condition` Prop A.1S (the Lyapunov-stochastic derivation underlying `#result-persistence-condition`'s scalar form) — the matrix derivation here is the analog at the matrix-Lyapunov level.

- `#deriv-fisher-local-update-gain` (matrix gain K as the per-channel primitive) and `#def-adaptive-tempo` (Tensor extension aggregating $\mathcal{T} = \sum_k \nu^{(k)}K^{(k)}$).

On the scope tag downstream. Until the broader promotion of `#result-persistence-condition`, `#result-adversarial-tempo-advantage`, `#form-composition-closure`, `#der-team-persistence`, `#deriv-critical-mass-composition` to invoke the matrix form directly, those segments continue to read scalar and inherit the "scalar / isotropic / nonredundant-channel scope" tag from `#def-adaptive-tempo`'s Epistemic Status. The per-direction primitive is in place; the downstream promotion is a follow-on cycle in `TODO.md`.

Working Notes.

- **Counterexample as load-bearing content.** Section 4's $\mathcal{T} = \begin{pmatrix} 1 & -0.9 \\ -0.9 & 1 \end{pmatrix}$ example is the segment's strongest content — it shifts the matrix-Loewner form from "interesting generalization" to "the safe condition; per-coordinate is unsafe under cross-dimensional correction." The example uses the smallest nontrivial dimension ($d = 2$), the simplest non-diagonal symmetric $\mathcal{T}$ structure, and isotropic disturbance / threshold parameters. Generalizations to $d > 2$, asymmetric $\mathcal{T}$, and anisotropic Σ_w/D_δ are mechanical; the qualitative phenomenon (off-diagonal correction → diagonal-direction variance concentration → per-coordinate false-pass) is robust.
- **Empirical validation.** A 2D simulation matching the §4 parameters would confirm the predicted false-pass behaviour of per-coordinate: run the linear SDE with $\mathcal{T}, \Sigma_w$ from §4 for N steps; observe that the per-coordinate marginals stay within ± 1.7 most of the time (per-coordinate "passes") while the diagonal-direction projection routinely exceeds ± 1.7 (matrix-Loewner "fails"). Mechanical simulation; would land cleanly under `obs-section-i-validation-simulations` as a new variant.
- **Open work — nonlinear matrix-sector.** The matrix-Loewner form is derived for linear correction. The nonlinear matrix-sector lift would replace the Lyapunov equation with the matrix-sector inequality $F(\delta) + F(\delta)^T \succeq 2\alpha\delta\delta^T/\|\delta\|^2$ (or a matrix-sector form derived from a quadratic Lyapunov function). Composes with `#deriv-sector-condition`'s nonlinear scalar machinery via the matrix template; not derived here.
- **Open work — adversarial extension.** The adversarial advantage exponent in `#result-adversarial-tempo-advantage` lifts to a matrix-eigenvalue-ratio exponent: an adversary controlling Σ_w^{adv} maximizes the bad-direction stationary variance via the matrix-Loewner form. Worth a follow-on derivation if `#result-adversarial-tempo-advantage` is promoted to matrix form.
- **Open work — composition lift.** Composite stationary covariance Σ_∞^c solves the Lyapunov equation with composite $\mathcal{T}^c$ and Σ_w^c (aggregated from sub-agents); composite Loewner condition $\Sigma_\infty^c \prec D_\delta^c$ governs composite persistence. Promoting `#form-composition-closure`, `#der-team-persistence`, and `#deriv-critical-mass-composition` to invoke the matrix form is the natural next AAT-1 follow-on cycle.
- **Landing context.** Landed in the 2026-05-12 AAT-1 follow-on cycle (matrix-Loewner persistence, succeed-beyond-claim); see CHANGELOG 2026-05-12 (late). The load-bearing derivation and counterexample are above; the follow-on extensions (Model D matrix lift, adaptive-gain matrix dynamics, variational matrix form) are flagged in the "Open work" notes here and in TODO. Originating spike is absorbed archaeology, not a live reference.

Appendix

Derivation: The Survival-Imperative Exploration Drive

The unified policy objective in `#disc-ciy-unified-objective` combines pragmatic value Q_O with an epistemic exploration term (Causal Information Yield, CIY) weighted by $\lambda_{\text{info}} \propto U_M$. The logic is intuitive: explore when uncertain.

However, AAT's underlying Lyapunov dynamics (`#result-persistence-condition`) force a *second, structurally distinct* exploration drive: the **Survival Imperative**. This segment derives that an agent must actively seek low-noise (high CIY) observations not just to learn, but to physically survive environmental drift when its model is *confident*.

The Persistence-Bounded Objective.

The agent's primary goal is to maximize pragmatic value: $\max_\pi \mathbb{E}_{a\sim\pi}[Q_O(a)]$

The agent is subject to the structural persistence condition ($\alpha R > \rho^{\text{eff}}$), meaning it only survives if its steady-state mismatch $R^* < R$. Under Model D dynamics, $R^* = \rho^{\text{eff}}/(\eta^* c_{\min})$. Using the optimal update gain $\eta^* = \frac{U_M}{U_M+U_o}$, we can rewrite the survival constraint exactly: $\frac{U_M}{U_M+U_o} > \frac{\rho^{\text{eff}}}{Rc_{\min}}$

Solving for U_o, we find the maximum observation noise the agent can tolerate without destabilizing: $U_o^{\max} = U_M\left(\frac{Rc_{\min}}{\rho^{\text{eff}}} - 1\right)$

This creates a strict constraint on the agent's action selection: $\mathbb{E}_{a\sim\pi}[U_o(a)] \le U_o^{\max}(M_t)$

The Tragedy of the Confident Agent.

Notice the proportionality: $U_o^{\max} \propto U_M$.

As $U_M \to 0$ (the agent becomes highly confident), $U_o^{\max} \to 0$. The survival constraint becomes infinitely tight.

If the environment is drifting ($\rho^{\text{eff}} > 0$), a confident agent inherently ignores noisy observations ($\eta^* \to 0$). To survive the drift, the agent *must* force an update. The only way to penetrate its own stubbornness is to generate an observation so pristine (low U_o) that the agent is forced to correct its course. Thus, a confident agent in a drifting world is *forced* to explore (seek low U_o) simply to survive.

Conversely, an uncertain agent (high U_M) has a loose constraint (large $U_o^{\max}$). It updates easily ($\eta^* \to 1$), so it can safely exploit noisy states without fear of structural collapse.

Deriving the Survival Lagrange Multiplier λ_{surv}.

Formulating the KKT Lagrangian for this constrained optimization: $\mathcal{L} = \mathbb{E}[Q_O(a)] - \lambda' \cdot (\mathbb{E}[U_o(a)] - U_o^{\max})$

Derivation (survival-exploration).

The shadow price λ' scales as $1/(U_o^{\max} - \mathbb{E}[U_o])$. At the boundary, $\lambda' \propto 1/U_o^{\max}$, yielding: $\lambda' \propto \frac{\rho^{\text{eff}}}{(Rc_{\min} - \rho^{\text{eff}}) \cdot U_M}$

To align this with the CIY objective, we invoke the structural assumption that Causal Information Yield is inversely proportional to observation ambiguity: $\text{CIY}(a) \propto 1/U_o(a)$. This scalar reciprocal mapping is a heuristic placeholder: the mathematically clean correspondence between the survival-imperative Lagrangian and the CIY/EIG objective is the LMI / matrix-valued lift in `#deriv-causal-ib-lmi`, where the Fisher Information Matrix and a positive-semidefinite Lagrange multiplier Λ replace the scalar substitution below. Readers tracking exactness rather than directional fidelity should treat the rest of this section as the scalar shadow of that LMI derivation; the *honest limit* sub-section below and `#deriv-causal-ib-lmi` together carry the formal load.

Transforming the Lagrangian penalty from U_o to $1/U_o$ (which introduces a non-linear mapping of the multiplier, operating-point dependent), the equivalent survival weight on CIY becomes: $\lambda_{\text{surv}}(M_t) \propto \frac{\rho^{\text{eff}}}{(Rc_{\min} - \rho^{\text{eff}}) \cdot U_M}$

Epistemic Status.

Derived (conditional). The derivation of the $U_o^{\max}$ bound and the KKT multiplier λ' is exact, conditional on the continuous-time Lyapunov bounds from `#result-sector-condition-stability`. The mapping to $\lambda_{\text{surv}} \cdot \text{CIY}$ is *heuristic*, conditional on the assumed structural relationship $\text{CIY}(a) \propto 1/U_o(a)$ and the non-linear transformation of the shadow price.

Discussion.

Two Parallel Exploration Drives. AAT dictates two correlated but distinct motivations for exploration, acting at opposite ends of the uncertainty spectrum: 1. **Epistemic Information Gain ($\lambda_{\text{info}} \propto U_M$):** Explored in `#disc-ciy-unified-objective`. The agent explores to reduce its model uncertainty. Dominates when U_M is high. 2. **Lyapunov Survival Imperative ($\lambda_{\text{surv}} \propto 1/U_M$):** Derived here. The agent explores to force high-fidelity corrections against environmental drift. Dominates when U_M is low.

Resolving the IB β vs ρ conflation. This derivation clarifies a structural tension in the standard IB formulation (`#form-information-bottleneck`). Standard IB natively discards old information as ρ rises because the joint distribution $P(\mathcal{C}_t, O_{t+1})$ decorrelates. The agent's *representational preference* (β) does not need to change. However, the agent's *action policy* is tightly coupled to ρ. As ρ increases, the survival constraint tightens ($U_o^{\max} \downarrow$), forcing the Lagrange multiplier λ_{surv} to spike. The environment dictates the representation natively; survival dictates the policy.

Connection to Active Inference. This derivation completely sidesteps the Active Inference "preferences-as-priors" assumption (the dark room problem). The survival exploration drive exists purely because failing to bound U_o mathematically guarantees structural collapse ($\alpha R < \rho^{\text{eff}}$). Exploration is derived as a physical survival imperative, not a psychological preference.

Honest limit: the scalar formulation admits a "blank wall" attack. The survival constraint as derived bounds the *scalar* observation-noise expectation $\mathbb{E}_\pi[U_o(a)] \leq U_o^{\max}$ — it cares only about the magnitude of U_o, not its alignment with the directions in which ρ^{eff} is acting. In multidimensional environments where some coordinates of the drift are inactive, an action that selects an observation channel orthogonal to the drift (a constant signal — the "blank wall") drives U_o to near-zero in that subspace and satisfies the scalar survival condition, while the agent's prediction error in the drifting subspace grows without bound because the observations carry no information about the drifting coordinates. The scalar reduction collapses the dimensionality that matters.

The scalar derivation here remains correct in its scope (1D environments and isotropic-disturbance multidimensional environments where any low-noise observation suffices), and the qualitative dual-drive picture survives the lift to higher dimensions. The full multidimensional resolution — upgrading the scalar inequality to a Linear Matrix Inequality on the Fisher Information Matrix, with a positive-semidefinite matrix Lagrange multiplier Λ that distinguishes by direction rather than magnitude — is in `#deriv-causal-ib-lmi`.

Findings.

Survival-Imperative Exploration as Lyapunov-Forced Drive. **Brief:** A confident agent in a drifting world is mathematically forced to seek high-fidelity (low-noise) observations not because exploration is informationally valuable but because failing to do so violates the structural persistence condition and guarantees collapse. The maximum tolerable observation noise $U_o^{\max}$ scales with U_M — meaning the more confident the agent becomes, the tighter the survival-noise constraint becomes, and the more the agent is forced to actively seek low-ambiguity observations to penetrate its own stubbornness. The result identifies a second exploration drive ($\lambda_{\text{surv}} \propto 1/U_M$) structurally distinct from and complementary to the standard epistemic information-gain drive ($\lambda_{\text{info}} \propto U_M$), with the two acting at opposite ends of the uncertainty spectrum.

Impact: Provides AAT with a derivation of *why exploration is required* that does not route through epistemic-value-of-information arguments and therefore does not inherit the dark-room-problem objection that has dogged active-inference accounts. The survival imperative falls out of the same Lyapunov persistence condition that grounds the rest of Section II — exploration is forced by the same mathematics that forces the agent's adaptive responsiveness, not added as a separate motivational layer. The dual-drive picture also resolves a structural tension in standard Information Bottleneck formulations between the representational β and the environmental drift rate ρ: the environment dictates representation natively (IB's joint distribution decorrelates as ρ rises); survival dictates policy (the constraint tightens as ρ rises). The scalar form admits a "blank wall" attack, resolved in `#deriv-causal-ib-lmi` by lifting the constraint to an LMI on the Fisher Information Matrix; the dual-drive picture survives the lift in matrix form.

Novelty Claim: *Claim differentiation* on the structural source of agentic exploration. Active-inference work derives exploration from epistemic value inside expected free energy; bounded-rationality and information-cost frameworks derive exploration from utility-shaped information processing. This finding derives exploration from Lyapunov-persistence collapse — a structural-stability argument rather than a value-of-information argument — and identifies a second exploration multiplier inversely proportional to model uncertainty, complementary to the epistemic drive that scales directly with it.

Related Work:

- Friston, Rigoli, Ognibene, Mathys, FitzGerald & Pezzulo 2015, "Active inference and epistemic value," *Cognitive Neuroscience* 6:187–214 (published 2015, found 2026-04 via Undermind Pillar 1 report) — *adjacent literature with structural contrast* — derives exploration from epistemic value inside expected free energy; the present finding's exploration drive is forced by Lyapunov bounds rather than preferences-as-priors, sidestepping the dark-room objection.
- Friston, Samothrakis & Montague 2012, "Active inference and agency: optimal control without cost functions," *Biological Cybernetics* 106:523–541 (published 2012, found 2026-04) — *adjacent literature* — broader active-inference framing within which the EFE epistemic-value account sits; same structural-vs-preference contrast.
- Ortega & Braun 2012, "Thermodynamics as a theory of decision-making with information-processing costs," doi:10.1098/rspa.2012.0683 (published 2012, found 2026-04) — *adjacent literature* — bounded-rationality framework deriving exploration-like behavior from information-processing costs; the present finding derives a structurally distinct second drive (Lyapunov survival) without inheriting the bounded-rationality objective form.

- Genewein, Leibfried, Grau-Moya & Braun 2015, "Bounded Rationality, Abstraction, and Hierarchical Decision-Making," *Frontiers Robotics AI* 2:27 (published 2015, found 2026-04) — *adjacent literature* — hierarchical extension of Ort12c; same structurally-different motivational source.
- Tishby, Pereira & Bialek 2000, "The information bottleneck method" (published 2000, found 2024) — *formal antecedent* — the standard IB formulation against which the β-vs-ρ tension is resolved; the survival imperative provides the missing policy-side coupling that standard IB does not natively carry.

Search Log:

- 2026-04 (*nominally comprehensive on the active-inference / bounded-rationality contrast*, via `ref/Novelty_defense_and_integration.md` Pillar 1): The Undermind report identified Friston 2015 as the canonical EFE-epistemic-value account against which this finding's structural-stability source for exploration differentiates, and the Ortega/Genewein bounded-rationality line as the closest information-cost-shapes-policy precursor. The verdict on Pillar 1 as a whole was *Conceptual Precursor* (High confidence) for the no-go half (`#der-causal-insufficiency-detection`); the survival-imperative half is the constructive complement, framed as differentiation from epistemic-value exploration rather than as a no-go contribution.
- 2025 (*intuition-only*): Pre-Undermind expectation was that the closest priors would be in active-inference (Friston) and Information Bottleneck (Tishby) literatures; this was confirmed and refined by the comprehensive search.

Working Notes.

- **Empirical validation.** The derivation is validated in `spikes/track-b-nonlinear-sims/variants/variant_causal_ib.py` (results: `variant_causal_ib_results.md`). In a highly volatile environment ($\rho = 0.5$) where exploitation produces ambiguous observations ($U_o = 100.0$), a purely greedy agent ($\lambda = 0$) suffers a 0% survival rate because its update gain collapses.
- **LMI generalization landed.** The multidimensional resolution is in `#deriv-causal-ib-lmi` (matrix Lagrangian on the Fisher Information Matrix; blank-wall attack resolved by complementary slackness on direction). Spike trail: `spikes/spike-causal-information-bottleneck.md` §7 (critique) and `spikes/spike-causal-ib-lmi-repair.md` (LMI derivation).
- **Cross-reference to NeurIPS Paper 1.** The survival-imperative drive is formalized at theorem grade in NeurIPS 2026 Paper 1 ("Tragedy of the Confident Agent: Forced Exploration for Survival in Drifting Environments", `~/src/neurips/01-tragedy-confident-agent/`): (i) §3 Lemma 2.1(ii) supplies the worst-case finite **exit-time bound** $T_{\text{exit}}(\delta_0) \leq \frac{1}{\alpha+\gamma} \log \frac{\rho/(\alpha+\gamma) - \|\delta_0\|}{\rho/(\alpha+\gamma) - R}$ that the controller-design overshoot is justified against; (ii) §5 Mechanism Step 4 + Appendix-KKT resolves the **shadow-price / controller-divergence puzzle** — the static-program LP multiplier is finite, while the survival-margin controller's λ_{surv} divergence is a controller-design overshoot ensuring loop closure inside the exit-time window (divergence is on the environment-side clock, not the controller-side multiplier); (iii) §App-A.7 / `#prop-fagp` names the **F-A-G-P enforcement framework** (strict-Feasibility / drift-Aligned bonus / Gain dominance / argmax-Presentable) as the four-condition operational decomposition for valid survival-margin controller family members. See `msc/neurips-back-integration-2026-05-08.md` §1 Paper 1 entries 1, 3, 4 for the cross-mapping.

Appendix

Derivation: Linear Matrix Inequality Form of the Causal-IB Survival Bound

In #deriv-causal-ib-exploration, the survival-imperative exploration drive was derived in scalar form: bound $\mathbb{E}_\pi[U_o(a)] \leq U_o^{\max}$ to keep the agent's mismatch within survivable bounds. The scalar bound admits a "blank wall" attack — actions that minimize U_o in a subspace orthogonal to the drift (e.g., observing a constant signal uncorrelated with the environmental drift) satisfy the survival math without probing the drifting coordinates, and the agent's prediction in those subspaces diverges unboundedly. This segment lifts the survival constraint to a Linear Matrix Inequality on the Fisher Information Matrix, replacing the scalar shadow price with a positive-semidefinite matrix Lagrange multiplier Λ that distinguishes by direction, not just by magnitude.

Formal Expression.

Multidimensional mismatch dynamics.

The mismatch state $\delta_t \in \mathbb{R}^n$ evolves under linear-Gaussian dynamics with action-dependent observation noise:

Definition (multidim-mismatch).

$$\delta_{t+1} = (I - K_t H)\delta_t + w_t - K_t v_t(a_t) \tag{.1}$$

where $w_t \sim \mathcal{N}(0, Q_\rho)$ is the environmental drift, $v_t \sim \mathcal{N}(0, R_o(a_t))$ is the observation noise for action a_t, and the optimal Kalman gain is $K_t = P_t H^T (H P_t H^T + R_o(a_t))^{-1}$ with P_t the agent's prior uncertainty covariance.

The steady-state mismatch covariance Σ_δ satisfies the discrete-time algebraic Riccati equation. *Survival* requires the spectral bound

$$\lambda_{\max}(\Sigma_\delta) < R^2 \tag{.2}$$

— the matrix lift of the scalar bound $R^* < R$ from #result-persistence-condition.

Fisher Information Matrix as matrix CIY.

For the linear-Gaussian observation model, the action-conditional Fisher Information Matrix is

Definition (matrix-ciy).

$$\mathcal{I}_o(a) = H^T R_o(a)^{-1} H. \tag{.3}$$

Where the scalar formulation in `#disc-ciy-unified-objective` defined CIY as a surrogate satisfying $\mathrm{CIY}(a) \propto 1/U_o(a)$, the multidimensional formulation identifies CIY directly with the action-conditional FIM. The CIY-U_o surrogate-mapping layer of approximation in the scalar derivation is removed at this layer; the residual CIY-vs-EIG concern (FIM equals EIG-rate only under Gaussian-linear conditions) persists as before.

The information-form Kalman update is additive in the FIM:

$$P_{t+1|t+1}^{-1} = P_{t+1|t}^{-1} + \mathcal{I}_o(a_t), \tag{.4}$$

where $P_{t+1|t} = AP_tA^T + Q_w$ is the prediction step that absorbs process noise. The simplification $P_{t+1}^{-1} = P_t^{-1} + \mathcal{I}_o(a_t)$ used in some passes refers to the information-update step alone; the full step adds the prediction contribution. The two coincide once the analysis is in steady state.

LMI survival constraint.

The discrete-time algebraic Riccati equation admits a stabilizing steady-state solution with $\lambda_{\max}(\Sigma_\delta) < R^2$ if and only if the action policy π provides Fisher information that dominates the unstabilizable modes of the system drift. Specifically, the policy must satisfy the Linear Matrix Inequality

Derived (lmi-survival).

$$\mathbb{E}_{a\sim\pi}[\mathcal{I}_o(a)] \succeq \mathcal{I}_{\min}(Q_\rho, A, R^2) \tag{.5}$$

where $\mathcal{I}_{\min}$ is the symmetric positive-definite lower bound on the FIM required to keep the steady-state covariance's largest eigenvalue below R^2. The closed form for $\mathcal{I}_{\min}$ as a function of (Q_ρ, A, R^2) is given by the steady-state DARE condition — standard control-theory machinery (Boyd, Ghaoui, Feron & Balakrishnan 1994 Ch. 3–5; Anderson & Moore 1979). AAT's contribution at this layer is the framing: $\mathcal{I}_{\min}$ enters as the *survival-imperative* Fisher-information floor, not as a controller-design parameter.

Tensor Lagrangian.

The agent maximizes pragmatic value under the LMI constraint:

Derived (tensor-lagrangian).

$$\max_\pi \mathbb{E}_\pi[Q_O(a)] \quad \text{s.t.} \quad \mathbb{E}_\pi[\mathcal{I}_o(a)] \succeq \mathcal{I}_{\min}. \tag{.6}$$

The Lagrangian for a matrix-inequality-constrained problem uses a positive-semidefinite matrix Lagrange multiplier $\Lambda \succeq 0$:

$$\mathcal{L}(\pi, \Lambda) = \mathbb{E}_\pi[Q_O(a)] + \mathrm{Tr}(\Lambda \cdot (\mathbb{E}_\pi[\mathcal{I}_o(a)] - \mathcal{I}_{\min}))\,. \tag{.7}$$

KKT conditions are primal feasibility ($\mathbb{E}_\pi[\mathcal{I}_o(a)] \succeq \mathcal{I}_{\min}$), dual feasibility ($\Lambda \succeq 0$), and complementary slackness ($\mathrm{Tr}(\Lambda \cdot (\mathbb{E}_\pi[\mathcal{I}_o(a)] - \mathcal{I}_{\min})) = 0$). The optimal action selection rule is

$$a_t \in \arg\max_a \left[\, Q_O(a) + \mathrm{Tr}(\Lambda \cdot \mathcal{I}_o(a)) \,\right]. \tag{.8}$$

The trace inner-product $\mathrm{Tr}(\Lambda \cdot \mathcal{I}_o(a))$ is the **matrix exploration bonus** — the multidimensional analog of $\lambda_{\mathrm{surv}} \cdot \mathrm{CIY}(a)$ from the scalar derivation.

Resolution of the blank-wall attack.

By complementary slackness, Λ has support only in eigendirections where the LMI constraint binds — i.e., directions where the agent's accumulated FIM sits at the threshold $\mathcal{I}_{\min}$. In non-drifting eigendirections of Q_ρ, $\mathcal{I}_{\min}$ contributes zero (no information needed there to survive), so Λ has zero weight in those directions.

Derived (blank-wall-resolution).

For a blank-wall action — high $\mathcal{I}_o(a)$ in a non-drifting eigendirection — the trace product $\mathrm{Tr}(\Lambda \cdot \mathcal{I}_o(a))$ evaluates to zero: the action's FIM eigenvalues lie outside Λ's support. The agent receives no exploration bonus for the blank-wall action, even though the action would satisfy the scalar magnitude bound. **The matrix Lagrangian distinguishes by direction, not just by magnitude**, mathematically forbidding the trivial-exploration solutions that the scalar form admitted.

Scalar reduction.

In the 1D case, H and $R_o(a)$ are scalars, $\mathcal{I}_o(a) = R_o(a)^{-1}$, and the LMI degenerates to the scalar inequality $\mathbb{E}_\pi[R_o(a)^{-1}] \geq \mathcal{I}_{\min}$ which, with $\mathcal{I}_{\min}^{-1} = U_o^{\max}$, recovers the scalar survival constraint of `#deriv-causal-ib-exploration`. The matrix multiplier Λ collapses to a scalar shadow price coinciding with λ_{surv}, and the matrix exploration bonus reduces to $\lambda_{\mathrm{surv}} \cdot \mathrm{CIY}(a)$. The multidimensional derivation is structurally consistent with — and properly subsumes — the scalar one.

Derived (scalar-recovery).

Tragedy of the Confident Agent — matrix form.

As the agent's prior covariance P_t shrinks in a particular eigendirection v, the agent's confidence in that direction grows. If v also has support in Q_ρ (a drifting eigendirection), the survival LMI requires $\mathcal{I}_{\min}$ to grow along v to compensate — a confident agent in a drifting direction must source information specifically along that direction, because its low P_t in v means its update gain is small there and slow to absorb new information. By complementary slackness, the matrix shadow price Λ develops a large principal eigenvalue along v, forcing the agent to choose actions whose $\mathcal{I}_o(a)$ has large support on v. The qualitative tragedy survives the lift: confident agents in drifting worlds are mathematically forced to probe the drifting eigendirections specifically, not just to minimize scalar noise.

Derived (matrix-confidence-tragedy).

Epistemic Status.

Derived (conditional). The structural derivation is exact under named conditions:

1. **Riccati stabilizability.** The discrete-time algebraic Riccati equation admits a stabilizing steady-state solution. Standard machinery: Boyd, Ghaoui, Feron & Balakrishnan 1994 §3–5 and Anderson & Moore 1979.
2. **Linear-Gaussian observation model.** The observation $o = Hx + v$ with $v \sim \mathcal{N}(0, R_o(a))$. Non-Gaussian or nonlinear models require local linearization or extended-Kalman treatment.

3. **Information-form Kalman in steady state.** The $\mathcal{I}_o$-additivity used in the FIM update reflects the information-update step alone; the prediction step's $AP_tA^T + Q_w$ contribution is absorbed into the steady-state Riccati analysis. Transient regimes admit the constraint shape but with $\mathcal{I}_{\min}$ a function of the transient covariance.

4. **DARE-imported $\mathcal{I}_{\min}$.** The closed-form expression for $\mathcal{I}_{\min}(Q_\rho, A, R^2)$ via the DARE is theorem-imported standard control theory. AAT's contribution is the framing — $\mathcal{I}_{\min}$ enters as the survival-imperative FIM floor, with Λ as the directional exploration-bonus shadow price — not the DARE solution itself.

The matrix-CIY mapping ($\mathcal{I}_o(a)$ as multidimensional CIY) is more rigorous than the scalar version: where the scalar formulation needed CIY $\propto 1/U_o$ as an assumed structural mapping, the matrix formulation directly identifies CIY with the action-conditional FIM, removing one of the two layers of approximation in `#disc-ciy-unified-objective`. The CIY-vs-EIG surrogate concern remains: FIM is the EIG-rate for Gaussian linear models but diverges from EIG under non-Gaussian or non-local conditions.

Discussion.

Connection to AAT's existing matrix-form machinery. The LMI repair sits naturally within several existing AAT components rather than being imported from outside:

- `#deriv-fisher-whitened-update-rule` derives Fisher-information-based credit assignment as AAT-internal under the (PI) parameterization-invariance axiom + Čencov 1982 uniqueness. The FIM appearing in the LMI is the same Fisher metric, used here for survival rather than for credit assignment — two roles for one geometric object.
- `#disc-additive-coordinate-forcing` lists Čencov-invariance as the 4th primary instance of the meta-pattern. The LMI's reliance on the FIM sits cleanly within that meta-pattern: the Fisher metric is forced by AAT-internal axioms, not adopted ad hoc.
- `#result-contraction-template` carries matrix-form sector machinery (Lohmiller-Slotine metric formulation) for Section III composition. The LMI's discrete-time Riccati is the control-theoretic analog of the contraction-metric framework — both are matrix-form sector conditions on multivariate dynamics.
- `#def-adaptive-tempo` will need a tensor extension to $\mathcal{T}$ for full alignment with this segment's directional structure; the LMI gives an explicit demand for tensor-valued adaptive rates.

Two parallel exploration drives — matrix lift. The scalar dual-drive distinction in `#disc-ciy-unified-objective` ($\lambda_{\text{info}} \propto U_M$ epistemic vs $\lambda_{\text{surv}} \propto 1/U_M$ Lyapunov-survival) generalizes to two matrix-valued drives that compose additively:

- Λ_{info}: epistemic information-gain matrix scaled by P_t. Dominates eigendirections where P_t is large (high uncertainty in some direction) — agent explores to learn that direction.
- Λ_{surv}: Lyapunov survival-imperative matrix derived above, with eigenvalue support where the LMI constraint binds. Dominates when P_t is small along a drifting eigendirection — agent explores to track the drift it's about to lose grip on.

The agent's total exploration shadow price is $\Lambda_{\text{info}} + \Lambda_{\text{surv}}$, with the relative direction-by-direction weights set by where the agent sits in (P_t, Q_ρ) space.

Connection to Active Inference. Same dark-room sidestep as the scalar version: the matrix-form survival imperative is forced by Lyapunov bounds, not preferences-as-priors. The matrix lift sharpens the argument by ruling out the non-exploratory "blank wall" trivial solutions that the scalar form admitted.

Empirical signature. A 2D drift environment with separable drifting/non-drifting subspaces is the simplest test: a scalar-Lagrangian agent succumbs to blank-wall traps; an LMI agent — with Λ developing weight specifically in the drifting direction — must select actions whose $\mathcal{I}_o$ aligns with the drift. The directional discrimination is empirically visible in the agent's action-selection histogram across the two subspaces.

Findings.

Matrix Lift of the Survival-Imperative Constraint via Fisher-Information LMI. **Brief:** Imagine an agent that's supposed to track something changing but only checks for new information by staring at a blank wall. A scalar version of the survival argument would let this work — the agent satisfies "I gathered enough information" without gathering information about the thing that's actually moving. This result fixes the math by making the argument directional: the agent has to gather information in the *directions* where the world is changing, not just gather some-information-anywhere. The scalar Causal-IB survival-imperative drive lifts to a Linear Matrix Inequality on the Fisher Information Matrix, with a positive-semidefinite matrix Lagrange multiplier Λ that distinguishes by direction. Complementary slackness mathematically forbids "blank wall" actions that satisfy the scalar bound by sourcing information in non-drifting subspaces. The fix uses Fisher-geometric machinery already AAT-internal via `#deriv-fisher-whitened-update-rule` and `#disc-additive-coordinate-forcing`'s 4th instance.

Impact: Closes a structural failure of the scalar derivation that admitted trivially-satisfying-the-math actions (constant signals, walls, etc.) without actually probing the drifting modes the agent must track to survive. The matrix lift demonstrates that the same Fisher geometric object plays multiple structurally-distinct roles — credit assignment via `#deriv-fisher-whitened-update-rule`, coordinate-forcing via `#disc-additive-coordinate-forcing`, and now survival via the LMI here. The "Tragedy of the Confident Agent" insight survives the lift in sharper form: the matrix shadow price's eigenstructure picks out the directions where confidence has outpaced incoming information, mathematically forcing the agent to probe specifically those directions. The scalar reduction recovers `#deriv-causal-ib-exploration`'s constraint as the 1D special case, establishing structural consistency.

Novelty Claim: *Claim differentiation* on the directional discrimination of the survival-imperative exploration drive. The scalar finding in `#deriv-causal-ib-exploration` differentiated AAT's structural-stability source for exploration from active-inference's epistemic-value source; this finding sharpens that differentiation by ruling out the trivial-exploration solutions that the scalar form admitted, demonstrating that the structural-stability source is genuinely directional rather than scalar. The DARE-imported $\mathcal{I}_{\min}$ and the matrix-Lagrangian KKT machinery are standard control-theoretic and convex-optimization apparatus; AAT's contribution at this layer is the framing that places $\mathcal{I}_{\min}$ as a survival-imperative Fisher-information floor and Λ as the directional exploration-bonus shadow price.

Related Work:

Table .1

ASF concern	*Prior-art language*	*Relationship / Positioning*
LMI / matrix-inequality formulation of stability constraints	Boyd, El Ghaoui, Feron & Balakrishnan 1994, *Linear Matrix Inequalities in System and Control Theory*, SIAM (published 1994, found 2025)	*formal antecedent* — adopted machinery; the LMI form, KKT for matrix-inequality-constrained optimization, and the DARE-imported stabilizability condition are standard control-theoretic apparatus
Discrete-time algebraic Riccati equation for steady-state Kalman survival	Anderson & Moore 1979, *Optimal Filtering*, Prentice-Hall (published 1979, found 2025)	*formal antecedent* — DARE provides the closed form for $\mathcal{I}_{\min}(Q_\rho, A, R^2)$
Fisher Information Matrix as the natural geometric object for inference	Čencov 1982, *Statistical Decision Rules and Optimal Inference* (published 1982, found 2024)	*formal antecedent* — Čencov uniqueness forces the Fisher-Rao metric under (PI) parameterization-invariance, integrated AAT-internally via `#disc-additive-coordinate-forcing`'s 4th instance

ASF concern	*Prior-art language*	*Relationship / Positioning*
Exploration as epistemic-value drive (alternative motivational source)	Friston, Rigoli, Ognibene et al. 2015, "Active inference and epistemic value" *Cognitive Neuroscience* 6:187–214 (published 2015, found 2026-04 via Undermind Pillar 1 report)	*adjacent literature with structural contrast* — the matrix-form survival imperative is forced by Lyapunov bounds, not preferences-as-priors. The matrix lift sharpens the contrast by ruling out the non-exploratory blank-wall solutions that the scalar form admitted
Information-form Kalman filter / additive FIM update	Standard Kalman filter literature; Anderson & Moore 1979 (published 1979, found 2025)	*formal antecedent* — the additivity $P^{-1}_{t+1\|t+1} = P^{-1}_{t+1\|t} + \mathcal{I}_o(a_t)$ used in the LMI derivation is the information-form update step
Scalar survival-imperative drive being lifted	`#deriv-causal-ib-exploration` (AAT-internal, 2025)	*upstream segment being matrix-lifted* — the LMI repair preserves the dual-drive structure in matrix form ($\Lambda_{\text{info}} + \Lambda_{\text{surv}}$) and resolves the blank-wall structural failure flagged in the scalar segment's Discussion

Search Log:

- 2026-04 (*nominally comprehensive on the exploration-source contrast*, via `ref/Novelty_defense_and_integration.md` Pillar 1): The Undermind report on Pillar 1 ("Causal insufficiency and forced exploration") established the active-inference EFE framing as the canonical alternative source for exploration drives. This finding's matrix-Lagrangian LMI is the directional sharpening of the scalar derivation in `#deriv-causal-ib-exploration` and inherits that segment's differentiation posture against EFE-style epistemic-value accounts.
- 2025 (*targeted*): Boyd et al. 1994 and Anderson & Moore 1979 confirmed as the formal antecedents for the LMI / DARE machinery; Čencov 1982 confirmed as the formal antecedent for the FIM-as-natural-geometric-object move via the (PI) axiom. AAT's contribution at this layer is framing rather than apparatus.
- 2025 (*intuition-only*): Pre-search expectation was that the LMI form and DARE machinery were standard and would not be novel apparatus; this was confirmed. The novel content is the placement — using the FIM as a survival floor rather than a controller-design parameter, and integrating directional exploration with the broader Fisher-geometric meta-pattern.

Working Notes.

- **Tensor adaptive tempo.** `#def-adaptive-tempo` now carries a Tensor extension sub-block ($\mathcal{T} = \sum_k \nu^{(k)} K^{(k)}$ with $K^{(k)} = (H_M + H_L^{(k)})^{-1} H_L^{(k)}$ from `#deriv-fisher-local-update-gain`) that gives the per-direction primitive this segment's LMI machinery needs. Promotion of the scalar persistence and adversarial-tempo results to invoke the tensor form directly (rather than per-coordinate scalars) remains queued — flagged in `TODO.md` under follow-on cycle work; the per-direction primitive is in place even if downstream summary results still read in scalar.
- **Worked 2D example.** A concrete 2D drift environment showing $\mathrm{Tr}(\Lambda \mathcal{I}_o)$ distinguishing wall-channel from drift-channel actions would sharpen the blank-wall resolution. Queued.
- **2D simulation.** Generalization of `spikes/track-b-nonlinear-sims/variants/variant_causal_ib.py` to 2D with separable drifting/non-drifting subspaces would empirically validate the matrix Lagrangian's directional discrimination. Queued.
- **EIG vs FIM gap.** For non-Gaussian or non-local observation models, FIM diverges from EIG. The CIY-vs-EIG concern in `#disc-ciy-unified-objective` Epistemic Status persists; the matrix lift removes the CIY-U_o surrogate but not the CIY-EIG surrogate.
- **Reasoning trail.** `spikes/spike-causal-ib-lmi-repair.md` records the spike that surfaced the LMI repair from the scalar segment's blank-wall critique.

- **Cross-reference to NeurIPS Paper 1.** The matrix-LMI lift is formalized at theorem grade in NeurIPS 2026 Paper 1 ("Tragedy of the Confident Agent: Forced Exploration for Survival in Drifting Environments", `~/src/neurips/01-tragedy-confident-agent/`). The paper carries: (i) **Per-action LMI strengthenings** at §App-A.6 (`#thm-per-action-lmi`, `#thm-per-action-lmi-iid`) — pathwise survival under arbitrary action schedules with parallel high-probability bound via matrix-Chernoff concentration (`#lem-fim-concentration`), closing the pathwise-vs-expectation gap that this segment's steady-state DARE form does not surface explicitly. (ii) **F-A-G-P enforcement framework** at §App-A.7 (`#prop-fagp`) — strict-Feasibility / drift-Aligned bonus / Gain dominance / argmax-Presentable, the four-condition operational decomposition for what makes a survival-margin controller a valid family member; the range-containment hypothesis range(Λ) $\subseteq$ range($\mathcal{I}_{\min}$) that denies blank-wall updates LMI-survival reward is the structural anchor. (iii) **Anisotropic-drift refinement** at §App-C — closed-form critical exponent p_{crit} for the $\Lambda \propto \mathcal{I}_{\min}^{p}$ rule, diverging as $\sigma_x/\sigma_y \to 1$; directional discrimination *rests on* anisotropic drift and the argmax flips at the isotropic limit. The scalar-isotropic case is the boundary where the matrix-LMI's distinguishing power degrades to the scalar form. (iv) **2D Gaussian empirical anchor** with the `wall_extreme` ε-blank-wall action and the three-controller comparison (greedy / scalar / LMI) — the matrix-LMI controller selects drift-aligned probing actions and survives at 100% across drift levels at zero pragmatic reward (a *bistability* of the discrete action set, explicitly characterized in §App-B to prevent overclaiming as a smooth survival-vs-reward Pareto frontier) where scalar surrogates collapse to 0% via blank-wall degenerate updates. See `msc/neurips-back-integration-2026-05-08.md` §1 Paper 1 entries 2, 3, 5, 6, 7 for the cross-mapping.

Appendix

Detail: Observation: Section I Validation Simulations

Six simulation variants validated, refined, and extended the analytical predictions from Section I, forcing regime splits and discovering scaling laws that the analytical derivations left ambiguous.

Overview.

The track-b simulation program tested Section I's analytical predictions about mismatch dynamics, adversarial coupling, observation noise, correction function robustness, and anisotropic persistence. The simulations modeled discrete-time mismatch dynamics as AR(1) processes with five correction functions (linear, saturating, threshold/dead-zone, sigmoid, structural breakdown), sweeping parameter spaces across gain, disturbance rate, coupling strength, observation noise, and dimensional structure.

The simulations were theory-shaping, not merely confirmatory. In particular:

- The adversarial tempo exponent turned out to be regime-dependent (deterministic drift vs. stochastic noise, coupling-dominant vs. non-coupling-dominant), forcing a regime split that the mismatch ODE left ambiguous.
- Observation noise was found to gate the adversarial advantage, partially recoverable by optimal gain – a finding that had no analytical treatment before the simulations.
- The scalar persistence condition was shown to overestimate adaptive capacity by 72% in anisotropic systems, motivating the per-dimension persistence condition.

Variant Summary.

Table .1

Variant	*Disturbance Model*	*Tested*	*Key Finding*	*Theory Impact*	*Promoted Segment*
A	**D** (deterministic drift)	Deterministic drift coupling; coupling-dominance sweep	Exponent $b \to 2.0$ in coupling-dominant limit (confirmed at 1.999)	Validates Model D derivation ($b = 2$)	14.3

Variant	*Disturbance Model*	*Tested*	*Key Finding*	*Theory Impact*	*Promoted Segment*
B	**D+S** (interpolation)	Drift-noise interpolation across correction functions	Smooth transition between drift ($b = 2.0$) and noise ($b = 1.5$) regimes; coupling dominance is the key qualifier, not drift vs. noise per se	Confirmed that the coupling-dominant condition is quantitatively load-bearing	14.3
C	**S** (stochastic AR(1))	Exponent vs. gain (η) in stochastic model	Exponent drops toward 0.5 as $\eta \to 0$ (away from coupling dominance at fixed q_{base}); discrete AR(1) exponent never exceeds 1.5	Validates Model S derivation ($b = 3/2$)	14.3
D	**D vs. S** (separated)	Exponent vs. base noise (q_{base}) at fixed η	Continuous ODE exponent → 2.0 (Model D), discrete AR(1) exponent → 1.5 (Model S), as $q_{\text{base}} \to 0$	Definitively separated the two asymptotes; validates both derivations	14.3
E	**S** (stochastic AR(1))	Observation noise; optimal gain validation	Observation noise collapses adversarial exponent from ~ 1.0 to ~ 0.2; Riccati-optimal gain restores it to ~ 0.4; 52% mismatch reduction at moderate noise	Observation quality gates tempo advantage; optimal gain (3.6) empirically validated	14.4
F	**S** (stochastic AR(1))	Multi-dimensional anisotropic correction; targeted adversarial attack	Per-dimension theory exact to 4 significant figures; scalar tempo overestimates by 72%; targeted attack amplifies advantage by 17%	Validates Model S per-dimension steady state	14.5
Hafez bridge	**S** (stochastic AR(1))	Bi-predictability P vs. AAT mismatch in adversarial and non-adversarial settings	P measures coupling architecture (scale-invariant); mismatch measures coupling performance; P is blind to adversarial dynamics	P and AAT mismatch are complementary diagnostics; H_b (agent opacity) has no direct AAT analog – potential gap for multi-agent work	–

Methodology.

Discrete mismatch dynamics. The environment follows a random walk $x_{t+1} = x_t + w_t$ with $w_t \sim N(0, q^2)$. The agent corrects via $\hat{x}_{t+1} = \hat{x}_t + \eta \cdot g(\delta_t)$, yielding the AR(1) mismatch process $\delta_{t+1} = (1 - \eta) \cdot \delta_t + w_t$ for linear g. This is the discrete-time analog of the mismatch ODE $d\|\delta\|/dt = -\mathcal{T} \cdot \|\delta\| + \rho$ from 3.9.

Parameter sweeps. Each variant swept its key parameter(s) across 7–20 values. Monte Carlo: 200 independent trials per parameter point, 10,000–20,000 timesteps per trial, with 2,000–5,000 step burn-in for steady-state convergence. Fixed random seeds for reproducibility.

Exponent fitting. Adversarial exponents were estimated by fitting $\log(\|\delta_B\|/\|\delta_A\|) = a + b \cdot \log(\mathcal{T}_A/\mathcal{T}_B)$ via weighted least squares, with weights inversely proportional to variance of each point's log-estimate. 95% confidence intervals via bootstrap (1,000 samples).

Correction functions. Five functions $g : \mathbb{R} \to \mathbb{R}$ were tested, all satisfying $g(0) = 0$ and $g'(0) = 1$: linear ($g(\delta) = \delta$), saturating ($g(\delta) = \delta/(1 + |\delta|/R)$), threshold ($g(\delta) = \delta \cdot \mathbf{1}[|\delta| > \epsilon]$), sigmoid ($g(\delta) = R \cdot \tanh(\delta/R)$), and structural breakdown ($g(\delta) = \delta \cdot \mathbf{1}[|\delta| < R_{\max}]$).

Simulation code. All code is in `../../spikes/track-b-nonlinear-sims/`. Initial simulations: `sim1_nonlinear_mismatch.py` (single-agent), `sim2_adversarial_coupling.py` (two-agent). Variant extensions: `variants/variant_ab_drift.py`, `variants/variant_cd_regimes.py`, `variants/variant_ef_extensions.py`, `variants/variant_hafez_bridge.py`. Detailed result write-ups: `variants/variant_ab_results.md`, `variants/variant_cd_results.md`, `variants/variant_ef_results.md`, `variants/variant_hafez_results.md`.

Key Findings.

The adversarial exponent is regime-dependent (now derived). The theory now distinguishes two disturbance models (Model D: bounded deterministic, GA-2; Model S: stochastic zero-mean, GA-2S), each producing a different steady-state scaling and therefore a different adversarial exponent:

- **Model D, coupling-dominant:** $\|\delta\|_{ss} = \rho/\mathcal{T}$. Adversarial exponent $b = 2$ (derived from Prop A.1; Variant A confirms at 1.999).
- **Model S, coupling-dominant:** $\|\delta\|_{\text{rms}} = \sigma_w/\sqrt{2\mathcal{T}}$ (from Prop A.1S). Adversarial exponent $b = 3/2$ (derived; Variants C-D confirm at 1.481).
- **Non-coupling-dominant:** Exponent degrades smoothly toward 1.0 (Model D) or 0.5 (Model S) as base disturbance grows relative to adversarial coupling.

The original sim2 result ($b \approx 1.05$) was not a falsification of Corollary 11.2 but a measurement in the wrong regime – stochastic noise coupling at moderate coupling dominance. Variants A–D systematically mapped the full regime space and validated the derived exponents. See 14.3 for the full treatment.

Observation noise gates adversarial advantage. Adding observation noise σ_{obs} to the mismatch signal collapsed the adversarial exponent from ~ 1.0 to ~ 0.2 at $\sigma_{\text{obs}} = 10 \times q_{\text{env}}$ (fixed gain). The Riccati-optimal gain (3.6) partially restored the advantage to ~ 0.4, more than doubling it but not recovering the noise-free level. The optimal gain helps most in the moderate-noise regime, where it achieved a 52% mismatch reduction over fixed gain. See 14.4.

Per-dimension persistence is exact; scalar overestimates. In a 3-dimensional system with 5:1 gain variation across dimensions, the per-dimension AR(1) steady-state prediction matched simulation to 4 significant figures. The scalar persistence condition overestimated aggregate mismatch by 72%, because it averages across dimensions while the weak dimension dominates the L_2 norm. Isotropic gain allocation reduced overall mismatch by 13%; targeted adversarial attack on the weak dimension amplified the mismatch ratio by 17%. See 14.5.

Nonlinear correction creates thresholds, not lower exponents. Under deterministic drift, saturating, sigmoid, and breakdown correction functions did not simply reduce the adversarial exponent. Instead, they produced catastrophic divergence when disturbance exceeded the correction capacity ($\rho > \mathcal{T} \cdot R$). This is the persistence threshold failure from 4.4, observed directly in simulation. The measured "exponents" above 3.0 for these functions were divergence artifacts, not meaningful scaling laws.

Hafez bridge: architecture vs. performance. Bi-predictability P (Hafez et al.) measures the informational architecture of agent-environment coupling; AAT mismatch measures the operational performance. In adversarial settings, P remained nearly constant (~ 0.268) across a 10:1 range of opponent tempo, while the mismatch ratio varied from 0.54 to 5.95. P is scale-invariant after z-score normalization, making it insensitive to adversarial dynamics. The two metrics are complementary: P diagnoses whether the coupling structure is sound; mismatch predicts how well it performs.

Epistemic Status.

Empirical. Simulation results are reproducible (code in `../../spikes/track-b-nonlinear-sims/`, fixed seeds, all results documented in variant write-ups). The key results – regime-dependent exponents, observation noise gating, per-dimension exactness – have been promoted to first-class segments (14.3, 14.4, 14.5) with their own epistemic assessments. This appendix serves as reference for the simulation program as a whole.

Discussion.

Internal validation, not external. The simulations operate in the model's own terms: AR(1) processes with parameterized correction functions, Gaussian noise, and discrete-time dynamics. They validate the mathematical predictions within the model – confirming that the analytical steady-state formulas, the exponent regimes, and the per-dimension theory are correct as statements about these stochastic processes. The gap between "the math predicts X within its own model" and "real agents exhibit X" remains an empirical question for each domain.

What the simulations did not test. The simulations did not test whether the AR(1) mismatch dynamics are a good model for any particular real-world adaptive system. They also did not test cross-dimensional coupling (off-diagonal correction), non-Gaussian disturbances, non-stationary parameters, or multi-agent composition dynamics. The interaction between observation noise and the deterministic-drift exponent regime ($b = 2.0$) was not tested – Variant E used stochastic coupling only.

Theory-shaping role. The most important contribution of the simulations was not confirming predictions but forcing the theory to be more precise. The regime split in the adversarial exponent, the observation-noise gating mechanism, and the per-dimension bottleneck effect were all findings that the analytical derivations initially left ambiguous or unaddressed. The simulations forced the disturbance model split (Model D vs. Model S), which in turn enabled the analytical derivation of both exponents. The simulations now serve as validation of the derived results: Variant A validates $b = 2$ (Model D), Variants C-D validate $b = 3/2$ (Model S), Variant F validates the Model S per-dimension steady state.

Part V

Appendices: Operational Domains

Appendix

Detail: Operationalization — Estimation Procedures

Estimation recipes for core AAT quantities, bridging the measurement gap between formal objects and practical deployment.

Measurement Targets.

Table .1

Quantity	*Role in AAT*	*Typical unit*	*Minimum data needed*
U_M	Model uncertainty (3.6)	domain-specific variance/entropy	Predictive posterior or ensemble spread
U_o	Observation uncertainty (3.6)	sensor variance/noise scale	Channel calibration or residual variance
$\rho_{\text{det}}(t)$	Mismatch injection rate bound, Model D (3.9)	surprise per time	Time-series of mismatch magnitudes; worst-case bounds
σ_w	Disturbance noise scale, Model S (3.9)	surprise per $\sqrt{\text{time}}$	Mismatch innovation variance (residual after correction)
ρ_{delib}	Local mismatch drift during pauses (4.1)	surprise per time	Deliberation windows with no corrective action
α	Lower correction efficiency bound (4.3)	inverse time	Vector mismatch trajectories + correction term
R	Radius where local sector condition holds (4.3)	surprise magnitude	Same as α, plus breakdown detection
$\lVert\delta_{\text{critical}}\rVert$	Functional adequacy threshold (4.4)	surprise magnitude	Task-level performance curve vs mismatch

Estimator Cookbook.

Estimating U_M and U_o. Use the most native uncertainty representation available in the domain:

Table .2

Domain	*U_M estimator*	*U_o estimator*
Kalman / linear Gaussian	Prior predictive variance $P_{t\|t-1}$	Measurement-noise variance R_t (known or EM-estimated)
Conjugate Bayes	Posterior variance / inverse effective sample size	Likelihood variance (or precision inverse)
RL with ensembles	Across-head predictive variance $\mathrm{Var}_i[Q_i(s,a)]$	TD-target noise variance over replay batches
Neural net regression	Ensemble or Laplace posterior variance	Aleatoric head output $\sigma^2(x)$
PID / classical control	State-estimation covariance from observer	Sensor noise from calibration + residual PSD

For the scalar gain heuristic (3.6), normalize to common units and compute:

Operational Definition.

$$\hat{\eta}_t^* = \frac{\hat{U}_{M,t}}{\hat{U}_{M,t} + \hat{U}_{o,t}} \tag{.1}$$

Estimating Disturbance Parameters ($\rho_{\mathrm{det}}, \sigma_w, \rho_{\mathbf{delib}}$). The disturbance model split (Model D vs. Model S; see 3.9, 4.3) requires different estimation procedures depending on the disturbance character.

Identifying the disturbance model. The choice between Model D and Model S is domain-dependent: - **Model D** (bounded deterministic, GA-2): appropriate when the environment changes persistently and directionally. Estimate ρ_{det} from worst-case bounds on mismatch injection rate. Examples: adversary who maneuvers systematically, API that drifts, climate that shifts. - **Model S** (stochastic zero-mean, GA-2S): appropriate when the environment fluctuates unpredictably around a stable mean. Estimate σ_w from residual variance of mismatch innovations. Examples: market noise, sensor noise, random perturbations. - **Mixed environments** use both: estimate ρ_{det} from the persistent component (trend) and σ_w from the residual variance after detrending. The adversarial exponent interpolates smoothly between $b = 2$ (pure drift) and $b = 3/2$ (pure noise); see 14.3.

Let $s_t = \|\delta_t\|$ in surprise units (e.g., negative log-likelihood residual scale).

Model D: Global mismatch injection rate ($\hat{\rho}$_det):

Operational Definition.

$$\hat{\rho}_{\mathrm{det}}(t) = \left[\frac{s_{t+\Delta t} - s_t}{\Delta t} + \hat{\mathcal{T}}_t\, s_t\right]_+ \tag{.2}$$

where $[x]_+ = \max(x, 0)$ and $\hat{\mathcal{T}}$ evaluated at time t is estimated adaptive tempo. Take the supremum (or a high quantile) over the estimation window.

Model S: Noise scale ($\hat{\sigma}\_w$):

Operational Definition.

$$\hat{\sigma}_w^2 = \mathrm{Var}\left[\frac{s_{t+\Delta t} - (1-\hat{\eta}^*)s_t}{\sqrt{\Delta t}}\right] \tag{.3}$$

i.e., the variance of the mismatch innovations (residuals after subtracting the expected correction). For AR(1) processes, this is the innovation variance $\hat{\rho}^2$ directly.

Note on estimation sequencing. Both estimators require $\hat{\mathcal{T}}_t$ (or $\hat{\eta}^*$), estimated from $\hat{\nu}$ and $\hat{\eta}^*$. Estimate the gain and event rate first (from the agent's internal statistics and observation timing), then use these to extract the disturbance parameters from the mismatch trajectory. This sequential structure avoids circularity but introduces sensitivity: errors in $\hat{\mathcal{T}}$ propagate into $\hat{\rho}_{\mathrm{det}}$ and $\hat{\sigma}_w$.

Local pause-window drift for Derived 4.2 ($\hat{\rho}_$delib):

Operational Definition.

$$\hat{\rho}_{\mathrm{delib}} = \operatorname*{median}_{w \in \mathcal{W}_{\mathrm{pause}}} \frac{s_{w,\mathrm{end}} - s_{w,\mathrm{start}}}{\Delta\tau_w} \tag{.4}$$

using windows where corrective action is suspended or effectively delayed.

Estimating α (sector lower bound). A uses $\delta^T F(\mathcal{T}, \delta) \geq \alpha\|\delta\|^2$ for $\|\delta\| \leq R$. Operationally:

1. Estimate $\dot{\delta}_t$ (finite differences or filtered derivative).
2. Compute $\widehat{F}_t = -\dot{\delta}_t + w_t$ where disturbance proxy w_t is estimated from exogenous perturbation channels or residual balancing.
3. Form ratios $r_t = (\delta_t^T \widehat{F}_t)/\|\delta_t\|^2$ on bins of $\|\delta_t\|$.
4. Set conservative lower bound $\hat{\alpha}$ as a low quantile (e.g., 10th percentile) of r_t in the valid region.

Estimating R (valid-region radius). Estimate R as the largest radius for which sector inequality violations remain below tolerance:

Operational Criterion.

$$\hat{R} = \sup\left\{r > 0 \,:\, \Pr\left(\delta^T \widehat{F} < \hat{\alpha}\|\delta\|^2 \,\middle|\, \|\delta\| \leq r\right) \leq \epsilon\right\} \tag{.5}$$

with a chosen violation tolerance ϵ (e.g., 5%).

Estimating $\|\delta_{\text{critical}}\|$. Define a mission-level performance metric J (reward rate, tracking error, service SLA, etc.). Set the critical mismatch threshold where performance crosses a minimum acceptable level $J_{\min}$:

Operational Definition.

$$\|\hat{\delta}_{\text{critical}}\| = \inf\{d : \mathbb{E}[J \mid \|\delta\| = d] < J_{\min}\} \tag{.6}$$

This anchors 4.4 to real task outcomes.

Recommended Estimation Sequence.

1. Fix mismatch representation δ in one consistent unit system (prefer surprise-scale).
2. Estimate U_o from channel physics/calibration; estimate U_M from model uncertainty.
3. Validate gain behavior against 3.6 ($\hat{\eta}^*$ trend checks).
4. Estimate ρ_{delib} from pause windows (4.1) and $\rho(t)$ from full traces.
5. Estimate α and R from local correction dynamics (A).
6. Estimate $\|\delta_{\text{critical}}\|$ from task-performance degradation.
7. Compute derived diagnostics:
   - Tempo margin (Model D): $\hat{\mathcal{T}} - \hat{\rho}_{\text{det}}/\|\hat{\delta}_{\text{critical}}\|$
   - Tempo margin (Model S): $\hat{\mathcal{T}} - n\hat{\sigma}_w^2/(2\|\hat{\delta}_{\text{critical}}\|^2)$
   - Reserve: $\widehat{\Delta\rho^*} = \hat{\alpha}\hat{R} - \hat{\rho}_{\text{det}}$
   - Deliberation feasibility: $\Delta\eta^*(\Delta\tau)\|\delta_{\text{post}}\| - \hat{\rho}_{\text{delib}}\Delta\tau$

Decision-Theoretic Procedures.

Estimating λ (Exploration Price). The exploration weight $\lambda(M_t)$ in the unified policy objective (6.5) prices information in value-equivalent terms:

Table .3

Context	*λ estimator*	*Source*
Finite bandits	Gittins index from dynamic programming	Exact (Gittins 1979)
Linear-Gaussian	Probing cost in quadratic objective	Exact (dual control)
Discrete MDP	$(\text{VoI})^2$/info gain	Information-directed sampling (Russo & Van Roy)
General	$\hat{\lambda} = c \cdot \hat{U}_M/\hat{U}_o$	Heuristic: scale CIY weight by relative uncertainty

For the heuristic: when $U_M \gg U_o$ (highly uncertain model), exploration is cheap relative to exploitation risk, so λ should be large. When $U_M \ll U_o$ (confident model, noisy observations), exploitation dominates. The constant c is domain-specific.

Deliberation Stopping Policy. From 4.1, deliberation of duration $\Delta\tau$ is warranted when $\Delta\eta^*(\Delta\tau) \cdot \|\delta_{\text{post}}\| > \rho_{\text{delib}} \cdot \Delta\tau$. Operationally:

1. Estimate ρ_{delib} from prior pause windows.
2. Before each deliberation episode, estimate $\|\delta_{\text{post}}\|$ as current mismatch + $\rho_{\text{delib}} \cdot \Delta\tau_{\text{planned}}$.
3. Estimate $\Delta\eta^*(\Delta\tau)$ from the diminishing-returns profile of past deliberation episodes.
4. Stop deliberating when the marginal improvement rate $\partial\Delta\eta^*/\partial\Delta\tau$ drops below $\rho_{\text{delib}}/\|\delta_{\text{post}}\|$.

Structural-Switch Trigger. From 4.5, structural adaptation is indicated when parametric convergence leaves a mismatch floor. Operationally:

1. Estimate the current mismatch floor $\|\delta\|_{\text{floor}}$ from converged residual statistics.
2. Estimate post-switch expected mismatch as $\|\delta\|_{\text{new}} \approx \rho/\alpha'$ where α' is the sector bound under the candidate new model class.
3. Estimate transition cost C_{switch}: knowledge loss, retraining time ($\Delta\tau_{\text{switch}}$), and accumulated mismatch during transition ($\rho \cdot \Delta\tau_{\text{switch}}$).
4. Switch when: $(\|\delta\|_{\text{floor}} - \|\delta\|_{\text{new}}) \cdot T_{\text{horizon}} > C_{\text{switch}}$.

Estimator Uncertainty Guidance.

$\hat{\eta}^*$ **(gain estimate).** Approximate variance via the delta method:

$$\text{Var}(\hat{\eta}^*) \approx \hat{\eta}^{*2}(1-\hat{\eta}^*)^2 \left[\frac{\text{Var}(\hat{U}_M)}{\hat{U}_M^2} + \frac{\text{Var}(\hat{U}_o)}{\hat{U}_o^2}\right] \tag{.7}$$

Near 0 or 1, stable (dominated by one source). Near 0.5, most volatile.

$\hat{\rho}$ **(mismatch injection rate).** Finite-difference estimation amplifies noise. Recommend smoothing before differencing. Minimum ~20 observations for stable trend; ~50 for reliable variance. The $[\cdot]_+$ clipping introduces positive bias at small ρ.

$\hat{\alpha}$ **(sector lower bound).** The 10th-percentile approach gives approximately 90% confidence under stationarity. Report quantile level, bin count, and bin width.

$\hat{R}$ **(sector-condition radius).** A sharp drop-off in sector-condition satisfaction is a strong signal; gradual decay suggests the sector condition may not hold cleanly.

Nonstationarity caveat. All estimators assume approximate stationarity over the estimation window. When environment regime changes are suspected (4.5), re-estimate from post-change data only.

Epistemic Status.

This is a *procedures document* — estimation recipes, not theoretical claims. Each estimator inherits the epistemic status of the quantity it targets: the gain estimator's validity depends on 3.6's structural claim; the sector-bound estimator's validity depends on 4.3's assumptions. The estimation procedures themselves are *conditional* on these theoretical foundations and on the stationarity and data-sufficiency assumptions noted above.

Working Notes.

- End-to-end worked examples instantiating the full chain are in (exact) and (approximate).
- The $\hat{\rho}$ estimator's dependence on $\hat{\mathcal{T}}$ creates a sequential estimation chain. Errors compound. An alternative approach: estimate ρ directly from exogenous environmental change measurements when available, bypassing the mismatch trajectory entirely.
- The structural-switch trigger's T_{horizon} (expected time the new model class will remain adequate) is itself uncertain and difficult to estimate. In practice, this is where the agent's M_t does the real work — predicting environmental stability.

Appendix

Worked Example: 1D Active Tracking (Kalman Domain)

Every AAT quantity has an exact Kalman-filter counterpart. This is a *validation* of the formal chain — all quantities are computable in closed form.

System.

The agent tracks scalar state x_t and chooses sensor mode $a_t \in \{L, H\}$:

- Dynamics: $x_{t+1} = x_t + v_t, v_t \sim \mathcal{N}(0, q), q = 0.25$
- Observation: $y_t = x_t + n_t^{(a_t)}$
- Low mode noise: $r_L = 9$; High mode noise: $r_H = 1$
- Event rate: $\nu = 5$ Hz
- High mode has higher energy cost

Chain Instantiation.

Scope (1.5). *Mapping: exact.*

$\Omega_t = x_t$ is partially observed via noisy channel. $\mathcal{A} = \{L, H\}$ is non-empty and causally affects observation quality. Residual uncertainty persists due to process and sensor noise.

Causal Structure (1.8) + CIY (3.7). *Mapping: exact.*

Action precedes observation and changes $P(y_t \mid do(a_t))$ through r_{a_t}. Using low mode as comparator action, the CIY is the interventional KL divergence:

Worked Quantity.

$$\mathrm{CIY}(H) = D_{\mathrm{KL}}\big(P(y \mid do(H)) \,\|\, P(y \mid do(L))\big) \tag{.1}$$

$$= \frac{1}{2}\left[\log\left(\frac{P^- + r_L}{P^- + r_H}\right) + \frac{P^- + r_H}{P^- + r_L} - 1\right] \tag{.2}$$

With $P^- = 4.25$: $\mathrm{CIY}(H) \approx 0.161$ nats.

Model (2.1). *Mapping: exact.*

Model state $M_t = (\hat{x}_{t|t}, P_{t|t})$ — a compression of interaction history with recursive update.

Mismatch (3.4). *Mapping: exact.*

$$\delta_t = y_t - \hat{x}_{t|t-1} \tag{.3}$$

Update Gain (3.6). *Mapping: exact.*

Scalar Kalman gain:

$$K_t = \frac{P_t^-}{P_t^- + r_{a_t}} \tag{.4}$$

With $P^- = 4.25$: $K(H) \approx 0.810$, $K(L) \approx 0.321$.

The exact uncertainty ratio mapping: $U_M = P_t^-$, $U_o = r_{a_t}$, $\eta^* = K_t$.

Exploration (3.7).

Worked Objective.

$$a_t^* = \arg\max_a \left[\mathbb{E}[\mathrm{value}(a) \mid M_t] + \lambda_t\, \mathbb{E}[\mathrm{CIY}(a) \mid M_t]\right] \tag{.5}$$

When uncertainty is high (P^- large), CIY term favors high mode. As uncertainty falls, policy shifts toward low-cost L.

Deliberation Threshold (4.1). Suppose a planning pause of $\Delta\tau = 0.5$ s, with measured $\rho_{\mathrm{delib}} = 0.40$ surprise/s.

Cost during pause: 0.20 surprise units. If $\|\delta_{\mathrm{post}}\| = 0.70$, deliberation is worthwhile when:

$$\Delta\eta^*(0.5) \cdot 0.70 > 0.20 \implies \Delta\eta^*(0.5) > 0.286 \tag{.6}$$

Structural Adaptation (4.5). Assume maneuvering regime change introduces sustained residual autocorrelation and mismatch floor. If estimated valid radius drops to $R = 0.08$ while $R^* = \rho/\alpha = 0.12$, parametric adaptation is no longer adequate ($R^* > R$), triggering model-class change (e.g., constant-velocity → constant-acceleration process model).

Tempo + Persistence (3.8, 4.4). Using action mix 70%H, 30%L:

$$\bar{\eta}^* = 0.7(0.810) + 0.3(0.321) = 0.663 \tag{.7}$$

$$\mathcal{T} = \nu\bar{\eta}^* = 5 \cdot 0.663 = 3.315\ \mathrm{s}^{-1} \tag{.8}$$

With $\rho = 0.18$ surprise/s and $\|\delta_{\mathrm{critical}}\| = 1$:

$$\mathcal{T} > \frac{\rho}{\|\delta_{\mathrm{critical}}\|} \Rightarrow 3.315 > 0.18\ \checkmark \tag{.9}$$

Lyapunov Bounds (4.3, 4.2). *Mapping: exact (derived, not estimated).*

By 4.2, the sector parameter for a scalar Kalman filter is $\alpha = \mathcal{T} = \nu\bar{\eta}^*$:

$$\alpha = \mathcal{T} = 3.315\,\mathrm{s}^{-1} \tag{.10}$$

This is derived from the Kalman gain, not estimated from data — the correction function $F(e) = Ke$ is exactly linear, so $\alpha = K$ per observation and $\alpha = \nu\bar{K} = \mathcal{T}$ in continuous time. With $R = 1.4$ (model class capacity, a domain parameter) and $\rho = 0.18$:

$$R^* = \frac{\rho}{\alpha} = \frac{0.18}{3.315} \approx 0.054 < R \tag{.11}$$

$$\Delta\rho^* = \alpha R - \rho = 3.315(1.4) - 0.18 = 4.46 \tag{.12}$$

The agent is comfortably within its invariant region with substantial adaptive reserve.

Mapping Quality Summary.

Table .1

AAT Concept	*Kalman Mapping*	*Status*
Scope	Exact	Definitional
Causal structure + CIY	Exact	Closed-form KL
Model (M_t as sufficient statistic)	Exact	Kalman state + covariance
Mismatch (δ_t = innovation)	Exact	Standard Kalman innovation
Gain ($\eta^* = K_t$)	Exact	Kalman gain IS uncertainty ratio
Tempo ($\mathcal{T} = \nu\bar{\eta}^*$)	Exact	Closed-form
Persistence condition	Exact	Linear ODE solution
Sector parameter ($\alpha = \mathcal{T}$)	Exact	Derived from Kalman gain via 4.2
Lyapunov bounds (R^*, $\Delta\rho^*$)	Exact	From derived sector parameter + domain R

Epistemic Status.

This is a *worked instantiation*, not a theoretical claim. Every mapping is exact — the Kalman domain is the canonical case where AAT's formal chain has closed-form realizations. The example validates that the formal chain is internally consistent and instantiable.

Working Notes.

- This example uses a 1D system. The tensor-tempo extension (14.5) becomes visible only in multi-dimensional tracking where different state dimensions have different observability.
- The structural adaptation trigger (constant-velocity → constant-acceleration) is manufactured for the example. A more natural test would be a real tracking system with genuine regime changes.

- The Kalman filter is provably optimal for the linear-Gaussian case. Every AAT quantity has not just an analog but the *exact optimal* value. This makes it the strongest validation but also the easiest — the real test is non-Kalman domains (see).

Appendix

Worked Example: Nonstationary Bandit (RL Domain)

AAT's conceptual architecture applies beyond the conjugate-Bayesian regime. The mapping here is *approximate* — it shows that AAT's concepts organize RL phenomena, but the quantitative relationships are structural analogies, not derivations. Where the mapping is exact versus approximate is marked explicitly.

System.

A k-armed bandit ($k = 4$) with slowly drifting reward means:

Formulation.

$$\mu_i(t+1) = \mu_i(t) + w_i(t), \quad w_i(t) \sim \mathcal{N}(0, q), \quad q = 0.01 \tag{.1}$$

- Observation: Pulling arm $a_t = i$ yields reward $r_t = \mu_i(t) + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$, $\sigma^2 = 1.0$.
- Event rate: $\nu = 1$ pull per step.
- Agent: Q-learning with fixed learning rate α and UCB action selection.

Chain Instantiation.

Scope (1.6). *Mapping: exact.*

$\Omega_t = (\mu_1(t), \ldots, \mu_4(t))$ — not directly observable. $\mathcal{A} = \{1, 2, 3, 4\}$. Residual uncertainty persists: means drift continuously, rewards are noisy, and only one arm is observed per step.

Causal Structure (1.8) + CIY (3.7). *Mapping: exact for causal ordering; approximate for CIY formula.*

The action (arm choice) temporally precedes the reward observation. Under the Gaussian reward model, CIY for arm i:

Discussion — Bandit CIY, Gaussian Case.

$$\mathrm{CIY}(i;\, M_{t-1}) = \frac{1}{2\sigma^2}\mathbb{E}_{j\sim q}\big[(\hat{\mu}_i - \hat{\mu}_j)^2\big] \tag{.2}$$

This measures how *distinctive* arm i's predicted reward is relative to alternatives — distinctiveness of predictions, not uncertainty about predictions. A proper uncertainty-aware CIY would require maintaining posterior variances per arm, which the basic Q-learning model does not track.

Model (2.1) and Sufficiency (2.3). *Mapping: exact structural mapping; the model is deliberately impoverished.*

$$M_t = (\hat{\mu}_1, \dots, \hat{\mu}_4,\ n_1, \dots, n_4) \tag{.3}$$

Model sufficiency $S(M_t) < 1$ for two reasons: (1) No drift tracking — the model treats reward means as stationary. (2) No uncertainty representation — unlike a Bayesian agent that would maintain posterior variances.

Mismatch (3.4). *Mapping: exact.*

$$\delta_t = r_t - \hat{\mu}_{a_t} \tag{.4}$$

Decomposes exactly: $\mathbb{E}[\delta_t^2] = (\hat{\mu}_{a_t} - \mu_{a_t}(t))^2 + \sigma^2$.

Update Gain (3.6). *Mapping: approximate. This is where the bandit agent is most deficient relative to AAT-optimal behavior.*

The Q-learning update

$$\hat{\mu}_{a_t} \leftarrow \hat{\mu}_{a_t} + \alpha \cdot \delta_t \tag{.5}$$

uses a **degenerate constant gain**. It does not adapt to the agent's current uncertainty state.

Optimal gain via effective window. A fixed α is equivalent to exponential discounting with effective window $W = (1-\alpha)/\alpha$. The per-arm model uncertainty balances estimation variance and drift-induced bias:

$$U_M \approx \frac{\sigma^2}{W} + q \cdot W \tag{.6}$$

The optimal window $W^* = \sigma/\sqrt{q} = 10$ gives $\alpha^* \approx 0.091$, $U_M \approx 0.2$, and:

$$\eta^* = \frac{U_M}{U_M + U_o} = \frac{0.2}{1.2} \approx 0.167 \tag{.7}$$

The discrepancy between α^* and η^* arises because α operates on raw prediction errors while η^* accounts for the full uncertainty structure. The two converge when the model properly tracks its own uncertainty.

Exploration (3.7). *Mapping: approximate structural analogy.*

UCB: $a_t = \arg\max_i[\hat{\mu}_i + c\sqrt{\ln t/n_i}]$

Table .1

AAT component	*UCB analog*	*Quality*
$\mathbb{E}[\text{value}(a) \mid M_t]$	$\hat{\mu}_i$	Exact
$\lambda(M_t) \cdot \mathbb{E}[\text{CIY}(a)]$	$c\sqrt{\ln t/n_i}$	Approximate

UCB captures the CIY structure (rarely-visited arms are more informative) but depends on visit count, not observation content.

Tempo + Persistence (3.8, 4.4). *Mapping: approximate.*

Per-arm tempo. With approximately uniform exploration, each arm sampled at rate $\nu/k = 1/4$:

$$\mathcal{T}_i = \frac{\nu}{k} \cdot \alpha = 0.25 \times 0.091 = 0.023 \text{ step}^{-1} \tag{.8}$$

Per-arm drift rate: $\rho_i = \sqrt{q} = 0.1$ reward units $\cdot$ step^{-1}.

Persistence check: $\mathcal{T}_i = 0.023$ vs $\rho_i = 0.1$ — **fails**. Per-arm correction tempo is too low relative to drift. The agent cannot keep all arms' estimates current under uniform exploration.

Interpretation. This is expected and informative. AAT diagnoses exactly why: with 4 arms and one pull per step, each arm is visited too infrequently to track its drifting mean. Two remedies:

1. **Increase** η^* (raise α): Extract more per observation, but increase steady-state noise.
2. **Concentrate** ν: Abandon uniform exploration. Focus pulls on a subset, increasing per-arm ν/k_{active}. This is exactly what a good UCB policy does.

With focused exploration ($k_{\text{active}} = 2, \alpha = 0.2$): $\mathcal{T}_i = 0.5 \times 0.2 = 0.1$ — barely meets the threshold. The fundamental tension: with limited pulls and significant drift, the agent must accept either failing to track some arms or using high α that introduces noise.

Aggregate failure. $\mathcal{T} = \nu \cdot \alpha = 0.091$ vs $\rho = k \cdot \sqrt{q} = 0.4$. The agent's total adaptive capacity is outpaced by total environmental drift — a regime where model-based approaches (Bayesian bandits, Kalman bandit filters) with higher effective η^* have a structural advantage.

AAT Concept	*Bandit Mapping*	*Status*

Mapping Quality Summary.

Table .2

AAT Concept	*Bandit Mapping*	*Status*
Scope	Exact	Definitional
Causal structure	Exact	Structural
CIY	Approximate	Model lacks uncertainty representation
Model (M_t)	Exact	Definitional
Model sufficiency ($S < 1$)	Exact	Q-learner demonstrably insufficient
Mismatch (δ_t)	Exact	Standard prediction error
Gain (α as degenerate η^*)	Approximate	Fixed, does not adapt
Exploration (UCB as approximate CIY)	Approximate	Structural analogy
Tempo ($\mathcal{T} = (\nu/k) \cdot \alpha$ per arm)	Approximate	Assumes uniform allocation
Persistence condition	Exact (qualitative)	Threshold structure is robust

Epistemic Status.

This is a *structural mapping*, not a derivation. The mapping is *exact* for scope, causal ordering, model compression, and mismatch. It is *approximate* for gain, exploration, and tempo — precisely because the Q-learner does not represent its own uncertainty, which is what 3.6 requires for optimal behavior. The gap between the Q-learner's fixed α and the AAT-optimal adaptive η^* is precisely the gap between a fixed-gain PID controller and a Kalman filter.

The mapping status is *conditional* because the quantitative relationships depend on the Gaussian reward model and specific parameterization. The qualitative conclusions (persistence failure under uniform exploration, the concentration-vs-noise tradeoff) should be robust.

Working Notes.

- The per-arm analysis is a natural instance of the per-dimension tempo decomposition (14.5): each arm is an independent mismatch dimension with its own $\mathcal{T}_i$ and ρ_i. The aggregate tempo overstates effective adaptation along any individual arm's dimension — exactly the failure mode that the scalar-to-tensor generalization captures.
- A Bayesian bandit (maintaining per-arm posteriors with exponential discounting) would achieve higher 2.3 by representing its own uncertainty, yielding an adaptive η^* that matches the AAT-optimal form. Comparing Q-learner vs Bayesian bandit performance under nonstationarity is a direct test of 3.6's structural claim.
- Reward-unit and surprise-unit formulations coincide up to normalization by σ^2. For this example ($\sigma = 1$), they are identical.

Appendix

Worked Example: Nonstationary Bandit with Explicit Strategy (Section II Validation)

AAT's Section II machinery — objectives, strategy DAGs, the orient cascade, satisfaction gap, control regret, chain confidence decay, observability dominance, and edge update via gain — is exercised on a concrete system simple enough for closed-form analysis. The key payoff: Section II diagnostics tell the agent things that Section I quantities alone cannot.

System.

A 3-armed Bernoulli bandit with drifting success probabilities and an agent that maintains an explicit strategy DAG for arm selection.

Environment. Three arms with time-varying success probabilities:

Formulation.

$$\theta_k(t) \in [0,1], \quad k \in \{1,2,3\} \tag{.1}$$

At each step, pulling arm k yields reward $r_t \in \{0,1\}$ with $P(r_t = 1 \mid a_t = k) = \theta_k(t)$. The best arm changes slowly: at rate ρ_θ, a random arm's success probability shifts by a random perturbation clipped to $[0.1, 0.9]$.

Concrete parameterization. At $t = 0$: $\theta_1(0) = 0.7$, $\theta_2(0) = 0.5$, $\theta_3(0) = 0.3$. Drift rate: $\rho_\theta = 0.05$ per step (one arm shifts by ± 0.05 every 20 steps on average). Horizon: $N_h = 100$ steps.

Agent. Maintains Beta-Bernoulli beliefs per arm and an explicit strategy DAG encoding its plan for achieving a target reward rate.

Section I Chain Instantiation (Summary).

Section I quantities map as in . The brief summary here establishes the epistemic baseline; the novel content is Section II.

Scope (1.6). $\Omega_t = (\theta_1(t), \theta_2(t), \theta_3(t))$; $\mathcal{A} = \{1,2,3\}$; residual uncertainty persists. *Exact.*

Model (2.1). $M_t = (\alpha_k, \beta_k)_{k=1}^{3}$ — Beta posteriors per arm, with exponential discounting (effective window $W = 20$):

$$\hat{\theta}_k = \frac{\alpha_k}{\alpha_k + \beta_k} \tag{.2}$$

Exact structural mapping.

Mismatch (3.4). $\delta_t = r_t - \hat{\theta}_{a_t}$. *Exact.*

Update gain (3.6). The Beta-Bernoulli conjugate update with discounted pseudo-counts gives:

$$\eta_k^* = \frac{1}{\alpha_k + \beta_k + 1} \tag{.3}$$

For a well-observed arm with $\alpha_k + \beta_k \approx W = 20$: $\eta_k^* \approx 0.048$. *Exact for the pulled arm; zero for unpulled arms.*

Tempo (3.8). Per-arm tempo: $\mathcal{T}_k = \nu_k \cdot \eta_k^*$ where ν_k is the pull rate for arm k. Under uniform exploration: $\nu_k = 1/3$, so $\mathcal{T}_k \approx 0.016$ per step. *Approximate.*

Section II Chain Instantiation.

Objective (5.4). *Mapping: exact.*

The agent's objective: achieve a target average reward rate over the horizon.

Definition (bandit-objective).

$$V_{O_t}(\tau_{0:N_h}) = \frac{1}{N_h} \sum_{t=0}^{N_h-1} r_t \tag{.4}$$

$$V_{O_t}^{\min} = 0.6 \tag{.5}$$

This is a utility-type objective with a satisfaction threshold: the agent counts the mission as successful if its average reward rate meets or exceeds 0.6. The value functional evaluates trajectories on $[0, 1]$ (average reward).

Value Object (5.6). *Mapping: exact.*

Under the one-step improvement convention ($\pi_{\text{cont}} = \pi_{\text{current}}$):

$$V_O(M_t, \pi; N_h) = \mathbb{E}\left[\frac{1}{N_h} \sum_{s=t}^{t+N_h-1} r_s \,\middle|\, M_t, \pi \right] \tag{.6}$$

For a greedy policy that always pulls the estimated-best arm $k^* = \arg\max_k \hat{\theta}_k$:

$$V_O(M_t, \pi_{\text{greedy}}; N_h) = \hat{\theta}_{k^*} \tag{.7}$$

This is tractable: the agent's expected average reward under greedy play equals its belief about the best arm's success probability. With M_0: $V_O = \hat{\theta}_1 = 0.7$.

Action-value form:

$$Q_O(M_t, k; \pi_{\text{greedy}}, N_h) \approx \frac{1}{N_h} \hat{\theta}_k + \frac{N_h - 1}{N_h} \hat{\theta}_{k^*} \tag{.8}$$

The first pull contributes $1/N_h$ of the horizon; the rest follows the greedy policy.

Objective Attainability (7.4). *Mapping: exact.*

$$A_O(M_t; \Pi, N_h) = \sup_{\pi \in \Pi} V_O(M_t, \pi; N_h) \tag{.9}$$

For a bandit with Beta-Bernoulli beliefs, the optimal policy (Gittins index or Bayes-adaptive) achieves an attainability that is bounded:

$$\max_k \hat{\theta}_k \leq A_O \leq \max_k \hat{\theta}_k + \epsilon_{\text{info}} \tag{.10}$$

where ϵ_{info} accounts for the information value of exploration (pulling a suboptimal arm now to identify the true best arm later). The upper bound is tight when exploration can reveal a better arm. For our initial beliefs: $A_O \approx 0.70 + \epsilon_{\text{info}}$.

Satisfaction gap:

$$\delta_{\text{sat}} = V_{O_t}^{\min} - A_O = 0.6 - 0.70 \approx -0.10 \tag{.11}$$

$\delta_{\text{sat}} < 0$: the objective is **attainable**. The agent believes it can achieve a 0.6 average reward rate. The negative gap is margin — the agent has room to spare.

Control Regret (7.5). *Mapping: exact.*

Under an explore-then-exploit policy that spends 30 steps exploring uniformly, then exploits the estimated-best arm:

$$V_O(M_t, \pi_{\text{explore-exploit}}; 100) = \frac{30}{100} \cdot \bar{\theta} + \frac{70}{100} \cdot \hat{\theta}_{k^*(30)} \tag{.12}$$

where $\bar{\theta} = (\hat{\theta}_1 + \hat{\theta}_2 + \hat{\theta}_3)/3$ is the average expected reward during exploration and $\hat{\theta}_{k^*(30)}$ is the estimated-best arm after 30 observations. With initial beliefs: $V_O \approx 0.3 \cdot 0.5 + 0.7 \cdot 0.7 = 0.64$.

$$\delta_{\text{regret}} = A_O - V_O(\pi_{\text{explore-exploit}}) \approx 0.70 - 0.64 = 0.06 \tag{.13}$$

The agent is leaving 0.06 reward-rate points on the table with its current explore-then-exploit strategy — it explores too much. This is the signal for strategy revision (9.1 step 3).

The 2x2 diagnostic (7.5 Discussion):

Table .1

	$\delta_{sat} \leq 0$	$\delta_{sat} > 0$
$\delta_{\text{regret}} \approx 0$	Success	Capability limit
$\delta_{\text{regret}} \gg 0$	**Strategy problem**	Both

Our agent is in the **Strategy problem** cell: the goal is achievable, but the current policy is suboptimal. Section I alone (which reports $\hat{\theta}_k$, η_k^*, and $\mathcal{T}_k$) cannot distinguish "good strategy, hard goal" from "bad strategy, easy goal." The 2x2 diagnostic is Section II's value-add.

Strategy DAG (7.3, 7.2). *Mapping: exact for structure and propagation; the DAG is simple enough for closed-form status.*

The agent's strategy for achieving $V_{O_t}^{\min} = 0.6$:

```
[root: "achieve ≥0.6 avg reward"] (AND) / \ [Phase 1: "identify best arm"] (OR) [Phase 2: "exploit best arm"]
/ | \ | [arm 1 [arm 2 [arm 3 [pull identified is best] is best] is best] arm reliably]
```

Nodes: - v_{root}: AND-node. Both phases must succeed: identify the best arm AND exploit it. Terminal satisfaction condition: average reward ≥ 0.6. - v_{id}: OR-node (Phase 1). Identifying the best arm succeeds if ANY one arm is correctly identified as best. - v_1, v_2, v_3: leaf condition nodes. "Arm k is the best arm." Base credence: $p_k(M_t) = P(\theta_k > \theta_j \; \forall j \neq k \mid M_t)$. - v_{exploit}: leaf action node (Phase 2). "Pulling the identified arm yields reward ≥ 0.6." Base credence: $p_{\text{exploit}}(M_t) = P(\theta_{k^*} \geq 0.6 \mid M_t)$.

Edge credences. The edges from leaf nodes to their parents carry causal credences: - $p_{v_k, v_{\text{id}}}$: "correctly identifying arm k as best enables Phase 1 success." These are structural (close to 1.0) — if the agent correctly knows which arm is best, Phase 1 succeeds by definition. Set $p_{v_k, v_{\text{id}}} = 1.0$ for all k. - $p_{v_{\text{id}}, v_{\text{root}}}$: "having identified the best arm advances the root goal." Set to 0.95 (slight uncertainty about whether identification translates to exploitation success). - $p_{v_{\text{exploit}}, v_{\text{root}}}$: "successful exploitation advances the root goal." Set to 0.90 (the identified arm might drift below threshold during exploitation).

Leaf base credences from M_t. At $t = 0$ with $(\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3) = (0.7, 0.5, 0.3)$ and moderate pseudo-counts ($\alpha_k + \beta_k \approx 5$) from prior experience:

$$p_1(M_0) = P(\theta_1 > \theta_2, \theta_1 > \theta_3 \mid M_0) \approx 0.75 \tag{.14}$$

$$p_2(M_0) \approx 0.20, \quad p_3(M_0) \approx 0.05 \tag{.15}$$

$$p_{\text{exploit}}(M_0) = P(\theta_{k^*} \geq 0.6 \mid M_0) \approx 0.70 \tag{.16}$$

Status propagation (7.3, 7.2):

Phase 1 (OR-node):

$$s_{v_{\text{id}}} = 1 - \prod_{k=1}^{3} (1 - p_{v_k, v_{\text{id}}} \cdot p_k) = 1 - (1 - 0.75)(1 - 0.20)(1 - 0.05) \tag{.17}$$

$$= 1 - (0.25)(0.80)(0.95) = 1 - 0.19 = 0.81 \tag{.18}$$

Phase 2 (leaf): $s_{v_{\text{exploit}}} = 0.70$.

Root (AND-node):

$$s_{v_{\text{root}}} = (p_{v_{\text{id}}, v_{\text{root}}} \cdot s_{v_{\text{id}}}) \cdot (p_{v_{\text{exploit}}, v_{\text{root}}} \cdot s_{v_{\text{exploit}}}) \tag{.19}$$

$$= (0.95 \times 0.81) \times (0.90 \times 0.70) = 0.770 \times 0.630 = 0.485 \tag{.20}$$

Plan-confidence score: $\hat{P}_\Sigma(M_0) = 0.485$. The strategy assigns about 49% confidence to achieving the objective. This is distinct from A_O (which optimizes over all policies) — $\hat{P}_\Sigma$ is the DAG's self-assessment of *this specific plan*.

Chain Confidence Decay (7.1). *Mapping: exact (mathematical identity).*

The root node's confidence illustrates chain decay through an AND structure. The root requires two children to succeed, each with their own sub-chains:

$$\log \hat{P}_\Sigma = \log(p_{v_{\text{id}},v_{\text{root}}} \cdot s_{v_{\text{id}}}) + \log(p_{v_{\text{exploit}},v_{\text{root}}} \cdot s_{v_{\text{exploit}}}) \tag{.21}$$

$$= \log(0.770) + \log(0.630) = -0.261 + (-0.462) = -0.724 \tag{.22}$$

Each term is negative; the AND combination accumulates them additively in log-space. If the agent added a third AND-child (e.g., "maintain budget"), even with $s_{\text{budget}} = 0.95$, the strategy-plan confidence would drop to $0.485 \times 0.95 = 0.461$. The pressure toward shallow, narrow DAGs is structural.

Contrast with the OR-node. Phase 1 (OR-node) benefits from redundancy: three alternative arms provide resilience. The OR combination yields $s_{v_{\text{id}}} = 0.81$ even though no single arm has credence above 0.75. The asymmetry between AND (multiplicative decay) and OR (complementary resilience) is the core structural result of 7.1 combined with 7.2.

Orient Cascade (9.1). *Mapping: exact for ordering; the content of each step is tractable in this domain.*

Suppose at $t = 40$, the agent has been pulling arm 1 (the initially-best arm) and observes a run of failures: the last 8 of 10 pulls yielded $r_t = 0$. Meanwhile, arm 2's true probability has drifted to $\theta_2(40) = 0.8$ (unbeknownst to the agent) and arm 1 has drifted to $\theta_1(40) = 0.3$.

The orient cascade proceeds:

Step 1: Reduce $\delta_{\text{epistemic}}$. Update M_t from the recent observations. The Beta posterior for arm 1 shifts:

$$\hat{\theta}_1 : 0.70 \to 0.38 \quad \text{(after 8 failures in 10 trials with discounted prior)} \tag{.23}$$

Arms 2 and 3 are unobserved since the exploration phase ended. Their posteriors remain at approximately $\hat{\theta}_2 = 0.52$, $\hat{\theta}_3 = 0.28$ (slightly regressed toward the prior due to discounting). *The epistemic update proceeds without reference to O_t or Σ_t* — directed separation (5.3) holds because the Beta update depends only on the observation and the prior, not on the agent's goals.

Step 2: Evaluate δ_{sat}. With the updated M_t:

$$A_O(M_{40}) \approx \max_k \hat{\theta}_k + \epsilon_{\text{info}} = 0.52 + \epsilon_{\text{info}} \tag{.24}$$

$$\delta_{\text{sat}} = 0.6 - (0.52 + \epsilon_{\text{info}}) \tag{.25}$$

If $\epsilon_{\text{info}} \approx 0.05$ (exploration can reveal a better arm): $\delta_{\text{sat}} \approx 0.6 - 0.57 = 0.03 > 0$. The objective is now **marginally unmet** — the agent believes it may not be able to achieve 0.6 average reward. This triggers the disambiguation procedure (7.4): is the goal truly infeasible, or is M_t wrong about what's achievable?

Step 3: Evaluate δ_{regret}. Under the current greedy policy (still pulling arm 1 despite updated beliefs):

$$V_O(M_{40}, \pi_{\text{current}}) = \hat{\theta}_1 = 0.38 \tag{.26}$$

$$\delta_{\text{regret}} = 0.57 - 0.38 = 0.19 \tag{.27}$$

Control regret is high. The agent is pulling arm 1 out of inertia, but M_t now says arm 2 is better. *This is the Section II payoff*: Section I would report the mismatch δ_t on each pull and the updated $\hat{\theta}_1$, but it cannot diagnose that the *strategy* is the problem. The high δ_{regret} identifies the corrective action: revise Σ_t, not O_t.

Step 4: Evaluate $\delta_{\text{strategic}}$. The edge residual for the root AND-node's exploitation branch:

$$r_{\text{exploit}} = \mathbb{E}[\Delta V_O \mid \text{exploit arm 1}] - \Delta V_O^{\text{observed}} \tag{.28}$$

The agent predicted reward ≈ 0.70 per pull from arm 1; it observed ≈ 0.20. The residual is large and positive, localizing the problem to the exploitation edge: the agent's causal belief that "pulling arm 1 yields high reward" is wrong.

Simultaneously, the leaf credences update: $p_1(M_{40}) \approx 0.15$ (arm 1 is probably not the best), $p_2(M_{40}) \approx 0.60$ (arm 2 is probably best, by default from the prior). The Phase 1 OR-node status shifts: the identification sub-goal may need to be revisited.

Step 5: Revise Σ_t. The agent restructures its strategy: switch exploitation target to arm 2 (the currently estimated-best arm). In DAG terms: $p_1(M_{40})$ has dropped to 0.15 while $p_2(M_{40})$ has risen to 0.60. The OR-node now routes through arm 2. The exploitation leaf updates: $p_{\text{exploit}}(M_{40}) = P(\theta_2 \geq 0.6 \mid M_{40}) \approx 0.40$.

After strategy revision: $\hat{P}_\Sigma(M_{40}) \approx (0.95 \times 0.73) \times (0.90 \times 0.40) = 0.693 \times 0.360 = 0.249$. Plan confidence has dropped from 0.485 to 0.249 — the agent correctly recognizes that its situation has deteriorated.

What Section I alone cannot provide. At $t = 40$, Section I reports: $\hat{\theta}_1 = 0.38$, $\hat{\theta}_2 = 0.52$, $\hat{\theta}_3 = 0.28$, and the per-arm mismatch signals. This tells the agent what it believes about each arm. It does NOT tell the agent: - Whether its *goal* is still achievable (δ_{sat}) - Whether its *strategy* is the problem or the goal is (δ_{regret}) - Which *part* of the strategy is failing ($\delta_{\text{strategic}}$, edge residuals) - What corrective action to take (the cascade's ordering)

Observability Dominance (8.3). *Mapping: exact mechanism; approximate quantitative form.*

When the agent stops pulling arm 3 after the exploration phase (say at $t = 30$), the observation rate for arm 3 drops to $\nu_3 = 0$. By 3.6:

$$\eta_3^* = \frac{U_{\text{edge},3}}{U_{\text{edge},3} + U_{\text{obs},3}} \tag{.29}$$

With $U_{\text{obs},3} \to \infty$ (no observations): $\eta_3^* \to 0$. Arm 3's credence $p_3(M_t)$ is **frozen at its exploration-phase value**.

For the strategy DAG, the leaf node "arm 3 is best" has: - Nominal credence: $p_3(M_t) \approx 0.05$ (from $t = 30$ beliefs) - Path observability: $\sigma_3 = 0$ (arm 3 is not being pulled) - Observability-adjusted confidence: $\text{conf}_{\text{obs}}(\text{arm 3 path}) = 0.05 \times 0 = 0$

Arm 3's path through the strategy is **epistemically dead** — the agent cannot learn whether arm 3 has become the best arm. If θ_3 drifts to 0.9 (which it might, given nonstationarity), the agent will never discover this through the exploitation-only policy.

The observability-dominance diagnostic. Compare arm 2 and arm 3:

Table .2

	Arm 2	*Arm 3*
Nominal credence p_k	0.52	0.28 (frozen)
Observability σ_k	moderate (occasional pulls)	0 (never pulled)
Adjusted confidence	$0.52 \times \sigma_2$	0
Can be updated?	Yes	No

Arm 2 is epistemically alive; arm 3 is epistemically dead. The agent should prefer exploring arm 2 (where it can learn) over arm 3 — unless it invests in making arm 3 observable again (pulling it despite low expected reward). This is the strategic case for exploration: not just expected value, but *observability maintenance*.

Section I vs Section II diagnosis. Section I reports $\hat{\theta}_3 = 0.28$ with growing uncertainty (the posterior variance increases under discounting with no new data). Section II adds the diagnostic: this arm's strategy path is epistemically dead ($\sigma_3 = 0$), and restoring observability has value beyond the arm's nominal expected reward.

Edge Update via Gain (8.4). *Mapping: exact for the Beta-Bernoulli case.*

When the agent pulls arm 1 at $t = 35$ and observes failure ($r_t = 0$):

Leaf credence update. The Beta posterior for arm 1 updates:

$$\beta_1 \to \beta_1 + 1, \quad \hat{\theta}_1 = \frac{\alpha_1}{\alpha_1 + \beta_1 + 1} \tag{.30}$$

This is the standard conjugate update — the gain principle reduces to Bayesian updating in the binary case (8.4).

Strategy propagation. The updated $\hat{\theta}_1$ changes $p_1(M_t)$ (the credence that arm 1 is best). This propagates through the OR-node to the root:

$$s_{v_{\text{id}}}^{\text{new}} = 1 - (1 - p_1^{\text{new}})(1 - p_2)(1 - p_3) \tag{.31}$$

The OR-node's status is *resilient* to single-arm credence drops: even if p_1 drops substantially, p_2 can compensate. This is the OR-node's structural advantage — redundancy absorbs single-edge failures.

The exploitation edge. The edge $p_{v_{\text{exploit}}, v_{\text{root}}} = 0.90$ is a causal credence: "pulling the identified-best arm reliably yields reward ≥ 0.6." After observing repeated failures from arm 1:

$$\text{signal} = 0 \quad \text{(failure observed)} \tag{.32}$$

$$p_{\text{exploit}}^{\text{new}} = p_{\text{exploit}}^{\text{old}} + \eta_{\text{edge}} \cdot (0 - p_{\text{exploit}}^{\text{old}}) \tag{.33}$$

With $\eta_{\text{edge}} = U_{\text{edge}}/(U_{\text{edge}} + U_{\text{obs}})$, the exploitation edge's credence decreases. The rate of decrease is governed by the gain principle: well-established edges (many prior successes, U_{edge} small) are slow to revise; newly formed edges are fast.

Directed Separation (5.3). *Mapping: exact. The agent is Class 1 (Separated) by construction.*

The Beta-Bernoulli update for M_t processes (a_t, r_t) without reference to (O_t, Σ_t):

$$(\alpha_{a_t}, \beta_{a_t}) \to (\alpha_{a_t} + r_t, \beta_{a_t} + 1 - r_t) \tag{.34}$$

The update function f_M has no G_t argument. The strategy DAG then updates its leaf credences from the new M_t — this is $f_G(G_t, M_{t+1}, e_t)$. The agent architecture is modular: a separate estimator (Beta posteriors) and planner (DAG evaluation), connected through the state-estimate interface.

$\kappa_{\text{processing}} = 0$ — no goal information flows into the epistemic update. Section II results apply exactly.

Satisfaction Gap Dynamics. *Mapping: exact definition; approximate computation.*

Tracking δ_{sat} over time reveals the agent's evolving assessment of goal feasibility:

Table .3

Time	$\hat{\theta}_{k^*}$	A_O *(approx)*	δ_{sat}	*Interpretation*
$t = 0$	0.70	0.72	-0.12	Comfortably attainable
$t = 20$	0.68	0.70	-0.10	Still attainable
$t = 40$	0.52	0.57	$+0.03$	Marginally unmet
$t = 50$ (after exploring arm 2)	0.75	0.77	-0.17	Attainable again — M_t improved
$t = 70$	0.78	0.79	-0.19	Confidently attainable

The key transition is $t = 40 \to 50$: δ_{sat} goes from positive (unmet) to negative (attainable) — not because the goal changed or the environment improved, but because M_t improved. The agent explored arm 2, discovered $\hat{\theta}_2 \approx 0.75$, and its attainability assessment corrected upward. This matches the 7.4 disambiguation: the positive δ_{sat} at $t = 40$ was caused by a bad model, not an infeasible goal. The orient cascade's ordering (fix M_t first, then reassess δ_{sat}) produced the right diagnosis.

The Section II Payoff.

The purpose of this worked example is to demonstrate that Section II adds genuine diagnostic value beyond Section I. Here is the explicit comparison:

Scenario at $t = 40$: arm 1 failing, arm 2 actually best, agent pulling arm 1.

Section I diagnosis. The mismatch signal δ_t is large. The posterior $\hat{\theta}_1$ has dropped. Per-arm tempo $\mathcal{T}_1 \approx 0.016$ is adequate for tracking arm 1's drift. The agent knows arm 1 looks bad. Section I prescribes: update M_t faster (increase η^*) or observe more (increase ν).

Section II diagnosis. The orient cascade runs: 1. M_t updates correctly — arm 1's posterior drops to 0.38. (Directed separation: this happens goal-blindly.) 2. $\delta_{\text{sat}} = +0.03$ — the objective is marginally unmet. The agent's best-believed policy barely misses the target. 3. $\delta_{\text{regret}} = 0.19$ — the current policy (pulling arm 1) is far from the best available (pulling arm 2). 4. The 2x2 diagnostic says: "goal may be achievable, strategy is definitely suboptimal." Corrective action: revise Σ_t, then reassess δ_{sat}. 5. Edge residuals localize the problem: the exploitation branch's predicted-vs-observed value gap is large. The leaf credences identify arm 2 as the best candidate. 6. Observability check: arm 3 is epistemically dead ($\sigma_3 = 0$). The agent should invest in observing arm 2 (or arm 3) before committing.

Section II tells the agent: *switch to arm 2, but first explore to confirm — the goal is probably still achievable, the current strategy is the problem, not the goal.* Section I tells the agent: *arm 1 looks bad.* The difference is the difference between a diagnosis and a symptom.

Mapping Quality Summary.

Table .4

AAT Concept	*Bandit-Strategy Mapping*	*Status*	*Notes*
Scope (1.6)	Exact	Definitional	Same as
Model M_t (2.1)	Exact	Structural	Beta posteriors per arm
Mismatch δ_t (3.4)	Exact	Standard	Prediction error
Update gain η^* (3.6)	Exact	Conjugate	Beta-Bernoulli gain
Tempo $\mathcal{T}$ (3.8)	Approximate	Per-arm decomposition	Assumes known allocation
Directed separation (5.3)	Exact	Class 1 (Separated) by construction	Beta update is goal-blind
Objective O_t (5.4)	Exact	Utility with threshold	$V_{O_t}^{\min} = 0.6$
Value object V_O (5.6)	Exact	Closed-form (greedy)	$V_O = \hat{\theta}_{k^*}$
Satisfaction gap δ_{sat} (7.4)	Exact	Definition applied	$V_{O_t}^{\min} - A_O$
Control regret δ_{regret} (7.5)	Exact	Definition applied	$A_O - V_O(\pi_{\text{current}})$
Strategy DAG Σ_t (7.3)	Exact	AND/OR with 7 nodes	Small enough for manual propagation
AND/OR scope (7.2)	Exact	Natural fit	Phase 1 is genuinely OR; root is genuinely AND
Status propagation	Exact	Closed-form	Forward pass computable by hand
Plan confidence $\hat{P}_\Sigma$ (7.3)	Exact (computation)	From propagation; systematically optimistic under correlated failure	0.485 at $t = 0$
Chain confidence decay (7.1)	Exact	Mathematical identity	AND-node multiplicative; OR-node resilient
Orient cascade (9.1)	Exact (ordering)	All 5 steps exercised	Content is tractable in this domain
Observability dominance (8.3)	Exact (mechanism)	Unpulled arms frozen	$\sigma_3 = 0 \Rightarrow \eta_3^* = 0$
Edge update via gain (8.4)	Exact (binary case)	Beta-Bernoulli conjugate	Gain principle = Bayesian update
Strategic calibration (8.1)	Approximate	Credit assignment tractable here	Single-arm attribution is clean
Satisfaction gap dynamics	Exact	Time series of δ_{sat}	Model improvement resolves positive gap

Quantities that map cleanly. All Section II definitions (objective, value object, satisfaction gap, control regret, strategy DAG) have exact instantiations. Plan confidence $\hat{P}_\Sigma$ is exact as a computation (the AND/OR propagation is correct), but systematically overestimates actual success probability under correlated edge failures (7.3). In this toy bandit, edge independence is a reasonable approximation; in complex domains the overestimate may be severe. The orient cascade ordering is exact and all five steps are exercised.

Quantities that map approximately. Strategic calibration is approximate because even in this simple domain, attributing value changes to specific DAG edges requires assumptions about the agent's execution fidelity. Observability dominance is exact in mechanism but the multiplicative form $\text{conf}_{\text{obs}} = \text{conf} \cdot \text{obs}$ is a first-order approximation.

Quantities that don't map. No Section II quantity fails to map in this domain. The system is simple enough that every quantity is computable (at least approximately). This is the advantage of the bandit setting: it is the simplest system that exercises all Section II machinery.

Epistemic Status.

This is a *worked instantiation* — it demonstrates that the Section II formal chain is internally consistent and instantiable. The mapping quality is *conditional*: the quantitative relationships depend on the Beta-Bernoulli reward model, the specific DAG structure, and the parameterization. The qualitative conclusions — especially the diagnostic value of the 2x2 table and the orient cascade's ordering — should be robust across reward models and DAG topologies.

The status is *conditional* rather than *exact* because the attainability A_O is approximated (the exact Bayes-optimal policy for a nonstationary bandit is intractable), and the strategic calibration mapping inherits the discussion-grade status of 8.1. The core Section II definitions (satisfaction gap, control regret, strategy-plan confidence) are exact instantiations.

Working Notes.

- This example uses a deliberately simple DAG (7 nodes, depth 2). A richer example with depth 3+ would better exercise chain confidence decay's exponential penalty and the evidence-starvation effect from 8.3's quantitative discussion. A natural extension: add an intermediate "confirm arm k is best" node between identification and exploitation, creating a 3-level chain.
- The explore-then-exploit policy is deliberately suboptimal to demonstrate nonzero δ_{regret}. A Gittins-index agent would achieve $\delta_{\text{regret}} \approx 0$, demonstrating the "Success" cell of the 2x2 table. Comparing the two in the same environment would validate the control-regret diagnostic.
- The satisfaction gap dynamics table (Section "Satisfaction Gap Dynamics") would benefit from simulation confirmation. The specific numbers are plausible but hand-computed. A Monte Carlo study averaging over environment drift realizations would strengthen the claim.
- This example does not exercise strategy-level tempo (8.10) or the three-way tradeoff (9.2). A future worked example targeting these would need a system with genuine deliberation costs — simulation in `spikes/sim-three-way-tradeoff.py` provides a drifting-bandit version but found deliberation rarely dominates in simple settings.
- The DAG's AND/OR assignment is unambiguous in this domain (Phase 1 is genuinely disjunctive; the root is genuinely conjunctive). Domains where the AND/OR assignment is uncertain or updatable would be a stronger test of 7.2.

Appendix

Worked Example: L1 Augmented DAG with Common-Cause Node

A concrete instantiation of the Correlation Hierarchy (7.3) and Proposition B.6 (N). Two OR-alternatives sharing an infrastructure dependency are modeled at L0 (independence) and L1 (augmented DAG), with the sector condition verified for L1.

Setup.

Two alternative paths to a goal G, sharing a common infrastructure dependency C:

- Infrastructure up with probability $\theta_C = 0.8$
- Path 1 succeeds with probability $\theta_{1|C} = 0.9$ when infrastructure is up, fails when down
- Path 2 succeeds with probability $\theta_{2|C} = 0.7$ when infrastructure is up, fails when down
- Goal succeeds if at least one path succeeds (OR)

Actual plan success probability (from the generative model):

$$P(G) = \theta_C \cdot [1 - (1 - \theta_{1|C})(1 - \theta_{2|C})] = 0.8 \cdot 0.97 = 0.776 \tag{.1}$$

L0 Analysis: The Independence Model.

The L0 DAG has two leaf nodes with marginal success probabilities:

$$\theta_1 = \theta_C \cdot \theta_{1|C} = 0.72, \qquad \theta_2 = \theta_C \cdot \theta_{2|C} = 0.56 \tag{.2}$$

OR-propagation:

$$\hat{P}_{\Sigma}^{L0} = 1 - (1 - \theta_1)(1 - \theta_2) = 1 - (0.28)(0.44) = 0.877 \tag{.3}$$

Overestimation: $0.877 - 0.776 = 0.101$ (13% relative error).

The overestimation equals $\mathrm{Cov}(X_1, X_2)$ — the covariance induced by the shared infrastructure. The OR-node's redundancy is illusory: both paths fail together when infrastructure fails, so the "backup" provides less protection than the independence model predicts. This is the $-\rho$ bias from the Correlation Hierarchy's direction-of-bias formula.

Naive L1: Why Adding a Node Is Not Sufficient.

The obvious L1 construction adds C as a parent of both paths:

- $B_1 = \mathrm{AND}(C, A_1), B_2 = \mathrm{AND}(C, A_2)$
- $G = \mathrm{OR}(B_1, B_2)$

Status propagation: $s_{B_1} = 0.8 \cdot 0.9 = 0.72$, $s_{B_2} = 0.8 \cdot 0.7 = 0.56$, $s_G = 1 - (0.28)(0.44) = 0.877$.

Same answer as L0. The OR-propagation treats B_1 and B_2 as marginally independent, but they are correlated through their shared parent C. Adding the common-cause node does not fix the overestimation because the node is *inside* the correlated structure (below the OR-node), not factored above it.

Correct L1: Factor the Common Cause Above the Correlation.

The correct L1 restructuring places C as an AND-prerequisite *above* the OR-structure:

- $G_{\text{sub}} = \mathrm{OR}(A_1, A_2)$ — the sub-plan assuming infrastructure holds
- $G = \mathrm{AND}(C, G_{\text{sub}})$ — infrastructure must hold AND sub-plan must work

Status propagation:

$$s_{G_{\text{sub}}} = 1 - (1 - \theta_{1|C})(1 - \theta_{2|C}) = 1 - (0.1)(0.3) = 0.97 \tag{.4}$$

$$s_G = \theta_C \cdot s_{G_{\text{sub}}} = 0.8 \cdot 0.97 = 0.776 \tag{.5}$$

Correct. Matches the actual plan success probability exactly.

Why this works: C is above the OR-node, so its value is factored out before the OR-propagation occurs. Conditional on $C = 1$, the two paths A_1 and A_2 are genuinely independent (their exogenous noise terms are independent — the common cause has been conditioned away). The OR-propagation's independence assumption holds within G_{sub} because the correlation source is above, not among the siblings.

The L1 construction principle: *factor the common cause above the correlation it creates.* When a common cause correlates siblings under an OR-node, restructure so the common cause gates the entire OR-structure via an AND-relationship. This ensures conditional independence of the OR-children and makes standard propagation correct.

Leaf	*True value*	*Tested when*	*Expected correction*

Sector Condition Verification (Proposition B.6).

Three leaf nodes with Beta-Bernoulli dynamics and ε-greedy path selection:

Table .1

Leaf	*True value*	*Tested when*	*Expected correction*
C (condition)	$\theta_C = 0.8$	Every trial	$-\delta_C/(n_C + 1)$
A_1 (greedy)	$\theta_{1\|C} = 0.9$	$C = 1$ AND path 1 selected	$-\theta_C(1-\varepsilon)\,\delta_{A_1}/(n_{A_1}+1)$
A_2 (explore)	$\theta_{2\|C} = 0.7$	$C = 1$ AND path 2 selected	$-\theta_C\varepsilon\,\delta_{A_2}/(n_{A_2}+1)$

The sector parameter (full proof in Prop B.6 of N):

$$\alpha_\Sigma = \min\left(\frac{1}{n_C+1},\ \frac{\theta_C(1-\varepsilon)}{n_{A_1}+1},\ \frac{\theta_C\varepsilon}{n_{A_2}+1}\right) \tag{.6}$$

Three-way gating. This is the first verified mixed AND/OR topology, combining:

1. **Condition testing** — C is tested every trial, same rate as a single edge (B.1)
2. **Evidence starvation** — action edges are gated by θ_C (tested only when $C = 1$), same mechanism as B.2
3. **Exploration gating** — OR-alternatives compete for test opportunities (ε vs $1-\varepsilon$), same mechanism as B.4

The B.5b bridge applies (non-negative Jacobian, componentwise corrections): $\alpha_s = \alpha_\Sigma$ with no penalty.

L0 vs L1 Comparison.

Table .2

Quantity	*L0*	*L1 (correct)*
Leaf edges	2 (A_1, A_2 at marginals)	3 ($C, A_1 \mid C, A_2 \mid C$)
Φ at truth	0.877 (biased)	0.776 (correct)
Φ – actual success	+0.101 (overestimates)	0 (unbiased)
Bottleneck α_Σ (at $\varepsilon = 0.3$)	$0.3/(n_2+1)$	$0.24/(n_{A_2}+1)$
Persistence threshold	Lower (easier)	Higher (harder)

The tradeoff: L0 has a higher sector parameter (easier to maintain persistence) but its reference value Φ^{L0} is systematically wrong — it overestimates actual success by the covariance ρ. L1 has a lower sector parameter (the θ_C attenuation makes action calibration harder) but its reference value Φ^{L1} is correct.

Both models have $\delta_s = 0$ when credences are at truth. The difference is what $\delta_s = 0$ *means*:

- **L0**: well-calibrated within the independence model — which overstates plan success by ρ

- **L1**: well-calibrated to actual plan success

L1 persistence is harder but calibration is honest. The extra leaf (C) adds a degree of freedom, and the condition-gating slows action calibration. But the resulting target (Φ^{L1}) is the true plan success probability, not a biased surrogate.

When Correct L1 Construction Is Not Possible.

The factoring-above principle works under two conjoint conditions: (a) the common cause cleanly gates all correlated children through a single AND-relationship (*topological*), and (b) the common cause is a *strict prerequisite* — $\theta_{\text{child}|\neg C} \approx 0$, so children cannot succeed when the common cause is absent (*semantic*). Both conditions must hold. The example above satisfies both: infrastructure gates all alternatives, and absent infrastructure forces failure ($\theta_{i|\neg C} = 0$).

Topological failure: mixed gating. When some OR-alternatives share a common cause and others do not, the factoring-above construction cannot place C above the full sub-plan. Conditioning-based propagation:

$$P(G) = \sum_{c} P(C = c) \cdot P_{\Sigma}(G \mid C = c) \tag{.7}$$

where $P_{\Sigma}(G \mid C = c)$ is computed by standard AND/OR propagation with C fixed. Cost: $O(2^k \cdot (|V| + |E|))$ where k is the number of common-cause nodes. Tractable for small k; exponential in general.

Semantic failure: soft facilitators. When the common cause is a soft facilitator — $\theta_{\text{child}|\neg C} > 0$, so children can still succeed when C is absent, just less reliably (favorable market conditions, a supportive environment, an enabling but non-essential resource) — the AND-prerequisite construction mathematically forces $P(\text{sub-plan} \mid \neg C) = 0$, which strictly understates actual success probability. This is a real failure mode even when the topology admits factoring-above. The natural repair at $O(1)$ cost per soft-facilitator node is the **mixture form (L1')**:

$$\hat{P}_{\Sigma}^{L1'} = \theta_C \cdot P_{\Sigma}(G \mid C) + (1 - \theta_C) \cdot P_{\Sigma}(G \mid \neg C) \tag{.8}$$

where the two conditional sub-plans may share structure but carry separate edge credences $p_{ij|C}$ and $p_{ij|\neg C}$. The parametric cost per soft-facilitator-affected edge doubles; the propagation remains polynomial. This is the gap between L1 (strict prerequisites) and L2 (arbitrary joint): L1' handles soft facilitators without the $O(2^k)$ exponential blowup of L2's explicit conditioning. Sector-condition verification for L1' is derived as **Proposition B.7** in N *when C is observable per trial*: under Beta-Bernoulli updating, componentwise updates per conditional branch, and facilitator monotonicity ($P_{\Sigma}(G_{|C}) \geq P_{\Sigma}(G_{|\neg C})$), the sector condition holds globally with five-way gating

$$\alpha_{L1'} = \min\left(\tfrac{1}{n_C + 1},\ \min_{j \in \mathcal{J}_C} \tfrac{\theta_C \pi_{j|C}}{n_{j|C} + 1},\ \min_{j \in \mathcal{J}_{\neg C}} \tfrac{(1-\theta_C)\pi_{j|\neg C}}{n_{j|\neg C} + 1} \right) \tag{.9}$$

(condition testing; evidence starvation and exploration gating on each of the $C = 1$ and $C = 0$ branches). The B.5b transfer to plan value is lossless by facilitator monotonicity plus AND/OR monotonicity on each conditional sub-DAG. In the strict-prerequisite limit $\theta_{j|\neg C} \to 0$, B.7 reduces to B.6's three-way gating. **When C is not observable on a single observation channel**, the mixture is refuted as a no-go: the per-trial Fisher information matrix on (θ_C, $p_{j|C}$, $p_{j|\neg C}$) factors as rank 1 around the indeterminacy manifold $\{\hat{\theta}_C \hat{p}_{|C} + (1 - \hat{\theta}_C)\hat{p}_{|\neg C} = \mu_j\}$, so by the Cramér-Rao bound no unbiased online estimator on the joint conditional vector admits $\alpha > 0$. Repair routes when C is unobservable: (i) augment C-observability (recovers B.7); (ii) run $K \geq 2$ children jointly under the same C-realization so the joint Fisher matrix reaches rank $2K + 1$; or (iii) fall back to plan-level tracking on the marginal $\hat{\mu}_j$ (recovers B.1 at the scalar marginal, losing the per-conditional decomposition). See X for the meta-pattern shared with the on-policy detection no-go in 8.2.

For strategies with many mixed or soft common causes, the agent faces a practical choice: model the most important common causes at L1 (for strict prerequisites) or L1' (for soft facilitators), accept residual L0 bias from unmodeled causes, or invest in richer propagation algorithms. See 7.3 Correlation Hierarchy for the higher-level treatment of L1 / L1' / L2.

Detecting Latent Common Causes.

The detection of causal insufficiency and interventional localization of latent common causes is treated as a standalone result in 8.2. The key connection to this worked example: the L0 residual $\Phi^{L0} - \bar{y}_G$ converges to $+\rho$ (our example: $0.877 - 0.776 = 0.101$), providing a precise, quantitative detection signal. The agent does not need to know the common cause exists *a priori* — it discovers the need for L1 from persistent structured residuals after convergence.

Epistemic Status.

Conditional on Beta-Bernoulli model. The L0 overestimation, L1 correction, and sector condition verification are exact algebra. The direction-of-bias formula ($\pm\rho$) is exact for two binary siblings; for deeper DAGs with multiple common causes, the bias involves higher-order covariance terms. The L1 construction principle (factor common cause above the correlation) is general for single common causes gating OR-siblings; the conditioning-based extension for more complex topologies is standard (variable elimination in Bayesian networks).

Appendix

Worked Example: `#worked-example-cam` *(missing)*

Segment `worked-example-cam` *not yet written.*

Claim: Coevolving automata (Miller 2022): AAT ↔ Moore machine mapping, meta-machine as $\varepsilon^*=0$ composition, simplest adaptive agent